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Out of Chaos: The Critical Role of Generalists

ABSTRACTS OF SUBMISSIONS ACCEPTED FOR PRESENTATION

INNOVATIONS IN MEDICAL EDUCATION

A COLLABORATIVE MEDICAL AND NURSING MODEL FOR TRAINING TO CARE FOR HOMELESS WOMEN. R.H. Means¹; A.A. Atasoylu². ¹Harvard Medical School, Brigham and Women's Hospital, Wellesley, MA; ²Harvard Medical School, Cambridge Hospital, Cambridge, MA. (Tracking ID #132909)

STATEMENT OF PROBLEM OR QUESTION: Training medical students and residents to care for vulnerable populations requires a creative, collaborative community-based approach. Homeless women represent an especially vulnerable population with unmet healthcare needs. Traditional hospital based training and shortened hospital stays often limit trainee understanding of the issues surrounding homelessness. Training future health care providers to recognize homelessness and gain an understanding of the specific issues affecting the homeless are essential components of any training program.

OBJECTIVES OF PROGRAM/INTERVENTION: 1. Teach medical students and residents the cultural contexts of homelessness and approaches to caring for this vulnerable population by exposing them to a community-based model of episodic medical care designed to care for homeless women within area shelters 2. Incorporate an interdisciplinary experience to emphasize the strength of collaboration. 3. Equip trainees with portable resources that will improve their care of this population.

DESCRIPTION OF PROGRAM/INTERVENTION: Women of Means (WOM) is a non-profit organization of volunteer physicians and staff nurses providing episodic medical care to homeless women in Boston area shelters. WOM physicians and nurses come together at a homeless women's shelter, to offer a unique collaborative model for training. As part of a one month community medicine elective, medical students, nursing students, nurse practitioner students and medical residents learn how to best care for this vulnerable population. Trainees work side by side offering their different perspectives while taking care of patients at the shelter. They are introduced to the specific health care needs of this population, the barriers to receiving health care and strategies to overcome these barriers. Emphasis is also placed on recognizing homelessness in the hospital setting and how to develop appropriate discharge plans through an understanding of available area resources in collaboration with hospital social workers who also receive WOM resources.

FINDINGS TO DATE: Since 1999, 40 nursing and medical students as well as medical residents each year have shared in this exciting elective. Each trainee was asked to write a self-reflection at the end of his/her elective. They all commented on what a transformative experience it was for them to see patients at the shelter. Some specific comments emphasized the importance of having such an experience in order to truly appreciate the issues and struggles affecting the lives of homeless women. Many were surprised at the resilience of the women, the importance of listening, and the need to come up with creative solutions for complex problems.

KEY LESSONS LEARNED: Training socially aware and knowledgeable providers how to care for homeless women requires a community-based approach. This setting exposes trainees to the specific health care and social issues affecting homeless women which they might not otherwise be exposed to in the hospital setting. The WOM collaborative medical and nursing model allows for a unique training experience and successfully highlights the need for a multidisciplinary collaborative approach to caring for such a vulnerable population. **MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:** Poster, Pocket card, and Resource "toolbox".

A FOUR-WEEK LANGUAGE AND CULTURAL IMMERSION ROTATION FOR RESIDENT PHYSICIANS IMPROVES SPANISH LANGUAGE SKILLS. S.B. Glick¹; L. Stevak¹. ¹John Stroger Hospital of Cook County, Chicago, IL. (Tracking ID #133314)

STATEMENT OF PROBLEM OR QUESTION: Language discordance between patient and physician creates barriers to health care. Can a four-week immersion rotation meaningfully improve residents' Spanish language skills?

OBJECTIVES OF PROGRAM/INTERVENTION: 1) To improve Spanish language fluency; 2) to appreciate the reasons for patient-physician language discordance and its impact on healthcare; and 3) to encourage culturally appropriate strategies in patient care.

DESCRIPTION OF PROGRAM/INTERVENTION: In September 2004, five PGY 1 and PGY 2 primary care residents participated in a new required four-week Spanish language and Latino culture immersion rotation. The rotation has four components: 1) language instruction; 2) cultural competence seminars; 3) clinical experiences in medically underserved settings that care for large numbers of Latino patients; and 4) evening homestays with immigrant families. Native Spanish speakers instruct residents daily, emphasizing the use of healthcare vocabulary in the medical setting. Cultural competence seminars, held two half-days each week, include Latino health beliefs and practices, the use of interpreters, assimilation and acculturation, and community concerns. Clinical experiences, including participating in classes for patients with diabetes, and performing histories and physical examinations in Spanish, occur 1-2 half-days each week in underserved, inner-city community health centers that care for large numbers of Latino patients. Immigrant families host residents in their homes three evenings each week. Residents eat dinner with the host family, participate in the family's evening activities, and speak Spanish exclusively.

FINDINGS TO DATE: Spanish language skills were evaluated before and after the rotation using the Simulated Oral Proficiency Interview (SOPI) and a written language assessment. The SOPI, developed by the Center for Applied Linguistics, is a tape-mediated, performance-based test of oral proficiency. The written language assessment requires examinees to answer questions and translate phrases. Before the rotation, all five residents tested at the novice level (three novice-low, one novice-mid and one novice-high). Novice-level speakers are able to respond to simple questions with one or two word answers. After the rotation, all five residents improved at least one sub-level (from novice-mid to novice-high and from novice-high to intermediate-low); three residents improved two sub-levels (from novice-low to novice-high).

KEY LESSONS LEARNED: The improvement in Spanish language skills seen after a four-week immersion rotation is akin to 1-2 semesters of college-level instruction.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Discussion, slides and audiotapes.

A "HANDS ON" PATIENT SAFETY CURRICULUM FOR RESIDENTS. M.L. Plews-Ogan¹; J.D. Voss¹; M. Nadkarni¹; N. May¹. ¹University of Virginia, Charlottesville, VA. (Tracking ID #134025)

STATEMENT OF PROBLEM OR QUESTION: the 2000 IOM report "To Err is Human" raised awareness of medical errors as an important cause of adverse patient outcomes and challenged the medical community to improve patient safety. Medicine has traditionally taken a "blame and shame" approach to errors, admonishing the individual involved but failing to focus on the interaction between system factors and human factors which perpetuate these errors. Leaders in patient safety call for a new approach to errors, one which understands systems and human factors in medicine and strives to build safer, more fail-proof processes to redress errors. Physicians must take a leadership role in making health care safer, but need the tools and the knowledge to do so. The ACGME has recognized this need, creating new competencies in systems based care and practice based learning to stimulate curricular changes in residency training.

OBJECTIVES OF PROGRAM/INTERVENTION: This unique educational project is designed to train physicians in the scientific underpinnings of patient safety and to familiarize them with a rigorous method of analyzing errors in medical care. They then have the opportunity to use their knowledge to analyze an error in their own practice and to design, implement and evaluate an intervention to address the causal factors identified in their analysis. Their experience takes place in a "living framework" of an active ongoing error reporting, analysis and response program in their residency clinic.

DESCRIPTION OF PROGRAM/INTERVENTION: In PGY1 trainees participate in two interactive seminars on medical error during their ambulatory curriculum. They learn the basics about human factors, cognition, systems thinking and error prevention. In PGY2 trainees participate in a seminar on error analysis. They are then assigned an actual case (reported through the voluntary reporting program of the clinic or if they choose, can analyze and error of their own). Trainees are responsible for investigating the details of the case, drawing up an event flow diagram and constructing a causal tree. They then present their case to the interdisciplinary patient safety committee of the practice for discussion and design of interventions to address the root causes of the case. In PGY3 trainees will re-visit their cases to determine the effectiveness of their interventions.

FINDINGS TO DATE: The patient safety curriculum has been included in the overall ambulatory curriculum and is in year one of implementation. Thus far, all PGY 1, 2 and 3 participants have been included in the curriculum (approximately 80 residents), case summaries will be included in their portfolios. Online evaluations and narrative interviews are currently being conducted to assess the effects that the curriculum had on attitudes toward errors, attitudes toward reporting, how competent they feel in analyzing why things went wrong and how they felt about discussing their own errors. The residents who have done a case analysis have completed a written summary of the analysis which will be included in their portfolio.

KEY LESSONS LEARNED: residents are enthusiastic about the experience, participate fully in the investigations and the analysis of errors and are able to design feasible and creative interventions to address such errors in a "living framework" of the resident clinic patient safety program. They are able to utilize knowledge about systems and human factors to analyze why things went wrong and how to address problems effectively.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: poster or oral presentation (would prefer oral presentation to demonstrate case analysis more effectively).

A MANAGED CARE MINI-ROTATION ALIGNED WITH THE ACGME CORE COMPETENCIES PRACTICE-BASED LEARNING AND SYSTEMS-BASED PRACTICE. P.Y. Watson¹; K. Scher²; B. Pai²; J.L. Musial¹; E.J. Scher¹. ¹Henry Ford Hospital, Detroit, MI; ²Health Alliance Plan, Detroit, MI. (Tracking ID #133249)

STATEMENT OF PROBLEM OR QUESTION: The Accreditation Council for Graduate Medical Education (ACGME) has required more thorough instruction in areas related to quality improvement (practice-based learning) and the complexity of medical practice (systems-based practice). In addition, many residents currently train in a managed care environment and most will eventually practice in managed care necessitating a better understanding of managed care practices.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) Develop a brief orientation to managed care for residents at the start of residency. 2) Introduce residents to managed care from two perspectives: the physician and the managed care organization. 3) To introduce the ACGME core competencies practice-based learning and systems-based practice through managed care activities.

DESCRIPTION OF PROGRAM/INTERVENTION: A two-day immersion seminar was developed for 34 internal medicine interns between the residency program and a partner managed care organization. From September to November 2004, one group (10-11 interns each) participated in the rotation each month. Short didactics delivered by managed care managerial and clinical staff intercalated with "hands-on" activities reinforced the learning objectives. Residents were introduced to: medical management, patient rights, managed care and physician practice, clinical practice guidelines, physician report cards, pay for performance, credentialing, and population management. Mixed-method evaluations were used to assess resident attitudes and performance. Residents completed several different evaluations of the rotation. Finally, all resident participants

completed a self-reflection exercise which asked them to evaluate change in their own beliefs in areas related to systems-based practice.

FINDINGS TO DATE: Residents significantly improved on pre- and post-testing (mean scores 57.7% vs. 81.3%, $p < .01$) given during the rotation. Surveys before and after the rotation showed residents had significantly positive changes in: their perceived understanding of managed care, their beliefs that managed care improves access for patients, their ability to practice in a managed care environment, their beliefs that managed care helps improve quality, and attitudes regarding managed care business priorities (all domains, $p < .01$). On average, residents rated the rotation better than "Useful" (mean 4.15 on a 5-point Likert scale). Residents consistently favored "hands-on" activities over didactics. Residents especially found activities related to physician credentialing, patient grievance, and quality management "Useful" to "Very Useful" (mean 4.4 on a 5-point Likert scale). Resident responses to self-reflective questions indicated that they found the course useful and will make specific changes to their own practice. Many of the exercises completed by the residents are maintained as part of their portfolio.

KEY LESSONS LEARNED: Both residency programs and managed care organizations can gain from better resident education in managed care, practice-based learning and systems-based practice. Collaboration between training programs and managed care organizations can result in a successful educational intervention with dramatic changes in knowledge, attitudes, and perhaps behavior with respect to managed care.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Course outline, establishment of learning objectives, ACGME based activities, survey instruments, self-reflective exercise.

A NEW WARD-BASED PROGRAM TO TEACH MEDICAL STUDENTS END-OF-LIFE CARE COMMUNICATION AND ASSESSMENT. M.S. Ellman¹; J.R. Rosenbaum¹; M.J. Bia¹. ¹Yale University, New Haven, CT. (Tracking ID #132834)

STATEMENT OF PROBLEM OR QUESTION: Despite end-of-life (EOL) care education in the pre-clinical curricula of many medical schools, graduating medical students and housestaff often do not feel well prepared to care for patients at EOL.

OBJECTIVES OF PROGRAM/INTERVENTION: Utilizing information gathered from a needs assessment of our graduating medical students in 2004, we designed and piloted a novel ward-based EOL care education program for 3rd and 4th year medical students. Our primary objective was to assess the feasibility and effectiveness of a program to improve student level of preparedness for communication and clinical assessment of patients at EOL. In addition, we aimed to facilitate student self-reflection about caring for patients at EOL to increase student comfort with EOL clinical care.

DESCRIPTION OF PROGRAM/INTERVENTION: We developed and piloted an inpatient clerkship-based EOL educational experience that was designed to increase student awareness and ability to communicate about EOL issues. EOL care faculty partnered with faculty in psychiatry consult service and surgery to implement the program on these clerkships. With the assistance of their clerkship attending, students identified a patient facing EOL concerns. The students were directed to address various EOL issues requiring them to review the medical record and interview the patient, members of the medical and nursing team, and the patient's family when appropriate. Students were encouraged to initiate dialogue with patients about specific EOL issues in an open-ended, patient-centered manner. Their assessment included consideration of symptom management, sources of and response to suffering, and EOL care planning. Students prepared a 1-2 page report of their findings, including self-reflection of their experience and challenges encountered. At the end of each monthly rotation, students shared their findings at a faculty facilitated conference. Student presentations emphasized two key EOL issues or challenges in their case as well as personal self-reflection on their experience.

FINDINGS TO DATE: To date, 30 students have participated in the program. By the end of this academic year in June, we anticipate a total of about 60 students. Initial student feedback indicates that the program provides a challenging and meaningful educational experience. Citing previous exposures to patients where EOL issues were present but not addressed, students especially valued the expectation and guidance to communicate with their patients about EOL concerns and discuss this experience with other students and faculty. By integrating this program into ward-based patient care activities, students appreciated the relevance of improving their skills in EOL communication and care. Additional program evaluation will include assessment of student self-reported preparedness in EOL care at the time of graduation and comparison with previous graduates and non-participating students.

KEY LESSONS LEARNED: Integrating a new EOL communication and clinical assessment educational program in the acute inpatient setting is feasible and well received by participating students. A model of collaboration between faculty on an EOL core team and faculty in surgery and psychiatry resulted in effective integration of EOL clinical skill education for students on selected ward rotations.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Slide presentation or poster.

A NOVEL YEAR-LONG LEARNER-CENTERED TUTORIAL FOR THIRD-YEAR MEDICAL STUDENTS. P.A. Cohen¹; D.A. Hirsh¹; B.R. Ogur¹; D.M. Elvin¹; A.M. Antman¹; D.H. Bor¹. ¹Harvard Medical School and Cambridge Health Alliance, Cambridge, MA. (Tracking ID #136145)

STATEMENT OF PROBLEM OR QUESTION: Traditionally, third-year medical school teaching does not: 1) progress developmentally throughout the year; 2)

use the clerks' clinical experiences to teach core material; nor 3) integrate with the basic sciences.

OBJECTIVES OF PROGRAM/INTERVENTION: The tutorial was designed to complement a novel, year-long clerkship for third-year medical students: 1) to teach clinical reasoning in a developmental sequence; 2) to integrate learning amongst all required specialties; and 3) to use the clerks' clinical experiences to teach a core curriculum which integrates clinical practice with the basic sciences.

DESCRIPTION OF PROGRAM/INTERVENTION: Tutorials meet weekly for 44 weeks, last about five hours, and are attended by a faculty coordinator, all eight clerkship students, and additional selected faculty determined by topic. Tutorials are designed to address predetermined core topics and learning objectives which were developed from a literature review of national clerkship curriculum guidelines. These topics and objectives integrate all required specialties. Each tutorial focuses on a case chosen from amongst the clerks' clinical experiences. The clerk responsible for that selected patient then leads the case discussion with guidance from the faculty coordinator. Each faculty coordinator then selects the best resources, from among clinical specialists, basic scientists, radiologists, pathologists and patient simulator exercises, to ensure coverage of all learning objectives. All students must complete required assignments (e.g.: readings, web-based case discussions) prior to each tutorial. Thus, each tutorial combines informed student-led case discussions, clinical and basic sciences, and multi-disciplinary perspectives. As an example, the alcoholism tutorial is taught jointly by an internist, a psychiatrist and a basic scientist, focuses on a clerk's patient, and employs a patient simulator.

FINDINGS TO DATE: We have completed 21 tutorials of the clerkship's first year. To date, over 50 faculty members have taught representing all required third-year specialties as well as emergency medicine and pathology. Preliminary results from student evaluations demonstrate that this is a highly valued learning method (100% of tutorial sessions were recommended for next year's curriculum; the majority received a score of 5 out of 5 indicating "most strongly agree"). Faculty feedback also demonstrates high levels of satisfaction. The primary recommendation on the part of both students and faculty was "more time for discussion." Complete faculty and student evaluation of the didactics will be available for presentation.

KEY LESSONS LEARNED: Provided with sufficient clinical experiences and curricular structure, third-year students can learn core curricular material in a developmental fashion from case-based discussions of their own patients. The tutorial format is an excellent venue to integrate required disciplines and the basic sciences within the third-year of medical school training.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster or talk outlining details of the educational intervention.

A PUBLIC HEALTH CURRICULUM FOR MEDICAL RESIDENTS. L. Hu¹; V. Stone¹. ¹Massachusetts General Hospital, Boston, MA. (Tracking ID #13266)

STATEMENT OF PROBLEM OR QUESTION: Medical education programs, including primary care residency programs, can benefit from integrating public health principles in their effort to prepare residents to effectively care for diverse patients in the community and promote the health of these populations.

OBJECTIVES OF PROGRAM/INTERVENTION: We designed an innovative public health curriculum that integrated with an already existing primary care residency program. The objectives of the curriculum were: a) To improve residents' understanding of public health concepts and programs. b) To provide an opportunity for residents to look critically at the current health care delivery system, focusing on access to care and quality of care issues. c) To create a forum to discuss current health policy issues, health care topics in the news, and to explore avenues for physician activism.

DESCRIPTION OF PROGRAM/INTERVENTION: The public health curriculum spanned 3 years and was integrated into the ambulatory block of the primary care residency program. Each seminar was 2 hours long, held one day a week over one month, and was attended by first, second, and third year residents. The curriculum included didactic lectures, workshops, panel discussions, and a residency retreat highlighting a public health issue (physician activism). Topics that were covered in the curriculum included: 1) Introduction to medicare, medicaid and managed care systems- who pays, what's covered, medical economics; 2) Uninsured and underinsured- who and why and the health implications; 3) Occupational and environmental health- case discussion focusing on issues relevant to the primary care physician; 4) Violence- demographics of victims and perpetrators of intimate violence and how violence affects our patients' behaviors and health; 5) Newsworthy health topics- discussion of health care topics in the news in a small group setting; 6) Pharmaceutical industry- their influence on physicians and how they market their products; 7) Introduction to health law- didactic on medical-legal issues and discussion on patient's rights, malpractice suits; 8) Physician advocacy and activism- cases and discussion led by local and national physician advocates.

FINDINGS TO DATE: Residents were surveyed at the conclusion of each seminar and asked to rate the relevance and quality of the content of each session on a 5-point likert scale. Overwhelmingly, residents rated 4 and 5 (5=excellent) on the seminars on physician activism, U.S. healthcare insurance, and current healthcare issues in the news. Comments written-in by residents included: "we need more of this in residency", "outstanding, for the first time I understand medicaid/medicare", "each session opened my understanding of public health... this was great". In feedback meetings, residents confirmed the importance of the public health curriculum in their medical education and expressed interest in more sessions, particularly on physician activism.

KEY LESSONS LEARNED: A public health curriculum is perceived by many medical residents to be lacking in their medical residency education. Residents were very receptive to the public health curriculum and see these topics to be an

important component of their medical education. Further evaluation to assess residents' knowledge in public health concepts and change in attitude toward their patients after taking the course will be performed this spring and presented.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:

A RESIDENT-INITIATED GROUP INTERVENTION FOR OBESE WOMEN FOSTERS RESIDENT LEARNING AND PATIENT SATISFACTION. M. Grotz-Rhone¹; K. Ouchida¹; C. Boutin-Foster¹; J. Weiser¹. ¹Cornell University Medical College, New York, NY. (Tracking ID #136172)

STATEMENT OF PROBLEM OR QUESTION: Can primary care residents develop a group intervention for obese women in an academic resident practice in order to promote resident learning and encourage risk reduction in their patients?

OBJECTIVES OF PROGRAM/INTERVENTION: To develop, implement, and evaluate a resident-initiated group education program for obese women.

DESCRIPTION OF PROGRAM/INTERVENTION: Two primary care residents developed this intervention as their independent project. Prior to implementing this program the residents reviewed relevant literature on management of obesity, identified faculty advisors, developed a curriculum with appropriate materials, and fulfilled the necessary administrative requirements. Ambulatory practice physicians were educated about the program and encouraged to refer their non-diabetic female patients with a BMI of >30. The program was composed of six weekly sessions. Each session lasted two hours and had the following format: 1) brief didactic components dedicated to nutrition and physical activity, 2) self-reflection exercises during which participants talked about personal and environmental barriers to nutrition and physical activity, 3) group activities during which participants brainstormed solutions and shared experiences, 4) homework assignments such as food diaries and pedometer logs, and 5) stress reduction and relaxation exercises that participants were encouraged to practice at home. In order to continually assess and improve the program, written feedback was solicited from participants at the end of each session and a pre-post test survey of nutrition and physical activity knowledge was administered.

FINDINGS TO DATE: There were a total of six participants in the program, the majority of which attended at least four classes. In verbal and written feedback, participants expressed satisfaction with the program. They demonstrated understanding of key concepts and shared ways in which they felt more capable of improving their nutrition and lifestyle habits. The participants also consistently expressed enthusiasm for the group setting. The process of running a group provided resident facilitators with new insights into the challenges faced by their obese patients because of variations in knowledge, motivation to change, social support and cultural experiences. The residents gained skills in researching an evolving field of obesity management, developing a curriculum, leading groups, and addressing the administrative hurdles associated with a starting a new program.

KEY LESSONS LEARNED: It is feasible for motivated primary care residents to implement a group education intervention in an academic ambulatory practice that promotes resident learning and patient risk reduction. The interaction between patients and physicians in a setting outside of the traditional office environment improved residents' satisfaction with their provision of care and fulfilled patients' interest in additional nutrition and lifestyle interventions in a primary care ambulatory practice.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Modalities will include 1) poster, 2) binder containing the program curriculum and handouts 3) examples of participant feedback

A SOLUTION TO "DISCONTINUITY" IN A RESIDENT CONTINUITY PRACTICE. L. Yanni¹; B. Johnson¹; S.A. Call¹. ¹Virginia Commonwealth University, Richmond, VA. (Tracking ID #132204)

STATEMENT OF PROBLEM OR QUESTION: Timely administrative follow-up is challenging in a resident continuity practice. Traditionally, administrative follow-up (communicating diagnostic test results, responding to phone calls, and completing forms) has been the sole responsibility of the primary resident physician. Theoretically, this reinforces continuity as well as upholds professional responsibility in managing a panel of patients. However, barriers to timely completion include call schedules, duty hours, and in some instances, multiple training sites. Consequences of poor follow-up include compromised quality of care, increased burden on the clinic staff and system, and follow-up that subsequently falls to teaching faculty.

OBJECTIVES OF PROGRAM/INTERVENTION: The POD system was designed to promote teaching, reinforcement, and measurement of resident achievement in the ACGME Competencies. The objectives are: To provide residents with a "system" experience that promotes improved patient care delivery and outcomes (Systems-Based Practice); To allow the POD administrator time, without the demands of scheduled patients, to formulate and answer clinical questions on ambulatory issues (Practice-Based Learning); To encourage resident-patient communication using follow-up letters and telephone calls as well as resident-PODmate communication using the medical information system, POD listserv, and text-page system (Interpersonal and Communication Skills); To assess resident documentation (Professionalism) and achievement of standard chronic disease outcome measures (Patient Care) through peer chart review.

DESCRIPTION OF PROGRAM/INTERVENTION: Each POD is a partnership of 6-7 residents who assist each other in administrative and patient care duties in the continuity setting. This system is designed to support residents when physical and timely access to clinic is not feasible. Each month there is a designated resident POD administrator. While each POD member is encouraged to complete

administrative follow-up on their own continuity patients, the monthly POD administrator is the back-up for managing the flow of paperwork. Each POD is assigned an attending, a POD mailbox, and a listserv address for ease of communication. The POD attending and PODmates evaluate the POD administrator on relevant competencies. The POD administrator also performs peer chart review using a quality assessment tool. Resident-created POD guidelines reinforce administrative responsibility in a continuity practice as well as standard ambulatory care practice guidelines.

FINDINGS TO DATE: Observations indicate that residents have assumed ownership of the POD system and are setting even higher standards for the practice of ambulatory medicine. For example, they are reinforcing to their peers the importance of accurate medication lists, problem lists, and health maintenance summaries in order to improve administrative efficiency. Paradoxically, residents are now more aware of outcomes on their continuity patients through peer communication. A POD survey is being administered to assess the effectiveness of the system in reinforcing competencies and improving patient care outcomes. Additionally, the perceived value of a system approach to follow-up is being measured.

KEY LESSONS LEARNED: There are effective generalizable solutions to discontinuity in a resident continuity teaching practice. Resident involvement in system development can enhance patient care and their education. The ACGME Competencies must be considered when designing any new educational innovation.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: This abstract may be presented in poster format.

AN EASY AND INNOVATIVE HOME VISIT PROGRAM FOR HOUSESTAFF. A.M. Molnar¹; R. Cunningham²; J. Sillman¹. ¹Brigham and Women's Hospital, Boston, MA; ²Partners Healthcare, Boston, MA. (Tracking ID #136086)

STATEMENT OF PROBLEM OR QUESTION: Residents have little time during busy inpatient and outpatient rotations to familiarize themselves with the social context and home environment of their patients.

OBJECTIVES OF PROGRAM/INTERVENTION: The program is designed to help residents: · Attain competency in home visits. · Improve connection with patients. Identify social factors that impact illness and health. Gain familiarity with community resources available to patients outside the hospital setting. · Improve resident satisfaction with ambulatory patient care. · Increase likelihood of repeat home visits as a physician.

DESCRIPTION OF PROGRAM/INTERVENTION: During each outpatient block, primary care residents in their 2nd and 3rd years are asked to identify 2 of their continuity patients who might benefit from a home visit. Each resident makes home visits to 2 selected continuity patients during each ambulatory block. Visits are conducted in pairs, with a co-resident "partner." · A home visit didactic session is given to residents prior to the first set of home visits. · At the start of each ambulatory rotation, residents discuss with their continuity clinic preceptors potential patients who might benefit from a home visit and goals for each visit. · On the day of the visit, the home visit partner is given a worksheet designed to prompt a discussion of specific goals for the visit and ways to achieve those goals. · During the visit, the partner takes notes while the primary resident interviews and examines the patient, discusses health with family members, and explores the home environment. · Feedback is given by the partner during return travel from each home visit. · A faculty "Doc of the Day" serves as a consultant for help with any questions that arise during the visits and is paged after visits are completed to ensure that residents have returned safely. · The home visit is documented and discussed with the preceptor at the next continuity clinic session.

FINDINGS TO DATE: Participants rated the home visit experience as highly valuable (N=16). The improved connection with patients was the most highly rated portion of the home visit experience (1.3 on a Likert Scale of 1 to 5, 1 being most positive).

KEY LESSONS LEARNED: A home visit curriculum can be easily integrated into housestaff curriculum. The curriculum is very popular with housestaff who rate the program highly on a number of indicators.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster

AN EVIDENCE-BASED MEDICINE SEMINAR FOR MEDICAL RESIDENTS. C. Wong¹; M.F. Hollon¹; G. Terasaki¹; S. Dodson¹; A. Ryce¹. ¹University of Washington, Seattle, WA. (Tracking ID #135257)

STATEMENT OF PROBLEM OR QUESTION: Is a structured curriculum that emphasizes "hands-on" development of evidence-based medicine skills useful in the training of medical residents?

OBJECTIVES OF PROGRAM/INTERVENTION: The practice of Evidence-Based Medicine is an essential skill for physicians, and a requirement of the American Board of Internal Medicine. The objective of this curriculum is to teach medical residents evidence-based medicine using an approach that emphasizes collaborative, real-time practice while minimizing participants' external time commitment.

DESCRIPTION OF PROGRAM/INTERVENTION: First-year internal medicine residents participate in a weekly, two-hour seminar for 8 weeks. Faculty, chief residents, and health sciences librarians teach principles of evidence-based medicine using interactive lecture and practical skills sessions. Sessions are themed in four-week cycles of "Introduction to Evidence-Based Medicine", "Therapy", "Prognosis", and "Diagnosis". Participants develop potential questions based on their own clinical practice and as a group select and formulate an appropriate evidence-based medicine question that they then research collaboratively during each session using appropriate databases. Participants appraise their findings together and summarize the results as a Clinically Appraised Topic (CAT), which is posted on the course website. Brief pre-tests and post-tests are used to evaluate self-assessed comfort level with the material.

FINDINGS TO DATE: In the first four months of this program, residents who stated that they were comfortable teaching evidence-based medicine to their team increased from 20% before the course to 86% afterward. The percent who could successfully calculate a number needed to treat (NNT) increased from 40% to 86%. Those who felt comfortable using medical subject headings (MeSH) to search PubMed increased from 20% to 93%. Qualitative comments from participants indicated that the focus on both literature searches and critical appraisal was beneficial. Participants valued that the learning took place within the structured course and did not conflict with other time demands. Participants were enthused to use databases of which they were not previously aware, and, furthermore, found the presence of health sciences librarians valuable.

KEY LESSONS LEARNED: A 2 hour per week seminar over 8 weeks with an emphasis on the time-efficient, collaborative development of practical skills is a useful, dynamic model for teaching evidence-based medicine to medical residents. Key lessons learned included the following: (1) By using a collaborative approach rather than individual presentations, participants receive immediate feedback from each other and from facilitators in their evidence-based medicine skills; (2) Researching questions in "real time" more closely approximates clinical practice and minimizing outside time was valued by residents; and (3) The synergy between clinical faculty and health sciences librarians improved the efficiency and quality of the literature search.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Presentation or poster with laptop computer demonstration of web site and lecture sessions.

AN INNOVATIVE USE OF NIGHT FLOAT ROTATION TO ENHANCE EVALUATION OF ACGME CORE COMPETENCIES. J. Mariotti¹; W. Iobst¹; E.J. Gertner¹. ¹Lehigh Valley Hospital, Allentown, PA. (Tracking ID #134806)

STATEMENT OF PROBLEM OR QUESTION: The institution of a night float system into an Internal Medicine Residency creates a paradox for training programs. While the night float provides service and assists in compliance with work hour restrictions, creating a structured educational component is difficult. Further, evaluating residents' performance on the night float rotation can be challenging, especially within the structures of ACGME core competencies.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) Create a learner-generated Night Float Curriculum using Chart Stimulated Recall that will enable consistent evaluation of residents in the following core competencies: Direct Patient Care, Practice-Based Learning and Improvement, and Systems-Based Practice. 2) Provide immediate feedback to residents on Night Float Rotation to not only improve skills in patient care, but also to enhance the teaching of evidence-based medicine.

DESCRIPTION OF PROGRAM/INTERVENTION: The evaluation process was accomplished through two main interventions. First, a chart stimulated review exercise was performed with PGY-1 residents utilizing a random history and physical performed during their rotation. This document was presented at a small group morning report, and evaluated and scored using a checklist focusing on internal consistency and quality. This history and physical, along with the checklist, was subsequently placed into the resident's file. Second, an evidence-based medicine presentation was required of the senior resident. This presentation was based on a clinical question encountered during the night float rotation and included: (1) the clinical question searched, (2) the search engines utilized, (3) the evidence-based answer to the question, and (4) one reference article identified during the search. The completed presentations were compiled at the end of the year to create a Night Float reading list for the next year's night float residents.

FINDINGS TO DATE: Since the inception of the Night Float curriculum, the evaluation of residents working overnight has become much stronger and consistent. The review of charting using a checklist has facilitated measurable improvement in resident communication and documentation, and the evidence-based medicine presentations have promoted better clinical management and practice improvement. In addition, these modalities enable continued compliance with ACGME core competency evaluation.

KEY LESSONS LEARNED: Night float has traditionally been a service rotation, but it cannot continue as such given current teaching and work hour requirements. By establishing a structured curriculum through chart stimulated recall, and developing it within the framework of the ACGME core competencies, we have changed night float into an innovative and valuable rotation for both teaching and evaluation.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: The compilation of evidence based medicine presentations will be available to demonstrate the breadth of topics reviewed and benefit of this intervention. In addition, the checklist for evaluation of the history and physical documentation and the results of the review process will be available to provide a better understanding of this innovation.

CORE CLINICAL SKILLS IN RESIDENT EDUCATION: TEACHING AND ASSESSING ACGME COMPETENCIES USING PALLIATIVE AND END-OF-LIFE CARE MODULES. C.F. Gracey¹; T.E. Quill¹; A.J. Caprio¹; T. Caprio¹; S. Clemensen¹; J. Greenlaw¹; E. Naumburg¹. ¹University of Rochester, Rochester, NY. (Tracking ID #134528)

STATEMENT OF PROBLEM OR QUESTION: Many physicians feel unprepared to provide competent care to dying patients or to participate skillfully in end of

life decisions. The communication and relationship building skills involved in palliative and end-of-life care are also critical in a therapeutic relationship. Specific problem areas for many clinicians include communicating bad news, lack of knowledge about how to deal with family conflict, how to increase patient involvement in treatment decisions, lack of knowledge about pain control, and lack of skill discussing adverse clinical outcome.

OBJECTIVES OF PROGRAM/INTERVENTION: The goal of the project is to teach paradigmatic clinical skills to residents through the use of six teaching and assessment modules. The modules are relevant for all residents, regardless of clinical specialty. Each module teaches skills and provides objective assessment of these skills. The topics relate not only to palliative and end-of life care but also to the core clinical competencies, especially interpersonal and communication skills and professionalism. The modules are communicating bad news, assessing pain and formulating treatment, obtaining informed consent, addressing family conflict around cardiopulmonary resuscitation, increasing patient involvement in treatment, and explaining an adverse clinical outcome.

DESCRIPTION OF PROGRAM/INTERVENTION: Each module has four components: 1) Independent Self Study and Pre-test. Depending on the module, this may include critical reading, observation of a videotape, or completing a computer stimulation. 2) Skill Building Conference. A brief review of the critical skills is followed by a practice session using role-play or standardized patient simulation. 3) 360 Degree Evaluation. Feedback/evaluation is obtained from independent sources observing the resident's clinical interaction (patient or family member, supervising or peer physician, social worker, nurse) 4) Standardized Patient Evaluation and Post-test. A standardized patient, trained to assess the resident's skills, provides immediate direct feedback to the learner.

FINDINGS TO DATE: Since offering the initial 3 modules of the program at the University of Rochester, residency programs across disciplines (internal medicine, family medicine, pediatrics, surgery, obstetrics/gynecology) as well as fellowships (oncology, radiation therapy, vascular surgery, neurosurgery) have requested to be involved in the program. We are currently presenting the first 3 modules this academic year, and the next three will be introduced next year. We will have pretest data available on all participants for these three modules, and information from the 360 degree evaluation and the standardized patient evaluation will be available by the end of the academic year.

KEY LESSONS LEARNED: The ACGME requirements for training in communication, relationship and professionalism skills provide incentive to programs to have their residents learn core clinical skills. When these core skills are packaged in a self-contained format that fits into busy schedules, and when they are presented in an interactive format focused on core domains that cross specialties, there is potential to reach a wide range of residents across disciplines who ordinarily might not receive training in such areas.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: poster; computer to show portions of the curriculum, including video of simulated patient interaction, demonstration role-plays, readings, powerpoint presentations, evaluation material, standardized patient scenarios.

CREATING A DERMATOLOGY CONFERENCE FOR GIM FACULTY USING CASES FROM AN URGENT CARE CLINIC. R.V. Liddicoat¹; E. Gonzalez¹. ¹Massachusetts General Hospital, Boston, MA. (Tracking ID #132918)

STATEMENT OF PROBLEM OR QUESTION: General Internists working in Urgent Care settings frequently treat Dermatologic conditions. Internal Medicine training in Dermatology is often inadequate.

OBJECTIVES OF PROGRAM/INTERVENTION: A Dermatology conference for GIM faculty was developed to 1. Improve the clinical Dermatology skills of GIM faculty working in an Urgent Care setting. 2. Give feedback to GIM faculty on cases seen in the Urgent Care Clinic.

DESCRIPTION OF PROGRAM/INTERVENTION: We created a case-based Dermatology conference using clinical cases from the Urgent Care Clinic of an academic medical center. A digital camera was used by clinicians in the Urgent Care to photograph cases. These images were presented at the conference, along with a brief clinical presentation that included history as well as diagnosis and treatment offered to the patient at the time they were seen in Urgent Care. The conference was led by a Dermatologist and dermatology fellows for second opinion and was attended by GIM faculty from the Urgent Care Clinic. The images and discussion from the conference were then posted in the Urgent Care Clinic for the benefit of GIM faculty who were unable to attend the conference. The conference discussion and images were also emailed to the GIM faculty member who saw the patient in Urgent Care.

FINDINGS TO DATE: Six conferences were held over a 6-month period and 34 cases were presented. Between five and six cases were presented at each conference. The most common diagnosis was eczema (5 cases) followed by contact dermatitis (3 cases) and atopic dermatitis (3 cases). Two cases each of Furunculosis, Erythema Multiforme, hypersensitivity reaction and fungal infection were seen. In 7/34 cases (21%), the diagnosis made at the conference was different than that made in the Urgent Care Clinic. In 9/29 cases (28%), the treatment recommended at the conference was substantially different than that given at the Urgent Care Clinic.

KEY LESSONS LEARNED: A case-based Dermatology conference utilizing store and forward images, in a telemedicine medium for distance learning and service, can be designed around Dermatology cases collected from an Urgent Care Clinic. GIM faculty are willing to collect cases for such a conference and appreciate receiving feedback on their diagnoses and treatments. One-fifth to one-quarter of cases seen at the Urgent Care would have been diagnosed and/or treated differently by the conference Dermatologist, indicating a need for additional training in Dermatology among GIM faculty.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:

CREATING LEADERS: FOSTERING MEDICAL STUDENT INVOLVEMENT IN COMMUNITY HEALTH. J.W. Fisher¹; Q. Doan¹; K. Moseley¹; N. Saksena¹; M. Bate²; K. Peek². ¹Baylor College of Medicine, Houston, TX; ²Texas Woman's University, Houston, TX. (Tracking ID #136214)

STATEMENT OF PROBLEM OR QUESTION: Although physicians often are called upon to participate in community activities such as non-profit board membership and health education presentations to local community organizations, most medical school curricula do not emphasize the skills needed to facilitate responsible community involvement for our future physician leaders.

OBJECTIVES OF PROGRAM/INTERVENTION: (1) Medical students will demonstrate important leadership skills – focusing on community involvement and public speaking. (2) Medical students will design and implement an after-school health curriculum (to be used by future medical students teaching at the high school level). (3) Faculty will develop, evaluate, and refine a curriculum for teaching students advocacy skills – in particular how to provide community service through health presentations.

DESCRIPTION OF PROGRAM/INTERVENTION: The Baylor Longitudinal Ambulatory Care Experience (LACE) is a required third year medical school course that has been developed to increase physician trainees' awareness and abilities to manage the diverse health care needs of patients within the context of a complex health care system. Up to 12 students may choose the specialized Women's Health Pathway (WHP) as their LACE course. One of the unique features of the WHP is its focus on community involvement. Medical students attend workshops to learn key presentation skills and basic curriculum design. Over the past year, WHP students have been developing and implementing an after-school health curriculum for female students at a local high school in an underserved area. Two to three medical students spend 2 hours on Thursday afternoons leading activities designed to improve health attitudes and behaviors among approximately 25-30 high school students. The sessions are designed to be highly interactive to promote attendance and participation. The medical student-designed health curriculum is reviewed, modified, & then duplicated for distribution to current and future medical student instructors. Pre-test and post-tests are used to assess changes in knowledge and attitudes of high school students. Medical students' knowledge, beliefs, and skills are evaluated qualitatively and quantitatively.

FINDINGS TO DATE: Both qualitative and quantitative assessments indicate that (1) medical student and high school student enthusiasm for the program is high; (2) medical students gain confidence in their ability to present publicly; (3) participating high school students demonstrate an improvement in knowledge about important health issues.

KEY LESSONS LEARNED: To ensure the sustainability of this community program, the after-school high school education sessions must be built into the medical student's schedule. In addition to an initial needs-assessment of the high school students, ongoing dialogue between the high school and medical students is essential to providing important health information in an appropriate and stimulating manner. This community-oriented program is a model for student advocacy that can be replicated at relatively low cost and high yield at other medical schools.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster, handouts with sample "health education modules," computer based powerpoint presentation.

CREATION AND IMPLEMENTATION OF AN EDUCATIONAL CURRICULUM FOR TEACHING MEDICAL DISORDERS IN PREGNANCY TO MEDICAL RESIDENTS. P.S. Gibson¹. ¹University of Calgary, Calgary, Alberta. (Tracking ID #136009)

STATEMENT OF PROBLEM OR QUESTION: Upon completion of residency training, Canadian internal medicine trainees are expected to demonstrate "expertise in the management of medical complications of pregnancy", as outlined in the Objectives of Training and Specialty Training Requirements in Internal Medicine by the Royal College of Physicians and Surgeons of Canada. This requirement is to be achieved despite the lack of a formal or standardized framework for training in this domain as well as infrequent and regionally variable clinical experience in caring for pregnant women with medical disorders during residency training.

OBJECTIVES OF PROGRAM/INTERVENTION: The purpose of this project was to: evaluate the knowledge base of senior internal medicine residents with respect to medical disorders in pregnancy (obstetric medicine); create a curriculum of case-based educational modules addressing the major topics in obstetric medicine; evaluate the product in "real time" through involvement in the residents' academic half-day program at our university site; and assess the educational benefit to the trainees involved.

DESCRIPTION OF PROGRAM/INTERVENTION: A group of Canadian obstetric internists (internal medicine physicians with special interest in pregnancy-related medical disorders) convened in 2002 and identified important (core) obstetric medicine topics. Modules incorporating educational objectives, clinical case scenarios with evidence-based responses, "take-home points" and suggested reading sources were formulated for each topic. Slideshows integrating the clinical cases and reference material were also created to facilitate multimedia stimulation of the learners. The educational tool was subsequently implemented and evaluated during a Maternal Health Seminar Series during residents' academic half-day for second and third-year internal medicine residents at the University of Calgary - initially from September 2002 to April 2003 (8 sessions, 24 hours total) and currently from October 2004 until April 2005. Baseline knowledge was assessed via a 40-question multiple-choice pre-course examination. Upon completion of the seminar series a knowledge examination was repeated and learner satisfaction was assessed via an anonymous survey. **FINDINGS TO DATE:** Prior to participating in the Maternal Health course, the senior (second and third year) medical residents at the University of Calgary

achieved a mean score of 56.1% (in 2002) and 55.5% (in 2004) on the baseline knowledge examination. Following the 8 monthly 3-hour sessions the trainees completed a repeat evaluation in 2003, achieving a mean score of 74.5% (improvement of 18.4%, $p < 0.05$ versus pre-test). The trainees expressed a high level of satisfaction with the organization, style and content of the curriculum. **KEY LESSONS LEARNED:** Our evaluation to date demonstrates that Medical Disorders in Pregnancy can be taught effectively to internal medicine residents, with a high level of learner satisfaction, via a curriculum of educational modules utilizing case-based, multimedia-enabled, small group sessions. This valuable educational tool is now being made available for wider educational use and evaluation via download from our internet site, www.maternalhealth.ca. **MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:** Sample modules will be available in hard copy at the presentation. The multimedia aspects will be demonstrated including viewing of the educational slideshows, as well as real-time website viewing and navigation.

CULTIVATING THE MEDICAL EDUCATORS OF TOMORROW: A MEDICAL EDUCATION ELECTIVE AND PROGRAM FOR MEDICAL STUDENTS. H. Chen¹; D. Tuttle¹; E.M. Aagaard¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #134685)

STATEMENT OF PROBLEM OR QUESTION: Students contribute significantly to the teaching endeavor at the University of California, San Francisco School of Medicine. The Medical Education Elective, under the auspices of the Medical Education Area of Concentration (Med Ed AoC), was developed to nurture this interest and advance student teaching skills.

OBJECTIVES OF PROGRAM/INTERVENTION: The Medical Education Elective is the first course in a series of planned electives. The specific goals of the elective are to: 1) stimulate 4th year medical student interest in the field of medical education and 2) provide support for and advance the teaching skills of students who participate in medical student teaching.

DESCRIPTION OF PROGRAM/INTERVENTION: A 4-week elective course was offered in the fall of 4th year comprising of: 1) interactive sessions to increase knowledge of teaching skills and 2) teaching laboratories to practice the skills. Coursework involved lectures, small group activities, and independent learning exercises, covering learning styles, lecture and small group teaching, and group dynamics. Students were given a pre- and post-course inventory with 4 teaching scenarios (problem based learning, feedback in the small group setting, small group dynamics, and large group lecture) to assess their confidence in their teaching skills, using a 5-point Likert scale (1 = not at all confident, 3 = somewhat confident, 5 = very confident). At the completion of the course, students were asked to rate their acquisition of skills (1 = strongly disagree, 5 = strongly agree) and the value of the sessions (1 = not at all valuable, 5 = highly valuable). Results are reported as a mean $\pm$ standard deviation (SD) and percent change in mean scores.

FINDINGS TO DATE: In 2004, 10 students enrolled in the elective. All expressed an intent to teach medical students or residents in their future careers and all had prior teaching experience. Student confidence in their teaching abilities was high prior to the course with an overall mean confidence rating of 3.6 ± 0.4 and greatest levels of confidence in ability to set the learning climate (4.3 ± 0.5), communicate goals (4.3 ± 0.5) and prepare audiovisuals (4.0 ± 0.9). Despite this, confidence increased across all skill areas with a post-course overall mean confidence rating of 4.3 ± 0.3 with the biggest gains in the ability to evaluate learner needs (22%), give and elicit feedback (28%, 25%), improve group dynamics (27%), and increase audience participation (27%). Students agreed or strongly agreed that they learned important small group teaching skills (4.0 ± 0.7), strategies for dealing with difficult small group dynamics (4.2 ± 0.7), information to help them give a better lecture (4.7 ± 0.7), and skills to make them a better teacher (4.4 ± 0.5). Students felt that the course will help them in their future careers (4.4 ± 0.7) and that it met their goals for taking it (4.2 ± 0.6). The most highly rated sessions were the small group discussions (4.1 ± 0.5) and the lecture feedback sessions (4.7 ± 0.5).

KEY LESSONS LEARNED: We have successfully developed and implemented a Medical Education Elective in the context of a larger program that provides advising and mentoring to students considering careers in medical education. Next steps include the development of additional electives in bedside teaching and curricular development, and sustaining support for faculty time.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: 1) Color displays of course schedule and syllabus 2) Pre and post-course inventory 3) Poster or oral presentation of data obtained.

DEVELOPING A CURRICULUM IN EVIDENCE BASED MEDICINE IN A COMMUNITY HOSPITAL BASED RESIDENCY PROGRAM. P. Radhakrishnan¹; M. Lewis¹. ¹St. Joseph Hospital, Phoenix, AZ. (Tracking ID #134629)

STATEMENT OF PROBLEM OR QUESTION: While most physicians and residents practice evidence based medicine, perceptions of EBM vary. In a local survey done we found that most practicing doctors use expert opinion databases such as UpToDate and journal reviews as sole sources EBM. In community based residency programs without the academic hospitalist model, where the residents are supervised by several internists, practicing EBM is often challenging. **OBJECTIVES OF PROGRAM/INTERVENTION:** Develop a model where the admitting internists and the Internal Medicine residents learn to apply the principles of EBM at the bed-side.

DESCRIPTION OF PROGRAM/INTERVENTION: The primary objective of this curriculum is to change resident attitudes and develop a habit of interpreting and applying current literature. An initial needs survey conducted revealed that while most residents and faculty review literature frequently and are familiar

with the principles of EBM, they do not apply EBM frequently in direct daily patient care. The program will be developed in a phased manner. The development of the curriculum will be driven by the residents and teaching faculty. 1. During resident work rounds, the senior resident will be responsible for maintaining a log of patient cases and develop the habit of formulating the Clinical Question. 2. During the initial phase, residents and faculty will be assisted in conducting searches by the librarian. 3. The literature search will be reviewed by one of the teaching faculty. Studies will be analysed and the Clinical Question will be discussed. 4. Blackboard (an online resource) will be used to post the articles, initiate discussion and get help in the statistical analysis of the articles. 5. EBM calculators will be available for residents to download to their PDA's. **FINDINGS TO DATE:** The program is ongoing. Several difficulties have been identified. 1. The residents have identified their deficiencies in interpreting literature 2. Short sessions are more effective. 3. It is challenging to effectively apply this program to all the in-patient residents due to the restriction of the work hours and the nature of community based residency programs where different attending physicians admit to the resident team.

KEY LESSONS LEARNED: The development of this curriculum is in process. 1. The initial findings indicate enthusiasm among the residents in applying EBM. 2. Problems identified include difficulty in formulating the Clinical Question, lack of consistency in performing the searches. 3. Resident behaviors and attending physician behaviors are difficult to change, repeated re-enforcement is required in application of the curriculum.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: The use of Blackboard will be demonstrated. Examples of recent discussions will be available. Information about online EBM resources and PDA EBM downloads will be made available.

DEVELOPING A PATIENT SAFETY CURRICULUM FOR INTERNAL MEDICINE RESIDENTS. E.J. Rosenberg¹; M. Duerson¹. ¹University of Florida, Gainesville, FL. (Tracking ID #135978)

STATEMENT OF PROBLEM OR QUESTION: Medical residents underreport medical errors to faculty and receive inadequate training in patient safety.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) Ascertain residents' knowledge, attitudes, and skills regarding patient safety. 2) Determine whether case-based seminars and use of standardized patient scenarios can teach housestaff to systematically identify, self-report, prevent, and investigate errors. 3) Use results from this pilot project to develop comprehensive patient safety curricula at this institution.

DESCRIPTION OF PROGRAM/INTERVENTION: Internal Medicine residents voluntarily completed an anonymous, self-administered survey to assess baseline patient safety knowledge, attitudes, and prescription writing skills. First-year residents (PGY-1) separately attended two, 1-hour seminars focused on identification of hazardous practices and systems-based analysis of errors. Residents were then asked to evaluate and disclose medical errors to standardized patients and were observed while discussing patient safety issues on mock team "work rounds." Success of the intervention was measured by comparing performance by first-year residents to that of upper level residents (PGY-2 and PGY-3) by means of standardized patient ratings and responses to a post-intervention questionnaire.

FINDINGS TO DATE: Thirty residents (51% PGY-1, 22% PGY-2, 26% PGY-3) have completed the baseline survey to date using Likert scale responses. Only 4 residents included the purpose of a medication on prescriptions. Seventy percent agreed/strongly agreed that "residents should be permitted to make and learn from their own mistakes"; 63% reported that they had discussed an actual or averted error within the past six months with an attending physician; 74% agreed or strongly agreed that this discussion motivated them to learn about the factors that might have led to that adverse event. Seventy-seven percent agreed/strongly agreed that they would feel comfortable disclosing a medical error to faculty physicians with whom they have worked. Only 53% agreed that they would feel comfortable disclosing an error to a patient and only 43% felt prepared to disclose errors to patients after completion of the PGY-1 seminars. Residents consistently cited the need to personally review all orders and communicate care plans to patients and nurses as the most important ways in which they could personally improve patient safety.

KEY LESSONS LEARNED: Preliminary data show that prescription writing skills among internal medicine housestaff need improvement. Residents report that they discuss adverse events with faculty; however, they feel unprepared and uncomfortable disclosing errors to patients that have suffered adverse outcomes. Data from standardized patient evaluations of the pilot project will be available by the end of January 2005.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Handouts of material covered during seminars and standardized patient scenarios.

DEVELOPMENT OF A NEW AREA OF CONCENTRATION IN COMMUNITY HEALTH AND SOCIAL ADVOCACY FOR MEDICAL STUDENTS. S. Jain¹; N. Wortis¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #132706)

STATEMENT OF PROBLEM OR QUESTION: The Area of Concentration (AoC) program at UCSF was developed to provide curricular support and mentoring for students engaged in scholarly projects. The AoC in Community Health & Social Advocacy is designed to encourage students to undertake community health and advocacy projects during medical school, generally during their fourth year.

OBJECTIVES OF PROGRAM/INTERVENTION: The objectives of the AoC in Community Health & Social Advocacy are to (1) develop a longitudinal curriculum to support medical student interest in community health and advocacy

projects, (2) teach students core skills needed to carry out their projects, and (3) provide mentoring and support from academic faculty and community-based partners.

DESCRIPTION OF PROGRAM/INTERVENTION: There are three distinct phases to the AoC in Community Health & Social Advocacy: preparation, experiential, and legacy. During the preparation phase, students explore their interest in community health and advocacy through a variety of elective experiences; they enroll in a two-week intensive seminar course to learn basic skills necessary for project design, implementation, and evaluation. During the experiential phase, students conduct their projects under the guidance of academic and community mentors; during this period, they meet monthly to share their progress and to reflect on their work. For the legacy phase, students create an enduring product that describes what they have learned and that ultimately benefits the community partner with whom they have worked.

FINDINGS TO DATE: The AoC in Community Health & Social Advocacy was introduced this year, and seven students have enrolled and started to carry out projects. The two-week seminar course was developed and offered this year, and it received excellent reviews from the students. Further evaluation will include assessing student knowledge and attitudes about the material presented in the course, providing feedback on student projects, and ensuring uniformity with the other AoCs being offered at UCSF.

KEY LESSONS LEARNED: During its first year, the AoC in Community Health & Social Advocacy has been well received by the medical students, who have enjoyed the didactic curriculum and who are actively working on their projects under the guidance of academic and community mentors. Students report the value of learning skills that they hope they can apply to their future work, as well as working with and learning directly from community partners. Future work will involve identifying additional projects and mentors and refining evaluation instruments for students and for their projects.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:

DEVELOPMENT OF A NEW EXPERIENTIAL ELECTIVE ON SUPPORTIVE CARE FOR TERMINALLY ILL PATIENTS. M. Wilson¹, E.H. Chittenden¹, S. Jain¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #134119)

STATEMENT OF PROBLEM OR QUESTION: Students must provide care for terminally ill patients but often feel ill-equipped to do so. We developed an elective, Supportive Care for Terminally Ill Patients: An Experiential Approach, to help meet this need.

OBJECTIVES OF PROGRAM/INTERVENTION: The objectives of the elective are for students to (1) understand the physical and emotional changes in dying patients, (2) feel comfortable with dying patients and the dying process, (3) learn to discuss emotional and spiritual issues and to provide psychosocial support to terminally ill persons and their loved ones, and (4) discuss and cope with personal reactions to the dying process.

DESCRIPTION OF PROGRAM/INTERVENTION: Participating students complete a curriculum in which they learn about the dying process, the needs of terminally ill patients, and the role that they will serve in this process. After completing this curriculum, each student is assigned a terminally ill patient living in the community and visits this patient at least two times each month. The focus of these visits is to provide support and comfort and to learn about the patient's experiences with her illness, not to address the patient's medical needs. In addition, students meet monthly over six months for a facilitated discussion to reflect on their experiences and to share what they have learned. In order to evaluate the effectiveness of the elective, we distributed baseline and follow-up questionnaires asking students to rate their comfort with spending time and discussing psychosocial and spiritual issues with dying patients.

FINDINGS TO DATE: Six students enrolled when the elective started in October 2004. Thus, the first cohort of students is still completing the program. However, baseline data on the students' knowledge and comfort level with palliative care and patient communication issues has been collected. On a 4 point scale ranging from 'strongly disagree' to 'strongly agree', students, on average, fall between 'agree' and 'disagree' when responding to statements that they are comfortable spending time with terminally ill patients and with discussing spiritual, emotional, psychosocial as well as death and dying issues with terminally ill patients. These data are particularly striking given that 4 out of the 6 students previously helped care for a dying person and 5 out of the 6 have had someone close to them die.

KEY LESSONS LEARNED: During this first offering of the Supportive Care for Terminally Ill Patients elective, we have been gratified by the enthusiasm expressed by students and patients for the program and by the variety of patient and student issues that have arisen in our monthly meetings. These issues provide rich topics for discussion that allow students to learn not only from their own patients but from their classmates' experiences as well. In addition, students report feeling empowered and surprised by their ability to help their patients despite not being part of their medical team.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:

DISSEMINATION OF AN EFFECTIVE SMOKING CESSATION TRAINING PROGRAM FOR SWISS PRIMARY CARE PHYSICIANS. J. Humair¹, I. Jacot Sadowski², J. Zellweger³, B. Kuenzi⁴, A. Burkhalter⁵, J. Cornuz⁶. ¹Geneva University Hospitals, Department of Community Medicine, Geneva, ; ²Institute for Social and Preventive Medicine, Lausanne, ; ³University Hospital of Lausanne, Medical Outpatient Clinic, Lausanne, ; ⁴Swisspep, Institute for Health Quality and Research, Bern, ; ⁵Lungenliga Schweiz, Bern. (Tracking ID #135527)

STATEMENT OF PROBLEM OR QUESTION: Counseling and pharmacotherapy are effective smoking cessation interventions but physicians often lack the skills

to apply them in clinical practice. Though effective educational programs exist, we must disseminate them in order to improve physicians' skills, increase patients' smoking cessation and reduce smoking-related diseases.

OBJECTIVES OF PROGRAM/INTERVENTION: To disseminate an effective training program in smoking cessation among Swiss physicians to improve their practices and increase patients' cessation rate.

DESCRIPTION OF PROGRAM/INTERVENTION: First we developed and tested an effective program using active learning methods and the "stages of change model" to train physicians in providing counseling and pharmacological therapy for smoking cessation (Cornuz & al Ann Intern Med 2002;136:429). Then we developed a strategy, coordinated, supported and funded by the Swiss Lung Association, to disseminate this program nationally: recruitment and training of physicians to teach smoking cessation interventions to their peers. We selected 24 experienced clinicians from different disciplines and states, who were recognized by their peers, highly motivated in both continuing medical education and smoking cessation. After a 1½ day session, new trainers learnt to: (1) master objectives and content of the smoking cessation training program; (2) apply adult education principles and teaching skills; (3) organize training sessions for their peers; (4) build a network of physicians active in smoking cessation. We trained them to teach 2 different courses adapted from the original program to meet the needs of practicing physicians: (1) a 1.5-hour session to inform and sensitize physicians about smoking cessation; (2) a 4-hour session enabling physicians to learn and practice the basic skills in smoking cessation interventions. Trainers received certification after a training session supervised by an expert, a contract requiring at least 3 yearly training sessions and payment per session covering preparation, training and loss of income from practice.

FINDINGS TO DATE: We designed a randomized trial to assess the efficacy of the training program in an academic primary care center. Compared to the control group, trained clinicians provided smoking cessation interventions of higher quality (mean score: 4.0 vs. 2.7, p=0.01), while 1-year smoking abstinence doubled among smokers visiting trained physicians (13% vs. 5%, p=0.02). In 2 years, trainers organized 108 training sessions for 1880 physicians from 15 states, with an average of 17.4 participants per session. 1102 and 778 participants attended respectively 59 short and 49 long courses.

KEY LESSONS LEARNED: This training program was effective to improve physicians' practices and smokers' cessation rate. Recruiting and training physicians to teach smoking cessation interventions to peers from their network was a feasible and successful strategy to disseminate this program. We plan to increase the number of trainers and to assess the impact of his program on smoking prevalence.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: CD-Rom, videotape, observation checklist and role play script used during training. Educational material used during training of teachers.

ESTABLISHING A DIABETES DISEASE MANAGEMENT PROGRAM IN A RESIDENCY CLINIC. D.L. Kansagara¹, E. Holmboe², K. Carr¹, S.J. Huot¹. ¹Yale University, New Haven, CT; ²American Board of Internal Medicine, Philadelphia, PA. (Tracking ID #135416)

STATEMENT OF PROBLEM OR QUESTION: Chart reviews done by residents of their own patients found the quality of diabetes care in our residents' clinic to be well below national benchmarks. National data have shown poor adherence to recommended practice and a large proportion of poorly controlled diabetics. There is an urgent need for teaching and implementation of effective disease management programs in postgraduate internal medicine ambulatory education settings.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) To enhance disease management support for low-income, poorly-controlled diabetics by teaching self-management skills and improving adherence to recommended therapies. 2) To teach internal medicine residents the components of chronic disease management, the role of interdisciplinary care, and the importance of clinical microsystems by involving them in a disease management program.

DESCRIPTION OF PROGRAM/INTERVENTION: We implemented a diabetes disease management clinic for poorly controlled diabetics in an urban internal medicine residency clinic. The care team consists of 1) second year residents on a required ambulatory care month, 2) a general medicine preceptor, 3) a nurse practitioner/diabetes case manager, 4) a dietician, 5) medical assistants, 6) a social worker, 7) and an endocrinologist. The four week clinical experience is supplemented with weekly didactic sessions which cover general systems issues in chronic disease management and diabetes management issues. Residents are expected to be active team members through follow-up contact with the nurse practitioner and participation in interdisciplinary team meetings. Residents also audit the charts of their own diabetic patients as part of their own practice-based learning and improvement.

FINDINGS TO DATE: Because one of the paramount goals of the clinic is to improve patient care, process of care and clinical outcomes data are being collected. The first set of data should be available in March 2005. To assess educational impact, the twelve residents completing the experience thus far were surveyed at the end of the rotation. Seventy-five percent (9/12) rated their own clinics as poorly-equipped to manage chronic illness. On the other hand, ninety-two percent (11/12) felt they now understood the components of effective chronic illness care. Residents reported a variety of aspects of their own practice they would change, ranging from use of more follow-up phone calls to using a team-based approach to care. Ninety-two percent (11/12) felt they were "likely" or "very likely" to make these changes and 92% (11/12) felt they were more likely to make these changes because of their experience in the diabetes disease management clinic.

KEY LESSONS LEARNED: A comprehensive diabetes disease management program for low-income patients can be successfully implemented in a residency clinic. Internal medicine residents are aware of existing deficiencies in disease management systems. An educational curriculum combined with active participation in an innovative system for disease management may inspire systems-based changes in residents' own clinical practice.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster, notebook with educational curriculum and patient education tools, and computer demonstrating downloadable glucometer program.

FEED YOUR FACULTY: FACULTY DEVELOPMENT AROUND THE ACGME COMPETENCIES. L. Logio¹; M. Ciccarelli². ¹Indiana University School of Medicine, Indianapolis, IN; ²Indiana University, Indianapolis, IN. (Tracking ID #135995)

STATEMENT OF PROBLEM OR QUESTION: Faculty Development workshops provide a simple solution to the challenge of residency programs to educate and enlist clinical faculty on the meaning and daily reality of the six core competencies as defined by the Accreditation Council of Graduate Medical Education (ACGME). 1

OBJECTIVES OF PROGRAM/INTERVENTION: 1. To teach the clinical faculty about the ACGME core competencies. 2. To provide a forum for continuous professional development in clinical teachers. 3. To demystify the language of the ACGME competencies. 4. To encourage the faculty to recognize what they do everyday in clinical work and how those tasks fit within the framework of the core competencies.

DESCRIPTION OF PROGRAM/INTERVENTION: With the Chairman's support, we developed quarterly evening workshops on clinical teaching skills and the six ACGME core competencies. The Faculty Enrichment and Educational Development (FEED) series has provided a forum for self-selected faculty to understand these competencies as well as to hone clinical teaching skills. We have completed five different workshops in 18 months. The general structure of each begins with a basic introduction and mini lecture of the topic followed by a demonstration. The group of participants is then divided into random small groups and given a task. The participants generally reconvene for a wrap up where each small group reports its findings to the larger group. Coffee and dessert is served at the conclusion to encourage faculty to continue the conversation. To date, the following topics have been covered: 1) Giving Effective Feedback, 2) Doctor Patient Communication Skills: how to teach and how to evaluate, 3) Professionalism 101: What's it All About?, 4) Evaluating Learners: Defining Average, Below and Beyond, 5) Teaching Caring Attitudes: Practical Professionalism. In each, we have incorporated the ACGME core competencies in the examples and small group activities.

FINDINGS TO DATE: The workshops have been well received. The attendance at each workshop has ranged from 45 to 70 faculty members. For certain topics, specific faculty were targeted. The Introduction to Clinical Medicine (ICM) Faculty who teach history-taking skills were invited by the ICM course director to the workshop on Doctor Patient Communication Skill. We have expanded our invitation list to include clinician educators beyond the Department of Medicine with routine participation from other departments. The interdepartmental conversation has been powerful in finding common ground and promoting collaboration on our educational resources.

KEY LESSONS LEARNED: Our efforts in the Department of Medicine have facilitated the residency training program's adaptation to the ACGME competencies with gradual expansion of our assessment tools for seamless integration into clinical work. Over 18 months we have worked to develop a core group of interested faculty members as the design team for this effort as well as a team of trained faculty development facilitators for the small group activities. Through these efforts, we have also been invited to present workshops for other departments and divisional activities. We have also recognized special needs for select groups of faculty with next steps planned to develop workshops for our hospitalist group, our fellowship directors, our resident curriculum coordinators, and our continuity clinic preceptors.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Modality to be used to demonstrate innovation at Meeting We can provide video reproductions of our workshops as well as summary data of our evaluations and ongoing needs assessment.

FINDING MEANING IN THE MIDST OF CHAOS: INCORPORATING THE HUMANITIES INTO RESIDENCY TRAINING. A.A. Atasoylu¹; M. Batalden¹; E. Gaufberg¹. ¹Harvard Medical School, Cambridge, MA. (Tracking ID #134999)

STATEMENT OF PROBLEM OR QUESTION: Medical training has become ever more challenging, with the increase in paperwork associated with delivering medical care, and shortened hospital stays. Because of increasing time pressures to admit and discharge patients, and because patients are frequently whisked off to diagnostic studies, trainees find little time to spend at the bedside connecting with their patients. The negative impact of this cannot be overestimated. Residents often refer to the training process as dehumanizing. They comment on the loss of personal and professional balance, and at times the loss of self and the values that attracted them to the medical profession. Trainees regret the lack of adequate space to consider their patients beyond a set of diagnoses, and to consider the impact patient care has on them personally and professionally.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) To improve resident understanding of, and connection to, themselves, their colleagues, and their patients 2) To increase resident satisfaction with their work lives.

DESCRIPTION OF PROGRAM/INTERVENTION: The shaping of caring, connected physicians requires creative approaches to help trainees find meaning in the midst of all the chaos. Through the use of the humanities in medical training, we create a space for exploring the doctor-patient relationship, inviting

trainees to reconnect with themselves and their patients. The medical humanities initiative has included medical humanities events, a medical humanities elective, and a literary journal to enable residents to reconnect with the humanistic aspects of medicine. Monthly medical humanities events create the forum for residents to connect with each other and faculty facilitators around issues related to being a physician. These events have included a trip to a Healer's Art exhibit, as well as poetry and book readings by physician authors. The medical humanities elective gives residents the opportunity to do focused reading and writing about the experience of being a patient or physician. Auscultations, our annual literary journal, features poetry, prose and visual arts from our community of providers and staff. We celebrated our humanities initiative and the publication of the journal with an end of year hospital-wide gathering of providers and staff sharing their talent through live poetry readings, musical performances and displayed artwork.

FINDINGS TO DATE: The medical humanities initiative has been overwhelmingly well received. Residents have commented on how meaningful it has been for them to be given the opportunity to reflect on their experiences and to connect with each other about their lives as physicians.

KEY LESSONS LEARNED: Incorporating the arts and literature into residency training adds tremendously to the training experience. Giving residents the time and space to reflect on their experiences promotes their professional development and likely increases their work satisfaction.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster, medical humanities resource packet.

GENERALIST INTERNISTS AS E-LEARNING AUTHORS. J. Jevtic¹; G.C. Lamb¹; K. Denson¹; S. Denson¹; J. Hariharan¹; A. Maguire¹; K. Novoa-Takara¹; K.J. Pfeiffer¹; J. Shubin¹; M. Ziebert¹; L. Currey¹; T. Ottow¹; K. Poindexter¹; L. Meurer¹; J. Morzinski¹; D. Simpson¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #134553)

STATEMENT OF PROBLEM OR QUESTION: Increasing clinical and curricular responsibilities coupled with 80/30 work hour restrictions has hindered general academic internists ability to provide formal education to post-graduate trainees. In response, medical educators must learn alternative methods to deliver quality education in a time efficient and cost-effective manner. One such opportunity is to effectively utilize e-learning management systems (LMS).

OBJECTIVES OF PROGRAM/INTERVENTION: Prepare generalist clinician educators(CE) to: 1) author e-learning applications using evidence-based design and assessment strategies (i.e. incorporation of visuals, personalization), 2) facilitate e-learning interactive dialogues (e.g., threaded-discussions, chat rooms), 3) determine the return on investment (ROI) relative to CE authored e-learning modules.

DESCRIPTION OF PROGRAM/INTERVENTION: We created a 25 hour blended faculty development course (FDC) for generalist medical educators over 16 weeks. The FDC used synchronous and asynchronous e-learning units to provide medical educators with hands-on experience as an on-line learner. Face-to-face instruction provided evidence-based concepts emphasizing learner motivation, multi-media instructional design, and assessment strategies for on-line learning. Each faculty participant was required to author an e-learning module for medical student or resident education demonstrating the application of the evidence-based principles within their instruction. Each participant took a pre- and post on-line instruction knowledge test, as well as monitoring of total hours spent on and off line over the 16 week course. Module examples include "Geriatric Polypharmacy", "Chronic Pain Management", "DVT management" and "Palliative Care". Finally, faculty reviewed each others e-learning modules using standards developed by the American Society for Training and Development (ASTD).

FINDINGS TO DATE: Results of the pre- and post-test showed improvement in knowledge scores from 43% to 68%. CE's reported that on average time to develop an e-learning module was 29.8 hours were required to develop an e-learning module with 18.1 hours off-line and 11.7 hours on-line. Eighty-eight percent of CE's completed their authored modules on time. End-of-course ratings averaged >4.0 (1=Strongly Disagree to 5=Strongly Agree) for effectiveness/satisfaction with on-line interactions, satisfaction with small group/team approach, and instructor support. CE's strongly recommended that this FDC be repeated with other primary care faculty (4.6).

KEY LESSONS LEARNED: CE's limited experience as learners with on-line educational formats presented a challenge to FDC authors as CE's reported being "overwhelmed" simultaneously learning on-line and designing their own modules. To balance these challenges, CE's reported that the inclusion of dedicated F2F "study hall" time to work on their own modules with personal e-trainers, clear deadlines, and seeing other CE's modules "under construction" provided ideas and motivation to complete the course.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: FDC storyboard and syllabus will be displayed. Several of the participants e-learning modules will be demonstrated via computer.

HARRISON'S CLUB: PROMOTING THE ACGME CORE COMPETENCIES BY GOING BACK TO THE BASICS. A.B. Geller¹; J.M. Golbin¹; W. Mir¹; J.M. Bratcher¹; M. Evvin¹. ¹Lenox Hill Hospital, New York, NY. (Tracking ID #134249)

STATEMENT OF PROBLEM OR QUESTION: The medical textbook appears to have gone the way of the 8-track player. Senior attendings are often overheard bragging of having read Harrison's Principles of Internal Medicine during their residency. Today, residents are taught to answer their questions on the internet, on services such as "Up-To-Date.com" or "MD-consult.com". Internet searches do provide some benefit, with quick, directed searches of narrow topics. However

residents lose the immeasurable benefit that comes with reading a textbook of medicine and discovering that their question is not as simple as skimming a review article. Instead, it is a pathophysiologic process, an esoteric etiology, a molecular level explanation, and a biochemical review. This aspect is what separates a physician from any of the myriad of "physician extenders" in the 2005 medical world.

OBJECTIVES OF PROGRAM/INTERVENTION: To stimulate interest in medical textbooks, specifically Harrison's Principles of Internal Medicine, through the introduction of Harrison's Club.

DESCRIPTION OF PROGRAM/INTERVENTION: This is a resident-conceived, organized, and run endeavor. Residents voluntarily meet once per week, always at the same time and location, to offer a sense of importance to the Club. The Club will be structured to provide a complete reading of Harrison's once every 3 years, by splitting the text into approximately 150 mini-chapters. The text will be read straight through, allowing a new attendee to pick up wherever the Club last left off. One resident each week will be the preceptor, allowing all participants the opportunity to teach their peers.

FINDINGS TO DATE: Feedback has been extremely strong in support of the Club. Attendance is tracked to allow an objective basis for future outcome analysis. This can be accomplished through a variety of techniques: comparison of In-Training exam scores of Club versus non-Club members, assessment of pre- and post-Club attendance resident confidence levels on teaching rounds and morning report, or even a comparison of ABIM certification scores.

KEY LESSONS LEARNED: The relatively simple introduction of Harrison's Club has already impacted tremendously on resident education. Medical Knowledge and Patient Care are emphasized. Both Professionalism and Interpersonal and Communication Skills are promoted through the group teaching format. In addition, Evidence-Based Medicine and PBL skills are developed through the readings. Harrison's Club truly enhances today's ACGME Core Competencies by going Back to the Basics.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: We plan to demonstrate the relative change in resident interest, attendance and perception prior to initiation of the club and throughout the year. This will be accomplished through a variety of illustrative modalities, including pie charts, comprehensive surveys, video testimonials and biweekly quiz statistics.

IMPROVING CARE FOR GERIATRIC PATIENTS: MEETING THE CHALLENGE. E. Eckstrom¹; J.L. Bowen²; C. Tanner²; M. Ririe¹; C. Joseph³; S. Desai². ¹Legacy Emanuel and Good Samaritan Hospitals, Portland, OR; ²Oregon Health & Science University, Portland, OR; ³Portland VA Medical Center, Portland, OR. (Tracking ID #132827)

STATEMENT OF PROBLEM OR QUESTION: General internists care for older adults and teach geriatrics to residents, but may lack the necessary knowledge and skills to do these tasks well.

OBJECTIVES OF PROGRAM/INTERVENTION: We developed, implemented, and evaluated a curriculum to answer the following: 1) Can an innovative workshop improve faculty knowledge, skills, and confidence in caring for and teaching about elderly patients? 2) Will the workshop induce faculty behavior change? 3) Can we "geriatricize" our teaching environment?

DESCRIPTION OF PROGRAM/INTERVENTION: This project was a collaboration between the University, VA, and a community program in Portland, OR. We first elicited internists' stories about teaching geriatrics and providing care to the elderly, and geriatricians' perceptions of internists' care of older adults. This needs assessment defined the content of a one-day CME workshop for general internists at our institutions. We pilot tested the intervention with 12 "star educators" (leaders in education at each institution who clearly influence training), then gave it to 33 more faculty. The curriculum was 30% didactic (given by geriatric experts) and 70% experiential (scripted cases with "standardized learners" played by our "star educators" and geriatric fellows). We developed pocket cards to use as teaching tools. Evaluation tools included workshop evaluations; faculty pre-post knowledge, attitude, and teaching confidence surveys; faculty post semi-structured interviews; and resident pre-post attitude surveys.

FINDINGS TO DATE: We identified these gaps in our general internists' geriatric knowledge and skills: cognitive decline, functional assessment, care transitions, behavioral management, and determining decisional capacity. Final results are pending, but workshop evaluations rated the course at 5.2 of 6 (6 being "outstanding"), and geriatric knowledge improved in participants.

KEY LESSONS LEARNED: Our careful needs assessment informed our successful, innovative workshop. The scripted cases, standardized learners, and pocket cards supported transfer of newly learned geriatric knowledge from lecture to teaching situations. "Star educators" are important local champions for facilitating change at distant sites.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Scripted cases, evaluation tools, pocket cards, workshop content, and evaluation results.

INTERN CURRICULUM: THE IMPACT OF A FOCUSED TRAINING PROGRAM ON THE PROCESS AND CONTENT OF SIGNING OUT PATIENTS. M. Carly¹; C.C. Smith¹; J.L. Schnipper¹. ¹Brigham and Women's Hospital, Boston, MA. (Tracking ID #135579)

STATEMENT OF PROBLEM OR QUESTION: What is the impact of a focused curriculum on the content and process of patient sign-out and on the effectiveness of intern cross-coverage?

OBJECTIVES OF PROGRAM/INTERVENTION: 1. To qualitatively describe current sign-out practices of housestaff at our institution. 2. To assess impact of an intern curriculum on sign-out practices. 3. To assess impact of an intern sign-out curriculum on continuity of care.

DESCRIPTION OF PROGRAM/INTERVENTION: The curriculum was a 9-week course, written and taught by senior residents, held in the first 2 months of internship. Curricular materials included a written syllabus and weekly 1-hour small group sessions with 5-10 interns taught by 3 to 6 residents. The first session was dedicated specifically to sign-out and included the following: a skit illustrating the steps of the sign-out process, a discussion of the key components of written and verbal sign-out, and a dialogue on how to optimize the sign-out process. During the subsequent 8 sessions, the components of sign-out were reviewed for specific clinical scenarios. Seventy interns participated. Thus far, data have been measured at 2 time points: prior to initiation of the curriculum and 2 months after its completion (July and October, 2004, respectively). Sign-out and cross-cover practices were measured via 2 separate written surveys of interns and residents sent out at these 2 time points. The surveys addressed the content of written sign-out and measures of discontinuity as determined by cross-covering residents. Data from written intern sign-outs were also abstracted to measure explicit indicators of good sign-out.

FINDINGS TO DATE: After the curriculum, interns self-reported that their written sign-outs included the following components more often than prior to the curriculum: medical problem list (63% vs. 89%, $p=0.05$), baseline physical examination (60% vs. 92%, $p=0.01$), updated patient assessment (89% vs. 100%, $p=0.02$), pertinent procedure/test results (83% vs. 100%, $p=0.02$), treatments that were effective/not effective (78% vs. 92%, $p=0.04$), and difficult psychosocial or family issues (50% vs. 85%, $p<0.01$); self-reported notation by night-float of cross-cover events also increased (77% vs. 100%, $p<0.01$). However, only one objective measure improved: inclusion of intern contact information on the written sign-out (60% vs. 88%, $p=0.02$). Even in the post-curriculum period, 52% of sign-outs were not performed in a quiet place and 50% of night float residents could not find the information needed to care for a patient when paged.

KEY LESSONS LEARNED: These data suggest that at baseline, deficiencies exist in the content and process of sign-out, with corresponding problems in continuity of care. A curriculum focused on sign-out may help address these deficits. The fact that self-reported outcomes improved more than objective outcomes raises the possibility of social acceptability bias and suggests that it may be easier to change knowledge of how sign-out should be performed than it is to change actual behavior. Additionally, outcomes may have improved purely as the result of increased intern experience with time. Comparison of data from experienced interns before and after institution of the curriculum will help elucidate the impact of the curriculum itself. Further work is needed to improve the curriculum and design additional interventions focused on the process and goals of sign-out, the institutional culture, and effectiveness in improving measures of continuity.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Written syllabus. Video of a teaching session.

INTERNATIONAL GUIDELINES ON MENTAL HEALTH TRAINING OF PRIMARY CARE PROVIDERS FOR TRAUMA EXPOSED POPULATIONS. D.P. Eisenman¹; B.L. Green²; N. Rayburn³; J. De Jong⁴; S. Weiner⁵. ¹University of California, Los Angeles, Los Angeles, CA; ²Georgetown University, Washington, DC; ³RAND, Santa Monica, CA; ⁴Vrije Universiteit, Amsterdam, ; ⁵University of Illinois, Chicago, Chicago, IL. (Tracking ID #135228)

STATEMENT OF PROBLEM OR QUESTION: Primary care providers (PCPs) may be the sole source of mental health care in countries emerging from war. Frequently, experts from developed countries travel internationally to these post-conflict countries to train PCPs in trauma mental health ("trauma"). An internationally-developed, evidence-based template for teaching and evaluating trauma training is needed.

OBJECTIVES OF PROGRAM/INTERVENTION: 1. To systematically identify the content and best strategy for implementing trauma training. 2. To develop guidelines for the trauma training of PCPs in post-conflict, generally developing, countries. 3. To disseminate the Guidelines to international governmental, non-governmental, and academic institutions for comment.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: For Objective #1, we performed a systematic search using MEDLINE, PsychInfo, and Social Science Abstracts. Internet searches identified grey literature and international experts identified unpublished reports. For Objective #2, a 12-country international panel drafted the Guidelines using the evidence and clinical expertise. For Objective #3, the Guideline was disseminated to 24 governmental, non-governmental, and academic institutions for comment.

FINDINGS TO DATE: Re Objective #1&2: The resulting Guidelines, organized under 4 domains, include but are not limited to: VALUES. 1) Traumatic events from war and collective violence should be viewed within a human rights framework. This informs the roles of PCPs in: documentation; advocacy; identification of links to other systems (e.g., legal, police, educational); protecting against the risk of medicalizing survivors. 2) Trauma is a multi-faceted problem best understood through the perspectives of multiple disciplines and in relation to local understanding of traumatic events. The values of key stakeholders (community, patients, and providers) must be addressed in the training. 3) Training should involve collaboration between international and local representatives. CONTEXTUAL FACTORS. 1) Training programs should consider the community and sociopolitical contexts in which the traumas occurred. The first step is to identify local, representative partner(s) for determining the need for the program and identifying local resources for mental health provision. 2) A training needs assessment attends to cultural factors and the complementarity of the training with existing health systems. 3) Training should be integrated within existing systems to promote sustainability and avoid setting up parallel systems of care. CURRICULAR ELEMENTS. 1) The curriculum imparts both knowledge (mental health literacy; common traumas, consequent mental disorders, their treatment; confidentiality; self-care techniques; crisis intervention; referral to mental health professionals and skills) and skills (communication, documentation, ac-

cessing community resources) EVIDENCE BASED TRAINING. Training programs should: 1) utilize a multifaceted range of evidence-based training models and techniques. 2) identify appropriate participants, their needs, goals, and limitations; utilize appropriate trainers intensity, duration and trainer/student ratio; 3) evaluate outcomes. Re Objective #3: International dissemination revealed agreement with the Guideline's recommendations. The Guidelines are viewed as also useful for responding to or traumatized communities in the U.S. (such as immigrant communities) and post-disaster.

KEY LESSONS LEARNED: The Guidelines provide trainers and funders with an internationally-developed, evidence-based template for teaching and evaluating trauma training.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster and Relevant Materials.

LEADERSHIP IN RESIDENCY TRAINING: AN IMPERATIVE IN THE 21ST CENTURY. L. Logio¹. ¹Indiana University School of Medicine, Indianapolis, IN. (Tracking ID #136198)

STATEMENT OF PROBLEM OR QUESTION: In 2005, medical education has an imperative to teach residents leadership skills and to foster a sense of commitment within them for quality improvement of the healthcare system.

OBJECTIVES OF PROGRAM/INTERVENTION: 1. To measure the leadership qualities and characteristics of professionalism in Internal Medicine residents. 2. To provide a forum for trainees to discuss leadership in medicine and the role and responsibility society places on doctors. 3. To foster a culture within a residency program of resident leaders and change agents.

DESCRIPTION OF PROGRAM/INTERVENTION: During the first month of the PGY-2 year, residents participate in a retreat to orient them to their new job description as upper level residents, including workshops on clinical teaching skills. During the retreat, residents participate in an open discussion about leadership. Residents are then asked to complete a self-assessment survey on their personal qualities of leadership, before they have served in the capacity as leader. As upper level housestaff, residents are subjects of a 360-degree assessment survey specifically developed for internal medicine resident leading an inpatient team. The Medical Leadership Assessment Inventory (MedLAI) measures leadership skills within this role as measured by the other team members, i.e., faculty, interns, and students. Ratings are collected each time they rotate through the county or VA hospital where they serve as the primary team leader. The survey tools are collected within the online evaluation system. Each upper level resident is again asked to complete a self-assessment survey while his team members are asked to complete their assessment of his leadership skills. A PGY-3 retreat provides another forum to discuss leadership. Activities during this retreat include opportunities for residents to build a consensus and change the opinions of their colleagues. At the conclusion of this retreat, each resident is given his cumulative data as an aggregate "self" score and aggregate "other" score in each of five categories: professionalism, team builder, coping with stress, self-awareness, and teacher.

FINDINGS TO DATE: The leadership data is generally well received by the residents. A trend of improvement is seen in all areas as rated by "Others." Self-ratings improve over time as well, except in the areas of teaching and self awareness. Residents report their own performance in these areas lower over time. There are gender differences in both "Others" and "Self" rating trends.

KEY LESSONS LEARNED: Using a newly developed 360-degree tool, professionalism and leadership in internal medicine resident trainees can be measured with evidence of progressive improvement in most areas. Gender-related trends are interesting but warrant further investigation. Retreats and workshops that focus on leadership help foster a sense of ownership and continuous quality improvement within the training program. Residents show more problem-solving skills and novel ideas for patient care service and educational program improvements.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: We will demonstrate trends in the data as well as the online survey and the reports generated by the 360-degree tool.

LEARNER AS TEACHER – TEACHING STUDENTS TO TEACH. J.M. Cunningham¹. ¹University of South Alabama, Mobile, AL. (Tracking ID #135737)

STATEMENT OF PROBLEM OR QUESTION: PROBLEM: To establish a series to formally teach teaching skills to fourth year medical students at our institution. OBJECTIVES OF PROGRAM/INTERVENTION: 1) introduce key concepts in teaching, 2) develop students' skills in analyzing a teaching encounter, 3) gain proficiencies in the participants' ability to teach.

DESCRIPTION OF PROGRAM/INTERVENTION: The design of the course involves a series of interactive small group discussions that focus on several key elements of teaching. Participants include fourth year students enrolled in the Acting Internship in Internal Medicine (usually 3 or 4 per group). Each group of students is exposed to four 1 to 1 1/2 hour long weekly sessions each divided into the following parts: 1) mini-lecture – define educational concepts and name key components, i.e. discuss it, 2) video-tape review of sample teaching sessions – recognize key behaviors, i.e. see it 3) roleplay that is video-taped and reviewed, i.e. practice it. Outside of the teaching sessions, the students will keep track of their teaching by filling out teaching encounter cards. These cards are carried with the student as they are working on the wards and are filled out after any teaching encounter with a third year student, with a patient, or with a patient's family. The student categorizes the teaching encounter, rates the effectiveness of the encounter and their comfort level with the encounter. This endeavor will help to personalize the teaching concepts from the interactive sessions. Pre- and post-course questionnaires are completed to help determine the effectiveness of

the course, if the student has found critical analysis of teaching helpful and to find if students are more comfortable with their teaching abilities.

FINDINGS TO DATE: We are tabulating the data obtained from the teaching encounter cards, pre- and post-course questionnaires. After analyzing the trend thus far, students have found that the course made them more aware of their teaching strengths and weaknesses, improved their enthusiasm for clinical teaching, and improved their perceived clinical teaching ability.

KEY LESSONS LEARNED: Indisputable is the importance of teachers to learn how to teach better. What might not be as apparent is the importance of a student to learn how to teach. With this knowledge a student is able to not only recognize a good teacher but also recognize what key behaviors makes this person a good teacher. They are able to critically analyze a teaching encounter, detect both the good and bad teaching behaviors, and then adopt the positive behaviors to refine their own teaching. Despite the importance of learning how to teach, formal courses in teaching are lacking in most medical schools. Although it may seem medical students are too early in their training to begin to teach, it is vital to establish good teaching behaviors early on in their training. By the time of their fourth year, most students have made strides toward becoming an adult learner and are open to concepts about teaching. So far in our series, the fourth year students are very receptive to learning about and conceptualizing teaching. MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: A PowerPoint presentation will be used to illustrate methods and key concepts.

LONGITUDINAL CLINICS FOR FIRST YEAR MEDICAL STUDENTS. M. Mayer¹; D. Neides²; J. Isaacson¹; N. Mehta¹. ¹Cleveland Clinic Foundation, Cleveland, OH; ²Cleveland Clinic Foundation, Solon, OH. (Tracking ID #136021)

STATEMENT OF PROBLEM OR QUESTION: Reinforcement of Communication Skills (CS) and Physical Diagnosis (PD) during the early stages of medical school is a desirable but elusive goal. The lack of such reinforcement leads to poor retention of these skills, and the need to relearn them haphazardly during third year of medical school and beyond.

OBJECTIVES OF PROGRAM/INTERVENTION: While planning for the Cleveland Clinic Lerner College of Medicine (which opened July 2004), it seemed that learning objectives dealing with communication, physical diagnosis, professionalism, preventive care and continuity of care would be reinforced well in a clinical setting. We wanted an objective-driven, skills reinforcing pairing of classroom instruction with patient based experiences, so students could learn clinical skills early.

DESCRIPTION OF PROGRAM/INTERVENTION: We developed an alternating week format with a CS and PD session one week, and Longitudinal Clinic (LC) the following week. These began in September for first year students. On the first week, CS and PD are taught in small group sessions, utilizing standardized patients. On the second week, students go to LC. A cadre of primary care physicians serves as LC preceptors. A job description with teaching objectives and tasks of the preceptor was developed from our learning objectives, and made explicit prior to signing up preceptors. Templates for a reduced patient schedule were developed to foster incorporation of direct observation into teaching of clinical skills. The learning agenda for each student at LC is re-calibrated weekly, using tools on our web portal. Students and preceptors go over the skills list adapted from the prior week's classroom session. They may also review a cumulative skills checklist. Two faculty development sessions for LC preceptors were held. All preceptors were required to participate. These focused on direct observation of care and assessment and feedback skills. Reflection by students and feedback from the LC preceptor are strong features of this program. Reflection is captured in real time on patient logs that students construct on the web portal, with comments by preceptors. The preceptor does skills assessments at the end of each session. Periodic assessments are being done and are being incorporated into a learning portfolio.

FINDINGS TO DATE: Many students find LC to be the most rewarding part of their early medical education. They are demonstrating increasing competence and confidence in relating to patients and developing doctoring skills. It is hoped that with early and frequent exposure to patient based care in this objective-driven, skills-reinforcing setting, high levels of clinical competence and reflective ability will be demonstrated in later clinical work.

KEY LESSONS LEARNED: Involving students in negotiating individual learning agendas was important. The development of cumulative skills lists helped students and faculty re-calibrate learning agendas. Patient logs are a rich source for students to reflect on lessons learned in developing doctoring skills. Engaging faculty as active teachers, utilizing direct observation skills, was essential; faculty development played a vital role.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: A live demonstration of our web portal, with skills checklists, patient logs, weekly assessment, cumulative skills checklists, and patient log summaries can accompany a poster or oral presentation.

LONGITUDINAL FORMATIVE FEEDBACK. B.R. Oqur¹; D. Hirsh¹; D. Bor¹. ¹Harvard Medical School, Cambridge, MA. (Tracking ID #136053)

STATEMENT OF PROBLEM OR QUESTION: Formative feedback in clinical clerkships is lacking, most notably from faculty. Student assessment by faculty in traditional clinical clerkships is limited by inadequate preceptor contact, negligible cross discipline coordination or communication, and inadequate exposure over time to permit evaluation of students' responses to feedback.

OBJECTIVES OF PROGRAM/INTERVENTION: The longitudinal integrated clerkship provides a unique opportunity to assess students' clinical learning and ability across disciplines and over time. Harvard Medical School's Cam-

bridge Integrated Clerkship is an innovative re-design of the core clinical year, structured to teach core Medicine, Surgery, Pediatrics, Obstetrics and Gynecology, Psychiatry, Radiology, and Neurology in a longitudinal, integrated way. Students have year-long relationships with preceptors in each of the major disciplines. These relationships occur in many venues: through the co-management of a cohort of longitudinal patients in ambulatory settings, through emergency room experiences and inpatient teaching rounds with a small set of dedicated teaching attendings, through regular meetings with faculty mentors, and through weekly structured tutorials. This structure affords the luxury of serial formative engagements with students and insights into their developmental progression of knowledge, skills and attitudes. Most importantly, it allows faculty to monitor the effect of feedback on student learning and thereby enhance future learning. Our objectives were to create mechanisms to provide effective formative feedback to students serially, using the structures of longitudinal faculty-student interactions of the Cambridge Integrated Clerkship.

DESCRIPTION OF PROGRAM/INTERVENTION: We created a four-step model. First, individual preceptors in each discipline assessed their student's performance, using a standard set of criteria and based on clinical competencies, and students assessed their own progress in these realms and set goals for improvement. Next, each student's multi-disciplinary team of preceptors met to review the student's progress in each of the major dimensions. Students' abilities and learning needs were considered individually and in comparison to other students in the clerkship. In the third step, students met with one of the course directors to receive the final summary of recommendations, to review suggestions for areas for work over the succeeding 6 months. Finally, students met with individual preceptors to develop plans for implementing these changes in each discipline. Their progress will be re-evaluated by students with their preceptors throughout the year and in a collaborative summative assessment by the students' multi-disciplinary teams at the end of the year.

FINDINGS TO DATE: Initial response of faculty and students to the assessment process was universally positive. Examples of the instruments used and of the assessments created will be presented.

KEY LESSONS LEARNED: Students and faculty highly value thoughtful and effective formative feedback, but it is not easy to come by in traditional clerkship settings. The longitudinal nature of the Cambridge Integrated Clerkship allows faculty sufficient exposure to and impact upon students' development over time. The integrated multi-disciplinary structure establishes a nucleus of educators who can collaboratively evaluate each student to create a productive agenda for further learning and development. There is subsequent ample opportunity to reassess students' progress.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster or oral presentation.

MULTIDISCIPLINARY PRIMARY CARE MORNING REPORT. C.L. Collard¹; B.L. Bryant¹; E. Hanesworth¹; A. Wolinsky¹; S.U. Rehman¹. ¹Ralph H. Johnson VAMC, Charleston, SC. (Tracking ID #132180)

STATEMENT OF PROBLEM OR QUESTION: Multidisciplinary health care teams have been shown to improve patient care and decrease health care costs. The presence of a pharmacist has been proven to improve the efficacy and safety of pharmacological therapy. Despite the increasing numbers of health care professionals who recognize the benefits of a multidisciplinary approach to patient care, this philosophy is rarely taught to students in the medical professions. Physicians still teach physicians, pharmacists still teach pharmacists, and nurses still teach nurses.

OBJECTIVES OF PROGRAM/INTERVENTION: A multidisciplinary approach to medical education may benefit future healthcare professionals by offering a new perspective to a seemingly old topic. The exposure to the perspectives of other healthcare professionals may serve to enhance the quality of patient care. The value of knowing the cost of a medication or understanding the barriers to its administration, for example, may impact the clinical use of an agent by a provider.

DESCRIPTION OF PROGRAM/INTERVENTION: Primary Care Morning Report involves interactive learning sessions focusing on the multidisciplinary team approach in primary care. Participants include residents and interns in medicine, neurology, and psychiatry, pharmacy residents, and medical, pharmacy, physician assistant and nursing students. Hourly weekly sessions are moderated by attending physicians and PharmDs. The residents and students of the above disciplines present patient cases focusing on topics pertinent to primary care. A case based, interactive learning format is utilized and is often supplemented with brief didactic presentations. Physicians provide insight regarding physical assessment and differential diagnosis. Pharmacists provide insight regarding drug interactions, indications, contraindications, doses, and cost of medications as it relates to the clinical presentation. Social workers and nurses provide appropriate feedback and commentary accordingly.

FINDINGS TO DATE: The educational program was initiated 2002 and has been running successfully for two years. Feedback by pharmacy residents, medical students and resident has been very positive. Evaluations range from 3.5-5 with 1 being the least helpful to 5 being the most helpful.

KEY LESSONS LEARNED: Interactive, interdisciplinary learning sessions are beneficial to medical education.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Will Present in Any Format

ORIENTATION TO THE CLERKSHIPS: PARTNERING WITH FOURTH YEAR STUDENTS TO "QUELL THE NERVES" OF NEW CLERKS. S. Schwab¹; M. Grayson¹; K. Edwards¹. ¹New York Medical College, Valhalla, NY. (Tracking ID #132768)

STATEMENT OF PROBLEM OR QUESTION: Third year students beginning their clerkships have concerns regarding being a clinical clerk. An orientation

workshop addressing these concerns was designed and implemented with the collaboration of 4th year students.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) Identify concerns regarding roles as a clinical clerk. 2) Develop strategies for managing concerns. 3) Demonstrate basic skills in discussing bad news with patients.

DESCRIPTION OF PROGRAM/INTERVENTION: A 3 hour workshop was designed for 3rd year students during the week prior to starting clerkships. The workshop addressed concerns ranging from logistical and content issues to student's anxieties about fund of knowledge and impending performance. Students were also trained in delivering bad news, a communication issue often raised as anxiety-provoking by students. The workshop consisted of a 1 hour large group and a 2 hour small group (10-15 students) co-led by a faculty member and 4th year student. While faculty can address certain issues, it was felt that 4th year students would best be able to address student's concerns and offer practical "survival" strategies. In the large group a video was shown and a paradigm for communicating bad news was presented. During the 1st hour of the small group students discussed their experiences with delivering bad news and practiced this in a role-play. The focus of the 2nd hour was on addressing anxieties about entering the 3rd year. Students were asked to pair off and tell their partner what they were most concerned about regarding working with patients, with families and with residents/attendings. Examples were given: "I am worried I will hurt a patient", or "what if I am asked to do something by a resident that I don't know how to do?" The group then came together and each student presented their partner's concerns. The 4th year student then led a discussion during which faculty and students brainstormed strategies for managing their fears about clerkships. A wide range of concerns about clerkships emerged from these discussions.

FINDINGS TO DATE: An evaluation was completed by 136 out of 180 students at the end of the workshop. Eight statements were rated on a scale of 1-4 with 4 being the most favorable. The mean rating for the overall experience was 3.39. The statement "The small group discussion helped to alleviate some concerns" received a mean score of 3.32. Sample comments from students included: "I found it extremely helpful to have a 4th year student answer questions about the specific hospitals we are rotating through"; "Small group helped to show everyone has similar fears"; "Some of my anxiety was quelled; the student helper was very honest and helpful"; and "I really appreciate this opportunity to get 4th year feedback. I wish we had more".

KEY LESSONS LEARNED: This workshop was rated highly. In particular, it demonstrated that new clerks benefited greatly from discussing their concerns with 4th year students and would like more time dedicated to this.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Written guidelines for faculty and fourth year students leading the "discussion of concerns" exercise will be distributed.

PATIENT-CENTERED CARE: AN EXPERIENTIAL CULTURAL COMPETENCE CURRICULUM FOR MEDICAL STUDENTS. A.G. Gomez¹; P. Henderson¹; A. Chirra¹; G. Guiton¹; L. Wilkerson¹. ¹David Geffen School of Medicine at UCLA, Los Angeles, CA. (Tracking ID #132623)

STATEMENT OF PROBLEM OR QUESTION: Students must appraise diverse worldviews of health and illness and anticipate patient cultural diversity. Every patient-physician interaction has the potential to be cross-cultural. Understanding this and addressing cultural bias might address healthcare disparities. The Liaison Committee on Medical Education determined that cultural-competence is essential, its instruction expected for medical school accreditation. The David Geffen School of Medicine at UCLA introduced cultural-competence curricula with emphasis on patient-centered interactions.

OBJECTIVES OF PROGRAM/INTERVENTION: 1. To develop knowledge, skills, and attitudes to practice medicine in a patient-centered manner, with a biopsychosocial perspective and an appreciation for racial and cultural diversity. 2. To begin a process by which students a) understand the role culture plays in all doctor-patient relationships, b) appreciate the complexity of cultural awareness, c) use techniques to work effectively with culturally diverse populations, and d) become aware of health care disparities.

DESCRIPTION OF PROGRAM/INTERVENTION: Patient-Centered Care is taught with multiple modalities including standardized patients, artistic presentations, appraisal of literature, community service and clinical experiences. From day one students participate in a group exercise, "Your Genogram/First Memory of Uniqueness", following the presentation of a play "Slice of Rice, Frijoles and Greens" highlighting Los Angeles' cultural diversity. A three year "Doctoring" interviewing course focuses on ethnic, racial, language and spiritual diversity. Patient concerns are addressed using a "3 C's" mnemonic (Call, Cause and Cope) adapted from Arthur Kleinman's "8 questions". In an elective, students learn culturally appropriate ways to help minority youth protect their health. A literary exercise, or "book club", is required of Ambulatory Internal and Family Medicine. Students read and critique Anne Fadiman's "The Spirit Catches You and You Fall Down". Medical Spanish is offered with immersion experiences in Los Angeles and abroad. In fourth year, students enroll in "colleges" where evening discussions focus on integrating patient-centered care. These and other experiences are designed so students acquire the awareness, knowledge and skills to deliver culturally appropriate, patient-centered healthcare.

FINDINGS TO DATE: UCLA students will begin to shape the way health care providers address healthcare disparities by focusing on the needs of patients. After one year, seniors performed 50% of patient-centered items in a Clinical Performance Examination designed to assess culturally competent interactions. Our goal is that students perform more than 80% of the patient-centered care items on our Senior Clinical Performance Examination.

KEY LESSONS LEARNED: We have a long way to go; intermittent instruction has yet to change behaviors to our satisfaction. A clear model for Patient-Cen-

tered Care such as our "three C's" needs to be explicitly and frequently integrated and reinforced in all preclinical and clinical experiences. We believe that we will succeed by ensuring relevance, focusing on real life examples from today's practice of medicine.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: video, exhibit course materials.

STUDENT EVALUATION OF INTERACTIONS WITH PHARMACEUTICAL REPRESENTATIVES DURING AN AMBULATORY CLERKSHIP. M. Stellini¹. ¹Wayne State University, Detroit, MI. (Tracking ID #134586)

STATEMENT OF PROBLEM OR QUESTION: The effects on practice patterns, health care costs, clinical judgment and evidence based approach to practice are all potentially affected by interactions between representatives of the pharmaceutical industry and physicians. Industry representatives blatantly target physicians in training, including students. The lack of "academic detailing" in most institutions and inappropriate role modeling by academic and community preceptors can allow for unbalanced and potentially deleterious influence on young trainees.

OBJECTIVES OF PROGRAM/INTERVENTION: Our intervention was established to: 1) expose students to two sources of guidelines on physician-industry interactions, 2) determine the degree of exposure to industry detailing students experience when they are in community preceptors' offices and 3) to gauge students' opinion about several aspects of such encounters.

DESCRIPTION OF PROGRAM/INTERVENTION: During the year 3 Continuity Clinic Clerkship (CCC) students are placed in community physicians' offices one half day per week for six months. Students work on several exercises during CCC. One exercise asks students to read guidelines published by the American Medical Association and the Association of American Medical Colleges. They also respond to several questions about the frequency and nature of interactions with industry representatives, their preceptor's attitudes about these interactions and their own assessment of the accuracy and utility of information provided by industry representatives.

FINDINGS TO DATE: Most students were exposed and even targeted by industry representatives when in the office setting. Students report a wide variety of attitudes displayed by their preceptors. Their own attitudes vary widely and are often discordant regarding the belief in the accuracy of information provided and the benefit of having the interaction.

KEY LESSONS LEARNED: Assessing influence on young trainees by industry representatives is important. The targeting of students is widespread. An academic counterpoint to industry detailing out to be provided.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: We will display our tool and the detailed analysis of interpretation of student responses.

STUDENT PERCEPTIONS OF PREPAREDNESS FOR THE CLINICAL SKILLS ASSESSMENT AND STEP 2 CS. T. Uchida¹; A. Ekpennyong². ¹Stroger Hospital of Cook County/Rush Medical College, Chicago, IL; ²Rush University Medical Center, Chicago, IL. (Tracking ID #132502)

STATEMENT OF PROBLEM OR QUESTION: Beginning with the class of 2005, all medical students are now required to take Step 2 CS of the USMLE Boards in which students are evaluated in clinical encounters with standardized patients (SPs). Up until this point, current students at our medical school had very little experience with SP assessments.

OBJECTIVES OF PROGRAM/INTERVENTION: To evaluate students' perceptions of their own preparedness for Step 2 CS and to evaluate their perceptions of the usefulness of a new Clinical Skills Assessment (CSA) in helping them prepare to take Step 2 CS.

DESCRIPTION OF PROGRAM/INTERVENTION: In May of 2004, 118 current third year students participated in the mandatory CSA which consisted of 3 clinical encounters. The format of the CSA was designed to mirror the format of Step 2 CS as closely as possible. Students spent 15 minutes performing a focused history and physical on an SP followed by 10 minutes writing a focused note. Unlike Step 2 CS, the CSA also included 5 minutes of self-assessment and 5 minutes of individualized feedback from the SP both regarding communication/interpersonal skills. Afterwards, 115 students (97%) completed an exit questionnaire from which the following data were obtained.

FINDINGS TO DATE: When asked about their preparation prior to the CSA, 96 students (84%) replied that they had done "no specific preparation." Furthermore, when asked about their plans to prepare for Step 2 CS, 51 (44%) of students stated they planned "no specific preparation." Thirty-nine (34%) students planned to use a board review book specifically aimed at Step 2 CS or OSCE-style tests, 37 (32%) students planned to use discipline-specific board review books, 21 (18%) students planned to enroll in the NBME-sponsored Pre-Implementation Pilot Examination, 13 (11%) students planned to use general textbooks, and 3 (2.6%) students planned to enroll in a test-prep course. Students were also asked to rank their agreement with statements about preparation on a Likert scale from 1 (strongly disagree) to 10 (strongly agree.) When asked to rank "I feel adequately prepared to do well in performing a focused history and physical of a given chief complaint," the mean score was 7.7 (std dev 1.5). For "I feel adequately prepared to do well on the communication/interpersonal skills," the mean score was 8.3 (std dev 1.8). For "I feel adequately prepared to do well in writing a focused note," the mean score was 7.7 (std dev 1.6). The mean score for "This CSA helped me prepare for USMLE Step 2 CS" was 8.4 (std dev 1.8), and the mean score for "I feel adequately prepared to do well on USMLE Step 2 CS" was 7.5 (std dev 1.9).

KEY LESSONS LEARNED: Students are confident in their preparedness for the new Step 2 CS portion of the USMLE Boards including the history taking, physical examination, communication/interpersonal skills and note-writing components. The vast majority of students did no specific preparation prior to the CSA, and a significant minority planned no specific preparation prior to Step 2 CS. Despite their confidence, however, in this group with minimal previous experience with SP assessments, the students overwhelmingly felt that the CSA was beneficial in helping them prepare for Step 2 CS. We plan to continue to offer and expand the CSA experience for our students.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: poster.

SYSTEMS-BASED WHAT?! A RESIDENT-DESIGNED INTERNAL MEDICINE CURRICULUM ADDRESSING THE SIXTH CORE COMPETENCY, SYSTEMS-BASED PRACTICE. A.L. Misakian¹; J.E. Adams¹; A.J. Huang¹; D. Bilbao¹; C. Lai¹; K. Julian¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #132620)

STATEMENT OF PROBLEM OR QUESTION: In 1999, the Accreditation Council for Graduate Medical Education (ACGME) established six core competencies for residency education. The sixth competency, systems-based practice, is defined as "an awareness of and responsiveness to the larger context and system of healthcare, and the ability to effectively call on system resources to provide care that is of optimal value." While clinical practice is affected by these issues, residency programs struggle with designing curricula to meet this broad competency.

OBJECTIVES OF PROGRAM/INTERVENTION: (1) To raise resident awareness of the healthcare system and its impact on the management of patients. (2) To create a sustainable curriculum that addresses the ACGME competency, systems-based practice.

DESCRIPTION OF PROGRAM/INTERVENTION: Four University of California, San Francisco (UCSF) Primary Care Internal Medicine residents underwent a four-week web-based course on healthcare systems and curriculum design. The course was created by the Partners for Quality Education/Achieving Competence Today Initiative, a program funded by the Robert Wood Johnson Foundation. Prior to developing our new curriculum, we surveyed residents and faculty; both groups perceived a curricular deficiency in the area of healthcare systems. In particular, residents ranked healthcare financing, cost-control mechanisms, and regulation/legislation as high priority topics. Based on this needs assessment, we developed a series of seminars including: (1) an overview of the U.S. healthcare system; (2) a case-based discussion on the financial incentives and consequences of different payer mechanisms; (3) a review of health insurance plans available to medical residents; and (4) an interactive seminar on public healthcare payment systems emphasizing current legislative proposals and the providers' role in reform. Each seminar was developed and facilitated by an interested resident with a faculty expert. These sessions were incorporated into an interdisciplinary conference series on chronic disease management, with trainees from the UCSF nursing and pharmacy schools.

FINDINGS TO DATE: Preliminary data from this pilot curriculum suggests that participants (N=9) feel that knowledge of healthcare financing is important and relevant to their clinical practice. Based on a 4-point Likert scale, 89% of learners indicated that the sessions were a valuable use of their time, with 80% reporting that the sessions on healthcare financing should be a required element of residency training. Participants were enthused about learning the material in an interactive, case-based format, and requested further discussion on the history and current state of healthcare financing. Learners also reported that the review of health plans guided their choices regarding their own health insurance and created a relevant context in which to understand payer mechanisms.

KEY LESSONS LEARNED: When presented in an interactive, case-based format, participants consider healthcare financing to be an important part of clinical training.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Oral or poster presentation.

TALES OF THE CAMBRIDGE INTEGRATED CLERKSHIP. B.R. Ogur¹; D. Hirsh¹; N. Baumer¹; C. Casey¹; J. Epstein¹; D. Foster¹; N. Himes¹; J. Radesky¹; J. Siegel¹; J. Wright¹; D.H. Bor¹. ¹Harvard Medical School, Cambridge, MA. (Tracking ID #135375)

STATEMENT OF PROBLEM OR QUESTION: Traditional inpatient-based, third-year medical students' interactions with patients are bounded by organizational disconnects: divisions by specialty, patients' transitions between inpatient and outpatient care, with limited time for students to connect with patients. Nor does the structure support long or deep relationships with experienced faculty. Most learning experiences are ad hoc. Planned curriculae are discipline-specific and presented in an immersive and non-developmental manner.

OBJECTIVES OF PROGRAM/INTERVENTION: The goal is to provide the core clinical education essential to the professional development of every physician regardless of his or her eventual choice of specialty. These skills are developed through longitudinal connection of students with patients and faculty and through a planned, developmentally-progressive curriculum.

DESCRIPTION OF PROGRAM/INTERVENTION: The Cambridge Integrated Clerkship is an innovative re-design of the core clinical year, structured to teach core Medicine, Surgery, Pediatrics, Obstetrics and Gynecology, Psychiatry, Radiology, and Neurology in a longitudinal, integrated way. Students are connected with patients through "whole illness episodes". The student meets the patient early in the illness, thus is able to participate in initial evaluation, problem formulation, and differential diagnostic evaluation. Following patients longitudinally across disciplines and venues where the patient receives important aspects of care, the student is able to follow therapeutic decision-making and

the patient's experience over the course of the illness. Experiencing whole illness episodes powerfully motivates students to learn, teaches diagnostic reasoning in real time, and connects them deeply to their patients in a way which promotes understanding and professional growth. Students also have year-long connection with experienced preceptors in all of the major disciplines. They learn through the collaborative management of a carefully-constructed patient cohort. Over time, preceptors guide students' learning and skill acquisition in an individualized and developmentally-appropriate way.

FINDINGS TO DATE: 8 students have participated in the pilot project which began in July, 2004. Quantitative and qualitative evaluation of the program is ongoing. We will present several examples of student experiences in following patients which illustrate innovative components of the Cambridge Integrated Clerkship. These components include deep, longitudinal connection to patients and families, following patients through all venues and disciplines of care, and experiencing "whole illness episodes" from initial diagnostic reasoning through to follow-up. Our anecdotes will illustrate both the structural aspects of the program which facilitate following "whole illness episodes" and also students' reflections on the value of this innovative process. We will also present examples illustrative of the value of longitudinal mentoring and of the structured, developmental curriculum.

KEY LESSONS LEARNED: Fundamental clinical knowledge, skills and attitudes can be effectively learned in a longitudinal integrated structure. Students find that following patients through "whole illness episodes," longitudinally and across venues of care is highly-motivating and deeply-involving. Longitudinal mentoring provides the opportunity for creating a developmentally-progressive educational experience and is deeply satisfying to students and faculty.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Representative student stories will be presented to illustrate the major innovative aspects of the program.

TEACH THE TEACHER: SUCCESSFUL DISSEMINATION OF A RESIDENT TEACHING SKILLS CURRICULUM. L. Bensinger¹; Y. Meah¹. ¹Mount Sinai School of Medicine, New York, NY. (Tracking ID #132689)

STATEMENT OF PROBLEM OR QUESTION: Resident as Teacher programs have been rapidly gaining the interest of residency program directors and faculty at academic medical centers that do not already have an established teaching curriculum for their residents. This is primarily due to the advent of the ACGME competency requirements and the recognition that residents are often the primary teachers at of interns and medical students. Our goal is to bring a 7-hour multi-specialty resident teaching skills curriculum to all residents at The Mount Sinai School of Medicine and its 12 affiliate institutions. The Teach the Teacher program was designed to successfully implement the curriculum on this large scale.

OBJECTIVES OF PROGRAM/INTERVENTION: Our primary objective is to design a faculty development program that enables faculty at MSSM and affiliate institutions to implement, effectively teach, and sustain a resident teaching skills curriculum for their own housestaff. The program was also designed to provide ongoing teaching skills training, resources, evaluation and mentoring for the involved faculty.

DESCRIPTION OF PROGRAM/INTERVENTION: In February 2004, we piloted the first Teach the Teacher course at MSSM. This intensive, 2-day faculty development program included 13 participants, representing 6 institutions and 7 different specialties. Faculty were placed in the roles of "residents" on the first day to experience the Resident Teaching Development Program (RTDP). On day 2, the faculty reviewed effective teaching methods, practiced teaching, received feedback, and created concrete implementation plans. They were also provided with resources such as a detailed teacher's guide, a group list-serve, and an extensive selection of web-based references.

FINDINGS TO DATE: We used qualitative and quantitative research techniques to evaluate the efficacy of the program. After completing the course, 100% of participants felt prepared to teach the RTDP to their residents. Six months after the course, 4 faculty had taught the 7-hour curriculum to their residents in its entirety while 5 others had taught a shortened version. Overall, 9 of the 13 faculty (69.2%) had taught some version of the course. Two more faculty had plans to begin teaching by 8 months post-course.

KEY LESSONS LEARNED: Resident teaching skills courses are a rapidly growing and vital component of graduate medical education, yet few faculty have the training to implement and teach such programs. The Teach the Teacher program is proving to be an effective means of training faculty to establish a resident teaching skills program in their own departments. In the next year we will conduct site visits to directly observing the faculty teach the program at their own institutions. This program will be continued yearly with the hope of developing a cadre of medical educators dedicated to enhancing teaching excellence at MSSM.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Trigger tapes used in the course displayed via laptop, written teacher's guide provided for course participants.

TEACHING MEDICAL STUDENTS TO INCREASE FEEDBACK FROM INSTRUCTORS: A BOTTOM-UP COMMUNICATION INTERVENTION. F. Milan, L. Dyche¹. ¹Albert Einstein College of Medicine, Bronx, NY. (Tracking ID #135026)

STATEMENT OF PROBLEM OR QUESTION: Timely and specific feedback is essential for skill building. However, despite aggressive faculty development on the topic, inadequate feedback remains a primary student concern about clerkship

rotations at Albert Einstein. Can students learn to help their teachers give them more feedback?

OBJECTIVES OF PROGRAM/INTERVENTION: 1. Provide students an opportunity to discuss the feedback experience. 2. Improve student attitudes toward receiving feedback. 3. Increase the frequency of feedback seeking behavior.

DESCRIPTION OF PROGRAM/INTERVENTION: One and one half hour interactive discussions with small groups of students early in their medicine rotation. The facilitators help students conceptualize the attributes of useful feedback, identify the psychological barriers that can arise for both the teacher and the learner, and develop approaches to obtaining feedback in the often harried environment of the ward. The exercise includes a role play. Specific topics include: 1. Properties of useful feedback (timeliness, balance, specificity) 2. Psychology of receiving feedback (fear of being criticized, of being a pest, of adversely affecting grade) 3. Psychology of giving feedback (fear of hurting feelings, difficulty formulating impressions) 4. Logistics of feedback on wards (rushed pace, lack of privacy) 5. Strategies for obtaining feedback on wards (feel entitled, ask, have questions ready, give teachers homework).

FINDINGS TO DATE: Evaluation method: controlled intervention study and class evaluations. Subjects: AECOM 3rd year class of '04 (n=116), 1/2 intervention group, 1/2 control. Outcome measure: Feedback Assessment (attitudes toward feedback, frequency of seeking feedback). Also student evaluations of class clarity and relevance on 5 item Likert scale with 5 as the highest rating. Findings: While attitudes were not significantly changed by intervention, attitudes did show a significant association with feedback-seeking from faculty and house staff (p = <.01). Students reporting never having asked for feedback went from 39% to 3.5% pre and post intervention (p = <.01). At the end of the rotation, the control group reported less feedback seeking behavior (6.4%) than the intervention group (19.7%) which trended toward significance. Class evaluations were strongly positive (mean = 4.67).

KEY LESSONS LEARNED: 1. Students respond energetically to the opportunity to discuss the topic. 2. Class associates with significant increases in feedback-seeking behavior. 3. Class is favorably evaluated by students.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster and edited video tape of the class.

TEACHING PULMONARY AUSCULTORY SKILLS TO MEDICAL STUDENTS IN INTRODUCTION TO PHYSICAL EXAM: A COMPUTER AND CASE BASED CURRICULUM. J. Jevtic¹; D. Torre¹; J.L. Sebastian¹; D. Bragg¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #133032)

STATEMENT OF PROBLEM OR QUESTION: Successfully teaching the skill of pulmonary auscultation in the classic lecture-discussion (LD) and small group model is associated with multiple barriers including patient recruitment issues, changing auscultatory findings, and the inability to objectively assess the knowledge and skills attained by student learners.

OBJECTIVES OF PROGRAM/INTERVENTION: To improve the knowledge and skills of second year (M2) students in pulmonary auscultation and assess their ability to accurately identify normal and abnormal lung sounds using PneumoSim[®], a computerized lung sound simulator.

DESCRIPTION OF PROGRAM/INTERVENTION: Following a one hour lecture on pulmonary exam and a small group session with patient interactions, we used the PneumoSim[®] to introduce a case-based curriculum that highlighted abnormal pulmonary auscultatory findings associated with six common pulmonary diseases (asthma, pneumonia, pleural effusion, pneumothorax, heart failure and COPD). The PneumoSim[®] produces computer generated auscultatory sounds that are heard with infrared stethoscopes and in our module was used synchronously with the case based curriculum. Students (n=196) underwent a knowledge quiz to test their ability to accurately identify normal and abnormal lung sounds prior to and after the module. M2 students also completed a pre/post self assessment rating their confidence in their ability to identify lung sounds (1 = not confident; 6 = very confident). Paired t-tests were used to examine differences between pre and post student ratings and knowledge quizzes.

FINDINGS TO DATE: Student's mean score on the knowledge quiz was 70% (+/-23) prior to exposure, rising to 94.6% (+/-11.5) after the PneumoSim[®] curriculum (p < .001). M2 students reported that their ability to identify normal breath sounds, wheezes, crackles, pleural friction rub, rhonci and egophony all significantly improved after their exposure to the PneumoSim[®] curriculum (p < .001). The majority of students felt that this portion of the curriculum contributed positively to their ability to accurately identify common lung sounds and should be used in future teaching sessions beyond the physical exam course. This correlates with our previous findings using the PneumoSim[®].

KEY LESSONS LEARNED: Results of our pre/post knowledge exam indicated that M2 students significantly improved their ability to accurately identify normal and abnormal lung sounds after exposure to the PneumoSim[®] curriculum. As well, students perceive their exposure to the computer based PneumoSim[®] curriculum to be a positive experience despite the lack of face-to-face patient contact.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:

TEACHING RESIDENTS ACGME COMPETENCIES: A CURRICULUM BASED ON CHRONIC ILLNESS CARE, PATIENT SAFETY, AND HEALTH ECONOMICS. J. Voss¹; M.L. Plews-Ogan¹; M. Nadkarni¹; A.M. Wolf¹; N. May¹; J.B. Schorling¹. ¹University of Virginia, Charlottesville, VA. (Tracking ID #136241)

STATEMENT OF PROBLEM OR QUESTION: One of the many challenges faced by US residency programs as they develop methods to teach the ACGME com-

petencies is making the new topics relevant for learners who are focused on medical knowledge and patient care skill acquisition.

OBJECTIVES OF PROGRAM/INTERVENTION: As part of our internal medicine residents' ambulatory training, we are implementing a curriculum to address systems-based practice, practice based learning, interpersonal and communication skills, and professionalism.

DESCRIPTION OF PROGRAM/INTERVENTION: Our program is grounded in three content areas: chronic illness care, clinical health economics, and patient safety. Learners participate in didactic and experiential workshops and then independent projects: 1) a root cause analysis of a medical error, an appropriate intervention, and a follow-up assessment of its effectiveness; and 2) an analysis of one aspect of their practice followed by an intervention and evaluation. Residents also create videotapes documenting their increased competency in motivational interviewing with chronically ill patients. One goal of the curriculum is to encourage residents to participate meaningfully in systems-based approaches to a range of clinically relevant problems.

FINDINGS TO DATE: To date, all categorical and primary care track residents have participated in the prescribed hands-on exercises. For example, all 2nd-year residents have successfully conducted a root cause analysis of a medical error, presented their findings to the Patient Safety Committee, and designed an intervention implementation plan. Residents have also completed chart reviews and presented findings in preparation for their individual quality improvement projects. Learners have responded enthusiastically to the opportunity to conduct real-time studies that address important patient care issues.

KEY LESSONS LEARNED: Using content areas (i.e., patient safety, chronic illness, health economics, and quality improvement) that reflect faculty interest and expertise allows for 1) meaningful, relevant learning experiences; 2) resident exposure to systems-based practice in a real setting; 3) use of experienced, engaged faculty; and 4) a synergistic effect among faculty and learners. Areas that need to be addressed are 1) faculty development to expand core of teaching faculty and role models in practice; 2) on-line database to allow residents easy access to data on their patient populations; and 3) a quality method of competency assessment.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster and laptop.

TEACHING RESIDENTS TO PARTNER WITH PATIENTS TOWARD EFFECTIVE BEHAVIOR CHANGE. S. Marcotullio¹; J. Tomaro¹; C.E. Reese¹. ¹Mercy Hospital of Pittsburgh, Pittsburgh, PA. (Tracking ID #134293)

STATEMENT OF PROBLEM OR QUESTION: Residents did not demonstrate awareness of, or comfort or confidence in using, effective behavior change strategies to help patients modify risk behaviors.

OBJECTIVES OF PROGRAM/INTERVENTION: 1. Prepare residents to quickly and accurately identify patients' stage of readiness to change risk behaviors. 2. Prepare residents to utilize effective behavior change strategies to partner with patients toward behavior change. 3. Create a "culture" of positive attitudes among residents and faculty regarding the possibility of change and self-determination in an inner-city clinic population.

DESCRIPTION OF PROGRAM/INTERVENTION: We implemented a 6-hour Behavior Change skills curriculum within our Ambulatory Medicine block rotation (1 1-1/2 session per week, for 4 weeks, in each of 3 years of training). Sessions include Introduction/Refresher about Communications Skills, Principles of Behavior Change, and Behavior Changes Tools and Strategies. Residents learn about effective behavior change strategies (determination of readiness to change as well as tools to promote change) through readings, assignments, discussion, and individual/group exercises including role play. Residents apply these strategies to themselves (having self-identified a risk behavior) and their patients in the Health Center (low-income, minority and marginalized patients disproportionately affected by abuse of alcohol, tobacco and drugs; obesity; depression; domestic violence; chronic disease and nonadherence to medications). Senior residents fill important leadership roles in the training program.

FINDINGS TO DATE: Over 70 residents have completed the Behavior Change training modules. Results from a Behavior Change Skills Inventory administered before the training sessions and a post-training survey have identified key areas of strength (using patient-friendly behaviors; checking comprehension) and weakness (structuring time agenda w/patients; asking patients' permission to discuss change) in using behavior change strategies with patients.

KEY LESSONS LEARNED: 1. Fostering behavior change in patients is a teachable skill. 2. It is often difficult for residents to really accept the slow pace of change. 3. Although change strategies sound simple, their successful application takes practice. 4. Leadership, encouragement and role modeling by seniors and preceptors are essential for residents to continue to practice behavior change skills. 5. Review skills periodically to maintain commitment. 6. Celebrate successes.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: - Very brief PPT presentation (<4 slides) - Hand out curriculum description - Use sample role play to structure discussion.

TEACHING THE CHRONIC CARE MODEL: DEVELOPING AN INTERDISCIPLINARY CURRICULUM FOR MEDICAL RESIDENTS, NURSE PRACTITIONER STUDENTS, AND PHARMACY STUDENTS. C. Lai¹; K. Julian¹; E. Wade¹; S. Janson¹; L. Kroon¹; A. Leeds¹; R.K. Oka¹; R.B. Baron¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #132665)

STATEMENT OF PROBLEM OR QUESTION: Given that chronic illness has replaced acute disease as the major cause of disability, the Institute of Medicine and academic societies, such as SGIM, have emphasized training medical residents to provide high-quality chronic care in an interdisciplinary setting.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) To teach and raise awareness of the six themes of the Chronic Care Model (Edward Wagner), including health care delivery system, decision system design, decision support, community resources, self-management support, and clinical information systems; 2) to teach and practice chronic care skills in an interdisciplinary clinic; and 3) to establish a Web-based reference site that includes the syllabus and teaching materials.

DESCRIPTION OF PROGRAM/INTERVENTION: In July 2004, a group of faculty leaders from the Schools of Medicine, Nursing, and Pharmacy developed a year-long, weekly interdisciplinary conference focused on teaching the principles of the Chronic Care Model, followed by a weekly continuity clinic. Primary care internal medicine residents rotate onto ambulatory care rotation every two months, for a total of six months per year, and are joined by nurse practitioner and pharmacy students. Each two-month block is focused on one of three chronic diseases: chronic lung disease, depression, and chronic cardiac disease. Each hour-long interactive conference is based on one of the six Chronic Care Model elements and is led by faculty or residents. Examples of seminars have included the following: 1) goals of the Chronic Care Model, 2) interdisciplinary team care, 3) skills workshops for smoking cessation and inhaler use, 4) evidence-based management of asthma, and 5) role of behavioral health care organizations.

FINDINGS TO DATE: Overall quantitative and qualitative feedback about the interdisciplinary chronic disease curriculum has been uniformly positive. On a 1 (poor) to 5 (outstanding) Likert scale, learners rated the seminars as being "excellent" to "outstanding" by all measures. Specifically, average ratings were the following: 4.26 in "success in achieving seminar objectives," 4.20 in "relevance of topic to interdisciplinary chronic disease care," and 4.36 in "quality of presentation."

KEY LESSONS LEARNED: Based on preliminary data, a well-structured, learner-centered curriculum can increase health professional trainees' awareness of chronic care principles.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster, syllabus, static Web display.

TEACHING TRANSLATION: FORMALLY INCORPORATING BASIC SCIENCE INTO A THIRD-YEAR MEDICAL SCHOOL CURRICULUM REINFORCES LESSONS LEARNED AT THE BENCH. D.M. Elvin¹; D.A. Hirsh¹; B.R. Ogur¹; P.A. Cohen¹; D.H. Bor¹. ¹Harvard Medical School and Cambridge Health Alliance, Cambridge, MA. (Tracking ID #135914)

STATEMENT OF PROBLEM OR QUESTION: Third-year students enter an exciting phase of training which is based on bedside learning. Currently, basic science curricula establish a sound foundation upon which clinical knowledge can be built. However, for some students the basic sciences seem a "prelude" to the clinical rotations and not an ongoing longitudinal focus. Scientific progress continues at a remarkable pace, and future medical breakthroughs will be facilitated by dialogues between basic scientists and clinicians. Can the values and culture of each discipline be inculcated both in students and basic scientists by modeling successful integration in small-group facilitated tutorials in the third year?

OBJECTIVES OF PROGRAM/INTERVENTION: To create a basic science curriculum to support and guide the tutorials of Harvard medical school's novel Integrated Clerkship. This curriculum enhances learning of core clinical topics and enables students to acknowledge basic science as practical, relevant, and applicable to their medical career regardless of their future specialty.

DESCRIPTION OF PROGRAM/INTERVENTION: The tutorial meets one half-day each week and is attended by all eight clerkship students throughout the year (44 weeks). For each tutorial, predetermined topics and objectives are established based on a literature review and clerkship curriculum guidelines from national organizations representing each required specialty. The basic science correlates are selected based on relevant pathophysiology, promising research directions in the field, a review of problem-based learning cases from years 1 and 2, and interest on the part of basic sciences faculty. Prior to the session a scientist is invited to attend as an expert and tailors their contribution based on details of the clinical case being presented by the student. During the tutorial, the scientist is given protected time to present material and discuss questions. For example, an expert in the genetics of murine lymphatic development participated in the tutorial covering lower extremity edema.

FINDINGS TO DATE: Preliminary results demonstrate a high degree of satisfaction with this learning method. Basic scientists express enthusiasm for the format and student participation is excellent. Quantitative and qualitative evaluation of the program is ongoing and will be presented.

KEY LESSONS LEARNED: Basic science naturally favors depth over breadth. By committing to a scientist from one field, others from equally relevant fields are excluded. The drive of the clinical student to gain a "general understanding" of disease pathophysiology contrasts with efforts of scientists who specialize in specifics. However, both sides of the equation share the goal of serving the needs of our community through systematic, thoughtful exploration. Through integrating basic science into the third year of medical school, future clinicians will better understand mechanisms of disease and be prepared to seize future opportunities to participate in translational science.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster or talk outlining details of results and course.

THE CULTURALLY COMPETENT HISTORY AND PHYSICAL EXAM. E. KröGer¹; G.L. Iacono Russell¹; M. Harris²; Z. Piotrowski¹. ¹West Suburban Medical Center, River Forest, IL; ²West Suburban Medical Center, Oak Park, IL. (Tracking ID #135722)

STATEMENT OF PROBLEM OR QUESTION: The cultural context of a patient strongly influences response to illness and subsequent medical care. Under-

standing a patient's cultural milieu often helps to delineate barriers to medical care. Resident physicians are often confronted with patients from cultures different from their own. As a result, improving their ability to appreciate cultural influences on chronic disease is essential.

OBJECTIVES OF PROGRAM/INTERVENTION: Our goal is to enhance and expand the resident's interviewing skills to allow them to more comfortably and confidently talk with their patients about their cultural/ethnic/religious background and its impact on illness and medical care. Our learning objectives are to positively impact the doctor-patient relationship in a way that: (1) enhances interviewing skills; (2) fosters a stronger doctor-patient bond and (3) encourages a more in-depth understanding of that patient's perspective.

DESCRIPTION OF PROGRAM/INTERVENTION: As part of a Health Resources and Service Administration (HRSA) grant designed to improve resident cultural competence, we have devised a practical methodology that incorporates the expansion of a standard History & Physical (H & P) with information of cultural relevance in the form of a culturally competent document. We have developed a standard template that includes questions which promote insight into the following key dimensions: (a) impact of the illness on the patient's real or perceived functional limitations; (b) impact of the patient's illness in relationship to his/her family or other relevant group and (c) patient's perception of illness in relationship to larger racial, religious or ethnic groups to which he or she belongs. In addition, we assess possible barriers to care including language, religion, financial status and literacy. At the same time, necessary historical and physical information is acquired.

FINDINGS TO DATE: We actively utilized resident and physician monitor feedback throughout the process of developing and revising our culturally sensitive H & P. This allowed for significant discussion by both groups and greater buy-in and acceptance of the process. Our H & P is designed so that the resident physician gives ratings to inquiries on a likert-type scale. In order to make those ratings, the resident must ask appropriate questions of the patient. We designed a set of questions to serve as a flexible guide in order to accommodate the individual style of the interviewer rather than rigid questions that might stifle natural flow. In order to help the resident with these interviewing skills we developed a training tape of a doctor-patient interview utilizing the new H & P format. The residents watched and critiqued this tape prior to using the new format with their patients.

KEY LESSONS LEARNED: Expanding a familiar tool in medicine to ask key cultural information can enrich the depth of the standard H & P. By doing so, cultural barriers that separate resident physician and patient can be bridged. This approach incorporates the process of cultural competency into every day care rather than marginalizing its importance.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Our presentation will include hanouts of the H & P format, the training curriculum for enhancing resident interviewing skills and a video of a mock interview.

THE USE OF TISSUE MODELS FOR ULTRASOUND-GUIDED VASCULAR ACCESS TRAINING. B.T. Rosen¹; M.J. Ault¹; B. Ault¹. ¹Cedars-Sinai Medical Center, Los Angeles, CA. (Tracking ID #136269)

STATEMENT OF PROBLEM OR QUESTION: Despite the widespread recognition that complications from all vascular access procedures could be minimized with the use of ultrasound-guidance and wide sterile barriers, no standardized or convenient mechanism currently exist for how to teach procedure proper procedural techniques to physicians.

OBJECTIVES OF PROGRAM/INTERVENTION: 1. To display a non-human tissue model which can be used for vascular access and wide sterile barrier training; 2. To allow participants to practice cannulating a simulated vessel under direct ultrasound visualization; 3. To review proper wide sterile barrier techniques.

DESCRIPTION OF PROGRAM/INTERVENTION: The tissue model is constructed from a whole plucked chicken, obtained from a local grocery store, into which we insert several strips of 0.2mm thickness rubber tubing that runs lengthwise through the chicken. Each length of tubing, or "vessel", is sealed with valves at both ends and filled with pressurized, colored tap water to simulate blood. A fully operational, portable ultrasound machine will be positioned at the station with a vascular access probe. Participants will be able to visualize the simulated vessels with ultrasound and to actually practice cannulating vessels under direct ultrasound visualization. Lastly, we will review techniques associated with establishing and maintaining a wide sterile barrier, including the placement of a sterile sleeve over the ultrasound probe so it can be used during the sterile procedure.

FINDINGS TO DATE: 136 individuals thus far have participated in the vascular access educational sessions that we have conducted at our Institution, Cedars-Sinai Medical Center. The majority (90%) of participants were residents in the Internal Medicine training program, while 10% of participants were attending physicians from a variety of specialties. 25.4% of the participants had used ultrasound guidance for central line placement at least once, and only 4% of participants use ultrasound consistently. Comparing mean pre-course and post-course comfort levels for the residents, comfort level with wide sterile barriers (WSB) increased from a baseline of 3.2/5.0 to 4.4/5.0 ($p < 0.0001$), and comfort level with the use of ultrasound for the placement of CVCs improved from 1.8/5.0 to 3.8/5.0 ($p < 0.0001$). Attending physicians' comfort level with WSB improved from a pre-course value of 2.8/5.0 to 4.5/5.0 ($p < 0.002$), while their comfort level with the use of ultrasound for the placement of CVCs improved from 2.4/5.0 to 4.1/5.0 ($p < 0.001$). The overall image quality was rated 3.8/5.0 and the model effectiveness for practicing wide sterile barrier was rated 4.8/5.0. The value of the educational session was rated 4.9/5.0, which was independent of the participant's level of training.

KEY LESSONS LEARNED: 1. Our tissue model is an effective way to introduce physicians to the use of ultrasound and proper wide sterile barrier techniques; 2. Participants appreciate an opportunity to practice cannulating vessels under direct ultrasound guidance; 3. Our educational sessions have been rated as highly worthwhile irrespective of the participants' level of training or the number of central lines they had placed prior to the sessions.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: 1. Non-human tissue models, in the form of whole plucked chickens that have been modified (as described above); 2. Portable ultrasound machines; 3. Central venous catheter kits; 4. An assortment of materials with which participants can review and practice wide sterile barrier techniques.

TOWARDS AN INFORMAL CURRICULUM THAT TEACHES PROFESSIONALISM: A TWO-YEAR PROGRESS REPORT ON CASCADING CHANGE AT IUSM. A. Suchman¹; P.R. Williamson²; R.M. Frankel³; D.K. Litzelman³; D.L. Mossbarger⁴; T.S. Inui⁴. ¹Relationship Centered Health Care, Rochester, NY; ²Relationship Centered Health Care, Baltimore, MD; ³Indiana University Purdue University Indianapolis, Indianapolis, IN; ⁴Regenstrief Institute, Inc., Indianapolis, IN. (Tracking ID #135650)

STATEMENT OF PROBLEM OR QUESTION: The most carefully-constructed formal curriculum on professionalism can be undermined if the informal curriculum (the social environment of the school) does not consistently exhibit appropriate professional values. But how can schools foster change in the relationships and interactions of thousands of people?

OBJECTIVES OF PROGRAM/INTERVENTION: The Relationship-Centered Care Initiative (RCCI) is a three-year effort at the Indiana University School of Medicine (IUSM) to initiate self-sustaining culture change and align the informal curriculum with the competency-based formal curriculum in domains of professionalism, communication, ethics and self-awareness by fostering mindfulness of the values being conveyed in everyday personal interactions and organizational behavior.

DESCRIPTION OF PROGRAM/INTERVENTION: In approaching such large-scale organizational change, we could not design and impose a culture, nor could we "process" every member of the organization through an intervention to change their behavior. We abandoned such engineering metaphors adopting instead one of "ripples spreading in a pond". We could design a first step – the stone dropped in the water – but only after it was completed could we discern the next steps, based on who became involved and what ideas they developed. As a first step (described elsewhere) 12 student and faculty volunteers conducted 80 appreciative interviews to elicit stories of IUSM's culture at its best. These stories were analyzed, and the themes presented back to the community in a number of venues. As a fresh and positive image of IUSM emerged, more people volunteered to bring the RCCI to their departments, committees, offices or projects. This group of voluntary internal change agents meets monthly for peer-coaching and instruction on change agency in support of dozens of projects they have initiated. We have also initiated activities to engage more medical students (with a chain of student-to-student interviews), residents (using noon conferences on professionalism as a venue for interviews) and alumni (with planned alumni-student interviews at an upcoming reunion).

FINDINGS TO DATE: The "rippling" of engagement has been striking, with the volunteers now numbering > 130. Examples of cascading change include: (1) new criteria and interviewing methods to select relationship-oriented applicants, (2) feedback on the work environment added to performance reviews of department chairs, (3) presentation of a student-published book of student stories to incoming students at the White Coat Ceremony, (4) replacement of a form letter with a more relational approach in response to poorly-rated courses, (5) adoption of practices to "humanize" committee meetings (e.g.: checking-in and appreciative debriefings).

KEY LESSONS LEARNED: Storytelling can increase mindfulness and inspire individual behavior change. The strategy of facilitated but non-designed cascading change appears to be succeeding. Since the initial interviews, most subsequent projects have been initiated by volunteers – the result of serendipitous events and not a priori design. Many changes (eg: student selection, department chair reviews and committee procedures) may be self-sustaining and have effects that amplify over time.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Dissemination-of-change map (a hyperlink-annotated PowerPoint presentation).

USE OF THE INTERNET AND THE POWER OF REPETITION TO TEACH CARDIAC AUSCULTATION. T. Seto¹; M. Barrett²; K. Thomas²; A. Rodd-Mackenzie². ¹Olive View - UCLA Medical Center, Sylmar, CA; ²Drexel University College of Medicine, Philadelphia, PA. (Tracking ID #132741)

STATEMENT OF PROBLEM OR QUESTION: The ability of medical students and residents to recognize heart sounds is poor (21%). This study was designed to test a better method of teaching cardiac auscultation, and see if it could be used over the internet.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) demonstrate an effective method of teaching cardiac auscultation; one that would result in improved proficiency in recognizing basic heart sounds 2) deliver this training to students in a simple and cost-effective method via the internet.

DESCRIPTION OF PROGRAM/INTERVENTION: Seventy-nine first and second-year medical students at an urban east coast medical school were randomized to one of 3 groups. Using simulated heart sounds, the groups received auditory training on 5 basic left-sided cardiac murmurs (AS, AR, MS, MR, Innocent Murmur) and 2 extra heart sounds (S3, S4) in the following manner: 1) Group A was given a one-hour audio CD with 250 repetitions of each of the 7 basic heart

sounds. 2) Group B was given the internet address to download a single digital audio file (.mp3) with content identical to the CD given to Group A. Members were instructed on downloading the audio file. 3) Group C was given a CD containing only the didactic material found in the audio content from Groups A and B. No heart sounds were included. Students were instructed to listen to their CDs or digital audio files exactly twice. Third-year students were selected as controls and did not receive either intervention. For Groups A through C, proficiency in cardiac auscultation was assessed before and after the intervention using a computer program which displayed a graphic of a stethoscope placed on a human torso and played 10 human heart sounds in random order. The control groups were only tested once since there was no intervention.

FINDINGS TO DATE: At completion of the study 154 students were available for analysis. Twenty-two students were assigned to Group A, 41 were assigned to Group B, and 16 students were assigned to Group C. Seventy-five students were available as controls. Group A improved from a pre-test score ($M \pm SD$) of $21.8\% \pm 11.4$ to a post-test score of $60.9\% \pm 23.7$. Group B improved from $30.0\% \pm 13.6$ to $62.0\% \pm 22.4$. Group C had a mean score of $39.4\% \pm 18.4$. Repetitive heart sounds were used in Groups A and B, and these two groups were not statistically different. Furthermore, there was a statistically significant increase in scores in both groups ($p < 0.001$ and 0.005 , respectively). Groups C and the control group both did not use repetitive heart sounds and were not statistically different. Group C, however, showed no significant increase in scores. Finally, groups that used repetitive heart sounds were statistically different from not only the control group, but also the combined groups that did not use repetitive testing. ($p < 0.01$).

KEY LESSONS LEARNED: 1) Cardiac auscultation is a clinical skill that can be mastered through use of intensive repetition of simulated heart sounds 2) Improvement in ability to recognize heart sounds was similar among students using a web-based digital audio file and an audio CD in an unmonitored setting 3) The use of simulated heart sounds in training translates into ability to recognize human heart sounds.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: A display allowing guests to try the computerized testing used in the study and compare their scores with others. A 10 minute presentation using didactic material and repeated heart sounds to demonstrate our method using a basic cardiac murmur (best with a small group in a quiet room and an adequate sound system to reproduce low-frequency sounds).

USE OF TRIGGER CASES AND PEER TEACHING TO DISSEMINATE SKILLS TO IMPACT HEALTH DISPARITIES. S. Chheda¹; C.L. Bell²; L.C. Dast²; B. Schwanke²; J. Banning²; S. Dotti¹. ¹University of Wisconsin-Madison, Madison, WI; ²University of Wisconsin Medical School, Madison, WI. (Tracking ID #135670)

STATEMENT OF PROBLEM OR QUESTION: The Institute of Medicine in 2002 proposed that healthcare provider factors contribute to health disparities and recommend that medical educators implement and evaluate curriculum to address these issues. Challenges to successful implementation of curriculum on healthcare disparities include: establishing relevance to medical students; need for small groups to enable meaningful discussion; and lack of multiple faculty "experts" to teach these issues.

OBJECTIVES OF PROGRAM/INTERVENTION: The objective of the program was to provide an interactive workshop to allow third year medical students to build skills in identifying and responding to issues that arise in the clinical setting that may contribute to healthcare disparities.

DESCRIPTION OF PROGRAM/INTERVENTION: We conducted a needs assessment of our 150 third-year medical students to determine their knowledge and comfort level regarding health disparities. One hundred thirty six (90%) of students completed the needs assessment. Based on these findings, we developed specific learning objectives and conducted and evaluated a three-hour workshop on skills to impact health disparities. Student learning objectives included: 1) openly acknowledging a patient's concerns regarding trust; 2) re-establishing rapport with a patient after an unintentional "faux pas"; 3) suggesting a patient change a cultural practice that may impair his/her health; 4) responding appropriately to demeaning comments from a patient; and 5) working effectively with patients with limited English proficiency. One hundred and fifty students participated in the three-hour workshop that consisted of a 45-minute didactic session, a 60-minute student-only small group (6 students), and a 75-minute peer-teaching small group (12 students) with faculty present. The student-only group session consisted of three brief-trigger scenarios where students interacted with standardized patients and interpreters. Each trigger scenario was followed by a worksheet-based discussion. The trigger scenarios were developed from actual patient encounters and were used to establish relevance to students and stimulate discussion. In the peer-teaching small group, two student-only small groups that had different trigger scenarios joined to instruct each other and discuss important learning points.

FINDINGS TO DATE: Students rated the curriculum highly on a 7-point Likert scale (mean 2.05, SD 1.11). The use of trigger cases (mean 1.65, SD 0.96) and student-only groups (mean 1.65, SD 0.96) received the highest satisfaction ratings. The didactic presentation received the lowest satisfaction rating (mean 2.89, SD 1.52). Most student comments were positive and included: "This was the best activity I experienced in medical school". Students' comments also revealed that worksheets were cumbersome. Pre- and four-month post-workshop knowledge assessment and skill comfort ratings will be presented.

KEY LESSONS LEARNED: Trigger scenarios followed by worksheet based student-only discussions and subsequent dissemination of learning points to peers with faculty present composed a well-received workshop on skills to impact health disparities. Worksheets should be revised to contain fewer questions. This format allows for establishing relevance of skills to students, small group discussions and the use of faculty to facilitate process without being required to assume an "expert role".

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: In addition to a poster presentation, copies of the needs assessment tool, trigger scenarios and worksheets will be displayed.

USE OF VISUAL ART TO ENHANCE INSTRUCTION OF THE GENERAL APPEARANCE COMPONENT OF THE PHYSICAL EXAM. R. Vedanthan¹; A. Antman²; J. Katz¹; E. Alexander¹. ¹Brigham and Women's Hospital, Boston, MA; ²Harvard University, Boston, MA. (Tracking ID #134709)

STATEMENT OF PROBLEM OR QUESTION: The General Appearance portion of the physical exam is an important diagnostic tool for the experienced clinician. The analysis of visual art utilizes many of the same visual observation skills that are required to gain expertise in the General Appearance exam. Thus, this study aims to determine if the use of visual art, as a pedagogical tool, can enhance the instruction and performance of the General Appearance portion of the physical exam.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) Utilize visual art as a pedagogical tool to enhance visual observation skills of 3rd-year medical students 2) Enhance instruction of the General Appearance portion of the physical exam 3) Increase students' descriptiveness and accuracy of General Appearance statements, and their comfort level and confidence in describing the General Appearance portion of the physical exam.

DESCRIPTION OF PROGRAM/INTERVENTION: 3rd-year medical students enrolled in the Core 1 Medicine Rotation at the Brigham and Women's Hospital are offered this innovative curriculum. The structure of the 3-month rotation is such that each batch of students can be divided into 2 groups: those who receive the intervention in the first month ("early" intervention group), and those who receive the intervention in the second month ("late" intervention group). The intervention consists of a visual art-based lecture that utilizes visual art—specifically, portraits from different time periods, media, and artists—to highlight the importance of visual observation skills, and to enhance students' ability to draw connections between specific visual findings and a larger theme or meaning to the image in question. The final portion of the intervention involves going to the patient's bedside to apply the newly enhanced skills of visual observation to the General Appearance of the patient. Throughout the rotation, students' admission notes are routinely collected. General Appearance statements from these admission notes have been extracted. They are analyzed according to a content analysis-based scoring system for descriptiveness. In addition, all students are exposed to "standardized patient images," and the General Appearance statements from these exercises are also collected and analyzed per the same content analysis-based scoring system for descriptiveness and accuracy. The structure of the course allows us to compare the "early" and "late" intervention groups in a manner that enhances the evaluation of the efficacy of the intervention (i.e. at the midpoint of the rotation, the "late" intervention group serves as a "control group").

FINDINGS TO DATE: Preliminary qualitative analysis of the General Appearance statements has demonstrated the following preliminary findings. First, baseline General Appearance statements are characterized by relatively poor descriptiveness. Second, the General Appearance statements improve in descriptiveness and accuracy over time. Third, at the midpoint, the "early" intervention group's General Appearance statements are generally more descriptive and accurate when compared with the "late" intervention group.

KEY LESSONS LEARNED: 1) Use of a visual art-based curricular innovation does enhance the performance and reporting of the General Appearance portion of the physical exam. 2) This curriculum is well-liked and very popular among the 3rd-year medical students. 3) This curriculum is easy to implement and is additive to the existing physical exam curriculum for 3rd-year medical students.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Oral or poster presentation

USING AN OBJECTIVE STRUCTURED TEACHING EVALUATION (OSTE) FOR FACULTY DEVELOPMENT. M.A. Wamsley¹; K. Julian¹; M.H. Vener¹; E.H. Morrison². ¹University of California, San Francisco, San Francisco, CA; ²University of California, Irvine, Orange, CA. (Tracking ID #135884)

STATEMENT OF PROBLEM OR QUESTION: The quality of clinical faculty teaching is traditionally measured by learner written evaluations. However, evaluations may relate more to a teacher's charisma or communication style than actual teaching skills. Objective Structured Teaching Evaluations (OSTE) allow faculty to practice teaching skills and get immediate feedback on their performance.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) To develop a set of common teaching scenarios and train students to enact these scenarios 2) To provide key clinical faculty an opportunity to practice their teaching skills and receive direct feedback from student participants 3) To determine whether there is a correlation between OSTE performance and teaching evaluations 4) To determine whether faculty participation in an OSTE leads to improvement in teaching evaluations.

DESCRIPTION OF PROGRAM/INTERVENTION: The OSTE is a half-day workshop that consists of four teaching scenarios (orienting a learner, outpatient precepting, bedside teaching, and feedback to a learner in difficulty). Fourth year medical students were recruited and trained to enact a teaching scenario, rate the instructor and provide face-to-face feedback after the encounter. Faculty from Internal Medicine, Pediatrics and Family Medicine were recruited to participate in the exercise. All encounters were videotaped and faculty received direct feedback after each encounter. A one-hour debriefing session emphasized key points and highlighted important aspects of each case. Each faculty participant developed a personal teaching action plan.

FINDINGS TO DATE: Participants completed evaluations rating the OSTE experience. Faculty ($n=8$) reported feedback from the standardized students was helpful (mean = 5.0; 1 = strongly disagree, 5 = strongly agree), felt they were likely to change their teaching practices as a result of the OSTE (mean = 4.75) and rated the overall experience highly (mean = 5.0). Students ($n=8$) reported they learned useful information about teaching through participation in the OSTE (mean = 4.75) and rated the overall experience highly (mean = 4.5). Results from faculty teaching evaluations will be forthcoming.

KEY LESSONS LEARNED: Developing and implementing an OSTE is time intensive, but provides faculty participants with the unique opportunity to practice their teaching skills and receive direct, immediate feedback from students. Faculty believe that OSTE participation leads to changes in their teaching behaviors. Student participants also derive benefit from OSTE participation.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:

USING DOCUMENTARY VIDEOS: TEACHING AND PEOPLE MANAGEMENT SKILLS FOR INTERNS. T. Tanabe¹; C. Goldberg¹. ¹University of California, San Diego, San Diego, CA. (Tracking ID #133595)

STATEMENT OF PROBLEM OR QUESTION: Interns in the clinical training programs play a significant role in teaching of medical students on inpatient rotations. Several studies have shown that graduating medical students feel ill prepared for internship in important areas such as managing conflicts with colleagues, teaching, handling of end-of-life care situations. Additionally, most students receive little formal education in teaching and people management skills before graduation from medical school.

OBJECTIVES OF PROGRAM/INTERVENTION: Documentary videos are particularly powerful tools for allowing medical students to prospectively consider many of the challenges that they will encounter during their Internship. "Teaching and People Management Skills for Interns" is designed for fourth-year medical students. The mission of "Teaching and People Management Skills for Interns" is to create a realistic view of 1) conflict management/professional behavior, 2) giving feedback, 3) facilitating student involvement on ward teams, 4) setting expectations and goals, and 5) communication skills. We developed a facilitator's guide to help educators lead meaningful, reflective, and innovative discussions among the students. The specific learning objectives are: 1) learn to be calm and rational in your responses with colleagues and staff, 2) recognize that feedback is an ongoing process and should be two-way and specific, 3) understand there are many things conspiring against student participation in ward based care-although they are modifiable, 4) make expectations of both teacher and student be clear to both parties, 5) realize the importance of bedside teaching.

DESCRIPTION OF PROGRAM/INTERVENTION: The videos are brief (2-5 minutes) and provocative, designed to capture specific interactions between students, interns, other clinicians, and patients. For example, one scenario shows surgical and medical intern involved in a heated dispute over the appropriate service for a particular patient (conflict management). The videos are embedded within PowerPoint presentation and we present six case scenarios. Subsequent slides pose basic questions that prompt appropriate, open ended discussion about each of the cases. Following the discussion questions, a "Take Home Message" slide reinforces key points identified by the authors. This one-day course is incorporated into a required component of the fourth-year student's curriculum. Six faculty serve as group facilitators.

FINDINGS TO DATE: We have successfully used a similar format (trigger videos followed by small group discussions) to cover clinical care issues. The videos were very effective in making clinical scenarios seem real and applicable to the students' future lives as interns. This is the first year that we will be using this same technique to address communication related issues.

KEY LESSONS LEARNED: 1. "The Teaching and People Management Skills For Interns" is an innovative program designed to prompt thought and discussion about these important and yet overlooked issues. 2. Documentary video is a useful medium to deliver the core message to the fourth-year medical students on topics they most likely would encounter as an intern. 3. This new curriculum's structure grows directly from our recognition that the fourth-year students are poorly prepared to address many of the interpersonal and teaching challenges that they will face as interns. By presenting this material in a non-threatening environment, we expect that our students will be better prepared for these new responsibilities.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Video, poster, and printed material.

USING STANDARDIZED PATIENT WORKSHOPS TO TEACH PARTICULAR HISTORY-TAKING AND COUNSELING SKILLS. A.R. Hoellein¹; J.F. Wilson¹; C. Griffith¹; S.A. Haist¹. ¹University of Kentucky, Lexington, KY. (Tracking ID #133534)

STATEMENT OF PROBLEM OR QUESTION: The University of Kentucky has a long history of incorporating standardized patients (SP) into routine history-taking and physical examination education. However, medical students rarely have the opportunity to learn and practice history-taking and counseling skills specific for other important and perhaps higher stake clinical scenarios.

OBJECTIVES OF PROGRAM/INTERVENTION: The purpose of this project is to enhance the curriculum of the required third-year four-week Internal Medicine Primary Care Clerkship at the University of Kentucky. Several important areas with unique history-taking and counseling skills are currently not specifically or comprehensively addressed yet are prevalent in our society, costly to our health-care system, or otherwise vital to the general internist. Such domains include sexual history-taking/HIV risk reduction (SHHIV), domestic violence (DV), adolescent medicine (AM), geriatric medicine (GM), complementary and alternative medicine (CAM), and nutrition/physical well-being (NPWB).

DESCRIPTION OF PROGRAM/INTERVENTION: Four novel SP Workshops (AM, GM, CAM, NPWB) were developed as part of a new curriculum for a required third-year medical school four-week primary care internal medicine clerkship. These are added to the three validated workshops already in place: preventative care, SHHIV, and DV. The four-hour workshops are interactive with students participating in four SP cases representing different clinical challenges. A faculty preceptor facilitates discussion of the issues raised and consideration of sensitive approaches to the problems instigated by the cases. Students receive the three original workshops and are randomized to participation in two of the new workshops (5 total). All students in every rotation group are assigned the same readings. At the end of the four weeks, students take a 100-item written exam and nine-station SP exam which include open-ended written exercises after the SP encounter. Scores on the domain-specific written exam items, domain-specific clinical exam SP checklists, and open-ended clinical question of new workshop participants and non-participants are analyzed with simple means, standard deviations, and multiple regression approaches solving for USMLE Step I scores and preventative care checklist items.

FINDINGS TO DATE: The AM and CAM workshops have been delivered to three while GM and NPWB delivered to two of the first five rotation groups of the 2004-2005 academic year. AM workshop participants are performing significantly better than non-participants on the SP checklist (89.28 ± 4.98 vs. 85.76 ± 4.42 , $p=.024$) and on the accompanying written exercise (85.0 ± 4.78 vs. 81.24 ± 4.25 , $p=.005$). GM participants also perform significantly better on the SP checklist (91.0 ± 7.53 vs. 80.28 ± 8.12 , $p=.000$). CAM and NPWB display promising trends in the clinical realm but only NPWB exhibits significantly better written exam scores ($60 \pm 20\%$ vs $87 \pm 14\%$, $p<.001$).

KEY LESSONS LEARNED: Even at this early stage of analysis, students participating in a four-hour SP workshop display superior domain-specific clinical skills as assessed by a SP clinical exam and open-ended written exercise. Therefore, simulation with SPs appears to be an ideal medium to teach more specialized history-taking and counseling skills.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: SP blueprints, scripts, and background data will be available for review. Also on display will be videotape of a sample SP encounter.

WEB-BASED INNOVATIONS IN MEDICAL EDUCATION

A RESIDENT-INITIATED WEB-BASED RESOURCE FOR EVIDENCE-BASED MEDICINE. J. Jacobsen¹; S. Mahajan¹; S. Chae¹; B. Lebowitz¹. ¹Columbia University, New York, NY; ²Cambridge University, Cambridge, United Kingdom. (Tracking ID #134545)

WEBSITE URL: <http://www.columbia.edu/~jj2085/residency/>

BACKGROUND: Residents are limited in the practice of Evidence Based Medicine (EBM) by lack of time, failure to remember unanswered questions, variations in faculty training, inadequate computing resources, and informational nihilism. How best to incorporate EBM into residency training remains unclear. A group of residents set out to collect seminal papers in internal medicine for rushed housestaff who may not know how to do complete EBM searches or which questions to ask when confronted with clinical scenarios.

CONTENT: A resident-managed website makes original research papers quickly available through full-text links. Important articles were cataloged by subject and grouped into clinical questions on the website. Print syllabi previously created by faculty, but not easily accessed by residents on-the-go, were incorporated. Links to EBM resources, including guides to critical appraisal and application of the literature, were also added. Residents are encouraged to use an upload option to add additional helpful articles. Articles can be found through an index or via a search engine. The "evidence-based search strategies section" saves questions and search strategies and re-runs the search in PubMed each time a user clicks on the question link, providing an updated list of relevant articles that address the clinical question.

DESIGN: Bibliographic data is stored in a database. This allowed for easy revisions to the website and eliminated the hand-editing and "bit rot" that might result if an HTML-writing program such as Microsoft's FrontPage or Adobe's PageMill were used. With this database approach, article information is stored in only one place; scripts use the database to generate the section page and the index page, and the two pages are automatically consistent. In addition, the database stores PubMed article ID's. Full information (title, journal, etc.) is downloaded from PubMed, ensuring its accuracy. Each article then has a link to the PubMed page, which makes the website useful to users without full-text subscriptions; the PubMed page can display an abstract and list articles that cite the article. The website is maintained using free software (Perl, Python, MySQL, and m4).

EVALUATION: The site was launched in April 2003, and by November 2004 averaged 350 hits per month: roughly 175 hits from inside the Columbia University network and 175 hits from outside. Because of browser caches and intervening web caches, these counts underestimate the true usage. Informal feedback and data from a pilot survey ($n=50$) confirmed that housestaff find the website useful. On a scale from 1 to 5 (5 = most useful), 81% of residents rated the website 4 or 5 for educating themselves, and 73% rated it 4 or 5 for educating other residents or interns.

SUMMARY: This Web-based "electronic file cabinet" of useful questions and articles helps disseminate clinical research among residents and encourages their interest in EBM. The strengths of this project are that it draws from the resources of residents, encourages them to participate in its design, and re-

sponds to their needs and interests. The project offers quick and easy access to articles that residents have found useful and has been popular with residents.

DESIGN OF A LONGITUDINAL, FOCUSED TUTORIAL SYSTEM (LOFTS). D.S. Bell¹; C.E. Harless¹; J. Higa¹; C.M. Mangione¹. ¹University of California, Los Angeles, Los Angeles, CA. (Tracking ID #136239)

WEBSITE URL: http://online.cme.ucla.edu/lofts_demo/

BACKGROUND: Physicians often have difficulty keeping up with guidelines. However, the methods they traditionally use produce poor long-term retention of learning and little transfer of learning into practice. We sought to develop an online tutorial system that can efficiently focus physicians on principles that they need to learn, display authoritative content to improve their understanding of these principles, and reinforce newly-learned principles after an appropriate time delay.

CONTENT: We authored 20 learning objectives covering principles embodied in the American Diabetes Association (ADA) guidelines on blood pressure and lipid management. For each learning objective, we identified one or more specific passages in the guidelines that provided authoritative or evidence-based support for the principle. We wrote 2 multiple-choice quiz items for each learning objective, with some based on patient cases and others being more factual.

DESIGN: We constructed a web software system, the Longitudinal Focused Tutorial System (LoFTS), to deliver quiz-based tutorials linked to authoritative passages. Learners are invited to specific modules with an email message containing an identifying link. Tutorials begin with an overview of the learning objectives and source document(s). They then proceed through a randomly-selected quiz question for each learning objective. When the user selects an answer, the linked passage is immediately displayed. If correct, the user can move to the next question. Otherwise, the user is advised to read the document passage and then choose a different answer. Users must actively choose the correct answer before moving on. After completing a module, learners can be automatically emailed an invitation to a reinforcement exercise, which covers the same learning objectives using quiz questions that the user hasn't already answered. LoFTS content modules are authored in the eXtensible Markup Language (XML). The system software is written in Java and uses MySQL to store usage data.

EVALUATION: We evaluated preliminary versions of LoFTS in cognitive interviews with 12 general internists and family physicians sampled from the AMA Masterfile. Subjects were audiotaped as they completed tutorials while "thinking aloud." Tapes were transcribed and coded in Atlas.ti following a grounded theory approach. Themes were extracted independently by two investigators. We found that the quiz questions were motivating and the ADA guideline passages were seen as authoritative. The system was valued for its efficiency, but pop-up windows were disorienting for some. Though all had Internet access in their offices, many had small monitors, making screen "real estate" scarce. Design changes were made to address these issues. We are now evaluating the LoFTS diabetes content with medicine and family practice residents. Among 31 residents who have participated to date, 19 have completed a follow-up exercise after an average of 4.9 days. On average, 51% of questions were answered correctly on the initial quiz and 63% were answered correctly on delayed follow-up. SUMMARY: An online tutorial with challenging quiz questions and authoritative content passages held appeal for efficient provider education. It was helpful to keep all content within the same window and to remove decorative elements that took up screen space. Residents' knowledge scores increased after the LoFTS tutorial; assessment of their long-term retention is pending.

ELECTRONIC PORTFOLIOS: PROMOTING TIMELY FEEDBACK ON A MEDICINE CLERKSHIP. H.E. Harrell¹; R.J. Rathe¹; S. Grek¹. ¹University of Florida, Gainesville, FL. (Tracking ID #136063)

WEBSITE URL: https://medinfo.ufl.edu/cgi-bin/eval.cgi?demo_portfolio

BACKGROUND: Previously we demonstrated that the introduction of educational portfolios into a medicine clerkship significantly improved students' perceptions of feedback by faculty. However, some students criticized the time commitment and logistics of this paper-based portfolio. Also when students rotated at distant clerkship sites, this format did not allow effective communication with the faculty advisor. To address these concerns, we designed an electronic version of the portfolio to streamline the portfolio process, to allow long-distance feedback and advising, and to add features to potentially further enhance feedback and promote reflective learning.

CONTENT: The portfolio is organized according to the six ACGME competencies and contains descriptions of the required content as well as suggestions for other educational activities that could be included in this student-generated portfolio. Evaluation criteria are linked to each submission type. Prompts are being introduced to many of the submissions to promote student reflection.

DESIGN: Our Web-based electronic portfolio can currently accommodate submissions as text, powerpoint, jpeg, pdf, or mpg video. Each student submission generates an automatic e-mail notification to their advisor with a direct link to the portfolio evaluation tool. Likewise, as soon as the faculty has evaluated the submission, the student receives an e-mail summary with a link back to their portfolio. Students have the option to archive any work that they do not want included in the final portfolio. They may also append text to earlier submissions to provide reflection or journal-style entries. The system continuously updates an assessment summary to provide overall "how am I doing" feedback for students and their advisors.

EVALUATION: Beginning in July 2004 we randomized students to electronic vs. paper portfolio and anonymously surveyed them about their satisfaction, feedback perceptions, and perceptions of time usage and efficiency. We also tracked the overall clerkship evaluations to assess for global changes in satisfaction and comments specific to the portfolio. Initially the electronic portfolio was rated significantly lower than paper ($p < .05$) and there was a trend toward lower feed-

back perceptions in the electronic group. Once the automatic e-mail feedback feature was added, students' perceptions of the electronic portfolio were more favorable than for the paper. Initially, students did not optimally use the electronic portfolio features to enhance the long-distance advising and comments reflected their dissatisfaction at not being able to meet face-to-face with the advisor. This too improved as students gained familiarity with the electronic system. The overall assessment of the clerkship and portfolio experience was not affected by this piloting of an electronic portfolio. In 2005 randomization ended and when given a choice, all twenty of this cohort opted for the electronic version.

SUMMARY: The electronic portfolio is rapidly becoming successful at promoting an interactive learning experience for students with timely feedback. When given a choice, students opt for the electronic version. However further data is needed to determine whether feedback perceptions will improve significantly.

THE ALCOHOL CLINICAL TRAINING PROJECT: A FREE ONLINE CURRICULUM FOR USE BY GENERALIST CLINICIAN EDUCATORS. D. Alford¹; S.E. Chapman¹; C. Dubé²; N. Freedner¹; R. Schadt³; R. Saitz¹. ¹Boston University Medical Center, Boston, MA; ²Brown University, Providence, RI; ³Boston University School of Public Health, Boston, MA. (Tracking ID #131776)

WEBSITE URL: www.mdalcoholtraining.org

BACKGROUND: Physicians receive little effective education regarding alcohol problems. The web-based Alcohol Clinical Training (ACT) curriculum is patient-centered and evidence-based and is intended for use by generalist clinician educators. The curriculum addresses the clinical skills needed for screening and brief intervention for culturally diverse patients with alcohol problems. The intended audience for this curriculum is generalist clinician educators; the materials are to be used with their learners: students, residents and practicing clinicians.

CONTENT: The purpose of this free, online curriculum is to offer generalist clinician educators a practical approach to teaching screening and brief intervention for alcohol problems in medical settings. Emphasizing cross-cultural efficacy, the curriculum teaches skills to: ask about alcohol use; assess severity and readiness to change; advise cutting down or abstinence, and assist in goal setting and further treatment when necessary; arrange follow-up to monitor progress; and, assure cross-cultural efficacy. The website includes a PowerPoint slide presentation with trainer notes, expert audio narrative of the slide presentation, and 3 embedded physician/patient video cases demonstrating different levels of alcohol severity. Materials are adaptable to varied settings and audiences (grand rounds, morning report, preclinic conferences). Pre- and post-tests are also provided.

DESIGN: The main components of the curriculum are the PowerPoint (PPT) slide presentation [which is available in several forms: native PPT (with and without audio), slides as web pages with video links, or videos available as streaming and as downloadable files for presentation] and 3 streaming video cases; videos and slides are integrated with direct hyperlinks within the file. Videos depict simulated patient/physician interactions, followed by the patient's reactions to the encounter. Faculty feedback about the curriculum is encouraged via an open-ended online feedback form and a structured evaluation form which can also be completed online. Authoring software used to develop the curriculum included: Final Cut, Adobe Premiere, Real Media streaming software, Adobe Audition software, Macromedia Dreamweaver, DVD studio and Roxio Toast.

EVALUATION: Of physician educators who were trained to use the curriculum, 100% rated the design and content of the curriculum as very good to excellent, and many indicated that they would use it in varied settings, including resident and medical student conferences, CME courses, precepting, and inpatient attending rounds. Visitors to the website completing online evaluations have also rated the design and content of the curriculum as very good to excellent, and indicated that they would use these materials in resident conferences, CME courses, and inpatient attending rounds.

SUMMARY: The ACT website offers generalist clinician educators training tools for teaching alcohol screening and brief intervention with a focus on clinical skills and cross-cultural efficacy. The availability on the web, the flexibility to use video and audio with and without slides for varied audiences and teaching settings, and the tailoring to the generalist clinician educator audience allow widespread use.

THE PROFESSIONAL DEVELOPMENT PORTFOLIO: REFLECTION-IN-ACTION. A.L. Kalet¹; J.J. Sanger¹; S. Hubbard¹; J.M. Chase¹. ¹New York University, New York, NY. (Tracking ID #136142)

WEBSITE URL: tools.med.nyu.edu/profportfolio

BACKGROUND: The ACGME and LCME have challenged us to evaluate professionalism but more sophisticated assessment strategies are needed. We designed, implemented, and evaluated an on-line Professional Development Portfolio (PDP) for all entering medical students (class '07 N=160), explicitly aimed at preparing future physicians to engage in self-reflection on professional behaviors.

CONTENT: After logging on, students view their individualized portfolio matrix displaying all required 1st year submissions. Students upload the appropriate document, write a brief 2-3 sentence reflection and submit. PDP content includes: standardized materials (e.g. essays, standardized patient interactions checklists, self, peer, patient, and faculty assessments) as well as individualized materials (e.g. evidence of community service and leadership activities) and comment cards, filed by others, either praising or expressing concern. There is

as least one PDP submission for each basic science course (e.g. Anatomy students assessed their peer dissection partners' professional conduct) as well as early clinical experiences. At the end of the year students wrote a 1-2 page self-assessment then met face-to-face meeting with their Faculty-Partner (FP) to receive individualized feedback, develop a learning plan for the coming year, and agree on a professionalism grade. Only students, assigned FP's, and PDP administrators have full access to the students PDP. FP's are prompted to review the PDP only when a comment card or the final assessment is filed. Faculty Development sessions focused on reviewing the self-assessment and giving effective feedback. Students are email reminded starting 2 weeks prior to a submission due date and then every 30 days until the submission is recorded. PDP material will be considered toward AOA and other honors eligibility.

DESIGN: The web-based portfolio was written in-house using Macromedia ColdFusion 5.0 middleware accessing an Oracle9i database. It is integrated with, and access is authenticated by PIMS, our school-wide personnel database. **EVALUATION:** End of year assessments were due by June 1, 2004 and compliance with the submissions was up to 100% by mid July with faculty being the main cause of the delay. Significant changes were made to the PDP as a result of student feedback. The PDP took a mean of 2 hours per student annually. Data from IRB approved controlled pre/post evaluation of the PDP is being analyzed. Measures include: Professionalism Attitudes and Beliefs Questionnaire and a measure of moral reasoning, student/faculty satisfaction and educational impact questionnaires.

SUMMARY: Collaboratively designed on-line portfolio systems can support professional development through structuring a habit of self-reflection and provide detailed documentation of professionalism competence.

UNDERSTANDING PRISON HEALTH CARE: FOSTERING COMPETENCE AND COMPASSION IN TREATING PRISONERS. M.A. Minor¹. ¹Brigham and Women's Hospital, Boston, MA. (Tracking ID #136238)

WEBSITE URL: <http://movementbuilding.org/prisonhealth/>

BACKGROUND: It has been said that to understand society, one needs to look within its prisons. Within prison walls one can see up front and center people who are afflicted by racism and poverty, as the majority of prisoners are people of color and underrepresented minorities, and those from economically impoverished communities. Prisoners thereby represent some of the most marginalized populations in the United States, in one structural and cultural institution. The poor health care status of prisoners is not surprising, given that poverty and race status are known to hasten the spread of communicable diseases and to contribute to the development of mental health disease. Within prisons one can observe patients who have lacked access to health care and who often present to medical attention in later stages of disease. Furthermore, many Americans believe that prisoners, by nature of committing crimes, have relinquished their rights to health care. Few would argue that prisoners should be medically neglected; however, the general temper of public opinion toward prisoners enables the public to turn their heads and forget those behind bars. Medical education modalities that address issues in prison health care in have been scarce. Physicians often do not understand the historical context of prisons and prison health care, the health care delivery system within prisons, the barriers to improving prisoner health care, the most common medical issues of prisoner populations, and the issue of medical neglect within correctional centers. Furthermore, during medical training physicians are not challenged to consider personal biases toward caring for prisoner patients. Without an understanding of these issues, it is difficult to provide competent and compassionate care. As such, physicians may be less able to meet the emotional and medical needs of this discrete cultural group, even though physicians in the public sector across all medical specialties treat prisoners or ex-prisoners. Medical internist, by virtue of managing the medical care of adults, are the best prepared to meet this growing care need.

CONTENT: This project uses audio and video narratives from prisoner advocate and health care provider interviews, a collection of artwork from prisoners and prisoner advocates, and summations of historical reviews from medical and periodical literature.

DESIGN: This didactic, on-line educational module is organized into eleven web pages, each which addresses a specific aspect of prison health. Each page includes a literature review, artwork from prisoners or advocates, and "voices of thought" - Quicktime audio/ videos that present expert opinions from leaders in the field of prison health care. The on-line module concludes with links to community organizations that address issues in prison health care. The curriculum can be completed in one sitting or over time.

EVALUATION: Includes e-mail response to the curriculum author.

SUMMARY: Addressing the health care needs of prisoners is paramount. If aspiring and practicing internists are able to develop a contextual understanding of the health care of prisoners, and can translate knowledge about prisoner health care issues into insight and empathy in treating this population, there is greater potential to learn the core skills necessary to improve interactions with and medical care of other underserved populations. This on-line curriculum begins to address these issues in a unique presentation.

USING THE WEB TO ENHANCE MEDICAL INTERVIEWING SKILLS AND INTRODUCE DIAGNOSTIC REASONING. C.G. Kreger¹, D. Knutson¹. ¹Ohio State University, Columbus, OH. (Tracking ID #135388)

WEBSITE URL: <https://medicine.osu.edu/physicaldiagnosis/>

BACKGROUND: Prompted by positive evaluations of our interactive web-based physical examination resource for first and second year students (2003), we began to enhance our interviewing course with similar interactive technology. This course had been taught in small groups with more than 40 facilitators, relying on the availability of volunteer patients and a small cadre of standardized

patients. The standardization of such activities was challenging with a class size of 210 students. A web-based format enabled us to present patient interviews through streaming video. Additional interactive tools were developed to illuminate the physician's thought processes, show that not all information is "equal" in diagnostic reasoning, and allow students to create a hierarchical differential diagnosis based on the patient's history. Lastly, this platform proved an ideal way to demonstrate more difficult interview situations.

CONTENT: Twenty cases, divided into three tiers, are in development. Tier 1 cases teach introductory interviewing skills (e.g. basic information gathering skills, the role of the past medical history, family, social, and sexual history in patient care etc.). Tier 2 cases, intended for second year students, reinforce data gathering skills, and also encourage students to bear in mind possible syndromes during the interview process as information gathered relates to the epidemiology, temporal pattern, and key features of those hypothesized diseases or syndromes. Students then create a hierarchical differential based on information presented. Tier 3 cases focus on advanced patient communication skills (e.g. how to deliver bad news; how to identify and discuss drug seeking behavior; how to discuss responsible sexual behavior etc.). Pedagogical tools include patient interviews, written histories, physical exam findings, selected diagnostic studies, and links to a glossary of terms as well as clinical pearls.

DESIGN: Macromedia Flash and video streaming are used to bring to life patient interviews, selected physical exam findings (e.g. heart and lung sounds), and teaching points. Students can provide on-line feedback and use a "notebook" to record their problem solving strategies which can be accessed by faculty facilitators. Content is entered through a custom-built content management system (CMS) and then stored in a SQL database, enabling content authors, who may lack specialized technical skills, to enter content independently of application developers.

EVALUATION: Evaluations based on three piloted cases demonstrated that a majority of students could navigate the site without difficulty, felt the cases helped them see how to use or perform a specific portion of the interview, and felt that the video and text were effective teaching tools. Occasional technical difficulties however may have reduced the number of students logging into the site over time, and lessened the value of the experience for others.

SUMMARY: With the development of these on-line readily available patient interviews and cases, our medical interviewing course now has a greater degree of standardization across student groups. In addition, we have been able to address more challenging interview scenarios than had been possible previously. With the positive feedback obtained thus far, we are continuing to expand our Tier two cases which delve more into the diagnostic process and thus require a greater degree of interactivity.

INNOVATIONS IN PRACTICE MANAGEMENT

AN INTERACTIVE COMPUTER KIOSK MODULE FOR THE TREATMENT OF RECURRENT UNCOMPLICATED CYSTITIS. E.M. Aagaard¹, P. Nadler¹, J. Adler¹, R. Gonzales¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #134683)

STATEMENT OF PROBLEM/QUESTION: Telephone management algorithms for uncomplicated urinary tract infections (UTIs) have been found to be safe, effective, and well received by patients. However, many women either have no primary care provider or have inadequate access to their provider for urgent telephone management of UTIs. We developed a program for expediting the management of women presenting to our urgent care clinic with uncomplicated UTIs.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: (1) To implement a voluntary, kiosk-based computerized module that facilitates treatment of women with uncomplicated UTIs without a provider examination. (2) To assess patient satisfaction with this module

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: We recently developed and validated an interactive computer kiosk module (via a touch-sensitive screen) based on previously described telephone management algorithms for UTIs. On December 15, 2004, the UTI Self-Management Kiosk Program was launched to the public. Women presenting to our urgent care clinic with symptoms suggestive of a UTI are referred directly to the computer kiosk. They receive a video introduction to the program, followed by a series of audiovisual questions to determine if the patient has an uncomplicated UTI, medication/ allergy history, and a brief satisfaction survey. Patients meeting criteria for treatment without a provider's examination receive a printout that includes their symptoms, current medications, medication allergies, and an antibiotic prescription with various options for antibiotic therapy. The printout is immediately reviewed by the next available urgent care provider, who then selects a specific antibiotic and signs the prescription. Patients who do not meet criteria for an uncomplicated UTI (which include those with prolonged symptoms, symptoms suggestive of pyelonephritis, relevant comorbid illnesses, or risks for antibiotic-resistant UTIs) are advised to wait to be evaluated by a provider as usual. Satisfaction data is collected by the computer program at the time of the interaction on all patients completing the module and includes ease of use, interest in other similar programs, and willingness to refer family and friends to use the program.

FINDINGS TO DATE/EVALUATION OF WEB SITE: To date, 27 women have accessed the UTI kiosk, and 9 were treated by the computer for uncomplicated UTI (33%). Of the 18 women ineligible for treatment, 1 was greater than 65 years of age, 6 had no prior UTI, 7 had symptoms longer than one week, 3 had new back/flank pain and 1 had severe vomiting. Patients completing the module were highly satisfied with it; all 9 women found the program easy to use; 8 of 9 answered that there should be computerized pathways for other types of common illnesses; and 7 of 9 stated that they would recommend this program to friends or other family members with bladder symptoms.

KEY LESSONS LEARNED: (1) Successful implementation of the program required significant education and acceptance by front desk staff and faculty. (2) Computer-based self-management programs are feasible and have the potential to improve patient satisfaction, decrease physician visits, and decrease costs. **MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:** (1)Lap-top computer simulation of the program (2)Copies of printout given to patients.

CAN DEPRESSION SELF-MANAGEMENT GROUPS ENGAGE OLDER PATIENTS WITH DEPRESSION? S.L. Swenson¹; M.D. Feldman¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #135781)

STATEMENT OF PROBLEM/QUESTION: The Institute of Medicine has designated patient self-management and major depression as two of its top priority areas for transforming health care. Although studies document the importance of patient-oriented components in successful chronic disease management, the effectiveness of self-management strategies for treating depression remains largely untested. We have developed a group intervention for patient activation and self-management in older primary care patients with depression.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: The CALM-D (Coping and Life Maneuvers for Depression) project aims to (1) demonstrate the feasibility of depression group visits in community- and hospital-based general internal medicine practices, (2) ascertain patient and physician attitudes toward and acceptance of the intervention, and (3) explore the impact of these groups on depression outcomes and patient self-efficacy.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: We are recruiting older (age > 50), English-speaking continuity patients with major depression or dysthymia as evidenced by baseline PHQ-9 scores of 10 or more. We are excluding those who require acute psychiatric hospitalization or have severe dementia. We initially performed a needs assessment of key stakeholders (medical directors, clinician, nurses, and administrative staff) regarding how best to integrate the program into the clinical sites. The intervention is based on the Chronic Disease Self-Management Program (CDSMP) developed by Lorig and colleagues and consists of eight weekly two-hour groups that meet in patients' primary care clinics. The CDSMP improves self-care attitudes and behaviors, functional status, and health care utilization among patients with a variety of chronic illnesses but has not previously been evaluated for depression treatment. A physician with depression expertise and CDSMP training and a care manager facilitate the groups and provide advice on medication management. **FINDINGS TO DATE/EVALUATION OF WEB SITE:** Eighteen clinicians have referred 27 patients; 5 were ineligible and 7 declined to participate. Reasons for declining included scheduling conflicts, concerns regarding language skills, unwillingness to engage in a group format, and other more compelling health or life problems. To date, we have 10 program participants. They are ethnically diverse (30% African-American, 20% Latino) with a median age of 65 (57-75). They are moderately to severely depressed (mean baseline PHQ-9 = 16, 12-20), and many have struggled with depression for decades. All have at least two medical or psychiatric co-morbidities, and many report significant pain (mean score = 5.4/10). Although some joined the program because they were not interested in specialty mental health services, half report barriers accessing specialty mental health care. Eighty percent have missed less than 2 sessions, and all generate weekly action plans. Participants express that they feel less isolated, have learned effective coping strategies, and benefit from sharing their experiences with others. Clinicians appreciate the detailed clinical assessment and assistance in co-managing these medically and psychosocially challenging patients. We will analyze depression and self-efficacy data on completion of the project.

KEY LESSONS LEARNED: Depression self-management groups can engage and retain older patients with depression and integrate easily into primary care settings. They may augment usual depression care by enhancing medication management, providing social support and teaching coping strategies, symptom management and problem-solving skills.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:

CENTRAL LINE ASSOCIATED BLOOD STREAM INFECTIONS ON GENERAL MEDICINE FLOORS. A. Raina¹; H.S. Sheth¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #136066)

STATEMENT OF PROBLEM/QUESTION: Often patients with central lines which include central venous lines (CL) or peripherally inserted central venous catheters (PICCs) are managed on medical floors. There is paucity of data pertinent to risk factors associated with central line-associated bloodstream infections (CLABs) on general medicine floors.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: The current study was aimed at identifying risk factors for CLAB in the setting of general medicine floors with a plan to design strategies to identify those at risk earlier and develop preventive measures.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: University affiliated hospital infection control committee had identified 35 patients on general medicine floors with CLABs from January - December, 2002. We compared these 35 patients (CLAB group) to randomly chosen 56 patients on general medicine floor from May - December 2002 who had central lines and did not develop blood stream infections. No other variables were matched when choosing the control group. All pertinent demographic and clinical data were collected from electronic medical records and archived hospital laboratory database (MARS). These groups were compared with respect to age, race, gender, survival, albumin level, steroid use, history of co morbid illnesses (Diabetes mellitus, congestive heart failure/cerebro-vascular disease, respiratory disease, renal failure, cancer, liver disease), being a nursing home resident (NH), transfer from other hospital, primary purpose for using the central line (medicines/Fluids,

blood product or Total Parental Nutrition (TPN)) and presence or absence of other nosocomial infection such as Vancomycin Resistant Enterococci (VRE) infection/colonisation, stool positive for C. Difficile Toxin or Methicillin Resistant Staphylococcus aureus (MRSA) anywhere other than bacteraemia. Logistic regression analysis was used to compare the groups.

FINDINGS TO DATE/EVALUATION OF WEB SITE: The CLAB and control group patients had mean age of 54 and 57 years respectively (p=0.5). Two groups were similar for race and gender. The logistic regression analysis for CLABs was completed using age, other nosocomial infections, renal failure, being a nursing home resident and primary purpose of central line. This showed that other nosocomial infections (OR 4.2, P=0.019) and being a nursing home resident (OR 6.3, p=0.027) were significant risk factors. Renal disease (p=0.07) and using CL for TPN (p=0.06) were marginally significant factors.

KEY LESSONS LEARNED: On general medicine floors patients with central lines have odds of 4.2 to develop CLABs if they also have VRE, C. Difficile or MRSA (Non-blood stream) infections. They are 6 times more likely to be NH residents. Patients with renal failure (acute and/or chronic) as well as those in whom central lines were used for TPN were at higher risk to develop CLABs.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Use of antibiotic coated catheters, enforcing strict barrier precautions and additional precautions used for CL care in the patients with above risk factors may be helpful in reducing the incidence of CLABs.

COORDINATING A COMMUNITY WIDE INTEGRATED DELIVERY SYSTEM USING INFORMATION TECHNOLOGY- LESSONS LEARNED FROM THE WATERBURY HEALTH ACCESS PROGRAM. K.D. Carr¹. ¹Waterbury Hospital, Waterbury, CT. (Tracking ID #135801)

STATEMENT OF PROBLEM/QUESTION: Waterbury Health Access Program (WHAP) is federally funded to help improve the access to and quality of health care for the underserved in Waterbury, CT through increased collaboration among Waterbury's health care provider network and a state-of-the art information technology system.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: -Improving access to primary care, specialty care, and pharmaceuticals for the uninsured and the underinsured through integrated Case Management, Disease Management, and Project Access, which is facilitated by a common Electronic Medical Record (EMR) and other systems. -Installing a common EMR in each of the consortium clinics, allowing for common access to patient information and scheduling from all locations including the emergency room.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: 1. Case Management System: We are currently testing an internet based case management system to facilitate enrollment into Federal and State funded health insurance plans, enrollment into pharmacy assistance programs (PAP), and scheduling appointments with primary and specialty care providers. 2. Disease management program a. Diabetes: The APRN Diabetic Case Manager is currently testing disease management softwares that produce reports on command. b. HIV/AIDS: A one-stop care system providing patients access to social worker, nutritionist, case manager, massage therapist, treatment adherence nurse, and providers, all at one location. c. Mental health: Screening through SF-12 survey. d. Cancer treatment/ prevention: Each consortium member help improving cancer prevention by having screening programs for underserved. 3. Waterbury Project Access Network: As part of the American Project Access Network, the system relies on equitable distribution of charity care by physicians. The case management system is designed to select and assign specialists at the time when eligible patients need referral. 4. WHAP Common EMR: Currently with 100,000 completely electronic charts, we expect to have over 150,000 electronic charts by end of 2005. The information network is currently being designed by Amicore. The database will support HIPAA-compliant sharing of information, and will interface with other billing systems.

FINDINGS TO DATE/EVALUATION OF WEB SITE: Since implementation in July 2004: 321 individuals were assisted in Case Management 64 of 118 patients approved for Medicaid 74 patients received free medication through PAP Over 50 physicians have pledged to support Project Access 274 patients enrolled in HIV Disease Management program 41 patients enrolled in Diabetes Management Program.

KEY LESSONS LEARNED: Building partnerships: The success is largely attributed to the approval, dedication, and enthusiasm by members and partners. Some accomplishments are: -Waterbury Medical Association agreed to become the organizational home for Project Access, provisional on hiring an Executive Director -Pharmaceutical companies have donated significant number of glucometers and testing strips -Amicore is donating few licenses for their software Training: Case Managers performed significantly better after training in financial eligibility screening. Learning from other communities: Ten representatives visited Asheville NC to gather information on Project Access. That visit provided a "critical mass" of leaders, knowledge and enthusiasm that helped fuel our successes. Consulting: with approved and other non-profit consultants, including attorneys, provided exposure to information and helped lay a foundation for some of the complex processes.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Computer USB port Internet access MS PowerPoint LCD projector.

CREATING A FIRM CHIEF SYSTEM IMPROVES WARD MEDICINE PATIENT CARE. D. Baruch-Bienen¹; L. Leykum²; E. Mortensen¹; G. Crawford¹; J. Patterson¹; S. Berrigan¹. ¹University of Texas Health Science Center at San Antonio, San Antonio, TX; ²Columbia University, New York, NY. (Tracking ID #135928)

STATEMENT OF PROBLEM/QUESTION: According to the Institute of Medicine, healthcare organizations should pursue six major aims: provide care that is

safe, effective, patient-centered, timely, efficient, and equitable. Improvement in inpatient ward medicine is critical for achieving these aims, yet most faculty physicians that spend only one to two months as ward attendings do not participate in ongoing activities that improve these areas.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: 1. Improve the communication and interaction of the medicine teams with other disciplines, including nursing, nutrition, social work, pharmacy, lab, radiology, information technology, and hospital administration 2. Monitor the impact on the firm using patient and staff satisfaction surveys, and through documentation of process improvements 3. Oversee the teaching of systems-based practice.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: On July 1 2004, four firm chiefs were each assigned to four VA ward teams. Responsibilities of the firm chiefs include ward attending orientation, organization of intern multidisciplinary rounds (twice a week), development of a systems-based curriculum, weekly meetings with the ward team and ward attending to discuss issues concerning systems-based practice, and attendance at a multidisciplinary logistics meeting twice a month. This meeting includes representatives from nursing, nutrition, social work, pharmacy lab, radiology information technology, the residency, and the hospital administration with the goal of problem-solving to overcome obstacles to patient care and housestaff training.

FINDINGS TO DATE/EVALUATION OF WEB SITE: Medicine inpatient surveys of overall rating of care from quarter four 2003 and quarter four 2004 show an increase in positive scores from 68.9% (n=90) to 83.3% (n=66) (p<0.001). Attendance at multidisciplinary rounds has increased; the four teams missed a total of three discharge planning meetings Sept-Nov 2004, compared to 18 meetings the year before (same time period). Feedback indicates that these meetings have been helpful to both housestaff and other services in discharge planning. Faculty representation at department service logistics meetings has also increased. Process improvements directly related to Firm activities include positive changes in the discharge appointment process, contingency plans for computer downtimes, improved computer order entry, decreased time for rehabilitation consults, increased compliance with timely attending notes and completion of consent forms, and timely completion of radiology tests. Nursing leadership reports that the addition of Firm Chiefs has resulted in more expedient resolution of patient problems and more effective interdisciplinary team relationships. A staff survey is planned for February 2005.

KEY LESSONS LEARNED: In order for physicians to participate in systems-based practice, they must be actively and consistently involved in improving the healthcare system. The traditional academic ward attending paradigm did not ensure such participation. Introduction of attending physicians with commitment to the organizational process has improved interdisciplinary communication, resident and medical student teaching of systems-based practice, and addresses the six major aims of health care organizations as defined by the Institute of Medicine.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: poster/oral presentation.

DEVELOPING AND EVALUATING A CLINICAL PHARMACY DISEASE MANAGEMENT PROGRAM IN AN INTERNAL MEDICINE PRACTICE. R.L. Connors¹; M.J. Mcquire²; J.M. Brewer³; N. Nkansah⁴; G.J. Noronha⁵. ¹Johns Hopkins University, Baltimore, MD; ²Johns Hopkins Hospital, Baltimore, MD; ³Johns Hopkins Community Physicians, Baltimore, MD (Tracking ID #136084)

STATEMENT OF PROBLEM/QUESTION: Can addition of a clinical pharmacy program improve outcomes for diabetics and other patients with cardiovascular disease in a general internal medicine practice?

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: 1) Develop and integrate a clinical pharmacy program to enhance care of patients with cardiovascular disease and diabetes into an existing internal medicine clinic. 2) Evaluate impact of the intervention on hemoglobin A1c, systolic and diastolic blood pressure (SBP and DBP), weight, smoking, and other measurable indicators of disease control. 3) Determine cost effectiveness of the intervention.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: We developed a clinical pharmacy program targeting cardiovascular risk factors and diabetes in an existing internal medicine practice with 35,000 visits per year. Patients were scheduled for 30 - 60 minute visits with clinical pharmacists who consulted internists as needed for clinical issues and medicine dose changes; all pharmacy notes were reviewed and co-signed by the patient's internist. Pharmacy visits were conducted by pharmacists with special training and interest in clinical medicine, collaborative practice and bidirectional communication with both referring physicians and patients. Outcomes were determined for a running report of all patients seen March 2002 to August 2003 using a pre-post intervention analysis. Data on show-rate, referring providers, practice revenues, diagnoses (ICD-9), and payors was also collected for use in financial and utilization analyses.

FINDINGS TO DATE/EVALUATION OF WEB SITE: 1) A statistically significant (P<0.001) improvement in A1C from 8.7 to 7.9 was noted in patients referred to pharmacy program after 6 months of intervention. 2) An improvement in SBP (P=0.024) was noted. 3) Pharmacists have integrated well into the practice with increasing utilization of the service by both physicians and patients. 4) Collections support only about 1/3rd of the cost of this service but the financial impact of improved outcomes needs to be determined.

KEY LESSONS LEARNED: Addition of a clinical pharmacy program improved outcomes of patients with diabetes and cardiovascular disease in a large general medicine practice. The program has been accepted well by internists and patients. The cost-effectiveness of this program has not been fully evaluated.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster or presentation of findings.

EMERGENCY DEPARTMENT PAIN MANAGEMENT OF OLDER ADULTS WITH HIP FRACTURES. U. Hwang¹; L.D. Richardson¹; T.O. Sonuyi²; R.S. Morrison¹. ¹Mount Sinai School of Medicine, New York, NY; ²University of Michigan, Ann Arbor, MI. (Tracking ID #135660)

STATEMENT OF PROBLEM/QUESTION: Disparities in Emergency Department (ED) analgesia due to patient race/ethnicity and age exist. ED crowding is a growing problem that may also impact pain management. Do other patient-related risk factors and ED crowding result in poorer quality pain management for older adults with hip fractures?

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: Our objectives were to evaluate patient-related and ED crowding factors associated with quality of ED pain management for older adults with hip fractures.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: We reviewed hospital registration data and ED records from a previously conducted (1997-98) prospective cohort study of patients >50 years of age hospitalized with hip fracture for patient-related risk factors: age, triage status, presence of dementia and delirium, complaint of pain; and ED crowding factors: ED census and mean length of stay (LOS). Outcomes were documentation of pain evaluation, time to pain assessment (ttPA), ordering of analgesic, type of medication, time to pain treatment (ttRx). Pearson correlation, t-test, chi-square, and logistic regression were conducted.

FINDINGS TO DATE/EVALUATION OF WEB SITE: 158 patients in the original study were admitted from the ED. Mean age was 83 years (±9); 82% complained of pain; mean ttPA was 40 minutes (±79); ttRx was 141 minutes (±120); mean delay to treatment (DtPA-ttRx) was 122 minutes (±125). Of those with complaints of pain, 57% received opioids, 36% received no analgesic, and 7% received non-opioids. Of the opioid patients, 61% received intravenous medication, 53% as meperidine. Patients with delirium were less likely to receive opioid analgesic: RR=0.82 (C.I. 0.69,0.97, p=0.03). Age, triage status, dementia, and complaint of pain were not associated. Of ED factors, only LOS was associated with quality of pain management. Correlation coefficient for mean ED LOS vs. ttRx was R=0.19 (p=0.10).

KEY LESSONS LEARNED: In this study, older adults with hip fracture were at risk for undertreatment of pain, considerable delays in analgesic administration once pain was identified, and treatment with inappropriate analgesics (i.e., meperidine) in the ED. Factors associated with inadequate pain treatment included the presence of delirium and ED crowding as measured by high mean ED LOS.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:

EVALUATION OF UNDERUTILIZATION OF STATIN THERAPY IN DIABETIC PATIENTS IN A GENERAL MEDICINE PRACTICE. J.B. Reilly¹; G.S. Fischer¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #134715)

STATEMENT OF PROBLEM/QUESTION: An automated report from the electronic health record (EHR) showed that only 46% of the diabetic patients (pts) in the University of Pittsburgh Physicians General Internal Medicine clinic were on statin drug therapy at the end of the 1st quarter, 2004. It is unclear why such a high percentage of diabetics were not on statin therapy despite the known benefits.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: 1. To increase the percentage of diabetic pts taking statin drugs. 2. To determine the physicians' reasons for not placing pts on statins.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: At the end of the 2nd quarter, 2004, MDs were given a list of all of their diabetic pts not on statins, excluding pts whose most recent LDL was < 100 mg/dl (based on American College of Physicians' guidelines). MDs were asked to indicate whether they would institute statin therapy for each pt, and if not, to provide a reason for not starting it. MDs were given a choice of responses and the opportunity to write in reasons not listed. Patient lists were re-distributed for the 3rd quarter, with a revised list of choices based on the previous survey write-in responses. Included in this list were 218 pts for whom there was no LDL value on record in the EHR for the previous 12 months.

FINDINGS TO DATE/EVALUATION OF WEB SITE: 1418 diabetics were seen between September, 2003, and August, 2004. The proportion of diabetic pts on statin therapy rose to 53% by the end of the 3rd quarter. Of those not on a statin (671), 206 (31%) had LDL < 100, so that a statin might not be indicated. MDs gave responses for 269 of the remaining 465 pts (58%). For 51 (19%) of these pts, MDs indicated that they had already been on non-statin lipid-lowering therapy (LLT), were started on lipid-lowering therapy (LLT) since the report had been generated, or had appointments planned at which LLT would be suggested. For pts not to be started on LLT, MDs gave the following explanations: nonadherence or multiple missed appointments (17%), lipid-lowering not a priority (i.e. poor life expectancy) (12%), adverse reaction or contraindication to statin use (10%), LDL insufficiently high to warrant therapy (8%), repeat LDL less than 100 mg/dl (4%), pt declined statin for financial reasons (3%) or for other reasons (9%), pt had no other cardiac risk factors (1.5%), and MD chose to recommend therapeutic lifestyle changes first (1.5%). 15% of pts were noted by MDs not to be diabetic, and 4% were no longer pts of the practice. 6% of pts were diabetic and part of the practice but were managed by endocrinologists. 2% had other comments and 7% acknowledged the pt but still gave no comment. Of the 465 eligible patients, 218 (47%) did not have an LDL level on file for the previous 12 months. MDs gave no comment for 121 (55%) of these pts. Seventy-two (33%) were noted to be nonadherent, non-diabetic, diabetic but managed by subspecialists (LDL might not be in EHR), or pts for whom LLT was not a priority. This left eighteen (15%) pts for which yearly lipid monitoring was appropriate but not conducted.

KEY LESSONS LEARNED: 1) Providing MDs with names of diabetic pts not on statins increased appropriate prescribing of statins. 2) MDs identify non-adher-

ence as a significant barrier to LLT. 3) Failure to measure LDL regularly in diabetics likely leads to underutilization of statin therapy. 4) Use of automated reports to judge 'quality' is limited by inaccurate coding in EHR. MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:

GENERALIST/SPECIALIST COMANAGEMENT OF CHRONIC KIDNEY DISEASE. B.J. Lee¹. ¹Kaiser Permanente, Hawaii Region, Honolulu, HI. (Tracking ID #135530)

STATEMENT OF PROBLEM/QUESTION: There are large numbers of patients with chronic kidney disease (CKD), that generalists manage on their own without the assistance of nephrologists.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: 1. Improve the management of patients with CKD. 2. Improve generalists' knowledge of current strategies for managing CKD, using their own patients as examples.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: Proactive "curb-side" consultation of patients with CKD. These interactions are initiated by a nephrologist, who screens an electronic database to find CKD patients at risk. The resulting "E-consult" is sent via email to the generalist, giving specific advice with regards to actual patients whose management can be improved. Such advice includes recommendations with regard to blood pressure and proteinuria management, cessation of nephrotoxins and screening and management of renal anemia and bone disease. The generalist decides how to best make use of these recommendations in managing his/her CKD patients. The nephrologist follows up with the generalist as necessary for the more difficult cases. The nephrologist keeps track of the average "effectiveness score" over time for the population, and if desired, tracking scores for the panels of each generalist or generalist group in order to give feedback with regards to overall effectiveness of CKD management.

FINDINGS TO DATE/EVALUATION OF WEB SITE: Over 12,000 patients in our HMO population of 230,000 have at least stage 3 chronic kidney disease. 8% of pre-ESRD patients have at least moderate (1 gm) proteinuria, and of these 45% are not on optimal doses of ACE inhibitors or ARBs. At least 35% are on medications that are contraindicated or require adjustment in renal insufficiency. Only 43% of systolic BP's are within target range (<130). Screening for bone disease is recommended at stage 3, but is very rarely done. The E-consult process is now in pilot phase. Generalists have been receptive to efforts by specialists to help manage their patients.

KEY LESSONS LEARNED: Case-specific advice from specialists can improve generalists' expertise in managing kidney disease. Nephrologists have the ability to, and should share responsibility with generalists for kidney disease on a population level. Doing this requires a high degree of specialist-generalist communication and cooperation.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Computer-based demo of the E-consult.

GROUP MEDITATION CLASSES FOR CHRONIC BACK PAIN. H. Fischer¹; S. Fischer²; L. Corbin²; T.D. Mackenzie¹. ¹Denver Health and Hospital Authority, Denver, CO; ²University of Colorado Health Sciences Center, Denver, CO. (Tracking ID #134840)

STATEMENT OF PROBLEM/QUESTION: Both patients and physicians express discontent with conventional medical therapy for chronic pain disorders despite \$18 billion spent annually for low back pain alone. Can we incorporate meditation classes into the management of patients with chronic back pain?

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: 1) Measure the rate of referral to the classes by providers and by post-visit chart review 2) Enroll 14 patients in 6 weekly meditation classes to determine the a) rate of participation in classes and b) rate of completion of surveys measuring functional status, self-efficacy, and symptom scores 3) Evaluate participants' perceived benefit of meditation through focus group discussions.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: Our federally qualified community health center has approximately 20,000 visits annually by Denver residents. The patient population is 80% Latino, 10% Caucasian, 5% African-American, and 5% other ethnicity. We discussed with our 9 providers the potential benefits of meditation for chronic low back pain. We asked them to refer to the classes over a 6-week period all chronic back pain patients over age 18 who were interested in meditation therapy and able to read at a 6th grade level. We excluded patients with a history of psychosis, bone metastases, or an inflammatory arthritis. We then examined a) the rate of provider-initiated referral and b) the percent of referrals who desired participation after learning more about the classes. We then retrospectively reviewed all charts for patients presenting over a subsequent 2-week period to determine the rate at which eligible patients present to our clinic. Telephone contact with the eligible patients determined the percent who desired participation in the classes. The primary author learned and practices meditation techniques described by Jon Kabat-Zinn and Herbert Benson, leaders in the use of meditation for medical therapy. He taught the first 14 willing referrals meditation techniques through 6 weekly, 1-hour classes. Subjects were asked to fill out surveys assessing functional status and symptom scores (Roland-Morris Disability Questionnaire and SF-36) and self-efficacy (Pain Self-Efficacy Questionnaire), at the first and last classes. Recruits also participated in a focus group at the end of the meditation classes.

FINDINGS TO DATE/EVALUATION OF WEB SITE: Provider-initiated referral yielded 3.8 eligible referrals per week and 87% or 3.3 per week, desired to participate in the classes. Post-visit chart review yielded 16 eligible patients per week and 47%, or 7.5 per week, desired to participate in the classes. Seven of the 14 patients enrolled in the classes attended 2 or more classes. All 7 of these patients filled out the initial and follow-up surveys. Five of the 7 participants

attended the focus group and expressed a perceived benefit in terms of well-being and pain control. They recommended more frequent classes over a longer span of weeks.

KEY LESSONS LEARNED: 1) There is a sufficient interest in meditation in our primarily indigent and Latino population to warrant further study of the impact of meditation classes on utilization, patient satisfaction, functional status, and pain control 2) Post-visit chart review with telephone contact yields the highest referral rate 3) About 50% of patients enrolled in meditation classes participate 4) Participants may benefit from more frequent classes over a longer span of weeks 5) Providers and other clinical staff were very supportive of this model of care for patients with chronic pain.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster or power point presentation.

HOW GOOD ARE INTERNISTS AT READING X-RAYS COMPARED TO RADIOLOGISTS? J.E. Cho¹; J. Blank¹; A. Cooperman¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #133358)

STATEMENT OF PROBLEM/QUESTION: In the outpatient setting, physicians often request an imaging study to guide medical diagnosis and treatment. However, a radiologist's interpretation is often not available at the time of the patient visit. Primary care physicians initially formulate clinical decisions and treatment plans based on their own impression of the radiograph reading. Although radiologists' readings usually occur within 24 hours, discrepancies in the interpretation may not be conveyed to the clinician for days or even weeks. We lacked a mechanism which allowed for timely feedback regarding clinically significant discrepancies to clinicians.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: 1) Facilitate timely communication of the radiologist's reading to the ordering clinician when a clinically significant discrepancy is identified. 2) Track discrepancies in the interpretation of imaging studies between primary care physicians and radiologists.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: We developed a system for timely feedback of discordant radiological findings to the primary care physician. When a radiograph was obtained the provider's interpretation was attached to the film and made available to the radiologist. The radiologist immediately notified the provider if there were any significant discrepancies between their reading and the initial reading.

FINDINGS TO DATE/EVALUATION OF WEB SITE: We reviewed 829 radiographic interpretations performed from May 2003 to January 2004. Of these films, 376 were chest radiographs and 453 were musculoskeletal radiographs. Of the chest radiograph interpretations, 79% (296) were concordant between primary care physicians and radiologists. Analysis of the discrepant chest radiograph readings identified only five cases (1.3% of the total number of films investigated) with a clinically significant disagreement. An 83% (375) concordance rate was found in musculoskeletal radiograph interpretations. Of the 453 musculoskeletal cases, we found twelve cases (2.6% of total) with a clinically significant difference in interpretation.

KEY LESSONS LEARNED: The accelerated feedback led to timely medical management and improved patient care. Although there was a notable discrepancy rate between radiographic readings made by primary care physicians and radiologists, clinically significant discrepancies were not common. Furthermore, our feedback system improves patient safety and is an educational tool for the primary care physician.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Posters and figures.

IMPLEMENTATION OF AN ELECTRONIC TEST RESULT MANAGEMENT SYSTEM TO IMPROVE THE QUALITY OF ABNORMAL TEST RESULT FOLLOWUP. E.G. Poon¹; D.W. Bates¹; G.J. Kuperman²; Q. Li²; P. Carchidi²; S.J. Wang²; B. Middleton²; T.K. Gandhi¹. ¹Brigham and Women's Hospital, Boston, MA; ²Partners Information Systems, Wellesley, MA. (Tracking ID #133645)

STATEMENT OF PROBLEM/QUESTION: Failure to review and follow-up on outpatient test results in a timely manner represents a patient safety and malpractice concern. Prior research suggests that paper-based systems currently used by many physicians to manage the large volume of outpatient test results may be inadequate, resulting in delays in test result review and poor physician satisfaction.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: 1) To improve the efficiency and reliability of test result management in the outpatient setting. 2) To improve adherence to established guidelines for the management of abnormal test results. 3) To improve patient satisfaction regarding communication of test results.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: Our healthcare system internally developed a comprehensive electronic results manager to help physicians review and act upon ambulatory test results in a safe, reliable and efficient manner. The application, called the Results Manager (RM), collects all tests ordered by each physician in an electronic 'inbox' to facilitate review. RM also prioritizes results based on degree of abnormality, presents guidelines to help clinicians manage abnormal results, allows clinicians to generate result letters to patients with predefined, context-sensitive templates and prompts physicians to set reminders for future testing.

FINDINGS TO DATE/EVALUATION OF WEB SITE: Rollout for 20 adult primary care practices affiliated with 2 academic medical centers started in Spring 2003. Training requirements varied considerably, depending on the presence of champions within the practice and physicians' dissatisfaction with their baseline paper-based systems. Hands-on training was in general more effective than

demonstrations. Training also needed to focus on workflow patterns for the practice and limitations of the application. As of August 2004, there were 165 active primary care physician users of RM. More than 6000 test result letters were written per month using RM. A written user satisfaction survey was administered in the Spring of 2004 to assess physicians' perception of this intervention. Physicians were asked to grade on a 5-point Likert scale the degree to which they agreed with the following statements (1 = Strongly Agree, 3 = neither agree nor disagree, 5 = strongly disagree): 1) RM improves quality of care, 2) RM decreases my malpractice risk, 3) RM is easy to use, 4) RM is useful to me, 5) Using RM to manage test results takes more time than before. Among the 59 respondents (response rate = 51%), physicians thought that RM improved quality of care (mean = 1.83, SD = 0.95) and decreased their malpractice risk (mean = 2.1, SD = 0.95). They also thought that RM was easy to use (mean = 2.3, SD = 1.07) and useful (mean = 1.9, SD = 0.93). Overall, they neither agreed nor disagreed as to whether using RM2 took more time than previous methods (mean = 3.2, SD = 1.4).

KEY LESSONS LEARNED: 1) Use of systems similar to RM has significant potential to improve quality of care and patient safety. 2) Primary care physicians were enthusiastic about the application. 3) Personnel involved in implementation must spend time before and after go-live to understand the workflow and attitudes of physicians to maximize their acceptance of the intervention. 4) Introduction of new systems to improve patient safety can also create opportunities for new errors. Users of information technology must understand its limitations and design appropriate fail-safe mechanisms, especially during the transition from paper-based to electronic systems.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster, slides, handouts, and live demonstration (if high-speed internet available).

IMPROVING PNEUMOCOCCAL VACCINATION RATES AMONG AN ELDERLY UNDERSERVED POPULATION. J.P. Deluca¹; J. Podrobinok¹. ¹Montefiore Medical Center, Bronx, NY. (Tracking ID #136212)

STATEMENT OF PROBLEM/QUESTION: National health objectives are to achieve 90% pneumococcal vaccination coverage of noninstitutionalized adults aged 65 years and older by the year 2010. Pneumococcal vaccination rates remain low nationally in this patient population, a problem magnified in underserved communities.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: To improve pneumococcal vaccination rates among urban, underserved patients over 65 years of age who are being followed at a community health center in the South Bronx.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: Beginning in September 2002, a two-year intensive quality improvement initiative was undertaken to improve pneumococcal vaccination rates at one large community health center in the South Bronx. The interventions included: (1) Physician Education: In September 2002 and September 2003 current pneumococcal vaccine recommendations were reviewed at the monthly staff meetings and hard copies were distributed to all 43 physicians in the practice. (2) Physician Reminders & Systems Changes: A physician's order for pneumococcal vaccine was added to an existing influenza vaccine order form used previously at our health center. Each time a physician ordered an influenza vaccine an assessment of pneumococcal vaccine status had to be made. Physicians had to "check off" one of the following: give pneumococcal vaccine today, previously vaccinated, no indication for vaccination, or patient declined. (3) Patient Education/Reminders: In both September 2002 and September 2003 post cards were mailed to all patients over age 65, reminding them to obtain their influenza vaccines. When patients arrived for their influenza vaccines, the physician and nursing staff educated patients about the importance of pneumococcal vaccine. (4) Access to care: Separate appointments for administration of influenza vaccines were made available each week during October-January 2002 and 2003 for patients who did not need a routine appointment with their individual physicians. Approximately three hours of physician and nursing time was set aside each week for these appointments.

FINDINGS TO DATE/EVALUATION OF WEB SITE: In September 2002, there were 740 patients enrolled at the health center who were 65 years of age or older. At present, 395 of 740 eligible patients' charts have been reviewed. Prior to the interventions in September 2002, 49.3% of the patients had received pneumococcal vaccines. By the end of year one (August 2003), 72.8% of the eligible patients had received pneumococcal vaccine, representing a 48% increase from baseline. By the end of year two (August 2004), 80.1% of the eligible patients had received pneumococcal vaccine, representing a 62% increase from baseline ($p < 0.05$). During these same two time periods, influenza vaccination rates among the same patient population were 62.9% and 65.2% respectively.

KEY LESSONS LEARNED: Pneumococcal vaccination rates among an urban, underserved elderly population can be significantly increased using a multifaceted approach that includes education and reminders for both physicians and patients, systems changes within a practice, and improved access to care.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:

INCREASING BUPRENORPHINE CERTIFICATION AMONG OFFICE-BASED PHYSICIANS IN MARYLAND. Y. Olsen¹; E. Gisriel²; M. Abraham²; M. Preston³; E.B. Bass¹. ¹Johns Hopkins University, Baltimore, MD; ²MedChi Foundation, Baltimore, MD; ³The Maryland Medical Society, Baltimore, MD. (Tracking ID #136148)

STATEMENT OF PROBLEM/QUESTION: Effective opioid addiction treatment with sublingual buprenorphine is now possible in office-based settings. How-

ever, utilization of the medication by potentially eligible physicians has been limited despite efforts by several organizations to promote the practice.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: We aimed to develop a year-long educational and outreach program targeted at office-based physicians to 1) increase awareness; 2) assess willingness to prescribe buprenorphine and the barriers that might prevent doing so; and 3) begin to develop and provide resources to assist physicians become certified buprenorphine providers.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: The Maryland State Medical Society, through its MedChi Foundation, worked with agencies in Baltimore City, an academic institution, and representatives from state and federal agencies to develop six buprenorphine-related activities. First, the MedChi Foundation mailed an informational letter on buprenorphine, its indications, benefits, physician training requirements, and available internet resources to over 20,000 physicians in Maryland. Second, the Foundation designed, produced, and disseminated informational posters to medical office to raise awareness among patients about the availability of buprenorphine. Third, collaborators delivered Grand Rounds presentations to providers at 15 different institutions across the state. Fourth, in conjunction with the Center for Substance Abuse Treatment (CSAT), the Foundation hosted two 8-hour physician buprenorphine training courses. Fifth, investigators at the Johns Hopkins University conducted a survey of primary care physicians, psychiatrists, and pain management specialists in Maryland to assess willingness to prescribe buprenorphine and identify resources required and barriers to adopting this practice. Finally, the Foundation held a panel presentation for interested physicians to discuss the current state of buprenorphine utilization and future strategies to move office-based opioid addiction treatment forward. There were two outcomes of interest for this project: 1) the number of participants in the physician buprenorphine training course and the final panel presentation; and 2) findings from the survey that would indicate future steps to promote adoption of buprenorphine.

FINDINGS TO DATE/EVALUATION OF WEB SITE: Over 120 physicians attended the buprenorphine training courses. Based on evaluations from course participants and information from CSAT representatives, this particular course exceeded previous sessions in both quality and quantity of attendees. Approximately 50 health care providers participated in the final panel presentation. Ideas brought up at this forum and findings from the physician survey indicate that future strategies should include: 1) advocating for appropriate reimbursement for buprenorphine-related services; 2) offering training and resources such as a directory for available treatment slots; 3) developing patient-specific information on buprenorphine; and 4) developing interactive educational programs that provide positive experiences for physicians in treating opioid dependent patients, including practical management tips.

KEY LESSONS LEARNED: Partnerships between academic institutions, state medical societies, and other issue-specific organizations are innovative ways of combining research, policy, and advocacy in efforts to change physician practice behavior.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: We will describe the program in a poster with supplemental materials such as copies of letters, presentations, and survey materials used in the project.

INTEGRATION OF PATIENT SELF-MANAGEMENT PROMOTION INTO PRIMARY CARE TEAMS: A PILOT STUDY USING STAGE-BASED MOTIVATIONAL COUNSELING FOR DIABETICS. J. Kimura¹; K. Dasilva²; S. Fowler²; R. Schneider²; D. Gilworth²; W. Nieves²; L. Schwab²; G.T. Moore¹. ¹Harvard Medical School, Boston, MA; ²Harvard Vanguard Medical Associates, Boston, MA. (Tracking ID #134058)

STATEMENT OF PROBLEM/QUESTION: Although patient self-management is the cornerstone of effective long-term chronic illness management, our current primary care delivery system is poorly structured to support the routine promotion of patient behavior change.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: 1) To use stage-based motivational counseling to promote better diabetes patient self-management 2) To redefine the advanced practice clinician (APC - nurse practitioner or physician assistant) role as the primary provider of chronic illness patient self-management support on the primary care team.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: Based at Harvard Vanguard Medical Associates, a large multispecialty group practice, the pilot involved four internal medicine primary care teams and focused on diabetic patients with a HgA1c > 8.0. Stage-based motivational counseling combines the Stages of Change assessment framework with motivational interviewing techniques to provide clinicians an organized approach to behavior change counseling. All pilot clinicians received motivational interviewing training in a one day retreat and several small group sessions. The APC promoted improved self-management around exercise, diet/nutrition, self-monitoring, appropriate medication use, and smoking cessation using stage based motivational counseling in a series of three planned patient visits. After each APC visit, patients received a follow-up call from the team RN. The mandatory use of templated visit notes focused patient visit agendas on self-management promotion and required APC documentation of collaborative action plans. To assess the pilot, self-reported patient self-management behaviors and key diabetic clinical parameters were collected. Clinical changes were compared to a usual care control group identified from the panels of nine non-participating physicians. Operational feedback was collected through standardized, semi-structured interviews with each participating clinician.

FINDINGS TO DATE/EVALUATION OF WEB SITE: 1) The top three patient-selected behavior change topics were diet/nutrition, exercise, and self-monitoring. In paired analysis, there were significant increases in Days Followed Recommended Diet [1.91; 95%CI: 1.04, 2.79], Days Checked Blood Sugar [1.63; 95%CI: 0.56, 2.01], with a positive trend for Days Exercised [0.93; 95%CI: -

0.06, 1.93]. 2) Comparing control to pilot groups, there were significant differences in mean change scores for diastolic blood pressure [5.38; 95%CI: 1.68, 9.09] and body mass index [0.79; 95%CI: 0.23, 1.35]. Systolic blood pressure, HgA1c, and LDL change scores also suggested improvement but were not significantly different from controls. 3) Qualitative post-pilot feedback noted positive experiences with team based patient self-management promotion.

KEY LESSONS LEARNED: 1) Team-based primary care with complementary provider roles allows the APC to spend time promoting better chronic illness patient self-management. 2) Stage-based motivational counseling is a useful approach to promote chronic illness patient self-management and to improve diabetes clinical parameters. 3) Templated electronic medical record notes are valuable operational tools that can standardize pilot implementation, provide point-of-service education on motivational interviewing for novice APCs, and allow rapid quality assurance audits through uniform documentation.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: 1) Stage-Based Motivational Counseling Matrix 2) APC Initial Visit Note Template.

INTRODUCTION OF A DEPRESSION SCREENING/MONITORING REGIMEN: SUCCESSSES AND CHALLENGES. M. Ong¹; M.D. Feldman²; D. Lee². ¹Stanford University, Stanford, CA; ²University of California, San Francisco, San Francisco, CA. (Tracking ID #135466)

STATEMENT OF PROBLEM/QUESTION: Primary care physicians do not adequately screen or treat patients for depression.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: The objective of this intervention was to provide primary care physicians trained in depression care with easy reminders to screen for depression and with a measure to follow severity of depression over time.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: The Depression in Primary Care project at UC San Francisco reorganizes primary care-based depression care along evidence-based best practices and provides payment to primary care physicians (PCPs) and practices for depression care. Following current U.S. Preventive Services Task Force recommendations, an annual depression screening procedure was introduced in conjunction with this project's coordinated program of depression care. PCPs underwent a depression skills training session which included sections on depression screening and the use of the PHQ-9 questionnaire, a validated depression measure that can be used to follow severity over time. An annual depression screen prompt was added to the automated PCP reminder prompts that accompany the health care maintenance data at every patient visit. This new depression screen prompt asks PCPs if the patient has been depressed or had anhedonia in the past month, and allows PCPs to enter yes, no, or free response feedback into the automated system. If a patient is identified by PCP entry as having depression or anhedonia, an additional depression prompt appears at the next visit. This new prompt asks PCPs to administer the PHQ-9 questionnaire and enter in the score. Results of depression screens and PHQ-9 scores are displayed in the health care maintenance data at all subsequent visits.

FINDINGS TO DATE/EVALUATION OF WEB SITE: Between April 2003 and August 2004, 2908 depression screens were entered, with an average of 171 per month. Overall, 3.4% of all patients seen in clinics received depression screens. Among depression screens, 18.1% (n=527) were noted to be positive. However, PHQ-9 score use was low. PHQ-9 scores were entered in only 2.2% (n=63) of all screens. Among those individuals whose depression screens were positive as of August 2004, only 11.4% (n=60) also had a PHQ-9 score entry. Among individuals with numeric responses (n=40), the average PHQ-9 score was 11.4.

KEY LESSONS LEARNED: Primary care physicians do use depression screens as part of a coordinated program of depression care when provided with a reminder and an easy place for entry. However, use of the depression monitoring measure was low. This may be due to primary care physicians not monitoring depression or low recognition of the PHQ-9 as a depression monitoring measure.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Depression screening and monitoring tools.

MANAGING THE CHAOS OF MEDICAL NEWS; MEASURABLE TRANSLATIONAL IMPACT OF A SYSTEMATIC EMAIL ALERT AND INTRANET APPROACH ON DIRECT LDL ORDERING. A.M. Davis¹; R.W. McLawhon¹; D. Lovinger¹. ¹University of Chicago, Chicago, IL. (Tracking ID #135095)

STATEMENT OF PROBLEM/QUESTION: Medical news in 2005 arrives on daily basis, commonly through press releases, media sound bites, and brief items in web portals. Physicians may need to react rapidly to patient concerns on the basis of information that is fragmentary, lacking context, and not infrequently alarmist in nature. The traditional means of updating physicians may not be up to the challenge of this accelerated news cycle. In other situations, authoritative new guidelines are published, but changes in individual physician practice are slow to occur. We describe an electronic 'clinical news' approach that has proved useful in a 35 member midwestern general internal medicine academic practice, and report evidence of its impact on group practice behavior using the example of direct LDL test ordering following the release of the revised NCEP cholesterol guidelines.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: Beginning in July 2000, an experienced faculty generalist has generated email alerts on roughly a twice weekly basis. Each alert includes a brief description of the relevant medical news along with a few sentences on context, editorial comment, and attached pdf or weblink. Topics were chosen if they were judged likely to 1) affect general internal practice and/or 2) generate extensive media comment and patient interest or concern. General categories of alerts include: 1) drug adverse

effects (e.g., WHI, cox-2); 2) infectious disease concerns (SARS, West Nile); 3) preventive and chronic disease guidelines, especially those with major revisions (NCEP); 4) provocative and 'landmark' studies (ALLHAT). Sources include: 1) early release articles; 2) FDA MedWatch; 3) CDC Health Advisories; 4) electronic table of contents scans of high-impact journals, and 5) New York Times and NPR. Alerts and links to articles are archived in the section intranet by content area for easy retrieval.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: Ready department-wide access to recent salient medical news for use in response to patient inquiries, for teaching purposes, and for more rapid translation of research into practice.

FINDINGS TO DATE/EVALUATION OF WEB SITE: The program has become widely valued in the section and has been sustained with protected time. A case example of its impact was demonstrated by the section's rapid adoption of the use of direct LDL (dLDL). Direct LDL compares favorably to the traditional calculated LDL, can be performed in non-fasting patients, and is compensated up to 6 times a year by Medicare, compared to only once yearly for the traditional lipid panel. While this test had been available for a year, its use did not expand until a 'clinical news' alert emphasized it during discussion of the July 2004 revised NCEP LDL targets. Ordering of dLDL as a percent of all lipid lab orders was 10.6% during the baseline Jan-June 2004, and rose to 37.5% during July-Nov 2004 (p < .0001) in general medicine, but was unchanged in cardiology clinics (6.1% vs. 4.7%, p=NS).

KEY LESSONS LEARNED: An email and intranet 'clinical news' system is feasible, readily accepted, and can promote rapid adoption of new clinical practices. This approach appears to be valued by faculty as complementary to traditional methods such as grand rounds, journal clubs, and newsletters. Further research is warranted to examine the program's potential impact on quality of care and cost-effective lab ordering.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster or Powerpoint.

MEDICAL JEOPARDY: EDUCATING HOMELESS WOMEN. E. Bocchioni¹; K. Alcarez¹; J. Andrieni¹; L. Sullivan¹. ¹Lenox Hill Hospital, New York, NY. (Tracking ID #135273)

STATEMENT OF PROBLEM/QUESTION: Identify the healthcare awareness level and the preventive health care maintenance among homeless women in a city shelter where the literacy rate is at the 6th grade level.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: 1. Increase healthcare awareness in a women's homeless shelter. 2. Educate homeless women about appropriate preventive healthcare maintenance. 3. Improve access to healthcare for this population of women.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: Internal Medicine residents voluntarily created a Women's Health Awareness day for a homeless women's shelter. This event was promoted by posting fliers at the shelter weeks in advance. The women were paired with Internal Medicine residents to complete a survey about their medical history to include preventive maintenance screening tests such as the dates of their last mammogram, pap smear and/or colonoscopy. The survey also identified if the women had a primary care attending. After the identification of the baseline medical health issues among these women, a Medical Jeopardy board game was constructed to educate this population. The board was divided into six medical categories which included: Diabetes, Heart disease, Depression, Sexually Transmitted Diseases, Smoking and Alcohol, and Cancer Screening. The Internal Medicine residents developed questions for each medical category. Each woman chose a medical category and proceeded to answer the question. Each woman received a prize from a grab bag for participation. The questions raised discussions between the women at the shelter and the Internal Medicine residents. By the end of the game, the women felt confident about participating and looked forward to the next Medical Jeopardy event.

FINDINGS TO DATE/EVALUATION OF WEB SITE: Most of these women who had a primary care physician were appropriately up to date with their health care maintenance. One third of the women did not have a primary care physician and were referred to the Internal Medicine Resident practice. Most of the women were motivated to play Medical Jeopardy and voiced that they wanted to learn about their health. These women were more knowledgeable about health topics than the Internal Medicine residents anticipated.

KEY LESSONS LEARNED: A compassionate relationship between the Internal Medicine residents and the women in the shelter developed and evolved through monthly Medical Jeopardy games at the shelter. Educating women at the shelter professionally gratified medical residents and empowered the women at the shelter to be proactive about their health. Several residents chose a Women's Health elective month at the shelter to advance educational initiatives with specialized workshops on obesity and smoking cessation for the women. Women at the shelter referred other homeless people to the medical resident practice.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster Presentation with Medical Jeopardy Board Demonstration of Health Topics and Questions

NOVEL CHRONIC CARE MODEL FOR UNINSURED DIABETIC PATIENTS WHO DO NOT HAVE PRIMARY CARE PHYSICIANS. M.A. Khan¹; A.T. Evans²; C. Kivlahan¹; S. Shah¹. ¹Cook County Hospital, Chicago, IL; ²Cook County Hospital and Rush Medical College, Chicago, IL. (Tracking ID #136161)

STATEMENT OF PROBLEM/QUESTION: Public hospitals in the U.S. are contending with an epidemic of uninsured patients with obesity, diabetes, and hypertension that is overwhelming their primary care capacity. We developed and

tested a novel model of providing care for these patients by incorporating chronic care principles into a special walk-in clinic for patients without primary care physicians.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: Our objectives for the open-access chronic care clinic were to: 1) recruit patients presenting to a public hospital's walk-in urgent care center who had diabetes and hypertension but no primary care physician; 2) aggressively optimize medications using evidence-based guidelines; 3) negotiate a brief educational program that focused on key self-management issues; 4) test innovative approaches for engaging patients in self care; and 5) critically evaluate process and health outcomes.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: Our clinic staff includes a nurse, a physician, a nurse assistant, and clerk. It is a paperless clinic, except for educational materials, with a detailed computerized patient registry. We designed a lifestyle modification school that includes a mock grocery store and have computer kiosks with interactive programs for assessment and education about cardiovascular risk and diabetes care. Currently, we see up to 30 patients a day. Clinic patients are socially disadvantaged: 50% uninsured and 38% Medicaid; 35% had not graduated from high school; 19% had difficulty reading a drug prescription label; 48% African American, 22% Latino/Hispanic, and 25% from other ethnic minorities. Most patients had multiple cardiovascular risk factors: 28% current smokers, 28% A1c > 10, 56% systolic BP > 140, 43% BMI > 30, 10% clinically depressed, and 31% ate vegetables less than 3 times weekly. More than a quarter of patients believed that high blood pressure and diabetes were not harmful.

FINDINGS TO DATE/EVALUATION OF WEB SITE: Greater than 73% of patients returned for a follow-up visit, and over 70% were prescribed aspirin, ACE inhibitor, and a statin. After the second return visit, there was an average decrease of 7 mmHg in systolic blood pressure. At follow-up, there was greater than 50 percentage-point absolute improvement in several key education goals: knowledge of target A1c and target blood pressure and appreciation that uncontrolled diabetes and hypertension were harmful. There was also an increase in eating vegetables and fruits. At follow-up, random blood sugar values were significantly improved but the effects on A1c were not yet evaluable because the clinic has been operating for only 3 months.

KEY LESSONS LEARNED: We learned that: 1) a chronic care clinic that emphasizes self management of diabetes, hypertension, and other cardiac risk factors can be feasibly introduced into a busy urgent care center at a public hospital with limited primary care capacity; 2) there is a high demand for these services among the uninsured population; 3) despite limited resources, limited patient contact, and a disadvantaged patient population, impressive improvements are possible for many process and intermediate outcomes. It is too early to assess impact on health outcomes. This model may be applicable to other settings where existing primary care capacity cannot meet the needs of patients with common chronic illnesses.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster and computer presentation.

OBESITY: IDENTIFYING PERCEPTIONS AND BARRIERS IN A HOMELESS WOMAN'S SHELTER. K. Seilhan¹, J.D. Andrieni², L. Sullivan¹. ¹Lenox Hill Hospital, New York, NY; ²New York University, New York, NY. (Tracking ID #135090)

STATEMENT OF PROBLEM/QUESTION: The prevalence of obesity is rapidly increasing with 65% of adult Americans being overweight and obese. The challenge of obesity is complicated in a homeless woman's shelter, which has limited resources and an underserved population with language and literacy barriers. **OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE:** Measure the prevalence of obesity in a homeless woman's shelter. Identify the perceptions of obesity and barriers to diet and exercise in a homeless woman's shelter. Introduce the homeless women to nutrition and exercise workshops.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: 71% of the women in the shelter voluntarily had their BMI calculated which demonstrated that 27% were overweight, 44% were obese, and 9% were morbidly obese. Participants completed a survey identifying beliefs about being overweight and whether obesity was perceived as bad for one's health. The survey also addressed the woman's perceptions and barriers to exercise and diet. The daily menu at the shelter was reviewed for content of food groups. A series of workshops were held to educate the women how to read nutritional labels and to analyze the food pyramid. The women also participated in "a walk in the park" exercise demonstration to understand exercise frequency and intensity.

FINDINGS TO DATE/EVALUATION OF WEB SITE: All women with BMIs consistent with obesity understood that they had a weight challenge, which could cause health-related problems. 61% of the women attempted losing weight by decreasing their food intake only. 78% of the women correctly identified food groups necessary for a balanced diet but only 22% thought that they ate a healthy diet. The major obstacle to eating a healthy diet was the menu choices at the shelter. At times, the meal was comprised of 100% carbohydrates. The menu at the shelter was changed to include healthier food choices. Nearly all the women acknowledged that exercise was important but only 50% of the women exercised. Only 1 woman performed a defined intensity and amount of exercise to influence outcomes.

KEY LESSONS LEARNED: Nutrition and exercise education within an underserved population's own environment is a useful strategy in a homeless shelter. The homeless women in the shelter were motivated to voluntarily participate in small groups workshops and exercise programs. In a transient population like a homeless shelter, educational initiatives must be repeated and maintained.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster Presentation and On-Site Discussion of Intervention and Handouts.

PATIENT SATISFACTION WITH SECURE, WEB-BASED ACCESS TO MEDICAL RECORD AND ELECTRONIC COMMUNICATION WITH OFFICE. G.S. Fischer¹, R.R. Wilson², C.S. Gadd¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #131594)

STATEMENT OF PROBLEM/QUESTION: What will patients think of a secure, web-based portal (Portal) which allows electronic communication with their primary care physicians' (PCP) office and review of portions of their electronic health record (EHR)?

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: 1. To demonstrate the feasibility of providing patients access to the Portal in a large, academic general internal medicine (GIM) practice. 2. To survey patient satisfaction with the portal.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: The implementation site is an academic GIM practice, with approximately 30 faculty physicians and 30,000 patient-visits a year. In September, 2003, we implemented a web-site through which patients could view portions of the EHR, request appointments, refills, and referrals, communicate with their physicians, and directly schedule appointments. Patients of 7 physicians were recruited to participate during their office visits or if they attempted e-mail contact with their physician. After one year, participants were asked to complete an on-line survey, responding on a 5-point Likert scale to statements about their satisfaction with the system, and to several open-ended questions regarding what they most and least like about the system.

FINDINGS TO DATE/EVALUATION OF WEB SITE: At the end of one year, 203 patients were enrolled. 86 completed the online survey (42%). 72% were "satisfied" or "very satisfied;" 76% found the Portal "very easy" or "easy" to use. 92% would continue to use the Portal, and 80% would recommend it to a friend, but only 55% felt they received adequate training to use the Portal. At least 75% of respondents were satisfied with prescription refills, medical advice requests, and review of EHR information. 33% were dissatisfied with scheduling and referral requests. 86% of respondents identified at least one positive feature of the system: 19 identified access, 14 convenience, and 12 ease of use. 15 specifically liked the ability to directly communicate with their PCP. 57% identified features that they disliked: 10 had experienced poor response from the office to messages, 9 found it difficult to access the site, and 12 felt there was inadequate information available on the site.

KEY LESSONS LEARNED: 1. Overall, patients were satisfied with the Portal. 2. Many patients value the convenience and easy access of the Portal. 3. Lack of response to patients' messages and difficulty accessing and using the site were the main causes of dissatisfaction. 4. Many patients need training in the use of the Portal.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: On-line demonstration of the system.

PILOTING AN OPEN ACCESS APPOINTMENT SYSTEM WITH FOUR PART-TIME PROVIDERS IN A CLINIC SERVING THE UNDERSERVED. S.L. Frelsen¹, R. Rodriguez², M. Holden², L.S. Sadowski³. ¹Stroger Hospital of Cook County/Rush Medical College, Chicago, IL; ²Ambulatory and Community Health Network, Cook County Bureau of Health Services, Chicago, IL; ³John H. Stroger Jr. Hospital of Cook County/Rush Medical College, Chicago, IL. (Tracking ID #133290)

STATEMENT OF PROBLEM/QUESTION: In our urban, ambulatory primary care clinic, patients, providers, and staff were dissatisfied with access to care for patients who had primary care providers. Internal medicine, family practice, and pediatric providers in our clinic see primarily uninsured Latino patients. For internal medicine, our patients routinely waited three months for a follow-up appointment. Missed appointments meant an additional three month wait. Patients with urgent issues waited one to four hours to be triaged or were sent to the ER for evaluation. Our no-show rate averaged 30%.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: 1. Provide patient-driven access to care 2. Increase satisfaction of patients, providers, and staff 3. Accommodate urgent visits minimizing nursing triage time 4. Decrease the no-show rate.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: Four primary care internal medicine providers (three MD's and one PA) formed a practice group with a combined equivalence of a 1.6 FTE. At least one provider covered 9 out of 10 half-day sessions the clinic is open. We entered our pilot with a defined capacity but without a quantifiable demand due to lack of IT support. We began with 50% pre-scheduled slots fillable up to 3 months in advance and 50% Open Access slots ("OA" slots) scheduled no more than 5 days in advance. Patients are encouraged to see another provider if their PCP is not in clinic within that 5 day period. Patients are encouraged to walk in or call on the day they would like to be seen and are offered the earliest available appointment with any OA provider. If all appointment slots are filled for a day when a patient would like to be seen, providers are asked to add on an appointment beyond the scheduled session. During the pilot, our practice group is closed to new patients. Once we went "live" with the OA pilot, patients could only receive pre-scheduled slots if requested by the provider. All other patients may receive an OA slot.

FINDINGS TO DATE/EVALUATION OF WEB SITE: Staff and providers were surveyed initially and at 3 months. After three months of the pilot, OA providers are more likely to agree that: Patients have appropriate access to their providers (5.25 on a 6-point likert scale compared to 1.75 prior to implementation); Their schedule accommodates patient needs (4.7 compared to 1.5); Their schedule accommodates walk-ins (5.25 compared to 2.0). (Note-6=strongly agree, 5=agree, 4=agree a little, 3=disagree a little, 2=disagree, 1=strongly disagree). Non-OA providers showed no significant changes in stress, satisfaction or in these variables. Stress level and satisfaction of nursing and front desk staff showed no significant change. Time to "third next available appointment," a common indicator of pre-scheduled appointment availability in the open access

literature, decreased from 3-4 months to 1-4 weeks. The no-show rate decreased to 10%. Overbooking decreased from 1-3 patients per hour down to nil.

KEY LESSONS LEARNED: You can modify open access to fit your practice and patients. Recommendations: Identify time for coordination, collaboration, and troubleshooting with all providers; Coordinate vacations, conferences, and in-patient time; Aim for buy-in among providers and staff; Balance response to flukes and adjustments for noted trends; If possible, calculate the backlog (demand) and accommodate those patients over the first 6-12 months of OA; Ideally, two providers should cover every half day session to accommodate the demand for same day appointments. We plan to decrease the proportion of pre-scheduled slots from 50% to 25% after three months of the pilot.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster, slides.

PRACTICE-BASED LEARNING AND IMPROVEMENT PROJECT IN PREVENTIVE MAINTENANCE FOR THE INTERNAL MEDICINE RESIDENTS. V. Nandwani¹; E.E. Baum¹; J. Fiorentino¹; L.M. Kirchner¹. ¹Canton Medical Education Foundation, Canton, OH. (Tracking ID #134854)

STATEMENT OF PROBLEM/QUESTION: As evidence of the effectiveness of preventive care accumulates, it is important for residency training programs to evaluate the adequacy of their training in preventive medicine. A random review of 20 charts at an outpatient general internal medicine clinic indicated that preventive maintenance care needed improvement.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: This project was designed to assess the effectiveness of a systems based practice intervention to improve the quality and documentation of preventive care delivery in an internal medicine clinic.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: A prospective clinic chart review was done for 18 internal medicine residents to evaluate their preventive maintenance practices at baseline, and six months after our systems based practice improvement intervention. The systems based practice improvement intervention included the following: The internal medicine residents attended an interactive clinic conference on preventive maintenance practices during which a new preventive maintenance clinic chart form was introduced. The current second and third year residents reviewed six charts of their clinic patients using a standardized form to record the adequacy of their preventive maintenance care. Clinic preceptors randomly reviewed some of the same charts with the residents for accuracy. Criteria for patient charts to be included in the study were the patients should be older than age 50 and seen at least three times by the resident being reviewed. In addition to the new preventive maintenance forms and educational intervention, the clinic preceptors and nursing staff continued to reinforce the importance of ordering preventive health services. Six months after the baseline chart review, the same residents again reviewed 6 charts that were randomly selected from those patients that fit the study criteria.

FINDINGS TO DATE/EVALUATION OF WEB SITE: Data collection is complete and a detailed analysis is in process. Preliminary data indicates 63.7% of the possible preventive interventions were done before our learning intervention, whereas 80.9% of all potential preventive interventions were done following our learning intervention.

KEY LESSONS LEARNED: Involving the internal medicine residents in their own chart review, rather than simply having the clinic preceptors or an independent researcher do the review was very effective. The residents were both surprised and disappointed that their baseline care was not better and this provided the impetus to improve their delivery of preventive care. An ongoing multidisciplinary approach to preventive services is the most effective way to assure adequate delivery of preventive health services. The residents who were most complete in their preventive care routinely listed preventive care as one of their problems on their assessment and plan.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Poster.

SYSTEMATIC APPROACH TO DIABETES CONTROL IN A LARGE POPULATION USING A WEB BASED DATABASE. M. Anderson¹; B.J. Estes²; R. Cooper¹; D. Peterson¹; D. Lohan¹; K. Melaniphy¹; C. Dimou¹. ¹Rush University Medical Center, Chicago, IL; ²Rush Health Associates, Chicago, IL. (Tracking ID #135269)

STATEMENT OF PROBLEM/QUESTION: As noted by the American Diabetes Association in their 2004 Standards of Medical Care in Diabetes "the implementation of the standards of medical care of diabetes is suboptimal" This Diabetes Care program of Rush Health Associates was developed to improve routine monitoring of patients with diabetes using simple office based techniques and a web based database.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: The objectives of the Diabetes Care program are to identify all patients with diabetes cared for by Rush Health Associates physicians and to improve the assessment of glycemic control as well as improve the monitoring and prevention of complications of diabetes.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: The Diabetes Care program at Rush Health Associates began in February 2003 after analysis of chart audits revealed that within a 12 month period only 85% of diabetic patients had a HbA1C level, 64% had an LDL level, and 35% had an assessment for nephropathy. The initiative was developed using multiple intervention strategies including a web based database, patient education, and physician feedback. Patients with diabetes were identified by using billing data and electronically loaded into the web based database. Laboratory and other test data are added to the database either by electronic downloads or hand entry after chart audits. An

online flowsheet was available to print and place in charts and physicians are required to use a flowsheet for all diabetic patients. Ophthalmologists are able to input limited exam data which is immediately available to primary care physicians. Using the information within the online database patients were targeted for invitation to group education run by a Certified Diabetic Educator. Patients were also sent reminders about needed interventions including influenza vaccinations, eye exams, etc. Physicians were given feedback using data stored in the online database and using chart reviews of patients that had been seen within the past 3 months.

FINDINGS TO DATE/EVALUATION OF WEB SITE: Over the course of 2 years 5000 patients have been added to the database. In that time yearly HbA1C testing has climbed to 92%, yearly LDL testing has increased to 78%, yearly nephropathy assessment has climbed to 55% and yearly eye exams have increased to 39%. Initial data analysis also showed that simple use of a flowsheet in the chart decreased average HbA1C from 8.5% to 7.8%

KEY LESSONS LEARNED: Diabetes is a complex disease which requires multiple interventions to decrease potential complications. Simple interventions are worthwhile in helping control the complexity of diabetes. However a systematic, multifactorial approach to management provides significant improvement in the care of diabetes.

THE ELECTRONIC PERSONAL HEALTH RECORD (EPHR): A NEW PARADIGM FOR OWNERSHIP AND USE OF HEALTH RECORDS. J.W. Humphry¹; S. Thomas²; M. Mok³; H. Rubin². ¹Hawaii Medical Service Association, Honolulu, HI; ²University of Hawaii, Honolulu, HI; ³People Chart, San Francisco, CA. (Tracking ID #135680)

STATEMENT OF PROBLEM/QUESTION: Health information is largely on paper, fragmented, and inaccessible to providers when needed, especially for complex patients seeing multiple providers. Safety net primary care practitioners often cannot afford EMR and even those who can, use systems that are generally incompatible with those of other providers who are caring for the same patients.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: To make information accessible using a web-based personal health records, especially to providers who do not have electronic medical records. To overcome barriers to implement electronic records in smaller and safety net practices that lack capital to invest in EMR. To supplement provider-controlled and authorized medical records needed for their business with patient-controlled and authorized medical records containing additional information.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: The US health system lags behind most developed countries in health information technology, leading to medical errors, duplication of services and excessive time in handling medical records. Most strategies for addressing these problems rely on interconnectivity between provider-based EMRs that currently contain less than 10% of all patient records and are unavailable to most providers. Therefore, an important supplement to interconnected provider-based electronic records may be a web-based personal health record that patients own and are responsible for. The ePHR is a complete record of an individual's health history, and accessible, all or in part, as desired and authorized by the patient, to providers, including nontraditional ones such as home care workers, case managers, care givers, and relatives and caregivers with health care backgrounds. The captured records are indexed, factored for authenticity, can be appended, and may contain comment sections for both patient and provider

FINDINGS TO DATE/EVALUATION OF WEB SITE: One proprietary ePHR vendor, Peoplechart, is currently in use by over 1,000 subscribers. An online survey of seniors conducted in 2002 showed that over 60% none or their medical records, or fewer than 10 pages. Over 20% had a medical emergency while traveling and over 70% saw a need for a personal health record. In a separate survey, 87% of adults with living parents felt that having copies of medical records is important. Further evaluation of the ePHR is planned in several settings and populations including disabled children, the homeless population, and community based elders. Impact will be measured for health outcomes, cost, efficiency and acceptability. The authors are conducting feasibility studies in Hawaii at two community health centers and in a palliative care setting. The study will provide ePHR to minority patients with complex chronic medical conditions. The clinics are assessing whether it has replaced the need for paper-based medical records, and saved costs of photocopies, faxes and record couriers to hospitals, ERs, and multiple clinic locations.

KEY LESSONS LEARNED: The office-based medical record is the current standard for longitudinal patient records. The ePHR may provide benefits for patients, physicians, and payers through improved coordination of care, lower costs, and better interactions between patients and physicians. Studies should examine whether coexistence of ePHR and EMR improves on having only one or the other.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: We are willing to do either a poster presentation or panel discussion. Our preference would be a poster presentation.

THE PROCEDURALIST: AN EMERGING SPECIALTY FOR GENERAL INTERNISTS. B.T. Rosen¹; P.K. Ng¹; M.J. Ault¹. ¹Cedars-Sinai Medical Center, Los Angeles, CA. (Tracking ID #136277)

STATEMENT OF PROBLEM/QUESTION: Physicians in all disciplines are certified in a group of core procedures during residency as a requirement for board certification. Once in busy clinical practices, however, these physicians perform these procedures infrequently.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: Given the wide variability in physicians' experience, background, and comfort in performing procedures, patient safety and standardization of procedural techniques is a

concern. Furthermore, scheduling and logistical constraints often force delays in patients' diagnostic evaluation and/or discharge from the hospital. To address these issues, the Division of General Internal Medicine at Cedars-Sinai Medical Center developed a Procedure Center in 1990. This department consists of General Internists who specialize in performing procedures on inpatients and outpatients and providing training opportunities for our resident physicians. These services have been analyzed to evaluate growth in volume, complication rates, and financial viability.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: The Procedure Center at Cedars-Sinai Medical Center is staffed by 4 General Internists who perform procedures 75% of their time and engage in Hospitalist responsibilities the remaining 25%. Meticulous data have been kept on all procedures performed by these physicians since 1990. A retrospective analysis was performed on this procedural data from 2000 to 2004 for the following procedures: Paracenteses, Thoracenteses, Bone Marrow Biopsies, and the placement of Central Venous Catheter/Dialysis catheter (non-tunneled catheter), Tunneled catheters, PICCs (Peripherally Inserted Central Catheters), and PAS-Ports (Peripherally Accessed Subcutaneous Ports).

FINDINGS TO DATE/EVALUATION OF WEB SITE: In the past 5 years we performed a total of 11,992 procedures, with an overall annual rate of growth (AROG) of 12.9% and an overall complication rate of 0.9%. We inserted 174 Central Venous Catheters in 2000 and this increased steadily to 546 in 2004, representing an AROG of 28.6%. Tunneled catheter insertions increased from 78 in 2000 to 102 in 2004 (AROG=15.5%), and PICC placement increased from 1430 lines in 2000 to 2391 in 2004 (AROG=11.0%). PAS-Port insertions numbered 65 in 2000 and increased to 145 in 2004 (AROG=24.7%). Paracenteses began in 2001 with 137 procedures and increased to 631 in 2004 (AROG=82%), while Thoracenteses began in 2002 with 57 procedures and increased to 211 in 2004 (AROG=114%). Bone Marrow Biopsies (under conscious sedation) began in 2001 with 8 procedures, and 65 were performed in 2004 (AROG=112%). From a financial standpoint, the Professional and Facilities fees generated from these procedures more than off-set the cost of the operation: Net operating profit in 2004 was an estimated \$12 million.

KEY LESSONS LEARNED: General internists are uniquely suited to regularly perform myriad procedures previously reserved for subspecialists. The steady growth in our Procedure Center suggests a successful model for centralizing procedural services so as to benefit patients (from a safety standpoint), referring physicians (from a time-saving standpoint), and hospitals (from a cost-savings and revenue-generating standpoint). Overall, this model offers a novel opportunity for General Internists looking to further expand clinical services.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:

UNIVERSITY OF PITTSBURGH HOUSESTAFFS' VIEWS ON PRE-PRINTED ORDER FORMS. T. Feinstein¹; H. Sheth¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133302)

STATEMENT OF PROBLEM/QUESTION: Many health institutes are implementing pre-printed order forms and guidelines on paper or computer to standardize patient care. In University settings, the junior housestaff manually enters most orders related to patient care. To date, there is little data concerning housestaffs' attitudes toward order entry.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: The purpose of this project was to evaluate the perceptions of University of Pittsburgh internal medicine residents regarding pre-printed order forms to determine housestaffs' 1) perceptions on impact of pre-printed order forms on their learning and patient care; and 2) preference of order entry.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: A self-administered survey was distributed to the housestaff at University of Pittsburgh, where the housestaff is familiar with pre-printed order forms used at the University affiliated hospitals and computerized physician order entry at the VA hospital. The surveys set forth six statements describing various experiences using pre-printed order forms, namely impact on efficiency, consistency, quality of care, medical errors, learning and ability to manage patients in future. Each housestaff member was asked to indicate agreement or disagreement on a 5-point Likert scale to evaluate his/her attitude toward pre-printed order forms.

FINDINGS TO DATE/EVALUATION OF WEB SITE: 36 surveys were completed, with two-thirds being preformed by interns. Only 2 house staff members had never used order forms. 84% believed that their current practice reflected 50-75% of evidence based medicine. 64% indicated that pre-printed order forms made housestaff more or much more efficient in managing their patients. 79% of housestaff stated that pre-printed orders made housestaff more or much more consistent in managing their patients. 73% agreed that pre-printed order forms were positive or very positive on improving the quality of patient care. 94% indicated pre-printed order forms reduced medical error. Only 59% of housestaff believed that pre-printed order forms had a positive or very positive impact on their learning and ability to manage patients in the future. Given a choice of order entry to manage their patients, 53% housestaff members favored pre-printed order forms and 38% favored computer entry.

KEY LESSONS LEARNED: The majority of housestaff favored pre-printed order entry. Housestaff most strongly favored pre-printed order forms for increased consistency and quality of patient care, and reduction in medical errors (73-94%). However, housestaff were most concerned that pre-printed orders forms would impact learning and ability to manage patients in the future (12% negative and 29% neutral). This concern may play role in compliance with use of order forms. It is hoped that learning can be enhanced by providing the EBM reference articles, and incorporating institutional guidelines and initiatives in existing training curriculum.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING:

USE OF TIME-MOTION ASSESSMENT TO IMPROVE EFFICIENCY OF PATIENT VISITS IN A PRIMARY CARE-BASED CHRONIC PAIN MANAGEMENT CLINIC. N.M. Potisek¹; R. Malone¹; T.J. Ives¹; P. Chelminski¹; M. Pignone¹. ¹University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #135034)

STATEMENT OF PROBLEM/QUESTION: Effective chronic pain management with opiates requires frequent return follow-up visits.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: 1. To determine time utilization using a time-motion assessment for chronic pain management visits within a structured, primary care-based chronic pain disease management program (PDM) 2. To test the effect of an intervention to improve efficiency and decrease the mean duration of patient visit while providing quality pain management.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: Time-motion assessments were completed at baseline and at the follow-up of interventions implemented to measure time utilization. Interventions were proposed after baseline data were reviewed by nurses, program assistant/case manager, and clinicians to improve visit efficiency. Interventions included using the program assistant to: 1) review and prioritize PDM visits before each clinic session to determine proper level of clinician and program assistant intervention with the patient, 2) Assist clinic nurses in completing initial pain assessments and placing patients in exam rooms, 3) Change clinic flow for collection of urine specimens for toxicologic screening, and 4) Use wireless technology for direct entry of clinical data on laptop computers into a clinical information system without traditional dictation.

FINDINGS TO DATE/EVALUATION OF WEB SITE: Baseline and follow-up time-motion data were collected on 28 and 29 consecutive patients, respectively. After excluding new patient visits or patients arriving greater than 45 minutes early for their appointment, 25 baseline and 19 follow-up patient encounters were analyzed. At baseline, visit duration was 65 minutes (+/- 20.8); 74% of this total visit time occurred before being evaluated by a clinician. The majority of the wait time occurred between check-in and being placed in an exam room (25 min.), and between being seen by the program assistant and the clinician (18 min.). After the interventions were implemented, visit duration decreased by 34% to 43 minutes (+/- 14 min, p<0.001); 63% of this total time took place before being evaluated by a clinician. Significant improvements were made in reducing the wait time by decreasing the time to be placed in an exam room from 25 minutes to 12.5 minutes, and the time between being seen by program assistant and clinician from 18 minutes to 4 minutes (+/- 1.1).

KEY LESSONS LEARNED: 1. Time motion assessments are an effective way to identify inefficiencies in a primary care-based clinic and to develop interventions in response to baseline data. 2. Time motion interventions can improve the efficiency of chronic pain management within a primary care-based disease management program, increasing the potential to serve a larger total volume of patients.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Modality to be used to demonstrate innovation at meeting: PowerPoint slides (LCD)

USE OF TIME-MOTION SURVEY TO EVALUATE THE EFFECT OF AN INTERVENTION TO IMPROVE EFFICIENCY OF PRIMARY CARE-BASED ANTICOAGULATION. R. Malone¹; B. Bryant¹; D. Wobbleton¹; T.M. Miller¹; P. Chelminski¹; M. Pignone¹. ¹University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #135093)

STATEMENT OF PROBLEM/QUESTION: Effective and safe anticoagulation requires frequent visits for monitoring. Inefficient anticoagulation visits place a substantial burden on patients and may inhibit adherence.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: 1. To determine, using a time-motion survey, the time usage for anticoagulation visits within a structured, primary care-based anticoagulation disease management program (ADM). 2. To test the effect of an intervention to improve anticoagulation efficiency.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: Baseline data was collected from 3/31/04 through 4/14/04. Ninety-three patients were seen by ADM and 26 (28%) completed the baseline survey, which included patient satisfaction questions. Time-motion surveys were completed by a combination of clinic staff and patient self-report. Baseline data was reviewed with key representatives from the clinic staff, laboratory services, ADM staff, and clinic providers. Interventions included 1.relocating point-of-care (POC) INR testing services closer to ADM, 2.simplifying the check-in process, 3.adding nursing support, and 4.educating clinic staff. Follow-up data were collected from 10/11/04 through 10/18/04. Ninety-six patients were seen by ADM and 48 (50%) completed the survey.

FINDINGS TO DATE/EVALUATION OF WEB SITE: At baseline, a full ADM clinic visit took a mean of 84 minutes to complete (+/- 49.9); 76% of this total visit time took place before being evaluated by an ADM clinician. The majority of wait time occurred during lab check-in, POC INR draw, administrative check-in (28.4 min), and waiting to be placed in an exam room (26.2 min). At baseline, 65% of patients ranked the ADM program as "very good", while 35% rated POC testing process as inconvenient. After the intervention, overall mean visit time decreased by 33.1 minutes or 39% (p<0.001). The ADM clinic visit took a total of 50.9 minutes (+/- 24.5), 60% of which took place before being evaluated by an ADM clinician. The most significant improvements were made in the time it took to check in and undergo POC INR testing, from 28.4 to 3.9 min. After the intervention, wait time for an exam room increased from 26.2 to 28.7 min, while volume increased from an average of 12 to 16 patients per day.

KEY LESSONS LEARNED: 1. Time motion studies are an effective way to identify inefficiencies and target interventions in primary care. 2. Clinic-based point of care testing can improve the efficiency of anticoagulation care within a primary care based disease management program.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: Modality to be used to demonstrate innovation at meeting: PowerPoint slides (LCD)

WEB-BASED REPORTING OF MEDICAL ERRORS IN AN AMBULATORY ACADEMIC GIM PRACTICE. D.P. Dunham¹; D. Liebovitz¹. ¹Northwestern University, Chicago, IL. (Tracking ID #132739)

STATEMENT OF PROBLEM/QUESTION: There has been considerable interest in identifying medical errors in the inpatient setting. There has been much less investigation in exploring patient safety in the ambulatory setting, despite more patient care occurring in this setting. We explored the utility of using a web-based reporting system to gain insight on the types and frequency of errors committed in an ambulatory setting.

OBJECTIVES OF PROGRAM/INTERVENTION/WEB SITE: The objectives of the reporting project: 1. Establish a taxonomy of medical errors in a university-based attending and resident ambulatory practice. 2. Determine differences in the perception of errors among health care providers and resident and attending physicians. 3. Develop a culture in which health care providers have a sense of ownership and responsibility for the safety of their practice.

DESCRIPTION OF PROGRAM/INTERVENTION/WEB SITE: A web-based questionnaire was made available to nursing staff, nurse practitioners, resident and attending physicians. An icon on the front page of the electronic medical record served as a link to the questionnaire. This questionnaire was to be completed by any health care team member, if anything happened in the practice that put patients at risk. The entries were evaluated by a group that included resident and attending physicians, nurses, a practice manager and a nurse practitioner. The entries were categorized by types as either process errors or knowledge and skill errors. The entries were also categorized as actionable or not.

FINDINGS TO DATE/EVALUATION OF WEB SITE: Over the past 16 months we have had 76 entries. Fifty-two entries came from attending physicians, 16 came from nurse practitioners, 6 came from staff members and 2 came from residents. Sixty of the entries were actionable, while 16 were not. Almost all entries (95%) described process errors: only 4/76 described knowledge errors. The most frequent process errors were in office administration (for example, registration problems or physical set-up of practice), (31/76, 41%). The next most frequent source of errors was miscommunication (25/76, 33%), followed by treatment errors (8/76, 10%), laboratory errors (7/76, 9%), knowledge errors (4/76, 5%), and errors in workforce management (1/76, 1%).

KEY LESSONS LEARNED: We found that a web-based error reporting system can be easily implemented within our existing electronic medical record and that the suggestions from this system were predominantly meritorious. Providers are much more likely to report process than knowledge based errors. Residents were less likely to utilize this system. Monthly review of submissions by a patient safety committee that includes nursing, physicians, and office administrators enabled effective and timely responses. We have continued to have staff and physicians use this system, and it has led to a greater appreciation and acceptance of the importance of safety in our practice.

MODALITY(IES) USED TO DEMONSTRATE INNOVATION AT MEETING: We will use a laptop computer with internet access to demonstrate how to enter and complete the questionnaires. We will also use the computer to demonstrate how categories of medical errors were assigned.

SCIENTIFIC ABSTRACTS

A CONTROLLED TRIAL OF A PRACTICE-BASED INTERVENTION TO IMPROVE PRIMARY CARE FOR FALLS, INCONTINENCE AND DEMENTIA. N.S. Wenger¹; C.P. Roth²; P.G. Shekelle³; R.T. Young¹; D.H. Solomon²; C. Kamberg³; J.T. Chang¹; T. Higashi¹; C. Maclean²; J.L. Adams²; R. Louie²; L. Chiang¹; K. Ransohoff⁴; M. Hoffing⁵; D.B. Reuben¹. ¹University of California, Los Angeles, Los Angeles, CA; ²RAND, Santa Monica, CA; ³RAND, Washington, DC; ⁴Sansum-Santa Barbara Medical Foundation Clinic, Santa Barbara, CA; ⁵Desert Medical Group, Palm Springs, CA. (Tracking ID #136042)

BACKGROUND: Medical care for age-associated conditions such as falls, urinary incontinence and cognitive impairment has been shown to be inadequate, with less than a third of recommended care rendered in one study. While many methods of improving geriatric care have been tested, few have been broadly implemented and no trial has tried to improve care by intervening at the level of the primary care clinician. We developed and tested a practice-based intervention to improve the process of care for falls, urinary incontinence and cognitive impairment performed in the primary care setting.

METHODS: We conducted a controlled trial in two large medical groups in Southern California, each with geographically distinct intervention and control clinics. The trial compared care received by 357 intervention group and 287 control group community-dwelling patients, 75 years and older, identified by the patient or a proxy as having difficulty with or concerns about falls, incontinence or cognitive impairment, between April 2002 and August 2003. Intervention and control practices both received case-finding for each condition, but only intervention practices received physician education and a multi-component practice-change intervention including structured visit notes and patient and physician condition-specific materials. The study aimed to improve the process of care for falls, incontinence and cognitive impairment, measured as the percentage of quality indicator recommended care received based on a 13-month medical record abstraction and patient (or proxy) interviews.

RESULTS: Before the intervention, process of care was the same in intervention and control groups. Overall care for the three conditions during the intervention was better in the intervention group than in the control group (47% [95% CI

44%-50%] v 32% [95% CI 29%-36%], $p < 0.001$). Intervention group patients, compared to control group patients, received better care for falls (44% v 23%, $p < 0.001$) and incontinence (49% v 36%, $p = 0.001$), but not for cognitive impairment (49% v 52%, $p = 0.66$). For example, intervention group patients, compared to control patients, were more likely to be examined if they fell or had gait concerns (45% v 12%) and were more likely to have a behavioral intervention tried first for bothersome incontinence (33% v. 4%).

CONCLUSIONS: A practice-based intervention can improve primary care provision of care for falls and urinary incontinence, but further improvement is needed because even with the interventions, patients received only half the recommended care for these conditions.

ADVERSE HEALTH EVENTS: HIDDEN BARRIERS TO AGING IN PLACE. K.G. Scandrett¹; D. Lau¹; K. Chin¹; E. Hagenmaier¹; T. Bandy¹; L. Emanuel¹. ¹Northwestern University, Chicago, IL. (Tracking ID #135757)

BACKGROUND: Many people desire to remain in their homes and communities as they grow older. While health problems, environmental hazards, poor social and financial resources, and medical system failures may all contribute to a loss of independence, the prevalence and root causes of these factors are not well-studied. Using the methods of the patient safety movement, some of these factors may prove to be preventable. To that end, we propose the need for a new concept in the literature on aging: "adverse health events." This term expands the concept of adverse medical events to include events not directly caused by medical management. We present results from a preliminary pilot study using qualitative methods to identify and analyze adverse health events occurring in the community setting.

METHODS: This is a cross-sectional pilot study of English-speaking individuals 65 years of age or older, and their accompanying caregivers. Over a five-weekday period between 10 a.m. and 4 p.m., patients eligible for study enrollment were those presenting for emergency care and subsequently admitted to Northwestern Memorial Hospital, Chicago, IL. Using an open-ended interview, three trained investigators collected data from each patient and caregiver in order to identify the event(s) leading to their visit. Employing qualitative methodology based on root-cause analysis, investigators met daily to identify the factors leading to these events, and to sort these factors into categories emerging from the data. Data on race/ethnicity, age, and gender of patient and caregiver were also obtained.

RESULTS: Twelve in-person interviews (four patient-caregiver dyads and eight interviews with patients only) averaging about 26 minutes each, were conducted at bedside in the emergency department. Average patient age was 78 years old (ranging from 65 to 94). Seven patients were female; six were white and six were black. Of the six patients who reported having a caregiver in the home, one had a paid home caregiver. Adverse health events in the home that resulted in hospital admissions included falls, stomach pain, constipation, edema, chest pain, medication problems, visual changes, and weakness. Preventable contributing factors to these events fell into four general categories: (A) lack of care coordination; (B) financial barriers to medication or medical care; (C) lack of knowledge regarding emergency services; and (D) an unsafe home environment.

CONCLUSIONS: Patient safety in the community-dwelling elderly is a significant public health issue. This study suggests that unintended, preventable adverse health events contribute to functional decline and morbidity in this population. As the national population ages, it will increasingly rely on personal and medical care given in the community setting. Therefore, future large-scale, representative epidemiological studies are needed to investigate the prevalence of adverse health events, identify the root-causes of the events, and analyze the relationship between causal factors and patient/caregiver characteristics. Once these are understood, interventions may be designed to improve the health and independence of this population.

AN ELECTRONIC MEDICAL RECORD (EMR)-BASED INTERVENTION TO REDUCE POLYPHARMACY AND FALLS IN AN AMBULATORY RURAL ELDERLY POPULATION. V. Weber¹; A. White²; R. McIvried¹; J. Sim¹. ¹Geisinger Health System, Danville, PA; ²Abt Associates, Cambridge, MA. (Tracking ID #133452)

BACKGROUND: Falls are the leading cause of injury-related deaths in the aging population, and account for fully 6% of US medical care expenditures in the elderly. Medication risk factors are a potentially reversible contributor to falls. Electronic Medical Record (EMR) systems can both identify at-risk patients and enable interventions to decrease risk factors for falls. The objectives of this study were to evaluate an EMR-based intervention to reduce medication use in an ambulatory elderly population at risk for falls, and to determine whether the intervention could reduce the number of falls and overall health costs in this population.

METHODS: Patients at risk for falls were identified using the GHS EMR database. All patients were community-dwelling, followed by a GHS primary care physician and had Geisinger Health Plan (GHP) Medicare + Choice insurance. 413 patients were randomized to the EMR intervention, with 207 assigned as controls. A medication review was conducted by either a pharmacist or geriatrician. Individualized suggestions regarding medications, often focused on the use of psychoactive medications were made, and a message was sent to the physician via the EMR. A link to a fall evaluation practice guideline was included as part of the message. Patients were contacted by telephone to obtain self-reports of falls at months 1, 3, 6, 9, and 12. Fall-related diagnoses and pharmacy data were collected through the EMR. Claims related cost data was collected via the GHP claims database. A combination of descriptive analyses and multivari-

ate regression models were used to evaluate differences between the two groups adjusting for baseline medication patterns and comorbidities.

RESULTS: The intervention group had lower utilization of psychoactive medications and fewer new psychoactive medications than the comparison group. Those differences were larger for those who were prescribed more psychoactive medications at baseline ($p < 0.01$). The intervention was also associated with a reduction in fall risk, with those in the intervention group only 0.38 times as likely to have one or more fall-related diagnoses ($p < 0.01$). There was no significant difference in health care costs seen between the two groups, however the lack of availability of nursing home cost data limited our ability to conduct cost effectiveness analyses.

CONCLUSIONS: The use of the EMR to guide best practice is promising but largely unproven. The current study suggests that using an EMR to assess medication use in the elderly can reduce the use of psychoactive medications and falls in a community dwelling, rural elderly population.

AN ELECTRONIC MEDICAL RECORD INTERVENTION INCREASED NURSING HOME ADVANCE DIRECTIVE ORDERS AND DOCUMENTATION. S.A. Lindner¹; J.B. Davoren²; A. Vollmer³; C. Landefeld³. ¹University of California, San Francisco, San Francisco, CA; ²San Francisco VA Medical Center, San Francisco, CA; ³UCSF, San Francisco VA Medical Center, San Francisco, CA. (Tracking ID #132650)

BACKGROUND: Optimal care of seriously ill patients requires orders based on well documented Advance Directives (AD). The purpose of this study was to develop and pilot test an Electronic Medical Record (EMR) intervention to increase the use of AD notes and the quality of AD orders.

METHODS: The study site is a VA Nursing Home, which has a long-standing EMR for all clinical information including physician orders. The intervention consisted of two changes in the EMR: first, the EMR required the physician to order admission with full resuscitation (Full Code), or admission with Do Not Resuscitate (DNR) and other treatment-limiting orders; second, 24 hours after admission, the EMR sent the admitting provider a templated AD Note to document life-sustaining treatment preferences and goals of care. We studied all patients admitted to the Nursing Home during six consecutive months, three before and three after the intervention. The primary outcome was completion of an AD Note by a patient's Primary Care Provider (PCP). Secondary outcomes included rates of physician orders about other life-sustaining treatments, and concordance between these orders and documented patient preferences, as ascertained from Admit or Advance Directive Notes.

RESULTS: The pre-intervention ($n=117$) and post-intervention ($n=107$) groups were similar (P.40) in mean age (71 years), gender (90% men), functional status (58% dependent in >1 Activity of Daily Living), dementia (25%), percent admitted for palliative care (14%), and patient preference for DNR status (38%). A PCP completed an AD Note for 5 patients pre-intervention (4%) and for 67 patients post-intervention (63%, $P < 0.001$); the rates of AD Note completion were 3%, 3%, and 6.7% in the 3 months pre-intervention and 47%, 68%, and 71% in the 3 months post-intervention. In a logistic regression analysis adjusted for age, race, gender, dementia, religion, admitting service, and DNR status, the intervention was associated with AD Note completion (OR 41.9, 95% CI 14.6-120.0). Consistent changes occurred in secondary outcomes: among patients preferring DNR, 86% had an accompanying DNR order pre-intervention vs. 98% post-intervention, ($P=0.065$); furthermore, orders to limit other life-sustaining treatments in these patients increased from 16% to 40% ($P=0.013$).

CONCLUSIONS: A targeted EMR intervention greatly increased completion of AD notes in seriously ill patients. This effect was sustained over three months, and orders were more often congruent with patient preferences to limit treatment.

BEING OLD AND DOING TIME: A DESCRIPTIVE STUDY OF THE FUNCTIONAL PROBLEMS OF OLDER WOMEN PRISONERS. B.A. Williams¹; K. Lindquist¹; H.M. Strupp²; D.J. Willmott³; R.L. Sudore¹; L.C. Walter¹. ¹University of California, San Francisco, San Francisco, CA; ²Legal Services for Prisoners with Children, San Francisco, CA. (Tracking ID #135714)

BACKGROUND: In the last decade the number of geriatric prisoners has increased by 450%, and women are the fastest growing subset of the prison population. The California Department of Corrections defines geriatric prisoners as older than 55 years. Few studies have examined functional problems in older prisoners and none have focused on women. Our objective was to describe the frequency of functional problems reported by older women prisoners and to explore how these difficulties interact with the prison environment.

METHODS: In August 2004 health surveys were sent to the estimated 270 prisoners older than age 55 living in all 3 California women's prisons. Subjects were identified by a legal advocacy group for prisoners. We performed descriptive analyses of the functional problems reported in the surveys. Functional problems were categorized into three groups: 1. Basic activities of daily living "ADLs" (bathing, eating, toileting, transferring and dressing) and mobility (walking and falls), 2. Prison activities of daily living "PADLs" (defined as activities specific to daily prison life including standing in line for medications, dropping to the floor for alarms, being strip-searched, getting on and off the top bunk, reading signs, hearing orders from staff) and 3. Safety (measured by reports of verbal harassment, physical abuse or feeling unsafe).

RESULTS: Presently, 115 surveys have been returned and respondents were evenly distributed from each prison. The mean age was 62 (range 55-82 years), 70% were white, 13% black, and 5% Latino. Three-quarters of sentences were either 15 years-to-life or life without possibility of parole. Nearly half of women had been in prison for more than 15 years. Difficulties in ADLs, Prison ADLs and safety concerns were common. At least one ADL impairment was reported by

14% of women and 21% reported needing help walking. Of those needing help with ADLs or walking ($n=29$), 61% received help from other prisoners, 4% from staff, and 35% reported that no one helped them. Falls were common with 50% reporting at least 1 fall in the past year, increasing to 69% in the subset of women with either walking or ADL difficulties. In contrast to ADL difficulties, nearly all women (98%) reported difficulties with at least 1 PADL. The most common difficulties included dropping to the floor for alarms (78%), standing in line for medications (75%), climbing on and off the top bunk (70%) (for the women assigned to a top bunk $n=33$), and hearing orders from staff (58%). Safety was a common concern with 70% of women reporting physical abuse or harassment by other prisoners and 20% feeling unsafe in their prison cells.

CONCLUSIONS: Functional problems and safety concerns are common among geriatric women prisoners in California. While some women have ADL limitations, many more have difficulties with the unique demands of daily prison life which we have termed "Prison ADLs." In the next decade, the population of geriatric prisoners is expected to grow exponentially. As it grows, it is important to either modify the prison environment to maximize the function and safety of geriatric prisoners or to find an alternative to incarceration for this population.

BODY WEIGHT PERCEPTION IN ELDERLY INDIVIDUALS. E.E. Tung¹; P. Targonski¹. ¹Mayo Clinic, Rochester, MN. (Tracking ID #136177)

BACKGROUND: Nutritional status is an important determinant of health among elders. Previous studies have demonstrated the negative consequences associated with malnutrition, including increased rates of hospitalization and mortality. Dieting and weight loss among elders may pose an important health threat. Studies examining body dissatisfaction have traditionally been written about young women. Few have examined the existence of distorted body image in late life. This study examines risk factors for inappropriate weight perception among elderly individuals.

METHODS: We utilized data from the Third National Health and Nutrition Examination Survey (NHANES III), conducted from 1988-1994. NHANES III is a cross sectional population based health survey. Our analysis was limited to those aged 60 and older, yielding 4,980 weighted records. The association between participants' characteristics and weight perception was examined. Exposure variables with a $p \leq 0.10$ in the univariate analysis were included in the final multivariate models.

RESULTS: Among individuals with body mass indices less than 25, male sex (OR=3.59, CI 2.46-5.24); non-white race (OR=3.71, CI 2.33-5.90); unmarried status (OR=1.55, CI 1.20-2.00); higher educational attainment (OR=1.40, CI 1.37-1.43); and utilization of an assistive device (OR=2.15, CI 1.11-4.18) were associated with increased likelihood of inappropriately classifying oneself as overweight. Among overweight elders (BMI > 25), use of an assistive device (OR=0.76, CI 0.65-0.89) and increased age (OR=0.92 per year of age, CI 0.91-0.93) made elders less likely to appropriately describe themselves as overweight. With each year of age, elders became less likely to consider themselves overweight, independent of true BMI.

CONCLUSIONS: The present study findings indicate that nontraditional risk factors may contribute to inappropriate weight image in later life. These results may represent the impact of societal pressures or fear of obesity-related illness. This analysis indicated a significant interaction between use of an assistive device and weight perception. Individuals with pathologic body perception may be at risk for disordered eating, weight loss, and malnutrition. Future studies examining the consequences of body weight misperception are necessary. Recognition of this trait is an important factor in the nutritional assessment of elderly patients.

CARING FOR THE DYING IN PRIMARY CARE: IS "CRADLE TO GRAVE" AN EMPTY PROMISE? M.J. Silveira¹; S. Raju¹; F. Jane¹. ¹Ann Arbor VAMC, Ann Arbor, MI. (Tracking ID #132914)

BACKGROUND: Good end-of-life care is consistent with the mission and purpose of good primary care. Nevertheless, efforts to improve care of the dying have examined how sub-specialists in palliative care, rather than primary care providers (PCPs), can meet the growing demand for end-of-life care. To date, such efforts have had limited success. Presumably, providing end-of-life care through primary care could expand access to end-of-life care more quickly and cost-effectively than would creating sub-specialty palliative care services. Thus, we asked, "What does it take to care for the dying in primary care?"

METHODS: We conducted 15 focus groups with PCPs and their nursing staff at 7 sites throughout Southeast Michigan to explore their care of patients with progressive, life-limiting conditions. PCPs and nursing staff were interviewed separately. Interviews were taped, transcribed, and analyzed using cross-case comparison. Transcripts were reviewed by multiple readers to: 1) develop a coding scheme, 2) identify themes, and 3) design an interpretive model.

RESULTS: Primary care providers and their nursing staff feel they have the knowledge and desire to provide competent end-of-life care to patients with progressive, life-limiting conditions. Their ability to do so, however, depends upon the following care delivery issues that are often beyond their control: 1) A system to maintain continuity with patients referred to subspecialists or admitted to inpatient services; 2) Scheduling flexibility to promote prompt, thorough, and frequent follow up with complex patients; 3) Care coordination to facilitate communication between PCPs, patients, and subspecialists; to provide a point of contact for patients; and to identify and access to community resources; 4) Clear role delineation among providers caring for mutual patients; and 5) Adequate information from subspecialists or inpatient services to participate in meaningful planning discussions with patients. Subjects feel that palliative care experts

are needed only for the most challenging of cases; otherwise, they present an obstacle to continuity with primary care. Clinics in which long-term continuity is not established between PCPs and patients (e.g., resident-run clinics), however, may benefit from the availability of dedicated palliative care services.

CONCLUSIONS: For PCPs to provide end-of-life care, they need better support within primary care rather than more education or palliative care services to refer patients to. Support tailored to the specific needs of each primary care practice may allow patients to have greater access to end-of-life care through their established primary care source.

CAUSE AND PLACE OF DEATH: THE FRAMINGHAM HEART STUDY EXPERIENCE.

E.P. McCarthy¹, M.J. Pencina², A.C. Bourland¹, M. Kelly-Hayes², R.B. Burns¹, J.M. Murabito². ¹Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA; ²NHLBI's Framingham Heart Study, Framingham, MA, Boston University School of Medicine, Boston, MA. (Tracking ID #135134)

BACKGROUND: The Framingham Heart Study (FHS) has followed a well-characterized cohort, with biennial exams from 1948 until their death. As the majority of the cohort has now died, we describe cause and place of death for men and women separately in a subset of the cohort who survived to at least 1980 (the date when place of death started to be recorded on death certificates) and then died.

METHODS: At FHS inception, the 5,209 original cohort participants ranged in age from 28 to 62 years. We studied participants who died from 1980 to 2003 (n=2,422). We reviewed all death certificates to classify place of death as home, other outpatient setting (e.g., emergency rooms), inpatient acute care hospital, and nursing home/other extended care facility. Cause of death was adjudicated by a panel of three FHS investigators using standardized criteria after reviewing all available medical information such as hospital and nursing home records, death certificate data, and next-of-kin interviews. Cause of death data were classified as cardiovascular disease, cancer, other, and unable to be determined. Demographic data from 1948 included age, sex, and education level. The most recent FHS exam data available prior to death was used to ascertain the presence and number of chronic conditions. We ascertained marital status at time of death from death certificates. We performed sex-specific multivariable logistic regression to examine the association between cause of death and 1) dying at home and 2) dying in an acute care hospital. Models were adjusted for age at study entry, age at death, education, marital status, and number of chronic conditions. We excluded 401 participants who did not attend at least one of the last three exams prior to death (6 years).

RESULTS: Of 2,021 participants in our study, 57% were women, 71% had a high school education or less, 52% were widowed, and 51% had 2 or more chronic conditions. At the time of their death women were older than men (mean age 85 vs. 81 years, respectively), but had a similar number of chronic conditions (mean 1.6 vs. 1.8, respectively). Women were more often widowed at the time of death than men (70% women vs. 30% men), whereas men were more often married than women at the time of death (65% men vs. 17% women). Overall, 35% of men and 29% of women died from cardiovascular disease; 23% of men and 19% of women died from cancer; 31% of men and 36% of women died from other causes; and a cause of death could not be determined for 11% of men and 16% of women. Men died more often than women in acute care hospitals (56% vs 47%), whereas women died more often in nursing homes/extended care facilities (19% vs. 36%). Men died at home as often as women (15% vs. 13%). In multivariable analyses, being married at the time of death was associated with dying at home for men [adjusted odds ratio (aOR) 1.7, 95% CI (1.1-2.9)], but not for women [aOR 0.8, (0.5-1.4)]. As expected, dying from cancer (compared to dying from cardiovascular disease) was strongly associated with dying at home in multivariable analyses for both men [aOR 4.9, 95% CI (2.8-8.5)] and women [aOR 4.3, (2.4-7.6)]. Conversely, dying from cardiovascular disease (compared to dying from cancer) was associated with dying in an acute care hospital in multivariable analyses for women [aOR 1.9, (1.3-2.8)], but not for men [aOR 1.4, (0.92-2.0)].

CONCLUSIONS: In this elderly subset of long term survivors of the FHS cohort who died from 1980 to 2003, women who died were older, widowed, and died more often in nursing homes or extended care facilities compared to men. These factors may help identify persons at risk for placement in extended care facilities prior to death and direct resources to this vulnerable group.

DEPRESSION AND DECLINE IN FUNCTIONAL STATUS AMONG PATIENTS WITH PERIPHERAL ARTERIAL DISEASE. B. Luo¹, K. Liu¹, L. Tian¹, J. Tan¹, M.M. McDermott¹. ¹Northwestern University, Chicago, IL. (Tracking ID #134936)

BACKGROUND: Depression is prevalent among patients with peripheral arterial disease (PAD) and is associated with greater impairment of lower extremity functioning. However, the effect of depression over time on objective functional status after adjustment for disease severity, co-morbidities, and recent interim events is unknown.

METHODS: To address this question, we performed a prospective cohort study of 417 patients with peripheral arterial disease (PAD) who were followed annually for 2 years as part of the Walking and Leg Circulation study. Depression was measured annually using the Geriatric Depression Scale (GDS-S). A GDS-S score of >5 was considered positive for depression. Our primary predictor variable, depression, was categorized as persistent, new, resolved, and no depression based on the current year and the prior year scores. Our outcome variables assessing functional status included change in 5 minute walk test distance and 4 meter fast walking velocity. In order to include repeated measures of variables, we performed a mixed effects linear regression to examine the effect of the var-

ious depression categories on functional status after adjusting for baseline demographics, ankle brachial index, leg symptoms, co-morbidities, and recent interim events.

RESULTS: The baseline prevalence of depression was 25%. Of those with depression, approximately 1/3 had resolution of their symptoms by the next annual visit whereas 2/3 had persistent depression. New depression was independently associated with negative change in 5 minute walk test distance (-59ft, 95%CI -120 to 0) and 4 meter fast walking velocities (-0.07m/s, 95%CI -0.12 to -0.01) compared with those of patients with no depression. Persistent depression was independently associated with negative change in 5 minute walk test distance (-47ft, 95%CI -90 to -5) and 4 meter fast walking velocities (-0.06m/s, 95%CI -0.10 to -0.02) compared with those of patients with no depression.

CONCLUSIONS: New or persistent depression was associated with worse functional status among patients with PAD independent of disease severity, co-morbidities, and recent cardiovascular events.

DEVELOPMENT AND VALIDATION OF A PROGNOSTIC INDEX FOR 4-YEAR MORTALITY IN COMMUNITY-DWELLING ELDERLY. S.J. Lee¹, K. Lindquist¹, M.R. Segal¹, K.E. Covinsky¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #135923)

BACKGROUND: Previous work has shown that both comorbid conditions and functional measures predict mortality in older adults. However, there have been few attempts at combining both classes of predictors into one prognostic index. The purpose of this study was to develop and validate such a prognostic index for 4-year mortality in community dwelling elders using information that can be obtained entirely from patient report.

METHODS: Using data from the Health and Retirement Study (HRS), we developed the prognostic index in 11875 community dwelling elders from the Eastern, Central, and Western regions of the country (mean age 67, 81% white, 57% female, 12% overall 4-year mortality). We validated the index on 8143 subjects from the Southern region (mean age 67, 71% white, 56% female, 13% overall 4-year mortality). Subjects were asked about their demographic characteristics, whether they had specific diseases, and whether they had difficulty with a series of functional measures. We used bivariate and multivariate analyses to identify variables independently associated with mortality and weighted the variables to create a risk index.

RESULTS: In the development cohort eleven independent predictors of mortality were identified and weighted to create a point (pt) scale. The eleven predictors included two demographic variables (age [60-64, 1pt; 65-69, 2pts; 70-74, 3pts; 75-79, 4pts; 80-84, 5pts; >85, 6pts]; male gender, 2pts), five comorbid conditions (diabetes, 1pt; cancer, 2pts; lung disease, 2pts; heart failure, 2pts; current tobacco use, 2pts), and difficulty with four functional variables (bathing or showering, 2pts; walking several blocks, 2pts; managing money, 2pts; pushing or pulling large objects, 1pt). We calculated a risk score for each subject summing the points for each risk factor present. The 4-year mortality rates by risk score and C-statistic for both the development and validation cohorts are shown in the table.

CONCLUSIONS: This prognostic index, incorporating age, gender, self-reported comorbid conditions and functional measures, accurately stratifies community-dwelling elders into groups at varying risk of mortality. This index may be useful to counsel patients, for risk adjustment, and for prognostic adjustment in epidemiologic studies.

4-Year Mortality Rates by Risk Score

Risk Score	Development Cohort	Validation Cohort
0-3 pts	83/4514 (1.8%)	76/3006 (2.5%)
4-5 pts	144/2798 (5.2%)	138/1817 (7.6%)
6-7 pts	268/2007 (13.4%)	200/1370 (14.6%)
8-9 pts	292/1194 (24.5%)	189/862 (21.9%)
10-12 pts	348/822 (42.3%)	281/647 (43.4%)
>12 pts	214/325 (65.9%)	179/278 (64.4%)
C-statistic	0.83	0.81

DIFFUSION OF STATIN USE IN ELDERLY PATIENTS FROM 1992-2000: THE EFFECT OF PRESCRIPTION DRUG INSURANCE. J. Tjia¹, J. Fu¹, J.S. Schwartz¹. ¹University of Pennsylvania, Philadelphia, PA. (Tracking ID #135931)

BACKGROUND: Statins have emerged as a key therapeutic agent to reduce cardiovascular morbidity and mortality in at-risk patients. Little is known about the use of these agents in the elderly and the factors leading to early adoption of use in this population. We studied national trends in statin use among older patients with cardiovascular disease in the 1990s to examine whether drug insurance coverage accelerates adoption of use.

METHODS: We used data from the inception cohorts of the Medicare Current Beneficiary Survey (MCBS), a national probability sample of Medicare beneficiaries, for subjects enrolling in 1992-2000. Our study population was community dwelling beneficiaries aged 65 and older with a history of cardiovascular disease identified by self-report or claims data. The main outcome variable was prevalence of statin use as determined from MCBS prescription medication files. The main predictor variable was outpatient prescription drug insurance. In each year, bivariate analyses compared the prevalence of statin use by drug insurance; multivariable logistic regression was used to estimate the independent effect of drug insurance on statin prevalence adjusting for age, sex, race, in-

come, geographic residence, functional status, health status and comorbid conditions.

RESULTS: Between 1992 and 2000, the percentage of older adults with cardiovascular disease using statins increased from 5.7% to 35.2% ($P < .001$). Overall, there was a small increase in statin use from 1992-1996 (from 5.7 to 11.6%; $P < .001$). From 1996-2000, the prevalence rose sharply (from 11.6% to 35.2%; $P < .001$). Increased use of these medications occurred across multiple age and race groups, and for those with and without drug insurance. However, those without drug insurance had a lower prevalence of statin use in each year. For example, statin use in 1992 was 4.4% versus 8.9% for those without and with drug insurance ($P < .001$). In 2000, statin use was 30.8% versus 38.0% ($P = .017$), respectively. After adjusting for potential confounders, the effect of drug insurance on the likelihood of statin use decreased between 1992 and 2000, from an adjusted odds ratio [AOR] of 1.94 (95% confidence interval [95% CI] 1.21-3.10) to an AOR of 1.29 (95% CI 0.89-1.87).

CONCLUSIONS: Drug insurance appears to contribute to the early adoption of statin utilization in older adults with cardiovascular disease. Although the effect appears to diminish over time, these data suggest that drug insurance is an important factor affecting the adoption and diffusion of emerging pharmacotherapy in chronic conditions.

DOES OUTPATIENT PRESCRIPTION DRUG INSURANCE CONTRIBUTE TO THE PRESERVATION OF FUNCTIONAL INDEPENDENCE IN OLDER ADULTS? J. Tjia¹, D.P. Egan¹, J.S. Schwartz¹. ¹University of Pennsylvania, Philadelphia, PA. (Tracking ID #136259)

BACKGROUND: Although there is increasing evidence that restricting prescription medication use because of cost increases the risk of adverse events such as hospitalizations, and hospitalizations in the elderly are often associated with unrecoverable declines in functional status, little is known about the association between outpatient drug coverage and preservation of functional status in older adults.

METHODS: We analyzed data from a prospective cohort study of older adults who reported taking any prescription medication in the 1998-2000 Medicare Current Beneficiary Survey (MCBS), a national probability sample of Medicare beneficiaries. We used multivariable logistic regression to assess the independent effect of prescription drug coverage in 1998 on change in functional status over 2 years of follow up. The two outcome variables included development of an impairment in (1) any activity of daily living (ADL) and (2) any instrumental activity of daily living (IADL). ADLs include any difficulty toileting, bathing, dressing, eating, transferring and walking. IADLs included any difficulty using the telephone, preparing meals, managing money, shopping, housekeeping, and doing laundry. After adjusting for differences in sociodemographic characteristics, health status, smoking, and comorbid chronic conditions, we determined the risk of a significant functional status decline compared to baseline among respondents with cardiovascular disease and arthritis.

RESULTS: Between 1998 and 2000, 9.0% of all respondents who lacked prescription drug coverage reported development of an ADL impairment compared to 8.2% of those who had drug coverage ($P = .20$); for IADL impairment, the difference was 7.0% for those who lacked prescription drug coverage versus 8.8% for those who had drug coverage ($P < .01$). Among respondents with cardiovascular disease, those without drug coverage had higher rate of ADL loss compared to those with coverage (15.5% versus 11.9%; $P = .03$) but no difference was observed for IADL loss (10.5% versus 9.1%; $P = .33$). Among respondents with arthritis, those without drug coverage reported a higher rate of ADL loss (12.4% versus 10.5%; $P = .05$) but not for IADL loss (8.6% versus 8.7%; $P = .48$). After adjusting for potential confounders, we found no statistically significant difference in ADL loss among respondents without drug coverage compared to those with drug coverage in the setting of cardiovascular disease (adjusted odds ratio [AOR] 0.66; $P = .07$) or arthritis (AOR=0.86; $P = .35$). Non-significant results were also found for IADL loss in both cardiovascular disease (AOR=1.01; $P = .98$) and arthritis (AOR=0.98; $P = .91$).

CONCLUSIONS: Our study fails to demonstrate a significant effect of prescription drug coverage on the preservation of functional status among older adults with cardiovascular disease or arthritis over a two-year follow-up period. Further investigation is necessary to explore whether drug coverage affects functional maintenance in other chronic conditions or over a longer period of observation.

ELDERLY WOMEN WITH RECENT MAMMOGRAMS: WHO WOULD CONSIDER STOPPING? M.B. Barton¹, G. Polk¹, S. Wang¹, S.W. Fletcher¹. ¹Harvard University, Boston, MA. (Tracking ID #133722)

BACKGROUND: Mammography screening in the elderly should be an individual decision based on a woman's health status and life expectancy. Recent data suggest that a significant number of elderly women in poor health continue to undergo mammography screening. We undertook a survey in order to understand what influences were important to elderly women in decisions about mammography.

METHODS: Women aged 70 to 84 enrolled in a large health plan in Massachusetts were eligible. Women with a personal history of breast cancer and those who had not had a mammogram in the last fifteen months were identified from insurance databases and excluded. Women were contacted by mail and then telephone; the telephone interviewer assessed each woman's ability to speak English and confirmed all eligibility criteria and invited eligible women to complete a twenty-minute telephone survey. The main outcome measure was intention to continue screening mammography.

RESULTS: 155 women agreed to participate, for a response rate of 40%. The mean age was 77.2 (range 70-84), 95% of respondents were white and 44% had household incomes below \$25,000. Ninety-one percent planned to get another screening mammogram in the next 12 months, 5% were undecided or planned a longer interval, and 4% had no plans to get another screening mammogram in the future. The mean age of women planning to continue screening (76.8) was younger than that of women not planning to continue (79.9) ($p = .01$), while other demographic (race, living situation, education level and income) and clinical (family history of breast cancer, prior history of false positive mammograms) variables did not significantly differ between the two groups. Burden of comorbid illness as reflected by the Charlson score was also not associated with women's plans ($p = .09$, Mantel-Haenszel). Women with a less than 50% ten-year expected survival based on age-comorbidity (Charlson) scores of 6 or higher strongly endorsed mammography: 86% said they planned to continue screening. Of women who planned to continue screening, 68% would not consider skipping or stopping screening even if their primary doctor advised them to. Women who intended to continue annual screening reported the following factors to be influential in their decision: "It feels great when mammogram results are normal" (rated extremely influential by 42%) and "It is important to me to practice good health behavior" (extremely influential: 41%). Of the women who were undecided or had decided to stop screening, the following factors were listed as moderately or more influential in their decision: "I am past the age where mammograms are useful" (46%) and "Mammograms are stressful" (46%). **CONCLUSIONS:** A great majority of women aged 70 to 85 who have undergone recent screening mammography in one insured population plan to continue to undergo regular screening; including those with potentially serious medical problems. A large proportion of these (68%) would apparently cling to the practice against their doctor's advice. Educational efforts directed at increasing individual decision-making regarding screening in the elderly may face barriers based on this enthusiasm for screening.

EQUAL ACCESS TO POOR PAIN CONTROL AT THE END OF LIFE. B.A. Williams¹, K. Lindquist¹, S.Y. Moody-Ayers¹, L.C. Walter¹, K.E. Covinsky¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #135009)

BACKGROUND: Minority patients of advanced age are more likely to have sub-optimal pain control in nursing homes, emergency rooms, hospitals and community settings compared to white patients. However, it is not known if minority elders are more likely to have pain at the end of life.

METHODS: Our cohort consisted of 4,103 proxies of decedents from the health and retirement survey (HRS), a national cohort study of people over age 50. Proxies whose family members or friends had died during 1996-2002 were interviewed in survey waves 1998, 2000 or 2002. Proxies were interviewed about the subject's end of life experience. Our primary predictor variable was ethnicity, based on the subject's pre-death interview stating with which group they most identified. Our primary outcome, pain, was defined as a proxy report that the subject was often troubled by moderate or severe pain in the last year of life. We performed bivariate and multivariate analysis to determine if ethnicity was associated with moderate to severe pain at the end of life. We also performed multivariate logistic regression to identify independent predictors of pain at the end of life.

RESULTS: The median age at death was 80 for whites and 77 for African American and Latinos; 50% of subjects were female. Of proxy respondents, 34.5% were spouses, 45.6% were children of the subjects and 19.9% were other relatives or friends. Ethnicity was not associated with moderate to severe pain at the end of life. Proxies identified moderate to severe pain occurring in the last year of life in 47% of whites, 44% of blacks and 52% of Latinos ($p = 0.14$). After adjustment for age, comorbidities, gender, education, time between death and interview and place of death there were still no significant associations between ethnicity and moderate to severe end of life pain (African American ethnicity OR=0.85, CI 0.67-1.06 and Latino ethnicity OR=0.94, CI 0.60-1.30 compared to whites). There were no interactions between ethnicity and age, gender or education. Factors that were independent predictors of pain in the last year of life were female gender (OR=1.22), ADL dependency (OR=1.45), fair or poor self-rated health (OR=1.39), cancer (OR=1.54), dying in the hospital (OR=1.20), absence of dementia (OR=1.56), and symptoms including dyspnea (OR=1.52), frequent vomiting (OR=2.36) and anorexia (OR=1.49) (all $p < 0.05$).

CONCLUSIONS: Although studies in other settings have found disparities in pain control in older patients from ethnic minorities, we found no evidence of ethnic disparities in pain symptoms at the end of life. In contrast, we found that proxies perceived high rates of inadequate pain control across all ethnic groups. Problems with inadequate pain control in the last year of life in elderly patients transcend ethnic differences.

HEALTH OUTCOMES IN OLDER MEN WITH LOCALIZED PROSTATE CANCER. R.M. Hoffman¹, M.J. Barry², M. McNaughton Collins². ¹University of New Mexico, Albuquerque, NM; ²Massachusetts General Hospital, Boston, MA. (Tracking ID #131012)

BACKGROUND: The benefit of treating localized prostate cancer in older men is uncertain. We compared health-related quality-of-life (HRQOL) outcomes and survival between older men who received aggressive treatment (radical prostatectomy or radiation therapy) versus those receiving conservative management (hormone therapy or no treatment).

METHODS: We conducted an observational study using a population-based cohort of men who were 75 to 84 years old when diagnosed with a clinically localized prostate cancer in 1994 or 1995. We used medical record abstractions and patient surveys to obtain clinical and HRQOL data at diagnosis and 24-

month follow-up. We assessed disease-specific HRQOL for the domains of urinary, bowel, and sexual function and bother. We assessed general HRQOL with the general health, role-physical, role-emotional, pain, vitality, and mental health domains of the Medical Outcomes SF-36 instrument. We used a propensity score method to adjust for baseline differences between men who received aggressive treatment and men who were conservatively managed. Propensity scores were used in logistic and linear regression analyses comparing HRQOL outcomes between treatment groups. Overall and disease-specific survivals through five-years of follow-up were estimated with multivariate proportional hazards models also adjusted for propensity scores.

RESULTS: At baseline, aggressively treated men ($n=175$) were characterized by younger age, white race, being married, higher income, a lower PSA level, a lower prevalence of chronic lung disease or heart attack, and being less bothered by urinary or sexual problems than men who were conservatively managed ($n=290$). We found no statistically significant associations between patient characteristics and treatment after adjustment for the propensity score. At 24 months following diagnosis, men who received aggressive treatment were more likely to report daily urinary leakage (OR=2.9, 9% CI 1.2–7.0) and to be bothered by urinary problems (OR=5.1, 1.3–9.1) and sexual problems (OR=2.8, 1.2–6.3). Men who received aggressive treatment reported higher scores in the general health (66.8 vs. 59.2, $P<0.04$) and role physical (60.8 vs. 47.7, $P<0.04$) scales. The disease-specific mortality hazard ratio was 0.43 (95% CI 0.15, 1.28), favoring aggressive treatment. However, the absolute 5-year disease-specific survival difference was only 6% (98% vs. 92%). Overall, 82% (127/155) of all deaths were from other causes.

CONCLUSIONS: Aggressively treated older men were healthier at baseline and had more favorable tumor characteristics than those who were managed conservatively. By 24 months following diagnosis men who received aggressive treatment had suffered more urinary dysfunction and were more bothered by urinary and sexual problems. General health and physical function were higher for aggressively treated men but this was likely due to residual selection bias. Aggressive treatment minimally reduced the adjusted risk of dying from prostate cancer; disease-specific survival, though, was relatively high in both groups. Physicians and older patients should consider these outcomes in making decisions about prostate cancer treatment and screening.

HEALTHCARE PREFERENCES AMONG COMMUNITY-DWELLING ELDERLY: THE FRAMINGHAM STUDY. E.P. McCarthy¹, M.J. Pencina², E.J. Oberacker², M. Kelly-Hayes², R.B. Burns¹, J.M. Murabito². ¹Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA; ²NHLBI's Framingham Heart Study, Framingham, MA, Boston University School of Medicine, Boston, MA. (Tracking ID #132692)

BACKGROUND: Little is known about advance care planning (ACP) or healthcare preferences (HCP) in community-dwelling elders. We describe self-reported ACP and HCP for surviving participants of the original Framingham Heart Study (FHS) cohort who have been followed for 5 decades.

METHODS: We are administering a HCP questionnaire to original cohort participants who are free of dementia, as part of the routine 28th FHS examination. We report on the first 137 interviews conducted between 2/1/2004-9/30/2004 [~ 68% of the projected 200 potentially eligible participants who may attend the 28th examination, which will be completed 12/2005]. The HCP questionnaire consists of 13 items, including whether participants have had discussions about end of life care, preferences for care, documentation of advance directives, and health perceptions.

RESULTS: Of 137 eligible participants, 3 refused and 2 did not complete the survey, yielding a 96% participation rate. Among the 132 respondents, 67% were women, 64% were widowed, and their mean age was 88 years (range 83-100). Overall, 9 (7%) respondents were upset or bothered by one or more questions and 35 (26%) had difficulty understanding one or more questions. Seventy-one respondents (54%) believed that they had better than 90% chance of being alive in 12 months and 67 (51%) believed that they would be able to care for themselves in 12 months. Most respondents (61%) preferred to die at home. Eighty-five respondents (64%) reported discussing their wishes for medical care at the end of life with someone, but only 16 (12%) with a physician or other healthcare provider. Eighty respondents (61%) had identified a healthcare proxy, and 60 (45%) completed a living will. Three-quarters of respondents preferred comfort care rather than life-extending care if they were to become seriously ill. Most respondents were very unwilling or would rather die than live out their life in a great deal of pain (70%), be attached to a ventilator (80%), fed through a tube (82%), or be unconscious or in a coma (92%) all the time. Only 64% of respondents would be very unwilling or rather die if they became forgetful or confused.

CONCLUSIONS: This very elderly community-dwelling population was willing to answer questions about ACP and state their preferences for care at the end of life. Most community-dwelling elders studied to date preferred measures directed toward providing comfort rather than extending life. Although many of these elders have completed advance directives by specifying a health care proxy and completing a living will, few have discussed their wishes with their physician or health care provider.

IMPROVING ELDERLY RECOVERY AFTER MAJOR ABDOMINAL SURGERY. V. Lawrence¹, J. Cornell¹, H.P. Hazuda¹. ¹South Texas Veterans Health Care System and University of Texas Health Science Center at San Antonio, San Antonio, TX. (Tracking ID #135123)

BACKGROUND: Elders undergo >400,000 major abdominal operations annually; that number will double in 20 years. We previously showed that after major

abdominal surgery, elders require, on average, 3 months to recover to preoperative function in Activities of Daily Living (ADL) and 3–6 months to recover in Instrumental Activities of Daily Living (IADL). Little is known of elders' coping strategies to deal with and perhaps speed recovery after major abdominal operations. Evidence of successful strategies would provide important information to develop interventions to improve recovery and reduce postoperative disability.

METHODS: In a prospective cohort study, 187 patients ≥ 60 years old undergoing major elective abdominal operations were interviewed with a semi-structured, open-ended questionnaire to elicit patient-perceived factors that helped with recovery at 1, 3, 6, 12, and 24 weeks postoperatively. Two coders, a clinician and a social scientist, reviewed and identified themes in the subjects' volunteered responses and, through a consensus-building process, jointly coded the themes in each subject's responses.

RESULTS: Mean age was 68.6 ± 6.38 with 53% men, 30% nonHispanic Caucasian, 57% Mexican American, and 9% other ethnic/racial groups; 59% had not completed high school, 16% had completed high school, and 25% more than high school education. Major themes for coping strategies were: family/friends, social support (especially tangible support for ADL and IADL), cognitive strategies (e.g., determination, humor), spirituality, medical advice (e.g., abdominal splinting, walking or exercise), medical device (e.g., abdominal binder). The three most often reported themes were family, tangible social support, and medical advice. The most frequently cited medical advice theme was walking or exercise (43% of patients, 54% of responses). Patients citing walking or exercise as helpful (walkers) did not differ from nonwalkers in age, ethnic distribution and marital status, but there was a marginally significant trend toward higher educational level among those reporting exercise as an aid to recovery ($p=0.052$). Walkers recovered significantly faster in IADL than nonwalkers (OR 2.5, 95% CI 1.1-5.9, $p=0.03$); there was no difference in ADL recovery. Multivariable logistic regression was limited by the small sample size; after adjusting for either age and gender or preoperative functional status and postoperative complication, ORs for walkers remained approximately 2 but were no longer statistically significant.

CONCLUSIONS: Patients value a variety of supports as helping recovery after major abdominal surgery; family, tangible social support, and medical advice from providers are most prominent. Although type II error was likely present due to small sample size, the results confirm the importance of provider information, support suggestions from prior research that patients with higher educational status have better disability coping skills, and suggest that the simple medical advice to walk may improve postoperative recovery. Future research should test interventions to enhance providers' information transfer, to translate coping skills of those with more education to those with less, and to compare simple advice to walk with structured rehabilitation programs.

INAPPROPRIATE GERIATRIC PRESCRIBING IN AN ACADEMIC INTERNAL MEDICINE PRACTICE: CUTTING DOWN ON THE BEERS. A.M. Wolf¹, J.M. Schectman¹, J.D. Voss¹, M.L. Plews-Ogan¹, D.M. White¹, D.L. Hill¹, J.M. Evans¹. ¹University of Virginia, Charlottesville, VA. (Tracking ID #135559)

BACKGROUND: Inappropriate prescribing and resultant adverse drug events remain a major threat to the health and well-being of our elderly patients. Nowhere is the risk for adverse drug events higher than in medically indigent patients with multiple chronic illnesses. We conducted a quality improvement (QI) intervention in a university-based internal medicine resident-faculty practice, both to quantify and to reduce inappropriate prescribing in this high risk population.

METHODS: A QI team comprised of four general internists, a geriatrician, a pharmacist, and two medical residents developed criteria for safety and appropriateness in prescribing based on the 2003 Beers criteria for potentially inappropriate prescribing, elements of the Medication Appropriateness Index, drug-drug interactions identified by Micromedex as moderate or severe, and dosing parameters defined by the 2003 Geriatric Prescribing Handbook. The QI team applied these criteria to their review of abstracted chart and pharmacy data from a sample of 143 patients greater than 74 years old. The team employed a consensus approach to assess the potential prescribing errors detected for clinical relevance and to issue written recommendations to the patients' providers.

RESULTS: The mean number of prescription medications per patient was 7.8. The QI team found potential prescribing errors in 78% of the 143 patients: 39% had single prescribing errors, 26% had 2 potential errors, and 12% had 3 or more errors. Forty percent of patients were on Beers list medications (33% on one, 7% on more than one). Other potential prescribing errors prompting QI recommendations included inappropriate dosing (22%), discrepancies between chart and pharmacy records or inadequate prescribing documentation (19%), drug-disease interactions (8%), inadequate safety monitoring (6%), drug-drug interactions (5%), and therapeutic duplications (3%). The most frequent recommendation made by the QI team was to discontinue a potentially inappropriate medication (31%), followed by substituting a safer drug (21%), decreasing medication dosage or changing dosing interval (20%), clarifying documentation (8%), and monitoring for toxicity (6%). Of the recommendations that remained applicable at the time that physicians received feedback, 81% were accepted by the physicians.

CONCLUSIONS: Potential geriatric prescribing errors were common in an academic practice setting, but physicians were highly receptive to recommendations to improve their prescribing practices. Systematic feedback based on established criteria is one promising method to reduce prescribing errors in this high-risk population and to inculcate physicians-in-training with safe prescribing practices.

INFLAMMATORY BLOOD MARKERS PREDICT MORTALITY IN PATIENTS WITH PERIPHERAL ARTERIAL DISEASE. H. Vidula¹; M.M. McDermott¹; K. Liu¹; J. Zheng¹; L. Ferrucci²; J.M. Guralnik³; P. Greenland¹; D. Green¹; M.H. Criqui⁴; J. Schneider¹; L. Tian¹. ¹Northwestern University, Chicago, IL; ²National Institute on Aging, Bethesda, MD; ³National Institutes of Health, Bethesda, MD; ⁴University of California, San Diego, La Jolla, CA. (Tracking ID #133967)

BACKGROUND: Associations between inflammatory marker levels and mortality in patients with lower extremity peripheral arterial disease (PAD) are not well studied. The objective of this study was to determine a) whether elevated inflammatory blood marker levels are associated with increased total mortality rates in persons with PAD and b) whether inflammatory blood marker levels are more predictive of near-term than later mortality.

METHODS: Participants in this longitudinal prospective cohort study were 377 men and women with PAD (ankle brachial index (ABI) <0.90). Serum amyloid A (SAA) and high sensitivity C reactive protein (hsCRP) were determined at baseline and annually. Patients were followed for a mean of 3.4 years for all-cause mortality. Cox proportional hazard analyses were performed to relate baseline and time-dependent blood marker levels with mortality, adjusting for age, sex, race, ABI, cholesterol levels, smoking, body mass index, and comorbidities.

RESULTS: Overall, 94 patients (25%) died during follow-up. Higher baseline levels of SAA (p trend=0.004), but not hsCRP were associated with greater mortality. In time-dependent analyses, increased levels of SAA (p trend=0.002) and hsCRP (p trend=0.001) were associated with higher mortality rates (see table). In the table, * denotes P<0.05 and ** denotes P<0.01 compared to the reference group.

CONCLUSIONS: These data suggest that baseline SAA levels and annual levels of SAA and hsCRP may be useful prognostic indicators in patients with PAD.

Adjusted associations between inflammatory blood marker levels and mortality in PAD patients

	Hazard ratios for baseline blood levels	Hazard ratios for time-dependent blood levels
1st hsCRP quartile	1.0 (Reference)	1.0 (Reference)
2nd hsCRP quartile	1.40	1.82
3rd hsCRP quartile	1.45	2.33*
4th hsCRP quartile	1.86	3.18**
1st SAA quartile	1.0 (Reference)	1.0 (Reference)
2nd SAA quartile	1.30	1.17
3rd SAA quartile	2.04	1.24
4th SAA quartile	2.45*	2.74**

LIMITED LITERACY IS ASSOCIATED WITH MORTALITY IN THE ELDERLY: THE HEALTH, AGING, AND BODY COMPOSITION STUDY. R. Sudore¹; D. Schillinger¹; S. Satterfield²; T.B. Harris³; K. Mehta⁴; E. Simonsick⁴; A. Newman⁵; C. Rosano⁶; R. Rooks⁶; S. Rubin¹; H. Ayonayon¹; K. Yaffe¹. ¹University of California, San Francisco, San Francisco, CA; ²University of Tennessee, Memphis, TN; ³National Institutes of Health, Chevy Chase, MD; ⁴NIA, Bethesda, MA; ⁵University of Pittsburgh, Pittsburgh, PA; ⁶Kent State University, Kent, OH. (Tracking ID #133833)

BACKGROUND: Though limited literacy has been associated with poor health outcomes and worse self-rated health, no prospective studies have assessed whether limited literacy is associated with mortality in older adults.

METHODS: We studied 2,512 well-functioning, community-dwelling elders in the Health, Aging, and Body Composition Study who completed a literacy evaluation with the Rapid Estimate of Adult Literacy in Medicine (REALM) in 1999-2000. REALM scores were categorized into limited literacy (0-8th grade) or adequate literacy (≥ 9 th grade). We assessed the association of limited literacy to all-cause mortality over the subsequent 5 year period. We calculated Kaplan-Meier survival curves and used Cox proportional hazard modeling. In multivariate analyses, we adjusted for patient characteristics associated with limited literacy or mortality at p<0.1, including: demographic characteristics (age, race, gender, education, income); co-morbidities (hypertension, diabetes, cardiac disease, stroke, obesity); self-rated health status; health-related behaviors (current or former smoker or ≥ 1 alcoholic drink per day); health care access (having a primary doctor, a flu shot in the past 12 months, or insurance to cover medication); and psycho-social status (depression, personal mastery: confidence or hopelessness in dealing with life's problems, or living alone). In addition, we assessed whether limited literacy was associated with modifiable underlying causes of death, such as cardiovascular disease or diabetes. We also performed sensitivity analysis excluding participants with >5 pt. decline on the Teng Modified Mini-Mental State Exam during the study period.

RESULTS: Participants had a mean age of 75.6 years, 48% were male, 38% were black, 24% had limited literacy, and mean follow-up time was 5.1 years. Compared to those with adequate literacy, those with limited literacy had a higher risk of death at 5 years [19.7% (n=117) versus 10.6% (n=203)] with a HR of

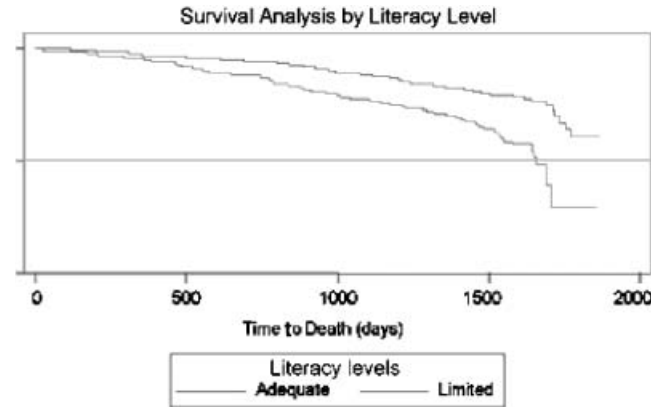


Figure 1: All Cause Mortality by Literacy Level

2.03; 95% CI, 1.62-2.55 (See Figure 1, below). After multivariate adjustment, limited literacy remained independently associated with mortality (HR: 1.70; 95% CI 1.26-2.30). Excluding 383 (15.2%) participants with incident cognitive decline did not change the adjusted association (HR 1.98; 95% CI, 1.43-2.74). Limited literacy was not statistically significantly associated with any underlying cause of death (≥ 0.2 for all).

CONCLUSIONS: Limited literacy is independently associated with a 2-fold increase in mortality in the elderly, even after adjusting for hypothesized mediators of the relationship between literacy and health. The pathways that might explain this association are not clear. Given the large public health impact of this finding, the mechanism by which limited literacy is associated with mortality in the elderly warrants further investigation. Supported by the NIA: N01-AG-6-2101, N01-AG-6-2103, and N01-AG-6-2106.

PHYSICIAN COUNSELING ABOUT EXERCISE TO OLDER WOMEN. M.A. Schonberg¹; E.R. Marcantonio¹; C.C. Wee¹. ¹Beth Israel Deaconess Medical Center, Boston, MA. (Tracking ID #135970)

BACKGROUND: Exercise can reduce a woman's risk of hip fracture, depression, arthritis pain, diabetes, and coronary artery disease, and it can improve longevity and cognitive function. However, few women over age 65 years engage in regular physical activity. Several organizations recommend that physicians counsel older patients about the importance of exercise. Little is known about physician counseling about exercise to older women or the potential impact of illness burden or functional impairment on exercise counseling.

METHODS: We studied 6,385 women age 50 years or older who saw a health care provider in the past year and responded to the 2000 National Health Interview Survey, a nationally representative survey of the non-institutionalized US population. We compared clinician counseling about exercise in the past year among women ages 50-65 (n=3023), 65-75 (n=1639), 75-85 (n=1321), and 85+ (n=402). We performed multivariable logistic regression and adjusted for demographic factors, body mass index (BMI), physical activity, smoking status, having a usual source of care, and number of doctor visits in the past year. To determine the influence of illness burden (comorbidities, self-reported health, hospitalizations in the past year) and functional status on exercise counseling to older women, we adjusted for these factors sequentially in additional models. All analyses were performed using SUDAAN and results were weighted to reflect national population estimates.

RESULTS: Of the 6,385 women, 58% were overweight or obese (BMI ≥ 25); 87% report seeing a primary care doctor for their usual care; and 28% reported that a clinician recommended that they begin or continue to do any type of exercise or physical activity during the past year. The Table below shows illness burden, functional limitations, activity level, and recommendations for exercise across the age groups. After adjustment, clinicians were less likely to counsel women age 75-85 (adjusted OR 0.77, 95% CI [0.58-1.03]) and women age 85+ (aOR 0.59 [0.37-0.94]). When we further adjusted for illness burden, clinicians were even less likely to counsel women age 75-85 (aOR 0.70 [0.53-0.93]) and women age 85+ (aOR 0.52 [0.33-0.81]) about exercise compared to younger women with comparable illness burden. Surprisingly, in stratified analyses the differences in exercise counseling across the age groups were more pronounced for women with less comorbidity than those with greater comorbidity. For example, compared to women age 50-65 with equal comorbidity, the aOR of exercise counseling among women age 85+ with <2 comorbidities was 0.37 (0.18-0.75) and for women age 85+ with 2 or more comorbidities was 0.66 (0.34-1.29). This trend was consistent for all age groups. Functional status did not attenuate these findings.

CONCLUSIONS: Counseling older women about exercise is low nationally. Despite known benefits of exercise, providers are less likely to counsel women age 75 and older about exercise than women age 50-65, with the greatest disparity among older women with less illness burden. Interventions should focus on increasing provider exercise counseling to the oldest women.

Illness Burden, Functional Status, Physical Activity, and Exercise Counseling by Age

	50-<65	65-<75	75-<85	85+
2 or more comorbid diseases	20%	36%	43%	52%
Functional limitations	3%	5%	12%	27%
At least moderate physical activity	58%	50%	39%	24%
Received exercise counseling	31%	29%	22%	14%

RANDOMIZED TRIAL OF MINDFULNESS MEDITATION IN OLDER ADULTS FOR THE TREATMENT OF CHRONIC LOW BACK PAIN. N.E. Morone¹; D. Weiner¹; C. Greco¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133921)

BACKGROUND: Conventional treatment of chronic low back pain (CLBP) in older adults is limited by increased susceptibility to side effects of medications, leaving older adults with few conventional alternatives for pain relief. A complementary therapy called mindfulness meditation offers a promising option. However, rigorous trials in older adults are lacking. We set out to study the feasibility of a mindfulness meditation program among older adults with CLBP and to develop initial estimates of treatment effect.

METHODS: For this pilot study, 36 community-dwelling older adults 65 years of age and older with CLBP were randomized to an 8-week program modeled on the Mindfulness-Based Stress Reduction Program or to a wait-list control group. Subjects met weekly for 90-minute sessions. CLBP was defined as moderate pain occurring daily or almost every day for at least the previous three months. Assessments of pain (McGill Pain Questionnaire-Short Form), disability (Roland Disability Scale) and quality of life (RAND-36) were done at baseline and 8 weeks. Subjects were required to meditate 6 of 7 days a week for 45 minutes a day. Meditation diaries recorded comments of subjects and time spent meditating daily. Effect sizes were determined using Cohen's d. There were no previous studies on which to base sample size, so we did not expect to find statistically significant differences in this small study, however, results will be used to inform future studies.

RESULTS: 36 subjects were recruited within a six-month period. There were no significant baseline differences between groups. 29 of the 36 participants completed the study (81%). Disability secondary to back pain showed a 2.2 point improvement on the Roland Disability Scale within the intervention group ($P=.03$), effect size .39. The Global Health Composite and Physical Health Composite summary scores of the RAND-36 showed 6.9 and 4.3 points improvement respectively ($P=.04$, $P=.03$) effect sizes .2 and .14. Among the individual scales, Pain improved 7 points ($P=.03$), effect size .48. The McGill Pain Questionnaire trended toward improvement but was not significant, effect size .14. No adverse events occurred during the intervention. The treatment diaries contained many positive comments on the effects of meditation. When the intervention group was compared to the control group for the above measures there was a trend toward improvement in all of them but this trend did not reach statistical significance.

CONCLUSIONS: It is feasible for community dwelling older adults to participate in a mindfulness meditation program. Since few alternatives exist for pain relief among older adults who have failed conventional therapy, the trend toward improvement in this pilot study suggests it may have potential benefit for this population. Larger studies are needed to establish the relationship.

RELIABILITY AND VALIDITY OF A STEADINESS SCORE. D.O. Clark¹; C.M. Callahan¹; S.R. Counsel². ¹Indiana University Center for Aging Research, Indiana University School of Medicine, Regenstrief Institute, Indianapolis, IN; ²Indiana University School of Medicine, Indiana University Center for Aging Research, Regenstrief Institute, Indianapolis, IN. (Tracking ID #132978)

BACKGROUND: Self-reported mobility has proven to be a valuable resource in predicting disability among older adults. Identifying additional self-report measures that capture disability risk early in the disability process, capture variance distinct from that captured by mobility, and can identify persons at risk for falls or health service use would represent an important advance for disability prevention and policy efforts. In this report, we present the internal consistency and construct and predictive validity of three survey questions on steadiness in a sample of community-dwelling, lower income older adults.

METHODS: Cohort study with 6-month follow-up of 357 adults aged 65 years or over. Measures used in the analyses include age, gender, ethnicity, education, number of chronic conditions, and steadiness. Measures used for construct and predictive validity included difficulty in mobility, activities of daily living (ADL), instrumental activities of daily living (IADL), falls and hospitalization. The steadiness questions were worded as follows: "we are also interested in your steadiness when walking, standing, and getting up from a chair. At this time, do you feel very steady, slightly steady, slightly unsteady, or very unsteady when standing? Walking? Getting up from a chair?"

RESULTS: The three steadiness questions showed good internal consistency (Cronbach's alpha=0.88), construct validity in Pearson correlations with mobility (0.57), ADL (0.53), and IADL scores (0.41), and predictive validity. With regard to the latter, controlling for age, gender, ethnicity, education, and number of chronic conditions, baseline steadiness was predictive of falls and hospitalization, and ADL and IADL decline. With mobility included as a covariate, however, the beta coefficients for steadiness were reduced. Steadiness remained a statistically significant predictor of falls, but not hospitalization, ADL, or IADL decline. Finally, analyses of each individual steadiness item alone indicated that

each question has a predictive validity that is a little less than that of the three-items together, but steadiness when walking predicted falls, hospitalization, and ADL and IADL decline.

CONCLUSIONS: Steadiness was predictive of 6-month falls and hospitalization, and 6-month decline in ADL and IADL function. Steadiness questions are a potentially valuable addition to both survey research and clinical screening to identify patients with current impairment status, and falls and disability risk. In regard to survey research, steadiness questions had considerably less missing data than mobility questions and represent a potentially complementary measure of impairment. As a clinical screening tool, steadiness when walking may be the best single steadiness question.

SATISFACTION AMONG MEDICALLY COMPLEX ELDERLY VETERANS ON SILDENAFIL. C. Peterson¹; K. Kelly²; G.Y. Ngeny³; A. Cafiero⁴. ¹Veterans Administration/Philadelphia, Philadelphia, PA; ²University of Pennsylvania, Philadelphia, PA; ³Morehouse School of Medicine, Atlanta, GA; ⁴University of the Sciences, Philadelphia, PA. (Tracking ID #133824)

BACKGROUND: BACKGROUND: Sildenafil has been claimed to be effective in elderly men with erectile dysfunction (ED), but few studies have looked at medically complex elderly over the age of 75. If a patient is unable to have an effective erection he is likely to be dissatisfied. Non response (dissatisfaction) may be related to physical and or emotional health but also to poor understanding of proper use. This study was designed to determine the prevalence and predictors of sildenafil dissatisfaction in an elderly veteran population.

METHODS: METHODS: Veterans 75 years or older who had received sildenafil were identified from pharmacy records at the Philadelphia VAMC. A scripted phone interview was conducted with 60 veterans consisting of a modified Sexual Health Inventory in Men (SHIM) and questions regarding recall of education about the use of sildenafil (SEd). A five-point Likert scale was used for the SHIM, while SEd collected dichotomous data. Sexual satisfaction data was determined by using the sexual intercourse satisfaction question in the SHIM. Patients were deemed to be dissatisfied with their sexual intercourse if they reported satisfaction less than 50% of the time (1 or 2 on the Likert scale). Demographic data included age, co-morbid disease states, duration of ED, duration of sildenafil use and number of medications.

RESULTS: RESULTS: 60 patients were interviewed. Mean age was 80 years; mean duration of ED was 6 years, mean duration of sildenafil use was 2.5 years. Mean number of medications was 6.25. 40% of patients were dissatisfied with sildenafil. Factors associated with sildenafil dissatisfaction include diabetes (Odds Ratio (OR)=4.00, 95% Confidence Interval (CI)=1.05-5.31), depression (OR=4.00, CI=1.05-5.31) and shorter duration of use (OR=0.66, CI=0.44-.98). SEd was not associated with dissatisfaction (OR=1.67, CI=0.58-.78), although overall understanding of sildenafil use was low. For example, only 23% recalled knowing several trials of sildenafil may be needed, 60% denied being told that it should not be taken with a heavy meal and 57% recalled needing to inform ER staff of sildenafil use within previous 24 hours in the event of a visit.

CONCLUSIONS: CONCLUSION: In an elderly, veteran population with multiple co-morbidities and concomitant medications, there was a high prevalence of sildenafil dissatisfaction. Diabetes and depression were identified as predictors of lower satisfaction. It is of concern that so few men understood how to properly use sildenafil including that it should not be taken with a fatty meal and that its use should be reported to the ER. Improved understanding about its proper use may increase clinical response and lead to greater satisfaction. Additional studies are needed to further identify predictors of dissatisfaction with sildenafil use in the aging population and to address measures which might improve satisfaction in this medically complex population. In an elderly, veteran population with multiple co-morbidities and concomitant medications, there was a high prevalence of sildenafil dissatisfaction. Diabetes and depression were identified as predictors of lower satisfaction.

THE EFFECTS OF SELF-MANAGEMENT AND HOME HELP ON OLDER ADULTS' HOSPITAL UTILIZATION PATTERNS. A.I. Arbaje¹; J. Wolff¹; N.R. Powe¹; G.F. Anderson¹; C. Boul¹. ¹Johns Hopkins University, Baltimore, MD. (Tracking ID #135927)

BACKGROUND: Older adults who transition from hospital to home may experience lapses in continuity of care and subsequent adverse events. Their ability to self-manage their health conditions and obtain help with their functional deficits may favorably affect their hospital utilization patterns and minimize the risk of discontinuity of care. We examined whether self-management skills and lack of help with functional deficits are associated with the length of stay (LOS) and the likelihood of readmission among hospitalized community-dwelling Medicare beneficiaries.

METHODS: We designed a cohort study in which beneficiaries were interviewed at home and monitored through Medicare claims for hospital use the following year. The sample consists of community-dwelling beneficiaries who participated in the Medicare Current Beneficiary Survey (MCBS) during 2000-2002 and who were hospitalized and discharged home ($n=1351$, representing 2,883,726 persons after weighting). Measures include self-reported ability to manage one's health conditions (4-point scale), unmet need for functional deficits (lack of personal assistance for difficulty with 6 ADLs and 6 IADLs), LOS of first admission, and non-elective readmission within 60 days of hospital discharge.

RESULTS: Self-management ability was associated with significantly reduced LOS (1.6-2.0 days, $p<0.05$) in a multivariate model that adjusted for age, gender, race, and self-reported health status. Unmet need was not associated with LOS in similar models. Fifteen percent of the participants who were hospitalized were readmitted within 60 days of discharge ($n=202$); participants who were

readmitted were less likely to be married, more likely to live alone, and had a greater number of self-reported chronic conditions. Self-management was not associated with readmission, but having an unmet need and being unmarried increased the odds of readmission significantly (ORs=1.48 and 1.99, respectively, $p<0.05$), after adjustment for age, gender, race, self-reported health status, and LOS of the initial admission.

CONCLUSIONS: Improving older adults' ability to self-manage their health conditions may decrease the duration of their stays in hospitals. Similarly, access to personal assistance for functional deficits after hospital discharge may decrease the likelihood of hospital readmissions. These findings could help health care providers target a vulnerable group of beneficiaries for intensive discharge planning and post-discharge assistance to improve continuity of care. These data also suggest that self-management skills may be relevant to Medicare coverage policy.

THE IMPACT OF OBESITY ON HEALTH CARE EXPENDITURES IN ELDERLY AMERICANS. D.E. Arterburn¹; M. Maciejewski²; L. Lawson³; L. Chen³; J. Tsevat¹. ¹Cincinnati VA Medical Center, Cincinnati, OH; ²University of Washington, Seattle, WA; ³Cincinnati Children's Hospital, Cincinnati, OH. (Tracking ID #133790)

BACKGROUND: Over 10 million elderly Americans were overweight or obese in 2000. Excess body weight is associated with increased risk of morbidity, mortality, and reduced quality of life; however, less is known about the impact of obesity on health care expenditures in the elderly. Thus, we sought to examine the impact of obesity on health care expenditures using a nationally representative sample of elderly Americans.

METHODS: We performed a cross-sectional analysis of 9,472 adults over age 65 using linked data from the 1996 through 2001 Medical Expenditure Panel Surveys and the 1995 through 2000 National Health Interview Surveys, two series of nationally representative surveys of the noninstitutionalized civilian population of the United States. Per capita health care expenditures were calculated for National Institutes of Health body mass index (BMI) categories based on self-reported height and weight, using a two-part, multivariable model adjusted for age, gender, race, income, education level, type of health insurance, and marital status. In MEPS, healthcare expenditures include direct payments for office and hospital-based care, home healthcare, dental services, vision aids, prescribed medicines, and out-of-pocket payments. All expenditure estimates were adjusted to 2001 dollars using the Consumer Price Index. We used logistic regression to predict the probability of incurring any health care expenditure. Next, we used linear regression on log-transformed expenditures and Duan nonparametric retransformation to predict adjusted costs among those who incurred any expense. Confidence intervals were generated via bootstrapping. We estimated aggregate national expenditures associated with excess body weight using the population-attributable costing methods of Wolf 2001 and Finkelstein 2004.

RESULTS: 3,673 adults over age 65 (40.2%) had normal body weight (BMI 18.5 to 24.9), 3,497 (37.1%) were overweight (BMI 25 to 29.9), 1,407 (13.8%) had class I obesity (BMI 30 to 24.9), 368 (3.5%) had class II obesity (BMI 35 to 39.9), and 287 (2.7%) had class III obesity (BMI ≥ 40). Ninety-eight percent of elderly Americans reported having health insurance coverage by Medicare, Medicaid, or other federal programs. When compared with normal-weight elderly individuals, the odds of incurring any health care expenditure were two-fold greater among those with class I obesity and more than three-fold greater among those with class II or III obesity ($P<0.005$ vs. normal weight group). Per capita health care expenditures for overweight elders and those with class I, II, and III obesity were 10.6% (95% confidence interval [CI]: 3.4% to 21.5%), 38.2% (95% CI: 24.3% to 51.6%), 50.5% (95% CI: 22.5% to 70.2%), and 68.1% (95% CI: 42.0% to 108.1%) higher than normal-weight elders (\$4,886), respectively. Excess costs among obese elders primarily resulted from greater per capita expenditures for outpatient care and prescription drugs. Finally, aggregate U.S. health care expenditures associated with excess body weight among the elderly exceeded \$17 billion in 2001.

CONCLUSIONS: The economic burden of obesity among elderly Americans adults is substantial. Further research is needed to identify interventions to reduce the incidence and prevalence of obesity in the elderly and improve the health and economic outcomes of obese elderly individuals.

VARIATION IN MAMMOGRAPHY IN THE ELDERLY BY HEALTH STATUS. M.B. Barton¹; R. Li¹; M. Kuldorff¹. ¹Harvard University, Boston, MA. (Tracking ID #133732)

BACKGROUND: Recent publications have suggested that many elderly women with poor health or functional status continue to undergo mammograms with little or no expected benefit. These reports have relied on women's self-report of recent mammography use, while the reliability of self-report in an aged population has not been confirmed. We undertook a claims-based assessment of mammography utilization in an aged cohort to assess the independent impact of age and health status on mammography use.

METHODS: Cohort study of elderly women aged 70 to 90 years enrolled in a large health plan in Massachusetts. Eligible women were continuously enrolled for 3 years and did not have a previous diagnosis of breast cancer according to claims data. Our main outcome was use of mammography in the latter two years of the study period. We determined health status using pharmacy data from the first year of the study period to construct the Chronic Disease Score (CDS), a comorbidity measure found to be predictive of mortality in a similar population. We used logistic regression to assess the independent effects of age and CDS on the probability of receiving a mammogram.

RESULTS: We studied 11,743 women with a mean age of 75.7 years. The mean CDS score was 2866 (range 1044 to 26922). Three quarters of the women had a

mammogram during the study period. In unadjusted analysis recent mammography use varied by age, with 86% of women aged 70 and 50% of women aged 88 having had a recent mammogram. In a multivariable logistic regression model controlling for age, worse health status (higher CDS) was significantly negatively associated with mammography ($p=.0045$) but the impact was limited (an estimated 0.4 percentage point drop in the odds ratio of a mammogram with each increase of 100 points in the CDS). Projected mammography use by an 85 year old woman varied only 4 percentage points from the healthiest quintile (56.0%) to the poorest health quintile (52.2%); for an 88 year old woman the projected difference was similar in magnitude (44.9% for healthiest quintile to 41.1% for women in the poorest health quintile.) In the multivariable model age was associated with larger changes in mammography utilization: mammography use was predicted for 69.3% of 80 year-olds versus 43.4% of women aged 88 years.

CONCLUSIONS: Our study confirms the survey-based findings of other investigators in showing the relatively small impact of health status on mammography use among elderly women. These findings suggest that significant numbers of women in poor health, who are unlikely to benefit from screening, are being subject to the risks and harms associated with mammography.

WHAT MIGHT MAKE YOU STOP CANCER SCREENING? A QUALITATIVE ASSESSMENT OF OLDER ADULTS VIEWS. C.L. Lewis¹; C. Kistler¹; H. Amick¹; L.C. Watson¹; D. Bynum¹; M. Pignone¹; L. Walter². ¹University of North Carolina at Chapel Hill, Chapel Hill, NC; ²University of California, San Francisco, San Francisco, CA. (Tracking ID #133117)

BACKGROUND: Evidence supporting cancer screening in adults over age 70 is sparse, therefore, older adults are encouraged to participate in decisions about continuing cancer screening. A better understanding of factors important to older adults deciding whether to continue cancer screening will help guide physicians in their discussions with older patients.

METHODS: We interviewed a convenience sample of 116 adults age 70 and over from two local Continuing Care Retirement Communities. All the participants were white, well educated (83% had college degrees), predominately women (67%), and healthy (66% reported their health as excellent to very good). One hundred thirteen participants responded to the question, "What might make you stop cancer screening?" The answers were transcribed and evaluated for recurring themes, using a grounded theory approach. Three of the investigators coded transcripts independently and disagreements were resolved by consensus.

RESULTS: Thirty-two of the respondents could not think of any conditions under which they would stop cancer screening. Fear of cancer was important for continued screening: "Cancer just gets worse. Get it as soon as possible." The remainder acknowledged that at some point cancer screening may no longer be desirable and many would stop if their health deteriorated, or if they were diagnosed with cancer or another serious illness. Most reported that this time would be very near the end of life: "if I thought I was on death's door", "if I had a terminal illness" or "if I were bed-ridden and out of it." Others reported that they would no longer get screened if it would not change their behavior: "If I had a mammogram that turned out to be positive, I would not have treatment. Therefore, why have the mammogram?" For some a recommendation was an important factor: "As far as I know I should continue screening, so unless my doctor says to stop, I will continue."

CONCLUSIONS: Factors important to older adults' decisions to stop cancer screening are similar to factors that physicians consider in these decisions. However, older adults may want to continue screening beyond a point when it could potentially prolong their lives. Fears about cancer may be an important determinant for continued cancer screening and need to be acknowledged in discussions about whether to stop cancer screening.

"I HELP YOU, AND YOU HELP ME": THE FEASIBILITY AND ACCEPTABILITY OF INTERACTIVE VOICE RESPONSE (IVR)-BASED TELEPHONE PEER SUPPORT AMONG PATIENTS WITH DIABETES. M. Heisler¹; J.D. Piette¹. ¹Ann Arbor VA Center for Practice Management and Outcomes Research/University of Michigan, Ann Arbor, MI. (Tracking ID #132930)

BACKGROUND: Group visits and other strategies to increase peer support are a low-cost way to improve chronically ill patients' self-care behaviors and health outcomes. Many patients, however, face barriers to participating in face-to-face programs. Novel delivery mechanisms are needed to translate clinic-based programs into interventions that are effective and accessible for large, diverse patient populations. We thus developed a theoretical rationale for and sought to determine the feasibility and acceptability of using an Interactive Voice Response (IVR)-based platform for a telephone-based peer support program among patients with diabetes.

METHODS: We designed a prototype IVR peer support system and tested it with diabetes patients with poor glycemic control receiving care at one Veterans Affairs (VA) facility in a single-group study with a pre-post design. Forty-two of 107 eligible patients agreed to participate, and 38 completed the six-week intervention (36% of eligible patients). Participants completed a baseline survey, received rudimentary training, and were matched based on their diabetes-related self-management needs. Participants were asked to contact their partner weekly using the toll-free IVR calling line that protected their anonymity and provided automated reminders if contacts were not made. We used the Internet to monitor the calling process and capture data on participants' contacts. At the completion of the 6-week period, participants completed follow-up surveys and brief telephone interviews.

RESULTS: The mean age of participants was 64, and 57% were using insulin. Over 80% of the pairs spoke at least once a week for two of the six weeks of the intervention, with an average call duration of 24 minutes. 79% of the participants responded that the IVR system was easy to use, and 90% stated that they would be more satisfied with their VA care if such peer support services were available. 70% found the calls helpful in managing diabetes symptoms, 73% reported that their partner helped them do things to stay healthy, and 70% stated that they helped their partner do things to stay healthy. There were significant improvements in participants' reported diabetes care self-efficacy between baseline and follow-up assessments ($p < 0.001$). Qualitative assessment suggested that participants found meaning and positive reinforcement for their self-care through also supporting their partner's efforts to manage diabetes.

CONCLUSIONS: An IVR peer support intervention is feasible, acceptable to patients, and may have positive effects on patients' diabetes self-management and health outcomes that warrant more rigorous evaluation in a randomized trial.

A PHARMACY-BASED INTERVENTION REDUCES EMERGENCY-DEPARTMENT VISITS IN PATIENTS WITH HYPERTENSION. W. Tu¹; M. Weiner¹; J. Wu²; M.D. Murray³.

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BACKGROUND: Adherence to prescribed pharmacotherapy is a key component in the management of hypertension (HTN). Patients with poorly controlled blood pressure (BP) risk disease exacerbation marked by emergency-department (ED) visits and hospital admissions. Pharmacy-based interventions may help patients to improve adherence to drug therapy (Rx) for HTN, thus reducing costly ED and inpatient encounters.

METHODS: A pharmacy-based intervention aimed at improving adherence to HTN Rx is being tested in an ongoing randomized clinical trial at an urban, academic medical center. The intervention included oral patient education, written medication information designed specifically for patients of low health literacy, close monitoring of patients' HTN Rx adherence, and coordination between pharmacists and physicians on matters of adherence. Consenting patients were enrolled in the study if they had a clinical diagnosis of HTN and received at least one prescription for HTN. Patients seeking primary care or medications outside of this public system were excluded. Comprehensive enrollment interviews provided detailed information on patients' demographic, socio-economic, and health-related information. Utilization data and blood pressure measurements were extracted from a comprehensive electronic medical records system. Log-linear regression analysis was used to model the numbers of ED and hospital encounters during the follow-up period. The length of follow-up was adjusted in the analysis as an offset parameter.

RESULTS: Participants (N=328) had a mean age of 58 years, and most (71%) were African-American. Patients had an average of 2.9 (SD = 1.34) prescriptions for HTN. Multivariable log-linear models showed that hospital admission was associated with baseline complications of HTN, such as prior myocardial infarction ($\beta = 0.56$, $p = 0.0001$), number of HTN Rx ($\beta = 0.26$, $p < 0.0001$), and eulipidemia ($\beta = 0.48$, $p = 0.0061$); absence of hospital admission was associated with better visual acuity ($\beta = -0.01$, $p < 0.0001$), absence of evidence of chronic heart failure ($\beta = -0.39$, $p = 0.0064$), self-reported adherence to medications ($\beta = -0.69$, $p = 0.0061$), years of education ($\beta = -0.06$, $p = 0.0166$), and higher average systolic BP within one year of follow-up ($\beta = -0.028$, $p < 0.0001$). Log-linear analysis of ED visits showed that increased ED visits were associated with control-group assignment ($\beta = 0.16$, $p = 0.0082$), complicated HTN ($\beta = 0.29$, $p < 0.0001$), younger age ($\beta = -0.0323$, $p < 0.0001$), number of HTN Rx ($\beta = 0.13$, $p < 0.0001$), and eulipidemia ($\beta = 0.19$, $p = 0.0061$); decreased ED visits were associated with better visual acuity ($\beta = -0.01$, $p < 0.0001$), absence of evidence of chronic heart failure ($\beta = -0.45$, $p < 0.0001$), self reported adherence ($\beta = -0.73$, $p < 0.0001$), cognitive function via ability to recognize letters and patterns ($\beta = -0.02$, $p < 0.0001$), and higher average systolic BP during a year of follow-up ($\beta = 0.013$, $p < 0.0001$).

CONCLUSIONS: A pharmacy-based intervention reduced ED utilization. Complicated HTN and concurrent CHF increase the risk of ED and hospital visits. Cognitive function was also an important predictor of ED and hospital admissions.

A RANDOMIZED CONTROLLED TRIAL OF THE EFFECT OF HOME BLOOD PRESSURE MONITORING VERSUS USUAL CARE ON MEDICATION ADHERENCE IN AMBULATORY HYPERTENSIVE PATIENTS. G. Ogedegbe¹; W. Chaplin².

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BACKGROUND: Home blood pressure monitoring (HBPM) improves blood pressure (BP) control in hypertensive patients. A putative mechanism through which HBPM exerts its effect on BP is via increased medication adherence (ADH); however there is no data to support this assertion. Among 312 ambulatory hypertensive patients, we compared the effects of HBPM versus usual care (UC) on self-reported ADH to prescribed antihypertensive medications.

METHODS: As part of a larger RCT designed to improve medication adherence in Community Health Centers, 220 patients were randomly assigned to HBPM and 92 to UC for a period of 12 weeks. Patients in the HBPM group used the life-link monitoring system to assess their HBP, while those in UC group received standard care. The well-validated Morisky questionnaire was used to assess ADH at baseline (Visit1) and 12 weeks (Visit2). Patients were categorized as nonadherent if they responded "yes" to any of the following questions: "Do you forget to take your BP medication? Are you careless about taking your BP medication?"

Do you stop taking your BP medication when you feel better or feel worse?" Because ADH changes over time in any one individual, we assessed the change in ADH status for each patient from Visit 1 to Visit 2, and then used McNemar's test of dependent proportions to assess if the change was different for the whole group; the HBPM group; and the UC group with a significance level set at $p < 0.05$.

RESULTS: Of 312 patients who took part in this study, 52% were black, 16% were whites, 31% were Hispanics, and 81% were female. Mean age was 50 yrs with a mean BP of 146/88 + 15/9 mm hg. Overall, 57% of the patients were adherent at Visit1, with a marginally significant difference between UC vs. HBPM (60% vs. 49% respectively, $p = 0.08$). Of the 178 patients with ADH data for both time points, 61% stayed the same, while adherence status changed for 39% (with 25% becoming adherent at Visit2 and 14% becoming nonadherent). Results of McNemar's test for the whole group suggest that the change in ADH status from Visit1 to Visit2 was significant for the 25% that became adherent than the 14% that became nonadherent ($p = 0.03$). This indicates a general tendency for the number of adherent patients to increase over the course of the study. This increase in adherence appears to be almost entirely due to the effect of the HBPM. Among the group assigned to HBPM, ADH status changed for 42% with 31% becoming adherent at Visit2 and 11% becoming nonadherent. The McNemar's test of this increase in adherence was statistically significant ($p < 0.003$). Whereas, for the UC group ADH status changed for 31% with only 13% becoming adherent at Visit2 and 18% becoming nonadherent. The McNemar's test of the increase in adherence in the UC group was not significant ($p = 0.64$).

CONCLUSIONS: Our results suggest that HBPM significantly improves ADH to prescribed medications compared to UC among ambulatory hypertensive patients. This mode of disease self-management should be encouraged in this patient population.

ACHIEVING NCEP GOALS IN PATIENTS WITH DIABETES: RESULTS FROM THE VDIS TRIAL. C.D. Maclean¹; A.G. Kennedy¹; B. Littenberg¹; R. Pinckney¹; P.A. Ades¹.

¹University of Vermont, Burlington, VT. (Tracking ID #131897)

BACKGROUND: Cholesterol management is essential in the preventive care of patients with diabetes. Recently, the National Cholesterol Education Program (NCEP) released a report providing updated treatment recommendations that includes an optional LDL goal of < 70 mg/dL for patients who are at very high risk for coronary heart disease (CHD). It is unknown how feasible this may be for very high risk patients with diabetes. Our goal is to assess the feasibility of attaining both the standard and the new optional LDL goals using currently available medications.

METHODS: We performed a cross sectional analysis of laboratory data from the Vermont Diabetes Information System (VDIS), a large randomized trial of a decision support system for patients with diabetes. Medication lists were obtained by direct observation during a home interview. Patients were categorized into high and very high risk CHD status, based on NCEP criteria. For patients not at LDL goal we estimated the changes in therapy necessary to achieve goals, based on best estimates of medication effectiveness from the literature. We used optimistic assumptions regarding adherence and tolerability of medications. The study population is adults with diabetes under care of a primary care provider in one of 47 practices in Vermont and northern New York enrolled in VDIS (N=650).

RESULTS: The median age of the subjects was 66 years (range 25-93), the median duration of diabetes was 7.4 years (SD 10.1), and the mean A1C 7.1% (SD 1.3). Of the entire cohort, 49.4% (321/650) had an LDL < 100 mg/dL. 29.4% (191/650) of patients were at very high risk for CHD and, according to the NCEP, have an optional LDL goal of < 70 mg/dL. Only 15.7% (30/191) of these very high risk patients had an LDL < 70 mg/dL. Of high risk patients, 17/459 (3.7%) would require more than two lipid-lowering drugs to achieve an LDL < 100 mg/dL. In the very high risk group, we estimate that 26.2% (50/191) of patients would require more than two drugs to reach an LDL < 70 mg/dL.

CONCLUSIONS: In many patients with diabetes at very high CHD risk, it will be difficult to attain an LDL goal of < 70 mg/dL. Approximately one quarter of patients will require more than two lipid lowering drugs at maximal doses to attain this goal, even assuming 100% tolerability of, and adherence to, lipid therapy. Given that three-drug lipid lowering therapy has not been tested in clinical trials, the latest recommendations from the NCEP regarding lipid lowering therapy for patients with diabetes may be overly aggressive.

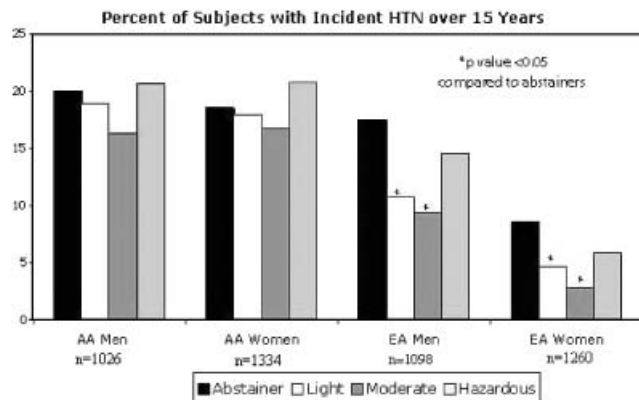
ALCOHOL CONSUMPTION AND INCIDENT HYPERTENSION OVER 15 YEARS IN AFRICAN AND EUROPEAN AMERICANS: THE CARDIA STUDY. J.H. Halanach¹; J. Bian¹; S.G. Kertesz¹; M.J. Pletcher²; M.M. Safford¹; C.E. Lewis¹; C.I. Kiefe¹.

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BACKGROUND: Alcohol consumption may have complex effects on health, some beneficial (decreased cardiovascular risk) and some deleterious (liver disease). The effect of alcohol on hypertension (HTN) is unclear since alcohol acutely increases blood pressure (BP), but moderate chronic consumption is associated with decreased risk of incident HTN in young European American (EA) women. We used data from the Coronary Artery Risk Development in Young Adults (CARDIA) study to examine the associations of light, moderate, and hazardous alcohol consumption with incident HTN over 15 years.

METHODS: The biracial CARDIA cohort was recruited in 1985 (aged 18-30), balanced by design on gender and race/ethnicity (African American (AA) and EA), and re-examined at 2, 5, 7, 10, and 15 years (retention 74%). A total of

4,718 participants without HTN at baseline were used in this analysis. At each exam, alcohol consumption was categorized into light (men <7; women <4 drinks/week), moderate (men 7–13; women 4–6 drinks/week), and hazardous (men ≥14; women ≥7 drinks/week). We used mean alcohol consumption across 15 years to classify subjects as abstainers (21%), light (51%), moderate (15%), and hazardous (13%) drinkers. Incident HTN was defined as, at any follow-up exam, taking antihypertensive medications, or systolic BP ≥140 mmHg, or diastolic BP ≥90 mmHg. Bivariate comparisons of alcohol consumers and abstainers were performed within each race/sex group. Discrete time hazard analysis with repeated observations at the participant level and incident HTN as dichotomous outcome variable was used to account for exam year, sex, race, age, body mass index, smoking, and education in multivariable models. RESULTS: Among EA, moderate drinkers had the lowest incident HTN, followed by light drinkers (FIGURE represents unadjusted percent of incident HTN by sex and race). The relationship between alcohol consumption and incident hypertension followed a similar pattern in AA, but the association was not significant. An overall model for the entire group of 4,718 participants and using abstainers as referent showed odds ratios (95% confidence intervals) of 0.61 (0.66, 0.83) for light drinkers, 0.58 (0.41, 0.82) for moderate drinkers, and 0.74 (0.53, 1.02) for hazardous drinkers. CONCLUSIONS: Light or moderate alcohol consumption is associated with decreased incident HTN among EA in our sample, but not among AA, who were at generally higher risk of developing HTN than EA. Levels of alcohol intake common in young adults are not associated with increased HTN risk.



ARE RESIDENTS LESS SUCCESSFUL THAN ATTENDING PHYSICIANS OR MID-LEVEL PRACTITIONERS IN ATTAINING QUALITY GOALS IN PRIMARY CARE? D.G. Federman¹; R. Krishnamurthy¹; A.C. Justice¹; S. Kancir²; J. Goulet¹. ¹Yale University, West Haven, CT; ²VA Connecticut Health Care System, West Haven, CT. (Tracking ID #132853)

BACKGROUND: Background: Cardiovascular disease is leading cause of death in the United States and efforts aimed at targeting cardiovascular disease risk factors have been promulgated by major organizations. Little is known regarding the relationship between provider type (e.g., resident, mid-level practitioner, attending physician) and attainment of goals with respect to the treatment of dyslipidemia, diabetes mellitus, and hypertension.

METHODS: Methods: Using the electronic medical record, we identified all patients seen in the Veterans Affairs Connecticut Health Care System in a 6-month period with an ICD-9 code corresponding to a diagnosis of coronary artery disease, diabetes mellitus, and hypertension. We recorded the most recent LDL-cholesterol for patients with diabetes or coronary artery disease, hemoglobin A1c for those with diabetes, and blood pressure for those with hypertension. We then examined the relationship between these measures and provider type.

RESULTS: Results: A total of 19,660 unique patients met entry criteria and were included in the analyses. Attending physicians saw 12,682 unique patients, mid-level practitioners 5,572 patients, and residents 1,406 patients. A total of 15,893 patients had a diagnosis of hypertension 6,296 a diagnosis of coronary artery disease, and 5,830 a diagnosis of diabetes mellitus, while 7,277 patients had at least two of these diagnoses. Residents were significantly less likely to attain patient blood pressure goals than attending physicians (59% vs. 54%, $p=0.0017$). After controlling for patient age and practice site, there were no significant differences between attending physicians and residents in attaining dyslipidemia (OR 1.15, 95% CI 0.96–1.38) or diabetes (OR 1.05, 95% CI 0.82–1.33) goals. However, attending physicians were significantly more likely to attain blood pressure goals than were residents (OR 1.23, 95% CI 1.08–1.39). Controlling for additional patient characteristics did not alter the findings.

CONCLUSIONS: Conclusions: Internal medicine residents may quickly develop expertise in managing dyslipidemia and diabetes mellitus. Residents in our sample were less likely than attending physicians to reach goal blood pressure for patients with hypertension. Educational efforts aimed at housestaff targeting improvement in the treatment of hypertension are warranted.

Patient characteristics, site of care, and diagnosis, by provider type, N=19,660 male veterans

	Attending Physician	Mid-level practitioner	Resident	p
Patient age, median (range)	n=12682 73 (23-95)	n=5572 72 (22-94)	n=1406 74 (32-93)	0.0001
site of care, n (%)				0.0001
West Haven	4,490 (36)	1,498 (27)	1,014 (72)	
Newington	4,525 (36)	2,722 (49)	392 (28)	
Community Based Outpatient Clinic	3,667 (29)	1,352 (24)	0 (0)	
Diagnosis, n (%)				
HTN, n=15,893	10,178 (80)	4,571 (82)	1,144 (81)	0.0167
CAD, n=6,292	4,147 (33)	1,729 (31)	416 (30)	0.0110
Diabetes, n=5,830	3,889 (31)	1,538 (28)	403 (29)	0.001
Any co-occurrence, n=7,277	4,803 (38)	1,985 (36)	489 (34)	0.0030
co-morbid conditions				
Alcohol disorder	653 (5.2)	362 (6.5)	66 (4.7)	0.0004
Drug disorder	237 (1.9)	140 (2.5)	27 (1.9)	0.0174
Severe mental illness	923 (7.3)	467 (8.4)	95 (6.8)	0.0172

ATHEROSCLEROTIC RISK FACTOR CONTROL IS ASSOCIATED WITH FEWER ADVERSE LIMB OUTCOMES IN PERIPHERAL ARTERIAL DISEASE. T.C. Collins¹; N.J. Petersen¹; R.L. Bush¹; M. Suarez¹; M. Ku-Goto¹; R.J. Beyth². ¹Baylor College of Medicine, Houston, TX; ²North Florida/South Georgia Veterans Health System, Gainesville, FL. (Tracking ID #133498)

BACKGROUND: We sought to determine the association of atherosclerotic risk factor control with adverse limb events in patients with peripheral arterial disease (PAD).

METHODS: We conducted a retrospective cohort study of patients with PAD as per an ankle-brachial index of <0.9 obtained in the Michael E. DeBakey VA Medical Center vascular laboratory from 1995 to 1998 (index date). To ascertain risk factor control, we performed a chart review from three years prior to the index date until the final visit; the final visit was defined as the first limb event (i.e., lower extremity bypass surgery or amputation), death, or the close of the study (i.e., December 31, 2001). The risk factors of interest were smoking, diabetes mellitus, hyperlipidemia, and hypertension. From the time that a patient had a documented risk factor until the final visit, we determined the proportion of laboratory results or clinic visits during which a given risk factor was controlled. Levels of control were based on national guidelines. Using a t-test, we determined the association of the level of risk factor control with an adverse limb event.

RESULTS: A total of 799 patients (mean age 64.8, SD 9.9 years) were diagnosed with PAD and 230 (28.8%) experienced an adverse limb event. Among these 799 patients and, as documented anytime during the observation period, 434 (54.3%) were smokers, 424 (53.1%) had diabetes mellitus, 333 (41.7%) had an elevated low-density lipoprotein (LDL), 343 (42.9%) had an elevated total cholesterol, and 552 (69.1%) had hypertension. For patients without an adverse limb event (n=569), glucose was controlled in 61.5% (95% CI 57.9, 65.2) of visits compared to control in 45.9% (95% CI 40.3, 51.6; $p<0.001$) of visits in patients with an adverse limb event. Similarly, hemoglobin A1c was controlled for 30.7% (95% CI 25.4, 35.9) of visits in patients without a limb event compared to 14.4% (95% CI 7.0, 21.7; $p=0.001$) of visits in patients with a limb event. LDL was controlled for 41.5% (95% CI 36.7, 46.2) compared to 24.0% (95% CI 15.8, 32.3) of visits for patients without versus with a limb event, respectively, $p=0.001$. Total cholesterol was controlled in 64.2% (95% CI 59.9, 68.7) of visits for patients without a limb event compared to 48.1% (95% CI 39.1, 57.1) of visits for patients with a limb event, $p=0.001$. There were no differences in the proportion of visits with documented control of smoking or hypertension and the proportion of adverse limb events.

CONCLUSIONS: Overall, among this cohort of patients with PAD and at-risk for adverse limb outcomes, risk factors were controlled in fewer than 50% of patients who suffered a major limb event. However, appropriate control of diabetes mellitus, LDL, and total cholesterol did appear to have a favorable impact on limb outcomes in patients with PAD.

CARE FRAGMENTATION AND PSYCHIATRIC ILLNESS IS ASSOCIATED WITH INCREASED EMERGENCY DEPARTMENT USE AMONG COMPLEX DIABETIC PATIENTS. C. Liu¹; A. O'Connor²; D. Eindstadter²; R.D. Cebul¹. ¹Case Western Reserve University, Cleveland, OH; ²MetroHealth Medical Center, Cleveland, OH. (Tracking ID #135495)

BACKGROUND: The care of complex diabetic patients often is fragmented among multiple specialists. The objective of the current investigation is to determine if the presence of psychiatric (Psych) comorbidity among diabetic patients aggravates their already fragmented care and results in excessive use of the Emergency Department (ED).

METHODS: This is a retrospective analysis of ED use among diabetic patients with chronic kidney disease (CKD, estimated GFR 20–60) seen at 10 primary care practices in a large urban health care system over a 2-year period. Diabetes, CKD, Psych and non-Psych comorbidities, visit adherence, A1c levels and

ED visits were identified from visit records, recorded lab values and ICD-9-CM codes from the system's electronic medical record. Psych comorbidities were classified as mood/anxiety, psychotic, or "other" disorders. "Fragmentation" was defined as the number of separate non-psych specialty clinics visited. A negative binomial model was used to estimate the effect of Psych disorders and Fragmentation on the number of ED visits after adjusting for age, sex, ethnicity, number of non-Psych comorbidities, diabetes control (average a1c level over 2 years), adherence (proportion of "kept" primary care appointments), and insurance (commercial, Medicare, Medicaid, uninsured) status.

RESULTS: Of 3873 patients with diabetes seen during 2002-2003, 623 (16.1%) had CKD, of whom 241 (38.7%) had one or more Psych co-morbidities. Psych patients were younger (66.7 yrs vs. 69.7 yrs, $p < 0.0005$), had more non-Psych comorbidities (2.92 vs. 2.56, $p < 0.0003$), poorer adherence (84.7% vs. 87.8% kept primary care visits, $p < 0.004$), and a greater level of Fragmented care (1.77 vs. 1.47 specialty clinics visited, $p < 0.0009$) than non-Psych patients. On bivariate analysis, the 141 (22.6%) diabetic CKD patients with mood and anxiety disorders were more likely than those without to have at least one ED visit (63.1% vs. 44.6%; OR 1.42, 95% CI, 1.21 to 1.66). After adjusting for age, sex, ethnicity, non-Psych comorbidities, diabetes control, adherence, and insurance, the presence of a mood/anxiety disorder increased the predicted number of ED visits by a factor of 1.50 (95% CI: 1.14 to 1.98), while greater Fragmentation increased the number of ED visits by a factor of 1.32 (95% CI: 1.19 to 1.47) for each additional clinic visited. Psych diagnoses other than mood or anxiety disorders were not associated with increased number of ED visits.

CONCLUSIONS: Mood and anxiety disorders are prevalent among patients with complicated diabetes, and are associated with increased fragmentation of care and increased use of the ED. We believe that fragmentation of non-psychiatric care further exacerbates the discontinuous nature of psychiatric care, and may contribute to poorer self-management of diabetes and its complications.

CARE MANAGEMENT DOSAGE. D.A. Dorr¹; A. Wilcox²; L. Burns³; P.D. Clayton³. ¹Oregon Health & Science University, Portland, OR; ²University of Utah, Salt Lake City, UT; ³Intermountain Health Care, Salt Lake City, UT. (Tracking ID #136109)

BACKGROUND: Care management programs for chronically ill patients can bring order to chaos in primary care settings. However, services delivered and effectiveness vary widely among programs. Robust measures to more easily compare different modalities of care management would be useful for evaluation. This study creates a dosage measure designed to be extensible across a broad range of settings.

METHODS: This study occurred at a care management program at Intermountain Health Care involving 7 primary care teams augmented by care managers and information technology. Patients were referred based on perceived need by physicians for a variety of chronic illnesses and social issues, and receive individualized care in a multicondition care management program. Data collected by team members during daily activities yielded a comprehensive database of services provided. We used this information to create an expanded framework of our multimodality care management system. A modified Delphi method was used to generate and validate the categories for breadth of services provided. The amount, duration, frequency, and breadth of services for care managed patients were tabulated and correlated. Unsupervised k-means cluster analysis was completed to generate types of doses. A multivariable model was used to compare dosage amounts and types with subsequent adherence to guidelines for diabetes care.

RESULTS: Dosage was calculated for 2,356 patients provided 10,194 care management services in 7 clinics in 2003. Conditions given as the reason for treatment included diabetes (866; 36.8%), mental health needs (774; 32.9%), social needs, hypertension, and asthma. Multiple reasons were given in 19.7% of patients. Average amount of direct (with patient) and indirect (away from patient) time were 54.6 ± 109.2 and 92.8 ± 130.4 minutes per patient, respectively. Average frequency of services was 4.3 ± 6.4 per patient per year. Average duration of care was 140 ± 88.2 days. Categories for breadth of care were 1) general education (part of 27.7% of services), 2) guideline adherence (51.7%), 3) motivational (25.6%), 4) community advocacy (35.6%), 5) coordinating activities (16.2%), and 6) communication/collaboration (15.5%). Comparisons of reasons for care and dosage indicated multiple pathways of care only partially dependent on reason. Cluster analysis found 5 statistically separate groups of dosage, which were categorized as long intense (high indirect amount, high breadth), active disease management (high direct, long duration), brief intense (short duration, high amount), resource aid (very brief for advocacy), and maintenance (long duration, low direct). Higher dosage amounts as well as the brief intense and active disease care types were associated with better guideline adherence for persons with diabetes.

CONCLUSIONS: This framework for care management dosage showed a wide range and amount of services provided in our generalist system. Rigorous measurement techniques allowed for a detailed examination of types of care given. Higher dosage and type of dose given were positively related to process based outcomes.

CHRONIC KIDNEY DISEASE: RELATIONSHIP OF NEPHROLOGIST INVOLVEMENT AND DISEASE COURSE. L.A. Orlando¹; D.B. Matchar¹. ¹Duke University, Durham, NC. (Tracking ID #132573)

BACKGROUND: Chronic kidney disease (CKD) affects 11% of the U.S. population, about 20 million people. Improving the management of CKD increases survival in those with CKD, prolongs development of end-stage renal disease

(ESRD), and improves morbidity and mortality once ESRD develops. Slowing progression and delaying ESRD are now more important than ever as the incidence of CKD is increasing, in large part due to an increasing prevalence of CKD risk factors such as age, hypertension, and diabetes. These concerns have lead investigators to retrospectively evaluate the timing of nephrology referral on the morbidity and mortality of dialysis populations. Based upon the results of these preliminary studies, experts have recommended early referral to nephrology specialists. However, widespread implementation has been limited by the capacity of currently practicing nephrologists. To better evaluate the exact role and timing of nephrology intervention we examined the relationship between the severity of CKD, the presence of subspecialty care, and progression in a pre-ESRD population.

METHODS: We retrospectively identified a cohort of veterans with primary care providers at the Durham, NC VA hospital who had CKD, defined as two abnormal outpatient serum creatinines (>1.3 mg/dL) at least 3 weeks apart between January 1, 1998 and December 31, 1999. We gathered data from cohort inception until December 31, 2004 on disease course, CKD-related complications, medications, comorbidities, and mortality. Disease course, the time spent in each CKD stage, was evaluated using a Cox proportional hazards model adjusted for demographics, comorbidities, medications, and a propensity score for two groups: those followed by a primary care provider only and those followed by a nephrologist. We used the propensity score to adjust for bias in patients referred to nephrologists.

RESULTS: We found 1,553 veterans with CKD, a prevalence of 13%. The mean age was 70 years, all were male, 33% were African-American, $>90\%$ had hypertension and $>50\%$ had diabetes and coronary artery disease. 50% used angiotensin converting enzyme inhibitors (ACEI) and less than 3% used erythropoietin. The median number of days spent in each CKD stage was as follows: stage 1-1,149, stage 2-1,206, stage 3-1,158, stage 4-794, and stage 5-709. There was no difference in survival between the two groups during stage 1 or 2; but during stages 3 through 5 individuals in the nephrology group spent an average of 152 days more in each stage. ACEIs and statins slowed progression while HgbA1c values $>7.0\%$ and vascular disease accelerated it.

CONCLUSIONS: These data suggest that an appropriate time for referral to a nephrologist for patients with CKD may be around stage 3. Prospective studies are needed to further clarify the role and timing of nephrologists in the early stages of CKD.

CHRONIC PAIN MANAGEMENT GUIDELINES: A SYSTEMATIC REVIEW OF CONTENT AND STRENGTH OF EVIDENCE. M.J. Bair¹; K.M. Richardson¹; S.K. Dobscha²; E.J. Yi²; M.S. Gerrity²; K. Kroenke³. ¹Indiana University School of Medicine and Roudebush VA Medical Center, Indianapolis, IN; ²Oregon Health and Science University and Portland VA Medical Center, Portland, OR; ³Indiana University School of Medicine and Regenstrief Institute, Indianapolis, IN. (Tracking ID #134186)

BACKGROUND: Although numerous practice guidelines have been developed to assist clinical decision-making in chronic pain management, a critical assessment of these guidelines has not been done. Our objective was to compare key management recommendations and strength of evidence among published chronic pain guidelines.

METHODS: Guidelines were identified by searching MEDLINE, EMBASE, CINAHL, and National Guideline Clearinghouse databases from 1966 through June 2004 using the following terms: guideline, practice guideline, consensus or position paper, position statement, pain, and analgesics. Guidelines were included if they related to one or more of five topics: (1) chronic non-cancer pain; (2) musculoskeletal pain (low back, osteoarthritis, etc.); (3) neuropathic pain; (4) opioid use for chronic pain, and (5) headaches. Guidelines addressing cancer and acute pain were excluded. Recommendations were categorized within four broad areas: (1) initial assessment; (2) non-pharmacologic treatment; (3) pharmacologic treatment (i.e. non-opioid and opioid); and (4) treatment goals and outcomes. Selected guidelines and recommendations were assessed independently by two reviewers for variation in content and strength of evidence.

RESULTS: We identified 20 guidelines and 53 unique recommendations. Twenty-one recommendations focused on pharmacologic treatment and 14 on initial assessment. Only 14 recommendations were endorsed by more than 50% of guidelines. The five most frequently mentioned recommendations were to perform a physical exam (17/20, 85% of guidelines), assess pain history (16/20, 80%), assess psychosocial history (15/20, 75%), offer behavioral/psychological therapies (15/20, 75%), and provide patient education (14/20, 70%). Only 6 of 20 guidelines (30%) recommended using oral, long-acting agents while opioid therapy was unmentioned in the others. Rating schemes for strength of evidence were included in 11 guidelines. Of the five most common recommendations, only "provide patient education" had level 1 (i.e. meta-analysis or >1 RCT) evidence support.

CONCLUSIONS: There is considerable variation across guidelines for chronic pain management and the most common recommendations were not consistently supported by the strongest evidence. Future efforts should focus on improved standardization of guideline development and reporting/grading of evidence. Our work has the potential to identify evidence gaps and inform development of a pain research agenda.

CORRELATES OF COMPLEMENTARY AND ALTERNATIVE MEDICINE USE IN A NATIONAL SAMPLE OF ADULTS WITH DIABETES. D. Garrow¹; L.E. Egede¹. ¹Medical University of South Carolina, Charleston, SC. (Tracking ID #136160)

BACKGROUND: Previous studies have yielded conflicting estimates of complementary and alternative medicine (CAM) use in persons with diabetes. We analy-

zed data from the 2002 National Health Interview Survey (NHIS) to provide an update of patterns and correlates of CAM use among persons with diabetes.

METHODS: The 2002 NHIS is the most comprehensive nationally representative survey on CAM use (included data on 19 different types). We created eight CAM use categories including dietary, herbal, chiropractic, yoga, relaxation, vitamin, prayer and other (acupuncture, ayurveda, biofeedback, chelation, energy healing or Reiki therapy, hypnosis, massage, naturopathy, and homeopathy). We also created an overall CAM use category that excluded vitamins and prayer. Overall CAM use and the 8 categories of CAM were compared in 2,474 diabetics and 28,625 non-diabetics. Among diabetics, multiple logistic regression was used to determine independent correlates of CAM use controlling for relevant covariates. STATA was used for analysis to account for the complex survey design.

RESULTS: Overall CAM use was not significantly different between diabetics and non-diabetics (47.5 vs. 47.9%, $p=0.81$). Diabetics were significantly less likely to use yoga (4% vs. 9.1%) and herbs (22.3% vs. 24.9%) but more likely to use prayer (66.5% vs. 50%) compared to non-diabetics. Among diabetics, pattern of use was: vitamins 67%, prayer 66%, herbs 22%, chiropractic care 21%, relaxation 17%, other types 14%, diet 7%, and yoga 4%. Independent correlates of CAM use among diabetics included non-black race, women, >high school education, living in Midwest region, and having functional limitations.

CONCLUSIONS: Unlike previous studies, there is no significant difference in the use of CAM between diabetics and non-diabetics. There is dramatic increase in overall use of CAM and patterns of use among persons with diabetes. Correlates of CAM use seem consistent with earlier studies. Internists need to probe more often about CAM use in diabetics because of increasing prevalence of use.

DEPRESSION AND DIABETES: DOES TREATING DEPRESSION IMPROVE GLYCEMIC CONTROL IN PATIENTS WITH DIABETES: SYSTEMATIC REVIEW AND META-ANALYSIS. R.O. Murdoch¹; P. Chelminski²; R.M. Malone²; D.A. Dewalt²; M. Pignone². ¹University of North Carolina at Chapel Hill, School of Medicine, Chapel Hill, NC; ²University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #133189)

BACKGROUND: Depression and diabetes commonly co-exist. It is unclear whether treating depression improves glycemic control in patients with diabetes. We performed a systematic review and meta-analysis of existing trials examining the effect of depression treatment on glycemic control in order to address this question.

METHODS: We searched PubMed for relevant articles using the keywords "Glycosylated Hemoglobin" OR "Glycated Hemoglobin" OR "Glycemic Control" AND Depression OR "Depressive Disorders." We also searched the references cited in several meta-analyses and review articles for potential articles to include in this analysis. We limited our analysis to randomized trials comparing treatment for depression against no treatment, placebo, or usual care. Studies recruiting patients with Type I and/or Type II diabetes were eligible. All studies had to utilize hemoglobin A1C as an outcome measure. Secondary inclusion criteria were: All participants over 18 years of age, publication in the English language, a description of selection criteria and the randomization process. We did not exclude studies that reported trials involving a small number of participants ($n < 25$). We performed meta-analysis using the DerSimonian and Laird random effects model in RevMan software to examine the net effect of interventions aimed at reducing depressive symptoms on hemoglobin A1C levels in samples of depressed diabetics.

RESULTS: Our PubMed search initially identified 100 articles. No further articles were identified from the references section of articles consulted. Out of this initial pool of articles, we identified 6 randomized trials that examined the effect of treating depression on glycemic control and met out inclusion criteria. The two largest and best designed studies were conducted by Katon et al ($n=329$) and Williams et al ($n=417$). Both compared a depression management intervention to usual care. Three other smaller trials compared pharmacologic intervention to placebo. One study compared diabetes education to cognitive behavioural therapy. For the main outcome of change in A1C, the weighted mean difference with treatment for depression was -0.10 (95% CI $-0.34, +0.15$). **CONCLUSIONS:** Available evidence does not suggest that treatment of depression in patients with diabetes has important effects on A1C. Screening and treatment of patients with diabetes for depression is warranted, however, because it can improve depressive symptoms.

DETERMINANTS OF DIABETES EDUCATION AND ASSOCIATION WITH DIABETES OUTCOMES. S. Kim¹; J.A. Shea²; J. Long². ¹University of Medicine and Dentistry of New Jersey, New Brunswick, NJ; ²University of Pennsylvania, Philadelphia, PA. (Tracking ID #136129)

BACKGROUND: Diabetes education is an essential component of diabetes care, but only 40% of patients receive it. While certain social and demographic factors have been associated with receiving diabetes education, these may be confounded by issues related to access to care. In addition, receiving diabetes education is not consistently associated with improved long term outcomes. The objectives of this study were to assess determinants and pattern of diabetes education among patients with similar access to diabetes care, and to determine if receiving diabetes education is associated with improved outcomes.

METHODS: We conducted a cross-sectional study of 458 patients at the Philadelphia VA Medical Center with a history of diabetes for at least 2 years who resided in the city of Philadelphia and received diabetes medications from the VA. All diabetic patients at the VA have access to diabetes education classes.

Study participants were identified by pharmacy records. Demographic information and medical history were obtained by telephone interview and chart review. Categorical variables were analyzed with chi-square. ANOVA and stepwise regressions were used to examine relationships among continuous variables and diabetes education.

RESULTS: Of the 656 eligible patients contacted, 458 participated in the study (70%). Mean age was 63.2 (SD=10.5). Sixty-nine percent were black and 25% were white. All patients had type 2 diabetes, with mean HbA1C level of 7.7 (SD=1.74). Sixty-seven percent of patients had received diabetes education; 50% had taken one class and 17% had taken more than one. Younger age was associated with receiving diabetes education ($p=0.0028$), as was black race ($p<0.0001$), and lower income ($p=0.038$). Education, insurance status, or insulin use was not associated with receiving diabetes education ($p>0.05$). Mean number of years since receiving diabetes education was 10.2 (SD=8.64). Stepwise regression models including age, years since receiving diabetes education, race, education, income, insurance status and number of diabetes education courses showed that receiving diabetes education was not associated with fewer diabetes complications, glycemic control, or receipt of appropriate diabetes related care such as influenza and pneumococcal vaccinations, or blood pressure control ($p>0.05$). It was a significant predictor of self-perceived health behaviors ($p=0.02$), having a total cholesterol of less than 200 at the most recent visit ($p=0.03$), and being on an angiotensin converting enzyme (ACE) inhibitor or angiotensin receptor blocker (ARB) if proteinuric ($p=0.05$). Of all patients, 82% reported that they would be interested in receiving diabetes education if their health care provider suggested it.

CONCLUSIONS: In this study of patients with adequate access to diabetes care, the percentage of patients receiving diabetes education was significantly higher than the national average. At a mean of 10 years after diabetes education, diabetes education status did not predict improved diabetes outcomes, though it was a significant predictor of some diabetes related care. Future research examining the impact of an on-going diabetes education program may help optimize the role of diabetes education in producing sustainable improvement in diabetes outcomes.

DIABETES DIAGNOSTIC CODES ASSOCIATED WITH PRIMARY CARE VISITS: INTERPRET WITH CAUTION. S.A. Rose¹; S.G. Kertesz¹; M.M. Safford¹; K.P. Palonen¹; J.H. Halanych¹; C.I. Kiefe¹. ¹Birmingham VA Medical Ctr. & U. Alabama Birmingham, Birmingham, AL. (Tracking ID #135494)

BACKGROUND: International Classification of Diseases-9th edition (ICD-9) codes applied to individual clinic visits represent a potential asset to large database studies of primary care (PC) for chronic disease. Clinicians may code only a subset of diagnoses addressed in a visit, leaving it unclear whether visit-based ICD-9 codes accurately reflect the diseases addressed in care, especially in settings where diagnoses are not tied to reimbursement. Because of uncertainty concerning the reliability of coding behavior for research purposes, we examined the sensitivity (Sn) and specificity (Sp) of a visit-based diabetes (DM) ICD-9 code for the provision of DM care in a Veterans Health Administration (VHA) primary care context.

METHODS: Patients from a single VHA Medical Center with >1 ICD-9 diabetes codes (250.xx) or any glycemic control medication in 2002-2003 were considered to have DM. From among this group (5,895 unique patients with 32,755 visits in 2002-2003), we selected a random sample of 96 primary care clinic visits with an associated ICD-9 250.xx code, and another random sample of 92 without the code. Using chart review as the gold standard, we evaluated whether DM care occurred at each visit, based on whether any 1 of 4 basic aspects of DM care were addressed: foot care, eye care, blood sugar control, or kidney function. From these data we calculated positive and negative predictive values (PPV and NPV, respectively). Separately, we ascertained from the source cohort the total number of primary care visits, and the proportion with a 250.xx code. Then, extrapolating from these data, we generated a 2 x 2 table to estimate the Sn and Sp of a visit-based 250.xx code for the provision of DM care during that visit.

RESULTS: Of the 96 visits with a 250.xx code, 92 met criterion for DM care (PPV=95.83%). Of the 92 visits without a 250.xx code, 43 met criterion for DM care (NPV=53.26%). In the source population, 20,893 PC visits had a 250.xx code, 64% of the 32,755 total. Based on our chart review, 0.9583 (PPV) x 20,893 (250.xx coded visits)=20,022 visits with a 250.xx code estimated to have included diabetes care. Similarly, 0.5326 (NPV) x 11,862 (non-250.xx coded visits)=6,318 visits without a 250.xx code estimated to have lacked diabetes care. This extrapolation resulted in the 2 x 2 table below, leading to Sn=78% and Sp=88%. Notably, DM care likely occurred at 78% of all visits, regardless of ICD-9 code.

CONCLUSIONS: We documented moderate Sn and Sp of a 250.xx code for DM services in primary care. While clinicians provided DM care in almost 80% of visits, diabetes was not always coded. Researchers should be cautious in attempting to identify visits at which no diabetes care occurred on the basis of ICD-9 codes alone.

Diabetes Care and Presence of Visit-Based Diabetes ICD-9 Codes (250.xx)

	Diabetes Care Present	Diabetes Care Absent	
250.xx present	20,022	871	20,893
250.xx absent	5,544	6,318	11,862
	25,566	7,189	32,755

DISCORDANCE BETWEEN CARDIAC PATIENTS AND THEIR PHYSICIANS IN ASSESSMENT OF FUNCTIONAL STATUS, PARTICULARLY AMONG FEMALE PATIENTS.

K. Bibbins-Domingo¹; D. Schilling¹; A. NàPoles-Springer¹; G. Somma¹; L. Karliner¹; V. Grubbs¹; E.J. Perez-Stable¹. ¹Department of Medicine and Medical Effectiveness Research Center, University of California, San Francisco, SF, CA. (Tracking ID #132561)

BACKGROUND: While diagnostic and treatment guidelines for most cardiac disease are based on patient functional status, it is not known whether patients and physicians agree in their ratings of functional status or what characterizes patients who differ with their physicians in these assessments.

METHODS: In an ongoing study, we enrolled patients from the cardiology clinic of an urban public hospital in California. Immediately following a visit, we interviewed patients and physicians regarding patients' functional status using measures of functional limitations, physical limitations and cardiac symptoms. We asked about functional limitations using the New York Heart Association (NYHA) Classification (1-4). We asked about physical limitations and cardiac symptoms using the Seattle Angina Questionnaire and the Kansas City Cardiomyopathy Questionnaire (0-100, collapsed into quintiles 1-5). Discordance was defined as any difference between patients and physicians in mean scores on these scales. Clinically significant discordance was defined as an absolute difference ≥ 2 units on each of the scales, or failure of the physician to recognize functional or physical limitations or cardiac symptoms if reported by the patient. **RESULTS:** As of 11/04, we had enrolled 143 cardiac outpatients and had completed physician questionnaires for 86 of these patients. The mean NYHA class reported by patients was 2.1 ± 1.0 , compared to 1.7 ± 0.7 reported by physicians ($p < 0.001$). Half (43/86) of the patient-physician pairs were concordant in NYHA class. In 36% (31/86) of patient-physician pairs, patients rated their NYHA class as worse than their physician, while in 14% (12/86), physicians rated patient NYHA class worse than the patient. Clinically significant discordance between patients and physicians was frequent for NYHA class (28% of pairs), physical limitations (36%), and symptoms (38%). In multivariate analysis adjusting for age, sex, race, language of the patient, number of comorbid conditions, and depression, female patients were more likely than males to have clinically significant discordance in NYHA class (adjusted OR 3.5, 95% CI 1.0-12.2) and physical limitations (adjusted OR 3.1, 95% CI 1.0-9.4), but not symptoms (adjusted OR 1.2, 95% CI 0.4-3.7).

CONCLUSIONS: Cardiac patients and their physicians are frequently discordant in their assessment of functional status, with physicians frequently underestimating the severity of these limitations, particularly for female patients. Future work should explore whether this discordance is a result of different frames of reference or poor communication, and how discordance affects clinical decision-making and outcomes.

DISPERSION OF CARE AND CARDIOVASCULAR OUTCOMES: DOES MORE MATTER? B. Watts¹; D.G. Litaker². ¹Louis Stokes Cleveland VA, Cleveland, OH; ²Case Western Reserve University, Cleveland, OH. (Tracking ID #134227)

BACKGROUND: Dispersion of care (DOC) is viewed as a potential barrier to delivery of coordinated, comprehensive primary care. DOC has become even more salient given recent mandatory restrictions on the workweek of resident physicians. Disruptions introduced by these restrictions may result in unanticipated and/or undesirable effects on patient care, especially in the long-term risk management of chronic diseases. The purpose of this study was to examine the relationship of DOC to cardiovascular treatment outcomes in VA patients assigned a resident as their primary care provider.

METHODS: In this retrospective cohort study, we assessed whether patients had attained nationally recommended risk-factor specific treatment goals for blood lipids, blood pressure ($< 140/90$ or $< 130/80$ in diabetics), and glycemic control (hemoglobin A1C $< 7\%$). We also assessed presence of blood pressure 160/100. Using the clinical database at the study site we identified 473 patients with a lipid profile obtained between January 2003 and October 2004 whose PCP was a 1st, 2nd, or 3rd year resident. DOC was determined using system-assigned numbers to identify the provider at each visit in the 12 months preceding the lipid profile. An established index of DOC was dichotomized to reflect either complete continuity vs. some degree of DOC. Multivariable logistic regression adjusted for patient demographics, clinical characteristics, and the nesting of patients within resident practices to assess the independent association of DOC with each outcome.

RESULTS: Of the 473 patients, 363 patients (76.7%) had DOC (range 2-5 providers). Although the DOC group included more women (5.8% vs. 0.9%; $p = .04$) and had more PCP visits (mean 3.3 ± 1.3 vs. mean $2.40 \pm .63$; $p < .01$), no other significant differences in patients' age, cardiovascular disease, hypertension (HTN), disease burden, current tobacco abuse, and diabetes mellitus (DM) were present. Attainment of LDL goal was similar in the two groups (65.8% vs. 64.6%; $p = .80$), yet adequate blood pressure control in those with HTN or DM was lower in the DOC group (42.6% vs. 56.7%; $p = .01$) and the proportion of patients with poor blood pressure control (160/100) was greater (14.6% vs. 6.4%; $p = .02$). In 148 DM patients, significantly fewer patients in the DOC group achieved desirable glycemic control (44.6% vs. 71.4%; $p = .01$). Using multivariate models, DOC remained associated with less frequent attainment of appropriate HTN control (OR 0.51; CI 0.30-0.87), an increased likelihood of having poor HTN control (OR 2.52; CI 1.11-5.75), and a lower likelihood of glycemic control in DM patients (OR 0.33; CI .12-.93). We found no significant association between DOC and lipid control.

CONCLUSIONS: The association between DOC and treatment outcomes does not appear uniform and may more negatively affect outcomes such as HTN and glycemic control that involve complex management and a greater familiarity with a patient's comorbidities, medical history, and compliance. In residents' clinics where DOC is common, developing systems to coordinate efforts at risk

factor management may be useful. Our data also suggest that caution is needed when selecting outcomes to examine the impact of continuity on patient care.

DO INTERNAL MEDICINE RESIDENTS FIND PAIN MEDICATION AGREEMENTS USEFUL? J. Chen¹; M.J. Fagan¹; S.E. Reinert²; J.A. Diaz³. ¹Brown University, Providence, RI; ²Lifespan Information Services, Providence, RI; ³Brown University, Pawtucket, RI. (Tracking ID #133928)

BACKGROUND: While pain medication agreements (PMA's) are widely recommended when treating chronic nonmalignant pain (CNMP) patients with long-term opioids, the utility of PMA's is unknown. This study examined internal medicine residents' attitudes regarding the usefulness of PMA's in a hospital-based, primary care clinic.

METHODS: From May to June, 2004, we surveyed all 124 residents in the Internal Medicine Residency program at Lifespan/Brown Medical School. The survey included Likert type items to assess how frequently residents saw patients with CNMP, what percentage of their patients taking chronic opioids had a signed PMA, their confidence levels in managing CNMP compared to diabetes, and how helpful PMA's were for managing patients taking chronic opioids. Respondents were asked if they had ever had a patient leave their practice, discharged a patient, or been threatened by a patient due to conflict over pain medication. We used Spearman correlation to determine a relationship between reported use of PMA and reported confidence in managing CNMP and adverse experiences with patients regarding pain medication.

RESULTS: Of 124 surveys distributed, 110 were returned for a response rate of 88.7%. 45% and 9% of respondents reported seeing patients with CNMP "frequently" (5-9 times/month) or "very frequently" (> 10 times/month), respectively. 50% of respondents indicated that 60-100% of their CNMP patients had signed a PMA, and 90% of respondents felt that PMA's were helpful for managing patients with CNMP. Respondents indicated that PMA's were somewhat or definitely helpful in reducing early refill requests for opioids (67%), reducing multiple prescribers of pain medication (76%), reducing calls or pages from patients for pain medication (57%), discussing potential problems of opioid treatment with patients (73%), and identifying patients who are abusing pain medication (66%). There were no statistically significant correlations between greater reported use of PMA's and the following: reported confidence in managing CNMP (r [Spearman] = 0.18; $p = 0.06$); reported experience of having had a patient leave a resident's practice ($r = -0.11$; $p = 0.26$) or discharging a patient ($r = -0.07$; $p = 0.47$) due to conflict over opioid prescribing; or having been threatened by a patient over pain medication prescribing ($r = -0.09$; $p = 0.40$).

CONCLUSIONS: Although use of PMA's was not associated with greater resident self-confidence in managing CNMP, internal medicine residents in this setting found PMA's to be useful in managing their CNMP patients. No association was noted between use of PMA's and increased adverse patient interactions regarding the prescribing of pain medication.

EARLY STRUCTURED EXERCISE TRAINING AMONG PATIENTS HOSPITALIZED WITH HEART FAILURE: A RANDOMIZED CLINICAL TRIAL. M.P. Kosssovsky¹; M. Louis-Simonet¹; M. Rabaeus¹; A. Righetti²; M. Bettoni³; P. Sigaud¹; T.V. Perneger⁴; J.M. Gaspoz¹. ¹Department of Internal Medicine and Groupe de Recherche et d'Analyse en Systèmes et Soins Hospitaliers (GRASSH), Geneva University Hospitals, Geneva, ; ²Department of Internal Medicine, Geneva University Hospital, Geneva, ; ³Cardiofour, Hôpital de la Tour, Meyrin, Geneva; ⁴Quality of Care Unit and Groupe de Recherche et d'Analyses en Systèmes et Soins Hospitaliers (GRASSH), Geneva, . (Tracking ID #132490)

BACKGROUND: Exercise training (ET) in patients with congestive heart failure (CHF) has been shown to reduce hospital readmissions for heart failure, to improve quality of life and to reduce all cause mortality. However, prior studies only enrolled patients in stable condition for at least 3 months. We sought to determine if a structured exercise training program performed early after a hospital stay for CHF was both secure and beneficial.

METHODS: Design: randomized clinical trial. Eligibility: all patients admitted for CHF with a left ventricular ejection fraction (LVEF) $\leq 45\%$, able to perform a moderate exercise and willing to participate. Intervention: random allocation to usual care or to a structured 7-month ambulatory exercise program consisting in at least 3 weekly sessions during the first month, followed by 2 weekly sessions for the remaining 6 months. Outcomes: time to readmission for any cause or CHF related, quality of life, exercise tolerance, survival.

RESULTS: From January 2002 to August 2004, 112 patients were randomly allocated in the control group (57) and in the ET group (55). Both groups were similar in terms of age (Mean age: 69 $\pm$ 12), gender, CHF symptoms severity (NYHA class on admission: 3.5 $\pm$ 0.6), comorbidity (Charlson comorbidity index: 2.5 $\pm$ 1.9) and medications (admission and discharge). ET was started within a mean of 20 $\pm$ 15 days after hospital discharge. Among patients who followed at least 1 ET session (45/55), 6 were readmitted within 120 days (13%) vs. 17 (30%) in the control group ($p = 0.05$; log rank test). Patients in the ET group showed a greater improvement in 6-minute walk test performance at the end of the program (102 vs. 39 m; $p = 0.02$) and in quality of life (24.3 vs. 11.6 percentage gain on the Kansas City Cardiopathy Questionnaire; $p = 0.05$). No significant difference was observed in terms of mortality (4 in ET group vs. 6 in control group; $p = 0.42$).

CONCLUSIONS: Early exercise training among unselected, aged and polymorbid patients discharged from general internal medicine wards for systolic CHF is both secure and beneficial in terms of hospital readmissions, exercise tolerance and quality of life. Structured exercise training should be incorporated in comprehensive CHF management programs.

EFFECT OF COMPLEMENTARY AND ALTERNATIVE MEDICINE USE ON RESOURCE UTILIZATION AND PREVENTIVE CARE PRACTICES AMONG ADULTS WITH DIABETES. D. Garrow¹, L.E. Egede¹. ¹Medical University of South Carolina, Charleston, SC. (Tracking ID #136169)

BACKGROUND: There is concern that complementary and alternative medicine (CAM) is used as an alternative to conventional therapies leading to decreased use of conventional services. We used data from the 2002 National Health Interview Survey (NHIS) to assess the relation among CAM use, resource utilization, and preventive care practices in adults with diabetes.

METHODS: The 2002 NHIS is the most comprehensive nationally representative survey on CAM use and included data on 19 different types. We analyzed data on 2,474 adults with diabetes. We created an overall CAM use category based on use of any of the following: diets, herbs, chiropractic care, yoga, relaxation, acupuncture, ayurveda, biofeedback, chelation, energy healing, Reiki therapy, hypnosis, massage, naturopathy, and homeopathy. We excluded vitamin use and prayer. We compared resource utilization (number of primary care and emergency department visits) and preventive care practices (receipt of flu and pneumonia vaccines) in user and non-users of CAM. STATA was used for statistical analysis to account for the complex survey design.

RESULTS: Overall, 48% of diabetics used at least one CAM therapy. Compared to non-user, CAM users were more likely to have multiple primary care visits (48.7% vs. 32%, $p < 0.001$) and emergency department visits (50.7% vs. 45.1%, $p = 0.01$). CAM users were also more likely to receive influenza (48.9% vs. 44.8%, $p = 0.04$) and pneumonia (50.3% vs. 44.6%, $p = 0.007$) vaccinations.

CONCLUSIONS: In contrast to the findings of previous studies, CAM use appears to be associated with increased primary care and emergency department use and increased likelihood of receipt of preventive care services. Future studies need to assess the implications of this finding.

EFFECTIVENESS OF ROSUVASTATIN COMPARED TO ATORVASTATIN ON NCEP ATP-III LDL-CHOLESTEROL GOAL ATTAINMENT IN A USUAL CARE SETTING. M.F. Bullano¹, K.L. McDonough², S.A. Kamat¹, D.A. Wertz¹, M.J. Cziraky¹, V.J. Willey¹. ¹HealthCore, Inc., Wilmington, DE; ²AstraZeneca Pharmaceuticals, LP, Wilmington, DE. (Tracking ID #131771)

BACKGROUND: Current data are limited regarding the effectiveness of rosuvastatin compared to atorvastatin on NCEP ATP-III LDL-C goal attainment outside of randomized controlled trials. The objective of this study was to evaluate the effectiveness of rosuvastatin compared to atorvastatin on LDL-C goal attainment using medical records and claims data to reflect a usual care community setting.

METHODS: Patients newly-initiated on rosuvastatin or atorvastatin between 8/1/03 and 6/30/04 were identified from pharmacy claims data in a West Coast health plan for this retrospective, longitudinal cohort study. Patients were excluded if they had any dyslipidemic therapy 12 months preceding their initial statin fill. Propensity score matching on baseline characteristics was employed to minimize selection bias between groups. Clinical data was collected on lipid levels, blood pressure readings, alcohol consumption, diet, smoking status, and family history of heart disease by abstracting patients' medical charts from both groups. Administrative claims and medical records were then utilized to calculate individual Framingham Scores and assign patients a NCEP risk status and corresponding LDL-C goal. Adjusted goal attainment odds ratios were calculated using regression techniques.

RESULTS: A total of 337 patients (166 rosuvastatin/171 atorvastatin) met the study inclusion/exclusion criteria. Mean age of the cohort was 55 ± 12 years. Patients in the rosuvastatin group compared to atorvastatin were found to have similar smoking status (10% vs. 9%), alcohol consumption (36% vs. 43%), mean blood pressure readings (130/80 mmHg vs. 129/80 mmHg), and Framingham 10-year risk scores (9.5 ± 8.4 vs. 7.5 ± 6.5); all $p > 0.05$. Mean dose for rosuvastatin was 11 mg compared with 16 mg for atorvastatin. Percent change required to reach LDL-C goal was higher for rosuvastatin ($26.3\% \pm 13.9\%$) compared to atorvastatin ($23.5\% \pm 13.7\%$); $p < 0.05$. The percentages of patients achieving their LDL-C goal with rosuvastatin and atorvastatin were 74% and 65%, respectively. After adjusting for age, gender, percent change required to reach goal, NCEP risk status, and therapy duration, significantly ($p < 0.05$) more patients were observed to achieve their LDL-C goal with rosuvastatin (odds ratio = 1.87; 95% CI: 1.10-3.17).

CONCLUSIONS: Rosuvastatin patients were nearly twice as likely to attain their NCEP ATP-III LDL-C goals compared to atorvastatin. These data, from a usual care community setting, confirm previous findings from randomized controlled trials and are among the first to document the effectiveness of rosuvastatin over atorvastatin outside of randomized controlled trials.

EVALUATION OF WOMEN PRESENTING WITH GASTROESOPHAGEAL REFLUX OR ATYPICAL CHEST PAIN IN AN URBAN CLINIC. T. Lingenfelter Pierce¹, S. Ali¹, M. Lim¹. ¹University of Connecticut, Farmington, CT. (Tracking ID #134328)

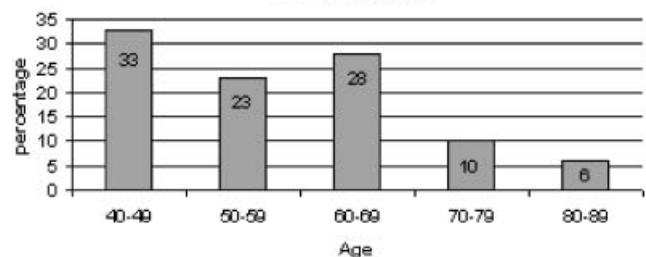
BACKGROUND: Coronary artery disease (CAD) is the leading cause of death in women, accounting for 35% of all-cause mortality in 1995 and for more deaths each year than all other causes combined. Although the incidence of myocardial infarction (MI) in women is lower than that in men, mortality after a MI is higher in women compared to men. One explanation for this is that women with CAD are about 10 years older than men at the time of presentation and carry a greater burden of risk factors: diabetes, hypertension and hyperlipidemia. Review articles in the New England Journal of Medicine have stated that women who present with new-onset chest pain are evaluated and managed less aggressively

than men. This may play a role in the increased mortality seen in women with CAD. Further, 40% of women with CAD present with atypical symptoms. In a review of over 3100 patients, women were less likely than men to have typical angina (28% versus 55%), which may in part explain the delays in seeking and receiving care. Our objective is to determine the frequency of cardiac work-up done in women older than 40 who presented with gastroesophageal reflux (GERD) symptoms or atypical chest pain.

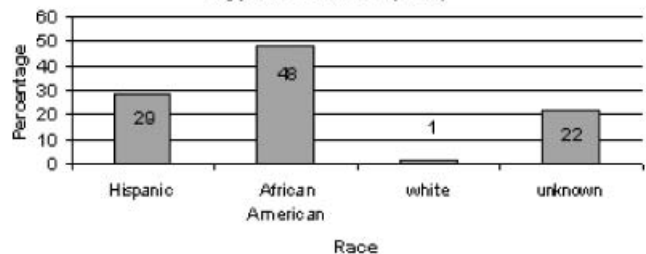
METHODS: This is a retrospective chart review of all women older than 40 seen in an urban teaching clinic with atypical symptoms. Patients should have seen their medical provider at least twice from July 1, 2003 to June 30, 2004. Outcome measures were a cardiac work-up, a gastrointestinal (GI) work-up or symptom resolution after symptom-specific medical management. Cardiac work-up was defined as a stress test/stress echo or Cardiology referral. GI work-up was defined as an upper endoscopy or Helicobacter pylori testing or Gastroenterology referral. Charts reviewed were those patients with an International Classification of Disease Ninth Revision (ICD-9 code) of GERD or atypical chest pain. Risk factors were collected from review of information documented in the progress notes rather than evaluation of recorded vital signs or laboratory values. A total of 74 patients' charts were reviewed. Of these, 5 were excluded from the study because the patients were only seen once in a walk-in clinic.

RESULTS: Demographic data from the charts reviewed are demonstrated in Graphs 1-8. No work-up was done in 43% of women who presented with atypical chest pain. Of these, however, 6 have symptoms that resolved with NSAID use ($n=1$) or proton-pump inhibitor/H2-blocker ($n=5$). The remaining 24 (35%) had persistent GERD or atypical chest pain symptoms without cardiac evaluation (Graph 9), despite the fact that 16 (68%) have 2 or more cardiac risk factors (Graph 10).

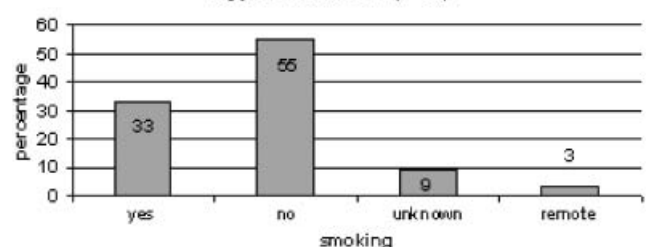
Graph 1: Age Groups of Patients with Atypical Chest Pain (n=69)



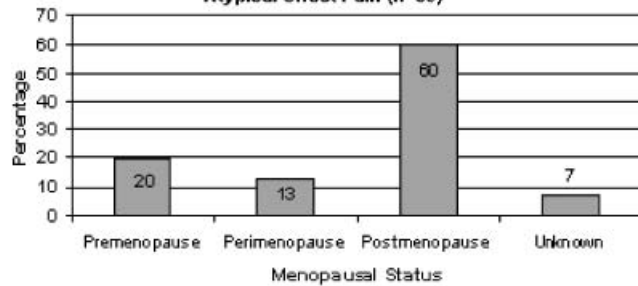
Graph 2: Race Distribution of Patients with Atypical Chest Pain (n=69)



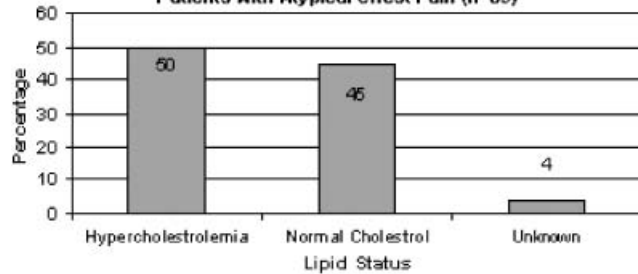
Graph 3: Prevalence of Smoking in Patients with Atypical Chest Pain (n=69)



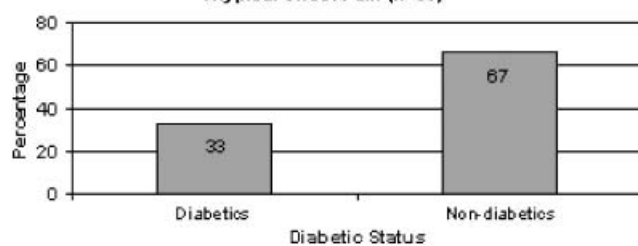
Graph 4: Menopausal Status of Patients with Atypical Chest Pain (n=69)



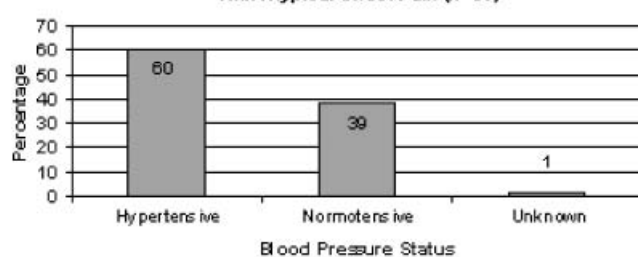
Graph 5: Prevalence of Hyperlipidemia in Patients with Atypical Chest Pain (n=69)



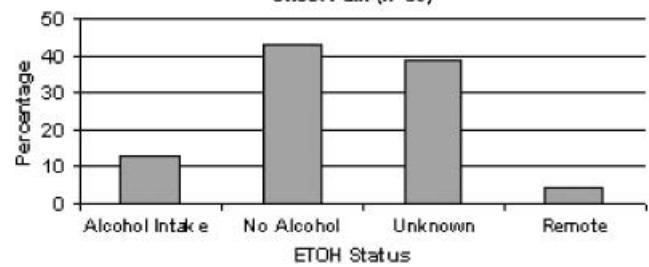
Graph 6: Prevalence of Diabetes in Patients with Atypical Chest Pain (n=69)



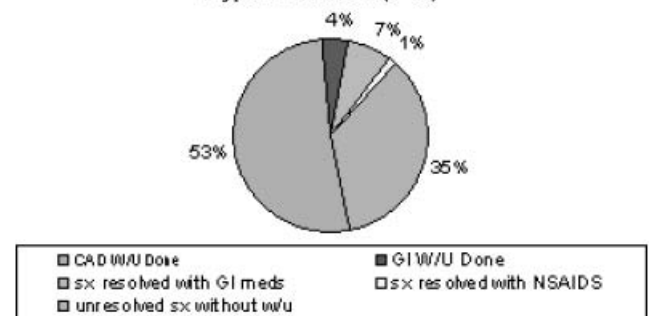
Graph 7: Prevalence of Hypertension in Patients with Atypical Chest Pain (n=69)



Graph 8: Alcohol Intake in Patients with Atypical Chest Pain (n=69)

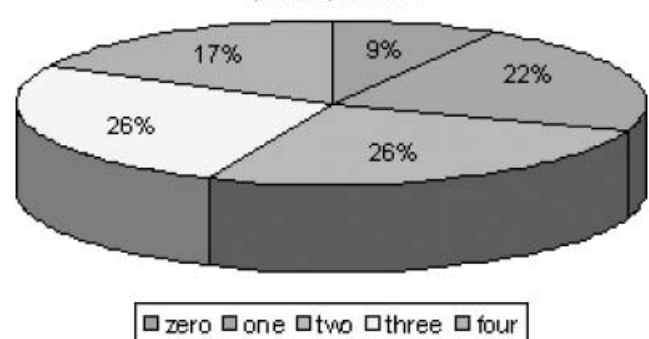


Graph 9: Work-up Done on Patients with Atypical Chest Pain (n=69)



Graph 10: Number of Cardiac Risk Factors* in Patients without Cardiac Evaluation (n=24)

*risk factors defined as smoking, diabetes, hypertension, hypercholesterolemia and postmenopausal state



CONCLUSIONS: CAD is a major cause of morbidity and mortality in women. This is influenced by multiple factors including having atypical symptoms and less aggressive medical management. In our review, almost half of the women with atypical chest pain were not evaluated. Of these, most had cardiac risk factors warranting further evaluation. Thus, we need to work aggressively in our clinic and within the medical community to evaluate women with chest pain even if they have atypical symptoms and especially if they have cardiac risk factors.

EVIDENCE BASED TREATMENT GUIDELINES POORLY FOLLOWED IN PATIENTS UNDER REGULAR CARE FOR HYPERTENSION. J. Witter¹; A. Davidson². ¹Denver Health and Hospital Authority, Denver, CO; ²Denver Public Health, Denver, CO. (Tracking ID #136020)

BACKGROUND: The Sixth Report of the Joint National Committee on Detection, Evaluation, and Treatment of High Blood Pressure (JNC-6), in effect from 1997 through 2003, recommended initial therapy of hypertension (HTN) with a thiazide diuretic or a beta-blocker, unless a compelling indication for use of an

other drug class was present. Compelling indications included use of beta-blockers with a history of myocardial infarction (MI), and use of angiotensin converting enzyme inhibitors and later angiotensin receptor blockers (ACE-inhibitor/ARB) with either congestive heart failure (CHF) or diabetes (DM). Furthermore, use of a thiazide was recommended for patients who need more than one medication to control their hypertension. While concordance with initial therapy recommendations has been documented, prescription patterns for patients treated over time have not been described.

METHODS: Using administrative files, we examined prescription patterns from 1997 through 2003 in a cohort of 6281 patients under regular care for hypertension in Denver Health outpatient clinics. Denver Health is an integrated safety net provider with hospital, primary care, and specialty services. The patients had at least 2 hypertension visits (identified by ICD-9 code 401.XX) over at least one year, and were restricted to those with Medicare or enrollment in our indigent care program. Greater than 95% of these patients filled their medications at Denver Health pharmacies due to significant price discounts. We linked medication use along with demographic variables and comorbid medical conditions to adherence with JNC-6 guidelines.

RESULTS: ACE inhibitor/ARB use increased over the study period; 32% of HTN patients were prescribed these medications in 1997 and 64% were prescribed them in 2003. Beta-blocker use fell from 48% in 1997 to 8% in 2003. Thiazide use remained stable (51% in 2003), while use of calcium channel blockers fell to 16% in 2003. Overall, 48% of patients were prescribed medications in concordance with JNC-6 guidelines in 1997 and 54% were in concordance in 2003. In 2003, 57% of patients with HTN and no history of DM, MI, or CHF received JNC-6 concordant medications, while 54% of DM patients, 34% of CHF patients, and 38% of MI patients received JNC-6 concordant medications. Males were less likely to receive concordant medications (OR=0.63, 95%CI 0.53-0.73), while African American race (OR=1.21, 95% CI 1.03-1.42) and number of visits (OR=1.02, 95% CI 1.01-1.03 for each subsequent visit) increased the likelihood of concordant prescription. Involvement of cardiology or nephrology specialists was associated with decreased adherence to guidelines (OR=0.77, 95%CI 0.69-0.92).

CONCLUSIONS: Previous research has found that JNC-6 guidelines are not closely followed with initial antihypertensive prescriptions. This study find that concordance with guidelines does not substantially improve when a patient is treated over time. Investigation into determinants of hypertension medication prescription patterns are needed in order to guide interventions that encourage evidence based treatment.

Proportion of Hypertensive Patients Receiving a Class of Medication, By Year

Medication Class	1997 (%)	1998 (%)	1999 (%)	2000 (%)	2001 (%)	2002 (%)	2003 (%)
Thiazide	49	50	50	51	52	51	51
B-blocker	48	31	27	25	21	15	8
ACE/ARB	32	38	43	49	53	59	64
Ca Channel Blocker	28	21	19	18	16	14	16
Other HTN drug	14	15	16	19	20	22	22

Proportion of HTN Patients Receiving JNC-6 Concordant Medications

HTN and Comorbid Disease	1997	1998	1999	2000	2001	2002	2003
All patients	48	47	46	47	48	49	54
HTN, No CM, CHF, MI	69	55	53	51	52	53	52
Diabetes	40	40	41	40	43	46	54
CHF	34	35	37	33	35	33	34
MI	44	44	40	36	33	35	38

FACTORS ASSOCIATED WITH MEDICATION ADHERENCE AND BLOOD PRESSURE IN HYPERTENSIVE PATIENTS. A.B. Morris¹; J. Li²; K. Kroenke³; T. England⁴; J. Young⁴; M.D. Murray⁵. ¹Purdue University School of Pharmacy & Pharmacal Sciences, Indianapolis, IN; ²Indiana University Purdue University Indianapolis, Indianapolis, IN; ³Indiana University Purdue University Indianapolis, Greenwood, IN; ⁴Wishard Health Services, Indianapolis, IN; ⁵University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #134194)

BACKGROUND: Adherence is important in chronic disease management, yet incompletely understood. Medications are the cornerstone of the management of chronic diseases such as hypertension. Adherence to antihypertensive medications is approximately 50 percent. Because of the relationship between poorly controlled blood pressure and future cardiovascular comorbidity, identification of remedial factors affecting adherence is critical. Patients whose adherence is poor could be provided with training and other interventions to help promote their ability to take their medications. Therefore, we studied the patient characteristics associated with self-reported medication adherence and pharmacy refill adherence, and the relationships between blood pressure and each of the adherence metrics.

METHODS: This is a secondary analysis of baseline data from a randomized controlled clinical trial of a pharmacy intervention conducted at a large urban

medicine practice. We studied 492 patients, 18 years old, diagnosed with hypertension and taking at least one antihypertensive medication. Trained interviewers assessed self-reported adherence (yes/no) using validated instruments. We further extracted prescription refill records data from an electronic medical record system. Adherence using pharmacy refill data was defined as obtaining 80 to 120 percent of prescribed antihypertensive medications. Depression was defined as a PHQ-8 score 10. First, we used univariate logistic regressions to identify the patient characteristics associated with each of the adherence metrics. Then we entered the significant factors from the univariate analysis into multiple logistic regressions to determine which characteristics predict each of the adherence metrics. Finally, multiple linear regression was used to determine the independent correlates of systolic and diastolic blood pressure. Self-reported and pharmacy refill adherence were evaluated separately in models for systolic and diastolic blood pressure.

RESULTS: Agreement between self-reported and refill adherence was poor ($\kappa=0.21$). Four patient variables were associated with adherence in multiple logistic regressions, increased age ($P < .002$) and being married ($P=.03$) were independent predictors of better self-reported and refill adherence, while depressed patients had worse self-reported adherence ($P=.005$) and African-Americans had worse refill adherence ($P < .0001$). Adherent patients had lower systolic (5.0 to 5.4 mm Hg lower) and diastolic (2.7 to 3.0 mm Hg lower) blood pressure ($P=.02$). Age was the only other variable strongly associated with systolic and diastolic blood pressure in both metrics of medication adherence ($P=.001$), with roughly a 3 mm Hg increase in systolic and 3 mm Hg decrease in diastolic blood pressure with each decade of age. Depression, race, and gender were associated with blood pressure in certain models.

CONCLUSIONS: Demographic characteristics and depression are adversely associated with both medication adherence as well as blood pressure control in hypertensive patients. Self-reported and refill measures provide different and complementary perspectives on adherence.

FACTORS WHICH AFFECT RETURN TO WORK FOR PROSTATE CANCER SURVIVORS. R.R. Bouknight¹; C.J. Bradley¹. ¹Michigan State University, East Lansing, MI. (Tracking ID #130865)

BACKGROUND: Cancer of the prostate is the most common malignancy of American men and the second leading cause of cancer deaths. In addition to losses due to mortality, prostate cancer and its treatment may compromise quality of life, especially for employed survivors who cannot return to work (RTW). The ability to re-enter the workforce is an indicator of recovery for these patients and physicians and employers can facilitate their workforce re-entry. The purpose of this research was to identify demographic, clinical, and employment factors which affect RTW for employed prostate cancer patients 12 and 18 months following diagnosis.

METHODS: An inception cohort of working men newly diagnosed with prostate cancer were identified and interviewed for a period corresponding to 3 months prior to, and 12 and 18 months following diagnosis. Questions regarding demographic, clinical and job characteristics were asked. With RTW at 12 and 18 months as the outcome, logistic regression analysis was used to determine the odds ratios (OR) and 95% confidence intervals (CI) for predictors of RTW.

RESULTS: We enrolled 267 men who were working at the time of prostate cancer diagnosis and 252 of these subjects participated in the 12- and 18-month post diagnosis interviews. Approximately 25% had not returned to work 12 and 18 months following their prostate cancer diagnosis. In the 12-month univariate analysis, factors associated with a lower likelihood of RTW were older age ($p \leq .0001$) and being married ($p=.03$). Having dependent children at home ($p=.04$), sick leave ($p \leq .0001$), and employer accommodation for cancer illness and treatment ($p \leq .0001$), were associated with a greater likelihood of RTW. In the 12-month multivariate analysis, older age (OR, .92 [95% CI .85-.99]) and having a job in which the subject was required to keep up with others (OR, .47 [95% CI .22-.99]) were associated with a statistically significant lower likelihood of RTW and employer accommodation for cancer illness and treatment (OR, 5.6 [95% CI 2.4-12.7]) was associated with a greater likelihood of RTW. Employer accommodation was also a strong independent predictor of RTW 18 months following a prostate cancer diagnosis (OR, 4.8 [95% CI 2.0-11.5]).

CONCLUSIONS: In this study, the willingness of the employer to accommodate their workers' cancer needs was an important factor in RTW. The Americans with Disabilities Act requires that employers provide reasonable accommodations for impaired workers including those recovering from malignancies. Physicians can facilitate RTW for their working prostate cancer patients by communicating with employers about their patients' treatment and accommodation needs.

FAILURE TO INTENSIFY ANTIHYPERTENSIVE THERAPY. M. Weiner¹; W. Tu¹; J. Wu²; Y. Ding²; M.D. Murray³. ¹Indiana University Center for Aging Research, Indianapolis, IN; ²Regenstrief Institute, Inc., Indianapolis, IN; ³University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #136186)

BACKGROUND: Drug therapy (Rx) for hypertension (HTN) includes the initiation of HTN Rx and then intensification, by increasing the dose of a drug or prescribing an additional drug. Little is known about frequency of HTN Rx intensification and factors related to it. We studied outpatients with HTN Rx, hypothesizing that clinicians do not intensify HTN Rx at most visits at which blood pressure (BP) remains elevated.

METHODS: At an urban public system on a midwestern, academic medical campus, we studied patients enrolled since 2002 in a randomized, controlled trial of a pharmacist-based intervention to improve patients' adherence to HTN

Rx. Patients were included in the study if they had a clinical diagnosis of HTN and received at least one Rx for HTN. Patients seeking primary care or medications outside the of this public system were excluded. Clinical encounters, prescriptions, and other data were extracted from a comprehensive electronic medical records system for a 22-month period. We used a validated method involving a Medication Event Monitoring System, to record dates when participants obtained new Rx bottles, reflecting an increase in the dosing frequency of an existing HTN Rx or the addition of a new HTN Rx. Visit-specific intensification of HTN Rx was conservatively defined as receipt of a new Rx bottle occurring any time from the index visit to the subsequent visit. Using chi-square and t-tests, we assessed associations between intensifying HTN Rx, demographics, and elevated BP (systolic BP ≥ 130 mmHg or diastolic BP ≥ 90 mmHg). We considered a p-value <0.05 to be statistically significant.

RESULTS: Participants (N=328) had a mean age of 58 years, and most (71%) were African-American. Elevated BP was observed in 47% (154/327), with 62% of these (95/154) having isolated systolic HTN. Neither age nor race was significantly associated with intensification of HTN Rx. HTN Rx was intensified in 26% of participants. Of the 2,146 outpatient primary-care visits with BP measurements, BP was elevated in 46% (980/2146). Although a visit with elevated BP was associated with intensification of HTN Rx ($p<0.01$), this occurred in only 6% (61/980) of visits (1.6% for non-loop diuretics, 1.4% for angiotensin-converting enzyme inhibitors, 1.3% for calcium-channel antagonists, 0.8% for beta-adrenergic receptor antagonists, and 0.6% for angiotensin-receptor antagonists). Even for 182 visits with highest BP (systolic BP ≥ 140 mmHg and diastolic BP ≥ 90 mmHg), HTN Rx was intensified in only 7% (13/182).

CONCLUSIONS: In an ongoing trial within an urban public outpatient setting, participants with HTN Rx and elevated BP seldom receive intensification of their therapy. The use of drugs to treat HTN among all participants suggests that this finding is not related to physicians' reluctance to use HTN Rx or patients' limited access to drugs. Although the methods were not designed to detect dose changes without increasing frequency, failure to intensify regimens may have many explanations, such as "white-coat" HTN, contraindications to certain drugs, side effects, and patients' preferences.

GROUP VISITS – IF YOU BUILD IT, WHO WILL COME? L. Kawasaki¹; P. Muntner¹; K. Hampton¹; S. Sanne¹; K.B. Desalvo¹. ¹Tulane University, New Orleans, LA. (Tracking ID #136048)

BACKGROUND: Group visits are an interactive, patient-centered model of care with demonstrated benefit in the management of chronic disease. The success of this model is dependent upon patient recruitment, yet little is known about the general acceptance of this model and potential predictors of willingness to participate.

METHODS: Objective: We surveyed urban, disadvantaged patients to evaluate correlates of willingness to attend group visits for the treatment of hypertension. Methods: A telephone survey of 244 patients eligible for a randomized trial comparing group visits versus individual care for hypertension management was conducted by one trained interviewer. The validated survey included questions about demographics, medical history, hypertension knowledge, compliance, psychosocial parameters and experiences with health care.

RESULTS: The majority of participants were female (79%) and black (92%) with a mean age of 55 years (SD +10.3) and an income of less than \$1000/month (77%). 93% reported a family history of hypertension and 30% had a diagnosis of hypertension for more than 10 years. 59% had hyperlipidemia; 18% currently smoked; 33% had diabetes; and fewer than 10% had a history of MI or CVA. 70% reported a willingness to attend group visits for their hypertension care. In a multivariable logistic model adjusting for age, gender, and race/ethnicity, significant predictors of willingness to attend group visits included: an income $>$1000 (OR 2.5, CI 1.2-5.2); diabetes (OR 1.9, CI 1.0-3.7); longer duration of hypertension (OR 1.9, CI 1.0-3.7); and no history of MI (OR 2.8, CI 1.0-6.4). Individuals reporting poor health were also more likely to be willing to attend (OR 2.3, CI 1.0-5.1). Those not willing to participate indicated that a reduced wait time to see the physician (33%), more time with their physician (25%) and parking/transportation subsidies (22%) would provide incentives to attend a group visit.$

CONCLUSIONS: The vast majority of these socio-economically disadvantaged patients expressed a willingness to participate in group visits for their hypertension treatment. Simple incentives and targeted promotion may enhance recruitment. Group visits appear to be an acceptable and viable alternative model of chronic disease care in vulnerable populations.

HEALTH AND HEALTH CARE NEEDS OF DEPRESSED PATIENTS DETECTED IN PRIMARY CARE. E. Yano¹; E.F. Chaney²; B. Simon³; L. Bonner⁴; A.B. Lanto¹; D. Campbell²; L.V. Rubenstein⁵. ¹VA Greater Los Angeles HSR&D Center of Excellence, Sepulveda, CA; ²University of Washington, Seattle, WA; ³VA Greater Los Angeles, Los Angeles, CA; ⁴VA Puget Sound Healthcare System, Seattle, WA; ⁵University of California, Los Angeles, Los Angeles, CA. (Tracking ID #135979)

BACKGROUND: Despite its prevalence, depression remains often undetected, under-diagnosed and under-treated. Performance measures designed to monitor and incentivize screening and diagnosis in primary care settings have been introduced, but little is known about the types of patients that are ultimately identified. We evaluated the health and health care needs of patients diagnosed with depression in diverse primary care (PC) settings in order to better understand the outcome of practice-wide screening efforts.

METHODS: As part of a group-randomized trial, we randomly sampled and interviewed patients with a scheduled PC appointment (n=13,392) in 10 VA pri-

mary care clinics in 3 VA networks across Ohio, Texas, Florida, South Dakota and Wisconsin. We screened for depression using the 2-item CIDI and diagnosed depression using the PHQ9 among positive-screens, resulting in a baseline cohort of 768 PC patients diagnosed with depression. We then collected baseline sociodemographics, health status, utilization, treatment history and patient preferences, all using computer-assisted telephone interviewing (CATI).

RESULTS: Overall, 17.3% of patients scheduled for a PC appointment screened positive for depression; 56.9% of positive-screens were subsequently diagnosed with depression using the PHQ9 (9.8% of all sampled PC patients). Among depressed PC patients, comorbid mental illness was common, with 69.0% having experienced recurring nightmares from a traumatic event (nearly two-thirds with traumatic thoughts/nightmares occurring in the past month) and 63.7% with comorbid anxiety (one-quarter reporting a panic attack within the past month). Medical comorbidities were also common, with histories of chronic lung disease (36.6%), pneumonia (35.9%), diabetes (34.5%), MI (28.0%), CHF (17.8%), cancer (17.6%) and stroke (15.1%). Sexual dysfunction was common (69.3%), being both longstanding (two-thirds for >6 months) and commonly undiscussed and untreated (53.6%). 44.8% had had at least one visit with a mental health (MH) specialist in the past 6 months, while 84.8% had had one or more PC visits.

CONCLUSIONS: Practice-wide depression screening in primary care settings yields a broad range of incident and prevalent cases, many of which are already in care. While PC visit rates present more opportunities for depression management, the degree of comorbid physical and mental illness among depressed PC patients precludes solo management by either PC generalists or MH specialists, rendering guideline-adherence at the provider level problematic without systems-level solutions that help PC providers adequately assess, treat and manage their patients with appropriate supports. While collaborative care models have been advocated, clearer evidence is needed on how to translate their potential benefits into realities for the practicing clinician.

IMPACT OF INCREASED ANTIDEPRESSANT ADHERENCE ON COMORBID MEDICATION USE AND RESOURCE UTILIZATION. W.J. Katon¹; C.R. Cantrell²; M.C. Sokol³; J.M. Gdovin⁴; E. Chiao⁴. ¹University of Washington, Seattle, WA; ²GlaxoSmithKline, Research Triangle Park, NC; ³GlaxoSmithKline, Philadelphia, PA; ⁴Applied Health Outcomes, Palm Harbor, FL. (Tracking ID #134950)

BACKGROUND: Patients with depression have been associated with non-adherence to antidepressant therapy and therapy prescribed for common chronic comorbid conditions, such as coronary artery disease and diabetes, which has led to poor clinical outcomes. Although the effect of depression on concurrent chronic medication adherence and clinical outcomes has been established, it remains to be determined if better adherence to antidepressant therapy would translate into improved adherence to medication for comorbid conditions and economic outcomes. Therefore, the purpose of this study was to determine if improved adherence to antidepressant medication is associated with 1) an increased likelihood of being adherent to comorbid chronic disease medications, and 2) a decrease in total and comorbid disease-specific medical costs.

METHODS: Patients on antidepressant therapy with evidence of coronary artery disease (CAD), diabetes (DM) or both between July 1, 2001 and December 31, 2002 were identified via medical and pharmacy claims obtained from the IHCIS National Managed Care Benchmark database (Waltham, MA). Patients were required to be aged 18 years and older, continuously eligible 6 months prior and 12 months after initiating antidepressant therapy, and have comorbid medication of interest prior to and after initiating antidepressant therapy (statin and/or oral diabetes therapy). Compliance, measured by medication possession ratios (MPR), was defined as 1) 80% MPR over 1 year for comorbid medications, and 2) 80% MPR over 180 days without a 15-day gap in therapy prior to 90 days for antidepressant therapy. Multivariate logistic regression and log-linear models were used for inferential analyses, controlling for age, gender, plan type, utilization of psychiatric specialty care services, antidepressant therapy change rates, antidepressant titration rates, obtaining a 90-day supply of comorbid medication, and pre-period comorbid medication adherence rates.

RESULTS: There were 8,040 patients included in this study. Patients who were adherent to antidepressant medication in the 180-day period were significantly more likely to be adherent to comorbid therapy over a 1-year timeframe when compared to patients who were not adherent to their antidepressant therapy. This effect was consistent across comorbid cohorts, CAD odds ratio (OR)=2.13 ($P<0.0001$); Diabetes OR=1.818 ($P<0.0001$); CAD/Diabetes OR=1.453 ($P=0.0002$). When analyzing disease-specific charges, CAD patients adherent to antidepressant therapy incurred 17% lower charges than did non-adherent patients ($P=0.017$); in the diabetes and CAD/diabetes cohorts, antidepressant-adherent patients incurred 8% and 14% lower charges, respectively, than did non-adherent patients, but differences were not statistically significant. When evaluating total medical charges, which included both disease- and non-disease specific charges, but did not include pharmacy costs, CAD and diabetes patients who were adherent to antidepressant therapy incurred lower medical charges (6.4% lower for CAD cohort [$P=0.048$]; 11.8% lower for diabetes cohort [$P=0.040$]) than did patients who were non-adherent. Antidepressant-adherent CAD/diabetes patients incurred almost 20% lower medical charges ($P=0.029$). **CONCLUSIONS:** Increased antidepressant adherence was associated with increased adherence with comorbid disease medication and a reduction in total medical costs. Future studies should investigate if increased adherence and reduced costs exist in disease states other than CAD, diabetes and CAD/diabetes, while expanding the timeframe of analysis beyond 1 year.

IMPLEMENTATION AND EVALUATION OF A LOW-LITERACY DIABETES EDUCATION COMPUTER MULTIMEDIA APPLICATION. B.S. Gerber¹; I.G. Brodsky²; K.A. Lawless¹; L.I. Smolin¹; A.M. Arozullah¹; E.V. Smith¹; M.L. Berbaum¹; P.S. Heckerling¹; A.R. Eiser³. ¹University of Illinois at Chicago, Chicago, IL; ²Maine Medical Center, Portland, ME; ³Drexel University, Philadelphia, PA. (Tracking ID #134308)

BACKGROUND: Low health literacy poses a major barrier to education and self-management. Despite increasing concern about the impact of low health literacy on diabetes care, there are few interventions available that address low literacy. This study evaluated a clinic-based multimedia intervention for diabetes education targeting individuals with low health literacy levels in a diverse population.

METHODS: Five public clinics in Chicago participated in the study with computer kiosks installed in waiting room areas. Two hundred and forty-four English and/or Spanish-speaking individuals with diabetes enrolled. Health literacy was assessed using the short form of the Test of Functional Health Literacy in Adults (S-TOFHLA). Participants were randomized to receive either supplemental computer multimedia use (intervention) or standard of care only (control). The intervention included bilingual audio/video sequences to communicate information, provide psychological support, and promote diabetes self-management skills without extensive text or complex navigation. The program was available on touch screen computers in clinical waiting areas for patients to utilize before appointments. Hemoglobin A1c (HbA1c), body-mass index (BMI), blood pressure, diabetes knowledge, self-efficacy, self-reported medical care, and perceived susceptibility of complications were evaluated at baseline and one year. Computer usage patterns and implementation barriers were also examined.

RESULTS: Complete one-year data were available for 183 subjects (75%). Fifty-five percent of the participants had low health literacy (S-TOFHLA score between 0 and 22). After one year, there were no significant differences in change in HbA1c, BMI, blood pressure, knowledge, self-efficacy, or self-reported medical care between intervention and control groups. Analysis of low literate subjects with poor glycemic control (baseline HbA1c greater or equal to 9.0% in 26 subjects), revealed a greater decrease in HbA1c in the intervention group than control group (-2.1% vs. -0.3%, $p=.036$). For higher literate subjects with poor glycemic control, there was no significant difference (-0.9% vs. -1.3%, $p=.548$). In addition, there was an increase in perceived susceptibility to diabetes complications in the intervention group. This effect was greatest among subjects with lower health literacy. There was also a trend towards greater self-efficacy among these participants. However, overall time spent with the intervention was greater for those with higher health literacy (Spearman's $\rho=0.284$, $p=.005$).

CONCLUSIONS: Access to multimedia lessons resulted in an increase in perceived susceptibility to diabetic complications, particularly in subjects with lower health literacy. Despite measures to improve informational access for individuals with lower health literacy, there was relatively less use of the computer among these participants. However, the intervention did result in significant improvement in HbA1c among low-literate subjects with poor glycemic control. Additional personal and organizational barriers must be addressed in conjunction with provider involvement to improve usability.

IMPLICATIONS OF CHANGING NATIONAL CHOLESTEROL EDUCATION PROGRAM GUIDELINES FOR THE TREATMENT AND CONTROL OF HYPERCHOLESTEROLEMIA IN THE UNITED STATES. S.D. Persell¹; D. Lloyd-Jones¹; D. Baker¹. ¹Northwestern University, Chicago, IL. (Tracking ID #134191)

BACKGROUND: Modifications to the National Cholesterol Education Program (NCEP) guidelines have lowered goal low-density lipoprotein cholesterol (LDL-C) levels and the thresholds for medical treatment. The nationwide implications of these changes in recommendations have not been fully evaluated. We sought to estimate the prevalence of LDL-C levels exceeding various NCEP targets among the US adult population in 1999 to 2000 and examine the implications of recent optional NCEP goals for guideline concordance and anticipated need for medical treatment.

METHODS: This retrospective study used questionnaire, examination, and laboratory data collected in 1999 and 2000 from 1,498 fasting adults age 20 to 79 years participating in the U.S. National Health and Nutrition Examination Survey. We assigned LDL-C goals according to the methods suggested by the NCEP using self-reported diagnostic information and measured cardiovascular risk factors. Outcomes of interest were the frequency of fasting LDL-C levels exceeding NCEP goals based on recommended targets from guidelines published in 1993, 2001, and 2004. We also examined the proportion of the adult population not currently using cholesterol-lowering therapy who would require greater than a 40% LDL-C reduction to reach NCEP goals (a reduction greater than those observed in long-term clinical trials of standard-dose statins). We used the survey sampling weights to estimate the size of the U.S. adult population meeting these criteria.

RESULTS: LDL-C values exceeded 1993 NCEP goals for 26.7% of adults; 30.5% had levels above the 2001 goals and 35.5% had levels above the optional goals described in 2004. An estimated 13,300,000 individuals (8.9%) were above the 1993 thresholds for drug therapy, 21,200,000 (14.3%) were above the 2001 thresholds for drug therapy, 39,400,000 (26.5%) were above the optional 2001 thresholds for drug therapy and 46,900,000 (31.5%) would have been above the optional 2004 thresholds for drug therapy. Of the 7.2% (95% CI, 5.4–9.0%) of adults who used cholesterol-lowering medication, 45% were above their 1993 goal and 66% were above the optional 2004 goal. Adoption of lower optional targets (<70 mg/dL for highest risk group and <100 mg/dL for individuals with a 10-year predicted risk of 10–20%) would mean nearly all persons in these two highest-risk groups would have LDL-C levels exceeding their goals, despite the fact that 23% of this population already used cholesterol-lowering medication.

Of adults not currently using cholesterol lowering medication, few required >40% LDL-C reduction to achieve 1993 or 2001 goals but 7.6% of adults required >40% LDL-C reduction to reach the optional 2004 goals.

CONCLUSIONS: LDL-C levels above the 1993 NCEP goals were common, especially for intermediate and high-risk individuals, and cholesterol-lowering medications were underused. Optional goals promulgated by the NCEP in 2001 and 2004 moderately increase the number of adults with LDL-C above their goal, and greatly increase the number of low, intermediate, and high-risk adults who exceed LDL-C thresholds for cholesterol-lowering medication. Our data suggest that many patients would be unable to reach the newest LDL-C targets using one of the therapeutic regimens tested in the long-term clinical trials of statins.

IMPORTANCE OF PATIENT-PROVIDER RELATIONSHIP TO RECEIPT OF AND ADHERENCE TO ANTIRETROVIRAL THERAPY. M.C. Beach¹; J. Keruly¹; R.D. Moore¹. ¹Johns Hopkins University, Baltimore, MD. (Tracking ID #134360)

BACKGROUND: Many aspects of patient-provider relationships are associated with positive patient outcomes; in particular, patient perception that their provider knows them 'as a person' is significantly related to adherence in primary care. The purpose of this study is to examine the association between patient perception of being 'known as a person' and receipt of and adherence to HAART among patients with HIV.

METHODS: Patients were interviewed while awaiting appointment with their primary care provider at an urban clinic specializing in HIV care. We used a cross-sectional analysis of 4694 patient interviews to evaluate the association between patient reports that their HIV provider 'knows me as a person' and three outcomes: receipt of HAART, adherence to HAART, and absence of detectable serum HIV RNA.

RESULTS: Patients who reported that their provider knows them 'as a person' were more likely to receive HAART (60% vs. 47%, $p<0.001$), be adherent to HAART (76% vs. 67%, $p<0.001$), and have undetectable serum HIV RNA (49% vs. 39%, $p<0.001$). Patients who reported their provider knows them 'as a person' were also older (mean 38.0 vs. 36.6 years, $p<0.001$), reported higher quality of life (mean LASA score 71.1 vs. 64.8, $p<0.001$), had been followed in clinic longer (mean 1933 vs. 1852 days, $p=0.002$), missed fewer appointments (mean proportion missed appointments 0.124 vs. 0.144, $p<0.001$), reported more positive beliefs about HAART therapy (39% vs. 28% strongly believed HIV medications could help them live longer, $p<0.001$), reported less social stress (50% vs. 62% did not eat regular meals, $p<0.001$) and were less likely to use illicit drugs or alcohol (36% vs. 50% used drugs, $p<0.001$; 42% vs. 53% used alcohol, $p<0.001$). Controlling for patient age, sex, race/ethnicity, quality of life, length of time in clinic, missed appointments, health beliefs, social stress, and illicit drug and alcohol use, patients who reported their provider knows them 'as a person' had higher odds of receiving HAART (OR 1.50, 95% CI 1.22–1.85), adhering to HAART (OR 1.48, 95% CI 1.07–2.05) and having undetectable serum HIV RNA (1.29, 95% CI 1.04–1.61).

CONCLUSIONS: Patients' perception of being 'known as a person' is significantly and independently associated with receiving HAART, adhering to HAART, and having undetectable serum HIV RNA. Efforts to improve health care quality, adherence and health of persons with HIV should focus on improving the patient-provider relationship.

NATIONAL CHOLESTEROL EDUCATION PROGRAM RISK ASSESSMENT METHOD AND THE MISCLASSIFICATION OF SOME ADULTS AS LOW-RISK. S.D. Persell¹; D. Lloyd-Jones¹; D. Baker¹. ¹Northwestern University, Chicago, IL. (Tracking ID #135641)

BACKGROUND: The National Cholesterol Education Program (NCEP) Adult Treatment Panel report from 2001 (ATP III) recommends risk assessment using risk factor counting for adults without established cardiovascular disease (CVD) or diabetes mellitus to help clinicians decide on low-density lipoprotein cholesterol (LDL-C) treatment goals. Clinicians are advised to estimate 10-year coronary risk using tables derived from the Framingham Heart Study only for adults with 2 or more risk factors; adults with only one risk factor or with 2 risk factors and high HDL cholesterol are classified as low risk by this method. This approach may falsely classify some individuals as low risk.

METHODS: We conducted a cross-sectional study using questionnaire, examination, and laboratory data collected in 1999 and 2000 from 1244 fasting adults age 20 to 79 years without diagnosed CVD or diabetes who participated in the U.S. National Health and Nutrition Examination Survey. We assigned LDL-C goals according to the methods suggested by the ATP III using risk factor counting followed by risk calculation for adults with 2 or more risk factors. We determined the proportion of the population classified as low risk using the ATP III method who had an estimated 10-year risk of death or myocardial infarction of 10% using only the Framingham method ("direct" method). We used the survey sampling weights to estimate the size of the U.S. adult population meeting these criteria.

RESULTS: Of U.S. adults aged 20 to 79 years without CVD or diabetes, 99,100,000 (75.9%) were assigned the lowest risk group when the risk factor counting method was used but only 92,900,000 (71.1%) were classified as lowest risk when the direct Framingham method was used. Applying the direct Framingham method to all adults resulted in 2,300,000 (1.7%) more adults being above their 2001 NCEP goal LDL-C and 3,300,000 (2.5%) being above the 2001 NCEP threshold for drug therapy. When judged according to the more aggressive "optional" LDL-C goals described by the NCEP in 2004 (<70 mg/dL for highest-risk and <100 mg/dL for moderately-high risk patients), applying the direct Framingham method to all adults resulted in 4,200,000 (3.3%) more adults being above both their LDL-C goal and the threshold for drug therapy.

Compared to low-risk persons whose classification did not change, those who were misclassified as low risk using the risk factor counting method were much more likely to be men (86% vs. 40%, $P < .001$) and over age 50 years (77% vs. 18%, $P < .001$).

CONCLUSIONS: Relying on the ATP III risk factor counting method rather than directly determining risk using the Framingham method misclassified 4.8% of the total population who were at moderately-high risk of CVD as low risk. Middle-aged/older adults and men were especially likely to be misclassified as low risk using the risk factor counting method. The NCEP ATP should advise physicians to use widely available computerized risk assessment tools or multivariable tables to estimate CVD risk and base their treatment decisions on these results in order to prevent the misclassification and under-treatment of some patients.

OBESITY MANAGEMENT: RESIDENT PRACTICE PATTERNS AND PATIENT PREFERENCE. N.J. Davis¹; A. Emerenini²; J. Arnsten¹; J. Wylie-Rosett¹. Montefiore Medical Center, Bronx, NY; ²Yeshiva University, Bronx, NY. (Tracking ID #134575)

BACKGROUND: 64 % of Americans are obese or overweight, and general internists play a pivotal role in identifying and treating obesity. The United States Preventive Services Task Force (USPTF) recommends screening all adult patients by calculating BMI, and treating all patients with BMI ≥ 25 . How well resident physicians comply with these recommendations, and whether patients prefer physician intervention in weight management, are not currently known. **METHODS:** We conducted a cross-sectional survey of consecutive patients attending an outpatient resident continuity clinic in the Bronx, NY. Prior to their physician visit, participants completed a 30-item questionnaire, including a 17-item Quick Weight, Activity and Excess Screener (WAVE) for adults, and 13 questions addressing patient preferences for physician intervention. Compliance to USPTF recommendations were determined by chart review. All internal medicine visits within 1 year prior to survey completion were included in the chart review. Patient BMI was calculated using the most recent height and weight measurements documented in the chart, and overweight and obesity were defined as BMI ≥ 25 and ≥ 30 respectively. Associations between categories of BMI and documented weight loss recommendations were tested using chi square analysis.

RESULTS: Among 84 survey respondents, mean age was 46.6 (SD = 14.9). 81% were female, over 50% were Black or Hispanic, and 79% were obese or overweight. Most respondents, including 86% of obese participants and 76% of overweight participants, wanted to lose weight. In addition, 48 % of obese participants and 64% of overweight participants desired a nutritionist referral. 67% of respondents believed their doctor could help them with weight loss, and most (70%) wanted their doctor's encouragement. Only 18% reported not wanting to discuss weight issues with their physician. Although 50% of participants were obese and 29% were overweight, BMI had been calculated in only 7% of charts (all of which recorded a BMI ≥ 30). Only 44% of patients with BMI ≥ 25 had weight loss recommendations documented in their charts, and only 21% had been referred to a nutritionist. Obese patients were more likely to receive weight loss recommendations than overweight patients (OR = 6.38, 95% CI 1.3, 30.9, $p = .012$), but were not significantly more likely to receive referral to nutritional services (OR = 4.8, 95% CI .55, 41.6, $p = .24$).

CONCLUSIONS: Resident physicians underutilize an important and simple screening tool in the diagnosis of obesity. Weight loss recommendation and referrals for treatment are offered to only a minority of obese or overweight patients, with obese patients being more likely to receive recommendations. Patients want physician intervention in the management of their weight issues and believe that their doctors can help. Resident physicians may need further education regarding the importance of calculating BMI and appropriate management strategies for patients with BMI ≥ 25 .

PATIENT SELF-EFFICACY, PROVIDER SELF-MANAGEMENT COUNSELING AND HEALTH STATUS AMONG PATIENTS WITH CHRONIC HEART FAILURE. U. Subramanian¹; J. Sutherland²; F. Hopp³; J. Lowery²; B.N. Doebbeling¹. ¹Indiana University Purdue University Indianapolis, Indianapolis, IN; ²Indiana University, Indianapolis, IN; ³VA Ann Arbor Healthcare System, Ann Arbor, MI. (Tracking ID #134939)

BACKGROUND: The Institute of Medicine lists patient Self-Management [SM] as a priority area for improving chronic illness care. The USPSTF guidelines for chronic heart failure [CHF] recommend that all patients should be provided with SM education and assistance. However, patients report receiving little or no assistance with SM from their providers. Objective: The objectives are two fold: first, describe the relationship between patients' self efficacy and health status of patients. Secondly, to relate the occurrence of SM education among veterans with CHF and their health status.

METHODS: Cross-sectional survey of veterans with CHF at 2 tertiary Veterans Affairs hospitals. Independent variables were patient's self efficacy and providers' provision of SM education. Patient self efficacy was assessed with 6 questions in domains ranging from managing CHF, overcoming barriers and coping with CHF. Self-efficacy items were scored on a 10 point Likert-type scale, with 1 being "not at all confident" and 10 being "totally confident." SM education from providers was assessed with 4 questions regarding diet, exercise, symptom, and medication management. Answers were coded as binary variables depending on whether patients reported counseling from providers. The dependent variable was perceived health status assessed by a previously validated single question. Multivariate regression assessed the effects of the independent variables on health status.

RESULTS: 133 patients completed SM questionnaire. There was wide variability in reported rates of SM education; ranging from 23% for exercise counseling to 85% for salt restriction and medication management. Approximately 15% of patients reported not receiving any SM education from their providers. Patients counseled about exercise was positively related to health status [$P < 0.01$]. Patients' confidence in managing their CHF and running their life was related to improved health status [$p < 0.01$; $p < 0.01$ respectively].

CONCLUSIONS: SM education from providers is associated with improved health status among CHF patients. However, counseling occurs less than half the time for most aspects of SM. This finding suggests further work to explore the relationship between provider SM counseling and patients' self-efficacy. Impact: SM education is a critical aspect of evidence-based practice for CHF patients. These findings call for interventions to improve SM counseling in chronic care.

PATIENT-PROVIDER COMMUNICATION BEYOND THE FACE-TO-FACE VISIT AMONG PATIENTS WITH DIABETES. W.M. Gregg¹; R.S. Dittus¹; T.A. Elasy¹. Vanderbilt University, Nashville, TN. (Tracking ID #136202)

BACKGROUND: Improving the quality of care for patients with chronic diseases often requires fundamental change in the ways care is delivered. Several studies have described the typical face-to-face clinical encounter, but little is known about the communications that take place outside of the traditional visit. Patient calls, emails and similar communications are often unplanned, sporadic and undocumented. Rigorous scientific analysis of their volume, content or frequency is almost nonexistent. Vanderbilt University Medical Center (VUMC) has a unique messaging system, used primarily within our primary care clinics that is used to document encounters that are not face-to-face (e.g. patient calls/requests, results communication and other contacts). This system, used in VUMC's paperless primary care clinics, allows characterization of such encounters.

METHODS: As part of an existing quality improvement initiative, we identified 7,880 patients with diabetes mellitus having two or more billable encounters at VUMC (inpatient or outpatient) during the two year period, 9/1/2002-8/31/2004, containing the ICD-9 codes 250.2-250.9, 357.2, 362.0, or 366.41. The study sample was further limited to the 2,244 patients seen at anytime in the VUMC primary care clinics since 1999 (primary patient lists) and who have had any communications saved to their chart during the two year period. Clinical data, including insulin use, frequency of communications and lab testing was extracted from the electronic chart as was total cost, number of encounters, total admissions. We randomly sampled 100 patients from this population and then randomly selected one communication from each patient for content analysis. Messages were classified using four content areas: contact type (phone, email, etc.), contact initiator (patient, provider, etc.), primary message subject (diabetes, other medical problems, etc.) and action taken (order given, results given, appointment scheduled, etc.).

RESULTS: The study sample had a mean age of 60.1 years, was 67% white, 27% African American and 55% female. During this two year period, the 2,244 patients generated 52,333 discrete communications compared to 48,069 face-to-face outpatient encounters. The median number of messages and face-to-face encounters for these patients over the two years, was 19 and 17 respectively. In univariate analysis, patients using insulin had an average of 24.6 communications in the two years, versus 22.7 in non-insulin users ($p < 0.02$). Female patients also had more communications than male patients, 25.5 vs. 20.7 ($p < 0.001$). The communication content analysis revealed several trends: contact type (phone - 49%, letter - 25%), contact initiator (provider - 57%, patient - 35%), primary message subject (result communication - 29%, general medical problems - 28%, diabetes - 13%) and action taken (order given - 34%, results given - 24%).

CONCLUSIONS: Patient-provider communications outside of the face-to-face visit represent a substantial portion of the total contacts between patients and their health care providers. Specifically, women and patients using insulin were more likely to be the subject of these recorded communications. The classification system developed in this study can describe and capture the variation in non-face-to-face encounters.

PHYSICIAN PROCESS MEASURES FOR CHRONIC HEART FAILURE CARE: RELATIONSHIPS TO ORGANIZATIONAL CHARACTERISTICS. U. Subramanian¹; K.F. Welke²; K. McCoy¹; T.E. Vaughn³; M.M. Ward⁴; B.N. Doebbeling⁵. ¹HSR&D Center of Excellence on Implementing Evidence-based Practice, Richard L. Roudebush VA Medical Center, Indianapolis, IN, Indianapolis, IN; ²Oregon Health & Science University, Portland, OR; ³Iowa City VA Medical Center, Iowa City, IA; ⁴University of Iowa, Iowa City, IA; ⁵Indiana University Purdue University Indianapolis, Indianapolis, IN. (Tracking ID #134377)

BACKGROUND: Variation in practice data suggests that there may be opportunities to improve the quality of care for people with chronic heart failure (CHF) in the U.S. Purpose: To explore the degree to which providers in the Veterans Health Administration (VHA) national health care system adhere to CHF guidelines, and to identify organizational predictors of adherence.

METHODS: Design: Secondary data analysis of provider adherence measures from chart reviews, along with surveys of facility quality managers. Data Sources: Multiple datasets were linked, including our VHA Quality Manager Survey (QMS), the 2001 External Peer Review Program Dataset, 2000 American Hospital Association Annual Survey, the 1999 Veterans Health Survey, and the 2001 Veterans Satisfaction Survey. Sample: 416 key informants at 143 VHA hospitals with ambulatory care clinics. Dependent Measure: A summary measure of 6 binary inpatient provider process adherence measures, including: 1)

Left ventricular ejection fraction (LVEF) documented, 2) Angiotensin converting enzyme inhibitor prescribed if LVEF <40%, 3) Beta blocker prescribed if LVEF <40%, 4) Anti-coagulant prescribed if diagnosed with atrial fibrillation, 5) Diet, weight, and medication instructions given at hospital discharge, and 6) Tobacco cessation counseling given at least once while an inpatient. These process measures were coded as a 1 if the intervention was provided and 0 if not. Independent Variables: Predictors were clustered into seven domains of conceptually-related survey items. These categories included: CHF guidelines implementation approaches, level of support for the guideline efforts, regional network leadership for guidelines, hospital use of guideline performance data, hospital culture, hospital structure, and patient population characteristics. Statistical Analysis: We constructed hierarchical logistic regression models using generalized estimating equation (GEE) methods, examining six provider process measures for each VA Medical Center (VAMC).

RESULTS: Limiting facilities to those with adequate complete data, 100 facilities were included in the analysis. Chart audit estimates of provider adherence measures were higher than expected (85% or more on average) for most measures, ranging from 14% for diet, weight and medication instructions at discharge to 91% for ejection fraction documentation and beta-blocker prescription. In GEE models, VAMCs with higher levels of adherence ($P < 0.05$) had: 1) changed the responsibilities of non-physicians to support adherence, 2) physicians that believed the guidelines were applicable to their practice, and 3) a lower proportion of their patient population that was Hispanic.

CONCLUSIONS: Adherence to most inpatient CHF quality of care measures overall was above that reported in other healthcare systems. Modifiable factors that healthcare organizations can adopt to enhance the delivery of chronic disease care in their settings include building provider support for the clinical practice guidelines and changing the responsibilities of non-physicians to support adherence.

PREDICTORS OF SCREENING FOR HEPATOCELLULAR CARCINOMA AMONG PATIENTS WITH HEPATITIS C IN PRIMARY CARE. K.J. Kim¹; M.J. Pletcher²; R.K. Fox¹. ¹UCSF Division of General Internal Medicine, San Francisco, CA; ²UCSF Department of Epidemiology and Biostatistics, San Francisco, CA. (Tracking ID #131982)

BACKGROUND: Hepatitis C virus (HCV) infection affects 1.8% of the U.S. population and is the leading cause of the rising incidence of hepatocellular carcinoma (HCC). Practice guidelines by the NIH and CDC recommend bi-annual screening for HCC in patients with cirrhosis due to HCV using serum alpha-fetoprotein (AFP) and abdominal imaging, although there are few data to support these recommendations. Primary care physicians perform most cancer screening, yet it is unknown whether or how appropriately they implement HCC screening recommendations in their patients with HCV. The objective of this study was to determine HCC screening patterns for patients with HCV in an academic primary care practice, and to identify significant predictors of screening.

METHODS: Using a computerized clinical database, we identified all patients with HCV (via serology or ICD-9 code) receiving care at one of three UCSF general medicine (GM) practices during a three-year period (1/1/01-12/31/03). Patients were excluded if they had a diagnosis of HCC or liver transplant, or had fewer than two GM clinic visits during the study period. Demographics, GM clinic visit frequency, gastroenterology (GI) specialty clinic visits, serologic studies, liver biopsies and radiographic studies were abstracted from the clinical database. The presence of cirrhosis was defined by liver nodularity reported on abdominal imaging (ultrasound, CT, or MRI), or stage 4 fibrosis (Metavir histological scoring) on liver biopsy if such studies were performed. HCC screening was defined by having at least one AFP as well as one form of abdominal imaging performed during the study period. We used logistic regression to identify predictors of HCC screening separately in subjects with cirrhosis (appropriate screening) and without cirrhosis (not appropriate screening).

RESULTS: Among 458 patients with HCV (mean age 54.8 years $\pm$ 11.8, 52% male, 46% non-White, 13% non-English speaking, 25% married, 71% with public insurance), the overall rate of HCC screening was 41% (95%CI: 37-46%). For patients with cirrhosis, 73% (37/51) were screened compared to 37% of non-cirrhotics (152/407, $p < .001$). Among cirrhotics, having public insurance was the only significant predictor of HCC screening (adjusted OR 11, 95% CI: 1.4-95). Screening rates were not much higher in the subset of cirrhotic patients seen by the GI clinic (76%). Non-cirrhotics, on the other hand, were more likely to be unnecessarily screened if they had been to GI clinic (47%; adjusted OR 4.77, 95% CI: 2.75-8.28), or had more frequent GM visits (adjusted OR 1.10 per additional visit, 95% CI: 1.05-1.16). Race/ethnicity, language, gender, marital status, and age did not significantly influence screening rates in either subset of patients.

CONCLUSIONS: Our findings show some cirrhotics are not getting screened for HCC, while many patients without cirrhosis are getting unnecessary screening. Educating providers about targeting only cirrhotic patients for screening should improve the consistency of care for patients with HCV and make more appropriate use of critical resources.

QUALITY IMPLICATIONS FOR DIABETES CARE OF A CHANGE FROM MULTISPECIALTY GROUP TO INDIVIDUAL PROVIDER DIRECT CONTRACTING: RESULTS FROM THE TRIAD STUDY. O. Duru¹; J. Fish¹; A. Brown¹; R. Brusuelas-James¹; W. Steers¹; C.M. Mangione¹. ¹University of California, Los Angeles, Los Angeles, CA. (Tracking ID #136200)

BACKGROUND: Managed care contracting arrangements between organized medical groups and health plans have several potential advantages to patients

over direct contracts between plans and individual providers. Ongoing quality improvement efforts within the group structure and streamlined coordination of care of a group may lead to improved processes and outcomes for patients with diabetes. Few studies have examined whether the quality of diabetes care within a single population might be affected from a change in contracting arrangements.

METHODS: Data are from the Translating Research into Action for Diabetes (TRIAD) study, a multicenter longitudinal cohort study of diabetes care in managed care. For this analysis, we examined a single health plan, in which 1,138 plan members with diabetes from 24 multispecialty groups were enrolled during 2000-01. Between baseline data collection and follow-up (2002-03), some group contracts were terminated in response to local market forces. Our primary predictor was whether the participant was cared for in a multispecialty group ($n=785$) or by an individual provider ($n=353$) at follow-up. We examined several dependent variables: 1) processes of care, 2) intermediate outcomes, including hemoglobin A1c, blood pressure, and low density-lipoprotein cholesterol (LDL), and 3) whether those patients with poorly controlled intermediate outcomes were managed with aggressive medication regimens. We conducted these tests with multivariate models that controlled for patient demographics and clinical status. Using these models, we obtained predicted probabilities and tested for differences between the two settings.

RESULTS: No differences were found on any of the measured processes of care between patients who remained in multispecialty groups and those whose providers switched to direct contracting arrangements. Predicted rates of process delivery generally exceeded 70% in both settings. 47% of patients in multispecialty groups at follow-up were predicted to have poorly controlled blood pressure as compared to 33% of those in the setting of direct contracting ($p=.04$), but no differences were seen in the predicted rates of aggressive blood pressure management. A1c and LDL levels were equally likely to be controlled within the two settings, but enrollment in a group was associated with a higher predicted likelihood of aggressive management of elevated LDL cholesterol (64%) than receiving care from a direct-contract provider (48%, $p=.003$).

CONCLUSIONS: This analysis found that patients with diabetes whose providers leave a group still receive similar processes of care. While patients in multispecialty groups have higher blood pressures, this is likely due to patient-related factors as elevated blood pressure is managed equivalently within both multispecialty groups and direct contracting arrangements. Conversely, patients in multispecialty groups receive more aggressive therapy for poorly controlled lipids, which could reflect increased attention to the macrovascular complications of diabetes within medical groups and health plans. Future in-depth studies examining shifts in financing arrangements may provide causal explanations for these findings.

REDUCING CARDIOVASCULAR RISK IN DIABETES: ROOM FOR IMPROVED INTENSITY OF HYPERTENSION MEDICATION MANAGEMENT. M.M. Safford¹; K. Kirk¹; C. Kiefe¹. ¹Birmingham VA Medical Center, Birmingham, AL. (Tracking ID #136118)

BACKGROUND: Hypertension (HTN) is a dominant risk factor for cardiovascular disease, especially among people with diabetes (DM). Yet, large numbers of individuals with DM have uncontrolled blood pressure (BP). Past reports suggest considerable room for improvement in intensifying BP medication regimens as a strategy to improve BP control. We studied recent trends in BP medication management in a Veterans Administration (VA) medical center.

METHODS: We used the Veterans Health Information Systems and Technology Architecture (VISTA) to identify patients in 2001-2, who were considered to have DM if they had >1 DM diagnostic code over the 2 years, or any antidiabetic medication. This cohort was then evaluated twice for HTN management, in 2001-2, and 2003-4. At each evaluation, HTN was defined to be present if they had >1 diagnostic code for HTN, or >2 systolic BP (SBP) ≥ 140 mmHg at least 14 days apart, or any SBP > 180 mmHg over the 2 year observation period. BP medication was "appropriately intensified" if dose was increased or a new BP medication class added after the last eligible primary care visit with SBP ≥ 140 mmHg (each subject assessed at most once). Separate logistic regressions for each observation period included age, sex, race/ethnicity, and SBP level.

RESULTS: The 6810 DM patients had mean (standard deviation) age 64.7 (10.8) years, 2.2% were women and 19.4% were African American. In 2001-2, 5190 patients had DM and HTN; in 2003-4, 3711 patients had DM and HTN, representing those who remained in care. The mean number of primary care visits was 6.1 (3.7) in 2001-2 and 6.3 (3.6) in 2003-4, and the mean number of BP medications at last measure was 2.2 (1.4) in 2001-2 and 2.7 (1.9) in 2003-4. At last measure, 59.0% had SBP ≥ 140 mmHg in 2001-2, and 50.6% in 2003-4. Of the 2074 who were uncontrolled in 2001-2 and who remained in care in 2003-4, 57.3% remained uncontrolled. HTN management in those with SBP ≥ 140 mmHg in the 2 assessment periods is presented in the Table. Intensity of BP medication management was considerably greater in 2003-4, although intensification occurred in fewer than 50% of visits with SBP ≥ 140 mmHg. In multivariable analysis, people of different ages, race/ethnicity or sex had similar patterns of medication intensification at both observation periods.

CONCLUSIONS: Two thirds of intensification opportunities were missed in 2001-2. While those who remained in care experienced improvements in intensity of HTN management in 2003-4, that care still fell considerably short, regardless of age, sex or race/ethnicity. Although we report some improvement compared to published reports, efforts to maximize medication intensification in hypertensive patients with DM are warranted, and enrolling individuals in ongoing care may be an important step toward that goal.

Table. VA users with DM in 2001-2: Intensity of BP medication management among those with HTN and SBP ≥ 140 in 2001-2 and 2003-4.

	2001-2 (N=3064)	2003-4 (N=1880)
On no BP meds	39.1%	18.9%
On 1 BP med	22.7%	20.1%
On 2 BP meds	17.2%	20.4%
On >2 BP meds	20.8%	40.3%
% of visits with medications intensified	33.7%	49.3%

RELATIVE CONTRIBUTION OF PARTICIPATORY DECISION-MAKING ON DIFFERENT ASPECTS OF HYPERTENSION CARE. A. Cho¹; C. Voils¹; Y.S. William¹; O.Z. Eugene¹; B. Hayden¹. ¹Center for Health Services Research in Primary Care, Durham, NC. (Tracking ID #135581)

BACKGROUND: Participatory decision-making (PDM) is a style of care in which providers share control of treatment decisions with their patients. PDM style has been shown to influence healthcare outcomes in various chronic diseases. However, the mechanisms by which PDM style affects health outcomes are not well understood. Our objective was to explore the impact of PDM style on distinct aspects of hypertension care, in order to examine how PDM style could influence hypertension outcomes.

METHODS: This cross-sectional study used baseline data from the Veterans' Study to Improve the Control of Hypertension (V-STITCH), which enrolled 588 randomly selected male veterans with hypertension from the Durham VAMC Ambulatory Care clinics. All participants were asked to complete in-person interviews at study entry. Their ratings of their providers' PDM style were measured using three validated questions, transformed to a 100-point scale. Outcomes were blood pressure control in accordance with JNC-7 guidelines, self-reported adherence to antihypertensive medications, presence of a home blood pressure monitor, frequency of home monitor use, and hypertension knowledge. Bivariate relationships were examined with Chi-square tests. Logistic and linear regression models included PDM and covariates age, race, marital status, comorbid diabetes, literacy, and education.

RESULTS: Average PDM score was 85.4 ± 20.7 SD; median PDM score was 96.3 (interquartile range 22.2). Higher PDM style was associated with a lower likelihood of baseline blood pressure control. For each 10-point increase in PDM, there was a nine percent decrease in the odds of having blood pressure under control (OR=0.91; 0.83 to 0.99, 95% CI). Higher PDM style was also associated with increased odds of having a home monitor. For each 10-point increase in PDM, there was an 11 percent increase in the odds of having a monitor (OR=1.11; 1.02 to 1.21, 95% CI). Higher PDM score predicted greater hypertension knowledge in African American patients ($b=0.13$, $P=0.047$ for the interaction), but not in white patients ($P=0.38$), illustrated in the Figure. PDM style was not significantly associated with either self-reported adherence to antihypertensive medications or frequency of home monitor use.

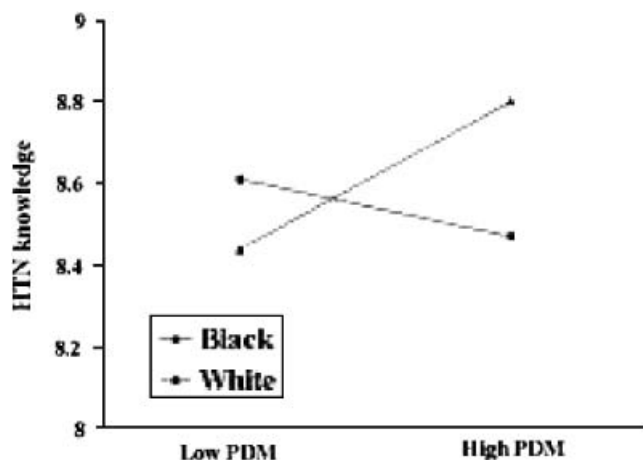


Figure. Interaction between PDM and race in their effect on hypertension knowledge.

CONCLUSIONS: This sample of male veterans with hypertension rated their providers as highly participatory. A discernible pattern was observed in the influence of provider participatory decision-making style on hypertension care. PDM style was related to certain aspects of care (presence of a home blood pressure monitor, hypertension knowledge) that could be considered 'provider-delivered,' but less related to 'patient-performed' activities (frequency of monitor use, adherence to antihypertensive medications). This distinction could be interpreted as reflecting the limits of providers' reach, and the need for supporting

interventions. Surprisingly, higher PDM style was associated with worse baseline blood pressure control. This finding may be the result of more participatory providers giving patients the latitude to reject changes in management, even ones that could improve control. Alternatively, patients with more severe hypertension may see their providers more frequently, and develop closer and more participatory relationships with them.

SELF-REPORT COMORBIDITY DATA AND FUNCTIONAL OUTCOMES IN ACUTE CORONARY SYNDROME PATIENTS. A.B. Olomu¹; W.D. Corser¹; M.M. Holmes-Rovner¹. ¹Michigan State University, East Lansing, MI. (Tracking ID #134952)

BACKGROUND: The influence of self-rated patient comorbidity levels on later functional outcomes has not been widely tested. This project investigated the influence of self-rated comorbidity levels and demographic characteristics on post-discharge: a) physical activity, b) quality of life, and c) depression levels, for recently hospitalized Acute Coronary Syndrome (ACS) patients at baseline (7-10 days post-discharge), and at 3 months.

METHODS: Data concerning self-rated comorbidity (generated by a previously developed scale: Katz, et. al. 1996), and post-discharge activity status (Duke ASI), depression (CES-D), quality of life (QLMI), were collected from 527 ACS patients by trained interviewers during structured phone interviews. A total of 492 patients provided complete comorbidity data at baseline. The effect of composite comorbidity levels (no. of comorbidities = 1-2; 3-4; 5+) on post-discharge outcomes was evaluated through one-way univariate ANOVAs procedures at the two post-discharge timepoints using SPSS 11.5 software.

RESULTS: Composite comorbidity scores were significantly associated with outcome scores in both baseline and 3 months ASI levels ($p < .001$ at both time-points), and QLMI scores (Baseline $p < .001$, Three-Month $p = .046$). Post-discharge depression scores were significantly associated with comorbidity at baseline ($p < .001$), but not at three-months ($p = .241$). Study outcome levels were also significantly influenced by: a) age, b) education, and c) primary insurance.

CONCLUSIONS: Along with the influence of major demographic characteristics on patients' later functional outcomes, the additional use of self-report comorbidity survey data to more fully investigate the occurrence of subsequent post-discharge health outcomes for ACS patients is promising. Further refinements of survey instruments to measure self-report comorbidity may be useful as both a health status assessment and clinical outcome tool.

SYSTEMATIC REVIEW OF INSTRUMENTS USED TO MEASURE THE IMPACT OF INTERVENTIONS IN END-OF-LIFE CARE. S.M. Dy¹. ¹Johns Hopkins University, Baltimore, MD. (Tracking ID #133349)

BACKGROUND: For the Southern California Evidence-Based Practice Center. Assessing the value of interventions in end-of-life care requires the use of comparable and high-quality measurement tools across different studies. As part of a systematic review on end-of-life care and outcomes, we abstracted published instruments that were used to measure outcomes in intervention studies in 5 domains of end-of-life care.

METHODS: We conducted a comprehensive search of Medline and the Database of Reviews of Effects (DARE), supplemented with hand search of the bibliography of the National Consensus Project for Quality Palliative Care and input from a technical expert panel. We augmented our search with relevant literature in congestive heart failure, lung cancer, and dementia. Eight reviewers selected post-1990 end-of-life-related intervention studies using pre-specified inclusion criteria. These criteria included exclusion Studies were classified into 1 or more of 5 domains, based on the intervention and outcomes: satisfaction, family and caregiver concerns, advance care planning, continuity of care, and symptoms (pain, dyspnea, depression and anxiety, and behavioral symptoms of dementia). For each study, 2 reviewers abstracted the outcomes and measurement tools used to demonstrate the impact of the intervention.

RESULTS: Our main search identified 21,745 articles. We abstracted data from 21 intervention studies related to advance care planning, 20 to continuity, 18 to symptoms, 13 to caregiver burden, and 12 to satisfaction. We identified 97 separate published instruments that were used to measure patient-centered outcomes in these studies. There was little overlap between studies in the tools that were used. 80/97 were used only once, and only 5 tools were used in 4 or more studies. Many of the instruments related to 3 of the 5 domains, including satisfaction, caregiver concerns, and symptoms, but few were related to continuity or advance care planning. Other common domains for instruments included quality of life and functional status. Few of the instruments used had been developed or tested in palliative care populations. Many of the interventions related to continuity in heart failure did not use instruments for patient-centered outcomes such as symptoms.

CONCLUSIONS: We found little overlap in the instruments used to measure patient-centered outcomes of interventions related to the end of life. The absence of measures tested in this population and of consensus on which domains and measures to use limits the quality of intervention studies and the ability to synthesize this literature. Further guidelines, research, and measure development should focus on high-quality measurement of specified outcomes in end-of-life care intervention studies.

THE EFFECTS OF OUTPATIENT MANAGEMENT ON HOSPITALIZATION FOR AMBULATORY-CARE SENSITIVE CONDITIONS ASSOCIATED WITH DIABETES MELLITUS. E.A. Warner¹; A.U. Huttenhower². ¹University of South Florida, Tampa, FL; ²Pfizer, Inc., Bradenton, FL. (Tracking ID #134604)

BACKGROUND: Diabetes mellitus is a chronic illness requiring complex management, provided primarily in outpatient settings. The Agency for Healthcare Research and Quality has defined certain hospital diagnoses related to diabetes

mellitus, including uncontrolled diabetes and short-and long-term complications of diabetes, as ambulatory care sensitive conditions (ACSC). These hospitalizations can potentially be prevented by good quality outpatient care. The purpose of this study was to evaluate which characteristics of ambulatory patients with diabetes were associated with hospitalizations for ambulatory-care sensitive conditions (ACSC).

METHODS: We conducted a retrospective chart review of adult patients with diabetes mellitus who were treated for at least one year in two clinics serving urban, low-income patients. We identified 65 patients over a 5-year time frame from 10/1/98 to 9/30/03 that were hospitalized with an ACSC associated with diabetes and compared them to 117 age- and gender-matched controls with diabetes who had no hospitalizations for an ACSC related to diabetes. The primary outcome measure was the difference in A1c between groups. We also compared demographic characteristics, the presence of comorbid conditions, the receipt of timely and appropriate services, achievement of goals of therapy, and drug therapy management.

RESULTS: There was no significant difference in ethnicity or smoking status between the two groups. Patients with an ACSC hospitalization were more likely to have a history of CHF, but there was no difference in hypertension, CAD, stroke, or depression. Patients with an ACSC hospitalization were less likely to have had an A1c within the last 12 months (69% versus 96%, $p < 0.001$), although there was no difference in having a lipid panel (77% versus 83%, $p = 0.326$) or microalbumin test (60% versus 74%, $p = 0.06$) within the last year. Blood pressure control was similar in both groups (ACSC mean 141/79 versus control 138/79, $p = 0.35$). Patients with an ACSC hospitalization had poorer glucose control (mean A1c 9.6 versus 7.68, $p < 0.001$) and worse renal function (mean creatinine 1.81 versus 1.09, $p = 0.039$), although there was no difference in lipid control (mean LDL 98.1 versus 105.6, $p = 0.33$) or presence of microalbuminuria (44% versus 33%, $p = 0.234$). Patients with ACSC hospitalization were more likely to be taking insulin (60% versus 34%, $p = 0.0001$). ACSC patients were less likely to have received angiotensin converting enzyme inhibitor or angiotensin-receptor blocker (ACE/ARB) therapy (63% versus 77%, $p = 0.023$) or statins (26% versus 43%, $p = 0.026$).

CONCLUSIONS: Other than underlying renal insufficiency and CHF, comorbid diagnoses had little effect on hospitalizations for ACSC. A1c was the major predictor of potentially preventable diabetes hospitalizations. Although blood pressure and lipid control were not directly related to ACSC hospitalization, the use of ACE/ARBs and statins was associated with reduced ACSC hospitalizations. Further work is needed to determine the roles of patient behavior and clinician practicing in achieving optimal outpatient care for patients with diabetes.

THE IMPLICATIONS OF CHOICE: DOES PRESCRIBING GENERIC OR PREFERRED PHARMACEUTICALS IMPROVE MEDICATION ADHERENCE FOR CHRONIC CONDITIONS? W.H. Shrank¹; T. Hoang²; S. Ettner³; P.A. Glassman⁴; K. Nair⁵; R. London⁶; D. Delapp⁶; J. Dirstine⁶; J. Avorn⁷; S.M. Asch⁸. ¹UCLA/Greater Los Angeles VA Healthcare System, Los Angeles, CA; ²West Los Angeles Veteran Affairs Healthcare System, Los Angeles, CA; ³University of California, Los Angeles, Los Angeles, CA; ⁴VA, West LA, Los Angeles, CA; ⁵University of Colorado Health Sciences Center, Denver, CO; ⁶Anthem Blue Cross Blue Shield, Denver, CO; ⁷Brigham and Women's Hospital, Boston, MA; ⁸RAND, Los Angeles, CA. (Tracking ID #135159)

BACKGROUND: A large proportion of Americans with prescription drug coverage are now enrolled in three-tier pharmacy benefit systems. These plans offer substantial choice to physicians and patients when prescribing, and require lower copayments for "preferred" formulary drugs. We studied whether patients enrolled in three-tier pharmacy benefit systems who receive generic or preferred brand-name agents when initiating chronic therapy were more likely to be adherent with treatment than those who received non-preferred brand-name medications requiring higher co-pays.

METHODS: We analyzed 2001-2003 pharmacy claims from a large health plan in the Western U.S. for six classes of chronic medications: HMG CoA reductase inhibitors, calcium channel blockers, oral contraceptives, orally inhaled steroids, angiotensin receptor blockers and angiotensin converting enzyme inhibitors. We evaluated rates that patients switched from generic to preferred or non-preferred medications, and vice-versa, after filling the initial prescription. Adherence was measured in terms of the proportion of days covered (PDC) in each drug class during the first year of therapy. We used multivariate linear regression to evaluate how the formulary status of the initial prescription (generic, preferred, non-preferred) influenced PDC. We defined adequate adherence as PDC >80 percent over the subsequent year, and used logistic regression to evaluate the relationship between formulary status of initial prescriptions and the likelihood of adequate adherence.

RESULTS: 7532 new prescriptions were filled in one of the classes evaluated, 1408 (23.2%) for non-preferred branded medications, 4376 (58.1%) for preferred branded drugs and 1408 (18.7%) for generic drugs. Those who received generic medications switched to preferred or non-preferred medications at less than half the rate of patients who received non-preferred medications, and patients who received preferred brand-name drugs were 30% less likely to switch to a medication in another tier than patients who received non-preferred drugs. After controlling for patient age, gender, income and drug class, PDC was 6.6% percent greater for patients who filled prescriptions for generic medications, ($p < .0001$) compared to patients who received non-preferred medications. PDC was 4.6% greater for patients who filled prescriptions for preferred brand-name medications ($p < .0001$) compared to patients who received non-preferred medications. Patients who received generic medications had 62% greater odds (95% confidence interval 1.39-1.89) of achieving adequate adherence, and patients who received preferred brand-name medications experienced 30% greater odds (1.15-1.47) of achieving adequate adherence, when compared to those who received non-preferred brand-name medications.

CONCLUSIONS: In three-tier pharmacy benefit plans, prescribing a generic or preferred medication within a therapeutic class can significantly improve a patient's adherence to that medication. Physicians who aim to improve patient adherence to chronic medications must be aware of patients' formularies. The Medicare Modernization Act includes incentives for health plans to deliver prescription drugs to seniors, likely in the form of tiered benefit plans. Attention must be paid to seniors' ability to navigate complex formularies and receive the least expensive of similarly effective medications, or risk non-adherence to potentially critical medications.

THE USE OF INFORMATION-TECHNOLOGY IN THE MANAGEMENT OF PATIENTS WITH DIABETES MELLITUS. A. Chou¹; J. Sutherland²; K. McCoy³; T. Vaughn⁴; B.N. Doebbeling¹. ¹Indiana University Purdue University Indianapolis, Indianapolis, IN; ²Indiana University, Indianapolis, IN; ³HSR&D Center of Excellence on Implementing Evidence-based Practice, Richard L. Roudebush VA Medical, Indianapolis, IN; ⁴University of Iowa, Iowa City, IA. (Tracking ID #136294)

BACKGROUND: The Institute of Medicine's Quality Chasm report identified 6 dimensions of information technology (IT) as critical components for quality improvement in clinical practice. The VA has been a leader in widely adopting IT for health care applications throughout its delivery systems. However, the utility of decision support systems (DSS) in chronic disease management has been infrequently addressed. This study evaluates the impact of IT use on the management of patients with diabetes mellitus (DM).

METHODS: A national sample of 2438 VA primary care providers (PCP), stratified by facility, was surveyed. Ratings assessed the utility, availability, and effectiveness of IT tools on care and reported practices surrounding DM management. Factor analysis yielded an IT factor assessing the extent to which the facility provided IT for computer-assisted DSS, computerized patient data, decision automation, on-line literature access, and electronic provider-to-provider communication. Chart audits assessed adherence with DM performance measures; they include appropriate HgA1c, dilated fundoscopy exam (DFE), and urine protein checks. Multiple regression analyses estimated DM performance measures as a function of availability of clinical reminders (CR), feedback of performance data, provider electronic communication, and IT use.

RESULTS: At the facility level, availability of performance data ($P = 0.03$), CR ($P < .0001$), and the IT factor ($P < .0001$) were positively associated with HgA1c checks. Appropriate DFE was positively related to the availability of performance data ($P = 0.04$), CR ($P < .0001$), and IT ($P = 0.0002$). Appropriate urine protein checks were positively associated with feedback of performance data ($P = 0.1$), CR ($P < .0001$), and the IT factor ($P = 0.001$).

CONCLUSIONS: These findings suggest that sharing performance data, CR use, and IT tools facilitated appropriate management of DM. IT encourages information sharing and use, integration of systems and functions, and linking with various environments. IT tools hold promise for facilitating decision-making and closing knowledge gaps between evidence-based and modal care for chronic disease management. Further understanding of IT's impact on care can identify conditions conducive to IT utilization, focus and refine efforts in IT implementation, to enhance clinical practice for generalists.

THREE YEAR UTILIZATION AND MORTALITY AFTER BARIATRIC SURGERY. D. Zingmond¹; M. McGory²; C. Ko¹. ¹University of California, Los Angeles, Los Angeles, CA; ²UCLA, Los Angeles, CA. (Tracking ID #135836)

BACKGROUND: Obesity is prevalent in the United States and associated with significant morbidity, mortality, and costs. Bariatric surgery successfully reduces weight and obesity-related morbidity over the long term. Population-based short- and intermediate-term utilization and mortality after bariatric surgery have not been described.

METHODS: Retrospective study of inpatient utilization three years pre- and post-surgery of persons having bariatric surgery in California (1995-1999). Bariatric surgery cases were identified from hospitalization abstracts from the California Office of Statewide Health Planning and Development OSHPDdischarge database by identifying all records with DRG equal to 288 (Operating room procedure for obesity) with primary procedure of obesity reduction (ICD9-CM codes 44.xx or 45.xx). Non-California residents were excluded. Number of admissions and total lengths of stay (LOS) excluding the index hospital stay by year preceding and following surgery were calculated. Income by zipcode was obtained from 1990 U.S. Census. Deaths were identified via the state death file. Hospital characteristics were obtained from annual OSHPD hospital reports and membership in the Council of Teaching Hospitals. We performed bivariate and multivariate linear and logistic regression analyses comparing predictors (patient demographics, severity, type of surgery, concurrent appendectomy or cholecystectomy, and hospital characteristics) to patient outcomes. Robust regression and clustering by hospital were performed.

RESULTS: 11,490 patients were identified for study. Patients were primarily female (84%), White (75%) or Latino (13%), and privately insured or self-pay (86%). Medicare and Medicaid paid for 11% of procedures. Laparoscopic approach (10%) and concurrent cholecystectomy (29%) were common. Cholecystitis was reported in 11% of patients. Concurrent appendectomy was occasional (3%). Mean Charlson score was 0.55. Mean surgical LOS was 4.3 days. Admit rates for the 3 years prior to surgery were 6.6%, 8.1%, 8.6% compared with 21%, 21%, and 18% during the 3 years after surgery. Similarly, mean LOS was 0.35, 0.45, 0.46 before surgery and 1.57, 1.13, and 1.01 after surgery; admits per year were 0.09, 0.11, 0.12 before surgery and 0.33, 0.29, and 0.26 after surgery. Mortality at 1 and 3 years was 0.9% and 1.6%. In multivariate logistic regression models of readmission at three years, Medicare and Medicaid insurance, greater

Charlson score, greater number of prior hospitalizations, concurrent appendectomy, and surgeries at smaller hospitals were associated with statistically significant greater odds; and lower income, self-pay, and laparoscopic surgery had lower odds of readmission. Models predicting number or summary LOS of readmission were similar. In mortality models, being male, older, having lower income, Medicare or Medicaid, higher Charlson Index, and greater admits in the prior year predicted higher odds of death. Laparoscopic surgery predicted lower odds of death.

CONCLUSIONS: As currently practiced, bariatric surgery has low mortality. Previous research has shown that bariatric surgery provides long-term reduction in obesity-related comorbidities. However, we have demonstrated increased hospital utilization immediately after bariatric surgery, and greater attention to this potential morbidity must be addressed by patients, providers, and payers.

TREATMENT OF EXCESSIVE ANTICOAGULATION WITH VITAMIN K: A META-ANALYSIS. K.J. Dezee¹; W.T. Shimeall²; K. Douglas²; N. Shumway³; P.G. O'Malley⁴. ¹Walter Reed Army Medical Center, Bethesda, MD; ²Uniformed Services University of the Health Sciences, Bethesda, MD; ³Walter Reed Army Medical Center, Washington, DC; ⁴Walter Reed Army Medical Center, Chevy Chase, MD. (Tracking ID #134608)

BACKGROUND: Patients who take oral anticoagulants and have an INR > 4.0 are at increased risk for bleeding. We performed a meta-analysis to assess the safety and efficacy of vitamin K.

METHODS: A systematic search (without language restrictions) using PubMed, EMBASE, the Cochrane Library, and relevant bibliographies from 1985 to present (the INR era) identified 758 papers, which were reviewed in duplicate for inclusion. Studies were included if they were randomized controlled trials (RCTs) or prospective, non-randomized trials and also used vitamin K to treat patients without major hemorrhage with an INR>4.0 but <10 due to oral anti-coagulant use. Included articles were independently reviewed in duplicate for quality and data abstraction. The main outcome was the proportion of participants with a therapeutic INR (1.8 to 4.0) at 24 hours.

RESULTS: Ten RCTs and 12 prospective trials were included. For the two RCTs that had acenocourmarol patients (n=169), there was no difference, compared to placebo, in proportion reaching the main outcome (p=0.48). Eight RCTs had patients that took coumadin (n=321). Among oral vitamin K treatment arms (4, n=75), the proportion with a therapeutic INR was 80% (95%CI 71-90%). This was similar to IV vitamin K treatment arms (6, n=69, 74%, 95%CI 64-84%). Treatment arms of subcutaneous vitamin K (3, n=58, 29%, 95%CI 18-41%) and placebo/observation (2; n=27, 24%, 95%CI 7-40%) were much less likely to achieve therapeutic values. The test for heterogeneity between the oral, IV, and subcutaneous subgroups was significant (p<0.001). Over half of subcutaneous vitamin K patients had a 24-hour INR above 4.0 (69%, 95%CI 57-81%). Only oral vitamin K patients had a 24-hour INR at or below 1.5 (8%, 95%CI 3-14%). Only one trial was of sufficient quality to abstract adverse outcomes, so summary estimates for hemorrhage and thrombosis could not be calculated.

CONCLUSIONS: Limited evidence suggests that oral and IV vitamin K are more effective treatment for excessive anticoagulation than simply withholding coumadin, but subcutaneous vitamin K is not. Whether or not treatment with vitamin K results in less hemorrhagic events among patients with excessive anticoagulation cannot be determined from the published literature.

Percentage of Patients Achieving INR 1.8-4.0 at 24 hours

Vitamin K	# Arms	N	% 24 hour INR 1.8-4.0	95% CI
Oral	4	75	80	71-90
IV	6	69	74	64-84
Sc	3	58	29	18-41
Placebo	2	27	24	7-40

TRENDS IN THE COMPLEXITY OF MEDICATION REGIMENS AND TEST ORDERING FOR PATIENTS WITH DIABETES FROM 1996 TO 2000. E.S. Huang¹; A. Basu¹; W. Manning². ¹University of Chicago, Chicago, IL; ²The Harris School of Public Policy, Chicago, IL. (Tracking ID #134594)

BACKGROUND: In the past decade, diabetes practice guidelines have called for lower risk factor target levels, which require more intensified medication regimens and more frequent laboratory testing. We assessed recent trends in the complexity of medication regimens and test ordering for patients with diabetes enrolled in a large managed care plan.

METHODS: Using the HEDIS and Hebert criteria, we identified non-pregnant adults living with diabetes from the claims data of health plans affiliated with United Health Care (UHC) for the period 1996 to 2000. This organization includes over 30 separate managed care plans and spans the Midwest and Southern regions of the country. No organized effort to improve diabetes care occurred during this time. We limited our patient sample to those who had at least one claim of any prescription drug or any laboratory test and were eligible for at least one year from their first utilization date (N=288,109). We grouped patients into calendar year cohorts based on their first utilization date and followed them for one year. We studied trends in the 1) the proportion of patients receiving different diabetes-related medications, 2) the average number of diabetes-related medications, 3) the proportion of patients receiving diabetes-related tests (glycosylated hemoglobin (HbA1C), urine microalbumin, and serum cholesterol), 4)

and the average number of tests ordered in one year of enrollment. We account for variability in trends across care plans.

RESULTS: The proportion of diabetes patients on any diabetes-related medication rose significantly from 53% to 65% (p<0.01), as was the case for glucose lowering medicines (31% → 48%, p<0.01) and cholesterol lowering drugs (12% → 23%, p<0.01) but not for blood pressure lowering drugs (38% → 48%, p=0.06). For patients who were prescribed medications, the average total number of diabetes-related medications rose from 2.56 medications to 3.34 medications (p<0.01) with smaller increases seen for glucose lowering (1.35 → 1.62, p<0.01) and blood pressure lowering regimens (1.94 → 2.12, p<0.01). For laboratory tests, we found a significant increase in the proportion of patients receiving any diabetes-related test (42% → 60%) as well as for each of the test categories (HbA1C (23% → 46%), cholesterol tests (33% → 49%), and urine microalbumin tests (2% → 10%) (all p's<0.01)). The average total number of tests ordered per year increased significantly from 2.55 → 3.60 (p<0.01). More modest increases were observed for HbA1C (2.07 → 2.26, p=0.03) and cholesterol tests (1.73 → 1.99, p<0.01) and no increase was seen for urine microalbumin testing. Trends in medication prescribing differed from plan to plan depending on the baseline level of prescribing, while laboratory test ordering trends were similar across health plans. Overall, the proportion of patients receiving both medications and laboratory tests related to diabetes grew (33% → 44%, p<0.01), while proportions receiving only medications or only laboratory tests remained stable.

CONCLUSIONS: Diabetes care grew more complex from 1996 to 2000 for patients enrolled in a large managed care organization. The most prominent change was an increase in the number of patients receiving medications or laboratory tests for multiple components of diabetes care, rather than changes in the intensity of individual care elements.

WHEN DO I TELL MY DOCTOR WHAT I SAW ON THE INTERNET? THE INTERACTION BETWEEN DOCTOR COMMUNICATION STYLE AND CANCER PATIENTS' SELF-EFFICACY TO COMMUNICATE. C.E. Golin¹; N. Arora²; M. Bowling¹; T. Strigo¹; B. Rimer¹. ¹University of North Carolina at Chapel Hill, Chapel Hill, NC; ²National Cancer Institute, Rockville, MD. (Tracking ID #136079)

BACKGROUND: Using the Internet for information and support is a potential means to enhance cancer patients' participation in their health care but little is known of cancer patients' Internet use or its effect on the doctor-patient relationship. Understanding the interplay between cancer patients' characteristics, their Internet use and aspects of the doctor-patient relationship could inform and enhance the health care of cancer patients.

METHODS: From April - Nov 2004, we invited all new subscribers of 15 Association of Cancer Online Resources mailing lists to complete web-based surveys. We asked them whether they had ever discussed Internet-obtained cancer-related information (INT-INF) with their doctor, and if so, whether it had worsened, improved or made no change in their relationship with their doctor. We used a validated 5-item 5-point scale to assess their self-efficacy to communicate with the doctor. We assessed a published 6-item 5-point scale of their view of their doctor's communication style. We also assessed clinical and demographic features. We analyzed baseline data from the first 546 respondents, conducting bivariate analyses and multivariate logistic regression to assess associations between patient and doctor characteristics and 2 outcomes: 1) whether patients discussed INT-INF with a doctor and 2) whether sharing INT-INF improved OR worsened/did not change the relationship with the doctor. Based on an a priori model, we assessed the effects of the interaction between patients' self-efficacy to communicate with the doctor and: a) the doctor's communication style and b) age.

RESULTS: Fifty-three percent of participants were male, 98% white, 58% had > 4 yrs of college; 17% were <40yrs, 21% were >60yrs old; mean illness duration was 3.9 yrs (SD 6.8). Most people (67%) preferred to make medical decisions jointly with the doctor. Eighty-three percent had talked to a doctor about cancer-related information obtained from the Internet. Of these, for 48% it made their relationship with their doctor better; for 48% it made no change; and for 4% it made it a little worse. In multivariate analyses, controlling for age, gender, education, disease status, illness duration, and preferences for participating in decision-making, when doctors had higher quality communication styles (above ≈ 15.1 [μ of scale=17.5, SD 2.6]), the more confident patients were to communicate with their doctor, the **more** likely they were to share INT-INF with their doctor (p=0.018). At the same time, at low levels of the doctors' communication quality, more confident patients were significantly **less** likely to have shared INT-INF with the doctor. Among those who shared INT-INF, for patients > 51 yrs old, the more self-efficacious they were, the **more** likely sharing INT-INF was to improve their relationship with their doctor (p=0.016). However, for those ≤ 51, the more efficacious they were, the **less** likely sharing INT-INF was to improve that relationship (p=0.016).

CONCLUSIONS: Few cancer patients' relationships with doctors worsened from sharing Internet-obtained information; for nearly half, it improved that relationship. When doctors were receptive to involving patients in the medical visit, confident patients were more likely to have shared information obtained on the Internet with their doctor.

ADDING FRACTURE RISK ESTIMATES TO BMD REPORTS: EFFECT ON PHYSICIAN TREATMENT DECISIONS. J.M. Neuner¹; P.W. Laud¹; M.M. Schapira¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #134411)

BACKGROUND: Guidelines for screening and treatment for postmenopausal osteoporosis have been created for primary care physicians, and there is grow-

ing interest in helping physicians tailor their care to patients' fracture risk. We evaluated the effect on osteoporosis treatment decisions of adding lifetime and five-year quantitative fracture risk estimates to a bone mineral density (BMD) report.

METHODS: A cross-sectional survey targeting primary care physicians was mailed to a random sample of U.S. physicians stratified to include nationally representative proportions of general practitioners, family practitioners, general internists and obstetrician-gynecologists. Survey components included: a) Likert-scaled items measuring osteoporosis knowledge, attitudes and screening, b) four vignettes that varied systematically with regard to patient age, weight, and hip BMD (reported as g/cm² with T-score and Z-score). Physician participants were randomized to receive standard or augmented BMD reports for each vignette. Augmented reports included quantitative estimates of lifetime and five-year hip fracture risk derived from published data from the Study of Osteoporotic Fractures. For each vignette, participants were also asked to estimate the hip fracture risk compared to average risk women of the same age and race (below average, average, greater than average). The effect of augmented BMD reports on prescription drug recommendations was examined using the Chi square statistic. A logistic regression model was also developed to examine the effect of augmented BMD reports on prescription drug recommendations when adjusted for physician specialty, physician demographics, and physician estimates of relative fracture risk for a below-average-risk patient vignette.

RESULTS: There were 287 primary care respondents, including 28% females, in a physician cohort with a median age of 49. There were no differences between respondents (overall response rate 52%) and nonrespondents in age or gender. Gynecologists were more likely to respond than other groups. In unadjusted analysis, physicians randomized to receive augmented BMD reports were no more or less likely to recommend prescription medications for any of the four vignettes (table). In the logistic regression model predicting the decision to treat the 70 year old average weight woman with T-score -1.01, physicians given augmented BMD reports were less likely to recommend prescription drugs (AOR 0.57 [0.33, 0.99]), as were physicians who correctly identified the woman as below-average risk (AOR 0.38 [0.21, 0.70]). There was no effect of physician gender, age, or specialty upon treatment recommendations.

CONCLUSIONS: An augmented BMD report that included quantitative risk estimates decreased physician use of pharmacologic therapy for osteoporotic fracture prevention for a woman of below-average risk. Comparative risk perceptions also affected physician treatment decisions. The addition of quantitative fracture risk estimates to BMD reports for lower-risk women has potential to change physician prescribing behavior.

Effects of Absolute Fracture Risk Estimates on Physician Prescription Osteoporosis Treatment Recommendations

Vignette Characteristics	% Treatment with Augmented BMD Report (n=138)	% Treatment with Standard BMD Report (n=141)	P value
70 year old T-score - 2.5	95	96	0.53
70 yr old T-score - 1.01	25	36	0.053
70 yr old low weight, T-score - 1.01	44	45	0.89
50 yr old T-score - 1.01	38	43	0.37

ANASTRAZOLE IS AN EXCELLENT CANDIDATE DRUG FOR BREAST CANCER CHEMOPREVENTION: AN ECONOMIC ANALYSIS. S. Cykert¹; N. Phifer². ¹University of North Carolina at Chapel Hill, Chapel Hill, NC; ²Moses H. Cone Memorial Hospital, Greensboro, NC. (Tracking ID #135965)

BACKGROUND: The Breast Cancer Prevention Trial demonstrated that tamoxifen (tam) effectively reduces breast cancer risk in high-risk women. However, tam's side-effect profile has severely constrained its use, especially for women above the age of 50. For tam, data describing contralateral breast cancer rates have served as an accurate barometer for its primary prevention effect. A recent breast cancer treatment report (N=6186) demonstrated that anastrozole reduces cancer occurrence in the contralateral breast 48% further than tam and has a more favorable side-effect profile. This result from the Arimidex, Tamoxifen Alone or in Combination Trialists' Group (ATAC), coupled with their 68 month follow-up, suggests that anastrozole is a good option for primary breast cancer chemoprevention. However, the use and cost-effectiveness of anastrozole might be limited by its high price. Therefore, the purpose of our study is to evaluate the viability of anastrozole as a primary, chemoprevention agent compared to "no treatment" and tam alternatives.

METHODS: We performed decision analysis using a Markov state transition model. Cost-effectiveness is reported from the payor's perspective. Patient characteristics and the annual rates of breast cancer, stroke, venous thromboembolic disease, endometrial cancer, and fracture were derived from the reports of the Breast Cancer Prevention Trial of the National Surgical Adjuvant Breast and Bowel P-1 Project (NSABBP) and the ATAC group. Life expectancy for each age

group was obtained from the National Vital Statistics Report. Mortality attributable to pulmonary embolus, stroke, breast cancer, and endometrial cancer as well as the costs of these complications were derived from appropriate literature and administrative data. Utility scores were measured by questioning 106 women. We searched the "Red Book" and Drugstore.com for drug prices. Sensitivity analyses were performed.

RESULTS: Using the 3.4%, 5-yr average breast cancer risk for patients enrolled in the NSABBP study combined with contralateral breast cancer data from the ATAC group, we determined that women aged 50-55 yrs should gain an average of 117 days of quality-adjusted life (QAL) by taking anastrozole compared to no treatment. The cost per QALY gained was \$33,000. At age 60, the predicted QAL gained was 73 days and the associated cost, \$52,900 per QALY. Although anastrozole is priced 4 times greater than tam, the incremental QAL gains over tam that range from 62 days at age 50 to 40 days at age 65 occurred at the costs of only \$23200 and \$46400 per QALY, respectively. If the anastrozole price (current cost \$225/month) is lowered to \$119/month or less, it dominates tam for breast cancer chemoprevention in all age groups.

CONCLUSIONS: From the above analyses, we anticipate anastrozole to be an excellent and cost-effective candidate drug for the primary prevention of breast cancer. This treatment should be particularly useful in older patients who are at excess risk of tam associated side-effects. Lower pricing would make the anastrozole chemoprevention strategy dominant.

ASSOCIATION OF BODY WEIGHT WITH PREFERENCE-BASED QUALITY OF LIFE IN U.S. ADULTS. D.E. Arterburn¹; L. Lawson²; L. Chen²; J. Tsevat¹. ¹Cincinnati VA Medical Center, Cincinnati, OH; ²Cincinnati Children's Hospital, Cincinnati, OH. (Tracking ID #133786)

BACKGROUND: Obesity has important effects on health and health care costs, and it has been linked to multidimensional impairments in health-related quality of life and psychosocial well-being. To facilitate future economic evaluations of weight loss interventions, we estimated the impact of body weight on quality of life in a nationally-representative sample of U.S. adults.

METHODS: We performed a cross-sectional analysis of data from 27,332 adults in the 2000 and 2001 Medical Expenditure Panel Surveys, two nationally representative surveys of the noninstitutionalized civilian population of the United States. Height and weight were self-reported, and quality of life was assessed using the EuroQol EQ-5D, a generic, preference-based instrument that has 5 dimensions (mobility, self care, usual activities, pain/discomfort, and anxiety/depression), each with 3 levels (no problems, some problems, extreme problems), generating 243 health states. Health state values (EQ-5D utility scores) range from no problems on any dimension (value = 1) to extreme problems in all (-0.594). The EuroQol also asks subjects to rate their overall current health using a visual analog scale (VAS), where 0 represents the worst imaginable health state and 100 indicates perfect health. We used multivariable linear and logistic regression to assess the associations between National Institutes of Health body mass index (BMI) categories and responses on the EuroQol. Analyses were statistically controlled for sociodemographic characteristics (age, gender, race, income, education, marital status, region of country, insurance type), smoking status, and 6 chronic health conditions (hypertension, diabetes mellitus, coronary heart disease, stroke, asthma, and emphysema) using STATA SE 8.0 survey commands.

RESULTS: Fifty-eight million US adults (39.3%) had normal body weight (BMI 18.5-24.9), 53 million (35.9%) were overweight (BMI 25.0-29.9), 22 million (14.8%) had class I obesity (BMI 30.0-34.9), 8 million (5.2%) had class II obesity (BMI 35.0-39.9), 4 million (2.9%) had class III obesity (BMI ≥ 40.0), and 3 million (2.0%) were underweight (BMI < 18.5) in 2000-2001. Obese adults and those who were underweight reported the worst quality of life. For example, when compared to normal weight adults, obese adults were 2 to 3 times more likely to report any limitations in mobility, self care, and usual activities. Unadjusted EQ-5D and VAS utility scores for BMI categories are presented in the Table; multivariable adjusted results were substantively similar. Above a BMI of 25, we observed a linear, inverse relationship between BMI and EQ-5D utility scores, where a five point increase in BMI was associated with an absolute 2.7% decrease in utility (95% CI: 0.024 to 0.031; adjusted R-squared = 0.15).

CONCLUSIONS: Body mass index is strongly associated with preference-based quality of life. Further research is needed to test the cost-effectiveness of weight loss interventions in U.S. adults using health state preferences derived from the general population.

Unadjusted EuroQol EQ-5D and VAS Utility Scores for BMI Categories

BMI Categories	EQ-5D Utility Score	VAS Utility Score
Underweight	.825	76.8
Normal	.864	81.8
Overweight	.845	80.6
Class I Obesity	.811	76.9
Class II Obesity	.777	73.6
Class III Obesity	.734	69.9

ASSOCIATION OF SOCIO-DEMOGRAPHIC FACTORS WITH UTILITIES FOR BREAST CANCER: IMPLICATIONS FOR DISPARITY. M.D. Schleinitz¹; D.M. Depalo²; J. Blume¹; M.D. Stein¹. ¹Brown University, Providence, RI; ²Rhode Island Hospital, Providence, RI. (Tracking ID #135604)

BACKGROUND: Women from certain socio-demographic groups are less likely to receive adjuvant therapy after a diagnosis of breast cancer than white or more affluent women. Whereas preference contributes to disparity in other health care scenarios, it is unclear if preference explains differential rates of breast-related health care. Utilities, a measure of the relative desirability of, or preference for, breast cancer health states are lacking for women of different backgrounds and ages. Our objective was to ascertain utilities for the different stages of, and treatments for, breast cancer from a sample of women of varied racial and socio-economic backgrounds, and to determine if differences in utilities between socio-demographic groups might explain disparity in breast cancer care.

METHODS: We assessed utilities for eight breast cancer related health states from a convenience sample, stratified by age and race, of 156 English-speaking women over 25 years old not currently undergoing treatment for breast cancer. We used standard gamble to determine utilities for five stages of breast cancer, and time trade-off for three therapeutic modalities. We used paired T-tests to compare utilities for different disease stages and for different treatment modalities. We developed a Markov model depicting the decision to accept chemotherapy for a current, hypothetical, diagnosis of surgically resected stage 2 breast cancer, and incorporated each subject's utilities and age to determine if her quality adjusted life expectancy would be maximized with or without chemotherapy. We used multivariate logistic regression to determine if age, race, education, socio-economic status, breast cancer risk and family history were associated with this optimal strategy.

RESULTS: Median utilities for the five disease stages were: stage 1 disease, 0.90 (inter-quartile range 0.50-1.00); stage 2, 0.75 (0.26-0.99); stage 3, 0.51 (0.25-0.94); stage 4 (estrogen receptor positive), 0.33 (0-0.75); stage 4 (estrogen receptor negative), 0.40 (0-0.75). With the exception of the two stage 4 health states, utilities were statistically distinct ($p < 0.01$). Utilities for treatment modalities were: chemotherapy 0.50 (0-0.92); hormonal therapy 0.58 (0-1) and radiation therapy 0.83 (0.13-1). Utilities for chemotherapy and hormonal therapy were both distinct from radiation therapy ($p < 0.05$), but not from one another. One hundred twenty two of 156 subjects maximized quality adjusted life expectancy given stage 2 breast cancer by accepting chemotherapy. In multivariate logistic regression the odds of maximizing quality adjusted life expectancy with chemotherapy were 0.01 (0.002-0.06) for women over 50, 0.19 (0.05-0.69) for black women, 0.06 (0.01-0.32) for women of low socio-economic status, and 4.1 (1.2-14) for married women and those with significant others. Neither level of education, breast cancer risk nor interaction terms were significantly associated with the optimal strategy.

CONCLUSIONS: Differences in utility for breast cancer health states may in part explain the lower rate of adjuvant therapy for black, older and less affluent women. Further work must clarify whether these differences are solely a result of preference or reflect limitations in health knowledge or in obtaining social and workplace support during therapy. Should this be true, interventions that alleviate the perceived social or economic hardship of adjuvant therapy may resolve disparity.

BREAST CANCER RISK PERCEPTION AMONG RELATIVES OF WOMEN WITH BREAST CANCER. M. Gilligan¹; M. Schapira¹; N. Lu¹; C. Boudreau¹; A.B. Nattinger¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #135005)

BACKGROUND: Breast cancer risk among first-degree relatives of women with breast cancer (FDRs) can range from low to high depending on details of the family history. Studies suggest that women overestimate risk of breast cancer compared to Gail model risk estimates. However, the Gail model will underestimate risk in women from families with a genetic predisposition. The purpose of this study was to explore factors related to perceived risk and accuracy (concordance between perceived risk and actual risk). We hypothesized that 1) Information processing style ("High Monitors" on Miller Behavioral Style Scale (MBSS)) would be associated with more accurate perceived risk, and 2) Accuracy would improve when women were given information aimed at clarifying their personal risk.

METHODS: An educational booklet was developed based on previous focus group analysis. FDRs completed a baseline survey and follow-up survey one month after receiving the booklet. Baseline information included risk factor information to determine actual risk; the MBSS; a 3-item numeracy scale; and socio-demographics. Pre- and post-intervention surveys assessed knowledge of breast cancer risk (25 true-false questions); and perceived risk of breast cancer (0-100% visual analog scale and "Low, Moderate, or High"). Women were classified as Accurate (A), Underestimator (U) or Overestimator (O) depending on concordance between perceived risk and both Gail risk and risk category as determined by clinical criteria. Multivariable logistic regression was performed to determine predictors of accuracy. A total of 118 women completed the baseline survey (total cohort) and of those, 57 completed the follow-up survey (subcohort).

RESULTS: No statistically significant differences were found between the total cohort and subcohort for any variables except age (mean 46.9 yrs and 49.8 yrs respectively, $p = 0.02$) and income ($\geq \$100,000$, 23% and 32%, respectively, $p = 0.01$). The cohort was predominantly white (98%) and well educated (65% with $\geq$ college degree). Over 60% answered all 3 numeracy items correctly. Gail risk estimates ranged from 8.6-44.1%. Baseline mean knowledge score was 71%, range 25-92%; median perceived risk was 50% with 29% choosing 50% as their lifetime risk. The majority of women overestimated their risk when com-

pared to either Gail risk (80%) or categorical risk (52% total cohort; 54% subcohort). After the intervention, knowledge scores increased to a mean of 83% (95% C.I. 81-19%). There was no significant decrease in perceived risk, and no significant increase in accuracy (from 32% to 33%). Using stepwise selection (among income, age, education and "Monitor" status) the only predictor of accuracy as determined from logistic regression was education ($\geq$ college compared to $\leq$ some college, OR 2.9, 95% C.I. 1.3-6.7).

CONCLUSIONS: In a well educated, "numerate" population, women significantly overestimated their perceived risk whether compared to Gail or categorical risk. The degree of overestimation did not improve despite an intervention aimed at improving accuracy of perceived lifetime risk. Accuracy was associated with higher educational status but not information processing style. A predilection toward selection of perceived lifetime risk of 50% suggests an attitude of "maybe I'll get it, maybe I won't" rather than an estimate of true statistical risk. Inasmuch as perceived risk is an important factor to study in health behavior, improved methods to measure it are needed.

BROADENING THE TEAM: A MULTIDISCIPLINARY APPROACH TO COMMUNITY-ACQUIRED PNEUMONIA GUIDELINE IMPLEMENTATION. J.T. Hagaman¹; G.W. Rouan¹. ¹University of Cincinnati, Cincinnati, OH. (Tracking ID #133576)

BACKGROUND: In recent years, much attention has been paid to community-acquired pneumonia (CAP) and reducing associated hospital stays. We have previously shown that by adopting an institutional guideline for CAP including criteria for early iv-to-oral antibiotic "switch therapy", length of stay (LOS) for CAP patients can be reduced by 0.5 days. This reduction in LOS was largely due to physician education and guideline awareness and usage. We hypothesized this was not the only obstacle to successful guideline implementation, and the etiology of such was multi-factorial.

METHODS: Five hundred patients with CAP admitted to University Cincinnati Medical Center (UCMC) were identified with DRG codes. The charts of 25% of these, were retrospectively reviewed to discern if, 1) early-switch therapy was applied by physicians in an appropriate manner; and 2) in what proportion of patients did institutional barriers exist that prevented the discharge of these patients in the timeliest fashion possible. Additionally, we sought to define the number of days that may be saved by correcting such institutional barriers.

RESULTS: Following guideline implementation, physician orders for early-switch therapy were written at the appropriate time for 80% of patients. Despite this, delays in order entry, med administration, and discharge processing resulted in fewer than 52% being switched and discharged as soon as intended. This resulted hospital stays unnecessarily lengthened by 0.6 days per patient.

CONCLUSIONS: Our previous efforts for guideline implementation have focused on physician education and behavior. Barriers to guideline implementation, however, extend to ancillary medical staff (nursing, pharmacy, etcetera). In turn, a multidisciplinary approach to guideline implementation is needed to optimize effectiveness. Despite improving physician guideline usage, as many as 0.6 additional days per patient could be saved. Educating ancillary medical staff on CAP guidelines, and implementing a "team" approach to guideline usage may further reduce LOS. A multidisciplinary approach to guideline implementation may further reduce LOS for hospitalized CAP patients.

CHOOSING WHERE TO HAVE HIGH RISK SURGERY: WHO DECIDES? C.T. Wilson¹; L.M. Schwartz²; S. Woloshin¹. ¹Dartmouth Medical School, White River Junction, VT. (Tracking ID #135718)

BACKGROUND: Medicare is contemplating providing hospital performance data to patients to help them decide where to have high risk surgery. However, the extent to which physicians - not patients - make this decision is not known. We sought to determine how often patients report the physician was the main decision maker in where to have surgery.

METHODS: We interviewed 510 randomly selected Medicare patients who had undergone 1 of 5 elective, high risk operations approximately 3 years earlier: abdominal aneurysm repair ($n = 103$), heart valve replacement surgery ($n = 96$), or resections for bladder ($n = 119$), lung ($n = 128$) or stomach ($n = 64$) cancer. The response rate was 68% among eligible survivors for whom contact information was available and were able to be interviewed and participate. Most respondents were male (67%) with a mean age of 71 years. The main outcome measure was the proportion of patients who responded that the main decision maker in where to have surgery was the physician.

RESULTS: Thirty-one percent of patients said the physician was the main decision maker about where to have surgery. This proportion was similar across varying patient age, income, or educational attainment. Men were more likely to say the physician was the main decision maker (34% vs. 24% for women, $p = 0.02$), as were patients in "poor to fair" health (37% vs. 28% for those in "good to excellent" health, $p = 0.049$). The physician was reported to be the main decision maker about where to have surgery in 39% of cardiovascular operations (cardiac valve-43%, aortic aneurysm repair-37%) as compared to the 26% of cancer operations (gastrectomy-30%, cystectomy-26%, pulmonary lobectomy-24%); the difference between these two categories of operations was statistically significant ($p = 0.001$).

CONCLUSIONS: Though most patients felt substantially involved, one-third said their physician was the main decision maker in choosing where surgery should take place. It is unclear whether this is the level of involvement patients want. Policy makers interested in empowering patients to participate in decisions about where to have surgery will need to further explore the preferences of this group of patients.

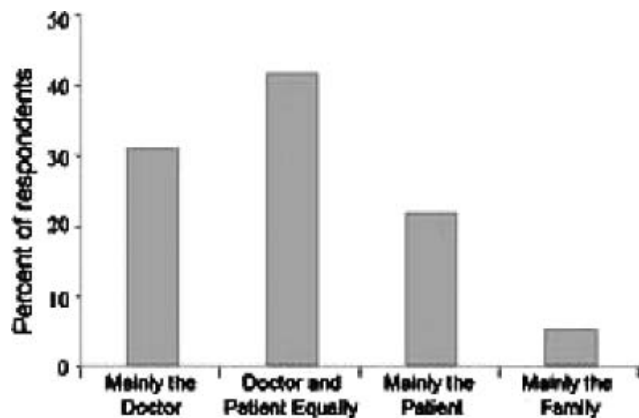


Figure 1 Responses to the survey question: "Who made the decision about going to your hospital?"

COST-EFFECTIVE ANALYSIS OF ALBUMIN FOR THE FLUID RESUSCITATION OF SEVERELY SEPTIC PATIENTS. E.A. Butcher¹; S. Cykert². ¹Internal Medicine Program, Moses Cone Hospital, Greensboro, NC; ²University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #135932)

BACKGROUND: The optimal intravascular fluid for volume resuscitation in the intensive care unit has yet to be determined. Meta-analyses have yielded conflicting results. A recent randomized control trial demonstrated similar mortality rates for albumin and saline. However, there was a trend toward improved survival with the use of albumin in patients with severe sepsis ($p=0.06$). The purpose of this study is to determine the cost-effectiveness of albumin for intravenous fluid resuscitation in patients with severe sepsis.

METHODS: We conducted a decision analysis to compare 4 percent albumin to normal saline for fluid resuscitation in patients with severe sepsis in the intensive care unit. results are reported from the societal perspective. Data on patient survival were derived from the recent trial of albumin versus normal saline for fluid resuscitation. Data on life expectancy were obtained from the National Vital Statistics Report. Information on long-term survival of patients discharged from the intensive care unit and post discharge care costs were obtained from the most relevant published literature. Multiple sensitivity analyses were performed.

RESULTS: The absolute survival benefit of 4.6 percent attributable to albumin results in a favorable cost-effectiveness ratio of \$19,634 per quality adjusted life year (QALY) gained. The sensitivity analysis showed that albumin remains a cost-effective strategy even if the survival benefit fell to four thousandths of a percentage point compared to saline (\$48,775 per QALY). Albumin remained cost-effective if we estimate the actual cost of nursing home placement to be three times the published cost (\$49,014 per QALY). If the actual cost incurred prior to a premature death is ten times the published cost, albumin rather than saline remains a cost-effective strategy (\$48,766 per QALY).

CONCLUSIONS: The use of 4 percent albumin intravenous resuscitation fluid for patients with severe sepsis is cost-effective even if the survival benefit is small.

DECISION ANALYSIS OF SECONDARY PREVENTION OF VENOUS THROMBOEMBOLISM. S.C. Handeyside¹; Y.S. Kwok¹. ¹University of Michigan, Ann Arbor, MI. (Tracking ID #133247)

BACKGROUND: The risks and benefits of extended treatment of idiopathic venous thromboembolism (VTE) are unclear. New strategies for extended treatment carry different risks and benefits.

METHODS: We compared the incidence of recurrent VTE in 4 cohorts of patients with idiopathic VTE over a 2-year period and a 5-year period using Markov decision-analysis models. The model was designed as a simulated secondary prevention trial. Enrollment in the simulated trial began after 6 months of standard therapy with warfarin. The 4 cohorts in our model were treated with the following therapies: conventional-intensity warfarin (INR 2.0 to 3.0), low-intensity warfarin (INR 1.5 to 2.0), ximelagatran (a direct thrombin inhibitor administered in fixed doses), and no treatment. We obtained outcomes probabilities, including recurrent VTE, major bleeding, and death, from the three large randomized controlled trials identified with a Medline literature search.

RESULTS: After 5 years treatment, recurrent non-fatal VTE occurred in 3% of patients treated with conventional dose warfarin, 6% of patients treated with ximelagatran, 10% of patients in the low-intensity warfarin arm, and 31% in the no treatment arm. Ximelagatran had favorable outcomes for bleeding events, with 5-year incidence of 3% Both warfarin treatment arms had a 4% bleeding rate. Bleeding events in the no treatment arm were 2%. Mortality over 5 years was lowest in the ximelagatran arm, with a rate of 3%. This was followed by the conventional-intensity warfarin and no treatment, each at 4%. Low-intensity warfarin had the highest mortality rate. Overall, the model indicated that treatment with conventional-intensity warfarin and with ximelagatran had the highest percentage (88%) of "well" patients after 5 years. 79% of patients in the low intensity warfarin cohort were well after 5 years, and 62% in the no treatment

arm. The results after 2 years of treatment showed similar trends. Sensitivity analysis did not significantly affect outcomes.

CONCLUSIONS: For secondary prevention of recurrent events in patients with idiopathic VTE, treatment with conventional intensity warfarin is the most effective over an extended treatment period. However, ximelagatran also is effective, and this treatment is associated with lower incidence of bleeding and death. Both treatment arms had a high percentage of "well" patients after 5 years.

2 and 5 Year Results

Assigned Therapy Arm	Well	Recurrent DVT/PE	Death	Major Bleeding
2 year results				
Conventional intensity warfarin	95%	1%	2%	2%
Low intensity warfarin	91%	4%	3%	2%
Ximelagatran	95%	3%	1%	1%
No anticoagulation	83%	14%	2%	1%
5 year results				
Conventional intensity warfarin	88%	3%	4%	4%
Low intensity warfarin	79%	10%	6%	5%
Ximelagatran	88%	6%	3%	3%
No anticoagulation	62%	31%	4%	2%

DOES LOCAL HEALTH CARE SPENDING REFLECT THE DECISIONS MADE BY INDIVIDUAL PHYSICIANS? B.E. Sirovich¹; P. Gallagher²; D.E. Wennberg³; E.S. Fisher⁴. ¹VA Outcomes Group, Dartmouth Medical School, White River Junction, VT; ²Center for Survey Research, University of Massachusetts Boston, Boston, MA; ³Maine Medical Center, Portland, ME; ⁴Center for the Evaluative Clinical Sciences at Dartmouth, Hanover, NH. (Tracking ID #135987)

BACKGROUND: Primary care physicians are faced with countless diagnostic and therapeutic decisions each day - decisions that often revolve around whether or not to intervene (for example, order a diagnostic test) for a specific patient. Because deciding to intervene inevitably leads to further health care utilization, we sought to determine whether the wide observed variation in health care spending across the United States reflects the underlying tendency of physicians in different areas to intervene for individual patients.

METHODS: We conducted a mail survey of a nationally representative random sample of primary care physicians (internists, family practitioners, and general practitioners) in 2004 (response rate 62%). Based on practice location, we assigned each participating physician ($n=965$) to one of 306 U.S. Hospital Referral Regions, which were in turn assigned to quintiles of spending based on a previously derived per capita Medicare spending measure. Physicians' tendency to intervene was measured by responses to clinical vignettes, in which physicians were asked how often they would order a diagnostic test, specialist referral, procedure, or hospital admission for the patient described. All tests for trend are based on logistic regression in which the individual physician's response (whether or not they would recommend an intervention at least "most of the time") was the dependent variable and the independent variable was quintile of spending of the physician's region.

RESULTS: Three detailed clinical scenarios offered physicians a total of 13 decisions about whether or not to intervene. In 9 out of 13 cases, higher local health care spending in a physician's region was associated with a greater likelihood of the physician ordering the intervention. (In the remaining 4 cases, there were no differences across quintiles of spending.) Spending was positively correlated with physicians' decisions to order 2 out of 5 diagnostic tests, 2 out of 2 referrals, 3 out of 3 procedures, and 2 out of 3 hospital admissions. For example, for an 85 year old man with endstage congestive heart failure and acute decompensation despite maximal medications, 37% of physicians in the highest spending quintile would recommend admission to the CCU compared with 18% in the middle quintile, and 12% in the quintile of lowest spending ($p_{trend} < 0.001$). For a 75 year old woman with typical heartburn symptoms not fully resolved on over the counter H2 blockers, 39% of physician in the highest spending quintile would recommend referral to gastroenterology, as would 21% of those in the middle quintile, and 10% of those in the lowest spending quintile ($p_{trend} < 0.001$). **CONCLUSIONS:** Local health care spending across the United States reflects the tendency of local physicians to recommend interventions for their patients. This is particularly true for high cost interventions such as referrals, procedures, and hospital admissions.

FACTORS INFLUENCING PATIENT INTEREST IN SCREENING CT SCANS FOR LUNG CANCER. M. Calixte¹; C. Carosella²; W. Lee¹; E. Wold¹. ¹New York Medical College, Valhalla, NY; ²New York Medical College, White Plains, NY. (Tracking ID #136188)

BACKGROUND: Using CT scans to screen smokers for lung cancer is not currently recommended as a part of standard medical practice. While there is no

evidence that screening decreases mortality the procedure is increasingly marketed to patients. However, the magnitude of patient demand for the procedure is not clear. Factors that may influence patient use of these scans are also not known.

METHODS: We performed a cross-sectional survey of smokers who were seen at 3 primary care medical practices. A convenience sample from each of the three practices was obtained. During recruiting sessions, all patients presenting for medical care were screened for inclusion in the study. Patients were included if they were at least 45 years old and had at least a 10 pack year smoking history. Patients were excluded if they were unable to read or complete the survey or if they had a history of lung or metastatic cancer. Patients who agreed to participate in the study answered basic demographic and clinical questions and then were given a scenario briefly describing CT screening for lung cancer. They were then asked to indicate whether the scan would be "worth it" under varying conditions of test accuracy and of extent of work-up for a positive result. The primary outcome was a comparison of patient perception of the scan's worth for inaccurate scans and surgical work-up compared to more accurate and less invasive scans. Secondary outcomes included the difference between the average number of "worth it" responses among patients with differing responses to questions concerning motivating factors for obtaining CT scans.

RESULTS: The average age of patients was 58 years old. Forty three percent of patients were currently smoking. Sixty six percent of patients indicated interest in having a CT scan. Interest declined to 57% and 43% when patients were told that the CT might not help them live longer or is "probably wrong" if cancer were diagnosed. If surgery were required to confirm the diagnosis, only 33% of patients expressed interest in screening.

CONCLUSIONS: The level of baseline interest in CT screening for lung cancer is high, but decreases substantially if patients are told of inaccuracy and the potential for surgical work-ups of positive results.

IMPLANTABLE OR EXTERNAL DEFIBRILLATORS: WHERE COST-EFFECTIVENESS CONFRONTS FISCAL REALITY. P. Cram¹; D.A. Katz¹; S. Vijan²; D. Kent³; K.M. Langa²; A.M. Fendrick². ¹University of Iowa, Iowa City, IA; ²University of Michigan, Ann Arbor, MI; ³Tufts-New England Medical Center, Boston, MA. (Tracking ID #134489)

BACKGROUND: Implantable cardioverter defibrillators (ICDs) are the most effective technology for preventing sudden cardiac death (SCD), but their use is limited by high cost [\$50,000]. In-home automated external defibrillators (AEDs) are less effective than ICDs, but also less expensive [\$2,000]. We estimated the incremental cost-effectiveness of prophylactic ICD implantation and in-home AEDs for reducing SCD mortality as compared with emergency medical services equipped with defibrillators (EMS-D), and used these analyses to quantify the aggregate societal benefits of alternative defibrillator allocation strategies.

METHODS: A Markov decision model was created using the societal perspective to evaluate the clinical benefits and resource use of three alternative strategies to reduce SCD mortality in a cohort of adults with congestive heart failure [annual SCD event rate: no ICD 5.7%; with ICD 2.2%]; Strategy 1 - individuals suffering SCD were treated by EMS equipped with defibrillators; Strategy 2 - all individuals received an AED, individuals suffering SCD at home were treated by in-home AEDs followed by treatment with EMS; Strategy 3 - all individuals received a prophylactic ICD. The initial annual probabilities of SCD and non-SCD mortality were derived from the recently published MADIT 2 defibrillator trial; subsequent mortality was adjusted upward based upon life-table estimates. Other clinical probabilities including SCD survival with and without an in-home AED, costs, and utilities were derived from the published medical literature.

RESULTS: Under base-case assumptions, providing all individuals with in-home AEDs produced a gain of 0.05 quality adjusted life-years (QALYs) when compared with EMS-D at an incremental cost of \$5,800, resulting in a cost of \$116,000 per QALY gained. Providing all individuals with ICDs produced a gain of 0.92 QALYs at a cost of \$147,900 compared with EMS-D, resulting in a cost of \$161,000 per QALY gained. Offering an AED to an individual who qualified for an ICD led to a savings of \$163,000 per QALY foregone - a measure of "decremental" cost-effectiveness. Alternatively, our results suggest that 25 individuals could be given AEDs for the lifetime cost of providing one person with an ICD. For every million dollars spent on defibrillators, 2.4 additional QALYs are produced if AEDs [8.6 QALYs/\$million] are purchased instead of ICDs [6.2 QALYs/\$million]. results were most sensitive to differences in defibrillator effectiveness, relative cost, and patient risk of SCD.

CONCLUSIONS: As financial constraints lead to rationing of effective interventions, it may ultimately become preferable to provide relatively more people with a less effective and less expensive intervention (in-home AEDs) instead of relatively fewer people with the most effective intervention (ICDs). Current coverage schemes -based strictly on clinical effectiveness- may need to be altered to reflect budgetary pressures.

INCREASING INFLUENZA VACCINATION AND REDUCING MORTALITY AMONG THE ELDERLY THROUGH DIRECT-TO-CONSUMER ADVERTISING. M.S. Patel¹; M.M. Davis¹. ¹University of Michigan, Ann Arbor, MI. (Tracking ID #136281)

BACKGROUND: Influenza-related mortality predominately and disproportionately impacts the elderly. Rates of annual influenza vaccination among the elderly are approximately 60%-65%, far below the Healthy People 2010 target of 90%. We examined the potential cost-effectiveness of a federal program of direct-to-consumer (DTC) advertising for influenza vaccine, as a method of increasing vaccination rates among persons aged 65 and older.

METHODS: Published data regarding influenza-associated mortality rates and vaccine efficacy among the US elderly were used to calculate the number needed to vaccinate (NNV) to prevent one all-cause death due to influenza. The NNV was then used to calculate the reduction in mortality expected from a 1% increase in the national elderly vaccination rate. Data regarding DTC advertising costs and effects for pharmaceuticals were used to inform estimates of DTC advertising costs necessary to increase vaccination rates by 1%. Published estimates of the value of one quality-adjusted life-year (QALY) and the utility of life for the average elderly person, in combination with the elderly population distribution from the Census and age-specific life expectancy for elderly persons from life tables, were used to calculate the average net present value of a life lost due to influenza mortality, discounted annually at 3%. This permitted calculation of the net present value of deaths prevented (lives saved) as a result of decreased influenza-related all-cause mortality in the setting of a 1% increase in the vaccination rate. Costs of the vaccine and vaccine administration were assumed to be covered by health insurance; adequate vaccine supply was also assumed.

RESULTS: The US elderly population in 2003 was 35.9 million. For influenza among the elderly, the all-cause mortality rate was .001325 and vaccine efficacy against death was 68% (95% CI: 56%-76%). The NNV was 1110 (993-1348), resulting in 321 (264-359) elderly lives saved with a 1% increase in the vaccination rate. Based on base-case parameters of \$200,000/QALY and utility = 0.9 for the average elderly person, we estimated an average net present value of \$1.9 million for an elder's life lost to influenza. The total savings expected with a 1% increase in the influenza elderly vaccination rate were \$610 million (\$502m-\$682m). With a more conservative QALY valuation, the expected mortality savings were more modest: at \$100,000/QALY, \$305m (\$251m-\$341m); at \$50,000/QALY, \$153m (\$126m-\$170m). At the most conservative parameter estimates of \$50,000/QALY and utility = 0.7 for a community-dwelling elder, the mortality savings were \$118m (\$98m-\$133m). In comparison, DTC advertising would be expected to cost \$30.9m to achieve a 1% increase in the elderly vaccination rate; i.e., DTC advertising for elderly flu vaccination was cost-saving.

CONCLUSIONS: Influenza-related mortality among the elderly may be reduced through better vaccination efforts. DTC advertising that targets the elderly population may represent a not only cost-effective but cost-saving strategy for the federal government to pursue as a means of increasing vaccination rates.

LIFETIME COST-EFFECTIVENESS OF AGGRESSIVE LIPID LOWERING THERAPY FOR SECONDARY PREVENTION IN POST-MYOCARDIAL INFARCTION PATIENTS. A. Karne¹; M. Dalal¹; S. Natarajan². ¹New York University School of Medicine, Section of Primary Care, New York, NY; ²VA New York Harbor Healthcare System and New York University School of Medicine, New York, NY. (Tracking ID #136043)

BACKGROUND: Recent data indicate that intense lipid lowering treatment using HMG CoA inhibitors reduces recurrent cardiovascular events in post-myocardial infarction patients. We evaluated the lifetime costs and health benefits of aggressive lipid lowering with 80 mg atorvastatin versus usual care with 20 mg simvastatin for secondary prevention of recurrent cardiovascular events.

METHODS: We used a Markov decision analytic model with cohort simulation to evaluate the lifetime cost-effectiveness of the 2 strategies. The recommendations of the Panel on Cost-Effectiveness in Health and Medicine were followed. The primary outcome is a composite of cardiovascular death, MI, unstable angina requiring hospitalization and/or revascularization. Effectiveness data are from the Pravastatin or Atorvastatin Evaluation and Infection Therapy - Thrombolysis in Myocardial Infarction 22 study that enrolled patients within 10 days of acute MI in stable condition. That study had a relative risk reduction in the primary outcome of 16% (95% confidence interval: 5%-26%) with aggressive therapy. We assume that usual care with 20 mg simvastatin is equivalent to 40 mg pravastatin. Mortality data are from published CDC vital statistics. Life-years were converted into quality-adjusted life-years (QALYs) using established methods. We used published cardiovascular disease cost data and drug costs from the Veterans Health Administration System. All costs are in 2002 US dollars.

RESULTS: The reference-case analysis yields an incremental cost-effectiveness ratio (ICER) for aggressive therapy with atorvastatin of \$44,185 per QALY gained. Raising the price of atorvastatin above \$1650 for two years of therapy (112% of the reference-case price) makes the ICER >\$50,000/QALY. While aggressive therapy is not cost effective below a relative risk reduction of 11.5%, it is highly cost-effective at higher estimates. When the cost of treating coronary heart disease is varied from 50% to 150% of the reference-case, aggressive therapy remains cost effective. Varying the discount rate from 0% to 7.5% yields results consistently in favor of aggressive therapy.

CONCLUSIONS: A strategy of aggressive lipid lowering provides more QALYs at a modest cost. Except in cases when the relative risk reduction by aggressive therapy is small, aggressive lipid lowering therapy is cost effective. If the relative risk reduction by aggressive therapy is confirmed, health care systems and policy makers should institute a policy of aggressive lipid lowering in all post-myocardial infarction patients.

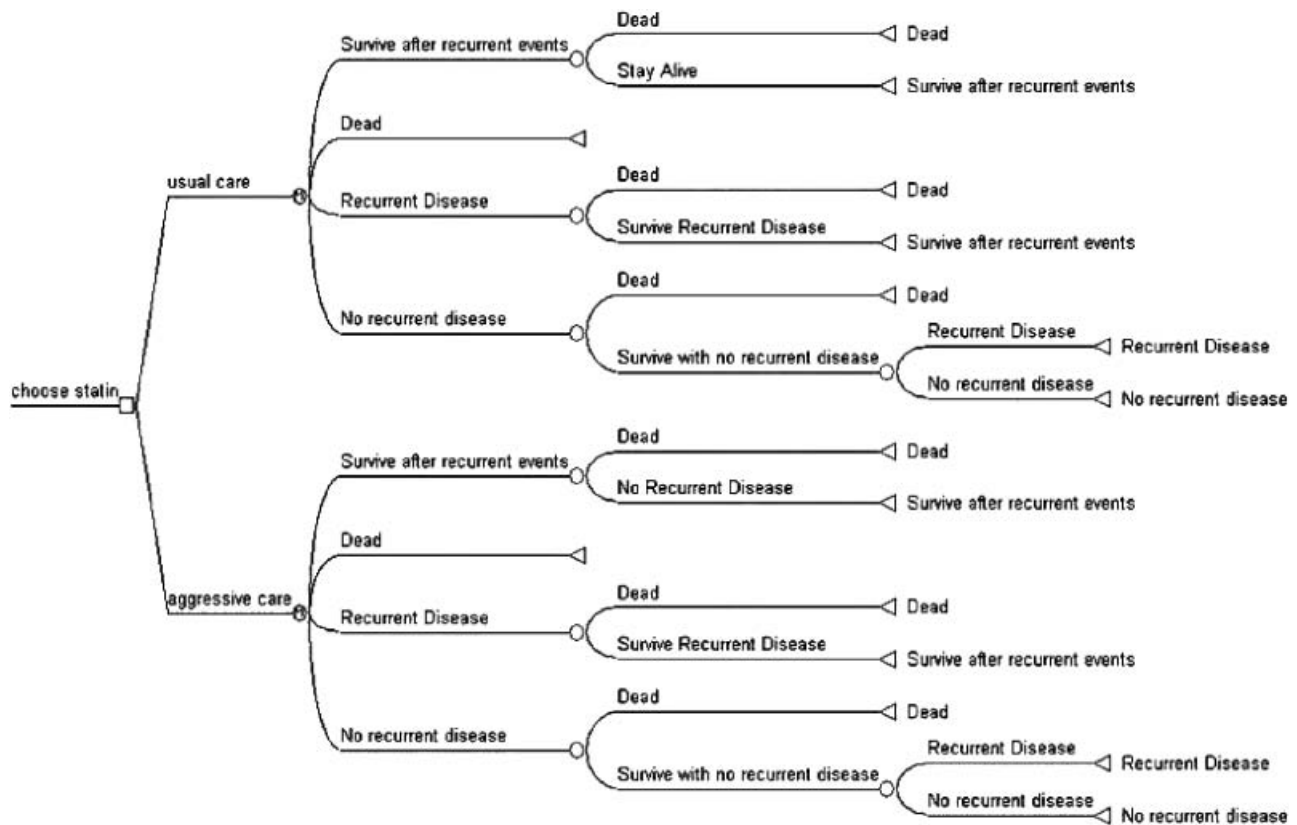


Figure 1: Decision Tree

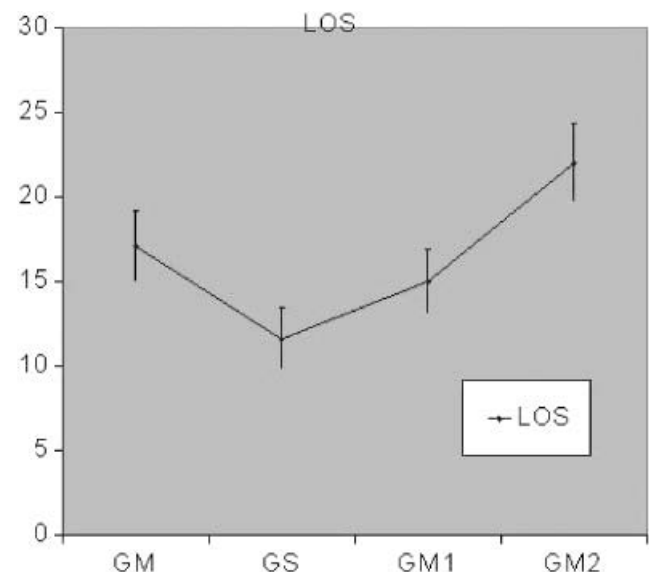
MEDICAL VERSUS SURGICAL MANAGEMENT OF EMPYEMA: A 7-YEAR EXPERIENCE IN A COMMUNITY HOSPITAL. S. Parikh¹; O. Schwetz¹; D. Panova¹; B. Kiesel¹; H. Friedman¹. ¹St. Francis Hospital, Evanston, IL. (Tracking ID #134413)

BACKGROUND: Complicated parapneumonic effusions (CPE) and empyema continue to account for significant morbidity and mortality, and uncertainties remain regarding their optimal management. The various treatment options are broadly categorized into: medical and surgical (either primarily or following failed medical therapy). Medical management involves the use of chest tube and/or fibrinolytics, whereas surgical management involves video-assisted thoracoscopic surgery (VATS) and if necessary, decortication with open thoracotomy. We chose to compare these two modalities of treatment.

METHODS: A retrospective review was performed on all adult patients discharged with a diagnosis of CPE and empyema from our hospital between January 1997

and December 2003. Data gathered included patient characteristics, co-morbid conditions, length of stay, treatment and outcomes. All patients received appropriate treatment for pneumonia, including systemic antibiotics. Forty-six patients were included in the study. They were grouped into medical management (GM, N=32, 70%) versus primary surgical management (GS, N=14, 30%). GM was further subdivided into successful medical (GM1) and failed medical (GM2, secondary surgical) groups. The baseline demographics and pleural fluid analysis of all the patients included in the study are shown in the table.

RESULTS: The average age of patients in GM was 61.6 years and in GS was 62.2 years. The number of males in GM was 23 (71%) and in GS was 10 (71%). The overall success rate for GM was 87.5% and for GS was 100%. There were 4 deaths



in GM compared to none in GS (mortality, 12.5% versus 0%).The length of stay (LOS) was significantly higher in GM compared to GS (17.1 days versus 11.6 days, $p=0.03$, student's t -test). Subgroup analyses showed that 12 patients in GM had successful resolution without surgical therapy (GM1, 37.5%) and 16 patients required surgical therapy (GM2, 50%). There were no deaths in GM2 and the average LOS for this group was 22 days. The average number of days after which VATS was done following failed medical therapy was 8.2.

CONCLUSIONS: There was a trend towards increased mortality, increased LOS and decreased success rate with medical management alone (GM). All patients who underwent secondary surgical therapy (GM2) as a rescue measure had an increased LOS but there was no difference in mortality. We conclude that primary surgical therapy (GS), because of higher efficacy, shorter LOS and less cost, may be the preferred modality of management. Large-scale randomized trials are warranted to confirm these observations and to determine the factors that predict a poor outcome and need for early surgical referral in these patients.

	GM	GS	Total
Number of patients, N	32	14	46
Average age, yrs	61.6	62.2	61.9
Male/ Female	23/9	10/4	33/13
Glucose (mg/dl)	48 $\pm$ 13.8	38 $\pm$ 9.2	43 $\pm$ 11.8
pH	7.28 $\pm$ 0.31	7.22 $\pm$ 0.14	7.25 $\pm$ 0.22
Total WBC (/cmm)	18800 $\pm$ 1500	17200 $\pm$ 2500	17650 $\pm$ 2130
LDH (IU/dl)	820 $\pm$ 270	1308 $\pm$ 700	1064 $\pm$ 462

MOVING BEYOND THE COST PER QALY: MODELING THE BUDGETARY IMPACT OF DRUG-ELUTING STENTS ON HEALTH SYSTEM PAYERS. F. Shrive¹; W.A. Ghali¹; S. Lewis¹; C. Donaldson²; M.L. Knudtson¹; B.J. Manns¹. ¹University of Calgary, Calgary, Alberta; ²University of Newcastle, Newcastle Upon Tyne, . (Tracking ID #135780)

BACKGROUND: Sirolimus-eluting stents have been shown to be efficacious but are also associated with a cost five times that of uncoated stents. We recently undertook an economic evaluation comparing sirolimus-eluting and uncoated stents in patients undergoing percutaneous coronary intervention (PCI). Overall, sirolimus-eluting stents were associated with an incremental cost per QALY of US \$48,139. However, cost-effectiveness varied substantially among certain patient subgroups. Moreover, in an attempt to further assist decision-makers with the decision as to whether, and for whom, to fund sirolimus-eluting stents, we modeled the impact of various funding scenarios on clinical outcomes and costs.

METHODS: Utilizing the Alberta Provincial Project for Outcomes Assessment in Coronary Heart disease (APPROACH) database, we are able to project the potential budgetary implications of funding sirolimus-eluting stents on a province-wide scale. Considering the number of PCIs performed in this province of approximately 3 million population, the average number of stents used per procedure, the cost of sirolimus-eluting stents, the projected efficacy of those stents in terms of deaths and second procedures avoided and lastly, the costs saved, we considered the implications of four funding scenarios; an across-the-board funding decision, restrict funding to patients with diabetes, restrict funding to those patients over 75 and restrict funding patients with complex lesions.

RESULTS: A fund-all scenario would require an additional US \$7.6 million (M) for the cardiac care budget in Alberta. This increased expenditure would be associated with a reduction of 6 deaths, 301 second revascularization procedures resulting in 2.0 M in averted health care costs over 1 year. Thus, the overall increase in health care costs for 1 year would be 5.6 M. Restricting funding to patients with diabetes would cost an additional \$1.1 M, avoid 2 of 6 of the potentially avoidable deaths (33%) and avoid 118 of 301 (39%) potentially preventable second procedures. Restricting funding to those over the age of 75 would cost an additional 0.9 M; prevent 2 deaths but only 39 (13%) second procedures. A funding strategy confining sirolimus-eluting stent use to patients with complex coronary lesions, using type "C" of the American College of Cardiology classification scheme as one indicator of complexity, prevents 5 of the 6 potentially avoidable deaths (83%) and 157 of 301 the preventable second procedures (52%) for only a \$0.3 M increase in expenditures.

CONCLUSIONS: Our findings indicate that the "economic case" for funding drug-eluting stents across-the-board is modest at best and that restricted funding scenarios (especially ones focusing on high risk lesions) are perhaps more reasonable. However, many complex issues remain. When considering the impact of restricted funding scenarios, a clear understanding of the trade-offs is important. Therapies currently being funded, the size of the local budget and, if necessary, what programme resources could be cut or scaled back to fund drug-eluting stents must all be explicitly considered.

PROVIDING PHARMACEUTICAL SAMPLES: MORE HARM THAN GOOD? M. Guidry¹; A. Toprani¹; J. Wiese¹. ¹Tulane University, New Orleans, LA. (Tracking ID #136173)

BACKGROUND: The pharmaceutical industry spends over 16 billion dollars per year on promotional activities. Providing free pharmaceutical samples is one of

the industry's methods to influence physician-prescribing behavior. This tactic capitalizes on physicians' desire to decrease costs to patients, and physicians' reluctance to stop medications one they have been started. Many of these sample medications do not meet first-line recommendations for the treatment of hypertension. We hypothesized that patients who were started on free samples would continue on these medications, even after the samples had been exhausted, resulting in an increased long-term cost to the patient.

METHODS: All samples dispensed at the Medical Center of Louisiana at New Orleans clinics are entered into a logbook. We reviewed the medical record of 171 consecutive patients that had received free anti-hypertensive samples in the previous three years. The charts were evaluated for two parameters: whether the patient could have received an alternative, less expensive, first-line medication as defined by JNC VII recommendations, and the duration the patient remained on the sample medication. Patients with a contraindication to recommended first-line therapies were excluded from the analysis. Two cost scores were calculated for each patient. The alternative cost (AC) was the cost expected if the patient had been on an alternative first-line medication (cost per day x number of days the patient was in the study). The sample cost (SC) was the cost the patient paid to remain on the sample drug after the patient ran out of samples (cost per day x number of days the patient was in the study minus the days samples were received). Costs for sample and alternative drugs was calculated using the Red Book. The total cost or savings to the patient was calculated by subtracting the alternative cost (AC) from the sample cost (SC).

RESULTS: Five classes of anti-hypertensive sample medications alone and in combination were dispensed. Of the 171 patients in the study, 109 patients had a complete medical record available for analysis. No patient had a contraindication to all first-line recommended therapies. Patients spent on aggregate 22,280 days on free samples. These same patients remained on these medications for a total of 46,718 days. The mean cost to the patient receiving free sample medications was \$228/year in excess of what they would have paid had they been prescribed a first-line alternative medication without samples.

CONCLUSIONS: Dispensing pharmaceutical samples in lieu of prescribing less expensive alternatives that meet first-line recommendations results in higher expenses to the patient, even after adjustment for the free sample period.

PUTTING THEIR MOUTH WHERE THEIR MONEY IS: ARE PATIENTS ENROLLED IN INCENTIVE-BASED PHARMACY PLANS MORE LIKELY TO TALK TO THEIR PHYSICIANS ABOUT OUT-OF-POCKET COSTS? W.H. Shrank¹; S.M. Asch²; S. Ettner³; A. Kirk⁴; P.A. Glassman⁵; S.A. Fox³. ¹UCLA/Greater Los Angeles VA Healthcare System, Los Angeles, CA; ²RAND, Los Angeles, CA; ³University of California, Los Angeles, Los Angeles, CA; ⁴UCLA, Los Angeles, CA; ⁵VA, West LA, Los Angeles, CA. (Tracking ID #133866)

BACKGROUND: Multi-tiered, incentive-based formularies have been widely instituted to control the rising costs of prescription drugs. To work properly, such formularies rely on patients to be aware of incentives and communicate their preferences with prescribing physicians. The impact of financial incentives on patient awareness of out-of-pocket costs and physician-patient communication about those costs is unknown. We studied whether patients enrolled in incentive-based pharmacy benefit plans are aware of out-of-pocket cost requirements when prescriptions are written, and whether incentives influence rates of patient-physician communication about out-of-pocket costs.

METHODS: We performed a matched telephone survey of primary care physicians in Southern California and their patients. A random sample of 2626 patients from the panels of 111 primary care physicians in Southern California were surveyed. 1917 patients responded (73% response rate). We gathered descriptive information about demographic and practice characteristics of physicians as well as their patients' health, socioeconomic status and other demographic information. We identified patients' reported enrollment in tiered pharmacy benefit structures, knowledge of out-of-pocket costs, and rates of discussion of out-of-pocket costs with physicians.

RESULTS: 57% of patients with prescription drug coverage reported being enrolled in multi-tiered plans. The majority of these (54%) were "never" or "sometimes" aware of their out-of-pocket cost requirements at the time of the physician visit. 68% of all patients, regardless of benefit structure, reported they never discuss out-of-pocket costs for prescription drugs with physicians. Patients enrolled in incentive-based pharmacy benefits were 81% ($p<.01$) more likely to discuss out-of-pocket costs with physicians compared to patients enrolled in single-copayment or no co-payment plans. However, even among patients enrolled in multi-tier plans, almost 66% reported "never" discussing out-of-pocket costs with their physicians, with this percentage rising to 79% for patients paying a single copayment for all drugs.

CONCLUSIONS: Multi-tier pharmacy benefit plans increase rates of discussions about out-of-pocket costs. Nonetheless, most multi-tier enrollees are unaware of out-of-pocket costs when prescriptions are written and most never discuss out-of-pocket costs with their physicians, casting doubt on the effectiveness of financial incentives to guide decision-making as intended. Interventions to improve patient access to out-of-pocket cost information and the frequency of patient-physician discussions about costs are needed.

THE ASSOCIATION OF PATIENT ATTITUDES TOWARD HEALTH CARE SCREENING WITH PREVIOUS SCREENING. S.D. Ellis¹; D. Bonds¹; S.L. Palla¹; E. Weeks¹; P. Lichstein¹. ¹Wake Forest University, Winston-Salem, NC. (Tracking ID #135234)

BACKGROUND: Routine screening for, and subsequent counseling to change, risky health behaviors, is frequently recommended to reduce the risk of morbidity and mortality and to increase quality of life. However, healthcare providers are often reluctant to ask about the more sensitive health behaviors, such as

domestic violence, drug use and sexual functioning, for fear of offending the patient. The objective of this study was to determine if recent screening by health care providers for sensitive health behaviors increased patients' acceptance of routine screening for that behavior.

METHODS: In a telephone survey of women in established relationships with primary care providers, patients were asked if they had been screened for 8 different health behaviors (exercise, smoking, use of alcohol or drugs, excessive stress, sexual functioning concerns, safety/violence in the home, guns in the home) in the past year and their attitude toward healthcare providers' routine screening for that risk factor. We collapsed response categories into dichotomous variables of pro-Routine Screening—belief that patients should be screened annually or at every visit—and anti-Routine Screening—belief that women should not be screened, or should only be screened if there is reason. Using these variables, we calculated the odds ratio of favoring routine screening among patients who had been screened for that behavior in the last year.

RESULTS: The analysis sample includes 1,588 women. The median age was 48 years; 63% were white; 48% married; 49% had household income less than \$25,000 per year; 87% had medical insurance. Fifty-two percent had a female PCP; 60% were seeing a family medicine physician; and 43% had seen a provider five or more times in the past 12 months. Seventy-three percent had been screened in the last 12 months for exercise and 57% for smoking, 41% for alcohol/drugs, 41% for stress, 21% for sexual functioning, 16% for safety/violence, and 6% for guns. There was high patient acceptance of routine screening for exercise (76%), smoking (70%), alcohol/drugs (71%), and stress (62%), but less for sexual functioning (37%), safety/violence (37%), or guns (23%). For all health behaviors measured, there was a higher likelihood of agreeing with the practice of routine screening if the participant had been screened in the past year: exercise (OR 2.8, 95% CI 1.9-4.2), smoking (OR 1.3, 95% CI 1.0-1.8), alcohol/drugs (OR 2.3, 95% CI 1.8-2.9), stress (OR 1.8, 95% CI 1.3-2.6), sexual functioning (OR 2.7, 95% CI 1.9-3.7), safety/violence (OR 3.8, 95% CI 3.0-4.9), and guns (OR 5.2, 95% CI 3.4-8.1).

Screening Question:	Unadjusted OR 95% CI	Adjusted OR* 95% CI
Exercise	2.8 1.6-2.9	2.8 1.9-4.2
Smoking	1.3 1.1-1.9	1.3 1.0-1.8
Use alcohol or drugs	2.3 1.4-2.4	2.3 1.8-2.9
Are under excessive stress	1.8 1.3-2.0	1.8 1.3-2.6
Have sexual functioning concerns	2.7 2.2-3.5	2.7 1.9-3.7
Have concerns about safety or violence in the home	3.8 3.0-6.0	3.8 3.0-4.9
Have a gun in the home	5.2 3.9-8.6	5.2 3.4-8.1

*Adjusted for patient demographics (age, marital status, race, income, insurance), provider characteristics (PCP gender, PCP type, number of visits), patient satisfaction (privacy, comfort talking about sensitive issues, patient trust), and clustering by clinic.

Table: Odds ratios of agreeing with routine screening for a particular health behavior if the participant had been screened for the behavior in the past year

CONCLUSIONS: In this analysis previous screening predicted a positive attitude toward routine screening. This relationship was especially strong for more sensitive health behaviors. Further research is needed to determine causality. However, health care providers can have a powerful influence on health behaviors. If our finding is substantiated, it provides evidence that health care providers can also affect patients' perceptions of what healthy behaviors are.

THE EFFECTS OF RECOMMENDED THERAPY VERSUS REAL WORLD THERAPY ON THE DIRECT MEDICATION COSTS OF TYPE 2 DIABETES. C. Criviera¹, D. Suh², E. Huang³, E. Cagliero⁴, R. Grant⁵, L. Vo⁶, H. Shin⁷, J. Meigs⁸, ¹sanofi-aventis, Bridgewater, NJ; ²Ernest Mario School of Pharmacy, Rutgers University, Piscataway, NJ; ³University of Chicago, Chicago, IL; ⁴Massachusetts General Hospital, Boston, MA; ⁵Massachusetts General Hospital, Boston, MA; ⁶Ernest Mario School of Pharmacy, Piscataway, NJ; ⁷Harvard Medical School and Massachusetts General Hospital, Boston, MA. (Tracking ID #133669)

BACKGROUND: Many patients with diabetes in community practice do not receive optimal medical therapy. Treatment with antihyperglycemic, antihyperlipidemic, and antihypertensive medications can be optimized by changing medication dosages or adding additional agents to reach target treatment goals. The purpose of this study was to estimate the incremental medication cost of providing recommended therapy vs. usual diabetes medication therapy to achieve treatment goals for diabetes in a primary care setting.

METHODS: Six hundred and one patients with type 2 diabetes receiving care in the outpatient clinics of Massachusetts General Hospital between March 1, 1996 and August 31, 1997 were randomly selected for medical record data abstraction. Treatment algorithms were applied based on 2004 practice recommendations for hyperglycemia, hyperlipidemia, and hypertension targets in order to move patients from actual metabolic control to target metabolic control. Four physician reviewers independently applied the algorithms and con-

sensus was obtained on recommended therapies for all patients. Mean incremental medication costs of changes to recommended from "real world" therapy were calculated with 95% bootstrap confidence intervals. We used analysis of covariance to test the significance of differences in incremental costs after adjusting for baseline differences in clinical characteristics.

RESULTS: Almost all patients (98%) were recommended to have changes in some component of their medication regimens. Among patients who were recommended to have a change in therapy, 83% of patients had an increase in dose of one of their medications, 78% had at least one addition of new medication, and 4% discontinued one of their current medications. Change recommendations resulted in the following mean yearly medication cost increases: 1) For aspirin, baseline treatment costs of \$1/year increased by \$3 (95% CI \$3-4) if generics were used for regimen adjustment or by \$14 (95% CI, \$13-15) if brand originators were used; 2) For antihyperglycemics, baseline costs of \$587/year increased by \$168 (95% CI, \$133-206; generics) or by \$192 (95% CI, \$154-230, brand originators); 3) For antihypertensives, baseline costs of \$531/year increased by \$75 (95% CI, \$57-93; generics) or by \$150 (95% CI, \$128-173, brand originators); and 4) For antihyperlipidemics, baseline costs of \$406/year increased by \$392 (95% CI, \$354-434; generics) or by \$528 (95% CI, \$487-573, brand originators). Overall, yearly medication cost per patient increased by \$639 (95% CI, \$584-694) from \$1525 to \$2164 if generics were used and by \$884 (95% CI, \$825-949) from \$1,999 to \$2,884 if brand name originators were used.

CONCLUSIONS: Changing "usual care" to recommended care would result in a modest cost increment for most patients. This increment is substantially reduced by the use of generic medications. These results provide clinicians, payers, health plans, and other healthcare decision makers valuable economic data abstracted from 'real world' clinical conditions to estimate the impact of fully implementing optimal medical therapy to reach treatment targets for patients with type 2 diabetes.

THE IMPACT OF ADVERSE EVENTS ON WARFARIN PRESCRIBING IN ATRIAL FIBRILLATION: A MATCHED-PAIR ANALYSIS. N.K. Choudhry¹, G.M. Anderson², A. Laupacis³, D. Ross-Degnan³, S. Normand⁴, S. Soumerai¹. ¹Harvard Medical School, Boston, MA; ²University of Toronto, Toronto, Ontario; ³Harvard Medical School and Harvard Pilgrim Health Care, Boston, MA; ⁴Harvard University, Boston, MA. (Tracking ID #133034)

BACKGROUND: Warfarin is received by only 30-60% of appropriate patients with atrial fibrillation (AF). Physicians' experiences with warfarin-associated adverse events may be an important determinant of warfarin use. It has been suggested that some adverse events (i.e., major hemorrhage in an AF patient treated with warfarin) may be more influential than others (i.e., thromboembolic stroke in an AF patient who did not receive warfarin) in determining how physicians use warfarin.

METHODS: Using linked administrative data for 116,200 elderly patients with non-valvular, non-transient AF, we identified 3921 patients who were hospitalized for major hemorrhage while on warfarin and 8720 hospitalized for thromboembolic strokes while not on warfarin. We identified the physicians responsible for the care of these patients and selected pairs of newly discharged patients treated by that physician, one patient before and one patient after the adverse event (i.e., exposure). Using this paired analysis we compared the odds of warfarin receipt for pre- and post-exposure patients after adjusting for stroke and bleeding risk factors that also might influence warfarin use. We also evaluated a comparison outcome - the odds of angiotensin converting enzyme (ACE) inhibitor prescriptions before and after exposure - that should not have been influenced by warfarin-associated adverse events.

RESULTS: For the 530 physician who were exposed to an adverse bleeding event and treated other AF patients during both the 90-day period before and the 90-day period after exposure, the odds of warfarin prescribing was 21% lower for post-exposure patients (adjusted OR 0.79, 95% CI: 0.62-1.00). Greater reductions in warfarin prescribing were found in analyses using patients for whom more time had elapsed between physician exposure and patient treatment. There were no statistically significant changes in warfarin prescribing after a physician had a patient experience a stroke while not on warfarin or in the use of ACE inhibitors by physicians who had patients with either bleeding events or strokes.

CONCLUSIONS: These results suggest that a physician's experience with warfarin-associated bleeding events can influence warfarin use. In contrast, adverse events associated with underuse of warfarin may not affect subsequent prescribing. Our findings highlight the need for strategies to modify physicians' perceptions of the risks associated with warfarin use and to encourage warfarin prescribing for patients with AF.

THE IMPACT OF PATIENT PREFERENCES ON THE COST-EFFECTIVENESS OF INTENSIVE GLUCOSE CONTROL IN OLDER PATIENTS WITH NEW ONSET DIABETES. E.S. Huang¹, M. Shook¹, L. Jin¹, M.H. Chin¹, D.O. Meltzer¹. ¹University of Chicago, Chicago, IL. (Tracking ID #134220)

BACKGROUND: Utilities are quantitative measures of preference that are used in the calculation of quality-adjusted life years (QALYs) in cost-effectiveness analyses (CEA). CEA in diabetes care have repeatedly reported that intensive glucose control is not cost-effective in older patients with new-onset diabetes. However, these analyses had limited data on diabetic health state utilities and have not evaluated the sensitivity of results to assumptions about complication and treatment utilities.

METHODS: We examined the sensitivity of a published model of the cost-effectiveness of intensive glucose control (the NIH Eastman Model) to alternative utility assumptions. The published model used complication utilities from the

literature based on diverse methods and assumed all treatment utilities were 1. From 519 patients over 65 living with diabetes, we systematically collected health state utilities for multiple complication (amputation, blindness, end-stage renal disease) and glucose control treatment states (conventional glucose control, intensive control with insulin, intensive control with oral medications). We then compared incremental CEA ratios (ICER)(\$/QALY) for hypothetical patients over 60 to 90 with new onset diabetes when 1) using all original utilities, 2) when using patient-derived complication utilities while retaining original treatment utilities, and 3) when using patient-derived utilities for complications and treatments in combination.

RESULTS: Incorporating our patient-derived complication utilities while maintaining baseline assumptions about treatment utilities improved ICERs for intensive glucose control (e.g. patients 60-65 with new diabetes, \$136,422/QALY $\rightarrow$ \$77,845/QALY) but intensive therapy was still not cost-effective at older ages. When treatment utilities were also incorporated, the cost-effectiveness of intensive treatment depended on the method of glucose control. Intensive control with insulin generated fewer QALYs than conventional control. However, intensive oral medication was beneficial at all ages and had an ICER < \$100,000 to age 70. The findings with patient-derived treatment utilities did not change significantly when using the original model utilities or patient-derived utilities for complications.

CONCLUSIONS: The cost-effectiveness of intensive glucose control in older patients with new-onset diabetes is sensitive to assumptions regarding quality of life with complications and highly sensitive to assumptions regarding quality of life with treatments. CEA of diabetes care should consider the sensitivity of results to alternative utility assumptions.

TRUSTWORTHINESS OF DIFFERENT SOURCES OF HEALTH INFORMATION. J.P. Mettlay¹; J. Maselli²; C.A. Camargo, Jr³; R. Gonzales². ¹University of Pennsylvania and VA Medical Center, Philadelphia, PA; ²University of California, San Francisco, San Francisco, CA; ³Massachusetts General Hospital, Boston, MA. (Tracking ID #136068)

BACKGROUND: Health information can be delivered to patients from a variety of sources but it is unclear which sources are most effective at conveying messages. As part of a national study to change patient demand for antibiotics, we interviewed patients seeking urgent care services in academic-affiliated acute care settings to 1) describe the spectrum of health information sources accessed in the past year, and 2) estimate the relative rank order of trustworthiness of various health information sources.

METHODS: An interviewer-administered survey was conducted between July and August 2004 in waiting rooms of three emergency departments (VA hospitals in San Francisco and Philadelphia and the Massachusetts General Hospital) and one adult urgent care clinic (UCSF). Adults (age > 18 years) were excluded if they did not speak English, were cognitively impaired, or were presenting with severe illness. A convenience sample of 100 subjects were enrolled from each of the four study hospitals. Respondents rated the trustworthiness of health information sources using a 5-point Likert scale. All sources were ranked from most to least trustworthy. The Wilcoxon rank sum test and ordinal logistic regression analysis were used to measure differences in trustworthiness between groups.

RESULTS: The sources of health information most frequently accessed during the past year, in descending order, were: physician (90%), nurses (75%), newspapers (62%), internet (52%), pharmacist (50%), friend or family (48%), telephone nurse (25%), and health fair (18%). After adjusting for age and sex, there were no differences between VA and non-VA patients in the frequency with which these sources were accessed. Among the different types of health information sources: "Your Own Doctor" rated the highest in trustworthiness, followed (in order) by the respondent's specific hospital, a government source (Centers for Disease Control and Prevention), and the respondent's local health department. An internet source (WebMD) and national television programs (Oprah Winfrey and Dr. Phil) were ranked lowest. Non-VA patients rated their own hospital an average of 0.5 points higher on the trustworthiness scale compared with VA patients (p=0.02).

CONCLUSIONS: Patients seeking care in urban emergency departments and urgent care clinics identify local sources of health information as the most trustworthy. Health educators should recognize the importance of providing local branding of messages when implementing health education campaigns.

UNDERSTANDING PATIENT FLOW - MEDICAL RESIDENTS' PERSPECTIVES ON OPTIMIZING INPATIENT ROUNDS. R. Tamler¹; J. Dedier². ¹Mount Sinai School of Medicine, New York, NY; ²Boston University, Boston, MA. (Tracking ID #135239)

BACKGROUND: Resident physicians play an important role in the discharge process at academic medical centers. Hoping to achieve earlier discharge times and improve the flow of admissions from the Emergency department (ED), hospital administrators have encouraged residents to see potentially dischargeable inpatients first on rounds. However, little is known about residents' preferred rounding patterns and whether they believe that rounding on potentially dischargeable inpatients first would actually lead to earlier hospital discharge times.

METHODS: All 46 2nd-year and 44 3rd-year medical residents at a large, urban academic medical center were surveyed by email about their preferred rounding strategy. They also were asked whether they believed that rounding first on inpatients considered to be potentially appropriate for discharge that day would improve overall hospital discharge times.

RESULTS: 58 of 90 email recipients responded to the survey (64%). Most residents (60%) favored rounding geographically by inpatient ward with all mem-

bers of the team present at all times (54%). 24 respondents (41%) believed that prioritizing potential discharges might lead to earlier discharge times. However, only 4 residents (7%) followed that approach on rounds.

CONCLUSIONS: Despite encouragement by hospital administrators, few 2nd- and 3rd-year medical residents at a large teaching hospital believed that seeing potentially dischargeable patients first would lead to earlier discharge times. Fewer, still, used this rounding approach. These results suggest that an understanding of resident physicians' interests, objectives and priorities for rounds is required to better engage them in efforts to improve patient flow.

WHO MAKES THE DIAGNOSIS AT AN ACADEMIC MEDICAL CENTER: THE ROLE OF CLINICAL SKILLS AND DIAGNOSTIC TEST RESULTS. D. Wahner-Roedler¹; S. Chaliki²; B. Bauer³; J.B. Bundrick¹; L.R. Bergstrom¹; M.C. Lee¹; P.L. Elkin¹. ¹Mayo Clinic, Rochester, MN; ²Mayo Clinic, Scottsdale, AZ. (Tracking ID #132664)

BACKGROUND: Advances in medicine have lead to the development and availability of a multitude of diagnostic tests and subspecialty evaluations. The contribution of the general internist in the context of all these advances appears to be undervalued. This study was inspired by the seemingly conflicting anecdotal data that many services are claiming responsibility for making more than fifty percent of the diagnoses in hospitalized patients. The objective of this study was to evaluate the contribution of clinical skills and diagnostic test results in arriving at the final diagnosis in this patient population.

METHODS: The records of all patients (n=248) admitted to a general internal medicine hospital service during a 3 months period in 2000 were reviewed by a board certified internist (SSC). The following information was available to the reviewer: inpatient, and outpatient record, laboratory results, radiology and pathology reports and the final diagnosis for the episode of care listed on the Mayo Master Sheet Index. All modalities that yielded the correct final diagnosis were recorded along with the date and time the study was performed. For example diagnostic credit would be given to an admitting clinician who diagnoses a pneumonia based on history and physical examination, although a confirmatory chest x-ray may have been performed. If the clinician said "pneumonia vs. congestive heart failure" and the chest x-ray showed an infiltrate consistent with pneumonia, credit for the diagnosis was awarded to the radiologic study. We subcategorized between "exact" and "close to exact" diagnosis. An example for the later would be if the clinician suspected anemia and the patient was later diagnosed with anemia due to a GI bleed.

RESULTS: All cases had a firm diagnosis by three months following hospitalization. Of the 248 cases, there was an "exact" in hospital diagnosis made in 215 cases, the diagnosis was "close to exact" in 31 cases and no diagnosis was made in 2 cases. Of the 246 patients in whom a final "exact" or "close to exact" diagnosis was made during their hospitalization the diagnoses were made by clinical judgment of the general internist in 50.4%, by a radiologic study in 31.7%, by a blood test or culture result in 9.4%, by biopsy in 3.3% and by various other modalities (endoscopy, echocardiogram, EMG, EEG, etc) in 5.2%. Of the 215 cases where an "exact" diagnosis in the hospital was identified 48.4 % of the diagnoses were made by clinical judgment of the general internist, 33.5% by a radiologic study, 9.8 % by a blood test or culture result, 3.7 % by biopsy and 4.6 % by various other modalities (endoscopy, echocardiogram, EMG, EEG, etc.).

CONCLUSIONS: This study shows that clinical expertise and skills are still significant factors with respect to making the correct diagnoses in hospitalized patients on a general medicine service at our institution.

A SPATIAL ANALYSIS OF STROKE MORTALITY IN NORTH CAROLINA. J.D. Joines¹. ¹Moses Cone Health System and the University of North Carolina at Chapel Hill, Greensboro, NC. (Tracking ID #135984)

BACKGROUND: High rates of stroke mortality have been observed for decades in the coastal plain of North Carolina and other southeastern states, an area known as the "stroke belt". The goal of this study was to examine county-level stroke mortality rates using exploratory spatial analysis and spatial regression models.

METHODS: Age-adjusted county-level stroke mortality rates for decedents aged 35 years and older were obtained for the 10-year period 1989 through 1998 using compressed mortality files from the Centers for Disease Control and Prevention (CDC), available through the CDC WONDER Online Database. County-level explanatory variables included socioeconomic predictors (poverty rate, percent urban population, percent nonwhite population, percent college graduates) and health resource predictors (hospital bed density and primary care physician density).

RESULTS: County-level stroke mortality rates ranged from 77 to 239 (mean 155) deaths per 100,00 for all decedents combined. Significant spatial autocorrelation was present (p-value < 0.001 for Moran's I), indicating spatial clustering of high-rate counties. A nonspatial (ordinary least squares) regression model explained 45% of the variation in stroke mortality rates; however, residuals showed spatial dependence, indicating that the clustering of high-rate counties was not accounted for by the model. In a spatial lag model, stroke mortality rates increased with higher percent nonwhite population and decreased with higher percent college graduates; other predictors were not significant.

CONCLUSIONS: North Carolina counties with high rates of stroke mortality are spatially clustered, and this clustering is not explained by the socioeconomic and health resource predictors considered. Stroke mortality rates are lower in counties with higher educational levels. As expected, stroke mortality rates are higher in counties with higher percent nonwhite population.

A SYSTEMATIC REVIEW OF THE EXCLUSION CRITERIA OF RANDOMIZED CONTROLLED TRIALS PUBLISHED IN MAJOR MEDICAL JOURNALS. H.G. Van Spall¹; A. Toren¹; R. Fowler¹. ¹University of Toronto, Toronto, Ontario. (Tracking ID #136201)

BACKGROUND: The under-representation of certain patient populations in clinical trials has come under scrutiny in recent years. This issue is of special significance to general internists, who must make sense of the chaos; they must interpret and apply the results obtained from carefully-selected participants in clinical trials to the wide spectrum of patients they see in clinical practice. The goal of our study was to determine whether the exclusion criteria in clinical trials play a role in the under-representation of specific populations, and if so, to offer recommendations that would make trials more generalizable to the population seen in clinical practice.

METHODS: We searched the MEDLINE database for randomized controlled trials (RCTs) published in *Annals of Internal Medicine*, *British Medical Journal*, *JAMA*, *New England Journal of Medicine*, and *Lancet* between 1998 and 2003. 50 of the 1621 resulting articles were selected in series, in a manner that prevented over-sampling from any one journal. Details regarding the accrual, exclusion, and follow-up of subjects, as well as the inclusion and exclusion criteria were extracted. Each of the exclusion criteria were graded as "strongly justified", "potentially justified" or "unjustified" according to published guidelines as well as consensus between the authors.

RESULTS: Only 72% of trials outlined exclusion criteria within the manuscript. Conditions related to female gender were grounds for exclusion in 42% of all trials. Subjects were excluded due to age, most commonly age less than 18 years, in 68% of all trials. Common medical conditions formed the basis for exclusion in a large proportion of trials; these included renal insufficiency (28%), cardiac disease (22%), unspecified medical illness (22%), liver disease (20%), neurological disease (18%), hypertension (10%) malignancy (10%), HIV infection (10%), and psychiatric illness (10%). 48% of trials excluded subjects for medication-related reasons, and 24% excluded on the basis of abnormalities in complete blood count. Other common exclusions were physical or cognitive impairment (14% of trials), social circumstances (14%), communication barriers (10%), limited life expectancy (8%), and race (4%). Only 51% of all exclusion criteria were deemed strongly justified in the context of the individual RCT; 39% were graded as unjustified. Nearly 75% of all trials contained at least one unjustified exclusion criterion. There were significantly greater numbers of unjustified exclusions among RCTs evaluating medications as opposed to other interventions (4.7 vs. 2.5 exclusions per trial; mean difference 2.8; 95% confidence interval 0.04-4.5; $p=0.04$).

CONCLUSIONS: Our study indicates that RCTs published in major medical journals, and particularly those RCTs evaluating medication therapies, may systematically exclude women, children, and subjects with common medical conditions, often without valid justification. We recommend careful selection of exclusion criteria in order to improve RCT generalizability, and full reporting of all inclusion and exclusion criteria. We envision a new "table 0", in which authors clearly outline the trial's inclusion and exclusion criteria, and specify the number of subjects that were excluded on the basis of each criterion. When exclusions are necessary, we recommend that each criterion be justified.

AN ADMINISTRATIVE DATA MERGING SOLUTION FOR DEALING WITH MISSING DATA IN A CLINICAL REGISTRY: ADAPTATION TO ICD-10. D. Southern¹; C.M. Norris²; H. Quan¹; F. Shrive¹; P. Galbraith¹; K. Humphries³; M. Gao³; M.L. Knudson¹; W.A. Ghali¹. ¹University of Calgary, Calgary, Alberta; ²University of Alberta, Edmonton, Alberta; ³University of British Columbia, Vancouver, British Columbia. (Tracking ID #132532)

BACKGROUND: We have previously described (J Clin Epi, 2000; 53: 377-383) a method for dealing with missing data in clinical research that involves merging administrative data coded in the 9th revision of the International Classification of Diseases (ICD-9-CM) with cardiac registry data to fill missing data 'holes' in the latter. Much of the world, including Canada, has now transitioned to the 10th revision (ICD-10) - a change that challenges researchers to derive new administrative data coding algorithms to replace widely-used ICD-9-CM algorithms. Responding to this challenge, we conducted a multi-step ICD-9-CM to ICD-10 translation process to derive an ICD-10 version of our missing data methodology, and then assessed its performance relative to that of our original method.

METHODS: We used cardiac registry data from the Alberta Provincial Project for Outcome Assessment in Coronary Health Disease (APPROACH) for this methodological research. Data for patients undergoing cardiac catheterization in fiscal years 1995-2003 were used. We also obtained corresponding administrative hospital discharge records for the same clinical care episodes, coded in ICD-9-CM for 1995-2001 and in ICD-10 for 2002-2003. We convened groups of clinical and coding experts, and through an iterative process derived a comprehensive listing of ICD-10 codes to represent each of the 18 clinical diagnoses (e.g., congestive heart failure, renal disease, hyperlipidemia) that need to be defined in administrative data for our data merging process. We then linked individual administrative data records to the corresponding records in the cardiac registry, and individual data fields were coded as 'present' when a clinical diagnosis was coded as present in either the original cardiac registry record or in administrative data. The resulting 'merged' database - devoid of missing data fields - was then evaluated for its ability to predict the death at one year based on its constituent variables. We derived two logistic regression models, the first for ICD-9-CM data (1995-2001) and the second for ICD-10 data (2002-2003), and assessed their performance using the c-statistic and a decile-of-risk table.

RESULTS: The old and new coding algorithms will be presented in detail. The prevalence of the individual clinical variables subjected to merging increases gradually across years, indicating a gradual increase in the average severity of illness. There was, however, no evidence of either an abrupt drop or rise in prevalence of any of the variables. The performance of the new data merging

model was strong, and comparable to that of our previously reported ICD-9-CM methodology: c-statistic=0.801 (95% confidence interval [CI] 0.792-0.810) for the ICD-10 model and c-statistic=0.783 (95% CI 0.778-0.788) for the ICD-9-CM model. The ICD-10 data model separated low risk from high risk cases just as well as does the ICD-9-CM model. The risk of death across deciles of expected risk of death ranged from 0.6 to 19% for the ICD-10 method and from 0.7 to 20% for the ICD-9-CM method.

CONCLUSIONS: The ICD-10 implementation of our data merging method performs at least as well as the previously-validated ICD-9-CM method. Such methodological research is an essential prerequisite to future health services research now that health systems are transitioning to the ICD-10 coding system.

APOLIPOPROTEIN E (APOE) GENE POLYMORPHISM AND CAROTID ATHEROSCLEROSIS IN THE NORTHERN MANHATTAN STUDY. K. Anand¹; R. Tadjana²; A. Bewtra¹; R. Sacco³. ¹Creighton University, Omaha, NE; ²Columbia University, Omaha, NE; ³Columbia University, New York, NY. (Tracking ID #136011)

BACKGROUND: Atherosclerosis is characterized by a complex interaction between genetic and environmental factors. Previous studies have suggested association between Apolipoprotein E (APOE) gene polymorphism and atherosclerosis. Maximal carotid plaque thickness (MCPT) is an ultrasound surrogate marker for atherosclerosis. Our aim was to investigate the association between APOE gene polymorphism and MCPT.

METHODS: A population-based cohort of 3298 stroke-free subjects was prospectively followed for stroke, myocardial infarction, and death in the Northern Manhattan Study (NOMAS). All the subjects had a baseline evaluation and information was collected on demographics and risk factors. We analyzed a sample of 621 subjects who had a carotid ultrasound examination and APOE genotyping completed. The APOE polymorphism was grouped into APOE4+ (E2/E4, E3/E4, E4/E4 genotypes) and APOE4- (E2/E2, E2/E3, and E3/E3 genotypes). Mean MCPT was calculated for sociodemographic factors, vascular risk factors and APOE groups. The continuous MCPT variable was dichotomized at the 75th percentile (1.9mm). The odds ratio (OR) for MCPT > 1.9 mm was calculated across the sociodemographic factors, vascular risk factors and APOE groups using univariate and multivariate logistic regression models.

RESULTS: The mean age was 69 +/- 10 years; 61% females; 27% African-American, 42% Hispanics, and 32% Caucasian. The mean MCPT was 1.08 +/- 1.19 mm and median was 0.92 mm. Distribution of APOE was as expected from Hardy-Weinberg proportions. Age > 65 (OR 5.71; 95% CI, 3.25-10.04), hypertension (2.10, 1.38-3.20), C-LDL (1.47, 1.01-2.14), cardiac disease (2.10, 1.38-3.19), former (1.75, 1.16-2.65) and current smoking (1.77, 1.02-3.06) were significantly associated with MCPT > 1.9 mm. The Hispanic race-ethnicity (0.43, 95%CI=0.27-0.68) was associated with MCPT less than 1.9mm. In a univariate analysis, APOE4+ was associated with an increase risk of MCPT > 1.9 mm among men (1.86, 1.02-3.40), but not among women (0.68, 0.38-1.21). APOE4+ was not associated with an increase risk of MCPT > 1.9 mm when stratified age and race-ethnicity. In the multivariate analysis, APOE4+ was significantly associated with an increased risk of MCPT > 1.9 mm among men even after adjusting for sociodemographics and other conventional vascular risk factors (2.46, 1.24-4.86).

CONCLUSIONS: Although the polymorphism of APOE is an important factor for development of cardiovascular diseases, our results showed that APOE4+ was only marginally related to carotid artery atherosclerosis. There is a deleterious effect of APOE4+ for atherosclerosis among men. Further studies regarding gender effect of APOE on atherosclerosis are needed.

ASSOCIATION OF THE K/DOQI CLASSIFICATION SYSTEM OF CHRONIC KIDNEY DISEASE AND MORTALITY IN A HEART FAILURE POPULATION. N.A. Khan¹; I.W. Ma¹; C. Thompson¹; K. Humphries²; D. Salem²; M.J. Sarnak²; L. Adeera¹. ¹University of British Columbia, Vancouver, British Columbia; ²Tufts University, Boston, MA. (Tracking ID #135037)

BACKGROUND: The Kidney Disease Outcomes Quality Initiative (K/DOQI) classification system, developed using general population data from NHANES III, is the new standard for staging severity of chronic kidney disease. The utility of this staging system to predict outcomes has not been validated in a high-risk patient population. As kidney disease is common among patients with heart failure, we sought to determine the utility of the K/DOQI classification system to predict mortality in a heart failure population.

METHODS: We conducted a retrospective analysis of the 6640 participants with asymptomatic and symptomatic heart failure from the Studies of Left Ventricular Dysfunction (SOLVD). Glomerular filtration rates (GFR) were categorized according to the K/DOQI classification system into 4 categories: ≥ 90 ml/min per 1.73 m², 60-89 ml/min per 1.73 m², 30-59 ml/min per 1.73 m² and 15-29 ml/min per 1.73 m². To determine the independent association of these categories of GFR and total mortality, we adjusted for age, gender, race, medication use, ejection fraction, baseline NYHA functional class, history of diabetes, and history of hypertension using logistic regression analysis.

RESULTS: Of the 6640 patients, 957 had a GFR ≥ 90 ml/min per 1.73 m², 3562 had a GFR of 60-89 ml/min per 1.73 m², 2081 had a GFR of 30-59 ml/min per 1.73 m² and 40 patients had a GFR of 15-29 ml/min per 1.73 m². Patients with lower GFR levels had a higher prevalence of anemia, hypertension, diabetes mellitus, stroke and angina compared to those with higher GFR. There was no difference in total mortality between those with mildly depressed GFR (60-89 ml/min per 1.73 m²) and those with normal range GFR (≥ 90 ml/min per 1.73 m²). Independent of baseline differences, lower levels of GFR were associated with a higher total mortality compared to those with GFR ≥ 90 ml/min: (30-59

ml/min per 1.73m²; HR: 1.32 (95% CI: 1.10-1.59), p=0.004; 15-29 ml/min per 1.73 m²: HR: 2.54 (95% CI: 1.54-4.19), p<0.001).
CONCLUSIONS: The levels of GFR from the K/DOQI classification system are strongly associated with mortality in a heart failure population. This classification system further defines <60 ml/min per 1.73 m² as a particularly high-risk group among those patients with systolic dysfunction.

BEHAVIORAL AND SOCIAL RISK FACTORS AMONG PATIENTS WHO USE VA HEALTH SERVICES AND THOSE WHO OBTAIN CARE ELSEWHERE. R.J. Larson¹; H. Welch².
¹VA Outcomes Group, White River Junction, VT; ²VA Outcomes Group White River Junction VT, White River Junction, VT. (Tracking ID #136203)

BACKGROUND: There has been growing interest in evaluating the quality of care delivered by the Veteran Health Administration (VHA) system. Unfortunately, because patients cared for in VHA systems differ in many important ways from patients cared for in other settings, efforts to compare the health outcomes of the two groups have been limited by the ability to account and control for these differences. While comparisons can adjust for measured differences, other important but unaccounted for risk factors may remain—particularly such powerful predictors as tobacco and alcohol use.
METHODS: We used the 2000 Behavioral Risk Factor Surveillance System (a national population-based cross-sectional survey), to examine differences in important demographic, behavioral, and social risk factors among patients who use VA health services and those who obtain care elsewhere. The sample was restricted to male respondents who reported using health care in the last 12 months. Crude and age-adjusted means and proportions were calculated for each variable of interest.
RESULTS: VA users were older (mean age 58 years versus 47 years), and more likely to be over 65 years of age (37% versus 20%). Adjusting for age, they were also more likely to be black (18% versus 10%), to be divorced (22% versus 8.7%) or widowed (6.2% versus 3.6%), to have less than college degree level education (88% versus 68%), and to earn less than \$20K/year (31% versus 15%). Additional findings are presented below. Excluding BMI, all crude differences are statistical significance at p<0.05.
CONCLUSIONS: In addition to age, sex and race, VA users differ significantly from non-VA users with regard to multiple important health risk factors including marital status, socioeconomic status, chronic health conditions, tobacco use, alcohol use and physical activity level. Efforts to compare health outcomes between these groups should account for these significant risk differences.

	Crude		Age-adjusted	
	VA User	Non-VA User	VA User	Non-VA User
BMI, % >25kg/m2	68	68	65	68
Do you have diabetes? %Y	16	9	11	9
If diabetic, do you use insulin? %Y	36	25	43	25
Ever told BP high? %Y	50	27	40	27
Ever told cholesterol high? %Y	47	33	30	33
Current tobacco user? %Y	34	23	36	23
Any alcohol use in last month? %Y	42	60	39	60
Physical activity regularly? %Y	38	46	47	46
Self-reported general health, %fair or poor	38	15	26	15

BODY MASS INDEX IS ASSOCIATED WITH INCREASED RISK OF CHRONIC KIDNEY DISEASE IN APPARENTLY HEALTHY MEN. R.P. Gelber¹; T. Kurth²; A.T. Kausz³; J.E. Manson²; J.E. Buring²; A.S. Levey²; J.M. Gaziano¹.
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BACKGROUND: Overweight and obesity are well-established risk factors for cardiovascular disease and for decline in kidney function among individuals with existing kidney disease. In contrast, their association with the development of chronic kidney disease is less clear. We therefore evaluated the association between body mass index (BMI) and the development of chronic kidney disease.
METHODS: Prospective cohort study of 11,104 initially healthy men (mean age 52.9 ± 8.5 years) who participated in the Physicians' Health Study and who provided a blood sample after an average of 14 years. Baseline BMI was calculated as self-reported weight in kilograms divided by height in meters squared. We estimated the glomerular filtration rate (GFR) using the abbreviated equation developed by the Modification of Diet in Renal Disease Study and defined chronic kidney disease as GFR<60 mL/min/1.73 m². We used logistic

regression to evaluate the association between BMI and chronic kidney disease.
RESULTS: After an average 14-year follow-up, 1377 (12.4%) participants had a GFR<60 mL/min/1.73 m². Higher BMI was consistently associated with increased risk of chronic kidney disease. Compared with participants in the lowest BMI quintile (<22.7 kg/m²), those in the highest BMI quintile (>26.6 kg/m²) had an odds ratio (OR) of 1.45 (95% CI, 1.19-1.76; *P* for trend<0.001) after controlling for age, smoking, alcohol intake, physical activity, and parental history of myocardial infarction. Additional control for the potential biological mediators hypertension, diabetes, elevated cholesterol, and the development of cardiovascular disease attenuated the OR in the highest BMI quintile to 1.25 (95% CI, 1.03-1.54, *P* for trend 0.007). We found similar associations when using different categories of BMI.
CONCLUSIONS: In this large cohort of initially healthy men, BMI was significantly and independently associated with increased risk of chronic kidney disease after 14 years. The association was partly explained by potential biological mediators but remained statistically significant. Strategies to reduce chronic kidney disease risk might include prevention of overweight and obesity.

C-REACTIVE PROTEIN LEVELS IN ADULTS WITH OUTPATIENT-TREATED COMMUNITY-ACQUIRED PNEUMONIA. R. Gonzales¹; S. Bukofzer²; T.A. Busman².
¹University of California, San Francisco, San Francisco, CA; ²Abbott Laboratories, Abbott Park, IL. (Tracking ID #135270)

BACKGROUND: C-reactive protein (CRP) is an acute phase reactant associated with atherosclerotic heart disease at relatively low levels<8 mg/L. Whereas CRP levels rise, often dramatically, in response to acute bacterial infections, and quickly return to normal in response to therapy. As a result, CRP has been advocated in the management of community-acquired pneumonia (CAP). The preponderance of studies measuring the accuracy of CRP in detecting CAP has been performed in hospitalized patients. We analyzed CRP levels in a recent clinical trial of a new antibiotic for the outpatient treatment of CAP.
METHODS: Prospective cohort study of 519 adults with CAP based on clinical symptoms and positive chest xray, and eligible for outpatient treatment. Follow-up visits occurred 2-4 weeks following study entry. We restricted our analysis to patients lacking specific comorbidities or active conditions known to stimulate CRP production (final n=318). Microbiology testing included sputum culture (aerobic), polymerase chain reaction on throat swabs, serology, and urinary antigen for legionella. Tests of statistical significance including the chi-square test and multivariable logistic regression analysis.
RESULTS: The median age of our study population was 43 years (interquartile range=32, 57), 60% were male, and 23% had chronic lung disease. Thirty percent had a bacterial etiology identified, of which 58% were *S. pneumoniae*, 23% *Mycoplasma*, 12% *Chlamydia*, and 7% *Legionella*. CRP was increased (>10 mg/L) in 85% of all patients, and in 92% of confirmed bacterial cases; whereas percent neutrophil count was increased (>70%) in 63% of all patients, and in 66% of confirmed bacterial cases. *S. pneumoniae* and *Legionella* had greater CRP elevations than *Mycoplasma* and *Chlamydia* (for CRP >100 mg/L: p=0.01) (see Table below). In multivariable regression analysis, confirmed bacterial etiology (but not patient age, sex, race, or chronic lung disease) was significantly associated with CRP level (OR=2.33, 95% CI: 1.03, 5.25). Clinical failure occurred in 9% of patients with follow-up (n=286). 50% of clinical failures had CRP >20 mg/L at follow-up compared with only 5% of clinical cures (p<0.001). There was no difference in clinical failure rates among those with and without increased CRP at baseline (9% vs. 10%) (P=0.90).
CONCLUSIONS: In the largest cohort to date of CRP levels in patients with radiographically-confirmed CAP using modern microbiological techniques, this study confirms that CRP is a sensitive marker of outpatient-treated pneumonia, particularly in patients with a confirmed bacterial etiology (92%). These findings suggest that a low/normal CRP level may be useful in ruling-out CAP (and reducing unnecessary antibiotic use) in settings where chest xray services are not readily available, as well as in monitoring response to therapy. The microbial etiology of the majority of cases of outpatient-treated CAP remains unknown.

Baseline CRP Levels in Relation to Microbiological Test Results		
	> 10 mg/L	> 100 mg/L
Negative (n =105)	82%	38%
Indeterminate (n =117)	84%	34%
Mycoplasma or Chlamydia (n =34)	88%	29%
S. pneumoniae or Legionella (n =63)	94%	57%

DECLINE IN KIDNEY FUNCTION IS ASSOCIATED WITH INCREASED MORTALITY AMONG PATIENTS WITH LEFT VENTRICULAR SYSTOLIC DYSFUNCTION. N.A. Khan¹; I.W. Ma¹; C. Thompson¹; K. Humphries¹; D. Salem²; M.J. Samak²; L. Adeera¹.
¹University of British Columbia, Vancouver, British Columbia; ²Tufts University, Boston, MA. (Tracking ID #135118)

BACKGROUND: Small studies have shown that acute worsening of kidney function is associated with increased mortality among hospitalized patients with decompensated heart failure. Therefore, we hypothesized that progressive kidney dysfunction would be independently associated with mortality in a large ambulatory heart failure population. We studied the rate of decline in kidney function in an ambulatory heart failure population, factors associated with rapid decline in kidney function and the impact of this decline on mortality.

METHODS: To study these objectives, we used data from the randomized controlled trials (the Studies Of Left Ventricular Dysfunction: SOLVD) that contain serial serum creatinine levels for 6640 ambulatory heart failure patients followed over 33 months. Reduction in glomerular filtration rate (GFR) from baseline was categorized as <5 ml/min per 1.73 m² per year (conservatively estimated as normal), $5-10$ ml/min per 1.73 m² per year, $11-15$ ml/min per 1.73 m² per year and >15 ml/min per 1.73 m² per year. To determine the independent association of worsening GFR and total mortality, we adjusted for age, gender, race, diuretic use, baseline ejection fraction, baseline NYHA functional class, history of diabetes, and history of hypertension, baseline GFR and change in NYHA functional class using logistic regression analysis.

RESULTS: Loss in GFR was common. Of the participants, 30% had no decline in kidney function, 34% had a decline of <5 ml/min per 1.73 m² per year, 19% declined $5-10$ ml/min per 1.73 m² per year, 5% declined at $11-15$ ml/min per 1.73 m² per year and >15 ml/min per 1.73 m² per year. Factors independently associated with development of a rapid decline in kidney function (i.e. >15 ml/min per 1.73 m² decline per year) included: age, female gender, non-white race, lower ejection fraction, poor baseline NYHA functional class (III or IV vs. I or II), and lower hematocrit. Enalapril assignment, diuretic and antiplatelet use was not associated with rapid decline in kidney function. Among those with rapid decline, >15 ml/min per 1.73 m² per year, mortality rates were considerably elevated compared to those with slower decline, <5 ml/min per 1.73 m² per year (369/756 (49%) vs. 879/4475 (20%), $p < 0.001$). Even after adjusting for baseline kidney function, baseline heart function, worsening heart function (any increase in NYHA functional class), as well as other cardiovascular risk predictors, rapid decline in kidney function was independently associated with a substantially increased mortality (Odds Ratio: 5.63, 95% CI: 4.90 to 6.46, $p < 0.0001$).

CONCLUSIONS: Rapid decline in GFR is common among patients with systolic dysfunction and is a significant marker for death regardless of baseline kidney function, heart failure class, or progression of heart failure. Additional mechanisms must be invoked to explain the excess mortality risk, and efforts should be made to identify factors responsible for the rapid decline in GFR.

DIETARY INTAKE OF WHOLE AND REFINED GRAIN BREAKFAST CEREALS AND WEIGHT GAIN IN MEN. L.A. Bazzano¹; Y. Song²; V. Bubes³; C. Good³; J.E. Manson²; S. Liu². ¹Beth Israel Deaconess Medical Center, Boston, MA; ²Brigham and Women's Hospital, Boston, MA; ³General Mills Bell Institute of Health and Nutrition, Minneapolis, MN. (Tracking ID #135612)

BACKGROUND: Prospective studies have suggested that substituting whole-grain for refined-grain products may lower the risk of overweight and obesity. Breakfast cereal intake is a major source of whole and refined grains, and has also been associated with having a lower body mass index (BMI). The aim of this study was to prospectively assess the association between whole and refined grain breakfast cereal intakes and risk of overweight (BMI ≥ 25) and weight gain.

METHODS: We examined 17 881 US male physicians aged 40 to 84 years in 1982 who were free of cardiovascular disease, diabetes mellitus, and cancer at baseline and reported measures of breakfast cereal intake.

RESULTS: Over 8 and 13 years of follow-up respectively, men who consumed breakfast cereal, regardless of type, consistently weighed less than those who consumed breakfast cereals less often (P -value for trend < 0.0001). Whole and refined grain breakfast cereal intake was inversely associated with body weight gain over 8 years, after adjustment for age, smoking, baseline BMI, alcohol intake, physical activity, hypertension, high cholesterol, and use of multivitamins. Compared with men who rarely or never consumed breakfast cereals, those who consumed ≥ 1 serving/d of breakfast cereals were 22 and 12% less likely to become overweight during follow up periods of 8 and 13 years (relative risks [RR], 0.78 and 0.88; 95% confidence intervals [CI], 0.67 to 0.91 and 0.76 to 1.00, respectively).

CONCLUSIONS: Body mass index and weight gain were inversely associated with intake of breakfast cereals independent of other risk factors.

EFFECTIVENESS OF INFLUENZA VACCINATION IN A YEAR OF POOR MATCH BETWEEN VACCINE AND CIRCULATING STRAINS OF INFLUENZA: A CASE-CONTROL STUDY. C.B. Holmes¹; D.E. Singer¹; M.J. Barry¹; S.M. Modermott¹; S.J. Atlas¹. ¹Massachusetts General Hospital, Boston, MA. (Tracking ID #133730)

BACKGROUND: The 2003-2004 influenza season was notable for a suboptimal match between the vaccine and circulating strains of influenza, and an early start to the season. A small CDC case-control study among persons aged 50-64 years showed a protective effect (Odds ratio 0.48-0.62). Our objective was to assess the effectiveness of the 2003-2004 vaccine against laboratory-confirmed influenza in the outpatient setting.

METHODS: Study cases were identified from cases of laboratory-confirmed influenza at our hospital, not hospitalized during the influenza season, and with a primary care provider (PCP) in our system. Controls were drawn from outpatients with a PCP at our institution, and were matched 1:1 to cases on age, gender, a clinic visit at the time of the influenza diagnosis, PCP location and site of

residence. Clinical and demographic data including vaccination status and date of vaccination were collected using electronic medical and billing records.

RESULTS: Influenza vaccination status was obtained for 116 (85.3%) of 136 eligible influenza cases and for 106 (77.9%) of controls. There were no significant differences in age, sex or the proportion of cases and controls that smoked or had chronic lung disease, although cases had a higher Charlson Morbidity Score (Table). 31.2% of cases, and 54% of controls had received TIV >2 weeks prior to illness, and the odds ratio based on matched pair analysis was 0.67 (95% CI 0.31 to 1.38), $p = 0.24$.

Characteristics	Influenza cases	Controls	P value*
Mean Age (SD)	44.0 (17.7)	44.0 (17.4)	0.23
% Female	52.0%	52.0%	-
Asthma COPD	13.9%	10.9%	0.43
Charlson Score	0.6	0.4	0.03
Current smokers	10.2%	9.7%	1.00

*Calculated using McNemar's test for matched pairs

CONCLUSIONS: TIV in a year with a poor match with circulating strains showed a modest but not statistically significant benefit in preventing laboratory-confirmed influenza in relatively young, healthy adults. These results generally confirm those of the CDC case-control study and support ongoing influenza vaccination efforts. More precise estimation of vaccine effect could be obtained by pooling of results across institutions. Future work needs to assess whether TIV offers protection in higher risk populations during mismatch years.

ESTROGEN METABOLISM AND BLOOD PRESSURE IN A POPULATION-BASED SAMPLE OF POST-MENOPAUSAL WOMEN. C. Masi¹; L. Hawkey¹; J. Berry²; J. Cacioppo¹. ¹University of Chicago, Chicago, IL; ²Northwestern University, Chicago, IL. (Tracking ID #132710)

BACKGROUND: Age, BMI, ethnicity, and hormone replacement therapy (HRT) are all significantly associated with cardiovascular disease outcomes. Recent animal studies have shown that 2-methoxyestradiol (2MeOE), an endogenous metabolite of 17 beta-estradiol, inhibits vascular smooth muscle proliferation and attenuates angiotensin II-induced hypertension and cardiac hypertrophy. The objective of this study was to evaluate the relationship between 2-hydroxyestrone (2OHE), which is a precursor of 2MeOE, and blood pressure in a population-based sample of post-menopausal women.

METHODS: A multistage probability sampling design was used to recruit women aged 50-65 years as part of the Chicago Longitudinal Study. Using enzyme immunoassay, we analyzed overnight urine samples for 2OHE, 16 alpha-hydroxyestrone (16OHE), and the 2/16OHE ratio. Seated beat-to-beat measures of systolic and diastolic blood pressure (SBP and DBP) were obtained in the laboratory during two 4-minute epochs. We then modeled SBP and DBP using 2OHE, 16OHE, and the 2/16OHE ratio as predictors and age, BMI, and ethnicity as covariates. The group taking HRT ($n = 30$) was evaluated separately from group not taking HRT ($n = 63$).

RESULTS: Age, BMI, SBP, and DBP did not differ based upon HRT status, although 2OHE and 16OHE levels were significantly higher among women taking HRT. Among women not on HRT, a standard deviation (SD) increase in 2OHE was associated with a 12.5 mm Hg reduction in SBP ($p < .05$) after controlling for 16OHE and all covariates. In the same group, a SD increase in the 2/16OHE ratio was associated with a 6.6 mm Hg reduction in SBP ($p < .05$) and a 3.6 mm Hg reduction in DBP ($p < .05$). Among women on HRT, a SD increase in 2OHE was associated with a 7.3 mm Hg reduction in SBP ($p < .05$) while a SD increase in the 2/16OHE ratio was associated with an 8.7 mm Hg reduction in SBP ($p < .05$).

CONCLUSIONS: The endogenous estrogen metabolite, 2OHE, is a significant negative predictor of blood pressure in a population-based sample of post-menopausal women, regardless of HRT status. The mechanism of this association is unknown but may involve 2MeOE, a metabolite of 2OHE and an inhibitor of vascular smooth muscle proliferation. Understanding the determinants of 2OHE production may lead to useful therapies for hypertension and cardiovascular disease.

HEPATITIS C MARKS A HIGHER RISK OF DEATH AMONG PATIENTS ADMITTED TO THE HOSPITAL WITH ALCOHOL-RELATED PROBLEMS. J.L. Tsui¹; M.J. Pletcher¹; R. Gonzales¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #135490)

BACKGROUND: Risk factors for infection with Hepatitis C Virus (HCV) and alcoholism overlap, and patients who are affected by both are at greater risk for liver fibrosis, cirrhosis and cancer. How often HCV complicates alcohol-related hospital admissions, and whether it impacts in-hospital mortality are unknown.

METHODS: We identified all admissions to an urban, public hospital between 7/96 and 1/02 that were labeled with an alcohol-related diagnosis by discharge ICD-9 code. Information on age, sex, race, labs, other co-morbidities (including hepatitis C), medical service and disposition was abstracted from electronic records. The Charlson comorbidity index was calculated using ICD-9 codes and the Model for End-Stage Liver Disease (MELD) score was calculated using laboratory data. The MELD score is a disease severity index that has been shown to

predict mortality in cirrhotic patients. We used logistic regression to model the risk of dying in the hospital.

RESULTS: Of 6354 alcohol-related admissions, 984 (15%; 95% CI: 14%-16%) were complicated by a concurrent diagnosis of HCV. Patients with HCV were significantly more likely (p -value<0.05) to be female (27% vs. 18%), African-American (38% vs. 32%), between the ages of 40-50 (49% vs. 39%), have more comorbidities (mean Charlson score 2.2 vs. 1.4) and more hepatic dysfunction (mean MELD score 5.8 vs. 3.0). In-hospital deaths occurred in 2.4% of non-HCV related admissions, versus 4.4% of HCV related admissions (unadjusted OR=1.8; 95% CI: 1.3-2.6), and HCV remained associated with higher in-hospital mortality even after adjustment for sex, age, race, medical service and Charlson score (OR 1.49; 95% CI: 1.01-2.18). When we additionally adjusted for MELD score, HCV was no longer independently associated with mortality (OR 0.8, 95% CI: 0.5-1.2). Patients with both HCV and high MELD scores (≥ 20) were at particularly high risk for dying in the hospital (33% mortality; 95% CI: 20%-46%).

CONCLUSIONS: Hepatitis C infection is common in hospitalized patients with alcohol-related diagnoses and marks a higher risk of in-hospital mortality that appears to be mediated by more severe hepatic dysfunction. Special attention should be paid to patients admitted for alcohol-related reasons who have HCV as they have a greater risk of death.

IMPLICATIONS OF EXPANDING THE DEFINITION OF OSTEOPOROSIS. M.B. Herndon¹, L.M. Schwartz², S. Woloshin², H.G. Welch². ¹Dartmouth Medical School and Department of Veterans Affairs, Lebanon, NH; ²Dartmouth Medical School and Department of Veterans Affairs, White River Junction, VT. (Tracking ID #133316)

BACKGROUND: The current bone mineral density definition of osteoporosis (i.e. a T-score of ≤ -2.5) fails to identify most women who will eventually develop a hip fracture. To better identify these women, some experts have suggested changing the bone mineral density threshold to a T-score of ≤ -2.0 . The number of American women who would be affected by the proposed changes is unknown. The objective of this analysis is to determine the number of American women who could be diagnosed as having osteoporosis using each definition. For context, we calculate the 10-year risk of hip fracture for women in both groups.

METHODS: The distribution of bone density for US women is from the 1988-94 National Health and Nutrition Examination Survey (NHANES III). We used the SVY series of commands in STATA 8.0 (College Station, TX) to calculate the proportion of women 50 and older who have osteoporosis according to two different definitions of the disease: 2.5 and 2.0 standard deviations below the mean femoral neck bone mineral density for white women ages 20-39 (i.e. T-score ≤ -2.5 and T-score ≤ -2.0). We applied this proportion to 2000 US Census data to estimate the number of American women 50 and older labeled with osteoporosis using the two different disease definitions. Data on the age-specific risk of hip fracture were calculated from the 2000 Nationwide Inpatient Sample (NIS). We constructed a life table to calculate the 10-year risk of fracture according to age. Data on the proportion of hip fractures occurring at different T-score thresholds are from the National Osteoporosis Risk Assessment Study (NORA), a longitudinal study of 200,160 postmenopausal women.

RESULTS: 2.4 million American women 65 and older meet the current definition of osteoporosis (bone mineral density T-score ≤ -2.5). Of these, 19% are expected to have a hip fracture in the next 10 years. Lowering the threshold to a T-score of ≤ -2.0 would label an additional 3.4 million women with osteoporosis. Of these, only 9% are expected to have a hip fracture in the next 10 years. Even at the lower threshold, more than half of hip fractures will occur in women not diagnosed with osteoporosis.

CONCLUSIONS: Broadening the definition of osteoporosis by lowering the T-score threshold would label millions of additional women with osteoporosis but would still fail to identify most women destined to have a hip fracture.

INTRACRANIAL HEMORRHAGE IS THE DOMINANT DETERMINANT OF DEATH AND DISABILITY FROM WARFARIN-ASSOCIATED HEMORRHAGE: THE ANTICOAGULATION AND RISK FACTORS IN ATRIAL FIBRILLATION (ATRIA) STUDY. M.C. Fang¹, A.S. Go², Y. Chang³, E.M. Hylek⁴, L. Henault³, N. Jensvold², D. Singer³. ¹University of California, San Francisco, San Francisco, CA; ²Kaiser Permanente Division of Research, Oakland, CA; ³Massachusetts General Hospital, Boston, MA; ⁴Boston University School of Medicine, Boston, MA. (Tracking ID #135842)

BACKGROUND: Warfarin therapy reduces the risk for atrial fibrillation (AF)-related ischemic stroke, but raises the risk for hemorrhage. Appropriate clinical decision-making requires better information about outcomes after warfarin-associated hemorrhages. Prior studies have lacked sufficient power to assess mortality and disability from hemorrhage, especially intracranial hemorrhage. We compared the mortality and disability resulting from intracranial and extracranial hemorrhages in a large, contemporary cohort of patients with AF.

METHODS: We prospectively studied 13,559 adults with AF enrolled in Kaiser Permanente of Northern California. The median follow-up was 2.4 years. Patients hospitalized for hemorrhage were identified from clinical databases using validated algorithms and confirmed using chart review. Major hemorrhages were categorized as intracranial (ICH) or extracranial (fatal, transfusion of 2 units packed blood cells, or into a critical anatomic site). We compared the discharge disability and 30-day mortality of ICH and major extracranial hemorrhage, adjusting for patient characteristics using multivariable Poisson regression.

RESULTS: We identified 72 ICHs and 98 major extracranial hemorrhages on warfarin out of over 15,300 person-years of follow-up. There were 40 hemorrhage-related deaths (annual mortality rate 0.26% [0.2-0.4]), with the majority of

deaths being from ICH (35 patients). Only 8% of ICH patients sustained no disability at discharge; in contrast, 91% of patients with extracranial hemorrhage were discharged with no deficits ($P < 0.001$). ICH continued to be the dominant determinant of death (adjusted odds ratio = 32 [9.4-111]) after adjusting for age, sex, anticoagulation intensity, and comorbid conditions.

CONCLUSIONS: ICH causes nearly 90% of the deaths from warfarin-associated hemorrhage among patients with AF and most of the disability among survivors. Previous studies generally combined both ICH and extracranial hemorrhages in their analyses; our work clearly shows that the long-term negative consequences of warfarin-associated hemorrhage are overwhelmingly the result of ICH. Thus the anticoagulation decision in AF largely depends on the balance of ischemic strokes prevented versus ICHs induced. Better predictors of ICH are crucially needed to improve the safe use of warfarin therapy.

LACK OF AN ASSOCIATION BETWEEN DIABETES AND AN INCREASED RISK OF IN-HOSPITAL DEATH AMONG PATIENTS WITH PNEUMONIA. K.E. Ray¹, M.J. Barnett¹, P. Kaboli¹, G.E. Rosenthal¹. ¹University of Iowa, Iowa City, IA. (Tracking ID #135051)

BACKGROUND: While conventional wisdom suggests diabetes should be an important risk factor for mortality in patients with pneumonia, prior research using administrative data has found similar or lower risks of death in diabetics, relative to non-diabetics. Because the reliability of diabetes as captured by administrative data is uncertain, the current study compared the risks of death associated with diabetes as measured by diagnosis codes and medication use. The study further assessed the agreement between diagnosis codes and medication use in defining diabetes.

METHODS: We used administrative and pharmacy data files to identify 14,057 consecutive patients who were: 1) admitted to 119 VA acute care hospitals in 2003 with a primary diagnosis of pneumonia; and 2) who had one or more outpatient prescriptions for oral hypoglycemics or insulin in the prior year. Diabetes was defined on the basis of: 1) secondary ICD-9 diagnosis codes using the algorithm of Elixhauser et al; and 2) a prior outpatient prescription for any oral hypoglycemic medication or insulin. Agreement between the two definitions was measured by the kappa statistic. The risk of in-hospital death associated with the two definitions of diabetes was determined using logistic regression analysis, adjusting for demographics, type of pneumonia, admission source, and comorbidities.

RESULTS: 27% of patients had diabetes as defined by inpatient ICD-9 codes, while 25% of patients had a prior outpatient prescription for insulin (12%) or an oral hypoglycemic (19%). There was substantial agreement between the two approaches to defining diabetes ($\kappa = 0.78$). Of the 14,057 patients in the cohort, 22% of patients had both a diagnosis of diabetes and a prescription for diabetes medication, 5% had a diagnosis but not a prescription, 3% had a prescription but not a diagnosis, and 70% had neither. Unadjusted mortality was lower for diabetics than non-diabetics, as assessed by diagnosis codes (6.6% vs. 7.7%, $P = 0.02$), but was similar when diabetes was assessed on the basis of medication use (7.1% vs. 7.4%, $P = 0.52$). Adjusting for demographic and clinical factors, the odds of death in diabetics and non-diabetics was similar when diabetes was defined by diagnosis codes (OR=0.94, 95%CI 0.80-1.01) or diabetic medication use (OR=1.03, 95%CI 0.88-1.21). Odds of death were also similar among patients taking insulin (OR=1.00, 95%CI 0.80-1.25) or oral hypoglycemics (OR=0.99, 95%CI 0.83-1.19).

CONCLUSIONS: Among hospitalized patients with pneumonia, ICD-9 codes appear to be relatively reliable for identifying patients with diabetes. Somewhat surprisingly, diabetes as identified by diagnosis codes or by the use of insulin or oral hypoglycemic agents was not associated with an increased risk of in-hospital death after adjusting for other covariates. The lack of an association may be unique to pneumonia or may reflect residual confounding (i.e., higher unmeasured severity in non-diabetics) in administrative data.

MRI IS SUPERIOR TO OTHER CONVENTIONAL DIAGNOSTIC TESTS FOR DIAGNOSING ACUTE OSTEOMYELITIS OF THE FOOT AND ANKLE - A META-ANALYSIS. A. Kapoor¹, M.P. Lavalley¹, D.T. Felson¹. ¹Boston University, Boston, MA. (Tracking ID #136035)

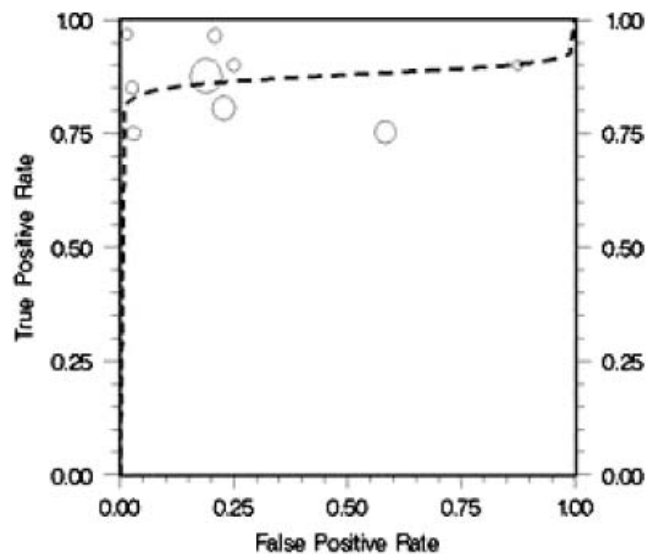
BACKGROUND: There is no uniformly followed algorithm for diagnosing osteomyelitis. Plain film radiography has been reported to have poor sensitivity (usually less than 60%). A meta-analysis of technetium bone scanning in JGIM reported a sensitivity of 85% could only be attained by accepting a false positive rate of approximately 40%. We conducted a meta-analysis to determine the test performance of magnetic resonance imaging (MRI) in making the diagnosis of acute osteomyelitis of the foot and ankle in adults with diabetes, peripheral neuropathy, or peripheral vascular disease.

METHODS: Published studies were identified by a structured search strategy in MEDLINE and EMBASE. Search for unpublished conference abstracts and bibliographies of the included articles is ongoing. Original studies were selected if they met the following criteria: 1) subjects were a non-surgical cohort of adult patients with a history of diabetes, peripheral vascular disease, or peripheral neuropathy included due to suspicion of acute osteomyelitis of the foot or ankle 2) absolute numbers of true positive, false-negative, true-negative, and false-positive results for the diagnosis of osteomyelitis were available at the patient level or could be derived from the presented data; 3) both T1 and T2 pulse sequence MRI data were recorded. The MRI performance was analyzed using the true and false positive rates of diagnosis and by constructing summary receiver-operating characteristic (SROC) curves. The SROC method uses linear regression of the log of the diagnostic odds ratio (odds of a positive test in subjects with osteomyelitis divided by the odds of a positive test in subjects without) with a

predictor based on the proportion of positive tests. We chose a sensitivity of 85% as a reasonable cutpoint to identify the associated false positive rate.

RESULTS: At the time of abstract writing, nine studies including a total of 258 subjects met inclusion criteria. In six studies, patients were selected prospectively. Most studies concerned exclusively diabetics with foot disease. Studies varied most in the way osteomyelitis was confirmed or excluded. Five studies confirmed positive scans with histologic analysis at least 50% of the time. By contrast, no study confirmed negative scans with histology more than 50% of the time; usually healing with a short-term antibiotic course was used to exclude osteomyelitis. True positive rates ranged from 0.75 to 0.97 and false positive rates ranged from 0.02 to 0.87. The pooled diagnostic odds ratio was 23.9 (95% CI 8.0 to 71.9). True and false positive rates for the nine studies are shown in the figure below along with the SROC curve. A false positive rate of 11% was identified at the 85% sensitivity level.

CONCLUSIONS: In a preliminary analysis of an ongoing meta-analysis, we found that MRI performs well in diagnosing acute osteomyelitis of the foot and ankle. It appears to have better sensitivity and specificity than plain film and technetium bone scanning in the diagnosis. In future analysis, we plan to compare directly the diagnostic test performance of MRI with other imaging modalities.



Note: bubble size equals sample size

NON-HDL VERSUS LDL CHOLESTEROL AS A RISK FACTOR FOR MYOCARDIAL INFARCTION IN WOMEN. W.R. Farwell¹; H. Sesso²; J.E. Buring²; J.M. Gaziano¹. ¹Massachusetts Veterans Epidemiology Research and Information Center; Brigham and Women's Hospital, Boston, MA; ²Brigham and Women's Hospital, Boston, MA. (Tracking ID #132007)

BACKGROUND: Whereas low-density lipoprotein cholesterol (LDL-C) requires a fasting sample, non-high-density lipoprotein cholesterol (non-HDL-C), which includes LDL-C as well as other triglyceride (TG) rich particles, does not require a fasting sample for calculation. Few studies have compared LDL-C with non-HDL-C for their ability to predict first nonfatal myocardial infarction (MI), particularly in women.

METHODS: We analyzed data from a cohort of 18,947 women from the Women's Health Study (WHS) who neither carried a diagnosis of hyperlipidemia nor took cholesterol lowering medications. At baseline, we collected blood samples and performed direct-measurement assays for lipid parameters. We collected baseline self-reported risk factors at enrollment. Each MI was self-reported and confirmed against WHO criteria. Cox proportional hazards models were constructed to compare the independent associations of non-HDL-C and LDL-C with the risk of MI. Adjustments were made for cardiovascular risk factors: age, hypertension, alcohol consumption, BMI, diabetes, smoking, family history of MI, physical activity, hormone use, and high-density lipoprotein cholesterol. The Akaike Information Criterion (AIC) for each model was computed.

RESULTS: During 9.9 mean years of follow-up, 118 cases of first nonfatal MI occurred. Mean values ($\pm$ SD) of LDL-C and non-HDL-C were 116.3 ($\pm$ 30.7) and 147.5 ($\pm$ 36.5) respectively. Tertiles of LDL-C were defined as <102.1, 102.1-126.6, and >126.6 mg/dL; non-HDL tertiles were defined as <130.1, 130.1-159.4, and >159.4 mg/dL. Hazard ratios (HRs) for each tertile of each lipid parameter compared to its first are listed in the table below. The P for trend and AIC of each multivariate model is listed in the table below.

CONCLUSIONS: Our study suggests that non-HDL-C may be better than LDL-C as a primary screen of women for risk of first nonfatal MI, given that non-HDL-C accounts for LDL-C plus other TG-rich atherogenic particles and does not require a fasting sample.

Tertiles of Lipid Parameter	1	2	3	P for Trend	AIC
LDL-C HR (95% CI)	1.00	0.92 (0.53-1.62)	1.51 (0.91-2.48)	0.06	1816.52
Non-HDL-C HR (95% CI)	1.00	1.81 (0.92-3.56)	2.91 (1.54-5.49)	<0.01	1807.49

Multivariate HRs and AIC of LDL-C and non-HDL-C for first nonfatal MI

PREOPERATIVE PULMONARY RISK STRATIFICATION FOR NON-CARDIOPULMONARY SURGERY: A SYSTEMATIC REVIEW. G.W. Smetana¹; V. Lawrence²; J. Cornell². ¹Beth Israel Deaconess Medical Center, Boston, MA; ²University of Texas Health Science Center at San Antonio, San Antonio, TX. (Tracking ID #132775)

BACKGROUND: In contrast to perioperative cardiac risk assessment, there have been no previous systematic reviews in the area of perioperative pulmonary risk assessment. We performed a systematic review of the evidence regarding risk factors for postoperative pulmonary complications (PPC) after non-cardiopulmonary surgery.

METHODS: We performed a systematic literature search from 1980-2002, and abstracted eligible studies using standardized forms. Inclusion criteria were primary data analysis and explicit definitions of pulmonary complications. Exclusion criteria included non-English studies, <25 subjects/group, physiologic outcomes only, use of administrative data to determine PPC, developing countries, and pediatric or immunosuppressed patients (pts). Meta-analytic methods included adjustment for bias in univariate analyses, assessment of publication bias, heterogeneity and outliers, and random effects models.

RESULTS: Of 15,499 citations, 120 were eligible for detailed abstraction; 88 studies provided unadjusted data on predictors for 11,851 PPC among 173,500 pts and 32 studies reported multivariable analyses of predictors for 10,960 PPC among 321,819 pts. Among studies with multivariable analyses, the 3 largest used patient subsets from the Veterans' Affairs National Surgical Quality Improvement Project and accounted for 91% of pts and 85% of PPC. PPC rates varied systematically with study size; studies with <500 pts had highly variable and higher PPC rates (median rate 14%, 95% CI 8.7-25) than studies of >500 pts (median rate 4%, 95% CI 2.6-6.3). Significant risk factors varied with sample size and surgical populations. Good evidence supports the following risk factors for PPC: COPD, advanced age, poor functional status, ASA class 3-5, prolonged, abdominal, aortic, noncardiac/nonpulmonary thoracic, or emergency surgery, and albumin <3 gm/dl. Fair evidence suggests the following may be important: obstructive sleep apnea, impaired sensorium, congestive heart failure, abnormal chest exam, poor exercise capacity, cigarette use, general anesthesia (versus spinal/epidural); neck, esophageal, and neurosurgery; pancuronium (versus shorter-acting agents), significant blood loss, increased BUN or creatinine, and abnormal chest radiograph. Good evidence indicates the following are not important risk factors for PPC: obesity, asthma, and hip, genitourinary, and gynecologic surgery. Evidence is insufficient regarding corticosteroid or alcohol use, arrhythmia, weight loss, diabetes, perioperative transfusion, vascular or laparoscopic surgery, transverse abdominal incisions, and preoperative spirometry.

CONCLUSIONS: PPC rates and important risk factors vary with study size and surgical population. Studies with >500 patients provide the most stable estimates of PPC incidence. We have confirmed the importance of several clinical risk factors for PPC while other, traditionally held, factors, including obesity and asthma, do not appear important. Among laboratory tests, we found no evidence that preoperative spirometry predicts PPC risk to a greater degree than clinical evaluation alone. Serum albumin <3 gm/dl was an important predictor. For clinicians, this first systematic review summarizes the evidence regarding preoperative pulmonary risk stratification before non-cardiopulmonary surgery. For clinical and health services researchers, it provides direction regarding appropriate sample size and the necessary minimum set of covariates when designing trials of interventions to prevent PPC.

PREVENTION OF POSTOPERATIVE PULMONARY COMPLICATIONS: A SYSTEMATIC REVIEW. V.A. Lawrence¹; G.W. Smetana²; J.E. Cornell¹. ¹Audie L. Murphy VA Hospital and The University of Texas Health Science Center at San Antonio, San Antonio, TX; ²Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA. (Tracking ID #132685)

BACKGROUND: In contrast to perioperative cardiac risk management, there have been no previous systematic reviews on perioperative pulmonary risk management. We performed a systematic review of the evidence for interventions to prevent postoperative pulmonary complications (PPC) after non-cardiopulmonary surgery.

METHODS: We performed a systematic literature search from 1980-2002 and abstracted data from eligible studies using standardized forms. Inclusion criteria were published randomized trials or systematic reviews that provided explicit definitions for PPC. Exclusion criteria were non-English studies, <25 patients

per arm, physiologic outcomes only; use of administrative data to determine PPC; pediatric or immunosuppressed patients; and studies from developing countries. We did not perform meta-analyses due to insufficient data or trial heterogeneity.

RESULTS: Of 15,499 citations, 30 randomized trials and 11 systematic reviews were eligible for detailed abstraction. Good evidence indicates that postoperative lung expansion (e.g., incentive spirometry, deep breathing exercises) and selective, rather than routine, nasogastric decompression after abdominal surgery prevent PPC. Fair evidence suggests that at least two months of smoking cessation before surgery and short-acting neuromuscular blockade reduce risk. Preliminary evidence suggests immune-enhancing enteral formulations may prevent postoperative infection, including pneumonia. Several meta-analyses indicate that epidural anesthesia or analgesia reduces PPC but recent large trials do not confirm benefit. Right heart catheterization, standard nutritional support in malnourished patients, and limiting transfusions or using leukocyte-depleted blood do not prevent PPC. Evidence is insufficient regarding prophylactic corticosteroids, transverse versus open abdominal incisions, and laparoscopic versus open operations. Many trials are flawed due to inadequate randomization, lack of blinding, inadequate sample size for expected rates of PPC, and unproven surrogate outcome measures.

CONCLUSIONS: Proven interventions to reduce PPC include postoperative lung expansion and selective nasogastric decompression. Smoking cessation for at least two months before surgery, short-acting neuromuscular blockade, and immune-enhancing enteral nutrition may be beneficial. Right heart catheterization, routine nutritional support for malnutrition, and transfusion do not prevent PPC. Clinicians can use this first systematic review to select perioperative interventions to reduce pulmonary complications after non-cardiopulmonary operations. For clinical and health services researchers, this review identifies strategies to improve design and methods for future randomized trials.

PRIMARY ALDOSTERONISM IN DIABETIC SUBJECTS WITH POORLY CONTROLLED HYPERTENSION. P. Cantey¹; G. Umpierrez². ¹Emory University, Atlanta, GA. (Tracking ID #135386)

BACKGROUND: Primary aldosteronism (PA) is a potentially curable form of hypertension whose prevalence according to current data lies between 5% and 23% in hypertensive populations. The prevalence of the disorder remains undefined in the diabetic hypertensive population. Diabetic subjects often go unscreened for PA, as data from the United Kingdom Prospective Diabetes Study indicate that a large percentage of diabetics will require 3 or more medications to control their blood pressure.

METHODS: This study was designed to determine the prevalence of PA in patients who attend the Diabetes Clinic. Subjects with a blood pressure 140/90 and who are taking ≥ 3 medications at maximal doses have a screening plasma aldosterone concentration to plasma renin activity ratio (PAC/PRA) performed. If the ratio is ≥ 30 ng/dl/ng ml⁻¹ h⁻¹, the PAC is ≥ 12 ng/dL, and the potassium is ≥ 3.5 meq they qualify for a confirmatory salt loading protocol. A positive confirmatory test is defined as a PAC ≥ 8.5 ng/dL or urine aldosterone concentration ≥ 12 mcg/d and a urine sodium concentration ≥ 200 mEq/24h.

RESULTS: As of this submission 29 subjects have completed the protocol. Four subjects (13.8%) have been diagnosed with PA by saline suppression testing. The final results for another 17 subjects who meet criteria in the Diabetes Clinic are currently pending. Five of these subjects (29.4%) have had a positive screen for PA. Extrapolating results based on prior performance of the screening PAC/PRA, we expect a final prevalence of PA between 12.9% and 20.6%.

CONCLUSIONS: Our preliminary data show that PA contributes to the inability to control blood pressure in 13.8% of diabetics on three or more medications for their pressure. These results indicate that just as one would screen non-diabetic patients on 3 or more antihypertensives for PA, it should be standard of care to screen similar diabetic patients with a screening PAC/PRA. More studies are needed to define the optimal screening protocol and the optimal treatment of PA in diabetic patients.

RED FLAG OR WHITE: A SYSTEMATIC REVIEW OF AMBULATORY BLOOD PRESSURE MONITORING FOR WHITE COAT HYPERTENSION. J.A. Tice¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #136231)

BACKGROUND: Blood pressure fluctuates widely within a single day and over time making accurate classification of an individual's blood pressure difficult. Furthermore, the blood pressure measured in the doctor's office may not represent the usual blood pressure of an individual outside of the medical setting. Identification of white-coat hypertension with ambulatory blood pressure monitoring may benefit patients by sparing them the expense and side effects associated with treatment of hypertension. Patients may also avoid the ill effects of being labeled as having the hypertension. On the other hand, patients with white-coat hypertension may benefit from treatment to lower their blood pressure.

METHODS: The Medline database, Cochrane clinical trials database, Cochrane reviews database, and the Database of Abstracts of Reviews of Effects were searched using the key words ambulatory blood pressure monitoring, 24 hour blood pressure monitoring. These were cross-referenced with the keyword "white coat". The search was performed for the period from 1966 through August 2004. The bibliographies of systematic reviews and key articles were manually searched for additional references. Studies comparing outcomes in patients with white coat hypertension to those with hypertension or normal blood pressure were included. The abstracts of citations were reviewed for relevance and all potentially relevant articles were reviewed in full.

RESULTS: The initial search found 393 articles. One randomized clinical trial, two cohort studies, and 21 cross-sectional studies met the inclusion criteria. The lack of a commonly accepted definition of white coat hypertension makes any attempt to synthesize the literature extremely difficult. Fifteen different definitions were used in the reviewed literature. Case-control studies generally reported that the risk of target organ damage associated with white coat hypertension fell between that of hypertension and normal blood pressure. Of the two prospective cohort studies comparing patients with white coat hypertension to patients with normal blood pressure, one reported that cardiovascular event rates were similar in the two groups, while the other reported that the event rates were significantly higher for patients with white coat hypertension. A subgroup of 695 patients randomized in the Systolic Hypertension in Europe Trial had ambulatory blood pressure monitoring. Although the study was underpowered to give a definitive answer in this subgroup, treated patients with white coat hypertension had no strokes during follow-up (versus two in the untreated group, $p=0.16$) and had one-third the number of cardiovascular events (estimated RR 0.31). Although not statistically significant, the absolute risk reduction in cardiovascular events for treatment of white coat hypertension (14.5/1000 person-years) was larger than that for hypertensive treatment of all patients with isolated systolic hypertension in the full trial (10.4/1000 person-years).

CONCLUSIONS: Current evidence suggests that patients with white coat hypertension should be treated similarly to patients with hypertension. Ambulatory blood pressure monitoring to identify white coat hypertension is not indicated.

RISK FACTORS FOR REPEAT SYPHILIS INFECTION, SAN FRANCISCO, 2001-2002. W.T. Phipps¹; C. Kent²; R. Kohn²; J.D. Klausner¹. ¹University of California, San Francisco, San Francisco, CA; ²San Francisco Department of Public Health, San Francisco, CA. (Tracking ID #135473)

BACKGROUND: After declining to historically low rates in the mid-1990s that led the United States Centers for Disease Control to announce a national plan for syphilis elimination, syphilis incidence has increased dramatically. Nationally, annual rates of syphilis increased 9.1% in 2001 and 12.4% in 2002, with outbreaks reported in several major US cities. The resurgence of syphilis has occurred primarily among men who have sex with men (MSM), disproportionately affecting those with human immunodeficiency virus (HIV) infection. Despite ongoing public health efforts to eliminate syphilis, infection rates among MSM have persisted. New prevention strategies are needed. We undertook a study to examine the rate of repeat syphilis infection in San Francisco and to identify risk factors associated with syphilis re-infection that may inform additional secondary prevention strategies and decrease the overall incidence of infection.

METHODS: The study population included all cases of primary and secondary syphilis diagnosed in San Francisco in 2001 and 2002. We defined a repeat case of syphilis as a new infection that occurred within one year of a prior syphilis infection. The control group was all other cases of syphilis in San Francisco in 2001 and 2002 without a repeat infection. Syphilis infection was identified by testing patients' sera with rapid plasma reagin and confirming reactive specimens by treponemal specific particle agglutination. Confirmed reactive titers were compared to a patient's previous syphilis history to identify new infection, defined as a four-fold increase in RPR titer. A clinician made the designation of a new infection at the time of diagnosis.

RESULTS: During the study period January 2001 to December 2002, 933 patients were diagnosed with primary or secondary syphilis. Of these, 54 patients (5.8%) had a repeat syphilis infection within one year. No differences in gender, age, race, stage of syphilis, number of period sexual partners, illicit drug use, or partner meeting venue were observed between cases with and without repeat infections. HIV infection was associated with an increased risk of repeat syphilis infection (OR=3.5; CI 1.9-6.4), which remained significant when adjusted for sexual orientation (OR=3.2; CI 1.5-6.8) and number of sexual partners (OR=1.6; CI 1.4-1.8). Gay or bisexual orientation was also associated with repeat syphilis infection (OR=2.8; CI 1.2-6.6), but it was not independently associated with risk of repeat infection when adjusted for HIV status (OR=1.5; CI 0.5-4.0).

CONCLUSIONS: We found that 5.8% of patients had repeat infection within one year of a prior syphilis infection. While these cases may not represent a core group responsible for sustaining the current epidemic, they are an at-risk group that may merit targeted public health interventions. Our finding that HIV-infection is independently associated with repeat syphilis infection is notable given the high rates of HIV-infection among those with syphilis, suggesting there is a large population potentially at risk for repeated syphilis infections. This result further highlights the important association between HIV and syphilis and validates current CDC recommendations for routine syphilis screening in all sexually active HIV-positive MSM, particularly if they have had a prior syphilis infection.

SMOKING INCREASES THE RISK OF HYPERTENSION. R. Halperin¹; H. Sesso²; M. Gaziano². ¹Harvard University, Boston, MA; ²Brigham and Women's Hospital, Boston, MA. (Tracking ID #132989)

BACKGROUND: Smoking acutely raises blood pressure; however, the relationship between cigarette smoking and chronic hypertension is less well defined with inconsistent results reported. Some studies actually suggest an increased risk of hypertension after quitting smoking, which may be confounded by subsequent weight gain that tends to be greater in women.

METHODS: We analyzed 13,437 middle-aged and older men from the Physicians' Health Study who were free of hypertension, cardiovascular disease and cancer. Subjects provided self-reported baseline smoking status (defined as either current smoker, past smoker or never smoker) and other clinical and lifestyle risk factor information. Incident hypertension was defined as either initiation of anti-hypertensive medication, systolic blood pressure >140 mmHg or diastolic blood pressure >90 mmHg as determined from self-reports on questionnaires during follow-up. We used Cox proportional hazards models to assess the risk of developing hypertension.

RESULTS: The mean follow-up period was 14.1 years during which 4,963 men developed hypertension. At baseline smokers were heavier, exercised less and drank more alcohol, generally reflecting a less healthy lifestyle. Compared with never smokers, past and current smokers had corresponding age-adjusted relative risks (RRs) of developing hypertension of 1.16 (95% confidence interval (CI): 1.09, 1.23) and 1.24 (95% CI: 1.12, 1.36). In a multivariate model additionally adjusted for body mass index, diabetes, alcohol use, exercise and parental history of myocardial infarction, the RRs of developing hypertension were 1.11 (1.04, 1.18) for past smokers and 1.15 (1.05, 1.27) for current smokers, as compared to never smokers.

CONCLUSIONS: In this cohort of men, smoking is independently associated with an increased risk of developing hypertension. Because smoking is known to have deleterious effects on the endothelium, this association may be biologically plausible and offers potential in the primary prevention of hypertension.

SPECTRUM OF INFLUENZA DISEASE AT A LARGE MEDICAL CENTER DURING A YEAR OF VACCINE MISMATCH. C.B. Holmes¹; D.E. Singer¹; M.J. Barry¹; S.M. McDermott¹; S.J. Atlas¹. ¹Massachusetts General Hospital, Boston, MA. (Tracking ID #133701)

BACKGROUND: The 2003-2004 influenza season was marked by an antigenic drift in which the A/Fujian influenza strain became the dominant circulating strain in the United States (U.S.), although the vaccine was formulated with the A/Panama (H3N2) strain. The clinical epidemiology of influenza in years of poor match between the vaccine and circulating strains of influenza is not as well characterized as it is in years with a good match. Our objective was to describe the sociodemographic characteristics, clinical features and outcomes, and vaccination status of confirmed adult cases of influenza, in order to better understand the impact of influenza during a year of vaccine mismatch.

METHODS: All adult patients (≥ 18 years) with a positive influenza A swab recorded in our institution's virology lab between October 1, 2003 and February 29, 2004 were included in the study. Demographic and clinical data were collected using electronic medical and billing records through the hospital's information system. Pre-specified influenza complications were assessed by a detailed review of patient records. When not available, influenza vaccination status was obtained from a mailed questionnaire or by telephone interview.

RESULTS: There were 371 adult cases of influenza, with 90% of cases diagnosed from December 1, 2004 to January 9, 2005. Cases had a median age of 46 (IQR 25.5-66.5) and 53% were women. 37% were admitted to the hospital for influenza-related illness. Admitted cases were significantly older than those treated as outpatients (mean ages 67 versus 39; $p<0.0001$) and were significantly more likely to suffer from asthma or other chronic respiratory illness (37% versus 12%; $p<0.0001$). The median length of stay for hospitalized patients was 4 days (IQR 1.5-6.5) and 7% of inpatients required ICU level care during their hospitalization. The hospital mortality rate was 3% (4 patients). 47.2% of influenza cases had been vaccinated >2 weeks prior to the diagnosis of influenza (65% of inpatients and 36% of outpatients). Cases of influenza with a history of vaccination were older (63 versus 43; $p<0.0001$), and were more likely to have a chronic respiratory illness than those who had not been vaccinated ($p<0.0001$).

CONCLUSIONS: Our institution had a large number of laboratory-confirmed influenza cases, with most cases occurring early in the season, a trend seen in other parts of the country. The rate of hospitalization was high, with the elderly and those with chronic lung disease admitted most frequently. Almost two-thirds of influenza cases requiring hospitalization were vaccinated at least 2 weeks prior to diagnosis, suggesting reduced effectiveness of the mismatched vaccine. Analytic studies are needed to directly address this question.

SUBCLINICAL HYPOTHYROIDISM AND THE RISK OF CORONARY HEART DISEASE: A SYSTEMATIC REVIEW AND META-ANALYSIS. N. Rodondi¹; D. Aujesky²; E. Vittinghoff¹; J. Cornuz³; D.C. Bauer¹. ¹University of California, San Francisco, San Francisco, CA; ²Department of Medicine, Lausanne University Hospital, Lausanne; ³University Hospital of Lausanne, Lausanne. (Tracking ID #131797)

BACKGROUND: Subclinical hypothyroidism is common, particularly in older adults. Controversy persists as to whether screening and treatment of subclinical hypothyroidism is warranted, as current evidence about the risks is limited. It has been associated with elevated total and LDL-cholesterol, and enhanced atherosclerosis. Data on the risk of coronary heart disease (CHD) in subjects with subclinical hypothyroidism are conflicting. We performed a systematic review to determine whether subclinical hypothyroidism is associated with CHD in adults.

METHODS: We searched MEDLINE from 1966 to July 2004, and the bibliographies of key articles in the field and those included in this review. We included all studies that provided risk estimates for CHD or cardiovascular mortality in subjects with subclinical hypothyroidism compared to those with normal thyroid function. Two authors independently reviewed each potential study for eligibility, assessed data quality, and extracted the data. We assessed

methodological study quality and performed sensitivity analyses. When available, we used multiply adjusted results.

RESULTS: We identified fourteen observational studies that met eligibility criteria and included a total of 1,362 CHD outcomes and 10,540 subjects. Subclinical hypothyroidism increased the risk of CHD (summary odds ratio [OR]: 1.65, 95% confidence intervals [CI]: 1.28-2.12) using a random-effect model. We found weak evidence for heterogeneity among individual study findings (P for heterogeneity 0.12). In 9 studies with either statistical adjustment or matching cases with controls, the summary OR for CHD was 1.81 (95% CI: 1.38-2.39), and was 2.38 (95% CI: 1.53-3.69) after pooling the three studies that adjusted for most cardiovascular risk factors. Sensitivity analyses including only population-based studies, and those with formal outcome adjudication procedures or adjudication without knowledge of thyroid status yielded similar results. Subgroup analyses by type of study design showed a similar trend, but lower risk, in the five prospective cohort studies (OR 1.42, CI: 0.91-2.21), as compared to the case-control and cross-sectional studies (OR 1.72, CI: 1.25-2.38).

CONCLUSIONS: Our systematic review indicates that subclinical hypothyroidism is associated with an increased risk of CHD. Clinical trials are needed to assess whether thyroxine treatment reduces the risk of CHD in subjects with subclinical hypothyroidism and whether benefits outweigh harms.

THE RELATIONSHIP BETWEEN ALCOHOL AND SERUM INFLAMMATORY BIOMARKERS. M. Benedict¹; J. Mrus¹; J. Tsevat¹; M.H. Eckman¹. ¹University of Cincinnati, Cincinnati, OH. (Tracking ID #133476)

BACKGROUND: Light to moderate alcohol consumption has been associated with a reduced risk of cardiovascular morbidity and mortality. Recent evidence has indicated that inflammation is a predictor of cardiovascular risk and that alcohol may exert an anti-inflammatory effect. However, the effects of alcohol and alcoholic beverage type on inflammatory markers have not been adequately investigated.

METHODS: We analyzed data on 8,793 adult participants in the Third National Health and Nutrition Examination Survey (NHANES III). Alcohol intake was categorized into four groups: no intake (group1), 1-30 drinks/month (group2); 31-90 drinks/month (group3); and >90 drinks/month (group4). Beverage specific intake was categorized as a relative percent intake of wine, beer or liquor over the past month. Inflammatory markers consisted of C-reactive protein (CRP) using a cut point of >3.0 mg/L, and fibrinogen. We performed multiple logistic regression analyses to estimate the odds ratio (OR) of an elevated CRP or fibrinogen using SUDAAN to account for the survey's complex sampling design.

RESULTS: The mean age (SD) of subjects was 44 (19) years; 50% were female. The number of subjects (%) and the median number of drinks per month in each group were: Group1: n=4395 (50%) drinks=0; Group2: n=3050 (35%) drinks=9; Group3: n=1031 (11%) drinks=52; and Group4: n=317 (4%) drinks=121. In a univariate analysis, the odds (95% CI) of an elevated CRP in each group relative to non-drinkers were: Group2 OR=0.52 (0.45-0.59), Group3 OR=0.45 (0.37-0.56), and Group4 OR=0.46 (0.31-0.68). After adjusting for age, gender, body mass index, race, smoking, physical activity, high density lipoprotein cholesterol (HDL), education, hypertension, bilirubin, diabetes, cardiovascular disease, medication use and diet, the odds of an elevated CRP was significantly lower in drinkers as compared with non-drinkers: Group2 OR=0.81 (0.71-0.92), Group3 OR=0.72 (0.56-0.92), and Group4 OR=0.73 (0.46-1.17). In the multivariable model, drinkers also had significantly lower fibrinogen levels: Group2 OR=0.77 (0.57-1.06), Group3 OR=0.72 (0.43-1.21) and group4 OR=0.55 (0.29-1.05), (p for trend=0.007). Compared with non-drinkers, those consuming $>75\%$ of their total intake as a single type of beverage had an odds ratio for an elevated CRP that varied from, OR=0.68 (0.49-0.94) for wine, OR=0.79 (0.62-1.01) for beer, and OR=1.04 (0.77-1.41) for liquor. Specific beverage type was not significantly associated with fibrinogen levels.

CONCLUSIONS: Alcohol consumption is associated with lower rates of elevated CRP and lower fibrinogen levels. Among drinkers, those who drink wine primarily have the lowest rates of high CRP's.

THE RELATIONSHIP BETWEEN SERUM BILIRUBIN AND C-REACTIVE PROTEIN. M. Benedict¹; J. Mrus¹; M.H. Eckman¹; J. Tsevat¹. ¹University of Cincinnati, Cincinnati, OH. (Tracking ID #133488)

BACKGROUND: C-reactive protein (CRP) is a serum marker of inflammation/oxidative stress that is important in the development of cardiovascular disease. Bilirubin, the end product of heme catabolism, has been purported to have important antioxidant functions but the relationship between serum bilirubin and CRP is unknown.

METHODS: We analyzed data on bilirubin and CRP from 8,599 adult participants in the Third National Health and Nutrition Examination Survey (NHANES III). Participants with liver disease, elevated transaminases, or serum bilirubin >5.0 mg/dL were excluded. We categorized serum bilirubin levels by quartile separately for men and women. C-reactive protein level was considered to be elevated if it was >3.0 mg/L. We performed multiple logistic regression analyses to determine the odds of an elevated CRP by quartile of bilirubin using SUDAAN to account for the multilevel, complex sampling design of the survey.

RESULTS: The mean (SD) age of subjects was 44 (19) years; 49% were female. The CRP level was elevated in 28% of the population. For males, the mean bilirubin (mg/dL) in each quartile was: Q1 (0.37); Q2 (0.50); Q3 (0.64); and Q4 (1.0); for women it was: Q1 (0.27); Q2 (0.40); Q3 (0.54); and Q4 (0.91). In a univariate analysis, the odds (95% CI) of an elevated CRP were: Q4 (referent); Q3: 1.56 (1.25-1.94); Q2: 2.02 (1.67-2.44); and Q1: 2.69 (2.09-3.47). After ad-

justment for age, race, alcohol intake, body mass index, educational level, physical activity, smoking, diabetes, cardiovascular disease, hypertension, high density lipoprotein cholesterol, dietary intake and medication use the odds (95% CI) of an elevated CRP were: Q4: (referent), Q3: 1.29 (0.99-1.68); Q2: 1.45 (1.16 - 1.82); and Q1: 1.99 (1.49-2.67). results were similar for non-Hispanic whites, non-Hispanic blacks and Mexican Americans. In men, the association was significant for the highest versus the lowest quartile of bilirubin while, in women, the association was significant across all quartiles.

CONCLUSIONS: Serum bilirubin levels are inversely related to levels of CRP. Given the established link between CRP and the risk of cardiovascular disease, the observed relationship suggests that bilirubin may play a preventive role in the development of coronary artery disease. While the cross sectional nature of the data precludes inferring causality, the observed inverse relationship between bilirubin and CRP warrants further investigation.

USE OF SELECTIVE SEROTONIN REUPTAKE INHIBITORS IS ASSOCIATED WITH INCREASED RATE OF HIP BONE LOSS. S. Diem¹; T. Blackwell²; K. Yaffe³; D.C. Bauer³; K.E. Ensrud¹. ¹University of Minnesota, Minneapolis, MN; ²California Pacific Medical Center Research Center, San Francisco, CA; ³University of California, San Francisco, San Francisco, CA. (Tracking ID #132430)

BACKGROUND: Depression is associated with low bone mineral density (BMD) in some studies but the biologic mechanisms underlying this association are unknown. Serotonin transporters have been documented in bone but their role in bone metabolism is unclear. The use of antidepressant medications, which often block the serotonin transporter, are associated with an increased risk of fractures in older women. We tested the hypothesis that elderly women who use selective serotonin reuptake inhibitors (SSRIs) and tricyclic antidepressants (TCAs) have increased rates of hip bone loss.

METHODS: We assessed current use of SSRIs and TCAs using an interviewer-administered questionnaire and serial BMD measurements in a cohort of 2722 older women (mean age 78.5 yrs) participating in the Study of Osteoporotic Fractures, a prospective cohort study of community dwelling women. Hip BMD was measured at the 6th exam and an average of 4.9 years later at the 8th exam. We verified current antidepressant use (defined as daily or almost daily use in the preceding 30 days) by inspection of medication containers and classified type of medication from product brand or generic names using a computerized medication dictionary. We categorized women according to their reported drug use as non-users (used neither SSRIs or TCAs at either exam, n=2406), SSRI user (use of SSRI at either exam but no use of a TCA, n=198), or TCA user (use of TCA at either exam but no use of a SSRI, n=118). Users of other classes of antidepressants at either visit were excluded. The mean annual percentage change in total hip BMD and 3 subregions of the hip were calculated for non-users, SSRI users, and TCA users. Depressive symptoms were identified using a cutoff score of ≥ 6 on the Geriatric Depression Scale as measured at the 6th exam. All results were adjusted for the following characteristics measured at the 6th exam: age, race, health status, functional status, walking speed, cognitive function, calcium supplement use, vitamin D supplement use, estrogen use, thiazide use, weight change, total hip BMD at the 6th visit, and depressive symptoms.

RESULTS: Mean total hip bone loss was -0.47%/year among non-users vs. -0.79%/year among the SSRI users (p=.0012). Mean total hip bone loss among TCA users (-0.46%/year) was not significantly different from non-users (p=.91). Findings were similar at the subregions of the hip and when women with depressive symptoms were excluded from the analysis.

CONCLUSIONS: Use of SSRIs in elderly women in this cohort is associated with increased rates of hip bone loss, while use of TCAs is not. Further work is needed to determine if SSRIs, a widely-used class of drugs, have a direct effect on bone metabolism.

VALERIAN FOR SLEEP: A SYSTEMATIC REVIEW AND META-ANALYSIS. S. Bent¹; A. Padula¹; D.H. Moore¹; M. Patterson¹; W. Mehling¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #135110)

BACKGROUND: Insomnia affects approximately one-third of the adult population and contributes to increased rates of absenteeism, health care utilization, and social disability. Current medical treatments have limited efficacy and troubling side effects. Extracts of the roots of valerian (*Valeriana officinalis*) are widely used for inducing sleep and improving sleep quality. We performed a systematic review to evaluate the evidence for the safety and efficacy of valerian for improving sleep quality.

METHODS: Studies were identified through a structured search of Medline (1966 to September 2004), EMBASE, IBIDS, BIOSIS, and the Cochrane Library using the keywords valerian, valeriana, and baldrian with no language restrictions. Randomized, placebo-controlled trials of valerian that reported a measure of sleep quality were included in the analysis. Two authors independently abstracted data regarding study eligibility, quality, treatment, blinding, withdrawals, adverse events, and sleep-related outcome measures.

RESULTS: Fifteen studies examining a total of 1152 patients satisfied all inclusion criteria. Valerian doses, preparations, and length of treatment varied considerably. Outcome measures were not consistently reported and limited the ability to pool results. A dichotomous outcome of sleep quality (improved or not) was reported by 6 studies and, when combined, showed a statistically significant benefit (RR of improved sleep=1.8, 95% CI: 1.2 to 2.9). The observed relationship between study size and outcome suggests that publication bias was present.

CONCLUSIONS: Despite methodological limitations, the available evidence suggests that valerian may be an effective treatment for insomnia. Larger studies with sleep quality measures are needed to further assess the safety and efficacy of valerian.

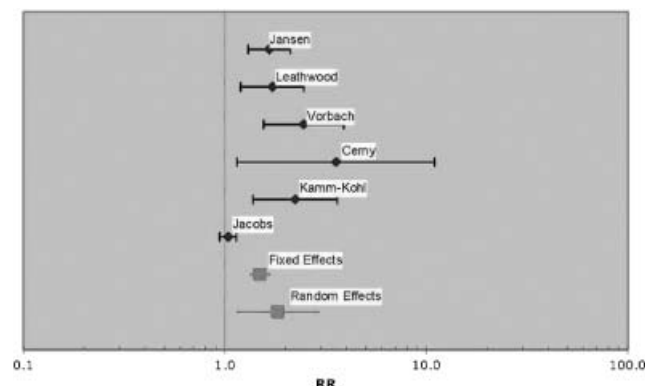


Figure: Meta-analysis of 5 Studies Reporting Dichotomous Outcomes for Sleep Quality (sleep improved or not)

WAIVER OR ALTERATION OF CONSENT FOR CLINICAL RESEARCH. B. Littenberg¹; C.D. Maclean¹. ¹University of Vermont, Burlington, VT. (Tracking ID #134206)

BACKGROUND: Federal laws and regulations, including the Health Insurance Portability and Accountability Act of 1996 (HIPAA) and the "Common Rule" governing human subjects protection, generally require that all research subjects provide consent. These provisions, intended primarily to protect individuals, have been described as significant barriers to the use of clinical registries and other population-based tools for health care research. However, the regulations do allow for the waiver or alteration of usual consent procedures when the research meets certain criteria: "(1) The research involves no more than minimal risk to the subjects; (2) The waiver or alteration will not adversely affect the rights and welfare of the subjects; (3) The research could not practically be carried out without the waiver or alteration; and (4) Whenever appropriate, the subjects will be provided with additional pertinent information after participation." Nonetheless, waivers and alterations are rarely used in health care research. Our objective is to describe the successful use of an altered consent process in community-based research.

METHODS: The Vermont Diabetes Information System collects laboratory data and patient identifiers from 13 hospitals around the region for a controlled trial of a quality improvement intervention. It includes 7,758 adults with diabetes receiving Primary Care from 146 providers in 71 practices in Vermont or nearby New York or New Hampshire. Randomization occurs at the practice level. Potential subjects (adults with diabetes) are identified by laboratory data and mailed a letter describing the study. They are asked to call a toll-free number (or their provider's office) to withdraw from the study. If they do not withdraw in ten working days, they are enrolled.

RESULTS: Thorough legal review by the University's Institutional Review Board, the National Institutes of Health, and legal representatives for the state-wide Quality Improvement Organization, hospital laboratories, and practices confirmed that the design meets all federal and state laws and regulations. Two-hundred-ten recipients requested withdrawal. An additional 31 patients called for more information but remained in the study. The overall participation rate was 97.3% (7548/7758).

CONCLUSIONS: HIPAA and other federal regulations raise challenges to the use of clinical registries in research, but alterations to the consent process are useful tools that allow low risk research to proceed under the new laws. It is possible to enroll a broad and representative population under current law while maintaining appropriate protections for research subjects.

A MULTIDIMENSIONAL INTEGRATIVE MEDICINE INTERVENTION TO IMPROVE CARDIOVASCULAR RISK. D. Edelman¹; E.Z. Oddone¹; R. Liebowitz¹; W.S. Yancy¹; M.K. Olsen¹; A.S. Jeffreys²; S.D. Moon¹; A.C. Harris²; L. Smith¹; R. Quillian-Wolever¹; T. Gaudet¹. ¹Duke University, Durham, NC; ²Durham Veterans Affairs Medical Center, Durham, NC. (Tracking ID #133082)

BACKGROUND: Integrative medicine is an individualized, patient-centered approach to health combining a whole person model with evidence-based medicine. Our objective was to determine if Personalized Health Planning (PHP), an intervention based on the theories and principles underlying integrative medicine, reduces 10-year risk of coronary heart disease (CHD).

METHODS: We conducted a randomized, controlled trial of a PHP intervention designed to help patients identify their own most important health problems and correct them. We enrolled 154 outpatients age 45 or over from local primary care practices with a history of at least one of these cardiovascular risk factors: hypertension, diabetes, obesity, or tobacco abuse. Following a health risk assessment, each patient in the intervention worked with a health coach and a medical provider to construct a personalized health plan. The plan identified specific

health behaviors important for each patient to modify; the choice of behaviors was driven both by cardiovascular risk reduction and the interests of each individual patient. The coach then assisted each patient in implementing her/his health plan. Techniques used in implementation included mindfulness meditation, relaxation training, stress management, motivational techniques, health education, and health coaching. Patients randomized to the comparison group received usual care without access to the intervention. The primary outcome was 10 year risk of CHD as measured by a standard Framingham-derived risk calculator, measured at baseline, 5, and 10 months. We performed linear mixed effects modeling for the primary analysis, with log-transformed 10-year CHD risk as the outcome and time and study arm as independent variables. All subjects, including drop-outs, were included in the model.

RESULTS: Baseline ten-year risk of CHD was 11.1% for patients randomized to usual care (UC, $n=77$), and 9.3% for patients randomized to PHP ($n=77$). Over the 10 months of the intervention CHD risk fell to 9.8% for usual care patients and 7.8% for intervention patients. Based on a linear mixed effects model, there was a statistically significant difference in the rate of CHD risk improvement between the two arms ($p=0.006$ at 5 months and $p=0.04$ at 10 months). In secondary analyses, patients in the PHP arm were found to have increased days of exercise per week compared with usual care (3.7 vs. 2.4, $p=0.002$); among patients overweight at baseline, those in the PHP arm had a greater weight loss compared with usual care (BMI of 32.1 vs. 33.4, $p=0.06$).

CONCLUSIONS: A multi-dimensional intervention based on integrative medicine principles reduced risk of CHD, possibly by increasing exercise and improving weight loss.

A RETROSPECTIVE ANALYSIS OF FACTORS ASSOCIATED WITH INCREASED LOCAL COMPLICATIONS FOLLOWING CARDIAC CATHETERIZATION. D. Zalenski¹; K. Nashar²; M. Aboyouseff²; M. Elnicki³. ¹University of Pittsburgh-Shadyside Campus, Pittsburgh, PA; ²University of Pittsburgh, Pittsburgh, PA; ³University of Pittsburgh Medical Center, Pittsburgh, PA. (Tracking ID #134238)

BACKGROUND: Over 800,000 cardiac catheterizations are performed annually in the United States. While major complications, such as stroke and death, are sited at <0.1%, local complications are much more common. Despite this fact, clinicians have few clinical predictors to help determine who will tolerate this procedure best. This study retrospectively investigates patient factors associated with cardiac catheterization complications. The objective was to identify variables that increased the likelihood of adverse outcomes, particularly bleeding risk.

METHODS: The records of 387 consecutive patients who underwent inpatient cardiac catheterization at a single institution were collected and retrospectively analyzed. We abstracted 16 variables as potential risk factors, which were evaluated in logistic regression models to derive their association with complications. Complications, recorded till hospital discharge, included excess bleeding at the site of entry and retroperitoneal extension of hematoma.

RESULTS: The mean age of the patients was 70.6 years old and 46% of the study population was male. The complication rate was 5.9% ($N=23$). The mean body mass index was 28.1 (SD 6.8). Complications were more likely in patients taking aspirin ($X^2=7.4$, $p<0.01$) or antiplatelet agents ($X^2=3.9$, p value=0.05), prior to the intervention and in those with known coronary artery disease ($X^2=4.3$, $p=0.04$). Other clinical characteristics analyzed: catheterization site, age, sex, body mass index, reason for catheterization, hypertension, diabetes, hypercholesterolemia, and tobacco use did not predict local complications. Steroids, heparin and warfarin use also had no association with catheterization complication rate.

CONCLUSIONS: In this retrospective analysis, local cardiac catheterization complications were significantly associated with anti platelet agents and a previous history of coronary artery disease. Unfortunately, these factors are not easily modified in patients presenting with indications for cardiac catheterization. Emphasis on frequent post catheterization monitoring for their development in this population is warranted. Contrary to popular perception, obesity and age did not lead to an increase in adverse outcomes.

A SYSTEMATIC REVIEW OF THE RELATIONSHIP BETWEEN INTERPERSONAL CONTINUITY AND PATIENT AND PHYSICIAN SATISFACTION. R.S. Adler¹; A. Vasilaidis¹; N.A. Bickell¹. ¹Mount Sinai School of Medicine, New York, NY. (Tracking ID #133485)

BACKGROUND: Interpersonal continuity between physicians and patients is a core tenet of primary care practice and likely affects patient and physician satisfaction. Studies on this subject have been published prior to 1984, but because of changes in the US health care environment and disruptions in the physician-patient relationship over the last 20 years, we sought to assess current evidence for the impact of continuity on patient and physician satisfaction.

METHODS: We systematically searched the English language Medline from 1984-2003 utilizing the MeSH terms: primary health care, continuity of patient care, patient and physician satisfaction, and all adult: 19+ years. Bibliographies of relevant articles were searched and additional unpublished data was obtained. Included studies met the following criteria: 1) study reported a measure of continuity and of patient or physician satisfaction 2) the continuity measure had to reflect direct continuity between a patient and a physician practicing internal medicine or family medicine. The review process and data extraction for each study were performed by two reviewers.

RESULTS: The initial search for continuity and patient satisfaction yielded 129 references and 35 relevant abstracts were reviewed. Based on abstract reviews, 12 full text articles were evaluated and 7 met inclusion criteria. Five additional articles that met inclusion criteria were obtained through review of article bibliographies for a total of 12 articles that we summarize. Continuity, patient satisfaction, and their relationship were measured and reported in multiple ways. There were 8 different measures for continuity and 10 for satisfaction. Nine studies showed a positive correlation, 2 studies showed mixed results, and 1 study showed no correlation. The magnitude of the association was highly variable and overall modest to weak ($r=0.08-.54$). The initial search for continuity and MD satisfaction yielded 13 references, 5 full text articles were selected for further review, and 2 met inclusion criteria. The same continuity measure was used in both studies and each measured physician satisfaction using a different instrument. There was a strong positive association between continuity and physician satisfaction ($r=0.54-.74$).

CONCLUSIONS: Continuity has a variable effect on patient satisfaction and the heterogeneity of the results may be due to the variety of continuity and satisfaction measures. For physician satisfaction, a greater and more consistent effect of continuity was found although the number of studies on continuity and physician satisfaction is limited.

APPROPRIATENESS OF ANTIVIRAL PRESCRIBING FOR INFLUENZA IN PRIMARY CARE. J.A. Linder¹; J.C. Chan¹; D.W. Bates¹. ¹Brigham and Women's Hospital, Boston, MA. (Tracking ID #135344)

BACKGROUND: Antiviral medications cost-effectively reduce influenza-related morbidity and potentially mortality. However, they receive relatively little use: nationally, physicians only prescribe antivirals to 19% of patients diagnosed with influenza. The appropriateness of this prescribing is unknown. We sought to 1) evaluate the appropriateness of antiviral prescribing for influenza and 2) identify factors associated with antiviral prescribing for influenza in primary care.

METHODS: We performed a retrospective analysis of visits by adults to Brigham and Women's Primary Care Practice-Based Research Network Clinics during influenza season from October 1, 2000 to May 31, 2004 with a primary claim diagnosis of influenza (International Classification of Disease-9 [ICD-9] code, 487; $n=535$). We used the visit note as the gold standard to verify the diagnosis of influenza. We also identified visits associated with electronic antiviral prescribing without a claim diagnosis of influenza, but with a chart-verified diagnosis of influenza ($n=25$). We determined if antiviral prescribing was appropriate based on information in the visit note. We defined appropriate antiviral prescribing as the patient having a) symptoms for 2 or fewer days, b) fever (patient-reported or temperature >100.5 degrees F), and c) any 2 of headache, sore throat, cough, or myalgias. We evaluated continuous variables with the Wilcoxon rank sum test, dichotomous variables with Fisher's exact test, and categorical variables with the Chi-squared test.

RESULTS: Among 535 visits with an ICD-9 code for influenza, the physician diagnosed the patient with influenza in 102 (19%) visits. Physicians prescribed antivirals in 15 (15%) of these visits. After adding an additional 25 visits associated with electronic antiviral prescriptions, we had a sample of 127 visits with a mean age of 41 (standard deviation ± 14) years old that was 76% female, 36% White, 32% Latino, 13% Black, and 19% other race/ethnicity. Of the 40 antiviral prescriptions, 28 (70%) were appropriate. Among patients who did not receive antivirals, 21 of 87 (24%) met criteria for appropriate antiviral prescribing. Antiviral prescribing was associated with a shorter symptom duration (median 2 days for those receiving antivirals versus 3 days; $p<0.001$), higher median temperature (100.0 degrees F versus 98.5 degrees F; $p<0.01$), and performance of an influenza test (20% versus 5%; $p<0.01$). Physicians prescribed antivirals more frequently to Blacks (44%) and patients with other race/ethnicity (67%) than to Whites (20%) and Latinos (20%; $p<0.001$).

CONCLUSIONS: An ICD-9 code for influenza had a poor positive predictive value for identifying physician-diagnosed influenza visits. Physicians prescribed antivirals in 15% of visits with a chart-confirmed diagnosis of influenza. Antiviral prescribing was inappropriate 30% of the time. In addition, antivirals may have been appropriate for 24% of patients diagnosed with influenza who did not receive them. Antiviral prescribing was associated with both clinical and non-clinical factors. To improve antiviral prescribing for influenza in primary care, interventions need to target the accurate identification of influenza visits, undertreatment, and also inappropriate treatment.

ARE PATIENT PSYCHOLOGICAL ASSESSMENT TOOLS APPLICABLE IN EVALUATING SURROGATES? W.A. Ury¹; J.R. Sood¹; D.P. Sulmasy¹. ¹New York Medical College, New York, NY. (Tracking ID #135721)

BACKGROUND: Surrogates give consent for most Do-Not-Resuscitate (DNR) orders, and anecdotal evidence suggests that end-of life care decisions are associated with high levels of stress. Although prolonged and extreme stress have each been associated with depression and anxiety, little is known about what psychological instruments would be appropriate to use with surrogates, since most instruments have only been developed and validated for use with patients.

METHODS: We assessed the stress level and psychological function of surrogates using the 15 item Horowitz Impact of Events Scale (IES), and the 28 item General Health Questionnaire (GHQ) during periods before and one year after instituting a change in DNR policy designed to minimize surrogate stress levels. Surrogates were interviewed an average of 8 days after consenting to a DNR order (pre=39, post=80). The IES has 2 subscales for intrusion and avoidance, and the GHQ consists of four subscales: somatic symptoms, anxiety, social dysfunction and depression. The chi-square statistic and T-tests were utilized in the analysis.

RESULTS: Surrogates ($n=119$) were mostly female (73.5%), caucasian (57.3%), non-catholic (54.7%), with a mean age of 55. The following table shows the pre

and post results for total IES and GHQ scale scores, and selected subscales. The subscale and total scale scores of each of the two instruments were all highly correlated (all scores, $p < .001$). Almost all (20/22) of the subscale and total scale scores from each instrument were correlated with the subscales of both of the instruments (all 20 significant scores, $p < .007$). We found no associations between either the IES, or GHQ, and sex or age of the surrogate, or the race, diagnosis, educational level, or insurance type of the patient.

CONCLUSIONS: These results indicate high levels of stress, equivalent to survivors of major earthquakes and fires, among the surrogates we studied. It appears that the IES scale might be a sensitive and responsive way of measuring stress level in surrogates. Further research that explains our findings, and establishes the validity, sensitivity and responsiveness of these, or other psychological assessment tools would seem warranted. Earlier detection of stress, depression and anxiety among surrogates may provide an opportunity for earlier intervention that could prevent or decrease the morbidity and psychosocial dysfunction associated with prolonged grief, major depression, and anxiety disorders.

	Pre	Post	p-value
IES Total	23.6	17.3	$p = .02$
IES Intrusion	16.2	8.4	$p < .0001$
IES Avoidance	7.4	8.9	$p = .35$
GHQ Total	28.8	26.1	$p = .23$
GHQ Social Dysfunction	9.4	7.9	$p = .002$

ASSESSMENT OF UNDERSTANDING: A QUALITY INDICATOR FOR COMMUNICATION BEFORE ADULT CANCER SCREENING. M.H. Farrell¹, P. Kuruvilla², R.S. Brienza³. ¹Medical College of Wisconsin, Milwaukee, WI; ²Contra Costa Regional Medical Center Family Practice Residency Program, Martinez, CA; ³Fairfield, CT. (Tracking ID #135865)

BACKGROUND: Screening for prostate and breast cancer have advantages, but both have been controversial enough in certain populations that clinicians are advised to discuss pros and cons before offering tests. Even if this communication takes place, it is unclear how effective it will be for the average patient, let alone those with limited health literacy. We are therefore developing communication quality indicators for use in efforts to improve communication services over entire health systems. One of our simplest indicators so far is whether clinicians include assessments of understanding (AUs) in their counseling. The objective of this study was to quantify the use of AUs by medicine residents in counseling prior to screening via prostate-specific antigen (PSA) or mammography.

METHODS: As part of a series of workshops, we transcribed 85 audiotapes of medicine residents counseling a 50-year-old male simulated patient about PSA screening and a 40-year-old female simulated patient about mammography screening. We used an explicit-criteria abstraction method to look in each transcript for 5 types of AUs: (1) *request for a "teach-back"* (eg, tell me in your own words what I just said) (2) *advance requests for questions* (eg, interrupt me if I say something you don't understand), (3) *open-ended AUs* (eg, what questions do you have), (4) *close-ended AUs* (eg, do you have any questions), and (5) the common but relatively ambiguous *"OK?" question* that is as much a request to proceed as an AU. The data dictionary also allowed for a *"partial"* AU designation if there was a leading syntax or no pause for a patient response. We calculated reliability via duplicate abstraction of 38% of transcripts and Cohen's method (corrected for chance, $k = 0.97$).

RESULTS: Only 215 full and 280 partial AUs were identified across the 85 transcripts; most were the ambiguous OK? question (126 full, 219 partial). Two transcripts (2.4%) contained a request for a teach-back; one other transcript contained a partial request with no pause for response. Only 1 transcript included an advance request for questions. Open-ended AUs were also rare; 2 transcripts each contained a single open-ended AU. Close-ended AUs were more common: 49 transcripts (57.6%) contained at least 1; 8 transcripts (50.8%) contained three or more. Using the quality indicator letter-grade approach described elsewhere, this analysis would have resulted in only 2 of 85 A grades (2.4%) and 2 B grades (2.4%). 81 transcripts would have qualified for our "targeted" intervention by virtue of C or lower grades: 47 C grades (55.3%), 19 D grades (22.4%), and 15 F grades with no AUs at all (17.7%).

CONCLUSIONS: The scarcity of AUs and the high prevalence of close-ended and partial AUs suggest that *communication quality before adult cancer screening is suboptimal*. Without actively trying to assess understanding, clinicians will not be able to tell whether their patients comprehend their risk communication, especially since many patients feel comfortable asking questions. In the next analysis we are studying how clinicians place AUs within counseling. This quantitative method should be useful in providing feedback and assessing for change in future studies.

ASSOCIATION BETWEEN EVIDENCE-BASED CARE AND WOMEN'S SATISFACTION WITH CARE. S.D. Ellis¹, D. Bonds¹, S.L. Palla¹, R.T. Anderson¹. ¹Wake Forest University, Winston-Salem, NC. (Tracking ID #136088)

BACKGROUND: The Institute of Medicine (IOM) and others call for healthcare delivery that is effective and evidence-based. At the same time, the IOM defines

quality care as that which is patient-centered, i.e., "respectful of and responsive to individual patient preferences, needs, and values." The measurement of patient satisfaction is one way to assess whether the health care delivered is patient-centered, especially when the patient satisfaction tool has been developed by asking patients their preferences for healthcare. However, little is known regarding whether patients perceive care that is evidence based to be satisfactory care; we sought to assess this.

METHODS: In a telephone survey of women in established relationships with primary care providers, patients were asked if they had received any of 16 different types of clinical or health behavior screenings or preventive counseling in the past year. Patient satisfaction was assessed using a summed score of items from the Primary Care Satisfaction Survey for Women. Based on U.S. Preventive Service Task Force recommendations, tests and screenings were coded as evidence based or non-evidence based, and within the evidence based category, coded as a physical test or verbal screen. The categories included: 1) evidence-based tests (blood pressure, mammogram, pap smear, STD screen, and colon cancer screen); 2) evidence-based verbal screening (screening for smoking, substance abuse and guns in the home, and family planning/contraception counseling); and 3) non-evidence based screening (physical exam, clinical breast exam, exercise, relationship concerns, safety/violence in the home, sexual functioning concerns, and stress management). For each of the three categories of health service we calculated the ratio of age-recommended tests received to those available, as well as the overall ratio. Using linear regression models we determined the association between the percentage of tests received to average satisfaction with care overall and by health service category.

RESULTS: Among 1636 women included, median age was 48 years; 63% were white; 48% married; 49% had household income less than \$25,000 per year; 87% had medical insurance. Fifty-two percent of patients had a female PCP; 60% saw a family medicine physician; and 43% had seen a provider five or more times in the past 12 months. Seventy-five percent of women reported receiving at least half of the evidence-based tests or screens, but only 20% reported receiving all of the age-appropriate evidence based care. Average satisfaction was 3.79 on a 1-5 likert scale in which 5 is extremely satisfied. The percentage of all 16 age-appropriate tests received was significantly and positively associated with average satisfaction ($p < 0.000$). In separate models distinguishing verbal versus physical evidence based tests, the receipt of physical tests was significantly associated with higher satisfaction ($p = 0.004$), but when non-evidence based tests are included with both types of evidence based tests, only receipt of non-evidence based screenings are significantly associated with increases in patient satisfaction ($p < 0.000$).

CONCLUSIONS: The number of non-evidenced based services provided appears to predict high satisfaction with care, an indicator of patient-centered care. We hypothesize that these services enhance the patient-physician relationship, despite their lack of proven effectiveness in improving health outcomes.

ASSOCIATION OF PHYSICIAN CULTURAL COMPETENCE AND DIVERSE LIFE EXPERIENCES WITH THEIR PATIENT'S FUNCTIONAL STATUS, BLOOD PRESSURE AND GLYCEMIC CONTROL. Y.M. Diaz¹, J. Bussey-Jones², M.A. Shaw³, E. Liles⁴. ¹University of Miami, Miami, FL; ²Emory University, Atlanta, GA; ³University of Louisville, Louisville, KY; ⁴University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #135423)

BACKGROUND: Physician cultural competence has been proposed as a means to address health care disparities in the United States. The purpose of this study is to examine the association between physician cultural competence and patient clinical outcomes.

METHODS: The CCTOP study (Cultural Competency Training and Outcomes in Patients) is a multicenter, cross-sectional study of patients presenting for care at 11 geographically diverse outpatient clinics and their physicians. Physician cultural competence was scored using the Inventory for Assessing the Process of Cultural Competence Among Healthcare Professionals (IAPCC), a validated scale (Campinha-Bacote, 1999). Physicians were also asked questions about cultural knowledge, diverse life experiences, and demographics. Patients were verbally administered a 76-item survey regarding their health and demographics. We examined the association between the IAPCC score, physician cultural knowledge, and diversity experiences with subscales of the SF36 Health Survey, blood pressure, and self-reported glycemic control. Multivariate regression analyses were performed controlling for age, gender, race, education and income.

RESULTS: We present preliminary results from 6 participating centers. 290 patients responded (203 women; 87 men) with a mean age of 56.5 years. 29% of patients reported their race as white, 44% black, 24% hispanic, and 3% other. One hundred eighteen physicians were residents and fifty-seven were attendings. Multivariate analysis revealed that, overall, patients cared for by physicians with more diverse life experiences scored higher on the SF-36 vitality subscale ($p = 0.013$). Physician diverse life experience was also associated with higher SF-36 vitality scores ($p = 0.03$) in a subset analysis of patients with hypertension and higher SF-36 role-physical scores ($p = 0.038$) in patients with diabetes. No association was found between cultural competence scores (IAPCC) or physician cultural knowledge with other subscales of the SF36, blood pressure in hypertensive patients, or self-reported glycemic control in diabetics.

CONCLUSIONS: Physician diversity life experiences may be associated with improvement in some patient outcomes. However, overall patient health and outcomes appears to be minimally impacted by physician cultural competence or knowledge, suggesting that physicians with similar backgrounds, attitudes, and cultural experience and knowledge have similar practice patterns. Further investigation is needed to determine the role of cultural competence in reducing health disparities.

CAN PATIENTS CONTRIBUTE ACCURATE INFORMATION TO THEIR ELECTRONIC HEALTH RECORD? L. Wuerdeman¹; L.A. Volk¹; L. Pizziferi¹; R. Tsurikova¹; C. Harris¹; R. Feygin¹; M. Epstein¹; J. Wald¹; D.W. Bates¹. ¹Partners HealthCare System, Inc., Wellesley, MA. (Tracking ID #135802)

BACKGROUND: Accurate and complete data are critical for patient care and it is also becoming increasingly important to support quality measurement data for pay for performance initiatives. Electronic health records (EHRs) currently rely primarily on medical practitioners and staff for review of accuracy. Valuable medical information is often missing from EHRs, which may compromise patient care and quality measurement. To address this issue we offered an online survey to patients to assess the extent to which patient reported information could enhance medical record and quality measurement data.

METHODS: This study involved 6 practices, 44 physicians and a total of 140 office staff. Patients from these practices were invited to complete one or more online surveys through the Partners Healthcare System's secure web-based patient portal, Patient Gateway. The surveys were developed by the Foundation for Accountability (FACCT). All participating patients were offered a general health survey. Each patient's answer was compared to current information in his or her ambulatory EHR.

RESULTS: In this study, 1094 out of 6008 possible patients participated. Respondents were predominately white (86%), 55% female, with a mean age of 47.4 years, similar to the invited population. Patients were better able, on average, to report only whether or not they had a screening test without indicating specific results. Information missing from the EHR regarding the presence or absence of tests was often provided by answers in patient surveys. Using the EHR as the gold standard, whenever there were valid measurements in both the EHR and patient surveys, patients' ability to report their values within specified criteria ranged from 70-88% with an exception of height, which matched 100% of the time. For health maintenance screenings such as Pap smear and blood in stool requiring only a yes or no answer, patients' answers were missing less than 20% of the time and matched the LMR for 67-90% of patients.

CONCLUSIONS: Patient reported data will be of variable value in enhancing quality measurement data depending on the health measure of interest. Patients can often report valuable information on the presence of tests which will make the EHR more complete. However, not surprisingly, patients' cannot always provide test result information (such as cholesterol levels) beyond data in the EHR. With the increase in attention given to Personal Health Records and patient portals, patient reported data should be considered a viable method of enhancing documentation but will not likely substitute for more comprehensive data-exchange between providers.

Table 1. Selected Health Measures – Absence in Electronic Health Records and Patients' Ability to Report (only a subset of tests studied are displayed)

Health Measure	% missing in LMR	% missing in patient survey	% missing in LMR that patient provided	% valid values in both	% match when 2 values available	Match Criteria
Height	64%	3%	95%	33%	100%	within 2 inches
Total Cholesterol	40%	71%	6%	26%	88%	within 20 points
Pap Smear (Women Only)	4%	2%	100%	94%	86%	ever had yes/no
Blood in Stool (patients ≥ 50 yrs)	74%	12%	82%	24%	90%	ever had yes/no

CLINICIAN ASSESSMENT OF FAMILY CANCER HISTORY AND COUNSELING ABOUT HEALTH-RELATED BEHAVIORS AMONG WOMEN WITH A HISTORY OF SCREENING MAMMOGRAPHY. L. Karlner¹; A. NaPoles-Springer¹; J.S. Haas²; K.M. Kerlikowske¹; G. Gildengorin¹; C.P. Kaplan¹. ¹University of California, San Francisco, San Francisco, CA; ²Brigham and Women's Hospital, Boston, MA. (Tracking ID #134981)

BACKGROUND: A woman's family history of cancer and her health-related behaviors (HRB) both affect her cancer risk; clinicians' knowledge of these factors is necessary for appropriate breast cancer risk management. We examined the association between a woman's perceived risk of and worry about breast cancer, history of preventive medical care, and clinician-patient communication, with the likelihood that her clinicians asked about her family's general cancer history, or ever counseled her about HRB and breast cancer risk.

METHODS: Cross-sectional telephone survey about breast cancer prevention. Women were recruited in a stratified manner based on their five-year risk of developing breast cancer (Gail Model - high or average/low) and their ethnicity (African American/Asian Pacific Islander/Latina/White). They were eligible if they had a screening mammogram between 2001-2002 at 1 of 5

facilities that participate in the San Francisco Mammography Registry, no personal history of breast cancer and were age 40-70. Predictors included perceived ten-year risk of developing breast cancer (high or average/low), worry about developing breast cancer (higher score=more worry), experience of attentive communication (higher score=better communication) and dismissive communication (higher score=worse communication) with their clinicians, and having had at least one well-visit in the past year. Primary outcomes were whether a woman reported that 1) clinician had asked in past year about her family's general cancer history, and 2) clinician had ever discussed breast cancer risk in relation to at least one HRB (fatty foods, alcohol, regular physical activity, bodyweight). Multivariate models controlled for age, ethnicity, education, language of interview, insurance/mammography facility, ever having a biopsy and/or follow-up mammography, Jewish heritage, family history of cancer, body mass index, and Gail Model risk category.

RESULTS: Among 1700 women who participated in the survey (63% response rate), the majority (87%) had at least one well-visit in the past year. Only 971 (58%) reported a clinician had asked about her family's cancer history in the past year; only 582 (34%) said a clinician had ever discussed HRB in relation to breast cancer risk. On average, women were minimally worried about breast cancer (range 1-4; mean 1.3+/-0.4). Report of a well-visit in the last year, higher score on attentive communication scale, lower score on dismissive communication scale, and higher score on worry scale were all associated with increased likelihood that a clinician had asked about family cancer history. Only less dismissive communication, higher worry and high 10-year perceived risk of breast cancer were independently associated with a clinician ever discussing HRB in relation to breast cancer risk.

CONCLUSIONS: Clinicians could substantially increase their rates of assessing family cancer history and counseling about health-related behaviors in relation to breast cancer risk. Enhanced opportunity for preventive visits, attentive communication, and evaluation of a woman's level of worry about breast cancer may improve those rates.

Predictor	Clinician asked about general family cancer history Odds (95% CI)	Clinician ever discussed health-related behavior in relation to breast cancer Odds (95% CI)
At least one well-visit in past year	2.8 (2.0-3.9)	1.2 (0.9-1.8)
Clinician more attentive	1.3 (1.0-1.8)	1.3 (0.9-1.8)
Clinician more dismissive	0.8 (0.7-1.0)	0.8 (0.7-1.0)
Higher worry about breast cancer	1.4 (1.0-1.8)	2.1 (1.6-2.8)
High 10-year perceived risk of breast cancer	NS	1.5 (1.2-2.0)

COMMUNICATION ABOUT SYMPTOMS IN WALK-IN CLINICS. J.L. Jackson¹. ¹Uniformed Services University of the Health Sciences, Bethesda, MD. (Tracking ID #131612)

BACKGROUND: In previous studies of patient-doctor communication, we found disagreement between patients and clinicians about whether a discussion of the symptom's cause or likely duration occurred. Our purpose was to explain this discrepancy.

METHODS: 200 consecutive adults presenting with a physical symptom were audiotaped. Post-visit patient surveys assessed unmet expectations, trust and satisfaction. Clinicians were surveyed for diagnosis and what was done for the patient. Audiotapes were reliably coded (Spearman: 0.81) for utterances on symptom cause or duration into one of three categories: explicit, implicit or none.

RESULTS: 200 patients (age: 54.5, 45% Black, 48% White) were seen by 28 clinicians. Diagnosis results: In 93% of the encounters, clinicians reported telling the patient their diagnosis, and recorded a specific diagnosis in 98% of charts. In contrast, 69% of patients reported being given a diagnosis. 55% of audiotapes had an explicit statement regarding diagnosis, while 22% contained implicit statements, such as "I don't think it's anything serious." 23% of tapes had no statement of symptom cause. There was no difference in patient's perception of whether they were given a diagnosis between encounters with implicit or explicit statements (77% vs 72% p=0.61). Patients were more likely to report being told the symptom cause if there was any diagnostic statement on the audiotape (RR: 1.4, 95% CI: 1.1-1.9). However, even when no statement was heard, patients recalled receiving a diagnosis 53% of the time. Visits with any diagnostic statements had higher patient satisfaction (p=0.03) and the clinician was more likely to be trusted by their patients (RR: 1.75, 95% CI: 1.1-2.1), than encounters without such statements. Prognosis results: Conversations about duration were less frequent. Clinicians reported providing duration information 64% of the time, while patients recalled receiving it in 50% of encounters. Most audiotapes (72%) had no statements about duration. Explicit duration statements were heard in 11%, and implicit ones, such as "I don't think this will last very long," in 17% of tapes. When an explicit statement was made, 93% of patients reported hearing one, compared to 47% and 40% of patients from encounters with implicit or absent duration statements. Encounters with explicit duration discus-

sions were less likely to report unmet expectations after the visit (RR: 0.24, 95% CI: 0.1-0.9).

CONCLUSIONS: It appears that some communication about symptoms is non-verbal, as both parties report its occurrence in the absence of objective audiotape evidence. When present, statements regarding the specific diagnosis and symptom duration are associated with increased trust and satisfaction and fewer unmet expectations. The infrequency of this discussion offers a possible focus for improvement in relationship-centered care.

CONTENT OF PATIENT MESSAGES SENT VIA A SECURE WEB PORTAL DESIGNED FOR PATIENT-CLINIC COMMUNICATION. L. Pizziferri¹; M. Lippincott¹; L.A. Volk¹; C. Wambsgans¹; J. Wald²; D.W. Bates¹. ¹Partners HealthCare System, Inc., Wellesley, MA; ²Partners HealthCare System, Wellesley, MA. (Tracking ID #136255)

BACKGROUND: Patient interest in electronic communication with physicians is growing. However, few physicians currently exchange e-mail regularly with patients, in part because of concerns regarding security, workload, and inappropriate use. Secure, triage-based web portals are now being developed to help manage these concerns. Our objective in this research was to evaluate the content of patient messages on a secure web portal, and assess the potential risk and benefits of using a patient-portal, in particular the frequency with which patients inappropriately report urgent issues using this modality.

METHODS: A qualitative review was performed in July 2003 of messages exchanged via Patient Gateway (PG), a secure web portal which allows patients and clinic staff to exchange messages online. Messages were categorized to determine the type of issues patients commonly write about when sending messages to the clinic, the average number of issues contained in patient messages, the rate of messages requiring physician effort, and the extent of inappropriate messages, in particular how often they were too urgent to be sent in this way. **RESULTS:** Of the 3190 messages reviewed, 1650 messages were generated by 688 patients. The remaining messages were generated by staff. There were 1463 message "threads" or back-and-forth replies between patients and staff. Overall, the average number of issues per message was 1.20 (SD=0.50). When patients sent messages via PG, they chose one of four "desks" as the recipient: Appointment Desk, Message Desk, Medication Desk, or Referral Desk and could use request forms for appointments, medication, and referral approval requests. Overall, free-text messages to the Appointment Desk had the greatest mean number of issues per message (1.40, SD=0.69). Appropriate clinic staff were assigned to handle particular message desks. The content of messages sent to each of the message desks is described in Table 1. Although the number of threads involving physicians sending messages to patients was relatively low (3.8%), messages sent to the Message Desk categorized as medication (10.4%), clinical (14.1%) or results (11.1%) related may have involved physicians outside of PG. Only 1.5% of all messages indicated a need for a 24-hour turnaround. There were only 2 messages which were clinically urgent.

CONCLUSIONS: Overall, patients appropriately used the messaging feature of the Web portal to communicate with their clinic; generally only one issue was contained per message and the overwhelming majority did not send time-sensitive or clinically urgent messages. However, reminders about appropriate use should be incorporated within the tool and messaging features to instill that electronic communication is not suitable for all issues. Most messages can be appropriately handled by administrative staff, though some messages do require clinician involvement.

CURRENT TREATMENT OF ADULT OUTPATIENT URINARY TRACT INFECTIONS. A.J. Kallen¹; B.E. Sirovich¹. ¹VA Outcomes Group, White River Junction, VT; Dartmouth Medical School, Hanover, NH. (Tracking ID #135584)

BACKGROUND: Urinary tract infections (UTI) are a common problem encountered in the primary and acute care setting. Due to their low cost and broad activity, sulfa antibiotics such as trimethoprim-sulfamethoxazole have traditionally been the drug of choice - and as of at least the mid 1990's they were the most commonly prescribed antibiotic for UTI. However, increasing resistance among *Escherichia coli* to sulfa antibiotics has prompted some to recommend that quinolones be used as first line agents for UTI in certain settings. We reviewed a large national database of outpatient visits to estimate the number of adult urinary tract infections and the prevailing therapeutic approach.

METHODS: We used the 2002 National Ambulatory Medical Care Survey and the National Hospital Ambulatory Medical Survey to identify all outpatient visits

by adults (> 17 years old) who received the diagnosis of urinary tract infection or cystitis. Visits were excluded if the patient was admitted to the hospital or was diagnosed with an upper urinary tract infection or an additional concurrent infection. The final sample included 1,142 visits. All analyses incorporated the appropriate sampling weights and accounted for a multistage clustered sampling strategy.

RESULTS: An estimated 11.8 million adult outpatient visits resulted in the diagnosis of urinary tract infection in 2002. Eighty-six percent of these visits were in women, 48% involved patients younger than 46 years old and 15% involved those 25 years old or younger. Most visits took place in physician offices (77%) rather than the emergency room (17%) or hospital clinics (7%). A urinalysis was ordered in 75% of visits; however, a urine culture was ordered in only 21%. Quinolones were the most commonly prescribed class of antibiotic (48%) followed by sulfas (33%, $p < 0.001$ for difference). Quinolones were more likely to be prescribed in episodes occurring in the Northeast (64%) compared to the Midwest (39%), South (47%) or West (42%). Excluding episodes in which more than one quinolone was prescribed, ciprofloxacin (61%) was the most often used followed by the more broad spectrum agents levofloxacin (30%) and gatifloxacin (5%).

CONCLUSIONS: The majority of visits for UTI occur in physician offices and do not result in a urine culture. Quinolones are now the most common antibiotic class used for the treatment of UTI, a change that has developed over the last decade. Increasingly common use of quinolones - including newer broader spectrum agents - may lead to the development of new patterns of resistance, including increasing resistance to this antibiotic class.

DEFICIENT HYPERTENSION MANAGEMENT OF PATIENTS WITH DIABETES IN PRIMARY CARE CLINICS. M.G. Weiner¹; C.A. Umscheid¹; C. Hollenbeck²; S. Virabhak³; Y. Lin¹; R. Gross¹; B.J. Turner¹. ¹University of Pennsylvania, Philadelphia, PA; ²Pennsylvania State University, Hershey, PA; ³Pfizer, Inc., New York, NY. (Tracking ID #135697)

BACKGROUND: Standards for hypertension (HTN) control in patients with diabetes (DM) have become increasingly strict under the JNC VI guidelines. We examined HTN control and management in primary care practices (PCPs) to evaluate: 1) if HTN is being more aggressively controlled in patients with than without diabetes; 2) predictors of HTN control in patients with diabetes at the JNC VI recommended level; and 3) predictors of inappropriate HTN management in patients with diabetes.

METHODS: We studied a retrospective cohort of HTN patients aged ≥ 18 and had at least three visits from 1/1/03 to 8/01/04 in one of 8 practices with a total of 258 clinicians (including nurses and residents) in Philadelphia urban and suburban settings. Demographic, clinical, laboratory, pharmacological and health care utilization data were obtained from electronic medical records and system-wide administrative data. DM was based on an ICD-9-CM diagnosis or HbA1C > 7 . HTN was defined using hierarchical criteria: 1) a prescribed BP medication (med); 2) ICD-9-CM diagnosis of HTN and ≥ 1 visit with a high BP defined as systolic BP (SBP) ≥ 140 or diastolic BP (DBP) ≥ 90 ; or 3) high BP on ≥ 2 visits. For each prescribed BP med, we determined the total daily dose and duration of treatment. High BP based on JNC VI for patients without diabetes was SBP ≥ 140 and DBP ≥ 90 and for patients with diabetes, a more stringent threshold of SBP ≥ 130 or DBP ≥ 85 was set. At each visit, management was considered appropriate if BP was normal or, if high, the dose of a current BP med was increased, a new BP med was started, or a med was renewed if expired > 1 month. The unit of analysis was the visit so random effects multiple logistic regression was used to adjust for clustering of patient and physician.

RESULTS: Our cohort included 3,597 patients with diabetes who accounted for 22,472 primary care visits while 10,086 patients without diabetes made 48,718 primary care visits. Using a less stringent criterion for high BP (i.e. either SBP ≥ 140 or DBP ≥ 90) in all study patients, visits of patients with and without diabetes were equally likely to have uncontrolled HTN (38.4 vs 38.7%, respectively). Adjusting for demographic, clinical, and health care factors, the adjusted odds ratio (AOR) of inappropriate HTN management at this less stringent BP level was lower for patients with diabetes (0.86, CI 0.81, 0.91). In analyses of patients with diabetes only and using a more strict high BP criterion of SBP ≥ 130 or DBP ≥ 85 , 57.0% of visits had uncontrolled HTN while 45.8% had inappropriate HTN management. In patients with diabetes only, the AORs of high BP at this level were increased ($P < 0.001$) by: two-fold for age 70+ yrs vs < 50 yrs, 57% for Black race, 36% for > 1 year since last visit, and 24% for morbid obesity (> 240 lbs) while the AOR for coronary artery disease (CAD) was reduced by 30%. Among diabetics, the AORs of inappropriate HTN management were increased ($P < 0.001$) for increasing age, Blacks, morbid obesity but reduced for CAD and > 1 year since last visit.

Table 1: Number and Percentage of Message Issues Sent by Message Desk

Desk	Type	Appointment	Medication	Referral	Clinical	Results	Registration	Other	PG Tool	Total
Appointment Desk	Form	136 (82.4%)	5 (3.0%)	3 (1.8%)	14 (8.5%)	3 (1.8%)	0 (0%)	1 (0.6%)	3 (1.8%)	165 (8.3%)
Medication Desk	Form	4 (0.9%)	451 (96.6%)	0 (0%)	5 (1.1%)	1 (0.2%)	3 (0.6%)	2 (0.4%)	1 (0.2%)	467 (23.6%)
Referral Desk	Form	0 (0%)	0 (0%)	263 (92.0%)	1 (0.3%)	0 (0%)	20 (7.0%)	0 (0%)	2 (0.7%)	286 (14.4%)
Appointment Desk	Free Text	37 (51.4%)	3 (4.2%)	3 (4.2%)	8 (11.1%)	9 (12.5%)	1 (1.4%)	5 (6.9%)	6 (8.3%)	72 (3.6%)
Medication Desk	Free Text	6 (3.4%)	109 (62.6%)	3 (1.7%)	19 (10.9%)	0 (0%)	5 (2.9%)	13 (7.5%)	19 (10.9%)	174 (8.8%)
Message Desk	Free Text	48 (8.0%)	63 (10.4%)	26 (4.3%)	85 (14.1%)	67 (11.1%)	246 (40.8%)	43 (7.1%)	25 (4.1%)	603 (30.4%)
Referral Desk	Free Text	16 (12.3%)	1 (0.8%)	68 (52.3%)	32 (24.6%)	1 (0.8%)	3 (2.3%)	7 (5.4%)	2 (1.5%)	130 (6.6%)
Other	Free Text	6 (7.1%)	8 (9.4%)	7 (8.2%)	23 (27.1%)	18 (21.2%)	4 (4.7%)	6 (7.1%)	13 (15.3%)	85 (4.3%)
Total		253 (12.8%)	640 (32.3%)	373 (18.8%)	187 (9.4%)	99 (5.0%)	282 (14.2%)	77 (3.9%)	71 (3.6%)	1982 (100%)

CONCLUSIONS: Primary care providers do appear to be more aggressive in trying to control BP in patients with diabetes at a less stringent level of 140/90 but are not attempting to achieve HTN control at the stricter standard for control recommended by JNC VI. Greater age, Black race, and morbid obesity were all independently predictive of both uncontrolled HTN and inappropriate HTN management in patients with diabetes.

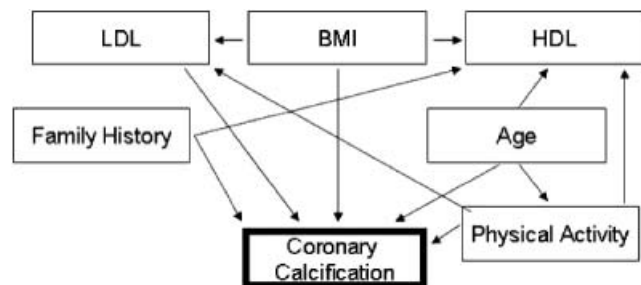
DETERMINANTS OF SUBCLINICAL ATHEROSCLEROSIS USING STRUCTURAL EQUATION MODELING. K. Degee¹; P.G. O'Malley¹; A.J. Taylor¹; J.L. Jackson². ¹Walter Reed Army Medical Center, Washington, DC; ²Uniformed Services University of the Health Sciences, Bethesda, MD. (Tracking ID #135712)

BACKGROUND: Coronary artery disease (CAD) risk factors and markers have been associated with subclinical atherosclerosis, but no reports describe the amount of variance these factors explain, or the relative direct and indirect effect of these associations. We used structural equation modeling (SEM) to model the complex web of interactions among conventional, emerging and behavioral risk factors and markers on coronary calcification (CAC), a valid surrogate marker of atherosclerosis burden.

METHODS: We used data from the Prospective Army Coronary Calcification project, a prospective study of 2000 participants between the ages of 39 and 50 presenting for a periodic physical. All underwent an assessment for coronary atherosclerosis using EBCT. Laboratory data included c-reactive protein, fibrinogen, homocysteine, LDL, HDL, lipoprotein(a), HgbA1C, fasting glucose and insulin. Demographic data included age, BMI, blood pressure and family history. Behavioral data included tobacco and alcohol use, physical activity, hostility, depression, somatization and functional status using validated instruments. Women were excluded from our analysis as >90% had no CAC. We subsequently built a model, using SEM, to explain the log of the CAC scores.

RESULTS: There were 1558 men with complete data for analysis. The mean age was 43 (sd 2.7), 6% were current smokers, and 22% had coronary calcification. A best fitting model explained only 5.2% of the variance of coronary calcification (CAC) scores (figure shows simplified model, without depression scale, insulin level, HgbA1c, systolic blood pressure, or smoking status). Emerging risk factors (C-reactive protein, Lipoprotein (a), and homocysteine) were not significantly associated with CAC. The most important risk factor was age, followed by family history, LDL, and BMI. Nearly half of BMI's effects on coronary calcifications were indirect via LDL and HDL.

CONCLUSIONS: In this study of young to middle aged men, the vast majority of the variance in coronary atherosclerosis is unexplained. BMI's effects on coronary calcification are significantly mediated through cholesterol levels.



Simplified SEM Model of Coronary Calcification

Standardized total effects on coronary calcification, n = 1558, 11 variables total

	Total Effect	% Direct Effect
Age	.16	94
LDL	.10	100
Family History	.09	99
BMI	.06	58

DIAGNOSING OVERACTIVE BLADDER USING THE OVERACTIVE BLADDER-VALIDATED 8 IN A PRIMARY CARE POPULATION IN THE UNITED STATES. V. Elinoff¹; T. Zyczynski²; T. Bavendam². ¹Regional Clinical Research, Inc., Endwell, NY; ²Pfizer, Inc, New York, NY. (Tracking ID #135606)

BACKGROUND: The Overactive Bladder-Validated 8 (OAB-V8) is a highly predictive screening tool for overactive bladder (OAB) in the primary care setting. To further examine the impact of considering both clinical questions and screener scores when diagnosing OAB, additional analyses were performed. Our aim was

to gain additional insight in determining an OAB diagnosis in the primary care setting.

METHODS: The OAB-V8 was evaluated as an OAB screener. The question, "How bothered are you. . . ?" by 8 symptoms was answered on a 6-point scale ranging from 0 (not at all) to 5 (a very great deal). Patients were recruited from 12 primary care practices and 1 gynecology practice as they presented for regularly scheduled appointments. After completing the screener, all patients were asked 4 questions by the clinician regarding the urinary symptoms of frequency, urgency, nocturia, and urge incontinence (UI). If the patient responded affirmatively to these questions or had a positive screener score (≥ 8), additional questions regarding lifestyle/coping issues were asked. Upon completion of the clinician questions and based upon clinician opinion, the patient was diagnosed as "No OAB," "Possible OAB," or "Probable OAB." Logistic regression models predicting "No OAB" versus "Probable OAB" were performed to include frequency, urgency, nocturia, and UI as covariates in addition to age, sex, and screener score.

RESULTS: 1260 patients (mean age, 52 ± 17 y) were included; 62% were women, and 89% were white. Patients experienced a mean of 5.7 ± 3.2 urinations/d; 1.3 ± 1.2 nocturia episodes/night. 22% experienced urinary urgency; 18% experienced UI. The model was highly predictive with a c-index of 0.97. The following variables were significant in the model: age ($p=0.008$); screener score ≥ 8 ($p<0.0001$); number of urinations/d ($p=0.002$); urgency ($p<0.0001$); and UI ($p=0.03$). Odds ratios and 95% confidence intervals predicting "Probable OAB" are presented in the table.

Table. Summary of Results

Variable	OR	95% Wald CIs
Age	1.03	1.01 1.05
Sex	0.96	0.50 1.82
OAB V-8 screener score ≥ 8	38.76	11.37 132.2
Urinations/d	1.19	1.07 1.33
Nocturia episodes	1.29	0.99 1.67
Presence of urgency	4.84	2.56 9.14
Presence of UI	2.32	1.09 4.91

CIs= confidence intervals; OR= odds ratio; UI= urinary incontinence

CONCLUSIONS: The OAB-V8 screener score and presence of urinary urgency are the 2 strongest predictors of OAB diagnosis in a primary care setting.

DISAGREEMENT OF PATIENT AND PHYSICIAN PERCEPTIONS CONCERNING THE SAME PRIMARY CARE ENCOUNTER. L.D. Herman¹; C.M. Bright²; J. Bussey-Jones³. ¹University of Tennessee at Chattanooga, Chattanooga, TN; ²Duke University, Durham, NC; ³Emory University, Atlanta, GA. (Tracking ID #136041)

BACKGROUND: Health disparities resulting in substandard care for minorities are documented. Clinical encounters have been postulated as a potential source of these disparities. Physicians' perceptions of patients have been shown to be negatively affected by race (van Ryn & Burke 2000). Little is known about the concordance of physicians' and patient's views on the primary care encounter. The CCTOP study examined patients' perceptions of treatment by their physician compared to physicians' perceptions of their patients.

METHODS: The CCTOP study is a multi-center cross-sectional survey of patients and physicians in primary care settings. The questions addressed attitudes on culture and bias and their effects on healthcare. A 76-item survey administered to patients included questions on how they felt they were viewed and treated by their doctor. Following the visit, the doctor was asked to rate the patient with a 14-item survey on the patient's demeanor, intelligence, and adherence to medical treatment. Responses were dichotomized for bivariate analyses.

RESULTS: A total of 291 established patients were surveyed. We present preliminary results from 250 matched patient-physician surveys. Patient's self-reported races were 44% Black (B), 29% White (W), 24% Hispanic (H), and 3% other. Physician's races were 57% W, 6% B, 10% H, and 23% other. Physicians rated their minority and non-minority patients equally on being pleasant and cooperative. Physicians viewed Black and Hispanics as more rational (93% B, 93% H, 82% W, $p=0.045$) and self-controlled (89% B, 92% H, 77% W, $p=0.031$). Blacks and Whites were rated equally on intelligence and Hispanics more intelligent (82% B, 83% W, 98% H, $p=0.018$). There was no association between patient's race and perceived likelihood to adhere to medical treatment. Nor was there an association between race and feeling like their doctor viewed them as smart. However, Black patients felt they were treated with less courtesy and respect by their doctor (22%B, 11% H, 8% W, $p=0.005$). More Hispanics and Blacks felt their doctor did not care for them (22% H, 13% B, 8% W, $p=0.013$)

nor listen to them (11% B, 7% H, 1% W, $p=0.012$). Overall more Blacks and Hispanics felt they received poorer care than Whites (22% B, 12% H, 5% W, $p=0.005$). There was no significant difference among patients by ethnicity in trusting their doctor to tell them if a mistake was made, or in preferring a physician of the same race. All ethnic groups reported following their doctor's advice.

CONCLUSIONS: Despite physicians' rating their minority patients equal or better than whites on multiple personality traits, minority patients felt they received inferior care. Yet they trusted their doctor, followed their advice and did not wish to seek another doctor of the same race. Perhaps the factors that contribute to health care disparities and patients' feelings of bias arise from experiences outside the medical encounter itself. The variance in physician and patient answers suggests that the effect of prejudice on patient interactions is not well recognized by physicians. More in-depth study of physician behavior and patient reactions is needed to exclude the medical encounter as a source of health disparities.

DISEASE SEVERITY AND BURDEN OF ATHEROSCLEROSIS AS RISK FACTORS FOR ADVERSE LIMB EVENTS IN PATIENTS WITH PERIPHERAL ARTERIAL DISEASE. T.C. Collins¹; P. Krueger¹; M. Ku-Goto¹. ¹Baylor College of Medicine, Houston, TX. (Tracking ID #136153)

BACKGROUND: We sought to determine the association of disease severity and the number of atherosclerotic risk factors with major limb events in patients with peripheral arterial disease (PAD).

METHODS: We conducted a retrospective cohort study of patients diagnosed with PAD (ankle-brachial index [ABI]<0.9) from 1995 to 1998 (index date). To identify the presence of an atherosclerotic risk factor, we performed a chart review, from three years prior to the index date until the first limb event (i.e., lower extremity bypass surgery or amputation) or the close of the study (i.e., December 31, 2001). The risk factors of interest were smoking, diabetes mellitus, hyperlipidemia, and hypertension. We categorized patients' disease severity based on their ABI with a level of 0.70 to 0.89 indicating mild disease, 0.41 to 0.69 indicating moderate disease, and 0.4 or less indicating critical disease. We also categorized patients based on their maximal number of risk factors (i.e., 0, 1-3, or 4) as documented anytime during the observational period. Patients were grouped, based on their ABI and risk factor profile, into nine categories. Using a chi-square test, we determined the percentage of patients with and without an adverse limb outcome based one of the nine categories.

RESULTS: We identified 799 patients with PAD and 230 (28.8%) experienced a major limb event following the index date. The mean age of the cohort was 64.8 SD 9.9 years. Within the cohort, the greatest number of patients had moderate disease and 1 to 3 risk factors (325 total [40.7% of the cohort]). Among the nine groups, the greatest risk for an adverse limb event was among patients with critical disease and no risk factors at 55.6%. Among patients with critical disease and 1 to 3 risk factors, the risk of an adverse limb event was 50.6%. In comparison, the risk for an adverse limb event was 35.7% for patients with critical disease and 4 risk factors. The risk for an adverse limb event was 42.2% for patients with moderate disease and no risk factors compared to a risk of 35.7% in patients with moderate disease and 1 to 3 risk factors; for patients with moderate disease and 4 risk factors, the risk for an adverse limb event was 11.2%. There were no adverse limb events in patients with mild disease and no risk factors; the level of statistical significance was $p<0.0001$ for comparing all nine groups.

CONCLUSIONS: Our results suggest that variables other than disease severity and the number of risk factors increase the risk for a major limb event in patients with PAD. Additional research is needed to address the association of level of atherosclerotic risk factor control with both the severity of disease at the time of diagnosis and the progression to a major limb event in patients with PAD.

DOES FEAR OF MALPRACTICE INFLUENCE CLINICAL AND TEACHING BEHAVIORS? D.A. Reed¹; D.M. Windish¹; R. Levine²; S.J. Kravet²; L. Wolfe²; S.M. Wright². ¹Johns Hopkins University, Baltimore, MD; ²Johns Hopkins Bayview Medical Center, Baltimore, MD. (Tracking ID #135934)

BACKGROUND: The increasing frequency of malpractice litigation is a source of concern for academic physicians. Fear of litigation may influence clinical and teaching behaviors including the practice of "defensive medicine." Defensive practice may adversely affect physician-patient relationships, quality of care, and health care costs. The objectives of this study were to: 1) characterize the attitudes of medical faculty toward malpractice, and 2) identify changes in clinical practice and teaching associated with fears of malpractice litigation.

METHODS: We surveyed 270 faculty physicians in the Department of Medicine at a large academic institution. We asked them about their attitudes toward malpractice and changes in clinical practice and teaching behaviors in response to the threat of litigation. Fear of litigation was measured using a previously validated instrument. Faculty were categorized as having high fear or low fear relative to the mean value on this scale. We performed comparisons by age, academic rank and percent effort in clinical activities.

RESULTS: 215 faculty responded (80%). Mean age was 46 years. Seventy percent were male and 52% were Associate or Full Professors. Respondents dedicated an average 37% effort to direct patient care and 40% effort to research. Most physicians (71%) believed that they would be involved in a malpractice case sometime in the next 10 years. Eighty-six percent acknowledged that being

named in a lawsuit or malpractice claim has a lasting negative impact on a physician's career. The mean score on the fear scale was 24.5 (possible range: 6-42). There was no difference by age, gender, academic rank, or clinical effort between those with scores above and below the mean. Faculty with high fear scores (>24.5) reported greater agreement that concerns about lawsuits and claims made against physicians negatively influence physician-patient relationships compared to faculty with low fear scores (74% vs. 60%, $p=0.04$). They also more frequently agreed that concerns about lawsuits negatively impact the teaching of medical learners (42% vs. 18%, $p=0.001$). As such, they limit learner autonomy (56% vs. 33%, $p=0.001$), procedure attempts by trainees (46% vs. 26%, $p=0.004$), and learner opportunity to give bad news to patients (44% vs. 27%, $p=0.02$). Younger faculty were significantly more likely to be concerned about future involvement in lawsuits than older faculty (81% vs. 59%, $p=0.001$). They also reported more frequently requesting consultations primarily to reduce the risk of being sued (50% vs. 29%, $p=0.002$) and a reluctance to rely on clinical judgment over technology due to perceived legal risks (80% vs. 69%, $p=0.03$). Compared to senior colleagues, young faculty reported giving medical learners (50% vs. 36%, $p=0.05$) and mid-level providers (50% vs. 31%, $p=0.01$) less autonomy due to prevalence of law suits.

CONCLUSIONS: Malpractice negatively influences clinical and teaching behaviors, particularly among younger faculty with lower academic rank and those with high malpractice fear scores. High fears of malpractice lead to defensive behaviors that may adversely impact patient care and education.

EFFECT OF A CLINICAL TRIAL ALERT SYSTEM ON INTERNIST PARTICIPATION IN TRIAL RECRUITMENT. P.J. Embi¹; A. Jain²; J. Clark²; S. Bizjack²; R.W. Hornung¹; C.M. Harris². ¹University of Cincinnati, Cincinnati, OH; ²Cleveland Clinic Foundation, Cleveland, OH. (Tracking ID #133263)

BACKGROUND: Clinical trial success depends on the recruitment of an adequate number of eligible subjects in a timely manner, but difficulties achieving such goals are common. Despite their importance to the process, few physicians contribute to recruitment efforts, and the vast majority of eligible patients are never offered the opportunity to participate in trials. To address this problem, we developed an Electronic Health Record (EHR)-based Clinical Trial Alert (CTA) system and evaluated its effect on physicians' recruitment of subjects to a Type 2 diabetes mellitus clinical trial.

METHODS: We performed a pre-post study of our CTA intervention at a large academic healthcare system with a fully implemented, commercial ambulatory EHR. Our subjects were the 10 endocrinologists and 104 general internists who practiced in main-campus or community-based clinics during both phases of our study. Following a 12-month control phase that included traditional efforts to encourage physicians' recruitment of patients to the associated trial, we activated the CTA for a 4-month post-intervention phase. When a patient's EHR-based data met selected trial criteria, the CTA notification screen reminded the physician of the clinical trial, prompted further consideration and discussion of eligibility with the current patient, and facilitated sending a secure referral message to the trial coordinator. We monitored the number of physicians participating and their referral and enrollment rates during both phases of our study. Differences in the number of physicians contributing to recruitment were tested using McNemar's exact test, and analyses of referral and enrollment rates were based on an assumed Poisson distribution of counts.

RESULTS: All 114 physicians received CTAs during the intervention period, and 42 (37%) referred at least one patient to the trial coordinator. Compared with pre-intervention recruitment, physician-generated recruitment improved during the post-intervention phase, with a 10-fold increase in the referral rate (5.7/month pre, 59.5/month post; rate ratio-10.44; 95% CI 7.98-13.68; $p<0.001$) and a doubling of the enrollment rate (2.9/month pre, 6.0/month post; rate ratio-2.06; 95% CI 1.22-3.46; $p=0.007$). Also increased were the number of physicians who generated referrals (5 pre, 42 post, $p<0.001$) and enrollments (5 pre, 11 post, $p=0.031$). Recruitment by both endocrinologists and general internists increased during the post-intervention phase, and general internists constituted the major referral source post-intervention.

CONCLUSIONS: An EHR-based Clinical Trial Alert system significantly increased physician participation in recruitment as well as physician-generated referral and enrollment rates to an ongoing clinical trial. Given the trend toward implementation of comprehensive EHR systems in academic health centers, this CTA approach may represent a much-needed solution to an important problem facing clinical trials and should be studied further.

EFFECT OF ALCOHOL CONSUMPTION ON BONE MINERAL DENSITY AND OSTEOPOROTIC FRACTURE: A SYSTEMATIC REVIEW. K.M. Berg¹; H.V. Kunins¹; R. Malik¹; K.A. Harris¹; S. Nahvi¹; A. Chaudhry¹; J. Arsten¹. ¹Albert Einstein College of Medicine, Bronx, NY. (Tracking ID #136008)

BACKGROUND: Low bone mineral density and osteoporotic fractures are highly prevalent in the United States, but their association with alcohol use is not well understood. Because alcohol consumption is also highly prevalent, understanding this association is necessary to help clinicians advise patients. We conducted a systematic review of the effect of alcohol on bone mineral density (BMD) and osteoporotic fracture in the general population.

METHODS: We identified studies published in English by searching MEDLINE, PsychINFO, Cochrane Controlled Trial Registry, and Current Contents, and searched reference lists of included articles and key review articles for additional

citations. To select studies for data synthesis, 2 independent reviewers assessed 702 citations and identified studies of adults that assessed the effect of alcohol on BMD or low-impact fracture of the hip, vertebrae or distal forearm. Methodological content was independently evaluated by 2 reviewers using established criteria, and data were extracted including measures of alcohol use and outcomes, confounders addressed, and magnitude of the associations.

RESULTS: Twenty-two studies met inclusion criteria. Three studies examined the effect of alcohol use on BMD in women, and found that among drinkers, BMD was greatest at moderate levels of drinking (? 1.4 drinks/day, > 1 drink/day, and > 7oz/week), compared to lower levels of drinking. This relationship was not consistently found in the 2 studies of the effect of alcohol use on BMD in men. Among studies that examined change in BMD over time, 4 studies in women found that bone loss over time was least among women with moderate alcohol consumption (1.4 drinks/day, 1.1 drinks/day, > 1.4 drinks/day, and 1-3 oz/week). Two studies of change in BMD over time among men had similar results, with the least bone loss occurring among those with moderate drinking (0.7-1.4 drinks/day and 3-7 oz/week). An additional 4 studies of change in BMD over time found that abstinent or low level drinkers had greater bone loss compared to moderate drinkers, and 3 of these studies reported more bone loss at higher levels of drinking compared to moderate levels. These data suggest a U-shaped relationship between alcohol use and bone loss over time, with the least bone loss occurring at moderate levels of alcohol consumption. Among studies that examined the association between alcohol use and hip fracture, 6 found a linear increase in hip fracture risk with increased alcohol use, 4 found a linear decrease in hip fracture risk with increased alcohol use, and 3 found no association. Similarly, no consistent association was found between alcohol consumption and fracture of the vertebrae or distal forearm. A limitation of most studies (21/22) is that they did not include a category of true heavy drinkers (> 3 drinks/day), making it impossible to assess the effect of very heavy alcohol use on BMD or fractures.

CONCLUSIONS: Moderate alcohol consumption (1-2 drinks/day) is associated with increased BMD among women. In addition, among both men and women, there is a U-shaped relationship between alcohol use and bone loss, with moderate alcohol consumption associated with less bone loss than either no/low or heavier alcohol use. Further studies are needed to assess BMD and bone loss in very heavy drinkers (> 3 drinks/day). Despite common beliefs, evidence for a pattern of association between alcohol use and osteoporotic fracture is lacking.

EFFECTS OF EARLY DOSE OPTIMIZATION AND CUSTOMIZED INSTRUCTIONS ON VIAGRA (SILDENAFIL CITRATE) TREATMENT SATISFACTION AND NONRESPONDER RESCUE. C.P. Steidle¹; A. Crowley²; H. De Riesthal²; L.J. Tseng². ¹Northeast Indiana Urology, PC, Fort Wayne, IN; ²Pfizer, Inc., New York, NY. (Tracking ID #132861)

BACKGROUND: Viagra[®] (sildenafil citrate) is an effective treatment for erectile dysfunction (ED) with high patient satisfaction rates. To achieve optimal success rates and multiple attempts, dose optimization and customized counseling are ideal. However, health providers frequently rely on the instructions provided by the sample packets with little follow-up, leading to possible suboptimal results. Retrospective studies have demonstrated the value of reeducation in enhancing Viagra treatment satisfaction. We prospectively investigated the effect of the package instructions alone followed by further customized instructions on the treatment satisfaction rates of men with ED who received Viagra.

METHODS: This was a multicenter, open-label, flexible-dose study of Viagra (25, 50, or 100 mg prn), conducted in two 4-week phases. Efficacy measurements included the Erectile Dysfunction Inventory of Treatment Satisfaction (EDITS), the International Index of Erectile Function (IIEF), and a patient-re-

corded log of sexual activity. In phase 1, patients were given 50 mg Viagra and instructed to follow the instructions in the sample pack with no verbal clarification. In phase 2, they received customized use instructions, and the Viagra dose could be adjusted to 100 mg to optimize efficacy or to 25 mg to enhance tolerability. Satisfaction was determined from responses to question 1 of the EDITS, which asked, "How satisfied are you with this treatment?"

RESULTS: Of 1124 men (mean age 54 ± 13 y) enrolled, 902 completed the study. At the end of phase 1, 40% of men were "very satisfied," and 36% were "somewhat satisfied" with Viagra. The mean baseline IIEF erectile function (EF) domain score of 14.3 increased to 23.5 (maximum score, 30; scores > 25 defined as no ED). At the start of phase 2, 52% increased their Viagra dose to 100 mg, and 2% decreased their dose to 25 mg. In phase 2, after dose optimization and customized instruction, 62% and 24% were "very satisfied" or "somewhat satisfied," and the mean EF domain score increased to 25.7. The percentage of attempts at which an erection lasted long enough for successful sexual intercourse increased from 79% to 91%. Of the men not initially satisfied during phase 1 who completed phase 2 (n=196), 64% were satisfied with treatment at the end of phase 2. For those who became satisfied during phase 2, mean EF domain scores increased from 16.6 to 26.0, and successful intercourse attempts increased from 51% to 90%.

CONCLUSIONS: Although initiating Viagra therapy with a 50-mg dose following sample package instructions resulted in high patient satisfaction rates, early follow-up for counseling on dose optimization and use instructions rescued patients who initially did not respond to ED treatment and further enhanced treatment satisfaction and improvement in EF.

ELEVATED LEVELS OF URIC ACID INDEPENDENTLY PREDICT HEMODYNAMIC COMPROMISE IN HEART FAILURE PATIENTS. V. Bead¹; M. Kittleson²; J.M. Hare². ¹Johns Hopkins Bayview Medical Center, Baltimore, MD; ²Johns Hopkins University, Baltimore, MD. (Tracking ID #134203)

BACKGROUND: Congestive heart failure is a leading cause of morbidity and mortality in the United States, and the ability to predict prognosis is essential for patient education and for optimal allocation of therapies. Biomarkers such as the natriuretic peptides, B-type natriuretic peptide (BNP) and N-terminal pro-BNP (NT-proBNP) offer valuable prognostic information in heart failure patients. The degree of elevation of BNP and NT-proBNP correlates with cardiac filling and worse outcomes in patients with heart failure. Uric acid also predicts poor prognosis in heart failure but the pathogenic mechanisms of hyperuricemia and its relation to hemodynamic parameters is unclear. The objective of this study was to determine if elevated uric acid levels are associated with hemodynamic decompensation in heart failure independently of natriuretic peptide elevations, thereby reflecting independent mechanisms in the pathogenesis of heart failure. **METHODS:** Levels of uric acid and NT-proBNP were measured during 138 right heart catheterizations performed in 117 patients between January 1, 2003 and July 1, 2004 as part of the evaluation or management of cardiomyopathy in patients with congestive heart failure at the Johns Hopkins Cardiovascular Diagnostic Laboratory. Patients who were status-post heart transplantation or receiving synthetic human BNP at the time of catheterization were excluded from the study. Pearson pairwise correlations were computed for uric acid, NT-proBNP levels, the logarithm of NT-proBNP levels, and hemodynamic parameters obtained during right heart catheterization. Patients were divided into groups based on the median serum uric acid and NT-proBNP levels. Because some individuals contributed more than one observation, groups were compared using linear regression models with generalized estimating equations and robust variance estimation assuming an independence working model for the within-cluster correlation.

Table 1: Adjusted effect of uric acid and NT-proBNP levels on hemodynamic parameters controlling for age, renal function, and diuretic and allopurinol use.

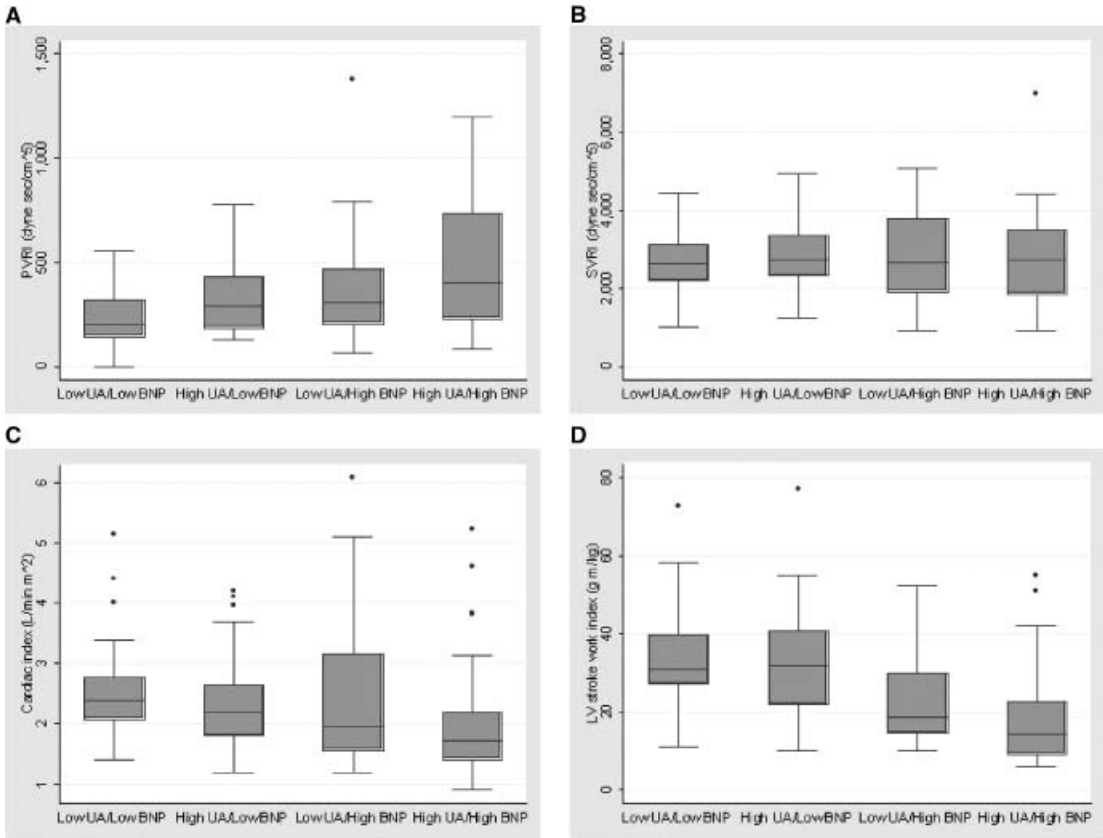
	Low NT-proBNP	Low NT-proBNP	High NT-proBNP	High NT-proBNP
	Low Uric Acid*	High Uric Acid*	Low Uric Acid*	High Uric Acid*
RA pressure (mm Hg)	6.8 (0.3) [†]	8.0 (0.2) [†]	7.9 (0.3) [†]	14.9 (0.7)
RV systolic pressure (mm Hg)	34.5 (0.2) [†]	37.2 (1.0) [†]	37.1 (0.7) [†]	50.6 (1.0)
RV diastolic pressure (mm Hg)	6.3 (0.01) [†]	9.0 (0.1) [†]	8.2 (0.3) [†]	14.6 (0.4)
PA systolic pressure (mm Hg)	34.8 (0.04) [†]	35.9 (2.3) [†]	37.4 (0.4) [†]	51.0 (1.0)
PA diastolic pressure (mm Hg)	16.9 (0.4) [†]	17.1 (1.3) [†]	18.2 (0.1) [†]	26.4 (0.4)
PA mean pressure (mm Hg)	22.9 (0.1) [†]	24.0 (1.2) [†]	25.2 (0.6) [†]	34.0 (0.7)
PCW pressure (mm Hg)	15.5 (0.4) [†]	14.5 (0.2) [†]	16.3 (0.2) [†]	23.9 (0.5)
PVRI (dyne*sec/cm5)	247 (5) [†]	330 (42) [†]	365 (40)	487 (37)
SVRI (dyne*sec/cm5)	2546 (140)	2759 (59)	2813 (113)	2870 (152)
LVSWI (g*m/kg)	30.8 (1.7) [†]	32.5 (3.1) [†]	22.8 (0.4) [†]	18.8 (1.4)
RVSWI (g*m/kg)	7.1 (0.5)	6.8 (0.9)	5.9 (0.3)	6.4 (0.8)
CI (L/min*m2)	2.5 (0.2)	2.5 (0.1)	2.4 (0.03)	2.0 (0.1)

*Values are mean (standard error) adjusted for age, renal function, diuretic use, and allopurinol use.

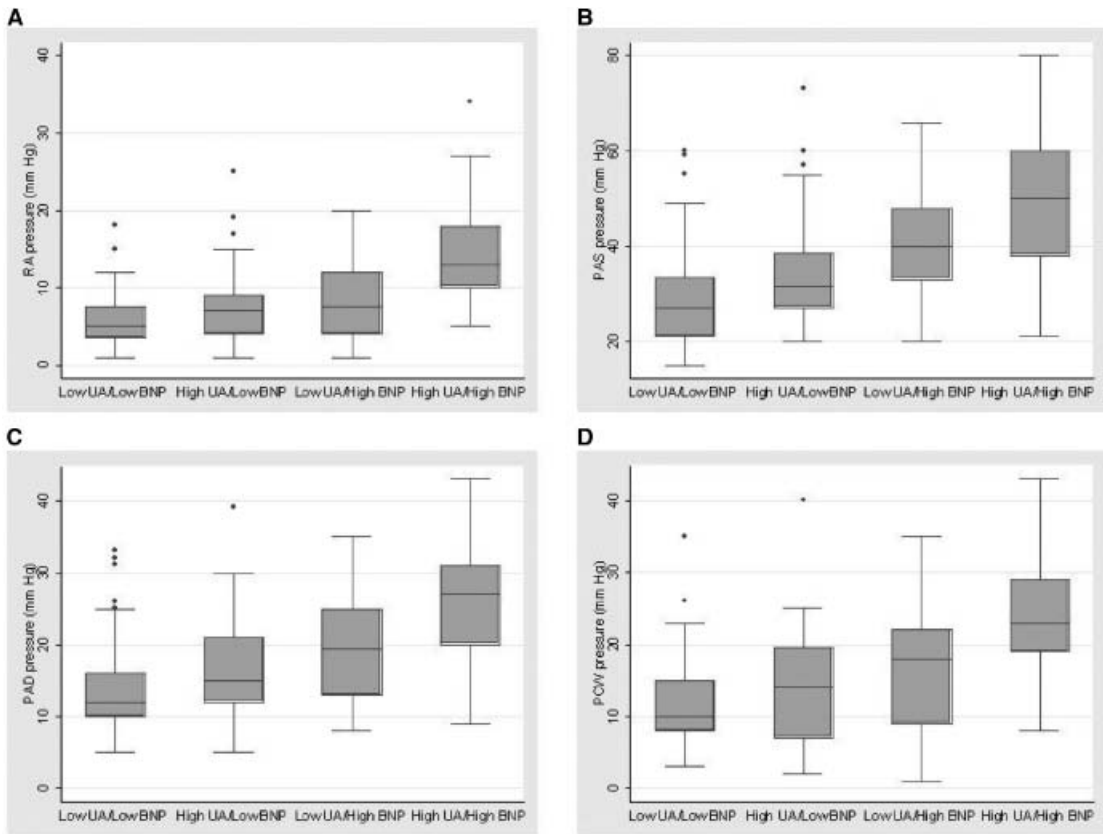
[†]p < 0.05 for comparison with high uric acid/high NT-proBNP group

CI is cardiac index, LVSWI is left ventricular stroke work index, NT-proBNP is N-terminal pro-B-type natriuretic peptide, PA is pulmonary artery, PCW is pulmonary capillary wedge, PVRI is pulmonary vascular resistance index, RA is right atrial,

RV is right ventricular, RVSWI is right ventricular stroke work index, SVRI is systemic vascular resistance index. High uric acid is > 476 micromol/L (8.0 mg/dl). High NT-proBNP is > 1500 ng/L.



Boxplots figure 1



Boxplots figure 2

RESULTS: Uric acid and NT-proBNP levels did not correlate with each other ($R=0.03$, $p=0.74$). To determine if uric acid levels offered additional clinical information independent of NT-proBNP, we divided the study population into four groups based on the median uric acid and median NT-proBNP levels. As the boxplots in Figure 1 indicate, higher uric acid levels were associated with higher filling pressures, including RA pressure, PAS pressure, PAD pressure, and PCW pressure, regardless of NT-proBNP levels (Figure 1; $p<0.05$ for the trend in all comparisons). In addition, higher uric acid levels were associated with higher PVRI, lower CI, and lower LVSWI in both high and low NT-proBNP groups (Figure 2; $p<0.05$ for the trend in PVRI and LVSWI). These trends persisted after adjusting for age, renal function, diuretic use, and allopurinol use (Table 1). An isolated elevation in uric acid was still associated with higher cardiac filling pressures and PVRI and lower LVSWI regardless of NT-proBNP levels. In addition, an elevation in both uric acid and NT-proBNP was associated with a greater rise in filling pressures, PVRI, and LVSWI than an increase in either uric acid or NT-proBNP alone.

CONCLUSIONS: Elevations in uric acid and NT-proBNP are associated with additive elevations in cardiac filling pressures and pulmonary vascular resistance. These findings demonstrate that uric acid and NT-proBNP have independent and complementary roles in heart failure pathogenesis. This study illustrates the value of uric acid, in addition to NT-proBNP, as a biomarker to assess hemodynamic decompensation in heart failure patients.

EVALUATION OF THE SINGLE ITEM HEALTH LITERACY SCREENER. N. Morris¹; C.D. Maclean¹; B. Littenberg¹. ¹University of Vermont, Burlington, VT. (Tracking ID #134252)

BACKGROUND: Reading ability, an aspect of health literacy, is important to knowledge about health care: health information, management of one's health, utilization of services, and outcomes. A full assessment of health literacy is time-consuming, expensive, and rarely done in clinical practice. A short, simple measure to screen for limited health literacy would allow for targeted interventions to enhance communication and optimize patient understanding when appropriate. Our aim was to assess the diagnostic accuracy of the Single Item Health Literacy Screener (SILS) as compared with the Short Test of Functional Health Literacy in Adults (STOFHLA).

METHODS: We did in-home interviews of adults with diabetes recruited from Primary Care practices participating in the Vermont Diabetes Information System. The STOFHLA is a valid, reliable and widely used 7-minute timed test of reading comprehension to measure health literacy. Results are categorized into inadequate (0-16 correct answers), marginal (17-22 correct answers), and adequate health literacy (23-36 correct answers). The SILS is a single brief question that can be asked by a health care professional or clerk: "How often do you need to have someone help you when you read instructions, pamphlets, or other written material from your doctor or pharmacy?" Responses of "sometimes," "often," or "always" are considered a positive result and suggest further assessment of health literacy is warranted. A "never" or "rarely" response is negative.

RESULTS: Of the 766 persons screened, 142 (18%) had limited health literacy (marginal or inadequate (STOFHLA score of 0-22), blind, or otherwise unable to read). Six-hundred (78%) screened negative on the SILS and 546 of these subjects had adequate health literacy (negative predictive value = 91% (95% confidence interval 88%, 93%)). Of the 166 who scored positive on the SILS, 73 had limited literacy (positive predictive value = 44% (.36, .52)). The sensitivity of the SILS was 54% (95% confidence interval 45%, 62%) and the specificity was 85% (82%, 88%). The area under the Receiver Operative Characteristic Curve (ROC) was 0.74 (0.69, 0.78). Performance was better for subjects with inadequate (rather than marginal) literacy: 69% (56%, 78%).

CONCLUSIONS: The SILS is a simple means to discriminate between limited and adequate health literacy in ambulatory adults with diabetes. Although not highly sensitive, the SILS can easily exclude patients who do not need further assessment. This can allow providers to target additional assessment of health literacy to the subset of the population most in need.

HEALTH LITERACY: PATIENTS' PERFORMANCE AND PHYSICIANS' PERCEPTIONS. P.A. Kelly¹; P.M. Haidet¹. ¹Michael E. DeBakey VA Medical Center/Baylor College of Medicine, Houston, TX. (Tracking ID #136039)

BACKGROUND: The relationship of general literacy with observable indicators such as verbal communication skill, age, and socioeconomic status is usually strong, but this may not be true of health literacy. Physicians, using only observable indicators, may overestimate patients' health literacy, resulting in communication "disconnects" that foster physician use of medical jargon and contribute to poorer patient physiological health outcomes. The objective of this project was to measure the prevalence of physician misjudgment of patients' health literacy and attempt to identify causes of misjudgment among veterans, a patient population of predominantly high-school graduates with a high prevalence of chronic illness.

METHODS: Twelve physicians were recruited at the Michael E. DeBakey VAMC to participate in the project. We randomly selected 8-12 patients who were scheduled for a visit with a participating physician to complete a written general literacy test (Test of Adult Basic Education, 17 items), a written health literacy test (Short Test Of Functional Health Literacy for Adults, 36 items), a verbal health literacy test (Rapid Estimate of Adult Literacy in Medicine, 66 items) and a brief demographic survey (12 items). All three literacy tests had been validated in a variety of patient populations and medical settings.

Physicians completed a demographic survey at the beginning of the project. After each patient visit, physicians completed a survey of their perceptions of the patient's literacy level, general health awareness, effectiveness as a communicator of their own health, and adherence to previous treatment recommendations.

RESULTS: We collected data from approximately 100 patients. Using our single-item patient health literacy measure, the physicians rated health literacy at the highest level, "able to read most patient health materials unassisted," for 75 of the 100 patients they saw. However, 30 of these highest ratings, and 48 of all ratings, overestimated the patient's actual health literacy level as determined by patients' scores on the tests. In contrast, only 16 underestimates of patients' health literacy were made. No physician exhibited statistically significant tendency to over- or underestimate more than others; frequency of over- and underestimation was not statistically significantly associated with any patient demographics.

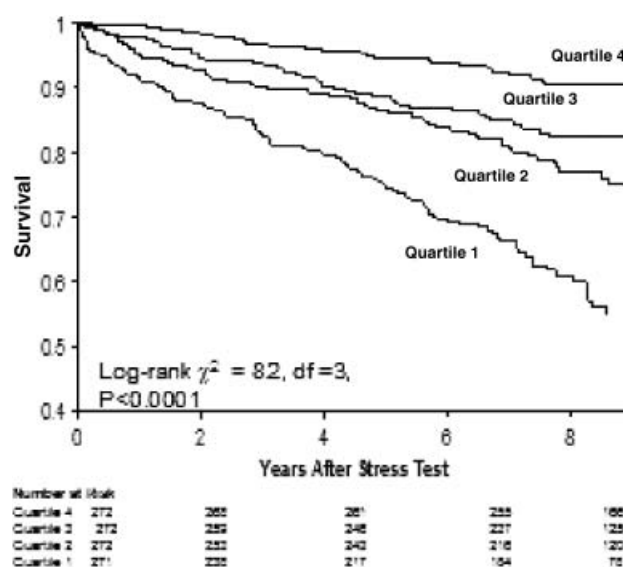
CONCLUSIONS: Physicians do indeed make on-the-spot estimates of patients' health literacy, but they are often overestimates. By investigating the prevalence of and demographics associated with physician overestimation of patient health literacy in other populations, we intend to move the medical profession towards better identifying those cases in which problems with health literacy are most likely to arise. This research may lead to a technique for reducing the number of low-literacy cases that go undetected, without the prohibitively expensive requirement that every patient receive a low-literacy intervention.

HEART RATE RESPONSE DURING DIPYRIDAMOLE STRESS AS A PREDICTOR OF MORTALITY IN PATIENTS WITH NORMAL MYOCARDIAL PERFUSION AND NORMAL ELECTROCARDIOGRAMS. R. Bhatheja¹; G. Francis¹; C. Pothier¹; M. Lauer¹. ¹Cleveland Clinic Foundation, Cleveland, OH. (Tracking ID #134984)

BACKGROUND: While it is well established that a blunted chronotropic response to exercise is associated with a higher risk of death, recently some data have emerged suggesting a similar association linking mortality risk to a blunted heart rate response to vasodilatory stress. We investigated the heart rate response to dipyridamole-induced stress as a predictor of death in the setting of normal myocardial perfusion and a normal electrocardiogram.

METHODS: We followed 1087 patients for 8 years (range 5.7-11.8) who underwent dipyridamole vasodilator stress and had normal perfusion scans and electrocardiograms. None had heart failure, known left ventricular systolic dysfunction, pacemaker implantation or valve disease. Heart rate response was assessed as the ratio of heart rate at peak stress to heart rate at rest. The primary endpoint was all-cause mortality.

RESULTS: There were 246 deaths. Quartile values for the peak to rest heart rate ratio were <1.19, 1.19-1.30, 1.31-1.44, and >1.44. Death rates according to quartiles of heart rate ratio were 103/271 (38%), 64/272 (24%), 52/272 (19%), and 27/272 (10%). After adjusting for age, gender, resting heart rate, blood pressure response, standard cardiovascular risk factors, and other confounders, a blunted heart rate response remained predictive of death (adjusted hazard ratio for lowest quartile compared to highest 3.3, 95% confidence interval 2.1-



Overall survival according to quartile of ratio of peak to resting heart rate. Values of the peak to rest heart rate ratio is quartiles 1, 2, 3, and 4 were <1.19, 1.19-1.30, 1.30-1.44, and >1.44. Curves were generated using the Kaplan-Meier technique.

5.1, $P < 0.0001$). When considered as a continuous variable, the logarithm of the heart rate ratio was the strongest predictor of death excepting age. The heart rate ratio predicted death equally well in patients taking and not taking beta-blockers.

CONCLUSIONS: A blunted heart rate response to dipyridamole stress is suggestive of increased risk of death even in the absence of left ventricular systolic dysfunction or myocardial ischemia. Whether this is primarily a manifestation of autonomic dysfunction or a manifestation of under-diagnosed severe coronary artery disease remains to be determined. Patients with normal vasodilator myocardial perfusion scans should be considered low risk only if they mount an appropriate heart rate response to the vasodilator stimulus.

Results of multivariable stepwise proportional hazards analysis for prediction of death where heart rate ratio is considered as a continuous variable.

Variable	Hazard ratio (95% P-value CI)	
Age (10 years)	1.6 (1.4-1.9)	<0.0001
Log of Heart rate ratio (1 standard deviation increase)	1.5 (1.3-1.7)	<0.0001
Referral for pre-operative evaluation	1.8 (1.4-2.4)	<0.0001
History of stroke	2.4 (1.5-3.8)	0.0001
Insulin treated diabetes	1.9 (1.3-2.8)	0.0008
Chronic obstructive lung disease	1.9 (1.1-3.1)	0.014
Male gender	1.5 (1.2-1.9)	0.0018
Prior Coronary Angioplasty	0.6 (0.4-0.9)	0.017
Peripheral vascular disease	1.6 (1.1-2.2)	0.0059
Statin	0.5 (0.3-0.9)	0.017

IMPROVING ELECTRONIC HEALTH RECORD (EHR) ACCURACY AND INCREASING COMPLIANCE WITH HEALTH MAINTENANCE CLINICAL GUIDELINES THROUGH PATIENT ACCESS AND INPUT. M. Staroselsky¹; L. Volk²; R. Tsurikova²; L. Pizziferri²; J. Wald²; D.W. Bates³. ¹Partners HealthCare System, Inc., Needham, MA; ²Partners HealthCare System, Inc., Wellesley, MA; ³Brigham and Women's Hospital, Watertown, MA. (Tracking ID #135688)

BACKGROUND: Health maintenance is crucial for preventing morbidity and premature mortality but many patients do not receive preventive services at recommended intervals. One reason is the lack of up-to-date information accurately reflecting patients' history. Electronic health records (EHRs) can provide decision-support, which has been demonstrated to improve provision of preventive services, and they are more easily shareable than paper records, but are often incomplete. Patient input has the potential to improve the accuracy of this information. In this study, we assessed the current state of EHR completeness for preventive measures and the added value of patient reported information. We also evaluated the impact of allowing patients to view and amend the health maintenance information in their EHRs.

METHODS: Participants were sent a survey, pre-populated with preventive screening information from their EHRs. They were asked to review this information and indicate whether it was accurate or if they had had a screening test done more recently. Of 1098 patients recruited from a primary care practice, 163 returned the survey. When a patient reported a more recent test than was noted in the EHR, researchers updated the EHR to reflect the additional information.

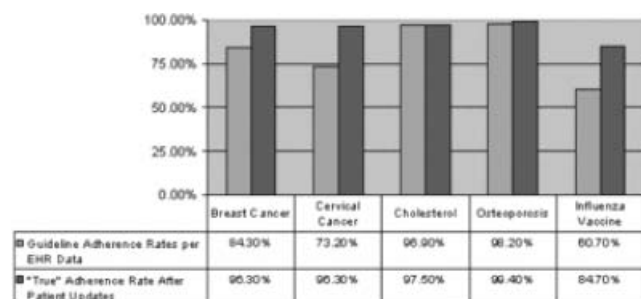
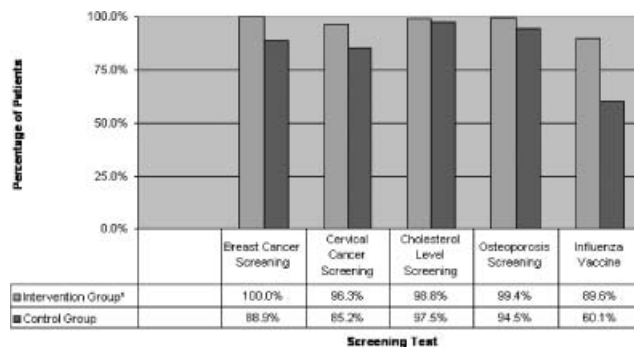


Chart 1: Health Maintenance Procedure Adherence Rates



*Intervention group statistics include patients who said they had done procedure elsewhere and those who had them done after receiving the survey information.

Chart 2: Percent Patients Considered Up-to-Date After Survey

Information was also gathered from the EHR 6 months after surveys were completed to analyze whether providing overdue test information encouraged patients to get tested and vaccinated. A review of medical records was performed on a control group to analyze differences in adherence to preventive guidelines between those that were notified of their overdue status and those who were not notified.

RESULTS: The EHR was frequently incomplete when compared to patient report. In particular, many patients were misidentified as being overdue for tests when they had had them in other places. This was especially true for cervical cancer screening and influenza vaccines where adherence rates increased from 73.2% and 60.7% to 96.3% and 84.7% respectively (Chart 1). Showing patients their information resulted in performing only a few additional tests compared to a control group and this had little impact on overall adherence. However, when the cumulative effects of additional patient-reported tests and tests performed after the survey were counted, intervention patients had higher adherence rates for every procedure (Chart 2).

CONCLUSIONS: Prevention data in EHRs was often incomplete, and patients were often able to provide useful information, demonstrating the value of patient contributions in keeping records up-to-date. Given the opportunity to review their medical record including information on preventive services, it appears that patients do not subsequently schedule more of their overdue procedures than if they had not been informed. However, by updating the records with more recent tests than reflected in the EHR, higher compliance rates and more accurate records were achieved. In today's pay-for-performance environment, there are both financial and quality reasons for doctors to have records that reflect accurate adherence rates of their practices. Engaging patients in the review and electronic submission of health maintenance information can contribute to a more complete EHR, but does not eliminate the need for direct routine data exchange across providers. Patient portals could be an important mechanism for patient input and review.

INFLAMMATION AND VENOUS THROMBOEMBOLISM IN HOSPITALIZED MEDICAL PATIENTS. J.Q. Usman¹; M. Young²; M. Brophy²; R. Lew²; F. Louis². ¹Boston University, Boston, MA; ²Boston VA Medical Center, Boston, MA. (Tracking ID #133067)

BACKGROUND: The risk of venous thromboembolism (VTE) associated with inflammation has been well demonstrated in lab studies but not in population studies. We used Veterans Administration (VA) administrative databases to compare the risk of subsequent VTE in patients discharged from the hospital following a medical admission for inflammatory and non-inflammatory conditions.

METHODS: Subjects were male patients who had an index hospitalization of >3 days duration between 4/1/97 and 9/30/02. The 100 most common discharge diagnoses for the index hospitalization were stratified a priori by experts as inflammatory (n=19), non-inflammatory (n=50) or conditions known to be associated with VTE ('high risk', n=25). Six ambiguous diagnoses were excluded. The outcome of interest was re-hospitalization for VTE. Subjects were censored at VTE, next hospitalization following the index hospitalization or death. VTE rates for each of the categories were compared using Cox proportional hazard models.

RESULTS: VTE occurred during the follow up period in 301 (0.47%) of the 64,055 inflammatory, 361 (0.36%) of 99,221 non-inflammatory and 296 (0.43%) of the 69,496 'high risk' subjects. The risk for subsequent VTE was greater for patients hospitalized for inflammatory conditions compared to non-inflammatory conditions after adjusting for age and length of index hospitalization (hazard ratio 1.4, $p < 0.0001$). There was no difference in the risk of VTE between groups hospitalized for inflammatory conditions compared to 'high risk' conditions.

CONCLUSIONS: Our findings suggest an elevated risk of VTE in patients discharged from the medical service following inflammatory conditions compared to non-inflammatory conditions. This risk is comparable to that of 'high risk' conditions. VTE prophylaxis which is typically administered to patients in the 'high risk' group maybe extended to patients admitted with inflammatory conditions.

IS ADHERENCE A PROBLEM FOR PATIENTS OR PROVIDERS? RESULTS FROM A FACTORIAL EXPERIMENT. K.M. Freund¹; C. Link²; L. Marceau²; A. O'Donnell¹; J. McKinley². ¹Boston University, Boston, MA; ²New England Research Institutes, Watertown, MA. (Tracking ID #134800)

BACKGROUND: While adherence usually refers to actions by a patient with regard to medical recommendations, less is known about how closely physicians adhere to guidelines regarding minimally appropriate practice performance.

METHODS: We conducted a factorial experiment to assess clinical practice decisions to common medical scenarios. We developed 2 clinically authentic scenarios of "patients" presenting with either coronary heart disease (CHD) or depression. We videotaped 16 versions of each scenario, varying dichotomous combinations of "patient" age, gender, race, and economic status (SES). 128 randomly sampled primary care physicians throughout Massachusetts were shown (in random order) one version of each scenario. Following the viewing, subjects were asked to indicate a range of actions they would take, including physical examination, prescriptions, tests, life-style recommendations, referral and follow-up. These actions were compared with major clinical management guidelines. Comparisons were made across the 4 "patient" demographic variables in the experiment, and comparisons were made based upon years of clinical experience of subjects.

RESULTS: While 73% reported they found clinical guidelines useful in patient care, only 25% said knowledge of guidelines contributed to their decision making. Some guidelines are followed by a majority of physicians (e.g., 88% and 75% would order a ECG/EKG and stress test for CHD and 82% would obtain a medical history for depression), however, other guidelines are not adhered to (e.g., less than a quarter would inquire about mood, sleep disturbance, suicidal ideation or medication for the depressed "patient"). Patient gender and age were associated with adherence to guidelines: female CHD "patients" were significantly less likely to receive a cardiac, pulmonary or vascular examination. This pattern was reversed for depression, with women receiving more evaluation than men. 40% of younger "patients" received advice on smoking compared with 23% of older "patients." Years of clinical experience significantly influenced the management of both medical conditions.

CONCLUSIONS: Despite assertions regarding their usefulness, many primary care providers do not adhere to clinical guidelines in the management of CHD or depression. Physician adherence to clinical guidelines is independently associated with both patient attributes (gender and age) and with the length of clinical experience.

LDL GOAL ATTAINMENT IN HIGH RISK PATIENTS – DOES AN INTERVENTION TARGETING PATIENTS AND PROVIDERS WORK? N.M. Afonso¹; G. Nassif¹; A.N. Aranha¹; B. Delor². ¹Wayne State University, Detroit, MI; ²Pfizer, Inc., Hartland, MI. (Tracking ID #135193)

BACKGROUND: Although physicians are aware of the National Cholesterol Education Program (NCEP) cholesterol guidelines and attempt to achieve these goals with their dyslipidemic patients, the majority of patients fail to attain cholesterol goals. This study was conducted to determine whether a combination of a provider awareness and patient education intervention program can improve the NCEP goal attainment among patients at high-risk for cardiovascular events.

METHODS: Methods: All high-risk patients visiting an Internal Medicine faculty practice were included in the study. Data collected included demographics, cardiovascular risk factors, laboratory information and treatment details. Patients also completed a Morisky questionnaire to evaluate medication adherence. They were educated about risk factors and provided information about cholesterol reduction and target LDL levels. Physicians were then given the information provided by their patients not at target and requested to evaluate the dyslipidemia management on these patients. Lipids were reevaluated 8-12 weeks after the intervention.

RESULTS: Results: 141 high risk patients were initially evaluated and 106 patients who had follow up lipids after the intervention were considered in the final analysis. The (mean±SD) age was 59.6±14.2 years, 76% were female. 89.9% were African American. 39.4% of the patients were at target LDL at the start of the study and 59.4% achieved target after the intervention. Pre-intervention cholesterol and LDL levels were 202.4±/68.5 mg/dl and 121.3±/53.8 mg/dl respectively. Post-intervention cholesterol and LDL levels were 183.5±/60.8 mg/dl and 106.7±/50.2 mg/dl respectively (p<0.001). No statistically significant changes were observed in the HDL and triglyceride levels. Based on the Morisky questionnaire 82% of patients were classified as high adherence and 18% as low adherence. In patients with low adherence, LDL levels increased after the intervention from 127.2±/80.9 to 142.1±/87.1 mg/dl, as compared to the high adherence patients who decreased their LDL from 114.3±/43.5 to 98.6±/35.2 mg/dl after the intervention (p<0.002).

CONCLUSIONS: Conclusions: This study demonstrates that an intervention simultaneously targeting patients and providers is successful in improving goal attainment among high risk patients. The Morisky questionnaire is good predictor of a successful outcome in this intervention.

PATIENT AND PHYSICIAN EXPECTATIONS FOR TIMELINESS OF CARE. D.W. Barry¹; T.V. Melhado¹; K.M. Chacko¹; R.S. Lee¹; J.F. Steiner¹; J.S. Kutner¹. ¹University of Colorado Health Sciences Center, Denver, CO. (Tracking ID #132784)

BACKGROUND: Timeliness of access to care is one of the six domains of health care quality identified in the Institute of Medicine's report, Crossing the Quality Chasm. Timeliness of care can improve patient outcomes, decrease emergency department utilization, and improve patient satisfaction with care. We compared

patient and physician perceptions of the appropriate timing of visits for common medical problems.

METHODS: This study was conducted in the waiting rooms of two general internal medicine clinics of the University of Colorado Health Sciences Center. A convenience sample of adult patients and adult companions was surveyed. We also surveyed general internists in the Division of General Internal Medicine via email and at a weekly conference. The instrument consisted of 11 brief vignettes designed to reflect clinical problems of varying urgency. Respondents were asked to indicate how soon the patient in the vignette should be able to see a physician. Each scenario had five response options: that day, tomorrow, 3-5 days, 1-3 weeks, and 1-3 months. To reflect clinical practice where patients are booked for urgent or non-urgent appointments, we collapsed the responses to 2 categories: today or tomorrow vs. less urgent responses. These dichotomized data were analyzed using Fisher exact test.

RESULTS: 262 patients completed the survey. 46 of 61 physicians responded (75.4% response rate). Seven of the 11 scenarios revealed no significant differences between patient and physician expectations of the timing of visits. Significant differences were found between patients and physicians for four scenarios. Patients were more likely than physicians to respond that an urgent appointment was appropriate for the following four vignettes: chronic stomach pain (61.7 vs. 4.3%, p<0.001), chronic knee pain (22.0 vs. 0%, p<0.001), routine diabetes follow-up (13.7 vs. 0%, p=0.004), and elevated cholesterol (11.7 vs. 0%, p0.012). The chronic stomach pain scenario demonstrated the largest difference between the groups. Both groups agreed on the 4 most urgent scenarios: wheezing in an asthmatic, pharyngitis, a twisted ankle, and a new rash.

CONCLUSIONS: Patients and physicians have different expectations regarding the timing of appointments for common chronic, but not acute, medical conditions. Understanding patient expectations may help physicians understand requests for urgent evaluation of chronic conditions or complaints. Failure to meet these requests may lead to patient dissatisfaction and increased use of health-care resources outside of their primary care physician.

PHYSICIAN BELIEFS AND PRACTICES REGARDING INFLUENZA DIAGNOSIS AND TREATMENT. M.B. Rothberg¹; A.B. Bonner²; M.H. Rajab²; H.S. Kim²; B.W. Stechenberg¹; D.N. Rose¹. ¹Baystate Medical Center, Springfield, MA; ²Scott and White Hospital, Temple, TX. (Tracking ID #133099)

BACKGROUND: Although influenza strikes 10-20% of the population each year, little is known about physicians' approach to this common illness. Options for diagnosis and treatment include rapid tests and antiviral medications. The purpose of our survey was to determine if physicians' approach to diagnosis and treatment of influenza is influenced by their specialty, individual beliefs and/or local practice patterns.

METHODS: At the end of the 2003-4 influenza season, we conducted a cross sectional on-line survey of physician knowledge, attitudes and practices regarding rapid diagnostic testing and use of antiviral therapy for influenza at two large academic medical centers in Massachusetts and Texas. E-mail invitations were sent to all general internists, pediatricians, emergency medicine and family practitioners at both medical centers. Results of the self-administered 41-question survey were automatically entered into an electronic database. Univariate and multivariate analyses were performed using antiviral prescribing as the dependent variable.

RESULTS: The survey was sent to a total of 579 physicians (277 MA and 302 TX). Three hundred thirty-seven physicians (177 MA and 160 TX) completed the survey (58% response rate). Sixty-one percent of the respondents reported having prescribed antiviral therapy during the past year, and 62% utilized rapid diagnostic testing for influenza. Respondents' beliefs concerning antiviral therapy included: reduces duration of illness by 1 day (87%); prevents serious complications (28%) and decreases mortality (28%). Physicians reported the following reasons for not prescribing antiviral agents: patients present too late (60%), influenza is self-limited (40%), uncertain of the diagnosis (36%), antiviral agents are too expensive (28%) and are not very effective (28%). No one cited concern for antiviral resistance as a reason for not prescribing these agents. In univariate analysis, physicians were more likely to report prescribing antivirals if they lived in TX vs. MA (76% vs. 48%, p<.0001), were attendings vs. residents (67% vs. 50%, p<.01), saw more patients/week (p<.0001), or utilized rapid testing (66% vs. 53%, p=.02). Internists were least likely (48%) and family practitioners most likely (84%) to prescribe antiviral therapy (p<.0001). Believing antivirals prevent complications or reduce mortality was associated with antiviral prescribing (p<.01). There were no differences in beliefs between the two locations. Results of multivariate analysis appear in the Table.

CONCLUSIONS: Physician prescribing of antiviral therapy for influenza is associated with local practice patterns, use of rapid testing and individual beliefs. Physicians cited patients presenting too late for treatment as the chief reason for not prescribing antiviral therapy.

Multivariate Predictors of Antiviral Prescribing for Influenza

Variable		Odds Ratio (95% CI)
Location	Texas vs. Massachusetts	3.5 (2.1-5.9)
Patient Volume	High vs. Low	2.0 (1.5-2.6)
Utilizes Rapid Testing	Yes vs. No	1.8 (1.1-3.0)
Beliefs	Shortens Illness by 1 day	4.9 (1.8-13)
	Decreases Mortality	3.0 (1.7-5.5)

PHYSICIANS ATTITUDES REGARDING AUTOPSY. A. Reddy¹; L. Cation¹. ¹University of Illinois at Peoria, Peoria, IL. (Tracking ID #132615)

BACKGROUND: Background: Historically, autopsies have provided significant and valuable medical information. Despite recommendations by the AMA to continue performing autopsies, there has been a decline of autopsy rates in America. The purpose of this study is to investigate and analyze the attitudes and practices of attending physicians and residents towards autopsies.

METHODS: Methods: This IRB approved study was conducted at a community teaching hospital. A voluntary, confidential 27-item questionnaire was sent to 200 attending physicians and 65 resident physicians in the internal medicine department.

RESULTS: Results: Fifty-four (27%) of the attending physicians and 51 (78%) of the resident physicians returned the survey (total response rate 40%). The demographics of the survey participants were: average age 37.8 (range 26-70), 70.5% male, and an average of 12.5 years from completion of medical school. Physicians rated the following statements regarding autopsies on a 5 point Likert scale (1 =strongly disagree, 5 =strongly agree): contribute to medical research and discovery (4.30), cost/benefit does not justify routine use (2.73), irrelevant because of current technologies (1.95), attendance is educational (4.13), received formal training in requesting autopsy (2.12), comfortable asking family members (3.83), ability to determine cause of death on clinical grounds alone is excellent (3.21), cause of death is unclear (3.54), unexpected findings at autopsy (3.17), answers difficult clinical problems (3.25), improved care provided to other patients (3.23), increase litigation (2.40), most families don't want autopsy (3.76), and had family refuse when indicated (3.59). Physicians rated the most common reason why families refuse autopsies as "they were emotionally too upset".

CONCLUSIONS: Conclusions: Our study reveals that community teaching hospital physicians feel that autopsies continue to play an important role in medical discovery and that current medical technologies did not preclude the use of autopsies. Fear of litigation was not a significant concern but family wishes may be important in the low rate of autopsies. Further study is needed to confirm our findings.

PHYSICIANS' ATTITUDES TOWARDS CULTURAL COMPETENCY AWARENESS AND PATIENT SELF REPORTED HEALTH STATUS. C. Hernandez¹; R. Brown²; E. Manjarrez³. ¹University of Florida, Gainesville, FL; ²George Washington University Medical Center, Washington, DC; ³University of Miami, Miami, FL. (Tracking ID #136101)

BACKGROUND: Cultural competency education has been suggested as a way to address aspects of health care disparities. Little data exist concerning physicians' attitudes regarding the perceived value of cultural competency awareness and its impact on patient care and health outcomes. The purpose of this study is to correlate physicians' attitudes towards cultural competency awareness, as reflected by an arrogance score and patient health and quality of life outcomes- specifically physical function, vitality and pain scores.

METHODS: The CCTOP (Cultural Competency Training and Outcomes in Patients) study is an ongoing multi-center, cross-sectional study of patients' and their physicians' attitudes regarding culture, bias and health. Preliminary data from six of the participating institutions is reported here. The patient survey includes information on demographics, preventive screening, diabetes control, quality of life, trust in their physician, and patient's reports of adherence to treatment plans. The physician survey gathered information concerning attitudes, knowledge, and previous cultural competency training during medical school. We asked physicians to agree or disagree on a 5-point likert scale with the statement: "Patients' health outcomes are improved if the physician has outstanding clinical skills irrespective of cultural awareness". We then examined the association between physicians' dismissive attitude (or arrogance) towards cultural competency awareness with patients' perception of their overall health status, including physical functioning, vitality, and pain scores. A linear regression analysis was performed to control for age, gender, race, education and income.

RESULTS: We found no difference in patients' diabetes control, preventive health counseling and screening (as determined by counseling on tobacco and alcohol cessation, yearly ophthalmologic examination screening in diabetics, cholesterol and colon cancer screening) regardless of physician's arrogance score. However, bivariate analysis revealed that patients' self report of poor quality of life was associated with physicians' attitudes that clinical skill rather than cultural competency awareness is associated with patient outcomes ($p=0.07$). In contrast, patients whose physicians disagreed with the physician arrogance question had higher physical function scores, 58.9 vs. 52.1 ($p<0.05$), and better health vitality scores 53.4 vs. 48.4 ($p=0.04$).

CONCLUSIONS: Patients whose physicians dismiss the value of cultural competency awareness are more likely to report poorer quality of life as reflected by self-reported level of physical functioning and health vitality. Educational programs need to address physicians' knowledge and attitudes concerning the role of cultural competency awareness in improving patients' health care status, as well as in eradicating health care disparities.

PREDICTORS OF PHYSICIANS' ATTITUDES TOWARDS DIRECT-TO-CONSUMER ADVERTISING AND RELATIONSHIP TO PRESCRIBING. S. Krishnan¹; C. Franz²; H.N. Young¹; J. Song¹; R.L. Kravitz¹. ¹University of California, Davis, Sacramento, CA; ²University of California, San Diego, La Jolla, CA. (Tracking ID #135789)

BACKGROUND: Direct-to-consumer advertising (DTCA) of prescription drugs promotes expanded patient involvement in care but may strain the patient-physician

relationship. We examined primary care physicians' (PCPs) views of DTCA in the context of a randomized trial involving Standardized Patients (SPs) presenting with depressive symptoms.

METHODS: Eighteen SPs made 298 visits to 152 PCPs in 3 U.S. cities. Six roles were created by crossing 2 conditions (major depression vs adjustment disorder) and 3 request types (brand-specific, general, or none). Antidepressant prescribing was ascertained via visit audiorecordings and chart review. PCP characteristics and attitudes were derived from self-administered post-study questionnaires. Physicians' views of DTCA were assessed using 12 Likert-type items (1=agree to 5=disagree) measuring perceived impact of DTCA on consumers (CONSUMER, $K=3$, $\alpha=.66$), on the health care marketplace (MARKET, $K=3$, $\alpha=.68$), and on the MD-pt relationship (MDPT, $K=6$, $\alpha=.83$); a composite score (TOTSCL) was generated as the mean of all 12 items ($\alpha=.84$).

RESULTS: On a 1-5 scale, mean values for TOTSCL, CONSUMER, MARKET, and MDPT were 2.6, 3.7, 1.9, and 2.3, respectively. PCPs agreed that DTCA alerts consumers to new therapies (73%) and encourages people to seek advice for conditions that might otherwise be neglected (63%) but also confuses patients (81%), promotes patient-physician conflict (59%), and leads to higher drug prices (86%). Using multiple regression, Caucasian ethnicity and HMO affiliation (but not age, gender, or specialty) were associated with significantly more negative DTCA attitudes. In a random effects logistic regression that accounted for clustering, the odds of prescribing antidepressants were lower among HMO physicians ($p=.04$) and higher in visits where SPs requested medication ($p<.001$); DTCA attitudes had no independent effect.

CONCLUSIONS: Primary care physicians acknowledged the benefits of DTCA for consumers while questioning its impact on the market and on the MD-patient relationship. HMO physicians appear to represent a relatively hard target for DTCA-associated appeals. More research is needed to elucidate the relationship between physician ethnicity and DTCA attitudes.

PREVALENCE OF INAPPROPRIATE USE OF NATURAL PRODUCTS. A. Sood¹; D.L. Wahner-Roedler¹; R. Sood¹; L.L. Loehrer¹; S. Bagniewski¹; S. Cha¹; B. Bauer¹. ¹Mayo Clinic, Rochester, MN. (Tracking ID #133579)

BACKGROUND: Most natural products are available over-the-counter and are not FDA regulated, thus increasing the potential for adverse effects. The present study was designed to determine the prevalence of inappropriate use of natural products (herbal products and dietary supplements, excluding vitamins and minerals) in a broad spectrum of patients seen at a tertiary care center.

METHODS: An identified, point of service survey was administered to 1,818 patients in six different specialties at the Mayo Clinic. The survey comprised of 85 questions asking patients about their CAM use over the previous 6 months. The response rate was 98.8%. Of the 1,795 respondents, 710 patients reported use of herbal products. The medical records of these patients were reviewed to abstract information regarding underlying medical conditions and concurrent use of prescription medications during the same 6 months period. Inappropriate use of natural products was defined as use in one of the following two situations: 1). Natural product taken in the presence of a known contraindication related to patient's medical conditions; 2). Possibility of interaction of the natural product with a prescription medication concurrently taken by the patient. The information about adverse effects of the natural products and potential drug interactions was obtained using Natural Medicines database and PubMed. Statistical analysis was performed using Chi-square and t tests (SAS software).

RESULTS: The mean age of the patients using natural products was 57.7 years (SD 14.9). Of the 710 patients using these products, 65% were females. Natural products were inappropriately used by 237 patients (33.4%) according to the criteria defined above. The most common reason for inappropriate use was a potential for drug interactions in 161 patients (22.7%). The 4 most commonly used natural products with a potential for drug interaction were echinacea, ginseng, glucosamine, and green tea. The most common prescription medications with a potential for interactions with herbal products were warfarin, hepatotoxic medications, anti-hypertensives, anti-epileptics and diabetes medications. Medical conditions that could be aggravated by the use of a specific natural product were documented in 154 patients and included GERD, bleeding disorders, diabetes mellitus, preop-status, asthma, migraine, hypertension and seizures.

CONCLUSIONS: Inappropriate use of herbs/dietary supplements is common in clinical practice. Fortunately, life-threatening sequelae due to the use of these supplements are rare, and none occurred in the population studied. A history of intake of natural products with a focus on potential for drug interactions should be included in most clinical encounters. Patients and providers need to be educated about the adverse effects of these agents.

PROSPECTIVE VALIDATION OF A SCORE IDENTIFYING PATIENTS WHO NEED TRANSFER TO A REHABILITATION FACILITY AFTER HOSPITALIZATION. M. Louis-Simonet¹; M.P. Kossovsky¹; P. Chopard¹; T.V. Perneger²; J.M. Gaspoz¹. ¹Department of Internal Medicine and Groupe de Recherche et d'Analyse en Systèmes et Soins Hospitaliers (GRASSH), Geneva University Hospitals, Geneva; ²Quality of Care Unit and Groupe de Recherche et d'Analyses en Systèmes et Soins Hospitaliers (GRASSH), Geneva. (Tracking ID #132494)

BACKGROUND: We previously derived in a cohort of 299 patients, a score predicting, on the 3rd hospital day, patients' need of transfer to a rehabilitation facility at the end of their hospital stay. It was based on 5 variables significantly associated with transfer to a rehabilitation facility: number of active medical problems on admission; no informal help from spouse; dependency for bathing,

for transfers from bed or chair and for preparation of medications at home. We recruited a patient cohort to prospectively validate the score.

METHODS: 161 patients (validation cohort) were recruited in October and November 2003. The score capacity to predict need of transfer to a rehabilitation facility was measured by means of receiver operating curves (ROC) in the two cohorts and the areas under ROC curves (AUC) were compared.

RESULTS: The score capacity to predict transfer to a rehabilitation facility was similar in the derivation and the validation cohorts (AUC=0.82 vs. 0.77; $p=0.31$). Associations of score categories with length of stay (LOS) and with inappropriate hospital days followed the same trend in both cohorts (Table).

Derivation cohort (n=299) Score	Patients transferred to a rehabilitation facility n (%)	LOS in days (mean $\pm$ SD)	Inappropriate hospital days (mean $\pm$ SD)	Inappropriate hospital days due to transfer delays (mean $\pm$ SD)
Less than 8 points n=141 (47%)	12 (9%)	11.4 ($\pm$ 6.6)	2.8 ($\pm$ 3.5)	0.2 ($\pm$ 1.1)
8 to 15 point n=108 (36%)	43 (40%)	12.6 ($\pm$ 6.2)	3.6 ($\pm$ 4.1)	1.5 ($\pm$ 2.9)
16 points and more n=50 (17%)	40 (80%)	14.5 ($\pm$ 7.0)	4.8 ($\pm$ 4.8)	3.6 ($\pm$ 3.6)
p	<0.001*	0.005**	0.03**	<0.001**

Validation cohort
(n=161)

Less than 8 points n=58 (36%)	7 (12%)	8.6 ($\pm$ 5.9)	3.0 ($\pm$ 3.7)	0.7 ($\pm$ 2.3)
8 to 15 point n=59 (37%)	24 (41%)	12.6 ($\pm$ 7.2)	5.0 ($\pm$ 5.5)	2.6 ($\pm$ 4.6)
16 points and more n=44 (27%)	27 (61%)	14.1 ($\pm$ 9.7)	5.3 ($\pm$ 6.5)	3.5 ($\pm$ 5.6)
p	< 0.001*	<0.001**	0.08**	<0.001**

* Fisher's exact test ** Kruskal-Wallis test

CONCLUSIONS: A simple score, easily computed on the 3rd hospital day, predicts a transfer to a rehabilitation facility with reasonable accuracy. Early transfer planning among patients with a score > 8 points could reduce inappropriate hospital use.

RESIDENTS' RESPONSES TO A SYSTEMS-BASED, CHRONIC ILLNESS CARE EDUCATION PROGRAM. N. May¹; M. Nadkarni¹; J.D. Voss¹. ¹University of Virginia, Charlottesville, VA. (Tracking ID #136222)

BACKGROUND: Our residency program is implementing a new curriculum that includes systems-based competency education and training in the care of chronically ill patients. The development of new, more relevant, learner-centered approaches to medical training requires close examination of the impact of these methods on the trainees themselves.

METHODS: We developed a new Diabetes Care Program (DCP) in our resident practice with the goal of educating learners about managing clinical care for chronically ill patients. A core of 12 learners—internal medicine residents, pharmacy residents, and family nurse practitioner students—worked with faculty experienced in chronic illness care and quality improvement to develop the program. As teams, they developed an integrated program designed to improve health outcomes for medically indigent patients with diabetes. While planning, implementing, and evaluating the DCP over two years, learners simultaneously received didactic and experiential instruction in the chronic care model and systems-based learning and improvement. All five third-year internal medicine residents (3 women and 2 men) who had participated for two years in the DCP were interviewed for 60-90 minutes. Respondents discussed many aspects of the program, including how the project had changed their practice of medicine; the project's value to their medical training; and how they would have improved the experience.

RESULTS: Residents agreed to participate in the DCP because it supported their professional needs and interests, including the desire to improve their pa-

tient care skills. Learners valued the sense of professional camaraderie, attending national meetings and conferences, and working with other disciplines. They reported learning an improved ability to cope with lack of control; new appreciation for the complexity and multidisciplinary aspects of care; increased knowledge about diabetes; increased openness to disease registries and other tools; and improved relationship with patients. Residents said that motivational interviewing training had the most significant impact on their practice. They spoke articulately and passionately about the importance of systems-based improvements in care, and they made numerous suggestions on how to improve the training program.

CONCLUSIONS: These residents welcomed clinically relevant, high-quality training experiences that helped them better care for their patients, even if they required additional time and effort. Innovative programs such as the DCP can provide valuable learning opportunities for residents with diverse needs and interests. Given the opportunity, learners can be a valuable resource as we examine new curricular innovations to address the ACGME competency requirements. We have used many of their suggestions to further develop our curriculum to teach systems-based care, practice based learning, professionalism, and interpersonal and communication skills.

RISK ASSESSMENT BASED ON PATIENT INPUT OF STRUCTURED FAMILY HISTORY INFORMATION COMPARED TO RISK IDENTIFIED BY ELECTRONIC HEALTH RECORD DATA. L.A. Volk¹; M. Staroselsky¹; L. Pizziferri¹; T. Wang¹; R. Tsurikova¹; J. Wald¹; D.W. Bates¹. ¹Partners HealthCare System, Inc., Wellesley, MA. (Tracking ID #136217)

BACKGROUND: Family history data often goes unrecorded or is available only in free text in the electronic health record. Patient provided family history information may provide incremental value if one can learn of increased risks that may influence management. Although sophisticated family history assessment is most expertly done by those trained in this area, much can be learned about a patient's risk from whether a relative with a condition is a primary or secondary relative, how many relatives have the condition, and the approximate age at which they were diagnosed. In this study, we assessed the availability of family history information in electronic health records and the extent to which structured input from patients would identify those at higher risk than determined by information readily available in their EHR alone.

METHODS: Patients were asked to provide family history information in a structured format that required patients to specify the number of family members by degree of relationship and age category that had been diagnosed with a condition. In addition to requesting information on six specific conditions (glaucoma, osteoporosis, coronary artery disease, breast cancer, and colon cancer), patients were given the opportunity to complete the form for other conditions for which they had a positive family history.

RESULTS: Among 1098 patients solicited, a family history survey was completed by 163 patients. Fifty-three percent (53%) had no family history information readily available in the electronic health record either on the patient's problem list or within a templated family history note. For glaucoma and osteoporosis, over 90% of the patients determined to have a positive family history were identified based solely on new information provided from the patients (Table 1). A family history of diabetes, breast cancer, or coronary artery disease was more frequently readily available within the EHR, although no more than 41% of patients with a positive family history for these conditions were identified this way. New information provided by patients led to an increased risk level for 50% of patients with positive family history of colon cancer, and the majority of patients with a positive family history for diabetes (74%), osteoporosis (95%), and glaucoma (95%), and increases in risk were found for 32% of patients with a positive family history of breast cancer and 42% with coronary artery disease.

CONCLUSIONS: Clinically relevant family history information is frequently missing or not readily available in electronic health records. For all diagnoses, patients provided family history data that might influence use of screening tests. With the increased emphasis on family history as the field of genomics becomes more sophisticated, these results demonstrate the potential benefits of soliciting information from patients in a structured format outside of visits. Further analyses are underway to evaluate whether providing this patient information electronically to physicians results in an increase in appropriate screening.

Table 1. Changes in Family History Risk Assessment Based on Patient Input Compared to Electronic Health Record Data

Condition	N pts with Family History Related to Risk Level for the Condition	N pts providing new information	N pts for whom new info REDUCED risk level	N pts for whom new info DID NOT CHANGE risk level	N pts for whom new info INCREASED risk level
Glaucoma	20	19	0 (0%)	1 (5.3%)	18 (94.7%)
Osteoporosis	22	20	0 (0%)	1 (5.0%)	19 (95.0%)
Diabetes	51	42	1 (2.4%)	10 (23.8%)	31 (73.8%)
Breast Cancer	47	40	0 (0%)	27 (67.5%)	13 (32.5%)
Colon Cancer	68	66	0 (0%)	33 (50.0%)	33 (50.0%)
Coronary Artery Disease	82	67	0 (0%)	40 (59.7%)	27 (40.3%)

SCIENTIFIC SUPPORT FOR NUTRITION FACTS IN A POPULAR DIET BOOK: CAVEAT EMPTOR. S.L. Goff¹; J.M. Foody¹; S.E. Inzucchi¹; D.L. Katz²; S.T. Mayne¹; H.M. Krumholz¹. ¹Yale University, New Haven, CT; ²Yale University, Derby, CT. (Tracking ID #133652)

BACKGROUND: Diet and nutrition books dominated the NYT Best Seller Advice Book List for much of 2003-2004. Consumers cite such books as an important source of nutrition information, yet these books may not be subject to the scientific standards of peer reviewed journals. Despite the popularity of books as a source of nutrition information, the extent to which assertions cited in nutrition books are supported in peer reviewed journals is not known.

METHODS: We chose the popular South Beach Diet as a case study of the scientific evidence supporting nutrition information in a popular diet book. Three independent reviewers identified nutrition related assertions in the text of the book. After a comprehensive list of assertions was agreed upon by reviewers, a fact checking team performed structured literature searches to assess support for each assertion. We adapted the search strategy used by the British Medical Journal's Clinical Evidence, searching for systematic reviews published prior to the publication of the South Beach Diet, followed by RCTs, then observational studies if necessary. Based on the review findings, each fact was assigned to one of four categories: 1) Data supports assertion 2) Data contradicts assertion 3) Data both supports and contradicts assertion 4) No direct data. Expert reviewers from the fields of preventive cardiology, nutrition, endocrinology, and preventive medicine then examined the category assignment for each assertion and indicated whether they agreed with the categorization based on their knowledge of the literature in the field. Disagreements with categorizations were reviewed by the primary author and re-assigned or maintained based on the reviewer's comments and recommended citations.

RESULTS: A total of 45 nutritional assertions were included in the review. Fourteen of the assertions were supported in the scientific literature, while 10 were contradicted, 6 found to have no direct data and 15 found to have data both supporting and contradicting the assertion.

CONCLUSIONS: Consumers obtain nutrition information from lay media such as diet books. Our findings indicate that over 2/3 of statements presented in a factual manner in a diet book with wide readership may not be supported in the scientific literature. These findings have important implications for educating consumers about sources of nutrition information.

THE CORRELATION OF PIT RECOVERY TIME OF PERIPHERAL EDEMA WITH ALBUMIN CONCENTRATION. D. King¹; D. Rozan²; A. Khan³; J. Wiese¹. ¹Tulane University, New Orleans, LA; ²Wake Forest University, Winston-Salem, NC; ³Mayo Clinic, New Orleans, LA. (Tracking ID #136134)

BACKGROUND: Edema is commonly classified using a 1 to 4 scale. The lack of consensus as to the meaning of this scale makes it of little clinical utility. In a survey, 74% of residents used the scale to indicate the height to which edema extended on the patient combined with an anatomical landmark (e.g., "2+ edema to the knees"). Starling's law suggest edema is due to either inadequate oncotic pressure or increased hydrostatic pressure. We investigated the utility of using the pit recovery time (the time required for a maximal pit in edematous tissue to recover to normal) as the measure for assessing peripheral edema.

METHODS: Seventy-three patients with peripheral edema of less than six months duration extending at least to 5 cm above the maleolus were enrolled in the trial. Participants must have had a serum albumin assessed within one week of study entry. A depression was made in the edema five centimeters above the maleolus for five seconds. The time for the depression created by digital pressure to recover to the same level as the skin surrounding the initial depression was recorded. The patient's albumin concentration was recorded after the examination.

RESULTS: There was a correlation between the albumin concentration and the pit recovery time (r^2 0.50). 93% of patients with a pit recovery time of less than 20 seconds had an albumin of less than 2.0 g/L. All of those patients with a pit recovery time of greater than 60 seconds had an albumin of greater than 3.5 g/L.

CONCLUSIONS: The pit recovery time is a useful predictor of the albumin concentration in patients with peripheral edema of less than six months of duration. An edema classification scale of 1+ for each 20 seconds of pit recovery time is a more useful method of describing peripheral edema, providing information as to the potential cause of the edema.

THE DIFFICULT DOCTOR? CHARACTERISTICS OF PHYSICIANS WHO EXPRESS FRUSTRATION WITH PATIENTS. E.E. Krebs¹; J.M. Garrett¹; C.Q. Porter¹; T.R. Konrad¹. ¹University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #135874)

BACKGROUND: The difficult patient-physician relationship is an enduring feature of clinical practice. Literature on the topic has focused primarily on the nature of the "difficult patient," rather than on physician and practice characteristics associated with these frustrating relationships. Our objective was to determine physician and practice characteristics associated with greater physician-reported frustration with patients.

METHODS: The Physician Worklife Survey surveyed a national physician sample by mail to evaluate multiple aspects of physician job satisfaction. Our analysis included physicians in family medicine, general internal medicine, and medical subspecialties. The survey assessed physician demographics, stress, burnout, anxiety, depression, perceived health status, and satisfaction with job, career, and specialty. Practice factors included case-mix, hours worked, and time allotted for patient visits. Physicians were asked to estimate the percentage of patients in their practice who were generally frustrating to deal with. We categorized physicians by quartile of reported frustrating patients. We then used Pearson's chi-square tests to compare characteristics of physicians in the top

quartile (highly frustrated physicians) to those in the other three quartiles. Logistic regression was used to model characteristics associated with greater physician-reported frustration with patients. All analyses were adjusted for weights and stratification included in the sample design.

RESULTS: Of the 1391 non-pediatrician respondents, 502 were family physicians, 451 were general internists, and 438 were medicine subspecialists. The mean age was 47 years, and 77% were male. On average, physicians reported that 12% of their patients were frustrating. Physicians reporting the most frustrating patients ($\geq 15\%$ of their patients) were younger ($p < 0.001$); worked more hours per week ($p = 0.041$); and had worse perceived health status ($p < 0.001$); and more symptoms of depression, stress, anxiety or burnout ($p < 0.004$ for all). Doctors with lower job, career, or specialty satisfaction were also more likely to be highly frustrated ($p < 0.005$ for all). Physicians who reported at least moderate exposure to primary care practice during their training were less often highly frustrated ($p = 0.018$). In the adjusted model, factors independently associated with high physician frustration included age < 40 years, work hours > 60 per week, less training in primary care practice, poorer perceived health status, greater burnout, poorer job satisfaction, and greater proportions of patients with complex psychosocial problems or substance abuse.

CONCLUSIONS: While nearly all physicians feel frustrated by some patients, a number of physicians report much higher levels of frustration. Physician factors associated with greater frustration include job dissatisfaction, poorer mental and physical health, longer work hours, and less training in primary care practice. These factors may decrease doctors' threshold for feeling frustrated when confronted by challenging patient-care situations. Additional research is needed to determine the relative importance of patient, physician, and practice factors in the formation of difficult patient-physician relationships and whether modification of physician and practice factors can improve physician frustration and patient satisfaction.

THE IMPACT OF CRP AS A CRITERION FOR CHOLESTEROL LOWERING TREATMENT: A POTENTIAL DISTRACTION FROM HIGH-RISK PATIENTS. K. Kerin¹; L. Schwartz¹; S. Woloshin¹; H. Welch¹. ¹VA Outcomes Group, White River Junction VT, White River Junction, VT. (Tracking ID #135702)

BACKGROUND: There is currently no evidence that cholesterol-lowering treatment benefits those whose only indication for treatment is elevated C-reactive protein (CRP). Nonetheless, some have suggested that treatment is warranted based on observational data showing that CRP predicts cardiovascular outcomes independent of LDL cholesterol. Little is known about how many people would be affected by adding a CRP criterion to current guidelines. We sought to determine the additional number of Americans for whom cholesterol-lowering treatment would be recommended if an elevated CRP level were adopted as another indication for treatment. To put this into context, we also determined how well lipid-lowering guidelines are being implemented currently - particularly for those at highest risk, patients with established coronary heart disease or equivalent.

METHODS: We used the 1999-2000 National Health and Nutrition Examination Survey (1750 adults over age 20 with complete data on risk factors, fasting LDL and CRP) to determine the additional number of Americans for whom cholesterol-lowering treatment would be recommended if a CRP level of more than 3 mg/L were adopted as an additional indication for treatment. In addition, we determined the number of people who meet current National Cholesterol Education Program Adult Treatment Panel (NCEP ATP III) guidelines for treatment (based on a combination of risk factors and LDL thresholds) - yet are not receiving therapy.

RESULTS: 47.8 million of the 201 million US adults meet criteria for using cholesterol-lowering medications according to the current guidelines. Adopting the proposed CRP criterion would make an additional 51.9 million adults eligible for cholesterol lowering treatment. The majority (90%) of those newly eligible based on the CRP criterion are those with the lowest predicted risk of heart disease according to traditional risk factors. Presently, only 30% of the highest risk patients—those with CHD or its risk equivalent—are being treated.

CONCLUSIONS: If a CRP level of > 3 mg/L were used as an additional treatment criterion approximately half of all US adults would be considered eligible for cholesterol lowering therapy. Before expanding the treatment criteria to more low risk patients in which the benefit is not established, we should focus on those patients at the highest risk where the benefit of treatment is clear.

THE INCIDENCE OF SYMPTOMATIC VENOUS THROMBOEMBOLISM IN THE ABSENCE OF ANTICOAGULANT PROPHYLAXIS IN HOSPITALIZED GENERAL MEDICAL PATIENTS. A. Dunn¹; A. Federman¹; M. Skamagas¹; M. Osofsky¹; J. Boxer¹; H. Patel¹; S. Goldberg¹. ¹Mount Sinai School of Medicine, New York, NY. (Tracking ID #134403)

BACKGROUND: Studies of general medical inpatients utilizing routine screening tests to assess for asymptomatic VTE have found the incidence to be 10-15% without prophylaxis; however, there is little data examining the incidence of symptomatic VTE in the absence of anticoagulant prophylaxis in this population. As newer and more expensive anticoagulant choices become available and as anticoagulants may be considered for patients with bleeding risk factors, awareness of the incidence of symptomatic events without prophylaxis will be necessary to allow clinicians to compare the potential benefits with the risks and costs. Therefore, we sought to determine the incidence of and risk factors for symptomatic VTE in hospitalized medical patients.

METHODS: We performed a retrospective review of inpatients from randomly selected admissions to the medical wards of the Mount Sinai Hospital between July 1, 2002 and December 31, 2002. Patients were excluded if they received

unfractionated heparin (UFH), had suspected VTE at admission, were admitted for a surgical procedure, or were at highest risk for VTE (defined a priori as either prior VTE or known thrombophilia, active cancer, or surgery within 3 months). For patients with multiple admissions, one admission was selected randomly for inclusion. Charts and test results were reviewed to identify whether VTE occurred during the hospitalization. VTE occurring more than 1 day after admission were considered to be hospital-acquired. Patients were divided into subgroups with 0 or ≥ 1 VTE risk factors. VTE risk factors were defined as age > 60 years, CHF, CVA, or obesity.

RESULTS: A random sample of 1140 admissions was selected. Of these, 503 admissions were excluded based on a review of hospital databases indicating that the patient received prophylaxis-dose UFH, the reason for admission was surgery, or for being a repeat admission. Of the remaining 637 admissions, 591 (93%) were available for review. Of these, 214 were excluded; 108 due to being at high-risk for VTE and 106 for meeting other exclusion criteria. The remaining 381 admissions were included in the analysis. The baseline characteristics are shown in the Table. Risk factor assessment revealed that 181 patients (48%) had 0 VTE risk factors and 200 patients (52%) had ≥ 1 risk factors. One hospital-acquired VTE (0.3%, 95% CI .05–1.0%) was noted. VTE incidence for patients with 0 and ≥ 1 VTE risk factors were 0% and 0.5%, respectively. The only hospital-acquired VTE was noted in a 79 year-old female admitted for a flare of ulcerative colitis.

CONCLUSIONS: The incidence of VTE in hospitalized medical patients who had no history of VTE/thrombophilia, active cancer, or recent surgery was low in the absence of anticoagulant prophylaxis. Though the examination of patients who did not receive anticoagulant prophylaxis may have led to the study population being at lower risk than patients for whom clinicians chose to administer prophylaxis, over one-half of the study population had 1 or more commonly cited risk factors for VTE. Prospective studies are needed to better determine risk factors for symptomatic hospital-acquired VTE for general medical patients.

Baseline Characteristics

Characteristics	N=381
Age, mean (SD)	59 years (21)
Female, %	190 (51%)
Length of Stay, median (range)	4.0 days (1–152)

THE INFLUENCE OF GENDER ON SEXUAL HISTORY TAKING AND HIV RISK REDUCTION COUNSELING. A.R. Hoelstein¹; S.R. Kesavalu¹; C.H. Griffith¹; J.F. Wilson¹; S.A. Haist¹. ¹University of Kentucky, Lexington, KY. (Tracking ID #135715)

BACKGROUND: It has long been known that the gender of the physician and of the patient can impact variables of medical communication. In this experiment, we study the effect of both student and standardized patient (SP) gender as it relates to the sexual history taking and HIV risk reduction counseling.

METHODS: At the University of Kentucky College of Medicine, the third-year class must adequately complete a 13-station standardized patient clinical exam to matriculate into the fourth year. Each station is graded by a checklist of salient items completed by the SP. One of the stations, the HIV risk reduction counseling case, was developed to be portrayed by both a male and female SP and contained gender-specific items on the checklist. There were eighty total items on the female checklist and sixty-nine total on the male checklist. The male checklist contained eleven and the female seventeen sexual history items and both contained ten HIV counseling items. One-half of all the students interviewed a man at risk for HIV and the other students encountered a woman. Every student had an equal chance to encounter either gender. A gender-neutral anxiety case was used as the control. Multiple regression approaches solving for measures of prior clinical performance and using the general linear model were used to analyze the data with SAS.

RESULTS: Ninety-seven students completed the CPX. Forty-nine encountered a female SP and forty-eight a male SP. There were twenty-two women in both groups. The performance of the female students was superior to that of the men in regards to both the elements of the sexual history taking and the HIV risk reduction counseling (62.8 $\pm$ 13.7 vs. 56.4 $\pm$ 15.2, $p < .001$ and 69.4 $\pm$ 6.9 vs. 67.4 $\pm$ 9.2, $p = .009$, respectively). Furthermore, when a woman encountered a female SP, the mean checklist score for sexual history items was 63.6 $\pm$ 15.9 and for HIV counseling was 72.2 $\pm$ 6.5 compared to male students interviewing men, 51 $\pm$ 13.1, and 63.0 $\pm$ 7.5, respectively, $p < .001$ for both groups.

CONCLUSIONS: Female medical students score better than their male counterparts on sensitive elements of a standardized patient encounter, sexual history taking and HIV risk reduction counseling. In fact, even when there are more items to ask, i.e., in an encounter with a female SP, the female students continue to excel. Limitations to our study include single institution and study of only one SP case. However, despite these limitations, our findings suggest both areas of future focus for clinical medical education and possible gender influences in actual patient-physician encounters.

TOWARDS IMPROVING PATIENT SAFETY: THE USE OF TISSUE MODELS FOR VASCULAR ACCESS TRAINING. B.T. Rosen¹; M.J. Ault¹; B. Ault¹. ¹Cedars-Sinai Medical Center, Los Angeles, CA. (Tracking ID #136288)

BACKGROUND: Following the Institute of Medicine report addressing medical errors, the Agency for Healthcare Research and Quality identified proper central venous catheter insertion techniques and the use of wide sterile barriers as

major quality indicators for patient safety. However, no standard has been established for how to teach proper procedural techniques to physicians, and patient safety continues to be a legitimate source of concern.

METHODS: The authors organized educational sessions at Cedars-Sinai Medical Center to teach the proper use of wide sterile barriers, ultrasound for central venous catheter placement, and sharps safety to medical professionals. Participants practiced these skills on non-human tissue models under the direct supervision of attending physicians with procedural experience. Surveys were distributed to assess participants' reactions to the training.

RESULTS: Participants had statistically significant improvements in mean comfort levels with the use of ultrasound for central venous catheter placement and the creation of wide sterile barriers. Comfort level in central venous catheter placement using ultrasound improved from 1.86/5.0 to 3.82/5.0 ($p < 0.001$), and comfort level in the creation of wide sterile barriers improved from 3.22/5.0 to 4.38/5.0 ($p < 0.001$). Image quality of the model was rated 3.8/5, and the overall worthiness of the session was rated 4.9/5.0.

CONCLUSIONS: Patient safety is an ongoing concern in the hospital setting. Given the participants' significant improvement in comfort levels with wide sterile barriers and the use of ultrasound for central venous catheter placement, the use of non-human tissue models and the implementation of procedural training sessions for physicians should be seriously considered.

TREATMENT NIHILISM: EXPLORING ATTITUDINAL FACTORS THAT MAY CONTRIBUTE TO DISPARITIES IN HEALTH CARE. D.G. Safran¹; S.L. Thier²; I. Wilson¹; W. Bender³; T. Sommers³; H. Chang¹; W.H. Rogers¹. ¹Tufts University School of Medicine, Boston, MA; ²Jefferson University, Philadelphia, PA; ³Tufts-New England Medical Center, Boston, MA. (Tracking ID #135117)

BACKGROUND: An extensive research literature documents treatment differences that are associated with patients' socioeconomic characteristics – most notably, race and ethnicity. However, little is understood about the mechanisms by which these treatment differences arise, and in particular, the extent to which they represent clinicians' deference to patient preferences as opposed to other factors. We explored attitudes of "treatment nihilism" expressed by Medicare beneficiaries aged 65 and older, and examined the extent to which these attitudes differ by race and ethnicity.

METHODS: The sample included Medicare beneficiaries from 13 states nationwide (MA, NY, PA, FL, IL, MN, TX, NM, AZ, CO, CA, OR, WA) who are part of a cohort established in 1998 to study primary care relationships and health outcomes, and who completed the 2002 study questionnaire ($n = 3,449$). In 2002, participants answered a series of questions concerning their likely actions if told by their primary physician that they had a "serious medical illness such as cancer or heart disease." Analyses presented here evaluate responses to an item asserting: "I would seek as little medical care as possible and let nature take its course." We term this "treatment nihilism" (TN). Logistic regression analyses modeled TN as a function of patients' age, sex, race (white, African-American, Asian, other), Hispanic ethnicity, years of education, household income, functional health status, number of chronic conditions, and trust in their primary physician.

RESULTS: Overall, 10.5% of seniors endorsed the assertion of TN. African-American and Hispanic elderly were more likely to report TN than non-Hispanic whites (23.9%, 22.5%, 9.7%, respectively, $p < .001$). Among those with less than a high school education, 24.0% expressed TN, compared with 8.4% among high school graduates ($p < .001$). In multivariate models controlling only for age, sex and race/ethnicity, TN was significantly associated with older age (OR = 1.07 per year, $p < .001$), African-American race (OR = 3.60, $p = .001$) and Hispanic ethnicity (OR = 3.06, $p < .001$). However, after controlling for socioeconomic status (education, income) and health (functional status, condition count), the effects of race and ethnicity were substantially mitigated. In fully-loaded models, low educational attainment and low household income were significantly associated with TN ($p < .001$), as was low mental health status ($p = .001$). Patients with more chronic conditions were less likely to express TN ($p = .013$), as were those with more trusting primary care relationships ($p < .001$). Neither African-American race ($p = 0.058$) nor Hispanic ethnicity ($p = 0.186$) were significantly associated with TN in these models.

CONCLUSIONS: With extensive evidence of treatment disparities in health care, attention must turn to understanding the mechanisms by which these occur. Our findings suggest that patient characteristics such as lower educational attainment, lower income and depressed mental health are significantly associated with a stated preference for "letting nature take its course" rather than seeking medical treatment. However, when these variables are masked or ignored, the attitudes can appear to be simply an attribute of African-American race and Hispanic ethnicity. Trusting primary care relationships appear to work against attitudes of TN. For clinicians, the findings underscore the importance of ascertaining – patient-by-patient – values and preferences regarding care and treatment, and carefully considering when it is appropriate to defer to these attitudes and when it is appropriate to attempt to influence them.

VALIDATING A COMPUTER MODULE FOR MANAGEMENT OF RECURRENT UNCOMPLICATED CYSTITIS IN WOMEN. P. Nadler¹; E.M. Aagaard¹; J. Maselli¹; J. Adler¹; R. Gonzales¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #134677)

BACKGROUND: Many women do not have access to telephone management of uncomplicated UTI, and present acutely to physician offices and urgent care clinics. Based on previously described telephone management algorithms, we validated an interactive computer kiosk module for the diagnosis and treatment of uncomplicated UTI.

METHODS: Prospective study of women age 18-64 years presenting to an urgent care clinic between October 2003 and March 2004. Women were referred by the triage nurse if they reported symptoms of dysuria, urinary frequency/urgency, or if the patient believed that she had a UTI for any other reason. Informed consent was obtained by the computer kiosk program. Using the kiosk touch screen, participants completed the interactive computer module in English. All women entered in the study then received usual care by a physician who was blinded to the computer information. Urine cultures were obtained as the referent standard. Women were deemed appropriate for computer-based, 3-day antibiotic treatment if they had a previous UTI, and denied having complicating features including specific comorbidities, vaginal discharge, back pain, emesis and fever/chills.

RESULTS: From a study population of 68 women, 18 (26%) met computer-criteria for treatment of uncomplicated UTI. Of the 18 computer-eligible cases, physicians diagnosed 17 with an uncomplicated UTI and 1 with pyelonephritis. On physician evaluation, however, none of these 18 cases had documented fever, tachycardia, costovertebral angle tenderness, nor vaginitis (by history or examination). Based on urine culture results, 12 of 18 (67%) women meeting criteria for computer treatment had a confirmed infection, representing 55% of all culture confirmed uncomplicated UTIs (total n=22). Among the 50 women meeting computer-algorithm criteria for further physician evaluation, the large majority reported concurrent history of either back pain (n=30) or vaginal discharge (n=24); and, two thirds of these women were diagnosed with uncomplicated UTI.

CONCLUSIONS: In our urgent care setting, an interactive computer module accurately identifies approximately half of women with confirmed uncomplicated UTIs for antibiotic treatment. Further clarification of back pain and vaginal discharge as either new or different could significantly increase the proportion eligible for computer treatment.

A BILINGUAL INTERACTIVE COMPUTERIZED EDUCATIONAL MODULE IS RATED HIGHLY IN A POOR, MEDICALLY UNDERSERVED PATIENT POPULATION. B. Leeman-Castillo¹; K. Corbett²; E.M. Aagaard³; J. Maselli³; T.D. Mackenzie⁴; R. Gonzales³.

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BACKGROUND: Computer health information has been shown to be an acceptable and effective means of educating mainstream patient populations about health care conditions. However, little is known about the acceptability and effectiveness of interactive computerized educational modules in poor and underserved communities with limited computer literacy. In a poor urban population, we evaluated the effectiveness of a module relating to appropriate antibiotic use.

METHODS: The study was conducted in a busy urgent care clinic in Denver between 1/00 and 4/00. Adults seeking care for acute respiratory tract infections (ARIs) were referred to a free-standing computer kiosk that housed the module. In English and Spanish, the audiovisual module collected patient demographics, knowledge and attitudes about antibiotics and ARIs, reasons for seeking care for their illness, and a symptom inventory. Based on each patient's symptoms, the computer suggested a likely diagnosis and offered tailored information regarding the illness and its appropriate treatment. Upon completion of the program, participants were asked to rate the module on several qualities known to enhance diffusion of innovations (low complexity, compatibility and relative advantage). The main outcomes for this study were whether the patient "learned something new about colds and flu", and whether they "trusted the computer information". Chi-square tests and multivariable logistic regression analysis were used to assess the significance of associations between predictor and outcome variables. Results are presented as adjusted odds ratios (OR) and 95% confidence intervals.

RESULTS: 296 persons completed the computer module: 81% were age 18-44 years, 59% were female, 54% were Hispanic, 16% completed the Spanish-language version, and 50% had annual household income <\$10,000. Spanish-language respondents (vs. English respondents) were much less likely to report prior computer experience (33% vs. 80%), more likely to need help from staff with the computer (83% vs. 26%), and strongly prefer answering to a person (60% vs. 20%) (all p-values <0.01). Differences between Spanish and English language respondents were smaller in rating the computer as easy to use (71% vs. 84%, p=0.06) and understanding computer messages (81% vs. 87%, p=0.04). After adjusting for patient demographics and computer module qualities (above), Spanish language users remained more likely than English language users to report learning something new (OR=5.0; [2.0, 12.4]), and trusting the information (OR=2.5 [1.0, 6.0]). Prior computer experience was a strong negative predictor of learning something new (OR=0.43 [0.22, 0.84]) and trusting the information (OR=0.31 [0.16, 0.60]).

CONCLUSIONS: Interactive computer-assisted health information programs were well received by patients in an urban urgent care clinic. They appear to have their greatest benefits among populations that have the least exposure to and experience with this medium, and may represent an important tool for programs aimed at reducing health disparities. Our finding that prior computer experience was associated with less learning and trust deserves further exploration.

A CLOSING GAP: RACIAL DISPARITIES IN THE QUALITY OF CARE FOR MEDICARE MANAGED CARE BENEFICIARIES BETWEEN 1997 AND 2003. A.N. Trivedi¹; A. Zaslavsky¹; E.C. Schneider¹; J.Z. Ayanian¹. ¹Harvard University, Boston, MA. (Tracking ID #133558)

BACKGROUND: Since 1997, all Medicare managed care (MMC) plans have been required to publicly report performance data using Health Plan Employer Data and Information Set (HEDIS) indicators. We studied whether disparities in these

quality indicators for Blacks and Whites in the MMC population have changed over time.

METHODS: We linked individual-level Medicare HEDIS data to the Medicare Enrollment file for each year from 1997-2003. We analyzed 2.8 million patient observations among MMC enrollees who were eligible for one or more HEDIS measures described in the table below. To assess time trends in racial disparities, we fit regression models predicting adherence to 10 HEDIS measures, including terms for race, year, and a race-year interaction. To assess the impact of health plan turnover, in the primary analysis we excluded 659,218 enrollees from health plans with fewer than 5 years of MMC participation, and in a secondary analysis we included these enrollees.

RESULTS: During the 7-year period, clinical performance improved on all measures for both Blacks and Whites (p<0.001 for all time-trends). The Black-White gap narrowed for all 6 process measures and 2 outcome measures (BP control and LDL control for diabetes), but widened for 2 outcome measures of A1c control for diabetes and LDL control after MI (p<0.001 for all race-year terms). The secondary analysis of patients enrolled in all plans that participated in MMC revealed similar results.

Eligibility	Measure (Initial Year)	Initial Rates(%)			2003 Rates (%)			Δ in Gap, 1997-2003
		White	Black	Gap	White	Black	Gap	
Diabetes	Eye Exams (1997)	51	43	8	72	70	2	-6
	A1c Test (1999)	73	67	6	90	88	2	-4
	LDL Test (1999)	70	60	10	94	92	2	-8
	LDL<130 (1999)	31	19	12	73	66	7	-5
	A1c<9% (1999)	71	69	2	82	75	7	+5
Recent MI	Beta Blocker (1997)	74	51	13	94	93	1	-12
	LDL Test (1998)	58	41	17	84	75	9	-8
	LDL<130 (1998)	13	11	2	68	51	17	+15
Hypertension	BP<140/90 (1999)	12	9	3	52	49	3	0
Women age 65-69	Mammogram (1997)	71	64	7	77	75	2	-5

CONCLUSIONS: A number of key HEDIS quality indicators have improved substantially since 1997 for both Whites and Blacks in MMC with corresponding reductions in racial gaps, but racial disparities in two important metabolic outcome measures have widened. Future research should examine factors that contributed to narrowing racial gaps and focus on interventions to eliminate persistent racial disparities in quality of care.

ABNORMAL PAP SMEAR RESULTS IN AN INCARCERATED POPULATION. J.G. Clarke¹; M. Hebert²; J. Rose². ¹Brown University, Providence, RI; ²Rhode Island Hospital, Providence, RI. (Tracking ID #136242)

BACKGROUND: Risk factors for cervical cancer include: early age of first intercourse, number of life time sexual partners, smoking tobacco and a history of sexually transmitted infections. The majority of women in prisons and jails are there as a result of drug use and/or sex trade, both of which are associated with many of the risk factors for cervical cancer. Follow-up of an abnormal pap smear after release is complicated for several reasons including transfer of information to new medical providers, lack of health insurance, high rates of homelessness and substance abuse issues. Guidelines for following Pap smears with atypical cells of undetermined significance (ASCUS) recommend repeating in 4-6 months or HPV testing. Given the setting and social situations of many of the patients it has been a long standing practice to send all women with abnormal Pap smears for colposcopy.

METHODS: A chart review was conducted to determine the prevalence and follow up of ASCUS Pap smears at the Adult Correctional Institute in Rhode Island. The medical records of all women who had been incarcerated between 1999 and 2004 were reviewed for Pap smear results. In cases with abnormal Pap smear results, an in-depth chart review was conducted abstracting risk factors for cervical cancer and follow-up evaluations as well as dates of incarceration and release.

RESULTS: Of the 4675 medical charts reviewed 785 women had a total of 3024 pap smears during periods of incarceration. Abnormal pap smears (ASCUS, AGUS, LGSIL or HGSIL) were documented in 196 (25%) women. Histological results were available for 51%. Of the 141 (18%) women with ASCUS Pap smears 65 had colposcopy (inflammation or normal 69%, CIN I 10%, CIN II 16%, CIN III 6% and carcinoma in situ 3%), 38 left the system without known follow up, 33 had normal Pap smears and 5 had abnormal repeat Pap smears.

CONCLUSIONS: The prevalence of ASCUS Pap smears in the general population is approximately 5% while in our sample we encountered a higher prevalence of 18%. Given the high prevalence of risk factors for cervical cancer among incarcerated women and problems with follow up it is reasonable to refer directly for colposcopy after an initial ASCUS pap smears. Providing colposcopy immediately after the first ASCUS Pap smear could greatly decrease the time to definitive diagnosis and treatment in this transient and high risk population. More studies are needed on post-release community follow-up and future risk factors.

ACCESS TO AMBULATORY CARE SERVICES AND CARE SETTING FOR MINORITIES IN MEDICAID MANAGED CARE. H.K. Seligman¹; A. Chattopadhyay¹; A.B. Bindman¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #133561)

BACKGROUND: Medicaid beneficiaries enrolled in managed care (MC) plans report greater access to ambulatory care services and private practice settings than those enrolled in fee-for-service (FFS) plans. We evaluated whether MC

enrollment has differential effects on access to ambulatory care services or private practice settings for racial and ethnic minorities.

METHODS: We analyzed 2157 white, 1878 Latino, 507 African-American, and 323 Asian Medicaid beneficiaries ages 18-65 who participated in the 2001 California Health Interview survey, a probability sample of the state population. We evaluated two outcomes: access to ambulatory care services and care setting. Access was defined as having a usual place to go when sick or needing medical advice. Care setting was defined as "safety-net" for those whose usual source was a clinic, health center, or hospital clinic, and "private practice" for those whose usual source was a physician's office or HMO (eg Kaiser). Those reporting the emergency room as their usual source were categorized as having no usual source. We first analyzed the effect of race and ethnicity on having a usual source of care and care setting. We then analyzed how MC (compared to FFS) enrollment affected these two outcomes after controlling for race and ethnicity. Finally, we analyzed whether the differences in these outcomes between MC and FFS differed by race and ethnicity (difference in the differences). We used multivariate logistic regression for all analyses and controlled for age, gender, and health status (5-point Likert scale, excellent to poor).

RESULTS: Among California's Medicaid beneficiaries, Latinos were less likely to report having a usual source of care than whites (75% vs 88%; OR 0.49, $p < 0.001$), while African-Americans (88%; AOR 1.08, $p = 0.74$) and Asians (86%; AOR 0.99, $p = 0.98$) reported similar rates. Compared to whites, Latinos sought care less often in private practice settings (44% vs. 67%; AOR 0.41, $p < 0.001$); African-Americans sought care equally as often in private practice settings (65%; AOR 0.93, $p = 0.64$); and Asians sought care more often in private practice settings (81%; AOR 3.36, $p < 0.001$). After controlling for race and ethnicity, MC enrollees were significantly more likely than FFS enrollees to report a usual source of care (AOR 2.24, $p < 0.001$) and to seek care in private practice settings (AOR 1.70, $p < 0.001$). The difference in access rates between MC and FFS was statistically different for Latinos compared to whites (=18% vs. 6%; AOR = 0.87; $p < 0.001$), but not for African-Americans or Asians when compared to whites. The difference in care setting between MC and FFS was similar in all racial and ethnic groups, but white MC enrollees remained significantly more likely than Latino MC enrollees to receive their usual care in private settings (73% vs. 49%).

CONCLUSIONS: Regardless of race or ethnicity, Medicaid MC enrollment is associated with a greater likelihood of having a usual source of care and of seeking care in private practice settings. While Latinos in Medicaid are less likely than whites to report a usual source of care, this disparity is erased in Medicaid MC. Medicaid MC facilitated the mainstreaming of Medicaid beneficiaries into private practice settings for all racial and ethnic groups, but did not eliminate disparities in Latinos' use of private care settings compared to whites.

ACCURACY OF PHYSICIAN SELF-REPORT OF SPANISH LANGUAGE PROFICIENCY. A. Rosenthal¹; F. Wang²; E.J. Perez-Stable¹; D. Schillinger¹; A. Fernandez¹. ¹University of California, San Francisco, San Francisco, CA; ²University of California, San Francisco, Orinda, CA. (Tracking ID #136010)

BACKGROUND: As health systems strive to meet the needs of linguistically diverse patient populations, determining a physician's non-English language proficiency is becoming increasingly important. However, brief, validated measures of physician language proficiency are lacking. We compared physician self-report of Spanish language fluency with patients' reports of physician proficiency to determine if any of four self-reported measures of physician Spanish language proficiency are useful measures of fluency in Spanish.

METHODS: We compared physician self-report of Spanish proficiency to Spanish-speaking patients' report of their physicians' language proficiency. Study participants were 110 Spanish-speaking patients and 46 physicians in public hospital clinics with professional interpreters available. Physicians rated their Spanish fluency with four items: one general fluency question, two clinically specific questions (confidence conducting a history and physical, and confidence conducting a sensitive discussion), and one question on interpreter use. Patients were asked if their doctor speaks Spanish ("yes/no"). Concordance, sensitivity, specificity, and positive and negative predictive values (PPV, NPV) were calculated for each of the items, and receiver operating (ROC) curves were used to compare performance characteristics. We dichotomized physician responses at two different points so as to compare the effect of grouping "fair" with "excellent/good" speakers versus with "poor/none" for the fluency question; and grouping "somewhat confident" with "extremely/very" versus with "not confident" for the two clinically specific questions. Finally, to distinguish patient assessment of physician Spanish proficiency from report of Spanish language use without fluency, or use of interpreter, we compared patient report to post-visit physician reports.

RESULTS: Patients had a mean age of 59 years (SD = 12) and 71% were women. Study physicians (N = 46) were residents and faculty in family medicine (N = 18) or internal medicine (N = 28). Most were women (N = 28) and 9 self-identified as Hispanic/Latino. Twenty-six (57%) physicians reported speaking "excellent" or "good" Spanish, while 20 (43%) reported "fair," "poor" or "no" Spanish. Concordance between physician and patient reports of physician Spanish proficiency ranged from 84 to 91%, regardless of how item responses were dichotomized. The PPV for each of the four items ranged from 91-99%, the NPV from 60-90%, and the area under their ROC curves from 90-95%. The general fluency question gave the best combination of PPV and NPV, but physicians who answered "fair" were equally likely to have patients report they did not speak Spanish as that they did. Combining a general fluency question with either of the clinically specific questions did not improve the predictive value of a "fair" response. Comparison of patient report with post-visit physician report indicated good agreement. In 41/46 (89%) encounters, patients and physicians agreed that the physician spoke Spanish during the encounter and the physician reported

not wishing for an interpreter. In 3/46 (6%) encounters the patient reported the physician "spoke Spanish" yet the physician reported use of an interpreter.

CONCLUSIONS: Physician self-report of language proficiency is highly correlated with patient report. While our results are limited by use of a dichotomous "gold standard", they suggest that in settings where no incentives are linked to language skills, simple questions can be used to assess physician language proficiency. These findings may allow Spanish-speaking patients to choose physicians, or by health systems to predict interpreter needs. As the number of LEP patients increases, we need better ways of ensuring physicians and patients can communicate effectively. Measuring the language capabilities of physicians is an important first step in this process.

ACCURATELY MEASURING AFRICAN AMERICANS TRUST IN HEALTH CARE. E. Jacobs¹; I. Rolle²; C. Ferrans²; R. Warnecke². ¹Rush University Medical Center, Chicago, IL; ²University of Illinois at Chicago, Chicago, IL. (Tracking ID #133543)

BACKGROUND: One of the many factors that likely contributes to health disparities among African Americans is distrust of physicians and health care institutions. The purpose of this study was to develop a culturally appropriate measure of general trust in physicians and health care institutions for use in research and clinical practice.

METHODS: The African American Trust in Health Care Scale was developed in a three step process: (1) Initial survey items were generated from focus groups findings (n = 25); (2) Items were vetted, focused, and refined using cognitive interviews (n = 25); (3) The psychometric properties of the resulting instrument were established using data collected from 265 respondents. We conducted a maximum likelihood factor analysis, established the instrument's internal consistency and reliability by calculating Cronbach's alphas, and evaluated its construct validity by measuring the correlation between our instrument and a the Measure of AIDS-related Distrust. Items referred to either physicians or hospitals and clinics in general and respondents used a 5-point likert scale from "never true" to "always true." All research participants were self identified African American adults seeking care at a large public hospital clinic or who were affiliated with a community based organization on the Westside of Chicago.

RESULTS: Eighty-three items were generated from analysis of the focus group data. Thirty-one items remained after culling items determined to be conceptually redundant, unimportant, or poorly worded based on cognitive interviews. The factor analysis revealed 4 factors: (1) Racism in Medicine (example item "The front desk staff often assume black patients don't have insurance"), (2) Physician Distrust (example item "Doctors think of themselves before their patient's well-being"), (3) Physician Trust (example item "Doctors care about their patients as people"), and (4) Respect (example item "Sometimes doctors treat black patients with disrespect"). Seven items did not load on any of the factors and were removed from the final instrument. Internal consistency of the entire scale (alpha = 0.85) and for each factor (0.85, 0.69, 0.74 and 0.85 respectively) was high. Convergent validity of the instrument was supported by its positive correlation with the Measure of AIDS-related Distrust (0.37; $p < 0.001$).

CONCLUSIONS: Based on these findings the African American Trust in Health Care Scale appears to be a culturally appropriate, reliable and valid instrument specifically developed for and validated for use with African Americans. This new instrument promises to contribute to a better understanding of health care disparities by serving as a useful and accurate tool for exploring the relationship between trust and health care in this population.

ADULTS WITH DISABILITIES PRIORITIZE THEIR MEDICAL BENEFIT OPTIONS. M. Ginsburg¹; S.D. Gould²; M. Danis³. ¹Sacramento Healthcare Decisions, Rancho Cordova, CA; ²University of Michigan, Ann Arbor, MI; ³National Institutes of Health, Bethesda, MD. (Tracking ID #134228)

BACKGROUND: Faced with growing budget crises, many states consider reducing Medicaid services. Ascertaining the benefit priorities of Medicaid enrollees could facilitate cuts that are the least disruptive to them. We conducted a study to determine the priorities of disabled adults enrolled in California Medi-Cal.

METHODS: Disabled adults enrolled in California's Medi-Cal Program (n = 131 in 12 groups) were recruited in urban, suburban, and rural locations throughout the state. Most participants were recruited from local Independent Living Centers and represented different age groups, ethnicities, years with Medi-Cal coverage, and types of disabilities. Individual participants and each group designed a package of Medi-Cal benefits using the CHAT exercise, in which a pie chart presented 14 categories of possible services to choose from, each with two or three tiers of benefits. While each category included a benefit tier that reflected current services, other tiers offered lower or higher benefits. One category (Enrollment) presented options to tighten eligibility by adding premium cost-sharing for some beneficiaries. Participants used a fixed number of markers (resources) to select coverage options. The number of markers for each benefit and tier reflected the cost of that service in the Medi-Cal budget, derived from medical claims data provided by the California Department of Health Services. Choosing all current services required 114 markers, choosing all options required 123 markers, while participants had 100 markers to use in designing the benefits. Thus, participants were asked to prioritize services they regarded as most vital. Pre- and post-CHAT survey questions measured demographic characteristics, opinions about coverage options, and assessed their views of the CHAT process. We describe benefit choices of individuals and groups and survey responses, and analyzed verbatim transcripts of group dialogue to learn about the views and values of participants.

RESULTS: Respondents were largely female (69%) and single (82%), about 1/3 were over 50, 42% described their health as fair or poor, and >1/2 had been enrolled in Medi-Cal for over 10 years. Dominant themes in group discussions included the importance of maintaining a full range of Medi-Cal services, and the need for sufficient choice and availability of providers (especially physicians) for beneficiaries to feel secure about health care services and quality. Despite reduced resources, participants opted to increase three categories relative to current services: Doctor Care, Dental Care, and Equipment. Participants most often opted to limit the scope of Medi-Cal relative to current coverage for three categories: Drugs (brand), Enrollment (eligibility), and Personal Care. Most (84%) were very or somewhat satisfied with their group's decision, and 91% agreed "If Medi-Cal coverage is reduced, it is important for Medi-Cal recipients to have a role in deciding what is covered."

CONCLUSIONS: As state legislators and agency personnel consider redesigning Medi-Cal in ways that are both fiscally and ethically responsible, the central issues and concerns of disabled Medi-Cal beneficiaries should be taken into account. This means that maintaining the full range of Medi-Cal services should be a priority, even if this necessitates greater restrictions in how services are used.

ALCOHOL AND DRUG USE DISORDERS AMONG PATIENTS WITH MYOCARDIAL INFARCTION: ASSOCIATIONS WITH ACCESS TO CARE AND MORTALITY. C.A. Beck¹; D.A. Southern¹; R. Saitz²; M.L. Knudsen¹; W.A. Ghali¹. ¹University of Calgary, Calgary, Alberta; ²Boston University, Boston, MA. (Tracking ID #135808)

BACKGROUND: Alcohol- and drug use disorders can influence other health outcomes through a variety of mechanisms, including suboptimal quality of care. In particular, one study reported that for those over age 65 in the United States, substance use disorders were associated with reduced access to cardiovascular procedures following acute myocardial infarction (AMI). However, it is not known whether this holds for a younger age group or in other health care environments. Our primary objective was to compare rates of catheterization, revascularization, and in-hospital mortality post-AMI among adults with and without a substance-related disorder, adjusting for other potential explanatory variables.

METHODS: This cohort study employed hospital discharge data from Calgary, Alberta to identify all adult AMI admissions (ICD-9-CM code 410) between April 1, 1996 and December 31, 2001. Each patient was classified as having alcohol and/or drug abuse or dependence (not nicotine) using an intentionally broad ICD-9-CM coding definition constructed by expert consensus. Information on catheterization and revascularization was obtained by linkage with a clinically detailed cardiac registry containing data on all catheterizations and subsequent revascularizations. Bivariate and multivariable analyses (controlling for age, sex, smoking, comorbidity, and disease severity) were used to compare patients with and without substance use disorders for occurrence of catheterization and in-hospital mortality post-AMI, as well as revascularization at one year post-AMI.

RESULTS: Of 7,876 admissions for AMI, 203 (2.6%) had a substance use disorder. Those with substance use disorders were significantly more likely to be male and under age 65, to smoke, and to have several relevant comorbidities. Bivariate analysis demonstrated no significant difference between those with and without substance use disorders in catheterization (52.7% vs. 58.0% respectively, $p=.13$) or mortality (8.9% vs. 7.1%, $p=.34$). However, in adjusted analyses the odds of mortality were significantly higher for those with substance use disorders, while the odds of catheterization were lower but of borderline statistical significance (see Table). Among the subset who underwent catheterization (73 with substance use disorders, 3,375 without), the adjusted hazard ratio for one-year revascularization was 0.85 (95%CI: 0.65-1.11) for those with a substance use disorder compared to those without.

Table: Odds ratios for in-hospital mortality and catheterization for those with a substance use disorder compared to those without

	Unadjusted OR (95% CI)	Adjusted OR* (95% CI)
Mortality	1.27 (0.78, 2.07)	2.02 (1.10, 3.69)
Catheterization	0.81 (0.61, 1.07)	0.75 (0.55, 1.01)

* Adjusted for age, sex, and relevant comorbidities

CONCLUSIONS: Alcohol and drug disorders are associated with significantly higher odds of in-hospital mortality following AMI, and may be associated with decreased access to catheterization and subsequent revascularization. Higher mortality in the face of poorer access to procedures suggests that these vulnerable individuals are under-treated following AMI, and that targeted efforts may be required to enhance the quality of care.

ARE LATINOS AND ASIAN AMERICANS/PACIFIC ISLANDERS USING FEWER MENTAL HEALTH CARE SERVICES? FINDINGS FROM THE CALIFORNIA HEALTH INTERVIEW SURVEY. S.A. Mohanty¹; M. Alegria²; A.S. Lau³; S. Asch⁴. ¹University of Southern California, Los Angeles, CA; ²Cambridge Health Alliance, Cambridge, MA; ³University of California, Los Angeles, Los Angeles, CA; ⁴VA Greater Los Angeles Healthcare System, UCLA and RAND, Los Angeles, CA. (Tracking ID #135521)

BACKGROUND: Mental illness is the most prevalent health problem in America. Depression, in particular, is a major public health problem that calls for awareness on the part of physicians and health care policymakers alike. The

Supplement to the Surgeon General's Report on Mental Health documents that race, ethnicity and culture are linked to the use of mental health (MH) services and the receipt of quality of MH care. California is unique given that it has one of the highest percentages of both Latino and Asian American/Pacific Islander (AA/PI) ethnic minorities. Given the paucity of population-based data on the rates of MH care by ethnic minorities in California, we sought to investigate depressed patients and the impact of ethnicity on utilization of MH services.

METHODS: We used the California Health Interview Survey (CHIS) adult sample of 55,248 subjects representing the state's non-institutionalized household population. We looked at the following outcomes among those who screened positive for a major depressive episode in the past 4 weeks: 1) use of a psychiatrist, psychologist, social worker, or counselor for emotional or MH problems in the past year; 2) use of prescription medications for MH in the past year; 3) whether a primary care professional asked the person about emotional or MH problems. We examined the impact of race/ethnicity on receipt of MH services among these depressed patients. We also looked at important potential predictors of use of MH care such as income, insurance status, education status, language proficiency, employment status, immigrant status, and citizenship status, and usual source of care using multivariate logistic regression analysis.

RESULTS: Approximately 4% of the California population had a positive depression screen in the past 4 weeks. Compared to non-Latino Whites (NLW), Latinos and AA/PIs with positive depression screen had much lower rates of being asked about MH by a primary care provider (NLW 59.0% vs Latino 27.4%, AA/PI 16.2%). Thirty-four percent of depressed NLW had seen a MH specialist. In contrast, only 9.7% of Latinos and 7.5% of AA/PIs had seen a specialist ($p<0.0001$ NLW vs. Latinos and AA/PIs). Sixty-one percent of NLW depressed patients received a prescription vs. 28% for Latinos and 20% for AA/PIs ($p<0.0001$). When adjusted for age and gender, Latinos and AA/PIs with a positive depression screen still had much lower odds of seeing a specialist compared to NLW (Lati-nos: Adjusted OR (AOR) 0.19, 95% CI 0.12, 0.28; AA/PI: AOR 0.15, 95% C.I. 0.08, 0.29). Similarly, Latinos and AA/PIs who screened positive were much less likely to receive a medication for a MH problem than NLW (Latinos: AOR 0.28, 95% C.I. 0.20, 0.40; AA/PIs: AOR 0.16, 95% C.I. 0.08, 0.33). We also found that depressed Latinos and AA/PIs were less likely to receive these MH services even after controlling for such factors as income, insurance status, education, and language proficiency.

CONCLUSIONS: There remain ethnic disparities in utilization of MH care independent of such factors as income, insurance status, education, and language proficiency. These disparities in use of MH services suggest a strong need for public policy, community action, and culturally sensitive outreach and education in California. Policies should be adopted that allocate resources to new, innovative strategies for MH care or insure expansion of existing programs proven to be effective in providing MH services to the Latino and AA/PI community.

ARE LIPIDS MANAGED DIFFERENTLY IN WOMEN AND MEN? AN EXAMINATION OF VETERANS AND MANAGED CARE ENROLLEES WITH DIABETES. C. Kim¹; E.A. Kerr²; S.J. Bernstein²; S.L. Krein². ¹University of Michigan, Ann Arbor, MI; ²VA Ann Arbor Healthcare System, Ann Arbor, MI. (Tracking ID #130880)

BACKGROUND: Several studies suggest that men at high risk for cardiovascular events have low-density lipoprotein (LDL) cholesterol managed more aggressively than women at high-risk. Our objective was to examine gender differences in lipid management using several types of quality measures, before and after adjustment for covariates.

METHODS: We conducted an observational cohort study among patients with diabetes receiving care in the Veterans Health Administration ($n=26,712$) in a midwestern geographic region or in a single managed care plan ($n=2,589$) in 2000-2001. Using automated data, we compared men and women within each healthcare organization using 4 measures of lipid management: 1) measurement of LDL, 2) among those with an LDL, an LDL $>=130$ mg/dl, 3) an LDL $>=130$ mg/dl without high-dose statin treatment, and 4) an LDL $>=130$ mg/dl without appropriate monitoring or adjustment ("inadequate management"). Inadequate management was defined using the following criteria, in this order: 1) no high dose statin within 100 days prior to the index LDL OR 2) no statin started within 6 months of the index LDL if no statin was already prescribed OR 3) no increase in statin dosage within 6 months if statin was already prescribed OR 4) no repeat LDL <130 mg/dl within 6 months of the index LDL. Comparisons were adjusted for age, insulin use, number of visits, hypertension, ischemic heart disease, congestive heart failure, stroke, and clustering of patients within site.

RESULTS: Among veterans, men and women had similar odds of having their lipids measured. More women had LDL $>=130$ mg/dl than men (32% vs. 25%, $p<0.0001$), but were less likely to receive a high-dose statin (14% vs. 20%, $p<0.0001$) before and after adjustment for covariates. Before adjustment, women veterans were more likely to receive "inadequate management," but this result did not persist after adjustment for covariates. In managed care, women were less likely to have their lipids measured (24% vs. 32%, $p<0.0001$) and reported higher rates of LDL $>=130$ mg/dl (26% vs. 22%, $p=0.047$), but once measured, women were as likely as men to be prescribed a high-dose statin and to receive adequate management, before and after adjustment for covariates.

CONCLUSIONS: Women with diabetes have higher lipid levels than men with diabetes in both the VA and managed care. Women veterans' lower receipt of adequate management may be related to gender differences in covariates for CVD risk. In managed care, women's lower rate of lipid measurement is not explained by gender differences in covariates for CVD risk. Quality initiatives in the VA should target advancement of therapy in all patients with diabetes, while managed care initiatives should focus on initial measurement of lipids in patients with diabetes.

ARE MEDICAL STUDENTS PREPARED TO CARE FOR LESBIAN, GAY, BISEXUAL, AND TRANSGENDERED PATIENTS?: A CURRICULUM NEEDS ASSESSMENT. N. Sanchez¹; J.S. Rabatin¹; J.P. Sanchez²; S. Hubbard¹; A.L. Kalet¹. ¹New York University, New York, NY; ²Albert Einstein College of Medicine, Bronx, NY. (Tracking ID #136090)

BACKGROUND: The literature suggests that medical schools inadequately address the health care issues specific to lesbian, gay, bisexual and transgender (LGBT) people who make up about 2-10% of the U.S. population. To inform curriculum development we assessed medical students' ability to identify and care for patients at one medical school in New York City.

METHODS: Between March 1 and April 15, 2004, third and fourth year medical students were sent an e-mail to participate in a confidential online survey of 64 quantitative questions designed to assess their attitudes about, knowledge of, and clinical skills needed for the health care of LGBT patients.

RESULTS: Two hundred and forty eight students responded (248/320, 77.5%). Medical students who reported frequent sexual history taking ($F=12.5, p<0.001$), held more positive attitude scores ($F=5.53, p<0.001$), and possessed higher knowledge scores ($F=3.57, p<0.015$) were more likely to inquire about sexual orientation, sexual activities, and family structure and perform a more comprehensive history and physical examination on LGBT individuals. Of the 248 respondents, 25.6% of respondents screen for same sex activity "rarely" or "never" when taking a sexual history. 53.6% "rarely" or "never" unveil a patient's sexual orientation. Overall, on the 13-item attitude survey the mean was 4.15 (5=most positive, SD 0.55, range 1.86-5.00), indicating a strong desire and willingness to provide health care to LGBT patients. The mean score on the 14-item knowledge test was 60 (SD 0.12) correct, with greatest deficiencies found in the areas of cancer risk, mental health, and nutrition.

CONCLUSIONS: Medical students at a New York City medical school serving a large LGBT population screen for LGBT patients when they master the communication skills to elicit a comprehensive history, hold positive attitudes toward LGBT patients, and possess knowledge of LGBT health care concerns. Medical students desire additional knowledge and clinical skills training, in the form of course lectures, clinical vignettes, and direct patient encounters, to better serve LGBT clients.

ARE REVERSE DISPARITIES LIMITED TO AGE-DEFINED SUBGROUPS OF THE VA PATIENT POPULATION? K. Volpp¹; R.A. Stone²; A. Jha³; J.R. Lave²; M.V. Pauly⁴; H. Kulusariz¹; H. Chen²; D. Polsky¹. ¹CHERP - Philadelphia VA Medical Center, Philadelphia, PA; ²CHERP, Pittsburgh VA Medical Center, Pittsburgh, PA; ³Boston VA Medical Center, Boston, MA; ⁴The Wharton School, University of Pennsylvania, Philadelphia, PA. (Tracking ID #135807)

BACKGROUND: Several studies have shown that Blacks hospitalized within the VA system have lower 30 day mortality than Whites. We undertook this study to examine whether these differences exist in six conditions for which hospital admission is generally not discretionary, whether such differences hold true for both patients over and under 65, and whether observed differences are a function of area-level socioeconomic characteristics.

METHODS: We conducted an observational study using data in the National Patient Care Database on 340,982 Black or White patients hospitalized with one of the following conditions at VA hospitals nationwide between fiscal years 1996 and 2002: acute myocardial infarction (AMI), congestive heart failure (CHF), gastrointestinal bleeding (GI bleed), hip fracture, pneumonia, or stroke. We used random effects logistic regression to assess differences in 30-day mortality between Black and White patients hospitalized with each of these conditions, with hospital as a random effect and covariate adjustment for age, gender, fiscal year, and 30 comorbidities. We estimated the models separately for patients over and under 65, with and without adjustment for area-level socioeconomic characteristics. We assessed whether the magnitude of the observed disparity differed by age group by examining an age by race interaction in a combined model.

RESULTS: For each of these six conditions, Black patients over age 65 had significantly lower mortality than for Whites, with an estimated odds ratio (OR) of 30-day mortality of 0.76 (95% CI: 0.68-0.85) for AMI, OR=0.70 (95% CI: 0.65-0.76) for CHF, OR=0.88 (95% CI: 0.79-0.99) for GI bleed, OR=0.67 (95% CI: 0.53-0.85) for hip fracture, OR=0.89 (95% CI: 0.84-0.95) for pneumonia, and OR=0.82 (95% CI: 0.75-0.91) for stroke. However, among patients under age 65, mortality for Black patients was not lower for 4 of these conditions: OR 1.14 (95% CI 0.95-1.37) for AMI, OR 1.0 (95% CI 0.85-1.18) for GI bleed, OR=1.10, 95% CI 0.99-1.21 for pneumonia, and OR 1.09 (95% CI 0.93-1.28) for stroke. Mortality was lower for Blacks under age 65 for CHF, OR=0.71 (95% CI 0.62-0.81) and not significantly lower for hip fracture, OR=0.50 (95% CI 0.21-1.15). The Black ORs were significantly lower for over age 65 compared to under age 65 patients hospitalized with AMI, stroke, and pneumonia (p-values < .01). Adjustments for area-level socioeconomic factors and individual income did not qualitatively affect the differences in the degree of disparity observed over and under age 65; neighborhood characteristics were similar in the older and younger populations.

CONCLUSIONS: Although the VA provides an equal access system, the reverse disparity in 30-day mortality occurs within the over-65 population for all six conditions and within the under-65 population only for CHF. Relative to comparably aged Whites, Blacks over age 65 fare relatively better than Blacks under age 65, although the reasons for this are not clear.

ASSOCIATION OF AGE AND RACE WITH ANALGESIC PRESCRIBING IN OUTPATIENT SETTINGS: ANALYSIS OF NATIONALLY REPRESENTATIVE DATA. A.D. Federman¹; A. Litke¹; R.S. Morrison¹. ¹Mount Sinai School of Medicine, New York, NY. (Tracking ID #133098)

BACKGROUND: Local and regional studies suggest that elderly and minority patients experience disproportionate under treatment of pain. Less is known

about national patterns of analgesic use in outpatient settings. We undertook the current study to examine the prevalence of analgesic use for painful conditions among adults in outpatient settings in the U.S., and to determine how age and race affect these rates.

METHODS: We conducted cross-sectional analyses using 1999-2002 data from the National Ambulatory Medical Care Survey (NAMCS) and the National Hospital Ambulatory Medical Care Survey (NHAMCS), which are nationally representative, cross-sectional chart abstractions of outpatient visits. The data include up to 3 reasons for the patient's visit, 3 provider-assigned diagnoses, and up to 6 medications provided, ordered or continued. We studied adults (≥ 18) with a complaint of pain and a diagnosis of cancer or inflammatory or mechanical disorders of the joints and spine. We excluded emergency department patients. Analgesics were assigned as: none, acetaminophen (APAP) or non-steroidal anti-inflammatory drugs (NSAID), or opiates. We estimated the association of age and race with NSAID/APAP or opiate use in separate logistic regression models, adjusting for demographics and health status, physician and practice types, and the survey's multilevel sampling design. Analyses were conducted separately for joint/spine disorders and cancer. Due to multiple comparisons, we defined statistical significance as $p<.005$ and calculated 99.5% confidence intervals for the odds ratios.

RESULTS: 7,673 outpatient visits, representing 142 million visits over 4 years, occurred for adults with pain and spine/joint disorders (95%) or cancer (5%). For spine/joint disorders, non-inflammatory conditions of the spine, including degenerative disk disease and sciatica, occurred in 58% of visits. Osteoarthritis occurred in 23%. The most common cancer diagnoses were breast (30%) and prostate (11%). Among patients with spine/joint disorders, 33% were >65 , 62% were women, 87% were white, and 59% of visits were for an acute problem. Only 43% of patients received pain medications: APAP, 1%; NSAIDs, 25%; opiates, 17%. NSAID/APAP use increased with age: age ≥ 75 , 32% vs. age 18-54, 21% ($p=.002$). In contrast, opiate prescribing decreased with age ($p<.001$), and prescribing rates for patients ≥ 75 were nearly half those of patients 18-54 (12% vs. 21%, $p<.001$). In multivariate analyses, patients ≥ 75 were less likely than those 18-54 to use opiates (adjusted OR 0.46, 99.5% CI 0.23 to 0.93), whereas NSAID/APAP use was higher (1.59, 1.00 to 2.53). A similar effect was observed when APAP use was removed from the model. Race had no significant association with NSAID/APAP and opiate prescribing. No significant regional variation in prescribing was observed. Among patients with cancer neither age nor race was significantly associated with NSAID/APAP and opiate use.

CONCLUSIONS: Analgesics are often not prescribed to patients with painful conditions in outpatient settings. Elderly patients, in particular, are less likely to use opiates, the most potent analgesics. Yet, they are more likely to use NSAIDs, a class of medications that presents a greater risk of potentially life threatening complications for older patients. Future research should address physicians' reluctance to use opiate analgesics in the elderly and develop interventions and policy to promote optimal use of opiates and NSAIDs in outpatient settings.

ASSOCIATION OF PHYSICIAN CULTURAL COMPETENCE AND LIFE EXPERIENCES WITH PREVENTION AND SCREENING PRACTICES. J. Bussey-Jones¹; Y. Diaz²; M.A. Shaw³; E. Liles⁴. ¹Emory University, Atlanta, GA; ²University of Miami, Miami, FL; ³University of Louisville, Louisville, KY; ⁴University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #134376)

BACKGROUND: Cultural competence training has been proposed as a possible means to improve disparities and clinical outcomes. Other work by our group suggests an association between diverse life experiences and cultural competence regardless of prior training. Little is known, however, about the relationship of physicians' cultural competence and diverse life experiences to clinical care. The purpose of this study is to examine the association between physician cultural competence and prevention and screening practices.

METHODS: The CCTOP study (Cultural Competency Training and Outcomes in Patients) is a multicenter, cross-sectional study of patients presenting for care at 11 geographically diverse outpatient clinics and their physicians. Physician cultural competence was scored using the Inventory for Assessing the Process of Cultural Competence Among Healthcare Professionals (IAPCC), a validated scale. (Campinha-Bacote, 1999). Physicians were also asked about life experiences (including socializing and living in culturally and ethnically diverse environments), cultural knowledge, and demographics. Patients were verbally administered a 76-item survey regarding their health and demographics. We examined the association between the IAPCC score, physician cultural knowledge, and diversity experiences with patient-reported preventive health measures such as counseling behaviors (e.g. clinically indicated tobacco, alcohol and hypertension education), diabetic eye examinations, and vaccinations. We also examined cholesterol, mammogram, and colon cancer screening. Multivariate regression analyses were performed controlling for age, gender, race, education and income.

RESULTS: We present preliminary results from 236 matched patient-physician surveys at six study sites. 291 patients responded (203 women; 88 men) with a mean age of 56 years. One-hundred and seventy five physicians (118 residents, 57 attendings) participated in the study. No association was found between eye examinations among diabetics, cholesterol, mammogram, and colon cancer screening ($p>0.05$) with physicians cultural knowledge, IAPCC score, nor diverse life experiences. Similarly, no association existed between physician counseling behaviors (smoking cessation, hypertension education, pneumococcal vaccination) with the aforementioned cultural competence surrogates ($p>0.05$).

CONCLUSIONS: Physician cultural competence, diverse life experiences, attitudes and cultural knowledge appear to have no association with prevention and screening practices. While further work is needed to determine the optimal strategies to diminish health disparities, these data suggest that physicians regardless of cultural experiences, knowledge, and attitudes have similar practice

patterns and may equally benefit from interventions that translate cultural competence to clinical practice.

ASTHMA OUTCOMES AND PATIENT LANGUAGE IN AN URBAN, MINORITY COHORT. L. Diamond¹; O. Susan¹; A.R. Montero¹. ¹Columbia University, New York, NY. (Tracking ID #136185)

BACKGROUND: African-Americans suffer disproportionate morbidity and mortality from asthma. African-Americans are more likely to be hospitalized for asthma and 4 to 6 times more likely to die from asthma when compared to whites. Asthma outcomes among Hispanics are less well characterized but also appear to be worse when compared to whites. We carried out this study to evaluate what impact possible language barriers might have on asthma outcomes in an urban cohort of Hispanic and African-American patients.

METHODS: Subjects were drawn from a larger study of 1,157 patients randomly selected from an urban, general medicine group practice. Our retrospective cohort analysis linked responses from that study's questionnaire with data from our practice's clinical information system. Our sample consisted of the 158 patients in the parent study who were asthmatic (based on ICD-9 code or discharge DRG codes) and either Hispanic or African-American. Our independent variable was preferred patient language: Spanish-speakers (n=120) and English-speakers (n=38). Our outcome variable was ER visit or hospitalization for asthma within 2 years of the study visit. We used logistic regression to adjust for those covariates which in bivariate analysis were correlated with ER visit/hospital admit.

RESULTS: We found that 47% of our English-speaking cohort had an ER visit or admission for asthma during follow-up compared with 28% in our Spanish-speaking cohort ($p < 0.01$). Other correlates of ER visit/hospital admit for asthma ($p < 0.10$) included presence of Medicaid insurance and younger age. In multivariate models, English-speaking remained independently associated with likelihood of ER visit/hospital admit for asthma (OR=2.89; $p < 0.02$).

CONCLUSIONS: In our sample of urban, minority patients we found an independent, positive association between English-speaking and likelihood for ER visit/hospital admit for asthma. Since all English-speaking patients were African-American and all Spanish-speaking patients were Hispanic, this suggests that despite potential language barriers, Hispanic patients are achieving better asthma outcomes when compared to African-Americans in our setting. However, the high absolute proportion of ER/Hospital admit for both groups suggests that both are still receiving suboptimal asthma care.

ATTITUDES, SKILLS AND THE NEED FOR CULTURAL COMPETENCE - WHAT DO INTERNAL MEDICINE RESIDENTS THINK ? O. Quirindongo-Cedeno¹; K.G. Thomas¹; J.M. Bowen¹. ¹Mayo Clinic, Rochester, MN. (Tracking ID #136019)

BACKGROUND: Achieving cultural competency in learners is an expectation for all residency programs. As stated in the Accreditation Council for Graduate Medical Education (ACGME) Professionalism Competency, "Residents are expected to demonstrate sensitivity and responsiveness to patients' culture, age, gender, and disabilities." In response to this expectation, residency programs are developing and implementing curricula to address the understanding, incorporation, and application of social and cultural factors in health, medicine, and patient care. This study aims to determine Internal Medicine Residents' attitudes and perceived skills regarding cultural competency in preparation for the development of an integrated longitudinal cultural competency curriculum in our Internal Medicine Residency Program.

METHODS: A sixty-three-item web-based cultural competency survey, modeled after previously studied instruments, was developed and administered to PGY-1, 2, 3 and preliminary residents in our Internal Medicine Residency Program. Residents were asked to rate the relevance of sociocultural issues in clinical practice and to rate their competence in effectively managing these sociocultural issues. Frequency of use of cross-cultural communication techniques and available educational resources were also assessed. Finally, residents were asked to identify preferences for specific learning formats for cultural competency education. Survey responses were collected using a five-point Likert-type scale.

RESULTS: The survey was distributed electronically to 170 residents. Sixty-two residents (36.4% response rate) completed the survey. Residents rated ethnic/racial factors (Mean 3.85, SD=1.02), sociocultural factors (Mean 3.88, SD=0.97), and socioeconomic factors (Mean 3.67 SD=1.1) as somewhat relevant in their clinical practice. Residents rated sexual preference factors as the least relevant to their clinical practice (Mean 2.75, SD=1.3). Overall, residents perceived themselves to be at least "a little competent" (Range 3.08-3.93) for all aspects of cross-cultural communication described in the survey, except as related to managing sexual (Mean 2.8, SD=0.90) and mental health issues (Mean 2.98, SD=0.85) with patients of different cultures. Residents rated small group discussion (Mean 3.5, SD=0.98) and one-to-one observation (Mean 3.48, SD=1.18) as the preferred learning formats for cultural competency education. Web based learning was rated as the least desirable (Mean 2.24, SD=1.00) learning format.

CONCLUSIONS: These survey results provide baseline information for the development and implementation of cultural competency curricula in Internal Medicine Residency Programs. Guided by these results, we suggest that cultural competency curricula should integrate multiple sociocultural issues, with special attention to topics related to sexuality and mental health. Consideration should be given to incorporating small-group discussion and one-to-one observations with feedback into cultural education initiatives. Future directions include rigorous outcome assessment to evaluate the impact of cultural competency curricula.

CAM USE IN A RACIALLY-BALANCED, INDIGENT INTERNAL MEDICINE CLINIC IN A SMALL URBAN AREA. A.R. Sarma¹; C. Collins¹; D.C. Parish¹; D. Mathis²; B.E. Smith¹; R. Devereaux¹. ¹Mercer University, Macon, GA; ²Mercer University, Georgia, Macon, GA. (Tracking ID #134616)

BACKGROUND: Complementary and Alternative Medicine (CAM) use is common in the United States; however, research has been limited to predominantly Caucasian populations in the large cities. Very few studies have investigated the use of CAM in the southern United States, and none of these have been properly randomized. We present our investigation of CAM use in a Southeastern small-urban, indigent population.

METHODS: We conducted a prevalence study employing a randomized sample of all patients visiting an Internal Medicine Clinic during the month of September 2004. An investigator verbally administered a survey with each individual in order to prevent bias with literacy, with a acceptance rate of 98%. Prevalence, disclosure of alternative therapies to physicians, as well as satisfaction and reason for use were outcome measures.

RESULTS: Overall, our patient population (N=198; ages 19-87; 83 males; 115 females; 85 Caucasian; 112 African-American; 1 Asian) most commonly used prayer as a form of CAM (81% in last 12 mos., 80% lifetime). Megadose vitamins (40% in last 12 mos.; 51% lifetime), herbal supplements (25% in last 12 mos., 30% lifetime), and mental healing (26% in last 12 mos., 26% lifetime) were the next most commonly used. 75% of participants stated they did not disclose their use of CAM to their physician and 84% reported that a physician had never asked them about CAM use. However, most agreed they would feel comfortable discussing this topic with their physicians. 80% of participants stated these therapies worked at least sometimes. Overall, the participants mentioned that their main reason for CAM use was heritage/cultural beliefs and practices inherited from family/friends. Overall, in this study population, African-Americans used more types of CAM than Caucasians over their lifetime. The substantial interracial differences were prayer (85% vs. 75%) and folk remedies (66% vs. 45%).

CONCLUSIONS: This small-urban, indigent, biethnic population has higher CAM use, especially prayer and folk remedies, than noted in previous studies. Factors favoring reliability of these findings include: 1) a randomized study design 2) removal of literacy bias by interview 3) 98% acceptance 4) and explanation of multiple CAM synonyms to enhance interviewee comprehension. This study reconfirms that physicians are rarely aware of the type or extent of CAM use among their patients; therefore, medical practitioners can miss dangerous drug-herb interactions. In addition, significant portions of patients are going outside of mainstream medicine for treatments based on their deeply rooted cultural/familial belief system; further studies are needed to better understand these belief systems and how they may be harmonized with the current biomedical paradigm.

CAN PATIENT-CENTERED ATTITUDES REDUCE RACIAL/ETHNIC DISPARITIES IN CARE? M.C. Beach¹; M. Rosner¹; L.A. Cooper¹; P.S. Duggan¹; J. Shatzer¹. ¹Johns Hopkins University, Baltimore, MD. (Tracking ID #134336)

BACKGROUND: Patient-centeredness has been advocated to reduce racial/ethnic disparities in health care quality, but there are no empirical data. The purpose of our study was to determine whether students with patient-centered (PC) attitudes were less likely to demonstrate disparities in clinical performance with African-American and white standardized patients (SPs).

METHODS: We trained one African-American (AA) and one white SP to each of four case scenarios (back pain, abdominal pain, chest pain and fatigue) used during exams for third-year medical students from two mid-Atlantic medical schools. Each of 177 students completed all four case scenarios. For each case, students were randomly assigned to interact with the AA or white SP. Just prior to the exam, students were surveyed about their attitudes towards patient-centered medicine. In analyses, students were divided at the median attitude score to compare performance with AA and white SPs for students with and without PC attitudes. Main outcome measures were student performance scores in interpersonal skill, history-taking, physical exam, and counseling.

RESULTS: Students were white (47%), African American (37%), Asian (13%) and Hispanic (3%). Most students were male (55%) and were between 21-30 years of age (86%). When interacting with white SPs, students with PC attitudes performed similarly to students without PC attitudes. With AA SPs, students with PC attitudes performed significantly better than students without PC attitudes in counseling (92.1 vs. 88.5, $p=0.024$) and tended toward better performance in interpersonal skill (82.1 vs. 79.9, $p=0.063$), history-taking (65.0 vs. 61.8, $p=0.079$), and physical exam performance (73.6 vs. 68.6, $p=0.079$). Students with PC attitudes scored lower in interpersonal skills for AA compared to white SPs (82.1 vs. 89.2, $p < 0.000$), but did not demonstrate significant disparities in performance of history (65.0 vs. 66.3, $p=0.492$), physical exam (73.6 vs. 76.8, $p=0.207$) or counseling (92.1 vs. 92.3, $p=0.894$). By contrast, students without PC attitudes demonstrated significant disparities in several skill areas by performing more poorly with AA compared to white SPs in interpersonal skill (79.9 vs. 88.6, $p < 0.000$), physical exam performance (68.6 vs. 76.4, $p=0.006$), and counseling (88.5 vs. 92.1, $p=0.020$) and had a trend towards poorer performance on history-taking (61.8 vs. 65.4, $p=0.079$). These findings persisted after controlling for student race/ethnicity.

CONCLUSIONS: Patient-centered attitudes may be more important in predicting physician behaviors with African American patients than with white patients, and may play a role in reducing disparities.

CERVICAL CANCER SCREENING AMONG HOMELESS WOMEN. M. Bharel¹; O. Blackstock²; J. O'Connell³. ¹Massachusetts General Hospital/Harvard Medical School, Jamaica Plain, MA; ²Harvard Medical School, Boston, MA; ³Massachusetts General Hospital/Harvard Medical School, Boston, MA. (Tracking ID #135942)

BACKGROUND: The Pap smear is an effective screening test for cervical cancer. Low socioeconomic status is associated with lower rates of Pap smear screening and higher rates of mortality from cervical cancer. Few studies have addressed cancer screening practices among homeless women, who carry a disproportionate burden of disease and a high mortality rate. We sought to investigate the prevalence of risk factors for cervical cancer and Pap smear status among a cohort of homeless women.

METHODS: Eighty consecutive women admitted to a 90-bed inpatient medical respite unit for homeless individuals were entered into the study. Each woman was asked to complete a questionnaire regarding previous Pap smear history and risk factors for cervical cancer. Pap screening was offered to women eligible according to national guidelines.

RESULTS: Sixty-six of eighty women agreed to participate and completed a questionnaire. Risk factors for cervical cancer were common in this cohort: 80.3% were smokers, 10.6% were HIV positive, 43.9% had a history of a sexually transmitted disease, and 10.6% had a family history of cervical cancer. Despite these risk factors, 35 women (53.0%) had not undergone Pap smear screening in the previous year. Of these 35 women eligible for Pap smear, 23 (65.7%) refused screening. The remaining thirteen consented to a pap smear and none was found to be abnormal. There were no significant differences in the characteristics of women according to screening status.

CONCLUSIONS: Over 50% of homeless women in this high risk cohort were in need of Pap smear screening at the time of interview. Given the well documented obstacles that homeless individuals face in accessing care, physicians should consider inquiring about Pap smears for homeless women seen in any medical arena, not just primary care settings. Future research should focus on the reasons for inadequate cervical cancer screening as well as reasons for the high observed refusal rate for screening.

COMPARISON OF CONSTRUCT VALIDITY OF THE PATIENT HEALTH QUESTIONNAIRE DEPRESSION MODULE (PHQ-9) IN AFRICAN-AMERICAN, LATINO AND NON-HISPANIC WHITE PRIMARY CARE PATIENTS. F.Y. Huang¹; H. Chung²; K. Kroenke³; R.L. Spitzer⁴. ¹Rutgers, The State University of New Jersey, New Brunswick, NJ; ²Charles B. Wang Community Health Center, New York, NY; ³Indiana University Purdue University Indianapolis, Greenwood, IN; ⁴New York State Psychiatric Institute, New York, NY. (Tracking ID #134219)

BACKGROUND: In 2003, the United States Preventive Services Taskforce recommended screening for depression in medical practices that are prepared to deliver guideline-based care. A self-administered version of the depression module of the PRIME-MD, the Patient Health Questionnaire-9 (PHQ-9), has become widely accepted as a screening and diagnostic depression measure in primary care settings. No studies, however, have analyzed the function and reliability of the PHQ-9 in different racial/ethnic populations despite growing evidence of racial disparities in access to and adequacy of treatment. Therefore, we (1) compared the self-reported severity of depression by PHQ-9 score and (2) analyzed the construct validity of the PHQ-9 by comparing the association of PHQ-9 scores to functional scales between different racial/ethnic groups.

METHODS: Secondary data analyses were performed on the original 6,000 subjects in the PHQ Primary Care Study and PHQ Obstetrics-Gynecology Study. In these two studies, 13.7% (819) of subjects were African-American, 21.7% (1,302) Latino, and 56.6% (3,396) non-Hispanic white. Sociodemographic measures, PHQ-9 scores, and multiple measures of construct validity, including the six subscales of the SF-20 health status measure, disability days and health utilization (i.e., clinic visits in the past 3 months) were collected. Analysis of variance was used to compare mean PHQ-9 scores and chi square compared rates of depression severity between different ethnic groups. Pearson correlation coefficients between PHQ-9 scores and functional scales such as the SF-20 Mental Health were compared using the Z test between African-American, Latino, and white patient populations. Hierarchical linear regression was used to control for age, gender, and language in these analyses.

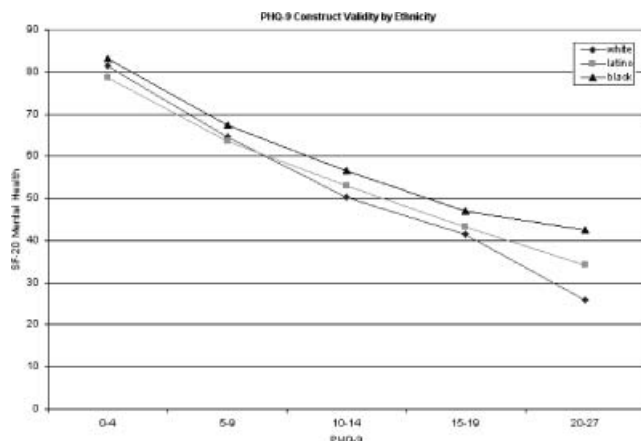


Figure 1

RESULTS: Mean PHQ-9 scores—4.4 for African-American, 4.7 Latino, and 4.8 non-Hispanic white—were not significantly different ($F=1.3$, $p=0.273$). Levels of depression severity were also not significantly different between the three different racial/ethnic populations. Similar strong associations between increasing PHQ-9 and worsening function on the SF-20 Mental Health were seen in all three groups. (Figure 1) Results were similar for the other five SF-20 scales, disability days, and clinic visits. These associations between PHQ-9 scores and functional scores were essentially unchanged after controlling for age, gender and language in these analyses with hierarchical linear regression.

CONCLUSIONS: The PHQ-9 displays similar construct validity across a variety of domains in African-American, Latino and white populations. Racially diverse primary care settings should consider the PHQ-9 a viable depression screening instrument when implementing guidelines based care for major depression.

CULTURAL COMPETENCE AMONG HEALTH PROFESSIONALS: A TAXONOMY. S. Saha¹; E. Morse²; J. Jimison². ¹Portland VA Medical Center, Portland, OR; ²Oregon Health & Science University, Portland, OR. (Tracking ID #136251)

BACKGROUND: Cross-cultural education has been recommended as a vehicle to reduce racial and ethnic disparities in health care. The last few years has seen a proliferation of programs training health professionals in "cultural competence." There is wide variation, however, in the use and conceptualization of this term and in the knowledge, attitudes, and skills thought to comprise cultural competence. To build an evidence base addressing the effectiveness of cultural competence training, including the measurement of training outcomes, it is necessary to develop a taxonomy of constructs representing the broad concept of cultural competence.

METHODS: We conducted a systematic review to identify conceptual models of cultural competence among health professionals. We first searched the MEDLINE, HealthSTAR, and CINAHL databases using strategies tailored according to the search terms available in each database. Two reviewers screened abstracts for articles whose primary focus was to describe models, definitions, dimensions, or components of cultural competence, cultural sensitivity, cultural awareness, cross-cultural health care, or cross-cultural communication among health care providers. A third reviewer adjudicated disagreements. We excluded articles that primarily addressed organizational or institutional cultural competence, as well as articles focusing on specific populations or disease entities. We supplemented our database search by reviewing reference lists from included articles and by searching websites related to cross-cultural health care. Two reviewers qualitatively analyzed each included article or web-based document for distinct constructs included as dimensions of cultural competence. A final taxonomy was established through discussion and consensus between the reviewers.

RESULTS: From 770 abstracts, we identified 26 unique articles describing models of cultural competence. From reference lists and web sites we identified 9 additional articles, monographs, and book chapters. Reviews of these models revealed four principal dimensions of cultural competence, each with two or more subcategories. Sociocultural Awareness describes an understanding of the meanings of race, ethnicity, culture, and social class and their influences on illness, health, and health care. Cultural Sensitivity encapsulates non-ethnocentric attitudes such as acceptance of different cultural perspectives (cultural humility) and willingness to adapt behavior to accommodate different preferences (cultural flexibility). Cross-Cultural Communication encompasses the skills needed to effectively engage and partner with patients from different backgrounds, including assessing beliefs (sociocultural assessment) and negotiating different interpretations of illness and preferences for therapy (cultural negotiation). Patient-Centeredness describes the attitudes and skills associated with valuing patients' perspectives and engaging patients as partners in decision making and management. Additional dimensions not directly relevant to the concept of culture, such as knowledge of epidemiological patterns of disease by race/ethnicity and ethnopharmacology, were included in some models.

CONCLUSIONS: Current models of the concept of cultural competence are highly variable but share several common themes. Developing a taxonomy of these themes is an important first step towards measuring outcomes to assess the effectiveness of different cross-cultural educational programs and the specific content elements they include.

DETERMINANTS OF QUALITY OF DIABETES MANAGEMENT AMONG COMMUNITY HEALTH CENTERS IN THE U.S. L.S. Hicks¹; J. O'Malley²; B.E. Landon¹; T. Lieu³; N. Cook⁴; T. Keegan⁵; B.J. McNeil¹; E. Guadagnoli¹. ¹Harvard University, Boston, MA; ²Harvard Medical School, Boston, MA; ³Harvard Pilgrim Health Care, Boston, MA; ⁴Massachusetts General Hospital, Boston, MA. (Tracking ID #135429)

BACKGROUND: Community health centers (CHCs) are responsible for providing care for many of the over 40 million uninsured Americans, many of whom are members of groups that have been previously documented to receive poorer care. In 2004, President Bush announced plans to expand the number of CHC sites during 2005 to address the health care needs of the uninsured. To date, however, there has been little examination of the quality of care delivered in CHCs or the patient and CHC characteristics that may be associated with quality.

METHODS: We examined medical records of 1,909 patients with diabetes receiving care from January 1, 1999 to August 1, 2001 in 42 publicly funded CHCs from throughout the U.S. and surveyed administrators at the CHCs about

their center's organizational characteristics. We used previously validated quality of care indicators based on national guidelines and tabulated the number of eligible patients meeting each quality indicator during the study period. For each patient, we created a summary scale for quality of care received, standardizing the indicators to have mean 0 and variance 1 and computed the mean of the standardized scores for each patient. We then re-scaled the score so that the mean represents the average percentage of times each indicator was met. Higher scores represent higher quality of care. We then averaged the scores across patients within a center to compute a center level quality score. We compare patient demographic and CHC organizational characteristics by patient and center level scores.

RESULTS: Of the 1,909 patients, 713 (37.3%) were white, 737 (38.6%) were male, and 460 (24.1%) were uninsured. The total percentage of patients meeting each quality indicator is presented in the table below. The mean diabetes quality score for the population was 33.5%. Black and Hispanic patients received significantly poorer quality (scores 29.9% and 29.7%, respectively) compared to whites (score 37.4%) ($P < 0.001$). Uninsured patients received significantly poorer care (score 28%) compared to those with insurance coverage (score 35.3%) ($P < 0.001$). Centers that were <30 years old had better quality (score 36.2%) than centers that had been in existence for 30 or more years (score 30.4%) ($P = 0.1$), centers that had computerized notes or medication lists available had better quality (score 38.4%) than those without computerized notes or medication lists (score 31.9%) ($P = 0.09$), and those centers whose annual expenses exceeded their revenue had poorer quality (score 27.4%) compared to those whose expenses were covered by their revenue (score 37.1%) ($P = 0.07$).

CONCLUSIONS: Overall quality of care for diabetics receiving care in CHCs is poor, although rates of vaccination and blood pressure control are similar to those reported in national survey data. Furthermore, we found racial and ethnic minority groups and the uninsured received poorer quality care than other groups, and that certain CHC characteristics may be associated with better care. Policymakers and CHC directors should carefully consider which center characteristics are most important in determining high quality diabetes care and efforts within CHCs should be made to further improve quality of care for minority and uninsured patients.

Percent of Eligible CHC Patients Meeting Quality Indicators for Diabetes

Indicator	% Met	Indicator	% Met	Indicator	% Met
Smoking cessation advice	42	Annual dilated eye exam	27	≥ 2 HbA1c assessments	38
Documented self-management goals	31	Annual Foot Exam	29	Most recent BP $\leq 130/80$	41
Influenza vaccination	37	ACE inhibitor use	58	Most recent HbA1c $\leq 7.0\%$	27
Assessment of nephropathy	48	Daily aspirin use	29	Most recent LDL ≤ 100 mg/dL	16

DETERMINANTS OF QUALITY OF HYPERTENSION CARE AMONG COMMUNITY HEALTH CENTERS IN THE U.S. L.S. Hicks¹; J. O'Malley²; B.E. Landon¹; T. Lieu³; N. Cook⁴; T. Keegan²; B.J. McNeil¹; E. Guadagnoli². ¹Harvard University, Boston, MA; ²Harvard Medical School, Boston, MA; ³Harvard Pilgrim Health Care, Boston, MA; ⁴Massachusetts General Hospital, Boston, MA. (Tracking ID #136278)

BACKGROUND: Community health centers (CHCs) are responsible for providing care for many of the over 40 million uninsured Americans, many of whom are members of groups that have been previously documented to receive poorer care for cardiovascular disease. In 2004 President Bush announced plans to expand the number of CHC sites during 2005 to address the health care needs of the uninsured. However, there has been little examination of the quality of care for cardiovascular disease delivered in CHCs or the patient and CHC characteristics that may be associated with quality.

METHODS: We examined medical records of 1,639 patients with hypertension receiving care from January 1, 1999 to August 1, 2001 in 41 publicly funded CHCs from throughout the U.S. and surveyed administrators at the CHCs about their center's organizational characteristics. We used previously validated quality of care indicators based on national guidelines and tabulated the number of eligible patients meeting each quality indicator during the study period. For each patient, we created a summary scale for quality of care received, standardizing the indicators to have mean 0 and variance 1 and computed the mean of the standardized scores for each patient. We then re-scaled the score so that the mean represents the average percentage of times each indicator was met. Higher scores represent higher quality of care. We then averaged the scores across patients within a center to compute a center level quality score. We compare patient demographic and CHC organizational characteristics by patient and center level scores.

RESULTS: Of the 1,639 patients, 758 (46.2%) were white, 664 (40.5%) were male, and 382 (23.3%) were uninsured. A comorbid diagnosis of coronary artery disease (CAD) was present in 174 (10.6%) of the patients. The total percentage of patients meeting each quality indicator is presented in the table below. The mean hypertension quality score for the population was 50.8%. Uninsured patients received significantly poorer care (score 47.4%) compared to those with

insurance coverage (score 51.8%) ($P = 0.01$). Patients with CAD received significantly better care (score 55.7%) than those without CAD (score 50.3%) ($P = 0.003$). Centers that had computerized notes or medication lists available had better quality (score 56.7%) than those without computerized notes or medication lists (score 49.1%) ($P = 0.03$).

CONCLUSIONS: Overall quality of care for hypertension patients receiving care in CHCs is similar to those reported in national survey data and that patients with previously diagnosed CAD received better care. Furthermore, we found the uninsured received poorer quality care than other groups, and that the use of computerized notes or medication lists by CHCs may be associated with better care. Policymakers and CHC directors should carefully consider which center characteristics are most important in determining high quality hypertension care and efforts within CHCs should be made to further improve quality of care for uninsured patients. Percent of Eligible CHC Patients Meeting Quality Indicators for Hypertension

Indicator	% Met	Indicator	% Met	Indicator	% Met
Smoking cessation advice	45	ACE-inhibitor/ AII-blocker use	54	1 Creatinine measurement during year	75
Documented self-management goals	29	Beta-blocker use	43	Most recent blood pressure control < 140/90 or < 130/80 for diabetes or renal failure	49

DEVELOPMENT AND PILOT EVALUATION OF A MOTIVATIONAL INTERVIEWING INTERVENTION TO PROMOTE SAFER SEX PRACTICES AMONG PERSONS LIVING WITH HIV. C. Golin¹; S. Patel¹; K. Paulovits¹; B. Quinlivan². ¹University of North Carolina at Chapel Hill, Chapel Hill, NC; ²University of North Carolina, Chapel Hill, NC. (Tracking ID #135499)

BACKGROUND: About 40,000 Americans still become infected with HIV each year. At least one-third of people living with HIV/AIDS continue to engage in risky sexual practices; this proportion appears to be increasing. As people with HIV live longer, healthier lives, interventions to help them practice safer sex are critically needed yet few studies have evaluated such programs among individuals already infected with HIV. Motivational Interviewing (MI) has been used to effectively improve safer sex practices among HIV-negative persons and has changed other health behaviors among HIV-infected people. We sought to develop a theory-based MI program to enhance safer sex practices among people living with HIV and pilot-test it for feasibility, usefulness, acceptability and content.

METHODS: Using a conceptual model of factors that influence the sexual behaviors of HIV-infected people and adapting a previously successful MI antiretroviral adherence program, we created a comprehensive safer sex MI intervention guide and manual tailored to guide MI sessions that address HIV risk behavior. The Motivational Interviewing Network of Trainers (MINT)-trained MSW conducting the MI completed 5 days of local training and formal coursework in the program. She then administered the program to 33 patients (21 African American; 17 women, 8 MSM, 8 heterosexual men). The MI counselor completed a standardized MI data recording sheet following each session to record its content. We conducted in-depth phone interviews among a subsample of 15 clients about their experiences with the MI and content-analyzed 53 MI data recording sheets to assess session length, content and flow. The MI tool: 1) provides a menu of topics related to safer sex practices from which the patient can choose; 2) provides a menu of standardized counseling strategies from which the counselor can choose 3) tailors the session to fit patients' individually assessed needs; 4) explicitly assesses a patient's relationship status and sexual activities; 5) assesses the patients' motivation and self-efficacy to address the chosen behavior and 6) delivers standardized informational messages, as needed.

RESULTS: Among the 15 in-depth interviews conducted, 93% of participants felt comfortable (73% very) with the MI, 100% felt the counselor "understood them a lot," and 80% related "a lot" to the topic menu. Sixty-six percent learned new information; 100% of those felt the information would help them a lot. Many participants reported having felt respected and "really listened to" about salient topics they otherwise had little opportunity to discuss. Ninety three percent felt likely (46% very) to change their behavior; 80% set goals to change behaviors; 86% felt more confident to change their behaviors; 90% of these developed specific strategies to change behaviors; Among the 33 receiving between 1-3 MI sessions (mean length 39 minutes), based on the MI data recording sheets, 66% chose a topic from the menu. Thirty-three percent chose to discuss condoms or other safer sex methods; 13% chose alcohol use during sex; 8% chose other ways to be intimate; 8% chose STIs; Other topics chosen to discuss included: serostatus disclosure with partners, spousal relationships, risk perceptions, and self-esteem.

CONCLUSIONS: We have developed an acceptable, useful, feasible method to conduct safer sex counseling routinely in the clinical setting for people living with HIV. More studies to test its effect on behavioral outcomes and HIV transmission are needed.

DIFFERENCES BETWEEN INFECTIOUS DISEASE AND GENERAL MEDICAL PROVIDERS IN COMFORT WITH PROVIDING PRIMARY CARE. S.L. Fultz¹; J. Goulet¹; S. Weissman²; D. Rimland³; D. Leaf⁴; C.L. Gilbert⁵; M. Rodriguez-Barradas⁶; A.C. Justice¹. ¹Yale University, West Haven, CT; ²Hospital of Saint Raphael, New Haven, CT; ³Emory University, Decatur, GA; ⁴University of California, Los Angeles, Los Angeles, CA; ⁵George Washington University Medical Center, Veterans Affairs Medical Center, Washington, DC; ⁶VA Medical Center and Baylor College of Medicine, Houston, TX. (Tracking ID #135078)

BACKGROUND: As HIV-related mortality has decreased through the use of highly active antiretroviral therapy (HAART), patients are living longer making comorbid disease management more important. We sought to examine Infectious Disease (ID) providers' self-reported comfort with prescribing non-HIV-related medications including lipid-lowering medications, antihypertensives, antidepressants, and hypoglycemic medications compared to General Medicine (GM) providers.

METHODS: As part of the Veterans Aging Cohort Study (VACS), patient-reported prevalence of conditions was reported. In addition, providers in infectious disease (ID) and general medical (GM) clinics completed surveys reporting their demographic, training, and certification characteristics, along with self-reported comfort prescribing antihypertensives, lipid-lowering agents, antidepressants, and hypoglycemic medications. Comfort was rated on a 1-5 Likert scale with scores of '4' or '5' classified as 'Comfortable.'

RESULTS: Using ID clinic patient-reported diagnosis, all four conditions were frequent: hyperlipidemia (24%), hypertension (33%), depression (43%) and diabetes (17%). Of 150 GM or ID certified attending physicians, 84 were in the ID clinic, and 66 were in the GM clinic. Overall, providers reported they were the primary care provider for 84% of their patients, with no significant difference between the clinics. GM providers were more likely than ID providers to report being comfortable with prescribing lipid-lowering medication (98% of GM providers vs 71% of ID providers), antihypertensives (98% vs 75%), antidepressants (79% vs 37%), and hypoglycemic medications (99% vs 58%, $p=0.0001$ for all comparisons). Multivariate analyses revealed that provider sex, race, and years in practice had minimal influence.

CONCLUSIONS: While ID and GM clinic providers are equally likely to classify themselves as the primary care provider, ID providers report less comfort prescribing common primary care medications than GM providers. Given the high frequency of these comorbidities, comfort with treating these conditions is essential. While this may not portend lower quality of care, it does warrant further investigation and attempts to increase providers' comfort.

DIFFERENCES BY RACE IN THE ETIOLOGY OF UPPER GASTROINTESTINAL HEMORRHAGE AND THE PRE-ADMISSION USE OF POTENTIALLY PREVENTABLE THERAPY. C.T. Whelan¹; P. Kaboli²; Q. Zhang³; D.O. Meltzer³. ¹University of Chicago, Oak Park, IL; ²Iowa City VA Medical Center, Iowa City, IA; ³University of Chicago, Chicago, IL. (Tracking ID #136095)

BACKGROUND: Erosive disease (ED) and peptic ulcer disease (PUD) are common causes of upper gastrointestinal hemorrhage (UGH). One prior study suggests that blacks are at an increased risk for ED, particularly from the use of non-steroidal anti-inflammatory drugs (NSAIDs). It is unclear why blacks are at higher risk. While biological factors may contribute to this increased risk, the use of potentially preventive therapies such as proton pump inhibitors (PPI) or selective cox-2 inhibitors (COX-2s) may be lower in blacks. We examine differences by race in the etiology of UGH and pre-admission use of these agents in patients with UGH.

METHODS: We identified and abstracted the charts of 230 consecutive patients with UGH admitted to 2 academic hospitals, 1 urban ($n=119$) and 1 rural ($N=111$) to assess UGH etiology based on EGD report. Risk factors for UGH identified included NSAID or aspirin (ASA) use, previous UGH, and PPI or COX-2 use. Differences among etiologies of UGH among blacks versus non-blacks were assessed. Among patients who could have been given a treatment known to decrease the rate of UGH (i.e. patients on any NSAID (selective or non-selective) who could have received a COX-2 or PPI; or patients on ASA or with a previous UGH who may have benefited from a PPI), differences by race in preadmission use of PPIs or COX-2s were analyzed. Differences were analyzed with the chi-square statistic.

RESULTS: Mean age was 58 years, 49% were male, and 40% were white. ED (43%), peptic ulcer disease (PUD) (33%), and variceal bleeds (21%), were the most common etiologies for UGH. However there were differences in rates between blacks and non-blacks in each of these etiologies: ED (54% vs. 34% respectively, $p=0.002$); PUD (39% vs. 28%, $p=0.08$); and varices (7% vs. 33%, $p<0.001$). Rates of prior UGH; NSAID use; or NSAID+ASA use were similar among blacks and non-blacks. However, more blacks (32%) used ASA than non-blacks (20%), $p=0.01$. There were no significant differences in PPI use between blacks and non-blacks in any sub-groups who may have benefited from a PPI. COX-2 use was higher among blacks (7%) than non-blacks (2%), $p=0.09$. Among patients who may have benefited from a PPI or a COX-2, 29% were taking one prior to admit and there was no difference between use in blacks (31%) and non-blacks (27%), $p=0.60$.

CONCLUSIONS: Significantly higher rates of ED and PUD were seen in blacks. However, preadmission use of preventive therapy was similar among at-risk blacks and non-blacks. These findings suggest that factors other than reduced use of preventive pharmacology may play a role in the higher risk of ED and PUD among blacks. Understanding these factors may provide insights into the risks of ED and PUD. A limitation to this study is the racial distribution by site; almost all blacks were at 1 site, which limits our ability to account for other site-specific factors, such as urban vs. rural. However, this increased risk of ED and PUD in blacks is consistent with prior studies. Further study clarifying the impact of race on the risk for ED and PUD may better define which patients may benefit from preventive therapy.

DISPARITIES IN THE MANAGEMENT OF EARLY BREAST CANCER AMONG ASIAN AMERICANS AND PACIFIC ISLANDERS. R.P. Gelber¹; J.W. Davis²; T.B. Seto³. ¹Massachusetts Veterans Epidemiology Research and Information Center, VA Boston Healthcare Center, Harvard Medical School, Boston, MA; ²Hawaii Medical Service Association, Honolulu, HI; ³University of Hawaii John A. Burns School of Medicine, Honolulu, HI. (Tracking ID #133941)

BACKGROUND: Despite consensus guidelines for the management of breast cancer, many patients do not receive standard therapy. Although this discrepancy may be partly related to patient ethnicity, little is known about breast cancer care among Asian Americans and Pacific Islanders (AAPIs). Objective: To evaluate the management of early stage breast cancer among AAPI women.

METHODS: Retrospective analysis of the first course of therapy for 2038 women (937 Japanese, 144 Chinese, 236 Filipino, 295 Hawaiian, 426 white; mean age $\pm$ SD, 59 $\pm$ 13 years) diagnosed with early stage breast cancer (stages I, IIA or B, IIIA) in Hawaii from 1995-2001. We linked data from the Hawaii Tumor Registry, a SEER population-based registry, to administrative data from Hawaii's largest health insurer. We evaluated the use of standard-of-care treatments as defined by NCI/NIH guidelines, including (1) breast-conserving surgery (BCS); (2) radiation therapy following BCS; and (3) chemotherapy for women with nodal spread. We used logistic regression to examine the association between ethnicity and quality of care.

RESULTS: Overall, 1227 (60.2%) women had stage I disease, 752 (36.9%) stage IIA or B, and 59 (2.9%) stage IIIA. Only 1623 (79.6%) of all women received BCS, and 64.9% of these received radiation therapy. Of those with node-positive disease ($n=526$), 82.7% received chemotherapy. Japanese and Hawaiian women were significantly less likely than whites to receive BCS after controlling for age, year of diagnosis, rural residence, tumor size, grade, nodal status, estrogen receptor status, history of prior cancer, Charlson comorbidity index, health plan type (HMO, fee-for-service, Medicare), and income (OR, 95%CI: 0.61, 0.42-0.88, for Japanese; 0.63, 0.40-0.98, for Hawaiians; 0.76, 0.42-1.35, for Chinese; 0.68, 0.42-1.10, for Filipinos). Filipino women were significantly less likely than whites to receive radiation after BCS (OR, 0.60; 95%CI, 0.39-0.91), while other AAPIs were treated similarly to whites (OR, 95%CI: 0.84, 0.61-1.16, for Japanese; 1.06, 0.69-1.62, for Hawaiians; 1.46, 0.83-2.56, for Chinese). AAPI women were as likely as white women to receive chemotherapy for node-positive disease (OR, 95%CI: 1.94, 0.84-4.48, for Japanese; 0.92, 0.36-2.33, for Hawaiians; 0.64, 0.19-2.12, for Chinese; 0.81, 0.32-2.10, for Filipinos).

CONCLUSIONS: We found significant disparities in the management of early stage breast cancer among AAPI women, with Japanese, Hawaiians, and Filipinos being particularly less likely to receive standard-of-care treatment, as compared to white women. Further study is needed to determine the reasons for the observed differences in care and to examine the impact of these disparities on health outcomes.

DO YOUNGER US STROKE SURVIVORS HAVE REDUCED ACCESS TO CARE AND MEDICATIONS? D.A. Levine¹; C.I. Kiefe¹; T. Houston¹; J.J. Allison¹; E.P. McCarthy²; J.Z. Ayanian³. ¹University of Alabama at Birmingham, Birmingham VA Medical Center, Birmingham, AL; ²Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA; ³Brigham and Women's Hospital, Harvard Medical School, Boston, MA. (Tracking ID #133925)

BACKGROUND: Millions of US stroke survivors require comprehensive care for risk factor modification and secondary prevention. Younger stroke survivors may have reduced access to care because they do not qualify for Medicare unless disabled for at least 2 years. We assessed age-related differences in access to care among stroke survivors.

METHODS: We examined data from 3,681 stroke survivors aged 45-85 years who responded to the 1998-2002 National Health Interview Survey (NHIS). We compared access to care within the last 12 months by age (45-64 vs. 65-85) using questions regarding health insurance, ability to afford medications, general doctor and medical specialist visits, and neurological disability (difficulty with functional activity due to stroke). We used multivariable logistic regression models to examine the association between access measures and age after adjusting for sex, race, income, hypertension, coronary heart disease, diabetes mellitus, and neurological disability. All analyses used SUDAAN and results were weighted to reflect national population estimates.

RESULTS: The NHIS data estimate 4.1 million stroke survivors aged 45-85 years in the US, of whom 1.3 million are aged 45-64 years. Younger stroke survivors ($n=1172$) were more likely than older ones ($n=2509$) to be male (52% vs. 47%, $p=0.01$), of non-white race (25% vs. 14%, $p<0.0001$), and report annual incomes $\geq$ \$20,000 (55% vs. 46%, $p<0.001$). Coronary heart disease was reported more commonly by older stroke survivors (39% vs. 30%, $p<0.0001$), but diabetes (25%), hypertension (70%) and neurological disability (40%) had similar prevalence by age. Younger stroke survivors reported worse access to care than older ones (Table). After adjustment, younger age remained independently associated with no general doctor visits, adjusted odds ratio (OR) 1.5, 95% CI (1.1, 2.1), no general doctor or specialist visits, OR 2.1, (1.4, 3.3), and inability to afford medications, OR 3.1, (2.3, 4.2). After further adjustment for health insurance, these odds ratios became 1.4(0.9, 2.3), 1.5(0.8, 2.8), and 3.2(2.1, 4.9) respectively.

CONCLUSIONS: Younger stroke survivors report reduced access to care compared to older stroke survivors. Lack of health insurance may play a role in physician visits but perhaps not in medication affordability. This disparity may lead to inadequate risk factor modification and recurrent stroke in this younger high-risk population.

Access to Care Measures among Stroke Survivors by Age (all p-values < 0.003)

	Younger Stroke Survivors	Older Stroke Survivors
Access to Care Measure	Sample n (weighted %)	Sample n (weighted %)
No health insurance	133 (11)	13 (0.4)
Inability to afford medications	194 (15)	152 (6)
No general doctor visit	167 (14)	262 (10)
No general doctor or specialist visit	101 (8)	135 (5)

DOES REALM LITERACY SCREENING TEST PREDICT MEDICATION COMPREHENSION. M.L. Plews-Ogan¹; J.H. Groninger¹; J.M. Schectman¹; M. Donald¹. ¹University of Virginia, Charlottesville, VA. (Tracking ID #134003)

BACKGROUND: Among potential risk factors for medication misunderstanding and medication errors is low health literacy. Low literacy has been linked to increased risk for hospitalization, poor diabetes control and worse outcomes for HIV care. Few studies have evaluated whether low health literacy puts patients at risk for poor understanding of and incorrect use of medications. With the known threat to patient safety posed by the use of medications, identifying risk factors for poor comprehension and incorrect use of medications is crucial. The Rapid Estimate of Adult Literacy in Medicine (REALM) has been validated as a brief, simple test to predict health literacy. We conducted a pilot study to evaluate the predictive value of the REALM score for medication knowledge.

METHODS: Patients at an academic internal medicine clinic who had brought their medications to their visit who agreed to participate were surveyed regarding level of education, self-reported literacy level, and number of medications. They were then administered the REALM test as well as a survey of medication comprehension (name of medication, dose, indication and known side effects) for each of their medications. The answers on the medication comprehension survey were compiled into a medication comprehension "score" (MCS).

RESULTS: A total of 72 patients were surveyed with a mean age of 61 (range 27-90) mean years of schooling completed 9.7 (range 1-18). For self-reported literacy level, 88% reported being able to read and understand their medication. Participants were able to provide the name for only 194 (50.1%) of medications taken, a known side-effect for only 49 (11.7%). The patients identified a correct indication for 81% of their medications and correct dosage taken for 92%. REALM score ($R=0.27$ $p<0.001$) were strong predictors of MCS and patients over age 65 scored lower on MCS than patients under 65 (2.11 vs 2.59 $p=0.02$). Race, gender and number of medications taken were not predictive of MCS. On multivariate analysis, REALM score ($p=0.0003$) and last grade completed ($p=0.039$) but not age, remained independently predictive of MCS.

CONCLUSIONS: This survey pilot study suggests that REALM score and last grade completed may independently predict medication comprehension as assessed by the Medication Comprehension Score. Therefore, the REALM may be an accurate screening tool for patients at risk for poor medication comprehension. This tool could enable physicians to identify patients at risk and intervene with brown bag reviews and specific communication techniques that have been demonstrated to improve comprehension and outcomes.

DRIVING DISTANCE AND DIABETES CONTROL. K.R. Strauss¹; B. Littenberg¹; A.R. Troy¹. ¹University of Vermont, Burlington, VT. (Tracking ID #134770)

BACKGROUND: Diabetes mellitus is one of the most common chronic diseases in America, affecting approximately 10% of the adult population. Despite advances in the treatment and control of diabetes, many barriers to good diabetic control remain. One potential barrier is the driving distance from a patient's home to his or her site of care. Our goal in this study was to identify and describe the role that driving distance to site of care plays in glycemic control.

METHODS: We studied randomly selected adults enrolled in the Vermont Diabetes Information System (VDIS). All subjects had the diagnosis of diabetes confirmed by their Primary Care Provider (PCP) and participated in an in-home interview. Data collected included social and economic variables, home address, and duration of diabetes. Glycemic control was measured by recording a recent A1C test. Using a geographic information system (ArcView 3.3), the addresses of these subjects and their PCPs were geocoded (matched) to a commercially available geographic data set. A networking script was then used to determine the driving distance from each subject's home to the PCP's office. Subjects were broken into groups according to the driving distance: near (<5km from his or her PCP; $n=274$) or far (5 km or more; $n=401$). We used linear regression to estimate the effect of driving distance on glycemic control.

RESULTS: Subjects who live within 5 km of their PCP had slightly better glycemic control (7.0 vs. 7.2; $P=0.19$ by t-test). Age was an important modifier of this relationship. In seniors (age 65 and older), after adjusting for sex, marital status, duration of diabetes, education, income, and health insurance coverage, the effect of distance was large and statistically significant (regression coefficient = -0.32; 95% CI -0.58, -0.05; $P=0.02$). In younger subjects (under age 65), the association between distance and glycemic control was weaker (coefficient = +0.28; 95% CI -0.06, +0.62; $P=0.11$).

CONCLUSIONS: Longer driving distances are associated with poor glycemic control in diabetic seniors in this largely rural population. This may be due to decreased mobility or sight, poor access to automobiles, or a number of other

factors. Because many younger diabetics are employed and may travel from work to their PCP, our estimates in subjects under age 65 must be viewed with caution. While further investigation is needed, especially in a more urban setting, PCPs should be aware that driving distance represents a potential barrier to good diabetic control.

EFFECT OF RACE, ETHNICITY AND SOCIOECONOMIC STATUS ON BONE DENSITY TESTING AFTER HIP FRACTURE. J.M. Neuner¹; X. Zhang¹; R. Sparapani¹; P.W. Laud¹; A.B. Nattinger¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #135906)

BACKGROUND: Osteoporotic fractures are known to be less common in members of some racial and ethnic groups than in white Americans, but the effects of this upon osteoporosis identification and care are not well-described. We hypothesized that even after hip fracture, nonwhite race and lower socioeconomic status are associated with lower rates of investigation of bone density in female Medicare recipients.

METHODS: A cohort of women Medicare enrollees aged 67-91 from three states (FL, NY, IL) who had a hip fracture in 2001 was identified by adapting a published fracture algorithm (previously reported positive predictive value 98%) using 100% of each state's Medicare claims. As bone density tests are Medicare-reimbursed every 2 years, appropriate osteoporosis testing was defined as any bone density test (ultrasound or X-ray absorptiometry) performed between 2 years before and 6 months after the fracture. Race and ethnicity were measured as designated by Medicare. Income was assigned based on the mean per capita income in the subject's zip code, and then categorized by terciles. Logistic regression was used to examine the association of bone density testing with race, ethnicity and zip-code level measures of socioeconomic status during the 30-month period with adjustment for age, comorbidity, urban/rural residence and state.

RESULTS: The overall rate of bone density testing among 17,934 hip fracture patients was 23%. In unadjusted analyses, the rate of testing in white women was 23%, in Hispanic women 10% and in black women 10%. In a logistic regression model adjusted for age and comorbidity, black women (adjusted odds ratio 0.38 [95% Confidence Interval 0.28, 0.50]) and Hispanic women (AOR 0.45 [95%CI 0.32, 0.62]) were less likely than white women to receive bone density tests. Compared with women residing in zip codes in the highest tercile of income, those in the lowest tercile were 23% less likely (AOR 0.78 [95%CI 0.71, 0.85]) and those in the middle tercile 11% less likely to receive testing (AOR 0.89 [95%CI 0.82, 0.97]). Compared with New York residents, Florida residents were more likely (AOR 1.50 [95%CI 1.38, 1.62]) to be tested, while Illinois residents were less likely (AOR 0.86 [95%CI 0.78, 0.94]). Urban residence and educational attainment did not affect testing and were removed from the model.

CONCLUSIONS: In a large population-based cohort, women of black race and Hispanic ethnicity, as well as those residing in lower-income zip codes, were much less likely to receive bone density testing. As all women were insured and at high recurrent fracture risk, barriers to osteoporosis recognition such as physician overapplication of population statistics to individual patients deserve further study.

ELIMINATING DISPARITIES IN DIABETES CARE: THE IMPACT OF DISEASE MANAGEMENT STRATEGIES WITHIN TRIAD. O. Duru¹; C.M. Mangione¹; A.J. Karter²; D.S. Kountz³; M.M. Safford⁴; C. Tseng⁵; B. Waitzfelder⁶; R. Gerzoff⁷; S. Huh¹; W. Steers¹; A. Brown¹. ¹University of California, Los Angeles, Los Angeles, CA; ²Kaiser Permanente Division of Research, Oakland, CA; ³University of Medicine and Dentistry of New Jersey, New Brunswick, NJ; ⁴University of Alabama at Birmingham, Mountain Brook, AL; ⁵Pacific Health Research Institute, Honolulu, HI; ⁶Centers for Disease Control and Prevention (CDC), Atlanta, GA. (Tracking ID #135935)

BACKGROUND: Minorities with diabetes generally receive lower quality care than whites after controlling for insurance status, and have higher rates of some, but not all, complications. Disease management programs to improve quality of care for all patients may indirectly reduce racial/ethnic disparities in diabetes care.

METHODS: Data come from 6,251 patients from 10 managed care plans and 68 provider groups included in the Translating Research into Action for Diabetes (TRIAD) study, a multicenter longitudinal cohort study of diabetes care in managed care. Race/ethnicity (white, African American, Latino, Asian/Pacific Islander) was the primary predictor variable. We examined three sets of dependent variables: processes of care from self-report and chart review, intermediate outcomes (hemoglobin A1c, blood pressure, and low-density lipoprotein/LDL cholesterol), and whether patients with poorly controlled intermediate outcomes were managed with aggressive medication regimens. Models were stratified by intensity of use of four provider group-level disease management strategies: use of a diabetes registry, physician reminders, performance feedback, and care management. Multilevel models were then used to generate predicted probabilities for each ethnic group at the 20th percentile (low-intensity) and 80th percentile (high-intensity) of each disease management strategy.

RESULTS: In low-intensity groups, disparities in LDL screening, A1c screening, and flu vaccine use were observed for African Americans relative to whites (ranging from 4% to 10%). Only the disparity in flu vaccine use remained in the high-intensity groups. Regardless of disease management intensity level, Latinos and Asians/Pacific Islanders had equal or superior predicted processes of care compared to whites. All minority groups had poorer predicted control of intermediate outcomes than whites within both low and high-intensity groups. African Americans had lower predicted values of aggressive therapy for elevated hemoglobin A1c values than whites in groups with low intensity of care management (disparity of 8%) and use of a diabetes registry (disparity of 7%). These

differences were not observed within groups intensively implementing these two disease management strategies. In contrast, among persons with poor control, Latinos and Asians/Pacific Islanders received the same level of aggressive therapy as did whites.

CONCLUSIONS: In a setting of uniform access to care, process delivery and aggressive management of poor risk factor control was comparable between Latinos and whites and Asians/Pacific Islanders and whites. However, in low-intensity groups disparities remained for African Americans compared to whites, with some attenuation in the high-intensity groups. All minority groups had worse intermediate outcomes than whites, suggesting that disease management programs as presently implemented are not eliminating disparities in risk factors. Research identifying mutable factors closely related to intermediate outcomes in diabetes might provide strategies to address the issue of disparities in diabetes complications.

EXAMINING THE LINK BETWEEN COMMUNICATION AND MEDICATION ADHERENCE AMONG AFRICAN AMERICANS AND LATINOS. A. Ang¹; A. Brown¹; D. Morisky¹. ¹University of California, Los Angeles, Westwood, CA. (Tracking ID #135871)

BACKGROUND: Hypertension disproportionately affects minority populations in part due to many different socio-cultural factors that affect adherence to anti-hypertensive drug therapies. In multicultural minority populations, issues dealing with patient communication may play a significant role because of linguistic and contextual barriers that inhibit effective provider-patient communication. In this study, we examine the relationship between provider-patient communication and medication adherence.

METHODS: 1,367 African American (75%) and Hispanic American (25%) adults participated in the study at a large teaching hospital on the West Coast. Participants are representative of the surrounding, predominantly low-income minority community. The patients were randomly assigned to health providers, and received one of three types of intervention designed to enhance provider-patient communication: Individualized patient counseling sessions (CS) following clinic visits, a computerized appointment (CAPT) reminder patient tracking system, home visits by community health workers (CHW) to encourage family members to support patient's management of lifestyle and medication adherence, or the usual care (UC). Patient adherence was measured at baseline and six months after the intervention using the Morisky adherence measure, which is scaled from 1 (low) to 3 (high). We used multilevel ordered logistic models to evaluate the effects of the intervention on patient adherence, after adjusting for individual health beliefs, social support, patient satisfaction, complexity of medication regimen, BMI, age, baseline adherence and ethnicity.

RESULTS: After the intervention, patients who received counseling sessions (odds ratio (OR) 1.23, (95% CI 1.07-1.84), $P=.02$) or home visits from community health workers (OR 1.31, (95% CI 1.12-1.95), $P<.001$) were more likely to adhere to their medication treatment. There was no significant difference between those who received computerized appointment reminders (OR .99, (95% CI .86-1.25), $P=.34$) and those who received the usual care. Patient adherence was also associated with health beliefs (OR 1.15, (95% CI 1.04-1.28), $P=.03$), patient satisfaction (OR 1.07, (95% CI 1.03-1.11), $P=.04$), and complexity of medication regimen (OR=.55, (95% CI .38-.79), $P=.01$).

CONCLUSIONS: In this randomized intervention study, African American and Latino patients who received personalized communication from health providers or community health workers had better medication adherence after the intervention. These providers may influence the adherence behavior of their patients by supporting and reinforcing current positive behaviors or clarifying misunderstandings and incorrect beliefs.

FACTORS MEDIATING ETHNIC DIFFERENCES IN GLYCEMIC AND CARDIOVASCULAR RISK FACTOR CONTROL IN DIABETES. L.H. Miller¹; S.R. Lipsitz²; S. Natarajan¹. ¹New York University, New York, NY; ²Medical University of South Carolina, Charleston, SC. (Tracking ID #136246)

BACKGROUND: Ethnic differences in cardiovascular risk factor levels and glycemic control persist in US adults with diabetes. While these disparities have been demonstrated, little is known about what patient and system characteristics are responsible for these differences. Identifying patient and system characteristics that mediate these differences may be important to developing interventions that target these characteristics in order to ameliorate these disparities.

METHODS: We analyzed the 1999-2000 National Health and Nutrition Examination Survey to evaluate if hemoglobin A1c % (HbA1c) and cardiovascular risk factors such as systolic blood pressure (SBP), low-density lipoprotein (LDL) and high-density lipoprotein (HDL) cholesterol, and triglycerides (TG) varied by ethnicity. Ethnic categories analyzed were non-Hispanic whites (NHW), non-Hispanic blacks (NHB) and Mexican-Americans (MA). Initially, multi-variate linear regression was used to test if HbA1c, SBP, LDL cholesterol, HDL cholesterol and TG differed by ethnicity. Further mediation analyses evaluated if education level, access to care, duration of diabetes and insulin use were responsible for the variation in cardiovascular risk factors and HbA1c among diabetics by R-square change analyses (ΔR^2). All analysis incorporated the complex survey design effects to determine population estimates.

RESULTS: Significant ethnic differences were noted in HbA1c, HDL cholesterol and triglyceride levels. NHW had significantly lower HbA1c levels (7.48%) compared to NHB (8.31%) and MA (8.26%). In contrast, NHB had significantly higher HDL cholesterol (50.5 mg/dl) and lower TG (177.9 mg/dl) than NHW (HDL 42.4 mg/dl, TG 243.5 mg/dl) and MA (HDL 43.6 mg/dl, TG 311.9 mg/dl). Education level was the single largest determinant of variability in HbA1c (ΔR^2 2.80,

$p<.05$), with more modest effects ($p>.05$) of access to care (ΔR^2 1.10), duration of diabetes (ΔR^2 1.18) and use of insulin (ΔR^2 1.62). Though the magnitude was small, duration of diabetes was significantly responsible for the variation in SBP (ΔR^2 change 0.27) and TG (ΔR^2 1.69), and access to care for variation in LDL (ΔR^2 0.05). None of these variables were significantly responsible for variability in HDL cholesterol.

CONCLUSIONS: Ethnic differences between HbA1c, HDL cholesterol and TG levels continue in US adults with diabetes. Education level is the predominant patient characteristic mediating differences in HbA1c. Eliminating disparities in education may decrease ethnic differences in glycemic control. Future research should evaluate other behavioral, socioeconomic and biologic factors as mediators of ethnic differences in order to effectively develop interventions to eliminate disparities.

GENDER AND TOTAL KNEE/HIP ARTHROPLASTY UTILIZATION RATE IN THE VA SYSTEM. S. Borrero¹; K. Kwoh¹; J. Sartorius²; S.A. Ibrahim². ¹University of Pittsburgh Medical School, Pittsburgh, PA; ²Center for Health Equity Research and Promotion, VA Pittsburgh Healthcare System, Pittsburgh, PA. (Tracking ID #133091)

BACKGROUND: Studies have reported that women are less likely than men to undergo certain medical procedures including cardiac catheterization, coronary revascularization, and renal transplant. Limited data suggest that gender differences exist in the utilization rate of elective total knee/hip joint arthroplasty, a cost-effective treatment option for end-stage knee/hip osteoarthritis (OA). OA is a leading cause of disability in the elderly and is more prevalent in women. A population-based study in Canada found women to be less likely to undergo the treatment compared to men. A similar study has not yet been done in the US. Therefore, we examined gender differences in utilization rates of total knee/hip joint arthroplasty among US patients who are potentially at risk for the procedure and have equal access to the treatment.

METHODS: The study sample included all Veterans Administration (VA) patients (total cohort) during fiscal year 1999 who were 50 years of age or older. To narrow the sample to those at risk for the procedure, we identified a sub-sample (OA sub-cohort) consisting of patients with a diagnosis of lower extremity OA. The primary outcome of the study was receipt of total knee or hip joint arthroplasty within two years (FY 2000 and 2001). Baseline comparisons were performed using chi-square tests for categorical variables and t-tests for normally distributed continuous variables. For multivariable modeling, we used random effects logistic regression with hospital site as the random effect to account for clustering effects. The final model outcomes were adjusted for age and the number of comorbidities. Analyses on the total cohort adjusted for OA in order to take into account gender differences in OA prevalence, and race and region were eliminated from the final models after determining that these variables did not impact the outcome.

RESULTS: A total of 1,968,093 VA patients were identified (total cohort) of whom 329,461 patients had lower extremity OA (OA sub-cohort). In the total cohort, 1,923,524 (97.7%) were male and 44,569 (2.3%) were female, which are reflective of the national VA patient population. In the OA sub-cohort, 319,924 (97.1%) were male and 9,537 (2.9%) were female. The mean age was 65 years in the total cohort and 66 years in the OA sub-cohort. Women had a significantly higher rate of rheumatologic disease than men ($p=0.001$). In the total cohort, 5,370 (0.3%) had total knee arthroplasty and 2,709 (0.1%) had total hip arthroplasty within the two-year follow-up period. In the OA sub-cohort, 4,791 (1.5%) had total knee arthroplasty and 2,220 (0.7%) had total hip arthroplasty. For the total cohort, the two-year adjusted odds of women undergoing total knee or hip arthroplasty were 1.02 (0.88 to 1.20) and 1.10 (0.89 to 1.36), respectively. For the OA sub-cohort, the two-year adjusted odds of women undergoing total knee or hip arthroplasty were 0.97 (0.83 to 1.14) and 1.00 (0.79 to 1.27), respectively.

CONCLUSIONS: Among patients at risk for total knee/hip joint arthroplasty, men and women who receive care in the VA system were equally likely to undergo the procedure.

GOOD DOCTOR-PATIENT RELATIONSHIP CAN OVERCOME LANGUAGE BARRIERS. Q. Ngo-Metzger¹; D. Sorkin¹; R.S. Phillips²; M. Massagli³; B. Clarridge⁴; S. Greenfield¹; S. Kaplan¹. ¹University of California, Irvine, Irvine, CA; ²Harvard University, Boston, MA; ³Dana Farber Cancer Institute, Boston, MA; ⁴University of Massachusetts, Boston, MA. (Tracking ID #133484)

BACKGROUND: Previous research has shown that language concordance between doctors and patients is related to higher patient satisfaction with care. To what degree, if any, can a good-doctor patient relationship overcome barriers imposed by doctor-patient language discordance? We sought to examine the independent effects of the quality of the doctor-patient relationship and language discordance on patients' ratings of care.

METHODS: We surveyed 3,258 (74% response rate) Chinese and Vietnamese patients who had an office visit in the last month at one of 11 health centers across the U.S. In a mail survey in multiple languages, we asked patients to report on different aspects of their visit and to rate the quality of overall care. We studied patients whose doctors spoke their native language (concordant group) vs. those who did not (discordant group). We used patient-level multivariable logistic regressions to determine the effects of language concordance and the quality of the doctor-patient relationship on patients' ratings of the quality of care (including rating of the overall quality of the last visit, rating of doctor, and recommendation of the clinic). We adjusted for patient demographic characteristics (including age, education, primary language, English proficiency, time in

the U.S.) and utilization factors (including clinic site, reason for visit, seeing usual clinician, interpreter use).

RESULTS: Approximately two-thirds of the sample was female, and the mean age was 52.90 years (sd = 17.07). The majority (55%) of patients had less than a high school education, and 90% spoke limited or no English. There were 2371 patients in the concordant group and 749 patients in the discordant group. Ratings of the last visit as fair/poor were similar for concordant (16.23%) and discordant patients (21.47%, chi square = 3.21, $p = 0.07$). Compared to those who rated the doctor-patient relationship higher, patients who gave worse ratings (mean difference 25.66, $t = -15.55$, $p < 0.001$, effect size of mean difference = 1.56) were more likely to rate the last visit as fair/poor. Multivariable analyses adjusting for demographic and utilization variables yielded similar results. Specifically, language concordance remained an insignificant predictor of overall visit rating (adjusted Odds Ratio 1.90; 95% Confidence Interval 0.90, 4.03). Low ratings of the quality of the doctor-patient relationship were significantly related to fair/poor ratings of overall visit quality (aOR 2.42; 95% CI 1.81, 3.23). In the restricted model with only demographic, utilization, and language concordance variables, the estimated R squared was 0.068 (-2Log-Likelihood 134.51) vs. 0.247 (-2Log-Likelihood 536.38, $p < 0.001$) in the full model that included the quality of the doctor-patient relationship. This pattern of findings was similar for multivariable regression models using other ratings of care (overall rating of the doctor and recommendation of the clinic) as dependent variables.

CONCLUSIONS: Positive patient experience of the doctor-patient relationship may be more important than doctor-patient language concordance for patients' ratings of the quality of care they receive. While it is important to minimize language barriers between doctor and patient, focusing on improving the doctor-patient relationship is very important for achieving optimal patient care.

HISPANIC/LATINO PERCEPTIONS ABOUT COLORECTAL CANCER AND ASSOCIATED SCREENING. G. Makoul¹; L. Francis¹; K.A. Cameron¹; M.S. Wolf¹; D.W. Baker¹.

¹Northwestern University, Chicago, IL. (Tracking ID #135899)

BACKGROUND: In the Hispanic/Latino population, there is a relatively high incidence of colorectal cancer (CRC) coupled with low utilization of proven screening tests. Past interventions tended to have marginal impact. As a precursor to developing a multimedia CRC-education program designed specifically for Hispanic/Latino patients, we conducted this study to assess perceptions regarding CRC and CRC screening, as well as factors perceived to promote or impede screening acceptance.

METHODS: We developed two versions of a structured interview, both of which include items regarding knowledge and experience regarding CRC and CRC screening, language preference, use and ability, and demographics. One version asked specific questions about fecal occult blood test (FOBT) and flexible sigmoidoscopy; the other focused on FOBT and colonoscopy. Questionnaires were translated into Spanish, back-translated to English, and revised to ensure parallel content. Four bilingual promotoras recruited adults in the target age range (50-80 years) at three sites in the community. Participants could choose whether to speak English or Spanish.

RESULTS: Promotoras completed interviews with 234 participants. The mean age was 62 years (sd = 7.6); 60% were female. All interviews were conducted in Spanish. In response to questions about language use, 85% of participants said that their preferred language is Spanish, but only 37% rated their Spanish reading ability as very good or excellent. Overall, 68.0% of participants indicated that they had heard of colon/rectal/colorectal cancer, and 75.0% reported a high level of interest in getting information about preventing CRC. 57.7% said that they know what the colon is, though only 32.1% of these correctly identified it in a diagram. Large intestine appeared to be a more common term: 78.2% reported knowing what it is, and 78.7% of these correctly identified it. When asked which term they prefer, 25% said colon, 29% said large intestine, and 46% had no preference. Less than one-quarter of participants knew what a polyp was. People were unfamiliar with the terms for screening tests. When asked "Do you know what a Fecal Occult Blood Test is? Sometimes it's called an FOBT or Hemoccult Test" 21.5% said yes, but 54.3 said they had completed an FOBT after hearing a description of the test. Similar results were seen for flexible sigmoidoscopy and colonoscopy. Participants perceived an FOBT as easier, safer, less painful, less embarrassing, and less scary than either of the invasive tests ($p < .001$). 68.3% of the people who had not done an FOBT and 60.6% of people who did not have flexible sigmoidoscopy or colonoscopy said screening had never been suggested; 26.0% and 21.2%, respectively, said they felt fine or were not worried. Once each test was explained to the participants, the vast majority indicated they would consider getting FOBT (84.6%), flexible sigmoidoscopy (74.4%), and colonoscopy (79.5%). Virtually 100% said they would get screened if a doctor suggested it.

CONCLUSIONS: In general, patients see some advantages to FOBT, but express a willingness to undergo any of the screening procedures, particularly if recommended by a physician. Results from this study suggest that physician recommendation is not enough, and reinforce the importance of using tools such as multimedia to enhance understanding by addressing concerns such as limited literacy and limited familiarity with relevant terms.

HOSPITAL UTILIZATION OF ADULT SURVIVORS OF CHRONIC CHILDHOOD ILLNESS IN THE UNITED STATES. M.J. Okumura¹; M.M. Davis¹. ¹University of Michigan, Ann Arbor, MI. (Tracking ID #132928)

BACKGROUND: Although there are increasing numbers of individuals with childhood chronic illness who survive to adulthood, little is known about the

care adult survivors receive, such as where adult survivors are hospitalized and the financial implications of their hospital care. A study of 10 children's hospitals had shown that 4.9-7% of discharges were adults with childhood chronic illness. Therefore, we examined national patterns of hospital care and related economic implications for adult survivors in adults with cystic fibrosis (CF) and sickle cell disease (SCD). We used asthma as a "common adult illness" comparison group.

METHODS: Using the 2002 National Inpatient Survey, a nationally representative dataset containing hospital discharge data from 995 hospitals, we identified patients > 18 years old with diagnoses of CF, SCD, and asthma. Outcome variables were length of stay (LOS) and mean hospital charges. Independent variables were age, gender, payer type, in-hospital death, volume, income, procedures and hospital type; adult ($> 75\%$ of all discharges for patients > 18 yo), children's ($> 75\%$ of all discharges for patients < 18 yo), and mixed (all other). We conducted case-mix adjusted, multivariate, weighted regression analyses with the primary comparison measure between adult, mixed and children's hospitals.

RESULTS: The sample included 10,863 weighted discharges with CF, 72,800 with SCD, and 1,449,199 with asthma nationwide. The mean age and 2.5th-97.5th percentile range of age by condition were: CF (29; 18-58), SCD (33; 18-58), asthma (54; 21-87). The majority of adults hospitalized with CF (67%) and SCD (73%) were inpatients in predominantly adult hospitals, compared to 86% of admissions for asthma; 6% of CF admissions and 2% of SCD admissions were to children's hospitals. Adjusted mean LOS was significantly longer in children's hospitals vs. adult hospitals and mixed hospitals for CF and SCD but not asthma (Table). Adjusted mean hospital charges were significantly higher for adults with CF and SCD admitted to children's vs. adult and mixed hospitals despite adjusting for LOS and co-morbidity (Table). For CF, 57% of charge difference between adult and children's hospital is accounted for by longer LOS at children's hospitals, whereas charge differences in SCD or asthma were not due to LOS.

CONCLUSIONS: The vast majority of adults with CF and SCD are admitted to predominantly adult hospitals, rather than pediatric facilities. Despite adjustment for LOS and case-mix, children's hospitals had dramatically higher charges than the adult hospitals as well as longer lengths of stay for CF and SCD. The reasons for these differences will need to be investigated further. Given that most of the inpatient care for adults with childhood chronic illness is in adult hospitals, the role of internists in their inpatient and outpatient management is important in their care.

	Mean Age in Years: Adult vs. Mixed vs. Children's Hospitals	Mean LOS: Adult vs. Mixed vs. Children's Hospitals	Mean Charges: Adult vs. Mixed vs Children's Hospitals
CF	30 vs. 28 vs. 20**	1.8 vs. 2.1 vs. 2.3 days**	\$17,438 vs. \$18,000 vs. \$40,506**
SCD	34 vs. 31 vs. 20**	1.8 vs. 2.0 vs. 1.9 days*	\$16,162 vs. \$17,255 vs. \$26,331**
Asthma	55 vs. 53 vs. 26 **	1.4 vs. 1.5 vs. 1.7days	\$15,755 vs. \$17,849 vs. \$25,528**

* $p < .01$; ** $p < .001$

HOUSING INSTABILITY AND FOOD INSECURITY AS BARRIERS TO HEALTH CARE: RESULTS FROM THE NATIONAL SURVEY OF AMERICAN FAMILIES. M. Kushel¹; R. Gupta²; L. Gee¹; J.S. Haas³. ¹University of California, San Francisco, San Francisco, CA; ²Harvard Medical School, Boston, MA; ³Brigham and Women's Hospital, Boston, MA. (Tracking ID #133744)

BACKGROUND: Competing demands on money and time may interfere with receipt of effective health care. For people living in poverty, obtaining adequate housing and food compete with obtaining health care as patient priorities. Housing instability and food insecurity describe having difficulty maintaining adequate housing or food. It is not known whether they are associated with poor health outcomes. We hypothesized that, in low-income housed persons, housing instability and food insecurity are independently associated with 1) poor access to ambulatory health care and 2) high rates of acute health care utilization.

METHODS: We conducted a secondary data analysis of factors associated with access to care and receipt of acute health care in the 1999 National Survey of American Families (NSAF), a nationally representative household survey of the adult non-institutionalized population ages 18-64. We restricted our analysis to households with income less than 200% of the poverty level. We constructed multivariate models to examine the association between housing instability (defined as having had difficulty paying one's rent or mortgage) or food insecurity (worrying food would run out, running out of food, or cutting out meals due to lack of money) and self-report of three access measures: 1) no usual source of care, 2) postponing needed medical care, 3) postponing needed medication, and two acute care measures 1) number of Emergency Department (ED) visits and 2) inpatient hospitalization.

RESULTS: 23.6% of subjects had housing instability and 42.7% had food insecurity. In models adjusted for age, gender, race, region, income, education, number of children, immigration status, receipt of public income support, insurance status and health status, housing instability was independently associated with all measures of poor access: having no usual source of care (AOR 1.38, 95% CI 1.15-1.64), postponing needed medical care (AOR 1.84, 95% CI 1.46-2.31) and medications (AOR 2.21, 95% CI 1.74-2.80) and both acute care measures: increased ED use (AOR 1.43, 95% CI 1.20-1.70) and hospitalizations

(AOR 1.30, 95% CI 1.01-1.67). Food insecurity was independently associated with postponing needed medical care (AOR 1.74, 95% CI 1.38-2.21) and medications (AOR 2.20, 95% CI 1.65-2.93) and increased ED use (AOR 1.39, 95% CI 1.17-1.66) and hospitalizations (AOR 1.42, 95% CI 1.09-1.85).

CONCLUSIONS: Among low-income housed adults, housing instability and food insecurity were independently associated with poor access to health care and high rates of acute health care. Even among non-homeless adults, having to struggle to obtain sufficient housing and food create competing demands that may contribute to poor health outcomes.

IMPACT OF HOSPITAL VOLUME ON RACIAL DISPARITIES IN CARDIOVASCULAR PROCEDURE MORTALITY. A.N. Trivedi¹; T.D. Sequist¹; J.Z. Ayanian¹. ¹Harvard University, Boston, MA. (Tracking ID #133278)

BACKGROUND: Hospital volume is a powerful predictor of mortality for many procedures. We assessed the relative contribution of hospital volume to overall disparities in cardiovascular procedure mortality.

METHODS: We analyzed 719,679 hospital discharges from the Nationwide Inpatient Sample from 1998 to 2001 with principal procedure codes for abdominal aortic aneurysm (AAA) repair, coronary artery bypass graft (CABG), percutaneous transluminal coronary angioplasty (PTCA), or carotid endarterectomy (CEA). For each procedure we compared in-hospital postoperative mortality rates for Blacks, Hispanics, and Whites, adjusting for comorbid illness using the Elixhauser method. The adjusted relation of race to receiving care in low-volume hospitals (below median) was assessed in logistic regression models controlling for age, sex, median income, location (urban/rural), payer, admission type (elective/urgent/emergent), and comorbidity. We analyzed the impact of hospital volume on racial disparities in risk-adjusted outcomes by predicting mortality with and without adjusting for hospital volume. All analyses used SAS-callable SUDAAN to account for the complex sampling and clustering of patients by hospital.

RESULTS: Blacks had greater risk-adjusted mortality than Whites following elective AAA repair (OR 1.84, 95%CI 1.20-2.84), CABG (OR 1.19, 95%CI 1.06-1.33), and CEA (OR 1.56, 95%CI 1.07-2.27), but not PTCA. Hispanics had greater mortality following AAA repair (OR 1.43, 95%CI 1.08-1.89), but not other procedures. After adjusting for multiple demographic and clinical characteristics, Blacks were significantly more likely than Whites to receive AAA repair (OR 1.78, 95%CI 1.47-2.17), CEA (OR 1.77, 95%CI 1.60-1.97) and CABG (OR 1.18 95%CI 1.12-1.24) in low-volume hospitals. Blacks were not more likely to receive PTCA in low-volume hospitals. Hispanics had significantly increased odds of receiving all four procedures in low-volume hospitals relative to Whites with adjusted ORs ranging from 1.72 (95%CI 1.67-1.78) for PTCA to 2.64 (95%CI 2.40-2.90) for CEA. Asians had the highest use of low-volume hospitals with an increased odds relative to Whites ranging from 1.96 (95%CI 1.84-2.08) for PTCA to 3.28 (95%CI 2.76-3.89) for CEA. Hospital volume was a significant predictor of mortality with hospitals in the lowest volume quartile having a 28% (CABG) to 130% (AAA) increase in risk-adjusted mortality relative to high-volume hospitals. Adjusting for hospital volume, however, had little effect (<6% change) on the relative odds of death for Blacks and Hispanics versus Whites.

CONCLUSIONS: Though minority patients are more likely to undergo cardiovascular procedures in low-volume hospitals, this racial difference in hospital volume is unlikely to explain a large portion of the racial and ethnic gap in post-procedure outcomes. Additional research is needed to determine why minorities are more likely to receive care in low-volume hospitals and experience higher mortality independent of hospital volume.

IMPROVED ANTIRETROVIRAL ADHERENCE AND HIV TREATMENT OUTCOMES AMONG HIV-INFECTED DRUG USERS ENROLLED IN METHADONE MAINTENANCE THERAPY. A. Palepu¹; M.W. Tyndall¹; R. Joy²; T. Kerr¹; N. Press¹; E. Wood¹; J.S. Montaner¹; R. Hogg¹. ¹University of British Columbia, Vancouver, British Columbia; ²British Columbia Centre for Excellence in HIV/AIDS, Vancouver, British Columbia. (Tracking ID #134525)

BACKGROUND: Since active drug use is a barrier to adhering to antiretroviral therapy (ART) and achieving optimal treatment outcomes, we examined the association of methadone maintenance therapy (MMT) on ART adherence and HIV treatment outcomes among a cohort of HIV-infected drug users.

METHODS: We obtained the demographics, drug and alcohol use, and addiction care history from the Vancouver Injection Drug User Study which is an open cohort study of injection drug users. The questionnaires were longitudinally linked to the British Columbia HIV/AIDS Drug Treatment Program to obtain ART adherence, clinical and HIV-1 treatment outcome. There were 278 VIDUS participants who accessed antiretroviral therapy through the Drug Treatment Program from August 1, 1996 to November 24, 2003. We constructed longitudinal logistic models using generalized estimating equations to examine the association of MMT with the following outcomes: adherence (95% as per pharmacy refill data); viral suppression (achieving at least two consecutive HIV-1 RNA levels below 500 copies/mL); CD4 cell rise of 50 cells/mm3. We adjusted for heroin use (at least weekly), gender, age, time on antiretroviral therapy and baseline viral load for the adherence model and included adherence as a co-variable in the HIV-treatment outcome models.

RESULTS: We found that MMT was positively associated with adherence (adjusted odds ratio [AOR] 1.49; 95% CI 1.12-1.98) whereas at least weekly heroin use, younger age were negatively associated with adherence. For the viral suppression analysis, we found a significant interaction between heroin use and MMT ($p=0.048$) and present the following AORs with the referent category being (heroin use and not on MMT, i.e. active drug use not in treatment): No heroin use

and on MMT (AOR 2.75; 95% CI 1.39-5.47); No heroin use and not on MMT (AOR 2.94; 95% CI 1.62-5.32); and Heroin use and on MMT (AOR 2.25; 95% CI 1.25-4.06). MMT was also positively associated with an increase in CD4 cell count (AOR 1.70; 95% CI 1.34-2.15).

CONCLUSIONS: Among HIV-infected injection drug users on ART, enrolment in MMT is associated with better adherence, HIV viral suppression and CD4 cell count response even after controlling for active heroin use. The integration of health care services that address addiction and HIV care may provide improved health outcomes for this vulnerable population and should be further explored.

INAPPROPRIATE HYPERTENSION MANAGEMENT OF BLACK PATIENTS IN PRIMARY CARE CLINICS. C.A. Umscheid¹; M.G. Weiner¹; C. Hollenbeak²; Y. Lin¹; S. Tang³; B. Turner¹. ¹University of Pennsylvania, Philadelphia, PA; ²Pennsylvania State University, Hershey, PA; ³Pfizer, Inc., New York, NY. (Tracking ID #136135)

BACKGROUND: Adverse health consequences of hypertension (HTN) disproportionately affect Black Americans. A key opportunity to reduce this risk occurs when Black patients visit their primary care physicians. In this study, we examined the effects of race on blood pressure (BP) control and HTN management in primary care clinics (PCC) affiliated with an academic health care system.

METHODS: We studied a retrospective cohort of adult HTN patients seen >3 times from 1/1/03 to 8/01/04 in one of eight primary PCC in urban and sub-urban Philadelphia. Demographic (i.e. sex, age, race, insurance, income), clinical (i.e. diabetes, CAD risk factors, morbid obesity), laboratory (i.e. creatinine) and health care utilization data (i.e. time since last visit, adherence to physician visits) was obtained from the PCC's electronic medical records (EMR) and system-wide administrative data. Patients met one of the following hierarchical HTN criteria: 1) a prescribed BP medication (med); 2) an ICD-9 diagnosis of HTN with at least one visit with a high BP; or 3) high BP on at least two visits. Race was self-designated or, when missing (8.4%), imputed from patient zip code. For each prescribed BP med, we determined the total daily dose and duration of treatment. High BP was based on JNC VI criteria and was defined as a systolic BP ≥ 140 or a diastolic BP ≥ 90 in non-diabetic patients and ≥ 130 or ≥ 85 in diabetic patients. For each visit, management was classified as appropriate if the BP was normal or, if high, the dose of a current BP med was increased, a new BP med was started, or a med was renewed if it had been expired for >1 month. Because the unit of analysis was the patient visit, random effects multiple logistic regression was used to adjust for clustering within each patient and physician. Separate models were estimated for diabetic and non-diabetic patients as well as visits with and without imputed race.

RESULTS: Our cohort of 13,683 HTN patients had 71,190 PCC visits to 258 clinicians. Black patients had 55% of all visits and diabetic patients had 32%. High BP was more common ($p<0.0001$) for Black patients than non-Black patients among the diabetic (59.3 vs. 52.5%) and non-diabetic (42.2 vs. 35.3%) cohorts. The adjusted odds ratio (AOR) of high BP for diabetic Black patients was 1.56 (95% confidence interval [CI] 1.34 to 1.80, $p<0.0001$) and non-diabetic Black patients was 1.63 (CI 1.46 to 1.81, $p<0.0001$). Inappropriate HTN management was also more common ($p<0.0001$) for Black patients among the diabetic (47.6 vs. 41.3%) and non-diabetic (32.3 vs. 27.3%) cohorts. The AOR of inappropriate management for diabetic Black patients was 1.33 (CI 1.19 to 1.50, $p<0.0001$) and non-diabetic Black patients was 1.31 (CI 1.21 to 1.42, $p<0.0001$). Race effect was robust in models that clustered on patient as well as physician and excluded visits with imputed race.

CONCLUSIONS: Not only are Black patients more likely to have uncontrolled HTN, they are also significantly more likely to have inappropriate HTN management than other patients in this cohort. Although these results come from a single health care system and BP measurement was not standardized, these data highlight important quality of care disparities in the management of Black patients with HTN.

INDIVIDUAL AND ZIP CODE LEVEL CHARACTERISTICS ASSOCIATED WITH TRUST OF THE HEALTH CARE SYSTEM. L. Morales¹; P. Robinson²; H. Liu¹; R. Hays¹. ¹University of California, Los Angeles, Los Angeles, CA; ²Charles R. Drew University of Medicine and Science, NA, CA. (Tracking ID #133191)

BACKGROUND: Differences in trust of health care providers and institutions have been linked to health disparities. Previous studies have been limited by the use of small sample sizes and a focus on the associations between trust and individual characteristics. By contrast, the present study uses a large nationally representative sample to examine the associations of zip code-level and individual characteristics with trust of the health care system.

METHODS: Responses by 143,723 individuals in 21,897 zip codes were analyzed using multilevel linear regression models. Sixty-nine percent of the study sample were female, 44% were between 45 and 65 years of age, 29% had a high school education or less, and 84% were white, 9% black and 4% Hispanic. The main outcome was a composite measure summarizing five questions about trust in hospitals, health plans, physicians, nurses and pharmacists (alpha=0.86). The individual level explanatory variables included age, gender, race/ethnicity, health status and educational attainment. The zip code level explanatory variables included percent black, percent Hispanic, percent Asian, percent foreign born, population density (persons per square mile), percent unemployed, percent white-collar workers, and median family income.

RESULTS: Among the individual-level explanatory variables, greater age, non-Hispanic black race/ethnicity (versus white) and less than high school, high school and some college (versus college or more) were associated with significantly greater trust. On the other hand, male gender and poor, fair, good or very

good health (versus excellent health) were associated with less trust. Controlling for all individual level variables, only percent black was associated with greater trust whereas percent Hispanic, percent Asian, percent foreign born, greater population density, percent white-collar workers and higher median family income were associated with less trust. All reported results were significant at $p < 0.05$.

CONCLUSIONS: Both individual and zip code level characteristics are significant independent predictors of trust of health care providers and institutions. Unexpectedly, we found that both black race/ethnicity at the individual level and percent black at the zip code level were associated with greater trust. It is important to examine whether this finding generalizes in other studies. Policy makers, health care providers and researchers should be aware of trust as a phenomenon that is shared among residents of neighborhoods. More research is needed to understand the determinants of trust at the neighborhood level so that interventions to reduce mistrust can be developed.

INDIVIDUAL-LEVEL TERRORISM PREPAREDNESS IN LOS ANGELES COUNTY. D.P. Eisenman¹; C. Wold²; A. Long³; J. Fielding⁴; L. Gelberg⁴. ¹University of California, Los Angeles & RAND Corporation, LA, CA; ²Los Angeles County Department of Health Services, Los Angeles, CA; ³Los Angeles Department of Health Services, Los Angeles, CA; ⁴University of California, Los Angeles, Los Angeles, CA. (Tracking ID #133597)

BACKGROUND: Knowing how household terrorism preparedness differs among vulnerable groups informs public health efforts to improve preparedness. Our study describes demographic variations in preparedness in Los Angeles County (LAC).

METHODS: We analyzed the LAC Health Survey, a random-digit telephone survey of LAC's adult population. 8,167 (58%) of 15,262 contacted households participated, answering 120 core questions. A random sub-sample of 1,041 answered 12 additional terrorism items. Interviews were conducted October, 2002–February, 2003 in English, Spanish, Mandarin, Cantonese, Korean, or Vietnamese. Outcome variables were responses to the items "In the past year, has anyone in your household done any of the following in response to the possibility of terrorism: 1) purchased or maintained emergency supplies of food, water or clothing? 2) Developed an emergency plan for you and your family?" Allowed responses were yes, no, don't know, refused. Covariates were: gender, age, race, education, income, language, U.S./non-U.S. born, disability status, household dependents $\leq$ age 18 and perceived likelihood of a terrorist event in LAC in the next 12 months. Univariate and bivariate analyses characterized the sample and the relationships between covariates and preparedness. Multivariate logistic regression generated adjusted odds ratios (AOR) and 95% confidence intervals (95% CI). The Behavioral Model for Vulnerable Populations was the framework for modeling preparedness, a personal health practice.

RESULTS: Most participants were ≥ 30 years old (75.36%), U.S. born (58.5%), reported no household dependents (57.31%), and believed an attack likely in the next year (59.03%). Overall, 28% purchased or maintained emergency supplies, 17% had an emergency plan, and 35% did either or both. 37% of Latinos and 31% of African Americans (AA) had emergency supplies compared to 21% of Whites and 19% of Asian/Pacific Islanders (API) ($p < 0.001$). More AA reported an emergency plan (28.3%) than any other group (Latino 15.6%, White 14.4%, API 17.0% $p = 0.02$). Persons with household dependents compared to persons without were more likely to report supplies (35.3% vs 23.5%; $p < 0.0001$) and having a plan (21.7% vs 12.9%; $p = .0001$). Non-U.S. born compared to U.S. born persons were more likely to report supplies (35.3% vs 23.4%; $p < .0001$) but not an emergency plan (15.9% vs 17.1%; $p = .619$). Physically disabled persons compared to non-disabled were more likely to have an emergency plan (22.3% vs 15.6%, $p = 0.027$). In multivariate analyses, factors associated with emergency supplies include AA (AOR 1.8; 95% CI 1.1, 3.1), household dependent (AOR 1.4; 95% CI 1.0, 2.0), non-U.S. born (AOR 1.9; 95% CI 1.3, 2.9), and high perceived likelihood of a bioterrorist attack (AOR 2.2; 95% CI 1.6, 3.0). Factors associated with an emergency plan included AA (AOR 2.6; 95% CI 1.5, 4.6), household dependent (AOR 2.4; 95% CI 1.6, 3.5), and physical disability (AOR 1.7; 95% CI 1.1, 2.7).

CONCLUSIONS: A year after the September 11th and anthrax attacks, most residents believed a terrorist attack in LAC was likely but only 35% had emergency supplies or plans. Low levels of emergency plans are especially concerning. AA, Latinos, persons with disabilities or dependents ≤ 18 , and non-U.S. born persons reported preparedness the most, indicating that some vulnerable groups (disabled, immigrants) may be particularly interested in improved preparedness. Our results inform LAC's household disaster preparedness interventions.

INHALED CORTICOSTEROIDS AND ASTHMA TREATMENT - MODIFIABLE BARRIERS TO MEDICATION ADHERENCE AMONG URBAN ADULTS WITH ASTHMA. M. Sajous¹; C. Mancuso²; C. Boutin-Foster¹. ¹Cornell University Medical College, New York, NY. (Tracking ID #131582)

BACKGROUND: Despite the known benefits of inhaled corticosteroids in the management of asthma, primary care physicians often find it challenging to foster adherence among their patients. The objective of this cross-sectional study is to identify potentially modifiable psychosocial variables associated with adherence to inhaled corticosteroids among adults with asthma.

METHODS: 101 adults with persistent asthma previously prescribed an inhaled corticosteroid were recruited from an urban general medicine outpatient practice. Comorbidity was assessed with the Charlson Comorbidity Index. The modifiable variables assessed were quality of life, stress, and coping style. Per-

ceived quality of life was measured with the Mini-Asthma Quality of Life Questionnaire, 15-items with a mean score of 1-7. Stress was measured with the Perceived Stress Scale, 10-items with a summed score of 0-40. Coping was measured with the Ways of Coping with Asthma in Everyday Life Scale, which includes six dimensions of coping: positive reappraisal, information seeking, restricted lifestyle, worry, hiding and ignoring asthma; this scale has 30 items with a mean score of 1-4 for each component. The primary outcome, adherence, was measured with a modified Morisky Medication Adherence Scale for patients with asthma, 6-items with a summed score of 0-24 (lower score indicates greater adherence). Logistic regression was used for data analysis. Univariate regression was used to determine the variance in adherence explained independently by each psychosocial variable and covariate control variables. The variables found to have statistically significant association with adherence were analyzed with multivariate regression.

RESULTS: Mean age was 42; 92% were female; 45% were African American, 42% were Hispanic; 50% had not completed High School; 81% were unemployed; 95% were on Medicaid; 54% reported an annual household income of less than \$10,000, and 90% reported less than \$30,000; 83% had a comorbidity score of two or less. Scores on the adherence scale had a bimodal distribution (Mode: 0 and 24; SD: 8.2) with 14% reporting complete adherence and 17% reporting complete non-adherence. In univariate analysis, for every unit increase in hiding as a coping mechanism (Mean: 1.9, SD: 0.8, higher score indicates more hiding), the odds of being adherent decreased by 0.51 times (95% CI: 0.29 - 0.86). In multivariate analysis, hiding remained significant, and for every unit increase in worry as a coping mechanism (Mean: 2.4, SD: 0.9, higher score indicates more worry) the odds of being adherent increased by 1.7 times (95% CI: 1.07 - 2.93). Better perceived quality of life (Mean: 3.9, SD: 1.3, higher score is better) was associated with decreased adherence to inhaled corticosteroids but was not statistically significant. Perceived stress (Mean: 27.8, SD: 8.2, higher score indicates more stress) was not associated with adherence.

CONCLUSIONS: The findings of this study indicate that patients who have a tendency to hide or to worry less about their asthma are less likely to be adherent with their medication regimen. In addition, patients who perceive themselves as being not ill are less likely to take their medications. Future studies should further investigate why some patients are more likely to dismiss their asthma and how perceived quality of life influences adherence. Patients should be educated on the importance of taking their inhaled corticosteroid in order to prevent exacerbations despite feeling well.

IS PERCEPTION OF CORONARY HEART DISEASE RISK ASSOCIATED WITH HEALTH BEHAVIORS IN AN INNER-CITY POPULATION? J.M. Barnhart¹; O. Cohen²; A. Alvarez¹; S. Sanchez¹. ¹Albert Einstein College of Medicine, Bronx, NY; ²Ferkau Graduate School of Psychology, Yeshiva University, Bronx, NY. (Tracking ID #134288)

BACKGROUND: Little is known about risk perception of coronary heart disease (CHD) among inner-city populations. The main objective of this study was to pilot test a novel scale designed to measure personal risk perception of CHD and examines its relationship to health behaviors.

METHODS: Subjects from 3 ambulatory clinic centers in the South Bronx were administered the Risk Perception Survey for Developing CHD. The 16 risk items were rated on a Likert scale with 1 = strongly disagree to 6 = strongly agree and scored so that higher values implied heightened perceived risk. Factor analyses confirmed 4 subscales: CHD worry, perceived health status, perceived CHD vulnerability, and self-efficacy (I don't mind exercising). Using self-reported data on diet and exercise, a composite health behavior variable was created to determine its relationship to risk perception. This composite variable was assigned points so that a higher score indicated healthier behaviors. Here we report data on the first 100 respondents recruited (50% of sample).

RESULTS: The internal consistency of the scale was acceptable (Cronbach's $\alpha = 0.72$; inter-item total correlation = 0.32). The mean age of the first 100 respondents was 56.6 yrs. (9.9); 63% were Hispanic and 27% were Black. The mean summary score for CHD risk perception was 54.1 (SD = 9.9; range 44-72) and the mean score for health behaviors was 27.4 (SD = 5.8; range 11-40). Only half of the respondents (50.5%) knew CHD was the number cause of death in adults. There was an inverse association of perceived risk of CHD and actual health behaviors ($r = -0.20$; $p = 0.04$). There were no differences in risk perception scores by age, race/ethnicity, sex, knowledge, or marital status. However, women and persons with hypercholesterolemia were more likely to engage in healthy behaviors than men and persons with normal cholesterol (mean scores, 28.4 vs. 25.4; $p = 0.012$ for sex; 29.3 vs. 25.5; $p = 0.002$ for cholesterol), respectively. In multiple linear regression models, after controlling for demographics, CHD knowledge, and risk perception, sex ($p = 0.026$) and hypercholesterolemia ($p = 0.008$) remained significant predictors for health behaviors.

CONCLUSIONS: In this sub-sample, there was low-moderate perception of coronary risk, which was weakly related to health behaviors. Sex and a clinical diagnosis of hypercholesterolemia appear to be stronger predictors for health behaviors. A culturally-sensitive risk communication intervention is needed to improve CHD knowledge and health behaviors among inner-city residents, especially men.

IS SELF-EFFICACY CONSISTENTLY ASSOCIATED WITH SELF-MANAGEMENT AMONG DIVERSE PATIENTS WITH DIABETES? U. Sarkar¹; D. Schillinger¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #135387)

BACKGROUND: While prior research demonstrates that improving diabetes self-efficacy can improve self-management behavior, little is known the applicability of this research to ethnically diverse, and low-income patients with di-

abetes. We examined the relationship between diabetes self-efficacy, or patients' confidence in their ability to manage their disease, and self-management behavior, in an urban, diverse population.

METHODS: We identified patients with type 2 diabetes from the clinical database of 2 primary care clinics at a university-affiliated public hospital in San Francisco. Bilingual trained interviewers administered an oral questionnaire to 408 patients in English or Spanish. We measured diabetes-specific self-efficacy using a validated eight-item scale in which patients rate their confidence in their ability to perform recommended self-management routines. To measure diabetes self-management, we used an established instrument that asks patients to report their behavior for the prior 7 days in five domains: diet, exercise, self-monitoring of blood glucose (SMBG), foot care, and medication adherence. Respondents self-reported their ethnicity. We administered an abbreviated form of the short-form Test of Functional Health Literacy in Adults (s-TOHFLA), in English or Spanish. We separately analyzed the relationship between self-efficacy score and the proportion reporting optimal performance in each self-care domain. We then examined whether the observed relationships varied by ethnicity and literacy level.

RESULTS: The study participants were ethnically diverse (18% Asian, 25% Black, 42% Latino, 15% White), and 48% had adequate functional health literacy (s-TOHFLA score >23). The mean self-efficacy score was 74/100 (SD 18). The internal consistency-reliability for the self-efficacy scale was high (standardized Cronbach alpha 0.78) and was highly consistent across ethnicity and literacy levels. The proportion of patients who reported optimal self-management over the prior week ranged from 33% for diet to 64% for medication adherence. Diabetes self-efficacy was associated with four of the five domains of self-management ($p < 0.01$). With each 10% increase in self-efficacy score, patients were more likely to report optimal diet (0.13 days more per week), exercise (0.09 days more per week), SMBG (OR 1.15), foot care (OR 1.20), but not medication adherence (OR 1.00, $p = 0.40$). The associations between self-efficacy and self-management were consistent across adequate (s-TOHFLA >23), marginal (s-TOHFLA 17-22), and inadequate (s-TOHFLA 0-16) literacy levels for diet, exercise, foot care, and SMBG. Similarly, self-efficacy was consistently associated with self-management across all 4 ethnic groups, with the following exceptions. Among Asian participants, we did not detect an association for diet; among African Americans, we did not observe an association for exercise; among Latinos, we did not observe an association for SMBG. For medication adherence, which was not associated with self-efficacy score in the population overall, a significant association was present in the white group (OR 1.60, $p = 0.02$), and a borderline association was seen among African Americans (OR 1.03, $p = 0.06$).

CONCLUSIONS: In a diverse urban population of low-income patients with type 2 diabetes, self-efficacy was a robust predictor of self-care, both across domains of self-care and across literacy and ethnicity. While further research should explore the ethnic variation in the relationship between self-efficacy and specific domains of self-management, self-efficacy appears to represent an appropriate target for interventions among diverse patients with diabetes.

IS WHAT WE HAVE HERE A FAILURE TO COMMUNICATE? A STATEWIDE EVALUATION OF THE ADEQUACY OF HOSPITAL INTERPRETER SERVICES FOR PATIENTS WITH LIMITED ENGLISH PROFICIENCY. D. Salas-Lopez¹; G. Flores²; S. Torres²; L. Holmes³; S. Toman². ¹University of Medicine and Dentistry of New Jersey, Newark, NJ; ²Medical College of Wisconsin, Milwaukee, WI; ³Office of Minority & Multicultural Health, New Jersey Dept of Health & Senior Services, Trenton, NJ. (Tracking ID #134225)

BACKGROUND: Forty-seven million Americans speak a non-English language at home and 21 million (including 4 million children) are limited in English proficiency (LEP). With 11% of its population LEP, New Jersey is a model state for studying interpreter issues. The objective of this study is to assess the adequacy of interpreter services in NJ hospitals meeting LEP patients' needs.

METHODS: Cross-sectional survey in which representatives at all 122 NJ hospitals were contacted to answer 37 questions on the characteristics of hospitals and their patients, interpreter services, and perspectives on resources/policies needed to provide quality services to LEP patients.

RESULTS: Sixty-seven hospitals completed the survey (55% response rate). A median of 9% of staff and 33% of physicians are bilingual. A median of 93 patients/hospital needed interpreters (range = 0-15,000), and the median interpreter services budget was \$10,063 (range = \$0-458,000). Ninety-seven percent of hospitals use phone interpreter services. Approximately 87% of hospitals have no interpreter services department, 19% offer no written translation services, and 31% lack multilingual signs. Only 3% of hospitals have a full-time interpreter, indicating a ratio of 1 interpreter per 235,769 LEP persons in NJ. Eighty percent of hospitals offer no staff training on working with interpreters. Three hospitals had more extensive interpreter services considered to be model programs. Hospitals stated that the following would improve interpreter services at their institutions: more funding; federal/state interpreter certification/guidelines; more MD/staff education on interpreter use; listing locally available interpreters; and identifying bilingual staff. Most hospitals said third-party reimbursement for interpreter services would be beneficial, by reducing costs, adding full-time interpreters, freeing staff to provide other services, meeting future population growth, and improving communication and education.

CONCLUSIONS: Most New Jersey hospitals have no formal interpreter services department; 97% have no full-time interpreter, 80% provide no staff training on working with interpreters, and deficiencies were noted in hospital signage and translation services. Most NJ hospitals stated that third-party reimbursement for interpreter services would benefit their hospitals.

LANGUAGE CONCORDANCE AND DIABETES CARE IN MANAGED CARE. A.F. Brown¹; D.M. Zulman¹; W.N. Steers¹; M. Liu¹; C.M. Mangione¹. ¹University of California, Los Angeles, Los Angeles, CA. (Tracking ID #134571)

BACKGROUND: Inadequate access to Spanish-speaking providers or translation services may contribute to health disparities among Latinos with diabetes. We evaluated whether patient-provider language concordance influences diabetes care in a setting where access to Spanish-speaking physicians is high.

METHODS: We analyzed data from the 23 provider groups in the Texas site of Translating Research into Action for Diabetes (TRIAD), a multicenter study of diabetes care in managed care. Patient information was obtained from surveys and medical records and provider information from surveys and administrative data. The Reference Group, Latino Spanish-speaking patients seen by providers who speak Spanish (SP PT/SP MD), was compared to Latino English-speaking patients seen by physicians who do not speak Spanish (LE PT/ENG MD), Latino English-speaking patients seen by providers who speak Spanish (LE PT/SP MD), and white patients (WHITE PT). There were too few Spanish-speaking patients seen by providers who did not speak Spanish to include in these analyses. Multivariate models were adjusted for patient age, sex, income, education, and duration of diabetes; physician age, specialty, and sex; and clinician-level clustering. Dependent variables included satisfaction with getting needed care, provider communication, and the courteousness and helpfulness of the office staff from the Consumer Assessment of Health Plans Survey (CAHPS); process of care (rates of HbA1c, lipid profile, and nephropathy assessment, dilated eye examinations, aspirin use, and foot examinations); and outcomes (glycemic, lipid, and blood pressure control).

RESULTS: The response rate was 69% for the patient survey. We had data on 99% of the physicians. Among the 363 physicians, 48% were fluent in Spanish. Only 28 Spanish-speaking patients were cared for by physicians who did not speak Spanish. In adjusted analyses, SP PT/SP MD had the highest adjusted rates of satisfaction with getting needed care, physician communication, and the office staff (see Table). They also had lower rates of HbA1c assessment than LE PT/SP MD and higher HbA1c levels than whites. There were no differences in other processes or outcomes.

CONCLUSIONS: Spanish-speaking Latino patients with diabetes generally selected bilingual providers. Although these patients were more satisfied with their care than other Latinos or whites and had comparable processes and outcomes across most indicators, they had lower rates of HbA1c measurement and poorer glycemic control. Further work is needed to identify reasons for persistent disparities among Spanish-speakers despite language concordance. Differences in Satisfaction and Quality by Patient/Provider Language: Mean or % (P)

	SP PT/SP MD (Reference Group)	LE PT/SP MD	LE PT/ENG MD	WHITE PT
N	184	627	249	757
Getting Care (%)*	86	75 (0.005)	73 (0.007)	73 (0.003)
Communication (%)	68	61 (NS)	53 (0.005)	54 (0.002)
Office Staff (%)*	89	79 (0.006)	70 (0.0001)	74 (0.003)
HbA1c assessed (%)	84	90 (0.05)	84 (NS)	83 (NS)
HbA1c (mean %)	8.05	7.80 (NS)	7.84 (NS)	7.34 (<0.0001)

*% w/highest score

NS=Not Significant vs. Ref. Group

LANGUAGE OF RESPONSE TO A HEALTH INTERVIEW SURVEY AND ACCESS TO CARE AMONG NEW YORK CITY LATINOS. S.Y. Angell¹; C.E. Brown¹; T. Matte¹. ¹New York City Department of Health and Mental Hygiene, New York City, NY. (Tracking ID #135330)

BACKGROUND: Measures of "acculturation", such as English language proficiency and, for the foreign-born, duration of residence in the US, have been associated with disparities in health care access, but few population-based studies have evaluated these associations within the Latino population. We assessed the relation of indices of acculturation to measures of limited health care access among Latino adults residing in New York City (NYC).

METHODS: We analyzed data for adult respondents to the 2003 NYC Community Health Survey who identified themselves as Hispanic, were aged 18-64 years, and were not covered by Medicare (N=1,943). Two indicators of limited access to care were used as dependent variables: usually going to the emergency department when sick or in need of health advice (ED), and having no primary care provider (NOPCP). Two independent variables were constructed to reflect acculturation among foreign-born Latinos relative to US-born: language in which the individual chose to be interviewed (English vs. Spanish) and duration of residence in the US (<4 years, 4-10 years, and 10 or more years). Separate logistic models were fit for each of the two access (dependent) variables and acculturation (independent) variables. Crude and adjusted odds ratios were compared to assess the extent to which observed disparities in access measures among Latino subgroups were explained by demographic factors associated with access to care, including age, gender, household income, and type of health insurance.

RESULTS: In our sample of Latino respondents, greater than 50% of respondents were between the ages of 25-44, just over one third were male, and almost

half had a household income less than \$25,000. Nineteen percent were uninsured. Thirty-two percent of the respondents reported NOPCP and 17% reported ED. Compared with US-born Latinos, foreign-born Latinos interviewed in Spanish were more likely to report NOPCP (OR=2.9, 95% CI 2.2-3.9) and ED (OR=2.0, 95% CI 1.4-2.9) (table). These differences were attenuated somewhat, but persisted with adjustment for demographics and health insurance (adjusted ORs 2.2, 95% CI 1.5-3.1 and 1.5, 95% CI 0.9-2.3, respectively). The increased odds of NOPCP and ED among the foreign-born Latinos interviewed in Spanish remained significant when compared with foreign-born Latinos interviewed in English (table). There were no significant differences in ED and NOPCP between foreign-born and US born Latinos interviewed in English in the crude or adjusted models. When the models were assessed using the independent variable reflecting time of residency in the US, significant crude associations with NOPCP and ED were largely explained by adjusting for insurance status and demographics.

Odds of No Primary Care Provider and Emergency Department Use by Place of Birth and Language

	No Primary Care Provider		Use Emergency Department	
	Crude OR (95% CI)	Adjusted OR (95% CI)	Crude OR (95% CI)	Adjusted OR (95% CI)
Place of Birth and Language of Interview				
US born/English	1.0	1.0	1.0	1.0
Foreign born/English	1.0 (0.7-1.5)	1.1 (0.8-1.6)	0.8 (0.5-1.2)	0.8 (0.5-1.2)
Foreign born/Not English	2.9* (2.2-3.9)	2.2* (1.5-3.1)	2.0* (1.4-2.9)	1.5* (0.9-2.3)

Adjusted for age, gender, household income and insurance status

* Significant difference between Foreign-born English vs. not English ($p < 0.01$)

CONCLUSIONS: Among NYC Latinos, those who were foreign born and chose to respond to this health survey in Spanish instead of English were more likely to report having no primary care provider and relying on the ED as a usual source of care. This suggests that within the foreign-born NYC Latino population, being less able or comfortable using English to discuss health topics may be a marker of language or cultural barriers to accessing care.

LATINOS, LANGUAGE AND TRUST IN HEALTH CARE. E. Jacobs¹; G. Juarez Dominguez²; R. Warnecke²; C. Ferrans². ¹Rush University Medical Center, Chicago, IL; ²University of Illinois at Chicago, Chicago, IL. (Tracking ID #133503)

BACKGROUND: An abundance of research has documented disparities in health and health care between Latinos and Caucasians. One possible reason for these disparities is that Latinos are less trusting of physicians and health care institutions and are therefore less likely to seek or accept care.

METHODS: Six focus groups (n=49) of Spanish-speaking Latinos living in Chicago were conducted. Participants were recruited from a public hospital clinic and a non-profit community based organization. A bilingual moderator facilitated the group discussions in Spanish using a semi-structured moderator guide. It included questions about trust and distrust in physicians, trust and distrust in health care institutions such as clinics and hospitals, and how trust and distrust influences health care decision making. Discussions were audiotaped, transcribed, imported into Atlas.ti and coded for themes by two independent raters using grounded theory analysis.

RESULTS: Several major themes were identified. Regarding physicians, trust is determined by the interpersonal and technical competence of physicians. The race/ethnicity of the physician and fears of discrimination based on immigration status rarely played a role in determining trust. Regarding institutions, trust is determined by the technical competence of the institution's operation and the interpersonal competence of its staff. Mistreatment by clinic and hospital staff, particularly African American staff, was commonly reported. For both physicians and institutions, perceptions of language based discrimination and problems accessing health care due to difficulties speaking English contributed to distrust. Consequences of distrust included reduced adherence to recommended treatment and desire to change physicians and health care institutions.

CONCLUSIONS: Distrust among Latinos arises from lack of linguistically accessible, culturally and technically competent care. Reducing language barriers and improving the cultural competence of care at both the physician and institutional level would likely reduce current health care disparities in the Latino Community through enhanced trust and improved access to care.

LOW RATES OF COLORECTAL, CERVICAL, BREAST, AND PROSTATE CANCER SCREENING IN ASIAN AMERICANS COMPARED TO NON-HISPANIC WHITES: CULTURAL FACTORS OR ACCESS TO CARE? N. Kandula¹; M. Wen²; E. Jacobs³; D. Lauderdale⁴. ¹Northwestern University, Chicago, IL; ²University of Utah, Salt Lake City, UT; ³Rush University Medical Center, Chicago, IL; ⁴University of Chicago, Chicago, IL. (Tracking ID #132889)

BACKGROUND: Asian Americans have lower rates of cancer screening compared to non-Hispanic Whites. Little is known about the mechanisms underlying racial disparities in cancer screening. The objective of this study was to determine whether cultural factors, such as nativity, duration of residence in the U.S., and English language proficiency, explain disparities in colorectal, breast, cervical, and prostate cancer screening in Asians and non-Hispanic Whites after accounting for socioeconomic status and access to care. In addition, we compared reasons that Asians and non-Hispanic Whites reported when asked why they did not obtain cancer screening.

METHODS: This was a cross-sectional population-based study in California. Study respondents were 36,729 non-Hispanic Whites and 3,956 Asians who participated in the California Health Interview Survey between November 2000 and October 2001. We compared self-reported completion of colorectal, breast, cervical, and prostate cancer screening among Asians and non-Hispanic Whites using multivariate logistic regression.

RESULTS: Foreign-born Asians were significantly less likely to report colorectal (adjusted odds ratio [AOR]=0.71, 95% confidence interval [CI] 0.58, 0.88), breast (AOR=0.58, 95% CI 0.49, 0.70), cervical (AOR=0.28, 95% CI 0.24, 0.34), and prostate cancer (AOR=0.42, 0.31, 0.55) screening compared to non-Hispanic Whites after adjusting for SES, insurance, access to care, and English proficiency. US-born Asians were only less likely to report cervical cancer screening. When asked what the most important reason was for not having each test, Asians were more than twice as likely than non-Hispanic Whites to report, "haven't had problems/symptoms." ($p < 0.01$ for each screening test).

CONCLUSIONS: Cultural factors may be important mediators of cancer screening disparities in Asians and non-Hispanic Whites. Asian Americans may believe that cancer-screening tests are in response to specific symptoms, rather than tests that are used prior to development of symptoms. On an individual level, health care providers may need to ask Asian American patients about their beliefs regarding cancer screening and counsel them about the potential benefits of cancer screening even in the absence of symptoms. Education and outreach aimed at increasing cancer screening rates in Asian Americans must consider how to effectively communicate the message that, "cancer-screening tests are needed, despite the absence of symptoms."

MEDIATORS OF THE RELATIONSHIP BETWEEN LITERACY AND DIABETES OUTCOMES. D.A. Dewalt¹; R.S. Boone¹; M.P. Pignone¹. ¹University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #134817)

BACKGROUND: One previous study demonstrated an association between low literacy and worse glycemic control among patients with diabetes, but the mechanism by which low literacy affects glycemic control is not well understood. We sought to evaluate the relationship between literacy and glycemic control among patients with type 2 diabetes and to explore the role of four potential mediators: trust, self-efficacy, desire to participate in medical decision-making, and diabetes related knowledge.

METHODS: We performed a cross-sectional study of adults with type 2 diabetes in the general internal medicine practice at the University of North Carolina. We identified consecutive patients who had not been participants in earlier studies. We measured literacy using the Rapid Estimate of Adult Literacy in Medicine (REALM) and defined low literacy as less than a 7th grade reading level. For outcomes, we measured most recent glycosylated hemoglobin (A1C) and had patients complete a validated diabetes related quality of life (DRQOL) instrument (scale 1-4). For potential mediators, we measured diabetes knowledge (scale 0-10), diabetes self-efficacy (scale 1-10), desire to participate in medical decision-making (scale 25-100), and trust in physician (scale 10-50) using validated questionnaires. All questionnaires were administered verbally by a research assistant to mitigate the effect of literacy on measurement. We performed bivariate analyses of the relationship between literacy and diabetes outcomes, and the potential mediating variables.

RESULTS: We enrolled 216 patients. Mean age was 59 years (range 24-83), 51% were African-American and 46% were Caucasian. 62% reported an income less than \$15,000/year, 18% were uninsured, 29% had Medicaid, 58% had Medicare, and 24% had private insurance. 43% had not completed high school, and 18% had low literacy on REALM. Mean A1C was 7.6% (SD=1.6). We found no relationship between literacy and A1C (low literacy (LL) 7.6% vs. higher literacy (HL) 7.6%, $p=0.9$), or between literacy and DRQOL (LL 2.2 vs. HL 2.2, $p=0.85$). Literacy was also not related to self-efficacy (LL 8.0 vs. HL 7.7, $p=0.15$), or trust in physician (LL 46 vs. HL 46, $p=0.98$). Higher literacy was related to a greater desire to participate in medical decision-making (LL 80 vs. HL 85, $p=0.0006$), and more diabetes related knowledge (scale 0-10) (LL 4.6 vs. HL 5.9, $p=0.0002$). Of the measured mediator variables, only diabetes self-efficacy trended toward a relationship with A1C (Person's correlation -0.12, $p=0.07$). (Results were similar if low literacy was defined as less than 9th grade level). Those who were uninsured (18% of sample) had higher A1C (8.4% vs. 7.5%, $p=0.0017$), but years of education and income were not related to A1C.

CONCLUSIONS: Low literacy is not associated with poor glycemic control in this population. Literacy is related to desire to participate in medical decision-making and knowledge, but these variables are not related to glycemic control. Other factors, including insurance status, may better explain differences in glycemic control.

MENTAL ILLNESS AND DISPARITIES IN FOLLOW-UP TESTING FOR HYPERCHOLESTEROLEMIA. R.A. Kaplowitz¹, M. Horng², M. Gaziano¹. ¹Massachusetts Epidemiology Research and Information Center, Boston, MA; ²Brigham and Women's Hospital, Boston, MA. (Tracking ID #133454)

BACKGROUND: Previous studies have found that excess mortality from cardiovascular disease in people with mental disorders can be attributed to disparities in quality of care. We examined disparities in the follow-up evaluation of abnormal cholesterol levels in this vulnerable population.

METHODS: Using a retrospective cohort study design, we analyzed the administrative data of 32,405 U.S. veterans who met VA guidelines for follow-up cholesterol testing with a fasting lipid panel. Abnormal cholesterol levels requiring follow-up were a total cholesterol level ≥ 200 mg/dl or a high-density lipoprotein cholesterol (HDL-C) level < 40 mg/dl in the absence of cardiovascular disease (CVD), or a low-density lipoprotein cholesterol (LDL-C) level ≥ 100 mg/dl in veterans with known CVD. Veterans with an ICD-9 diagnosis code for a mood disorder, thought disorder, or an anxiety disorder concurrently treated with a psychiatric medication were compared with veterans who had neither a psychiatric diagnosis nor a psychiatric medication. Cox proportional hazards models determined the likelihood of follow-up testing after an abnormal cholesterol test associated with mental illness, adjusted for outpatient service utilization, age, gender, and the presence of an ICD-9 diagnosis code for CVD, tobacco use, diabetes mellitus, hypertension, alcohol abuse, and substance abuse.

RESULTS: 6,683 veterans (20.6% of the study population) had a treated mental illness and an initially abnormal cholesterol test. After a median follow-up time of 1.3 years, 45.4% of veterans with mental illness and 49.1% of veterans without mental illness received a fasting lipid profile ($p < 0.001$). The unadjusted hazard ratio (HR) for follow-up testing associated with mental illness was 0.89 (95% confidence interval 0.85, 0.92). The HR for follow-up testing associated with any mental illness, adjusted for outpatient service utilization, demographic data, and clinical covariates was 0.72 (95% confidence interval 0.68, 0.75).

CONCLUSIONS: Our findings suggest that veterans with mental illness receive less follow-up for abnormal cholesterol levels than veterans without mental illness. Further studies should examine the impact of incomplete follow-up for hypercholesterolemia on disparities in cardiovascular outcomes.

NON-RESIDENTIAL NEIGHBORHOOD EXPOSURES SUPPRESS NEIGHBORHOOD EFFECTS ON HEALTH. S. Inagami¹, D. Cohen², B. Finch³. ¹UCLA/WLAVA HSR&D, Los Angeles, CA; ²RAND Corporation, Santa Monica, CA; ³RAND Corporation/The Pardee RAND Graduate School, Santa Monica, CA. (Tracking ID #135319)

BACKGROUND: Researchers in the past decade have begun to examine residential neighborhood level predictors in order to explain the social and structural mechanisms that lie behind the geographic distribution of disease. But most neighborhood predictors have shown weak, inconsistent and variable results across specific causes of mortality and morbidity, across gender and race, and across different study samples. It is easy to dismiss these mixed results as an indication that contextual qualities of the residential neighborhood are poor and weak predictors upon an individual's health. But these mixed results may also reflect the variability of effect that contextual qualities of the residential neighborhood have on the individuals who reside there. Exposure of individuals to different environments may influence the benefits or the risks associated with the individual's residential neighborhood. To determine if other exposures modify residential neighborhood effects we examined the effect of other environments that figure in daily routines on adult self-reported health.

METHODS: We linked the 2000 US Census data with the Los Angeles Family and Neighborhood Study database consisting of 3407 adults sampled from neighborhoods and households in LA County. Neighborhood characteristics of census tracts where respondents lived, worked, obtained health care, shopped, and worshipped were obtained from the 2000 US Census. Multilevel linear regression and ordered logistic models were used to estimate associations between self-reported health and non-residential neighborhood exposures after adjustment for individual-level factors and exposure to residential neighborhoods.

RESULTS: Living in deprived neighborhoods was associated with worse self-reported health; models including exposure to non-residential neighborhoods increased this association. In a dose-response fashion, the greater the exposure to less deprived non-residential neighborhoods, the greater the magnitude in improved self-reported health.

CONCLUSIONS: People who spent time in less deprived neighborhoods had better self-reported health compared to those who stayed in deprived neighborhoods. Inclusion of non-residential neighborhood exposure in the model increases the predictive validity of residential neighborhood exposure, indicating its role as a suppressor variable. Non-residential neighborhood exposure may explain why previous studies may not have found robust associations between residential neighborhood predictors and health. Reform of clinical practice cannot be severed from the social structure and dynamics from which diseases arise. Understanding how social and structural geography contributes to health and disease can allow for more innovative and coherent ways to reforming clinical practice. **TABLE FOOTNOTES:** *Health is reverse coded, with 1 being Excellent and 5 being Poor. **Homeses: quartile of residential neighborhood disadvantage level (vlo being most disadvantaged, vhi being least disadvantaged); Ref category: Homeses vhi. \$P1-P3: graded doses of exposure to more advantaged neighborhoods. (P1=smallest; P3=greatest). §§N1: exposure to more disadvantaged neighborhoods. MODEL A: Model does not include exposure to non-residential neighborhoods. MODEL B: Model includes exposure to disadvantaged and graded doses of more advantaged neighborhoods; Reference category: those whose non-residential neighborhood exposure have similar disadvantage scores to their neighborhood of residence.

Table 1: Multivariate Hierarchical Linear Models of Non-Residential Neighborhood Exposures and Residential Neighborhood Exposures on Self-Reported Health*

	Total	Sample	(n =	3407)
	MODEL A	MODEL A	MODEL B	MODEL B
	β	P	β	P
Homeses** Vlo	.284	.000	.532	.000
Homeses Lo	.163	.029	.337	.000
Homeses Hi	.055	.451	.143	.000
P3§			-.241	.001
P2			-.151	.028
P1			-.071	.286
N1§§			.090	.164
Log Income	-.147	.000	-.141	.000
Car Ownership	-.126	.023	-.106	.057
Log Age	.704	.000	.703	.000
College	-.333	.000	-.324	.000
Employed	-.208	.000	-.187	.000
Female	.132	.003	.136	.002
Married	.031	.490	.028	.536
Latino	.327	.000	.274	.000
Black	.139	.101	.102	.228
Asian	.052	.565	.062	.492

OLD AND SLEEPING ROUGH: ELDERLY HOMELESS PERSONS ON THE STREETS OF BOSTON. J.J. O'Connell¹, J.S. Roncarati¹. ¹Boston Health Care for the Homeless Program, Boston, MA. (Tracking ID #136232)

BACKGROUND: Homelessness in the waning years of life is a personal tragedy and a societal disgrace. Little is known about older individuals struggling to live out their days in parks, down back alleys and under bridges. In order to improve our understanding of this sub-group of older homeless persons, we prospectively followed 30 elderly individuals living on Boston's streets during the 5-year period from 2000-2004.

METHODS: The Boston Health Care for the Homeless Program's (BHCHP) Street Team has cared for over 800 individuals living on the streets of Boston each year. In January 2000, a cohort of 30 persons age 60 or older was identified, including 8 (27%) women and 22 (73%) men. The ages ranged from 60-82 years, with an average of 67 years. 28 (93%) had been homeless for at least a decade, of whom 14 (47%) had been homeless for over two decades. The two remaining persons had been episodically homeless for five years. All suffered from multiple chronic medical illness. 22 (73%) had a documented DSM-IV diagnosis (7 (23%) schizophrenia, 6 (20%) bipolar illness, 1 (3%) with OCD, 8 (27%) with anxiety disorder or chronic depression) and 17 (57%) had active substance abuse. Contact data is recorded on each patient at a weekly meeting and other data sources are BHCHP's medical record, hospital records at MGH and BMC and a supplemental street database.

RESULTS: Ten (33%) individuals died during the five year observation period, nine men and one woman. Six of the decedents were men still living on the streets at the time of their deaths. Two deaths were traumatic: one suffered head trauma in an assault, while another was crushed by a car that over turned while he was sleeping next to a Jersey barrier. The other four men died of complications of chronic medical problems. One with advanced cirrhosis was found in a park after succumbing to a massive hemorrhage from esophageal varices, another with advanced diabetes was found in a park and later died of sepsis, another with COPD was found unconscious outside a shelter and later died in the ICU of aspiration pneumonia. The oldest member of this cohort died of heart failure at 82 years. The four other deaths occurred in persons who had been placed in nursing homes and assisted living facilities. One man and one woman died of respiratory failure caused by COPD. One man with mental retardation died of colon cancer, and another man with organic brain disease died of complications from a seizure. At the end of five years, 5 men and 3 women are in nursing homes. Only 5 (17%) have been successful in finding housing. Of the remaining 8 persons, 7 (23%) remain on the streets after 5 years and one has been lost to follow up.

CONCLUSIONS: The high morbidity and mortality among this cohort of thirty elderly rough sleepers poses significant challenges seeking to provide housing and health care services for this vulnerable sub-group of homeless persons and goals to continue to improve continuous primary care, despite marked difficulties in reaching out to this elusive population is imperative.

OUTPATIENT DIABETES CARE FOR HOMELESS AND HOUSED VETERANS: MORE EQUAL THAN NOT? S.G. Kertesz¹, M.M. Safford¹, S. Rose¹, K.P. Palonen¹, A.J. Gordon², J.H. Halanych¹, K. Kirk¹, C.I. Kiefe¹. ¹Birmingham VA Medical Ctr. & U. Alabama Birmingham, Birmingham, AL; ²Pittsburgh VAMC & U. Pittsburgh, Pittsburgh, PA. (Tracking ID #133383)

BACKGROUND: National data suggest limitations in homeless persons' access to ambulatory care. No studies, however, compare utilization of ongoing care for homeless and housed outpatients in health systems designed to serve both groups. We used Veterans Health Administration (VHA) data to compare primary care entry, and utilization of ongoing care for homeless and housed VHA users with diabetes.

METHODS: All patients in 2003 national VHA data with ≥ 2 diabetes-related diagnoses were considered to have diabetes ($n=735,897$). "Homeless" patients were those using VA homeless services or carrying an "absence of housing" diagnosis ($n=8855$, 1.2%). The proportion achieving ≥ 1 Primary Care Visit (PC1) reflected entry to care. Ongoing care endpoints were proportions with ≥ 4 Primary Care Visits (PC4), or ≥ 1 Eye Clinic Visits (Eye). The 3 multivariable analyses adjusted for age, race, illness burden (Diagnostic Cost Groups (DCG)), mental illness, alcohol and drug abuse.

RESULTS: Homeless persons were younger (mean (sd) age 54 (9) vs. 67 (11) years, for homeless vs. housed, respectively), had higher illness burden (DCG score 1.9 (1.6) vs. 1.0 (1.3)), and higher proportions with alcohol abuse (43% vs. 3%), drug abuse (36% vs. 1%) and mental illness (71% vs. 25%) (all $p<.0001$). While modestly fewer homeless (vs. housed) patients achieved primary care entry (PC1) (92% vs. 97%), more of the homeless obtained ongoing care: PC4 (59% vs. 47%), Eye (56% vs. 53%) (all $p<.0001$). After multivariable adjustment, the homeless had reduced odds ratios (OR) for PC1, and modestly reduced OR for Eye (Table). Models including all significant interactions did not qualitatively alter the findings.

CONCLUSIONS: Homeless and housed diabetic VHA users had high ($>90\%$) proportions entering primary care, although the homeless were moderately disadvantaged for this endpoint. In contrast, we did not find major disadvantages to homeless patients in measures of ongoing care. Our operational definition for homelessness probably selected for individuals with strong connections to VHA services. While national data indicate that not all homeless veterans obtain these services, our study suggests near-parity in utilization of ongoing care for those who do, and should reinforce efforts to connect homeless veterans to care.

Table: Utilization of Selected Outpatient VHA Services (Adjusted Odds Ratios for Homeless vs. Housed)

	1+PC visit	4+PC visits	Eye Clinic
OR(95% CI)	0.68(0.63,0.75)	1.05(1.00,1.10)	0.89(0.85,0.93)

PATIENT AWARENESS OF LANGUAGE LAW—DOES IT MATTER? V. Grubbs¹; A. Chen²; A.B. Bindman¹; E. Vittinghoff¹; A. Fernandez¹. ¹University of California, San Francisco, San Francisco, CA; ²San Francisco General Hospital, SF, CA. (Tracking ID #133138)

BACKGROUND: Federal statute mandates that health care facilities receiving federal funding make interpreters available at no cost to limited English proficient (LEP) patients who cannot communicate with their providers. Though increasing patient awareness is thought to be an effective means for maximal policy implementation, the extent and impact of patient awareness of language law (AOL) on language access is unknown.

METHODS: In June 2003, a telephone survey of 1200 Californians was conducted in 11 non-English languages. The survey included items on ability to speak English, awareness of language law, and means of communicating with health care provider. We analyzed the relationship between LEP participants' awareness of language law and the likelihood of having a language-concordant provider or receiving an interpreter.

RESULTS: Of the 1200 participants, 1006 (84%) were LEP. Among LEP respondents, 580 (58%) reported having a language-concordant provider, 236 (41%) of whom were aware of the language law (AOL). Among the remaining 426 (42%) respondents without a language-concordant provider, 136 (32%) were AOL and only 112 (26%) reported receiving an interpreter in the health care setting. AOL was associated with having a language-concordant provider in unadjusted bivariate analysis (OR 1.46; 95% CI 1.12, 1.90; $p=0.005$). After logistic regression controlling for age, gender, education, income, insurance status, and time lived in the U.S., AOL remained significantly associated with having a language-concordant provider (OR 1.37; 95% CI 1.04, 1.80; $p=0.025$). AOL was not associated with receiving an interpreter (adjusted OR 0.70; 95% CI 0.41, 1.18; $p=0.18$). Further analysis revealed no evidence of significant interaction between AOL and other predictors of receiving an interpreter.

CONCLUSIONS: Roughly one-third of participants were aware of the language law. Awareness of the law was predictive of having a language-concordant provider, but was not associated with receipt of interpreter services. However, nearly half of limited English proficient respondents did not have a provider who spoke their language—most of whom did not receive an interpreter in the health care setting. Even when patients are aware of their rights, policy mandates without resources are not sufficient to resolve language barriers in settings where bilingual provider supply cannot meet demand.

PATIENT-PHYSICIAN COMMUNICATION AND RACE AND LANGUAGE CONCORDANCE: IS IT MORE IMPORTANT TO LOOK OR SOUND LIKE YOUR PATIENT? A.A. Donato¹; R.V. Brown²; C. Bright³; J. Bussey-Jones⁴. ¹Reading Hospital and Medical Center, West Reading, PA; ²George Washington University, Washington, DC; ³Duke University, Durham, NC; ⁴Emory University, Atlanta, GA. (Tracking ID #134620)

BACKGROUND: The 2004 ACP position paper on Racial and Ethnic Disparities in Health Care states that "clear communication in clinical encounters is key to healthy patient outcomes, leading to greater patient satisfaction, and increased quality of care, which increases health-seeking behavior." Increasing racial and linguistic concordance has been proposed as a way to improving patient-physician communication and thereby positively impact health outcomes.

METHODS: The CCTOP Study (Cultural Competence Training and Outcomes in Patients) is a multi-center cross-sectional study of patients presenting for care at eleven outpatient medical centers. The study assessed physicians' attitudes regarding culture and bias, as well as patients' feelings regarding trust, communication and satisfaction with their physicians on five-point scales (1=strongly agree to 5=strongly disagree).

RESULTS: Preliminary results were available from 291 patients (70% non-white) and their 175 physicians (44 attendings and 131 residents, 47% non-white) at the first 5 sites. Patients with physicians of the same race were more likely to consider their doctor courteous and respectful (1.2 vs. 1.5, $p=0.000$), to understand their problems well (1.4 vs. 1.7, $p=0.000$), to communicate in words they understand (1.4 vs. 1.8, $p=0.000$), to listen to concerns and questions well (1.3 vs. 1.8, $p=0.000$), and to spend enough time with them (1.4 vs. 1.7, $p=0.000$). Patients with physicians of the same race were also more likely to say they were satisfied with their service (1.2 vs. 1.6, $p=0.000$). Patient-physician language concordance was not associated with improvement in any communication scores or overall satisfaction. Interestingly, neither racial nor linguistic concordance was associated with better patient-rated participatory communication scores.

CONCLUSIONS: In this study, race concordance had a strong association with improved patient-physician communication scores independent of participatory communication scores and language concordance. These findings suggest that physician and patient attitudes may influence the patient-physician relationship to a greater degree than language concordance and "teachable" skills in patient-centered communication. This work supports assertions that strategies to diversify the physician workforce are a vital part of the plan to improve minority patient outcomes.

PERCEIVED DISCRIMINATION IN HEALTHCARE: THE CALIFORNIA HEALTH INTERVIEW SURVEY. P. Johansson¹; D. Buchwald²; C. Jacobsen². ¹Agency for Healthcare Research and Quality, Rockville, MD; ²University of Washington, Seattle, WA. (Tracking ID #135853)

BACKGROUND: Racial and ethnic discrimination has been associated with psychological distress, low self-rated health, high blood pressure, smoking, and alcohol abuse. Most research on discrimination in the U.S. has focused on Whites and African-Americans, and found that 10-26% of African-Americans reported discrimination when receiving healthcare. American Indian/Alaska Native (AI/AN) perceptions of discrimination in healthcare have never been examined, even though their health outcomes are among the worst of any U.S. racial/ethnic group. Our objectives were to examine perceived discrimination among people identifying only as AI/AN, African-American, Asian-American, White, and people reporting both AI/AN and White ancestry.

METHODS: We used data on perceived discrimination collected by the 2001 California Health Interview Survey (CHIS). CHIS 2001 is one of the largest cross-sectional surveys ever conducted in the U.S. Respondents were recruited from every county in California using random digit dialing to assemble a representative sample of the state's non-institutionalized population residing in households. We compared the prevalence of and reasons for perceived discrimination in healthcare among people identifying only as AI/AN, African-American, Asian-American, or White, and people reporting both AI/AN and White heritage (AI/AN+White). We used chi-square tests to evaluate the prevalence of perceived discrimination and the reasons for perceived discrimination across racial groups. Whites were the reference group in logistic regression models that evaluated the odds of perceived discrimination by race after adjusting for relevant covariates.

RESULTS: Discrimination was perceived by 7.1% of the AI/AN group, 8.8% of AI/AN+White respondents, 5.6% of African-Americans, 4.3% of Whites, and 2.6% of Asian-Americans. Compared to Whites, only the AI/AN+White respondents had significantly higher odds of perceived discrimination (odds ratio=2.0, 95% CI=1.6-2.5), and only Asian-Americans had significantly lower odds (odds ratio=0.6, 95% CI=0.4-0.7). Among respondents citing only 1 reason for discrimination, race/ethnicity was most frequently cited by the AI/AN, African-American, and Asian-American respondents, and insurance type was the most cited reason among AI/AN+White and White respondents. The high proportion of AI/AN, African-American, Asian-American, and AI/AN+White respondents who reported only "> 1 reason" made it impossible to conclusively determine the most prevalent reason in these groups.

CONCLUSIONS: AI/ANs, and especially those who identify as both AI/AN and White, are the most likely among CHIS racial groups to report discrimination in healthcare. These findings invite further inquiry and may play a role in the striking health disparities experienced by AI/AN communities.

PERCEPTIONS OF PHYSICIAN-PATIENT COMMUNICATION EXPLAIN RACIAL DIFFERENCES IN TRUST. H.S. Gordon¹; R.L. Street²; P.A. Kelly¹; B. Sharf²; J. Soucek¹. ¹VA Medical Center and Baylor College of Medicine, Houston, TX; ²Texas A&M University, College Station, TX. (Tracking ID #132004)

BACKGROUND: Black patients have lower trust in physicians than white patients, a finding that has been observed in both primary care and specialty settings. However, reasons for this racial disparity is poorly explained. We propose that one reason that black patients may have lower trust in physicians is because of their perceptions of poorer quality communication with physicians.

METHODS: Eligible patients had pulmonary nodules or lung cancer and were seen in thoracic surgery or oncology clinic for initial treatment recommendations at a large southern VA hospital. Questionnaires were used to determine patients' self-identified race, health status, perceptions of physicians' commu-

nication (informativeness, support, and partnership-building), overall satisfaction with care, and trust in physician and hospital (VA). Trust in physician (5-items) and trust in hospital (4-items) were assessed before and after the visit. Responses were on a 10-point scale. Survey data were compared by patient race using t-tests or chi-square tests as appropriate. Multiple linear regression models to examine predictors of post-visit trust in physician and hospital were developed separately and independent variables were entered in groups.

RESULTS: Complete data were obtained for 100 patients (23% black, 77% white). Age, gender, marital status, clinic site, and health status did not differ by race, but black patients had less education ($P=0.01$) than white patients. Pre-visit trust in physician and hospital, and post-visit trust in hospital did not differ by race. Black patients had lower post-visit trust in physician (mean score 8.0 vs. 9.3; $P=0.02$) than white patients. Compared with whites, blacks judged physicians as less informative (7.3 vs. 8.5; $P=0.04$), less supportive (8.1 vs 9.2; $P=0.04$) and less likely to engage in partnership-building (6.4 vs. 8.2; $P=0.001$). In multiple regression analyses, pre-visit trust in physician and black race were significant ($P<0.001$) predictors of post-visit trust in physician, even after controlling for patients' characteristics. However, after controlling for physicians' informativeness, support and decision-making style, pre-visit trust in physician and race were not statistically significant predictors of post-visit trust in physician. Pre-visit trust in hospital significantly ($P<0.001$) predicted post-visit trust in hospital even after controlling for characteristics and communication. Patient satisfaction was correlated ($P<.001$) with post-visit trust in physician ($r=0.60$) and hospital ($r=.66$) and when added to the regression models was a statistically significant predictor of post-visit trust, but adding satisfaction to the final models did not change the relationship of previously entered covariates with post visit trust in physician or hospital.

CONCLUSIONS: Perceptions that physicians were less informative, less supportive and less likely to share decision-making explained black patients lower trust in physicians. Black patients' perceptions of physicians' communication could promote a cycle of less trust and less communication with profound implications for utilization of health services and adherence to physicians' recommendations.

PHYSICIAN-PATIENT COMMUNICATION IN THE CHINESE IMMIGRANT COMMUNITY.

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BACKGROUND: The Chinese immigrant community may be receiving sub-standard health care, and with its rapid growth this problem is magnified. Race, language barriers, and cultural sensitivity are some of the many factors that affect the doctor-patient relationship and may play a role in the Chinese immigrant health disparity. This study examined, from the patient's perspective, how a physician's ethnicity and language skills affect communication between Chinese immigrants and their doctors.

METHODS: Questionnaires were administered at the New York City Chinese Planning Council to Chinese patients ($n=40$) who had seen, within the past 5 years, both a dialect-concordant Chinese doctor and a non-Chinese doctor who did not speak Chinese. The survey was developed from a review of the Commonwealth Fund's Health Care Quality and Minority Health Survey. Each survey participant was asked x items about their most recent experiences with both a Chinese and a non-Chinese doctor. Descriptive statistics were used to summarize the data. In-depth post visit interviews were conducted at Gouverneur Hospital (New York, NY). Ten interviews were conducted with patients who saw a dialect-concordant Chinese doctor and 10 interviews were conducted with patients who saw a non-Chinese doctor who did not speak Chinese. All interviews were tape-recorded, transcribed, and translated. Data was evaluated using content analysis and grouped into three categories: (1) any phrases mentioned by more than 2 or 3 interviewees; (2) patterns of language problems; and (3) patterns of communication content problems.

RESULTS: The results of the questionnaires showed that Chinese patients perceived Chinese doctors as understanding them substantially more than non-Chinese doctors (85% vs. 47.5%) and Chinese patients reported understanding Chinese doctors substantially more than non-Chinese doctors (90% vs. 45%). However, Chinese patients rated both Chinese and non-Chinese doctors similarly on other aspects of the doctor-patient interaction: patient involvement (37.5% vs. 35%), time spent (60% vs. 50%), confidence in the doctor (72% vs. 60%), and respectful treatment (82.5% vs. 72.5%). Qualitative interviews indicated that better understanding was linked to concordance in language, rather than to differences in interpersonal communication style. Themes that emerged from patient interviews include: 1) Compassion: The doctor's manner of relating to the patient was perceived as important. Two interviewees referred to an apparent lack of compassion on the part of Chinese doctors; 2) Language: Concordance made communication so much easier that most patients preferred Chinese doctors. Several patients commented on problems with using translators, particularly regarding the accuracy and inconvenience of translation; 3) Ethnicity: There was no evidence that ethnicity and cultural compatibility was an issue beyond language concordance.

CONCLUSIONS: This study suggest that Chinese patients perceive a greater amount of communication errors when there is no language concordance. Chinese patients prefer Chinese doctors, and the preference for race concordant doctors is largely attributable to language concordance. Based on these findings, we recommend encouraging Chinese speaking doctors to serve the Chinese immigrant community and improving translation services, particularly increasing the accuracy and convenience of translation.

PREDICTORS OF ON-CALL SPECIALIST RESPONSE TIMES IN CALIFORNIA EMERGENCY DEPARTMENTS.

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BACKGROUND: Emergency departments (ED) overcrowding has raised increasing concern in the past decade. Overcrowding has resulted in ambulance diversion, long patient waits, worse health outcomes, and higher costs. ED overcrowding has become particularly significant in California due to the large number of hospital ED closures. Limited availability of on-call specialists in EDs has been thought to contribute to ED overcrowding. The Centers for Medicare and Medicaid Services (CMS) have applied a 30-minute standard to on-call physician response to EDs. We sought to describe facility and neighborhood characteristics associated with ED specialist response.

METHODS: We used direct observational data from a random sample of 1,798 patients visiting 30 California EDs over a six month period during 2000-2001. Using multivariate logistic regression and survival analysis, we identified independent predictors of whether an on-call specialist did not respond to a call or responded after the 30-minute CMS threshold.

RESULTS: The average response time was 10.1 min (range 0 to 1 hour 44 minutes, median 5 minutes). However, 13% ($N=33$) of on-call specialists did not respond at all. Of the 87% ($N=227$) who did respond, 91% ($N=207$) responded in less than 30 minutes. In univariate logistic regression analysis, on-call specialists in hospital zip codes with incomes less than \$16,000 and those in areas that were less than 95% urban were less likely to either respond to a call or responded after 30 minutes. After controlling for hospital zip code income, hospital visits per year, percentage urban, nonurgent or critical, ownership status, teaching status, percentage over age 65, those near closure, and trauma designation, on-call specialists in hospitals with zipcode incomes less than \$16,000 had a lower odds of responding at all or responding within the 30 minute threshold (AOR 0.25, 95% C.I. 0.08, 0.79) compared to those in hospitals with zipcode incomes greater than or equal to \$16,000. Urban setting was not significant in multivariate analysis. Among teaching hospitals, on-call specialists were more likely to respond within 30 minutes vs. non-teaching hospitals (AOR 5.71, 95% C.I. 1.10, 29.50). In multivariate survival analysis, for every \$10,000 increase in hospital zipcode income, there was a 63% increased likelihood of an on-call specialist responding. (Hazard ratio (HR) 1.6, 95% C.I. 1.2, 2.2, $p=0.003$). Among specialists who did respond to an ED call, for every \$10,000 decrease in hospital zipcode income, patients waited 5.5 minutes longer (95% C.I. 2.5, 8.4, $p=0.001$).

CONCLUSIONS: The vast majority of on-call specialists met federal recommendations of a 30-minute response, but those in poor neighborhoods were much less likely to do so. Teaching hospitals were more likely to adhere to the 30 minute guideline probably due to more resident and fellow subspecialty training programs. As ED closures and limited on-call specialist reimbursement continue to afflict health care systems across the U.S., on-call response times will likely escalate, particularly for hospitals in poorer areas. State and federal policies should focus on providing more funding for on-call specialist panels in these poor areas.

PRESCRIPTION UTILIZATION IN AN URBAN HOMEBOUND POPULATION. I. Kronish¹; R.S. Morrison¹; A.D. Federman¹; J. Boal¹. ¹Mount Sinai School of Medicine, New York, NY. (Tracking ID #132851)

BACKGROUND: The number of medically homebound adults has grown with the aging of the US population, yet little is known about their health care utilization. We sought to characterize the health status and prescription utilization of urban homebound adults.

METHODS: We performed a cross-sectional analysis of all 415 patients in an urban primary care program for homebound adults in October 2002. Data were abstracted from medical records and included demographics, chronic diseases, goals of therapy, activities of daily living (ADL), and prescription medications used. We estimated out-of-pocket costs for patients without prescription drug coverage based on average wholesale price. We used multivariate linear regression to examine the association of demographics, health and functional status, and palliative care goals with volume of prescription utilization. The model included race/ethnicity and ADL score; other variables were included if $P<.05$ for their univariate association with number of prescriptions used.

RESULTS: The median age was 83 years (range 25-106), 13% were <65 years and 77% were female. 63% were non-white, 28% spoke Spanish and 31% lived alone. Medicaid covered 64% of patients. Although no patients lacked health insurance, 26% had no prescription drug coverage. Patients had a median of 4 comorbidities (range 0-13), and a mean ADL score of 8.7 (SD 5.4), indicating marked functional dependence. Palliative care was an important goal for 34%. The study population used a median of 8 (range 1-23) drugs per month. Patients who lacked prescription drug coverage used a median of 6 (range 1-17) drugs per month, with an estimated median monthly out-of-pocket cost of \$223 [range \$1-\$1512]. On univariate analysis, patients used fewer prescriptions ($P<0.05$) if they were younger, male, living alone, lacked prescription insurance, had a lower Charlson score, dementia, or a palliative care goal. In multivariate analysis, female gender ($\beta=1.7$, $P=0.004$), living alone ($\beta=1.4$, $P=0.03$), Charlson score ($\beta=1.0$, $P<0.001$), dementia ($\beta=-1.7$, $P=0.004$), and palliative care goal ($\beta=-1.4$, $p=0.009$) remained significantly associated with number of prescriptions used. Age and prescription insurance status were no longer significantly associated.

CONCLUSIONS: Medically homebound adults had a high level of prescription medication use that was strongly associated with their burden of comorbidities.

Even though older age and functional dependence are distinctive features of homebound adults, neither measure was associated with prescription utilization. For homebound patients who lack prescription drug coverage, prescription medication use may represent substantial financial burden. Additional research is needed to determine whether out-of-pocket costs represent a barrier to care for patients in this population.

PREVENTIVE SERVICE UTILIZATION AMONG FREE CLINIC PATIENTS. E.E. Tung¹; R. Stroebel¹; P. Targonski¹; S. Freytag²; R. Staver¹. ¹Mayo Clinic, Rochester, MN; ²Salvation Army Free Clinic, Rochester, MN. (Tracking ID #136077)

BACKGROUND: Despite national efforts to improve cancer-screening rates, underserved women remain less likely to receive preventive care. Poor, elderly, and minority women continue to experience disproportionate mortality from breast and cervical cancer attributed to excess presentation with late-stage disease. Mammography and Pap smear testing remain the best methods for reducing mortality. Little is known about the specific barriers, which prevent underinsured women from utilizing available cancer-screening modalities. Free clinics may provide a unique opportunity to reach a concentrated population of women lacking adequate screening services.

METHODS: Demographic and epidemiologic data was collected from 1,133 consecutive patients seen at the Salvation Army Free Clinic from 1999-2001. Persons seeking care completed a data collection instrument requesting self-reported health-related and demographic information at the time of registration. 397 women over the age of 18 who had completed the items inquiring about completion of mammogram and Pap testing were included in the study. Logistic regression was used to identify factors associated with non-completion of breast and cervical cancer screening.

RESULTS: Thirteen of 81 (16%) of women aged 50 and older had received mammography within two years preceding their visit. Forty-four of 397 eligible women had completed Pap testing. Nonwhite women (OR=3.01 (CI=1.38-6.55)) and women with lower educational attainment (OR=2.89 (CI=1.22-6.81)) were at increased risk of Pap test non-completion.

CONCLUSIONS: Dramatically low rates of screening were found. None of the variables traditionally associated with health disparities predicted non-completion of mammography. Non-white race and low education were significant risk factors for non-completion of Pap smears, but accounted for only 29% of the variance in regression models. Despite demographic variation among users of the free clinic medical services, these women are homogeneously at risk for breast and cervical screening non-completion. The lack of other identifiable risk factors and the underutilization of existing free screening programs suggest that additional barriers prevent optimal health maintenance. Future interventions targeting these women are necessary in the effort to achieve the screening objectives of Healthy People 2010.

PRIMARY CARE AND HEALTH INSURANCE AMONG WOMEN RELEASED FROM NEW YORK CITY JAILS. J. Lee¹; D. Vlahov²; N. Freudenberg³. ¹Cornell University, New York, NY; ²New York Academy of Medicine, New York, NY; ³Hunter College, New York, NY. (Tracking ID #132499)

BACKGROUND: Women in the criminal justice system have higher rates of infectious diseases, psychiatric diagnoses, addictions and social problems than non-incarcerated women, and for HIV infection, higher rates than incarcerated men. Incarceration and the discharge planning process represent an important opportunity to link this vulnerable population to primary care. To assess the correlates of primary care utilization (PCU) of women released from New York City jails, this report uses data from a recent evaluation of Health Link, a drug use and HIV risk reduction program enrolling incarcerated women and adolescent males as they leave jail and return to the communities of the South Bronx and Harlem.

METHODS: This analysis examined factors associated with primary care utilization and health insurance coverage among a prospective cohort of 511 women leaving jail in New York City from 1997-2001 and completing a single follow-up interview at one year. The main outcomes were health insurance status at follow-up and primary care utilization, defined as self-report of visiting a doctor's office, private clinic, hospital outpatient clinic, or, if HIV+, an HIV specialty clinic at least once during the year after release. Independent variables of interest were analyzed within six groupings of demographics, social support, risk behavior and illegal activity, medical conditions, and health care utilization using a change-in-estimate approach to multivariate logistic regression for the two dichotomous outcomes. All analysis was performed using Stata SE 8.0.

RESULTS: One year after release from jail, almost half (47%) of women reported primary care utilization and more (56%) reported health insurance coverage. Neither outcome was more likely among women reporting diabetes, asthma, or depression. PCU was more likely among those who reported receipt of public benefits at follow-up (Adjusted OR 2.33, 95% CI 1.38-3.93), health insurance (2.24, 1.31-3.83), at least a moderate level of social support since release (1.99, 1.02-3.88), or HIV seropositivity (3.54, 1.89-6.61). Any illegal activity (0.52, 0.29-0.93) or gonorrheal infection (0.10, 0.02-0.71) since release was inversely associated with PCU. Health insurance coverage was more likely if women received public benefits at follow-up (13.08, 7.37-23.21), at baseline (2.23, 1.33-3.73), or reported PCU (2.45, 1.44-4.18) or hospitalization since release (2.49, 1.15-5.42). Women were less likely to be insured if re-arrested (0.23, 0.12-0.41). No other self-reported medical conditions or sociodemographic variables correlated with primary care utilization or health insurance coverage.

CONCLUSIONS: This analysis of a large sample of women leaving jail show a lack of primary care follow-up among those with chronic medical conditions. In

addition to established predictors of primary care such as health insurance, moderate to high levels of social support and avoiding illegal activity were two-fold more likely if a woman visited a primary care provider. Efforts to increase health insurance coverage in the general population should have beneficial effects on those released from correctional settings. In addition, programs that foster social support and aid incarcerated individuals in obtaining health insurance and appropriate benefits upon release may facilitate indicated health service utilization and thus contribute to improved health.

PRIMARY CARE FOR LOW-INCOME POPULATIONS: A COMPARISON OF HEALTH-CARE DELIVERY SYSTEMS. E. Grossman¹; A.T. Legedza¹; C.C. Wee¹. ¹Beth Israel Deaconess Medical Center, Boston, MA. (Tracking ID #133562)

BACKGROUND: Delivering cost-effective care to low-income populations is becoming increasingly important as state Medicaid budgets are squeezed. In an effort to cut costs, some states have considered shifting primary care for low-income populations from hospital-based practices to community-based health centers (CHCs). To explore ramifications of these proposed changes, we examined three quality measures at different types of primary-care sites.

METHODS: We used data from the 2000-2002 National Health Interview Survey for adult participants with a family income of less than 200% of the poverty threshold (n=38,329). We described patients (pts) who receive primary care at hospital clinics, physician (MD) offices or HMOs, or CHCs, and compared 1) emergency-room (ER) visits; 2) delays in medical care due to office/clinic constraints (appointment availability, office wait time, and accessibility of scheduling staff), and 3) influenza and pneumococcal vaccinations. Multivariable logistic regression models adjusted for age, sex, race/ethnicity, education, language, region, employment, insurance, transportation accessibility, alcohol or tobacco use, and illness burden (self-reported health status, hospitalizations and office visits, comorbidities and disabilities, mood disturbances and need for mental health care). All analyses were performed using SUDAAN and were appropriately weighted to account for the complex sampling.

RESULTS: More than half of our sample (59%) received primary care at an MD office or HMO, while 20% went to a CHC and 3% to a hospital clinic; 18% did not identify any of these three sites as a source of primary care. There were significant demographic differences between the pts at the three site types (p<.001 for all comparisons). The mean pt age was 49 years (y) at MD offices/HMOs, 48y at hospital clinics, and 41y at CHCs; 31% of MD office/HMO pts were racial minorities, compared to 61% of hospital-clinic pts and 48% of CHC pts. Only 13% of MD office/HMO pts were uninsured - compared to 29% of hospital-clinic pts and 33% of CHC pts. Across multiple measures, hospital-clinic pts were more likely to have a higher burden of disease; hospital-clinic pts were also more likely to report fair or poor health status (33% vs. 23% of MD office/HMO pts and 20% of CHC pts). We found significant differences across sites of care for our quality measures of interest - even after adjustment for multiple sociodemographic and health-status factors (see Table). Although hospital-clinic pts did not report higher rates of delays in care due to office/clinic constraints, they did report more ER visits than patients at CHCs or MD offices/HMOs; however, they also had higher rates of adult vaccinations than pts at the other sites.

CONCLUSIONS: Our results suggest that no one type of primary care site delivers clearly higher quality care for low-income populations. Further studies will need to examine additional quality indicators among sites of primary care. Quality issues should be considered as we decide how best to deliver cost-effective care to vulnerable populations.

Multivariable Regression Results

	Clinic/health center	Hospital outpatient department	MD office/HMO
% > 1 ER visit in last year	11%	20%	10%
Adj OR (95% CI)	Reference	1.8 (1.4, 2.3)	1.0 (0.9, 1.1)
% with delays in care in last year	11%	14%	8%
Adj OR (95% CI)	Reference	1.2 (0.9, 1.6)	0.8 (0.7, 0.9)
% vaccinated for influenza in last year	23%	33%	32%
Adj OR (95% CI)	Reference	1.3 (1.1, 1.7)	1.0 (0.9, 1.1)
% vaccinated for pneumococcus lifetime	14%	23%	22%
Adj OR (95% CI)	Reference	1.4 (1.1, 1.8)	0.9 (0.8, 1.0)

RACE CONCORDANCE AND TRUST IN THE PATIENT-PHYSICIAN RELATIONSHIP. K.V. Mizelle¹; R.L. Johnson²; M.C. Beach²; L. Boulware²; L.A. Cooper². ¹University of Virginia, Charlottesville, VA; ²Johns Hopkins University, Baltimore, MD. (Tracking ID #135769)

BACKGROUND: Trust, a key component of relationships in healthcare, differs across patient racial and ethnic groups and may contribute to disparities in minority healthcare. Although patient-physician race concordance has been linked to better access to care, communication, and patient ratings of care, its association with trust has not been extensively studied. We studied whether race concordance is associated with higher levels of patient trust in physicians,

physician perceptions of patient trust, and accuracy of physicians' perceptions of patient trust.

METHODS: We used questionnaire data from two brief cohort studies conducted in 1998 and 2002. A total of 61 primary care physicians and 458 patients were recruited from offices in the Baltimore, MD, Washington DC, and Northern VA areas. Our main outcome variables, patients' trust in their physicians and physicians' perceptions of individual patients' trust in them, were measured with single items from the Trust in Physician Scale and the Physician Satisfaction with Primary Care Office Visits scale. We also examined the accuracy of physicians' perceptions of patient trust. We used logistic regression with generalized estimating equations to account for the clustering of patients within physicians and adjusted for patient and physician demographic and relationship factors in multivariate models.

RESULTS: In unadjusted analyses, the probability of patients reporting trust in their physicians was 12% greater in race-concordant [86% (95% CI: 76-92%)] vs. race-discordant dyads [74% (95% CI: 68-80%)], $p=0.04$. Similarly, the probability of a physician reporting trust from an individual patient was 10% greater in race-concordant [69% (95% CI: 61%-77%)] vs. race-discordant dyads [59% (95% CI: 48-69%)], $p=0.04$. After adjustment for potential confounders, the probability of patient self-reported trust remained 11% greater in race-concordant vs. race-discordant dyads, but the association was not statistically significant. The probability of physicians perceiving trust by patients was 18% greater in race-concordant [74% (95% CI: 66-82%)] vs. race-discordant dyads [56% (95% CI: 44-66%)], $p=0.004$. The accuracy of physician perception of patient trust was also greater in race-concordant vs. race-discordant dyads (80% vs. 59%, $p=.015$). Physicians were less likely to underestimate patients' trust in race-concordant vs. race-discordant dyads (12% vs. 28%, $p=.006$).

CONCLUSIONS: Patients' trust in physicians and physicians' perceptions of patients' trust are greater in race-concordant relationships. Physicians in race-discordant relationships are more likely to misperceive patients as distrusting of them. Further research is needed to understand the mechanisms and consequences of these misperceptions. Engendering trust between patients and physicians and improving the accuracy of physicians' perceptions of trust across racial and ethnic lines may be important strategies for improving race relationships in healthcare and reducing minority healthcare disparities.

RACE, PROXIES FOR LIFE EXPERIENCES, AND PHYSICIAN TRUST. D. Kuykendall¹, M. Kallen¹. ¹Baylor College of Medicine, Michael E. DeBakey VA Medical Center, Houston, TX. (Tracking ID #134232)

BACKGROUND: Patients' trust in their physicians can influence satisfaction with care, adherence to treatment and health outcomes. Trust itself can be influenced by a variety of factors associated with life experiences (e.g., type of illness, health beliefs, race). To the extent that certain groups have different life experiences, differing levels of physician trust might be expected. Current research shows an unclear relationship between race and physician trust. Some studies demonstrate trust to be lower among African American patients, while others fail to find this relationship. One limitation of such studies is that race is typically viewed as a main effect; interactions of race with proxies for other life experiences are not investigated. Our objective is to demonstrate how information about race can be used to better understand physician trust. We hypothesize that understanding of physician trust will be enhanced by considering the interaction of race with other proxies for life experiences (i.e., health status, income).

METHODS: Consecutive patients from outpatient clinics representing one public, one private, and one VA clinic were recruited from waiting rooms prior to appointments. One hundred fifteen African American and 148 White American patients participated, providing demographic information and completing a battery of scales about their health and health care. Multiple linear regression tested associations between physician trust and patient characteristics. Average score on the 11-item Trust in Physician Scale (1-low trust; 5-high trust) served as the dependent measure. Explanatory variables included clinic, age, race, health status (self-reported bodily pain in the last month (1-none; 5-very severe), and personal income. Interactions were created between race and all other explanatory variables.

RESULTS: No main effects were shown on physician trust due to clinic, age, reported pain, race, or income. After controlling for these factors, two significant interactions emerged ($p<.05$). Little-to-no pain (e.g., health status) had a significant enhancing effect on physician trust for White but not African American patients (Table 1). In contrast, reports of low income had a significant detrimental effect on physician trust for African American but not White American patients (Table 2).

CONCLUSIONS: Findings suggest that the association between race and physician trust is complex. Additional patient characteristics and experiences should be considered in conjunction with race to better understand pre-existing levels of physician trust. There are several implications. In studies of physician behavior and trust, interactions of race with other factors should be included when controlling for pre-existing levels of physician trust. From a policy perspective, there may be identifiable groups who are vulnerable to having low physician trust. Developing a model of physician trust based on the interaction of a variety of predisposing circumstances would help providers recognize potential communication challenges and inform educational initiatives.

Table 1. - Physician Trust (1 =low; 5 =high) by Race and Pain Level

	African American	White American
little-no pain	3.63	4.00
Moderate-severe pain	3.66	3.54

Table 2 - Physician Trust (1 =low; 5 =high) by Race and Income

	African American	White American
<=\$20k	3.42	3.82
>\$20k	3.86	3.70

RACE/ETHNIC DISPARITIES IN HEALTH AND UTILIZATION AMONG HUMAN IMMUNODEFICIENCY VIRUS (HIV) POSITIVE VETERANS. K.A. McGinnis¹, M. Skanderson², J. Conigliaro², C.L. Gibert³, N.R. Gandhi⁴, L.I. Backus⁵, A.C. Justice⁶. ¹Center for Health Equity Research and Promotion, VA Pittsburgh Healthcare System; ²University Center for Social and Urban Research, University of Pittsburgh, Pittsburgh, PA; ³Center for Health Equity Research and Promotion, VA Pittsburgh Healthcare System, Pittsburgh, PA; ⁴George Washington University Medical Center, Veterans Affairs Medical Center, Washington, DC; ⁵Division of Infectious Diseases, Emory University School of Medicine, Atlanta, GA; ⁶Center for Quality Management in Public Health Palo Alto VA, Palo Alto, CA; ⁷Yale University; Veterans Aging Cohort Study, VA Connecticut Healthcare System, West Haven, CT. (Tracking ID #133788)

BACKGROUND: While race/ethnicity associated health care disparities have been widely documented, the cause of these differences is unclear. Our goal was to compare health status at time of HIV diagnosis in the Veterans Affairs Healthcare System (VAHS) and utilization during the following year by race/ethnicity. **METHODS:** We identified veterans with a first HIV diagnosis in the VAHS from 10/1998 to 9/2003 using ICD-9 codes. Time in VAHS before HIV diagnosis was calculated. To evaluate health status at diagnosis, we searched for 15 comorbid, 15 HIV-related, and five mental health conditions, and drug and alcohol abuse/dependence occurring before HIV diagnosis, using ICD-9 codes. Analyses were limited to veterans in the Immunology Case Registry having CD4 and HIV RNA viral load (VL) data. We identified veterans who had four or more (4+) outpatient visits in the year following diagnosis. Age, gender, time in VAHS, number of comorbid conditions (0,1-2,3+), and presence of HIV-related conditions (0,1+) were compared using appropriate statistical tests. Logistic regression was used to determine whether race/ethnicity was associated with CD4<200 and presence of an HIV-related condition at the time of first diagnosis in univariate analyses and multivariate models adjusted for age, time in VAHS before first HIV diagnosis, comorbid and mental health conditions, and drug or alcohol abuse/dependence. Logistic regression was used to determine whether race/ethnicity was associated with having 4+outpatient visits in the year following HIV diagnosis in univariate analyses and a multivariate model adjusted for age, time in VAHS before diagnosis, comorbid, HIV-related, and mental health conditions, drug or alcohol abuse/dependence, CD4 and VL.

RESULTS: We identified 4,753 HIV+ veterans. Mean age was 47 years. 98% were male, 35% white, 56% black, and 9% Hispanic. Time in VAHS before diagnosis differed between groups ($p=.001$), with blacks having been in the VAHS the longest (5.6 years) followed by Hispanics (3.8 years) and whites (1.9 years). Blacks were more likely to have had 3+comorbid conditions than whites and Hispanics (11% vs. 8% and 8%; $p<.001$). In adjusted models, at time of diagnosis, compared to whites, blacks and Hispanics were more likely to have CD4<200 (OR=1.28, $p<.001$; OR=1.33, $p=.010$) and were more likely to have an HIV-related condition (OR=1.22, $p=.023$; OR=1.18, $p=.26$). Blacks and Hispanics were less likely than whites to have 4+outpatient visits in the year following diagnosis (OR=.71, $p=.006$; OR=.69, $p=.04$). Of note, some comorbid conditions were associated with CD4<200 and presence of an HIV-related condition at the time of diagnosis.

CONCLUSIONS: Black veterans were in the VAHS for the longest period of time prior to HIV diagnosis and were sicker from comorbid and HIV-related conditions at the time of diagnosis. Despite being sicker, black veterans were less likely to have 4+outpatient visits in the year following diagnosis before and after adjusting for comorbid, HIV-related, and mental health conditions. Future work must determine whether this disparity in follow-up care reflects fewer scheduled visits or poorer attendance.

RACIAL DIFFERENCES IN DOCTORS' INFORMATION-GIVING AND PATIENTS' PARTICIPATION. H.S. Gordon¹, R.L. Street², B. Sharf², J. Soucek¹, P.A. Kelly¹. ¹VA Medical Center and Baylor College of Medicine, Houston, TX; ²Texas A&M University, College Station, TX. (Tracking ID #135909)

BACKGROUND: Racial disparities in health care are widespread and poorly explained. Studies indicate that doctor patient communication often varies by patient race, especially with respect to information-exchange. Because communication is a process of personal and mutual influence, doctors may provide more information to some patients than to others as a matter of style or when patients ask questions, express concerns, and make assertions.

METHODS: Eligible patients had pulmonary nodules or lung cancer and were seen in thoracic surgery or oncology clinics for initial treatment recommendations at a large southern VA from 2001-2004. Doctor-patient consultations were audiotaped and patients completed questionnaires. Patients self-identifying as black (n=30) or white (n=107) are included in this analysis. Audiotapes were transcribed and unitized into utterances. Utterances were coded into categories including doctors' information-giving and patients' active participation (e.g., questions, assertions, concerns). These categories were further coded according to whether the communicative behavior was prompted (e.g., doctor provided information in response to a patient question) or self-initiated. Data were compared by patient race using t-tests or chi-square tests as appropriate. We used mixed linear regression to determine the independent predictors of doctor's information giving after controlling for clustering of patients by doctor.

RESULTS: Patient age, gender, marital status, clinical site, and mental and physical functioning were similar by race ($P>0.20$). Black patients had lower

income, less education ($P < 0.01$) and were somewhat less likely to bring a companion to the visit (47% vs. 65%; $P = 0.06$) than white patients. Doctors were generally less informative (49.3 vs. 87.3 mean utterances; $P < 0.001$), self-initiated less information-giving ($P = 0.02$), and when prompted responded with less information ($P < 0.001$) to black patients compared with white patients. When compared with white patients, black patients were overall less active participants (17.8 vs 27.2 utterances; $P = 0.01$) and self-initiated ($P = 0.008$) less active participation, but were similar in their participation when prompted by the doctor ($P = 0.73$). In visits where a companion was present, black patients' companions produced significantly fewer active participation utterances (7.6 vs. 15.4; $p = 0.03$) than white patients' companions. In mixed regression analyses, after adjusting for clustering by doctor, patient characteristics, and patient and spouse active participation, black and white patients received statistically similar numbers of information-giving utterances (96.9 vs. 100.3, $P = 0.64$). However, in stratified (prompted or self-initiated) regression analyses, after adjusting for clustering by doctor and patient characteristics, black patients received fewer prompted information-giving utterances ($P = .003$) and similar frequency of self-initiated information-giving utterances ($P = 0.21$).

CONCLUSIONS: Black patients' received less information from their doctors. These differences were due to the tendency for black patients to be less active participants than were white patients. Our findings reflect a cycle of communication that perpetuates patient passivity and limited information-exchange. The patient does little to prompt the doctor for information and the doctor in turn provides less information and does little to encourage active patient participation.

RACIAL DIFFERENCES IN RESPONSE TO BETA-BLOCKERS FOR HEART FAILURE THERAPY. C.O. Phillips¹; V.R. Bead²; R.C. Ziegelstein²; E.B. Bass²; H.R. Rubin². ¹The Cleveland Clinic Foundation, Baltimore, MD; ²Johns Hopkins University, Baltimore, MD. (Tracking ID #136024)

BACKGROUND: To investigate the association of racial differences in response to beta-blockers for heart failure (HF) therapy and racial disparities in clinical outcomes.

METHODS: We searched MEDLINE (1966 to Dec 2004), the Cochrane Library (Controlled Trials Registry and Database of Systematic Reviews), article bibliography and contacted experts for relevant studies. Inclusion criteria were English language studies, random allocation of subjects with chronic HF, and subgroup analyses defined by race. We extracted subgroup data defined by race for mortality with beta-blocker therapy and for individual agents studied.

RESULTS: We selected 4 trials which included 10,074 patients (10.3% black, predominantly African American). Mortality with beta-blocker therapy differed by race; relative risk [RR], [95% confidence interval [CI]] for mortality was 0.70 [0.64, 0.78] for non-black vs. 1.01 [0.81, 1.25] for black patients. Mortality with individual agents also differed by race; bucindolol 0.85 [0.75, 0.97] for non-black patients vs 1.15 [0.92, 1.44] for black patients, carvedilol 0.63 [0.51, 0.77] vs 0.76 [0.37, 1.54] and long acting metoprolol 0.67 [0.54, 0.82] vs 0.79 [0.36, 1.74], respectively.

CONCLUSIONS: The class and agent specific response to beta-blockers for HF therapy may not be equivalent in all ethnic groups, particularly African Americans. These findings underscore the need for further research to optimize therapy for ethnic minorities with HF.

RACIAL DISPARITIES IN HEALTH CARE GOALS FOR DIABETIC VETERANS. J.A. Long¹; J.P. Metlay². ¹Philadelphia VA CHERP, Philadelphia, PA; ²University of Pennsylvania, Philadelphia, PA. (Tracking ID #134061)

BACKGROUND: Racial disparities in achieving diabetic preventive care goals have been widely documented. In this study we evaluate whether racial disparities in achieving diabetic preventive care goals exists among veterans using the VA system.

METHODS: This project was a retrospective cohort study. Using hospital databases, we sampled veterans at the Philadelphia VAMC who were prescribed hypoglycemic medications and had an HbA1c drawn within 3 months of sampling. Each participant completed a telephone survey and had their VA chart reviewed for a period of 12 months preceding the date of interview. Our primary outcome of interest was achieving a HbA1c $\leq 8\%$. Secondary outcomes included achieving within the past year: an LDL of < 100 mg/dl; diastolic BP < 80 mm Hg; systolic BP < 130 mm Hg; having a foot exam; receiving an influenza vaccination; receiving a pneumonia vaccination; completing a dilated retinal exam; and having urine evaluated for micro-proteinuria. We performed bivariate analyses to determine if race was associated with the outcomes. For multivariate analyses, after taking into account clustering by provider, groups of variables were entered together and those significant at the $p < 0.10$ level were retained.

RESULTS: 433 AA and white veterans enrolled. 73% were AA. Whites were more likely to have a HbA1c $< 8\%$ (81% vs 60%, $p < 0.0001$); an LDL < 100 mg/dl (54% vs 42%, $p = 0.0259$), and a diastolic BP < 80 mm Hg (79% vs 69%, $p = 0.0445$). AAs were more likely to have a foot exam (86% vs 78%, $p = 0.0268$). For all other outcomes the rates for AAs were only marginally less compared to whites. After adjusting for clustering by provider, race was no longer significantly associated with diastolic BP. In multivariate analyses of those outcomes associated with race, inclusion of extensive measures of socio-demographic status, diabetic self-care, clinical status, and access to care did little to explain the association between race and the outcomes of interest. The odds for AAs compared to whites were: HbA1c $< 8\%$, unadjusted 0.36 (95% CI 0.21-0.60), adjusted 0.35 (95% CI 0.19-0.62); LDL < 100 mg/dl, unadjusted 0.62 (95% CI 0.41-

0.96), adjusted 0.71 (95% CI 0.44-1.15); and having a foot exam, unadjusted 1.83 (95% CI 1.06-3.14), adjusted 1.84 (95% CI 1.01-3.36).

CONCLUSIONS: Compared to whites, AA veterans with DM are at higher risk of failing to achieve goals for HbA1c and LDL cholesterol. We found little to explain disparities in HbA1c between AA and white diabetic veterans. However, AAs were at least as likely to receive services easily provided in an encounter, such as foot exams, while less likely to achieve outcomes that require greater patient participation. Interventions aimed solely at providers may be less effective in reducing disparities in diabetic outcomes than interventions aimed at patients and providers.

RACIAL DISPARITIES IN PRIMARY CARE AND HOSPITALIZATIONS FOR AMBULATORY SENSITIVE CONDITIONS (ACSC) AT THE END-OF-LIFE. A.C. Kronman¹; A. Ash¹. ¹Boston University, Boston, MA. (Tracking ID #133917)

BACKGROUND: End-of-life care is not always effective, or provided equitably. Minorities receive less aggressive procedures, yet costs are higher for minorities at the end of life. Disparities are known to exist in the quantity and quality of primary care received between races. The role of primary care in end-of-life care has not been well investigated. This study uses Medicare data to examine racial differences in primary care visits and hospitalizations for ACSC during the last 18 months of life.

METHODS: A random sample of 1 million Medicare beneficiaries stratified by decedent status and race was obtained from CMS. Beneficiaries excluded from the study were less than 66 years of age, not entitled to both Medicare A and B, and not enrolled in Medicare Fee for Service plan continuously. Utilization data was obtained from the MedPAR, carrier and hospice files of beneficiaries who died in the last 6 months of 2001, and linked to NDI records to determine cause of death. Demographics were obtained from the CMS 2000 Denominator file. Hospital admissions for two specific ACSC: diabetes (DM) and congestive heart failure (CHF) were drawn from the final 6 months of life, while outpatient visit frequencies were from the 12 months prior to that period. Univariate and bivariate analysis was used to assess differences between races.

RESULTS: Study sample ($n = 162,992$) characteristics: mean age 81 (range 66 - 98), female 56%, black 36%, Hispanic 10%, Other 14%. Racial differences were found in hospice use (24% white, 18% black, 20% Hispanic, $p < 0.0001$) and in-hospital deaths (38 % white, 46 % black, 52 % Hispanic, $p < 0.0001$). Outpatient visits differed significantly between races; particularly among those who had not had any ambulatory care visits (16 % white, 26 % black, 24 % Hispanic $p < 0.0001$) or primary care visits (30 % white, 39 % black, 37 % Hispanic, $p < 0.0001$). DM was the cause of 4% of the 49,598 beneficiaries with DM who were hospitalized. There were significant differences between races for those hospitalized for DM (whites 2.7%, blacks 5.1 %, Hispanic 5.0 %, $p < 0.0001$) and those hospitalized more than once (whites 7.7%, blacks 10.1 %, Hispanics 7.1 %, $p < 0.0001$). CHF was the cause of 18% of the 48,770 hospitalized beneficiaries with a CHF diagnosis. There were racial differences for those hospitalized for CHF (white 16%, black 18%, Hispanic 21%, $p < 0.0001$) and those with more than one admission (white 23 %, black 26%, Hispanic 26%, $p < 0.0001$).

CONCLUSIONS: There are significant disparities between races in frequency of primary care visits, hospice use, and hospitalizations for two ACSC toward the end of life. This suggests that increasing the provision of primary care to elderly minority groups could improve management of chronic disease, decrease costs, and improve care quality care at the end of life. Understanding the relationship between primary care and health care utilization at the end of life warrants further study.

REAL-WORLD EFFECTIVENESS OF WARFARIN THERAPY FOR STROKE PREVENTION IN MEDICARE BENEFICIARIES OF ALL RACES. E. Birman-Deych¹; M.J. Radford²; D. Nilasena³; B.F. Gage¹. ¹Washington University in St. Louis, St. Louis, MO; ²Yale University School of Medicine, New Haven, CT; ³Centers for Medicare and Medicaid Services, Dallas, TX. (Tracking ID #135087)

BACKGROUND: Without antithrombotic therapy patients with atrial fibrillation, have a stroke rate five-fold greater than age-matched patients. In randomized, controlled trials warfarin prevented 65% of ischemic strokes (hazard ratio [HR] 0.35; 95% CI 0.48-0.26) compared to no antithrombotic therapy. However, the real-world effectiveness of warfarin therapy is unknown, especially in black and Hispanic populations. Objective: To quantify use of warfarin therapy, quality of International Normalized Ratio (INR) monitoring, and effectiveness for stroke prophylaxis in Medicare beneficiaries with atrial fibrillation.

METHODS: Quality improvement organizations used standardized chart abstractions to determine the presence of stroke risk factors and prescription of warfarin therapy in 17,272 Medicare beneficiaries with chart-documented atrial fibrillation. We used Medicare claims data to determine rates of INR monitoring and hospitalizations for ischemic stroke based on validated ICD-9-CM codes.

RESULTS: After adjusting for antithrombotic therapy indication and risk factors, warfarin prescription was more frequent and monitoring more adequate in white than in black or Hispanic patients ($p < 0.0001$). The effectiveness of warfarin was lower in Medicare beneficiaries than in trial participants. In a race-stratified Cox model warfarin was protective in white Medicare beneficiaries (HR:0.62, 0.54-0.72), but not in Black (HR: 1.5, 0.83-2.7) or Hispanic beneficiaries (HR: 0.97, 0.47-2.02) (p for interaction = 0.02).

CONCLUSIONS: As compared to white beneficiaries with atrial fibrillation, black and Hispanic Medicare beneficiaries with atrial fibrillation receive warfarin therapy less commonly, their INR monitoring is less regular. In clinical practice, warfarin prevented only 38% of strokes in white patients and had no

significant benefit in black or Hispanic Medicare beneficiaries with atrial fibrillation. More careful monitoring of these vulnerable patients is necessary.

REGIONAL PATTERNS OF CARDIAC PROCEDURE USE FOLLOWING ACUTE MYOCARDIAL INFARCTION AMONG AMERICAN INDIANS. T.D. Sequist¹; A.M. Zaslavsky²; J.M. Galloway³; J.Z. Ayanian¹. ¹Brigham and Women's Hospital, Boston, MA; ²Harvard Medical School, Boston, MA; ³University of Arizona, Flagstaff, AZ. (Tracking ID #132935)

BACKGROUND: The prevalence of coronary heart disease is rising among American Indians (AI), but there is limited evidence describing processes of care for AI with acute myocardial infarction (AMI). We compared rates of cardiac catheterization, percutaneous coronary intervention (PCI), and coronary artery bypass graft surgery (CABG) between AI and whites with AMI.

METHODS: We combined data from the Nationwide Inpatient Sample and the Indian Health Service National Patient Information Reporting System to identify AI receiving care in both federal and non-federal hospitals. We identified 2,511 AI and 316,526 whites over the age of 30 years admitted with AMI during 1998 to 2001 as defined by a primary International Classification of Diseases 9th Revision (ICD-9) diagnostic code of AMI. Cardiac catheterization, PCI, and CABG were identified using ICD-9 major procedure codes. We fit multivariate logistic regression models to compare rates of cardiac procedure use between AI and whites after adjusting for age, gender, and Charlson comorbidity index. We constructed similar models stratified by geographic region and by the presence of diabetes, as well as restricted the analysis of rates of PCI and CABG to patients undergoing cardiac catheterization.

RESULTS: There were substantial admissions for AI in four geographic regions, including the Eastern region (n = 662), West North Central region (n = 447), West South Central region (n = 562), and the Mountain region (n = 840). AI were less likely than whites to undergo cardiac catheterization in 3 of 4 geographic regions, including the West North Central region [odds ratio (OR) 0.57, 95% confidence interval (CI) 0.44-0.76], the West South Central region (OR 0.32, 95% CI 0.24-0.43), and the Mountain region (OR 0.52, 95% CI 0.37-0.72). AI were less likely than whites to undergo PCI in these same regions, with odds ratios ranging from 0.43 (95% CI 0.31-0.57) in the West South Central region to 0.69 (95% CI 0.53-0.89) in the West North Central region. Nationally, AI were less likely than whites to undergo CABG (OR 0.76, 95% CI 0.60-0.95), however there was no regional variation in this difference. The presence of diabetes did not modify differences in cardiac catheterization or PCI, however AI were less likely than whites to undergo CABG among diabetic patients (OR 0.48, 95% CI 0.32-0.73), but not among non-diabetic patients (OR 0.90, 95% CI 0.72-1.12). There were no differences in rates of PCI and CABG between AI and whites among those receiving cardiac catheterization.

CONCLUSIONS: Differences in cardiac procedure use are concentrated in western regions of the country and are especially related to access to cardiac catheterization. Future research is needed to elucidate the mechanisms of these differences.

REGIONAL TRENDS IN USE OF MAJOR PROCEDURES AMONG BLACK AND WHITE ELDERLY: IS THE GAP NARROWING? A. Jha¹; E.S. Fisher²; Z. Li¹; E. Orav³; A.M. Epstein¹. ¹Harvard University, Boston, MA; ²Center for the Evaluative Clinical Sciences at Dartmouth, Hanover, NH; ³Brigham and Women's Hospital, Boston, MA. (Tracking ID #136006)

BACKGROUND: Racial disparities in major procedure rates are well known and data suggest that national gaps in care have remained stable. However, given substantial regional variation in disparities, we sought to identify local regions that have successfully eliminated gaps in care.

METHODS: We used 100% data file from the Medicare Part A program from 1992 through 2001 to calculate age-adjusted procedure rates for black and white enrollees 65 years of age or older. We calculated the white minus black gap for men and women separately for three procedures (coronary artery bypass graft surgery, carotid endarterectomy, and total hip replacement). We limited our analyses to hospital referral regions large enough to provide stable estimates.

RESULTS: During 1992-1994, compared with white enrollees, black enrollees received lower rates of all three procedures in all 158 HRRs examined (p < 0.05 in 152 HRRs). During the decade of monitoring, the white minus black gap widened in 89 HRRs (p-value for trend < 0.05 in 42 HRRs) and narrowed in 69 HRRs (p-value for trend < 0.05 in 22 HRRs). At the end of the study period, in 1999-2001, blacks still received less of each procedure in all 158 HRRs examined (p < 0.05 in 154 HRRs).

CONCLUSIONS: We found that while the white minus black gap narrowed in a few regions, the large gaps we found in the early 1990s persisted at the end of the decade. Despite the national attention on racial disparities, we found no evidence that any local regions have eliminated the racial gap in care for the three procedures examined.

REPORTING AND USE OF RACE/ETHNICITY IN GENERAL MEDICAL JOURNALS. I.W. Ma¹; N.A. Khan¹; A. Kang¹; N. Zalunardo¹; A. Palepu¹. ¹University of British Columbia, Vancouver, British Columbia. (Tracking ID #135966)

BACKGROUND: To assess health care disparities, accurate race/ethnicity descriptors of study participants are required. Methods of reporting and analysis of these factors in biomedical research are largely unknown. We sought to systematically examine the reporting practice and quality of race/ethnicity information in biomedical journals. Secondly, we sought to identify study

characteristics associated with the use of race/ethnicity data beyond baseline patient descriptors

METHODS: Observational study of all primary research articles that reported more than one racial/ethnic group, published between 1999 and 2003 in *Annals of Internal Medicine*, *JAMA*, *The Lancet*, and *The New England Journal of Medicine* (n=1152). Articles were reviewed for their use of race/ethnicity and socioeconomic variables. Inter-observer reliability was assessed by independent abstraction of 10% of study sample.

RESULTS: There were a total of 116 different terms used to describe various racial/ethnic groups. Assignment of race/ethnicity by self-report was stated in only 13% of papers; 52% of papers identifying race/ethnicity of study participants did not report any socioeconomic information. Factors independently associated with the use of race/ethnicity data beyond baseline descriptors were: studies with 1000 or more participants (adjusted odds ratio [OR], 3.81; 95% confidence interval [CI], 2.97-4.90), method of race/ethnicity assignment described (OR 2.37; 95% CI 1.61-3.48), and two or more socioeconomic categories described (OR, 1.78; 95% CI, 1.34-2.37).

CONCLUSIONS: Between 1999 and 2003, race/ethnicity information was sub-optimally reported in general medical journals. Terminology used was highly variable. Method of establishing racial/ethnic categories, rationale for collecting race/ethnicity data, and socioeconomic information were under-reported. Better data collection and reporting are necessary for describing participant characteristics and accurately studying the impact of race/ethnicity on health care disparities.

SELF-MANAGEMENT REDUCES SYMPTOMS OF DEPRESSION IN PATIENTS WITH DIABETES. J. Tong¹; E.J. Boyko¹; L. McFarland²; G.E. Reiber¹. ¹University of Washington, Seattle, WA; ²VA Puget Sound Health Care System, Seattle, WA. (Tracking ID #131657)

BACKGROUND: Depression is more prevalent in patients with diabetes and is associated with worse health outcomes and quality of life. The impact of a self-management program that teaches patients self-efficacy on reducing symptoms of depression in patients with diabetes has not been well studied.

METHODS: We conducted secondary data analysis of a randomized controlled trial of 146 patients with diabetes, aged 40-85 years old, on Medicaid SSI or dual enrolled Medicare/Medicaid, and living in Seattle, Washington. Subjects were randomly assigned to either a validated Health Enhancement Program (HEP) that promotes making patient-driven health action plans, or a control group that received usual care. Both groups were followed for one year.

RESULTS: A total of 146 patients were randomized (72 to intervention group and 74 to control group). 138 (94.5%) patients completed the study. The mean age of the study population was 60 years with 69% females. 71.2% of the patients were ethnic minorities, 38% completed less than high school education, and 30% were non-English speakers. The prevalence of symptoms of depression and anxiety was high in this population (47.3% and 58.2% respectively). Study participants had an average of 5 co-morbid conditions at baseline. The two groups were similar in diabetes and depression related measures at baseline. At 12 months, fewer patients in the intervention group had significant symptoms of depression or anxiety compared with controls [21 (30%) vs. 33 (47%), p = 0.04 for depression and 24 (34%) vs. 38 (55%), p = 0.01 for anxiety]. Receiving diabetes HEP intervention was associated with a lower depression score as determined by the Geriatric Depression Scale 15 (GDS15) ($\beta = -0.28, p = 0.023$) or a lower risk of being screened positive for depression (OR, 0.36, 95% CI, 0.16-0.82). Change in GDS score had a positive association with HbA1c% level at study exit ($\beta = 0.08, p = 0.042$) after adjusting for baseline GDS score and HbA1c% value.

CONCLUSIONS: A self-management program that promotes "self-efficacy" reduced symptoms of depression and anxiety after 12 months in a Medicaid population with diabetes. Improvement in depression symptoms was associated with lower HbA1c% levels at 12 months. These findings support the screening of mood disorders in patients with diabetes, particularly in the disadvantaged, high-risk population. Treating patients with symptoms of depression may lead to improvement of diabetes control. Future studies are needed to identify the specific components of the HEP program that lead to improvement of depressive symptoms.

SLEEPING ROUGH: OUTCOMES OF A FIVE-YEAR OBSERVATIONAL STUDY OF HOMELESS PERSONS LIVING ON THE STREETS OF BOSTON, 2000-2004. J.J. O'Connell¹; J.S. Roncarati¹; P.J. Perri¹; E.C. Reilly¹; C.A. Kane¹; K.E. Jones¹; J. Strupp Allen¹. Boston Health Care for the Homeless Program, Boston, MA. (Tracking ID #135231)

BACKGROUND: The obstacles to health care services faced by the sub-population of homeless persons who live on the streets ("rough sleepers") are daunting. A dearth of literature exists to document the burden of medical illness of urban street dwellers. Efforts to care for this elusive population have been limited to acute and episodic interventions. Lack of continuity has obviated quality primary and preventive care. Boston Health Care for the Homeless Program' (BHCHP) Street Team (two internists, a physician assistant, two psychiatrists, a therapist, and a nurse) utilizes a multidisciplinary team model of care that offers health care services directly on the streets of Boston. The Street Team works directly with day and night outreach teams from local agencies, conducts hospital-based clinics and inpatient care at Massachusetts General Hospital (MGH) and Boston Medical Center (BMC). Immediate access is available directly from the street to two public dual diagnosis detoxification units as well as BHCHP's 90-bed medical respite program.

METHODS: In January 2000, a cohort of 119 high-risk individuals was identified from over 800 street persons cared for by the Street Team. All persons in

this cohort were over age 18, had lived on the streets for at least six months, and meet one of several criteria for increased morbidity and mortality. The cohort does not differ demographically from the 8000 sheltered homeless persons receiving BHCHP services. However, 88% of the high-risk cohort suffer from the tri-morbidity of chronic medical illness, major mental illness, and active substance abuse. Contact data is recorded on each patient at a weekly Street Team meeting. Other data sources include BHCHP's electronic medical record, hospital records at MGH and BMC, and a supplemental street database maintained by the team.

RESULTS: After five years, the Street Team was able to identify the whereabouts of 114 (96%) of the 119 in the cohort: 33 (28%) have died, 7 (6%) are disabled or ill enough to have been placed in nursing homes, 2 (1.6%) are incarcerated, 10 (8.4%) are in emergency shelters, transitional shelters, or recovery programs, 38 (32%) individuals are housed in apartments or rooming houses, or living fulltime with families and friends, 24 (20%) remain on the streets, and 5 individuals (4%) have been lost to follow-up. Primary care outcomes are evaluated annually, and in 2004 included: 65% had PPD documentation within the past year; 75% received a Pneumovax within the last ten years; 38% received a yearly flu vaccine. For female patients, 8 (31%) of 26 women received a PAP smear and 5 (23%) of the 22 women received a mammogram in the past two years.

CONCLUSIONS: Mortality and morbidity are high among those living on the streets of Boston. BHCHP's service delivery model provided continuity of primary and preventive care to a high-risk cohort of rough sleepers, with primary care outcomes comparable to those achieved for the sheltered homeless population receiving care from BHCHP. The Street Team offers a model of street care for cities seeking strategies to reach this vulnerable and disenfranchised population.

STRESSED OUT: DEFINITIONS OF STRESS AMONG HYPERTENSIVE AFRICAN AMERICAN MEN. J.E. Ravenell¹, M.E. Charlson¹. ¹Cornell University, New York, NY. (Tracking ID #135860)

BACKGROUND: African American (AA) men have the highest age-adjusted mortality rate from hypertension in the United States. Stress is postulated to be a major etiologic factor in the development of hypertension, while reducing stress has been shown to improve blood pressure. Though evidence suggests AA men have higher levels of stress than other groups, few studies have specifically examined AA men's perceptions of stress. The objective of this study is to explore and illuminate perceptions of stress among AA men with hypertension.

METHODS: Open-ended questions in qualitative one-to-one interviews were used to assess how hypertensive AA men define stress and its relationship to worry, fear, and perceived racism. Twenty hypertensive AA male participants were recruited from an adult internal medicine practice at an urban academic medical center. Patients were identified from review of an electronic medical record system using ICD 9 codes for diagnosis of HTN. Patients were eligible if they were self-identified AA males, age 18 or older, and fluent in English. Interviews were audio taped, transcribed verbatim, coded and analyzed using grounded theory.

RESULTS: Among the 20 AA male participants, the mean age was 62 with a range of 32-89 years old. Definitions of stress included "unusual circumstances in one's life," "anticipation of what is going to happen", and "a prolonged state of dissatisfaction." Stress was perceived as being similar to worry and fear, distinguished only by duration and intensity. Perceived racism in employment settings and in health care settings was an important source of stress, and often transformed benign activities (i.e. going shopping) into "stressful situations" for AA men. 65% reported being worried about racism at work, and 35% reported at least one experience of perceived racism involving a physician.

CONCLUSIONS: African American men perceive stress as a key influence on their health. Understanding how African American men conceptualize stress may be important in tailoring interventions for stress reduction among hypertensive African American men. Additionally, perceived racism in everyday life and in the medical encounter is prevalent, and may have important implications for the health of hypertensive African American men.

THE ASSOCIATION OF DISTANCE TO CARE WITH SEVERITY OF DISEASE, LENGTH OF STAY AND CHARGES FOR PERSONS HOSPITALIZED WITH HIV/AIDS. R.E. Vargas¹, P. Robinson², S. Ettner¹, W.E. Cunningham¹. ¹University of California, Los Angeles, Los Angeles, CA; ²Charles R. Drew University of Medicine and Science, NA, CA. (Tracking ID #136180)

BACKGROUND: Our study examines the association of patient residential distance to the nearest HIV/AIDS medical provider and nearest ancillary services site with severity of disease, length of stay and hospital charges for patients with HIV/AIDS admitted to acute care hospitals.

METHODS: We combined data from the California Office of Statewide Health Planning and Development 2000 discharge files, the AIDS Project Los Angeles list of HIV services, and the 2000 Census. Our main predictors were distances from the patient's population-based zip code centroid to the nearest HIV/AIDS medical provider site and nearest ancillary care service site. Our outcomes were dichotomized severity of disease indexes for conditions present at the time of hospitalization: 1) Deyo Charlson 2) Dartmouth Manitoba (DMI) Charlson scores and 3) the Severity Classification for AIDS Hospitalizations (SCAH); length of stay and hospital charges were additional outcomes. We used multiple logistic and linear regression to control for differences in patient age, gender, race, and insurance status; and levels of poverty, English language proficiency, racial and ethnic segregation, education level and number of hospital beds in the patient's

neighborhood of residence. Generalized Estimating Equations were used to adjust the variance for clustering by zip-code.

RESULTS: We identified 3543 hospitalizations for persons with HIV/AIDS in Los Angeles County in 2000. Patients were 85% male, 56% white, 29% black, and 37% Hispanic irrespective of race. Insurance coverage was 26% Medicare, and 47% Medicaid and other indigent care programs. Patients came from communities where, on average, 35% of people had less than a high school education and the ratio of households making less than \$20,000 to those making over \$60,000 was 2:1. Mean patient's distance from care was 0.68 miles (range 0.03-13.3; s.d.0.82) from ancillary care services and 1.81 miles (range 0.05-15.8; s.d.1.5) from medical care services. Multiple logistic regressions suggested that a greater distance to ancillary services was associated with a significantly lower disease severity index among patients admitted to acute care hospitals for both Charlson indexes, Deyo: Odds Ratio (OR) 0.84, 95% Wald Confidence Interval (CI) (0.73, 0.97) and DM: OR 0.86 CI (0.75, 0.99) but not for the SCAH index: OR 1.04 CI (0.93, 1.16). Multiple linear regressions suggested that greater distance from ancillary care services was associated with a significantly greater length of stay (estimate 0.078; p = 0.002) but not with hospital charges (estimate 0.044; p = 0.120). Regression models did not show a significant association between distance to medical care provider and length of stay, charges or any severity indexes.

CONCLUSIONS: Our results suggest that greater distance to ancillary, but not medical, HIV services is associated with a lower disease severity at admission and a longer length of hospital stay during admission, implying a lower clinical threshold for admission and a greater threshold for discharge for patients from communities that are further away from ancillary HIV care services.

THE EFFECT OF QUALITY IMPROVEMENT ON RACIAL DISPARITIES IN DIABETES CARE. T.D. Sequist¹, A.S. Adams², F. Zhang², D. Ross-Degnan², J.Z. Ayanian¹. ¹Brigham and Women's Hospital, Boston, MA; ²Harvard Medical School and Harvard Pilgrim Health Care, Boston, MA. (Tracking ID #133120)

BACKGROUND: Since 1997 Harvard Vanguard Medical Associates (HVMA) has implemented quality improvement (QI) programs for diabetes including a new electronic medical record (EMR), electronic physician reminders, and patient mailings. We evaluated whether these QI programs were associated with changes in racial disparities in care.

METHODS: We identified diabetic patients ≥ 18 years receiving primary care at HVMA, a large multispecialty group practice, during 1997 to 2001 using insurance claims and EMR data. We assessed 5 quality measures based on American Diabetes Association guidelines: annual retinopathy screening, annual HbA1c testing, annual LDL cholesterol testing, and rates of appropriate HbA1c control ($< 7.0\%$) and LDL cholesterol control (< 130 mg/dL). To assess changes in racial disparities for each indicator, we used logistic regression to adjust for race, year, race-year interactions, age, and sex, with general estimating equations to control for repeated measures among patients.

RESULTS: Among 7,216 diabetic patients, 5,180 were white and 2,036 were black, with a mean age of 58 years and 51% female. From 1997-2001, LDL testing and control improved markedly with reductions in racial gaps, but HbA1c testing and control were essentially unchanged by race and retinopathy screening declined similarly for whites and blacks (see Table).

	1997 Rate (%)		2001 Rate (%)		Δ in Racial Gap	P value for Δ
	White	Black	White	Black		
LDL testing	43	30	65	62	-10%	<0.001
LDL control	18	9	45	39	-3%	<0.001
HbA1c testing	76	78	76	77	+1%	0.06
HbA1c control	34	24	36	26	0%	0.34
Retinopathy screen	75	71	71	66	+1%	0.83

CONCLUSIONS: Racial disparities were diminished following successful QI efforts in cholesterol management, but persisted following comparable QI efforts in glycemic control that had limited impact. Generalized quality improvement programs can result in reduced disparities when overall quality of care improves rapidly, but reducing other persistent disparities may require a specific focus on minority health.

THE EVIDENCE BASE FOR DIVERSITY IN THE HEALTH PROFESSIONS. S. Saha¹, S. Shipman². ¹Portland VA Medical Center, Portland, OR; ²Oregon Health & Science University, Portland, OR. (Tracking ID #136263)

BACKGROUND: Increasing the racial and ethnic diversity of the health professions workforce is commonly cited as essential to reducing racial/ethnic disparities in the quality of health care. Programs promoting diversity, however, including affirmative action policies, have become vulnerable in the last decade. Sustaining such programs will increasingly require documented evidence of the benefits of health professions diversity.

METHODS: Based on a conceptual model describing the potential impact of health professions diversity on health care access, quality, and outcomes, we conducted a systematic review to identify studies addressing four hypotheses: 1) that minority and non-minority health care providers serve different populations; 2) that racial/ethnic and language concordance between patients and health care providers is associated with minority patients' use of services and adherence, quality of care, and health outcomes; 3) that institutional workforce diversity is associated with minority patients' trust in health care institutions and use of services; and 4) that minority and non-minority health care providers, leaders, and researchers advocate for and implement different programs, policies, and research agendas. We searched the MEDLINE, HealthSTAR, CINAHL, and PsycINFO databases using strategies tailored to each hypothesis. Two reviewers screened abstracts. Disagreements were adjudicated by consensus. We also reviewed reference lists for relevant articles. One reviewer read and qualitatively summarized each original study in terms of quality, characteristics (e.g., population, health profession category), and principal findings.

RESULTS: From 586 abstracts, we identified 41 original studies. Sixteen studies addressed the service patterns of physicians (15) and dentists (1). These studies, many of them rated as high-quality, uniformly found that minority providers disproportionately served minority, poor, Medicaid, and uninsured populations. Twenty-five studies addressed race and language concordance between patients and physicians (11), mental health providers (9), substance abuse counselors (4) and medical students (1). These studies were mixed in their findings but generally found that both race and language concordance were associated with higher interpersonal quality of care (usually assessed with patient satisfaction surveys), and to a lesser extent, utilization and adherence. No studies directly addressed our hypotheses about institutional diversity and patient trust, or about differential program, policy, and research priorities between minority and non-minority health professionals.

CONCLUSIONS: Current studies related to the impact of healthcare workforce diversity on health care access, quality, and outcomes are generally limited to those demonstrating that minority physicians disproportionately serve underserved populations, and that patient-provider race and language concordance are associated with patient satisfaction. Further research is needed to examine the service patterns of minority and non-minority health professionals other than physicians; the impact of race and language concordance on quality and outcomes of care; the association between institutional diversity and patient trust and use of services; and the programmatic and policy priorities of minority and non-minority health professional leaders. Such studies may play an important role in determining the fate of programs and policies aimed at increasing the diversity of the health professions workforce.

THE HIGH ACCEPTANCE RATE OF ROUTINE, VOLUNTARY HIV TESTING IN A MASSACHUSETTS PRISON. R.V. Liddicoat¹; H. Zheng¹; Y. Golan²; J. Internicola³; E.P. Rubinstein⁴; K.A. Freedberg¹; R.P. Walensky¹. ¹Massachusetts General Hospital, Boston, MA; ²Tufts University, Boston, MA; ³Suffolk County House of Corrections, Boston, MA; ⁴Massachusetts Department of Public Health, Boston, MA. (Tracking ID #132625)

BACKGROUND: U.S. prisons have a high prevalence of HIV infection among inmates and are ideal settings for routine, voluntary HIV counseling, testing, and referral (CTR) programs. However, few prisons in the U.S. offer routine HIV testing to all inmates at entry. We examined the impact of a routine, voluntary HIV CTR intervention on HIV test acceptance and HIV case detection in a Massachusetts prison.

METHODS: We provided HIV education and offered routine, voluntary HIV CTR in a Massachusetts prison for sentenced inmates. We compared the first month of the intervention, (11/15/04-12/14/04), to a control period, (1/1/03-12/31/03). During the control period, HIV education was provided to all sentenced inmates, but HIV testing was made available only upon inmate or physician request. We compared, by Fisher's exact test, the percentage of all inmates tested and percentage tested positive between intervention and control periods as well as known HIV prevalence during both periods. We also examined reasons for HIV test refusal during the intervention.

RESULTS: During the first month of the intervention, we offered routine, voluntary HIV CTR to 223 sentenced inmates who entered the prison and were not already known to be HIV-infected. Of those, 157/223 (70.4%) accepted and had an HIV test. During the 2003 control period, 310/1,757 (17.6%) inmates requested and received testing [OR 11.10 95% CI (8.12-15.18)]. During the intervention, 1/157 (0.7%) inmates tested was found to be HIV-infected, compared to 3/310 (1.0%) during the control period ($p = 0.71$). Of all inmates who entered the prison during the intervention period, 7/230 (3.0%) were previously known to be HIV-infected, similar to the control period, 55/1,757 (3.1%) ($p = 0.94$). The most common reasons for refusal during the intervention were "already tested" (53.2%), "not at risk" (33.9%), "afraid to know" (14.5%) and "don't want to find out while in prison" (9.7%). Of those who refused due to a previous negative HIV test, 20/62 (32.3%) reported HIV testing within the previous 3 months.

CONCLUSIONS: Over 70% of prison inmates in a Massachusetts prison accepted HIV testing when it was offered routinely, a dramatic increase over HIV testing initiated by inmate or physician request. Programs like this have the potential to identify substantial numbers of undiagnosed HIV patients.

THE RELATIONSHIP BETWEEN PATIENT RACE AND PATIENTS' PERCEPTIONS OF THEIR PHYSICIANS' CULTURAL COMPETENCE AND PATIENT-CENTERED COMMUNICATION SKILLS. J.J. Rencic¹; E. Liles². ¹Tufts-New England Medical Center, Boston, MA; ²University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #136124)

BACKGROUND: Patient-centered communication, defined as communication that promotes shared decision making between physician and patient, has been

shown to improve compliance and some health care outcomes. Recently, it has been suggested that the cultural competence of physicians can impact their ability to effectively communicate. Our study examines the relationship between patient race, and patients' perceptions of physicians' patient-centered communication skills and cultural competence.

METHODS: The CCTOP study (Cultural Competency Training and Outcomes in Patients) is an 11-center, cross-sectional study of patients' and physicians' attitudes regarding culture, bias, and clinical care. A 76-item questionnaire was administered to 291 patients. We performed a bivariate analysis of patients' perception of their autonomy in making medical decisions and their doctors' understanding of their culture. We then constructed regression models with patients' perceptions of their autonomy in medical decision making as the outcome with race as the independent variable of interest while controlling for age, gender, race, income, education, and perceptions of physicians' cultural awareness.

RESULTS: We report preliminary data based on 291 patients surveyed on the data of six centers. In terms of race, the patients were 30.2% white, 45.0% African-American, and 24.4% Hispanic. The majority of the patients surveyed felt that their doctors are aware of their cultural beliefs (65.5%) and that their doctors try to give them some control regarding treatment decisions (67.6%). Seventy-four percent of patients who felt that their doctors were aware of their cultural beliefs believed that their doctors gave them some control regarding treatment decisions as opposed to only 55.6% of those who felt their doctor unaware of their cultural beliefs (OR 2.26, CI 1.32-3.86). Logistic regression analysis adjusting for age, gender, race, income, and education, showed that patients who felt that their doctor understands their culture remained 2.4 times (CI 1.09-5.42) more likely to feel that their doctor gave them some control over treatment decisions. However, white patients were approximately 2.3 times more likely to believe that they have this autonomy compared to African-American (OR 0.41, CI 0.18-0.91) and Hispanic patients (OR 0.41, CI 0.17-0.99).

CONCLUSIONS: These preliminary findings demonstrate that minority patients feel that they have less control in making decisions regarding their treatment. Significantly, most of the minority patients in our study were in race-discordant physician relationships. Though patients who believed that their doctor understood their culture felt more autonomy in decision making, race was not an independent predictor for such a belief. These data suggest that focusing only on training and educating physicians in cultural competency is unlikely to be sufficient to improve patient-centered communication and healthcare outcomes in minority patients.

TRUST IN THE HEALTH CARE SYSTEM AND PROVIDERS AMONG HIV-POSITIVE AND -NEGATIVE WOMEN FROM THE BRONX, NY. C.O. Cunningham¹; N.L. Sohler¹; L. Korin¹; K. Anastos¹. ¹Albert Einstein College of Medicine, Bronx, NY. (Tracking ID #133057)

BACKGROUND: Trust in the medical system and providers is essential for patients to be active participants in their health care. Distrust in medical providers, which is frequently reported among minority patients, has been associated with a range of negative outcomes including lower patient satisfaction, poor treatment adherence, less use of preventive services, and poor overall health status. As HIV disease continues to devastate minority populations and requires strict adherence to complicated treatment regimens, physicians treating HIV-positive patients have realized that a trusting relationship with their patients is essential. This study examines whether HIV-positive women trust the health care system and their providers more than HIV-negative women from the same geographic region and socioeconomic background.

METHODS: A sub-sample of 106 women from the Bronx site of the Women's Interagency HIV Study (WIHS, a prospective cohort study of the natural history of HIV infection in women), were interviewed during a regularly scheduled follow-up study visit about sociodemographic characteristics, recent substance use behavior, physical and mental health, trust in the health care system, and trust in the primary health care provider. Laboratory data were collected to examine HIV status and severity. HIV-positive and HIV-negative women were compared on ratings of trust in the health care system and their providers. Linear regression analyses were used to compare the two groups on trust ratings, while adjusting for differences in sociodemographic characteristics and overall health ratings.

RESULTS: Of the 106 WIHS participants interviewed, the majority were black or Latino, single, unemployed, had not graduated from high school, and had public insurance. Nearly half had severe depressive symptoms, 12% were actively using heroin or cocaine, and most HIV-positive participants had CD4+ counts between 201-500 cells/mm³. Most participants reported a high degree of distrust in the health care system, with 50% agreeing that "there is a cure for AIDS, but the government is keeping it from us" and 67% agreeing that "medical scientists know more about HIV than they're letting on." There were no differences between HIV-positive and -negative participants in these items. However, HIV-positive participants reported a great deal of trust in their providers, and their overall rating was significantly higher than that of HIV-negative participants (Trust in provider scale: 8.38 vs. 7.28, $p < 0.05$; range 1-10; 10 most trust). In linear regression analysis, HIV-positive status continued to be a significant predictor of greater trust in one's provider.

CONCLUSIONS: Black and Latino women from the Bronx, NY distrust the health care system specifically regarding HIV disease. However, HIV-positive women trust their primary health care providers, rating trust in their providers significantly higher than those of their HIV-negative counterparts. These findings are likely to be explained by unique characteristics of HIV disease, HIV providers, and HIV-positive women. As HIV continues to affect vulnerable populations that historically have been less trusting of the health care system, it is important to recognize how trust may impact the treatment of HIV disease.

U.S. HOSPITALIZATIONS & COSTS FOR ILLICIT DRUG USERS WITH SOFT TISSUE INFECTIONS FROM 1998-2001. T.A. Takahashi¹; M. Maciejewski¹. ¹University of Washington, Seattle, WA. (Tracking ID #133359)

BACKGROUND: Up to 32% of injection drug users have soft tissue infections (STIs) at any given time. Injection drug use-related STIs have become the most common reason for admission to some urban, public hospitals because approximately 40% of injection drug users who seek emergency department care are admitted for treatment of their infections. The economic and health care burden of treating these infections has never been determined nationally, perhaps because no ICD-9 code exists for STIs related to injection drug use. This study estimates national inpatient health care utilization and costs for hospitalizations with illicit drug use and STI diagnoses as a proxy for injection drug use-related STIs and assesses for trends over time.

METHODS: We used the Health Care Cost and Utilization Project Nationwide Inpatient Sample, a national database of a 20% sample of non-Federal U.S. hospitals, for 1998-2001. Hospitalizations for illicit drug users with STIs were identified by discharge ICD-9 codes for all patients aged 11-65 years. We describe hospitalizations for illicit drug users with STIs, estimate their costs using charge data, and assess for trends over a four-year period from 1998-2001. All hospital charges were adjusted to 2001 dollars.

RESULTS: On average, there were an estimated 26,982 hospitalizations for illicit drug users with STIs in the U.S. for each year from 1998-2001, which comprised approximately 0.14% of all hospitalizations each year. Twenty-three percent of patients hospitalized for illicit drug use related STIs had no insurance (self-pay) and 1% died in the hospital. The median length of hospitalization for illicit drug users with STIs was 3.8 days, and the duration of hospitalizations remained stable from 1998-2001. However, median charges per hospitalization increased from \$7687 in 1998 to \$10,238 in 2001 due to an increase in the median charges per day of hospitalization. Estimated hospitalization charges for illicit drug users with STIs totaled over 276 million dollars in 2001. Although the proportion of hospitalizations that were for STIs in general increased linearly from 2.56% in 1998 to 2.81% in 2001 ($P < 0.0001$), hospitalizations for STIs related to illicit drug use remained stable. Therefore illicit drug users made up a decreasing proportion of all hospitalizations for STIs over the 4-year period (from 6.4% in 1998 to 4.9% in 2001; $P < 0.0001$). Approximately 4.4% of hospitalizations assigned an ICD code for illicit drug use also had a STI diagnosis each year. Overall, 68% of hospitalizations for illicit drug use-related STIs were in urban, teaching hospitals. However, there was a significant increase in these hospitalizations at non-teaching hospitals from 27% in 1998 to 39% in 2001 ($P < 0.0001$).

CONCLUSIONS: The proportion of hospitalizations for illicit drug users with STIs remained stable from 1998 to 2001. However, the estimated median charges for each hospitalization associated with such infections increased by 33% and totaled over a quarter of a billion dollars in 2001. Nearly one-fourth of hospitalizations for illicit drug use-related STIs were not covered by insurance. Interventions that facilitate early identification and outpatient treatment of STIs may aid in reducing hospital charges. Having a unique ICD-9 code for illicit drug use STIs would improve the accuracy of tracking these infections in the future.

USE OF ADJUVANT CHEMOTHERAPY AND RADIATION THERAPY FOR COLORECTAL CANCER AMONG ASIAN AMERICANS AND PACIFIC ISLANDERS. R.P. Gelber¹; J.W. Davis²; T.B. Seto³. ¹Massachusetts Veterans Epidemiology Research and Information Center, VA Boston Healthcare System, Harvard Medical School, Boston, MA; ²Hawaii Medical Service Association, Honolulu, HI; ³University of Hawaii John A. Burns School of Medicine, Honolulu, HI. (Tracking ID #134309)

BACKGROUND: Consensus guidelines and randomized trials support the use of chemotherapy for stage III colon cancer and chemotherapy with radiation therapy for stages II and III rectal cancer to improve survival. While the use of these treatments may be partly related to patient ethnicity, the management of Asian Americans and Pacific Islanders (AAPIs) with colorectal cancer remains largely unknown. Objective: To evaluate the use of chemotherapy (CTX) and radiation therapy (XRT) for colorectal cancer among AAPIs.

METHODS: Retrospective analysis of the first course of therapy for 602 patients (336 Japanese, 46 Chinese, 74 Filipino, 48 Hawaiian, 98 white; mean age 67 ± 13 yrs) diagnosed with stage III colon cancer ($n=347$) or stages II ($n=109$) or III ($n=146$) rectal cancer in Hawaii from 1995-2001. We linked data from the Hawaii Tumor Registry, a Surveillance, Epidemiology, and End Results Program population-based registry, to administrative data from Hawaii's largest health insurer. We evaluated the use of standard-of-care treatments as defined by NCI/NIH guidelines: (1) CTX for stage III colon cancer and stages II or III rectal cancer, and (2) XRT for stages II or III rectal cancer. We used logistic regression to examine the association between ethnicity and quality of care.

RESULTS: Overall, 71.6% ($n=431$) of patients received CTX. Only 57.6% ($n=147$) of patients with rectal cancer received XRT. CTX rates were similar among all ethnicities (from 67.6% of Filipinos to 77.1% of Hawaiians, $P = 0.8$). Adjusting for tumor stage and site, age, gender, year of diagnosis, rural residence, cancer history, grade, Charlson comorbidity index, health plan (HMO, fee-for-service, Medicare), and income, CTX was used significantly less often among older patients (OR, 0.02; 95%CI, 0.01-0.06, for age ≥ 85 vs. <55) and those with prior cancer (OR, 0.53; 95%CI, 0.29-0.99) or stage II rectal cancer (OR, 95%CI: 0.37, 0.21-0.65, for stage II rectal; 1.66, 0.92-3.01, for stage III rectal vs. stage III colon). XRT rates were low among all ethnic groups (from 52.9% of Chinese to 77.8% of Hawaiians, $P = 0.26$). Hawaiians were more likely than whites to receive XRT in adjusted analyses (OR, 95%CI: 4.61, 1.25-16.99, for Hawaiians; 1.30, 0.54-3.12, for Japanese; 0.91, 0.29-2.82, for Filipinos; 0.95, 0.25-3.57, for Chinese). Significant predictors of lower XRT use included advanced age (OR, 0.09; 95%CI, 0.02-0.47, for age ≥ 85 vs. <55) and prior cancer (OR, 0.28; 95%CI, 0.11-0.74).

CONCLUSIONS: We found low rates of standard treatments for colorectal cancer among all ethnic groups in our population, without significant disparities among AAPIs. Further study is needed to determine the reasons for the observed low use of CTX and XRT.

VALIDATION OF A STRUCTURED DIAGNOSTIC INTERVIEW FOR DETECTING POST-TRAUMATIC STRESS DISORDER AND MAJOR DEPRESSION IN ASYLUM SEEKERS. S. Durieux-Paillard¹; P.A. Bovier¹; B. Whitaker-Clinch¹; A. Eytan². ¹Geneva University Hospitals, Geneva, ²Geneva University Hospitals, Chene-Bourg, . (Tracking ID #134139)

BACKGROUND: The number of refugees, including asylum seekers and displaced persons, has increased world-wide. Post-traumatic Stress Disorder (PTSD) and Major Depressive Episode (MDE) are the most frequent psychiatric diagnoses found in refugees studies, resulting from exposure to war and organized violence. As PTSD and MD can lead to marked psychological distress, mental health has become a public health issue in refugees populations. Concerning asylum seekers in western countries, early screening of these disorders should be performed as soon as possible, as hard asylum procedures may be accompanied by an increase in psychopathology. Methodological problems such as transcultural validity and translation difficulties of existing instruments make the screening of these disorders difficult. In primary care setting, the cultural diversity of these populations is a particularly challenging situation. The Migrants Health Center is an ambulatory facility of Geneva University Hospitals devoted to asylum seekers, where about 800 newly-arrived persons, representing more than 50 countries, are screened every year. For 12 years, nurses systematically interview new asylum seekers, using a questionnaire designed to facilitate screening of health problems. The aim of the present study was to test the ability of a new diagnostic instrument to detect persons who need care for PTSD and/or MDE.

METHODS: Type: Prospective cohort study. Measure Instrument: PTSD and MDE sections of the Mini International Neuro Psychiatric Interview (MINI) were included in the usual questionnaire, in order to improve psychological screening. After a pre-test period among an adaptation process of the MINI questions was judged necessary, because the original version sound ill-adapted, culturally too direct and too complex for the target population. Procedure and reference measures: usual screening interview by nurses, and if necessary with interpreters, using the adapted questionnaire, followed by a clinical assessment by a psychologist (gold-standard). Presence or absence of PTSD and MDE diagnoses were recorded, according to the first and the second interview. Performances of the questionnaire were compared to the clinical assessment. Statistical analysis: sensitivity (se), specificity (sp), positive and negative predictive values (PPV and NPV) were calculated.

RESULTS: 101 persons tested mean age: 30 years, 74% men, origins: Africa 58%, Europe 37%, Asia 5%. Diagnoses (Q: questionnaire C: clinical assessment) MDE Q: 30% C: 33% PTSD Q: 28% C: 30% Both Q: 21% C: 24% Questionnaire performances MDE se 79%, sp 94%, PPV 86.5%, NPV 90% PTSD se 85.5%, sp 94.5%, PPV 86%, NPV 92% Differences according to geographical origins and use of interpreters

CONCLUSIONS: After adaptation to the life-context of asylum seekers, a generic and non culture-specific diagnostic instrument can be used to screen PTSD and MDE in a culturally diverse population.

VALIDATION OF RACIAL AND ETHNICITY CLASSIFICATION IN HOSPITAL ADMINISTRATIVE DATA. N. Laiteerapong¹; B.J. Sherman¹; K. Kim¹; K.M. Freund¹. ¹Boston University, Boston, MA. (Tracking ID #135477)

BACKGROUND: Hospital administrative databases have frequently been cited as having high rates of racial misclassification. This misclassification can be a major obstacle in the health surveillance of racial and ethnic minority populations and may obscure disparities in health status measures or health care. This study examined the validity of racial/ethnicity data in a hospital registration database and possible variables associated with misclassification.

METHODS: Race/ethnicity, birthplace, age, primary language, and insurance status data were gathered from a hospital registration database from patients seen in an urban medical center from July 2002 to September 2002. A second set of data, including race/ethnicity and educational status, were collected from a self-administered health history form completed by patients seen in one specific women's health primary care practice at the medical center. Self-identification of race/ethnicity was used as the gold standard of race/ethnicity categorization, based on accepted current clinical practice and the Office of Management and Budget's Statistical Policy Directive No. 15. We calculated concordance rates, kappa values, and confidence intervals between registration information and self-identified classification for each race/ethnicity group. Concordance between the two sources of data on race/ethnicity by other patient demographics was also analyzed. Fisher's Exact Test and chi-square analysis were used to evaluate the possible association of racial concordance to each demographic variable.

RESULTS: Of the 252 patients studied, 124 (49%) identified themselves as white, 96 (38%) as black, 27 (11%) as Hispanic origin, 5 (0.4%) as Asian or Pacific Islander (Asian). Concordance between self-identification and registration information was 94% for whites, 95% for blacks, 81% for Hispanic origin, and 100% for Asians. Kappa values were 0.90 (95% confidence interval [CI], 0.85-0.96) for whites, 0.90 (95% CI, 0.84-0.95) for blacks, 0.81 (95% CI, 0.69-0.93) for Hispanic origin, and 0.91 (95% CI, 0.73-1.00) for Asians. Analysis indicated that patients born in the United States or Canada were less likely to be misclassified in the registration database compared with foreign-born patients

($P < 0.0001$). Age, primary language, education level, and insurance status were not associated with race/ethnicity concordance.

CONCLUSIONS: Hospital administrative data can potentially be an accurate source of data to answer important health services questions about race/ethnicity and services. Validation against patient self-report is necessary to assess the validity of data at a specific institution.

WHAT IS "CULTURAL COMPETENCE?" AFRICAN AMERICANS' PERSPECTIVES ON PATIENT-PHYSICIAN INTERACTIONS. S. Saha¹; E. Morse²; T. Gipson³; R. Loudd²; C. O'Brien⁴. ¹Portland VA Medical Center, Portland, OR; ²Oregon Health & Science University, Portland, OR; ³University of Rochester, Rochester, NY; ⁴Portland State University, Portland, OR. (Tracking ID #136226)

BACKGROUND: Studies suggest that racial and ethnic discordance between patients and their physicians may contribute to disparities in the quality of health care. Cultural competence training has been advocated as a means to bridge the quality gap caused by racial and ethnic discordance. There is little empiric information, however, to guide the content of cultural competence training. Specifically, little is known as to how racial and ethnic discordance affects the patient-physician relationship, and what minority patients consider to be important deficits in the quality of their interactions with physicians.

METHODS: As part of a larger community-based participatory research project, we conducted eight focus groups (FGs) with African American patients with diabetes and/or hypertension, four each with men and women. Community members trained in FG facilitation led the groups, which explored good and bad experiences with physicians, perspectives on physician race/ethnicity, physicians' handling of sensitive topics (e.g., drug use, domestic violence), trust, respect, and adherence to physician recommendations. FG discussions were audiotaped, transcribed verbatim, and analyzed using a grounded theory approach, aided by NUD*IST software.

RESULTS: Several themes emerged from the FG data. Among the most prominent was a frustration with physicians' not validating patients' perspectives on their illnesses. As one woman articulated, "Sometimes I don't think they have a lot of respect for a woman knowing her own body." Participants also expressed the importance of physicians' stereotyping and bias with regard to African Americans. Bias was often related to drug use and ranged from subtle—"They say they understand and they're happy that you're in treatment and all that, but their actions show something different. . . Just the way he talks to you. . . you could tell when somebody is there for you. . . they're not looking at you, they're looking through you"—to overt—"I said, 'Read my lips! I told you I don't use drugs, I never use drugs, I don't even take medication (pounding table for emphasis), so don't ask me no more.'" Most participants denied the importance of physician race, indicating that communication and respect were the physician characteristics that mattered most.

CONCLUSIONS: FGs with African-Americans with diabetes and HTN revealed the importance of physicians' not discounting patients' explanatory models of illness, validating a central component of many cross-cultural training programs. Patients reported being able to discern subtle manifestations of physician bias, again validating a hypothesized source of racial disparities in the quality of patient-physician relationships. Patients explicitly denied the importance of physician race and suggested that interpersonal competence was of primary importance. Further research is needed to understand whether the effects of race concordance in the patient-physician relationship are mediated by different communication and interpersonal styles, or by subtle, perhaps unconscious, effects of race on patients' and physicians' perceptions.

WILLINGNESS TO PARTICIPATE IN CLINICAL TRIALS AMONG ELDERLY WHITES AND AFRICAN AMERICANS. R.W. Durant¹; R.B. Davis¹; E. Marcantonio¹; M.B. Freeman²; B.E. Landon¹. ¹Beth Israel Deaconess Medical Center, Boston, MA; ²Hebrew Senior Life Research and Training Institute, Boston, MA. (Tracking ID #134246)

BACKGROUND: Despite National Institutes of Health mandates for minority inclusion in clinical research, African Americans are underrepresented in clinical trials. Many have hypothesized that African Americans are less willing to participate in clinical trials partially due to higher levels of distrust in the health care system. We sought to determine the differences in willingness to participate in clinical trials among elderly whites and African Americans and the factors that influence their willingness to participate.

METHODS: We surveyed, via mail, 1412 eligible community dwelling elderly (age > 50) whites and African Americans in the Boston metropolitan area. We assessed willingness to participate in 3 hypothetical clinical trials with increasing levels of risks (diet trial for obesity, antihypertensive trial for HTN, chemotherapy trial for cancer) on an ordinal scale from 1 ("not willing") to 5 ("very willing"). We created a mean willingness to participate scale calculating the mean score for each respondent based on responses to the 3 vignettes (Cronbach's alpha 0.74). For analysis, each respondent was categorized as less (mean score < 4) or more (mean score > or = to 4) willing. We assessed trust in one's primary care provider (PCP) and trust in clinical research separately using two previously validated scales. We also assessed sociodemographic factors, comorbidities, health status, previous exposure to clinical research, personal experiences with discrimination, and attitudes toward potential barriers and incentives associated with trial participation. We performed bivariable analyses to examine the associations between these factors and willingness to participate. Those factors significantly associated with the outcome were used to create a multivariable logistic regression model.

RESULTS: We received responses from 755 eligible persons (53% response rate). The study population was 63% white and 37% African American. African

Americans were not significantly more distrustful of their care providers (46.3% vs. 44.6%, $p = 0.66$) or clinical research (3.2% vs. 3.0%, $p = 0.84$). In unadjusted analyses, whites were slightly more likely to be more willing (32.8% vs. 29.9% $p = 0.13$) to participate in clinical trials compared to African Americans. In the multivariable model, male gender (OR 2.40, 95% CI 1.58, 3.64), having dependent care responsibilities (OR 1.96, 95% CI 1.16, 3.32), previous trial participation (OR 1.84, 95% CI 1.24, 2.74), and the endorsement of the importance of participant benefit from trial participation (OR 2.16, 95% CI 1.24, 3.76) were all associated with an increased willingness to participate. Respondents who thought it would be important for their own PCP to ask them to be in a trial were less willing to participate (OR 0.60, 95% CI 0.37, 0.96). Neither race nor distrust was independently related to willingness to participate in the final multivariable model.

CONCLUSIONS: African Americans were not significantly less willing to participate in clinical trials compared to whites. Distrust was not an independent predictor of willingness to participate in clinical trials. Low representation of African Americans in clinical trials may be due to factors other than a lower willingness to participate.

A COMPARISON OF CARE QUALITY BETWEEN PATIENTS WHO RELY ON AN INTEGRATED HEALTHCARE DELIVERY SYSTEM AND THOSE WHO OBTAIN CARE OUTSIDE THE SYSTEM. D. Helmer¹; U. Sambamoorthi²; M. Rajan¹; A. Tiwari¹; L. Pogach³. ¹War Related Illness and Injury Study Center NJVAMC, East Orange, NJ; ²University of Medicine and Dentistry of New Jersey, Newark, NJ; ³VA New Jersey Health Care System, East Orange, NJ. (Tracking ID #135016)

BACKGROUND: Integrated health care delivery systems (IHDSs) theoretically provide higher quality of healthcare than the traditional fee-for-service approach by emphasizing continuity of care, a systematic approach to quality of care, efficiency of care, and preventive care/disease management. However, direct comparison of healthcare obtained in IHDSs and fee-for-service care is often difficult because of selection biases and inability to ascertain utilization. Although the Veterans Health Administration (VHA) has many attributes of an IHDS, VHA enrollees frequently pursue outside care. To test the hypothesis that reliance on care provided by an IHDS is associated with higher quality care than a patient-coordinated approach to health care including services from non-IHDS providers, we compare the quality of care measures of veterans with diabetes who rely completely on VHA to those who obtain care through both VHA and Medicare.

METHODS: Data from VHA and Medicare were merged for veterans with diabetes enrolled in VHA and Medicare Part B in 1998 and 1999 ($n = 108,027$). Patients were classified as 'single-system users' or 'dual-system users' based on the pattern of ambulatory visits to health care professionals in 2000. Quality of diabetes care was assessed by testing of hemoglobin A1c (HgbA1c) and low density lipoprotein (LDL) in 2000. Multivariate probit models, including an instrumental variable approach that corrects for endogeneity, socioeconomic status, demographics, and case mix adjustment using diagnostic codes, were used to assess the impact of single- versus dual-system use on diabetes quality processes measures.

RESULTS: Dual-system users (53.7%, $n = 58,104$) were more likely than single-system users (46.3%, $n = 49,923$) to be white, non-Hispanic, and married; live further from a VA facility; have a higher case-mix relative risk score; and have higher income. LDL was tested in 76.2% of dual-system and 63.2% of single-system users. HgbA1c was tested in 79.8% of dual-system and 77.1% of single-system users. Among dual-system users who had the quality of care measure performed, 66.5% had their LDL and 77.1% had their HgbA1c tested in VHA. After controlling for patient characteristics, dual-system users had a higher number of ambulatory visits (15.7 versus 13.9 visits), lower probability of HgbA1c testing (4.9%) and higher probability of LDL testing (4.8%). Two stage selection models using instrumental variables demonstrated the existence of endogeneity, but no adequate instrumental variables were available in the data set.

CONCLUSIONS: There are significant socioeconomic, demographic and clinical differences between veterans who use the VHA only and those who go outside the VHA, despite equivalent coverage. The results suggest that while the VHA provides most veterans with recommended diabetes services with fewer visits, in some circumstances the veterans who go outside the system may benefit from the additional care. Further research is needed to understand the implications of patient choice to receive healthcare outside of an IHDS upon quality of care and associated resource use.

A RANDOMIZED CONTROL TRIAL OF FINANCIAL INCENTIVES FOR SMOKING CESSATION. K. Volpp¹; A. Gurmankin²; A. Gomez³; D.A. Asch³; J.A. Berlin⁴; J.J. Murphy⁵; J. Zhu⁶; C. Lerman⁷. ¹CHERP - Philadelphia VA Medical Center, Philadelphia, PA; ²Dana-Farber Cancer Institute and Harvard School of Public Health, Boston, MA; ³Center for Health Equity Research and Promotion - Philadelphia VA Medical Center, Philadelphia, PA; ⁴Center for Clinical Epidemiology and Biostatistics, University of Pennsylvania, Philadelphia, PA; ⁵Primary Care and Consultative Medicine, Philadelphia VA Medical Center, Philadelphia, PA; ⁶Division of General Internal Medicine, University of Pennsylvania, Philadelphia, PA; ⁷Department of Psychiatry and Abramson Cancer Center, University of Pennsylvania, Philadelphia, PA. (Tracking ID #134761)

BACKGROUND: Approximately 70 percent of smokers want to quit smoking, but only about 3 percent of smokers quit each year. Smoking cessation programs are an underutilized cost effective way to help people quit smoking. Our objective was to determine whether modest financial incentives increase the rate of smoking cessation program enrollment, completion, and quit rates.

METHODS: 179 patients at the Philadelphia Veterans Affairs Medical Center who reported smoking at least 10 cigarettes per day were randomized into incentive and non-incentive groups. Both groups were invited to join a 5-class smoking cessation program at no out-of-pocket cost. In addition, the incentive group was offered \$20 for each class attended and \$100 if they quit smoking (biochemically confirmed). Subjects were blinded as to the existence of the other group. Intention to treat analysis was used for each of the outcome variables. Chi-Square tests were used to test for significant differences in enrollment and completion between the two groups.

RESULTS: Subjects had smoked an average of 29.8 years, with a mean consumption of 23.4 cigarettes per day. There were no significant differences between the incentive and enrollment groups in the number of cigarettes smoked per day, the length of time patients had been smoking, degree of addiction, quitting intentions, risk perception of smoking-related risks, sociodemographic characteristics, and distance from the VA hospital. The incentive group had higher rates of program enrollment (43.3% vs. 20.2%, $p < .001$) and program completion (25.8% vs. 12.2%, $p = .02$). Data on quit rates 30-days post-program completion showed significantly higher tobacco cessation rates in the incentive group than in the control group (16.3% vs. 4.6%, $p < .001$). Quit rates conditional on program enrollment were 23.5% in the control group and 39.5% in the incentive group. Effects of the incentives on 30-day quit rates were larger among heavy smokers ($p = .04$). At 6 months, quit rates in the incentive group were higher (6.5%) vs. the control group (4.6%) to a nonsignificant degree ($p > .20$).

CONCLUSIONS: Modest financial incentives are associated with significantly higher rates of smoking cessation program enrollment and completion and short-term quit rates. The use of financial incentives could be an effective strategy for increasing tobacco cessation rates, but optimal program design would include an incentive for longer-term cessation.

ACCESS TO CARE, HEALTH STATUS, AND RECEIPT OF PREVENTIVE SERVICES IN THE US AND CANADA: RESULTS OF A CROSS-NATIONAL POPULATION-BASED SURVEY. K.E. Lasser¹; D.U. Himmelstein¹; S. Woolhandler¹. ¹Cambridge Health Alliance and Harvard Medical School, Cambridge, MA. (Tracking ID #133008)

BACKGROUND: Canada, with a system of universal health insurance, spends about half as much on health care per capita as does the US, yet Canadians live 2-3 years longer. Few population-based data are available on health habits and processes of care in the 2 countries that might explain this paradox.

METHODS: We analyzed population-based data on 3,505 Canadians and 5,183 Americans from the Joint Canada/US Survey of Health. Conducted by the National Center for Health Statistics and Statistics Canada, the survey content was based on the US National Health Interview Survey and the Canadian Community Health Survey. We used the X² test to compare differences in health status and access to care between US and Canadian respondents. Controlling for sex, age, income level, race and immigrant status, we used multiple logistic regression to analyze country of residence as a predictor of access to care, quality of care and satisfaction with health care.

RESULTS: With the important exception of lower rates of cigarette smoking (16.8% vs. 19.0%), US respondents appear less healthy than Canadians, with higher rates of obesity (20.7% vs. 15.3%), physical inactivity (13.6% vs. 6.5%), diabetes (6.7% vs. 4.7%), hypertension (18.3% vs. 13.9%), arthritis (17.9% vs. 16.0%) and COPD (1.9% vs. 1.0%). American women had higher Pap smear rates (at 3 year intervals) than did Canadian women (88.9% vs. 79%). American women reported higher rates of mammography screening "within less than 2 years" (88.8% vs. 81.3%) but not within the past five years (96.3% vs. 94.9%). While more US respondents had unmet health care needs than did Canadians (13.2% and 10.7% respectively), their reasons for having such needs differed. Seven percent of US respondents (and less than 1% of Canadians) had unmet need due to financial barriers, while 3.5% of Canadians had unmet needs due to waiting times (versus less than 1% of Americans). All univariate comparisons were significant at $p < .05$. In multivariate analyses, Americans (as compared to Canadians) were less likely to have a regular doctor (odds ratio [OR], .66; 95% confidence interval [CI], .57-.75), more likely to have unmet health care needs (OR 1.27; 95% CI, 1.08-1.48), more likely to forgo needed medicines (OR 2.12; 95% CI, 1.73-2.59), and more likely to be very satisfied with their health care (OR 1.47; 95% CI, 1.32-1.63).

CONCLUSIONS: Americans are less able to access medical care than are Canadians. Yet, when they do receive medical care, Americans are more satisfied with such care. Providing health insurance coverage to all US residents would improve access to care, while adequately financing health care and adequately reimbursing services such as Pap smears might improve processes of care and satisfaction with health care in Canada.

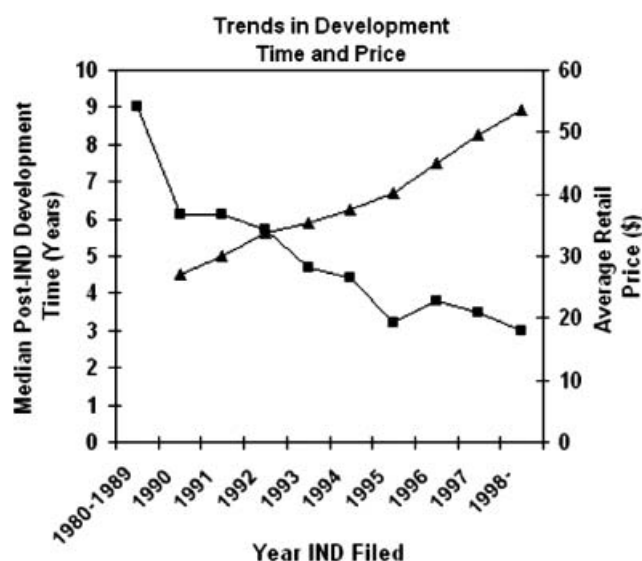
ARE DRUG DEVELOPMENT TIMES INCREASING OVER TIME AND LEADING TO HIGHER PHARMACEUTICAL PRICES? S. Keyhani¹; M. Diener-West²; N.R. Powe². ¹Mount Sinai School of Medicine, New York, NY; ²Johns Hopkins University, Baltimore, MD. (Tracking ID #133779)

BACKGROUND: Previous reports on drug development times have been based on proprietary data reported by the pharmaceutical industry; total development times up to 15 years have been described and increasingly longer clinical trial times have been alleged as one reason leading to higher drug prices.

METHODS: We conducted a retrospective study of 168 drugs. We assembled data on post-investigational new drug (IND) application development times, drug characteristics and government regulatory designations. There were 168 drugs with development data available in the Federal Register Government

Printing Office database approved between 1994-2002 by the Food and Drug Administration (FDA). From this database, we obtained data on length of post-IND drug development time, the IND filing date and the New Drug Application approval date. We also obtained information on FDA regulatory designations (Fast Track Status/Accelerated Review) for drug approval from the FDA. We calculated the separate time periods for clinical trial, regulatory review and total post-IND development for drugs classified into different clinical groups. We examined trends in length of post-IND-development, clinical trial and regulatory review periods over time and compared how changes in development over time varied by drug group and regulatory designations.

RESULTS: Median total post-IND drug development and clinical trial times for drugs approved between 1994-2002 was 6.3 years and 5.1 years respectively. The median clinical trial time was shortest for infectious disease agents (4.5 years) and longest for psychiatric drugs (7.2 years) (p -value < 0.05). Post-IND development time decreased on average by 244 days each year (p -value < 0.001). Average retail prescription drug prices increased from \$27 to \$53 in the same time period (see graph). Post-IND development time also decreased within each clinical drug group and was a result of decreases in both clinical trial and regulatory review periods. The median clinical trial period of drugs with fast track status was 4.4 years, while the median clinical trial period of drugs without this designation was 6.7 years (p -value < 0.05). Decreases in post-IND development time observed in the past decade were also apparent in drugs that did not receive fast track status or accelerated review.



CONCLUSIONS: Post-IND drug development times have decreased significantly in the past decade. FDA regulatory designations have led to rapid drug approval; however, development time has also decreased independent of FDA regulatory designations. Despite shorter post-IND development times, drug prices continue to rise making development time an unlikely cause of higher drug prices. To our knowledge, this is the first analysis of development time based on publicly available data.

ARE HEALTH DEPARTMENTS REALLY READY 24/7? N. Lurie¹; A. Diamond²; D. David³. ¹RAND, Arlington, VA; ²RAND Corporation, Arlington, VA; ³RAND Corporation, Pittsburgh, PA. (Tracking ID #135025)

BACKGROUND: Background: The ability to receive and respond to emergency case reports 24 hours/day, 7 days/week is a key goal of public health preparedness efforts. We tested the ability of health departments (HDs) to do this over a 10-month period in a convenience sample of 19 metropolitan area health departments across the US.

METHODS: Methods: We purchased disposable cell phones and programmed them to correspond to the local area code of the HD being tested. We placed 6-10 calls per HD (mean = 6.5) using a schedule that included days, evenings and weekends, purporting to be a physician with a "concerning case." We assessed how often we could locate a single HD number, and whether we were transferred directly to the appropriate "action officer" or were required to leave a call-back number. We also timed how long it took to reach the action officer. In 6 sites, we also described symptoms consistent with smallpox, anthrax, plague, or botulism and asked for advice and feedback. IRB approval and subsequent consent for unannounced was obtained from the director of each health department prior to testing. The fact that tests were occurring was not shared with HD staff.

RESULTS: Results: 55% of HDs did not have a single number or had an incorrect phone number on their websites. Response times for calls that were returned ranged from < 1 minute to 2470 minutes (mean 50 mins); 47% of HDs

returned all calls within 30 mins. The longest average response times were observed at the end of the workday, during the evening, and on the weekend (means were 88 mins, 57 mins, and 92 mins, respectively.) Across the 19 HDs there were no standardized procedures for triaging and handling calls, and the action officers varied widely in expertise. When we presented cases to action officers, results were mixed across and within HDs; in at least 2 instances, we were told not to be concerned about the case, given that no similar cases had been reported. We were rarely asked about isolation or personal protective equipment.

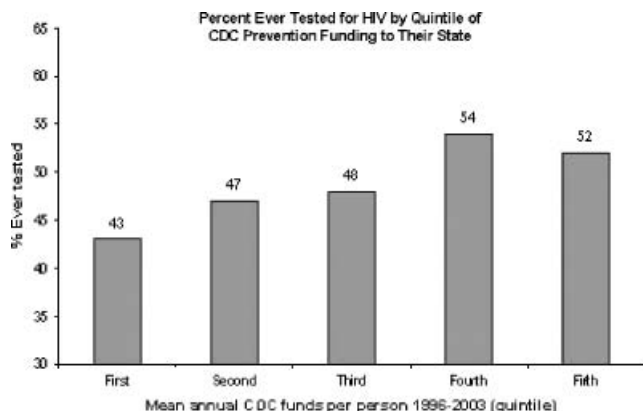
CONCLUSIONS: Despite the emphasis on public health preparedness, there is substantial variability in the ability of HDs to receive and respond to emergency case reports. Improving preparedness must involve improved access to front line contact with HDs.

ASSESSING THE IMPACT OF CDC HIV PREVENTION FUNDING ON HIV TESTING AND KNOWLEDGE. B.P. Linas¹; H. Zheng²; E. Losina³; R.P. Walensky¹; K.A. Freedberg¹. ¹Massachusetts General Hospital, Boston, MA; ²Harvard University, Boston, MA; ³Massachusetts General Hospital and Boston University School of Public Health, Boston, MA. (Tracking ID #132645)

BACKGROUND: The Centers for Disease Control and Prevention (CDC) annually distributes over \$300 million for HIV education/prevention. We sought to determine whether this funding increases HIV testing utilization and improves knowledge about prophylaxis of mother-to-child HIV transmission (pMTCT).

METHODS: We used respondent's state of residence to link data from the U.S. Behavioral Risk Factor Surveillance System (BRFSS) with Kaiser Family Foundation tracking of CDC HIV education/prevention funding. For each survey year 1997, 1999, 2001, and 2003, we determined the percent of respondents who reported having ever been tested for HIV by quintile of state per capita CDC prevention funding in preceding years. Using logistic regression of the 2003 dataset, we adjusted for the following confounders of the relationship between CDC funds and the odds that an individual had ever been tested for HIV: age, sex, race, income, education level, employment status, marital status, HIV risk behavior, and respondent's state AIDS prevalence. We then used 2003 data to report the percent of respondents that was aware of pMTCT by quintile of CDC funding, and used logistic regression to adjust for the same confounders.

RESULTS: There were 158,752 respondents in the 2003 BRFSS data set. Overall, 49% (range by state 32%-62%) had been tested for HIV and 56% (range by state 47%-64%) were aware of pMTCT. Median spending for HIV education/prevention was \$1.54 per person and ranged from \$0.73 per person (first quintile) to \$4.64 per person (fifth quintile). Percent of people ever tested for HIV increased as a function of per capita funding for HIV education/prevention (Figure; p for trend <0.0001). We observed a similar relationship in 1997, 1999, and 2001 (p for trend within each year <0.0001). In multivariate analysis, the odds ratio of ever having been HIV-tested for those living in second, third, fourth, and fifth quintile states compared to those in first quintile states were: 1.11 (95% CI 1.05-1.17), 1.11 (1.06-1.17), 1.32 (1.25-1.40), and 1.22 (1.15-1.31). In 2003, knowledge of pMTCT also increased as a function of CDC funding. In states in the lowest quintile of CDC funding, 51.9% were aware of pMTCT, as compared to 52.3%, 54.3%, 60.4%, and 60.3% in the second through fifth quintiles, respectively (p for trend <0.0001). In multivariate analysis, the odds ratio of being aware of pMTCT for those living in second, third, fourth, and fifth quintile states compared to those in first quintile states were: 1.01 (95% CI 0.95-1.06), 1.06 (1.00-1.11), 1.22 (1.15-1.29), and 1.19 (1.12-1.27).



CONCLUSIONS: Increases in CDC HIV education/prevention funding directly and independently correlate with increases in HIV testing and knowledge of pMTCT. Additional funding would likely increase the impact of efforts to improve HIV testing and knowledge in the United States.

BRAND AND GENERIC MEDICATION PRESCRIBING FOR MEDICARE BENEFICIARIES WITH HYPERTENSION. A.D. Federman¹; E.A. Halm¹; T. Hochman²; A.L. Siu¹. ¹Mount Sinai School of Medicine, New York, NY; ²Veterans Administration Medical Center, Bronx, NY. (Tracking ID #135446)

BACKGROUND: High costs have resulted in avoidance of medications by up to 40% of adults over 65. Opting for generic over brand agents can reduce out-of-pocket spending, but little is known about the relative use of generic and brand drugs by Medicare beneficiaries (MB). We compared brand and generic antihypertensive medication (AHM) use in older adults with hypertension (Htn), and focused on the effect of income and prescription (Rx) coverage.

METHODS: We conducted a cross-sectional analysis of 2001 data from the nationally representative Medicare Current Beneficiary Survey (MCBS) (n=12,864, 71% response rate). We included community-dwelling adults >65 with Htn who used ≥ 1 drug from 5 AHM classes: angiotensin converting enzyme inhibitors (ACEI, 2/10 agents available as generics), alpha-adrenergic receptor antagonists (AA, 3/3 generics), beta-adrenergic receptor blockers (BB, 9/9 generics), calcium channel blockers (CCB, 4/8 generics), and thiazide diuretics (3/4 generics). We excluded those with end-stage renal disease. MCBS interviewers record drug names verbatim from pill bottles or pharmacy receipts 3 times per year. If interviewers documented a multisource drug by its brand name we coded it as a brand agent. The primary outcome was use of a brand AHM in any of the 5 AHM classes. When respondents used both brand and generic agents the outcome was coded as generic, although coding as brand in such cases produced similar results in multivariate analyses. We estimated adjusted odds ratios (AOR) for the association of brand use with income and type of Rx coverage in logistic regression models. Income was coded as % of poverty (<100%, 100-135%, 136-150%, 151-200%, 201-300%, >300%). Rx coverage included none, Medicaid, managed care, self-purchased, employer-sponsored, and other. All analyses adjusted for age, sex, race, ethnicity, urban residence, Census region, no. chronic diseases, and no. AHM used. We performed the analyses in SUDAAN to correct standard errors for the MCBS multilevel sampling design.

RESULTS: The analysis included 4,639 adults, representing 14.9 million elderly MBs with Htn nationally. The mean age was 77, 39% were female, 55% had incomes ≤200% of poverty, 26% lacked Rx coverage, 36% had coronary disease, and median AHM use was 2, (range 1 to 8). Overall, ACEI were used by 45%, BB by 43%, CCB by 41%, thiazides by 27% and AA by 12%. The rates of brand AHM use in each drug class were: ACEI, 87%; BB, 25%; CCB, 69%; thiazides, 29%; and AA, 45%. In multivariate analysis individuals without Rx coverage and those with managed care were less likely to use brand drugs from the 5 drug classes compared to those with private coverage (AOR 0.74, 95% CI 0.61-0.89, and AOR 0.80, 0.64-0.99, respectively). The effect was driven by BB use: in a subgroup analysis of BB users, brand BB use was lower for those without Rx coverage (AOR 0.67, 0.49-0.91) and those with managed care (AOR 0.62, 0.43-0.91), whereas there were no significant differences by coverage type in the 4 other AHM subgroups. Income had no significant association with brand use in the primary or subgroup analyses.

CONCLUSIONS: Class of AHM and type of prescription coverage, rather than income, appear to drive brand AHM use in elderly MBs. Nonetheless, brand AHM use remains high for individuals with all types of Rx coverage, even among those with low incomes. These data suggest that opportunities may exist to increase use of generics and potentially decrease costs for MBs and Medicare itself.

CERTIFICATE OF NEED REGULATIONS DO NOT RESTRICT ACCESS TO CORONARY REVASCULARIZATION AFTER ACUTE MYOCARDIAL INFARCTION. I. Popescu¹; M.S. Vaughan Sarrazin¹; G.E. Rosenthal¹. ¹VA Medical Center and University of Iowa College of Medicine, Iowa City, IA. (Tracking ID #135102)

BACKGROUND: Opponents of certificate of need (CON) regulations argue that CON limits competition and decreases timely access to needed services, while advocates argue that CON may lead to higher volume clinical programs. However, few empirical analyses have directly examined the relationship between CON and the quality and accessibility of care. Thus, the current study evaluated the effect of CON regulations on rates of coronary revascularization and mortality following acute myocardial infarction (AMI).

METHODS: Medicare Part A administrative data was used to identify all patients hospitalized with a primary diagnosis of AMI during 1998-2000 (n=786,593) in 27 states with and 23 states without CON regulations for cardiac services during the study period. Outcomes, including revascularization with either coronary artery bypass graft surgery (CABG) or percutaneous coronary angioplasty (PCI) and 30-day mortality were adjusted for demographic and clinical (comorbidity, acuity) risk factors using logistic and Cox regression analysis. All analyses accounted for clustering.

RESULTS: Revascularization patterns after AMI differed in CON and non-CON states. Patients in CON states were less likely to have revascularization at the admitting hospital (22% vs. 26%, p<.001), but were more likely to be transferred (18% vs. 15%, p<.001) to another acute care hospital and undergo revascularization at the transfer hospital (12% vs. 9%, p<.001). Annual hospital procedural volumes (for Medicare patients) were higher in CON states for both PCI (428 vs. 280, p<.001) and CABG (313 vs. 192, p<.001). Of all revascularization procedures, 36% were CABG; proportions were relatively similar in states with and without CON (37% vs. 34%). Adjusting for patient risk factors using Cox regression, the likelihood of revascularization in CON states was lower during the first 2 days after AMI (HR 0.88, CI 0.81-0.96, p = 0.003), but was somewhat higher thereafter: HR 1.02 (CI 0.98-1.05, p = 0.3) for days 3-7; HR 1.28 (CI 1.23-1.34, p<.001) for days 8-14; and HR 1.10 (CI 1.05-1.15, p<.001) for days 15-30. Patients in states with CON had lower adjusted mor-

tality at both 7 days (OR 0.92, CI 0.89-0.95, $p < .001$) and 30 days (OR 0.94, CI 0.92-0.96, $p < .001$).

CONCLUSIONS: Patients with AMI were more likely to receive early revascularization in states without CON regulations. However, overall revascularization rates at 30 days were similar, and patients in states with CON regulations had somewhat lower mortality, which may be due to the higher hospital volumes in such states. CON regulations may be associated with beneficial patient outcomes.

CHANGES IN HEALTH FOR THE UNINSURED AFTER REACHING AGE-ELIGIBILITY FOR MEDICARE. D.W. Baker¹; J.M. Feinglass¹; R. Durazo-Arvizu¹; W.P. Witt¹; J.J. Sudano²; J. Thompson¹. ¹Northwestern University, Chicago, IL; ²Case Western Reserve University, Cleveland, OH. (Tracking ID #134178)

BACKGROUND: Uninsured adults in late middle age are more likely to have a health decline than the privately insured. It is unclear if this increased risk of adverse health outcomes persists for a period of time even after gaining insurance or whether gaining health insurance leads to improvements in health. This study examined these issues among a cohort of U.S. adults who gained Medicare coverage based upon age-eligibility.

METHODS: We used data from the Health and Retirement Study (HRS), a prospective study of U.S. adults aged 51-61 in 1992 with interviews every 2 years. We analyzed the '96, '98, '00, and '02 data files and identified the first interview at which participants reached age 65 and reported having Medicare (t0). Changes in health were defined for 1) the two-year period during which subjects transitioned to Medicare (t-2 to t0), and 2) the subsequent two year period (t0 to t2). Self-reported overall health was measured as excellent, very good, good, fair, or poor. We defined two dichotomous variables: 1) "major decline in self-reported health" (yes/no) as either a decline from a) excellent, very good, or good health at the start of a follow-up period to fair or poor health at the end of the follow-up period, or b) a decline from fair to poor health; and 2) "any improvement in overall health." Physical difficulties were measured with 4 dichotomous items assessing walking or climbing stairs ("Mobility") and 6 items measuring instrumental activities of daily living ("Agility"). We determined changes in the number of physical difficulties between interviews and categorized these as any improvement ($\Delta \leq 0$), no change ($\Delta = 0$), or any decline ($\Delta \geq 0$). We used multivariate logistic and multinomial regression to analyze whether being uninsured prior to Medicare was associated with worse outcomes after adjusting for baseline health, sociodemographics, and health behaviors.

RESULTS: During the transition period, the interval between the interview prior to age 65 (t-2) and the first interview after reaching age 65 (t0), previously uninsured individuals (N=441) were more likely than those who previously had private insurance (N=2978) to have a major decline in overall health (adjusted relative risk [ARR] 1.46; 95% CI 1.03-2.04) and a new physical difficulty affecting Mobility (ARR 1.24; 95% CI 0.96-1.56) or Agility (ARR 1.33; 95% CI 1.12-1.54). In contrast, during the subsequent two years after the first HRS interview when participants had gained Medicare coverage (t0 to t+2), the previously uninsured no longer had worse outcomes compared to those who previously had private insurance: ARR major decline in overall health 1.14 (95% CI 0.76-1.68; $p = 0.52$), ARRs new Mobility difficulty 0.93 (95% CI 0.67-1.26), and ARR new Agility difficulty 1.07 (95% CI 0.85-1.31). During both periods, adjusted rates of improvement in overall health and physical functioning were similar for the previously uninsured and those who were privately insured before gaining Medicare.

CONCLUSIONS: Gaining Medicare coverage does not lead to immediate health benefits for individuals who were uninsured prior to reaching age 65, but after two or more years of continuous coverage the increased risk for adverse health outcomes among the previously uninsured disappears. Lowering the age of eligibility for Medicare could yield significant health benefits that could partially offset the cost of expanding coverage.

COMPARING 4-YEAR HEALTH OUTCOMES OF ELDERLY ADULTS ENROLLED IN TRADITIONAL MEDICARE (FFS) VS. MEDICARE HMOs. D.G. Safran¹; W.H. Rogers¹; I. Wilson¹; H. Chang¹; A. Li². ¹Tufts University School of Medicine, Boston, MA; ²Tufts-New England Medical Center, Boston, MA. (Tracking ID #135116)

BACKGROUND: Since 1972, Medicare beneficiaries have had the option of enrolling in a Medicare-qualified HMO. Currently, 13% of beneficiaries are enrolled in a Medicare HMO, and while growth has been slow since 1999, the Medicare Modernization Act (MMA) is expected to re-energize the program and stimulate growth. Despite this long history, there is no conclusive information comparing the health outcomes achieved by Medicare HMOs with those of traditional Medicare (FFS). Available data derive largely from studies of select plans, in select locations, before the 1990s enrollment surge, and the studies yield conflicting results. This study compares functional health outcomes and mortality of seniors in Medicare FFS vs. HMOs over a 4-year period (1998-2002). It represents the longest running cohort with comparative health outcomes data and is unique in its monitoring of beneficiaries who switch systems.

METHODS: In 1998, we established a longitudinal cohort of non-institutionalized Medicare beneficiaries aged 65 and older residing in the 13 states with the largest and most well-established Medicare HMO systems (AZ, CA, CO, FL, IL, MA, MN, NM, NY, OR, PA, TX, WA). FFS and HMO enrollees were matched on age, sex and zip code. From 1998 through 2002, we monitored their primary care, health outcomes and enrollment status using a combination of annual surveys and administrative data obtained from the Centers for Medicare and Medicaid Services (CMS). Health outcomes included death and changes in functional status as measured by the SF-36 Physical Component Summary (PCS) and Mental Component Summary (MCS). Mortality analyses include all original

sample members (n=15,963), and functional health outcome analyses employ data on all study participants at all available study intervals (n=12,899), clustered to account for multiple observations per person. Models control for beneficiaries' sociodemographic characteristics, diagnoses, enrollment status, and enrollment changes (timing and type).

RESULTS: Analyses reveal no statistically significant differences in health outcomes between FFS and HMO enrollees after adjusting for population differences. Among beneficiaries stably enrolled in either FFS or an HMO, average 2-year changes in physical functioning (PCS) differed by 0.10-points (-1.30-points vs. -1.40-points, n.s.), and average mental health outcomes (MCS) differed by 0.15-points (-0.45-points vs. -0.60-points, n.s.). Mortality did not differ significantly by system, though it tended to be higher in FFS (Hazard Ratio [HR]=1.14, n.s.). However, there was compelling evidence that health status figures importantly into beneficiaries' system-switching decisions. Compared to those stably enrolled in either system, PCS declines were larger among those switching from HMO to FFS (-1.95) and smaller among those switching from FFS to HMO (-1.10). Similarly, recent switchers into HMOs had lower mortality risk than those remaining in FFS (HR=0.78); and recent switchers from HMO to FFS had 41% higher mortality risk than those remaining in HMOs (HR=1.41).

CONCLUSIONS: This 4-year follow-up of health outcomes among Medicare FFS and HMO enrollees finds no significant outcome differences between the systems, though it finds evidence of healthier beneficiaries opting into HMOs and less healthy opting out. Beneficiaries' ability to switch systems as they feel their health requires may have created a form of "self-correction" that contributes to the observed health outcomes equilibrium. However, it is also possible that, as Medicare HMOs have proliferated and matured, the systems have converged to where any remaining care differences are inconsequential. While we cannot say whether these findings would generalize to a system wherein beneficiaries were "locked in" after enrollment, we conclude that the current program, designed to allow ongoing choice, yields outcomes that do not favor either system over the other.

COMPARING PRIMARY CARE PRACTICE IN AUSTRALIA, NEW ZEALAND, AND THE UNITED STATES. A.B. Bindman¹; C.B. Forrest²; H. Britt³; P. Crampton⁴; A. Majeed⁵. ¹University of California, San Francisco, San Francisco, CA; ²Johns Hopkins University, Baltimore, MD; ³University of Sydney, Sydney, New South Wales; ⁴University of Otago, Dunedin; ⁵Imperial College of Science, Technology and Medicine, University of London, London. (Tracking ID #136268)

BACKGROUND: Few cross-national comparisons of healthcare systems have been done to explicitly examine primary care practice. Such analyses could provide new insights into strategies for primary care reform. We used nationally representative surveys to contrast patient mix, scope-of-practice, visit duration, and specialty referral rates among primary care physicians in Australia (AU), New Zealand (NZ), and the United States (US).

METHODS: Cross-sectional analysis of 3 comparable surveys of primary care visits: the BEACH program (AU), the NATMEDCA (NZ), and the NAMCS (US). Primary care physicians (general practitioners in AU and NZ; general internists, general pediatricians, and family practitioners in US) completed a questionnaire after an office visit on patient demographics, diagnoses, duration of visit and whether a referral was generated. Data were collected between 2000-2002. Information was recorded on 79,850 visits in AU, 10,064 in NZ and 25,838 in US. Diagnostic codes (ICPC in AU, READ in NZ, and ICD-9-CM in US) were mapped to the Johns Hopkins Expanded Diagnostic Clusters (EDCs) to create a uniform taxonomy for problems managed. As a measure of scope of practice, we determined the number of problems (EDCs) accounting for 75% of all problems managed (higher numbers=greater breadth). Referral rates were calculated as percentage of visits associated with a referral.

RESULTS: 20-23% of visits were made by those over age 65 in each country, but a higher share of visits was accounted for by children in the US (35% versus 17% AU and 26% NZ). Females constituted 56-58% of visits in each country. An average of 1.38 problems/visit was managed in AU, 1.41 in NZ, and 1.55 in the US. The average number of minutes spent per problem managed was 9.7 in NZ, 10.8 in AU, and 11.9 in US. In the US, 46 EDCs accounted for 75% of problems managed as compared with 47 in NZ, and 52 in AU. Correlations in the frequencies of managed health problems between countries were high (.89-.96 for pairwise comparisons). Overall, the referral rates were 10.6% in AU, 8.5% in NZ, and 7.4% in the US; differences persisted after controlling for age. In all three countries, less frequently managed problems were referred at higher rates (correlations ranged from -0.29 to -0.33).

CONCLUSIONS: Despite different national demographics and healthcare systems, the healthcare needs that present to primary care practice in Australia, New Zealand, and United States are similar. On average, US primary care physicians spend more time managing a health problem than their Australian and New Zealand counterparts and refer to a specialist less frequently. The high intensity of service provision in US primary care practice compared to New Zealand and Australia may be the result of differences in practice style or consumer expectations. Lower referral rates recorded in primary care in the US may reflect the unique ability of US patients to self-refer to specialists. Future studies should attempt to determine whether differences in primary care practice across countries are associated with variation in the quality of care.

COMPARISON OF MEDICARE PRESCRIPTION BENEFIT COSTS WITH A DISCOUNT INTERNET PHARMACY: WHAT'S A PATIENT'S BEST DEAL? J. Hayes¹; H. Walczak²; A.V. Prochazka². ¹Denver Veterans Affairs Medical Center, Denver, CO; ²University of Colorado Health Sciences Center, Denver, CO. (Tracking ID #133339)

BACKGROUND: The magnitude of Medicare patient savings with the Medicare Prescription Drug Improvement and Modernization Act compared with established discount pharmacies is not clearly known for any specific cardiovascular

regimen. Our goal was to estimate the savings for a typical cardiovascular patient in the Medicare benefit using thirty and ninety day supplies compared with a well known discount internet pharmacy site.

METHODS: We chose two equipotent evidence-based regimens, one brand-name and one generic, typical of a cardiovascular patient. (see table below) We used drugstore.com to obtain a discounted internet price for our selected regimens. We then used a Medicare (www.medicare.com) discount card plan and selected the same regimens. The least expensive Medicare card plan within zip code 80220, using the smallest possible radius, was chosen for both regimens to evaluate total annual costs for a non-married patient determined to be above 135% of the poverty level income threshold for 2004 (annual income greater than approximately \$12,662).

RESULTS: For our selected brand-name cardiovascular patient regimen, the Medicare prescription discount card program offered a 2.0% annual savings compared with drugstore.com for a thirty day supply. However, the same regimen if filled ninety days from the Medicare supplier offered a 11.8% annual savings compared with drugstore.com. For our selected generic cardiovascular patient regimen, the Medicare prescription discount card program offered a 31.3% annual savings compared with drugstore.com for a thirty day supply. However, the same regimen if filled ninety days from the Medicare supplier offered a 42.8% annual savings compared with drugstore.com.

CONCLUSIONS: In our brand-name cardiovascular regimen, using the new Medicare prescription discount cards offered a modest savings when compared to the internet discount pharmacy site while a generic regimen offered substantial savings when compared to the internet discount pharmacy site. We conclude that the Medicare Discount Cards when coupled with a generic regimen and ninety day supply can offer substantial savings to our elderly cardiovascular patients.

Total Annual Costs for the Two Regimens

	Drugstore.com 30 day supply	Medicare Card 30 day supply	Medicare reduction compared with Drugstore.com	Drugstore.com 90 day supply	Medicare Card 90 day supply	Medicare reduction compared with Drugstore.com
Annual Brand Name Regimen	\$1770	\$1734	2.0%	\$1680	\$1492	11.8%
Cost:-Atorvastatin						
10 mg qd-Ramipril						
10 mg qd-Metoprolol						
XL						
100 mg qd						
Annual Generic Regimen Cost:-	\$1056	\$648	31.3%	\$940	\$538	42.8%
Lovastatin						
40 mg qd-						
Lisinopril						
20 mg qd-						
Metoprolol						
50 mg bid						

EFFECTS OF FINANCIAL STRESS FROM PRICE COMPETITION ON HOSPITAL QUALITY OF CARE. K. Volpp¹; T. Konezka²; J. Sochalski³; J. Zhu³. ¹CHERP - Philadelphia VA Medical Center, Philadelphia, PA; ²University of Chicago, Chicago, IL; ³University of Pennsylvania, Philadelphia, PA. (Tracking ID #135837)

BACKGROUND: Hospital competition in the United States shifted from a quality/amenity basis to a price basis with the growth of managed care in the 1980s and 1990s. Research demonstrating that price competition lowers the rate of increase in hospital costs as well as profit margins has been based largely on data from the experience with selective contracting in California. The passage of selective contracting legislation in California in 1982 gave third-party payers new freedom to negotiate discounts with hospitals and to channel patients away from high-cost providers. Little is known about whether cost containment by hospitals was achieved through reductions in the quality of services provided or through greater efficiency. In this study, we examine whether financial stress in hospitals induced by price competition affected quality of care.

METHODS: Our analysis uses California OSHPD panel data from 1983 through 2001 linked with state death certificates to assess the effects of increasing price competition on 30-day mortality for AMI patients. We use hospital fixed effects to control for time-invariant differences between hospitals. Since selective contracting is expected to have a stronger effect in more competitive areas, our primary study variables are HMO penetration, which in the fixed effects specification represents increases in HMO penetration, and an interaction between HMO penetration and hospital market concentration. An instrumental variables approach is used to adjust for the potential endogeneity of HMO penetration. We test whether potentially more vulnerable populations such as the uninsured were adversely affected to a greater degree than the insured in terms of either the quality or quantity of services received. We test the robustness of our analyses to market definition and using model specifications ranging from linear probability models to conditional logits.

RESULTS: Overall 30-day mortality from AMI dropped from 14.9% to 8.3% from 1983 to 2001 in California. However, the rate of improvement varied significantly

across markets. We find that for every 1 percentage-point increase in managed care penetration there is .053 percentage point higher AMI mortality ($p < .01$). In less competitive markets, managed care penetration appears to lower AMI mortality to a non-significant degree. Using patient flow markets, the adverse effects of price competition are largest among the under 65 insured and effects are not significant among the uninsured. Using HSA-based markets, effects are largest among the uninsured. Adverse effects from price competition were strongest in the late 1980s and early 1990s.

CONCLUSIONS: Price competition among hospitals adversely affected AMI mortality. The degree to which mortality worsened needs to be considered in the context of reported cost savings from competition.

FACING THE MALPRACTICE CRISIS: ACADEMIC PHYSICIANS' WILLINGNESS TO ACCEPT QUICK FIX SOLUTIONS. R.B. Levine¹; S.J. Kravet¹; D.M. Windish²; D.A. Reed²; L. Wolfe¹; S.M. Wright¹. ¹Johns Hopkins Bayview Medical Center, Baltimore, MD; ²Johns Hopkins University, Baltimore, MD. (Tracking ID #134469)

BACKGROUND: Research, teaching, and patient care, encompass the core mission of academic medical centers. Increasing malpractice rates have threatened this mission. Current efforts to address the crisis have focused primarily on tort reform. This involves complicated legal processes, multiple stakeholders and will take time to result in meaningful change. In the interim, academic medical centers are faced with the challenge of paying rapidly escalating malpractice premiums. The purpose of this study was to determine the willingness of physicians at an academic medical center to accept various strategies designed to rapidly contain institutional malpractice costs.

METHODS: We surveyed all 270 full-time clinically active physicians in the Department of Medicine at a large academic center. Respondents were asked about their current knowledge of the cost of malpractice premiums and the most recent rate of increase. They were also queried about their willingness to reduce patient-care activities and accept decreases in salary and fringe benefits as potential mechanisms to deal with rising malpractice costs. Descriptive statistics were used to characterize physician demographics and responses. T-tests were used to compare physician responses with actual malpractice premium values.

RESULTS: Two-hundred and fifteen faculty responded (80%). Thirty percent were female. The mean age of respondents was 46 years (SD 10). Fifty percent were Associate or Full Professors. The mean amount of time spent in research, patient care without learners and patient care with learners was 39.5%, 21% and 18.3% respectively. Thirty percent of faculty spent 50% or more of their time on patient care. Respondents estimated the annual increase in malpractice premiums from 2004-2005 to be 29% and the current mean rate per physician at \$13,724.50. The true premium increase is 28% (95% CI 25.7-32.2; $p = 0.55$) and the current mean rate per physician at our institution is \$10,453 (95% CI 12,346-15,166; $p < .001$). Most faculty believe that the following factors place academic centers at a greater risk for malpractice claims: higher expectations from patients and society (81%), sicker patients (81%) and the presence of trainees (62%). When asked if they would be willing to give up patient care responsibilities completely, almost all physicians opposed or strongly opposed this measure (95%). A similar number opposed or strongly opposed the option of providing patient care every other year at double the effort and withdrawing from patient care on alternate years (97%). Seventy percent of respondents were willing to limit their clinical procedures to only those that they regularly performed. Almost all physicians opposed or strongly opposed salary reduction (97%) or decreases in fringe benefits (99%) as ways to help the institution curtail escalating malpractice costs.

CONCLUSIONS: While academic physicians are aware of the magnitude of the current malpractice crisis, very few are willing to limit or eliminate patient care activities or accept a decrease in salary or benefits in an effort to help recoup institutional malpractice costs.

HOW WELL HAVE THE DIFFUSION OF COX-2 INHIBITORS BEEN LIMITED TO SETTINGS OF MAXIMAL BENEFIT? G. Alexander¹; C. Dai¹; R.S. Stafford². ¹University of Chicago, Chicago, IL; ²Stanford University, Stanford, CA. (Tracking ID #134015)

BACKGROUND: Increased awareness of the cardiovascular risks associated with rofecoxib and celecoxib have prompted concerns regarding the safety of cyclooxygenase-2 inhibitors (Cox-2s). Previously, we found that marked increases in Cox-2 use have occurred among patients at low risk for adverse events from NSAIDs. However, little is known as to whether some physician or health system characteristics are associated with greater selectivity of Cox-2 use among patients who stand the most to gain from the pharmaceutical innovation.

METHODS: We analyzed the universe of patient visits where a Cox-2 was continued or prescribed using data from the National Ambulatory Medical Care Survey (1999-2002) and National Hospital Ambulatory Medical Care Survey (1999-2002), nationally representative surveys of community and hospital-based outpatient practices. Weighted logistic regression was performed to examine the association between patient, physician, and practice characteristics and the selective use of Cox-2 inhibitors among patients at high risk for adverse gastrointestinal events from non-selective non-steroidal anti-inflammatory drugs (NSAIDs). Risk was determined by the GI risk score developed by Singh and colleagues. Sensitivity analyses were performed to examine sensitivity of our findings to assumptions of the model and GI risk score.

RESULTS: Overall, preliminary analyses indicate that approximately 25.1 million (37%) of 67.7 million patient visits where Cox-2s were continued or prescribed were considered appropriate based on the GI risk score. On bivariate analysis, visits to general internists and family practitioners were more likely to be associated with selective use than those to other physician specialties (41%

vs. 32%, $p = 0.04$). The likelihood of receiving a Cox-2 appropriately was also greater for patient visits covered by Medicare or Medicaid rather than other forms of insurance (72% vs. 43% vs. 11%, $p < 0.001$). There were no statistically significant bivariate differences in the selectivity of Cox-2 based on any of the practice characteristics examined, including practice ownership (physician vs. hospital/HMO; private vs. public), size (solo vs. group), region of country, or setting (urban vs. rural). Selectivity of use also did not vary by year examined. Despite some of the bivariate associations described above, which may be in part due to patient selection and confounding, on multivariate analyses no physician or practice characteristics examined were independently associated with greater selectivity of Cox-2 use. These findings remained consistent in numerous sensitivity analyses.

CONCLUSIONS: These results indicate that few if any physician or practice characteristics have been strongly associated with a greater selectivity of Cox-2 use among patients believed to stand to benefit the most from their selective advantage over NSAIDs. This finding and the ubiquitous inappropriate use that we describe demonstrate the challenge of limiting innovative therapies to settings where they are maximally effective.

INFLUENCE OF PATIENTS' REQUESTS FOR DIRECTLY ADVERTISED ANTIDEPRESSANTS: A RANDOMIZED CONTROLLED TRIAL. R.L. Kravitz¹; R. Epstein²; M.D. Feldman³; C.E. Franz¹; P. Franks¹. ¹University of California, Davis, Sacramento, CA; ²University of Rochester, Rochester, NY; ³University of California, San Francisco, San Francisco, CA. (Tracking ID #132722)

BACKGROUND: Direct-to-consumer (DTC) advertising of prescription drugs in the United States is both ubiquitous and controversial. Critics charge that DTC advertising leads to over-prescribing, while proponents counter that such advertising helps avert under-use of effective treatments, especially for conditions that are poorly recognized or stigmatized. Depression is an example of such a condition.

METHODS: To ascertain the effect of patients' requests on physician decision making, we conducted a randomized controlled trial using Standardized Patients (SPs). Six roles were created by crossing two conditions (major depression vs adjustment disorder) with three request types (brand-specific vs general vs none). The SPs were randomly assigned to make 298 unannounced visits to 152 primary care physicians in 3 U.S. cities; each physician saw two SPs, and detection was infrequent (12.8%). Data on prescribing, mental health referral, and primary care follow-up were obtained from SP written reports, visit audio-recordings, chart review, and analysis of written prescriptions and drug samples. The effects of request type on prescribing were evaluated using contingency tables and confirmed in generalized linear mixed models that accounted for clustering and adjusted for site, physician, and visit characteristics.

RESULTS: In major depression, unadjusted rates of antidepressant prescribing were 53%, 76%, and 31% for SPs making brand-specific requests, general requests, and no requests, respectively ($p < .0001$). In adjustment disorder, antidepressant prescribing was 55%, 39%, and 10%, respectively ($p < .0001$). In multivariate analysis examining interaction effects, brand-specific requests were a more potent stimulus of prescribing in major depression than in adjustment disorder. "Minimally acceptable initial care" (any combination of an antidepressant, mental health referral, or follow-up within two weeks) in the major depression role was offered to 98% of SPs making a general request, 90% of those making a brand-specific request, and 56% of those making no request ($p < 0.001$).

CONCLUSIONS: Patients' requests have a profound effect on physician prescribing in major depression and adjustment disorder. DTC advertising may have competing effects on quality, potentially both averting under-use and promoting over-use.

INSURANCE COVERAGE FOR OBESITY IN PENNSYLVANIA. A.G. Tsai¹; D.A. Asch²; T. Wadden¹. ¹University of Pennsylvania, Philadelphia, PA; ²University of Pennsylvania, Gladwyne, PA. (Tracking ID #133634)

BACKGROUND: Obesity has reached epidemic proportions in the United States. Government authorities are increasingly calling for primary care physicians to help patients lose weight. One barrier to weight management in primary care is lack of reimbursement for treatment directed specifically towards weight loss. Anecdotal reports suggest that: 1) coverage is infrequent; and 2) coverage policies may be starting to change as the consequences of excess weight increase health care costs. The goal of this study was to quantify the reimbursement policies of insurers in the state of Pennsylvania. We anticipated that: 1) reimbursement policies would not reflect recent evidence from the literature regarding the effectiveness of weight loss counseling; and 2) coverage for bariatric surgery would be more frequent than for lifestyle modification therapy because of the large weight losses and mortality benefit associated with surgery.

METHODS: We developed a three-page questionnaire, based on previous surveys and on the input of experts in the field of obesity. Coverage was categorized into four areas: lifestyle modification; commercial programs; pharmacotherapy; and bariatric surgery. We inquired about details of coverage (amount reimbursed, clinical criteria required) as well as the number of enrollees in the plan and the department of the responding person. A list of Pennsylvania health plans was obtained from the State Department of Insurance, and these companies were contacted via their public relations department. Results were analyzed with two by two tables.

RESULTS: Of the 19 plans eligible to participate, 16 did so. The responding plans accounted for 55% of Pennsylvania's residents. Plan respondents represented a variety of departments, most commonly medical directors or policy offices. Health plans included: HMOs ($n=4$); Medicaid plans ($n=5$); indemnity ($n=1$); PPO/POS plans ($n=6$). Sixty-three percent of plans provided some cov-

erage for either individual or group counseling. Only 31% of plans provided coverage for intensive counseling, as defined by the U.S. Preventive Task Force Service. Few plans reimbursed for commercial programs or for pharmacotherapy. In contrast, every plan paid some of the cost of bariatric surgery. Surgery was covered more frequently than every other treatment modality ($p < 0.05$ for all comparisons), and most plans listed similar clinical criteria for reimbursement of surgery. Across all treatments, there were no differences in coverage based on plan size or type.

CONCLUSIONS: Insurance reimbursement for obesity is not consistent with recent evidence from the literature on the effectiveness of weight loss counseling (e.g., the Diabetes Prevention Program) and is not consistent with the U.S. Preventive Task Force Service recommendation for intensive therapy. However, weight loss surgery is consistently reimbursed with standard clinical criteria. Possible reasons for these results include: 1) evidence for the effectiveness of weight loss counseling is fairly recent; 2) no data show a long-term mortality benefit for non-surgical treatment; 3) companies may be concerned about the costs of reimbursing for obesity therapy; 4) obesity is still considered a personal health problem, not a medical condition. We anticipate that as the consequences of obesity increasingly affect population health and increase health care costs, coverage policies may begin to change.

KNOWLEDGE AND ATTITUDES REGARDING VACCINATION AGAINST HEPATITIS A AND B IN PATIENTS WITH CHRONIC HEPATITIS C VIRUS INFECTION: A COMPARISON OF FAMILY MEDICINE AND INTERNAL MEDICINE PHYSICIANS IN THE UNITED STATES. J.A. Foont¹; S. Chaudhari¹; C.T. Tenner²; E.H. Weinschel¹; E.J. Bini¹. ¹New York University, New York, NY; ²New York University School of Medicine, New York, NY. (Tracking ID #136208)

BACKGROUND: Although vaccination against hepatitis A virus (HAV) and hepatitis B virus (HBV) is recommended for all patients with chronic hepatitis C virus (HCV) infection, physician vaccination practices are suboptimal. Since residency training for Family Medicine (FM) and Internal Medicine (IM) physicians differs, we hypothesized that there are differences in knowledge and attitudes regarding vaccination against HAV and HBV in patients with chronic HCV infection between these two groups of healthcare providers.

METHODS: A 2-page questionnaire was mailed to 3000 primary care (FM and IM) physicians randomly selected from the AMA Physician Masterfile. The survey was pre-tested prior to mailing and included questions about physician demographics, knowledge, and attitudes regarding vaccination.

RESULTS: Among the 3000 physicians surveyed, 86 were undeliverable, 52 were not in practice, and 1209 (42.2%) of the 2862 eligible physicians returned completed surveys. There were no differences between respondents and non-respondents with regard to age, sex, geographic location, or specialty (FM vs. IM). Of the 1209 respondents (557 FM and 652 IM), FM physicians were less likely to see 3 or more HCV+ patients per week than IM physicians (29.3% vs. 34.8%, $P = 0.04$). The proportion of physicians who knew that the following HCV+ persons should be vaccinated against HAV or HBV is shown below. More FM than IM physicians stated that none of these HCV+ patients should be vaccinated against HAV (23.7% vs. 11.8%, $P < 0.001$) or HBV (21.9% vs. 10.6%, $P < 0.001$). FM physicians were less likely than IM physicians to agree/strongly agree that HCV+ patients should be tested for HAV antibodies (60.3% vs. 71.8%, $P < 0.001$), susceptible patients should be vaccinated against HAV (74.7% vs. 81.6%, $P = 0.004$), HAV antibody testing should be done prior to HAV vaccination (38.1% vs. 56.6%, $P < 0.001$), and that the HAV vaccine is safe (65.3% vs. 76.3%, $P < 0.001$) and effective (60.7% vs. 70.0%, $P = 0.001$) in HCV+ patients. In addition, FM physicians were less likely than IM physicians to agree/strongly agree that HCV+ patients should be tested for HBV antibodies (74.3% vs. 86.5%, $P < 0.001$), susceptible patients should be vaccinated against HBV (79.5% vs. 88.3%, $P < 0.001$), HBV antibody testing should be done prior to HBV vaccination (52.2% vs. 75.7%, $P < 0.001$), and that the HBV vaccine is safe (70.7% vs. 79.1%, $P = 0.001$) and effective (68.8% vs. 75.2%, $P = 0.01$) in HCV+ patients. **CONCLUSIONS:** Among FM and IM physicians in the U.S., there are substantial gaps in knowledge regarding HAV and HBV vaccination of patients with chronic HCV infection. Disparities between specialties may exist due to differences in patient volume, demographics, and emphasis placed on vaccination in their respective training programs. Regardless of the underlying cause, these results demonstrate that HAV and HBV vaccine education programs are needed.

Proportion of physicians who knew vaccination standards for various HCV+ patients

HCV+ Patients with:	Should Receive HAV Vaccine	Should Receive HBV Vaccine					
	FM	IM	P-Value	FM	P-Value	IM	P-Value
Normal ALT levels	57.3%	65.0%	0.006	61.2%		66.0%	0.09
Elevated ALT levels	62.7%	71.6%	0.001	65.5%		74.2%	0.001
HCV PCR positive	62.7%	76.5%	<0.001	65.4%		79.6%	<0.001
Compensated cirrhosis	66.1%	74.4%	0.002	66.8%		73.8%	0.008
Decompensated cirrhosis	35.5%	48.3%	<0.001	36.6%		52.9%	<0.001
HIV coinfection	56.4%	69.9%	<0.001	55.7%		72.9%	<0.001

LACK OF INSURANCE FOR MEDICATIONS, MEDICATION RESTRICTION, AND RISK OF HOSPITALIZATION. K. Bibbins-Domingo¹; G. Gildengorin¹; S. Gregorich¹; A.B. Bindman¹; E.J. Perez-Stable¹; J.S. Haas². ¹Department of Medicine and Medical Effectiveness Research Center, University of California, San Francisco, SF, CA; ²Brigham and Women's Hospital, Boston, MA. (Tracking ID #132526)

BACKGROUND: Patients who lack insurance for medications may restrict use of essential medicines, but resulting adverse health outcomes have not been well described.

METHODS: The Medicare Current Beneficiary Survey is a continuous survey of a nationally representative sample of Medicare beneficiaries. Using data from the 1998 survey, we identified beneficiaries aged ≥ 65 years who reported a history of heart disease or diabetes. We created a categorical measure of supplemental insurance for medications (no supplemental insurance, Medicaid, HMO, self-purchased, employer-sponsored). We used a nominal logistic model that accounted for sampling weights to determine the association between supplemental insurance and percent of the year on angiotensin converting enzyme inhibitors or angiotensin II receptor blockers (ACE-ARB) (None, $<25\%$, 25-49%, 50-74%, 75-89%, $\geq 90\%$). We also determined the association between ACE-ARB use and risk of hospitalization, and the cost savings associated with expenditures on ACE-ARBs. All models were adjusted for demographic and socioeconomic factors and co-morbid conditions.

RESULTS: 4,436 participants had heart disease or diabetes, representing 13,726,967 Medicare beneficiaries. Forty percent had no supplemental insurance for medications, 10% had supplemental insurance through Medicaid, 17% through an HMO, 4% through self-purchased insurance, and 29% through an employer-sponsored plan. Most beneficiaries (72%) were not taking ACE-ARBs. Beneficiaries who lacked supplemental insurance were more likely than those with employer-sponsored plans not to be taking ACE-ARBs (OR 1.7, 95% CI 1.3-2.3). Among those who had ever filled a prescription for ACE-ARBs, 30% were taking ACE-ARBs for $\geq 90\%$ of the year, while 17% were using ACE-ARBs for $<25\%$ of the year. Beneficiaries lacking supplemental insurance were more likely than those with employer-sponsored plans to be taking ACE-ARBs for $<25\%$ of the year (OR 2.4, 95% CI 1.4-4.2). Twenty-nine percent of ACE-ARB users were hospitalized in 1998. Those taking ACE-ARBs $<25\%$ of the year were more than twice as likely as those taking ACE-ARBs $\geq 90\%$ of the year to be hospitalized (OR 2.3, 95% CI 1.4-3.8). At an average cost per daily dose of ACE-ARB equalling \$1.02 and an average cost per hospitalization equalling \$7,582, expenditures for ACE-ARBs may be associated with an annual average net savings of up to \$1,101 per beneficiary in avoided hospitalizations.

CONCLUSIONS: Among Medicare beneficiaries with heart disease or diabetes, lack of supplemental insurance for medications is associated with less or no use of essential medications. This medication restriction may be associated with increased hospitalizations and costs.

LOCATION AND CHARACTERISTICS OF DIAGNOSTIC RADIOLOGY FACILITIES OFFERING SCREENING TOTAL BODY CT SCANS IN NEW YORK STATE. E. Wold¹; M. Reddy¹; W. Lee¹; D. Brand¹; D. Sulmasy². ¹New York Medical College, Valhalla, NY; ²New York Medical College, New York, NY. (Tracking ID #136072)

BACKGROUND: Despite concern regarding the impact of CT scans that screen for cancer on patient outcomes, public health and health expenditures, there are no data describing the prevalence of this practice. Such data may be an important aid to policy makers who must assess the potential impact of these tests on the healthcare system. We hypothesized that practices offering screening will comprise a larger percentage of all practices in areas closer to urban centers and with a smaller ratio of diagnostic CT practices to population.

METHODS: Using the American College of Radiology (ACR) online database, we identified practices in New York State that have applied for or are accredited to provide diagnostic CT scans, and conducted a cross sectional survey. A research assistant, who did not identify herself as a researcher, called each practice and asked whether any of three types of screening CT scans (Total body, Cardiac, Lung) were offered. This data was then matched to U.S. Census data for the county where each center was located and to center websites where available. Descriptive statistics were used to describe prevalence. Categorical variables were compared using Chi-Square tests. Continuous data were compared with Student's T-Test.

RESULTS: A total of 213 practices were identified. One hundred ninety five practices could be contacted. Thirty three of these (17%) practices offered total body screening (TBCT). Twenty-one percent of TBCT practices were located in New York City and all 33 practices were located either within New York City or its immediately surrounding counties. In some counties, up to 50% of CT practices offered TBCT. Thirty of 33 (91%) TBCT practices reside in counties in the top 50% of median household income compared with 100 of 162 (62%) non-TBCT practices, $p = .001$. In addition, the counties with at least one TBCT practice had a lower mean ratio of persons to CT practices, 64,500 compared to 192,369, $p < .05$. A total of 119 practices advertised on the internet. Nineteen of these (16%) practices offered either non-specific or TBCT screening on the web. Of those sites offering TBCT or non-specific screening, 13/19 (68%) also offered TBCT to the researcher over the phone. Practices that had offered TBCT over the phone who also had an identified web page offered a greater number of diagnostic modalities than practices who offered only diagnostic CT scans. Of the 119 practices identified on the internet who thus had information available regarding the types of procedures offered, those practices who offered at least 4 different diagnostic modalities (i.e. MRI, U/S, etc) were more likely to offer TBCT than smaller practices. Nineteen of 26 (73%) practices that offered TBCT over the phone and had a web page, also performed at least 4 different imaging modalities compared to 47 of 93 (50%) practices that did not offer TBCT but had a web page, $p < .05$.

CONCLUSIONS: Radiology practices that offer screening CT scans make up a significant percentage of radiology practices particularly in wealthier metropolitan areas of New York.

MEDICAID PATIENTS PERCEIVE MEDICATION COPAYS AS A BARRIER TO ACCESS TO HEALTH CARE. M.A. Mendiola¹; A.G. Pereira²; M. Kurvers²; T. Kurvers²; K. Larsen³. ¹Hennepin County Medical Center, Brooklyn Park, MN; ²Hennepin County Medical Center, Minneapolis, MN; ³University of Minnesota, Minneapolis, MN. (Tracking ID #135949)

BACKGROUND: In 2003, the Minnesota legislature passed legislation requiring Medicaid patients to pay a nominal copay for medications. Many physicians have anecdotes about Medicaid patients who have had poor outcomes because they have not been able to afford their medication copay. Recently, a class action lawsuit was brought against the state of Minnesota challenging the legality of this law. This study was designed to determine the prevalence of difficulty obtaining medications due to copays in the Medicaid population at an urban safety net hospital.

METHODS: 143 patients were selected from the waiting rooms of the general medicine clinic and medicine walk-in clinic at Hennepin County Medical Center in Minneapolis, MN for survey. They were anonymously interviewed by one interviewer. An interpreter was used as needed. They were asked questions about their type of insurance, the number of prescriptions, and how often they were unable to obtain their medications due to copays. They were also asked if their doctors were aware of their inability to get medications, and if not having medications due to copays had affected their utilization of emergency services.

RESULTS: 67 of 143 patients interviewed reported they had been on Medicaid in the last six months, and had at least one prescription. 32 (47%) of these people had at least one episode when they were not able to fill their prescription(s) in that time. 11(34%) of the people who were unable to fill reported going to the emergency room, for a total of 27 visits, for reasons they attributed to copays. 19 of 32 (57%) said that their doctor was not aware of the problem. 24 of 32 (75%) said that not having their medication because of copays had negatively impacted their health.

CONCLUSIONS: This was a pilot study designed to study how the new Medicaid copay system is impacting access to health care among patients in an urban safety net hospital. Nearly one half of the patients surveyed were negatively affected by copays. This was a convenience sample, and relies on patient self-report, but it suggests a significant problem. It remains to be seen if the copay system will be ultimately cost effective, or if it will prove penny-wise but pound foolish.

NIH RESEARCH FUNDING RANK AND RETURN OF INDIRECT COST REVENUES TO DEPARTMENTS AND INVESTIGATORS. V.A. Lawrence¹; J.A. Youngers¹; J.A. Pugh¹; R.A. Clark¹. ¹Audie L. Murphy VA Hospital and The University of Texas Health Science Center at San Antonio, San Antonio, TX. (Tracking ID #132611)

BACKGROUND: Investigators and their departments frequently lobby that returning a portion of grant revenues for facilities and administration services costs (F&A or indirect costs) helps incentivize the research enterprise. We tested the hypothesis that returning a portions of such funds to investigators and departments is therefore associated with higher rank in NIH research funding.

METHODS: Cross-sectional e-mail survey, 12/2002, to directors of sponsored programs offices at the top 167 U.S. academic institutions in total NIH research awards in FY 2001 (data from <http://grants.nih.gov/grants/award/award-tr.htm>). Queries included: if a portion of F&A revenues is distributed to generating investigators and departments; if so, the formula for the distribution; and if a share of the costs of F&A services is directly billed to investigators and departments. We stratified institutions by quintiles of NIH research funding rank. **RESULTS:** Response rate was 75% for all institutions (126 of 167) and those with medical schools (83 of 110). Most (80%) reported that some funds, formulaically based on F&A revenue, are returned to investigators and/or departments. Percent of all schools distributing funds, from highest to lowest rank quintiles, was 62%, 74%, 90%, 89%, and 91% ($p = 0.003$ for trend). Percent of medical schools distributing funds, from highest to lowest rank quintiles, was 61%, 78%, 90%, 86%, and 80% ($p = 0.09$ for trend). Most institutions (70%) did not bill the cost of F&A services to their schools. Percent of all schools and medical schools, respectively, NOT billed for a share of institutional F&A costs and reporting internal distribution of F&A revenues was 73% (73/101) and 59% (38/64). Percent of schools NOT billed for a share of F&A costs and internally distributing a portion of F&A revenues, from highest to lowest rank quintiles, was 11%, 16%, 26%, 22%, and 25% ($p = 0.026$ for trend).

CONCLUSIONS: These results do not indicate a simple, uniform, direct association between return of F&A revenues and NIH rank in research awards as a surrogate outcome for incentivizing research. The data hint at an inverse correlation, with schools of lower rank more often distributing such funds to investigators and departments. These data are only cross-sectional; however, this is the first analysis of this issue of which we are aware and it suggests a more complex relationship than simple direct correlation. More research is needed to understand actual causative relationships between research and local financial incentives, i.e., whether lower ranking schools use an explicit strategy to retain and nurture faculty, how changes in policy affect subsequent research productivity, and the importance of intangibles, such as institutional prestige, in the local research culture.

OUT-OF-POCKET DRUG COSTS BEFORE AND AFTER THE MEDICARE MODERNIZATION ACT: ARE THERE DIFFERENCES IN HOW PEOPLE FARE? W.F. Gellad¹; H.A. Huskamp²; K.A. Phillips³; J.S. Haas¹. ¹Brigham and Women's Hospital, Boston, MA; ²Harvard Medical School, Boston, MA; ³University of California, San Francisco, San Francisco, CA. (Tracking ID #134050)

BACKGROUND: The Medicare Modernization Act of 2003 (MMA) includes a prescription drug benefit that will take effect on January 1, 2006. Because this benefit contains gaps in coverage, many beneficiaries will have significant out-of-pocket drug costs. The purpose of this study is to evaluate whether an individual's sociodemographic characteristics and number of chronic illnesses are associated with projected drug expenditures in 2006, and to estimate how out-of-pocket drug costs may change with the MMA drug plan.

METHODS: A cross-sectional analysis was performed of adults over the age of 64 with an income at least 150% of the poverty level who are included in the nationally representative Medical Expenditure Panel Surveys (MEPS), 1996-2000 (n=6,085). The total and out-of-pocket drug expenditures for respondents were inflated by a factor to approximate their 2006 total and out-of-pocket expenditures. This inflation factor was based on data from the Centers for Medicare and Medicaid Services to account for the yearly rise in drug prices and utilization. Once the estimation of 2006 total drug expenditures was made, that value was inserted into the generally accepted formula for cost-sharing that most seniors will face once the MMA takes effect in 2006. Out-of-pocket drug costs prior to the MMA and projected out-of-pocket costs after the MMA benefit, both in 2006 dollars, are then compared using multivariate analyses.

RESULTS: In 2006, 30.5% of Medicare beneficiaries are estimated to have total drug expenditures over \$2,250, a level that will require significant cost-sharing under the MMA. 35% of individuals with three or more chronic conditions fall into the \$2,251-\$5,100 category, meaning if an additional medicine is prescribed for them, they will likely pay the full cost of that new medicine. Mean annual out-of-pocket drug expenditures will decline from \$1,219 per person prior to the MMA to \$974 after the MMA (i.e. average savings of \$245). In multivariate linear regression models of the difference in out-of-pocket costs before and after the MMA, African American and Hispanic seniors had smaller out-of-pocket savings on drugs under the MMA compared to whites (average of \$150 and \$152 less per year, respectively), after controlling for sociodemographic characteristics and number of chronic illnesses. Seniors who currently do not have supplemental insurance coverage experienced greater savings compared to individuals who currently have private coverage (average additional savings of \$288).

CONCLUSIONS: Many Medicare beneficiaries will have total drug expenditures that will require substantial cost-sharing under the MMA. Overall, the MMA may be associated with modest declines in out-of-pocket drug expenditures for seniors, although the savings in out-of-pocket drug expenditures may not be shared equitably for some vulnerable populations. Several studies have shown that as out-of-pocket drug costs increase, adherence to medication decreases and in some cases may be associated with adverse events. The government and the research community will have to pay very close attention once the Medicare drug bill is in place to ensure that seniors benefit equitably from this government program.

PUBLIC HOSPITAL SYSTEMS, DIABETES AND DISPARITIES IN GLYCEMIC CONTROL. D. Schillinger¹; D.S. Lessler²; B.M. Reilly³; D.H. Bor⁴; A.B. Bindman¹; M. Kleinpeter⁵; D.C. Ziemer⁶; I. Glazer⁷; J. Huang⁸; M. Regenstein⁹. ¹University of California, San Francisco, San Francisco, CA; ²University of Washington, Seattle, WA; ³Cook County Hospital, Chicago, IL; ⁴Harvard University, Cambridge, MA; ⁵Tulane School of Medicine, New Orleans, LA; ⁶Emory University, Decatur, GA; ⁷Memorial Health System, Jacksonville, FL; ⁸National Public Health and Hospital Institute, Washington, DC. (Tracking ID #133371)

BACKGROUND: Public hospitals provide care to many of the most vulnerable members of U.S. society, and share a commitment to eliminate barriers to care based on race, ethnicity, or ability to pay. While recent diabetes studies have profiled the performance of the VA and private managed care systems, little is known about the care of diabetes across public hospitals, or to what extent disparities in diabetes outcomes exist within these systems.

METHODS: In 2002-3, the National Public Health and Hospital Institute assembled a quality improvement consortium involving 7 public hospital systems (Atlanta GA, San Francisco CA, Seattle WA, Hollywood FL, New Orleans LA, Cambridge MA, and Chicago IL) that provide care to over 125,000 patients with diabetes on an annual basis. We obtained demographic, clinical, and utilization data over a 3-year period (Jan 1 2000-Dec 31 2002) on a random sample of 3,000 diabetes patients from each system. To be included, patients had to be ≥ 18 years old and have made 2 or more outpatient visits during the study period with an associated ICD-9 code of diabetes. Six of the 7 sites reported race/ethnicity data on $>98\%$ of patients and were included in this analysis. Using patients' last HbA1c value, we measured the proportion of patients with poor glycemic control (HbA1c $\geq 9\%$, a standard benchmark), and explored whether glycemic control varied by race/ethnicity, insurance status, or ambulatory use.

RESULTS: Overall, 69% of patients were non-White minorities, and 42% were uninsured. Over the 3-year period, 34% of patients rarely made outpatient visits (2-4 visits/3 yrs), 26% made occasional visits (5-10 visits/3 yrs) and 39% made regular visits (>10 visits/3 yrs). Overall, 26% of patients had poor glycemic control (HbA1c $\geq 9\%$). Compared to Whites, poor glycemic control was more common among Blacks (OR 1.73) and Hispanics (OR 1.84) but not Asians (OR 0.80) ($P < .001$), and was more common among the uninsured compared to Medicaid (OR 1.62, $P < .001$). Adjusting for age, sex, race/ethnicity, insurance, and outpatient utilization, Blacks and Hispanics had worse glycemic control as measured by mean HbA1c (8.3% and 8.2%) than Whites or Asians (7.8% and 7.9%), as did the uninsured (8.2%) compared to Medicaid (8.0%) ($P < .001$). In adjusted models, there was a graded relationship between extent of ambulatory use and glycemic control with rare, occasional, and regular users having mean HbA1c levels of 8.4%, 8.1% and 7.8%, respectively ($P < .001$).

CONCLUSIONS: Despite limited resources and the challenges posed by caring for diverse and vulnerable populations, public hospital systems perform at a level comparable to other health systems with regard to glycemic control. While the disparities based on race/ethnicity and insurance status mirror national trends, our results suggest that public hospital systems should target access for the uninsured, continuity of care for all patients, and quality of care for minority populations. Given their shared missions and the availability of electronic data systems, public hospital systems have the opportunity to collaborate in quality of care benchmarking, crafting innovative diabetes improvement initiatives, as well as implementing established strategies that have been successful in other health systems.

RACE AND LANGUAGE CONCORDANCE AND THEIR EFFECTS ON PATIENT-PHYSICIAN TRUST SCALES. A.A. Donato¹; R. Brown²; C. Bright³; J. Bussey-Jones⁴. ¹Reading Hospital and Medical Center, West Reading, PA; ²George Washington University, Washington, DC; ³Duke University, Durham, NC; ⁴Emory University, Atlanta, GA. (Tracking ID #134650)

BACKGROUND: The 2004 ACP position paper on Racial and Ethnic Disparities in Health Care recommends efforts to increase linguistic and racial concordance between patients and physicians as solutions to the healthcare gap that exists between minority and majority populations. Racial concordance has been positively associated with patient satisfaction and healthcare utilization, while linguistic concordance has been associated with improved physical functioning and well-being. The relative importance and interrelation of racial and linguistic concordance to patient trust and outcomes is unknown.

METHODS: The CCTOP Study (Cultural Competence Training and Outcomes in Patients) is a multi-center cross-sectional study of patients presenting for care at 11 outpatient medical centers. The study assessed physicians' attitudes regarding culture and bias, as well as patients' feelings regarding trust, participatory communication and satisfaction with their physicians on five-point scales (1 = strongly agree to 5 = strongly disagree).

RESULTS: Preliminary results were available from 291 patients (70% non-white) and their 175 physicians (44 attendings and 131 residents, 47% non-white) at the first 5 sites. Patients with physicians of differing race were more likely to doubt their physician cares about them (4.1 vs. 4.4, $p = 0.05$), and would fail to protect their health information (3.8 vs. 4.2, $p = 0.01$). Patients with physicians of differing race were less likely to trust the physician would put their needs first (1.9 vs. 1.6, $p = 0.03$), admit a mistake (1.8 vs. 1.6, $p = 0.02$), and were less likely to refer their doctor to a friend (1.6 vs. 1.3, $p = 0.02$). Language concordance was not associated with any higher trust scores as compared to discordant patient-physician pairs. No differences were seen in patients' participatory communication ratings between either language- or race-concordant and discordant pairs.

CONCLUSIONS: Race concordance was associated with significantly higher patient trust scores when compared to discordant pairs, while language concordance had no significant impact on trust scores. Results of this study suggest that the positive effects of race concordance do not seem to be associated with participatory communication or language concordance. Healthcare policymakers need to consider the beneficial effects of racial concordance that exist independent of language concordance and interviewing technique when developing policy for minority recruitment and retention in healthcare. Improving physician-patient trust when race is discordant remains a challenge.

REDUCING FINANCIAL BARRIERS TO TEST STRIPS IS NOT ASSOCIATED WITH BETTER GLYCEMIC CONTROL IN TYPE 2 DIABETES: RESULTS OF A RANDOMIZED CONTROLLED TRIAL OF REIMBURSEMENT POLICY. J.A. Johnson¹; S.R. Majumdar¹; S.L. Bowker²; E.L. Toth³; A.E. Edwards³. ¹University of Alberta, Edmonton, Alberta; ²Institute of Health Economics, Edmonton, Alberta; ³University of Calgary, Calgary, Alberta. (Tracking ID #134506)

BACKGROUND: Self-monitoring of blood glucose (SMBG) is a cornerstone of self-care for all patients with diabetes, although the practice is costly and the benefits to patients with type 2 diabetes are not clear. Therefore, we conducted a randomized trial to determine whether patients who received free testing strips would achieve better glycemic control than control patients.

METHODS: Patients with type 2 diabetes (excluding those with private insurance for SMBG supplies or those using insulin) were recruited from 38 community pharmacies. Patients were randomized to receive free testing strips for 6 months or not; all patients were provided with a free SMBG meter. The primary outcome was change in A1c over six months. Outcomes were ascertained in an independent and blinded fashion and all comparisons were based on intention-to-treat analyses.

RESULTS: We randomized 262 patients (131 Intervention and 131 Controls). Their mean age was 68.4 (SD 10.9) years, 48% were male, mean duration of diabetes was 8.2 (SD 7.2) years, 97% used oral hypoglycemic agents, and their mean entry A1c level was 7.4% (SD 1.2). Over 6 months, we observed no difference in the change in A1c between Intervention (0.24 decrease) and Control patients (0.16 decrease; $p = 0.43$). Similar results were observed in subgroup analyses based on the level of glycemic control at baseline (i.e., A1c < 8.0 [$p = 0.90$] or A1c > 8.0 [$p = 0.49$]) and in multivariable analyses that controlled for imbalances in baseline patient characteristics. Intervention patients reported testing, on average, 0.65 days per week more often than Controls ($p = 0.01$), although increased testing itself was not associated with better glycemic control.

CONCLUSIONS: Reducing financial barriers to SMBG by providing free testing strips did not lead to improved glycemic control in patients with type 2 diabetes. Our results may have significant policy implications with respect to reimbursement for testing supplies.

RIGOROUS DEVELOPMENT METHODS DO NOT CONSISTENTLY PRODUCE GUIDELINES THAT CLINICIANS FIND VALID. T.K. Nuckols¹; Y. Lim¹; B. Wynn²; S. Matthe²; C.H. Maclean³; P. Harber³; P. Wallace¹; S.M. Asch⁴. ¹RAND Corporation, Santa Monica, CA; ²RAND Corporation, Washington, DC; ³University of California, Los Angeles, Los Angeles, CA; ⁴RAND, Los Angeles, CA. (Tracking ID #132524)

BACKGROUND: Existing guideline assessment methods address technical quality, including rigor of development, editorial independence, and other dimensions. National Guidelines Clearinghouse lists guidelines meeting basic development standards. These methods and standards assume that rigorous development produces clinically valid guidelines; to date, no attempt has been made to confirm this. Further, methods for assessing guideline content validity have not existed. The objective of this analysis was to develop methods for assessing guideline content validity, and to determine whether good technical quality consistently produces guidelines clinicians find valid.

METHODS: As part of a project for the California workers' compensation system, this study evaluated the technical quality and clinical content of five selected guideline sets addressing work-related injuries. These five sets addressed common and costly therapies for injuries of the spine, arm, and leg; their development included systematic literature reviews and multidisciplinary clinician panels; and they were nationally recognized. Technical quality was evaluated using the AGREE Instrument, which produces domain scores ranging from 0.00 (lowest) to 1.00 (highest). Clinical content was evaluated by adapting components of the RAND/UCLA Appropriateness Method. A multidisciplinary clinician panel rated content in a two-round, modified-Delphi process, producing panel conclusions of: comprehensive, of intermediate comprehensiveness, or not comprehensive; and valid, of uncertain validity, or not valid. Panelists rated 16 selected topics in greater detail, any residual content in each guideline set, and each set overall. The selected topics addressed utilization decisions for surgery, physical therapy, and chiropractic manipulation provided for lumbar spine, shoulder, and carpal tunnel disorders. Panelists relied on prior understandings of the literature and clinical experience in judging comprehensiveness and validity.

RESULTS: Regarding technical quality, AGREE domain scores (see Table) were higher than for many guidelines in prior studies, particularly for rigor of development and editorial independence. Rigor-of-development scores were 0.81 to 0.88, compared with 0.05 to 0.92 in prior studies. Editorial independence was 0.75 for one guideline set and 0.92 to 1.00 for the other four, compared with 0 to 1.0 in prior studies. Regarding clinical validity, ratings were highly variable for the selected topics addressed by each guideline (i.e., topics of intermediate or better comprehensiveness). For four guideline sets, panelists judged the content valid for 36 to 50% of topics addressed, of uncertain validity for 50 to 64%, and not valid for 0%. Two of these sets were judged valid overall and two were of uncertain validity. For the remaining set, the content was valid for 7% of topics addressed, of uncertain validity for 60%, and not valid for 33%; overall, panelists judged this set not valid. National Guidelines Clearinghouse currently lists two of the guideline sets and will soon list a component of the set with the lowest validity scores.

CONCLUSIONS: Despite relatively rigorous development methods, clinicians felt that much of the content in these five guideline sets was of uncertain validity and that one set was not valid. Rigorous development methods do not appear to consistently produce guidelines that clinicians find valid.

Technical Quality and Clinical Validity of Selected Guidelines

Guideline Number	1	2	3	4	5
Technical Quality					
AGREE	Score	Score	Score	Score	Score
Domain					
Scope and Purpose	1.00	0.89	0.89	1.00	1.00
Stakeholder Involvement	0.54	0.79	0.79	0.88	0.79
Rigor of Development	0.81	0.88	0.83	0.88	0.81
Clarity and Presentation	0.96	0.88	1.00	1.00	0.96
Applicability	0.17	0.33	0.33	0.61	0.72
Editorial Independence	1.00	1.00	0.75	1.00	0.92
Clinical Validity					
16 Selected Topics	% (N)	% (N)	% (N)	% (N)	% (N)
Valid	40 (2)	50 (6)	7 (1)	43 (6)	36 (5)
Validity	60 (3)	50 (6)	60 (9)	57 (8)	64 (9)
Uncertain					
Not Valid	0 (0)	0 (0)	33 (5)	0 (0)	0 (0)
Total Topics Addressed	100 (5)	100 (12)	100 (15)	100 (14)	100 (14)
Residual Content	Not Addressed	Validity	Validity	Validity	Validity
Overall	Valid	Valid	Not Valid	Validity	Validity
				Uncertain	Uncertain

THE FINANCIAL IMPACT OF CURRENT DRUG PATENT POLICY ON MEDICAID DRUG SPENDING. A.S. Kesselheim¹; M.A. Fischer¹; J. Avorn¹. ¹Brigham and Women's Hospital, Boston, MA. (Tracking ID #135044)

BACKGROUND: The rising cost of prescription drugs has become a critical health policy issue, especially for public insurance programs such as Medicaid, and elevated drug prices have been linked to diminished adherence to treatment among other low-income patients. Substituting generic products with identical therapeutic effects can reduce costs, but a number of factors limit the success of this practice. First, brand-name drug manufacturers can obtain market exclusivity extensions beyond the initial terms of their patents. Second, reductions in generic drug prices can be slower than anticipated. Finally, physicians often do not prescribe generic alternatives even when such drugs are available. We sought to quantify the aggregate cost to Medicaid of market exclusivity extensions, artificially elevated generic prices, and low generic substitution rates.

METHODS: We identified three drugs—amoxicillin/clavulanate potassium (Augmentin), omeprazole (Prilosec), and metformin (Glucophage)—for which substantial market exclusivity extensions had occurred. Using data on all filled prescriptions from each state Medicaid program, we analyzed prescribing patterns through the second quarter of 2004. For each drug, we determined both the original expiration date of patent protection and the date on which a generic alternative actually became available. We then calculated what Medicaid would have paid for each drug if no market exclusivity extensions had been granted, if the initial generic price matched the final stabilized generic price, and if full generic substitution had occurred during the earliest quarter that the generic equivalent could have been available. By comparing predicted to actual expenditures, we determined the cost to Medicaid of these current distortions of "natural" patent expiration and generic implementation policies and practices.

RESULTS: For the three drugs studied, the average term of patent extension was 18 months (9 for amoxicillin/clavulanate potassium, 21 for omeprazole, and 24 for metformin). For the first two drugs, the final stabilized generic prices were 6% less than the initial generic prices, whereas for metformin the final stabilized price was 40% less. In each case, the rate of actual generic substitution started slowly and reached a plateau at less than full substitution (87% for amoxicillin/clavulanate potassium, 64% for omeprazole, and 87% for metformin). Adding together the total costs of these distortions and practices, we calculated that Medicaid expenditures were increased by \$501 million from April 2000 through June 2004 for just three commonly used medications.

CONCLUSIONS: Market exclusivity extensions of brand-name drugs, elevated generic prices, and under-utilization of generics lead to markedly increased drug costs for the Medicaid program and other payers, as well as for patients. These expenditures represent substantial potential savings, which could be achieved without affecting patients' clinical outcomes. Developing a more rational drug policy that reforms these features of generic implementation could help Medicaid programs protect their drug budgets without further reducing expenditures in other realms, such as limiting benefits or restricting eligibility qualifications.

THE IMPACT OF INCREASED MEDICATION CO-PAYMENTS ON BLOOD PRESSURE AMONG VETERANS. B.Y. Lee¹; K.G. Volpp¹. ¹University of Pennsylvania, Philadelphia, PA. (Tracking ID #136276)

BACKGROUND: Previous observational studies have shown that small increases in co-payments for prescription drugs significantly reduce utilization of medications, particularly within low socioeconomic status populations. Little is known about the health impact of co-payment increases. In February 2002, the Veterans Affairs (VA) system increased Veteran prescription co-payments from \$2 to \$7 (Public Law 106-117). The objective of this study was to examine the impact of this co-payment change on blood pressure control among patients of different income levels in the Philadelphia VA Medical Center (PVAMC) population.

METHODS: This was a retrospective observational study using the VA VISTA database. The two comparison time periods were a 24-month period (8/1/99-8/01/01), referred to as the Pre-Period, before Public Law 106-117 and a 24-month period (8/1/02-8/30/04), the Post-Period, after Public Law 106-117. Our study included patients who received blood pressure medications in the Pre-Period and had blood pressure readings and evidence of some type of clinical activity in the Pre- and Post-Periods. All patients were between 21 and 90 years of age and received care at the PVAMC. Patients who were exempt from co-payment charges were excluded. Patient household income was derived from zip-code matched census data. We constructed a linear regression model to see how household income correlated with changes in mean systolic blood pressure (SBP) and diastolic blood pressure (DBP) between the Pre- and Post-Periods.

RESULTS: Our inclusion and exclusion criteria yielded 15,758 patients (15,271 men). Of these, 3,443 (40.1%) were African American. The first, second, third, and fourth quartiles of annual household income were less than \$34,638, between \$34,638 and \$51,747, between \$51,747 and \$63,214, and greater than \$63,214. Patients in the lowest income quartile had on average a 2.57 mm Hg increase in mean SBP compared with a 0.89 increase in the second quartile, a 1.17 decrease in the third quartile, and a 1.65 decrease in the fourth quartile. African-American patients had on average a 3.25 increase in mean SBP compared with a 0.87 decrease for patients of other races. The regression model confirmed that changes in mean SBP was positively correlated with being African-American ($p < 0.001$) and negatively correlated with increasing income quartiles ($p < 0.001$).

CONCLUSIONS: Our results indicate that increases in medication co-payments may have negatively impacted blood pressure control in lower income patients and African-Americans. This raises the concern that co-payment increases may worsen disparities in cardiovascular outcomes between high and low income groups as well as between Whites and African-Americans.

THE IMPACT OF MEDICAID PROGRAM CHANGES ON A CHRONICALLY ILL, LOW INCOME POPULATION. R. Solotaroff¹; M. Carlson²; B. Wright³; J.E. Devoe⁴; J. Smith⁵; J. Boone⁴; T. Edlund⁶. ¹Portland VA Medical Center, Portland, OR; ²Portland State University, Portland, OR; ³Providence Health System, Portland, OR; ⁴Oregon Health & Science University, Portland, OR; ⁵Oregon Health Policy and Research, Portland, OR. (Tracking ID #134637)

BACKGROUND: In early 2003, faced with budget shortfalls, Oregon lawmakers implemented cost-sharing mechanisms (premiums and co-payments) and more stringent eligibility requirements in an effort to control costs associated with the state's Medicaid program, the Oregon Health Plan (OHP). Following this implementation, many beneficiaries, including large numbers of chronically ill persons, disenrolled from the program. We sought to determine the effects of disenrollment from Medicaid on health status, health care utilization, and financial strain among persons with chronic medical conditions.

METHODS: We conducted a mail survey of 1378 adults under 65 who were enrolled in the OHP prior to the 2003 policy changes. We collected information on demographics, presence of chronic illness (diabetes, asthma, hypertension, high cholesterol and congestive heart failure), OHP enrollment status, finances, health status, access and utilization of health care services. We categorized respondents as chronically ill or non-chronically ill. We then stratified these groups by insurance status – continuously enrolled in OHP; left OHP, other insurance now; left OHP, no insurance now – and compared these groups in terms of health status, health care access and utilization, and financial strain using 2-sided chi-square tests.

RESULTS: Of the 1378 respondents, 643 (46.7%) had at least one of the five chronic conditions queried. Of these 643, 183 (28.5%) had left OHP and not regained insurance. The chronically ill in all three insurance categories were more likely than their non-chronically ill counterparts to report poor health status, cutting back on food budgets to pay for medical bills, and inability to pay for needed medication ($p < .05$ for all comparisons). Among the chronically ill, the insured and uninsured were equally likely to have poor health status and to have cut back on food budgets to pay for health care (see Table). The majority of chronically ill with insurance coverage were unable to pay for needed medications, though this was significantly different from the chronically ill without insurance. The uninsured were twice as likely as the insured to report an inability to obtain care when needed, and half as likely to have had a primary care visit in the past six months. No significant difference was found in emergency department (ED) visits.

CONCLUSIONS: These results reveal the potential negative impact of an abrupt loss of health insurance on a chronically ill, low income population, as well as the difficulties faced by all low-income, chronically ill persons, even those who maintain insurance. Poor health status and financial difficulties were nearly as prevalent in the insured as the uninsured in this population. More dramatic effects of health insurance coverage were seen in measures of access and utilization. Policymakers should consider that in this vulnerable population, small policy changes in Medicaid that lead to disenrollment can have a large impact on access and utilization. This lack of access, in addition to the financial strain and health burden borne by many chronically ill and low income persons, may result in worsening health status over time.

Measures of Health Status, Access, Utilization and Finances, by Insurance Status (as % of All Chronically Ill Respondents)

	'Fair' or 'Poor' Health	Cut Back on Food Budgets For Medical Bills	Unable to Pay For Prescriptions*	Unable to Obtain Needed Care in Last 6 Months*	Obtained Primary Care in Last 6 Months*	ED Visit in Last 6 Months
Continuously Enrolled in OHP	57.2%	42.0%	55.8%	30.2%	86.7%	30.9%
Left OHP, Insured Now	58.1%	42.4%	55.3%	42.7%	82.4%	35.3%
Left OHP, Uninsured Now	54.7%	48.9%	77.5%	74.0%	47.5%	37.2%

*Indicates Statistically Significant Difference Between Insurance Groups, $p < .05$

THE POWER OF PAPERWORK: HOW PHILIP MORRIS NEUTRALIZED THE MEDICAL CODE FOR SECONDHAND SMOKE. D. Cook¹; E.K. Tong¹; S.A. Glantz¹; L. Bero¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #136207)

BACKGROUND: In 1994, an International Classification of Disease-9 (ICD-9) exposure code for secondhand smoke was created, but is presently an invalid entry on a common Medicare form. The federal regulatory process allows private organizations to respond to policy decisions like classification schemes and paperwork requirements. We examined the tobacco industry's response to the creation of a new ICD-9 code for secondhand smoke. The tobacco industry has disputed scientific evidence and government conclusions that secondhand smoke causes disease, as well as the establishment of smoke-free policies in workplaces and public places.

METHODS: We searched tobacco industry documents that were made public through litigation. These documents include internal corporate memos, budgets, strategic plans, and reports from companies such as Philip Morris, British American Tobacco, and R.J. Reynolds. The University of California San Francisco

Legacy Tobacco Documents website (<http://legacy.library.ucsf.edu>) contains 40 million pages that are publicly accessible. Searches using standardized techniques, such as narrowed searches using Boolean terms and snowball techniques of surrounding Bates identification document numbers, were completed between October 1, 2003 and March 1, 2004. Keywords included "ICD-9," "E-code," and names of consultants or organizations described. Documents were qualitatively analyzed by two of the authors for relevance and agreed upon by consensus.

RESULTS: Philip Morris (PM) tobacco company budgeted over two million dollars for an "ICD-9 Project" in the mid-1990s to counteract the secondhand smoke exposure code. The concern was that individual physicians were now "authorized" to attribute illnesses and deaths to secondhand smoke, which could be linked to national health costs. Consulting firm Multinational Business Services (MBS), whose founding director was a former official with the Office of Management and Budget, was hired with a contract stipulation that the relationship with PM not be disclosed. PM and MBS first tried to prevent the code's adoption by presenting third party objections to the National Center for Health Statistics and also challenging the code procedurally. When this initial effort failed, they successfully petitioned removing all exposure codes' entry on the Medicare 1500 form (the nationwide standard billing form) under the Paperwork Reduction Act. MBS argued that physicians do not provide enough information to coders, exposure codes were an undue diagnostic burden on physicians, secondhand smoke was a chronic cause of exposure and differed from acute exposure codes, and unreliable statistics would create a liability to the government. Finally, they advocated for an alternative European coding system, which would not include chronic exposures, that is still being discussed for incorporation into the revised ICD-10 codes.

CONCLUSIONS: PM's actions represent the first time an ICD-9 code has been successfully challenged, and undermine the collection of data on secondhand smoke's relationship to illness. The arguments made by PM's consultants to invalidate the code contradicted their private concerns that physicians could link secondhand smoke with disease. The process for creating and utilizing medical codes can be influenced by lobbyists with undisclosed private industry clients. The Medicare 1500 form will be renewed in March 2006, presenting an opportunity to readdress allowing all exposure codes.

VARIATION IN THE THRESHOLD TO INTERVENE AMONG PRIMARY CARE PHYSICIANS. B.E. Sirovich¹; D.J. Gottlieb²; E.S. Fisher². ¹VA Outcomes Group, Dartmouth Medical School, White River Junction, VT; ²Center for the Evaluative Clinical Sciences, Dartmouth Medical School, Hanover, NH. (Tracking ID #132523)

BACKGROUND: Previous research has documented dramatic variation in health care spending across the United States, with little relationship to health outcomes. Although high spending areas have more physicians per capita, it is not known whether this disparity fully explains the differences in spending or whether individual physicians in high spending regions have greater tendency to intervene for individual patients. We sought to measure physicians' tendency to intervene across different levels of local health care spending.

METHODS: We used data from the Community Tracking Study, a telephone survey including a nationally representative sample of 5571 primary care physicians providing care to adults in 1998 and 1999 (response rate 61%). Based on practice location, we assigned each participating physician to one of 306 U.S. Hospital Referral Regions, which were in turn assigned to quintiles of spending based on a previously derived per capita Medicare spending measure. Physicians' tendency to intervene was measured by responses to six clinical vignettes, in which physicians were asked how often they would order a test, referral or treatment for the patient described. All tests for trend are based on linear regression in which the individual physician's response (what percentage of the time he would recommend the intervention) was the dependent variable and the independent variable was spending in the physician's region.

RESULTS: In 5 out of 6 vignettes, physicians in high spending regions were more likely to recommend interventions than those practicing in low spending regions. For example, for a 35 year old man with back pain and foot drop, physicians in high spending regions would recommend an MRI 82% of the time, compared with 69% for physicians in low intensity regions ($p_{trend} < 0.001$). For a 60 year old man somewhat bothered by symptoms of benign prostatic hyperplasia, physicians in high spending areas would make a Urology referral 32% of the time, while those in low spending regions would do so only 23% of the time ($p_{trend} < 0.001$). Adjusting our analysis for physician specialty further reinforced our findings that physicians in high spending regions have a greater tendency to intervene.

CONCLUSIONS: Varying rates of health care spending across the United States reflect the underlying tendency of local physicians to recommend interventions for their patients.

WHAT DO PATIENTS UNDERSTAND ABOUT REGULAR AND "LOW-CARB" FOOD LABELS? THE ROLE OF LITERACY AND NUMERACY. R.L. Rothman¹; R. Housam²; H. Weis¹; D. Davis¹; R. Gregory¹; T.A. Elasy¹. ¹Vanderbilt University, Nashville, TN; ²University of Maryland at Baltimore, Baltimore, MD. (Tracking ID #132811)

BACKGROUND: Comprehension of food labels is important for patients to follow proper dietary recommendations. The FDA has recently discussed altering food labels to improve their utilization. However, patient understanding of current food labels has not been adequately evaluated.

METHODS: We performed a cross-sectional study of patients attending an academic primary care clinic. We developed and administered a 24-item survey to evaluate patient comprehension of food labels for both regular and "low-carb"

food products purchased from a local grocery store. Literacy was measured with the Rapid Estimate of Adult Literacy in Medicine (REALM), and numeracy with the Wide Range Achievement Test (WRAT3).

RESULTS: To date, 115 patients have completed the study. Mean age was 43 years, 68% were female, 21% were African American, and 74% were privately insured. Over 75% had at least a high school education and 80% had ≥ 9 th grade literacy skills (REALM), but 57% had < 9 th grade numeracy (WRAT3). Over 35% of patients reported a chronic illness (ex. hypertension, diabetes) requiring dietary restrictions, and 28% of patients reported following a specific diet plan. Over 90% of patients reported reading food labels. Most read food labels ≥ 3 times/week and reported that food labels were easy to read. However, on average, patients only answered 72% of the food label questions correctly. Only 37% of patients could calculate the number of carbohydrates in a 20-ounce bottle of soda that had 2.5 servings in the container. Only 23% of patients could determine the number of "net" carbohydrates defined on a "low-carb" food product. Common reasons for incorrect responses included misapplication of the serving size, confusion reading the food label, and incorrect calculations. Higher comprehension of food labels was significantly correlated with higher education ($r=0.44$), income ($r=0.56$), literacy ($r=0.50$), and numeracy ($r=0.69$).

CONCLUSIONS: While patients reported frequently reading food labels, their comprehension was suboptimal. Patient comprehension of food labels was correlated with their literacy and numeracy. Providers need to consider patient's ability to understand food labels when providing dietary recommendations to their patients. The FDA also needs to consider changes to food labels to make them more comprehensible—particularly with regard to serving sizes and "low-carb" products.

AN ASSESSMENT OF EMPIRIC ANTIMICROBIAL THERAPY WITH A BETA-LACTAM AND FLUOROQUINOLONE ON MORTALITY FOR PATIENTS HOSPITALIZED WITH COMMUNITY-ACQUIRED PNEUMONIA. E. Mortensen¹; M. Restrepo²; A. Anzueto²; J. Pugh¹. ¹University of Texas Health Science Center at San Antonio, San Antonio, TX; ²Audie L. Murphy VA Hospital and The University of Texas Health Science Center at San Antonio, San Antonio, TX. (Tracking ID #132803)

BACKGROUND: National clinical practice guidelines have recommended specific empiric antimicrobial regimens for patients with community-acquired pneumonia (CAP). However evidence confirming improved mortality with many of these regimens is lacking. Our aim was to determine the association between the empiric use of a beta-lactam with fluoroquinolone, versus other recommended antimicrobial therapies, and mortality in patients hospitalized with CAP.

METHODS: A retrospective, observational study was conducted at two tertiary teaching hospitals. Eligible subjects were admitted with a diagnosis of, chest x-ray consistent with, and a discharge ICD-9 diagnosis of CAP. Subjects were excluded if they were "comfort measures only," transferred from another acute care hospital, or treated with non-guideline concordant antibiotics. A subject was classified as having severe CAP by either being in pneumonia severity index (PSI) class V or by having been admitted to the intensive care unit within the first 48 hours of admission. A propensity score was used to balance the covariates associated with the use of the beta-lactam/fluoroquinolone combination as compared to other antimicrobial combinations. A multivariable logistic regression model was used to assess the association between 30-day mortality, and the use of a beta-lactam antibiotic with a fluoroquinolone, after adjusting for potential confounders including the propensity score and severity of illness.

RESULTS: Data was abstracted on 625 subjects at the two hospitals. Mortality was 8.2% at 30-days, and 25% of subjects were classified as having severe CAP. In subjects with severe CAP mortality was 32% for subjects who received beta-lactam/fluoroquinolone as compared to 16% who received other guideline-concordant combinations. After adjusting for potential confounders, and the propensity score, the use of a beta-lactam with a fluoroquinolone (odds ratio 2.8, 95% confidence interval 1.5-5.1) was associated with increased mortality.

CONCLUSIONS: Using initial empiric antimicrobial therapy with beta-lactam and a fluoroquinolone was associated with increased short-term mortality for patients with severe CAP. Further research is needed to determine what are appropriate empiric antimicrobial therapies for patients with CAP.

BURNOUT AND INTENT TO LEAVE IN ACADEMIC HOSPITALIST AND NON-HOSPITALIST GENERAL MEDICAL WARD ATTENDINGS. T. Wetterneck¹; M. Linzer¹; R.E. Gangnon¹; A.D. Auerbach²; P. Kaboli³; J.L. Schnipper⁴; V. Arora⁵; D.V. Gonzales⁶; R.M. Wachter⁷; D.O. Meltzer⁸. ¹University of Wisconsin-Madison, Madison, WI; ²University of California, San Francisco, San Francisco, CA; ³University of Iowa, Iowa City, IA; ⁴Brigham and Women's Hospital, Boston, MA; ⁵University of Chicago, Chicago, IL; ⁶University of New Mexico, Albuquerque, NM. (Tracking ID #135224)

BACKGROUND: Hospitalist physicians typically attend on inpatient wards more frequently than non-hospitalist physicians and it has been hypothesized that their increased inpatient exposure and experience are important to their effectiveness. However, the impact of these frequent duties on burnout is not well characterized. This multicenter study aims to assess worklife and factors related to burnout for general medical hospitalist and non-hospitalist ward attendings.

METHODS: General medical ward attendings at 6 academic medical centers were sent an end-of-the-year survey in years 1 and 2 of the study (Summer 2002 and 2003) assessing aspects of attending physician worklife, demographic information and number of weeks and weekends worked as ward attending. Work-home interference, satisfaction, stress and intent to leave were assessed on 5-point Likert scales (1=high interference, 5=low interference; 5=strongly satisfied). The stress score was the mean response of 4 of questions with a lower

score indicating high stress. Burnout was measured on a single item, 5-response question. A response of 3 or higher indicated definite burnout. Hospitalist ward attendings were identified by their institutional designation. Logistic regression controlling for age, sex, number of weeks and weekends worked, job experience, site, and year of response was used to identify factors associated with burnout in years 1, 2 and for the entire sample of physicians surveyed using the first year of response to the survey by the physician.

RESULTS: The attending physician response rate in Year 1 was 63% (172/272), and 19% were Hospitalists. 71% of physicians responded in Year 2 (200/280), with 17% hospitalists. There were 212 unique physician responses for both years with usable values for all data fields. No statistically significant differences were found between hospitalist and non-hospitalist ward attendings in mean work-home interference, career satisfaction, stress, or burnout in Years 1 or 2. In both years, hospitalist attendings were more likely to state the amount of call they performed was excessive, 14% of ward attendings met criteria for burnout in each year. When controlling only for age, sex, hospitalist designation, and number of inpatient weeks and weekends worked, female sex predicted burnout in year 1 ($p = 0.008$), year 2 ($p = 0.002$) and the entire sample ($p = 0.011$) and a greater number of inpatient weeks work predicted burnout in year 1 ($p = 0.014$) with a trend toward significance in the entire sample ($p = 0.085$). Predictors of burnout for the entire sample, also controlling for site and year, included high perceived work-home interference (OR 0.2, $p < 0.001$), low satisfaction (OR 0.1, $p < 0.001$) and high stress (OR 0.13, $p = 0.033$). Ward attendings experiencing burnout were more likely to state that they intended to leave their current practice situation ($p = 0.018$), and academic practice in the next two years ($p < 0.001$), decrease work hours within five years ($p < 0.001$) and leave patient care within five years ($p = 0.02$).

CONCLUSIONS: Burnout is common in medical ward attendings, including hospitalists. Strategies to reduce work-home interference, improve career satisfaction and reduce stress should be explored and implemented to prevent job turnover in academic medicine.

CARDIOPULMONARY RESUSCITATION OUTCOMES: WHEN SHOULD RESUSCITATION BE STOPPED? R.P. Ferguson¹; T. Haddad¹; A. Hinduja¹. ¹Union Memorial Hospital, Baltimore, MD. (Tracking ID #133759)

BACKGROUND: Cardiopulmonary resuscitation (CPR) is successful in up to 50% of inpatient arrests. However, hospital discharge and long-term survival rates following CPR are much lower. Most studies are not recent and many are retrospective chart reviews suffering from flawed methodology. The objective of this study was to determine under what conditions should CPR be continued. **METHODS:** Our prospective two-year study from October 1, 2001 to September 30, 2003, identified all inpatient CPR efforts on the Union Memorial inpatient medicine service. ER, cath lab, and surgical patients were excluded. CPR was defined as the utilization of chest compression and/or defibrillation for the loss of cardiac function. Reports of CPR were reviewed within 48 hours, and events were confirmed, when necessary, by interviewing resident participants. Variables analyzed included demographics, initial rhythms, monitoring, month of the year, length of CPR, and survival outcomes: immediate, discharge, and six-months post-CPR. Patients discharged were followed up by phone or through primary physicians.

RESULTS: There were 131 events that met the definition of CPR. The average age of patients was 65. 71 survived initially (54.2%) and 16 were discharged (12.1%). There are 8 who survived at least 6 months (6.1%) and are living at home and doing well. Initial rhythms of ventricular original ($p = 0.006$) were a positive predictor for discharge and 6 months survival. The presence of monitoring at the time of the arrest also appeared to be a positive predictor. Only one out of thirty-three individuals who received multiple CPRs survived to discharge. Similarly, no individuals who received more than 15 minutes of CPR ($p < 0.001$) were discharged. Although the numbers were insufficient to be statistically significant, it is noteworthy that no CPR efforts in July of either year lead to discharge.

CONCLUSIONS: Our survival percentages at all three endpoints are similar to prior studies, indicating a lack of improvement in CPR outcomes over the last 40 years. Monitoring, ventricular arrhythmias, and short codes (less than 15 minutes) are much more likely to end favorably. This study supports the belief that continuing CPR under certain conditions is generally inadvisable.

EFFECTS OF INPATIENT EXPERIENCE ON OUTCOMES AND COSTS IN A MULTICENTER TRIAL OF ACADEMIC HOSPITALISTS. D.O. Meltzer¹; V. Arora¹; J.X. Zhang¹; A.D. Auerbach²; J.L. Schnipper³; T. Wetterneck⁴; P. Kaboli⁵; D.V. Gonzales⁶; W. Levinson⁷; R.M. Wachter⁸. ¹University of Chicago, Chicago, IL; ²University of California, San Francisco, San Francisco, CA; ³Brigham and Women's Hospital, Boston, MA; ⁴University of Wisconsin-Madison, Madison, WI; ⁵University of Iowa, Iowa City, IA; ⁶University of New Mexico, Albuquerque, NM; ⁷University of Toronto, Toronto, Ontario. (Tracking ID #135504)

BACKGROUND: Some studies of hospitalists in academic medical centers have found that hospitalists improve costs or outcomes, but other studies have not found these effects. Most of these studies have nonrandom assignment of patients to hospitalists and none are multicenter studies. Inpatient experience is one factor that has been hypothesized to explain variations in the effectiveness of hospitalists. We studied patients assigned to hospitalist or non-hospitalist physicians based only on day of admission to determine the effects of hospitalists and physician experience on outcomes and costs in 6 academic medical centers.

METHODS: From July 2001 to June 2003, 31,891 general medicine inpatients were assigned to hospitalist or non-hospitalist physicians according to a pre-

determined daily call schedule. Physician experience was measured by age, number of inpatient months in the past 5 years, and accumulated total and disease-specific inpatient general medicine case counts to date during the study period. Patient interviews at admission and 1 month after discharge and administrative data were used to study effects on outcomes and costs.

RESULTS: 12,001 patients were cared for by hospitalists and 19,890 by non-hospitalists. There were no significant differences in age, race, gender, Charlson Index, or distribution of primary diagnosis between the two groups. There were no statistically significant differences in in-hospital mortality, 30-day readmission and emergency room use, 30-day self-reported health status, or patient satisfaction. Mortality data up to 1 year after admission is pending. Average length of stay was 0.05 days shorter for hospitalist patients but this difference was not statistically significant. Costs were also similar between the groups. This occurred despite the finding that length of stay and costs decreased with accumulated total case counts to date during the study period with maximum reductions in resource use of 4% after about 6 months inpatient experience over the study period. These results were not altered by controlling for age and number of inpatient months in the past 5 years. Holding inpatient experience constant, hospitalists had greater length of stay and costs than non-hospitalists, especially for weekend admissions, for which hospitalists were more likely to have coverage by other physicians.

CONCLUSIONS: Academic hospitalists had no statistically significant effects on outcome measures available to date and did not produce the large resource savings suggested by several earlier studies. Hospitalist outcomes improved with experience during the study period and decreased with weekend discontinuities. Continued analysis of these and other factors that may influence the effectiveness of hospitalists is important to maximize the efficacy of hospitalist programs and to predict whether the effectiveness of hospitalists is likely to increase if hospitalists become more experienced as this young profession matures. Given these effects of experience and discontinuities on the effectiveness of hospitalists, evidence that attending physician burnout may increase with greater time on service suggests an important challenge for hospitalist programs in balancing the needs of physicians and patients.

EVALUATION AND TREATMENT OF HIGH CHOLESTEROL DURING HOSPITALIZATION FOR ACUTE CORONARY SYNDROME. A.B. Olomu¹; W.D. Corser¹; Z. Yang²; A. Siddiqi¹; M.M. Holmes - Rovner¹. ¹Michigan State University, East Lansing, MI; ²University of Florida, Gainesville, FL. (Tracking ID #134846)

BACKGROUND: Recent clinical trials to evaluate the effects of modifying blood lipid levels in subjects with cardiovascular disease have demonstrated substantial reduction in risks of recurrent disease and extension of overall survival. Evidence suggests that HMG-CoA reductase inhibitors or "statin" medications administered early to patients with Acute Coronary Syndrome (ACS) may reduce the incidence of recurrent ischemic events. The purpose of this study was to determine the rate at which physicians evaluated and treated hypercholesterolemia in a sample of patients hospitalized for ACS, and identify patient characteristics more often associated with prescription of a statin or other lipid lowering medication.

METHODS: We enrolled and reviewed the charts of 621 patients prospectively identified with ACS in five mid-Michigan community hospitals during the period January 2002 to April 2003. Using data from medical records, patients were followed for the duration of their hospitalizations for the process outcomes of having cholesterol measured and having cholesterol-lowering medication prescribed during hospitalization and at discharge. We used logistic regression procedures to assess which patient characteristics and processes of care predicted cholesterol measurement and/or treatment.

RESULTS: Patients' mean age was 62.16 years (SD 13.01); 63.6% male and 82.0% white. Overall, 549 (88.4%) of these patients had total cholesterol measured, 553 (89.0%) had high-density lipoprotein (HDL) measured and 546 (87.9%) had low-density lipoprotein (LDL) calculated. Of those 457 patients with values measured, 143 (31.3%) had a total cholesterol >200 mg/dl, 316 (69.1%) had an HDL <40 mg/dl, and 302 (66.1%) had an LDL >100 mg/dl. Four hundred and twenty eight (68.9%) of total patients had a lipid-lowering medication prescribed. A total of 117 (81.8%) of patients with a total cholesterol >200 mg/dl, 234 (77.5%) with an LDL >100 mg/dl, and 221 (69.9%) with HDL <40 were discharged on a lipid lowering medication. Independent predictors of cholesterol treatment were: a) gender (61.6% females versus 73.3% males, $p = .002$); b) invasive cardiac procedure during hospitalization ($p < .001$), and c) an LDL >100 mg/dl ($p = .002$). Seventy-three (11.8%) of patients who received a lipid lowering medication before or during hospitalization, were not discharged home on any lipid lowering agent.

CONCLUSIONS: Despite widely published guidelines, many patients hospitalized for ACS may neither be screened, nor prescribed, lipid-lowering agents to lower the risk of subsequent acute coronary episodes. Women in our study were significantly less likely to receive these medication orders. Completion of a cardiac diagnostic procedure and the identification of an elevated LDL were more likely to result in the ordering of lipid lowering medications. Substantial opportunity to prevent recurrent acute myocardial infarctions through increased use of lipid lowering agents is still being missed.

IMPROVED USE OF PROPHYLAXIS FOR VENOUS THROMBOEMBOLISM FOLLOWING AN EDUCATIONAL INTERVENTION. S.L. Cohn¹; A. Adekile¹; V. Mahabir¹. ¹State University of New York Health Science Center at Brooklyn, Brooklyn, NY. (Tracking ID #132667)

BACKGROUND: Evidence-based guidelines on the prevention of venous thromboembolism (VTE) recommend the use of prophylaxis in medical patients

with risk factors for VTE.¹ However, recent studies suggest that implementation of such guidelines in clinical practice is suboptimal. In this study, we assessed the impact of a simple educational intervention on rates of VTE prophylaxis in the medical wards of our hospital.

METHODS: An informal survey indicated prophylaxis rates of approximately 50%, which prompted further study. Formal baseline data, including individual risk factors for VTE, contraindications to anticoagulation, and the use and appropriateness of prophylaxis, were collected retrospectively from the medical records of medically ill patients during October 2002. Patient risks of VTE (low, medium, or high) and appropriateness of prophylaxis were classified according to the American College of Chest Physicians (ACCP) guidelines.¹ Concurrently, an educational intervention was initiated which consisted of orientation of new medical housestaff to VTE risk factors by the chief resident; in-hospital education of nurses by the nurse educator; distribution of VTE-risk-factor pocket cards; VTE-risk-factor posters in nursing stations and off-duty rooms; and monthly audits of prophylaxis type and appropriateness by the Division Chief of General Internal Medicine. Three-month data were subsequently collected from 12 and 18 months following the intervention.

RESULTS: In total, 312 patients were included. The average age was 60.9 years and the most common VTE risk factors were age >40 years (84%, 262/312 patients) and non-ambulatory status (85.2%, 231/271 patients). Prophylaxis (mechanical or pharmacological) was prescribed to 46.9% of baseline patients, with only 42.8% receiving appropriate prophylaxis. A significantly higher number of patients received prophylaxis following the educational intervention at both 12 and 18 months compared with baseline (87.1% in both groups, $p < 0.001$). The use of suitable prophylaxis also improved over time ($p < 0.001$), with 68.1% and 85.0% of patients receiving appropriate prophylaxis at 12 and 18 months, respectively. Unfractionated heparin was the most frequently used pharmacological prophylaxis, followed by low-molecular-weight heparin.

CONCLUSIONS: Simple educational intervention can result in a marked increase in the proportion of medical patients receiving prophylaxis, as well as the proportion receiving appropriate prophylaxis, as recommended by evidence-based guidelines. ¹Geerts WH, et al. Chest 2001;119(3 Suppl):132S-175S. (updated in 2004)

INPATIENT NURSING BURNOUT: CAN PHYSICIANS HELP? T.B. Wetterneck¹; M. Linzer¹; F.A. Sobande¹; P. Carayon¹. ¹University of Wisconsin-Madison, Madison, WI. (Tracking ID #135156)

BACKGROUND: Workforce shortages are plaguing the nursing profession and hospitals, in particular, have felt the brunt of this shortage. Understanding nursing burnout, which has been associated with job turnover, is crucial.

METHODS: Inpatient nurses were surveyed as part of a study to evaluate the implementation of new technology at an academic medical center. Demographic information, job challenge, role ambiguity, uncertainty, workload, aspects of job control (overall, task, decision and resource control), satisfaction with care provided, time pressure and burnout were assessed. Burnout was measured on a single item, 5-response question. A response of 3 or higher indicated definite burnout. Multiple linear regression controlling for age, gender, level of education and job experience was used to model predictors of nursing burnout, satisfaction with care provided and time pressure.

RESULTS: The response rate was 32% (192/601 surveyed). 91% of nurses were women, 69% were under the age of 45 and the average length of time working for the institution was 7.4 years. Nurses reported high levels of workload, medium-high levels of role ambiguity, high challenge, medium job control and high levels of job uncertainty. 25% of nurses reported symptoms of burnout. Predictors of burnout included age, increasing levels of role ambiguity (beta-coefficient = .24, $p < .001$) and decreasing job control (beta-coefficient = -.21, $p < .01$; $R^2 = 23\%$; $p < .001$). Challenge and uncertainty were not associated with burnout. Predictors of satisfaction with care provided included lower levels of role ambiguity (beta-coefficient = -.22, $p < .01$), low uncertainty (beta-coefficient = -.19, $p < .05$) and high challenge (beta-coefficient = .20, $p < .05$; $R^2 = 17\%$; $p < .001$). Predictors of time pressure included role ambiguity, uncertainty and job control ($R^2 = 28\%$; $p < .001$). Decreasing levels of role ambiguity (beta-coefficient = .30, $p < .001$) and uncertainty (beta-coefficient = -.21, $p < .01$) and increasing job control (beta-coefficient = .26, $p < .001$) predicted more perceived time available to complete tasks safely.

CONCLUSIONS: Burnout is common amongst inpatient nurses. Role ambiguity and work control feature prominently as independent and strong predictors of nursing burnout. Physicians may help decrease burnout in inpatient nurses by clarifying roles and respecting a nurse's need for autonomy and control in the work environment. Further research should be directed at understanding the physician's role in nursing burnout.

MULTI-SITE EVALUATION OF THE PROCESS AND OUTCOMES OF CARE FOR UPPER GASTROINTESTINAL HEMORRHAGE. P. Kaboli¹; A.D. Auerbach²; J.L. Schnipper³; T. Wetterneck⁴; D.V. Gonzales⁵; D.O. Meltzer⁶. ¹University of Iowa, Iowa City, IA; ²University of California, San Francisco, San Francisco, CA; ³Brigham and Women's Hospital, Boston, MA; ⁴University of Wisconsin-Madison, Madison, WI; ⁵University of New Mexico, Albuquerque, NM; ⁶University of Chicago, Chicago, IL. (Tracking ID #135068)

BACKGROUND: Non-variceal upper gastrointestinal hemorrhage (UGIH) is a common reason for hospital admission and has known process of care measures that are associated with improved outcomes. We evaluated variations in the process and outcomes of care across multiple study sites to identify factors associated with higher quality care.

METHODS: The analyses included 420 consecutive UGIB patients admitted to 6 academic medical centers with one of four diagnoses: peptic ulcer disease (PUD), erosive disease (ED), angiodysplasia, or Mallory-Weiss tear (MWT). The primary measures evaluated were resource utilization, as defined by cost and hospital length of stay (LOS), complications, defined by need for emergency surgery, recurrent bleeding, transfer to intensive care unit, or repeat endoscopy (EGD), and outcomes, defined as inpatient mortality and 30-day re-admission rate. The process of care measures were EGD within 1 day of admission and use of a proton-pump inhibitor (PPI). EGD confirmed all diagnoses. Data were obtained from chart abstraction and administrative data files.

RESULTS: Mean age of subjects was 59.5 years; 57% were male, 45% white, and 22% black. Based on a validated UGIB re-bleeding risk score, 41.4% were low-risk for re-bleeding. Mean LOS was 5.2 days, total hospital costs were \$11,980, in-hospital mortality was 1.43%, and 30-day readmission rate was 5.5%. UGIB etiologies were ED 53%, PUD 38%, angiodysplasia 5.5%, and MWT 3.5%. There were considerable differences in LOS by study site with a range of 3.6 to 6.9 days ($P < .001$) and hospital costs of \$7,849 to \$15,227 ($P < .001$). The most significant predictor of reduced LOS and costs was EGD within 1 day of admission. If EGD was performed within 1 day of admission (56% of patients), hospital LOS was 4.06 days compared to 6.74 days ($P < .001$) and was also seen when stratified by low re-bleed risk (3.21 vs 4.49 days; $P < .001$) and intermediate/high re-bleed risk (4.67 vs. 8.32 days; $P < .001$). EGD in 1 day and PPI use were not associated with complications, 30-day re-admission, or re-bleed risk. However, having an EGD within 1 day varied by site (9.3% to 29.2%; $P < .01$) as did use of a PPI (6.6% vs. 89.4%; $P < .001$). There were no differences in 30-day readmission rate or mortality by study site, in spite of considerable differences in LOS, time to EGD, or PPI use. Neither weekend admission nor care by a hospitalist was significantly associated with any of the outcomes of interest. Discharge within 1 day of EGD did not lead to increased 30-day re-admissions, regardless of re-bleed risk.

CONCLUSIONS: There is considerable variation in the quality of care for UGIB, as measured by various process and outcome measures. However, early endoscopy and early discharge for low-risk patients remains one of the most critical factors leading to reductions in resource utilization with similar clinical outcomes. Healthcare systems should continue to work to develop innovative strategies to maximize the availability of endoscopy services for UGIB.

PATIENT SATISFACTION IN A HOSPITALIST MODEL OF CARE. W.D. Rifkin¹. ¹Yale University School of Medicine, New Haven, CT. (Tracking ID #132862)

BACKGROUND: Previous examinations of satisfaction within a hospitalist model of care have focused upon medical students, housestaff and non-hospitalist physicians. No previous examination has directly compared hospitalized patient satisfaction by whether they were seen in a hospitalist or non-hospitalist model of care. It has been suggested that patients may be less satisfied within a hospitalist model due to interruption in their usual doctor-patient relationships. This analysis compares patient satisfaction for hospitalist patients and all discharges from the Department of Medicine.

METHODS: During the first quarter of 2004 all patients cared for by 8 full-time hospitalists and 124 community-based physicians were asked to complete patient satisfaction surveys. Hospitalist patients were given a paper version of the survey to be mailed in from home, while Press Ganey, an independent survey enterprise, contacted all discharged patients via phone. Both surveys had identical questions pertaining to physician satisfaction. Only the five physician domain questions were included in this analysis. These include time spent, concern, kept informed, friendliness and skill. Hospitalist patients were also included in the hospital-wide survey. Response rates were 40% for hospitalist survey and 25% for the hospital-wide survey. Mean responses (scale 0-100) for each question and for all five combined were compared using the Wilcoxon two sample test. Press Ganey surveys from a floor which admitted only Department of Medicine patients were used to ensure only intra-departmental comparisons.

RESULTS: 57 completed surveys were returned by hospitalist patients. 52 patients (hospitalist and non-hospitalist) responded to the Press Ganey calls. Despite the cross-over design, for all five questions individually, mean satisfaction trended higher for hospitalist patients, reaching statistical significance for question 1 (time spent 86.0 vs. 79.8, $p = .05$) and question 3 (kept informed, 87.3 vs. 81.3, $p < .03$). When all five questions were examined collectively as an overall physician satisfaction rating, patients of hospitalists were more satisfied (89.0 vs. 83.0, $p < .0003$).

CONCLUSIONS: Satisfaction rates for hospitalist patients were higher in two of five individual domains and overall than they were for all medical inpatients. These findings were surprising in that the study design should have made it more difficult to demonstrate differences between groups, due to potential cross-over. It is consistent with suggested strengths of the hospitalist model that the domains of time spent and kept informed were most strongly positive for the hospitalists. Response bias may skew the results overall, but would be unlikely to favor one group over the other. While satisfaction surveys can suffer from ceiling effects, the magnitude of the difference between groups represents a large percentile variance in satisfaction.

PATTERNS OF STRESS ULCER PROPHYLAXIS IN NON-CRITICALLY ILL MEDICAL PATIENTS. K.O. Hwang¹; K. Sanja¹; D.R. Blumberg¹; R. Griffith¹. ¹Morristown Memorial Hospital, Atlantic Health System, UMDNJ-NJMS, Morristown, NJ. (Tracking ID #134111)

BACKGROUND: Previous studies have shown that 51 to 68% of prescriptions for acid-suppressive medications in hospitalized patients are unjustified. We

hypothesized that many non-critically ill medical patients in our hospital receive prophylactic agents to prevent stress ulceration of the gastrointestinal (GI) tract. The purpose of this study was to observe the patterns of stress ulcer prophylaxis (SUP) in non-critically ill medical patients.

METHODS: The study was a retrospective analysis of adults admitted to Morristown Memorial Hospital from August 2003 to July 2004. All patients were on the medical wards (not in the intensive care unit) and were under the care of medical residents and teaching faculty. Patients with GI bleeding at time of admission or hospitalization 2 days were excluded. The records were reviewed for risk factors for stress ulceration, use of SUP, and incidence of stress ulcer bleeding. Chi square analysis was used, with Yates corrections for 2 by 2 contingency tables.

RESULTS: The study included 545 patients with average age of 56 years and average length of hospitalization of 5.9 days. Of the 545 patients, 27% had no risk factors for stress ulceration, 39% had 1 risk factor, 26% had 2 risk factors, and 9% had 3 or more risk factors. 297 of 545 (54%) patients received SUP. Although use of SUP was directly related to number of risk factors ($P < 0.0001$), 37% of the patients without any risk factors still received SUP (Table). Furthermore, patients with zero or only 1 risk factor comprised 56% of all patients who received SUP. The most common risk factors were nonsteroidal anti-inflammatory drugs or aspirin (NSAIDs/ASA), anticoagulation, and systemic corticosteroids. Patients were more likely to receive SUP if they had the risk factors of NSAIDs/ASA ($P = 0.0002$), anticoagulation ($P = 0.0043$), hepatic failure ($P = 0.0032$), prior GI bleed ($P = 0.032$), or prior peptic ulcer disease/gastritis ($P < 0.0001$). The remaining risk factors that we analyzed (corticosteroids, COPD, sepsis, and renal failure) were not associated with SUP use. Although there were 5(1%) cases of GI bleeding, none were due to stress ulceration.

CONCLUSIONS: In non-critically ill medical patients, prescription of SUP was directly related to the number of risk factors. But SUP was also given to many low-risk patients. The incidence of stress ulcer bleeding was too low to be detected in this study. Randomized, prospective trials are needed to clarify the benefit of SUP in this population.

Table: Percent of patients who received SUP according to number of risk factors ($P < 0.0001$ for the distribution)

Number of risk factors	+SUP
0	37% (55/148)
1	52% (110/210)
2	70% (97/139)
3 or more	73% (35/48)

PREDICTING HOSPITAL READMISSION FOLLOWING GENERAL MEDICINE ADMISSION. A.M. Arozullah¹; J. Kim²; S.D. Lee³; R. Soltysik¹; T. Khan¹; S. Kurup¹; P.R. Yarnold⁴.

¹Jesse Brown VA Medical Center, University of Illinois at Chicago, Chicago, IL; ²University of Illinois at Chicago, Chicago, IL; ³University of North Carolina at Chapel Hill, Chapel Hill, NC; ⁴Northwestern University, Chicago, IL. (Tracking ID #135100)

BACKGROUND: Low health literacy and poor social support are associated with increased hospitalizations, but it remains unclear whether these factors are useful in predicting hospital readmission of individual patients. This prospective study determined predictors of readmission in the six months following General Medicine admission.

METHODS: Patients admitted to the General Medicine service between 8/1/01-4/1/03 at a University-affiliated VA Hospital were eligible. Patients with dementia, blindness, deafness, or VA care for <6 months were excluded. Patients were interviewed during initial hospitalization to assess health literacy, health status (SF-12), health habits (Health-Promoting Lifestyle Profile), medication compliance, social support (MOS Social Support Questionnaire), and prior healthcare utilization. Patients were followed prospectively for six months. A physician reviewer, blinded to questionnaire results, recorded VA hospital readmissions and all VA follow-up visits occurring within six months post-discharge using electronic medical records. Follow-up visits were classified as General Medicine versus Subspecialty. A non-linear multivariate model, developed using hierarchically optimal classification tree analysis, was used to identify predictors of hospital readmission.

RESULTS: There were 400 patients enrolled in the study, of whom 141 patients (35%) had hospital readmissions within six months post-discharge. Neither low health literacy nor social support were significant predictors of readmission. Significant predictors of readmission included using only VA healthcare services ($p < 0.035$), lower self-reported nutrition habits ($p < 0.013$), lack of General Medicine follow-up visits ($p < 0.002$), and greater than 3 outpatient visits in the year prior to initial admission ($p < 0.035$). Three-fourths (73%) of patients used only VA healthcare services, 74% reported poor nutrition habits, 28% lacked General Medicine follow-up visits, and 55% had greater than 3 outpatient visits in the prior year. The Table below displays patient subgroups (strata) classified by these predictors.

CONCLUSIONS: Assessing previous healthcare utilization and nutrition habits prior to discharge may detect patients who are at high risk for hospital readmission. Our results indicate that General Medicine follow-up visits were associated with lower hospital readmission rates, whereas subspecialty follow-up visits were not associated with readmissions.

N (%)	Using Only VA Care	Nutrition (4-24)	Post-Discharge GMC Visit	Prior Outpatient Visits in Past Year	Readmission Rate
60 (15%)	Yes	≤ 16	No	-----	61.7%
29 (6%)	No	-----	No	-----	48.3%
23 (6%)	Yes	> 16	No	-----	47.8%
88 (22%)	Yes	≤ 16	Yes	> 3 visits	46.6%
59 (15%)	Yes	≤ 16	Yes	≤ 3 visits	23.7%
79 (20%)	No	-----	Yes	-----	19.0%
60 (15%)	Yes	> 16	Yes	-----	15.0%

RESUSCITATION IN THE HOSPITAL: CIRCADIAN VARIATION OF CARDIOPULMONARY ARREST. J.L. Jones-Crawford¹; D.C. Parish¹; B. Smith¹; F. Dane². ¹Mercer University, Macon, GA; ²Saginaw Valley State University, Saginaw, MI. (Tracking ID #132845)

BACKGROUND: Background and Objectives. Although over 25 reports have found that outpatient frequency of sudden cardiac death peaks between 6 am and 12 noon, only a few studies, with inconsistent results, have examined potential circadian variation of death events in hospitalized patients. We address this knowledge gap.

METHODS: Methods. Retrospective analysis was completed for all admissions to a single acute care hospital over a 14-year period (1987-2000), during which a resuscitation was attempted. Simple Chi-square analysis for goodness of fit examined cardiopulmonary arrest by one-hour, two-hour, four-hour, and eight-hour time intervals. We analyzed frequency of events overall and by gender, age group, year of study, rhythm, cardiac disease versus other diagnoses, witnessed versus unwitnessed event, and survival to discharge.

RESULTS: Results. For the 4,692 admissions in which one or more resuscitations were attempted (only the first attempt is reported), we found circadian variation when assessing data by one and two hour intervals, with more events in the morning. Significant circadian variation was also found for organized rhythms and asystole. The frequency of asystole peaked between 3 am and 7 am, and survival to discharge was lowest during this time: 28.1% vs. maximum survival of 34% between 7 am and 11 am. In simple logistic regression, both time of day and rhythm were significantly related to survival to discharge, but after multivariable adjustment, time became insignificant, while rhythm remained significant ($p < 0.0001$).



Goodness of fit for frequency of events in hourly time periods

CONCLUSIONS: There is circadian variation of cardiopulmonary arrest in a hospital but patterns differ from population and outpatient settings. Late night variations in rhythm and survival suggest a delay between onset of clinical death and discovery, which may contribute to negative outcomes.

RISK ASSESSMENT MODEL FOR VENOTHROMBOEMBOLISM IN MEDICALLY TREATED POST-HOSPITALIZED PATIENTS. S.H. Yale¹; S. Medlin²; H. Liang¹; P. Tonia¹; J.J. Mazza¹. ¹Marshfield Clinic Research Foundation, Marshfield, WI; ²Marshfield Clinic, Marshfield, WI. (Tracking ID #135178)

BACKGROUND: Venothromboembolic (VTE) is an important condition in hospitalized medical patient extending into the post-hospital period. Risk factors in medically treated patients that contribute to deep venous thrombosis (DVT) subsequent to hospitalization have not been well described but represent a broad spectrum of conditions. The primary objective of this study was to identify characteristics among medically treated hospitalized patients who develop a VTE within 60 days after discharge and to use these factor(s) and/or categories of factors to create a risk assessment profile that can be used to identify patients who are at increased risk to develop a VTE after hospitalization.

METHODS: A retrospective study design was performed on the computerized medical records and charts of 380 patients (190 with VTE and 190 without a VTE) who received care at the Marshfield Clinic, St. Joseph's hospital or regional centers from 1995-2002 and were subsequently re-hospitalized within 60 days. Factors were identified in each of the following categories (1) current medical conditions, (2) preexisting medical conditions, (3) medical interventions, (4) drugs, and (5) physical factors. Univariate analyses and multivariate logistic regression were performed to identify the significant risk factors in association with having a VTE.

RESULTS: Multivariate logistic regression model with percent concordant = 80.5 and c-statistic = 0.820 revealed that patients with cancer receiving chemotherapy, varicose veins, non-insulin dependent diabetes mellitus, medications (oral contraceptives, tamoxifen, raloxifene, thalidomide, amphotericin B, clomiphene, cyclosporine, potassium chloride, and corticosteroids), nursing home residents, and age over 60 had a significantly positive effect on having a VTE, while angina had a significantly negative effect on having a VTE. A novel risk assessment model was created using risk factors from the multivariate regression model and two risks (paralysis and dexamethasone) from the univariate analyses. Using the new risk assessment model, the prevalence of VTE was found to be 91.4% (53/58) in the high-risk category, 54.5% (116/213) in the moderate-risk category, and 19.4% (21/108) in the low-risk category.

CONCLUSIONS: We developed a risk assessment profile that can be used to identify patients who may be at increased risk to develop VTE after hospitalization. Availability of risk factor assessment to guide anticoagulation therapy decisions will have important ramifications in terms of outcomes with respect to both cost and morbidity reduction.

THE PROGNOSTIC SIGNIFICANCE OF THE DIFFERENTIAL COUNT IN HOSPITALIZED PATIENTS. A.N. Kho¹; S.L. Hui¹; C.J. McDonald¹. ¹Regenstrief Institute, Inc. and Indiana University School of Medicine, Indianapolis, IN. (Tracking ID #133544)

BACKGROUND: The complete blood count and differential is one of the most commonly ordered tests in medicine, but information on the prognostic significance of the manual differential count is lacking. Our **objective** was to uncover independent predictors of high mortality within the automated and manual differential count. Determining clear markers of increased mortality from this commonly ordered test can guide the clinician to subsequent care and diagnostic studies.

METHODS: We conducted a retrospective cohort study. Using a comprehensive electronic medical record system, we extracted clinical and laboratory data on 115,278 adult admissions to an urban primary care hospital from 1993-2003. We defined the primary endpoint as death occurring within 30 days of the admission date as determined by the hospital records and the Indiana State death tapes. Excluding prisoners and pregnant patients, we analyzed 35,692 unique patients with an automated differential count at admission, of whom 1,442 died (4.0%). We entered age, gender, hematocrit, and all elements of the differential (automated and manual) into a multiple logistic regression model. Age and hematocrit were entered as continuous variables; all other variables were entered as either present (1) or absent (0). Using backwards elimination and a stepwise forward method to select variables with a P value < 0.005 yielded a similar model. The author manually reviewed a random sample of patient charts to confirm the clinical conditions associated with the presence of these cell types. **RESULTS:** Presence of nucleated RBCs (NRBCs), burr cells, or absolute lymphocytosis was associated with a three-fold increased risk of death within 30 days of admission (Table 1). Controlling for presence of comorbidities as measured by the Charlson Index yielded similar odds ratios. NRBCs were associated with a higher mortality rate across a wide range of diagnoses (average mortality rate = 25.5%), except in patients with sickle cell disease and leukemia. In a few cases, presence of NRBCs was an unrecognized marker of severe illness, and the patients returned shortly after discharge *in extremis*. Burr cells were primarily found in patients with acute renal failure, significant liver disease, or heart failure and their presence correlated with an average 27.3% mortality rate. Absolute lymphocytosis primarily appeared under significant physiologic stress and correlated with a higher mortality rate in trauma patients (9.9%) and patients with significant CNS pathology (17.9%).

Parameter	Number	Odds Ratio	95% CI	P value
Intercept				< .0001
Age (Continuous, Years)	35688	1.042	1.039-1.045	< .0001
Gender (M = 1)	19788	2.06	1.83-2.318	< .0001
WBC > 12,000/ μ L	11124	1.683	1.493-1.897	< .0001
Hematocrit (Continuous, %)	35691	0.969	0.961-0.976	< .0001
Lymphocytosis (ALC > 4000/ μ L)	3046	2.914	2.494-3.404	< .0001
Macrocytes	3475	1.615	1.392-1.874	< .0001
Dohle Bodies	524	1.744	1.307-2.327	0.0002
Sickle Cells	62	0.043	0.006-0.32	0.0022
Burr Cells	253	3.365	2.428-4.663	< .0001
Vacuolated Cells	897	1.524	1.19-1.951	0.0008
Band Forms	7594	1.704	1.496-1.942	< .0001
Nucleated RBCs	467	3.317	2.564-4.291	< .0001

Table 1. Multiple logistic regression model with a P value < 0.005 with death within 30 days of admission as the outcome (CI = confidence interval)

Table 1. Burnout, Depression, and At-Risk Alcohol Usage by Year in School

	Total	1st Year	2nd Year	3rd Year	4th Year	Trend P Value
Burnout						
Emotional Exhaustion						
% High (N)	34.7 (185)	29 (52)	51 (59)	28 (23)	33 (51)	0.9571
Depersonalization						
% High (N)	25.8 (137)	18 (32)	27 (31)	29 (24)	33 (50)	0.0024
Personal Accomplishment						
% Low (N)	42 (224)	35.2 (63)	38.8 (45)	42.2 (35)	52.6 (81)	0.009
Burnout						
% (N)	45 (239)	37 (65)	53 (61)	43 (35)	51 (78)	0.0299
Depression						
% Screen positive (N)	56 (296)	56.4 (101)	69.9 (80)	51.8 (43)	47 (71)	0.0255
At Risk Alcohol Usage						
% (N)	48 (264)	62.3 (109)	50.4 (58)	35.4 (29)	44.4 (68)	0.0003

CONCLUSIONS: Prior work has suggested an association between some elements of the manual differential count and high mortality. This work confirms that NRBCs, burr cells, and absolute lymphocytosis are strong independent predictors of 30 day mortality in hospitalized patients. Understanding the "red flags" within this common test can alert the clinician to patients at high risk of death.

"OH! I HAVE CHEST PAIN.": MEASURING RESIDENT COMPETENCE IN ACUTE CARE. S. Zabar¹; K. Hanley¹; A.L. Kalet¹; E.K. Kachur²; D.L. Stevens¹; E. Pearlman¹; M.D. Schwartz¹; K. Felix¹; M. Lipkin¹. ¹New York University, New York, NY; ²Medical Education Development, New York, NY. (Tracking ID #134105)

BACKGROUND: Residents must master the complex skill of working with other professionals to deliver care to acutely ill patients. We developed a performance-based assessment (PBA) to measure their interdisciplinary skills and competence in managing an acute medical emergency (aortic dissection or right sided myocardial infarction).

METHODS: We developed, piloted, and implemented a standardized patient (SP) case as part of an annual 10-station objective structured clinical examination (OSCE). The 10 minute station presented a 61 y.o. woman (SP) in the ER with severe chest pain. The residents' task was to assess and treat the patient. A standardized nurse (SN) gave medication and provided lab results if requested. The SN reported improvement or worsening of clinical condition based on the resident's actions, revealed x-ray and EKG results if ordered and responded to the leadership style of the resident. The SP was trained to prioritize requesting pain relief. The SP and SN were trained to evaluate residents' overall performance on a 4-point scale. Trained faculty observed residents, provided verbal feedback, and completed a 16-item behavioral checklist covering 5 domains (data gathering, rapport building, patient education, synthesis and knowledge (on a 1-4 scale) and four global assessments (overall performance, communication skills, knowledge, and professionalism on a 1-9 scale, 9 = highest). All participants completed a post-OSCE satisfaction survey.

RESULTS: 66 medical residents (8 PGY1, 50 PGY2, 8 PGY3) and 4 faculty participated in the case. Mean faculty ratings were: Overall Performance 6.5, Communication 6.6, Knowledge 6.3, and Professionalism 7.7 (ranges of 2 to 9). SP and SN satisfaction scores agreed the strongest (Spearman correlations $r = 0.491$, $p < .01$). SN satisfaction and faculty overall performance, communication and knowledge correlates consistently ($r = 0.434$, $r = 0.455$, $r = 0.338$ $p < .01$). 65% of residents reported at least some prior experience with a similar case, 76% rated the difficulty as just right, 72% reported it gave valuable feedback and 91% stated the educational value was high. This case's educational value was rated 20% higher than the 9 other cases. Common mistakes made by residents included: ignoring the patient and their pain, using jargon, and not collaborating with the nurse.

CONCLUSIONS: A complex PBA case provided meaningful feedback about core knowledge, communication, and leadership skills and higher educational value than simpler cases. While the significant agreement among multiple raters suggests reliability, there is likely to be unique information from the patient, nurse and faculty representing different expectations and views. PBA is a valuable formative assessment tool for residents' interdisciplinary skills.

A MULTICENTER STUDY OF DEPRESSION, ALCOHOL USAGE, AND BURNOUT IN AMERICAN MEDICAL STUDENTS. L.N. Dyrbye¹; M.R. Thomas¹; J.L. Huntington¹; K. Lawson²; P. Novotny¹; J. Sloan¹; T. Shanafelt¹. ¹Mayo Clinic, Rochester, MN; ²University of Minnesota, Minneapolis, MN. (Tracking ID #132900)

BACKGROUND: Practicing physicians and residents have been identified as being at risk for depression, problematic alcohol usage, and professional burnout including emotional exhaustion, depersonalization, and low personal accomplishment. These domains of burnout are markers for professional distress and may adversely impact professional development. Although extensive data on the prevalence of these symptoms exists in residents and practicing physicians, the prevalence has not been well studied among medical students.

METHODS: All 1,090 medical students in the state of Minnesota were surveyed electronically in April of 2004, after IRB approval. Participation was elective and all responses were anonymous. Medical students in Minnesota attend one of

three schools: a private medical school, a traditional public university, or a public university with a focus in primary care. The 2-item Primary Care Evaluation of Mental Disorders (PRIME MD) was used to screen for symptoms of depression. Burnout was measured using the Maslach Burnout Inventory (MBI), which assesses three domains of burnout: emotional exhaustion, depersonalization, and personal accomplishment. According to convention burnout was defined as a high score on the depersonalization or emotional exhaustion subscale. Items from the Alcohol Use Disorders Identification Test (AUDIT) measured at risk alcohol use and dependence. Simple binomial proportions and associated confidence intervals were used for the presence/absence of burnout dimensions. A sample of 546 individuals provides accuracy of these proportions to within three percentage points with 95% confidence. Logistic regression was used to examine the incidence of burnout over time.

RESULTS: Of the 1088 students whose e-mail could be confirmed 545 (50%) completed the survey. The demographic characteristics of the responders were consistent with the overall population demographics. Forty-five percent of the students met criteria for burnout with 35% having high emotional exhaustion, 26% having high depersonalization, and 42% having a low sense of personal accomplishment. Fifty-six percent of the students screened positive for depression and 48% reported at risk alcohol usage. The prevalence of a positive depression screen ($P = 0.0255$) and at risk alcohol usage ($P = 0.003$) decreased over the 4 years of training. In contrast, each additional year in training was accompanied by an increase in burnout with a rise in both depersonalization ($P = 0.0024$) and a decrease in personal accomplishment ($P = 0.009$). (Table 1). **CONCLUSIONS:** Burnout was common (45%) in medical students in this multicenter study and increased for each year in training. In contrast to this increase in professional distress, personal distress (depression and alcohol use) decreased by year in training. These results suggest the experience of distress varies at different points in the medical school experience with professional distress the dominant aspect of distress by the time of graduation. This finding supports the hypothesis that burnout begins in medical school. How burnout impacts the well-being, professionalism, and competency of medical students remains to be studied. Medical schools need to recognize burnout among their students, understand the potentially profound adverse impact of burnout on professional development, and implement programs to prevent and deal with student burnout.

A MULTI-PROGRAM EVALUATION OF READING HABITS OF PRIMARY CARE INTERNAL MEDICINE RESIDENTS WHILE ON AMBULATORY CARE ROTATIONS. C. Lai¹; E.M. Aagaard¹; S.L. Brandenburg²; M. Nadkarni³; H. Wei⁴; R.B. Baron¹. ¹University of California, San Francisco, San Francisco, CA; ²University of Colorado Health Sciences Center, Denver, CO; ³University of Virginia, Charlottesville, VA; ⁴Cornell University Medical College, New York, NY. (Tracking ID #134863)

BACKGROUND: Little is known regarding residents' educational reading habits during ambulatory rotations. We conducted a multi-program cross-sectional study of primary care internal medicine residents to determine (1) reading time during an ambulatory care rotation; (2) the most commonly used resources; and (3) motivating and inhibiting factors to reading during ambulatory block rotations.

METHODS: Eight Primary Care Programs distributed the anonymous survey to their 2nd and 3rd year residents during an ambulatory care rotation. Participants were asked Likert-scaled and multiple-choice questions relating to their ambulatory reading habits: (1) time spent reading in the past week, (2) sources commonly used, (3) factors that motivated or inhibited reading; and (4) curricular methods to stimulate resident reading. Descriptive statistics were performed to assess the factors that most motivate and inhibit resident reading habits.

RESULTS: The response rate was 70% (87/124). The respondents were 70% women with 68% reporting plans to pursue primary care careers after residency. Respondents reported reading medical topics for a mean of 4.2 hours per week ($SD = 3.0$), with about two-thirds of this time devoted to ambulatory topics. However, they spent more time (4.7 hours, $SD = 4$) on non-medical reading. Seventy percent of residents identified online-only sources as their most important reading resource for ambulatory education, with 97% of residents reporting regular use of Uptodate.com and 35% of Google.com or other search engines. Based on a 5-point Likert response scale (1 = never, 5 = very frequent), the most frequently utilized ambulatory reading sources were online-only sources

(4.16, SD = 0.88) and journals (3.43, SD = 0.93). Residents were most motivated to read by patient cases (4.38, SD = 0.65) and presentation assignments (4.09, SD = 0.89), and were most inhibited by family responsibilities (4.00, SD = 0.94), lack of motivation (3.94, SD = 0.82), and personal responsibilities (3.65, SD = 1.26).

CONCLUSIONS: Primary care internal medicine residents on ambulatory rotations are most motivated to read by patient cases or when they are directly responsible for leading a discussion, but feel inhibited by family/personal responsibilities and lack of motivation. To stimulate and reinforce resident reading, residency programs should consider (1) fostering patient-based and case-based learning; (2) assigning resident-led discussions and presentations; and (3) providing workshops on time management and work-life balance.

A NEEDS ASSESSMENT FOR DOMESTIC VIOLENCE EDUCATION: FIRST-YEAR MEDICAL STUDENTS' EMOTIONAL RESPONSES, ABILITY TO EMPATHIZE, AND COUNSELING SKILLS FOR VICTIMS OF ABUSE. D.W. Rudy¹; M.J. Lineberry¹; H. Garrett¹; M. Hoffman¹; V. Depp Cline¹; S.A. Haist¹; W. John¹. ¹University of Kentucky, Lexington, KY. (Tracking ID #135875)

BACKGROUND: In spite of the wide recognition of the prevalence and serious nature of domestic violence (DV), physicians often fail to identify the problem or appropriately intervene. Misguided advice may have disastrous consequences. Education in DV is needed throughout the medical curriculum. DV is a highly emotional and complex biopsychosocial problem that requires the physician to recognize their own biases and emotional responses, as well as demonstrate the ability to empathize with victims. To help guide the development of a DV curricula, we conducted a needs assessment of first-year medical students by assessing students' ability to identify the perspectives of a victim of DV, students' emotional responses to the situation, and what they would say to the victim.

METHODS: 95 first-year medical students viewed a 2 minute video monolog of a standardized patient portraying a DV victim. The patient is very distraught and fearful. She does not want her husband reported due her financial dependency and rationalizes the incident through self-blame. Immediately following the video students were asked to provide written responses to their ideas of the patient's perspective, how they would feel in the role of the physician, and what they would say to the patient. Written responses were analyzed for thematic categories by two reviewers in an iterative process. The two reviewers then coded the students' responses for the presence or absence of the themes. Discrepancies were resolved via a third coder (inter-rater reliability 0.93).

RESULTS: Students' feelings were directed at themselves (50% sorrow/sadness, 19% frustration) and the husband (51% anger towards husband) more so than the victim (25% compassion/sympathy). Students also identified their own concerns as worries of the victim (33% stigma of situation, 8% fear of addiction). Some students even admitted negative thoughts toward the victim (10%). Only 38% felt a responsibility to help the patient. The majority of students reported intent to convey concern for the patient (67%), the victims' need to seek counseling and support (50%), the victim's need to get away from her husband (45%), concern for the victim's children (36%), and statistics/information about DV (34%). While 82% of students recognized the victim's belief that she was responsible for her abuse, only 19% assured the victim that abuse was not her fault. Regarding reporting, only 16% of students would inform the patient of their obligation to report the abuse, 3% would report the abuse if there was a future episode and 10% only if the victim requested such.

CONCLUSIONS: The results demonstrate that victims of DV elicit emotions in medical students that may hinder treatment if not recognized. Although a majority students intend to convey concern for the patient, they are unaware of legal obligations to report DV (required in our state) and of the danger associated with advising the victim to leave her husband unprepared. In addition to addressing these important content areas, educational interventions should also incorporate exercises to help students identify their emotions and how they affect their treatment of victims of DV as well as exercises to facilitate students' empathic responses to victims.

A RANDOMIZED CONTROL TRIAL USING INSINUATED STANDARDIZED PATIENTS TO ASSESS RESIDENTS' DOMESTIC VIOLENCE SKILLS: A PILOT STUDY. S.A. Haist¹; J.F. Wilson¹; M.J. Lineberry¹; J.S. Gibson¹; C.H. Griffith¹. ¹University of Kentucky, Lexington, KY. (Tracking ID #134721)

BACKGROUND: An educational intervention was designed to teach internal medicine residents about domestic violence (DV). To study the intervention's effectiveness, DV clinical skills of residents participating in the DV workshop were compared to residents not participating.

METHODS: Twenty-seven internal medicine residents were randomized to receive either a two-hour DV workshop or a two-hour Chronic Pain (CP) workshop. The two workshops were given simultaneously on four nights. Participating residents were paid \$200. A pre-workshop questionnaire included demographics, self-assessed prior DV and CP training and perceived knowledge. The workshops were interactive using two standardized-patient (SP) cases portraying different aspects of DV or CP. Faculty-led discussion followed the SP interviews. Thirteen novel SPs never used at the study institution were trained to two cases (subtle DV, feeling tired and sad; obvious DV, shoulder injury with previous trauma). The two cases were randomized and insinuated into the residents' continuity clinics at either 1-3 months or 3-7 months after the workshops. The SPs completed a detailed checklist immediately after the encounter. Multiple regression techniques (GLM) were used to compare the results of DV SP checklist scores of DV workshop residents vs. CP workshop residents. Gender, prior

DV training, perceived knowledge and workshop participation were control variables.

RESULTS: Fourteen and 13 residents participated in the DV and CP workshops, respectively. Twenty-five and 26 residents encountered the subtle and the obvious DV cases, respectively. Fifteen of 26 SPs encountered by DV workshop residents were identified as DV victims vs. eleven of 22 SPs encountered by CP workshop residents. Scores on the subtle DV checklist items were 50% for DV workshop residents vs. 29% for CP workshop residents ($F = 4.0$, $P = .04$) controlling for gender. The DV workshop residents (subtle and obvious DV cases) scored 38% on the DV safety plan subscale vs. 19% for the CP workshop residents ($P = .10$, effect size 0.62 SD). For the subtle and obvious DV cases, 79% of the most important DV specific checklist item scores were higher for DV workshop residents than for the CP workshop residents (e.g. for "Asks if you have money hidden away" DV workshop residents scored 49% and 18% higher compared to CP workshop residents). For residents who identified the SP as a DV victim, 11/15 of the DV workshop residents received credit for at least 75% of the DV checklist items vs. 2/12 CP workshop residents, logistic Wald 7.30, $P = .006$; OR = 13.8, CI = 2.1-92.0. Self-reported previous DV knowledge and being a woman positively influenced performance. Time between workshop and the SP encounter did not affect the results.

CONCLUSIONS: Residents participating in a DV workshop demonstrated better clinical skills than a control group as assessed by an insinuated SP portraying a subtle DV case in the residents' continuity clinic. Also, DV workshop residents were more likely to obtain more information from the SP if the SP were identified as a DV victim. To our knowledge, this is the first randomized study using insinuated SPs demonstrating improved DV clinical skills following an educational intervention.

A RANDOMIZED CONTROLLED TRIAL OF NIGHT FLOAT: EFFECTS ON RESIDENT SLEEP, FATIGUE, AND PATIENT CARE. V. Arora¹; C. Dunphy¹; V. Chang¹; F. Ahmad¹; H.J. Humphrey¹; D.O. Meltzer¹. ¹University of Chicago, Chicago, IL. (Tracking ID #133574)

BACKGROUND: Night float (NF) is routinely used to alleviate sleep deprivation in residents. However, few studies have objectively assessed effects of NF on residents and the patients they care for. The aim of this study is to assess the effects of NF on resident sleep, fatigue, and patient care.

METHODS: From July 2003 to June 2004, we conducted a within-subject paired randomized controlled trial of NF on an academic general medicine service. For 2 weeks of each month, NF provided interns an opportunity to sleep for 7 hours (12am to 7am) during their call night. The remaining 2 weeks were a standard call (SC) schedule. To record sleep, interns wore Actiwatchs, which are wrist monitors that detect motion. Using Experience Sampling Method, random alerts prompted interns to rate real-time fatigue from 1 (most alert) to 7 (most tired) on a Pocket PC-based Stanford Sleep Scale. Fixed effects multivariate linear regression was used to assess the effect of night float on sleep time and fatigue. Hospital paging data confirmed the time an intern forwarded their pager to night float. T-tests were used to compare cost and length of stay for patients admitted to NF or SC services. Patients rated their satisfaction with their care from their doctors on a scale of 1 (extremely satisfied) to 4 (extremely dissatisfied) during a one-month follow-up phone interview. Chi-square analysis was used to compare satisfaction scores between patients admitted to NF and SC services.

RESULTS: 38/40 interns participated for a total of 119/178 (67%) of intern-months. With NF, interns received 44 minutes more sleep (187 min vs 143 min, $p < 0.001$) and were less fatigued than with SC (1.74 vs 2.26, $p < 0.001$). Not surprisingly, mean fatigue scores were higher on post-call days, but the effect of NF on post-call fatigue was greater. NF covered interns reported an almost one point decrease in fatigue than with SC (2.23 vs 3.16, $p < 0.012$). For reference, one point on this scale is the difference in fatigue between on-call and post-call interns on SC (2.06 vs. 3.16). Interestingly, interns on NF also reported less fatigue on their on call day (1.59 vs 2.06, $p < 0.007$). Paging data revealed that in 79% of instances, interns chose to only forward the care of their cross-cover patients to NF, and retain care of their own patients during their call night. Interns who forwarded care of all their patients received 66 minutes more sleep than when on SC (209 min vs 143 min, $p < 0.001$). For each hour that an intern signed over the care of all of their patients, they received 40 minutes of sleep ($r = 0.69$). In interviews, interns unanimously preferred NF coverage when on-call, citing benefits of increased sleep, but more often a greater ability to "focus on their own patients" and "get work done" without interruptions from cross-coverage. However, 98% of these same interns stated that when they were not on call, they preferred their patients cared for on by an intern on SC, citing concerns regarding double sign-out and poor communication. Analysis to date for 5001/5556 (90%) of inpatients that year confirms that NF had no effect on length of stay (4.93 days NF vs. 4.84 SC, $p < 0.39$), cost (\$9011 NF vs. \$9488 SC, $p < 0.88$), or patient satisfaction (1.42 NF vs. 1.49 SC, $p < 0.17$).

CONCLUSIONS: Night float can provide increased sleep and alleviate fatigue in residents, without adversely affecting patient care. However, concerns of discontinuity impair its' use.

ACCURACY OF SELF-AUDITS IN AN ACADEMIC TEACHING HOSPITAL. M. Panda¹; T. Ball²; M. Calloway³; R. Jones⁴; N.A. Desbiens³. ¹University of Tennessee at Chattanooga, Chattanooga, TN; ²University of Tennessee, Chattanooga, Chattanooga, TN; ³University of Tennessee, Chattanooga, TN; ⁴Chattanooga, TN. (Tracking ID #132781)

BACKGROUND: Medical errors cause nearly 44,000 deaths and lead to an enormous cost burden. One of the reasons cited for these errors is poor documentation by physicians. Proper documentation in a patient's chart is a skill

that is important to both the physician and the physician-in-training. Therefore, chart audits are a method of quality assurance to assure that physicians-in-training are adequately documenting necessary patient care information. There is substantial research proving the effectiveness of audits in a pediatric program. Purpose: 1. To compare the documentation skills of the first (PGY-1), second (PGY-2), and third (PGY-3) year residents through the use of a chart self-audit process. 2. To determine the accuracy of self-audits when compared to third party audits.

METHODS: A 39 item audit form based on the Health Plan Employer Data and Information Set (HEDIS) was developed. The audit form addressed various issues such as dictations, medications and preventive health. Each resident used the audit form to review five of their own medical charts. The same five medical charts of each resident were then reviewed by two independent researchers who were blinded to the resident audits and used the same audit sheet to independently determine the documentation. The audits completed by the two independent reviewers were compared and in areas where there was a discrepancy, an agreement was reached by reviewing the medical chart together. The audits done by the independent reviewers were considered to be the gold standard which was compared to the audits completed by residents to determine the accuracy. For the purposes of analysis the audit form was divided into 6 main categories: Dictated Note, Chart Components, Screening, Vaccinations, Medication list and presence of Narcotic contract if the patient is on narcotics.

RESULTS: Among the 27 residents that participated in the study, there were 10 first year, 9 second and 8 third year residents. The residents completed 135 audit forms. Three medical charts could not be found, while two of the audits completed by the residents were illegible and not included in the study. One audit form was illegible and was not included in the study. Therefore, for the study a total of 130 medical charts were examined by both the residents and the independent reviewers. The table below summarizes the agreement rate (%) between residents and gold standard. Component PGY-1 PGY-2 PGY-3 Overall Dictated Note 95 94 91 93 Chart Components 91 87 84 87 Preventative Health Screening 71 78 64 71 Diabetic Screening 88 89 84 87 Vaccinations 82 79 84 82 Medication List 79 78 71 76 Narcotic Agreement 94 68 97 86 The residents' performance as compared to the gold standard varies between each component and post graduate year for the study variables examined, the first year residents seemed to fare better. There were significant differences in accuracy between the resident audits and the gold standard in the areas of preventative health and updates to the medication list. In the categories of dictated note and diabetic screening there was a higher rate of agreement.

CONCLUSIONS: Self-audits are not appropriate as a tool for monitoring quality of care in an outpatient clinic at an academic teaching hospital.

ACGME DUTY HOUR REQUIREMENTS, WHAT DO FACULTY THINK? G.M. Talente¹; L.J. Staton¹; M.R. Carroll¹; M. Mumtaz¹; L.C. Larsen¹. ¹East Carolina University, Greenville, NC. (Tracking ID #135170)

BACKGROUND: As of July 2003, all residency programs in the United States were mandated to comply with new duty hour standards. The changes limit resident duty hours to an average of 80 hours a week and set other restrictions. The standards are proposed to help strengthen both patient care and education. We explored the attitudes of faculty regarding the impact of the changes on their quality of life, patient care, teaching, and on residency training as a whole.

METHODS: We conducted a cross sectional survey of physician faculty in all disciplines at a university based residency training site one year after the implementation of the new work hour rules. The survey was comprised of 36 items including demographics and a Likert-scale measure of attitudes and beliefs regarding the impact of the new duty hour guidelines.

RESULTS: 147/212 faculty (69%) responded. Sixty seven percent of the respondents were males, 61% were tenure track faculty. Responses were received from all disciplines, with Pediatrics (21%), and Internal Medicine (25%) having the highest percentage. Nearly all faculty were aware of the ACGME requirements (99%). Many felt problems existed with previous work hours (54%), and still existed with resident work hours (46%). In regards to quality of life respondents differed on whether the new guidelines worsened their own quality of life with faculty in family medicine feeling the guidelines had ($p < .001$). Most faculty agreed their quality of life had not improved (66%). Most reported increased patient care responsibilities (63%). In regards to patient care issues, most faculty felt patient continuity of care (69%) and resident time with their patients (73%) had decreased. Few (7%) felt that medical errors had decreased, one of the primary goals of the new guidelines. In regards to teaching, few faculty felt residency training had improved (13%). Teaching was most negatively affected according to Family Medicine faculty ($p < .01$). The majority agreed that resident quality of life had improved (80%), but most felt that residents would be less prepared to enter practice (66%). Estimation of higher average weekly personal work hours by respondents was correlated with responses that the new guidelines negatively impacted faculty quality of life, patient care and resident education ($p < .05$).

CONCLUSIONS: Faculty feel that the new guidelines are negatively affecting patient continuity of care but are not reducing medical errors as intended. They are concerned about the impact of the guidelines on teaching. While they feel the guidelines improve resident quality of life they do not feel that they have improved resident education and preparedness for practice, the other main goal of the guidelines. Faculty feel the new guidelines have increased their patient care responsibilities and in some groups that they have worsened quality of life for faculty. In the face of this perception by faculty that the new guidelines are not meeting their intended goals yet seem to be affecting faculty lifestyle and work responsibilities, an effort should be made to find objective evidence of the guidelines impact on patient care and resident education.

ARE STANDARD MEASURES OF KNOWLEDGE AND CLINICAL SKILLS ASSOCIATED WITH OBJECTIVE MEASUREMENT OF RESIDENT CLINICAL PERFORMANCE? L.L. Willett¹; G.R. Heudebert¹; C.I. Kiefe¹; T.K. Houston¹. ¹University of Alabama at Birmingham, Birmingham, AL. (Tracking ID #134597)

BACKGROUND: The implications for clinical care of standard approaches to housestaff evaluation are poorly understood. Therefore, we assessed the correlation between 1) standard evaluation measures of knowledge and clinical skills of internal medicine (IM) trainees and 2) the trainees' performance of preventive health services in a clinical setting.

METHODS: We abstracted 1,102 charts of patients followed in continuity clinic by IM residents to measure the percent of appropriate indicated preventive health services received. Resident-level performance audit data were linked to their individual evaluation measures including: selection scores for match list ranking, clinic faculty global evaluations on a five-point scale (1 = worst, 5 = best), USMLE Step-1 and In-Training Examination scores, and the mini-CEX (summary of 4 scores).

RESULTS: Of the 130 residents, 70% were male, 42% were interns, and 34% intended to practice primary care. Mean (standard deviation) clinic faculty rating was 4.25 (0.55), selection score was 4.2 (0.56), USMLE Step-1 was 220 (17), In-Training exam was 57 (8.2) and Mini-CEX was 7.1 (0.74). The Table summarizes the mean percent (SD) of patients seen by residents who received the preventive service when indicated, and linear correlation coefficients of individual audit performance with the standard evaluation measures (* = $p < 0.05$, ** = $p < 0.02$).

CONCLUSIONS: Standard measures of resident evaluation were not strongly correlated with actual performance of six preventive health services. The strongest correlation was seen between the mini-CEX and quit smoking advice. Thus, the performance audit appears to be a unique measure of evaluating resident performance.

Mean performance and Correlation coefficients between performance and evaluation measure

Preventive health measure	Mean % performance (SD)	Selection scores	Clinic ratings	USMLE-1 scores	In-training scores	Mini-CEX scores
Smoking screening	53% (24)	-0.05	0.05	-0.03	-0.03	-0.07
Quit smoking advice	62% (34)	0.03	0.00	0.01	0.03	0.24**
Colon ca screening	82% (20)	0.06	-0.10	0.11	0.00	0.17
Breast ca screening	89% (22)	0.10	0.04	0.22*	0.08	-0.01
Cholesterol screening	82% (18)	-0.02	-0.09	0.02	0.06	0.03
Pneumococcal vacc	61% (26)	-0.08	0.02	-0.10	0.01	0.13

ART AND EMPATHY: DO MEDICAL STUDENTS GET THE PICTURE? D.W. Rudy¹; D.W. Victor¹; J.F. Wilson¹. ¹University of Kentucky, Lexington, KY. (Tracking ID #136136)

BACKGROUND: The recent advances in technology in medicine along with forces that diminish the amount of time doctors spend with patients may be changing the characteristics of those students interested in pursuing medical school. There is evidence that the current applicant pool may be entering medicine primarily due to interest in science and technology. Does this occur at the expense of the interest in medicine for the humanistic factors? One of the most important aspects of humanism in medicine is empathy. This study was designed to establish a baseline of the empathetic nature of entering first year medical students as well as to act as learning point for the importance of observation.

METHODS: First year medical students were shown a copy of painting "The Doctor" (1887) by Sir Luke Fildes during a lecture importance of observation. The painting depicts a physician vigilantly attending over an ill child. The mother is seated with her head down and the father stands next to her with his hand on her shoulder. The picture is classically interpreted as depicting the dedication and empathy of the physician. Students were asked to "write down your observations." Responses were read independently by two reviewers in for emergent themes regarding the people depicted in the painting. The reviewers negotiated the final thematic categories and developed a coding strategy. The responses were then re-read independently and coded as to the presence or absence of each thematic category. Differences between reviewers were resolved by consensus.

RESULTS: Sixty-eight (72%) of the students completed the assignment. The themes included: the identification of the characters in the painting, their descriptions, actions, and feeling/emotions. (See table)

CONCLUSIONS: Students identified the situation and the setting more consistently than the emotional aspects of the painting. For instance, the fact the child was positioned on two chairs was mentioned in 47% of responses while they only identified the emotions of doctor 26% of the time. Only 59% commented on the emotional state of the mother (crying, praying, and distraught/worried). Students' comments in aggregate show that they viewed the doctor as working; observing and thinking after some treatment was given to the sick child. When emotions of the doctor were mentioned they were primarily negative emotions (confused/perplexed 13% and pensive/distressed 6%). Only 7% of the students

described the doctor as concerned/compassionate/empathetic. Sir Luke Fildes intended the painting to show the compassion and empathy of the physician. Our students missed this point. After the students completed the writing assignment the interpretation of the painting was discussed. It was emphasized that observing the emotions of others as well as one's own emotions are a crucial part of the diagnostic and therapeutic process. In the future, we plan on using the discussion of art and empathy in small group discussions.

Results

Results			
Doctor		Child	
Observing/watching	51%	On chairs	47%
Thinking/pondering	28%	Sick	68%
Gave medicine	40%	Sleeping	19%
Perplexed/confused	13%	Possible diagnosis	16%
Compassion/empathy	7%		
Pensive/distressed	6%		
Mother		Father	
Crying	31%	Comforting mother	49%
Distraught/grieving	51%	Observing	19%
Head on table	19%	Distraught/grieving	12%
Praying	18%	Being strong	9%

ASSESSING PHYSICAL EXAMINATION SKILLS OF INTERNAL MEDICINE RESIDENTS: IS SENIORITY BETTER? L.L. Willett¹; C. Estrada¹; A. Castiglioni¹; S. Massie¹; G.R. Heudebert¹; R.M. Centor¹. ¹University of Alabama at Birmingham, Birmingham, AL. (Tracking ID #134645)

BACKGROUND: Direct observation and assessment of physical examination skills of internal medicine residents are limited. We sought to determine whether departing postgraduate year (PGY)-3 residents have superior physical examination skills and comfort level compared to incoming PGY-1 residents.

METHODS: We used a five station Objective Structured Clinical Examination (OSCE) to assess physical examination skills of residents. The stations included cases of pneumonia, congestive heart failure (CHF), ascites, shoulder pain and back pain. Standardized patients scored performance with task-specific checklists (score range: 0-100%). Residents assessed their comfort level overall and for each individual station on a 5-point Likert scale (1 = low, 5 = high). We used nonparametric analysis.

RESULTS: PGY-3 residents (n = 8) had significantly increased overall performance compared to PGY-1 residents (n = 16) for all stations combined (P = 0.03). The median scores for individual stations were higher for PGY-3 vs PGY-1 residents, although was not statistically significant: pneumonia 65% vs 54% (P = 0.21), CHF 82% vs 73% (P = 0.14), ascites 85% vs 69% (P = 0.11), shoulder pain 68% vs 57% (P = 0.06) and back pain 84% vs 75% (P = 0.88). Additionally PGY-3 residents had statistically increased comfort level overall 4 vs 2 (P = 0.003) and for each individual station: pneumonia 4 vs 3 (P 0.001), CHF 4 vs 3 (P = 0.009), ascites 4 vs 3 (P = 0.001), shoulder pain 3 vs 2 (P = 0.01) and back pain 4 vs 2.5 (P = 0.05). We found no correlation between overall comfort level and overall performance (r = 0.30, P = 0.20).

CONCLUSIONS: Our study suggests that departing PGY-3 residents have superior physical examination skills and greater comfort level as compared to incoming PGY-1 residents, a reassuring finding for program directors. Whether the improved subjective comfort level exceeds objective performance is unclear.

Median performance of residents by year of training (25th, 75th percentile)

Overall	PGY-1	PGY-3	P-value
OSCE	66% (57,73)	76% (65,82)	0.03
Comfort level	2 (2,3)	4 (4,4)	0.003

ASSESSING SURGICAL AND MEDICAL HOUSE OFFICERS' GERIATRICS KNOWLEDGE. B.C. Williams¹; J.T. Fitzgerald¹. ¹University of Michigan, Ann Arbor, MI. (Tracking ID #133150)

BACKGROUND: As the population ages, assessing surgical and medical house officers' knowledge of the care of older patients is crucial to the improvement and evaluation of training programs. No reliable, valid instruments have been developed for this purpose in surgical specialties and medical subspecialties. Our objective was to design a brief written instrument to measure knowledge of clinical geriatrics among house officers in surgical specialties and medical subspecialties.

METHODS: A 30 item multiple-choice test was created by an expert panel of geriatricians, surgeons, and medical subspecialists using existing and generated items. Test items covered inpatient and common geriatric syndromes in 18 pre-defined topic areas. All items consisted of a brief patient scenario followed by clinical options with one best response. After pilot testing among 56 house officers at one institution, 7 items with limited reliability or very high (>95%) correct response rates were eliminated. The resulting 23-item instrument was administered to 301 house officers at 4 institutions in surgical disciplines (25% of respondents), emergency medicine (29%), medicine subspecialties (19%), internal medicine residency (12%), and other disciplines (15%). Internal reliability

was determined using Cronbach's coefficient alpha. Validity was examined by comparing test scores by level of training using one-way analysis of variance. **RESULTS:** Assessments were completed by 108 HO1s (36% of total), 71 HO2s (23%), 39 HO3s (13%), and 83 HO4s (28%). 3 items decreased internal reliability and were dropped. Among the remaining 20 items covering 17 topic areas, Cronbach alpha was 0.60, indicating an appropriate balance between inclusion of a variety of topics in clinical geriatrics while limiting response time to less than 30 minutes. Residents averaged 62% correct, with 58, 63, 62, and 69 percent correct responses among HO1, HO2, HO3, and HO4s, respectively. Analysis of variance showed that the graded increases in scores by year of training were statistically significant (p < .0001).

CONCLUSIONS: This reliable, valid 20-item multiple choice instrument can be used by surgical specialty and medical subspecialty house officer programs to gather baseline data to guide curriculum reform or evaluate program performance to meet certification requirements. The instrument is now available free on the Web to registered users at www.med.umich.edu/geriatrics/educationalprograms/gme.htm.

ASSESSING THE EDUCATIONAL VALUE OF AUTOPSY REVIEWS. K.J. Pfeifer¹; V. Bonne¹; D. Faust¹; M. Muntz¹; K. Wiger¹; S. Davids¹; M. Frank¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #135943)

BACKGROUND: In recent years there has been a marked decline in the number of autopsies performed at teaching hospitals. Despite this trend, the ACGME maintains autopsy review as an important educational requirement of Internal Medicine residency programs. Since work hour restrictions and other forces have decreased the amount of training time available, it is imperative that the instructional worth of all resident activities be scrutinized. The objective of this study was to determine residents' perceptions of the educational value of autopsy reviews.

METHODS: The Internal Medicine Residency Program collaborated with the pathology department to create a notification system that would assure that the Program's chief residents would be alerted to all scheduled autopsies from March 1, 2004 to December 31, 2004. The chief residents determined from the medical record which patients were cared for by Internal Medicine residents, and then notified these residents of the scheduled autopsy. Residents were expected to attend the autopsy unless it caused a serious disruption of patient care or work hour violations. Following the autopsy, residents completed a questionnaire that determined their perceptions of the educational and clinical value of the autopsy as well as what Program duties they missed during the autopsy. An associate program director then met with each resident and conducted a structured interview to elicit the resident's feelings regarding attending the autopsy.

RESULTS: Of 85 scheduled autopsies, a total of 24 autopsies were completed on Internal Medicine patients, with 42 residents involved in their care. Sixteen autopsies (67%) were attended by at least one resident, and a total of 22 residents (52%) were able to attend their patients' autopsies. On a 5-point Likert scale (1 = least value, 5 = greatest value), residents rated attendance at autopsies as providing valuable feedback regarding their patient care (mean 4.2) and as a quality educational experience (mean 4.6). While most residents (78%) reported expecting the information they received from the autopsy, only 52% felt that it provided an explanation of the patient's cause of death. Missing team rounds was the most frequently reported adverse consequence of attending autopsies (55%), but residents also listed interference with patient care (27%) and absence from educational conferences (9%). Themes extracted from interview data included collaboration and pathophysiology review with pathologists as greatly educational and personal observation of autopsies as helpful for renewing perspective on the gravity of physicians' management decisions.

CONCLUSIONS: Resident perceptions of the educational value of autopsies affirm ACGME requirements for incorporation of autopsy reviews into internal medicine training. However, even with a program to facilitate this, competing demands on resident training time create challenges to successful assimilation of autopsy review into resident education. Further studies of the specific instructional benefits of autopsy review are required to guide the development of programs that integrate resident autopsy education without compromising other training experiences.

BEDSIDE TEACHING FROM THE PERSPECTIVE OF THE LEARNER. K.N. Williams¹; S. Ramani¹; B. Fraser¹; J.D. Orlander¹. ¹Boston University, Boston, MA. (Tracking ID #135841)

BACKGROUND: Bedside clinical teaching has decreased since the 1960's despite the view held by many that it is essential to physician training. Few studies exist on the perspective of the learner. The objectives of this study were to explore learner opinions on the role of bedside teaching in medical education, their perceptions on barriers to bedside teaching, and their suggestions for improving the frequency and effectiveness of bedside teaching.

METHODS: Separate focus groups consisting of fourth-year medical students, first-year, or second-year internal medicine residents were recruited to discuss bedside teaching. Discussions were recorded and transcribed, and salient themes were coded independently by three investigators; disagreements were resolved through consensus. Focus groups were conducted until the data became saturated and no new themes emerged.

RESULTS: Preliminary analysis indicates that students believed that bedside teaching was a valuable component of their training, particularly in learning such skills as communication, history-taking, physical examination, and professionalism. However, they expressed reluctance to use it as a forum to discuss differential diagnosis and management. This reluctance related to concern for the patient in the context of self-perceived professional incompetence. Students

reported many barriers to bedside teaching categorized as either those related to the characteristics of and relationships between teacher, learner, and patient or those related to the medical environment. The former category includes barriers related to patient cooperation, communication with the patient, concern for the patient, learner and teacher initiative, teaching expectations, teaching skills, and clinical skills and knowledge and the latter includes those related to time, work volume, patient diversity, institutional priority, and technology. Students offered several strategies to improve bedside teaching: improved communication between the patient, teacher, and learner; increased teacher and learner initiative, clear expectations for teaching; and increased institutional support through protected time for teaching and teacher training for attendings and residents. The views of interns and residents, although similar to students, were influenced by work responsibilities and by their dual role as both learner and teacher. For example, they felt that institutional pressure for timely admission and discharge of patients conflicted with efforts to teach at the bedside and expressed concern that bedside teaching, if not done properly, could compromise the therapeutic relationship between physician trainees and the patient. Their dual role as learner and teacher provided insight when offering strategies to improve bedside teaching: increased flexibility when planning teaching sessions while accounting for work schedules, selectively and efficiently integrating teaching with work, establishing a positive learning climate, teacher training, and protected teaching time for attendings.

CONCLUSIONS: The preliminary results of this inquiry suggest that learners believe that bedside teaching is effective for learning certain aspects of clinical medicine, that teaching at the bedside requires effort and skill, and that this skill can be learned. By proposing specific enabling strategies, they acknowledged that effective bedside teaching could occur despite the barriers that impede it. This inquiry notes important differences in the opinions of students and housestaff reflecting the work responsibilities and dual role as learner and teacher of the latter. These results will guide efforts by medical educators to increase the frequency and effectiveness of bedside teaching.

BURNOUT AMONG THIRD YEAR MEDICAL STUDENTS. M. Mintz¹; S. Youcha¹; J. Twombly¹; M. Krason¹. ¹George Washington University, Washington, DC. (Tracking ID #133922)

BACKGROUND: Burnout is a syndrome characterized by emotional exhaustion, cynicism or depersonalization in relationships with coworkers or patients; and a sense of inadequacy or reduced personal accomplishment. Previous research has demonstrated that burnout is prevalent among resident and attending physicians. Physicians who meet criteria for burnout have been shown to provide suboptimal patient care. However, little is known regarding burnout in medical students. The purpose of the study was to evaluate the prevalence of burnout among medical students at the end of their third year.

METHODS: The Maslach Burnout Inventory (MBI) has been shown to be a reproducible and valid instrument to assess burnout in a number of fields including education and health care. The MBI is a 22 item questionnaire which asks respondents to indicate on a 7-point Likert scale the frequency with which they experience certain feelings related to their work. The MBI evaluates three domains of burnout: 1) emotional exhaustion (i.e. "I feel emotionally drained from my work"), 2) depersonalization ("I've become more callous toward people since I took this job"), and 3) personal accomplishment ("I feel I'm positively influencing people's lives through my work"). Scores for each domain are calculated and assigned tertiles (low, medium, high) based on previously reported data from healthcare professionals. The health professional MBI was administered anonymously to all third year medical students from The George Washington University School of Medicine (GWSOM) from the classes of 2004 and 2005 one month before the end of their third year. The survey was administered online as part of a course in professionalism which is required for all third year students and meets one day during each of six clinical clerkships. The only modifications to the survey were defining "work" as what students did on the wards during their clerkship and substituting the word "client" with "patient." Burnout was defined as scores in the highest tertile on either the emotional exhaustion or depersonalization subscale, or the low range of the personal accomplishment subscale.

RESULTS: 278 (142 class of 2004, 136 class of 2005) or 95% of third year students from GWSOM participated. 32% and 28% of the students (2004 and 2005 respectively) scored high for emotional exhaustion, 16% and 15% of the students scored high for depersonalization and 10% of both classes had a low sense of personal accomplishment, indicating burnout in all three areas. Combining scores for both the high and medium tertiles of both classes, 69% of students demonstrated emotional exhaustion, 49% demonstrated depersonalization, and 33% demonstrated low personal accomplishment.

CONCLUSIONS: Burnout was relatively common among third year medical students with burnout from emotional exhaustion present in almost one third of all third year medical students from two subsequent classes. Thus, burnout which is known to occur in residents and attending physicians appears to begin in medical school. Further investigation regarding why clinical clerks feel emotionally exhausted is needed in order to improve the quality of the ward experience and to diminish burnout. Areas unique to medical students including power differential, student mistreatment, and vulnerability of novice learners should be considered.

CAN RESIDENTS PREDICT THEIR STRONGEST AND WEAKEST SUBJECTS ON THE INTERNAL MEDICINE IN-TRAINING EXAM? J. Fischer¹; L. Cation¹; L. Meyer¹. ¹University of Illinois at Peoria, Peoria, IL. (Tracking ID #135792)

BACKGROUND: Medicine residents are not particularly skilled in predicting their overall In-Training Exam (ITE) performance; only half of the internal medicine residents studied by Hawkins et al were able to accurately predict the

tertile into which their overall ITE score would fall. A resident's ability to predict specific subject area performance on the ITE, however, has not been studied. We describe resident ability to predict strongest and weakest subject areas on the internal medicine ITE.

METHODS: The study was completed at a university-affiliated, community-based internal medicine residency program. All residents who took the internal medicine ITE in 2002 or 2003 were asked to complete a voluntary "Performance Prediction Survey." This survey, administered immediately prior to the ITE, asked residents to predict the one subject area (e.g. cardiology, pulmonary, general medicine etc.) that they felt would be their strongest and, likewise weakest, on the ITE. The effect of sex, program type (internal medicine, medicine-pediatrics, preliminary medicine), and career choice (primary care, fellowship, other) on the ability to predict strong and weak subject areas was analyzed using Chi square analysis.

RESULTS: Prediction data was available for 96 of the 114 residents who took the ITE during the time period studied. For all residents combined, the most frequent "strongest subject area", both predicted and actual, was general medicine. The most frequent "weakest subject area" predicted was hematology/oncology while actual was pulmonary. Overall, only 19% of residents could predict their strongest and only 20% could predict their weakest subject areas. Neither sex, program type nor career choice significantly impacted the frequency with which residents could correctly identify their actual strongest or weakest subject area. However, statistically significant differences were found between the expected and actual "strongest subject area" predictions when compared within both program type ($P = 0.015$) and career choice ($p < 0.0005$). Within program type, medicine-pediatric residents overestimated scores in general medicine and cardiology and underestimated their scores in gastroenterology. Medicine residents underestimated their scores in general medicine. Career choice comparisons showed that those pursuing fellowships overestimated their scores in endocrinology and pulmonary medicine but underestimated their scores in general medicine. Primary care bound residents underestimated their scores in endocrinology and overestimated their scores in general medicine.

CONCLUSIONS: Residents taking the internal medicine ITE have a poor ability to predict their strongest and weakest subject areas. While this ability may be influenced by program type and career choice, larger studies are needed to more precisely address this issue. Because efforts at continuing medical education can be challenging in today's time-limited education environment, it is imperative that physicians be able to accurately identify strong and weak knowledge areas.

CLERKSHIP DIRECTORS' OPINION REGARDING PREPARATION FOR BASIC CLINICAL SKILLS DURING YEARS 1 AND 2: RESULTS OF THE 2002 CDIM BASIC SCIENCE SURVEY. M.L. Cannarozzi¹; K. O'Brien¹; P.M. Wallach¹; L. Crespo¹; D.M. Elnicki². ¹University of South Florida, Tampa, FL; ²University of Pittsburgh, Pittsburgh, PA. (Tracking ID #134655)

BACKGROUND: At the completion of the first two years of medical school, medical students are traditionally considered prepared to begin clerkship education. The purpose of this study was to assess current views of clerkship directors regarding student overall preparation for medical clerkship training, as well as preparation in several core clinical skills, including physical examination, history taking, procedural skills, medical knowledge base, critical literature appraisal and clinical epidemiology.

METHODS: As part of the 2002 Annual CDIM Survey, seven questions regarding preparation for clinical clerkships were included. Respondents were asked to rate the adequacy of student preparation in certain basic clinical skills using a five-point Likert scale (1 = extremely well, 3 = satisfactory, 5 = extremely poor). Eighty-nine out of 123 CDIM members responded (72%).

RESULTS: Overall, respondents felt that students were well prepared for clinical training (mean 2.4 ± 0.7). However, when questioned regarding preparedness in each of the above mentioned basic clinical skills, specific areas of weakness/strength were identified. Although the majority of clerkship directors considered students to be well prepared in history taking skills (2.3 ± 0.8) and medical knowledge base (2.1 ± 0.8), there was less confidence in preparedness of students in skills of physical examination (2.9 ± 0.8), critical literature appraisal (2.8 ± 1.0), clinical epidemiology (3.1 ± 0.9), and procedural skills (3.8 ± 1.1).

CONCLUSIONS: Regardless of career choice, medical students need to demonstrate competency in a core set of basic clinical skills. This data suggests students are well prepared in terms of basic clinical knowledge and history taking. Current curricular structures should continue to focus on improving skills in physical examination, critical literature appraisal and clinical epidemiology. More study is needed to determine the best method(s) to teach procedural skills within an integrated four year curriculum.

COMMUNICATION SKILLS ARE HIGHLY CORRELATED WITH HISTORY CONTENT: FINDINGS OF A COMPREHENSIVE CLINICAL SKILLS EXAMINATION FOR MEDICAL STUDENTS. L.R. Tewksbury¹; S. Zabar¹; J.M. Chase¹; C. Gillespie¹; A.L. Kalet¹. ¹New York University, New York, NY. (Tracking ID #134653)

BACKGROUND: While superior communication skills have been associated with improved patient outcomes, limited data are available on the impact of student communication skills on other required competencies such as the ability to obtain an accurate and complete history. High quality clinical skills assessment provides important feedback to educators regarding student skills, including important associations which can drive curricular needs. Are there

correlation between medical student communication skills and accuracy of history as assessed by a high quality Comprehensive Clinical Skills Examination (CCSE)?

METHODS: A 6 station CCSE for all 4th year medical students was designed, through collaborative effort of all core clerkship directors at our institution, to measure communication skills (CS), accuracy and completeness of history content (HG=history gathering), and physical exam skills (PE). Standardized patients were trained to assess skills using carefully designed behavioral objective checklists. A previously validated CS checklist was used across all 6 cases. Case-specific checklists of HG and PE skills were based on content and quality of information gathered as determined by clerkship directors and other content experts. All checklists used a 3-point scale for each behaviorally anchored item: not done, done minimally, and well done.

RESULTS: 125/148(85%) of 4th yr medical students who participated in the exam consented to have their data analyzed anonymously. Internal consistency of checklists, as measured by Cronbach alpha, was best for CS (.91), followed by HG (.80), then PE (.60). Students mean (SD) performance, as measured by % items rated well done was: CS = 62.88%(11.21%), HG = 43.90%(11.22%), PE = 42.40%(8.17%). Across all cases, CS was highly correlated with HG($r = .47$, $p < .001$) but not with PE($r = .12$, $P = .17$). The correlation between CS and HG was significantly positive within all six cases, ranging from $r = .19$, $P = .022$ for a neurology-based case to $r = .58$, $p < .001$ for a pediatrics-based case.

CONCLUSIONS: Medical student performance on a CCSE suggests that excellent communication skills are associated with the gathering of an accurate and complete history. This supports the important need for communication skills curriculum.

COMPARISON OF INTERNAL MEDICINE RESIDENTS, ATTENDINGS AND NURSES PERCEPTIONS OF THE NIGHT FLOAT SYSTEM. E.A. Akl¹; A. Baig¹; E. Rich¹; J.L. Izzo¹; B.J. Grant¹; H.J. SchüNemann¹. ¹University at Buffalo, Buffalo, NY. (Tracking ID #135778)

BACKGROUND: Internal Medicine residency programs use the night float system increasingly to meet the new resident work hour regulations. This study compared residents, attendings and nurses perceptions of this system's impact on clinical, educational and personal outcomes.

METHODS: We used electronic and postal mail to survey 107 residents, 48 attendings and 69 nurses in a university-based Internal Medicine residency program in Buffalo, New York. The survey consisted of thirty-four positively framed statements about the impact of the night float system on patient care, and on residents' training and personal life. Participants used a 7 point Likert scale (with -3 = strongly disagree, 0 = neutral and 3 = strongly agree) to rate their agreement with these statements. We compared, for each group, the means (μ) of ratings to 0 (neutral answer) using one sample t-test. We then compared the means of the ratings among groups using the Mann-Whitney U test.

RESULTS: The response rate was 84% for residents, 69% for attendings and 64% for nurses. For almost all statements, residents had higher ratings than attendings and nurses, but there was no statistically significant difference between nurses and attendings ratings. Residents rated the majority of statements about the impact of the night float system on patient care positive (μ range 0.44-1.23, p values < 0.003), while attendings and nurses ratings were statistically neutral. All three groups gave their lowest ratings to the night house staff communication with patients ($\mu = 0.17$, -0.84 , -0.98 ; $P = 0.262$, 0.001 , < 0.001 respectively). In terms of the night house staff training, residents and attendings rated negatively the training quality ($\mu = -0.28$, -0.72 ; $P = 0.098$, 0.009), the daily feedback ($\mu = -0.54$, -1.34 ; $P = 0.001$, < 0.001), and the end of rotation evaluation of the night house staff ($\mu = -0.31$, -1.06 ; $P = 0.064$, < 0.001); they rated positively the adequacy of clinical burden ($\mu = 0.68$, 0.91), house staff independence ($\mu = 1.53$, 1.25) and confidence ($\mu = 1.07$, 0.78) (all $p < 0.001$); they disagreed however on the accountability of the night house staff ($\mu = 0.83$, -1.06 ; $p < 0.001$). Residents rated their personal life, physical wellbeing and social life positively for the day floor experience ($\mu = 0.58$, 0.62 , 1.09 ; all $p < 0.001$) and neutrally for the night floor experience ($\mu = -0.19$, 0.02 , 0.07 non significant p). While nurses had mostly statistically neutral rating for their interaction with the night house staff, they rated the night house staff familiarity with the patients' cases and management plans negatively ($\mu = -0.77$; $p < 0.001$).

CONCLUSIONS: Residents have more positive perceptions of the impact of the night float system on clinical, educational and personal outcomes than attendings and nurses. The night house staff communication with patients and their training quality should be the major areas of concern for medical educators in a residency program with a night float system.

CONSULTATIVE MEDICINE TRAINING IN INTERNAL MEDICINE RESIDENCY: A SURVEY OF TRAINING PROGRAMS. S. Gajula¹; J.D. Rogers¹; L.J. Cation¹. ¹University of Illinois College of Medicine, Peoria, IL. (Tracking ID #136038)

BACKGROUND: The 2004 Residency Review Committee (RRC) program requirements for internal medicine (IM) state: "Residents should have sufficient knowledge and experience to act, under supervision, as consultants to physicians in other specialties. It is desirable that the principles involved in consultation be included in the core conference series." No additional guidelines for consultative medicine (CM) training are offered, and there is little published literature in this area. This IRB-approved study attempts to add to the knowledge on CM training by determining the methods programs use to meet the RRC requirements.

METHODS: A 22 item survey regarding CM training was mailed to all IM program directors in the continental U.S. The survey asked what methods their

program used to train its residents in CM, and specifically whether the program had a separate general medical consult (GMC) rotation. For programs with such a rotation, questions regarding its structure and content were asked.

RESULTS: The preliminary response rate was 33% (129 of 386 programs surveyed). Methods used commonly to meet RRC requirements for CM training include didactic lectures (85%), a written curriculum (63%), and having residents perform consultations in outpatient continuity clinics (76%), inpatient general medical services (71%) and on subspecialty rotations (85%). Less than half (45%) of programs utilize case-based discussions. Sixty-five percent of programs have a separate GMC rotation. This rotation is required in 86% of these programs and elective in only 13%. Approximately half (51%) of GMC rotations are performed only on an inpatient service, while the other half (49%) include both an inpatient service and outpatient clinic. None are performed solely in an outpatient clinic. In only 15% of programs with GMC rotations are PGY-1 residents assigned to this service; the remainder use only senior level residents on this service. For programs that do not have a separate GMC rotation, the most common reasons (non-mutually exclusive) cited were: the perception that residents have adequate experience in consultative medicine on other rotations (60%); insufficient demand for medical consults to justify a separate rotation (57%); and inadequate number of residents to staff a separate rotation (45%). Only 36% of programs without a separate rotation reported an inadequate number of faculty for such a rotation. Fifty-two percent of responding programs strongly agreed that consultative medicine skills are important for general internists, but only 26% strongly agreed that their graduates received adequate training in consultative medicine.

CONCLUSIONS: Consultative medicine training is a required part of residency education, yet no specific guidelines exist to assist residency programs in how to meet this requirement. Our preliminary survey results show that most programs use didactic lectures and a written curriculum in consultative medicine. The majority of responding programs also require their senior residents to rotate on a separate general medical consultation service. Further study is needed to help determine the best methods to train IM residents to be effective medical consultants.

CONTINUING MEDICAL EDUCATION ONLINE: WHAT BARRIERS DO PROVIDERS IN PRACTICE PERCEIVE? T. Houston¹; R. Shewchuk¹; L.L. Casebeer¹; J.J. Allison¹; C.I. Kiefe¹. ¹University of Alabama at Birmingham and the Birmingham VA Medical Center REAP, Birmingham, AL. (Tracking ID #135955)

BACKGROUND: The increasing number of online courses for continuing medical education (CME) may enhance providers' ability to keep up-to-date. However, barriers to accessing CME online may exist. We conducted formative research to assess to primary care providers' perceived barriers to using online CME.

METHODS: Nominal Group Technique (NGT) meetings were convened with primary care providers to identify potential barriers to using the internet for CME delivery. The NGT is a highly structured, multi-step consensus building procedure that is often used in formative research to elicit and prioritize a group's responses to a specific question. Participants were each given 6 votes to assign anonymously to three barriers that they personally considered to be most significant. For this study two groups, each consisting of 12 community based providers, recruited from 3 states, participated in online, virtual NGT meetings. In addition, a group of 7 VA providers participated in a traditional, in-person NGT meeting. Participants in all three sessions were asked to respond to the same question: "What are the barriers to using the Internet for CME?"

RESULTS: The combined group of 24 providers who participated in the two on-line NGT meetings identified a total of 22 non-redundant barriers to using the internet for CME. The voting revealed considerable agreement in how the barriers were perceived. Of the 142 votes that could be allocated by providers in the two groups, 82 were assigned to four barriers: 1) Insufficient time; 2) Inadequate access to necessary computer resources; 3) Inadequate knowledge/Lack of confidence in using computers and internet technology; and 4) Concerns about the reliability of internet CME delivery. When further asked to define insufficient time, providers reported inadequate blocks of uninterrupted time, download time of web too long/lack of high-speed connection, and too many distractions in available time. General agreement also was observed in how the VA providers prioritized the 26 barriers that they identified. Of the 42 votes available to participants in the traditional in-person meeting, 35 were assigned to three barriers: 1) Insufficient time; 2) Concerns about the reliability of internet CME delivery, and 3) Costs associated with internet CME delivery relative to other forms of CME.

CONCLUSIONS: Primary Care Providers consistently reported insufficient time as the primary barrier to accessing online continuing education. Although prior research indicates that available computer technology and Internet access in primary care practices has greatly increased, technology concerns were also prominent. Online CME course developers should consider provider workflow and technology limitations as they craft their educational curricula.

CULTURAL COMPETENCY TRAINING OF INTERNAL MEDICINE PHYSICIANS DURING MEDICAL SCHOOL AND RESIDENCY. E.C. Manjarrez¹; C. Hernandez²; C.M. Bright³. ¹University of Miami, Miami, FL; ²University of Florida, Gainesville, FL; ³Duke University, Durham, NC. (Tracking ID #135854)

BACKGROUND: The Institute of Medicine's 2002 report, "Unequal Treatment: Confronting Racial and Ethnic Disparities in Healthcare", recommended that all U.S. medical schools begin cultural competency training for physicians as one of the ways to improve healthcare for ethnic minorities in the U.S. There is little

descriptive data available documenting when, how much, and which formats are used in medical school and residency. In addition, there are no surveys of physicians' opinions of such training on the benefit to their practices.

METHODS: The CCTOP Study (Cultural Competency Training and Outcomes in Patients) is an ongoing multicenter, cross-sectional study of patients' and physicians' opinions on the interaction between culture and healthcare. Preliminary data from six of the centers is presented here. Resident and attending internal medicine physicians self administered a 93 item survey. The questions asked covered format, duration, and timing of cultural competency training during medical school, residency, and the postgraduate years. Physicians were also asked about their opinions regarding the benefits of their training to their patients as well as their interest in attending a cultural competency training course for CME credit. Descriptive statistics were compiled.

RESULTS: A total of 175 internal medicine physicians (149 residents and 26 attendings) participated in our survey. During medical school, 79% received cultural competency training in the format of a lecture, 57% participated in small group workshops, and 36% participated in a cultural immersion experience. During residency, 43% received training in the format of a lecture, 24% participated in small group workshops, and 23% participated in a cultural immersion experience. The most commonly reported timing of cultural competency training was during the first two years of medical school (54 and 62%) respectively i.e. in the pre-clinical years. Seventy one percent of internists surveyed believe that cultural competency training improves their ability to care for patients. Seventy five percent of internists surveyed would attend continuing medical education on cultural competency training.

CONCLUSIONS: Medical schools use a variety of formats to teach physicians to become culturally competent. Physicians surveyed believe there is a benefit to such training. Further research by the CCTOP Study and others is aimed at evaluating the effect of cultural competency training on hard outcomes that benefit patients as well as identifying the best timing and format of cultural competency education.

CULTURAL COMPETENCY: COMPARING KNOWLEDGE, ATTITUDES, AND TRAINING OF INTERNATIONAL AND US MEDICAL GRADUATES. A. Sawh¹, J. Bussey-Jones², C.M. Bright³, C. Hernandez⁴. ¹Salem Veterans Affairs Medical Center & University of Virginia School of Medicine, Salem, VA; ²Emory University, Atlanta, GA; ³Duke University & Durham Veterans Affairs Medical Center, Durham, NC; ⁴University of Florida College of Medicine, Gainesville, FL. (Tracking ID #135990)

BACKGROUND: The 2002 Institute of Medicine Report, "Unequal treatment: Confronting Racial and Ethnic Disparities in Health Care", recommended cultural competence training as a means of improving health care disparities among minorities. Most US medical schools provide cultural competency training although curricula vary considerably. In 2003-2004, international medical graduates (IMGs) accounted for 26% of nearly 100,000 resident physicians in ACGME-accredited and combined specialty programs. Sparse data are available on the cultural competency, knowledge, medical school training and attitudes towards cultural competency of IMGs. This study seeks to examine these issues and compare results to US graduates (USGs).

METHODS: The Cultural Competency Training and Outcomes in Patients (CCTOP) study is an ongoing multicenter, cross sectional study of primary care outpatients and their physicians regarding culture, bias and healthcare. The "Inventory for Assessing the Process of Cultural Competence Among Healthcare Professionals" (IAPCC) scale is a validated measure of cultural competency. This score dichotomized physicians into "culturally competent" (higher scores) and "culturally aware" (lower scores). Questions of specific cultural knowledge, medical school diversity training (format and concepts addressed), and personal attitudes towards cultural competency were also asked. Bivariate analyses were performed.

RESULTS: Data from six of eleven participating centers is reported. 174 physicians responded, 138(79%) USGs and 36(21%) IMGs. 29.5% USGs and 25% IMGs were "culturally competent" on the IAPCC scale with 70.5% and 75% "culturally aware" respectively (p=0.61). Notably, 39.6% IMGs versus 70.6% USGs identified African-American distrust of the health care system as a consequence of the Tuskegee Study (p=0.001). Regarding format of diversity training, IMGs reported less lectures (69% IMGs vs. 82% USGs, p=0.83) and workshops (47% vs. 60%, p=0.15) but greater cultural immersion experience (53% IMGs compared to 32% USGs, p=0.022). Only 41.7% IMGs versus 63.8% USGs felt that their medical school training was adequate in addressing language (use of interpreters, nonverbal communication, word meanings) (p=0.016). IMGs, however, were more likely to agree that training on the healthcare impact of socioeconomic status was adequate (94% vs. 70%, p=0.003). There was no statistical difference on how well other concepts were addressed in medical school (specific cultural content, racism/stereotyping, access to healthcare or the doctor-patient relationship). Both IMGs and USGs agreed that diversity training is important (94% each) and improves patient outcomes (85% IMGs vs. 78% USGs). However, IMGs reported greater willingness to attend cultural competence/diversity training if offered for CME credit (88% vs. 71%, p=0.034).

CONCLUSIONS: Significant self reported differences in format of training, adequacies of cultural/diversity concepts taught and core cultural knowledge exist between IMGs and USGs. However, there was no statistical difference in cultural competency scores. The magnitude of IMG representation in the US workforce demands that these variables be examined as relates to patient perceptions and healthcare outcomes. Additionally, increased IMG interest in attending cultural competency CME suggests willingness to adapt to their new environment. Educators should take advantage of this motivation.

CULTURAL ISSUES IN HEALTH CARE: AN EVALUATION OF AN INTRODUCTORY CURRICULUM FOR FIRST YEAR MEDICAL STUDENTS. S.G. Chheda¹, S. Dotti¹, L. Zakowski². ¹University of Wisconsin-Madison, Madison, WI; ²University of Wisconsin-Madison, Deerfield, WI. (Tracking ID #136244)

BACKGROUND: The Institute of Medicine in 2002 proposed that healthcare provider factors contribute to health disparities and recommend that medical educators implement and evaluate curriculum to address these issues. We developed a two-hour small group curriculum for first year medical students to introduce the concepts of health disparities and to discuss differences between patient, physician and health care cultures. Students viewed two videotapes and discussed pre-determined questions in groups of 6 students and 1-2 faculty members. We also conducted a brief faculty development session prior to the small group sessions and provided a faculty facilitation guide to the pre-determined questions.

METHODS: We used a 21-item questionnaire to evaluate the student curriculum before and after the group, including knowledge (9 questions), comfort with cultural issues in healthcare (6 questions), and satisfaction (6 questions on post-intervention questionnaire only). Half of the students (N=68) completed the questionnaire prior to the curriculum and all of the students (N=147) completed it afterwards. We used a similar questionnaire to evaluate the faculty development session, and all faculty (N=22) who attended the session completed the questionnaire prior to the session and after facilitating student small groups.

RESULTS: Students improved significantly on seven of the nine knowledge items (Chi square, p<.05). They also reported a significant increase in comfort on four of six items (Paired-Samples t-test, p<.02). These included: "discussing biases I have about people of different backgrounds than mine"; "interacting with a patient who is mistrustful of me and/or the healthcare system"; "caring for a patient who chooses to use a complementary therapy over one I have recommended"; and "suggesting a patient changing a cultural practice that may impair one's health". There were no significant differences in post-test scores between students who did or did not complete the pretest (knowledge items: Mann-Whitney U, ns; comfort items: independent samples t-tests, ns). Overall 78% of students rated their session as going "very well" on a 3-point Likert scale. All individual components of this curriculum were rated highly. Faculty knowledge improved significantly on one of eight knowledge items. (Chi square, p=.03). Faculty reported a significant increase on five of six comfort-in-teaching assessment items (Paired-Samples t-test, p<.05).

CONCLUSIONS: A two-hour small group student curriculum consisting of video viewing and discussion of pre-determined questions was well received and improved students' knowledge and comfort levels regarding cultural issues in healthcare. A brief faculty development session followed by facilitation of small groups increased faculty comfort levels regarding teaching about these issues.

DEFINING PROFESSIONALISM FOR CLINICAL TEACHERS. G.C. Lamb¹, L. Currey¹, K. Marcdante¹, J. Morzinski¹, D. Simpson¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #136139)

BACKGROUND: Professionalism in medicine has emerged as a major area of emphasis in recent years. Brought to the forefront by the Medical Professionalism Project and the Charter on Medical Professionalism in 2002, the focus on professionalism pervades much of the activities in academic medicine. In particular it is one of six competencies mandated by the ACGME Outcomes project. Despite this emphasis, there is very little literature on the professionalism associated with teachers in the clinical setting.

METHODS: Subjects consisted of 20 physician educators from medicine, family medicine and pediatrics participating in a 5-month faculty development course on teaching. As a requirement of the course, participants completed a weekly "Critical Teaching Incident"(CTI) report and submitted that report electronically to the program coordinator. Each CTI contained a brief description of a teaching situation, the emotional response evoked in the teacher (e.g., anger, elation, frustration), the intended teaching objectives (framed specific to ACGME competencies), a critical analysis of interaction and a description of the elements of professionalism as an educator that were present (or should have been) in the interaction. During one course session, participants completed a worksheet in which they provided a specific behavioral example of teacher professionalism and discussed describe which, if any, of the ABIM professionalism elements (e.g., integrity, excellence, accountability, respect, duty) were present in the example. The professionalism attributes from the weekly CTI's and session worksheets were collated and subject to content analysis independently by two of the authors (GL and LC) to identify primary themes. The themes generated were compared and a common list generated by consensus among three authors GL, LC and DS. The consensus list was then edited independently for clarity and consistency by the remaining authors.

RESULTS: 91 specific behaviors were identified across the 6 elements of professionalism including 15 under Integrity/Honor, 18 under Excellence, 10 under Accountability, 11 under Altruism, 19 under Respect and 18 under Duty. Upon further analysis, these professionalism behaviors were organized into three categories: responsibilities to one's self, responsibilities to the learner, and responsibilities to the patient. Specific examples included: (under Integrity/Honor): self - "Acknowledge one's own limitations"; learner - "be fair, consistent and nonjudgmental"; and patients - "Be honest and truthful with patients". Responsibilities to patients were emphasized over other responsibilities in most reports.

CONCLUSIONS: In an era of greater emphasis on professionalism in medicine, it is important that there be an explicit consideration of a professional code of conduct for clinical teachers. We were able to identify specific behaviors relevant to teaching in the clinical setting, organized within the elements of the ABIM

professionalism elements. These findings contribute to a broader discussion regarding professionalism in medical education. Our work was limited to the perceptions of relatively junior faculty in a single institution, but the methods and structure can be used to determine the generalizability of these findings to a broader spectrum of settings and clinician educator experience levels.

DESIGNING A CULTURAL COMPETENCE OSCE EXPERIENCE: LESSONS LEARNED FROM STRUCTURED INTERVIEWS WITH MEDICAL STUDENTS. A.R. Green¹; E. Miller²; J.R. Betancourt²; D. Hirsh³; E. Krupat⁴; R. Wilson¹; W.C. Taylor¹. ¹Beth Israel Deaconess Medical Center, Boston, MA; ²Massachusetts General Hospital, Boston, MA; ³Harvard Medical School, Cambridge, MA; ⁴Harvard Medical School, Boston, MA. (Tracking ID #133583)

BACKGROUND: Cross-cultural interviewing skills are challenging to teach and to evaluate. Medical schools are starting to develop standardized patient (SP) experiences (or OSCEs) for this purpose, but little has been published describing students' perspectives on these. We designed an OSCE station focused on cross-cultural communication (ccOSCE) and interviewed students about the experience to learn how best to adapt this tool for cc education.

METHODS: We recruited a convenience sample of 22 second-year Harvard medical students by e-mail. Participants were 64% female and 21% underrepresented minority. All had completed the ccOSCE and 6 other OSCE stations in the past 4 weeks. The ccOSCE portrays 58 year-old Latina woman with poorly controlled hypertension in an outpatient setting. Among other skills, SPs and faculty observers evaluated students' ability to discuss the SPs' non-adherence to medication and to explore potential reasons. These included: different beliefs about hypertension (related to stress); pattern of medication use (when feeling anxious); and use of alternative herbal remedies. Two investigators conducted semi-structured interviews with students about their experiences with the ccOSCE. Three investigators developed a list of provisional themes from a random subset of five interview transcripts. They reviewed the remaining transcripts and added additional themes as these emerged, validating them through an iterative process of multiple reviews.

RESULTS: This study focuses on challenges involved in designing a ccOSCE. We organized themes into 4 categories: learning goals, logistical issues, faculty feedback, and SPs. Many students commented that the learning goals were too complex, unclear, and could promote stereotyping, but liked the focus on non-adherence. Logistical issues included: "separateness" of the ccOSCE, the suggestion of spreading cc skills over several stations, the perception of a "trick case," and confusion about whether to counsel the patient. Most students said faculty feedback was helpful in discussing their thought process and specifying issues they did or did not explore. Some felt it was non-specific, based on unclear expectations, and stressed medical over cc issues. Students considered the SPs realistic, but inconsistent in how easily they gave up information. Many felt that SP feedback was unhelpful (vaguely complimentary and sparse). Some found it useful, encouraging students to speak in clear terms and ask appropriate and sensitive questions.

CONCLUSIONS: Designing a ccOSCE involves many conceptual and logistical challenges. Our results demonstrate a tension between teaching cross-cultural skills as a focused experience (potentially marginalizing the topic and promoting stereotypes) and spreading it across stations (increasing the complexity of the learning goals and requiring students to integrate cc and other skills). Learning goals should be clear, concise, and effectively communicated to both faculty and SPs so their feedback can be standardized and specific.

DETERMINANTS OF MENTAL HEALTH AMONG INTERNS. M.R. Heller¹; E. Ziv¹; S. Wail¹; A. Vidyarthi¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #133940)

BACKGROUND: Recent studies have demonstrated an association between increased work hours and decreased intern productivity, and between increased burnout among residents and decreased quality of patient care. However, there has been little focus on factors which contribute to positive aspects of mental health during internship. We sought to identify such predictors, and hypothesized that interns who felt "supported" by their residency program (i.e. identified mentor relationships or support groups) would score higher on a positive affect scale.

METHODS: We surveyed 163 first year medicine, pediatric, psychiatry, family practice, and surgery residents at a large academic center. Outcomes were measured using the Positive and Negative Affect Scores (PANAS), a reliable and validated survey instrument used widely in mood assessment. We used linear regression models to adjust the positive affect (PA) scores for gender, race, marital status, and number of hours worked in the week prior to completing the survey.

RESULTS: Among all interns, there was a 55% response rate. The mean PA score was 31.0 ± 6.2 . Interns who reported that they felt part of a team "very often" or "all the time" had a multivariate adjusted PA score 3.6 (95% CI 0.5-6.8) points higher than those who felt part of a team "never," "a few times," or "fairly often" over the past month. Using the same frequency scale, interns who reported feeling a sense of camaraderie at work more frequently had a multivariate adjusted PA score 4.9 (95% CI 2.0-7.8) points higher than other interns. Finally, interns who reported more frequently laughing with colleagues had a multivariate adjusted PA score 4.1 (95% CI 1.2-6.9) points higher than others. Surprisingly, PA scores were NOT significantly different between interns who identified having a mentor or support groups as part of their program and with those who did not.

CONCLUSIONS: Positive well-being among interns across specialties at this academic training institution is significantly associated with "teamwork" factors. Surprisingly, formal structures like mentors or support groups, do not contribute to positive affect.

DO ATTENDING PHYSICIANS, NURSES, RESIDENTS, AND MEDICAL STUDENTS AGREE ON WHAT CONSTITUTES MEDICAL STUDENT ABUSE? E.H. Wu¹; D.M. Elnicki¹; P.E. Ogden²; The Pams Investigators¹. ¹University of Pittsburgh, Pittsburgh, PA; ²Texas A&M University, College Station, TX. (Tracking ID #132221)

BACKGROUND: National surveys administered to graduating medical students by the Association of American Medical Colleges (AAMC) continue to report abuse during medical training. It is not known if attending physicians, nurses, residents, and medical students agree on what constitutes abuse or whether they perceive abuse to be of similar severity. It is also not known what factors may influence perception of abuse.

METHODS: From June through September 2004 at 13 medical schools, internal medicine attending physicians, nurses, internal medicine residents, and third- and fourth-year medical students viewed five vignettes, previously filmed using actors and scripts depicting potential abuse of medical students. Each vignette contained one or more abusive behaviors identified in previous studies, including belittlement, performing inappropriate tasks, sexual harassment, and ethnic stereotyping. For each vignette, respondents anonymously answered questions on the nature of the encounter, including whether it constituted abuse and the severity (1=low, 7=high) of the abuse. Demographic information and past personal experience of abuse were collected. Analysis was done using ANOVA and Chi-square.

RESULTS: Of the 237 respondents, 52.2% were female, and there were similar proportions of attending physicians (24.5%), nurses (19.1%), residents (32.3%), and students (24.1%). At least 80% of each response group felt that three vignettes represented medical student abuse. Less than 25% of each group felt that the fourth vignette represented abuse. For the fifth vignette, in which a student is criticized by an attending physician for a poor patient presentation, 57.4% of attending physicians considered it abuse, compared to 21.4% of nurses, 40.8% of residents, and 28.3% of students. Attending physicians were more likely than nurses (RR 1.89, 95% CI 1.32-2.69) and students (RR 1.79, 95% CI 1.22-2.61) to consider this vignette as representing abuse. For one vignette in which an attending physician criticizes a student during rounds, students rated the severity significantly lower (4.38) compared to the other groups (5.27, 5.16, 5.19, all $p < .05$). Respondents were more likely to perceive abuse in three of the vignettes if they had past personal experience of abuse as a medical student (all $p < .05$). Perceptions of abuse did not differ by respondents' race or gender.

CONCLUSIONS: Attending physicians, nurses, residents, and medical students generally agree on what constitutes medical student abuse, but attending physicians may perceive more behaviors as abusive. Medical students may perceive negative feedback as less abusive. Those who felt they experienced abuse as a medical student may be more likely to perceive behaviors as abusive.

DO RESIDENTS RETAIN THEIR TEACHING SKILLS IN THE LONG RUN? A. Vanka¹; M. Ayier¹; L. Meyer¹; G. Lombard¹. ¹University of Illinois at Peoria, Peoria, IL. (Tracking ID #135782)

BACKGROUND: Resident physicians play a crucial role in the education of medical students. A recent study revealed a significant improvement in the teaching skills of generalists residents who participated in the residents' as teachers curriculum as assessed by student raters. However there is limited data regarding the long-term impact of the residents as teachers' workshop. A mandatory Residents as Student Teachers (RaST) program was implemented at our institution five years ago. Incoming residents from ten residency programs and Geriatric fellows participate in this workshop. The workshop focuses on topics such as orienting learners, microteaching skills, and feedback and evaluation. The purpose of this follow-up study was to assess the long-term impact of the RaST program on the residents' perceptions towards teaching and their teaching skills.

METHODS: A pre-RaST and post-RaST questionnaire was administered to all participating residents prior to and immediately following the one-day mandatory RaST program from 1999-2004. During December of 2004, a follow-up RaST questionnaire was sent to all participants. The questionnaires were comprised of 34 questions addressing topics such as ability to teach, use of clinical micro-teaching skills, preparedness to teach, setting expectations, and providing effective feedback. The residents completed the survey using a 5-point Likert scale. In addition, the follow-up questionnaire asked the residents to identify the three most important teaching skills they learned from the RaST workshop. Quantitative data was analyzed using SPSS. Descriptive statistics, T-test, and one-way ANOVA were used where appropriate.

RESULTS: 186 residents completed the pre-RaST, 192 completed the post-RaST, and 35 have completed the follow-up RaST questionnaire thus far. The time interval between the workshop and follow-up RaST ranged from 4 to 48 months. Residents perceived a statistical increase in their confidence regarding teaching ability, ability to set goals and expectations, preparedness to teach, and ability to understand the difference between feedback and evaluation at the completion of the RaST workshop. A significant increase in residents' ability to assume control of the team and explain each role of the team member to the team was noted in the follow-up RaST compared to the post-RaST (See Table 1). **CONCLUSIONS:** After completing the workshop, the residents are able to understand the difference between feedback and evaluation, incorporate micro-teaching skills into their daily teaching, assume leadership, and set expectations to their learners. The RaST program has a positive long-term impact on the residents' approach to teaching medical students.

DO WE PRACTICE WHAT WE PREACH? LOOKING FOR PREREQUISITES OF EVIDENCE-BASED PRECEPTING IN REAL-TIME AMBULATORY CARE. J. Tilburt¹; R. Mangrulkar²; N. Siddiqui³; S.D. Gooldd⁴; J.A. Carrese⁴. ¹Johns Hopkins University,

Table 7

Question	Mean			P-value
	Pre-RaST	Post-RaST	Follow-up RaST	
I am well prepared to teach	3.20	3.88	3.77	<0.0005
I have confidence in my teaching ability	3.40	3.78	3.62	<0.0005
I assume control of the team	3.02	3.35	3.91	<0.0005
I explain the role of each team member at the start of each rotation	2.77	3.39	3.91	<0.0005
Solicit learning goals from members of the team at the start of the rotation	2.65	3.59	3.29	<0.0005
Try to teach general principles so that the underlying principle is clearly communicated	3.64	4.01	3.88	<0.0005
Understand the difference between feedback and evaluation	3.75	4.19	4.17	<0.0005

Baltimore, MD; ²University of Michigan, Ann Arbor, MI; ³Case Western Reserve University, Cleveland, OH; ⁴Johns Hopkins Medical Institutions, Baltimore, MD. (Tracking ID #136057)

BACKGROUND: Many residency programs in internal medicine implement curricula in Evidence-Based Medicine (EBM), yet we know little about evidence-based behaviors in real-time clinic precepting encounters. The aim of this study was to describe the degree to which five basic prerequisite components of evidence-based precepting were present in internal medicine resident-attending encounters in resident continuity clinics.

METHODS: Using observational qualitative methods, we collected field notes and audio taped 95 resident-preceptor encounters (70 residents, 29 preceptors) in 4 ambulatory site mentor rooms affiliated with one large internal medicine training program that teaches EBM. We also asked residents to identify their most important question for each patient case that corresponded to that preceptor encounter. Clinic mentor rooms served as the setting for our study. Multiple doctors convene and discuss patient care in mentor rooms, often simultaneously. Due to challenges in audio taping in this setting, we selected the highest quality transcripts for further analysis (43). We then evaluated transcripts of each encounter using five parameters that represent basic prerequisites of evidence-based precepting: (1) presence of an explicit clinical question in the interaction (Y/N); (2) presence of an explicit EBM process (Y/N) – noticeable effort to ask, acquire, appraise, and apply scientific evidence; (3) quality of the precepting interaction, – preceptor's attention to resident concerns, quality care, and fostering of education; (4) resident ability to articulate a focused, detailed, specific clinical question when invited; and (5) relevance of the dialogue to the question identified by the resident as most important. Parameters 3–5 were scored on a 4-point scale (poor, fair, good, excellent). Two investigators scored a subset of transcripts for all five parameters (% initial agreement: 73, 86, 90, 94, 72 respectively; initial kappa: .49, .44, .29, .62, 0 respectively) and then together resolved scoring discrepancies. One investigator then scored all 43 transcripts. **RESULTS:** Of 43 analyzed encounters, 19 contained some form of explicit clinical question, while only 4 demonstrated an evidence-based process. Despite this lack evidence-based process, we rated the overall quality of precepting as "good or excellent" in 17 encounters, 17 as "fair," and 9 encounters as "poor." We rated the resident's ability to articulate a quality clinical question as follows: 21 encounters as "good or excellent," an additional 10 as "fair," and the remaining 12 as "poor." We also found that most transcripts contained dialogue that was highly congruent to the central question of the encounter as identified by the resident, reflected in ratings of "good or excellent congruence" in 33 of 43 transcripts.

CONCLUSIONS: For the observed encounters, evidence-based process rarely occurred explicitly in real-time precepting. However, doctors verbalized some type of explicit clinical questions in nearly half of precepting encounters. Furthermore, when prompted, residents can formulate at least fair quality clinical questions, and preceptors can conduct relevant dialogue to address resident questions. Therefore, in the program described many prerequisite components of evidence-based precepting seem to already exist even if the EBM process itself occurred rarely. If recognized as such, clinical questions could serve as triggers for initiating evidence-based process in precepting. Future interventions to improve evidence-based process might consider both prompting residents to articulate more structured clinical questions and training faculty to detect more subtle clinical questions in real-time.

DOES AN ADOLESCENT MEDICINE WORKSHOP USING STANDARDIZED PATIENTS INCREASE KNOWLEDGE AND IMPROVE SKILLS? A.R. Hoellein¹, C.A. Feddock¹, J.F. Wilson¹, S.A. Haist¹. ¹University of Kentucky, Lexington, KY. (Tracking ID #134169)

BACKGROUND: Medical students rarely have the opportunity to learn and practice the tactful history-taking and delicate counseling skills critical for the health of adolescent patients. The purpose of this study is to determine the impact of an adolescent medicine (AM) workshop (WS) using standardized patients (SP) on knowledge and clinical skills of third-year medical students.

METHODS: A four-hour AM WS was developed as part of a new curriculum for a required third-year medical school four-week primary care internal medicine clerkship. Along with three other novel interactive SP WS, the AM WS was randomized for delivery to one-half of the rotational groups. The WS is designed for one-on-one student-SP interaction in four SP cases representing different clin-

ical challenges (taking a sexual history, depression/suicide screening, risky behavior screening and counseling, and smoking cessation counseling). A faculty preceptor facilitates group discussion of the issues raised and consideration of sensitive approaches to the problems instigated by the cases. Participating students are also provided a 26-page AM reference. All students in every rotation group are assigned AM readings. At the end of the four weeks, all students take a 100-item written exam (six AM questions) and nine-station SP exam (one AM station) which includes open-ended written exercises after the SP encounter. Scores on the written exam AM items, AM SP checklist, and AM open-ended clinical question of workshop participants and non-participants were analyzed with simple means, standard deviations, and multiple regression approaches. **RESULTS:** The AM WS has been delivered to three of the first five rotation groups during the 2004–2005 academic year. Twenty-five students participated in the workshop and 17 did not. Workshop participants performed significantly better than non-participants on the AM SP checklist (89.3 ± 5.0 vs. 85.8 ± 4.4 , $F=5.5$, $p=.02$) and on the post-encounter written exercise ($85.0 \pm 4.8\%$ vs. $81.2 \pm 2.5\%$, $F=8.8$, $p=.005$) with a trend of better scores on the written exam AM items ($77.3 \pm 14.3\%$ vs. $69.6 \pm 21.4\%$, $F=2.0$, $p=.17$).

CONCLUSIONS: Even at this early stage of analysis, students participating in a four-hour SP workshop display superior AM clinical skills as assessed by a SP clinical exam and open-ended written exercise. All students can master the assigned readings for the purposes of the written exam, but in lieu of practice with actual patients, simulation with SPs appears to be an ideal medium. These findings lend additional support to the theory that there are unique aspects to caring for adolescent patients which might be better taught using an interactive pedagogy.

DOES DISASSOCIATING GRADES FROM COMPETENCIES REDUCE GRADE INFLATION? H.E. Harrell¹, J. Rodriguez¹. ¹University of Florida, Gainesville, FL. (Tracking ID #135204)

BACKGROUND: The University of Florida College of Medicine transitioned to a competency-based evaluation system in 1997 while also maintaining traditional grades. Our competency system has four levels of competence linked to descriptive anchors. Advantages of criterion-referenced competencies include improving discrimination in student assessment by linking it to observable behaviors (which theoretically should reduce inflated evaluations) and being able to track student growth throughout training. Most clerkship directors rely heavily on these competency assessment forms when assigning grades by artificially converting the ordinal competency system into a pseudo-numerical continuum. (A number is assigned to each competency level; evaluations are averaged and converted to a grade.) Not only is this statistically unsound but it may promote inaccurate competency assessment. Furthermore, on the Medicine clerkship we frequently observed evaluators disregard descriptive anchors in favor of selecting the competency level they thought correlated with the desired grade. We hypothesized that explicitly separating the competency evaluation from the grade calculation would promote more discriminating competency assessments. This would manifest as a broader use of the assessment scale and higher student competence as the year progressed.

METHODS: The Medicine Clerkship evaluates students' inpatient ward performance using a faculty facilitator who gathers the student's team members together at the end of a rotation and helps them come to consensus on the student's performance using a form that assesses the six core ACGME competencies. The form provides descriptive anchors for each competency level that are based on observable behaviors. In 2002–2003, the facilitators instructed all evaluators that competency ratings would not be used when calculating student grades. Evaluators were encouraged to focus solely on the descriptive anchors regardless of the time of year when assessing a student's competency level. At the end of the session the team had the opportunity to suggest a grade for the student that could take into account other factors that affect performance (like time of year, patient complexity). Previously, evaluators were encouraged to use the descriptive anchors but they were not asked to suggest a grade as the grade was linked directly to these evaluations. Competency ratings were compared overall and for each competency category using paired t-tests. Grade distributions were compared using a Chi-square test.

RESULTS: There were no significant differences in competency evaluations between the two years for combined or individual competency categories. However, the grade distribution shifted significantly toward higher grades ($p<0.001$) with greater than two-thirds of students receiving a grade recommendation of "A" after evaluators started recommending the grade.

CONCLUSIONS: Focusing evaluators on descriptive anchors and away from grades when assessing student competence did not result in better discrimination when using the competency assessment tool. Allowing evaluators to recommend grades magnified grade inflation. This suggests that evaluators are not directly linking student behavior to their global assessment of student performance. In fact, competency assessments that are linked to specific behaviors may help reign in some grade inflation. But other strategies to improve discrimination of student performance need further exploration.

DOES INTERNAL MEDICINE RESIDENT BURNOUT AFFECT SELF-REPORTED QUALITY OF CARE? R.K. Gopal¹, F. Carreira¹, W.A. Baker², J.J. Glasheen¹, T.J. Miyoshi¹, L.A. Crane³, A.V. Prochazka¹. ¹University of Colorado Health Sciences Center, Denver, CO; ²Denver Health and Hospital Authority, Denver, CO; ³Department of Preventive Medicine and Biometrics, University of Colorado Health Sciences Center, Denver, CO. (Tracking ID #135150)

BACKGROUND: Resident burnout is a major problem. Resident burnout association with self-reported sub-optimal quality of care has been reported at one academic center.

METHODS: We administered a postal survey of internal medicine residents at the University of Colorado Health Science Center, in May 2003 and May 2004. The survey contained the Maslach Burnout Inventory, a 22-item questionnaire organized into three subscales: emotional exhaustion (EE, high score ≥ 27), depersonalization (DP, high score ≥ 10), and personal accomplishment (PA, high score ≤ 33). We defined burnout as high EE or DP. We assess quality of care using hypothesis driven questions and questions developed by Shanafelt et al. Some residents may have responded both years.

RESULTS: The combined response rate for both years was 80.5% (227/282). One hundred and thirty-two residents (58.1%) met criteria for burnout. Demographics (age, post graduate year, training track, previous rotation) were similar between those residents with or without burnout except a trend towards more women without burnout (63.2% vs. 31.6%, $p=0.059$). More residents with burnout believe that the residency program views their purpose as patient care more than education (50.0% vs. 33.3%, $p=0.015$). Confidence doing bedside procedures (22.7% vs. 29.5%, $p=0.247$) and overall confidence in making important clinical decisions (26.0% vs. 26.3%, $p=0.903$) was similar among burned-out and not burned-out residents. Residents with burnout reported working 4.2 more hours per week (73.7 vs. 69.5, $p=0.083$). Both burned-out and non-burned-out residents reported spending similar amount of time of clinical activities ($p=0.559$) but burned-out residents reported spending more time doing clerical work ($p=0.036$). Burned-out residents reported when doing cross cover, deferring more decisions to the next day (61.4% vs. 19.1%, $p<0.001$) and deferring more family meetings and discussions to the next day (59.6% vs. 28.9%, $p=0.001$) at least monthly. Burned-out residents report, "worry that signing out post-call, that I am not giving the best care" more often than non-burned out residents (58.6% vs. 23.4%, $p=0.017$). More burned out residents reported "discharging patients to make the service manageable" (26.5% vs. 8.4%, $p<0.001$), "not perform[ing] a diagnostic test because of a desire to discharge a patient" (6.1% vs. 1.1%, $p=0.083$), and making "treatment or medication errors that were not due to lack of knowledge or experience" (10.8% vs. 4.2%, $p=0.081$). Burned out residents also reported "not fully discuss[ing] treatment options or answering patient's questions" (15.2% vs. 3.2%, $p=0.002$) or "order[ing] restraints or medications without evaluating the patient" (6.1% vs. 4.2%, $p=0.556$). Residents with burnout reported paying "little attention to social or personal impact of illness on my patient" (13.6% vs. 5.0%, $p=0.039$) and having "little emotional reaction to the death of one of my patients" (9.8% vs. 1.0%, $p=0.009$). Finally, burned out residents felt more "guilty about how I treated a patient from a humanitarian standpoint" (5.3% vs. 0.0%, $p=0.043$).

CONCLUSIONS: We confirm that residents with burnout report more sub-optimal quality of care. Burned-out residents also appear to defer clinical actions while doing cross-cover that may delay care. Interventions to decrease resident burnout may improve quality of patient care. Shanafelt TD et al. Burnout and self-reported care in internal medicine residency program. *Ann of Intern Med.* 2002; 136:358-67.

EBM USER AND PRACTITIONER MODELS FOR GRADUATE MEDICAL EDUCATION: WHAT DO RESIDENTS PREFER? E.A. Aki¹; N. Maroun¹; G. Neagoe¹; G. Gordon²; H.J. SchüNemann¹. ¹University at Buffalo, Buffalo, NY; ²McMaster University, Hamilton, Ontario. (Tracking ID #132533)

BACKGROUND: A common belief is that not all residency trainees are interested in attaining an advanced level of Evidence-Based Medicine (EBM) skills, and those who are have limited time for applying these skills. We investigated the hypothesis that not all medical residents are interested in attaining an advanced level of skills and explored the underlying reasons for their preferences.

METHODS: We surveyed residents from diverse specialties at the University at Buffalo attending a residency core curriculum EBM master session. The self-administered questionnaire focused on residents' preferences for the EBM practitioner model (aiming at a high level of EBM skills) over the EBM user model (aiming at a more restricted level of EBM skills). We then conducted a focus group discussion with 12 of these residents regarding their preferences. Two investigators independently coded and analyzed the content of the discussion transcripts. They used both a deductive process to test the study hypothesis and an inductive process to identify emerging themes.

RESULTS: Of 85 residents responding to the survey (69% response rate), 55 (65%) preferred the EBM practitioner model. Residents in early stages of their training were more likely to prefer the practitioner model. The analysis of the focus group discussion suggested that preferences for the practitioner model result from a number of perceived advantages: the opportunity to gain advanced EBM skills during residency, the ability to later apply any of the two models, and the gain of independent thinking and greater self confidence in critical appraisal skills. The EBM user model had the key advantage of perceived reduced time requirements.

CONCLUSIONS: Our results suggest that EBM curricula should aim to focus on EBM practitioner models. One caveat is that residents' preference for the practitioner model may represent an idealistic view that dims with increasing experience of the exigencies of practice. Novel educational interventions allowing a choice between the two models may respond to the varying preferences of medical residents.

EFFECT OF MEDICAL EDUCATION AND LIFE EXPERIENCES ON THE CULTURAL COMPETENCE OF PHYSICIANS. R.V. Brown¹; A.A. Donato²; A. Sawh³. ¹George Washington University, Washington, DC; ²Reading Hospital and Medical Center, West Reading, PA; ³Salem VAMC, Salem, VA. (Tracking ID #135221)

BACKGROUND: The 2002 Institute of Medicine Report, "Unequal Treatment: Confronting Racial and Ethnic Disparities in Health Care", recommended cul-

tural competence training as a means of improving health care disparities among minorities. Currently, most U.S. medical schools provide some form of cultural competency curriculum for their medical students, although the type and amount vary considerably. There is no current consensus on the best way to teach cultural competence, as outcomes on these curricula have not been measured.

METHODS: The CCTOP study (Cultural Competency Training and Outcomes in Patients) is a multicenter, cross-sectional study of patients presenting for care at 11 outpatient clinics and their physicians' attitudes regarding culture, bias, and healthcare. Cultural competence was scored using the Inventory for Assessing the Process of Cultural Competence Among Healthcare Professionals (IAPCC), an 80-point validated scale. The IAPCC scores were dichotomized into respondents who were "culturally competent" (score of 60-74) and "culturally aware" (score of 40-59). In addition to the IAPCC, physicians were asked questions of specific cultural knowledge, attitudes, life experiences, and demographics. Life experiences were analyzed individually and as a part of an additive score of number of diverse life experiences.

RESULTS: Preliminary results are available from one hundred and seventy-five physicians (113 residents, 62 attendings) from the first six reporting sites. Bivariate analyses revealed that cultural competence was significantly associated with receiving training in third year of medical school ($p=0.02$) and in the format of a workshop ($p=0.03$). Having dated a person of a different racial/ethnic/cultural group ($p=0.05$), and female gender ($p=0.03$) were also associated with cultural competence. Regression analyses revealed an association between IAPCC score and racial/ethnic/cultural diversity in college, medical school, and residency, but not in childhood. In addition, we found an association between a higher number of diverse life experiences and higher scores on the IAPCC ($p=0.02$). We detected trends toward significantly higher IAPCC scores in respondents who received training post-residency ($p=0.08$), traveled outside of the U.S. for greater than one month ($p=0.06$), and who served in the military ($p=0.09$). Cultural competence was not associated with training received in the first two years of medical education, race of physician ($p=0.56$), socioeconomic status in childhood ($p=0.69$), number of years out of medical school ($p=0.50$), or being born outside the U.S. ($p=0.59$).

CONCLUSIONS: Physicians who received cultural competence training in third year and in the form of a small group or workshop were more culturally competent as measured by the IAPCC. Cultural competence was also associated with diverse life experiences, and the more of these experiences, the higher degree of cultural competence. These data are preliminary, but the findings may be useful in the design and timing of cultural competency curricula, as well as help refocus admission processes in medical and graduate medical education. Future research should focus on other determinants of cultural competence, including whether certain life experiences impact cultural competence as much or more than curricular interventions.

EFFECTS OF A FACULTY DEVELOPMENT PROGRAM IN GERIATRICS FOR SURGEONS AND SUBSPECIALISTS. B.C. Williams¹; M.R. Bergen²; G.A. Stratos²; P. Pompei³; J.T. Fitzgerald¹; K. Skeff². ¹University of Michigan, Ann Arbor, MI; ²Stanford University, Palo Alto, CA; ³Stanford University, Los Altos, CA. (Tracking ID #132796)

BACKGROUND: Professional societies of surgical specialties and medical subspecialties (SSMS) increasingly emphasize the importance of education in the care of older patients. Previous faculty development programs in geriatrics have focused on primary care disciplines.

METHODS: As part of a program to develop faculty in clinical care, teaching, and house officer curriculum development related to the care of older patients, 13 SSMS faculty from 4 surgical (orthopedics, general surgery, plastic surgery, gynecology) and 3 related (emergency medicine, anesthesiology, physical medicine) specialties and 5 medical subspecialties (oncology, critical care, rheumatology, gastroenterology, endocrinology) participated in 7 weekly 2-hour seminars. 3 sets of seminars were held, each with 4-5 participants, and were largely based on the Stanford Faculty Development Center's (SFDC) Geriatrics in Primary Care program. A comparison group of 11 general medicine (GENMED) faculty participated in nearly identical seminars during the same period. Participants provided evaluations pre- and post-seminar using a validated, reliable survey instrument developed at the SFDC. Knowledge items were multiple choice; all other questions were rated on 5-point scales (relevance and usefulness ratings 1="low", 5="high"; attitudes questions 1="Strongly Agree, 5="Strongly Disagree" with higher values representing more positive attitudes).

RESULTS: SSMS faculty showed significant improvement in knowledge (17 items, 40% vs. 60% correct, $p<.001$) and self-rated ability to practice geriatrics (14 items, mean 2.37 vs. 3.40, $p<.001$) and teach geriatrics (14 items, mean 2.15 vs. 3.31, $p<.001$). Educational and clinical relevance of the seminars were perceived to be high (7 items, means 4.12 and 4.04, respectively). The content area rated as most relevant was assessing and promoting behavioral change among patients. Seminars were perceived as highly useful (1 item, mean 4.69). Attitudes towards care of older patients were initially high and showed no change (12 items, pre-post scores 3.85 vs. 3.87, $p=.85$). Compared to the SSMS group, the GENMED group had higher baseline knowledge and skills scores and lower attitudes scores, but showed changes similar in direction and magnitude.

CONCLUSIONS: To our knowledge this is the first report of the effects of a faculty development program in geriatrics aimed at faculty from surgical specialties and medical subspecialties. Among motivated faculty, education in clinical care and teaching related to older patients can have a substantial impact. Education in areas such as facilitating behavioral change is likely most different from traditional surgical and subspecialty training, and may have particularly large benefits.

EVALUATING EVIDENCE-BASED MEDICINE COMPETENCE: A SYSTEMATIC REVIEW OF INSTRUMENTS. T.M. Shaneyfelt¹; K. Baum²; D.S. Bell³; D.A. Feldstein⁴; S.S. Kaatz⁵; C.T. Whelan⁶; M.L. Green⁷; ¹University of Alabama at Birmingham, Hoover, AL; ²University of Minnesota, Bloomington, MN; ³University of California, Los Angeles, Los Angeles, CA; ⁴University of Wisconsin-Madison, Madison, WI; ⁵Henry Ford Hospital Detroit, Detroit, MI; ⁶University of Chicago, Oak Park, IL; ⁷Yale University, Waterbury, CT. (Tracking ID #136056)

BACKGROUND: Despite the widespread teaching of evidence-based medicine (EBM), educators struggle to evaluate trainees' competence. As members of the SGIM EBM Task Force, the authors performed a systematic review of currently available EBM evaluation instruments.

METHODS: Computerized databases, internet sites, reference lists of retrieved articles, education journal tables of content, and personal files were searched to locate EBM evaluation instruments. Articles were included if they reported an evaluation instrument or approach that 1) evaluated EBM knowledge, skills, attitudes, behaviors, or patient outcomes, 2) included psychometric data from trainees, and 3) provided sufficient description of the instrument or approach. Two reviewers independently abstracted each article for trainee characteristics, targeted evaluation domains, evaluation methodology, and psychometric testing data. Discrepancies were resolved by consensus.

RESULTS: Of the 235 articles identified, 84 met inclusion criteria. The interrater reliability of the inclusion process was good (Kappa 0.685, $p < 0.001$). Most EBM teaching evaluations targeted medical students (41.6%) or residents (28.6%) and were developed to evaluate locally developed EBM curricula (76.6%). Most (79%) targeted specific EBM skills including asking clinical questions (16.9%), acquiring evidence (32.5%) and critical appraisal (41.6%). 39% evaluated self-reported EBM knowledge, largely about critical appraisal. Few (36%) assessed actual EBM behaviors, attitudes toward EBM (25%) or patient outcomes (2.6%). The most common methodologies relied on written tests (35%) or surveys of self-reported EBM knowledge or skills (41.6%). A minority of instruments underwent psychometric testing (44%) and a few notable exceptions (16%) demonstrated sufficient construct validity and reliability to evaluate changes in EBM knowledge and critical appraisal skills.

CONCLUSIONS: The majority of EBM evaluation instruments lack the psychometric rigor to determine the effectiveness of educational interventions or the competence of individual trainees. Educators should look to the few valid instruments for evaluating critical appraisal skills and EBM knowledge while investigators develop robust strategies to evaluate other EBM behaviors.

EVIDENCE BASED MEDICINE AND COMPETENCY: DO INTERNAL MEDICINE RESIDENTS HAVE THE MEDICAL KNOWLEDGE? A. Djurich¹; M.J. Bair¹. ¹Indiana University School of Medicine, Indianapolis, IN. (Tracking ID #133459)

BACKGROUND: The Accreditation Council for Graduate Medical Education has called for residency programs to teach and assess evidence-based medicine (EBM). Demonstrating resident competence in EBM critical appraisal skills is a challenge for program directors. Therefore, our objective was to integrate the competencies of medical knowledge and practice-based learning and improvement (PBLI) by evaluating residents' knowledge and skills related to EBM core material.

METHODS: On a required one-month ambulatory block rotation, medicine and medicine-pediatric residents received case-based didactic sessions on a variety of outpatient topics such as common symptoms, preventive care, chronic disease management, and acute conditions. We integrated core EBM material into these sessions. Residents were given separate one-hour sessions on "EBM - Diagnosis" and "EBM - Therapy". The content considered essential to understanding EBM, focused on residents' ability to understand, interpret, and calculate important terms, including sensitivity, specificity, likelihood ratios, and differences between absolute/relative risk reduction, and number needed to treat/harm. Residents were given three ten-item tests: a PreTest (immediately prior to the session), a PostTest (immediately after the session), and a delayed final PostTest (approximately two weeks after the session).

RESULTS: For the diagnosis and therapy sessions, 57 and 122 residents took the PreTests and PostTests, respectively. Less than 15% of residents were absent at the delayed PostTests due to vacation or other clinical duties. Mean scores and standard deviations for all three tests for both sessions are shown in the table.

CONCLUSIONS: Resident PreTest scores were poor. After each session, resident PostTest scores improved significantly, with excellent retention two weeks later on the delayed PostTests. Given a brief didactic, case-based session on common EBM topics, residents can improve their knowledge of EBM core material. This mini-EBM curriculum assesses resident medical knowledge of the EBM portion of the PBLI competency.

Resident Mean Test Percent Scores and Standard Deviations

	PreTest	Immediate PostTest	Delayed PostTest
Diagnosis	45.1 (21.2)	88.6 (12.0)	85.1 (15.7)
Therapy	51.1 (22.0)	86.6 (19.2)	84.7 (18.9)

EXPLORING INTERNAL MEDICINE INTERN NIGHT FLOAT (INF) RESPONSIBILITIES. P. Olshausen¹; E. Ijo-Su¹; P. Charney¹. ¹Norwalk Hospital, Norwalk, CT. (Tracking ID #132716)

BACKGROUND: With implementation of the new resident work hour regulations, Internal Medicine Residency Programs have developed new systems to

redistribute hours and at the same time provide appropriate and safe patient care. Our INF provides cross-coverage overnight for all medical patients on the inpatient floors including the inpatient rehabilitation and the psychiatric units, but excluding the Critical Care Units. An "Intern Survival Guide" compiled by our second year residents addresses many common intern management issues faced, based on individual experiences rather than collective data. Only one small study to date has attempted to examine the specific management problems faced by an INF.

METHODS: Therefore, a resident-faculty research group initiated detailed data collection. The INF completed a form quantifying and categorizing cross coverage calls and whether or not issues had been anticipated by the primary resident team. The calls were categorized as pain, cardiac, endocrine/metabolic, GI, respiratory, psychiatric and miscellaneous. Resident sign-out sheets were collected and the patient load and type of sign-out requests were also recorded. The INF also completed a pre- and post-experience survey reviewing INF as a clinical and educational experience and providing comments on how to improve the experience. Five interns have participated, covering 53 shifts from September until December of 2004 in a 250-bed community based teaching hospital with over 3,000 medical admissions annually.

RESULTS: During the 53 shifts, a total of 3,218 patients were cross covered (mean 60.7/night) and 1,241 calls received (mean 23.4/night). INF reported that 92% of the calls were not anticipated by the primary resident. The most common management issues were: pain (13.5%), cardiovascular urgencies (12.5%), medication issues (10.8%), endocrine/metabolic (10.4%), anxiety/agitation (8.1%), insomnia (7.9%), fever (6.9%), GI (5.8%), respiratory (5%) miscellaneous 19% (including patient falls, death pronouncement, i.v. access, speaking with patients and their families, etc.). On average, the INF received 7.1 sign-out requests per night to follow up on lab values, etc. (total: 374). Many of the INF coverage issues are reviewed in our current intern survival guide.

CONCLUSIONS: The Night Float rotation exposes the interns repeatedly to common management issues in a busy hospital environment. With detailed exploration of the INF experience, additional patient care protocols may be appropriate as well as more focused educational sessions and further Intern Survival Guide revisions. These adjustments will hopefully evolve to better prepare interns for INF and decrease the number of unanticipated calls. An analysis of the full academic year and comparison of experiences at several institutions will be useful in generalizing conclusions.

FEEDBACK AFTER INTERNAL MEDICINE ROTATIONS: WHAT DO THE FACULTY SAY? P.B. Hasley¹; R.M. Arnold¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #134350)

BACKGROUND: Several authors have provided medical educators with guidelines on how to give feedback to learners. However there are no previous studies describing how faculty actually give feedback on the medical wards. The aim of this study is to describe how faculty give feedback after an internal medicine rotation; in particular: 1) the subjects discussed during an evaluation session; 2) the degree to which specific examples are given during feedback and 3) the frequency with which positive or corrective feedback is provided.

METHODS: Study participants were medical students, house staff and internal medicine faculty at the University of Pittsburgh Medical Center. At the end of a four week hospital rotation in internal medicine, a summative evaluative session is routinely scheduled between faculty and learner. After obtaining informed consent the sessions were tape recorded. Tapes were transcribed and all identifying information was removed. A coding scheme was developed based upon current guidelines for the provision of feedback. The coding scheme indicated if feedback was general (Example: "You were great!") or specific (Example: "Your fund of knowledge was good."). If specific feedback was provided, the subject of the feedback was coded into 13 mutually exclusive domains (eg: medical knowledge, efficiency). Specific feedback was also characterized regarding: (a) whether it was positive versus corrective, (b) whether there was elaboration of the feedback statement, (c) whether a specific example was given supporting the feedback and (d) for corrective feedback, if advice for improvement was discussed. Finally we coded whether self-evaluation or learner feedback was elicited. Ten percent of all transcripts were independently coded by both authors and disagreements resolved by consensus and the coding scheme refined to increase agreement. Subsequent coding was done by one author.

RESULTS: 51 evaluation sessions with 19 attendings, 10 medical students, 31 PGY I's and 10 PGY II's and III's were transcribed and coded. Faculty were on average 42 years of age, were 26% female and had a mean of 10 years of teaching experience (range 1 to 27 years). 74% of the faculty were generalists. 47% of the attendings asked the trainee for a self-evaluation, 76% asked for the learner's evaluation of the ward month and 49% asked for an evaluation of the faculty's performance as an educator. 90% of evaluation sessions included general, positive statements about the learner's performance. 100% of sessions had specific feedback; the most common domains discussed were: medical knowledge, interpersonal relationships and thoroughness. On average, 6 specific evaluative statements were made per feedback session: 4.1 were positive and 1.9 statements were corrective feedback. Roughly 53% (3.2) of the feedback statements were statements without examples or elaborations; ("I think you worked very well with the team."); 40%(2.4) included elaboration and 7%(0.4) included specific examples. In only 12% (0.7) of cases was the learner given advice on how they could correct deficiencies or improve their performance.

CONCLUSIONS: During a summative evaluation: 1) Frequent specific comments are made regarding the positive aspects of a medical trainee's performance. 2) Faculty do not elaborate or give examples from the learner's own behavior. This is particularly true when giving corrective feedback. 3) Only half of faculty asked for feedback from the learners or for the learner to perform a

self-assessment. The impact of these evaluative practices on learners needs to be examined.

FROM ROOKIE RESIDENT TO SAVVY SEARCHER: IMPROVING EVIDENCE BASED MEDICINE SEARCH SKILLS THROUGH EDUCATION. E.A. Sastre¹; R.B. Davis¹; R.A. Ramanan¹. ¹Division of General Medicine, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA. (Tracking ID #133755)

BACKGROUND: Evidence based medicine (EBM) is a core clinical competency of residency training. We developed a curriculum to teach MEDLINE search skills and studied the impact of the intervention on residents' attitudes, knowledge and skills.

METHODS: We developed a physician-led, hands-on curriculum to teach MEDLINE search techniques and taught all Internal Medicine residents in a large urban academic medical center who rotated through Ambulatory Medicine from September through November 2004. The three-session course provided an introduction to MEDLINE, described search techniques, and allowed residents to perform practice MEDLINE searches at individual computers. We designed a questionnaire to collect data regarding attitudes toward MEDLINE searches, MEDLINE usage patterns, and knowledge about search techniques. We developed an evaluation tool to assess skills by observing residents perform a MEDLINE literature search. Residents completed the questionnaire and performed the observed MEDLINE search both before and after the course. Using McNemar tests, we used each resident as their own control to measure change in attitudes, knowledge and skills.

RESULTS: Out of sixty-eight eligible residents, 46% were post-graduate year (PGY) II, 54% were PGY III, 55% were female, and the mean age was 29 years. Most had prior research experience and had received prior instruction on performing a MEDLINE search. Fifty-six residents (82%) completed the curriculum as well as both the pre- and post-intervention questionnaires and MEDLINE searches. Demographic characteristics of participants mirrored the entire residency program. Regarding attitudes measured following the curriculum, most residents estimated it would require less time to complete a MEDLINE search after the course as compared to before the course ($p < 0.0001$). In addition, resident's self-reported confidence in their ability to locate the best article to answer their clinical question rose from 32% to 75% ($p < 0.0001$). All residents reported that they would incorporate these skills into future clinical practice. In reference to knowledge, the percent of residents who could identify an established method to conduct a search increased from 21% to 86% ($p < 0.0001$). In addition, there was a significant increase in knowledge of MEDLINE search tools including MeSH terms (38% to 100%, $p < 0.0001$), setting of limits (79% to 100%, $p < 0.0001$), subheadings (54% to 89%, $p < 0.0001$) and MeSH Major Topic (27% to 98%, $p < 0.0001$). Regarding skills, residents reported increased use of MEDLINE following the course ($p < 0.01$). When observed performing a MEDLINE search at the completion of the course, residents demonstrated increased use of MeSH terms ($p < 0.0001$), limits ($p < 0.001$), subheadings ($p < 0.0001$), and MeSH Major Topic ($p < 0.05$). Residents were more likely to find the most relevant randomized controlled trials when performing an observed search after the course as compared to before the course (75% vs 46%, $p < 0.001$). Over 90% of the participants strongly agreed that the course was useful and valuable, and that they would recommend it to fellow residents.

CONCLUSIONS: Residents gained confidence following this curriculum and demonstrated marked improvement of their attitudes, knowledge and skills. Broader inclusion of curricula to improve search skills may facilitate the use of EBM in clinical practice.

FROM THE BLACKBOARD TO THE BEDSIDE: AN EMPIRIC ASSESSMENT OF CORRELATES OF THE HIDDEN CURRICULUM IN END-OF-LIFE CARE MEDICAL EDUCATION. M.W. Rabow¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #135514)

BACKGROUND: The "hidden curriculum" (HC) in medical education is characterized by informal influences including interpersonal relationships and student perceptions of institutional values. The HC is thought to impact what students actually learn (versus are taught) and how they behave. It is likely to be especially important in developing appropriate attitudes and professional behaviors, as well as in learning strategies for dealing with distressing emotions in clinical work. However, there is limited empirical data on discordances between the formal and hidden curricula, on the formation of the HC or its impact on professional development. We sought to assess perceptions of the HC in end-of-life care (EOLC) education by 3rd year medical students (MS3s).

METHODS: A cohort of students at a large, urban medical center completed a confidential, self-administered questionnaire twice, at the end of 2nd year coursework (MS2) and again 8 months into 3rd year ward rotations (MS3). In addition to posing questions about sociodemographics, the survey elicited responses about four psychological constructs: (1) personal and medical school EOLC experiences and relationships, (2) guilt and negative emotional reactions to EOLC, (3) perceptions of institutional EOLC values, and (4) perceptions of the HC, defined as discordances between what was formally taught and how students were really expected to behave on the wards. Scaled scores for these constructs were estimated by fitting a Rasch model to corresponding subsets of questions. MS2 and MS3 data for individual students were used for pre-post analysis, but combined for cross-sectional analysis. Correlations suggested relationships among the variables in the model and these inter-relationships were analyzed using multiple linear regression. Effect sizes (ES) were calculated using the standard deviation of the distribution of the outcome variable "Percep-

tion of a Discordant HC." Significance was taken as $p < 0.05$. The study received IRB approval.

RESULTS: Overall, 126 students completed both surveys (response rate 89.2%); 56.7% were female, average age was 25.5 years, and 51.2% were White, 28.3% Asian, 8.7% Hispanic, 2.4% Native American, 1.6% African American. Reliability of scales ranged from 0.53-0.77. Compared with Whites, African American and Latino students were more likely to perceive a discordant HC ($p = .027$, $ES = .98$ and $p = .029$, $ES = .56$, respectively). Between the first and second surveys, students' perceptions of a discordant HC decreased ($p < 0.001$), with a greater decrease among Asian students than Whites ($p = .034$). Perception of the discordant HC was significantly associated with having negative emotional reactions to EOLC ($p < .001$, $ES = .94$) and negative perceptions of institutional EOLC values ($p < .001$, $ES = .21$). Age, gender, personal EOLC experiences, patient interactions, and role modeling were not significantly correlated with perception of a discordant HC.

CONCLUSIONS: MS3s differed in perceptions of discordance in the EOLC hidden curriculum. Negative emotions and students' perceptions of institutional values strongly correlated with perceptions of a discordant HC, however role models and other interpersonal interactions did not. These preliminary data suggest that interventions to improve EOLC training address the presentation of institutional values and ways to moderate negative emotional responses to EOLC. Further research should explore the impact of student perceptions of a discordant HC on professional behaviors.

FUNDING MEDICAL EDUCATION RESEARCH. D.A. Reed¹; R.B. Levine¹; D.E. Kern¹; S.M. Wright¹. ¹Johns Hopkins Bayview Medical Center, Baltimore, MD. (Tracking ID #135701)

BACKGROUND: The Institute of Medicine has called for increased rigor of educational research, yet less than 0.001% of federal spending in graduate medical education is spent on educational research. How medical education research is currently funded and the costs of conducting studies in medical education are unknown. The objectives of this study were to: 1) determine how recently published medical education research studies were funded, 2) approximate the costs of conducting medical education research, and 3) understand the opinions of education researchers regarding funding for medical education research.

METHODS: We conducted a cross-sectional survey of first authors of published studies in medical education in the past year. Original medical education research studies performed at American institutions that had been published in 13 prominent peer-reviewed journals were included. The survey assessed demographic information about authors, pursuit and attainment of funding for the study, duration of the study and percent effort devoted by authors, resources used and their costs, and author opinions regarding educational research. Logistic regression was used to identify factors associated with the acquisition of funding. The cost of each study was calculated by multiplying the percent effort dedicated by each author by the national average salary for general internists according to academic rank for the duration of the study, and adding the cost of resources used.

RESULTS: Of 290 medical education studies, authors of 243 studies responded (84%). The 243 studies included a wide range of study designs: 30% program description, 30% survey, 13% qualitative, 8% evaluation of a scale/tool and 19% other. The majority of studies (64%) were completed within two years. The average percent effort invested by first authors was 12%. The median cost of conducting these medical education studies estimated by authors was \$10,000. The actual median calculated cost was \$20,126. Of the 243 studies, authors of 72 (29.6%) successfully obtained funding. Governmental grants were the most common funding source acquired (23.6%). The median amount of funding obtained was \$15,000. Of studies that were funded, the median cost was \$35,111. Two factors independently associated with the attainment of funding were author training in grant writing (OR 2.05, 95% CI 1.1, 3.7) and a greater number of medical education publications by the author (OR 2.4, 95% CI 1.2, 4.6). A majority (79%) of authors agreed that there is inadequate funding available nationally for medical education research. Eighty-eight percent felt increased funding would improve the quality of educational research. One-third of authors had considered changing their research focus to fields other than medical education based on the dearth of funding opportunities.

CONCLUSIONS: The vast majority of the medical education research that is published is not formally funded. The studies that are successful in obtaining funding are significantly under-funded. If the medical community hopes to increase the quality of educational studies, policy reform demonstrating greater appreciation for medical education scholarship through increased funding is needed.

FUTON BIAS: IS IT ONLY THE EARLY BIRD THAT GETS THE WORM? N.S. Murali¹; H.R. Murali¹; P. Auethavekiat²; P. Erwin¹; S. Cha¹; P.S. Mueller¹; A.K. Ghosh¹. ¹Mayo Clinic, Rochester, MN; ²Bumrungrad Hospital, Bangkok, (Tracking ID #135766)

BACKGROUND: Full Text on the Net bias (FUTON Bias) is the tendency to peruse what is more readily available on the Internet as Full text and ignoring relevant articles where abstracts are not available. The latter type of research although equally relevant and important could be overlooked and thus "invisible". By being more visible, FUTON articles are likely to be read or cited more often. We sought to determine if there was a change in the impact factor of Cardiology, Nephrology and Immunology journals relative to the time of being available FUTON.

METHODS: All Cardiology, Nephrology, Rheumatology and Immunology journals were identified using MEDLINE, PubMed, and EMBASE databases and

several search engines. The online status of these journals in MEDLINE (as FUTON, abstracts only, and no abstracts or citations) was ascertained. Impact factors for all available journals from 1990 to 2003 were abstracted. Journals that went online in the first half of the Internet era (prior to 1998), as opposed to those that went online later, were compared using sign rank and rank sum tests. RESULTS: In general, impact factors have increased with the availability of the World Wide Web. Of the 315 journals identified, the "prior to 1998 era" journals ($n=164$) tended to have higher mean impact factors than the "post-1998 era" journals ($n=149$) available as FUTON (2.08 ± 0.36 vs. 1.03 ± 0.10). The "post-1998" journals had remarkable improvement in their impact factors ($p=0.0006$) soon after they went online and caught up with pre-1998 journals in two years of such availability. Interestingly journals that provided only references or citations online failed to improve their impact factors with the availability of the internet. Further, FUTON journals continued to maintain higher impact factors than their peer journals available only as abstracts ($p<0.00001$) year 1 to 4 after they went online.

CONCLUSIONS: This suggests that although internet has increased awareness of the existence of a journal, those allowing FUTON access are likely to make substantial leaps in their impact factors and continue to outstrip their peers who fail to provide such access. Not recognizing this bias has extremely important implications that impede the progress of evidence based medicine since it curtails the necessary dissemination of knowledge by making research less visible to the community in general and readership in particular. Finally, even journals that provided FUTON access later in the internet era had significant improvement in their impact factors suggesting that even the late birds managed to feed heartily.

HOW INTERNAL MEDICINE RESIDENCY PROGRAMS TRAIN, TRACK, AND DETERMINE PROCEDURE COMPETENCE. D. Francois¹; J. Rogers¹; J. Fischer²; L. Cation¹. ¹University of Illinois at Peoria, Peoria, IL; ²University of Illinois, Peoria, IL. (Tracking ID #133315)

BACKGROUND: Both the American Board of Internal Medicine (ABIM) and the Accreditation Council on Graduate Medical Education (ACGME) require that internal medicine program directors designate whether graduating residents are considered "competent" to perform a core set of common internal medicine procedures. While program directors have long been certifying residents as "competent" in procedural skills, little is known regarding how procedure skills are taught, how they are tracked, and how competence is determined.

METHODS: A 20 item voluntary survey regarding procedure training and documentation was mailed to all internal medicine program directors within the continental United States in 2004. The survey asked the program director to state their program's current procedure training, procedure documentation/tracking, and procedure competence determination practices.

RESULTS: The response rate was 51% (202/398 programs surveyed). Procedure training practices used in residency programs primarily include didactic teaching conferences (80%) and procedure workshops (57%). Documentation practices vary: 28% of programs track only ABIM required procedures, 47% track ABIM plus selected other procedures and 25% track any procedure performed. Regarding procedure documentation, 25% of programs are using written logs only, 37% use written logs which are then entered into a computer database, and 38% of residents enter their procedure activity directly into a computer database. Only half (50%) of programs currently track procedure complication rates. A minority of programs (11%) utilized a special IM procedure clinical service. The majority (76%) of programs have written policies regarding how a resident is declared competent to perform procedures. Currently, four non-mutually exclusive means are used to determine procedural competence in medicine training programs: quantity of procedures done (85%), direct procedural observation by faculty (69%), written examination (11%) and resident self-declaration of competency (8%). Most programs allow residents who have been declared competent by the program to perform elective procedures without direct attending oversight (86%) and supervise other residents in lieu of attending oversight (92%). In IM residencies where there are Medicine-Pediatrics residents, only 36% of IM programs allow procedures performed on pediatric patients to count towards internal medicine procedure certification.

CONCLUSIONS: Internal Medicine training programs vary in how they train their residents in procedure competence, how they track procedure activity and how they determine whether their residents have attained "competence" in the procedural practice prior to residency completion. Nearly all programs provide some sort of didactic instruction. A "competency" designation is commonly arrived at via review of procedure quantity coupled with faculty observation. Training residents to competently perform common procedures is important for patient care and safety. Further study is needed to identify the best way to train, track and determine competence for these important skills.

IDEA IN EVOLUTION: AN ATTEMPT TO USE RIME TERMINOLOGY TO MORE ACCURATELY ASSESS MEDICAL STUDENTS' ADMISSION WRITE-UPS. E.A. Baker¹; J. Riddle². ¹Rush University Medical Center, Chicago, IL; ²University of Illinois at Chicago, Chicago, IL. (Tracking ID #134881)

BACKGROUND: The IDEA method was developed as a tool to reliably and reproducibly assess medical students' admission write-ups. This method uses the categories of reduced, dispersed and elaborated, as defined by structural semantic theory, to categorize write-ups with an emphasis on the assessment section of the write-up. Reliability was demonstrated in a preliminary study, but continued use of the method has revealed limitations. The terminology is unclear to some clinical faculty, and categories as defined fail to describe some

write-ups. An initial attempt to refine the categorizations using RIME (reporter, interpreter, manager, educator) terminology is described.

METHODS: Lack of agreement in the categorization of multiple write-ups was found when two evaluators trained to use the IDEA method reviewed a series of student write-ups. The evaluators found that they relied upon information not included in the defined criteria for their ratings, and that some write-ups did not fit into a category. A new categorization scheme that incorporated the structural semantic definitions of reduced, dispersed and elaborated with the RIME terms of reporter and interpreter was developed. Five categories were defined: early reporter, imprecise interpreter, solid reporter/early interpreter, excellent interpreter and superior interpreter. A revised rating form was developed which included a detailed set of criteria for each category.

RESULTS: Twenty new admission write-ups were categorized by the two evaluators. Despite the detailed set of criteria used, the evaluators agreed on the categorization in only nine of twenty write-ups (45%). In five write-ups, one evaluator categorized the write-up as either early reporter or imprecise interpreter, and the other categorized it as early interpreter. In four write-ups, one evaluator categorized the write-up as early interpreter and the other as excellent interpreter, and in two write-ups one evaluator categorized the write-up as excellent interpreter and the other as superior interpreter. Three characteristics were identified that lead to a disagreement in categorization: 1. write-ups with a discrepancy in the quality of the HPI and the assessment sections, 2. write-ups in which some but not all criteria for a category were met, and 3. write-ups for patients with firmly established diagnoses, where discussion of pertinent diagnostic possibilities was limited.

CONCLUSIONS: An attempt to refine the categorization of the IDEA method using RIME terminology did not improve its reliability. Three circumstances were identified that lead to disagreement in the categorization of multiple write-ups. Renaming categories, more clearly defining the explicit criteria needed for each category, and establishing criteria when diagnostic certainty exists should address the limitations identified in the current study. A simple and reliable assessment tool to assess students' write-ups continues to be needed. Future studies are planned, after further refining the IDEA method, to reevaluate its reliability, with the hope that it can eventually meet this need.

IMPROVING HOUSESTAFF CONFIDENCE IN PERFORMING ACLS USING HIGH FIDELITY SIMULATION. J.A. Watts¹; J. Jorgensen¹. ¹Madigan Army Medical Center, Tacoma, WA. (Tracking ID #132991)

BACKGROUND: Prompt initiation of appropriate ACLS techniques can improve patient outcomes, but effective leadership of a code team requires confidence and experience. Which experiences lead to improved house staff confidence and performance in ACLS scenarios has not been well described. The purpose of this study was to assess if house officer confidence correlates with level of experience, and if code leadership training using high fidelity patient simulators effectively improves house officer confidence.

METHODS: We constructed a closed postal survey to measure house officer experience and confidence in leading ACLS scenarios, measuring levels of confidence using a 5 point Likert scale. All subjects then underwent a series of 4 simulated clinical scenarios (ventricular fibrillation, stable narrow complex tachycardia, PEA, and acute respiratory failure) using the SIMAN (Laerdal Medical Corp, New York) patient simulator. Each house officer served as the code leader twice. House officer confidence was then reassessed using the same closed postal survey. We analyzed the relationship between learner's experience and confidence using the Pearson correlation coefficient (r) and pre and post test differences in confidence using the Wilcoxon Signed Rank Test.

RESULTS: Twenty four housestaff physicians completed the study. All 24 housestaff reported that their lack of experience in performing ACLS was directly related to their lack of confidence. Furthermore, 21 of 24 housestaff stated that a lack of confidence would cause them to hesitate in assuming the role of code leader during an ACLS resuscitation. The mean number of active ACLS resuscitations per learner at baseline was 2.3 ± 2 , and the number of passively observed resuscitations was 5.8 ± 3 . Mean baseline confidence level was 2.1 ± 1 . We found a strong correlation ($r=0.78$, $p<0.001$) between perceived confidence and active participation in ACLS resuscitations, and a weaker but significant correlation ($r=0.49$, $p<0.01$) between perceived confidence and resuscitation observation. Following simulator training we noted a mean increase in perceived confidence to 2.94 ± 1 ($p<0.001$).

CONCLUSIONS: This study demonstrates a strong correlation between active ACLS participation and perceived house staff confidence in leading ACLS scenarios. Our study also establishes an important relationship between confidence and assuming a leadership role in ACLS events. Our data suggests that code leadership training using a high fidelity human simulator significantly improves confidence after even limited exposure. These findings highlight the importance of providing well designed training opportunities to prepare residents for critical clinical roles throughout their graduate medical education curriculum.

INTERNAL MEDICINE RESIDENTS' COMFORT IN CARING FOR ADOLESCENTS AT A LARGE UNIVERSITY-BASED PROGRAM. A.R. Gonzaga¹; M.A. McNeil¹; M. Clifton¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133335)

BACKGROUND: Adolescence is associated with a high level of morbidity and mortality due to risk-taking behaviors. Unfortunately, adolescents are the largest group of people to underutilize preventive services, partly due to the inability to find physicians, including internal medicine trainees, who are knowledgeable about adolescent health issues. This study set out to identify the curricular gaps

and resident attitudes in adolescent medicine at a large university-based residency program.

METHODS: A survey was designed to identify residents' perceived comfort in managing and counseling on adolescent health issues, and performing procedures common in the care of adolescents. Topics were based on the current goals for adolescent medicine curricula as outlined by the Federated Council for Internal Medicine Task Force on the Internal Medicine Residency Curriculum, AMA, and American Academy of Pediatrics. Survey items asked respondents to indicate (on a Likert-type scale of 1 = very uncomfortable to 5 = very comfortable) how comfortable they feel managing or providing counsel on the medical issue, or performing the procedure described. Questions on attitudes toward adolescent care were included. Demographic information was requested. After obtaining IRB approval, the questionnaire was distributed to all 3rd year residents. Descriptive statistics were used to summarize the data.

RESULTS: Responses were obtained from 35 of a total 48 senior residents (73%). Fifty-four percent of respondents were male. Most respondents desire to care for adolescents in the future (mean = 3.57 ± 1.31). Most did not feel residency had prepared them to care for adolescents (mean = 2.77 ± 0.81). The mean ratings given to 50 of 53 items on the comfort in management of or counseling on the adolescent medical issue were below comfortable levels. The mean ratings for the 5 lowest rated items in each category are displayed (Table). Six of 11 items on comfort with procedural skills were ranked below the "can perform independently" level (Table).

CONCLUSIONS: Residents in a large university internal medicine residency program do not feel comfortable managing common medical problems, counseling on health related topics, and performing common procedures for adolescents. These results indicate that current training in adolescent medicine is insufficient to prepare residents to care for adolescents, particularly given most desire to care for adolescents in the future. Based on these results future curricular development should focus on improving the counseling skills and adolescent medicine knowledge base of residents, and in providing more opportunities to practice procedures common in the care of adolescents.

Residents' Lowest Rated Items on Comfort with Management, Counseling, and Performing Procedural Skills in Adolescents

Comfort in Managing	Mean (SD)	Comfort in Counseling	Mean (SD)	Comfort in Performing	Mean (SD)
Learning Disabilities	2.15 (0.91)	Date Rape Drugs	2.55 (1.20)	Prescription of Emergency Contraception	1.97 (0.92)
Surveillance after Treatment of Childhood Malignancy	2.19 (1.11)	Victim of Rape	2.58 (1.20)	Microscopic Interpretation of Vaginal Discharge	2.45 (0.79)
Delayed Puberty	2.33 (1.08)	Party Drugs	2.91 (1.16)	Back Exam to Screen for Scoliosis	2.61 (0.79)
Poor School Performance	2.36 (0.93)	Protective Sports Gear	3.06 (1.17)	Patient's First Pelvic Exam	2.61 (0.93)
Short Stature	2.55 (0.91)	Community Violence	3.09 (1.21)	STD Screening in Males — Collection of Urethral Swabs	2.63 (1.04)
				Interpretation of Pap Smear Results	2.67 (0.82)

INTERNAL MEDICINE RESIDENTS REJECT "LONGER AND GENTLER" TRAINING. F. Carreira¹; W.A. Baker²; R.K. Gopal¹; G.J. Jeff¹; L.A. Crane¹; T.J. Miyoshi¹; A.V. Prochazka¹. ¹University of Colorado Health Sciences Center, Denver, CO; ²Denver Health and Hospital Authority, Denver, CO. (Tracking ID #135996)

BACKGROUND: Increasing complexity of medical care, coupled with caps on resident work hours, has prompted consideration of extending internal medicine training to four or five years. While the current 80-hour/week cap on work hours has been embraced by many internal medicine residents, it is unclear whether further hour reductions, and extension of training beyond the current duration of three years, would be accepted.

METHODS: A postal survey was sent to all 143 internal medicine residents at the University of Colorado School of Medicine training program in May 2004. The survey dealt with a variety of issues related to house-officer duties, and opinions. Burnout was measured using the Maslach Burnout Inventory, a 22-item questionnaire organized into three subscales: emotional exhaustion (EE, high score ≥ 27), depersonalization (DP, high score ≥ 10), and personal accomplishment (PA, high score ≤ 33). We defined burnout as high EE or DP. Agreement scores between burned out and non-burned out residents were com-

pared by using the t-test for equality of means. House-staff were asked to respond to the following comments: (A) "I would be willing to train longer (4-5 years for internal medicine) if I could work fewer hours, such as 60 hours per week." and (B) "Overall, an 80 hour work week is still too much. A 60-hour limit would be better." Possible responses consisted of a 5-point Likert scale included (1) Strongly Agree, (2) Agree, (3) Neither Agree or Disagree, (4) Disagree, (5) Strongly Disagree.

RESULTS: The response rate was 74% (106/143) with slightly less second year residents responding (PGY1 37%, PGY2 26%, PGY3 37%, $p=.687$) Response rates for each training tract mirror the program as a whole (preliminary 4.2%, primary care 20.8%, and categorical 75.0%). Two-thirds of respondents were in the 26-30 year-old age group with 58% of respondents being female. The vast majority (84%) of residents disagreed with the notion of extending training to four or five years, including 51.9 % who disagreed strongly, while 2.8% were neutral, and 12.3 % agreed with extension of training. Residents that met criteria for burnout tended to be slightly less averse to extending training (strongly agree or agree, 18.9% vs. 4.3%, $p=0.044$) than residents with lower burnout scores. The majority of residents (68.9%) disagreed with establishing a 60-hours per week cap instead of the current 80 hours per week cap, while 14.2% were neutral, and 16% preferred changing to a 60-hours per week cap. Residents with higher burnout scores were not significantly differ from residents with lower burnout scores regarding the preference for an 80-hours per week limit (Disagree or strongly disagree, 72.4% vs. 65.4%, $p=0.522$).

CONCLUSIONS: Most internal medicine residents are strongly opposed to extending their training to four or five years and would prefer the current 80 hour/week work hours cap, compared to a 60 hour/week cap. A longer, less intense pace of internal medicine training seems to be less attractive in the eyes of prospective internists.

INTERNAL MEDICINE RESIDENTS' EXPERIENCES WITH AND ATTITUDES TOWARD CHRONIC PAIN MANAGEMENT. J. Chen¹; M.J. Fagan¹; S.E. Reinert²; J.A. Diaz³. ¹Brown University, Providence, RI; ²Lifespan Information Services, Providence, RI; ³Brown University, Pawtucket, RI. (Tracking ID #133915)

BACKGROUND: Despite calls for improved education of medical trainees in the treatment of chronic non-malignant pain (CNMP), little information exists about internal medicine residents' confidence in, and experiences with, managing CNMP in their outpatient practices, or their attitudes towards interactions with CNMP patients.

METHODS: We surveyed all 124 residents in the Internal Medicine Residency programs at Lifespan/Brown Medical School between May and June of 2004. The survey included demographic questions as well as Likert-type questions assessing residents' frequency of seeing patients with CNMP, self-confidence in managing CNMP compared to managing diabetes (1 = much less confident; 5 = much more confident), level of preparation in the management of CNMP compared to diabetes, and the impact of interactions with CNMP patients on the residents' overall clinic experience. Reponents were asked how much time patient visits for CNMP took and how difficult they were (1 = much easier; 5 = much more difficult) compared to visits for diabetes. We compared mean responses across residency year of training using one way ANOVA and Chi-square analysis.

RESULTS: Survey response rate was 88.7% (110/124). 55% of respondents reported seeing CNMP patients 5 or more times/month and 73% reported that visits with CNMP patients took "somewhat" or "much longer" than visits for patients with diabetes mellitus. 63% of respondents also reported feeling "somewhat" or "much less" confident in managing CNMP than in managing diabetes. Confidence did not increase with level of training (mean Likert response for PGY1, 2 and 3 = 2.3, 2.4 and 2.4, respectively; $p=0.72$). 63% rated their preparation for managing patients with CNMP as "fair" or "poor," and self-rated levels of preparation did not increase with level of training (mean response for PGY1, 2, and 3 = 2.1, 2.3 and 2.2, respectively; $p=0.71$). 48% of respondents reported that follow-up visits for their own patients with CNMP were "somewhat" or "much more" difficult than visits for patients with diabetes, and 91% reported that follow-up visits with other providers' patients were "somewhat" or "much more" difficult than visits for patients with diabetes (mean response 3.4 vs. 4.3, $p<0.001$; paired t-test). 38% of respondents reported having been threatened by a patient over the prescription of pain medication, and 68% reported that their experience managing CNMP had a "negative" or "very negative" effect on their overall clinic experience.

CONCLUSIONS: Although residents in this setting frequently encountered patients with CNMP, they found CNMP management to be difficult, lacked confidence in managing patients with CNMP, and did not gain confidence over the course of training. Substantial opportunities exist for improving resident education in CNMP.

LEARNING FROM MISTAKES: HOW STUDENTS AND RESIDENTS LEARN FROM MEDICAL ERRORS. M.A. Fischer¹; K. Mazor²; J. Bari²; E.J. Alper¹; D.M. Demarco¹; M. Pugnaire¹. ¹University of Massachusetts Medical School (Worcester), Worcester, MA; ²Meyers Primary Care Institute, Worcester, MA. (Tracking ID #135661)

BACKGROUND: Trainees are exposed to medical errors throughout medical school and residency, yet little is known about how they learn from these experiences. In order to limit errors and maximize learning from mistakes, educators must understand how students and residents learn from errors, what facilitates and limits learning, and the best methods of instruction. Recent studies of medical errors have examined health systems, attending physician, patient or pre-

ceptor response. We designed this qualitative study to better understand the factors involved in trainees' learning from medical errors.

METHODS: We developed an interview text based on review of the literature and discussion with leaders in patient safety and medical education. Following pilot testing, we revised the text and completed structured telephone interviews with 57 trainees from one academic medical center (20 medical students, years 1-4; 37 medicine and surgery residents PGY-1-4). Five authors reviewed transcripts of audiotaped interviews. We employed content analysis to identify major themes and areas of tension or conflict.

RESULTS: Trainees clearly stated that they were aware that medical errors occur from their first days of medical school. Many believed that errors were "inevitable" and in some cases they considered this before deciding to enter the medical field. Some thought about errors "all the time," and had an intense emotional response to the idea of committing errors in patient care. Students and residents noted a conflict between institutional recommendations to discuss errors with risk management, team members or patients, and the actions of some faculty or supervising residents. In many cases they were confused about whether, when and how to initiate discussion after errors occurred. Students in particular noted the inherent conflict between reporting errors (sometimes made by supervisors) to their seniors and those who were responsible for their own evaluation. Some trainees placed great faith in the medical system to protect them from committing errors, and from responsibility when errors occurred. No students or residents felt that they learned better from near misses than from actual errors, and in fact many believed that they learned the most when harm was caused.

CONCLUSIONS: Our findings suggest that awareness of medical errors has been successfully raised. This is just the start. Institutions can immediately address individual variability in faculty response and local culture by disseminating clear, accessible response algorithms to guide behavior when errors occur. Educators should develop longitudinal curricula that integrate actual cases and faculty disclosure. Future multi-institutional work should focus on major themes identified in our study such as teaching and learning in emotionally charged situations, learning from errors and near misses and appropriate levels of individual and systems responsibility.

LEARNING IN THE SHADOW OF DEATH: MEDICAL STUDENT TRAINING EXPERIENCES AND ROLE MODELING IN END-OF-LIFE CARE. M.W. Rabow¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #135507)

BACKGROUND: Little is known about the experiences of medical students learning how to provide care to dying patients. Prior research suggests that students have minimal contact with appropriate end-of-life care (EOLC) role modeling, but empiric assessments are limited, have not followed students over time, and have not compared the differential role modeling students receive from 2nd year (MS2) clinical preceptors vs. 3rd year (MS3) ward attendings vs. housestaff. We assessed students' EOLC experiences, attitudes and emotions over the course of their 3rd year.

METHODS: MS3s at a large, urban medical center completed a confidential, self-administered questionnaire. The questionnaire was reviewed by national experts, pilot tested by MS2s students at an affiliated medical school program, and approved by the institution's review board. Responses included Likert scale and yes/no options. Students completed the questionnaire just prior to the beginning of 3rd year and then again 8 months into 3rd year ward rotations. Analysis included simple frequencies and paired t-tests for changes over time for individual students, with p values <0.05 considered significant.

RESULTS: Overall, 126 students completed both surveys (response rate 89.2%). More students were female (56.7%), with an average age of 25.5 years. Student ethnic backgrounds included 51.2% White, 28.3% Asian, 8.7% Hispanic, 2.4% Native American, 1.6% African American. By the end of 2nd year, 76.4% of students had personally experienced the death of a loved one; 7.1% of students suffered such a loss during the 3rd year. Students were more likely to see housestaff than attendings perform EOLC tasks, such as breaking bad news (p=0.018), leading advance directive discussions (p<0.001), and managing physical (p=<0.001) and emotional symptoms (p=0.002). In fact, students did not observe attendings performing EOLC tasks during 3rd year any more than they had observed preceptors do so during their 2nd year. Students were more likely to discuss EOLC (p<0.001) or the supervisor's emotions (p=0.008) with housestaff compared to attendings. Over time, students were more likely to feel badly about the care they had witnessed for dying patients (p=0.001). MS3 attendings were less likely to discuss patient emotions than MS2 preceptors (p=0.006). Attendings were more critical than housestaff (p=0.024) or other students (p=0.013) of MS3's EOLC emotions. The most commonly cited sources of emotional support for MS3s were: other medical students (36.3%), family or friends (35.6%), and no one (17.0%). Over time, students became less supportive of each others' emotions (p=0.035).

CONCLUSIONS: As they begin ward medicine, medical students have limited observation of attending EOLC behaviors and students feel increasingly uneasy about EOLC. They also perceive limited support from attendings and decreasing support from fellow students. The importance of role modeling and support in EOLC training deserves further exploration.

LEARNING TO PERFORM GENDER-SPECIFIC EXAMINATION SKILLS: PREDICTORS OF OPPORTUNITIES, SUPERVISION, AND CONFIDENCE. H.S. Powell¹; J.D. Bridge¹; S. Eskesen²; F. Estrada³; M.B. Laya¹. ¹University of Washington, Seattle, WA; ²Valley Family Medicine, Renton, WA; ³University of Arizona, Tucson, AZ. (Tracking ID #134072)

BACKGROUND: The pelvic, breast, and male genital examinations are very challenging to teach and learn. We sought to evaluate the role of student gen-

der, completion of the ob-gyn clerkship, patterns of supervision, and number of examinations performed on self-rated confidence in performing gender-specific examinations.

METHODS: Third and fourth year medical students (N=404) were surveyed and asked to approximate the number of pelvic, breast, and male genital examinations they had performed and to estimate the percentage of these examinations that were observed and the percentage repeated by a supervising physician. They were also asked to rate their confidence in performing these examinations. Survey results were analyzed by gender, level of training and whether they had completed the ob-gyn clerkship. A chi-square analysis was performed for each of the questions regarding clinical experience and confidence. A two-sided p-value<0.05 was considered statistically significant.

RESULTS: Response rate was 48% and 54% of the respondents were female. Fourth year female students performed significantly more pelvic (p<.03) and breast examinations (p=.01) than fourth year male students. Eighty one percent of female students had performed more than 15 pelvic exams as compared to 59% of male students (p=.03). Eighty one percent of female students had performed greater than 15 breast examinations as compared to 49% of male students (p=.01). The percentage of examinations that were repeated by a supervising physician was low. Only 34%, 44%, and 25% of all students had greater than 75% of their pelvic, breast, and male genital examinations, respectively, repeated by a supervising physician. Male students were less confident in performing the pelvic exam (p<.01) and female students were less confident performing the male genital exam (p<.01). The only predictor of confidence in performing each of these examinations was the number of examinations performed (p<.001). Confidence did not correlate with the percentage of exams observed or repeated by a supervising physician, student gender, or completion of their ob-gyn rotation.

CONCLUSIONS: Student gender was a marker for suboptimal exposure for performing opposite sex gender-specific examinations. We found no evidence of a "preceptor effect" i.e. student confidence did not correlate with the number of examinations observed or repeated by a supervising physician. The only predictor of confidence in performing these examinations was the number performed. Special efforts should be made to eliminate gender disparity in opportunities to perform gender-specific examinations.

Fourth Year Medical Students with Self-Reported Confidence in Performing Gender-Specific Examinations.

Type of Examination	Female Students	Male Students	P value
Pelvic exam	88%	61%	P<.01
Breast Exam	79%	76%	P=.92
Male Genital Exam	50%	80%	P<.01

MEDICAL ERROR DISCLOSURE ATTITUDES IN INTERNAL MEDICINE RESIDENTS: CHARACTERISTICS AND COMPARISON TO OTHER TRAINEES. P.Y. Watson¹; C.M. Gyiraszin¹; J.L. Musial¹. ¹Henry Ford Hospital, Detroit, MI. (Tracking ID #134047)

BACKGROUND: Professional associations, ethical guidelines, and patient safety groups promote disclosure of medical errors to patients. Several studies have analyzed the behavior and attitudes of physicians and patients with respect to error disclosure. Few studies, however, have specifically investigated the attitudes of resident physicians.

METHODS: An anonymous questionnaire was distributed to residents in internal medicine and twelve other residency programs at a large, midwestern, urban, independent academic medical center. Residents were queried about their attitudes regarding medical error disclosure to patients, supervising physicians, and peers. The questionnaire consisted of clinical vignettes and attitude-based survey questions.

RESULTS: Of 113 internal medicine residents, 74 (65.4%) returned the survey. Nearly all residents (91.9%) felt ethically compelled to disclose medical errors. Most residents would disclose errors to a patient regarding a missed drug allergy with mild reaction (91.9%), improper inpatient medication orders leading to moderate complications (87.8%), patient misidentification resulting in improper radiology testing (91.9%), and a catastrophic error resulting in ICU transfer (82.4%). Only 52.7% of residents felt comfortable informing patients of their medical errors while few residents (37.8%) agreed they had adequate training in how to communicate with patients about medical errors. Some residents agreed that potential barriers to disclosing medical errors include: lack of support by colleagues and peers when disclosing errors (18.9%), causing more harm or anxiety for the patient (27%), fear of litigation (36.4%), and perception of skills by senior staff (59.4%). Overall, however, the majority of residents (89%) indicated that disclosing medical errors will improve patient care. Among internal medicine residents, few significant differences existed between interns and more senior residents. Junior and senior internal medicine residents were more likely to witness unreported errors (p<.01) and to have received adequate training in communicating errors to patients (p<.05) compared to interns. When all internal medicine resident responses were compared to resident responses from emergency medicine, family practice, general surgery, and obstetrics/gynecology few significant differences were found. Both general surgery (p<.01) and obstetrics/gynecology (p<.05) residents however, were less likely to inform patients of a medical error if it did not result in harm.

CONCLUSIONS: Overall, the majority of residents would disclose medical errors to patients in various hypothetical clinical situations. However, a significant

number of residents feel inadequately trained and have discomfort disclosing errors. According to the data, several barriers could potentially inhibit disclosure of errors. Despite preconceived notions about differences among training specialties, there is very little disparity in the attitudes of residents in other major training specialties when compared to internal medicine. In the setting of comprehensive patient safety instruction, these results may help guide the development of curriculum to better educate residents in medical error disclosure.

MEDICAL STUDENT INTEREST IN PURSUING A CAREER IN GENERAL INTERNAL MEDICINE: THE EFFECTS OF THE INPATIENT GENERAL MEDICINE ROTATION. V. Arora¹; T.B. Wetterneck²; J.L. Schnipper³; P. Kaboli⁴; A.D. Auerbach⁵; D.V. Gonzales⁶; W. Levinson⁷; H. Humphrey¹; D.O. Meltzer¹. ¹University of Chicago, Chicago, IL; ²University of Wisconsin-Madison, Madison, WI; ³Brigham and Women's Hospital, Boston, MA; ⁴University of Iowa, Iowa City, IA; ⁵University of California, San Francisco, San Francisco, CA; ⁶University of New Mexico, Albuquerque, NM; ⁷University of Toronto, Toronto, Ontario. (Tracking ID #133587)

BACKGROUND: Medical student interest in a career in general internal medicine (GIM) has declined. For this reason, it is important to examine factors that can increase entry into a career in GIM. Although it has been suggested that the third year inpatient medicine rotation can influence interest in a career in GIM, there is little evidence to date evaluating this hypothesis. The aim of this study is to assess the effect of satisfaction with the general medicine inpatient rotation on student interest in a career in GIM.

METHODS: From July 2001 to July 2003, third year medical students at six academic medical centers were asked to rate their satisfaction with the general medicine inpatient rotation (overall and with certain aspects, e.g. quality of attending rounds) using a 5 point Likert scale on an end-of-clerkship survey. Interest in pursuing a career in GIM was defined as a response of 4 (Very Likely) or 5 (Certain) to the question, "Upon completing this clerkship, how likely are you to pursue a career in general internal medicine?" We used multivariate logistic regression, controlling for site and initial interest in GIM, to determine the effect of satisfaction with the general medicine inpatient rotation (both overall and with certain aspects) on interest in pursuing a career in GIM.

RESULTS: Of the 777/856 (56%) student respondents, the majority (89%) rated their overall satisfaction as "Satisfied" (4) or "Very Satisfied" (5). 38% (180/474) of students intended to pursue a career in GIM. In multivariate analyses, students with higher overall satisfaction were more interested in pursuing a career in GIM (OR 2.88, $p < 0.001$). Specific aspects of the rotation (e.g. teaching, patient care, and characteristics of attending or resident) did not predict a student's intention to pursue a career in general internal medicine. However, the following aspects predicted a high degree of overall satisfaction with the rotation: better relationship with the attending (OR 2.36, $p < 0.001$) and resident (OR 1.81, $p < 0.044$); clinical excellence of the resident (OR 2.03, $p < 0.018$); quality of attending rounds (OR 1.79, $p < 0.019$); presence of a structured curriculum (OR 1.49, $p < 0.029$); and teaching about clinical topics (OR 1.84, $p < 0.015$) and cost-effectiveness (OR 1.46, $p < 0.05$). Nevertheless, the greatest predictor of interest in pursuing a career in GIM was a student's pre-clerkship interest in GIM. Students with low pre-clerkship interest were less likely to enter a career in GIM (OR 0.16, $p < 0.001$). Conversely, students who were initially "Very Interested" (4) or "Certain" (5) about a career in GIM were more likely to pursue a career in GIM after the rotation (OR 4.71, $p < 0.001$ and OR 8.20, $p < 0.020$, respectively). In open-ended commentary, students highlighted the importance of pre-clerkship interest: "While I have no interest in internal medicine as a career, the learning was good and appreciated on the rotation." A dissatisfied student similarly states, "My dissatisfaction with this rotation had nothing to do with the fact that I have no interest in general internal medicine as a career."

CONCLUSIONS: Students who are more satisfied with their inpatient general medicine rotation are more likely to pursue a career in GIM. However, pre-clerkship interest is a far more important predictor of intent to enter a career in GIM. Factors affecting interest in GIM prior to the clinical rotation may be an important target of efforts to increase entry into the field.

MONEY ISN'T EVERYTHING: WHICH CAREER VALUES DETERMINE WHETHER STUDENTS WILL BECOME GENERAL INTERNISTS? M.S. Grayson¹; D.A. Newton²; L.F. Thompson³. ¹New York Medical College, Valhalla, NY; ²Brody School of Medicine (East Carolina University), Greenville, NC; ³North Carolina State University, Raleigh, NC. (Tracking ID #131622)

BACKGROUND: Although general internists play a critical role in the US health care system, medical student interest in a career in General Internal Medicine (GIM), as opposed to Internal Medicine Subspecialties (SS), continues to decline. Information about the characteristics and values of students choosing both general and subspecialty medicine careers can be of value in addressing health workforce needs. The objective of the current study is to compare the career values of 4th year medical students planning to pursue GIM versus SS careers. **METHODS:** 1334 (73%) of the medical students graduating from the Brody School of Medicine (East Carolina University) and New York Medical College between 1998 and 2004 completed surveys immediately prior to graduation. The 132 students intending to pursue GIM were compared to the 157 intending to pursue SS. The analysis included demographics (gender, race, marital status, size of high school community), anticipated income 5 years after residency, and Likert scale responses to items gauging the degree to which students valued various career attributes. Factor analysis of these career attributes indicated that 29 of them clustered into 7 career values: 1) comprehensive patient care, 2) prestige, 3) lifestyle, 4) helping others, 5) working with the poor, 6) research, and 7) income.

RESULTS: Students choosing SS and GIM careers had similar demographic characteristics. Students interested in SS anticipated significantly higher incomes than those pursuing GIM careers (\$182,940 vs. \$129,402, $p < .001$). Five of the seven career values significantly ($p < .01$) differentiated GIM aspirants from their SS counterparts. Students planning SS careers were more likely to value prestige, research opportunities, and income. Those interested in GIM were more concerned with comprehensive patient care and working with the poor.

CONCLUSIONS: Career values, as well as anticipated income, play a significant role in determining whether students entering internal medicine will remain as generalists or choose to subspecialize. An awareness of these factors may play a role in defining selection criteria for medical students, assist internal medicine faculty in career advising, and facilitate the development of educational programs designed to meet national health care needs.

MONITORING OCCUPATIONAL STRESS DURING INTERNSHIP. S. Allen¹. ¹University of Connecticut, Farmington, CT. (Tracking ID #135950)

BACKGROUND: Stress is typical during residency training. The situational, professional, and personal sources of these stressors have been discussed in the literature. The Residency Review Committee - Internal Medicine states that program directors of internal medicine residency training programs are responsible for monitoring resident stress. No published reports have described such a monitoring program in internal medicine.

METHODS: The Occupational Stress Inventory - Revised (OSI-R) was administered to a group of 14 interns in the Primary Care Internal Medicine Residency Program at the University of CT. The interns completed the 20-minute, confidential inventory as a group during time reserved as part of an educational seminar. To date, the group has completed the OSI-R in August and November of 2004. The OSI-R has 14 subscales that are grouped into three questionnaires: the occupational role questionnaire (ORQ) which reflects occupational stress, the personal strain questionnaire (PSQ) which reflects psychological strain, and the personal resources questionnaire (PRQ) which indicates deficits in coping skills.

RESULTS: For the ORQ and PSQ, T-scores of 40-59 are within the normal range; scores > 60 indicate maladaptive stress, debilitating strain, or both. The median T-score for each subscale was between 40 and 60 except for: 1) the August subscale Role Insufficiency which was 38 indicating the interns felt a relatively clear sense of what they are expected to do and 2) the November subscale Responsibility which was 61. Three (21%) interns scored > 60 on at least one subscale of the ORQ in August and 9 (63%) did in November. Most of the November high scores (8/15 or 53%) were on the Responsibility subscale indicating stress related to the level of responsibility for work performance of others, the frequent need to respond to others' problems and poor relationships with people at work. Four (29%) scored > 60 on at least one subscale of the PSQ in August while 5 (36%) did in November. For the PRQ, T-scores of 40-49 are normal and < 39 indicate deficits in coping skills. Three of 14 (21%) scored < 39 on the PRQ in August and 4 of 11 (36%) did in November.

CONCLUSIONS: The OSI-R took little time and was easy to administer and score. No differences were found in the personal strain or personal resources domains between August and November. There was a low level of occupational role-related stress in August but a threefold increase by November suggesting the possibility of a seasonal variation. Additional surveys will be administered through the academic year to address this issue. Program directors can use the OSI-R to recognize and monitor stress in an effort to: 1) comply with RRC-IM guidelines and 2) develop programs to address resident stress.

ON THE ASSESSMENT OF CLINICAL TEACHING: A NEW METHOD FOR RATING THE VALIDITY EVIDENCE. T.J. Beckman¹; D.A. Cook¹; J. Mandrekar¹. ¹Mayo Clinic, Rochester, MN. (Tracking ID #132652)

BACKGROUND: Modern theory states that validity is a hypothesis, and that evidence is collected to support or refute this hypothesis. The five recognized sources of validity evidence are content, internal structure, relation to other variables, response process and consequences. Although experts emphasize the importance of utilizing a variety of validity evidence when evaluating assessment tools, a review of published clinical teaching assessments suggested that authors pursue a limited range of validity evidence. To further define the role of validity in clinical teaching, we evaluated the literature for evidence supporting the validity of clinical teaching assessments. Our objectives were to develop a method for rating the validity evidence, and to quantify the evidence supporting existing assessments of clinical teaching.

METHODS: Searching MEDLINE, EMBASE, PsycINFO, ERIC, and Social Sciences/Citation indices yielded over 330 articles on the assessment of clinical teaching. Twenty-two articles met predetermined search criteria. Authors TJB and DAC developed a method for rating the five categories of validity evidence reported in each article, and they independently rated the articles using the following system: N=the category of evidence was not discussed and no data was presented; 0=the category of evidence was discussed but data was not presented; 1=data weakly supported the evidence category; 2=data strongly supported the evidence category. We then quantified the validity evidence by summing the ratings for each evidence category. Finally, we calculated weighted kappa coefficients to determine inter-observer reliabilities for each category of validity evidence.

RESULTS: Content and internal structure evidence received the highest ratings (27 and 32 respectively, of 44 possible). Response process, relation to other variables and consequences received the lowest ratings (2, 9 and 2 respectively). Inter-observer reliability was good to excellent for content, internal structure

and relation to other variables (kappa range 0.52 to 0.96, all p-values <0.01), but poor for response process and consequences.

CONCLUSIONS: Evidence for content and internal structure is well represented among published assessments of clinical teaching. Evidence for relation to other variables, response process and consequences receives little attention. Future research should emphasize these lesser-used categories. The high inter-observer reliability for content and internal structure likely reflects the favorable interpretability of these categories. Conversely, the low inter-observer reliability for response process and consequences likely reflects the scarcity of evidence reported in the literature. Our method for rating the validity evidence should prove useful in all settings where validity evidence is utilized.

OUTCOMES OF A LONGITUDINAL FACULTY DEVELOPMENT COURSE IN EVIDENCE-BASED MEDICINE: KNOWLEDGE, SKILLS, AND RESOURCE USE. C.M. Warde¹; L. Nicholson²; J.R. Boker³. ¹University of California, Irvine, Long Beach, CA; ²University of California, San Diego, La Jolla, CA; ³University of California, Irvine, Irvine, CA. (Tracking ID #133935)

BACKGROUND: Faculty clinicians frequently lack essential EBM knowledge and skills that are necessary to teach their trainees. We implemented a longitudinal course for clinical faculty to improve the practice and teaching of EBM at the point of care, and evaluated the outcomes of this intervention. Content focused on 3 practical areas: 1) general understanding of the EBM approach to health care; 2) knowledge and use of EBM computerized resources; and 3) critical appraisal of common article types.

METHODS: Twenty-eight teaching faculty – 14 from a university hospital and 14 from a university-affiliated community hospital – were invited to participate in a yearlong course on EBM for the practicing physician. Participants completed a two-part pre-course needs assessment: the validated Fresno test of EBM skills and a newly developed questionnaire on demographics and EBM exposure. Curriculum followed the teaching model of the SGIM EBM Task Force and was presented in nine, 90-minute sessions in a computer lab setting. Following brief didactics by the facilitator, participants supplied case-based questions for interactive searches of internet EBM resources and critical appraisal of articles thus obtained. The online EBM resources were also made available in participants' practice settings throughout the year. At the end of the course, subjects repeated the two-part testing.

RESULTS: Evaluation data were obtained on twenty-six faculty. Comparison of pre- and post-course mean Likert scores (1=lowest; 5=highest) revealed a significant improvement in participants' self-rated EBM knowledge (2.54 to 3.38, $p < 0.01$, for community faculty; 2.62 to 3.31, $p < 0.01$, for university faculty) and self-described EBM resource use (2.73 to 3.60, $p < 0.01$, for community faculty; 3.56 to 3.98, $p < 0.05$, for university faculty). A ratio of intent to use experience-based versus evidence-based resources significantly changed for the community faculty (1.34 to 0.88, $p < 0.01$), and correlated directly with measured use. Both groups improved their literature appraisal skills as measured by Fresno test pre- and post-course percent correct (47.28 to 71.85, $p < 0.001$, for community faculty; 68.58 to 84.92, $p < 0.001$, for university faculty).

CONCLUSIONS: Despite important clinical and teaching roles, our faculty subjects were deficient in EBM skills and resource utilization at baseline. Combining a learner-centered, interactive course with consistent worksite availability of web-based EBM resources, participants improved in all three practical content areas – general understanding of EBM, use of EBM resources, and appraisal of common article types. These behavioral changes will allow point-of-care EBM to be practiced by faculty and modeled for their trainees.

PERFORMANCE ON A STANDARDIZED PELVIC EXAMINATION: DOES LEVEL OF TRAINING MATTER? F.S. Massie¹; C.D. Lorish¹; A.H. Wood¹; L.L. Willett¹; C. Estrada¹; A.R. Goepfert¹. ¹University of Alabama at Birmingham, Birmingham, AL. (Tracking ID #135062)

BACKGROUND: To compare performance on a standardized comprehensive pelvic examination across levels of training.

METHODS: During July and August 2004, as part of an Objective Structured Clinical Exam (OSCE), we tested the performance of PGY-1 residents (OB/GYN and Internal Medicine) and fourth year medical students (MS4) on 2 pelvic models (each model assessed a different aspect of the exam). Performance was assessed by 2 checklists derived from the American College of Physicians' Pelvic Examination training video. A trained nurse practitioner, who observed the exam in the room, completed a 24-item checklist focusing mostly on the "manual aspects" of the exam technique requiring close observation (Direct Score). Simultaneously, a trained nurse, who observed the encounter via a video monitor, completed a 15-item checklist focusing mostly on the "verbal aspects" of the exam (Viewing Score). Scores from both checklists were summed and re-scaled to a range 0-100%. We used the Kruskal-Wallis and the Chi-Square tests to compare groups.

RESULTS: Six PGY-1 OB/GYN residents, 30 PGY-1 IM residents, and 162 MS4's completed the comprehensive pelvic examination. The median score for all trainees ($n = 198$) was 69% for the Direct score and 60% for the Viewing score. The performances did not differ between groups for either score (Table). A rectal exam was performed by none of the PGY-1 OB/GYN residents, by 23.3% of the PGY-1 IM residents, and by 50.6% of the MS4's ($P = 0.001$).

CONCLUSIONS: Our study suggests that fourth year students and PGY-1 residents demonstrate similar performance on a comprehensive pelvic exam. Overall, their performance was judged suboptimal and may suggest a need for more structured curriculum on performing a comprehensive pelvic examination.

Performance by Group of Trainees (Values are Medians with Interquartile Ranges)

Score	PGY-1 (OB/GYN) (n=6)	PGY-1 (IM) (n=30)	MS4 (n=162)	P-value
Direct Score, %	73 (13)	63 (24)	69 (15)	0.31
Viewing Score, %	43 (38)	63 (8)	60 (22)	0.12

PRACTICE-BASED LEARNING IN MEDICINE RESIDENCY: A CONTROLLED TRIAL OF A AUDIT AND FEEDBACK TO IMPROVE PERFORMANCE OF PREVENTIVE SERVICES. T. Houston¹; G.R. Heudebert¹; L.L. Willett¹; K.P. Palonen¹; S. Massie¹; C.I. Kiefe¹. ¹University of Alabama at Birmingham, Birmingham, AL. (Tracking ID #134014)

BACKGROUND: Among practicing physicians, performance audit and feedback has been used for quality improvement, with varying success. However, these methods have not been evaluated extensively during medicine residency training. We developed and tested a curriculum using the Achievable Benchmarks of Care (ABC) method for individual performance feedback.

METHODS: To assess provision of six appropriate preventive services, we abstracted baseline charts of patients seen at least twice by Intervention ("I") residents ($N = 70$) in clinic (July 02-Sept 03). Definitions of the preventive services were based on U.S. Preventive Task Force Guidelines (eg: Breast Cancer Screening=For women ages 52-69, was mammogram obtained, scheduled, or offered in two years?). Each resident's baseline performance was calculated; then, using ABC methods, benchmark performance was calculated for each service. The curriculum intervention included: 1) lecture detailing aggregate residents' performance; 2) confidential graphic report for each resident comparing individual performance with benchmarks; 3) review of the report with program directors at mid-year review. To evaluate intervention's impact on performance, follow-up chart abstraction was then repeated six months post-intervention. To further isolate the curriculum effect versus other changes in performance as residents' training progresses, we also abstracted charts for 2 periods (baseline and follow-up) of a similar group of non-overlapping historical control ("C") residents ($N = 60$) who graduated 1-2 years prior to the intervention. Differences (I vs. C) in baseline vs follow-up were assessed using a Group*Time interaction term in a multivariable patient-level analysis adjusting for patient's number of visits, and race, and resident baseline level of training, and using generalized estimating equations to account for patient-within-resident clustering.

RESULTS: A total of 1,750 patients' charts were abstracted. Patients had a mean age of 56 (SD 9.5), 56% were female, 72% were African-American. Appropriate patients seen by "I" residents more frequently received services during follow-up, vs. baseline, (Table, $p < 0.04$ for all but Breast Cancer Screening). No significant improvements were seen among patients seen "C" residents. After adjustment, improvement was significantly greater for "I" vs. "C" for four of six services ($p \leq 0.03$ for Quit Smoking, Colon Cancer, Pneumovax, and Cholesterol).

CONCLUSIONS: Our results demonstrated improvements in the intervention group that were of greater magnitude and consistency than some previous audit and feedback studies for practicing physicians; perhaps because of the added intensity of the individualized performance review. Compared with practicing physicians, residents may also be more responsive to this feedback because it was framed as part of their training.

Baseline (B) vs. Follow-up (F/U) Proportion of Patients Receiving Appropriate Services by Resident Group (I vs. C)

Screening/Service	"I" Patients (B vs F/U)	"C" Patients (B vs. F/U)
Tobacco Use Screen	55% vs. 62%	49% vs. 44%
Quit Smoking Advice	65% vs. 86%	58% vs. 65%
Colon Cancer Screen	85% vs. 92%	81% vs. 81%
Breast Cancer Screen	89% vs. 94%	91% vs. 88%
Pneumonia Vaccination	59% vs. 69%	63% vs. 57%
Cholesterol Screen	86% vs. 93%	79% vs. 81%

PRECONCEPTION COUNSELING, INFERTILITY, AND THE MEDICAL COMPLICATIONS OF PREGNANCY: ARE WE ADEQUATELY PREPARING HOUSESTAFF IN THESE AREAS? C.L. Spagnoletti¹; M.A. Mcneil¹; D.M. Rubio¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133096)

BACKGROUND: The topics of preconception counseling, infertility, and medical complications of pregnancy are listed as competency requirements for internal medicine (IM) trainees by the American Board of Internal Medicine (ABIM). However, research has shown that residents and practicing internists feel that their training in these areas has been inadequate. This exploratory study sought to determine which factors are associated with residents' perceived preparedness to care for reproductive-age and pregnant women.

METHODS: A 62-item questionnaire was developed to assess residents' perceptions about their training in the aforementioned areas. Fifty-three Likert-type items pertained to the following categories: didactic educational experiences in residency, clinical experiences in residency, perceived importance of and attitudes towards these topics, and perceived preparedness to deliver care in these areas. Items focused on topics identified by the ABIM, such as preconception counseling and care, management of medical conditions and drug safety during pregnancy, and evaluation of infertility. Demographic items were also included. In November 2004, we administered the questionnaire to all senior residents and 2004 graduates of the University of Pittsburgh-affiliated IM training programs. The mean score for each category was calculated. T-tests and correlations were used to determine which factors were associated with perceived preparedness.

RESULTS: The overall response rate was 81% (85/105). Fifty-five percent of responders were men, 51% graduated in 2004, and 71% planned a subspecialty career. Eighty percent were in the categorical residency track and 20% were in non-traditional tracks (primary care, geriatrics, women's health, medicine/pediatrics). Ninety-four percent felt that learning about these topics during residency is important (mean score 3.5; 1=not at all important, 4=very important), 77% reported receiving little to no teaching on these topics (mean score 2.5; 1=no teaching, 5=a lot), and 85% reported caring for patients with these issues less than yearly or never (mean score 1.6; 1=never experienced, 5=weekly). Within the preparedness category, only 11% felt prepared to perform preconception counseling, 16% to address issues related to infertility, 27% to manage medical complications in pregnancy, and 38% to safely prescribe drugs to pregnant women (overall mean preparedness score 2.1; 1=not at all prepared, 4=very well prepared). Responders who cared for reproductive-age female patients in their continuity clinic felt more prepared to deliver care in these areas than those who did not (mean 2.3 vs. 2.0, $p=.008$). Future career choice, residency track, and gender did not affect preparedness. Significant correlations were found between perceived preparedness and perceived importance ($r=0.23$, $p=.05$), educational experience ($r=0.79$, $p=.01$), and clinical experience ($r=0.71$, $p=.01$).

CONCLUSIONS: Although trainees from a large university-based program think that learning about preconception counseling, infertility, and medical complications of pregnancy is important, they feel unprepared to care for women who have such issues. We found that the perceived preparedness to deliver care in these areas is strongly associated with exposure to didactic educational and clinical experiences during residency. In light of current competency requirements, IM residency programs would benefit from providing focused training in the care of reproductive-age and pregnant women.

PROCEDURAL SKILLS OF THE GENERAL INTERNIST 18 YEARS LATER: A RESURVEY OF ACP MEMBERS. R.S. Wigton¹; P.C. Alguire². ¹University of Nebraska Medical Center College of Medicine, Omaha, NE; ²American College of Physicians, Philadelphia, PA. (Tracking ID #134842)

BACKGROUND: In 1986, we found that general internist members of the ACP did a large number of procedures in their practice. Because of changes in the practice of general medicine, new regulatory requirements and increasing numbers of subspecialists, we resurveyed the ACP membership regarding the procedural skills they currently do.

METHODS: We surveyed a probability sample of ACP members similar in characteristics to the original survey group to determine the number and types of procedures they do. We replaced 6 infrequently done procedures of the 40 in the original survey with 6 outpatient procedures.

RESULTS: To date, 1180 of 2403 surveys have been returned (49%), of which 866 are general internists in active practice. There has been a dramatic decline in procedures done from 1986. On average, the percentage doing each procedure now is less than half what it was in 1986. The average number of different procedures done in practice has declined from 16 in 1986 to 7 now. Only two procedures (skin lesion biopsy and removal) showed an increase. In contrast to the previous results, the more recent graduates (after '82) report doing fewer procedures, while in the 1986 survey newer graduates did more procedures than older graduates. The number of procedures done remains high for those who practice in smaller cities or smaller hospitals.

CONCLUSIONS: The number and variety of procedures done by general internists have declined. These declines are likely related to changes in regulatory controls (e.g., gram stain), availability of ancillary services (ABGs), availability of subspecialists (cardioversion), and movement to outpatient practice. It is not known if these changes affect the quality of medical care. The changes are also likely to affect the availability of procedures during residency training. When procedures are no longer done by the majority of general internists, it is less clear whether they should continue to be required for all residents or become optional, separately credentialed parts of training.

Selected Results

Procedure	% Doing Procedure	
	1986	2004
Lumbar Puncture	73%	26%
Gram's Stain of Sputum	50%	5%
Thoracentesis	66%	24%

1 (continued)

Procedure	% Doing Procedure	
Micro Exam of Urine	70%	28%
ABGs	52%	17%
Joint Aspir. & Injection	72%	53%
CVP Placement.	39%	16%

RESIDENTS' PERCEPTION OF THE VALUE AND FREQUENCY OF FEEDBACK THEY RECEIVED DURING RESIDENCY TRAINING. G.M. Etherton¹; R.S. Wigton²; T.G. Tape². ¹University of Nebraska Medical Center, Omaha, NE; ²University of Nebraska System, Omaha, NE. (Tracking ID #133482)

BACKGROUND: Feedback is important to resident training, but there is little information on what feedback residents actually receive in their training or how it can be improved.

METHODS: We surveyed all 549 residents who graduated from our institution from 1999-2003 regarding the types of feedback they got and which types they valued most.

RESULTS: Of the 549 residents 221 (40%) completed the survey. A majority of all respondents rated the value of feedback highly and reported that the feedback they received was of good quality, but they desired more. Residents received feedback in some categories and not in others. Over 50% reported getting minimal feedback in 16 of 24 categories. For 21 of 24 categories, the value of feedback was high but the frequency of receiving that type of feedback was low. For example, 70% of respondents valued mid-rotation feedback but only 10% reported getting it; for face to face feedback 76% and 22%; and for patient feedback 61% and 12% respectively. Unexpected findings include high value rating for feedback coming from peers, supervising residents and patients. Many respondents reported receiving checklist and end of rotation feedback frequently, but did not value it highly. Analysis by specialty found that surgeons differed from non-surgeons in several categories. Surgeons reported receiving feedback that cited specific examples and identified areas of improvement more often than non-surgeons. More surgeons than non-surgeons valued feedback directed towards a common goal.

CONCLUSIONS: Overall, residents from our institution valued feedback highly but did not get feedback in some categories they valued most. Residents placed high value on feedback from peers, supervising residents and patients suggesting that residents would like input from non-traditional sources for feedback about their clinical performance. Differences exist among specialties regarding the value and types of feedback received. Our results suggest that resident training could be improved by providing more of the feedback residents value most, including increasing the feedback given before the end of rotations.

RESIDENTS' COMFORT PERFORMING INPATIENT MEDICAL PROCEDURES. G.C. Huang¹; C.C. Smith¹; C.E. Gordon¹; S.N. Weingart². ¹Beth Israel Deaconess Medical Center, Boston, MA; ²Dana Farber Cancer Institute, Boston, MA. (Tracking ID #135774)

BACKGROUND: Although medical residents perform a variety of bedside procedures, the traditional practice of "see one, do one, teach one" is neither standardized nor rigorous. Residents may be asked to perform procedures that are potentially dangerous and for which they feel poorly prepared. We hypothesized that a majority of residents would be uncomfortable performing these procedures, that comfort would increase with procedural experience, and that overall comfort would be associated with experience, faculty supervision, and elective procedures.

METHODS: In a twelve-month period from July 2003 to June 2004, we studied internal medicine residents at a Boston teaching hospital. Residents entered into an electronic log the following procedures: central venous catheter placement, lumbar puncture, paracentesis, and thoracentesis. The log elicited resident characteristics (training level, previous procedural experience) and procedural features (time of day, urgency of indication, location, number of passes, immediate complications). Using a five-point Likert scale, residents also rated their comfort with the following aspects of the procedure: knowledge of indications and contraindications, the consent process, sterile technique, anatomical landmarks, patient positioning, interpretation of results, level of supervision, and management of complications. We defined a comfort score based on a response of "extremely comfortable" or "somewhat comfortable" on every aspect of the procedure. Using Chi-square statistics, we analyzed residents' overall comfort performing procedures, stratified by training level, gender, 1-year or 3-year program, urgency of indication, time of day, and involvement of a medical procedure service. We also examined comfort with each aspect of the procedure and by training level. We created a step-wise multivariable logistic regression model with backward elimination ($p<0.05$), clustered by resident, to identify factors associated with overall comfort.

RESULTS: Out of 158 eligible residents, 106 residents performed 527 procedures: 268 central lines, 95 lumbar punctures, 81 paracenteses and 83 thoracenteses. Ninety-four (89%) were in the categorical 3-year training program. Fifty-five (52%) were first-year residents. Half of residents reported having performed one to five procedures previously. Half were performed in the intensive care unit, 57% had urgent indications, and faculty supervision via the Medical

Procedure Service was obtained in 39% of cases. Residents identified variable levels of comfort with different aspects of the procedure: 35% of procedures were associated with a lack of comfort in managing complications, and 24% with a lack of comfort in the level of supervision. Residents reported a lack of comfort in 9% of procedures related to equipment usage, 5% related to anatomical landmarks, and 3% related to patient positioning. Residents conveyed a lack of comfort with at least one aspect of the procedure in 51% of procedures. In the multivariable analysis, overall comfort was associated with use of the medical procedure service (OR 1.9, 95% CI 1.1-3.5), success on the first needle pass (1.5, 1.0-2.2), and the number of procedures performed during the year (1.1 for each additional procedure, 1.0-1.1). Factors inversely associated with comfort were status as a first-year resident (0.3, 0.1-0.5), first time doing the procedure (0.4, 0.2-0.8), thoracenteses (0.4, 0.2-0.7), and emergent procedures (0.6, 0.3-1.0). **CONCLUSIONS:** Many medical residents are uncomfortable performing common bedside procedures. Lack of comfort managing complications and lack of appropriate supervision contributed to resident discomfort, and these issues should figure prominently in curricula to address resident training in medical procedures.

PGY = post-graduate year

	PGY1	PGY2	PGY3
Indications and contraindications	10 (4%)	0 (0%)	1 (1%)
Obtaining consent	6 (2%)	1 (1%)	3 (3%)
Anatomy	14 (5%)	5 (4%)	9 (9%)
Equipment	36 (13%)	4 (3%)	5 (5%)
Sterile technique	6 (2%)	0 (0%)	1 (1%)
Patient positioning	8 (3%)	3 (2%)	3 (3%)
Level of supervision	101 (36%)	14 (10%)	9 (9%)
Interpreting results	20 (7%)	2 (1%)	3 (3%)
Managing complications	127 (45%)	29 (21%)	30 (29%)

Residents' Lack of Comfort with Procedural Aspects

SELF-ASSESSMENT QUESTIONS AND LEARNING STYLES IN WEB-BASED LEARNING: A RANDOMIZED, CONTROLLED, CROSSOVER TRIAL. D.A. Cook¹; W.G. Thompson¹; K.G. Thomas¹; M.R. Thomas¹. ¹Mayo Clinic, Rochester, MN. (Tracking ID #134513)

BACKGROUND: Logistic difficulties and demands for increasing faculty productivity challenge teaching in ambulatory settings. Simultaneously, academic health centers are charged to reform resident education and improve teaching and assessment. We previously demonstrated that Web-based learning (WBL) is effective in our resident continuity clinics. We next sought to optimize WBL through enhanced instructional methods. We hypothesized that internal medicine residents using WBL modules with self-assessment questions and feedback would have higher test scores than those using WBL modules without questions. We also sought to determine which format residents prefer and whether associations exist between resident learning styles and test score or format preference. To do this, we compared WBL modules with and without self-assessment questions and feedback in a randomized crossover trial.

METHODS: We developed four WBL modules in ambulatory internal medicine in both "standard" format and "question" format. Content and layout were identical in both formats except for case-based multiple choice questions and feedback embedded throughout the modules with questions. Consenting residents completed two modules using the standard format and two modules using the question format, with sequence randomly assigned. WebCT administered assessments including Felder's Index of Learning Styles, post-tests of knowledge, and a year-end questionnaire. Test scores were analyzed using mixed effects ANOVA. Other outcomes were analyzed using the Student's t, Wilcoxon rank sum, and Wilcoxon signed-rank tests.

RESULTS: Consent was obtained from 122 residents. Residents had higher test scores when using the question format than when using the standard format both before (78.7% versus 76.2%, $p=0.013$) and after (80.0% versus 77.3%, $p=0.009$) adjustment for study group, gender, clinic site, and post-graduate year. Learning styles did not affect scores except that visual-verbal "intermediate" learners did better than "verbal" learners (80.6% versus 70.1%, $p=0.005$). Sixty-five of 78 residents (83.3%) preferred the question format ($p<0.001$). Learning styles did not influence preference ($p>0.384$). Although residents felt that the question format took longer than the standard format to complete (60.4 versus 44.3 minutes, $p<0.001$), 55 of 77 (71.4%) reported that it was more efficient and 63 of 76 (82.9%) felt that it was more effective.

CONCLUSIONS: Self-assessment questions and feedback improved test scores for internal medicine residents using a Web-based course in ambulatory medicine. Residents preferred the format with questions and felt it to be more effective and more efficient even though it took longer. The difference in scores among verbal and intermediate learners needs confirmation by future research. These findings verify what theory has long suggested—namely, that instructional methods that actively engage learners are critically important. These findings hold implications not only for WBL, but also for "traditional" educational activities. As medical knowledge and clinical complexity grow and time for learning decreases, it will be increasingly important to identify most effective educational practices. Future research, both in WBL and in other teaching modalities, should focus on further defining the effectiveness of selected instructional methods in specific learning contexts.

SOURCES OF STRESS AND METHODS OF COPING WITH STRESS IN THIRD-YEAR MEDICAL STUDENTS. A. Dhurandhar¹; D.R. Reiffer¹. ¹Northwestern University, Chicago, IL. (Tracking ID #133486)

BACKGROUND: While medical students experience significant amounts of stress during the process of medical education, the third year of medical school appears to be a uniquely anxiety-provoking transition for many students. Several sources of stress exist during this period of clinical learning, including the constant need to adapt to different individuals and clinical teams, the expectations of superiors and patients, and the challenge of ethical dilemmas and difficult conversations.

METHODS: Data for this study were drawn from a survey administered to third-year medical students at Northwestern University Feinberg School of Medicine. Students were included in the study if they attended the session of Patient, Physician and Society III course during which the survey was administered. The survey included Likert-type questions, matching and open-ended questions. The questionnaire assessed the major sources of stress and methods for coping with stress during the third year.

RESULTS: 124/166 (74.7%) students responded to the survey. Of the 124 students, 110 (88.7%) indicated gender: 54.5% male and 45.5% female. 108 (87.1%) indicated age which ranged from 22 to 33 years with an average age of 24.9 years ($sd=2.1$). 115 (92.7%) listed at least one major stress in their lives. Those cited included: lack of time (48.7%), maintaining personal-professional balance (37.4%), choosing a residency (29.6%), lack of sleep (27.8%), evaluations (27.0%), performance anxieties (18.3%), lack of knowledge (16.5%), problems with family or significant other (16.5%), team's expectations (12.2%), responsibility of caring for patients (10.4%). Almost half of the students (47.6%) agreed or strongly agreed that they had difficulty coping with stresses in third year rotations. Close to two-thirds (62.1%) of students agreed or strongly agreed that coping with stress was more difficult in the third year than in the first 2 years of medical school. 104/124 (83.9%) students noted at least one method of coping with stress which included: exercise (52.9%), talking to family, friends and/or significant other (40.4%), spending time with family, friends and/or significant other (35.6%), watching TV/movies (33.7%), sleep (24.0%), drinking alcohol (12.5%), and eating (10.6%). 113/124 (91.1%) mentioned ways in which they had changed during third year which included feeling more cynical (17.7%), more confident (12.4%), more irritable (12.4%), and less satisfied with their career path (8.0%). Over half of students (53.2%) agreed or strongly agreed that they did not feel as if they were fulfilling their own expectations of themselves. 45.2% agreed or strongly agreed that they were not living up to others' expectations of them. 76.6% agreed or strongly agreed that these expectations created stress for them. 85/124 (68.5%) gave suggestions for improving the difficulties of third year which included allowing students more free time either in the form of shorter hours or more days off (35.5%) and having more understanding residents and/or attendings (11.8%). However 15.3% said they did not know or did not have any ideas.

CONCLUSIONS: The third year of medical school appears to be a particularly stressful transition for a large number of students. The stresses stem from performance anxieties, evaluations, lack of time, inadequate sleep, insufficient knowledge, and uncertainty about career path as well as personal expectations and perceived expectations of others. Although many students indicated that they had noticed an increase in negative attitudes during the third year, the majority of students still employed healthy coping mechanisms for stress including exercise, talking to and spending time with family, friends and/or significant other, entertainment and sleep. In addition, students felt that having more free time would improve their experience during the third year.

Major Sources of Stress

Major Stresses Mentioned	Percentage (%) of Students That Named Particular Stress
Lack of time	48.7
Personal-professional balance	37.4
Uncertainty about career path	29.6
Lack of sleep	27.8
Evaluations	27.0
Presenting and performance anxieties	18.3
Lack of knowledge	16.5
Problems with family or significant other	16.5
Team's expectations	12.2
Responsibility of caring for patients	10.4

Students' Coping Mechanisms for Stress

Coping Mechanisms Mentioned	Percentage (%) of Students That Named Particular Mechanism
Exercise	52.9
Talking to family, friends and/or significant other	40.4
Spending time with family, friends and/or significant other	35.6
Watching TV/movies	33.7
Sleep	24.0
Drinking alcohol	12.5
Eating	10.6

STRUCTURED PRACTICE OPPORTUNITIES WITH A MNEMONIC IMPROVE MEDICAL STUDENT INTERVIEWING SKILLS FOR INTIMATE PARTNER VIOLENCE. D.S. Morse¹; E. Edwardsen²; R.M. Frankel³. ¹University of Rochester School of Medicine, Rochester, NY; ²University of Rochester School of Medicine and Dentistry, Pittsford, NY; ³Indiana University Purdue University Indianapolis, Indianapolis, IN. (Tracking ID #133964)

BACKGROUND: Low rates of intimate partner inquiry, detection, and adequate management are reported in the medical setting.

METHODS: In this prospective, randomized trial, forty-three medical students were consented to be blindly assigned to either the intervention group (teaching module with guided discussion and practice highlighting use of a mnemonic) or the control group (general discussion and provision of the mnemonic at the end of the session). This exercise took place in the context of other interviewing skills training. These students subsequently interviewed standardized patients in an assessment format whose entire history and diagnoses were unknown to them prior to the encounter. The interviews were videotaped and then independently analyzed quantitatively by two investigators to assess detection and management content in the setting of intimate partner violence. Then the interviews were subjected to content analysis to assess what modes of questioning were successful in detecting the presence of intimate partner violence.

RESULTS: Seventy-five percent of the intervention group and sixty-two percent of the control group reported that the intervention was helpful. Sixty-eight percent of the intervention group and forty-five percent of the control group asked a direct question about intimate partner violence. Forty-five percent of the intervention group and 19% of the control group provided a referral. Statistically significant differences were noted in eliciting a history of prior abuse and use of empathic statements. (P values: .027) Times for interviews for both groups were about 6 minutes. Students who successfully obtained a history of abuse consistently asked direct non-judgmental questions.

CONCLUSIONS: Students are more likely to learn desired interviewing skills when they have the benefit of guided discussion, practice and memory aids. Detection of abuse was associated with direct questioning strategies.

STUDENTS TEACHING STUDENTS — DOING WELL BY DOING GOOD IN THE MUSC SUPPLEMENTAL INSTRUCTOR PROGRAM. J.G. Wong¹; T.D. Waldrep¹; T.G. Smith¹. ¹Medical University of South Carolina, Charleston, SC. (Tracking ID #136094)

BACKGROUND: Formal systems of peer teaching (such as Teaching Assistants) while common in many advanced degree graduate school programs, are considerably less prevalent in medical school. Medical students' may perceive that the time spent teaching others may be better spent in self-study activities. In 1997, The Medical University of South Carolina's (MUSC's) Center for Academic Excellence (CAE) created a supplemental instructor (SI) program whereby interested upper level medical students are hired to work closely with a small group of junior peers, primarily with basic science topics and USMLE Step 1 examination. This study examined if participation, as a teacher in the SI program, resulted in measurable academic improvement for those students serving as the supplemental instructors.

METHODS: Admission characteristics (grade point average [GPA], Medical College Aptitude Test [MCAT] score, age, year of enrollment and gender) of all SIs were compiled from the academic years 1996-2000. A second cohort of students, who shared the first groups' admission characteristics but who did not teach as SIs, was identified. Outcome measures for both groups included United States Medical Licensing Examination (USMLE) Step 1 and 2 scores, and final medical school GPA. Paired student two-tailed T-test statistics compared group means on all outcome variables.

RESULTS: There were 154 students who served as SIs during the time frame. Admission characteristics were available for 141 (92%) and an equal number of non-SI students were matched into the comparison cohort. There were no significant differences between the admission characteristics of the two cohorts (see table 1 below) The outcome measures in the SI group were significantly

higher than those in the non-SI group for all three variables — average USMLE Step 1 and Step 2 scores and final medical school GPA (see table 2 below)

Table 1 — Admission Characteristics

	GPA	MCAT Score	Age (in years)
Supplemental Instructors	3.51	28.2	24.4
Non-supplemental Instructors	3.52	28.8	24.2

Table 2 — Outcome Measures

	USMLE Step 1	USMLE Step 2	Final GPA
Supplemental Instructors	222.1	220.0	3.47
Non-supplemental Instructors	211.6	214.3	3.29
p-value	<0.001	<0.001	<0.032

CONCLUSIONS: Our findings suggest that teaching in the SI program was beneficial to the SIs' own academic success as measured by GPA and USMLE test scores. Establishing such a program may be one method other institutions could use to strengthen medical student performance.

TEACHING PRACTICE-BASED LEARNING AND IMPROVEMENT AND SYSTEMS-BASED PRACTICE USING A DIABETES REGISTRY - A RANDOMIZED CONTROLLED TRIAL. K.G. Thomas¹; M.R. Thomas¹; R.J. Stroebel¹; G.J. Hanson¹; F.S. McDonald¹; J.C. Kolars¹; J.M. Naessens¹; T.R. Huschka¹. ¹Mayo Clinic, Rochester, MN. (Tracking ID #135364)

BACKGROUND: Six core competencies, including Practice-Based Learning and Improvement (PBLI) and Systems-Based Practice (SBP), have been endorsed by the ACGME and are required components of residency program accreditation. This study aims to use an electronic chronic disease registry as a tool for teaching PBLI and SBP in an internal medicine resident clinic. Additionally, this study aims to assess the impact of this intervention using disease management outcomes and a web survey of resident knowledge and attitudes.

METHODS: This study was conducted in a general internal medicine continuity clinic during the 2003-2004 academic year. Continuity clinic firms were randomized into intervention and control groups using cluster randomization. Residents randomized to the intervention group received instruction on the use of a diabetes registry and automated systems for diabetes care in their continuity clinics. Residents assigned to the control group received usual continuity clinic education, which consisted of faculty discussion of patient care on a case by case basis. Resident skills were assessed using clinical data extracted from the diabetes registry and included monitoring for hemoglobin A1c (HbA1c), LDL cholesterol, and blood pressure. Resident knowledge and attitudes regarding PBLI/SBP were also assessed using a sixty-five item web survey constructed to allow self-assessment in the stated ACGME competency requirements for PBLI/SBP. Results were assessed using Chi-square tests with odds ratios as appropriate with statistical significance defined for p values of <0.05.

RESULTS: Clinical data was obtained on 483 patients, 252 cared for by 39 residents in the intervention group, and 231 cared for by 37 residents in the control group. All eligible residents participated in the study. Residents in the intervention group were 1.7 times more likely to have performed HbA1c tests within 6 months (p=0.003) and 1.9 times more likely to have performed HbA1c tests within 12 months (p=0.002) on their diabetic patients. Residents in the intervention group were 1.8 times more likely to have performed an LDL Cholesterol test within 12 months (p=0.005) on their diabetic patients. Successful management to clinical targets for HbA1c, LDL and mean arterial pressure was not different between groups either at the initiation of or at the end of the study. The post-study survey of residents demonstrated that intervention residents were more likely than control residents to endorse the following statements: "Practice-based improvement with respect to diabetes is most effectively performed using a systematic approach," (p=0.04), "Physicians must assist diabetic patients in dealing with system complexities," (p=0.005), "In my continuity clinic I am aware of (p=0.007) and respond to (p=0.014) the larger context and system of health care in which I care for diabetic patients," "In my continuity clinic I assist diabetic patients in dealing with complexities of the health care system," (p=0.046), and "In my continuity clinic I partner with health care managers and health care providers to assess, coordinate, and improve health care for diabetic patients," (p=0.032).

CONCLUSIONS: Use of an electronic chronic disease registry within internal medicine residency continuity clinics can improve compliance with chronic disease surveillance guidelines and can improve resident attitudes toward PBLI and SBP. Future directions include development of educational interventions aimed at improving disease control.

THE EFFECTS OF A DISCHARGE SUMMARY CURRICULUM ON THE QUALITY OF DISCHARGE SUMMARIES IN AN INTERNAL MEDICINE RESIDENCY PROGRAM. J.S. Myers¹; C.K. Jaipaul¹; J.R. Kogan²; S. Krekun¹; L.M. Bellini¹; J.A. Shea³. ¹University of Pennsylvania, Philadelphia, PA; ²University of Pennsylvania, Huntingdon Valley, PA; ³Society of Directors of Research in Medical Education, Philadelphia, PA. (Tracking ID #132873)

BACKGROUND: The hospital discharge summary serves as the gold standard document for retrieving the vital details of a patient's hospital course. Interns who have little or no experience with discharge summary skills or processes are often required to dictate this document in academic medical centers. Incomplete, inaccurate, or unclear discharge summaries may lead to potentially avoidable medical errors. The goal of this study was to determine the effectiveness of an intervention aimed at improving the quality of interns' discharge summaries at a single academic medical center.

METHODS: Twenty-five interns were prospectively randomized to one of three interventions: 1) curriculum only (CURR, n=9), 2) curriculum plus feedback from an attending who reviewed their discharge summaries (FDBK, n=7), or 3) no discharge summary teaching (CTRL, n=9). A nine-item discharge summary evaluation instrument was used to evaluate approximately ten discharge summaries for each intern pre and post intervention. Five items were scored yes/no (presence of discharge summary components, use of headings, accuracy of primary diagnosis, secondary diagnoses, and procedures). Four items were rated using a 7-point Likert scale (quality of the history of present illness, quality of hospital course, summary length, and overall readability). Data were nested within each intern and analysis of covariance (ANCOVA) was used to examine group differences controlling for baseline performance. Effect sizes were calculated for each item to compare the differences between group means relative to the pooled standard deviation.

RESULTS: Four hundred and thirty nine discharge summaries were evaluated. ANCOVA revealed significant differences among the three groups for two items: percentage using headings to identify summary components (FDBK=.78, CURR=.48, CTRL=.30, p=.001), and overall summary readability (FDBK=5.5, CURR=5.2, CTRL=5.0, p=.02). For 5 of the remaining 7 components, the aggregated group means were in the expected order, with scores being highest for the FDBK group and lowest for the CTRL group. The median effect sizes averaged over all items were large comparing FDBK to CTRL (.86) and FDBK to CURR (.78). There were few differences between the CTRL and CURR groups (.05). There was a significant decrease in absolute summary word count (FDBK 611, CURR=653, and CTRL=769, p<.0001). There was no difference between groups in days from patient discharge to summary dictation (p=.35).

CONCLUSIONS: Interns receiving discharge summary dictation instruction were found to have shorter, more organized, and more readable discharge summaries. The addition of individualized feedback on summaries further improved skills compared to the curriculum alone. Incorporation of a discharge summary curriculum into graduate medical education may improve discharge summary length, organization, and overall readability.

THE IMPACT OF RESIDENT WORK HOURS LIMITS ON INTERNAL MEDICINE RESIDENTS' CONTINUITY CLINIC EXPERIENCE. V.I. Parekh¹; B. Sharpe²; K.E. Fletcher³; P. Cornia⁴; A. Vidyarthi². ¹University of Michigan, Ann Arbor, MI; ²University of California, San Francisco, San Francisco, CA; ³Milwaukee VAMC, Milwaukee, WI; ⁴University of Washington, Seattle, WA. (Tracking ID #136250)

BACKGROUND: The Accreditation Council on Graduate Medical Education (ACGME) has restricted resident work hours to no more than 80 hours per week. In response to these rules that took effect July 1, 2003, internal medicine residency programs have been forced to reorganize schedules across the board. For many programs this has included night float or other similar inpatient rotations in an effort to maintain resident coverage of the inpatient services. While much discussion has focused on the impact of these rules on inpatient care and education, very little has been published on the impact on residents' outpatient longitudinal experience.

METHODS: We used a retrospective pre- and post-implementation design to examine the outpatient continuity clinic experience of a group of internal medicine residents at three large academic medical centers (University of California San Francisco, University of Washington and University of Michigan). We studied clinic experiences during the last 6 months of 2002 (July 1 to December 31), before the work hours rules were in place, compared to the last 6 months of 2003 after the rules took effect. We obtained detailed data from one major clinic site at each center, including number of clinics, patients per clinic and time interval between clinics. We examined categorical internal medicine residents only and included both scheduled continuity clinic visits and extra "make-up" clinics that were scheduled during the study period. We analyzed data by post-graduate year and clinic site.

RESULTS: We examined data from 236 residents at 3 medical centers. At each of the centers there was a significant decrease in the total number of patients seen by senior medical residents in their continuity clinics (PGY-2 or PGY-3) in the 6 months after implementation of the work-hours rules compared to the same 6 month period a year earlier (see table). Notably, the total number of patients seen by interns (PGY-1) did not show a similar trend and actually in-

creased at one site (Clinic A: 46.2 vs. 46.0; Clinic B: 30.1 vs. 40.4; Clinic C: 59.4 vs. 62.8; p<0.05 only for Clinic B). In addition, we also found an increase in the maximum interval between scheduled clinic sessions for residents regardless of PGY level, with some residents experiencing gaps greater than 1 month between sessions in the 6 months after the work-hours rules took effect.

CONCLUSIONS: In our study of 3 major academic centers, there was a significant decrease in the volume of patients in senior residents' continuity clinics in the 6 months after the implementation of the ACGME work hour rules. Notably, there was no such decrease in interns' patient volumes arguing against a secular trend as the explanation for this finding. In addition there was an increase in the maximum interval between continuity clinic sessions after the work-hours rules were implemented. We theorize that this differential impact on senior residents reflects the addition of night float and other inpatient rotations that were required to meet the work hours mandates. At most institutions, interns do not staff these new rotations and are therefore less likely to be impacted. Further study of these findings, including examination of patient characteristics such as age, diagnoses and proportion of new vs. continuity patients is warranted in order to better understand the impact of the work-hour rules on residents' continuity clinic experience.

Senior Resident Total Patient Volume (6 months)

Clinic	2002 (pre-rules)	2003 (post-rules)	p-value
A	62.2	33.9	<0.05
B	67.4	54.7	<0.05
C	98.5	87.7	<0.05

THE INTERPERSONAL AND COGNITIVE DOMAINS OF TEACHING: CONSTRUCT VALIDITY OF A MULTI-DIMENSIONAL SCALE. T.J. Beckman¹; J. Mandrekar¹. ¹Mayo Clinic, Rochester, MN. (Tracking ID #132519)

BACKGROUND: We are unaware of hypothesis-driven studies showing that clinical teaching assessments are comprised solely of interpersonal and cognitive domains. Additionally, our published review of clinical teaching evaluations revealed that most studies are biased by heterogeneous samples of evaluators. Our objective was to determine the construct validity of faculty assessments comprised of interpersonal and cognitive domains, utilizing evaluations obtained from resident physicians on a general internal medicine hospital service. We hypothesized that resident physicians would differentiate between interpersonal and cognitive teaching behaviors, and that this multidimensional model would have strong evidence for construct validity.

METHODS: One thousand inpatient evaluations were completed on 60 general internal medicine faculty members between March 2000 and January 2004. Education theory and expert review supported a two-dimensional, 14-item scale. Principal Components Analysis was used to explore the scale's dimensionality. Cronbach alpha was used to determine internal consistency reliability and intra-class correlations were used to determine inter-observer agreement. Relationships between domains and instructor characteristics were also examined.

RESULTS: Principal Components Analysis confirmed the existence of interpersonal and clinical-teaching domains. An efficiency domain was also discovered. Internal consistency reliabilities of all domains are high (alpha >0.90). Inter-observer agreement is very good (range: 0.64 to 0.83). A trend in the interpersonal domain is higher scores for lower-ranking professors (i.e., instructors and assistant professors). Significant findings are higher overall scores in the interpersonal domain (p<0.001), higher scores for assistant professors in the interpersonal domain (p=0.008) and higher scores for male than female faculty in the interpersonal (p=0.041) and clinical-teaching (p=0.008) domains.

CONCLUSIONS: Clinical teaching evaluations are reducible to interpersonal, clinical-teaching, and efficiency dimensions. Evidence for construct validity includes predicted domains, high internal and inter-observer reliabilities, and differential item functioning among faculty subgroups. Unlike previous studies, utilizing a homogenous sample of evaluators minimized construct-irrelevant variance. Interestingly, lower-ranking professors scored higher in the interpersonal domain, suggesting that lower-ranking professors may focus more attention on teaching activities than full professors. Future studies should determine whether the interpersonal and clinical-teaching domains remain stable among evaluations in different medical specialties, and whether scores in these domains predict meaningful faculty outcomes.

THE REPORTER-INTERPRETER-MANAGER-EDUCATOR (RIME) METHOD FOR EVALUATION OF MEDICAL STUDENTS: EXPERIENCE AND LESSONS LEARNED FROM ONE INSTITUTION. E.D. Brownfield¹; D. Cruz¹; B. Pettitt¹; J. Goedken¹; W. McDonald¹. ¹Emory University, Atlanta, GA. (Tracking ID #133004)

BACKGROUND: Variation and lack of standardization of clinical evaluation of medical students is an important problem in medical education. An alternative to the traditional grading system is the RIME method. The RIME evaluation system is a reliable and valid tool for the identification of students with insuf-

ficient knowledge and unprofessional behavior. Here we report our initial descriptive results from clinical clerkships at Emory University School of Medicine using the RIME method.

METHODS: Over a three-month period, four third-year clerkships phased in the RIME method of evaluation in addition to the usual clerkship evaluation. At the end of rotations, residents and faculty completed both a usual clerkship and a RIME evaluation form of medical students along with a survey assessing attitudes about the new evaluation system. In analysis, we assessed the RIME level of our students and compared it to Pangaro's matrix outlining at which level of training proficiency of the individual RIME components should be attained. We also compared the assigned RIME level to the letter grade assigned using the usual clerkship evaluation. Finally, we assessed how reliable the RIME method is in identifying students who demonstrate unprofessional behavior and faculty/resident attitudes towards the RIME method of evaluation.

RESULTS: A total of 132 RIME evaluation forms were completed across the four participating clerkships. The median RIME level assigned to students overall was Interpreter. Based on Pangaro's matrix, this correlates to a proficiency level of fourth-year students. 73% of students received a letter grade of A, 26% received a B, and 1% received a C on the usual clerkship evaluation. There was a correlation between median RIME levels and assigned letter grade, however the distribution of RIME levels was large within any one letter grade. Only 2% of evaluations expressed concerns about student professionalism. The students of concern received ratings from Reporter/Interpreter to Educator on the RIME scale, when according to the RIME, should have been rated Inadequate. 51% of evaluators believed the RIME system provided an accurate assessment of student performance compared to the usual evaluation, 10% did not, and 39% were unsure. Finally, 96% of evaluators believed they would not have graded the student differently if the RIME system was used as the standard evaluation process.

CONCLUSIONS: Our initial results suggest that evaluators tend to assign students at RIME levels higher than would be expected from the Pangaro matrix. This is supported also by the inflation of grades seen at our institution. Although RIME levels correlated with letter grade, our study demonstrates the wide variability in using an evaluation tool and assigning grades. It further suggests that even when unprofessional behavior is recognized, it is not necessarily reflected in the evaluation process. Over half of the evaluators thought RIME provided an accurate assessment of student performance, but close to half were unsure or non-believers. Finally, evaluators believed they would not change their grading criteria if the RIME system was used instead of the usual clerkship evaluation. Focused faculty and resident development on methods of evaluation are needed to address ongoing problems with validity and reliability in student assessments. Development in this area is needed to address both grade inflation as well as a standardized assessment of professionalism.

USMLE STEP 1 AND STEP 2 SCORES CORRELATE HIGHLY WITH INTERNAL MEDICINE NBME SUBJECT EXAMINATION SCORES. K.J. White¹; B.G. Dwinell²; L.J. Adams²; S.A. Paul². ¹University of Colorado Health Sciences Center, Aurora, CO; ²University of Colorado Health Sciences Center, Denver, CO. (Tracking ID #133738)

BACKGROUND: Residency program directors utilize students' US Medical Licensing Examination (USMLE) Steps 1 and 2 scores in deciding interview invitations and ranking decisions. Thus, performance on these examinations is an important component of the residency application. Previous studies have addressed the correlation between National Board of Medical Examiners (NBME) subject examination scores and licensure examination scores. None of these studies have specifically addressed a direct correlation between the internal medicine subject exam and USMLE scores. Such a comparison can provide information regarding students at high risk for poor performance on the internal medicine subject examination and on USMLE Step 2. Comparison of these subject exam scores with clinical scores from the medicine clerkship can offer further information about more subjective assessments of the students reflected in clinical evaluations.

METHODS: Data from 123 students completing their required 12 week internal medicine clerkship at the University of Colorado Health Sciences Center (UCHSC) during the academic year 2002-3 were assessed. This included the scores from USMLE Step 1 and Step 2, the medicine subject exam, and the clinical grade on the clerkship. Using a scatter plot graph, correlation coefficients were derived for each comparison.

RESULTS: A strong linear relationship was seen between the medicine subject exam scores and USMLE Step 1 scores (Pearson correlation coefficient, $r=0.64$), and Step 2 scores ($r=0.72$). A somewhat weaker linear relationship was seen between the clinical scores and USMLE Step 1 ($r=0.52$), and USMLE Step 2 ($r=0.55$). Comparison between clinical scores and the medicine subject exam also revealed a linear relationship ($r=0.51$).

CONCLUSIONS: This study demonstrates a direct correlation between the students' scores on the medicine subject exam and their scores on licensure examinations. Accordingly, we have demonstrated a technique for identifying students at risk of poor performance on the medicine subject exam. Those who perform poorly on the USMLE Step 1 examination are at risk of performing poorly on the NBME examination. This is especially important because this study demonstrates that the USMLE Step 1 score is a stronger predictor of subject exam scores than is the students' clinical grade. Identification of students at high risk for poor performance on the subject exam could lead to a proactive approach to improve medicine subject scores, which might include tutoring or identifying test skill resources. Finally, we have demonstrated a potential means for identifying students at high risk for poor performance on USMLE Step 2. Students who perform poorly on the internal medicine NBME subject examination are more likely to do poorly on USMLE step 2. In fact, all students who did not pass USMLE Step 2 had scored less than 70% on the subject exam. Based on

these findings, students who do poorly on the medicine subject exam are now required to retake the exam even if their clinical evaluations indicate a passing performance. Students who do poorly on this exam are also advised to develop a specific strategy for studying for USMLE Step 2 in order to avoid failing this high stakes exam.

VIEWS OF PATIENTS AND PHYSICIANS ON THE IMPORTANCE OF CULTURAL COMPETENCE TRAINING. J. Bussey-Jones¹; A.A. Donato²; R.V. Brown³; C. Hernandez⁴; Y.M. Diaz⁵; J. Rencic⁶. ¹Emory University, Atlanta, GA; ²Reading Hospital and Medical Center, West Reading, PA; ³George Washington University, Washington, DC; ⁴University of Florida, Gainesville, FL; ⁵University of Miami, Miami, FL; ⁶Boston University, Boston, MA. (Tracking ID #135096)

BACKGROUND: The 2002 Institute of Medicine Report, "Unequal Treatment: Confronting Racial and Ethnic Disparities in Health Care" proposed cultural competence education as a means to address health care disparities. While many legislative, accrediting, and educational institutions have implemented programs to address cultural competence education, little is known about patients and physicians views on the relevance of this education to clinical care.

METHODS: The CCTOP study (Cultural Competency Training and Outcomes in Patients) is a multicenter, cross-sectional study of patients presenting for care at 11 geographically diverse outpatient clinics and their physicians. Physicians were questioned about their views on cultural competence education. Patients were verbally administered a 76-item survey with questions pertaining to their views on cultural competence training of their physicians, as well as questions regarding their health and demographics.

RESULTS: We present preliminary results from 250 matched patient-physician surveys. One hundred and seventy-five physicians (118 residents, 57 attendings) completed the survey. Ninety-eight percent of physicians agreed or strongly agreed with the statement "Patients' health outcomes are improved if the physician understands the health beliefs and practices of the patient's cultural group". In the patient survey, 290 patients responded (203 women; 87 men) with a mean age of 56 years. Forty-four percent of patients reported their race as African American, 29% White, 24% Hispanic, and 3% other. Patients were asked to agree or disagree with the following questions "A doctor who understands my culture will take better care of me" and "I believe that training my doctor in my culture will improve my medical care". Bivariate analysis revealed forty-one percent of minority patients vs. 31% whites agreed with the first statement while 39% of minorities and 25% of whites agreed with the latter statement ($p=0.018$ and $p=0.01$ respectively).

CONCLUSIONS: Most physicians agree that knowledge about the culture, beliefs, and practices of their patients can improve clinical outcomes. Minority patients are more likely to believe that cultural training is important and to value a physician who understands their culture. These data suggest a general receptiveness to cultural education among physicians. This education may be particularly important for physicians caring for minority patients.

WHAT DO MEDICAL SCHOOLS TEACH ABOUT WOMEN'S HEALTH AND GENDER DIFFERENCES? J.B. Henrich¹. ¹Yale University, Hamden, CT. (Tracking ID #132926)

BACKGROUND: Despite considerable attention to women's health over the past decade, it is uncertain what progress has been made in integrating women's health and gender-specific content into medical school curricula. To obtain updated information, we examined the curricula of U.S. medical schools to assess the inclusion of these subjects into schools' curricula, and to identify factors that have facilitated change.

METHODS: We used data from a Curriculum Management and Information Tool (CurrMIT) managed by the Association of American Medical Colleges (AAMC). Between November, 2003 and February, 2004, we searched the curricula of the approximately 95 schools that entered required courses/clerkships into CurrMIT to identify interdisciplinary women's health courses/clerkships, or those with a gender-specific focus separate from traditional ob/gyn offerings. A subset of these schools ($N=60$) entered comprehensive information into CurrMIT. We searched their curricula to identify all entries that matched a specified list of 18 women's health topics (e.g., domestic violence, osteoporosis, menopause, eating disorders), or included gender-specific content on any topic (e.g., gender differences in heart disease, depression, and HIV infection, excluding basic reproductive-tract topics, such as normal menstruation and pregnancy, taught by 100% of schools in a 1994 AAMC survey). 24 additional gender-specific topics were identified, for a total of 42 topics. Outcome measures were: 1) range and mean number of topics taught by schools, adjusted for the amount of information entered, and 2) relationship between schools' characteristics and the number of topics taught.

RESULTS: Among the 95 schools with basic information on courses/clerkships, only 10 courses/clerkships at 9 schools met our criteria for an interdisciplinary women's health didactic or clinical experience. In the subset of 60 schools, the number of topics taught by each school varied from 0 to 26, with a mean of 11; no school listed more than 62% (26/42) of the topics. 50% or more of schools taught 11 of the 18 specified topics (61%), while 20% or fewer included gender-specific information on conditions that affect both women and men and where there are important gender differences. In the adjusted analysis, the number of topics listed, expressed as a proportion of the total number of course/clerkship sessions entered by each school, ranged from 0% to 15%, with a mean of 1%. In a weighted regression model, the presence of a National Center of Excellence in Women's Health ($p=0.01$) and of a female medical school dean ($p=0.14$) were positively associated with the mean number of topics taught.

CONCLUSIONS: While the majority of medical schools include important women's health topics in their curricula, few offer interdisciplinary women's health courses or clerkships, or include gender-specific information on conditions that cause the greatest morbidity and mortality in women; this pattern has persisted for almost a decade. The presence of a women's health program or center, or of senior female administrative leadership, may increase the women's health and gender-specific content of that school's curriculum.

WHAT MAKES ATTENDING ROUNDS SUCCESSFUL? A QUALITATIVE STUDY. A. Castiglioni¹; R.M. Shewchuk¹; L.L. Willett¹; G.R. Heudebert¹; R.M. Centor¹. ¹University of Alabama at Birmingham, Birmingham, AL. (Tracking ID #135053)

BACKGROUND: Ward Attending Rounds (AR) are a fundamental component of residency training. These daily sessions differ in style and effectiveness. Better understanding of interns' and residents' perceptions of what makes AR successful should aid training programs, inform attending physicians and enhance residents' educational experience.

METHODS: Nominal Group Technique (NGT) is a multi-step, structured group process used to elicit and prioritize answers to a specific question. We convened NGT meetings with two groups of medical interns (I1, I2) and two groups of medical residents (R1, R2), eliciting their perceptions on ward AR. We asked the first group of interns (I1) to identify factors that contribute to "successful" ward AR. We then had them prioritize these factors with respect to their importance. We had the second group of interns (I2) identify factors that "detract" from the success of ward AR and had them prioritize these with respect to their importance. We repeated the same process with the two groups of residents (R1, R2).

RESULTS: I1 (n=7) identified 27 factors that contribute to successful rounding experiences. This group expressed consensus in ranking: attending being approachable (not intimidating), attending showing enthusiasm, and making sure teaching is of high quality as the three factors most important for successful ward AR. I2 (n=6) identified 40 factors that detract from the ward AR experience. The results from I2's prioritization task indicated general agreement that: having mean attendings, receiving disrespectful comments from attendings and housestaff, and too long or too short rounds are the most significant detractors. R1 (n=9) identified 32 factors that contribute to successful rounding experiences. This group expressed considerable consensus in ranking: attention to length of rounds, permitting house staff autonomy (not micro-managing), and establishing goals/expectations at the start of rotations as the three factors most important to successful ward AR. R2 (n=6) identified 34 factors that detract from the ward AR experience. The results from R2's prioritization task indicated general agreement that: very long rounds, interruptions, time constraints and other responsibilities, and poor rapport between the attending and team members are the most significant detractors of successful ward AR.

CONCLUSIONS: The results of our study show that interns and residents differ in what they consider most important for successful ward attending rounds. Interpersonal factors such as attending approachability and respectful attitude are most relevant for interns while structural factors, including consideration for house staff time and autonomy, are most significant for residents. This difference in prioritization could represent a developmental or even a "culturalization" process residents go through. These findings should assist attending physicians improve the way they conduct rounds targeting the different needs of both interns and residents.

A SURVEY OF CLINICAL RESEARCH INVESTIGATORS AND CLINICAL RESEARCH COORDINATORS ABOUT THE PROCESS OF INFORMED CONSENT. N.J. Farber¹; J. Castellano¹; J. Leary-Prowse¹. ¹Christiana Care Health System, Wilmington, DE. (Tracking ID #131648)

BACKGROUND: Little research has been conducted on the actual process of informed consent in clinical research. Most studies have asked general questions rather than targeting specific aspects of the informed consent process. The purpose of this study therefore is to survey investigators and coordinators about their views on the conduct of the informed consent process in clinical trials.

METHODS: A survey asked respondents about their activities involving the informed consent process in clinical research. The survey was pre-tested among IRB members at a large tertiary care health system for face and content validity. Thereafter, the survey was sent to all research coordinators and clinician investigators at the same large, tertiary care health system as well as at a large children's hospital. All non-respondents were sent a second mailing and then contacted via e-mail. In addition to descriptive data, the effects of the demographic data on respondents' actions involving the informed consent process were analyzed via multiple logistic regression.

RESULTS: Of 250 individuals that received the survey, 135 (54%) returned completed questionnaires, and 89 (66%) of these respondents were involved in obtaining informed consent and make up the study group. Most respondents (54%) spend 30 minutes or less in the informed consent process, and only 47% check the understanding of the patient. A majority of respondents encourage questions (95%), encourage patients to take home the consent form (67%), and encourage discussion with family or friends (84%), but only 27% felt that the primary purpose of clinical research was to benefit future patients, and 36% would encourage subjects to enter a clinical trial. Respondents with training in the informed consent process and those having spent 12 or more years in research spent more time on average than those respondents who had no training or who spent less time in clinical research ($p=0.01$; $p=.003$). Clinical research coordinators were more likely than clinician investigators to encourage patients to take home the informed consent form ($p=0.044$) and to leave the decision to enter the trial up to the patient ($p=0.022$).

CONCLUSIONS: Although coordinators and investigators generally encourage patients to be involved in the informed consent process in clinical research, they spent little time informing patients and to some degree attempt to pressure patients to enter trials while refraining from checking patients' understanding of the research. Training and experience may have a positive impact on the informed consent process. Investigators and IRBs should ensure adequate training of investigators and coordinators, and the informed consent process should be left to the most experienced individuals with the most time available, often the research coordinators.

ARE PHYSICIANS' ATTITUDES OF RESPECT ASSOCIATED WITH MEASURABLE DIFFERENCES IN THEIR COMMUNICATION BEHAVIORS WITH PATIENTS? M.C. Beach¹; D.L. Roter²; L.A. Cooper². ¹Johns Hopkins Medical Institutions, Baltimore, MD; ²Johns Hopkins University, Baltimore, MD. (Tracking ID #134385)

BACKGROUND: While maintaining an attitude of respect towards patients might be considered a moral ideal, there are no data to suggest that physicians behave differently based on their attitudes towards particular patients. The purpose of our study was to determine whether higher levels of respect are associated with measurably more positive physician communication behaviors during encounters with patients.

METHODS: We audiotaped 215 patient-physician encounters with 30 different physicians in primary care. After each patient encounter, we asked the physician to rate the level of respect that s/he had for that patient using the following item: "Compared to other patients, I have a great deal of respect for this patient" on a 5-point scale between strongly agree and strongly disagree. We then analyzed the audiotapes using the Roter Interaction Analysis System (RIAS) to characterize communication behaviors. Our outcome variables were four physician communication behaviors: information-giving, rapport-building, global affect, and verbal dominance. We used linear and log linear regression with Generalized Estimating Equations (GEE) to account for clustering of patients within physicians and calculated incidence rate ratios (IRR) to compare physician communication behaviors in visits where the physician reported that s/he strongly agreed, agreed or did not agree (neutral or disagreed) that s/he had a great deal of respect for the patient.

RESULTS: Physicians strongly agreed that they had a great deal of respect for 73 patients (34%), agreed for 96 patients (45%) and were either neutral or disagreed for 46 patients (21%). The level of respect that physicians reported for individual patients was not significantly associated with that patient's age, gender, race, education, or health status; was not associated with the physician's gender, race, or number of years in practice; and was not associated with race concordance between patient and physician. Compared to visits in which the physician did not agree that they had a great deal of respect for the patient, physicians provided patients with more information during visits in which s/he agreed (IRR 1.32, 95% CI 1.09-1.59) or strongly agreed (IRR 1.57, 95% CI 1.04-2.35) that s/he had respect for the patient. Similarly, physicians offered significantly more rapport-building statements during visits in which s/he agreed (IRR 1.14, 95% CI 0.99-1.31) or strongly agreed (IRR 1.19, 95% CI 1.02-1.49) that s/he had respect for the patient. Physicians had a more positive affect score when interacting with patients for whom they strongly agreed that they had respect ($=+1.36$, $p=0.009$). Physicians were no more or less verbally dominant during encounters in which they strongly agreed, agreed, or did not agree that they had a great deal of respect for the patient.

CONCLUSIONS: Physicians who report having an attitude of respect towards particular patients provide more information, offer more rapport-building statements, and have a more positive affect in visits with those patients. Physician respectful attitudes may be important to target in improving communication with patients.

ATTITUDES TOWARD MEDICAL ERROR DISCLOSURE: RESULTS FROM A SURVEY OF FACULTY, RESIDENTS, AND STUDENTS. L.C. Kaldjian¹; E.W. Jones¹; B.J. Wu²; V.F. Hoffman¹; G.E. Rosenthal¹. ¹University of Iowa, Iowa City, IA; ²Hospital of St. Raphael, New Haven, CT. (Tracking ID #134415)

BACKGROUND: Voluntary disclosure of medical errors by physicians is an important part of patient safety, professional learning, and patient care, but the factors that motivate or impede disclosure are complex. We surveyed physicians, residents, and students to determine attitudes toward error disclosure and investigate motivating and impeding factors.

METHODS: Based on a previously developed taxonomy of factors that influence error disclosure, we conducted a survey of attending physicians, residents, and students at 3 geographic sites incorporating 2 medical schools, 2 university hospitals, and 1 community hospital. The survey instrument was pilot tested for validity and reliability and included items representing attitudes and beliefs measured on a 5-point Likert scale, disclosure vignettes, experience with errors, and demographic variables. Data collection is still in progress. Sample size to date is 420; participants thus far include 56 attending physicians, 164 residents, and 200 students, with response rates of 75%, 70%, and 83%, respectively.

RESULTS: Regarding disclosure of errors to institutions, 86% of respondents agreed that reporting errors to their institution improves the quality of care for future patients, 57% would be more likely to report errors if they knew they would receive feedback afterwards, and 34% knew what kinds of errors should be reported. Regarding disclosure of errors to patients, 87% agreed that disclosing errors to patients is the right thing to do even if it comes at a significant personal cost, 69% agreed that error disclosure to patients would help alleviate their feelings of guilt, and 56% agreed that disclosing errors to patients strength-

ens patients' trust in them as physicians. Regarding disclosure of errors to colleagues, 91% were motivated to discuss errors with colleagues to learn whether colleagues would have made the same decision, 79% were motivated by a desire for colleagues to learn from their mistakes, and 64% were motivated by a desire to unburden themselves. 54% of respondents agreed that physicians tend to expect perfect performance from each other and 73% believe that competition in education and training encourages students and trainees to keep silent about their mistakes. Regarding impediments to error disclosure, respondents were concerned about the following factors: negative patient/family reaction (89%), malpractice litigation (87%), professional discipline (73%), loss of reputation among colleagues (71%), blame from colleagues (67%), and negative publicity from news media (48%). In answer to a vignette involving disclosure of a mistake to a patient, 51% of respondents believed that disclosure would increase the risk of being sued for malpractice, 22% believed that disclosure would decrease the risk of a malpractice suit, and 27% believed it is not possible to predict the effect of disclosure on the likelihood of a malpractice suit. Preliminary chi square analyses have identified attitudinal differences based on gender and level of training. Further bivariate and multivariable analyses are pending.

CONCLUSIONS: Factors affecting the voluntary disclosure of medical errors are diverse and affect the domains of patient safety, professional learning, and patient care. Efforts to improve error disclosure in each of these domains will require multifaceted changes that reflect this diversity.

COMMON MISCONCEPTIONS ABOUT HIPAA PRIVACY REGULATIONS. G.K. Mello¹, B. Lo¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #134635)

BACKGROUND: Many physicians express that the HIPAA privacy regulations may impede patient care. We sought to examine how commonly physicians have concerns, what those concerns are, and whether they are based on misunderstandings of the regulations.

METHODS: We administered a survey to 38 attending and 75 resident physicians in a general internal medicine practice (total n=52, response rate 46%). The survey examined physician understanding of HIPAA privacy regulations and the impact of these regulations on clinical practice.

RESULTS: Of physicians surveyed, 67% reported formal HIPAA training. With regard to the impact of HIPAA on patient care, 21% thought patient care was better, 33% worse, and 46% were neutral. Twenty-seven percent reported a case within the past year where non-disclosure of information had an adverse effect on patient care. Of these cases, 79% related to a question about communicating with another health care provider and 64% to a particular mode of communication, such as sharing patient information via a fax machine. Providers had misconceptions that HIPAA forbids the following: communicating patient information via e-mail (53%) or fax (49%), disclosing prescription information to a family member (82%), or disclosing a patient's condition to a relative (92%). Most physicians understood that HIPAA provisions allow public health disclosures (83%) and verbal disclosure of patient information to another treating physician (55%).

CONCLUSIONS: Many physicians misunderstand the HIPAA privacy regulations. The misconception that HIPAA does not allow communication of information needed for patient care may have an adverse impact on patients.

CONTROLLED TRIAL OF A NEW POLICY ELIMINATING SIGNED CONSENT FOR DO NOT RESUSCITATE ORDERS. W.A. Ury¹, J.R. Sood¹, K. Teixeira¹, R. McAuley¹, J. McGugin¹, D.P. Sulmasy¹. ¹St. Vincent Catholic Medical Centers, New York, NY. (Tracking ID #132688)

BACKGROUND: Some institutions require patients and families to give signed consent for Do Not Resuscitate (DNR) orders, especially in NY, though not required by that state's DNR law. We conducted a controlled trial of a new policy that made witnessed verbal consent the preferred mode, reduced the number of DNR forms from 7 to 1, and specified on the new form decisions about treatments other than CPR (Concurrent Care Concerns or CCCs).

METHODS: A controlled trial at 2 NY teaching hospitals, changing the policy and educating the staff at Hospital A, while Hospital B served as a comparison site. At both hospitals, in the pre and post periods, we surveyed house officer (HO) attitudes and used a validated chart review method to assess number of CCCs addressed per DNR order (such as need for hospice or limits on intubation). At hospital A, we also measured pre and post stress levels of surrogates consenting for DNR orders using the Horowitz Impact of Events Scale (IES). Pre-post differences (after 1 year) were tested with Mann-Whitney-U, t-tests, or χ^2 , and were adjusted for covariates using appropriate techniques.

RESULTS: HO participation in the survey ranged from 80% to 96% across sites pre and post. HOs in Hospital B were more likely to be foreign-born and trained ($P < .001$). The confidence of HOs in their ability to discuss informed consent for general medical procedures was similar in both hospitals and did not change after the intervention. At Hospital A, the percentage of HOs reporting high confidence in their ability to obtain consent for DNR orders increased post-intervention (76% to 93%, $p = .002$, adjusted $p = .001$), while there were declines in percent reporting difficulty talking to patients (53% to 23%, $p < .001$, unadjusted & adjusted) and difficulty talking to families (57% to 32%, $p = .001$, unadjusted & adjusted) about DNR orders. There were no significant changes in these variables pre to post among HOs in Hospital B. A 98% of HOs thought the new policy was better for families and for staff. We reviewed 168 charts of patients with DNR orders pre and 101 post at Hospital A, and 80 pre and 80 post at Hospital B. The mean number of CCCs/DNR order increased (1.02 pre to 4.16 post) in Hospital A ($p < .001$ unadjusted & adjusted) and did not change significantly (1.24 pre to 1.35 post) in Hospital B. Participation rates for surro-

gates were 77% ($N = 40/52$) pre and 61% ($N = 80/132$) post. Participating surrogates did not differ from non-participants in age, sex, race, or relation to patient. Mean total IES score for surrogates at Hospital A declined post-intervention (23.6 to 17.3, $p = .02$, adjusted $p = .05$), indicating lower stress.

CONCLUSIONS: This policy change was associated with improved HO attitudes about DNR orders, more attention to the Concurrent Care Concerns of patients with DNR orders, and decreased surrogate stress. These results raise questions about the wisdom of hospital policies requiring signed informed consent for DNR orders.

DO RELIGION AND SPIRITUALITY INFLUENCE HEALTH? THE EFFECT OF PHYSICIANS' RELIGIOUS CHARACTERISTICS ON THEIR BELIEFS AND OBSERVATIONS. F.A. Curlin¹, S. Sellergren¹, J. Lantos¹, M.H. Chin¹. ¹University of Chicago, Chicago, IL. (Tracking ID #135901)

BACKGROUND: The collective experiences of physicians should provide informative insights into any interaction between religion/spirituality (R/S) and health, yet physicians' perspectives are also shaped by the patients they have cared for and by their own core values and beliefs. This study examines the ways physicians' religious characteristics and other pertinent physician and patient population characteristics predict physicians' beliefs and observations regarding the relationship between R/S and health.

METHODS: Mailed survey of a stratified random sample of 2000 US physicians from all specialties. Key outcomes include physicians' beliefs and observations regarding the extent and the mechanisms by which R/S influence health. Key predictors include physician religiosity/spirituality, prior training regarding R/S and medicine, region, and patient suffering.

RESULTS: Response rate was 63%. Overall, about half (56%) of physicians believe R/S has much influence on health, and almost all (85%) agree that the influence is generally positive. Most physicians (64%) believe the experience of illness often or always increases patients' awareness of and focus on R/S issues, but few (6%) believe that R/S often changes "hard" medical outcomes. Rather, physicians believe R/S often helps patients to cope (76%), gives patients a positive state of mind (74%), and provides emotional and practical support via the religious community (55%). In contrast, few physicians (7%) report that R/S often causes guilt, anxiety, or other negative emotions, leads patients to decline medically indicated therapy (2%), or is used by patients to avoid taking responsibility for their own health (4%). Physician religiosity is the strongest predictor of physicians' responses. In multivariate analysis, compared to those who are neither religious nor spiritual, religious physicians are substantially more likely to report that R/S strongly influences health ($OR = 17.1$ [9.7-30.2]), and that patients often mention R/S issues [2.2 [1.2-4.1]]. They are also much more likely to interpret the influence of R/S in positive rather than negative ways. Physicians with prior training regarding R/S and medicine were actually less likely to report that R/S has much influence (0.6 [0.4-1.0]), either positive or negative. Physicians who practice in the South, and those taking care of more patients with severe suffering, were generally more likely to endorse strong and positive influence of R/S.

CONCLUSIONS: Physicians' beliefs and observations regarding the influence of R/S on health appear to be influenced most by physicians' own religious characteristics. R/S and medicine appears to be a domain in which religious diversity among physicians leads them to interpret similar observations in markedly different ways. Medical educators and policy makers might seek to foster consciousness of and candor about the ways physicians' religious commitments shape their beliefs about this complex area.

HOW DO RESIDENTS RESPOND TO A PATIENT REQUEST FOR PRAYER? S. Green¹, M. Rhodes², M. Brann¹, J.F. Wilson¹, C. Griffith¹. ¹University of Kentucky, Lexington, KY; ²University of Michigan, Ann Arbor, MI. (Tracking ID #135895)

BACKGROUND: Doctors involved in care of patients at the end of life occasionally are asked to participate with family members or patients on a spiritual level. Studies generally do not observe how physicians respond to spiritual issues as they arise in a clinical encounter. Most residents receive little to no preparation or guidance for such encounters.

METHODS: Thirty-four PGY 2 and PGY 3 Internal Medicine residents were recruited to agree to be videotaped in a standardized patient encounter in a "challenging doctor-patient situation." The standardized patient scenario reflected an actor/actress playing the role of a spouse of a patient dying in the Intensive Care Unit with multisystem organ failure. The resident was introduced to the scenario as coming on service to the current situation and initiating a discussion of futility of care. The standardized patient script included a request at the close of the interview for the resident to pray with them. Two trained coders watched all the videotapes. A coding scheme was developed for verbal and nonverbal aspects of the interview which included descriptive and qualitative data.

RESULTS: Twenty-six of thirty-one residents agreed to pray with the spouse. Twenty-one agreed readily, three agreed but were judged uncomfortable, two initiated prayer prior to the spouse and four refused. Of the four who refused, three deferred to clergy and one ignored the request. Of the prayer types, three were resident-led. Ten were verbal prayers by the patient and fourteen were nonverbal. Twenty-four (70%) of the residents mentioned religious themes in the discussion independent of the prayer request.

CONCLUSIONS: Resident physicians responded respectfully to prayer requests. Several seemed uncomfortable. Responses vary greatly. Residents may benefit from educational interventions related to communication with patients and families about religious issues.

IS "INTEGRATIVE MEDICINE" POSSIBLE? AN ETHICAL FEASIBILITY ASSESSMENT. J. Tilburt¹. ¹Johns Hopkins University, Baltimore, MD. (Tracking ID #136287)

BACKGROUND: "Integrative medicine" refers to healthcare that combines the best of conventional and alternative healing practices in a holistic approach to care. In many ways this approach seeks to incorporate the values from what different stakeholders believe to be two popular approaches to clinical excellence: evidence-based medicine (EBM) and complementary and alternative medicine (CAM). Yet, potentially serious tensions may arise between ethical commitments of EBM and values of various CAM healing traditions. These potential value differences may have moral implications for shared decision-making and effective health communication that stretch beyond the usual ethical principles of medical ethics. Unless tensions between these differences can be better understood and adjudicated with greater coherence, ethical integrative medicine may not be ethically feasible. The aim of this analysis was to describe various answers to the question "Can evidence based medicine be integrated with complementary and alternative medicine?"

METHODS: This historical, ethical and cultural analysis examines several possible answers to the question "Can evidence based medicine be integrated with complementary and alternative medicine?" In it we examined the history of modern allopathic medicine's professionalization and concurrent marginalization of alternative healers, the re-emergent cultural attention to alternative remedies, and recent trends in intellectual history most relevant to medicine. From this examination we then outlined several possible answers to the above question. Each answer was examined for its ethical assumptions and then compared with other answers.

RESULTS: Our analysis examined four answers to the question "Can evidence based medicine be integrated with complementary and alternative medicine?" Answers included naïve optimist, allopathic positivist, pragmatist, and pluralist positions. (1) Naïve optimists assume that evidence must confirm their strongly held intuitions about the utility of CAM. This misconstrues the nature of scientific method foundational in evidence based medicine. (2) Allopathic positivists worry about suggesting that CAM has any verifiability that is scientifically testable and biologically plausible for fear that the "quacks" will usurp medicine's rightful healing authority. (3) Pragmatists maintain an agnostic equipoise regarding the possibility that CAM modalities "work," while at the same time embracing a narrow, hierarchy of scientific knowledge. And finally, (4) pluralists challenge the purported neutrality and universality of any methods of evaluation, and thereby struggle to make meaningful judgments about various healing approaches. Each of these positions has moral implications that we address.

CONCLUSIONS: Two popular approaches to clinical excellence, evidence-based medicine and complementary and alternative medicine, may struggle in combining to establish a truly ethical integrative medicine. Further examination of the assumptions and goals of both EBM and CAM as well as substantial accommodation on both sides may be necessary before the benefits of both can be realized in a truly integrative medical practice. In this author's opinion an ethical approach that balances the insights of both pragmatist and pluralist positions may be most fruitful in yielding a new ethic for integrative medicine.

PHYSICIANS' DECISIONS TO WITHHOLD AND WITHDRAW LIFE-SUSTAINING TREATMENT. N.J. Farber¹, P. Simpson¹, T. Salam¹, V.U. Collier², J.L. Weiner³, E.G. Boyer⁴.

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BACKGROUND: There are few data available about what decisions physicians would make when encountering patients who request that life-sustaining measures be withheld or withdrawn. We therefore studied internists about their views on withholding and withdrawing life-sustaining treatment.

METHODS: A survey asked respondents about cases of patients who ask that treatment be withheld or withdrawn. Thirty-two hypothetical cases were included. The survey was sent to a random sample of 1000 generalist and subspecialist internists. The effect of the demographic data on the number of scenarios in which care was likely to be withheld/withdrawn was analyzed via ANOVA and multiple logistic regression.

RESULTS: Of 971 surveys received by subjects, 407 (42%) were completed and returned. Although a majority of respondents were willing to withhold/withdraw in all 32 of the scenarios, 49% were unwilling to withhold or withdraw in at least 1 scenario. Respondents were less likely to withdraw treatments than withhold them ($p < 0.0001$), and were less likely to withhold or withdraw feeding tubes ($p < 0.001$) or antibiotics ($p = 0.001$) than ventilators or dialysis. Respondents were also less likely to withhold or withdraw treatments in non-terminally ill ($p = 0.02$) and alert patients ($p < 0.0001$) compared with terminally ill and demented patients, respectively. Training in palliative care ($p = 0.02$) was associated with a greater willingness to withhold/withdraw treatments.

CONCLUSIONS: A large percent of internists would be unwilling to adhere to at least some of patients' wishes to withhold or withdraw life sustaining treatment. The type of clinical scenario and type of treatment affect internists' decisions about whether they would withhold or withdraw such treatment. Training in palliative care can influence physicians' willingness to adhere to patients' wishes about life-sustaining treatment, and therefore educational programs should be provided to all students, residents and in CME programs.

TERMINAL ILLNESS ATTITUDES AND APPROVAL OF SUICIDE. J.L. Givens¹. ¹University of Pennsylvania, Philadelphia, PA. (Tracking ID #136106)

BACKGROUND: Society is wrestling with the acceptability of suicide in the setting of terminal illness. Although acceptance of suicide is known to vary by racial-

ethnic group and other patient demographics, little is known about the association of specific attitudes about end of life care and approval of suicide in terminal illness. Exploration of such attitudes can lead to the development of additional supports for the terminally ill.

METHODS: Study data come from the 1998 General Social Survey (GSS), a nationally representative, annual survey of US adults, administered by the National Opinion Research Center (NORC). The outcome variable is the response to the question: "Do you think a person has the right to end his or her own life if this person has an incurable disease?" Key independent variables are the following attitudinal questions about terminal illness, measured on a 5 point likert scale: 1) Worry about the economic burden of terminal illness; 2) Concern over the emotional burden one's family will face making decisions at the end of life; 3) Concern about pain at end of life; 4) Worry that running out of money or insurance will result in second class health care; and 5) Belief that a person's religious community will be helpful at the end of life. A multivariable logistic regression model adjusting for sociodemographic characteristics examines the association between these attitudes, analyzed as ordinal variables, and approval of suicide.

RESULTS: Of the 825 participants, most were white, female, less than 55, had less than a high school education, and income $> \$25,000$. The majority of respondents (63%) approved of suicide in the setting of terminal illness. In adjusted analysis, black race (adjusted odds ratio [AOR] 0.47, 95% confidence interval [95% CI] 0.30-0.73), age > 55 (AOR 0.54, 95% CI 0.37-0.79) and Republican affiliation (AOR 0.68, 95% CI 0.48-0.95) were associated with lack of approval, while those with more than high school education were more likely to approve of suicide (AOR 1.61, 95% CI 1.11-2.34). Two attitudes about terminal illness were significantly associated with approval of suicide: worry about the emotional burden of a person's family making medical decisions (AOR 1.30, 95% CI 1.11-1.53) and belief that a person's religious community will be helpful at the end of life (AOR 0.71, 95% CI 0.60-0.84). On the other hand, worries about economic burden (AOR 1.14, 95% CI 0.98-1.33), pain at the end of life (AOR 1.02, 95% CI 0.86-1.21), and receiving second class health care (AOR 1.01, 95% CI 0.87-1.18) were not associated.

CONCLUSIONS: In this national sample of community adults, worry about the emotional burden of family medical decision making at the end of life was associated with a 30% increase in the adjusted odds of endorsing suicide in the context of terminal illness, while belief in the helpfulness of a religious community was associated with a 30% decrease. These data support the promotion of interventions to help patients understand the role of advance directives, health care proxies, and religious communities in end of life care.

THE INFLUENCE OF THE PHARMACEUTICAL INDUSTRY ON THE SELECTION OF MEDICATIONS BY MEDICAL RESIDENTS. C.M. Vergara¹, M.C. Lindberg², F. Wang², C. Gruman³. ¹Hartford Hospital, Hartford, CT; ²University of Connecticut College of Pharmacy, Storrs, CT; ³BraceLand Center for Mental Health and Aging, Hartford, CT. (Tracking ID #135395)

BACKGROUND: Objectives: To observe trends in test answer responses to experimental test questions which were designed to assess the drug selection preferences of medical residents and to correlate these trends with a) self-reported interaction of medical residents with the pharmaceutical industry and b) sociodemographic variables of the test participants.

METHODS: All medical residents from a traditional, university-based internal medicine residency program were required to complete a 60-question, multiple choice test about ambulatory medicine. The test contained four experimental questions that were designed to assess drug selection preferences. Each experimental question provided a new and heavily advertised drug as one of the possible multiple-choice answers. Test participants were asked two additional non-test questions that surveyed how often they interacted with representatives of the pharmaceutical industry. **STATISTICAL ANALYSIS:** The responses to experimental questions were cross-tabulated to a) the responses to the two non-test questions regarding interactions with the pharmaceutical representatives, and b) selected demographic variables, such as age, sex, level of training, physician degree and medical school training.

RESULTS: 98% percent of the medical residents participated in the study ($N = 181$). Approximately 60% reported one or more interaction per month with pharmaceutical representatives. Cross-tabulation of the test responses to responses from the two non-test questions about interactions with the pharmaceutical industry revealed no statistical correlation. However, cross tabulation of the demographics showed statistical significance with respect to level of training to 2 of the 4 test questions. For one question, first year medical residents (PGY1s) were more likely ($p = .002$) to select the older and cheaper anti-hypertensive medications, such as diuretics and beta-blockers. Second (PGY2s) and third year residents (PGY3s) were more likely to select newer and more expensive antihypertensive medications, such as ACE inhibitors and calcium channel blockers. For the other test question, PGY2s and PGY3s were more likely than PGY1s ($p = .029$) to select a fluoroquinolone rather than a macrolide antibiotic for the treatment of outpatient community-acquired pneumonia. A third question, which showed no statistical correlations, demonstrated the preferential selection (42.1% overall) of fluticasone/salmeterol discus (Advair) over fluticasone alone for the treatment of mild persistent asthma.

CONCLUSIONS: These findings contradict the expected observation that upper level residents, who have had greater exposure to clinical medicine and expectedly more cognizant of clinical guidelines, should select the older, more evidenced-based medications. Although many factors may be involved in explaining these observations, the potential influence of the pharmaceutical industry on drug selection may be one of the more likely explanations Conclusion A. The use of experimental questions embedded into formal tests may be a novel idea to study the influence of the pharmaceutical industry on drug selection

preferences. B. The lack of correlation between self-reported interaction with representatives of the pharmaceutical industry and selection of newer more expensive drugs may be due to many factors. For example, the small sample size may lead to insufficient power to detect a real difference. Additionally, the experimental questions are constructed de novo and have not been validated for assessing drug selection.

A FIVE YEAR EVALUATION OF A METHADONE MEDICAL MAINTENANCE PROGRAM.

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BACKGROUND: Methadone maintenance is the most widely accepted and best studied treatment for opioid dependence. Federal and state regulations mandate initial supervised daily methadone dosing to minimize diversion, but take-home doses are permitted after treatment stability and safe methadone handling are demonstrated. Even extremely stable patients, however, are typically required to visit their methadone program two or three times per week. Though many patients are best served by long term methadone maintenance treatment for their opioid dependence, mandated frequent contact with their methadone program may in fact challenge or undermine sustained recovery. To better serve patients who have achieved long term recovery, physician's office-based methadone treatment (methadone medical maintenance) has been piloted in several states. We report on a program that began in the Bronx, New York in 1998, and describe patient characteristics and treatment outcomes through August 2004. Entry criteria include attendance at a traditional methadone program within the same treatment system, employment, and three years with no illicit substance use. The program's central pharmacy distributes monthly supplies of methadone in tablet form, prescribed by a physician who regularly meets with patients.

METHODS: Chart reviews of all patients enrolled in the methadone medical maintenance program were conducted, and characteristics of medical maintenance patients were compared to those of patients attending traditional methadone maintenance clinics in the same treatment system in the Bronx.

RESULTS: 127 patients were enrolled in our methadone medical maintenance (MM) program between January 1999 and July 2004. Compared to patients in our traditional methadone treatment programs, MM patients were older (52 vs. 44 years), and a larger proportion were male (72 vs. 59%) and Caucasian (50 vs. 17%). On admission to MM, the mean length of methadone treatment was 15 years. Among those in MM, 84% had a history of injecting drugs, 62% had a history of arrest, 92% were employed, and 71% had a high school diploma or its equivalent. Hepatitis C virus was highly prevalent (84%), but only 6% were HIV-infected. There was a strong correlation between hepatitis C exposure and injection history. Most patients (58%) were regular cigarette smokers. During MM treatment, mean methadone dose increased from 66 to 72 mg. Only two patients disenrolled from the program: one relocated, and one returned to a traditional methadone maintenance clinic. Nine patients died, with no deaths directly related to substance use. Only one patient reported heroin use since admission to medical maintenance. Urine toxicology results corroborated an extremely low level of illicit substance use: only 0.78% of the aggregate urine samples collected from medical maintenance patients were positive for non-prescribed opiates, and only 0.35% were positive for cocaine. To confirm proper methadone use and lack of diversion, 28% of patients had an unannounced recall of their monthly methadone supply, and there were no discrepancies at any of these recalls.

CONCLUSIONS: In its first five years, our methadone medical maintenance program has successfully allowed a selected group of patients to remain stable in recovery from opiate dependence. This and similar programs demonstrate that for eligible patients, methadone treatment can be expanded to more conventional and less regulated settings.

A LATENT CLASS ANALYSIS OF SMOKING-RELATED ATTITUDES AND BEHAVIORS AMONG YOUNG WOMEN.

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BACKGROUND: Young women who smoke face serious health risks yet rarely seek help. Despite aggressive anti-smoking campaigns, smoking rates are rising among young women, suggesting the need for new approaches to reach this population. Segmenting audiences using typologies based on cluster-analytic techniques can facilitate targeting interventions to specific populations, based on the association of behaviors with other health behaviors and psychological and social antecedents. We sought to profile patterns of behavioral, attitudinal and cognitive variables related to tobacco use among young women using latent class analysis.

METHODS: Participants were part of an ongoing Midwestern longitudinal study of the natural history of cigarette smoking from adolescence to adulthood. Women between the ages of 18-24 years at the time of the young adult follow-up survey (mean age 21 years; 95% Caucasian) who reported smoking at least monthly were selected for the current analysis (n=416). Variables included demographics, attitudes, social influences, enjoyment, risk perceptions, smoking behaviors, and intention to quit. Four latent class (LC) models, specifying 1-4 latent classes were tested. A comparison of model fit indices suggested that a three LC solution fit best. The average LC probabilities for most likely LC membership were 0.87, 0.90, and 0.88 for the 3 LCs, indicating good separation between classes. Cross-sectional external validation ANOVAs were performed on life satisfaction, reasons for smoking, and smoking onset. Logistic regression

analyses assessed smoking cessation (abstinent for at least 6 months) at five years follow-up.

RESULTS: Three distinct clusters emerged with the following general characteristics: 1) working class women with jobs who tended to smoke daily but reported high levels of positive affect and life satisfaction (n=203); 2) light-smoking college students who exercised regularly, began smoking after high school, planned to quit in the next year, and successfully quit at 5 years; (n=85); and 3) heavy smokers who were more likely to have children and smoking friends, did not intend to quit smoking, reported high levels of negative affect, and who were more likely to smoke for addictive reasons and to control affect (n=128). Differences in successful smoking cessation were statistically significant across these 3 clusters (19.5%, 34.5%, and 12.6%, respectively).

CONCLUSIONS: Our analyses identified 3 distinct clusters of young female smokers, clusters which were significantly associated with smoking patterns and quit rates. The psychosocial and behavioral profiles of these clusters can potentially be used to tailor smoking interventions more effectively to clusters within this population.

ADDITIONAL SILDENAFIL TREATMENT TO FURTHER IMPROVE SRI-ASSOCIATED SEXUAL DYSFUNCTION, MAINTAIN ANTIDEPRESSANT ADHERENCE, AND IMPROVE OUTCOMES IN MAJOR DEPRESSION.

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BACKGROUND: Serotonergic reuptake inhibitor antidepressant-associated sexual dysfunction (SRI-AASD) occurs in 25%-75% of patients with major depressive disorder (MDD) and can lead to early treatment discontinuation and failure. We hypothesized that open-label (OL) sildenafil treatment would improve SRI-AASD in partial and nonresponders to double-blind (DB) treatment, and that effectiveness, antidepressant adherence, and MDD in remission (MDD-R) would be maintained during subsequent OL extension treatment.

METHODS: Under DB, 90 men with MDD-R (17-item Hamilton Depression Rating Scale [HAM-D] scores ≤ 7) and SRI-AASD on a stable-dose antidepressant were randomized to sildenafil (50-100 mg adjustable) or placebo for 6 weeks, with full responders (Clinical Global Impression-Sexual Function [CGI-SF] score ≤ 2) eligible for 8 weeks of OL extension. Partial or nonresponders were offered 6 weeks of OL sildenafil (interim treatment) to continue to try to improve erectile function and become eligible for the 8-week extension: total study duration, ≤ 23 weeks. DB assignment remained concealed during subsequent OL treatment. Score changes on the CGI-SF, International Index of Erectile Function (IIEF), University of New Mexico-Sexual Function Inventory (UNM-SFI), and HAM-D were compared using paired *t*-tests, chi-square, and ANOVA.

RESULTS: DB sildenafil improved all sexual dysfunction domain scores (effect size, 1.07; 95% CI, 0.77-1.37); 55% (24/44) sildenafil vs 4% (2/45) placebo patients were full responders. Among DB phase completers, 91% (69/76) began OL sildenafil: the 26 full responders started extension treatment directly, while 43 of 50 partial- or non-responders underwent interim treatment; 35 of 43 (81%; 95% CI, 67%-92%) became full responders, with no significant difference ($P=0.4$) between those initially randomized to placebo (placebo/sildenafil group: 85% [23/27]; 95% CI, 66%-96%) or sildenafil (sildenafil/sildenafil group: 75% [12/16]; 95% CI, 48%-93%). Mean IIEF and UNM-SFI scores improved (total and most domains) in both groups ($P<0.03$ Bonferroni adjustment, *t*-test). Mean HAM-D scores improved (placebo/sildenafil group, $P<0.05$) or remained stable (sildenafil/sildenafil group). Overall, SRI-AASD fully improved in 86% (59/69) of sildenafil-treated men: 24 during DB and 35 during interim OL. Improvements were maintained during OL extension.

CONCLUSIONS: Sildenafil effectively reverses SRI-AASD, facilitating antidepressant adherence and maintenance of MDD-R. OL treatment may better reflect clinical practice results than do randomized controlled trials.

ADOPTION OF BUPRENORPHINE FOR OPIOID ADDICTION: ARE PHYSICIANS WILLING?

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BACKGROUND: Since passage of federal regulations that allow for office-based treatment of opioid addiction with sublingual buprenorphine, use of the medication has been slow. We conducted a survey of outpatient physicians in Maryland to assess baseline willingness to prescribe buprenorphine, and to obtain input from survey respondents on factors associated with willingness to do so.

METHODS: We mailed a survey to a sample of 344 primary care physicians, psychiatrists, and pain management specialists in Maryland randomly selected from the AMA master database. The outcome was level of willingness to prescribe buprenorphine to an opioid dependent patient. Other survey domains included sociodemographic characteristics, practice patterns, addictions experience and knowledge, attitudes toward addiction, and needed resources and barriers to prescribing buprenorphine.

RESULTS: Of 107 respondents, 25% were willing to prescribe buprenorphine to opioid dependent patients, 11% were unsure, and 64% were unwilling to do so. Only 9% of those willing to prescribe the medication had applied for a waiver to do so. Over 95% of willing physicians reported having great need for 1) adequate reimbursement for buprenorphine services provided; 2) quick access to referral sites for relapsed patients; 3) relationships with drug treatment specialists; and 4) patient literature on buprenorphine if they were to prescribe buprenorphine. The majority of willing physicians (63%) reported that \$30 to \$99 per follow-up visit would make it worthwhile for them to prescribe the medication. Using multiple logistic regression adjusting for sociodemographic characteristics, patient panel size, number of addicted patients, selected attitudes and barriers, and

presence of on-site substance abuse counseling, being less than 50 years of age (OR 5.5 [95% CI 1.1-27.2]), and having had a positive experience with an opioid dependent patient (OR 6.7 [95% CI 1.6-28.6]) were associated with willingness to prescribe buprenorphine.

CONCLUSIONS: Most office-based physicians in Maryland responding to this survey were unwilling to prescribe buprenorphine themselves. Strategies for increasing adoption of buprenorphine among willing physicians not yet prescribing the medication may include development and promotion of linkages with drug treatment specialists, and reasonable reimbursement rates for services. In addition, medical education efforts should focus on developing positive training experiences involving physicians and opioid addicted patients.

ADVERSE EVENTS DURING MEDICAL AND SURGICAL HOSPITALIZATIONS FOR PERSONS WITH SCHIZOPHRENIA. G.L. Daumit¹, D.M. Steinwachs¹, P.J. Pronovost¹, C.B. Anthony¹, E. Guallar¹, D.E. Ford¹. ¹Johns Hopkins University, Baltimore, MD. (Tracking ID #13561f)

BACKGROUND: Persons with schizophrenia comprise a vulnerable population with a mortality rate 2.5 times higher than the general population. Despite this, little is known about patient safety for persons with schizophrenia, and they may be at high risk for adverse events. Virtually all inpatient deaths in schizophrenia occur during admissions to medical or surgical (not psychiatric) services, where provider inexperience such as misinterpretation of somatic symptoms or antipsychotic drug errors could contribute to adverse events including sequelae from oversedation. We aimed 1) to estimate the prevalence of adverse events in medical and surgical hospitalizations for persons with schizophrenia compared to those without schizophrenia and 2) to examine the relation between adverse events and ICU admission, in-hospital death and increased length of stay (LOS) for persons with schizophrenia.

METHODS: We performed a cross-sectional study using data on discharges from all Maryland acute care hospitals' medical and surgical services in 2001-2002. For primary outcomes, we applied AHRQ's Patient Safety Indicators (PSIs), developed to detect adverse events in administrative data. The PSIs are validated and each has explicit inclusion/exclusion criteria. We studied PSIs with an adequate number of events for analysis. We compared PSIs for hospitalizations with a secondary diagnosis of schizophrenia to those without and performed multivariate logistic regression to determine the association between schizophrenia and each PSI adjusting for patient (age, race, gender, insurance, medical comorbidity) and hospital (teaching/trauma center) characteristics. For hospitalizations with schizophrenia, we examined similarly the association between each PSI and the risk for ICU admission and in-hospital death (not 'Failure to rescue' where death is an inclusion criteria), and we used median regression to examine the relation of each PSI to increased LOS.

RESULTS: There were 1746 medical/surgical hospitalizations with a secondary diagnosis of schizophrenia and 732,210 without schizophrenia. The mean age was similar at 55 years. Hospitalizations with schizophrenia had a higher prevalence of: non-whites (50% vs. 34%); men (48% vs. 38%) and public insurance (75% vs. 50%) than those without. CAD, COPD, DM, liver disease and substance abuse were more common for schizophrenia. The Table shows adverse event (PSI) prevalences and the adjusted relative odds (OR) of having a PSI. For hospitalizations with schizophrenia, having these PSIs showed a higher adjusted OR (95%CI) for ICU admission: resp. failure, OR=13.1 (12-14); DVT, OR=4.3 (3-4); sepsis, OR=7.2 (2-33). The adjusted ORs for death were higher with PSIs of resp. failure, OR=9.3 (3-26) and sepsis, OR=7.2 (2-32). The median adjusted increase in LOS in days (95%CI) for schizophrenia was elevated significantly for: infections, 18.1 (15-20); resp. failure, 14.1 (10-18); DVT, 10.7 (7-14); sepsis, 24.2 (21-27).

CONCLUSIONS: Medical and surgical hospitalizations for persons with schizophrenia had at least twice the odds of several types of adverse events than those without schizophrenia. Adverse events were associated with increases in ICU admission, length of stay and in-hospital death. The elevated odds of resp. failure and DVT in schizophrenia could be consistent with sedation due to drug errors leading to aspiration or reduced mobility. Understanding factors contributing to adverse events will be important to improving safety and potentially decreasing premature mortality in this vulnerable population.

Table: Adverse events (PSIs) for hospitalizations with and without schizophrenia

PSIs	Prevalence per 1000 admissions		Adjusted (95%CI) Relative Odds	
	Schizophrenia (n=1746)	No schizophrenia (n=732,210)		
Infections due to med. Care	13	5	2.5	(1.3-4.9)
Failure to rescue	141	150	1.0	(0.8-1.4)
Post-op resp. failure	109	45	2.1	(1.4-3.1)
Post-op DVT	35	13	2.0	(1.2-3.3)
Post-op sepsis	115	41	2.3	(1.5-3.5)
Post-op hematoma	47	37	1.1	(0.7-1.8)

AMONG INJECTION DRUG USERS, INTEREST IS HIGH, BUT ACCESS LOW TO HCV ANTIVIRAL THERAPY. K.H. Seal¹, A.H. Kral², J. Lorvick², L. Gee², J.I. Tsui¹, B.R. Edlin². ¹San Francisco VA Medical Center and University of California, San Francisco, San Francisco, CA; ²University of California, San Francisco, San Francisco, CA. (Tracking ID #13491f)

BACKGROUND: Disproportionate numbers of injection drug users (IDUs) are infected with hepatitis C virus (HCV). This study sought to ascertain interest in receiving HCV antiviral therapy among IDUs.

METHODS: From 2001-2002, 719 IDUs were recruited from street settings in San Francisco. IDUs were included in this analysis who reported that they had been infected with HCV. HCV antibody status was verified using third-generation enzyme immunoassay. Data on sociodemographics, risk behavior and health care utilization were collected. Participants were read a detailed uniform description of HCV antiviral therapy using pegylated interferon and ribavirin which included duration of therapy, the chances of clearing HCV, side-effects, and the odds of getting cirrhosis if untreated. Participants were asked to rate how interested they would be in receiving HCV treatment using a 5-point adjectival scale (definitely, probably, not sure, probably not, definitely not) and the frequencies of each response were determined. A dichotomous outcome variable, "interest in HCV treatment," was created for participants responding that they were "probably" or "definitely" interested in HCV treatment and compared in univariate analysis with the group of remaining responses. Multivariate logistic regression was used to determine independent predictors of IDUs' interest in receiving HCV treatment.

RESULTS: Among 719 IDUs, HCV antibodies were detected in 90%, although only 445 (62%) reported that they had been infected with HCV (p<0.0001). Of these 445 IDUs, over a third (31%) reported sharing injection equipment in the past 30 days. Most participants (84%) had received health care in the past year. While 71% reported having a primary health care provider, only 21% were aware of interferon as a treatment for HCV and 3% reported having received interferon. Nevertheless, after listening to a description of HCV treatment, 298 (67%) said they were either definitely (52%) or probably (15%) interested in receiving HCV treatment. Interest in receiving HCV antiviral therapy was independently associated with age >30 years (OR=4.95; 95% CI=1.22-14.3), having received healthcare in the past year (OR=1.9; 95% CI=1.13-3.29), and considering one's overall health to be poor (OR=2.3; 95% CI=1.4-3.7).

CONCLUSIONS: Despite the extremely high prevalence of HCV in this population of IDUs, few individuals were aware of treatment for HCV and fewer still had received it. Once informed about HCV antiviral therapy, the majority expressed an interest in receiving HCV treatment. Since some IDUs may be appropriate candidates for HCV antiviral therapy and the majority reported having a primary health care provider, efforts should be made to expand HCV testing and education about available HCV treatment options.

ANTECEDENTS OF PRESCRIPTION OPIATE ABUSE IN THE CARDIA STUDY. M.J. Pletcher¹, S.G. Kertesz², S. Sidney³, C.I. Kiefe³, S.B. Hulley¹. ¹University of California, San Francisco, CA; ²University of Alabama, Birmingham, AL; ³Kaiser Permanente Division of Research, Oakland, CA. (Tracking ID #133733)

BACKGROUND: Abuse of prescription opiates has increased rapidly during the past decade, but it is unclear what places people at risk for this potentially harmful behavior. We were particularly interested in whether use of medically prescribed opiates precedes non-medical use (abuse), and how often abuse of prescription opiates occurs before abuse of other substances.

METHODS: We identified Black and White men and women participating in the Coronary Artery Risk Development in Young Adults (CARDIA) Study who denied abuse of opioids (heroin or prescription opioids) at the Year 10 CARDIA examination in 1995-6, and then answered questions about abuse of prescription opioids 5 years later (2000-01). We estimated which Year 10 participant characteristics predicted initiating abuse of prescription opioids during these 5 years of follow-up using logistic regression.

RESULTS: Among 3204 participants (44% men, 47% Black, age 40 ± 4 years at follow-up), 23 reported initiation of prescription opioid abuse during the next 5 years (0.7%, 95%CI: 0.5%-1.1%). More than half of these were White men (n=12, OR 3.4 vs. all other race/gender groups, 95%CI: 1.5-7.6), and all 23 had previously reported marijuana use at Year 10 (OR infinite; p<.001). Thirty-seven participants (1%) overall had reported being prescribed opioids by a physician at Year 10; two of these (5.4%) reported subsequent abuse of prescription opioids at Year 15, indicating a positive association between medical prescription of opioids and subsequent abuse (OR=8.6; 95%CI: 1.9-38, exact p=.028). This association persisted after excluding non-users of marijuana, and then adjusting for age, race, gender, income, education, health insurance status, speed, cocaine, tobacco and alcohol use at Year 10 (OR=16; 95%CI: 2.8-94, exact p=.011). Speed use at Year 10 also remained a significant predictor in adjusted models (OR=11; 95%CI: 2.8-44), but cocaine, tobacco and alcohol use did not. The absolute risk of new prescription opioid abuse among users of both marijuana and speed (2.7%; 95%CI 1.6-4.7) increased to 25% (95% CI 3.2-65%) when such participants had also been medical users of prescription opioids at Year 10.

CONCLUSIONS: In this community-based cohort of young adults, abuse of prescription opiates was invariably preceded by abuse of other illicit substances, particularly marijuana, and was independently associated with previous medical use of opiates.

ARE PUBLIC/PRIVATE PARTNERSHIPS DELIVERING BETTER PUBLIC MENTAL HEALTH SERVICES FOR THE SAFETY NET POPULATION?. S.A. Mohanty¹, A.L. Diamant², I. Lagomasino³, S. Asch³. ¹University of Southern California, Los Angeles, CA; ²University of California, Los Angeles, Los Angeles, CA; ³VA Greater Los Angeles Healthcare System, UCLA and RAND, Los Angeles, CA. (Tracking ID #134663)

BACKGROUND: Although depression is common in primary care settings and may result in significant disability, it is frequently underdiagnosed and under-treated. Low-income patients may be susceptible to receive fewer mental health (MH) services due, in part, to their limited access to such care. We compared patterns of depression care delivered by county-contracted public private partnership (P/PP) primary care clinics and those owned and operated by the Los Angeles County Department of Health Services (LAC-DHS). The P/PPs are partnerships between the LAC-DHS and private, community-based providers committed to providing quality health services to low income and uninsured communities with limited access to primary care. The county-contracted clinics (Health Centers (HCs), Hospital Outpatient Clinics (HOCs) and Comprehensive Health Centers (CHCs)) vary with regards to volume and access to services. **METHODS:** We analyzed 165 depressed patients who made at least one visit to a doctor in the past 12 months and who completed the Patient Assessment Survey II (PAS II), a cross-sectional study of subjects surveyed from February-July 2002. A person was considered depressed if they reported ever being told that they had depression and also screened positive on the Rost-Burnam 3-item depression screener. Main outcome measures included: 1) primary care provider's suggestion of MH counseling; 2) patient attended at least one counseling session; 3) patient receiving a prescription medication for MH. Since either treatment modality (counseling or antidepressant medication) is accepted as first-line treatment for depression, we also analyzed those patients who were either suggested counseling or received medication as an indicator of the quality of care delivered by both the provider and facility. We also looked at those who attended at least one counseling session or received medication for depression treatment as an indicator of quality of care actually received. Using multivariable logistic regression adjusting for age, gender, and other sociodemographic characteristics, we compared the likelihood of receipt of these depression services by facility type.

RESULTS: In bivariate analysis, P/PPs and CHCs had higher rates in providing depression care compared to other county-owned facilities. After multivariate adjustment, county-owned HCs had the lowest odds of suggesting a counseling session (Adjusted OR 0.32, 95% CI 0.10, 1.00 vs. CHCs), of having patients who had a counseling session (AOR 0.22, 95% CI 0.07, 0.76 vs P/PPs), and prescribing a medication (AOR 0.17, 95% CI 0.05, 0.54 vs. P/PPs) for depression. Patients in P/PPs and CHCs were more likely to either be suggested counseling or receive a medication compared to HCs and HOCs. However, P/PPs patients were more likely to actually receive a counseling session or be administered medications for depression compared to the other County-owned facilities after multivariate adjustment (AOR: HOC 0.1, 95% C.I. 0.3, 0.8; HC 0.1, 95% CI 0.02, 0.3; CHC 0.2, 95% C.I. 0.04, 0.98. All AOR compared to referent P/PP).

CONCLUSIONS: P/PPs provide better overall depression care compared to the other facility types given the difficult social circumstances of public health delivery systems. The P/PP model appears to be a more effective model of primary care for depression. Policy makers should take this variation into account when working to improve depression services and determining allocation of resources for the two different delivery models.

BUPRENORPHINE REDUCES HIV RISK BEHAVIOR AMONG OPIOID DEPENDENT PATIENTS IN PRIMARY CARE. L.E. Sullivan¹, B.A. Moore¹, M.C. Chawarski¹, R.S. Schottenfeld¹, P.G. O'Connor¹, D.A. Fiellin¹. ¹Yale University, New Haven, CT. (Tracking ID #135998)

BACKGROUND: Buprenorphine is available to treat opioid dependent patients and can be provided through primary care physicians' offices. While methadone has been shown to be an effective means for reducing HIV risk behavior and HIV seroconversion, the effects of buprenorphine maintenance have not been systematically investigated. We evaluated whether physician prescribed buprenorphine treatment reduces HIV risk behavior, whether reductions persist over time, and whether abstinence achievement, use of other substances, or gender affects risk behavior.

METHODS: We performed a longitudinal analysis of patients enrolled in a 24-week clinical trial of buprenorphine maintenance in primary care, measuring changes in the AIDS/HIV Risk Inventory (ARI), a structured interview that assesses for high-risk drug and sexual behaviors. To evaluate the effects of abstinence achievement and gender, we stratified patients based on >6 consecutive weeks of opioid-free or cocaine-free weekly urine toxicology tests, and gender and included these variables in the analyses. We also examined the effects of cocaine or alcohol use at enrollment with mixed model ANOVA. Chi-square, t-tests, and linear regression analyses were performed.

RESULTS: Of the 166 subjects enrolled in the study, the mean age was 36 years (18-56) and 129/166 (78%) were male. The mean years of opioid dependence was 8 (1-35) with 52/166 (31%) reporting current injection use, 137/166 (83%) any current heroin use, 44/166 (27%) any current other (e.g. prescription) opioid use, 52/166 (31%) current cocaine use, and 68/166 (41%) with current alcohol use. In the patients with baseline (n=155), 12-week (n=91) and 24-week (n=75) ARI scores, treatment was associated with significant reductions in overall ARI score (18.5 vs. 12.8 vs. 14.2, p=.03), drug (6.5 vs. 2.3 vs 1.9, p<.001) and sexual (9.0 vs. 7.4 vs. 6.0, p=.04) behaviors subscores. Sixty-nine of the 155 patients (45%) were opioid abstinent for >6 weeks. After 12 weeks, opioid abstinent patients had decreases in their overall (-6.8) and sex-related (-2.8) HIV risk scores while non-abstinent patients had increases in their overall (+2.0) and sex-related (+1.4) scores (p=.03 and .07 for between group comparisons). Both groups had similar decreases in drug-related (-3.2 vs. -3.4, p=.8) HIV risk. When stratified by cocaine abstinence or gender, no significant differences were found (p>.20 for all comparisons). Similarly, when opioid abstinence, alcohol or cocaine use at enrollment, and gender were entered into a linear regression model, only opioid abstinence predicted change in overall ARI score (p=.04).

CONCLUSIONS: We conclude that buprenorphine reduces overall, drug-related and sex-related HIV risk and that this trend persists over time. Drug-related HIV risk behaviors are reduced regardless of whether patients achieve sustained opiate abstinence. Opioid abstinent patients also decrease sex-related and overall HIV risk behaviors, while those who do not become abstinent exhibit an increase in overall and sex-related HIV risk behaviors. We were unable to detect an association between cocaine or alcohol use or gender with changes in HIV risk behaviors over time.

COMBINATION AND EXTENDED DURATION MEDICATIONS FOR TOBACCO DEPENDENCE TREATMENT. M.B. Steinberg¹, J. Foulds¹, D. Richardson¹. ¹University of Medicine and Dentistry of New Jersey, New Brunswick, NJ. (Tracking ID #134643)

BACKGROUND: In recent years, tobacco dependence treatment has improved with the widespread use of nicotine replacement therapy and bupropion. However, current single drug regimens are often inadequate for heavily dependent smokers. The goal of this analysis was to evaluate the effect of combination and extended duration of cessation medications among smokers with significant tobacco dependence.

METHODS: We performed a cohort study in 790 smokers attending a tobacco treatment clinic from 2001 through 2003, who provided data at their 4 week follow-up. Patients underwent a standardized evaluation. The number of medications prescribed and duration of therapy was determined by the treating physician. Level of dependence was assessed by time-to-first-cigarette and cigarettes-per-day. Confidence and importance in quitting were measured on 10-point Likert scales. Medication usage data was determined from 4-week follow-up. Primary outcomes were carbon-monoxide confirmed and self-reported 7-day point abstinence at 4 weeks, and self-reported abstinence at 6 months. Those patients who could not be contacted at 6 months were considered still smoking. Chi-square and adjusted odds ratios with 95% confidence intervals were calculated for abstinence rates.

RESULTS: Patients ranged in age from 15-80 years, were primarily female (61%) and Caucasian (71%). Most (90%) had at least a high school degree. Patients were significantly dependent, with 83% smoking within 30 minutes of waking, and 68% smoking a pack or more per day. The overall abstinence rates at 4 weeks and 6 months were 59%(469/790) and 36%(281/790), respectively. Nearly 87%(686/790) of patients seen in the clinic used medications as part of their treatment plan. Combinations of medications were commonly used with 23%(186/790) of patients using one medication, 42%(335/790) using 2 medications, 17%(132/790) using 3 medications, and 4%(33/790) using 4 or more medications. Despite the fact that smokers who were prescribed more medications were more dependent, those patients who used no medications had a 31%(32/104) abstinence rate at 4 weeks, 52%(96/186) with one medication, 64%(216/335) with 2 medications, 74%(98/132) with 3 medications, and 82%(27/33) with 4 or more medications (p<.001). At 6 months, 20%(21/104) of patients using no medications reported abstinence, 37%(68/186) who used one medication, 37%(123/335) who used 2 medications, 42%(55/132) who used 3 medications, and 42%(14/33) who used 4 or more medications (p=.008). After controlling for dependence, demographics, confidence, and importance in quitting, the number of medications used predicted higher abstinence rates at 4 weeks, with adjusted odds ratios of 2.49(95% CI = 1.44-4.30) for 1 medication, 4.76(2.85-7.95) for 2 medications, 6.38(3.43-11.87) for 3 medications, and 12.16(4.37-33.85) for 4 or more medications (p<.001). A significant proportion (23%(183/790)) of smokers continued to use medications for 6 months post-quit-date. Those patients who were still using medications at 6 months were more likely to remain abstinent from tobacco (65%(119/183)) compared with those who had stopped their medications prior to 6 months (27%(162/607)) (p<.001).

CONCLUSIONS: These data suggest that simultaneous use of multiple medications and extended durations of medications may improve abstinence rates in highly dependent smokers. The findings are useful in planning future randomized trials of treatment protocols for dependent smokers.

Adjusted Odds Ratios for 4 week and 6 month abstinence rates

Characteristic	N	Adjusted OR - 4 week (95% CI)	p value	Adjusted OR - 6 month (95% CI)	P value
Age					
	701			1	
Under 25					
25-44	3422	1.18 (1.21-3.91)	.01	1.38	NS
45-64	3252	1.79 (1.52-5.09)	.00	2.18 (1.14-4.17)	.02
65+	53	3.87 (1.67-8.97)	.00	4.49 (1.94-10.40)	.00
Gender					
	4821			1	
Female					
Male	3081	1.11	NS	0.97	NS

1 (continued)

Characteristic	N	Adjusted OR - 4 week (95% CI)	p value	Adjusted OR - 6 month (95% CI)	P value
Race					
Caucasian	5621			1	
African-American	1540.85		NS	0.70	NS
Hispanic	49 0.58		NS	0.85	NS
Other	22 1.35		NS	1.82	NS
Education					
No high school degree	761			1	
High school graduate	1910.87		NS	1.95 (.98-3.87)	.06
Some college	2911.1		NS	2.29 (1.19-4.4)	.01
College graduate	1451.75 (.92-3.36)		.09	3.38 (1.68-6.82)	.00
Graduate degree	87 1.45		NS	2.66 (1.24-5.69)	.01
Time to 1st cigarette					
less than 5 min	3401			1	
5-30 min	3061.84 (1.30-2.62)		.00	1.35 (.95-1.91)	.09
more than 30 min	1302.31 (1.42-3.74)		.00	1.78 (1.13-2.82)	.01
Number of medications					
0	1041			1	
1	1862.49 (1.44-4.30)		.00	2.13 (1.17-3.88)	.01
2	3354.76 (2.85-7.95)		.00	2.30 (1.32-4.01)	.00
3	1326.38 (3.43-11.87)		.00	2.30 (1.22-4.31)	.01
4 or more	33 12.16 (4.37-33.85)		.00	3.05 (1.25-7.43)	.01

COMORBID DEPRESSION AND PAIN: IS THERE AN ASSOCIATION WITH MORTALITY IN MEDICAL OUTPATIENTS? M.J. Bair¹, H. Harris², K. Kroenke³, C.A. Mchorney⁴.

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BACKGROUND: While depressive disorders and painful symptoms commonly coexist and add substantially to patient suffering, their combined effect on mortality has not been studied. Therefore, our objective was to investigate the effect of comorbid depression and pain on mortality in medical outpatients.

METHODS: Data were analyzed from the Medical Outcomes Study (MOS), a four-year prospective observational study from 1986-1990 of adult medical outpatients. The primary outcome was mortality during the study period. Patients with depression were identified in a two-stage screening procedure. Current depressive disorder was based on Diagnostic and Statistical Manual (DSM) III criteria and included both major depression and/or dysthymia. Pain was assessed using the pain severity item from the MOS SF-36 bodily pain subscale. Pain severity was classified into four categories: none, mild, moderate, or severe. We categorized patients into four groups by combining depressive disorder and pain severity (none to severe). Multivariate logistic regression was used to explore the independent effect of comorbid depression and pain on mortality, while controlling for demographics and other medical conditions. An analytic sample of 2914 patients was included in this study.

RESULTS: The mean age of our sample was 54; included 62% women, 79% white, and 58% were married. Fifteen percent met criteria for depressive disorder. Half of patients were free of pain (49.5%), while 1 in 5 had either mild (19.7%) or moderate (22.7%) pain, and 1 in 12 patients experienced severe pain (8.2%). Back and musculoskeletal problems were frequently reported by MOS

participants. In a multivariable regression model, patients with depressive disorder and severe pain had a three-fold increase in the odds of mortality (OR=3.1, p=0.02) at four years compared to the reference group (i.e. those with uncomplicated hypertension), controlling for demographics, and other medical conditions and disease severity. This mortality risk was greater than that associated with coexisting depression and complicated diabetes (OR=2.5). **CONCLUSIONS:** Patients with depressive disorder and severe pain had significant increases in mortality, even after adjusting for demographics, chronic medical conditions, and disease severity. The increase in mortality observed in patients with depression and pain is comparable to disorders such as congestive heart failure and complicated diabetes. The biological pathways and plausibility to explain this relationship needs further study.

CO-OCCURRING INJECTION DRUG USE AND MENTAL ILLNESS ARE ASSOCIATED WITH POORER VIROLOGICAL SUPPRESSION AMONG HIV-INFECTED INDIVIDUALS ON HAART. G. Chander¹, S. Himelhoch², P. Korthuis³, R.D. Moore¹, K. Gebo¹. ¹Johns Hopkins University, Baltimore, MD; ²University of Maryland, Baltimore, MD; ³Oregon Health & Science University, Portland, OR. (Tracking ID #134823)

BACKGROUND: HIV disproportionately affects individuals with co-occurring mental illness (MI) and injection drug use (IDU), yet the impact of these co-occurring conditions on HIV treatment outcomes is understudied. We examined virological suppression among HIV positive individuals on highly active antiretroviral therapy (HAART) with co-occurring MI and IDU. IDU only, MI only, and neither.

METHODS: Using 2001 data from the HIV Research Network (HIVRN) we performed a cross-sectional study of four primary care sites with on-site mental health services located in 2 Eastern, 1 Southern, and 1 Western site. Clinical, demographic, and resource utilization data were abstracted from medical records. Our outcome of interest was virological suppression. We considered individuals to be virologically suppressed if all HIV-RNAs drawn during 2001 were <400 copies/ml. Our independent variables of interest were 1) Any MI+IDU 2) IDU only 3) Any MI only 4) Neither. We used chi-squared analysis for comparison of categorical variables, and logistic regression for multivariate analysis adjusting for age, race, sex, frequency of visits, and study site.

RESULTS: Of 4,336 individuals on HAART, 71% were male, 41% African American, 24% Hispanic, and 33% Caucasian. The median age was 41 (range 19-88). 7% had both IDU and MI, 21% had IDU only, 14% had MI only, and 57% had neither. Overall, 49% of the sample had an HIV-RNA <400. Those with co-occurring MI and IDU, and IDU alone were less likely to have suppressed HIV-RNA compared to those with neither (table 1). After adjusting for age, race, sex,

Table 1: Virological Suppression Stratified by Injection Drug Use (IDU) and Mental Illness (MI)

Category	Percent HIV-RNA <400	Adjusted Odds Ratios (95% CI)
Neither MI nor IDU (n=2466)	51.5% (Reference)	Reference
MI only (n=620)	47.9% (p=0.133)	0.87 (0.72-1.04)
IDU only (n=928)	47.3% (p=0.031)	0.82 (0.69-0.96)
MI and IDU (n=322)	42.2% (p=0.002)	0.66 (0.52-0.84)

number of visits and study site, individuals with co-occurring MI and IDU had the lowest odds of HIV-RNA suppression compared to those with neither (table 1).

CONCLUSIONS: Co-occurring mental illness and injection drug use is associated with decreased virological suppression among HIV + individuals on HAART. Expanded mental health and substance abuse treatment services and interventions targeted specifically towards those with co-occurring mental health and substance abuse disorders may improve their treatment outcomes.

COUNSELING AND ATTENDANCE REQUIREMENTS FOR BUPRENORPHINE TREATMENT IN PRIMARY CARE. D.A. Fiellin¹, M.V. Pantalon¹, M.C. Chawarski¹, L.E. Sullivan¹, D. Barry¹, P. O'Connor¹, R.S. Schottenfeld¹. ¹Yale University, New Haven, CT. (Tracking ID #134116)

BACKGROUND: Heroin and prescription opioid addicted patients can now be treated by primary care physicians using buprenorphine. The optimal duration of each counseling session or frequency of office visits for successful treatment has not been established. The purpose of this study was to investigate systematically the independent effect of two intensities of counseling and attendance requirements for patients receiving buprenorphine maintenance in a primary care.

METHODS: We conducted a randomized clinical trial of standard medication management (SMM) and enhanced medication management (EMM) in patients receiving primary care office-based buprenorphine treatment. One hundred and sixty-six participants in this 6-month trial were randomly assigned to one of

three treatments: EMM and thrice weekly attendance and medication pick-up (EMM3); SMM and thrice weekly attendance and medication pick-up (SMM3); or SMM and weekly attendance and medication pick-up (SMM1). SMM was a brief (15-minute) manual-guided, medically focused counseling treatment provided by trained nurses. EMM was a more extended (45 minute) manual-guided advice and counseling provided by the same trained nurses.

RESULTS: 56 patients were randomized to EMM3, 56 to SMM3 and 54 to SMM1. The groups did not differ with respect to mean age (35.1 vs. 35.9 vs. 37.1 years), % male (79% vs. 77% vs. 78%), % white (82% vs. 77% vs. 70%), % prescription opioid dependent (22% vs. 21% vs. 18%), years of opioid use (7.5 vs. 8.6 vs. 8.1), or % prior methadone treatment (71% vs. 59 vs. 67%); $p > 0.05$ for all comparisons. There were no differences between treatment arms regarding treatment outcomes: proportion completing 24 weeks of treatment (43% vs. 39% vs. 48%); the percent of opioid negative urine screens (45% vs. 44% vs. 43%); percent of cocaine negative urine screens (52% vs. 48% vs. 53%); proportion achieving 3 weeks of continuous opioid negative urine screens (64% vs. 68% vs. 61%), or achieving 6 weeks of continuous opioid negative urine screens (37% vs. 46% vs. 50%); $p > 0.25$ for all comparisons.

CONCLUSIONS: We conclude that brief weekly counseling and attendance with medication distribution has comparable efficacy as more intensive counseling and thrice weekly attendance for patients receiving buprenorphine in primary care.

DEPRESSED PATIENTS ARE LESS LIKELY TO FOLLOW RECOMMENDED RISK REDUCING BEHAVIORS AFTER ACUTE CORONARY SYNDROMES. L. Kronish¹, N. Rieckmann¹, E.A. Halm¹, K. Davidson². ¹Mount Sinai School of Medicine, New York, NY; ²Columbia University, New York, NY. (Tracking ID #132841)

BACKGROUND: The presence of depressive symptoms during hospitalization is a strong risk factor for mortality in post-Acute Coronary Syndrome (ACS) patients. We hypothesized that persistently depressed patients following an ACS would be least likely to adhere to recommended risk factor modification behaviors.

METHODS: 421 post-ACS patients at 3 university hospitals were enrolled within 7 days of their ACS. 355 patients completed 3-month follow-up (84%). We used the Beck Depression Inventory (BDI) to assess depressive symptoms in hospital and 3 months later. BDI score ≥ 10 indicates symptoms of at least mild to moderate depression. We assessed adherence to risk-factor modification behaviors by patient self-report at 3 months. We used χ^2 to compare differences in adherence among 3 groups: Never depressed (BDI < 10 at hospitalization and 3 months; 50% of patients); Remitted depressed (BDI ≥ 10 at hospitalization, < 10 at 3 months; 23% of patients); and Persistently depressed patients (BDI ≥ 10 at hospitalization and 3 months; 22% of patients). A small number of patients (3%) became newly depressed at 3 months and were not included in the analysis.

RESULTS: Patients were on average 61 years (range 32-93); 40% were female and 11% were Hispanic. 51% had ST-elevation MIs. Compared to Never depressed patients, Depressed patients (both Remitted and Persistent) reported significantly lower adherence to the following activities: quitting smoking ($p < .001$), taking medications ($p = .002$), exercising ($p = .001$), participating in cardiac rehabilitation ($p = .002$), and modifying diet ($p = .05$). Compared to Remitted depressed patients, Persistently depressed patients reported less adherence ($p < .05$) to quitting smoking, exercising, and cardiac rehabilitation [see Table]. Remitted and Persistently depressed patients had similar SES, cardiac severity, and comorbidities.

CONCLUSIONS: Depressed patients adhere to risk reduction behaviors less than non-depressed patients, and persistently depressed patients adhere less than those patients whose depression remits. These behaviors reduce the risk of subsequent cardiac events, and may explain in part why depression predicts mortality post-ACS.

Behavior	Never depressed	Remitted depressed	Persistently depressed
Quit smoking	59%	65%	25%
Took meds	86%	78%	69%
Exercised	62%	57%	38%
Cardiac rehab	32%	25%	12%
Modified diet	73%	69%	64%

DEPRESSION AND DIABETES: SEX DIFFERENCES. C. Laskowski¹, B. Littenberg¹. ¹University of Vermont, Burlington, VT. (Tracking ID #134234)

BACKGROUND: The incidence and prevalence of diabetes are higher in depressed adults. Further, more severe depression has been associated with poorer glycemic control in some populations. We sought to understand potential modifiers of the relationship between depression and glycemic control in patients with established diabetes.

METHODS: We surveyed 333 adults (182 women and 151 men) enrolled in the Vermont Diabetes Information System. All subjects were receiving outpatient services in Primary Care settings in Vermont or adjoining states and had been

diagnosed with diabetes by their Primary Care Provider. Depression was measured by the Patient Health Questionnaire-9 (PHQ), a validated measure of the presence and severity of depression in Primary Care. Diabetic control was measured by serum A1C. We used linear regression to control for age, education, marital status, duration of diabetes, and antidepressant use.

RESULTS: The mean A1C was 7.2% (median 7; Inter-Quartile Range 6.3, 7.9). The mean PHQ score was 3.8 (median 2; IQR 0, 5). 68% of subjects had no or minimal findings of depression (PHQ < 5 ; mean A1C 7.2%) 22% were mildly depressed (PHQ 5-9; A1C 7.3%). 11% were moderately or severely depressed (PHQ > 9 ; A1C 7.6%). Neither PHQ nor A1C varied significantly by sex. PHQ was directly associated with A1C by simple linear regression without adjustment (coefficient = +0.036; 95% CI +0.006, +0.065; $P = 0.015$). Exploratory analyses revealed that sex was an important modifier of this relationship. In linear regressions controlling for age, education, marital status, duration of diabetes, and antidepressant use, PHQ was independently associated with A1C in men (0.059; CI +0.008, +0.109; $P = 0.023$) but not women (-0.010; CI -0.051, +0.031; $P = 0.62$). The results were similar using other markers of depression such as self-report.

CONCLUSIONS: Depression is common in people with diabetes. In this study, a 5 point difference in PHQ scores (about one severity category) was associated with a clinically important 0.29% difference in A1C in men. No such relationship was observed in women. Possible explanations for these findings include differential measurement of depression in men and women (instrument bias), unmeasured confounders, or intrinsic psychological or biologic sex differences. The causal direction of the relationship between depression and diabetes, if any, is unclear. Further research is required to examine whether treatment of depression alters glycemic control (or vice versa) in men or women.

DEPRESSION AND MEDICATION ADHERENCE IN PATIENTS WITH HEART DISEASE: RESULTS FROM THE HEART AND SOUL STUDY. M.A. Whooley¹, S. Pipkin¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #133391)

BACKGROUND: Depression is associated with increased mortality among patients with coronary heart disease (CHD). Medication non-adherence may contribute to this association, but it is not known whether depression is associated with medication adherence in patients with CHD.

METHODS: To determine whether depression is associated with medication non-adherence, we performed a cross-sectional study of 1024 outpatients with stable CHD. Of these, 84 subjects were excluded because they were not taking a cardiac medication (beta blocker, renin-angiotensin system inhibitor, aspirin, or statin), leaving 940 participants for the analysis. We assessed current major depression using the Diagnostic Interview Schedule, and asked participants "Overall, in the past month, how often did you take your medications as the doctor prescribed?" We considered participants who responded "all of the time" or "nearly all of the time" to be adherent. Those who responded "most of the time," "about half the time," or "less than half the time" were considered non-adherent. We used logistic regression to examine the association between depression and medication non-adherence, adjusted for potential confounding variables.

RESULTS: Of the 940 participants, 204 (22%) had current depression. Of these, 14% (28/204) were non-adherent, compared with 5% (40/736) of the nondepressed participants [odds ratio (OR) 2.8, 95% confidence interval (CI), 1.7-4.7; $p < 0.0001$]. The relation between depression and medication non-adherence persisted after adjusting for potential confounding variables, including age, ethnicity, education, cognitive function, and measures of cardiac disease severity (OR 2.2, 95% CI, 1.2-3.9; $p = 0.009$). This association was similar in users and nonusers of aspirin, renin-angiotensin system inhibitors, and statins, but differed in users and nonusers of beta blockers (p for interaction = 0.0002). Depression was strongly associated with medication non-adherence among the 590 participants who reported taking beta blockers (OR 6.3, 95% CI, 2.9-13.7; $p < 0.0001$), but not among the 344 participants who were not taking beta blockers (OR 0.6, 95% CI, 0.2-1.5; $p = 0.25$).

CONCLUSIONS: Depression is associated with medication non-adherence in patients with coronary heart disease, particularly in those who are taking beta blockers. Medication non-adherence may contribute to adverse cardiovascular outcomes in depressed patients.

DEPRESSION, DIABETES, AND CLINICIAN-PATIENT COMMUNICATION. S.L. Swenson¹, A. Stewart¹, D. Schillinger¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #135415)

BACKGROUND: Both depression and poor clinician-patient communication have been associated with worse outcomes and lower levels of self-care among patients with diabetes, and some have hypothesized that poor communication may be an explanatory factor for the depression-diabetes outcome relationship. We investigated whether patients with diabetes and depressive symptoms report worse clinician-patient communication and which domains of communication are most affected.

METHODS: We employed a cross-sectional survey design to investigate the experience of care among 408 English- and Spanish-speaking patients with type 2 diabetes in family medicine and internal medicine clinics in a public hospital. We recruited participants from all patients in a comprehensive diabetes registry who had primary care appointments during the study interval. Using established criteria, we categorized depressive symptoms as severe if participants had a score of 15 or more on the CES-D 10 (Clinical Epidemiologic Services for Depression, short form). To evaluate communication, we used the Interpersonal Processes of Care in Diverse Populations (IPC) scale, a 20-item survey that has

been validated in ethnically-diverse populations. We evaluated overall communication and separately examined 7 specific communication sub-scales: general clarity; elicitation of patient problems; empowerment; decision-making; explanation of condition; explanation of process of care; and self-care. We dichotomized the resultant Likert-scale responses (1-5) into "optimal" (5) vs. "sub-optimal" (1-4) communication and performed logistic regression to examine the independent effects of depression on overall communication and communication sub-scales. We adjusted for variables that were significant at $p < 0.2$ on bivariable analysis or known to be associated with both depression and communication.

RESULTS: Participants were ethnically diverse (41% Latino, 24% African-American, 19% Asian-American) and had a median age of 59; 55% were high school graduates. Nearly two-thirds (64%) reported at least one diabetes complication, and 41% reported severe depressive symptoms. Forty-seven percent reported overall communication as suboptimal. Those with severe depressive symptoms were more likely to report sub-optimal communication than participants without severe depressive symptoms (58% vs. 38%, $p < .0001$). After adjusting for age, sex, ethnicity, education, language, and diabetes complications, the presence of severe depressive symptoms was an independent predictor of sub-optimal communication (AOR 2.17, 1.29-3.67). Examination of communication sub-scale scores revealed that the presence of depressive symptoms was independently associated with the domains of general clarity (AOR 1.96, 1.12-3.44), decision-making (AOR 1.77, 1.09-2.89), and explanation of process of care (AOR 1.54, .95-2.51), but not with the other communication sub-scales.

CONCLUSIONS: Diabetes patients with severe depressive symptoms are more likely to report sub-optimal clinician-patient communication. This effect was heterogeneous across domains of communication, with depressive symptoms having a greater impact on certain explanatory aspects of communication and participatory decision-making. Given the prevalence of depressive symptoms among diabetes patients, further work should explore whether these findings result from differences in patient perceptions, competing visit demands, or differences in the process or content of the clinician-patient encounter.

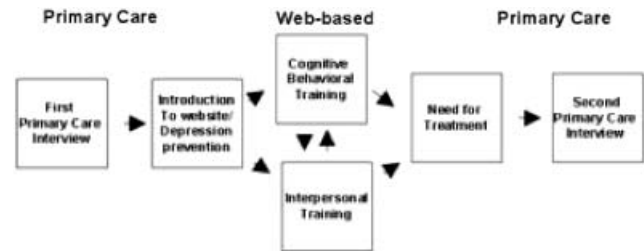
DEVELOPMENT AND PROCESS EVALUATION OF A PRIMARY CARE "BEHAVIORAL VACCINE" FOR DEPRESSION PREVENTION IN ADOLESCENTS AND YOUNG ADULTS (PROJECT CATCH-IT). B.W. Van Voorhees¹; J.M. Ellis¹; J. Gollan¹; D.E. Ford². ¹University of Chicago, Chicago, IL; ²Johns Hopkins University, Baltimore, MD. (Tracking ID #134291)

BACKGROUND: Twenty-five percent of all adolescents will experience a depressive episode during the important transition to adulthood with substantial adverse impact on future development. Although group cognitive-behavioral (CBT) and individual interpersonal psychotherapy (IPT) can reduce the risk of a depressive episode, five key dissemination barriers have prevented the development of a primary care based prevention strategy (loss of fidelity, poor motivation, inadequate dose, lack of socio-cultural relevance, and high costs). The purpose of this study was to develop a primary care/Web-based depression prevention intervention and conduct a process evaluation of the intervention with regard to these key dissemination barriers.

METHODS: We employed an interdisciplinary team of investigators in a multi-step development process: (1) initial translation by primary care physician of manuals for face-to-face preventive interventions, (2) Web-site design using an informal focus group, (3) serial fidelity reviews by a health psychologist, manual authors, and a practicing psychotherapist, and (4) socio-cultural review by an adolescent editor. Fourteen young adults recruited from two primary care settings rated each component of the intervention with regard to dissemination barriers: (1) fidelity (checklist, qualitative reviews), (2) motivation (ratings, 1-5 scale, strongly disagree to strongly agree, and themes), (3) dose (time on each component), (4) cultural acceptability (ratings themes) and (5) low potential costs (physician time, willingness to pay).

RESULTS: Based on a "vaccine" model, the intervention includes an initial motivational interview (MI) in primary care to engage the adolescent ("adjunct"), eleven Web-based modules ("active components") based on CBT (behavioral activation, counter pessimistic thinking), and Interpersonal Psychotherapy (activate social network, strengthen relationship skills), and a follow-up MI in primary care to enhance behavior change ("modified response to environmental challenge"). Fidelity check list and serial reviews were satisfactory. Key motivations for participation included: (1) reducing risk, (2) intervention effectiveness, (3) "resiliency", and (4) altruism. Thirteen of fourteen participants engaged the web-based components, spending an average of 15 minutes (SD=4.8 min) per module and completing a mean of 7.2 modules (SD=3.9) with high ease of reading scores (all scores >4.5). While all of the young adults found the "every-day" language easy to understand, some felt that the intervention was "too simple" to work (lack of cultural authority). In terms of socio-cultural relevance, participants rated negative feelings (sadness, anger) in the disagree range (mean scores from 1.6 (SD=0.53) to 2.2 (SD=1.16) on a 1-5 scale) and helpfulness in the agree range (mean scores ranging from 3.4 (SD=0.92) to 4.6 (SD=0.53)). With regard to cost, the mean durations of the two MIs were similar to a typical primary care visit (M=18.8 minutes, SD=6.4 and M=17.5, SD=3.5). Adolescents were willing to pay a mean of \$20.63 (SD=\$14.99) for the intervention.

CONCLUSIONS: Complex preventive mental health interventions can be translated into primary care settings with high fidelity and adequate motivation, dose, and socio-cultural relevance using multidisciplinary teams and incorporating the opinions of potential users. The adolescents' willingness to pay was similar to the co-payment required for the two Level III primary care visits needed for the intervention.



Intervention Sequence and Overview

DEVELOPMENT OF THE DEPRESSION PROGNOSIS INDEX (DPI) TO PREDICT SHORT-TERM OUTCOMES FOR PRIMARY CARE PATIENTS WITH DEPRESSION. N. Recker Rayburn¹; D.E. Ford²; C.D. Sherbourne³; K.M. Rost⁴; E. Keeler⁵; L.V. Rubenstein⁶. ¹RAND Health Program, Santa Monica, CA; ²Johns Hopkins University, Baltimore, MD; ³RAND Health, Santa Monica, CA; ⁴University of Colorado, Denver, CO; ⁵RAND, Santa Monica, CA; ⁶University of California, Los Angeles, Los Angeles, CA. (Tracking ID #134478)

BACKGROUND: Prognostic scales based on data gathered when an illness is first clinically detected have been widely used in clinical and research evaluations of medical illnesses. No such instrument has been validated for major depression in primary care settings. Our objective was to develop a Depression Prognosis Index (DPI) utilizing data on common medical, psychological, and social co-morbidities in predicting six-month depression outcomes among primary care patients with major depression.

METHODS: As part of the Quality Improvement in Depression (QID) study, we screened consecutive patients in 108 primary care practices for major depression excluding those with severe alcohol symptoms. At baseline, we assessed depression with a modified 23 item version of the Center for Epidemiologic Studies of Depression (CESD) and the Composite International Diagnostic Interview (CIDI). At baseline we collected data on extensive sociodemographic characteristics, duration of depression, recent depression treatment and physical, psychological, and social comorbidities (SF-36 and other measures). The CESD six months after baseline was our primary outcome. We used forward stepwise regression on a random 1/3 of the patients to develop two models: one that included baseline CESD as a predictor and one that did not. A series of stepwise regressions within each domain were developed before building the final model. We validated these models on the remaining 2/3 of the sample.

RESULTS: We enrolled 1499 patients with major depression. The DPI predicted 6-month symptom severity with an R2 of .40 on the training sample and at least .25 on the validation samples. Almost two-thirds of the patients classified in the "poorest prognosis" quartile on the DPI were likely to meet full diagnostic criteria for major depression at 6 months compared to 13% in the most favorable quartile. Social support and common physical symptoms contributed substantially to the prediction above baseline depression symptom severity and duration. Sociodemographic characteristics, number of chronic medical conditions, suicidality, alcohol quantity/frequency and severity of pain did not contribute added information. Anxiety symptoms were predictive only in models that did not include baseline depression symptom severity. The DPI did not perform differently based on intervention status, severity of depression at baseline, or whether depression treatment was initiated at the index visit.

CONCLUSIONS: The DPI allows reasonable prediction of short-term outcomes for primary care patients with major depression. The level of prediction compares favorably to similar prognostic indices for medical conditions such as congestive heart failure. This index may be of use to researchers as a case-mix adjuster and to clinicians to identify patients with depression who may need more intensive management.

DOES ADHERENCE TO DEPRESSION GUIDELINES IN PRIMARY CARE IMPROVE OUTCOMES? K.A. Hepner¹; L.V. Rubenstein²; M. Rowe³; K.M. Rost⁴; C.D. Sherbourne⁵; D.E. Ford⁶. ¹RAND Corporation, Santa Monica, CA; ²RAND Corporation; VA Greater Los Angeles at Sepulveda, Los Angeles, CA; ³University of Colorado, Denver, CO; ⁴Johns Hopkins University, Baltimore, MD. (Tracking ID #135960)

BACKGROUND: AHRQ national clinical guidelines for depression focus on primary care clinician assessment and follow-up as well as depression treatments, and are based on expert panel evidence review. No study, however, has evaluated whether adherence to the guidelines in real-world primary care is linked to better outcomes. The objective of this study was to determine whether a newly developed index for assessing guideline concordance of care for depression in primary care predicts depression outcomes.

METHODS: Three studies in the Quality Improvement for Depression (QID) collaboration collected data from 1131 representative patients from 45 primary care practices in 13 states. These patients were classified as having major depression using the full CIDI after screening positive for depression. Practices included staff model managed care, fee for service, and network style care. Patients completed computer assisted telephone interviews at baseline, six, twelve, eighteen, and twenty-four months. The newly developed Depression Quality Index (DQI) consisted of 20 guideline-based process measures assessing care received from baseline to 12 months. It was developed through a modified Delphi expert panel and is measured using patient self-report. Our outcome variables were CESD score and recovery at 6, 12, 18 and 24 months. We used multivariate regression analyses to assess whether overall depression care quality between

baseline and six or twelve months was associated with depression outcomes, controlling for sickness at baseline. Models controlled for sickness at baseline using a validated sickness index based on symptom severity, physical, psychological and social comorbidity that predicts outcomes at six months.

RESULTS: The DGI was significantly predicted the CESD, controlling for sickness at baseline, at six ($p < .001$), twelve ($p < .001$), eighteen ($p < .01$), and 24 ($p < .001$) months. The DGI was also significantly associated with recovery, controlling for sickness at baseline, at six ($p < .01$), eighteen ($p < .05$), and 24 ($p < .05$) months. The major quality impact on outcomes occurred between low quality and middle or high quality, as shown in the table below.

CONCLUSIONS: The quality of primary care for depression, as measured based on guidelines that focus significantly on primary care clinician assessment and follow up, has a strong effect on depression outcomes at all levels of sickness at baseline.

Percent Recovered by Sickness and Quality Tertiles

	Low Quality	Middle Quality	High Quality
Least Sick	74%	81%	82%
Middle Sick	49%	66%	65%
Sickest	25%	47%	44%

EFFECT OF DEPRESSION-TARGETED TELEPHONE-BASED OUTREACH ON MEDICALLY ILL INDIVIDUALS. M. Ong¹; M.D. Feldman²; E. Ciemins³; F. Azocar³. ¹Stanford University, Stanford, CA; ²University of California, San Francisco, San Francisco, CA; ³United Behavioral Health, San Francisco, CA. (Tracking ID #135474)

BACKGROUND: Medically ill patients often have co-morbid mental illness, particularly depression, but are less likely to use mental health services compared to other patients. Early detection and treatment of depression may minimize the health and economic costs of depression. Telephone-based outreach services to medically ill individuals not previously known to have mental illness could improve detection and treatment of depression in an outpatient setting.

METHODS: A pilot study of telephone-based outreach services was evaluated using a matched cohort of 41 enrollees and 48 controls. Individuals were enrolled if they incurred high medical utilization in the past year or had a recent hospitalization. Depression status was unknown prior to enrollment. Matching was based on age, sex, medical diagnosis and prior year medical claim costs. Both groups were surveyed one year after enrollees received telephone-based outreach services, and had their health care costs and utilization evaluated both pre- and post-intervention.

RESULTS: Enrollees were more likely to have poor self-reported health (19.5% vs. 4.2%, $p < 0.05$) and higher PHQ-9 scores than controls (8.1 vs. 4.0, $p < 0.01$). Among the 18 enrollees and 25 controls who were currently employed, enrollees worked more hours per week (39.7 vs. 32.1, $p < 0.10$) and were more likely to work full-time (77.8% vs. 48.0%, $p < 0.10$) compared to controls. Controls had no behavioral health utilization. Enrollees had more behavioral health utilization (2.1 claims vs. 8.0 claims, $p < 0.05$) and higher costs (\$16.73 per month vs. \$39.20 per month, $p < 0.05$) after the intervention. Both enrollees (17.3 to 26.4, $p < 0.05$) and controls (6.3 to 10.6, $p < 0.05$) had increased medical utilization after the intervention. Total costs did not significantly decline for enrollees (\$1129 per month to \$1028 per month) but did for controls (\$291 per month to \$254 per month, $p < 0.10$). Regression analyses showed no significant relationship between the intervention and total costs, but showed that the intervention added 9.16 hours to the total hours worked per week ($p < 0.05$).

CONCLUSIONS: Telephone-based outreach services appear to enhance the work productivity of those enrollees who continued to work one year after receiving outreach. Telephone-based outreach services do significantly increase behavioral claim costs. However, this increase is small in comparison to overall medical costs, which also are not significantly increased after the intervention.

EMPLOYMENT PATTERNS AMONG PRIMARY CARE PATIENTS WITH PANIC AND GENERALIZED ANXIETY DISORDER (PD/GAD). B.L. Rollman¹; B. Herbeck Belnap¹; S. Mazumdar¹; F. Zhu¹; H. Schulberg²; J.R. Lave¹. ¹University of Pittsburgh, Pittsburgh, PA; ²Cornell University, White Plains, NY. (Tracking ID #135583)

BACKGROUND: Anxiety disorders are prevalent in primary care practice and are often unrecognized and inappropriately treated by PCPs. Although anxiety disorders are associated with significant functional impairment, their impact on employment patterns and absenteeism has not been well characterized. We examined these questions using data collected as part of a clinical trial to improve the quality of primary care for PD/GAD.

METHODS: We enrolled patients from 4 Pittsburgh-area primary care clinics. All met criteria for PD and/or GAD on the PRIME-MD and had at least a moderate level of anxiety symptoms on the Hamilton Rating Scale for Anxiety (HRS-A > 13) and the Panic Disorder Severity Scale (PDSS > 6). We assessed self-reported current employment, days absent from work in the past month, and depressive symptoms over the telephone.

RESULTS: Of the 193 patients enrolled, 21 (11%) had PD, 81 (42%) GAD, and 91 (47%) had both PD and GAD. Their mean age was 44 (range 19-63), 81% were female, and 95% were Caucasian; these characteristics did not differ by type of anxiety disorder. Overall, 117 (61%) were employed either full- or part-time. While employment status did not differ by mean PDSS or HRS-A score, patients

with both PD and GAD were less likely to be employed than those with either PD- or GAD-alone (51% vs. 67% vs. 71%; $p = 0.03$). Those employed also worked a median of 40 hours/week (range: 2-60) and missed a median of 2 work days in the past month (range 0-20). These figures did not differ by type of anxiety disorder. Although anxiety symptom scores were not correlated with the number of hours worked/week, higher scores were correlated with more work days absent in the past month (HRS-A: $r^2 = 0.22$, PDSS: $r^2 = 0.26$; both $p < 0.001$).

CONCLUSIONS: Primary care patients affected by PD and/or GAD have high rates of unemployment that vary by disorder. When employed, these patients also miss work at rates that correlate with their anxiety symptoms. Although further study is necessary to better assess the impact of PD and GAD on worker productivity, raised awareness of the impact of anxiety disorders on employment patterns may result in increased attention devoted to these conditions by PCPs, health plans, and employers.

ETHNIC DIFFERENCES IN THE TIMING OF SMOKING CESSATION FOR PATIENTS IN ALCOHOL DEPENDENCE TREATMENT. S.S. Fu¹; M.L. Willenbring²; D.B. Nelson³; S. Nugent³; A.M. Joseph³. ¹University of Minnesota, Minneapolis, MN; ²National Institutes of Health, Rockville, MD; ³VA Medical Center, Minneapolis, MN. (Tracking ID #132908)

BACKGROUND: The Timing of Alcohol and Smoking Cessation (TASC) Study compared the effects of concurrent vs. delayed smoking intervention for patients receiving intensive treatment for alcohol dependence. Main results from this trial showed no difference between groups in smoking cessation rates, however, findings suggested that providing concurrent smoking cessation treatment might adversely affect alcohol treatment outcomes. The purpose of this analysis was to examine ethnic differences in smoking and alcohol outcomes.

METHODS: We analyzed data from this randomized, controlled trial, which was conducted at three substance abuse treatment sites in Minneapolis-St. Paul, Minnesota from 1997 to 2001. There were 1943 patients in intensive alcohol treatment who were screened for participation. Eligible smokers (N=499) were enrolled and randomized to concurrent (during alcohol treatment) or delayed (6 months later) smoking intervention. To examine ethnic differences, this analysis focused on smokers of Caucasian (n=381) and African American (n=78) ethnicity. The smoking intervention included nicotine replacement therapy and individual behavioral counseling. Smoking cessation outcomes were measured 6, 12, and 18 months after study enrollment and included 7-day point prevalence smoking abstinence. Alcohol outcomes were measured 6, 12 and 18 months after study enrollment and included 6-month prolonged alcohol abstinence and 30-day prolonged alcohol abstinence. Analysis used an intention-to-treat approach.

RESULTS: At 18-months after study enrollment, 7-day point prevalence smoking abstinence rates were 14.4% for Caucasians and 10.3% for African Americans, the differences were not statistically significant. Among Caucasians, 6-month prolonged alcohol abstinence rates were consistently worse in the concurrent group than the delayed group at 6, 12 and 18 months (43% vs. 60%, $P = 0.001$; 35% vs. 45%, $P = 0.045$; 42% vs. 51%, $P = 0.059$ respectively). Among African Americans, 6-month prolonged alcohol abstinence rates were not consistently different in the concurrent group than the delayed group at 6, 12, and 18 months (47% vs. 45%, $P = 0.861$; 33% vs. 31%, $P = 0.793$; 36% vs. 43%, $P = 0.544$ respectively). Similarly, at 6, 12 and 18 months, 30-day prolonged alcohol abstinence outcomes were worse for Caucasians (53% vs. 68%, $P = 0.004$; 47% vs. 56%, $P = 0.111$; 50% vs. 63%, $P = 0.008$ respectively) in the concurrent group than the delayed group at but not consistently different for African Americans (50% vs. 57%, $P = 0.528$; 42% vs. 45%, $P = 0.751$, 48% vs. 39%, $P = 0.438$ respectively).

CONCLUSIONS: There was no evidence of ethnic differences in the overall success of smoking cessation treatment for patients in intensive alcohol treatment. However, this analysis suggests that effect of the timing of smoking intervention on alcohol outcomes varies by ethnicity. Among African Americans, alcohol outcomes may not be adversely affected by delivery of concurrent smoking cessation treatment. Further research is required to confirm these findings.

FACTORS INFLUENCING MENTAL HEALTH REFERRAL IN PRIMARY CARE: RESULTS FROM A RANDOMIZED TRIAL USING STANDARDIZED PATIENTS. R.L. Kravitz¹; M.D. Feldman²; R. Epstein³; L. Meredith⁴; E.J. Callahan¹; P. Franks¹. ¹University of California, Davis, Sacramento, CA; ²University of California, San Francisco, San Francisco, CA; ³University of Rochester, Rochester, NY; ⁴RAND, Santa Monica, CA. (Tracking ID #133202)

BACKGROUND: Psychological treatments such as cognitive or interpersonal psychotherapy are of proven benefit in mild-moderate depression, but referral rates are modest. We sought to identify predictors of referral from primary care physicians (PCPs) to mental health (MH) professionals as part of a randomized trial involving first visits by unannounced standardized patients (SPs) with depressive symptoms.

METHODS: PCPs (n=152) participated in 298 visits by 18 unannounced SPs in 3 U.S. cities. Physicians were assigned to see one SP with major depression (MD) and one with adjustment disorder (AD); at each visit, SPs made a brand-specific request for Paxil®, a general request for medication, or no request (allocated at random). Data on MH referrals were obtained from SP reports and chart review; PCP characteristics and attitudes were derived from a written survey (n=151). Patient, physician and organizational effects on referral were evaluated using contingency tables and generalized linear mixed models that accounted for clustering and treated physicians as a random effect.

RESULTS: Physicians recommended that SPs seek care from a MH professional in 36% of visits. 53% of referrals were to non-psychiatric MH professionals, 2% to psychiatrists, and 44% to unspecified MH specialists. PCP's offices provided

varying levels of assistance with referrals, from recommending a specific therapist (27%) to telling patients to "call your health plan" (19%). In the bivariate analysis, significantly increased rates of referral were noted among physicians who were male ($p=.038$) or caucasian ($p=.048$), who practiced in academic settings ($p=.009$), and who had a personal or family history of depression treated with psychotherapy ($p=.002$). Higher referral rates were also observed in visits where SPs made a request for antidepressant medication ($p=.01$). Being able to obtain MH consultation "typically within two weeks" was of borderline significance ($p=.055$). In the multivariate analysis, significant predictors included academic practice, ability to obtain MH consultation within 2 weeks, personal/family history of psychotherapy, and SP antidepressant requests ($p<.05$).

CONCLUSIONS: Academic physicians, those practicing in settings where MH consultation is promptly available, those with a personal/vicarious history of psychotherapy, and those caring for medication-requesting SPs were more likely to recommend MH referral. While system-level efforts to enhance access to appropriate MH specialty care should continue, physician biographical factors and patient cues (both potentially modifiable) may be more important than previously realized.

HIV PHYSICIANS' PREPAREDNESS TO PRESCRIBE BUPRENORPHINE FOR OPIOID ADDICTION. L.E. Sullivan¹, J. Tetrault¹, D. Bangalore¹, D.A. Fiellin¹. ¹Yale University, New Haven, CT. (Tracking ID #132994)

BACKGROUND: Treatment of opioid addiction improves outcomes in HIV positive patients and can decrease HIV transmission. Trained and certified HIV physicians can now prescribe buprenorphine to their opioid addicted patients. Little is known regarding HIV physicians' preparedness to prescribe buprenorphine. This study was designed to assess HIV physicians' preparedness to prescribe buprenorphine, as compared to non-HIV physicians, following attendance at a federally required training on the use of buprenorphine.

METHODS: We conducted a survey of physicians completing one of six eight-hour training courses designed to educate physicians about the use of buprenorphine. We collected information on demographics, provider type, clinical experience and preparedness to prescribe buprenorphine. We compared HIV physicians to non-HIV physicians.

RESULTS: Of the 174 physicians completing the survey, 69 (40%) indicated that they provided HIV care. The HIV physicians reported providing a mean of 9 years of HIV care to a mean of 59 current patients. Fifty-two percent of their patients had intravenous drug use as their HIV risk factor and 71% of their patients were on highly active antiretroviral therapy. Fewer of the HIV physicians compared with the non-HIV physicians were male (57% vs. 70%, $P=.08$), more were black (19% vs. 9%, $P=.05$), and more had graduated from medical school within the last 10 years (26% vs. 15%, $P=.07$). While the two groups had similar lengths of experience treating opioid addicted patients and cared for similar numbers of these patients per month (40 vs. 33, $P=.7$), HIV physicians cared for a greater percentage of patients addicted to opioids other than heroin (33% vs. 25%, $P=.2$), and a greater percentage of opioid injecting patients (33% vs. 23%, $P=.02$). Following the training, similar proportions of the HIV and non-HIV physicians planned to submit the application for the DEA registration to prescribe buprenorphine (62% vs. 66%, $P=.6$). The majority of HIV and non-HIV physicians felt prepared to prescribe buprenorphine (92% vs. 87%, $P=.4$), but agreed that their comfort level would be increased if they could telephone an expert (83% vs. 79%, $P=.8$) or were able to visit a practicing expert's office (65% vs. 60%, $P=.5$). While the most common reason cited for not submitting or for being undecided in both groups was lack of experience (10% and 14%, $P=.5$), 28/69 (41%) of the HIV physicians stated they had concerns about prescribing buprenorphine to their patients with 12/28 (43%) reporting drug-drug interactions as their primary concern.

CONCLUSIONS: There are differences in demographic characteristics and clinical experience between HIV and non-HIV physicians and the clinical profile of the opioid addicted patients for whom they provide care. Following training, the majority of physicians planned to obtain the appropriate DEA registration and felt prepared to treat opioid addicted patients with buprenorphine. However, hesitant HIV physicians were primarily concerned about adequate experience and drug-drug interactions. These areas need additional attention in future educational efforts to expand access to buprenorphine through HIV physicians.

IMPACT OF PATIENT-PHYSICIAN COMMUNICATION ON TREATMENT OF PRESCRIPTION PAIN RELIEVER ABUSE. J.R. Kramer¹, L.D. Woodard¹, L. Capistrano¹, V.L. Cox¹, R.O. Morgan¹. ¹Houston Center for Quality of Care & Utilization Studies - Baylor College of Medicine, Houston, TX. (Tracking ID #135250)

BACKGROUND: Approximately 1.5 million individuals are dependent on or abuse prescription pain relievers in the United States. In our previous work, we identified barriers to patients raising the issue of their substance abuse with their physicians. The goal of this study was to determine whether discussing substance abuse problems with a physician leads to higher rates of treatment among individuals who abuse prescription pain relievers.

METHODS: Data were obtained from the Texas Commission on Alcohol and Drug Abuse's 2000 Texas Survey of Substance Use Among Adults. 10,227 adults participated in a telephone survey conducted from July 2000 to March 2001 regarding use of tobacco, alcohol, illicit drugs, and problems they experienced related to substance use, such as access to treatment. We restricted the analysis to respondents who reported prior non-medical use of prescription pain relievers. We calculated descriptive statistics for questions that addressed dis-

cussions with physicians regarding substance use and used survey logistic regression to determine predictors of receiving treatment, including age, race, gender, marital status, education, employment, household income, veteran status, history of incarceration, and co-use of other illicit substances or alcohol. Treatment was defined as attending a meeting or self-help group or receiving treatment or counseling for alcohol or drug use. Variables with a univariate p -value <0.15 were entered in the multivariable model.

RESULTS: Approximately 4% of Texans reported non-medical use of pain relievers. Of these, 66.4% were males, 41% were 18 to 24 years old, 24% were 25 to 34 years old, and 35% were 35 years and older. The majority of individuals were white (76%), while 6% were Black and 18% were in other racial/ethnic groups. 91.6% reported heavy alcohol, marijuana, and/or other illicit substance co-use. Significant predictors for treatment in this population included history of incarceration for at least one night (OR=3.4; 95% CI: 2.0-5.5; p -value <0.01), being 35 years or older compared to 18 to 24 year olds (OR=2.2; 95% CI: 1.2-3.9; p -value=0.01), and heavy alcohol, marijuana, and/or other illicit substance co-use (OR=3.2; 95% CI: 1.1-8.9; p -value=0.03). 67% of pain reliver users reported seeing a health professional in the past 12 months. Of those who had at least one alcohol and drug problem in the past 12 months, 34% reported discussing their use of drugs or alcohol with their doctor. Patients who reported discussing their substance abuse with a doctor were two times more likely to report ever receiving treatment than people who did not discuss their use with their doctor (OR=2.14; 95%CI: 1.2-3.8, p -value=0.01). Of those who discussed their substance use with their physician, 83% felt the doctor provided helpful advice regarding this issue.

CONCLUSIONS: Among patients who abuse prescription pain relievers, those who discuss their substance use with physicians are significantly more likely than those who do not to receive treatment for their condition. This underscores the importance of addressing the issue of substance abuse during the medical encounter. Further work is needed to help patients and physicians overcome barriers to addressing substance abuse problems.

LONG-TERM OUTCOMES AMONG HEALTHY YOUNG ADULTS WHO USE ILLICIT DRUGS (THE CARDIA STUDY). S.G. Kertesz¹, M. Pletcher², S. Person¹, J. Halanay¹, S. Taylor³, C.I. Kiefe¹. ¹University of Alabama at Birmingham, Birmingham, AL; ²University of California, San Francisco, San Francisco, CA; ³University of California, Los Angeles, Los Angeles, CA. (Tracking ID #133093)

BACKGROUND: While consequences of drug dependence are well-known, clinicians lack prognostic data for persons who use illicit drugs (but are not necessarily dependent). The longitudinal Coronary Artery Risk Development in Young Adults study (CARDIA) allowed us to test whether self-reported illicit drug use was associated with lower health-related quality of life (QOL) 13 years later in a community-based sample.

METHODS: A biracial cohort of young adults was recruited in 4 metropolitan areas, and assessed in 1987-88 and 2000-01. Cocaine, amphetamines ("speed"), opiates, marijuana and tobacco cigarette (for comparison) use reported in 1987-88 were categorized as Current (any in last 30 days), Past, and Abstinent. Physical and Mental QOL were assessed 13 years later using Short Form-12 Physical Component Summary, and Mental Component Summary scores. For each substance, a linear regression was prepared for each outcome (Physical QOL, Mental QOL), adjusting for age, sex, race, alcohol use and socioeconomic status (SES). All substances and covariates were combined in a final model.

RESULTS: For 3523 CARDIA subjects followed for 13 years, mean age (SD) in 1987-88 was 27(3.6) years; 27% were African-American (AA) women, 19% AA men, 28% European-American (EA) women and 25% EA men. Baseline prevalence of Current/Past use was: cocaine (8%/30%), amphetamines (1%/26%), opiates (0.3%/6%), marijuana (23%/50%) and tobacco (26%/14%). Current drug use was rarely >15 days/month. Opiates were more strongly associated with lower Physical QOL, and cocaine and amphetamines with lower Mental QOL 13 years later (Table). These associations were of a magnitude similar to those for tobacco. Though associations were strongest for Current use, they were also present for Past use. Linear trends (Current vs. Past vs. Abstinent), tested in separate models (not shown), were significant for Mental QOL for all drugs (all $p<.01$), and for Physical QOL for opiates ($p<.001$). Unadjusted analyses stratified by race/sex were mostly consistent with the overall results, although many associations lost statistical significance. Simultaneous adjustment for all substances in a single model revealed opiates and cocaine (Physical QOL) and cocaine (Mental QOL) as independent predictors.

CONCLUSIONS: Illicit drug use in young adulthood, even without dependence, appears to be associated with reduced health-related quality of life later in life.

Table: Regression coefficients for Current Use of substance (vs. abstinence) and Physical and Mental QOL 13 Years Later

	Physical QOL(95%CI)	Mental QOL(95%CI)
Marijuana	-0.5 (-1.2, 0.2)	-1.8 (-2.6, -0.9)
Cocaine	-1.1 (-2.0, -0.2)	-2.4 (-3.6, -1.3)
Speed	-2.6 (-5.0, -0.2)	-2.1 (-5.0, 0.7)
Opiates	-4.7 (-9.2, -0.1)	0.3 (-5.1, 5.7)
Cigarettes	-1.5 (-2.1, -1.0)	-0.8 (-1.5, -0.1)

MODERATE ALCOHOL USE AND HEALTH-RELATED QUALITY OF LIFE 13 YEARS LATER: A CLOUD FOR THAT SILVER LINING (THE CARDIA STUDY). S.G. Kertesz¹; M. Pletcher²; S. Person¹; J.H. Halanych¹; C. Kiefe¹. ¹University of Alabama at Birmingham, Birmingham, AL; ²University of California, San Francisco, San Francisco, CA. (Tracking ID #133615)

BACKGROUND: The salutary effects of moderate drinking on cardiac risk are well-known. However, clinical advice on alcohol use must consider 2 questions where data are less clear: (1) are there long-term benefits for broad measures of health-related quality of life, as opposed to effects for specific diseases? (2) are physical and mental health effects similar? We examined health-related quality of life (HRQOL) 13 years after alcohol use was assessed in a prospective community cohort.

METHODS: A biracial of young adults was recruited in 4 metropolitan areas, and assessed in 1987-88 and 2000-01 as part of the Coronary Artery Risk Development in young Adults (CARDIA) study. Alcohol use in 1987-88 was categorized as Non-Use, Moderate (Men: ≤ 14 drinks/week and ≤ 4 drinks/day; Women: ≤ 7 drinks/week and ≤ 3 drinks/day) and Risky (above either the per-week or per-day Moderate thresholds). Physical and Mental HRQOL was measured in 2000-01 with the physical and mental component summary scores (PCS and MCS, respectively) of the Short Form-12. A linear regression was prepared for each outcome (Physical and Mental HRQOL), adjusting for age, sex, race, education and socioeconomic status.

RESULTS: For 3523 CARDIA participants examined in 2000-01 (mean age in 1987/88: 27 years), prevalence of Risky, Moderate and Non-Use of alcohol in 1987-88 was 36%, 49% and 15%, respectively. Mean (SD) Physical HRQOL and Mental HRQOL in 2000-01 were 51.8 (7.5) and 50.4 (8.9), respectively. Moderate alcohol use was associated with better Physical HRQOL compared with Non-Use, but both Moderate and Risky drinking were associated with worse Mental HRQOL at 13 years' follow-up, even after adjustment for covariates (see Table). Unadjusted analyses stratified by race/sex were mostly consistent with the overall results, although many associations lost statistical significance.

CONCLUSIONS: Moderate alcohol use was associated with better physical, but worse mental HRQOL 13 years later. Small average HRQOL differences reflect the community-based (i.e. non-clinical) nature of this sample, but nonetheless hint at a downside to moderate drinking.

Table: Regression Coefficients for Moderate & Risky Alcohol Use (vs.Non-Use) & HRQOL 13 Yrs Later

	Physical HRQOL (95%CI)	Mental HRQOL (95%CI)
Moderate Use	+1.2 (+0.4, +1.9)*	-1.2 (-2.1, -0.4)*
Risky Use	+0.3 (-0.5, +1.1)	-1.5 (-2.4, -0.6)*

* $p < .01$ (vs Non-Use)

PERSPECTIVES OF PERSONS WITH OPIOID DEPENDENCE: AN ANALYSIS OF ON-LINE DISCOURSE. L. Capistrano¹; L.D. Woodard¹; P. Haidet¹; R.L. Street²; B.F. Sharf². ¹Houston Center for Quality of Care and Utilization Studies - Baylor College of Medicine, Houston, TX; ²Texas A&M University, College Station, TX. (Tracking ID #135700)

BACKGROUND: Efforts to integrate treatment of opiate addiction into medical practice have the potential to significantly expand access to care. However, studies indicate that physicians feel unprepared to treat patients with opiate addiction. We conducted this study to explore issues surrounding opiate addiction from the perspective of patients.

METHODS: We conducted a thematic analysis of postings by an Internet discussion group. "The Patient Buprenorphine Web Board" is for people actively using or recovering from opiate addiction. This web board consisted of 857 members, of which 665 participants posted information. We collected one week's worth of discourse, from July 13 to July 20, 2004, which generated 180 pages of text. Four investigators analyzed transcripts to identify key issues and themes. Our analyses were guided by a focus on the cause, control, severity, treatment, and meaning of opiate addiction from the patients' perspective.

RESULTS: Four themes emerged. First, participants discussed the meaning of addiction, which encompassed the reasons they used opiates and their feelings about their addiction. As one participant commented, "Addiction fills the void. When you get clean, you're like a doughnut with an even bigger void. You suck in meaninglessness, drown in a sea of futility. Unless you can replace it with something solid, it seems only a matter of time before your back where you started." Second, participants described the impact of addiction on their daily life. As one participant stated, "We are definitely not rich by any means (\$38k per year gross) and I have been spending \$1200 per month on my habit. It has put us in bad debt and I have to stop." Third, participants conveyed their experiences with recovery, stating that relapse is a common occurrence. One participant noted, "I have detoxed and relapsed before and I know the tricks, traps, and lies we fall into before we start getting high again. It is an ongoing process though and instead of being anxious and frustrated and impatient, try and look at the things that are better." Finally, patients perceived their addiction to be a health problem that is similar to other types of chronic conditions and described the impact of their addiction on interactions with physicians. As one participant stated, "Some people need an opiate to function, just as a diabetic needs insulin." Another commented, "I was at my doctor's office today for the second time in almost a year. The first time I went, I never told him I was on meth (methadone). Today, he asked me about three times if I was on any type of meds, and I told him no."

CONCLUSIONS: Our findings suggest that patients with opiate addiction often view their addiction as comparable to other types of chronic medical conditions.

Patients' views of opiate addiction reveal a complexity and depth that is often not solicited or understood by physicians. Because patients' perspectives often impact illness behaviors, such as disclosure during the medical interview, further work is needed to promote sharing and understanding of patient and physician perspectives.

PHYSICIANS-IN-TRAINING ATTITUDES TOWARD CARING FOR AND WORKING WITH PATIENTS WITH ALCOHOL AND DRUG ABUSE DIAGNOSES. C.M. Vergara¹; M. Lindberg¹; C. Gruman²; R. Wild-Wesley¹. ¹Hartford Hospital, Hartford, CT; ²Bracefield Center for Mental Health and Aging, Hartford, CT. (Tracking ID #135060)

BACKGROUND: Physicians in all specialties commonly encounter patients who abuse alcohol or illegal drugs. Working with these patient populations can be challenging and can potentially engender negative attitudes. This study is designed to identify the progression of attitudinal shifts over time of physicians-in-training toward caring for substance-abusing patients.

METHODS: A 31-item survey was designed to capture demographic information of participants, attitudes toward treating patients with substance abuse diagnoses, previous participant education, experience in and comfort with diagnosing and treating substance abuse, and satisfaction achieved in working with this patient population. Medical students in their third and fourth years of education as well as residents in training years one through four were surveyed. Responses to the survey's attitudinal items were analyzed across years of training looking for changes associated with time and experience.

RESULTS: Fifty-seven percent of eligible participants anonymously completed the survey. There was general agreement across all years of training that health care professions should be allowed continued employment in their professions when in recovery from alcohol abuse ($P=0.424$) and drug abuse ($P=0.409$). Across years of training there was agreement that patients can recuperate and provide meaningful contributions to society when recovering from alcohol ($P=0.847$) and drug ($P=0.859$) abuse. From medical school years through residence there was enhanced beliefs that alcohol abusing patients ($P=0.027$) and drug abusing patients ($P=0.009$) over-utilize health care resources. Most trainees, despite year of education, believe patients who abuse alcohol ($P=0.521$) and illegal drugs ($P=0.356$) have challenging medical and social issues from which they can learn. There was consistency across years in the perception that providing care to alcohol-abusing patients ($P=0.679$) and drug-abusing patients ($P=0.090$) is repetitive and detracts from the care of others. All felt their training was adequate to care for alcohol ($P=0.628$) and drug ($P=0.484$) -abusing patients. Satisfaction achieved in caring for alcohol ($P=0.017$) and illegal drug ($P=0.015$) -abusing patients consistently diminishes over years in training.

CONCLUSIONS: There are positive as well as negative aspects for physicians-in-training who care for patients with alcohol and illegal drug abuse problems. Combining effective education strategies with the needs of physicians at specific points in their education may be effective in reversing the negative trends seen in attitudes toward caring for patients with substance abuse problems.

POST-TRAUMATIC STRESS DISORDER (PTSD) IS CLOSELY ASSOCIATED WITH SUBSTANCE USE DISORDERS (SUD) IN PRIMARY CARE. J. Liebschutz¹; V. Brower¹; R. Saitz¹; J.H. Samet¹. Boston University, Boston, MA. (Tracking ID #133010)

BACKGROUND: PTSD, a psychiatric diagnosis resulting from trauma exposure, is prevalent among those in substance abuse treatment, although the nature of the association between PTSD and SUD is not well understood. The relationship between SUD and PTSD has not been studied among primary care patients. We sought to determine if a lifetime history of PTSD was more common among those with current SUD and whether patients attributed some substance use to self-medication for PTSD symptoms among primary care patients.

METHODS: English-speaking patients, 18-65 years, awaiting a primary care appointment were systematically recruited and interviewed. The dependent variable, Lifetime PTSD diagnosis, and the independent variables, current Alcohol and Drug Abuse/Dependence diagnoses, hazardous alcohol use, and use of individual substances, were assessed by the Composite International Diagnostic Interview. A subject's report that substances were used to treat PTSD symptoms was another variable of interest. All analyses are bivariable, unadjusted comparisons.

RESULTS: Subjects ($n=606$) were: 51% female; mean age 42 years; 59% Black, 19% White, 8% Hispanic; 34% high school graduates; and 50% with annual incomes $< \$20,000$ per year. Lifetime PTSD was identified in 34%. SUD included 12% with current drug abuse or dependence [DAD], 10% with alcohol abuse or dependence [AAD], and 12% with hazardous alcohol use. PTSD was present in 52-53% of all subjects with SUD compared to 32-33% among those without those disorders, $p < 0.005$ for all comparisons. Of illicit substances, use of sedatives, analgesics and heroin was significantly associated with PTSD ($p < .005$), whereas use of cocaine, marijuana and other substances were not associated with PTSD. Among those with a diagnosis of DAD or AAD, 83% used substances for PTSD symptoms vs. 42% of those without a substance dependence diagnosis ($p < .0001$).

CONCLUSIONS: More than half of primary care patients with SUDs have PTSD and the vast majority of these dually diagnosed self-medicate for PTSD symptoms. "Non-therapeutic" use of specific substances (i.e., sedatives and analgesics) and hazardous drinking in primary care patients is highly associated with PTSD. These findings suggest the need to consider PTSD and its symptom complex when dealing with substance use in primary care settings.

POST-TRAUMATIC STRESS DISORDER AND THE RISK OF CORONARY HEART DISEASE IN THE NORMATIVE AGING STUDY. G. Kim¹; E. Lawler¹; H. Sesso²; P. Schnurr²; M. Gaziano¹. ¹Massachusetts Epidemiology Research and Information Center, Boston, MA; ²Brigham and Women's Hospital, Boston, MA; ³Department of Veterans Affairs (VA) National Center for PTSD and Research, Hanover, NH. (Tracking ID #133499)

BACKGROUND: Post-traumatic Stress disorder (PTSD) has been associated with poor physical health including self-reported coronary heart disease (CHD). Few studies have examined the association of PTSD and clinical cardiac outcomes. We hypothesized that clinical PTSD is associated with increased risk of coronary heart disease (CHD).

METHODS: We examined this relation in the VA Normative Aging Study, an ongoing prospective cohort of older men with mean age of 65.2 years. In 1989, 791 men free of diagnosed CHD completed the Mississippi Scale for Combat-Related PTSD. Responses for the PTSD scale were categorized into high PTSD symptoms, which included men scoring in the upper 10th percentile of the Mississippi score distribution ($M > 73$) and low PTSD symptoms ($M \leq 72$). This cutpoint had .50-.57 sensitivity and .88-.91 specificity for a full or partial lifetime diagnosis of combat-related PTSD according to the Clinician-Administered PTSD Scale, a structured clinical interview administered to a subset of 436 participants. Baseline covariates included age, smoking, BMI, blood pressure, diabetes, alcohol, and serum cholesterol. The outcome was the combined endpoint of nonfatal myocardial infarction (MI), fatal CHD, or revascularization (CABG or angioplasty). We used multivariate logistic regression to assess the relationship between PTSD status and incident CHD.

RESULTS: At baseline, 76 men (9.6%) had high PTSD symptoms ($M > 73$). During an average of 10 years of follow-up, 96 cases of incident CHD occurred: 38 cases of nonfatal MI, 24 cases of fatal CHD, and 34 revascularizations. Compared to men with low PTSD, men with high PTSD had an age-adjusted relative risk of 2.23 (95% CI, 1.22-4.09) and a multivariate relative risk of 2.07 (95% CI, 1.10-3.87) for the combined endpoint (nonfatal MI, CHD death, or revascularization). Multivariate relative risks for nonfatal MI alone ($p = 0.17$) and combined nonfatal MI and CHD death ($p = 0.43$) were not significant. We found no significant dose-response relations between incident combined CHD and PTSD severity on a continuous scale ($p = 0.35$).

CONCLUSIONS: These results suggest that PTSD may be associated with an increased risk of CHD in older men. Further studies are needed to confirm these findings.

Table 1. Multivariate-Adjusted Relative risks of CHD by PTSD status, based on Mississippi score

Outcome		Mississippi PTSD Score		P value
		≤ 73 ($n = 715$)	> 73 ($n = 76$)	
Nonfatal MI, CHD Death, and Revascularization	Cases	80	16	
	Age-adjusted RR	1.0	2.23 (1.22-4.09)	0.009
	Multivariate* RR	1.0	2.07 (1.10-3.87)	0.023
Nonfatal MI and CHD Death	Cases	34	8	
	Age-adjusted RR	1.0	1.55 (0.70-3.42)	0.28
	Multivariate* RR	1.0	1.39 (0.61-3.16)	0.43
Nonfatal MI	Cases	32	6	
	Age-adjusted RR	1.0	1.95 (0.78-4.85)	0.15
	Multivariate* RR	1.0	1.88 (0.76-4.67)	0.17
Revascularization	Cases	26	8	
	Age-adjusted RR	1.0	3.65 (1.63-8.16)	0.002
	Multivariate* RR	1.0	3.27 (1.55-6.92)	0.003

CHD indicates coronary heart disease; RR, relative risk; MI, myocardial infarction; 95% confidence intervals are shown in parentheses.

* Adjusted for age, smoking status (never, any), systolic and diastolic blood pressure (mm Hg), serum total cholesterol (mg/dL), body mass index (kg/m²), history of diabetes, alcohol intake (g/dl drinks/d).

PRESCRIBING PATTERNS OF OPIOID PAIN MEDICATIONS AND BENZODIAZEPINES AMONG HIV+ AND HIV- PATIENTS IN THE NATIONAL VA SYSTEM. L.A. Chwastak¹; J. Goulet¹; K.J. Bryant²; J. Conigliaro³; S. Crystal⁴; N.L. Day⁵; D.A. Fiellin¹; P. O'Connor¹; D. Rimland⁶; M. Rodriguez-Barradas⁶; A.C. Justice¹. ¹Yale University, New Haven, CT; ²National Institutes of Health, Bethesda, MD; ³University of Pittsburgh, Pittsburgh, PA; ⁴Rutgers, The State University of New Jersey, New Brunswick, NJ; ⁵Emory University, Decatur, GA; ⁶Baylor College of Medicine, Houston, TX. (Tracking ID #135692)

BACKGROUND: The use of opioid medications for the treatment of chronic non-cancer pain is controversial. The safety and efficacy of this treatment approach has not been established, it is not clear whether these medications improve functional outcomes, and there is concern about the potential for abuse, particularly among patients with a history of substance use disorders. Recent studies suggest that concurrent benzodiazepine use is common among patients who are treated for non-cancer pain, especially among HIV+ patients. This study describes and compares the patterns and correlates of prescriptions for opioid pain medications and benzodiazepines among HIV+ patients in the national VA system, compared to an HIV- control group.

METHODS: We examined the prevalence and correlates of prescriptions for opioid pain medications and benzodiazepines among the ($n = 49,052$) HIV+ veterans in the national VA system, and an age-, race-, and gender- matched control group ($n = 49,052$) in general medical care. Data for the study came from Veterans Administration national outpatient pharmacy and clinical databases for the period between Oct 1, 1998 and Sept. 30, 2003. Prescription information was obtained from the VA's Pharmacy Benefits Management system, a compu-

terized record of all outpatient prescriptions issued from VA facilities. Medical and psychiatric diagnostic information was obtained from the VA's electronic medical records system. Chi-square and logistic regression analyses were used to determine whether prescription of opioid pain medications differed by HIV status and by the presence of substance abuse and psychiatric disorders, controlling for the potentially confounding covariates.

RESULTS: Overall, 32.9% of patients were prescribed an opioid pain medication. Opioid medication prescription rates were similar among HIV+ patients and HIV- patients (33.4% vs. 32.5%), but this difference was statistically significant ($p < 0.01$). HIV+ patients were more likely to be prescribed long-acting opioids (6.4% vs. 4.6%, $p < 0.01$). HIV+ patients were not more likely to be prescribed benzodiazepines compared to the general medical sample (15.5% vs. 15.3%), but they were statistically significantly more likely to be prescribed both opiates and benzodiazepines (11.1% vs. 9.5%, $p < 0.01$). Logistic regression analyses revealed that among both the HIV+ and HIV- patients, those with substance use disorders were more likely to be prescribed an opioid pain medication or benzodiazepine (OR = 1.13, 95% CI = 1.07-1.21, $p < 0.01$). Secondary analyses include evaluation of the impact of specific substance use disorders and methadone maintenance treatment, and comparisons of dose and duration of treatment in these subgroups.

CONCLUSIONS: Rates of opioid pain medication and benzodiazepine prescriptions are quite high among both HIV+ and HIV- patients, with a significant proportion of patients receiving both types of medications. In both HIV+ and HIV- patients, those with histories of substance use disorders are much more likely to receive treatment with these medications. Moreover, apparent differences in rates of prescriptions between the HIV+ and HIV- groups may largely be explained by differences in rates of substance use disorders.

PREVALENCE OF BURNOUT SYNDROME IN AMBULATORY CARE CLINICIANS. R. Mejia¹; M. Aleman¹; M. Aznar¹; M.P. Majdani¹; R. Fayanás¹; E.J. Pérez-Stable². ¹Programa de Medicina Interna General, Hospital de Clínicas, Universidad de Buenos Aires, Buenos Aires, ²University of California, San Francisco, San Francisco, CA. (Tracking ID #135777)

BACKGROUND: Burnout syndrome (BO) is a chronic adaptive disorder associated with inadequate reaction to psychological work demands that adversely affects quality of life of the person with BO and may impact the ability of clinicians to provide quality care for patients. We set to estimate the prevalence of BO among physicians who care for adult patients in Buenos Aires.

METHODS: Physicians listed in general medicine and family practice scientific societies and provide ambulatory care for adult patients in Buenos Aires were surveyed by e-mail between December 2003 and March 2004. We contacted 2577 physicians expecting a low response rate. The estimated sample size was 171 to ascertain BO prevalence. The Spanish version of the Maslach Burnout Inventory was used and is a self-administered questionnaire that addresses three independent domains of BO syndrome: emotional exhaustion (EE), depersonalization (DP) and personal accomplishment (PA). High scores in EE defined as $> 19/54$ or in DP defined as $> 6/30$ or a low score on PA defined as $< 39/48$ indicate a diagnosis of BO.

RESULTS: 464 physicians responded to the questionnaire (response rate 18%). 48% were women, mean age was 39, 73% had a live-in partner and 65% had children. Physicians had been in practice for a mean of 13 years and spent 33 hours a week seeing patients. Prevalence of BO was 42% for EE, 41% for DP and 63% for PA. Prevalence in the first and second domains increased significantly when physicians worked in clinic practice more than 20 hours a week (EE 34 vs 46%, DP 31 vs 63% $p < 0.05$ in both domains). No differences were found according the sex of the respondents. Having a live-in partner was associated with a significantly less prevalence of BO (EE 38 vs. 51%, DP 38 vs. 50%, PA 60 vs. 71%; $p < 0.05$ in all domains). BO prevalence was less in those physicians with children (EE 35 vs. 53%, DP 36 vs. 50%, PA 57 vs. 75%; $p < 0.05$ in all domains). Residents have more BO compared with physicians who have finished their training (EE 62 vs. 58%, DP 62 vs. 36%, PA 76 vs. 60%; $p < 0.05$ in all domains).

CONCLUSIONS: In this physician sample in Argentina there is a high prevalence of burnout. Additional research is needed to confirm these preliminary results and to develop strategies for promoting physicians' wellbeing.

PRIMARY CARE PATIENTS WITH POSTTRAUMATIC STRESS DISORDER (PTSD) AND TRAUMA EXPOSURE USE MORE HEALTH CARE SERVICES. A. Kartha¹; J. Samet¹; R. Saiz¹; T. Keane²; T. Brower²; J. Liebschutz¹. ¹Boston University, Boston, MA; ²Boston VA Medical Center, Boston, MA. (Tracking ID #132549)

BACKGROUND: PTSD confers increased risk for health care utilization in male veterans exposed to combat. In civilian populations, trauma exposure is associated with increased utilization of health services, but the role of PTSD in mediating that utilization has not been studied. We examined the independent associations of PTSD and Trauma Exposure with greater health care utilization among primary care patients and explored the role of PTSD in mediating the trauma-associated increase in utilization.

METHODS: English speaking patients, ages 18-65 years, waiting for their provider appointment in an academic, urban, hospital-based primary care clinic were systematically recruited and interviewed by trained research assistants. Independent variables included Current PTSD diagnosis [measured by the Composite International Diagnostic Interview (CIDI)] and Trauma Exposure [endorsement of any of 10 trauma items in the CIDI]. Outcomes included number of outpatient, inpatient, emergency department (ED) and mental health visits in the 12 months prior to interview obtained from the Electronic Medical Record. Analyses included unadjusted bivariate comparisons, logistic regression and

Table 1. Association of PTSD with 12-month Utilization (Poisson regression)

	Outpatient visits(mean#)	Inpatient stays(mean#)	Inpatient nights(mean#)	ED visits(mean#)	Mental health visits(mean#)
PTSD (n=96)	8.8	0.4	3	2	1.6
No PTSD (n=312)	8.2	0.1	0.8	1.4	0.5
P	0.5	<0.0001	<0.0001	0.05	0.0001

Poisson regression. Regressions were adjusted for age, gender, income, drug and/or alcohol abuse, depression and comorbid physical illness (Charlson Index).

RESULTS: 416 subjects had a mean age of 41 years, were 50% male with 60% black, 18% white and 8% Hispanic. Half had annual income <\$20,000, 16% had current substance abuse or dependence, 42% had depression and the mean Charlson Index score was =0.7 (range 0-9, possible scores 0-37). Prevalence of Current PTSD was 24% and Trauma Exposure was 79%. Subjects with PTSD had significantly more inpatient stays, spent a greater number of nights in the hospital, had more ED visits and had more mental health visits than subjects without PTSD (Table 1). Subjects with Trauma Exposure had significantly more outpatient visits, ED visits and mental health visits but no difference in inpatient visits than subjects without Trauma Exposure (all p values <0.05). After adjusting for PTSD, the effects of Trauma Exposure were reduced but still statistically significant except for outpatient visits which was no longer significant.

CONCLUSIONS: In a civilian, urban primary care population, PTSD was associated with greater use of both general and mental health care resources. Trauma Exposure also conferred an independent risk for increased use of health care resources, although some of the effects were mediated through PTSD. The majority of this care is delivered in non mental-health settings. Future research should determine if this excess use is appropriate.

SOMATIC SYMPTOMS OF ANXIETY WORSEN OUTCOMES IN A RANDOMIZED TRIAL OF TELEPHONE-BASED CARE MANAGEMENT IN PRIMARY CARE. R. Drayer¹; F. Zhu¹; S. Mazumdar¹; C.F. Reynolds¹; B. Herbeck Belnap¹; K. Shear¹; B.L. Rollman¹.
¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #131550)

BACKGROUND: Somatic symptoms are responsible for over half the primary care visits in the United States each year. They are frequently associated with treatment nonresponse, increased health services utilization, and decreased work productivity in patients with depression. We analyzed data from a randomized trial of telephone-based care management for anxiety disorders in primary care to determine if somatic symptoms of anxiety also predicted negative outcomes.

METHODS: Patients with generalized anxiety disorder (GAD) or panic disorder (PD) on the PRIME-MD were recruited from 4 Pittsburgh-area primary care practices. They were randomized to usual care or guideline-based telephone care management. We assessed self-reported somatic and psychic anxiety symptoms using the Hamilton Rating Scale for Anxiety (HRS-A) and recorded medical problems, health services utilization, functional status, and employment status.

RESULTS: The mean age of our 191 person sample was 44 (range 19-63), 81% were female, and 95% were Caucasian. 20 patients (10%) had PD, 80 patients (42%) had GAD, and 91 patients (48%) had both PD and GAD. Somatic subscale scores from the HRS-A did not differ among the three groups of patients. The mean number of medical comorbidities was 2.4 (range 0-9). Baseline somatic symptom scores correlated weakly with the number of medical comorbidities ($r=0.24$), the number of days missed from work over the last month ($r=0.23$), the number of days spent in bed over the last month ($r=0.30$), and the number of days over the last month when usual activities were cut back ($r=0.21$). Elevated baseline somatic subscale scores predicted decreased improvement in 12-month total HRS-A scores and bed days regardless of intervention status or number of medical comorbidities, but did not predict changes in health services utilization or days absent from work.

CONCLUSIONS: Elevated levels of somatic symptoms were associated with work absenteeism and presenteeism at baseline, and predicted decreased response to anxiety treatment. Although somatic symptoms were weakly associated with medical comorbidities, they independently predicted negative outcomes. Since misattribution of somatic symptoms can result in a missed anxiety disorder diagnosis, an extensive workup for physical illness, and continued lost productivity, more research is needed on ways to improve recognition of anxiety disorders.

THE DEVELOPMENT AND HEALTH CLAIMS OF ASIAN HERBAL CIGARETTES. A. Chen¹; S. Glantz¹; E.K. Tong¹.
¹University of California, San Francisco, San Francisco, CA. (Tracking ID #136249)

BACKGROUND: In recent years, the tobacco industry has been promoting "reduced harm" tobacco products, which propose to lower tobacco toxins or nicotine levels, but have not yet been demonstrated to be effective. Similarly, Asian tobacco companies have also been developing cigarettes containing both tobacco and Asian traditional herbs and making health claims. With Asia having the highest percentages of smokers in the world and tobacco companies expanding in Asia, such products may affect smoking rates. This is the first descriptive

analysis examining the development of Asian herbal cigarette products and their associated health claims.

METHODS: We first collected brand information about Asian herbal cigarettes from news accounts published between January 2000 and August 2004 through www.tobacco.org, a tobacco control news website. Other brand information was gathered from China's tobacco-related news website (www.tobaccochina.com) and Asian herbal cigarette company websites. To gather information tracking these products, we searched tobacco industry documents made public through litigation in the U.S. The University of California San Francisco Legacy Tobacco Documents website (<http://legacy.library.ucsf.edu>) contains 40 million pages that are publicly accessible. Searches using standardized techniques, such as narrowed searches using Boolean terms and snowball techniques of surrounding Bates identification document numbers, were completed between July 2003 and December 2004. Keyword searches included "herbal cigarette," specific brand names, cigarette companies, and herbs.

RESULTS: We found 22 brands of Asian herbal cigarettes: China (14), South Korea (6), Taiwan (1), and Thailand (1). China developed its first product in 1959, but most were produced from the 1980s to present. The three other countries produced their products around 2000. Most Chinese products contain both tobacco and herbs, often unnamed mixtures, except for 2 herb-only products that are advertised as substitutes for cigarettes. Health claims include 8 brands that relieve respiratory symptoms, 7 that reduce toxic substances like carcinogens or tar, 3 that protect the kidneys or lungs, and 2 that boost immunity. Seven Chinese cigarette companies supported their claims by reporting collaborations with research institutes. The Korean brands' ingredients also include unnamed herbal mixtures, but two products containing loess silt or green tea-soaked tobacco claim to reduce harmful tobacco components. Two Korean products contain herbs only: one product NosmoQ is described by a Korean medical publication that it was similarly effective to nicotine patches for cessation, and approved by the Korean Food and Drug Administration. Taiwan and Thailand both introduced a product that relieved respiratory symptoms, but stopped production; Japan imported products from China in the 1980s, but never developed their own. These discontinued efforts coincided with public objections against the products' health claims. Three brands have been sold within the US.

CONCLUSIONS: Asian herbal cigarettes incorporate Asian cultural health concepts, with many making claims of improving smoking-related symptoms or reducing harm. The herb-only products have been promoted as alternatives or aides to smoking cessation, but the effects of combusting herbal products needs to be further investigated. These products may create barriers in smoking cessation with culturally acceptable products that purport to improve health.

THE EFFECT OF MENTAL ILLNESS SEVERITY ON RECEIPT OF MAMMOGRAPHY. C.C. Doebbeling¹; L.E. Jones².
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BACKGROUND: The literature reports contradictory results regarding receipt of mammography by persons with mental health disorders (MHD). The objective of this research is to determine whether the severity of a MHD and type of MHD influenced receipt of mammography.

METHODS: We analyzed a 100% sample of Wellmark Blue Cross Blue Shield claims data, 1996-2001. Women were eligible for analysis if they were ages 40-64 and had filed at least one medical claim. Follow-up was discontinued prior to a diagnosis of incident or in-situ breast cancer as that would influence mammography receipt. Subjects were classified as having a MHD if at least one claim for a MHD was filed prior to the end of their eligibility period. A "low" severity MHD was noted if women had neither a dual diagnosis nor a MHD hospitalization. A "medium" severity MHD was noted if women had either a dual diagnosis or a MHD hospitalization. A "high" severity MHD was noted if women had both a dual diagnosis and a MHD hospitalization. The reference group was comprised of women without any MHD claims. The most clinically prominent MHD was used for analyses pertaining to specific MHD. Multivariate logistic regression was used to determine if receipt of mammography differed for women with any MHD, and 9 specific MHD, as compared to women without MHD based on the severity of the MHD. Analyses were adjusted for age, number of months of eligibility, rural residence, and number of non-mental health visits to primary care and ob/gyn providers.

RESULTS: 59,673 women with MHD (mean age 48.3 yrs) and 131,683 women without MHD (mean age 49.8 yrs) were analyzed. Receipt of mammography was dependent on the severity of the MHD. Women with low severity MHD were equally likely to have received mammography as women without MHD (OR=0.98; 95% CI: 0.95-1.01), but women with medium severity MHD (OR=0.62; 95% CI: 0.59-0.66) and high severity MHD (OR=0.38; 95% CI: 0.33-0.43) were less likely to receive mammography. Receipt of mammography was also influenced by the specific type of MHD. Women with low (OR=0.59; 95% CI: 0.45-0.78) and medium severity (OR=0.47; 95% CI: 0.33-0.67) psy-

chotic disorders were the least likely to have received mammography. For women with high severity MHD, diagnosis of a somatoform disorder predicted lowest receipt (OR=0.17; 95% CI: 0.04-0.62).

CONCLUSIONS: The presence and severity of a MHD significantly influences receipt of mammography for women with all types of MHD, with few exceptions. Further studies are warranted to determine the barriers (e.g., embarrassment) for receipt of mammography for insured women with high severity MHD who should have similar access to mammography facilities as women with low severity MHD or women without MHD. The fragmentation of the current mental healthcare system may be one potential barrier to receipt of mammography, as prior studies have noted that women with chronic MHD often have difficulty.

THE IMPORTANCE OF PRIMARY CARE PRACTITIONERS IN DEPRESSION QUALITY OF CARE. K.A. Hepner¹; L.V. Rubenstein²; M. Rowe¹; K.M. Rost³; C.D. Sherbourne¹; D.E. Ford⁴. ¹RAND Corporation, Santa Monica, CA; ²RAND Corporation; VA Greater Los Angeles at Sepulveda, Los Angeles, CA; ³University of Colorado, Denver, CO; ⁴Johns Hopkins University, Baltimore, MD. (Tracking ID #136197)

BACKGROUND: The Depression Quality Index (DQI) was recently developed to assess the quality of depression care in primary care patients. Based on the AHRQ national clinical guidelines for depression, the index was developed using patient self-report data from the Quality Improvement for Depression (QID) collaboration, which included 1131 representative patients diagnosed with depression from 45 primary care practices in 13 states. Analyses have demonstrated that the global Depression Quality Index (DQI) predicts depression outcomes at 12-, 18-, and 24-months. The DQI evaluates processes performed by both primary care and mental health practitioners. The objective of this study was to evaluate the unique effect of assessment and treatment from primary care clinicians on depression outcomes.

METHODS: The DQI includes 20 guideline-based process indicators that were used to create four subscores: Primary Care Assessment, Primary Care Treatment, Mental Health Specialty Care, and Cross-Provider Care. Primary Care Assessment included four indicators that evaluate detection of depression and assessment of suicide, alcohol abuse, and depression history. Primary Care Treatment included three indicators that assess whether the primary care clinician discussed depression with the patient or provided advice or counseling for depression including any elements of cognitive behavioral counseling. Mental Health Specialty Care included three indicators that assess the quantity and quality of visits with mental health providers. Cross-Provider Care includes ten indicators in which the process could be completed by either the primary care physician or mental health provider. Outcome variables included CESD score and recovery at 12, 18 and 24 months. We used multivariate regression analyses to determine the effect of each subscore on outcome, controlling for sickness at baseline. Models controlled for sickness at baseline using a validated sickness index based on symptom severity, physical, psychological and social comorbidity that predicts outcomes at six months.

RESULTS: Primary Care Assessment predicted CESD score at twelve ($p < .001$), eighteen ($p < .001$), and 24 ($p < .01$) months. In addition, Primary Care Treatment also predicted CESD score at twelve ($p < .001$), eighteen ($p < .05$), and 24 ($p < .01$) months. In predicting recovery from depression, Primary Care Assessment was only significant at eighteen months ($p < .05$), while Primary Care Treatment did not significantly predict recovery. Mental Health Specialty Care and Cross-Provider Care did not significantly predict depression outcomes. Of the DQI quality subscores, the quality of primary care clinician assessment and treatment had the largest and most consistent impact on outcomes.

CONCLUSIONS: Analyses have demonstrated that overall level of adherence to depression guidelines, as measured by the global DQI, predicts depression outcomes. The current analyses illuminate those results by demonstrating that primary care clinician assessment and follow-up treatment are the components of the global DQI that are most strongly and consistently related to outcome. These analyses highlight the importance of primary care clinician in successfully treating depression.

THE ROLE OF COOP/WONCA CHARTS IN PREDICTING PSYCHOLOGICAL DISTRESS IN PATIENTS WITH MEDICALLY UNEXPLAINED SYMPTOMS AND HIGH UTILIZATION OF HEALTH CARE. M. Yamada¹; Y. Ichiyama¹; Y. Oda¹; S. Emura¹; K. Shunzo¹. ¹Saga University Hospital, Saga, . (Tracking ID #132837)

BACKGROUND: Prior researches indicate that patients with medically unexplained symptoms (MUS) and high utilization of health care (HUH) are more likely to be have associated psychological distress (PD). The COOP/WONCA (C/W) charts consist of one chart each on physical fitness, feelings, limitation in daily activities, limitation in social activities, changing in health, and overall health. We hypothesized that the high score of C/W Charts (high-C/W) may have an important role in diagnosing PD, additional information to MUS and HUH.

METHODS: From November 2002 to March 2003, patients who presented themselves to the hospital for first time with somatic symptoms were enrolled in this study. To detect PD, we performed Mini International Neuropsychiatric Interview (M.I.N.I.) and followed patients up for more than 1-year. In this study, we defined psychological distress to include anxiety, depression, adjustment disorders, and somatoform disorders in DSM-IV. MUS was simply defined as complaints of patients with which was in physician's opinion unexplainable. We compared diagnostic accuracy between MUS or HUH alone and MUS or HUH plus high-C/W. We calculate sensitivity, specificity, positive likelihood ratio, and negative likelihood ratio in patients with both MUS and HUH, and with any 2 of

MUS, HUH, and high-C/W in diagnosing PD. Patients with both PD and organic disease were included in the group of patients with PD, when somatic symptoms were unexplained.

RESULTS: One hundred and six patients out of 496 were diagnosed to have PD. One hundred and thirty three patients out of 496 showed MUS. Patients with MUS included 93 patients with PD and 40 patients with organic diseases. One hundred and three out of 496 showed HUH. Patients with HUH included 49 patients with PD and 54 patients with organic diseases. Both MUS and HUH were related to the presence of psychological distress. With respect to C/W, high score of 2nd, 3rd, and 6th charts were associated with the presence of psychological distress. Sensitivity, specificity, positive likelihood ratio, and negative likelihood ratio of MUS, HUH, and each plus high-C/W in diagnosing PD were shown in Table. Combination of high-C/W with MUS elevate positive likelihood ratio. Addition of high-C/W to MUS and HUH elevate sensitivity if any 2 out of 3 items in MUS, HUH, and high-C/W.

CONCLUSIONS: MUS, HUH, and high-C/W were associated with psychological distress. The high-C/W score in addition to MUS and HUH increased the clinical usefulness in diagnosing PD.

Sensitivity, specificity, positive likelihood ratio, and negative likelihood ratio in each setting

	MUS	MUS and high-C/W	HUH	HUH and high-C/W	MUS and HUH	any 2 of MUS, HUH, and high-C/W
Sensitivity %	87.7	78.3	46.2	39.6	39.6	92.5
Specificity %	89.7	93.6	86.2	90.3	99.2	83.8
Positive LR	8.55	12.22	3.34	4.07	51.51	5.72
Negative LR	0.14	0.23	0.62	0.67	0.61	0.09

TREATING HOMELESS OPIOID DEPENDENT PATIENTS WITH BUPRENORPHINE IN AN OFFICE-BASED SETTING. D.P. Alford¹; C.T. Labelle²; J.M. Richardson²; J.J. O'Connell³; C.A. Hohl³; D.M. Cheng⁴; J.H. Samet⁵. ¹Boston University School of Medicine, Boston Medical Center, Boston Public Health Commission, Boston, MA; ²Boston Medical Center, Boston Public Health Commission, Boston, MA; ³Boston Health Care for the Homeless Program, Boston, MA; ⁴Boston University School of Public Health, Boston Medical Center, Boston, MA; ⁵Boston University Schools of Medicine and Public Health, Boston Medical Center, Boston, MA. (Tracking ID #132820)

BACKGROUND: Buprenorphine (BUP) treatment of opioid dependence in the primary care setting expands available medical care for this chronic disease. However, use of BUP in homeless individuals can seem daunting due to their unstable living environment and limited social support. We developed an interdisciplinary model of care to provide BUP treatment and assessed its feasibility and benefit.

METHODS: Between 8/03 and 10/04 collaborative BUP care between primary care physicians and a single nurse care manager (NCM) occurred in an urban Homeless Clinic (HC). Patient characteristics and BUP treatment outcomes were assessed and contrasted with a similarly run program in an urban Primary Care clinic (PC) at the same medical center during a period when federal regulations limited BUP program size.

RESULTS: HC and PC patient characteristics and outcomes are depicted in the following two tables.

Patient Characteristics	HC N=46	PC N=41
Male	59%	77%
White	43%	85%
Mean age (SD)	39 (8.5)	33 (10.6)
Self-reported psychiatric illness	90%	54%
Any identified social supports	6%	92%
HIV-infected	32%	4%
Hepatitis C-infected	91%	43%
Methadone Maintenance history	61%	9%
Median years drug use (range)	17 (3-30)	3 (1-15)
Median detoxification attempts (range)	17 (5-40)	4 (0-18)

Patient Outcomes	HC N=46	PC N=41
Retention Rates (BUP Successes):		
3 months	80% (37/46)	100% (41/41)
6 months	72% (33/46)	85% (35/41)
12 months	57% (26/46)	59% (24/41)
Clinically appropriate BUP terminations	50% (10/20)	65% (11/17)
Opioid negative urine tests over 12 months	91%	83%
Receipt of counseling and/or self-help involvement	85%	82%
Mean # NCM contacts (SD):		
1 st two weeks BUP	21 (7.8)	17 (5.8)
2 nd two weeks BUP	6 (2.7)	2 (0.8)

CONCLUSIONS: Despite homeless clinic (HC) patients' substantial comorbidities, poor social support, and longstanding opioid dependence, buprenorphine treatment outcomes were successful in a majority at one year and appeared comparable to other urban primary care (PC) patients, although HC patients required more clinical time by the nurse care manager.

DEVELOPING METHODOLOGY TO MONITOR THE ACGME DUTY HOUR REQUIREMENTS: A GRADUATE MEDICAL EDUCATION TOOL. A. Hunter¹; R.A. Harrison¹; S. Desai¹; J. Dickey¹; D. Choi¹; D. Girard¹. ¹Oregon Health & Science University, Portland, OR. (Tracking ID #135759)

BACKGROUND: Residency training institutions are required to monitor compliance with the ACGME duty hour regulations, as well as the impact of residency training on well-being. The purposes of this study are to understand the impact of these duty hour regulations on residents' perceptions of training and well-being, and to develop a survey for ongoing institutional monitoring.

METHODS: From analysis of focus groups involving residents from the major training programs (Family Medicine, Obstetrics-Gynecology, Surgery, Internal Medicine, Pediatrics) a 33 question, institution-wide survey instrument was developed to evaluate resident perceptions of the impact of duty hour implementation on training. The focus groups identified five domains from which questions were developed: clinical practice, well-being, education, scholarship and system's issues. Questions assessing compliance with the new duty hour restrictions were asked. Questions were piloted by a representative group of residents. Participants were asked to compare the current experience with that preceding the new duty hour restriction. In May 2004, the web-based survey was distributed to all upper level residents in the major training programs at our institution. Interns were excluded because of no experience prior to the new duty regulations. Follow-up distributions occurred over a two-month period. Data were analyzed by the descriptive analysis and compared between the major training programs. The study was anonymous and voluntary and was approved by the Institutional Review Board.

RESULTS: Ninety-three of 223 (42%) residents from the target training programs responded to the survey with the highest in second year residents (58%). Residents reported the following compliance with the ACGME duty hour restrictions; 80-hour work week (87.1%), one day off in 7 day's worked (94.6%), 30 hour on call shift (88%), and 0-hour duty free period (92.5%). The majority of respondents reported no significant impact on clinical practice and quality of patient care, an increase in the perception of workload, more patient hand-offs, and that the stress level at work is the same or increased. The majority of respondents reported improved well being, sleeping 5-8 hrs of a night, feeling more rested, having more time for family and other relationships, and overall improved personal well-being. The majority of respondents reported decreased time for teaching students but variability in responses for teaching junior residents, and attending teaching.

CONCLUSIONS: Despite a low response rate, preliminary data from our two institutions, suggest that implementation of the ACGME duty hour restrictions can result in an improved sense of personal well-being without a negative impact on patient care or professionalism. The perceived impact on education is variable. Additionally, we have developed a tracking tool to allow for ongoing institutional monitoring of the impact of resident work on well-being at our institution. Future study is warranted to validate this survey instrument and assess the applicability of these data to a broader population.

EVALUATION OF SGIM'S YEAR-LONG MENTORING PROGRAM. M.H. Chin¹; C.M. Mangione²; R.S. Phillips³; E.J. Thomas⁴; J. Tsevat⁵; D.S. Bell²; M. Zelder¹. ¹University of Chicago, Chicago, IL; ²University of California, Los Angeles, Los Angeles, CA; ³Harvard University, Boston, MA; ⁴University of Texas Health Science Center at Houston, Houston, TX; ⁵University of Cincinnati, Cincinnati, OH. (Tracking ID #133014)

BACKGROUND: Our aim is to evaluate SGIM's Year-Long Mentoring Program, an initiative designed to provide mentoring to faculty at institutions with few local mentors.

METHODS: In 2003, we matched 30 mentor-mentee pairs. Most met at the annual 2003 SGIM meeting and were given instructions to continue interacting over the ensuing year. In December 2004, we emailed a brief survey to each participant pair. With data collection ongoing, the current mentor response rate is 57% and the mentee response rate is 66%.

RESULTS: 84% of mentees were instructors or assistant professors, 63% were women, half were clinician-educators and half clinician-investigators. All mentors were associate or full professors. 69% were men. The overall program evaluation was positive, but about half of the participants raised concerns regarding the infrequency and limited quality of contact. 74% of mentees strongly agreed or agreed that the program was useful. 68% of mentees reported receiving advice not available at their own institution. 81% of mentees would sign up for the program again. However, half of the pairs contacted each other only every 2 months or less frequently. Mentors reported spending a mean (SD) of 1.5 (1.7) hrs/month with their SGIM mentees. 63% of mentees agreed that they would continue communicating with their mentors beyond the initial year. Mentees were more likely to agree that their goals in the program were general rather than specific, although individual mentees reported a broad range of actual activities and career/personal advising. About 20% of mentor-mentee pairs planned to write a paper together. Responses to open-ended questions showed variability in participants' goals and expectations. Many thought that the program would benefit from more structure, especially regarding reminders about contact. Major positives included working with an experienced mentor, receiving unbiased advice, networking, helping mentees, and having to commit only a limited

amount of time. Negative aspects included distance, difficulty connecting, uncertain goals, lack of structure, little incentive to follow-up, and limited ability to give material help.

CONCLUSIONS: The SGIM Year-Long Mentoring Program shows potential, but could benefit from regular reminders for contact and suggestions of possible activities. The program needs to be flexible to accommodate diverse participants, but even many experienced mentors might benefit from a menu of structured suggestions on the mentoring process.

PERCEIVED LEADERSHIP INTEGRITY AND PHYSICIAN STRESS, BURNOUT, AND INTENTION TO LEAVE PRACTICE. R.M. Poses¹; L. Manwell²; M. Mundt³; M. Linzer³. ¹Brown University, Pawtucket, RI; ²University of Wisconsin-Madison, Madison, WI; ³University of Wisconsin - Madison, Madison, WI. (Tracking ID #135076)

BACKGROUND: There are numerous anecdotes of questionable integrity of the leadership of health care organizations. We assessed physicians' perceptions of the leadership of their practice organizations, and the relationship of perceived leadership integrity to physicians' demographic characteristics, organizational characteristics, and physician stress, burnout, and intention to leave practice.

METHODS: MEMO (Minimizing Error, Maximizing Outcome) is a longitudinal cohort study of 420 primary care physicians, their office working conditions, and their patients. Physicians were surveyed two times, about one year apart. The first survey had assessed aspects of organizational climate based upon the work of Kralewski. In the second survey we asked 5 questions, each on a 5-point Likert Scale, about perceived attitudes of the leaders of the physician's practice organization to: physician's core values (with possible responses ranging from -2, oppose core values; through 0, neutral; to 2, actively support); controlling costs vs. quality (-2, costs come first regardless of quality to 2, quality comes first); physicians raising quality issues (-2, punish physician and/or hide problem to 2, praise physician and address the problem); balancing patient care and the leader's self-interest (-2, put self-interest ahead of good care to 2, put patient care ahead of self-interest); and honesty (-2, dishonest to 2, incredibly honest). We summed responses to the five questions to create the Perceived Leadership Integrity Index (PLII) with possible values ranging from -10 to 10. In this survey we also assessed work stress with a 4-item scale, and burnout and intention to leave practice on 5-point scales.

RESULTS: The first survey had 420 family medicine and general internal medicine respondents (61% of those surveyed, 84% of the target sample of 500) in four metropolitan areas from three states, plus an additional state-wide sample with rural physicians. We obtained responses from 287 of the 420 (68.3%) to the second survey. Mean PLII was 1.97 (SD = 4.06). The PLII had a Cronbach's alpha = 0.87. The PLII correlated positively with male physician gender ($p < 0.02$), physician ownership of the practice ($p = 0.005$), and physician income ($p = 0.005$), but not race or ethnicity. Organizational quality emphasis, cohesiveness, alignment with leadership values, trust, and communication and use of information technology, as assessed by the first survey, all predicted PLII ($p < 0.001$). In multiple linear regression analyses controlling for demographics, organizational climate, and office atmosphere, lower PLII predicted ($p = 0.0001$) physician stress. Similarly, lower PLII predicted burnout ($p = 0.0001$), and intention to leave ($p = 0.026$).

CONCLUSIONS: Our new PLII had good psychometric properties, and convergent validity as demonstrated by its associations with other measures of organizational climate. Negative perceptions about perceived integrity of clinical leaders correlated with physician stress, burnout, and intention to leave their practices. Although correlation does not prove causation, our data suggest that integrity of health care leaders is another working condition that can have a substantial impact upon important outcomes for physicians and, potentially, their patients.

PROGRESS REPORT: PART-TIME PHYSICIANS - PREVALENT, CONNECTED, AND SATISFIED. H.F. Meacham¹; R.B. Levine²; M. Mundt³; M. Linzer³. ¹University of Miami, Coral Gables, FL; ²Johns Hopkins Bayview Medical Center, Baltimore, MD; ³University of Wisconsin-Madison, Madison, WI. (Tracking ID #133052)

BACKGROUND: More physicians are choosing to work part-time. The impact of part-time arrangements on physician and patient outcomes is unclear. Workplace conditions may play an important role. The purpose of this study was to determine the relationship between workplace conditions, part-time status and physician outcomes.

METHODS: The MEMO (Minimizing Error, Maximizing Outcome) project surveyed 688 physicians in the midwest and NYC. Physicians were queried about their general well-being and health, depressive symptoms and burnout, job satisfaction, and their intent to leave their practice, as well as past errors, and the likelihood of future errors using the OSPRE (Occupational and Preventable Error) measure. The survey also assessed participants' perceptions of their workplace environment using Organizational Climate (OC) scales focusing on cohesiveness, organizational trust, and alignment with leadership principles. Respondents rated their sense of control over work environment, and self-reported their part-time status. T-tests were used to compare responses between part-time and full-time physicians. Regression analysis determined which workplace factors were associated with job satisfaction.

RESULTS: The response rate was 61% (n=425). Seventy-seven respondents (18%; 31% of women and 8% of men, $p < .0001$) defined their work as part-time. There were no significant differences in age, years in practice, marital status, and ethnicity between part-time and full-time physicians. Part-timers worked an average of 34.7 (SD 9.9) hours per week. Full-time physicians worked significantly more hours, 51.9 (SD 10.5, $p < .0001$). There was no significant dif-

ference in proportion of time spent seeing outpatients between part-time and full-time physicians (54% v 55%, $p=.64$). Part-time physicians spent significantly less time than full-timers seeing hospitalized patients (6% v 10%; $p=.0001$) and more time on teaching and research (14% v 10%, $p=.04$). There were no significant differences between part-time and full-time physicians in measures of well being, depression, general health, burnout, intent to leave practice, self-reported past errors or prediction of future errors. Part-time physicians reported higher job satisfaction (mean score 3.87 v 3.62, $p=.01$) than full-time physicians. There were no differences between part-time and full-time physicians' measures of workplace cohesiveness, sense of organizational trust, alignment with leadership values, or control over work. Regression analysis of workplace factors associated with job satisfaction revealed that for full-time physicians, work control ($p=.001$) and trust ($p=.001$) are key predictors of satisfaction. For part-time physicians, only trust in the organization was significantly associated with higher job satisfaction ($p=.03$).

CONCLUSIONS: Working part-time does not affect physician outcomes related to personal health, burnout, intent to leave one's practice or self-reported errors. However, participating part-time physicians were significantly more satisfied in their jobs. This difference may be related to the work environment itself, and a part-time physician's sense of trust in the workplace, encompassing organizational loyalty and sense of belonging. Given that trust is a critical factor for both part-time and full-time physicians, organizations may wish to explore methods for measuring and improving trust in the workplace in order to preserve and enhance physician job satisfaction.

REPRESENTATION OF FEMALE PHYSICIAN-INVESTIGATORS IN FIRST AND SENIOR AUTHOR POSITIONS OF PUBLISHED ORIGINAL RESEARCH: A 35-YEAR PERSPECTIVE. R. Jaggi¹; E.A. Guancial¹; C.M. Cooper¹; L. Henault¹; Y. Chang¹; R. Starr¹; N.J. Tarbell¹; E.M. Hylek². ¹Massachusetts General Hospital, Boston, MA; ²Boston University Medical Center, Boston, MA. (Tracking ID #133128)

BACKGROUND: Participation of women in the medical profession has increased dramatically over the past four decades. Women now represent 48% of medical students compared with 6% in 1960. Still, women comprise only 13% of full professors among medical school clinical faculty. Because academic advancement is largely driven by peer-reviewed original research, we sought to determine the representation of female physician investigators in scientific publications over the past 35 years.

METHODS: Original articles of selected journals were categorized by first and senior author gender—The New England Journal of Medicine, JAMA, Annals of Internal Medicine, Annals of Surgery, Obstetrics and Gynecology, and The Journal of Pediatrics. Gender was determined by analysis of first name and resolved in ambiguous cases via internet searches. Data were collected for 1970, 1980, 1990, 2000, and the first six months of 2004. Analysis was restricted to authors of original articles holding an MD degree and from US institutions.

RESULTS: Gender was determined for 96.4% of the 6549 US MD authors. Overall, 14.7% of the first authors and 9.3% of the senior authors were female. Significant trends of increased female representation were evident for each journal over the 35-year period (Table). By 2000, female first authorship had increased by 30% in Ann Intern Med, J Peds, and Obstet Gynecol. Positive trends of smaller magnitude were also noted in the senior position. Proportions of female authors increased most sharply in the specialty journals of obstetrics and pediatrics and remained extremely low in surgery. Journals with the highest impact factors had lower proportions of female authors. Female physicians served as senior mentor for 8.1% of the original research published in NEJM, JAMA, and Ann Intern Med during the first 6 months of 2004.

CONCLUSIONS: Significant strides have been made by female investigators over the past 35 years. Our data suggests however, that there continues to be a scarcity of women in senior positions. Efforts aimed at faculty development and retention are urgently needed to help ameliorate the dearth of female mentors and role models.

Table 1: Women Investigators in First or Senior Author Position*

	1970	1980	1990	2000	2004†
Overall					
1 st author, % female	6.1%	7.9%	17.2%	27.9%	26.4%
N, 1 st authors	906	796	792	596	288
Senior author, % female	3.2%	3.8%	10.3%	18.9%	17.3%
N, senior authors	722	689	679	560	277
NEJM					
1 st author, % female	4.9%	11.9%	16.9%	21.4%	14.3%
N, 1 st authors	142	118	142	98	42
Senior author, % female	3.6%	3.6%	9.8%	14.9%	12.8%
N, senior authors	111	110	123	94	39
JAMA					
1 st author, % female	5.9%	4.6%	20.3%	21.0%	23.3%
N, 1 st authors	222	151	123	119	60
Senior author, % female	2.4%	2.4%	12.7%	17.1%	6.6%
N, senior authors	170	127	102	111	61

Ann. Int. Med.					
1 st author, % female	4.8%	5.7%	12.3%	34.1%	32.0%
N, 1 st authors	105	122	106	44	25
Senior author, % female	4.4%	3.5%	4.3%	25.6%	4.3%
N, senior authors	90	113	92	43	23
Ann. Surg.					
1 st author, % female	1.8%	3.7%	5.3%	12.0%	10.2%
N, 1 st authors	165	164	131	108	49
Senior author, % female	0.7%	0.7%	0.9%	2.0%	5.9%
N, senior authors	148	149	117	101	51
Ob/Gyn.					
1 st author, % female	7.2%	7.7%	20.2%	38.3%	40.0%
N, 1 st authors	166	156	213	162	70
Senior author, % female	6.1%	5.2%	15.8%	29.0%	29.2%
N, senior authors	115	115	183	138	65
J. Peds.					
1 st author, % female	14.2%	20.0%	31.2%	46.2%	35.7%
N, 1 st authors	106	85	77	65	42
Senior author, % female	3.4%	10.7%	17.7%	27.4%	42.1%
N, senior authors	88	75	62	73	38
*M.D. only, United States only †Data through June 30					

ANTIOXIDANTS IMPROVE EXERCISE PERFORMANCE IN SUBJECTS WITH TYPE 2 DIABETES. S.L. Brandenburg¹; J. Weil¹; J. Reusch¹; C. Oliver-Pickett¹; J. Regensteiner¹. ¹University of Colorado Health Sciences Center, Denver, CO. (Tracking ID #133283)

BACKGROUND: Exercise performance and endothelial dilator function are impaired in persons with type 2 diabetes (T2DM). Impaired endothelial function may be one factor associated with impaired exercise performance in this population. Some prior studies have indicated that intravenous administration of ascorbate or L-arginine improves endothelial function in a wide array of cardiovascular disorders and risk conditions. A pilot study showed that oral administration of antioxidant vitamins or L-arginine improved brachial artery flow-mediated dilation in subjects with T2DM but not in control subjects. Therefore, we evaluated the hypothesis that in persons with T2DM, exercise tolerance as mediated by improved endothelial function would be improved by the administration of L-arginine or antioxidant vitamins.

METHODS: Forty-one nonsmoking, sedentary, overweight subjects with T2DM and no evidence of end organ complications were studied. Subjects taking insulin and/or thiazolidinediones were excluded. Subjects completed baseline evaluations of exercise performance, flow mediated brachial artery dilation, and forearm blood flow. VO2max was determined using a graded exercise protocol on a bicycle ergometer. Brachial artery flow mediated dilation measured by ultrasonography was used to assess nitric oxide mediated dilator activity. Forearm blood flow was determined by venous occlusion strain gauge plethysmography. Subjects were randomized to daily doses of: 9gm of L-arginine, Vitamin E (1800mg) and vitamin C (1000mg), or Placebo. Exercise performance, flow mediated brachial artery dilation and forearm blood flow were reevaluated after 8 weeks. The subjects in each group were of similar age, gender distribution, BMI and had similar HbA1C values at entry.

RESULTS: VO2max and endothelial function improved significantly after treatment with antioxidant vitamins ($p<.05$) (Table 1). In contrast, L-arginine failed to improve exercise capacity or endothelial function. Improved VO2max was correlated with improved forearm blood flow ($r=0.63$, $p<0.01$) and tended to be correlated with brachial artery reactivity ($r=0.58$, $p=0.08$).

CONCLUSIONS: Antioxidant vitamins improved exercise capacity in subjects

Table 1 - Results

	Placebo	L-arginine	Antioxidants
VO2max (ml/kg/min)	Pre: 21.6 ± 4.6 Post: 22.1 ± 5.0	23.4 ± 6.2 22.5 ± 5.9	25.2 ± 6.8 27.9 ± 8.6**
Brachial artery reactivity (cm)	Pre: 0.02 ± .02	0.02 ± .02	0.03 ± .02
Plethys-mography (ml/100ml/min)	Post: 0.03 ± .01 Pre: 15.4 ± 8.1	0.03 ± .03 19.1 ± 12.9	0.04 ± .02* 16.1 ± 6.8
	Post: 12.6 ± 6.1	17.7 ± 8.5	19.5 ± 5.9**

* $p<0.05$ difference within a group before vs after treatment,

† $p<0.05$ difference between groups

with T2DM. The data provides evidence of a relationship between improved exercise capacity and endothelial function. L-arginine did not improve exercise capacity or measures of endothelial function. An inferred mechanism by which impaired endothelial function may impact exercise performance is as follows: Reduced endothelial dependent vasodilatation in response to increased metabolic demand may lead to a decreased myocardial blood flow response and/or decreased blood flow to skeletal muscle during exercise. Substances which improve endothelial function may also improve exercise capacity in subjects with early T2DM. Substances which alone improve exercise performance may provide additional benefit in conjunction with exercise training.

ASSOCIATION OF PHYSICIAN CULTURAL COMPETENCY TRAINING AND PATIENT-REPORTED COMPLIANCE, PHYSICIAN COUNSELING, AND PREVENTATIVE SERVICES. E. Liles¹; M.A. Shaw²; J. Bussey-Jones³; Y.M. Diaz⁴. ¹University of North Carolina at Chapel Hill, Chapel Hill, NC; ²University of Louisville, Louisville, KY; ³Emory University, Atlanta, GA; ⁴University of Miami, Miami, FL. (Tracking ID #134762)

BACKGROUND: Physician cultural competency training has been proposed as a means to reduce health care disparities in the United States. The purpose of this study is to examine the association between physician cultural competency training and patient compliance, receipt of counseling and receipt of preventative services.

METHODS: The CCTOP study (Cultural Competency Training and Outcomes in Patients) is a multicenter, cross-sectional study of patients presenting for care at 11 geographically diverse outpatient clinics and their physicians. Physicians were asked about the format, duration and timing of their cultural competency training. Patients were verbally administered a 76-item survey regarding their health and demographics. We examined the association of the format (lecture, workshop, immersion), duration and timing of cultural competency training during medical school or residency with patient-reported compliance, receipt of counseling, and receipt of preventative services using bivariate analysis. Multivariate regression analyses were performed controlling for patient age, gender, race, education and income.

RESULTS: We present preliminary results from 250 matched patient-physician surveys. 291 patients responded (203 women; 88 men) with a mean age of 55.6 years. Only the timing of cultural competency training revealed any association with clinical outcomes. Not receiving training during fourth year of medical school was associated with lower rates of cholesterol screening (OR=0.11, $p=0.007$). No association was found between self reported medication compliance, eye examinations among diabetic patients and format, duration or timing of training ($p>0.05$). Similarly no association existed between physician counseling behavior (smoking cessation, hypertension education, or alcohol cessation), and other preventative services (mammography, pneumococcal vaccination, colonoscopy) with any training measure ($p>0.05$). These results remained consistent when we performed stratified analyses by race.

CONCLUSIONS: Physician cultural competency training appears to have minimal association with important patient outcomes. Cultural competency training, as it currently exists, may not have a significant role in reducing disparities in health care. Further investigation is needed into the optimal strategy to improve cultural competency among physicians. In addition, efficacy of initiatives such as cultural competency training to reduce disparities in health care should be evaluated.

BARRIERS TO VACCINATION AGAINST HEPATITIS A AND HEPATITIS B IN PATIENTS WITH CHRONIC HEPATITIS C VIRUS INFECTION: A NATIONAL SURVEY OF FAMILY MEDICINE AND INTERNAL MEDICINE PHYSICIANS. J.A. Foont¹; S. Chaudhari¹; C.T. Tenner²; E. Weinschel²; E.J. Bini¹. ¹New York University School of Medicine, New York, NY; ²New York University School of Medicine, New York, NY. (Tracking ID #136235)

BACKGROUND: Although vaccination against hepatitis A virus (HAV) and hepatitis B virus (HBV) is recommended for all patients with chronic hepatitis C virus (HCV) infection, physicians often do not follow these recommendations. The aim of this study was to determine Family Medicine (FM) and Internal Medicine (IM) physician practices and barriers to vaccination against HAV and HBV in this population.

METHODS: A 2-page questionnaire was mailed to 3000 primary care (FM and IM) physicians randomly selected from the AMA Physician Masterfile. The survey was pre-tested prior to mailing and included questions about physician demographics, knowledge, and attitudes regarding vaccination.

RESULTS: Among the 3000 physicians surveyed, 86 were undeliverable, 52 were not in practice, and 1209 (42.2%) of the 2862 eligible physicians returned completed surveys. There were no differences between respondents and non-respondents with regard to age, sex, geographic location, or specialty (FM vs. IM). Of the 1209 respondents (557 FM and 652 IM), FM physicians were less likely to see 3 or more HCV+ patients per week than IM physicians (29.3% vs. 34.8%, $P=0.04$). The proportion of FM and IM physicians who usually/always vaccinate eligible patients against pneumococcal pneumonia (90.3% vs. 89.9%, $P=0.67$) and influenza (94.1% vs. 95.6%, $P=0.25$) did not differ. In contrast, FM physicians were less likely than IM physicians to usually/always test HCV+ patients for immunity against HAV (33.9% vs. 48.6%, $P<0.001$), vaccinate susceptible patients against HAV (42.3% vs. 53.1%, $P<0.001$), and test for HAV antibody prior to vaccination (33.9% vs. 48.8%, $P<0.001$). Similarly, FM physicians were less likely than IM physicians to usually/always test HCV+ patients for immunity against HBV (50.8% vs. 68.0%, $P<0.001$), vaccinate susceptible patients against HBV (54.9% vs. 64.7%, $P=0.001$), and test for HBV antibody prior to vaccination (51.2% vs. 65.8%, $P<0.001$). Overall, there were numerous barriers to HAV and HBV vaccination, including patient refusal

(41.0%), cost (39.8%), lack of reimbursement (34.3%), compliance issues (28.3%), time constraints (20.8%), feel uncomfortable with vaccination (16.6%), lack of evidence regarding efficacy (10.0%) and safety (9.7%) of vaccination, and staffing issues (8.9%). The median number of barriers was 3 (interquartile range, 1 - 4) for FM physicians and 2 (interquartile range, 1 - 3) for IM physicians ($P<0.001$), and FM physicians were more likely to report that they did not feel comfortable with HAV/HBV vaccination as compared with IM physicians (23.3% vs. 10.9%, $P<0.001$).

CONCLUSIONS: Despite published recommendations for vaccination against HAV and HBV in patients with chronic HCV infection, physicians often do not test or vaccinate susceptible individuals. There is wide variability in vaccination practices and barriers to vaccination between FM and IM physicians, and interventions are needed to overcome these barriers and to improve vaccination rates.

BIOMEDICAL RISK ASSESSMENT AS AN AID TO SMOKING CESSATION: A SYSTEMATIC REVIEW. R. Bize¹; Y. Mueller²; B. Burnand²; J. Cornuz¹. ¹University Institute of Social and Preventive Medicine and Outpatient Clinic, Lausanne, ²University Institute of Social and Preventive Medicine, Lausanne, (Tracking ID #133234)

BACKGROUND: A possible strategy for increasing smoking quit rates could be to provide smokers with feedback on the biomedical or potential future effects of smoking, such as with measurement of their exhaled carbon monoxide (CO) level, lung functions, or genetic susceptibility to lung cancer. The objective of the study was to review systematically data on smoking cessation rates from controlled trials that used physiological feedback.

METHODS: We systematically searched the Medline (from 1966 to 2004) and Embase (from 1997 to 2004) databases, as well as the Cochrane's Tobacco Addiction Register. Methodological terms were combined with terms related to smoking cessation counseling and biomedical measurements. All retrieved references were considered for inclusion by two independent readers. Inclusion criteria were: a randomised controlled trial design; subjects participating in smoking cessation programs; interventions based on a biomedical test to increase motivation to quit; control groups receiving all other components of intervention; outcome being smoking cessation rate at least 6 months after the start of the intervention. Data extraction was conducted independently by two assessors, and results were standardized as odds ratios (OR) with 95% confidence intervals (95%CI). An OR > 1 favours the intervention group.

RESULTS: One hundred and seventy references were selected among 4118 for full text assessment. Nine trials were finally retained for data extraction and analysis. Three of them isolated the effect of exhaled CO measurement on smoking cessation rate. Odds ratios ranged from 0.73 (95%CI 0.38-1.39) to 1.18 (95%CI 0.84-1.64). Spirometry results were used in one other trial with an OR of 1.21 (95%CI 0.60-2.42). Exhaled CO measurement and spirometry were used together in three trials. Odds ratios still did not reach statistical significance and ranged from 0.6 (95%CI 0.25-1.46) to 3.50 (95%CI 0.88-13.92). Three trials used other motivational feedback techniques with an OR of 0.80 (95%CI 0.39-1.65) for genetic susceptibility to lung cancer testing, 3.15 (95%CI 1.07-9.34) for ultrasonography of carotid and femoral arteries performed in light smokers (12 cig/d), and 0.15 (95%CI 0.01-2.89) for the demonstration of child exposure to environmental tobacco smoke by measuring child's urinary cotinine level.

CONCLUSIONS: Due to heterogeneity between studies regarding recruitment, setting and intervention, we did not pool the data. This systematic review showed no evidence for effectiveness of biomedical risk assessment as an aid to smoking cessation.

CORRELATES OF COLORECTAL CANCER SCREENING AMONG U.S. WOMEN. N. Peterson¹; H. Murff²; B. Laffeur¹; R. Ness¹; R.S. Dittus¹. ¹Vanderbilt University, Nashville, TN; ²Tennessee Valley Healthcare System, Nashville, TN. (Tracking ID #133329)

BACKGROUND: Among women, colorectal cancer (CRC) is the third most common cause of cancer and the third leading cause of cancer death. One previous study showed that men were more likely than women to get screened for CRC. This study uses the National Health Interview Study (NHIS) to explore gender differences in utilization of CRC screening tests and gender-specific correlates of CRC testing to help identify at-risk subgroups.

METHODS: NHIS is a continuous, national household survey of civilian non-institutionalized persons using in-person surveys. The 2000 cancer control supplement was administered to 32,374 adults ≥ 18 years of age with a 72% response rate. Men and women age ≥ 40 were asked if they ever had a sigmoidoscopy, colonoscopy, or proctoscopy (hereafter all referred collectively to as lower endoscopy) and if they ever had a home fecal occult blood test (FOBT). No questions were asked about barium enema. Subjects who had not had recent screening were asked if a doctor or other health professional had recommended a lower endoscopy or home FOBT in the last year. Men and women ≥ 50 years were eligible for analysis. Subjects were considered to be current in testing if they reported lower endoscopy in the last 10 years or home FOBT in the last one year. The Survey Data Analysis (SUDAAN v.9) computer package was used to calculate variance estimates; and the results were weighted to reflect U.S. population estimates and to adjust for nonresponse (SAS v.9.1).

RESULTS: Female respondents were slightly older than males (65.8 vs. 63.8 years). Over 81% were non-Hispanic white, 9.4% non-Hispanic Black, 6.5% Hispanic, and 2.2% other. Among women, 45.5% had never had testing for colorectal cancer with either FOBT or lower endoscopy, and only 41% were current for testing with either test. In contrast, 72.8% of women reported having a mammogram in the last 2 years and 73.4% reported having a Pap smear in the last 3 years. Compared to older men, a greater proportion of older women were not

current for testing (55.9% for women vs. 48.5% for men in subjects ≥ 75 years). Proportionally, more non-Hispanic Black women than men were not current in testing (66.1% vs. 58.2%). Among women, younger age (50-64), being a minority, having public insurance, not having a usual source of care, and not having a recent mammogram were risk factors for not being current in CRC testing in multivariate analysis (adjusting for age, race, education, family income, insurance, having a functional limitation, having a usual source of care, first degree family history of CRC, and having a mammogram in the last 2 years). Among women who were not current for CRC screening and who had at least one outpatient visit in the last 12 months, over 87% reported that they had not received a recommendation for CRC screening.

CONCLUSIONS: Despite the significant reduction in CRC-related morbidity and mortality with CRC screening, it is underutilized among women. In order to improve screening rates among women, gender differences in CRC screening and risk factors for not being screened need to be further studied. In addition, primary care providers need to be encouraged to recommend CRC screening to all appropriate patients.

EFFICACY AND EFFECTIVENESS OF THE INFLUENZA VACCINE IN PATIENTS WITH HODGKIN'S AND NON-HODGKIN'S LYMPHOMA. J.J. Mazza¹; S.H. Yale¹; J. Arrowood²; C.E. Reynolds¹; I. Giurich¹; C. Po-Huang¹; R.D. Kurt¹. ¹Marshfield Clinic Research Foundation, Marshfield, WI; ²St. Joseph's Hospital, Marshfield, WI. (Tracking ID #135072)

BACKGROUND: There is limited knowledge regarding the timing of administration, efficacy and effectiveness of the influenza vaccination in patients with malignancies. The objective of this study is to determine the immunologic response to Fluzone® (A antigens (New Caledonia/20/99 [H1N1], A/Panama/2007/99 [H3N2], and B antigen [HongKong/1434/2002]) in patients with lymphoma receiving cytoreductive chemotherapy and/or radiation therapy and to evaluate the extent of clinical failure of vaccination as assessed by polymerase chain reaction (PCR).

METHODS: Eligible subjects at least 18 years of age received the trivalent influenza vaccine between October and December 2003. For patients in a cycle of chemotherapy, influenza vaccination was administered at least 2 weeks after the initiation of treatment or 5 to 7 days prior to the next scheduled treatment cycle. Hemagglutination (HA) antibody titers to the 3 vaccine antigens were obtained at baseline, 1, 3, and 6 months post vaccination. Beginning on day 8 through May 1st 2004 post vaccination subjects were instructed to report respiratory illness for PCR viral analysis and HA serology.

RESULTS: 29 patients (age range 32-81; median 68) and 29 age and gender matched controls (age range 35-82; median 61) were enrolled. A mean of 27 days occurred from the chemotherapeutic cycle start date to administration of the influenza vaccination. 3 (10%) cases and 12 (41%) controls had a mean increase in titers to one of the major antigens of influenza A (signed-rank test $p < 0.05$ at baseline, 1, 3 and 6 months for H3N2). A statistically significant 4-fold rise in antibody titers was found between controls and cases at baseline for H1N1 and baseline, 1 and 3 months for H3N2 antigen. 8 (28%) cases and 6 (21%) controls had a mean increase in titers to at least one of the major antigens of influenza B (signed rank test $p < 0.05$ at baseline, 1 and 3 months). There was no statistically significant difference in 4-fold rise in antibody titers to influenza B antigen. Only 4 subjects presented for a respiratory illness visit with PCR confirmed negative in all.

CONCLUSIONS: Compared to an expected 70% seroconversion rate, a suboptimal immunologic response was found to the Fluzone® vaccination among cases and controls. Immunosuppression from cytoreductive chemotherapy and proposed timing of the vaccine is sufficient to explain the response seen within cases at 1 and 3 months. Future studies are needed to evaluate timing, booster doses and/or enhanced immunogenicity of the influenza vaccination

HCV INFECTION (HCVI) PREVALENCE IN ARGENTINE AND IMPACT OF UNSAFE HEALTH CARE PARENTERAL PRACTICES (HC-UPP) : A SEROPREVALENCE SURVEY. J.T. Insua¹; M. O'Flaherty²; M. Fontera Vacca³; A. Camino¹; R. Maria Pia¹; M. Silva⁴. ¹Hospital Universitario Austral, Universidad Austral, Pilar, Buenos Aires; ²Universidad Austral, Pilar, Buenos Aires; ³Hospital Universitario Austral, Universidad Austral, Pilar, Buenos Aires; ⁴Hospital Universitario Austral, Universidad Austral, Pilar, . (Tracking ID #134725)

BACKGROUND: HCVI epidemiology and risk factor contribution in Argentina is poorly described, relying on special populations' data, blood bank/laboratory data and no community based studies about risk factor exposure. WHO Global Burden of Disease Study estimated that unsafe injections in health care settings cause 40% of new HCV infections worldwide; but Latin America lacks adequate data about unsafe injections, has serious inpatient safety problems and high nosocomial infection rates. Study purpose was: 1) to estimate a defined population prevalence of HCVI and 2) to assess risk factor exposure for HCVI, including potential impact of Health Care Unsafe Parenteral Practices (HC-UPP).

METHODS: A cluster-survey of HCVI seroprevalence in Derqui, a representative community of a large part of the Buenos Aires province, Argentina was designed. During November-December 2003, 1472 individuals aged 1 to 100 years old, selected by probabilistic polyetapic sampling, were surveyed by previously trained interviewers —audited by a community survey professional team, collecting demographic, socioeconomic (SES) and risk factor exposure data for HCVI. HCV antibodies were determined by MEIA-3 blood assay. HC-UPP was defined as: a) ever received an injection with a glass syringe or b) underwent surgery 3 or more times before 1990 (a period of heavy utilization of re-used equipment in health care facilities). Age-specific prevalence rates were obtained. The association of HC-UPP with HCVI, adjusted for known HCVI risk factors (I.V. illicit drug use [IV IDU], health care occupation, any surgery, transfusion history and imprisonment) was evaluated with a logistic regression model, and the pop-

ulation attributable fraction (PAF) was calculated (Levin's Formula). STATA 8.0 was employed using commands for complex sampling.

RESULTS: A young population (43% under 30 years old), with low SES (unemployment rate of 20.8%; 86% with 12 years or less of formal education), living a mean of 17.4 years in this community, was found. HCVI seroprevalence was 0.87%, (95% CI 0.31-1.18) with the highest rate among those aged 40-49 years (2.51%). Risk factor prevalence for HCVI was: 27.3% for any surgery; 15% reported an injection with a glass syringe; 7.7% were operated on 3 or more times; 7.5% received one or more blood transfusions, 7.37% worked in healthcare related activities. 1.2% of community-based individuals reported IVIDU and 1.7% individuals reported history of being in prison. HC-UPP occurred in 19.4% and was associated with HCVI (OR 4.34, 95%CI, 1.28-14.78, $p = 0.024$). IVIDU showed a trend to increased risk (OR 4.88, 95%CI 0.91-26.20, $p = 0.06$). PAF for HC-UPP was 15.3% (95% CI 8.8%-21.3%).

CONCLUSIONS: HCVI seroprevalence was somewhat lower than that reported for the US and the Americas. HC-UPP was associated to HCVI with a considerable population impact, as may also be to IV drug use in the community, after adjusting for other known risk factors in this population. Parenteral exposure through health care sources is less well known than IVIDU as a risk factor, and may be one to address more carefully in future HCV epidemiology studies in Latin America.

HOUSESTAFF ADHERENCE TO CERVICAL CANCER SCREENING GUIDELINES. J.S. Ross¹; B.A. Forsyth²; J.R. Rosenbaum¹. ¹Yale University, New Haven, CT; ²Mount Sinai School of Medicine, New York, NY. (Tracking ID #134919)

BACKGROUND: Post-graduate training is a period of time during which house-staff undergo intensive clinical education, characterized by long work hours with minimal flexibility. Prior research has demonstrated increased risks of psychiatric and pregnancy-related disorders during post-graduate training. Low utilization of preventive health care services may also be associated with post-graduate training. Our objective was to assess female housestaff adherence to United States Preventive Services Task Force (USPSTF) cervical cancer screening guidelines.

METHODS: We administered a cross-sectional survey to female housestaff at one university hospital examining cervical cancer screening behaviors, perceptions of personal adherence to guidelines, and barriers to preventive care. Primary outcomes included 1) time since most recent Pap smear, 2) adherence to USPSTF guidelines for cervical cancer screening, and 3) perception of personal adherence to screening guidelines. Participants who had a history of an abnormal Pap smear or who had been sexually active but had never had a Pap smear qualified for annual screening according to USPSTF guidelines. All other respondents qualified for triennial screening. Data were analyzed using descriptive statistics and Chi-squared tests for association.

RESULTS: Surveys were completed by 52% (204/393) of female housestaff, mean age 30.1 years. Fifty-three percent identified themselves as white and 51% were training in primary care specialties. Fifty-four percent reported having had a Pap smear in the past year, 85% in the past three years. Of participants who qualified for annual screening according to USPSTF cervical cancer guidelines, 54% were adherent, while 87% of those who qualified for triennial screening were adherent (OR=0.17, $p < 0.0001$). Eighty-four percent accurately perceived their own screening behavior to be either adherent or non-adherent ($\kappa = 0.58$). There was no difference in the accuracy of perceived personal adherence between adherent and non-adherent housestaff. Of the 43% of respondents who identified at least one reason for not having a Pap smear as often as they thought they should, the most frequently cited reasons were 1) not having time to schedule or keep appointments ($n = 72$), 2) not having a primary care physician or obstetrician-gynecologist ($n = 22$), and 3) not feeling comfortable having a Pap smear performed at their workplace institution ($n = 17$). Participants identifying themselves as Asian were significantly less likely to be adherent to USPSTF guidelines (67%) than participants identifying themselves as white (85%, $p = 0.009$).

CONCLUSIONS: Preventive health care utilization rates among female house-staff who qualified for triennial cervical cancer screening were adequate, though not optimal, and consistent with rates in comparable populations. In contrast, female housestaff who qualified for annual screening demonstrated inadequate utilization of preventive health care, despite accurate perceptions of their own need to undergo screening. While this study is consistent with the importance of having a regular source of care in obtaining preventive care, it also clarifies the continued need to improve the flexibility of post-graduate training and to encourage housestaff to obtain appropriate preventive health care.

INCREASED ADULT FAT INTAKE ASSOCIATED WITH THE PRESENCE OF CHILDREN IN THE HOUSEHOLD: RESULTS FROM NHANES III. H.H. Laroché¹; T.P. Hofer¹; M.M. Davis¹. ¹University of Michigan, Ann Arbor, MI. (Tracking ID #136023)

BACKGROUND: The impact of children on adults' dietary behavior is poorly understood. Previous investigators have postulated that having children in the household might increase fat intake and decrease fiber intake. However, this has never been investigated in a nationally representative sample. Given higher birth rates among African-American and Latino adults who are also disproportionately likely to be overweight compared to non-Latino whites, we also investigated whether the presence of children in the home has differential effects among different racial/ethnic groups.

METHODS: In public-use data from the National Health and Nutrition Examination Survey (NHANES) III, we identified adults aged 17-65 years living in households with and without children. NHANES oversamples African-Ameri-

cans and Mexican-Americans, permitting adequate sample size for comparisons among racial/ethnic groups. The principal outcome variable was total fat intake in grams based on a 24-hour dietary recall. The independent variable of chief interest was the presence of one or more children in the home or not. Other independent variables included respondents' education, race/ethnicity, age, presence of a spouse or spouse equivalent in the home, poverty income ratio, and gender. No information on age of child or relation to respondent was available. In secondary analyses, we added 24-hour total kilocalorie consumption as a control variable, and added interaction terms for race/ethnicity and child in order to investigate potential differences by race/ethnicity. STATA 8.0 was used for all analyses, accounting for complex survey design using NHANES-assigned weights.

RESULTS: The analytic sample included 7444 respondents; 4028 (54.1%) adult respondents reported one or more children in the household. 57.4% of African-American adults and 66.6% of Latino adults reported children in the household, compared to 39.0% of non-Latino white adults ($P < .001$). 56.9% of the sample was female. Overall, presence of children in the household was associated with increased adult fat consumption (mean = 86.9 g/24 hrs for adults with children versus 82.2 g/24 hrs for adults without children; $P < .05$). Interaction terms for race/ethnicity and child were not significant, suggesting no differential effect of children by race. With inclusion of total kilocalories in the model, the effect of children on adult fat intake showed the same trend but was no longer statistically significant.

CONCLUSIONS: The presence of children in the home is associated with higher fat intake among adults in a national sample; the 4.7 g difference is equivalent to over 7% of the recommended daily fat intake of 65 g in a 2000-kcal diet. The attenuation of this effect when considering total caloric consumption suggests that part of the association of children with total fat consumption is mediated though overall increase in caloric consumption while the other part may be driven by food choices with higher fat content. Internists and other health care providers should consider the potential influence of children in the home on adults' dietary behaviors.

INFORMING PATIENTS ABOUT PSA TESTING PRIOR TO SCREENING: DOES PRACTICE REFLECT RECOMMENDATIONS? I. De Alba¹; J.M. Sweningson¹; F.A. Hubbell¹.

¹University of California, Irvine, Irvine, CA. (Tracking ID #134378)

BACKGROUND: There is no scientific consensus on the role of prostate-specific antigen (PSA) testing in prostate cancer screening. However, there is unanimous agreement on the need for informed decision making prior to PSA testing. The United States Preventive Services Task Force, the American Cancer Society and other major medical organizations concur on recommending physicians to discuss with eligible patients the potential benefits and side effects of early prostate cancer detection and treatment. While close to 54% of men age 50 report having a PSA test in the last year, it is unclear what proportion of them received counsel on PSA testing by their physician. Lack of a discussion of PSA testing impairs patient informed decision making and may even have an impact on the patient's decision to undergo testing.

METHODS: This is a cross-sectional study using 2000 National Health Interview Survey data. The outcome of interest was patients' report of physician discussing the advantages and disadvantages of PSA testing prior to screening. We included men age 50 who reported a previous PSA as part of a routine physical exam or routine screening. Bivariate analysis was performed to ascertain variables associated with the outcome of interest. Logistic regression was used to adjust for potential confounders and to determine predictors of PSA discussion between physician and patient.

RESULTS: A total of 2,175 men met our inclusion criteria; 84% were White, 9% African American, 5% were Hispanic and 1% reported other or multiple races. Just 20% had less than high school education. Only 65% of those who had a prior PSA test reported having a discussion with their physician about advantages and disadvantages of testing prior to screening. In logistic regression, after adjusting for age, education, income, health status and having health insurance, we found that African American men (OR=1.54, 95% CI 1.03-2.31) as compared to White and men with a regular source of care (OR=2.73, 95% CI 1.45-5.11) were more likely to report having a discussion with their physician about pros and cons of PSA prior to testing. Other factors associated with having a discussion on PSA were test first suggested by physician as compared to patient's request (OR=2.00, 95% CI 1.52-2.64) and having 2 or more PSA measurements in the past 5 years (OR=1.52, 95% CI 1.20-1.92). Ever having an abnormal PSA test and positive family history for prostate cancer were not associated with having a conversation on PSA with their physicians.

CONCLUSIONS: A substantial number of men continue to be screened for prostate cancer with PSA without a prior discussion with their physician on advantages and disadvantages of PSA testing. This may impair patients' informed decision making and contravenes current guidelines. Men without a regular source of care or with less than two PSA measurements in the past 5 years the least likely to report holding this discussion with their health care provider. On the other hand, physicians who are perceived as proactive by their patients and suggest PSA testing are more likely to discuss the advantages and disadvantages of PSA testing.

IS THERE AN ASSOCIATION BETWEEN LIPID LOWERING DRUGS AND SATURATED FAT INTAKE AMONG US ADULTS? D. Mann¹; J. Lee¹; S. Natarajan². ¹Cornell University Medical College, New York, NY; ²New York University, New York, NY. (Tracking ID #133530)

BACKGROUND: Lipid lowering drugs are effective in reducing cardiovascular risk. Many physicians fear that the lipid lowering impact of these medications

may be misconstrued by the public as a license to abandon attempts to reduce saturated fat intake. This potential negative effect of lipid lowering therapy must be explored in order to address the concerns of healthcare providers as well as help guide their use in the US population. This analysis of cross-sectional national data evaluates the association between use of lipid lowering medications and dietary intake of saturated fat.

METHODS: Using National Health and Nutritional Examination Survey 1999-2000 data, variables for age, gender, saturated fat intake (g/day), total kilocalorie intake, history of high cholesterol, history of diabetes, history of cardiovascular disease and active lipid lowering medication use were created. After testing for differences between mean saturated fat intake in adults with normal cholesterol, lifestyle-treated high cholesterol and drug-treated high cholesterol [by Analysis of Variance (ANOVA)], multivariate linear regression models controlling for age, gender, kilocalories, BMI, diabetes and prior cardiovascular disease were used to test for an association between saturated fat intake in adults belonging to the 3 groups. All analyses incorporate the appropriate sample design weights and strata variables.

RESULTS: The sample included 3,191 people with self-reported information regarding cholesterol and diet, of whom 1,166 people reported high cholesterol (population estimate of approximately 46.4 million). Mean saturated fat intake and standard error (SE) among US adults was 22.4 g/day (SE=0.9) in drug-treated adults (n=441), 26.3 g/day (SE=0.7) in lifestyle-treated adults (n=725), and 27.3 g/day (SE 0.6) in those with normal cholesterol (n=1999). Significant differences among the means of the 3 groups were found by ANOVA ($F=12.4$, $p<0.0001$). The unadjusted linear regression model revealed a -2.3 g/day (C.I. -3.8, -0.9) lower saturated fat intake among hypercholesterolemics (drug-treated and lifestyle-treated) compared to adults with normal cholesterol. Among adults with high cholesterol, there was a -3.8 g/day (C.I. -6.8, -0.9) lower saturated fat intake among those on lipid lowering drugs compared to those not on drugs. The fully adjusted linear regression model controlling for age, gender, kilocalorie intake, BMI, diabetes, and cardiovascular disease demonstrated a -0.2 g/day (C.I. -1.0, 0.5) lower saturated fat intake among hypercholesterolemics compared to adults with normal cholesterol. Among hypercholesterolemics, there was a -0.2 g/day (C.I. -1.5, 1.2) lower saturated fat intake among lipid lowering medication users compared to non-users.

CONCLUSIONS: Our unadjusted analysis indicates that taking lipid lowering medications or having a diagnosis of high cholesterol is associated with a lower dietary intake of saturated fat, which is attenuated to a statistically insignificant level in the fully adjusted model. It appears that taking lipid lowering medications is not associated with higher saturated fat intake. However, to definitively evaluate the relationship between taking drugs to lower cholesterol and dietary saturated fat intake, a prospective study to collect detailed dietary and medication information along with biochemical measures is needed.

PATIENT DESIRE FOR PSA: WHAT MOTIVATES IT? R.E. Swaney¹; P.I. Simmons¹; T.V. Melhado¹. ¹University of Colorado Health Sciences Center, Denver, CO. (Tracking ID #136163)

BACKGROUND: Prostate cancer (PC) screening with PSA in common despite lack of evidence that screening reduces mortality and uncertain risk-benefit ratio of screening. We sought to identify patient factors, especially attitudes and beliefs, associated with desire for PSA. A better understanding of these factors would guide shared decision making as well as decision aid development.

METHODS: We distributed written questionnaires to consecutive men age 40 and older attending appointments at 2 academic internal medicine clinics, measuring attitudes and beliefs about PC and PSA screening. Men with histories of PC were excluded. We conducted a multivariate analysis, based upon a conceptual model, with desire for regular PSA testing as the outcome of interest. Chi-square testing was used for bivariate analyses, logistic regression for multivariate analyses.

RESULTS: Response rate was 71.3% (N=500, mean age 61). Seventy-four percent of respondents wanted regular PSA testing, while only 42% believed that their doctor thought they should be tested. Sixty-seven percent believed that PC is found earlier with PSA testing than without, and 56% believed that PSA testing reduces the likelihood of dying from PC. Respondents were not very concerned with potential disadvantages of screening: 40% believed that PSA testing can lead to additional testing that is uncomfortable, 20% that it can lead to unnecessary worry, and only 15% that PC treatment side effects are worse than the disease. A multivariate analysis indicated that a conceptual model including perceived severity of PC, symptoms of perceived concern for PC, worry about PC*, perceived risk of PC, receipt of a recommendation to have PSA*, knowing someone with PC*, family history of PC*, belief that PSA testing represents responsible health behavior*, wanting PSA for peace of mind*, perception that PSA testing is part of usual care*, desire for screening tests in general*, and perceived efficacy of PSA testing* was highly predictive of desire for PSA (*indicates significance at $p < .05$ in bivariate comparisons; model adjusted for demographics, health status, having had PSA, having had elevated PSA, and belief that PSA is covered by health insurance). Three factors were independently significant in the multivariate model: wanting PSA for peace of mind (OR 10.0, $p < .001$), belief that PSA testing represents responsible health behavior (OR 7.4, $p < .001$), and perceived efficacy of testing (OR 3.5, $p = .002$); c index for this model .864. Although 70% of men wanted PSA for peace of mind, only 18% believed that a normal PSA test meant they didn't have PC.

CONCLUSIONS: Improving knowledge about efficacy of screening and individual risk of PC alone will not address major attitudes and beliefs influencing desire for PSA.

PERCEPTION OF "HEALTHY" BODY WEIGHT BY PATIENTS WITH DIABETES. K. McQuigley¹, R. Hess¹, C.L. Bryce², G.S. Fischer¹, F. Katherine¹, E. Olshansky¹, D. Sacco¹. ¹University of Pittsburgh, Pittsburgh, PA; ²University of Pittsburgh Medical Center, Pittsburgh, PA. (Tracking ID #135098)

BACKGROUND: Weight management is an important part of diabetes prevention and treatment, yet obesity remains common. Ability to identify a healthy body weight may be a key part of preventing or recognizing excess body weight. We sought to determine whether patients with diabetes could identify a healthy body weight.

METHODS: As part of a clinical diabetes education initiative, we mailed surveys to 1350 patients in a large, urban, university-based general medicine practice. The survey included questions on knowledge (Diabetes Knowledge Test) and behaviors related to diabetes. Patients were asked to identify their current height and weight, as well as "the healthiest weight for your height." Patients with diabetes were identified anonymously from electronic medical record problem lists. All responses were anonymous; multiple mailings were precluded by the need to avoid undue burden on those not interested in participating. We calculated current body mass index (BMI, kg/m²) using reported current height and weight, and ideal BMI (calculated from height and "healthiest" weight), and assigned weight status using standard clinical cut-offs. Differences in accuracy of "healthy" weight identification in different groups were calculated using Chi-square and t-tests.

RESULTS: The survey was returned by 289 people (21%). Of respondents, 61% were male, 15% African American, 76% White, 6% Native American, and 4% Other race/ethnicity. Among men, the calculated current BMI was, on average, overweight [29.5 kg/m² (SD 5.3)]; with 17% reporting a healthy weight, 40% overweight and 43% obese. Among women, average BMI was obese [32.3 kg/m² (SD 8.1)], with 17% reporting healthy weight, 25% overweight, 58% obese. Both men and women reported an average ideal BMI in the upper end of the healthy range [24.9 (SD 3.1)]. However 42% of respondents reported a weight that was overweight, and 4% one that was obese as "healthiest". Accuracy of healthy weight identification was linked with sex (36% of men vs. 63% of women identified a "healthy" weight; $p < 0.005$) and body size. An overweight or obese weight was named as ideal by 9% of those respondents with a current healthy BMI, 42% of those who were overweight and 63% of those who reported obese measurements. Accuracy of healthy weight identification did not vary with race/ethnicity, age, diabetes knowledge, or years since diagnosis of diabetes.

CONCLUSIONS: Despite the importance of body weight in the prevention and control of diabetes, many patients (nearly half of this sample) overestimate the weight that would be "the healthiest weight for [their] height." Women were more likely to accurately name a healthy weight than men. Our findings do not support reports of larger body size preference in African Americans; this may reflect a focus on health rather than aesthetics, or a secular change in attitudes. As neither diabetes knowledge nor time since diagnosis influenced the accuracy of healthy weight identification, current diabetes education programs are likely to inadequately address weight in this population.

PHYSICIANS' ACTUAL USE AND THE POTENTIAL VALUE OF GLOBAL CHD RISK CALCULATION. S.L. Sheridan¹, G. Macdonald¹. ¹University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #134386)

BACKGROUND: New clinical guidelines recommend that all patients 40 years of age or older should have their global coronary heart disease (CHD) risk calculated every 5 years and know their risk of having a coronary event. Ideally, this would increase utilization of effective preventive therapies. However, little is known about whether or how physicians and patients use global CHD risk in practice. In this study, we examine physicians' use of CHD risk calculation in clinical practice and estimate the number of patients who might benefit from such calculation.

METHODS: Two investigators independently reviewed a random sample of charts from new patient visits (June 2002 to June 2003) at one university internal medicine practice to determine whether new patients who had no prior history of cardiovascular disease or other serious medical illness had their global CHD risk measured and addressed during the first year in the practice. We defined "addressed" as documentation of a plan to monitor or treat a patient's calculated global CHD risk or CHD risk factors. Reviewers additionally calculated a global CHD risk for each patient using a Framingham risk calculator to estimate the number of patients who may have benefited from knowing their CHD risk.

RESULTS: We reviewed charts for 197 new patient visits with 79 different physicians. Attending physicians comprised 24% of the physician group and were involved in 50% of new visits. Physicians had information available on all CHD risk factors for 72% of patients (mean age was 57.9; 40% were male; 37% had hypertension, 42% abnormal lipids, and 11% diabetes; 24% smoked) and addressed at least one CHD risk factor for 61%. Physicians calculated and recorded a global CHD risk for only 3%, however. Twelve months after the visit, accounting for risk factor treatment, we calculated that 10% of patients had a CHD risk $>20\%$ and 30% had a CHD risk of 10-19%, qualifying them for additional risk-reduction therapy by current evidence-based guidelines.

CONCLUSIONS: Few physicians are recording global CHD risk. With 40% of new patients remaining at high risk of CHD after one year of care, the potential value of global risk calculation should be further examined as a means to encourage physicians to utilize effective CHD prevention therapies.

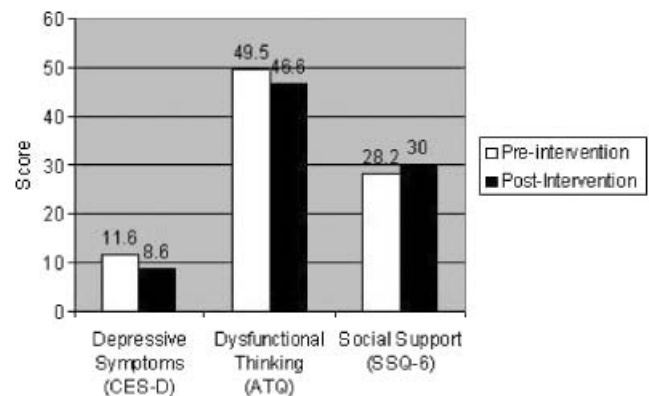
PILOT STUDY OF A PRIMARY CARE/WEB-BASED DEPRESSION PREVENTION INTERVENTION FOR YOUNG ADULTS (PROJECT CATCH-IT). B.W. Van Voorhees¹, J. Ellis¹, D.E. Ford². ¹University of Chicago, Chicago, IL; ²Johns Hopkins University, Baltimore, MD. (Tracking ID #134262)

BACKGROUND: One in four adolescents and young adults will experience an episode of depression before age 24 with significant medical, social and economic costs. While depressed mood, dysfunctional (excessively pessimistic) thinking and lack of social support are modifiable risk factors for depressive disorders, there is no primary care based approach for depression prevention in adolescents and young adults. In this study, we conducted an initial evaluation of a combined primary care/Web-based depression prevention intervention that targets these risk factors in a sample of young adults at risk for developing major depression.

METHODS: The intervention includes an initial motivational interview (MI) in primary care to engage the adolescent (GUARDS - Goals Understanding Adolescent Risk Depression Self-efficacy), eleven Web-based modules based on Cognitive-Behavioral and Interpersonal Psychotherapy, and a follow-up MI in primary care to enhance behavior change. In order to increase cultural accessibility, we used a 7-8th grade reading level with common examples and no technical terminology. We recruited 14 young adults from two urban primary care clinic settings who had at least one risk factor for a depressive episode (family or personal history of an episode), but who were unlikely to meet criteria for major depression (Center for Epidemiologic Studies Depression Score CES-D < 17). We evaluated three outcomes: 1) acceptability (performance, reasons for non-completion and satisfaction ratings), 2) adverse effects (negative emotions), and 3) evidence for benefit (pre/post comparison measures of mood (CES-D), dysfunctional thinking (Automatic Thoughts Questionnaire, ATQ), and social support (SSQ-6) including effect size (ES, change as a proportion of standard deviation).

RESULTS: This was a diverse study sample (7/14 male, 7/14 nonwhite) with a mean age of 20 and CES-D score of 13 (mild-moderate depressive symptoms). All participants completed the first motivational interview, 13/14 engaged the website, and 8/14 completed the entire intervention. Reasons for non-completion included medical illness (1), recommendation by counselor or family member (2), technical problems with site (1), lack of motivation (1) and preferred face-to-face approach for current problem (1). The mean number of modules completed was 7.2/11 (65%) and the mean time on site was 1.8 hours. On a 1-5 scale, completers reported high levels of readability ($M = 4.7$, $SD = 0.5$), and ease of understanding ($M = 4.7$, $SD = 0.4$) and low levels of negative emotions ($M = 2.0$, $SD = 0.8$). Global satisfaction (1-10 scale) was highest for the motivational interview ($M = 8.5$) and lower for the Web-based components ($M = 5.9$, $SD = 2.9$). For completers, favorable trends were noted for the targeted risk factors between the pre/post measures: depressive symptoms, CES-D -3, P -value < 0.15 ($ES = 0.43$); dysfunctional thinking, ATQ, -2.9, p -value < 0.51 ($ES = 0.17$); and low social support, SSQ-6 +1.8, p -value < 0.13 , ($ES = 0.27$).

CONCLUSIONS: A primary care/Web-based depression prevention intervention may be acceptable to young adults with minimal adverse effects. Effect sizes for measured risk factors were similar to other depression prevention interventions. Completers may experience favorable changes in known risk factors.



Pre/Post Measures of Risk Factors for Onset of Major Depression

PRACTICE ORGANIZATION DETERMINANTS OF COLORECTAL CANCER SCREENING RATES. E. Yano¹, L. Soban¹, D. Etzioni², P. Parkerton³. ¹VA Greater Los Angeles HSR&D Center of Excellence, Sepulveda, CA; ²David Geffen UCLA School of Medicine, Los Angeles, CA; ³UCLA School of Public Health, Los Angeles, CA. (Tracking ID #135762)

BACKGROUND: Colorectal cancer (CRC) imparts one of the highest levels of clinically preventable burdens of any disease process. Despite widespread recognition of the importance of early detection through screening, variation in CRC screening performance remains high. While some variation is patient-related (e.g., insurance, knowledge deficits, patient preferences), recent research has pointed to the contribution of practice-level factors. We explored the primary care practice-level determinants of CRC screening variations, adjusting for patient-level covariates, to inform improved structuring of primary care and referral practices.

METHODS: We merged patient-level CRC screening data from the Department of Veterans Affairs' Office of Quality and Performance (OQP) External Peer Re-

view Program (EPRP) for fiscal year 2001 with patient sociodemographics, comorbid conditions, and utilization data from the VA National Patient Care Database from 24 months prior to the index visit used for EPRP chart review. We clustered sampled patients within each VA medical center, confirming their primary care (PC) exposure using centralized utilization databases and visit patterns. We determined practice features (e.g., size, practice autonomy, resource sufficiency) to which individual veterans were exposed using the National VA Survey of Primary Care Practices and area characteristics (e.g., urban/rural location) from the Area Resource File ($n=140$ VAs). We used a random effects logistic model to evaluate the practice-level predictors of CRC screening adjusting for patient-level covariates among all guideline-eligible veterans, after eliminating cases with documented histories of CRC or irritable bowel syndrome ($n=37,935$ patients).

RESULTS: The strongest predictors of CRC screening performance are higher PC visit rates ($p<.0001$), smaller facility size ($p<.0001$), degree of primary care practice autonomy ($p<.02$), and greater sufficiency of clinical support arrangements ($p<.02$). Patient-level factors associated with higher CRC screening rates were more advanced patient age ($p<.0001$), greater income ($p<.0001$) and male gender ($p<.004$). Practice characteristics predicted a greater share of the variance than did patient-level predictors.

CONCLUSIONS: Adjusting for patient-level factors, the organization and management of primary care practice has substantial impact on the rates at which patients achieve guideline-concordant colorectal cancer screening. In closed care systems, where primary care programs have experienced resource drains in clinical support and declining autonomy over local primary care operations and referral procedures have significantly lower CRC screening performance. Large urban centers also face unanticipated challenges in assuring adequate screening penetrance despite better access to GI specialists in part because of less organized primary care delivery.

PREDICTORS OF NON-ADHERENCE TO SCREENING COLONOSCOPY. T.D. Denberg¹; T.V. Melhado²; J.M. Coombes³; K. Berman¹; T. Byers¹; A.C. Marcus²; J.F. Steiner¹; D.J. Ahnen¹. ¹University of Colorado Health Sciences Center, Denver, CO; ²AMC Cancer Research Center, Denver, CO; ³University of Colorado Health Sciences Center, Aurora, CO. (Tracking ID #131357)

BACKGROUND: Colonoscopy has become a preferred colorectal cancer (CRC) screening modality. Little is known about why patients who are referred for colonoscopy do not complete recommended procedures. Prior adherence studies have evaluated colonoscopy only in combination with flexible sigmoidoscopy, failed to differentiate between screening and diagnostic procedures, and have examined cancellations/no-shows, but not non-scheduling, as mechanisms of non-adherence.

METHODS: Sociodemographic predictors of screening completion were assessed in a retrospective cohort of 647 patients referred for colonoscopy at a major university hospital. Then, a sample of patients who never completed screening after referral ($n=52$) was interviewed by telephone and comparisons in reported reasons for non-adherence were made by gender.

RESULTS: Half of all patients referred for colonoscopy failed to complete the procedure, overwhelmingly because of non-scheduling. Female sex, younger age and insurance type predicted poorer adherence. Married men were the most likely to complete screening. Patient-reported barriers to screening completion included cognitive-emotional factors (e.g. lack of perceived risk for CRC, fear of pain, and concerns about modesty and the bowel prep), logistic obstacles (e.g. cost, other health problems, and competing demands), and health system barriers (e.g. scheduling challenges, long waiting times). Women reported more concerns about modesty and other aspects of the procedure than men. Only 40% of patients were aware of alternative screening options.

CONCLUSIONS: Adherence to screening colonoscopy referrals is sub-optimal and may be improved by better communication with patients, barriers counsel-

ing to help resolve logistic obstacles, and improvements in colonoscopy referral and scheduling mechanisms.

REACH-OUT DIABETES RISK REDUCTION PROGRAM FOR AFRICAN AMERICAN YOUTH: PILOT EXPERIENCE USING COMMUNITY LAY HEALTH LEADERS. D.L. Burnet¹; R. Das¹; A. Plaut¹; D. Schnobrich¹; N. Swamy¹; A. Jackson¹; M.H. Chin¹. ¹University of Chicago, Chicago, IL. (Tracking ID #134367)

BACKGROUND: Obesity rates are rising in all age groups, and type 2 diabetes incidence is increasing in youth, especially among minorities. The purpose of this study was to develop and pilot test a culturally sensitive, community-based family oriented intervention to decrease overweight and risk of type 2 diabetes in high risk inner-city African American youth.

METHODS: REACH-OUT is a family-based nutrition and exercise program for overweight African American youth aged 9-12 and their families, led by lay health leaders and conducted in a community setting on Chicago's South Side. Sessions alternate between a focus on nutrition (at a local grocery store) and physical activity (at a YMCA). Strategies include group problem solving and concrete goal setting. Twenty-nine families (62 subjects) participated in a pilot study of the REACH-OUT intervention. The first cohort of 10 families was led by the investigators; subsequent cohorts were led by lay health leaders recruited from the community. Lay leaders received 8 hours of training, including role play and practice sessions, prior to delivering the intervention. Lay-led sessions were observed by the investigators to assure consistency, and feedback was given at the end of each session. We conducted qualitative interviews with lay health leaders and with program participants in order to better understand their experience and refine the program prior to further dissemination.

RESULTS: Attendance and retention were stronger in the lay-led groups (70% vs. 82% attendance rates, respectively). Children's BMI's remained stable on average, and blood pressures declined over 14 weeks. Children's mean fasting glucose to insulin ratio remained stable, during a developmental phase when glucose tolerance commonly shows transient decline. Parents lost 4 kg on average. Families reported increased physical activity and improved nutrition behaviors. Health knowledge increased. The lay-led groups demonstrated better overall outcomes, with BMI z-scores and body fat for children in the lay-led groups decreasing somewhat over fourteen weeks. One-year follow-up data are pending. Qualitative interviews revealed insights regarding challenges (transportation, time limitations, safety concerns, lack of support from other family members) as well as facilitating factors for behavior change. Families reported that children often became active agents of change, asking for more physical activity and participating in healthy food choices. Families requested more hands-on activities and increased involvement by lay health leaders. These changes have been incorporated into the program.

CONCLUSIONS: This community-based, culturally-tailored health intervention proved acceptable and feasible as evidenced by low attrition rate and high attendance scores. Lay health educators can effectively engage participants, thereby improving outcomes. A full-scale, randomized controlled trial of the REACH-OUT nutrition and exercise program for families is currently underway.

RECEIPT OF PREVENTIVE HEALTHCARE SERVICES BY UNITED STATES MEN ACROSS THE LIFESPAN. A.J. Viera¹; J. Thorpe¹; J.M. Garrett¹. ¹University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #133662)

BACKGROUND: Life expectancy for men in the United States is six years less than for women. Across the lifespan, all-cause mortality is higher in every age group of men compared to women. Given the strong evidence that certain clinical preventive services reduce morbidity and mortality, a difference in receipt of preventive services may partially explain this disparity. However, the extent to which receipt of preventive services differs between men and women has not

Age (yrs)	Blood pressure check		Cholesterol measure		Colorectal cancer screen		Counseled to stop smoking		Checkup within 2 years		Dental checkup once per year	
	M	W	M	W	M	W	M	W	M	W	M	W
20 - 29	76.6	89.2	49.2	56.0	N/A	N/A	40.9	51.1	62.1	79.8	58.4	67.8
30 - 39	78.6	89.4	64.2	70.1	N/A	N/A	40.1	50.2	62.9	79.8	56.0	71.5
40 - 49	79.1	89.5	76.7	81.2	N/A	N/A	47.0	57.2	71.4	80.7	58.1	73.6
50 - 59	82.9	89.1	84.0	87.4	60.2	56.4	46.4	56.6	78.0	82.0	57.7	69.1
≥ 60	87.5	90.9	88.2	90.7	59.4	55.6	48.9	59.1	84.0	87.6	54.3	64.1
Interaction	Yes, $p=.021$		No, $p=.67$		No, $p=.82$		No, $p=.64$		Yes, $p<.001$		Yes, $p=.030$	
Odds ratio (95% CI)			.76 (.70-.84)		1.2 (.99-1.4)		.66 (.56-.79)					

*Accounting for interaction between sex*age and adjusted for education, insurance, usual source of care, marital status, perceived health status, and region of country. If no interaction was present, adjusted odds ratio is reported. N/A= not applicable.

Adjusted *percentages of Preventive Services Received by United States men (M) and Women(W), Stratified by Age Group

been adequately described. The objective of this study was to describe the receipt of basic preventive care services by adult men in the United States across the lifespan compared to women.

METHODS: We used data from the 2000 Medical Expenditure Panel Survey, a nationally representative survey of healthcare use in the United States, to compare adjusted and unadjusted weighted percentages of men and women who received selected preventive services stratified by age group. Outcome measures were receipt of six basic preventive health services. Over 14,900 men and women 20 years or older were included in the analyses (weighted $n=196,900,000$). Multivariable analyses were conducted using separate logistic regression models for each outcome, with sex as the main independent variable, adjusted for covariates. In addition, models included a variable to account for the interaction between sex and age in the age categories for each model. Covariates that did not confound the relationship between sex and each outcome were omitted from final models. Adjusted percents were calculated using the beta-coefficients from final models.

RESULTS: With the exception of colorectal cancer screening, the percentages of men who received selected services were significantly lower than women in all age groups (Table). In younger age groups, men were particularly less likely than women to have received certain services, whereas the difference in the older age groups was less pronounced. This interaction between sex and age group was present for blood pressure measurement ($p=.021$), routine checkups ($p<.001$), and dental visits ($p=.030$). For services unaffected by age group (cholesterol screening and smoking cessation advice), the odds that men received the service were significantly lower than that of women. These adjusted odds ratios for cholesterol screening and smoking cessation advice were 0.76 (95% CI .70-.84) and 0.66 (95% CI .56-.79), respectively.

CONCLUSIONS: With the exception of colorectal cancer screening (for which rates in both sexes are low), men in the United States receive less basic preventive healthcare services than women. This disparity is even greater in younger age groups for certain services. Potential exists for preventive services to have a more substantial impact on morbidity and mortality in men, and alternative methods of preventive service delivery to young and middle-aged men should be explored.

SCREENING FOR CHRONIC HEPATITIS B INFECTION IN CHINESE-AMERICAN IMMIGRANTS IN TWO ACADEMIC GENERAL MEDICINE PRACTICES. C. Lai¹. ¹University of California at San Francisco, San Francisco, CA. (Tracking ID #133294)

BACKGROUND: Chinese-American immigrants are frequently chronic carriers of the hepatitis B virus (HBV), which puts them at high risk of developing hepatocellular carcinoma and for transmitting the infection to others if they remain undiagnosed. Despite guidelines recommending screening for HBV in patients from endemic areas, little is known about academic general internists' practice patterns regarding HBV screening in Chinese-American immigrants.

METHODS: Using data from computerized medical records, we conducted a retrospective cohort study of HBV screening rates among Chinese immigrants in two academic general medicine practices. We included limited-English proficient (LEP) Chinese patients who had at least two visits to the general medicine practices from January 2001 to January 2004, and excluded patients who had a diagnosis of another condition that could have led to hepatitis B testing, including acute hepatitis (A, B, C), chronic hepatitis C, acute liver failure, HIV, and other conditions. LEP Chinese patients were defined by self-identification of Asian race and primary language as Cantonese or Mandarin. The primary outcome was receipt of a hepatitis B surface antigen (HBsAg) test at any time. We calculated rates of HBV screening for LEP Chinese compared to non-Hispanic Whites, and tested for bivariate associations between screening status and various sociodemographic variables.

RESULTS: Of the 878 Chinese patients eligible for analysis, 574 (65%) had a HBsAg test, compared to 2071 (21%) of 9687 non-Hispanic White patients ($p<0.001$). In the LEP Chinese, compared to unscreened patients, screened patients were more likely to be married (69% vs. 31%, $p=0.0009$), to have been patients in the general medicine practice longer (3.3 years vs. 2.7 years, $p=0.0003$), and to have made more visits to the practice (13.6 vs. 10.9 visits, $p<0.0001$). There were no significant differences in patients' age or insurance type between the screened and unscreened Chinese patients. Of note, 6% of the Chinese patients screened had positive HBsAg tests, compared to 1% of non-Hispanic White patients.

CONCLUSIONS: Limited-English proficient Chinese patients had a higher rate of HBV screening than did non-Hispanic Whites, but it was still less than the 100% screening goal. The findings also suggest that Chinese immigrants who receive continuous primary care are more likely to be screened. Clinic-based interventions to improve the detection and prevention of hepatitis B are needed in high risk populations.

SCREENING FOR DEPRESSION IN PATIENTS WITH CORONARY HEART DISEASE: DATA FROM THE HEART AND SOUL STUDY. D.D. McManus¹; S. Pipkin¹; M.A. Whoolley¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #134042)

BACKGROUND: Major depression is associated with adverse outcomes in patients with coronary heart disease (CHD). Treatment of depression improves patient outcomes, but how best to identify and treat depression in busy cardiology practices remains uncertain.

METHODS: We compared the test characteristics of four depression case-finding instruments with a gold standard Diagnostic Interview for Depression in 1024 adults with stable CHD who were recruited from outpatient clinics in the San Francisco Bay Area. The 4 case-finding instruments were the 10-item short

form of the Center for Epidemiologic Studies Depression Scale (CES-D), the 9-item Patient Health Questionnaire (PHQ-9), the 2-item Patient Health Questionnaire (PHQ-2), and a brief two-question screening instrument that asks (1) "During the past month, have you often been bothered by feeling down, depressed, or hopeless?" and (2) "During the past month, have you often been bothered by little interest or pleasure in doing things?" A "yes" answer to either of these two questions was considered a positive test.

RESULTS: Of the 1024 participants, 224 (22%) had major depression. Using standard cutpoints for identifying major depression, sensitivities of the instruments ranged from 39 to 90%, and specificities ranged from 69 to 92% (Table). A positive response to the two-question instrument had a sensitivity of 90%, a specificity of 69%, and a negative likelihood ratio of 0.14. Areas under the receiver operating characteristic curves were similar for all of the instruments (range 0.84 to 0.87).

CONCLUSIONS: The two-question instrument is a sensitive tool for identifying depression in patients with CHD from diverse outpatient care settings. A negative response to both questions effectively rules out depression.

TABLE. TEST CHARACTERISTICS OF CASE-FINDING INSTRUMENTS

Instrument	Sensitivity (95% CI)	Specificity (95% CI)	Positive Likelihood Ratio	Negative Likelihood Ratio
CES-D	76% (70-81%)	79% (76-82%)	3.6	0.30
PHQ-9	54% (47-61%)	90% (88-92%)	5.4	0.51
PHQ-2	39% (33-46%)	92% (90-94%)	4.9	0.66
Two Questions	90% (86-94%)	69% (66-73%)	2.9	0.14

SHOULD A COLON CANCER SCREENING DECISION AID INCLUDE THE EXPLICIT OPTION OF NO TESTING? A CONTROLLED TRIAL COMPARING TWO VIDEOS. J. Griffith¹; M. Kovacs²; F.J. Fowler²; C. Lewis¹; M. Pignone¹. ¹University of North Carolina at Chapel Hill, Chapel Hill, NC; ²Foundation for Informed Medical Decision Making, Boston, MA. (Tracking ID #134869)

BACKGROUND: The effect of including the explicit option of no testing in decision aids for colorectal cancer (CRC) screening is unclear.

METHODS: To compare, in a controlled trial, two CRC screening decision aids, each approximately 25 minutes in duration. The two decision aids differ only in the inclusion or exclusion of explicit information stating that deciding not to be tested is a reasonable option for average risk adults ages 50-75 because the potential benefits may be too small to outweigh the downsides. We developed two prototype colon cancer decision aid videotapes that were identical except that one videotape (Version A) contained information suggesting that deciding not to be screened was a reasonable choice to consider, while Version B did not contain this information. The effects of the different versions were compared in a controlled trial. Adults ages 50-75 were recruited from the community by telephone. Participants were asked to attend group sessions of 6-12 held in the evening at one of three sites, one in North Carolina and two in the Northeast. Six groups viewed Version A and six groups viewed Version B. Questionnaires were administered before and after the video. Main outcomes of interest, measured on 5-point Likert scales (higher scores reflected more positive ratings), included: subjective impressions about the amount and clarity of information presented, balance with respect to the benefits and downsides of screening, overall rating of the videotape, interest in being screened, and intent to ask one's provider for screening after viewing. Participants' knowledge of CRC was also assessed before and after the videos.

RESULTS: 106 adults participated: 57 viewed Version A and 49 Version B. Mean age of participants was 60 years (range 50-81), 53% were women, 76% were White, 22% African-American and 99% had graduated high school. 72% of participants were up to date with screening. There were few differences between groups at baseline. Analysis of the post-test questionnaire found:

CONCLUSIONS: Conclusions: A CRC screening decision aid video that includes explicit information on the option of not being tested (Version A) received somewhat less favorable subjective ratings than an otherwise identical decision aid

	Version A	Version B	p-value
Clarity of information on benefits	3.4	4.1	$p<0.01$
Amount of information on benefits	2.9	3.0	$p=0.34$
Clarity of information on downsides	3.2	3.6	$p=0.02$
Amount of information on downsides	2.7	3.0	$p=0.03$
Helped me sort what was important	3.5	3.8	$p=0.15$
Helped me prepare for making a decision	3.6	3.9	$p=0.11$
Balance	2.0	1.6	$p=0.05$
Overall rating of video	3.5	3.8	$p=0.16$
Interest in screening	3.1	3.0	$p=0.86$
Intent to be screened	3.0	2.7	$p=0.48$
Knowledge score	4.2	4.1	$p=0.93$

Table 1.

Navigation	Content	Screener	Audio
More Info on: • What to expect from the tool • How to use the buttons	More: • Information about substance use • Diversity • Peer voices • "Teasers" about supplemental buttons	• Activate text for each survey answer • Minimize clicks on small radio buttons	• Closer tracking to text on screen • Musical signal to advance webpage

without this information (Version B) with respect to amount and clarity of information. The version with the explicit option not to be tested was considered to be better balanced in presenting the benefits and downsides of screening. There were no important differences between the two videotapes in knowledge about screening, interest in screening, or intent to engage in screening.

SMOKING CESSATION WEBSITES AS AN APPROACH TO INCREASING PHYSICIAN VISITS FOR SMOKING CESSATION. T. Houston¹; D.E. Ford². ¹University of Alabama at Birmingham, Birmingham, AL; ²Johns Hopkins University, Baltimore, MD. (Tracking ID #136085)

BACKGROUND: Smoking cessation websites are increasing and provide help at a very reasonable cost. However, little is known about who will participate in an quit smoking website intervention, or how it will affect their subsequent behavior. We evaluated use of a quit-smoking website, and what services and treatments intervention participants used afterwards.

METHODS: We developed a National Cancer Institute-funded website (free2quit.com). Content, based on formative research with smokers, included three modules: 1) Self-help, 2) Talking to Your Doctor, and 3) Talking to Your Family. Content was tailored to the individual based on introductory survey questions including stage of change and prior quit history. For example, in the Self-help, participants who were ready to quit could complete a comprehensive quit plan and set a quit date; whereas those who were not ready to quit completed a decisional balance survey. The "Doctor" module included scenarios that role modeled asking the doctor for help, and a communication facilitator that would help patients write a "dear doctor letter" requesting treatment. Participants were recruited through the Internet using Google ads and were then randomized on-line. Control patients were referred to an NIH smoking education website (smoking.drugabuse.gov). We followed-up with participants by email survey at one-month. **RESULTS:** Participants (127 Intervention, 123 Control) were mostly female (73%), young (66% less than age 45), white (89%) and preparing to quit (88%).

RESULTS: Participants (127 Intervention, 123 Control) were mostly female (73%), young (66% less than age 45), white (89%) and preparing to quit (88%). Most (84%, n=210) reported a serious quit attempt, and of these: 59% reported prior nicotine replacement (NRT), and 44% reported bupropion use. The most common reason for relapse was depression or anger. Among 127 Intervention users, mean website time was 8.2 minutes, and 63% used the Self-help, 31% the Family, and 34% the Doctor module. Ninety-nine participants (39%) completed follow-up (52 intervention, 49 control). One-month cessation rates were similar among intervention (N=11 (21%)) and control (N=13 (27%)). However, among the 75 individuals who did not quit, intervention users more frequently reported a quit attempt (N=27 (67%)) compared with controls (N=14 (44%)), p=0.04). Follow-up use of bupropion and NRT did not differ for intervention versus control. Intervention and control users reported similar frequency of talking to their doctor about smoking during follow-up (39% vs. 42%).

CONCLUSIONS: Individuals who found our website had considerable past history with smoking cessation, although many had not used pharmacotherapy. Although preliminary results suggest an effect on quit attempts, one expressed purpose of the website, to encourage further treatment seeking, was not achieved.

START TALKING ABOUT RISKS: A QUALITATIVE ASSESSMENT OF PATIENT AND PROVIDER VIEWS INFORMING WEBTOOL DEVELOPMENT TO SUPPORT HIV "PREVENTION FOR POSITIVES" COUNSELING. C.E. Golin¹; M. Boland¹; S. Patel¹; C. Aspden¹; B. Fowler¹; L. Sutherland¹; B. Quinlivan². ¹University of North Carolina at Chapel Hill, Chapel Hill, NC; ²University of North Carolina, Chapel Hill, NC. (Tracking ID #135456)

BACKGROUND: New guidelines require providers to counsel all HIV-infected patients about safer sex yet few data suggest how to do this effectively. Webtools have been used to coach patients but have not been tested for HIV prevention. Therefore, we assessed patient and provider views of HIV-infected persons' prevention needs and used these data to develop and test a web-based screening and educational tool to support provider-delivered HIV prevention counseling.

METHODS: Using a theoretical model, we conducted in-depth, qualitative interviews with 20 patients and 19 providers at a large Infectious Diseases clinic in the southeastern U.S. Using a constructivist research paradigm, we coded interviews using AtlasTI and N6 software. Based on these results, we developed and tested, for content and usability, a brief screening and educational webtool to support provider counseling. The Start Talking about Risks tool includes: 9 screening questions; videos about practicing safer sex; videos coaching patients to discuss safer sex; and 13 supplemental buttons. We tested the tool among 6 users representative of our site, collecting: 1) users' aloud comments while em-

playing the tool, and 2) responses to open-ended questions of reactions to content.

RESULTS: Patients recommended a range of topics for safer sex counseling: "risks of sexual activities"; "safe sex choices"; "how to use condoms", "oral sex", "self-esteem", "telling others you're positive", "sex with positives." As one person put it, "You've got to go in-depth to the 'how-to' for it to be a learning experience." Another stressed the need for a positive message: "The emphasis should be that you can still . . . have a healthy sexual relationship and appetite, but . . . you have to play safer." Providers wanted to learn brief, effective methods to counsel and convey information. Patients cited improved patient-provider relationships, knowledge, and support as potential benefits. Patients and providers expressed concerns about lack of time and discomfort with discussing sensitive topics. Providers mentioned state health department regulations and patients' competing needs as barriers to prevention counseling. Facilitators were: strong patient-provider relationships, healthy patients, and conversation triggers. 4 men and 2 women tested the tool: 5 were African-American with mean age 39 and low to moderate computer skills. Participants were moderately to highly satisfied with webtool content. User input is summarized in Table 1 below. Modifications have been made based on testing results.

CONCLUSIONS: We have developed a user-friendly, informative, relevant webtool to support routine prevention counseling in a clinical setting for people living with HIV. Further studies are needed to assess its impact on provider and patient behavior.

THE EFFECT OF FIBER, FOLATE, AND EXERCISE ON THE RISK FOR COLON POLYPS IN A MULTIETHNIC COLON CANCER SCREENING POPULATION. R. Williams¹; R. Lin¹; G. Huang¹; H. Tran¹; M.A. Poles¹; F. Francois¹. ¹New York University, New York, NY. (Tracking ID #135747)

BACKGROUND: Dietary fiber, regular exercise, and daily folate supplementation have been shown to decrease the risk of colorectal cancer (CRC). Past studies have looked at these lifestyle modifications in white non-Hispanics. However, it is unclear if these factors will decrease the risk of CRC across multiple ethnicities. The aim of this study was to evaluate the relationship of fiber, folate, and exercise to the prevalence of polyps in a multiethnic urban screening population.

METHODS: Consecutive adults referred for screening colonoscopy in a municipal hospital were prospectively enrolled. A detailed questionnaire was administered to collect data on medical, dietary, and exercise history. Preventative lifestyle measures (PLMs) included ingestion of > 10gms of fiber daily, taking at least 1mg of folate daily, or exercising at least 1 hour/week. Endoscopy was performed and the number, size, and location of all polyps were documented. Advanced colonic neoplasms were defined as adenomas ≥ 10mm in diameter, any adenoma with villous histology (regardless of size), high-grade dysplasia, or adenocarcinoma.

RESULTS: A total of 206 patients were enrolled of which 93 (45.1%) were Hispanic, 68 (33%) were Asian, 28 (13.6%) were African American, and 15 (7.3%) were Caucasian. The overall prevalence of colon polyps was 17%. A total of 74 (35%) of individuals reported at least one PLM. The percent of individuals with at least one PLM did not differ between ethnic groups. The prevalence of polyps in patients without any PLMs was 19%, while it was 13% in those with one PLM, and 0% in those with two or more PLMs. In logistic regression analysis, having one or more PLMs was protective against colonic polyps compared to those without any PLMs, after controlling for age, sex, race and body mass index (OR 0.36; 95% CI 0.14 to 0.96, p=0.040). There was a trend towards finding more left sided polyps and more polyps with advanced histology in the group without any PLMs.

CONCLUSIONS: In this multiethnic population, having one or more PLMs was associated with a lower prevalence of colonic polyps compared to those without any preventative measures. Given these findings, the implementation of fiber, folate, and exercise in the primary care setting might help to reduce the occurrence of CRC. Further studies are warranted in ethnically diverse populations to evaluate the role of lifestyle changes in CRC prevention.

THE FAMILY MEDICAL HISTORY IN PRIMARY CARE: POTENTIAL LOST OPPORTUNITIES FOR SCREENING AND PREVENTION. H.J. Murff¹; S. Syngal². ¹Vanderbilt University, Nashville, TN; ²Dana Farber Cancer Institute, Boston, MA. (Tracking ID #132843)

BACKGROUND: Individuals at increased risk for breast or colorectal cancer based on their family history are recommended to initiate cancer-screening interventions at earlier ages than average risk individuals. Few studies have examined the comprehensiveness of the family medical history interview in primary care and the impact of incomplete information gathering on patient management.

METHODS: We compared pedigree information collected through a self-completed family medical history survey to that documented within the patients chart for 310 patients seen in two primary care practices in Boston. Information was collected for first- and second-degree relatives regarding breast, ovarian, and colorectal cancer. Using previously published models, we constructed risk scores for individual patients based on their family cancer history and categorized patients as either average risk or increased risk. We compared the percent of individuals identified as increased risk by survey to the percent identified through the chart review using McNemar's Test.

RESULTS: A total of 262 first- and second- degree relatives were identified as affected with either breast, ovarian, or colorectal cancer. Only 48% (125/262) of these affected relatives were identified through the chart review. For breast cancer, three percent of women were considered at increased risk based on chart review compared to 6.8% identified as increased risk through the survey ($P=0.002$). For colorectal cancer, 1.6% of participants were categorized as at increased risk based on chart review compared 8.4% identified through the survey ($p>0.0001$). Sixteen percent (1/6) of women who were potential candidates for BRCA1/2 testing had corresponding documentation of their increased risk pedigree within the medical record. A single individual met diagnostic criteria for HNPCC based on their self-reported family history collected by the survey. This individual had no corresponding documentation of their increased risk pedigree within the medical record. Although numbers were small, there was a statistically significant trend towards increased risk individuals with chart documentation of their pedigree to receive mammography screening ($P=0.0005$) and colorectal cancer screening ($P<0.0001$) at an earlier age when compared to increased risk individuals without their complete pedigree documented within the medical record or individuals with no family history recorded.

CONCLUSIONS: The family medical history exam is frequently incomplete and over half of individuals at increased risk for breast or colorectal cancer do not have documentation of this increased risk within their medical record. Inappropriate risk assessment may have implications for preventive services in cancer care.

THE IMPACT OF USING CORONARY HEART DISEASE RISK CALCULATION TOOLS WITH DIFFERENT PREDICTED OUTCOMES. S.L. Sheridan¹; G. Macdonald¹; M. Pignone¹; R. Simpson¹. ¹University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #134432)

BACKGROUND: Guidelines recommend that providers and patients use global coronary heart disease (CHD) risk as a foundation for primary CHD prevention, but provide little guidance on how risk should be determined. This leaves providers and patients to choose among risk calculation tools that differ on multiple factors, including predicted outcome. In this study, we estimate the impact of using Framingham risk calculation tools with different predicted outcomes on decision-making about CHD prevention.

METHODS: Two investigators independently reviewed a random sample of charts from new patient visits (June 2002 to June 2003) at one university internal medicine practice to obtain risk factor information on new patients who had no prior history of cardiovascular disease or other serious medical illness. Investigators then calculated a global CHD risk score for each patient using two Framingham CHD risk calculators: the National Cholesterol Education Program (NCEP) Risk Calculator (outcome: MI and CHD mortality) and the Heart-to-Heart (H2H) Calculator (outcome: MI, CHD mortality, and angina). Risk scores were subsequently applied to current risk-based guidelines for the primary prevention of CHD to determine the proportion of patients who would qualify for aspirin (i.e. CHD risk $>10\%$) and cholesterol lowering medication (i.e. high LDL by risk-based NCEP guidelines) using each risk calculator.

RESULTS: We reviewed the charts of 197 patients. Mean age was 57.9. 40% were male. 37% had hypertension, 42% abnormal lipids, and 11% diabetes. 24% smoked. 72% had documentation of all necessary information for global CHD risk calculation. For the 114 non-diabetic patients for whom risk could be calculated, the NCEP calculator designated 50% of patients as low risk ($<5\%$), 18% as moderate risk (5-9%), 27% as high risk (10-19%), and 4% as very high risk ($\geq 20\%$), while the H2H calculator designated 24% low risk, 32% moderate risk, 32% high risk, and 12% very high risk. Using the H2H Calculator, 14 additional non-diabetic patients would qualify for treatment with aspirin and 12 additional for treatment with cholesterol lowering medication. For those with diabetes, using the H2H calculator would qualify 3 fewer patients for aspirin, but result in an equal number of patients who qualify for cholesterol treatment.

CONCLUSIONS: The use of risk calculators with different predicted outcomes results in significantly different choices about primary CHD prevention, especially among non-diabetic patients.

THE QUALITY OLYMPICS: A SYSTEM-WIDE APPROACH TO IMPROVING DIABETES CARE. S.K. Ober¹; A. Caron²; M. Davidson³; D.C. Aron¹. ¹Case Western Reserve University, Cleveland, OH; ²Louis Stokes Cleveland Department of Veterans Affairs Medical Center, Cleveland, OH; ³Louis B. Stokes Cleveland VA Medical Center, Cleveland, OH. (Tracking ID #135851)

BACKGROUND: Unacceptable levels of adherence to recommended preventative care have prompted physician managers to search for interventions to improve outcomes, eg: clinical reminders, audit and feedback. Key components necessary for successful continuous quality improvement (CQI) include a strong and innovative quality leadership together with top hospital management and a current reporting system that can be accessed by local managers. The competition among Community-based Outpatient Clinics (CBOCs) termed the "CBOC Olympics," was designed to provide local managers with such a tool. This com-

petition provided broad dissemination of real-time data and was tied directly to network performance measures, as well as providing positive incentives to improve adherence and enhancing multidisciplinary teamwork. The goal of this project was to develop and evaluate a CQI intervention to improve adherence among CBOC primary care providers to clinical reminders for their patients with diabetes mellitus (DM).

METHODS: We created a inter-clinic competition to provide continuous feedback to clinicians and staff about every CBOCs level of adherence to clinical reminders. Specifically, we tracked 6 DM measures: eye exam, foot exam, lipid screening, hypertension screening, hemoglobin A1c monitoring, and hemoglobin A1c control. VHA network leadership determined target measures for satisfactory and exceptional performance. Data from primary care physicians from 13 VA CBOCs in Northeast Ohio were reviewed monthly from June 2002 through November 2004 to determine guideline adherence to diabetes clinical reminders. CBOCs were divided into 3 tiers in order to adjust for patient volume. Data was analyzed using Statistical Process Control (SPC). Site visits were performed monthly to all clinics by an interdisciplinary senior management team. Monthly Olympic winners from each tier were disseminated to all employees and were based solely on adherence to clinical reminders. Characteristics of successful CBOC Olympic winners are included.

RESULTS: Baseline results were high; every CBOC achieved satisfactory adherence in 2002. There was an overall trend of improvement; the average overall initial adherence to clinical reminders was 81% and increased to 83%. This increase has been maintained. There was also a decrease in overall variation to adherence rates. Between June 2002 and May 2003, the Upper and Lower Control Limits (UCL and LCL) were 0.955 and 0.693 respectively. This variation was reduced between July 2003 and October 2004: the UCL was 0.95 and the LCL was 0.71. However, there was considerable variation among the CBOCs. Of the 13 CBOCs, 7 showed significant, maintained improvement. The average adherence rate for the first and second years ranged from 76.6% to 85.0% and from 80.9% to 89.7% respectively. Examination of tier results indicates that the smallest and largest tiers made improvements, but the middle tiers adherence declined. Characteristics of successful CBOCs include enthusiastic physician managers, multiple opinion leaders, good interdisciplinary collaboration, and ownership of the data.

CONCLUSIONS: Successful implementation the CQI initiative depended on continuous reinforcement, clarity of objectives, and effective opinion leaders. The system as a whole reduced variation in adherence to clinical reminders, but the level of adherence desired was not achieved. This suggests that system redesign may be necessary to achieve optimal outcomes.

USE OF SCREENING MAMMOGRAPHY AMONG WOMEN ON DIALYSIS. L.C. Walter¹; K. Lindquist²; A. O'Hare¹; K.L. Johansen². ¹San Francisco VA Medical Center, San Francisco, CA; ²University of California, San Francisco, CA. (Tracking ID #131961)

BACKGROUND: There is much controversy over whether to screen women on dialysis with mammography. While life expectancy is reduced for women on dialysis, there is a subset of women with life expectancies greater than 4 years who may benefit from screening mammography. The objective of this study was to describe national patterns of mammography use in the dialysis population and determine if screening is targeted to the subset of women who live more than 4 years after starting dialysis.

METHODS: Using the U.S. Renal Data System we identified a cohort of 18,993 women aged ≥ 50 years (mean age = 70) who started dialysis in 1997 and lived ≥ 3 months. We tracked each woman over 4 years for the performance of screening mammography (Current Procedural Terminology code = 76092) or death. Four-year mortality and screening mammography rates were calculated for women according to advancing age and comorbid conditions. Mammography use was also analyzed as a time-dependent covariate to determine whether women who were less likely to die were targeted for screening.

RESULTS: 65% of women died within 4 years of starting dialysis (50% died of cardiovascular disease; 0.05% died of breast cancer). Over the same period, 20% of women had at least one screening mammogram. 11% of women who died received a mammogram compared to 38% of women who remained alive. Even women aged ≤ 60 years without cardiovascular disease ($n=1,711$) had a 4-year mortality rate of 38%; 35% received a mammogram. Women on the renal transplant list ($n=1,262$) had a mortality rate of 21%; 61% received a mammogram. At any point in time, women who had a screening mammogram within the past year were less likely to die than those who did not (hazards ratio = 0.47; 95% confidence interval 0.44-0.52).

CONCLUSIONS: Rates of screening mammography are appropriately low among women on dialysis given their high 4-year mortality rate, which is higher than that of average 95-year-old women in the United States. While screening is being targeted to women on dialysis who live longer, most women on dialysis are unlikely to benefit from screening mammography. Screening should be targeted primarily to women considered healthy enough for renal transplantation.

WHY DO WOMEN VISIT THE DOCTOR MORE OFTEN FOLLOWING A FALSE POSITIVE MAMMOGRAM? G.C. Lamb¹; R. Sparapani¹; J. Yauck¹; P. Laud¹; A.B. Nattinger¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #135497)

BACKGROUND: Following a false positive mammogram (FP), it has been demonstrated that previously healthy women may develop increased anxiety, concern about cancer and a change in health related behavior characterized by increased visits to the doctor. We hypothesized that the increase in physician visits results from a "labeling" phenomenon in which these individuals adopt a "sick" role associated with a perceived worsening of health and increased dis-

ability. To test this hypothesis we employed the Medical Expenditures Panel Survey (MEPS), a nationally representative in-depth survey of health care use, to determine the impact of a false positive mammogram on the patients' perception of their own physical and mental health.

METHODS: Information available in MEPS includes demographics, health conditions and use of medical services. The survey incorporates a number of questions regarding perceived health status, physical and social functioning, each rated yes/no or on a 5 point Likert scale. The survey uses an overlapping panel design in which each person identified undergoes 5 rounds of interviews, 6 months apart, over a 2 1/2 year period with a new panel of households/individuals incorporated each year. All women who received a bilateral screening mammogram in 1996-1998 were identified. Those with breast cancer and those with records that did not extend at least 6 mo prior to and 12 mo after the mammogram were excluded. An FP was defined as any mammogram followed by a breast related procedure within 3 mos or a repeat mammogram within 9 mo. All others were defined as a true negative (TN). Non breast related office visits for the 6 mo prior to and 12 mo following the index mammogram were tallied and compared using a multiple regression model adjusting for visit rate prior to the mammogram, age, race, perceived health status, census region and income. Values for perceived health status, perceived mental health status, limitations in ability to work, limitations in physical function, limitations in social function and days absent from work were compared before and after the mammogram using a similar multiple regression model adjusting for status of the variable prior to the mammogram, age, race, census region and income.

RESULTS: 1828 individuals met the inclusion criteria of whom 206 (11.3%) met the definition for a false positive. Average age for the group was 55 (range 17 - 90) and 86% were white. Prior to the index mammogram, the visit rate for both the FP and TN groups was 3.25 visits/yr. Following the mammogram, the mean visit rate for the FP group was 4.50 and for the TN group was 3.49 ($p < 0.0001$ in the adjusted model). Despite this significant increase in visits, there was no difference in the patients' ratings of their overall health (2.3 vs 2.2), mental health (1.8 vs 1.8), work limitations (14% vs 13%), social limitations (9% vs 6%), limitations in physical function (15% vs 14%) or absenteeism (1.5 days vs 2.5 days) following a false positive mammogram.

CONCLUSIONS: In this nationally representative sample, we confirm that women who have a false positive mammogram are substantially more likely to visit their physicians in the year following the test than women with a negative test. Yet, strikingly, this is not associated with a perception of worsening health or impairment. This argues against the hypothesis that false positive tests lead to a labeling phenomenon. Recent studies have suggested that women experiencing a false positive mammogram are grateful that they were tested and are more likely than their peers to have a repeat mammogram. Perhaps this suggests that women following a false positive mammogram have greater confidence in their physicians, are more likely to seek their help and are more compliant with recommendations. Further research is needed to elucidate these findings.

BIATRIC SURGERY OUTCOMES: A TAXONOMY OF PATIENT GOALS. S.L. Goff¹, R. Bell¹. ¹Yale University, New Haven, CT. (Tracking ID # 134498)

BACKGROUND: The number of bariatric surgical procedures has more than tripled in the past five years, requiring primary care physicians becoming increasingly involved in long term post-operative follow up. There is currently variation in the percent of excess weight lost and patient satisfaction with surgery, in part attributable to patient post-operative health behaviors and patient expectations. Explicit goal setting is a patient-centered method of enabling patients and members of their health care team to define and achieve successful outcomes. There is currently little data on bariatric patients' goals following surgery as described by the patients themselves. The objective of this study is to elicit and classify these goals.

METHODS: This qualitative study used patient focus groups to explore post-operative goals following bariatric surgery. Individuals were recruited from a support group open to the public. One focus group was held with women awaiting bariatric surgery, two with women who had undergone surgery, and one with men who had undergone surgery. A total of seventeen patients participated, and included participants who self-identified as black and white. All patients were within twenty-four months of surgery. A structured interview guide was used for each focus group, and sessions were recorded and professionally transcribed. The constant comparative method of qualitative data analysis was used, and thematic saturation was reached with the female groups. A structured interview was held with the surgeon associated with the support group to ascertain physician goals for patients. Transcripts were coded and categorization of identified goals was discussed by two authors.

RESULTS: Goals identified by patients had a limited number of attributes, resulting in a classification system for these goals. Goals were categorized by three axes: physical social and psychological. Each of these axes had three domains with a total of twenty-one associated themes. Examples of themes included a desire to reverse a related disease process, such as discontinuing diabetes medicine. A less expected theme was related to ambivalence about the role of food in one's life, with participants hoping to stop feeling distress about losing "food as a friend". Surgeon goals for patients were focused on the physical axis, including weight loss and resolution of co-morbid illnesses.

CONCLUSIONS: Our findings show that bariatric surgery patients have a broad spectrum of goals following surgery, and that the surgeon's goals for patients may be focused on a subset of these goals. This taxonomy may help inform development of a goal setting intervention that could then be assessed for impact on outcomes. Such a goal setting tool may assist the health care team in providing the necessary support for individual patients. Identifying goals and co-

ordinating needed services may best be carried out by the patient's primary care physician.

BARRIERS AND FACILITATORS OF PROSTATE CANCER SCREENING DISCUSSION. C.E. Guerra¹, S.E. Jacobs¹, J.A. Shea², E.A. Ohr¹, R.C. Hornik¹, D. Frosch¹, J.H. Holmes¹. ¹University of Pennsylvania, Philadelphia, PA; ²Society of Directors of Research in Medical Education, Philadelphia, PA. (Tracking ID # 134179)

BACKGROUND: Prostate cancer screening (PCS) is controversial. Ideally, patients should be aware of the potential risks and benefits related testing. This study was conducted to assess whether primary care physicians discuss PCS and to determine the barriers and facilitators of PCS discussions.

METHODS: Qualitative study involving in-depth, semi-structured interviews with 17 purposively-sampled, community and academic based primary care physicians. In addition, actual barriers of and facilitators to PCS discussions were evaluated using chart-stimulated recall. Analysis was performed using consensus conferences based on Grounded Theory technique.

RESULTS: All 17 of the participating physicians reported they discussed PCS with their patients. Six physicians reported that they sometimes conduct testing without discussion because of lack of time or patient inability to understand discussion. Physicians identified the following barriers to PCS discussion: patient (old age, low life expectancy, comorbidity, inability or unwillingness to give informed consent), physician (forgetfulness, negative attitudes about PSA), system (lack of time and reimbursement for counseling, visit type other than physical, conflicted guidelines, lack of reminder system), and test barriers (complexity of discussion about PSA). Reported facilitators to PCS discussion were: patient (patient requests screening or has interest in preventive health, high education or ability to understand a discussion, African-American race, family hx or symptoms of prostate cancer, friend or family member screened for or diagnosed with prostate cancer), physician (positive beliefs in screening, fear of liability, review of labs), system (annual physicals or dedicated time for health maintenance, public service initiatives and reminder systems), and test facilitators (ease of blood draw for PSA). Using chart-stimulated recall, 124 encounters were reviewed with physicians of which 44 met the exclusion criteria [history or symptoms of prostate cancer (13), age (12), BPH (9), no memory of visit (2), prostatitis (2), other (6)]. PCS was not discussed in 54 of the 80 (68%) encounters reviewed. Of the 54 encounters without a discussion, 32 had documentation of screening within the previous year and thus were not due for a discussion. In the remaining 22 encounters without a discussion in the preceding year, screening status was unclear in 7 subjects. In the 15 subjects where screening was clearly overdue, the reasons for not discussing screening were: acute or problem focused visit (5), forgetfulness (4), comorbidity (4), low life expectancy (3), lack of time (2), age (1), low education (1), and assuming patient would decline (1). In the 26 encounters where PCS was discussed, the facilitators for discussion were: annual exam (8), review of lab work (8), patient request (6), extra time (4), previously informing the patient they would be due for screening at the index visit (4), high education (2), family hx (2), monitoring for medication adverse effect (testosterone, immunosuppressives) (2), history of elevated PSA (2), reminder (1), African American race (1), and other (2). In 3 of these encounters, the extent of the discussion consisted of the doctor stating that a PSA blood test was included in ordered blood tests and in 5 encounters there was only a post-screening statement made by the physician about results of the PSA.

CONCLUSIONS: PCS was absent in approximately 19% of the encounters reviewed. When present, PCS was often limited in length and quality and sometimes did not occur before PSA testing. Important barriers to discussion are lack of dedicated time for health maintenance, physician forgetfulness and patient characteristics. Future research should be aimed at determining how educational and decision support interventions can be used to involve more patients in decisions related to PCS.

CARING TOGETHER: A QUALITATIVE STUDY OF THE CHALLENGES AND REWARDS OF FAMILY CAREGIVING IN PALLIATIVE CARE. J. Hauser¹, T. Bandy¹, L. Emanuel¹. ¹Northwestern University, Chicago, IL. (Tracking ID # 136296)

BACKGROUND: The magnitude of care given to patients by their family members is staggering in size and cost: recent estimates identify up to 27.6 million family caregivers who provide as much as \$196 billion in care when indirect costs are considered. Nonetheless, the needs and coping skills of family caregivers remain poorly understood and poorly supported. We undertook a qualitative study with current and recently bereaved caregivers in a palliative care program to determine the challenges and rewards they face and the coping strategies they utilize.

METHODS: We used a qualitative design that included in-depth interviews of 21 current and recently bereaved caregivers in palliative care. The interview protocol covered the caregivers' story of their loved one's illness, the difficulties or challenges they faced in caregiving, the fulfilling or rewarding aspects of caregiving and the coping strategies they employed. Interviews were transcribed verbatim and methods of grounded theory and iterative coding were used to identify salient themes.

RESULTS: The sample of 21 caregivers ranged in age from their early 20s to 70s. The length of time that they spent caring for their relatives ranged from several months to more than a decade. The diagnoses of the patients cared for were multiple cancers, heart disease, dementia and AIDS. The areas of challenge that caregivers identified included practical difficulties and emotional challenges. The practical aspects were a lack of training in the tasks of caregiving (such as medication administration, moving and bathing of patients) and difficulty coordinating among multiple family and professional caregivers. The emotional

aspects were feelings of "helplessness" and a lack of control over the illness. Rewarding aspects of caregiving divided into these two overall domains of practical activities and more emotionally-laden ones. For the former, caregivers found fulfillment when they were able to perform basic ADLs for relatives and ease their symptoms. More importantly, subjects identified intense moments of presence and closeness with their relative and a feeling of "reciprocity" of care. They also identified a process of reliving moments from before the illness as a fulfilling part of the experience of care. Caregivers employed a variety of coping strategies such as turning to spirituality and prayer as well organized and individual expression, that included both verbal and written expression in the form of lists and journals.

CONCLUSIONS: Caregivers identified distinct practical and emotional domains of caregiving as challenging and rewarding. In the emotional domain, both rewarding and challenging aspects of caregiving centered on the relationship with the person being cared for: a deep closeness that some caregivers felt and an intense helplessness at witnessing that disappear. In the practical domain, since many of the positive aspects of caregiving involve mastery of tasks, specific caregiver training might address these. In the area of coping, spirituality was vital for many of these caregivers as was expression in the setting of counseling and support groups. The less formalized means of expression that caregivers identified as helpful, such as informal peer support, using journals or making lists suggests another intervention modeling or training caregivers in expressive and narrative techniques. The next stage of this project will explore the feasibility of such an intervention.

GENDER DIFFERENCES IN RESIDENTS' STRATEGIES FOR ESTABLISHING MENTORING RELATIONSHIPS. M. Cunnane¹; M.A. Mcneil¹; J. Lakoski¹; J. Chang¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133087)

BACKGROUND: Mentoring relationships are important for residents' career development. These complex personal and professional interactions may be influenced by gender; female residents identify fewer and less successful mentoring relationships than male residents. The purpose of our study was to explore gender-specific differences in residents' individual strategies for establishing mentoring relationships.

METHODS: We conducted 2 focus groups with a total of 21 residents. Focus groups were organized according to gender; one focus group was composed of 12 female residents, and the other focus group was composed of 9 male residents. Participants represented a variety of specialties (Pediatrics, Internal Medicine, General Surgery, Family Practice, and Obstetrics/Gynecology) and training levels (PGY-1 to PGY-7). Discussions were audiotaped and transcribed. Two investigators coded the transcripts separately and then compared coding. We developed a final coding scheme and used it to identify emerging themes regarding assigned mentoring, finding a mentor, and barriers to mentoring.

RESULTS: Male and female residents had participated in mentoring relationships by being assigned a mentor through their residency program, or by finding a mentor on their own. Residents of both genders perceived assigned mentoring as ineffective, short-lived, and forced. While male and female residents agreed that finding one's own mentor was best, their approaches for doing so differed. Men's strategies were more numerous than women's and included identifying mentors through: 1) research ("So I would be able to spend time doing research and probably cultivate the relationship based on that."); 2) similar interests ("If you're in some sports you can have a mentor do the same activities at the same time."); 3) friendship ("The more people you're more friendly with you're just surrounded with a pool of eligible mentors."); 4) networking ("And sometimes the role of the mentor is just teaming you up with someone who is a better mentor assigned things can work as a launching block."). Female strategies were limited and included identifying mentors through: 1) word of mouth ("One of the upper level residents will recommend somebody to you."); 2) work experiences ("Just you're put together in a situation on a team or in a location."). Women were much more passive ("They asked me to choose someone."), than men ("Seize the moment.") in their approaches to finding a mentor. Barriers to mentoring cited by all participants included lack of time and lack of institutional support.

CONCLUSIONS: Residents of both genders feel that assigned mentoring is ineffective; successful relationships are "found." Women, in contrast to men, lack strategies and initiative for finding a mentor; this difference may translate into fewer mentoring relationships for women. Institutional resources should be invested in developing programs that promote strategies for finding effective mentors, with special attention paid to gender differences.

MAMMOGRAPHY SCREENING DECISIONS AMONG ELDERLY WOMEN. M.A. Schonberg¹; R.A. Ramanan¹; E.R. Marcantonio¹. ¹Beth Israel Deaconess Medical Center, Boston, MA. (Tracking ID #133676)

BACKGROUND: Despite uncertain benefit, many women over age 80 receive screening mammography. Little is known about physician counseling and decision-making of elderly women around mammography screening.

METHODS: We conducted semi-structured individual interviews with a purposeful sample of 23 women aged 80 and older (mean age 86 ± 4 years) without a history of breast cancer who received primary care at a large academic medical center (13 women had undergone mammography in the past 2 years, 10 had not). We also interviewed 16 physicians who care for elderly women at the same center (internists [6], geriatricians [5], gynecologists [3], mammographers [2]). We asked patients to describe factors influencing their decision whether or not to have a mammogram in the past 2 years. We probed patients about the influence of physicians, family, friends, and their health on the decision. We asked providers to identify factors that influenced elderly women's mammography de-

cisions and to describe their counseling about mammography screening to elderly women. Interviews were audio-taped and transcribed verbatim. Data were coded and analyzed independently by the 3 investigators. Participants were interviewed until no new major themes emerged.

RESULTS: Three types of elderly patients were identified: 1) women who were certain they wanted screening mammograms; 2) women who strongly opposed getting screening mammograms; and 3) women without a preference who followed their physician's recommendation. Both patients and providers identified the importance of physician influence, patient factors (preferences, habit of screening, personal/family history of breast disease, and perception of risk), system factors (mailed reminder cards, ease of getting a mammogram, and available treatments) and social influences (family or friend's experience or recommendations). In general, patients had limited knowledge of risks of mammography and their risk of breast cancer. Although several providers felt that patient's age, health, and functional status affected their screening recommendations, few patients felt that these factors influenced their decisions. Many providers recommend mammograms to all women over 80 years unless the patient's life expectancy is extremely compromised. Physicians reported that discussions about stopping screening are challenging because they require estimation of life expectancy, elicitation of end of life care preferences, and time during a visit. A long-standing doctor-patient relationship and/or patient's trust in her physician facilitated these discussions. Physicians requested more guidance and data about the risks and benefits of mammography in this age group.

CONCLUSIONS: Some elderly women have strong opinions about whether or not to get screening mammography while others are more influenced by their physicians. Providers are generally enthusiastic about mammography screening for elderly women. Discussing stopping screening is difficult for providers. Better data about the outcomes of breast cancer screening and treatment in this age group could inform provider recommendations and lead to more rational use of mammography.

OLDER ADULTS' RELUCTANCE TO USE ANTIDEPRESSANTS: A QUALITATIVE STUDY. J.L. Givens¹; K. Knott¹; K. Ruckdeschel¹; C. Zubritsky¹; C. Datto¹; D. Oslin¹; F.K. Barg¹. ¹University of Pennsylvania, Philadelphia, PA. (Tracking ID #136168)

BACKGROUND: Although depression is highly prevalent in elderly primary care patients and pharmacologic treatment for depression is efficacious, older adults frequently do not receive treatment for this disease. Because patient level factors may play a role in the underuse of effective treatments, this qualitative study in depressed, community-dwelling elders investigates reasons for resisting pharmacological treatment for depression.

METHODS: Participants were older adults with depression who participated in one of two quantitative, randomized treatment studies: The Prevention of Suicide in Primary Care Elderly: Collaborative Trial (PROSPECT) or the Primary Care Research in Substance Abuse and Mental Health for the Elderly (PRISM-E). Of 254 participants recruited from primary care clinics affiliated with the University of Pennsylvania, a purposive sample of 68 elders participated in semi-structured interviews in their homes. Interviews were audiotaped, transcribed, and entered into QSR N6 for coding and analysis. A multidisciplinary team of investigators coded the transcripts, and identified key features of narratives expressing reluctance or refusal to use pharmacologic treatment for depression.

RESULTS: Participants were mostly women (69%) with a mean age of 75 and equally distributed between African Americans and whites. Four themes characterizing resistance to pharmacologic treatment of depression emerged: (1) **FEAR OF DEPENDENCE ON MEDICATION FOR DEPRESSION:** Many expressed a fear of needing to take antidepressants for the rest of their lives or of becoming addicted. (2) **CONCERN THAT MEDICATIONS FOR DEPRESSION WILL CAUSE UNNATURAL HAPPINESS:** Some thought they needed to feel sadness, to "face reality", and did not want to mute this capacity. Others were afraid of being unnaturally happy. (3) **RESISTANCE TO MEDICALIZATION OF SYMPTOMS:** Participants who had suffered the loss of a loved one were resistant to treating symptoms of sadness, while others rejected medications because they ameliorated symptoms but did not address the underlying cause of depression. (4) **PRIOR NEGATIVE EXPERIENCES WITH PSYCHIATRIC MEDICATIONS:** Some participants rejected antidepressants because they had been treated in the past with other psychiatric medications, including tranquilizers, and were afraid of being sedated.

CONCLUSIONS: These qualitative interviews offer important insights into why depressed elders resist pharmacological treatments for depression. Older patients in our study expressed concerns that seem to reflect misconceptions about how antidepressants work: fear of addiction; fear of unnatural happiness; concern about the inability to feel grief and sadness; and fear of sedating side effects. These results highlight the need for patient education and dialogue regarding the characteristics of current antidepressant therapy.

OPERATIONALIZING PROFESSIONALISM IN MEDICINE. G. Makoul¹; M.M. Green¹; A. Zick¹. ¹Northwestern University, Chicago, IL. (Tracking ID #136289)

BACKGROUND: The competency of professionalism has gained increasing attention throughout the continuum of medical education and practice. Despite a large amount of published work in the area, there has been inadequate delineation and validation of behaviors associated with professionalism. The overall goal of this project is to systematically develop behaviorally-based measures of professionalism that are relevant across specialties.

METHODS: We conducted a series of 17 focus groups designed to explore behavioral signs of professionalism in medicine: a total of 15 groups with patients, inpatient nurses, outpatient nurses, resident physicians, and attending physi-

cians recruited in each of three subspecialties (internal medicine, general surgery, psychiatry), as well as two mixed groups of patients, one conducted in a rural setting. We convened a national advisory board of experts on professionalism in medicine to help guide this process, but started with a blank slate. In other words, while we were aware of previously identified definitions and domains of professionalism, we did not use them because we wanted to avoid biasing the process or outcome of the focus groups. Each focus group had an average of 7 participants as well as two trained facilitators. Discussions were videotaped to facilitate latent content analysis and constant comparative analysis; all three authors participated in a debriefing session after each group. Items generated, whether by patients, nurses or physicians, were included for discussion in all subsequent groups. We applied to each item a measure of reading-difficulty termed Lexile, and refined wording to meet the level at which people can read 8th grade texts with more than 80% comprehension. We then conducted pilot national telephone surveys before embarking on a national survey of 400 adults to measure the importance Americans attach to each item as a sign of professionalism, as well as to determine whether patients think they can know if a doctor exhibited the behaviors.

RESULTS: The focus groups generated a total of 71 behaviorally-based items ranging from the very basic (e.g., good hygiene) to communication-oriented behaviors, accountability, and service to the profession. Some of the domains outlined in prominent publications (e.g., The Physician Charter) were not evident in the set of behaviors that emerged in the focus groups. At least 75% of participants in the pilot surveys indicated that they could assess 40 of the behaviors, which were subsequently incorporated into the national telephone survey. We have completed 148 of the 400 national surveys, and it appears that approximately 15 items will meet the criteria set for inclusion in a patient-based assessment of professionalism (i.e., >80% deeming them very important and >70% indicating that a patient could gauge the behavior). Several of these highly rated items reflect communication (e.g., listens carefully, answers questions from patients and families, communicates clearly and effectively). About 10 additional items are likely to be dropped due to redundancy; another set of items appears to be well suited for peer-assessment and self-assessment.

CONCLUSIONS: It is possible to define professionalism in terms of tangible behaviors. Our next steps are to: (1) conduct focus groups with parents, nurses, and doctors of pediatric patients to learn if any specialty-specific items should be added for use in this context; (2) conduct surveys of American physicians to determine the value they attach to each behavior, retaining for peer-assessment and self-assessment those items deemed most important.

RELATIONSHIPS AS CULTURAL COMPETENCE: A QUALITATIVE STUDY OF COLON RECTAL CANCER SCREENING RECOMMENDATION STRATEGIES. N.J. Burke¹; C.P. Somkin²; J.M. Walsh¹; R. Otero-Sabogal¹; D. Cabral¹; G. Gao³; G. Joseph¹; R. Pasick¹. ¹University of California, San Francisco, San Francisco, CA; ²Kaiser Permanente Division of Research, Oakland, CA; ³San Jose State University, San Jose, CA. (Tracking ID #135278)

BACKGROUND: Rates of CRC screening are lowest among the underserved, those who are non-white or non-English speaking and/or of low socio-economic status. CRC mortality is highest among African Americans and rates of late stage diagnosis are higher for African Americans, Asians, and Hispanics than for non-Hispanic Whites. CRC screening depends on both clinician referral and patient follow-up. Effective communication is pivotal, and may require specific strategies when physician and patient are of different cultural backgrounds. One purpose of this study is to observe patient-provider communication across cultures to identify elements of strategies effective in promoting colorectal screening.

METHODS: Thirty-three video-taped direct observations of CRC recommendation discussions with ethnically discordant physician-patient dyads in an urban community clinic and urban HMO were conducted. Post-visit video recall interviews were conducted separately with physicians and patients. Observations and interviews were transcribed, and all qualitative data were coded and reconciled for trustworthiness.

RESULTS: Physicians employ a variety of culturally appropriate relationship-building techniques (story-telling, personal talk, shared experience) in order to positively influence patients' intention to get screened. Three cases illustrate this point. First, we examine a White physician who has been working for thirty years in an urban community clinic with primarily African-American patients. He adopts the cultural idiom of storytelling prevalent in his community. He listens to his patients' stories about cancer and screening tests, and then counters them with his own stories about his personal experience with screening and that of his other patients. This doctor's patients empathize with his personal experience and respond positively to his sharing of stories about other patients like them. The second case involves a White female physician at an urban HMO whose panel includes many monolingual Spanish speaking Latinos. This doctor adopts a cultural style of interaction involving physical proximity and extended eye contact in which she addresses personal and family concerns; draws upon knowledge of family dynamics to influence patient health behavior; and spends time explaining system navigation issues. As a result, her patients describe their relationship with her as personal, comfortable, and trusting: "like family." Our third physician is a young White man at an urban HMO whose panel includes many African American patients. His strategy allows his patients to direct the interaction; he listens to them and then integrates the experiences patients' have shared into an appropriate CRC recommendation. Given the doctor's youth, many of his relationships with his patients are relatively new. This approach allows him to learn about his patients, and they describe his style as caring, respectful, and humanizing.

CONCLUSIONS: Even in time-pressured environments, physicians can cultivate culturally appropriate communication strategies resulting in strong, trusting relationships with patients and subsequent patient intended compliance.

Quality of relationship, an essential element of cultural competence, facilitates effective communication about CRC screening.

TREATMENT DECISION-MAKING IN EARLY-STAGE PROSTATE CANCER. T.D. Denberg¹; T.V. Melhado¹; F.J. Kim¹; J.F. Steiner². ¹University of Colorado Health Sciences Center, Denver, CO; ²University of Colorado Health Sciences Center, Aurora, CO. (Tracking ID #134063)

BACKGROUND: In early-stage prostate cancer (PCa), treatment decision-making is complicated by informational complexity and multiple treatment options that have significant side effects and uncertain mortality benefits. Nonetheless, maximizing patient involvement in PCa decision-making has become a normative goal. Little is understood, however, about how men and their spouses experience the decision-making process following diagnosis.

METHODS: We carried out in-depth, semi-structured individual interviews of 22 men with recently-diagnosed early-stage PCa and (separately) 8 spouses. Inclusion criteria included men <75 years with T1 or T2 disease, Gleason grade >5, and PSA <20. Interviews were transcribed verbatim and thematic content analysis centered on knowledge of PCa and treatment options, emotional repercussions, decisional preferences, social supports, and interactions with urologists.

RESULTS: Several themes summarized in the Table have important implications for the quality of patient-involvement in treatment decision-making. Examples are given in the text.

CONCLUSIONS: Unlike previous studies of treatment decision-making in early-stage PCa, we interviewed patients and their spouses individually during the actual decision-making period. Traditional decision aids meant to enhance the quality of decision-making emphasize objective facts and numerical risk information. In so doing, they may give short shrift to the (often profound) emotional and social influences on decision-making that were identified in this study. For men with early-stage prostate cancer and their significant others, enhanced decision tools and systems of care are needed to improve treatment decision-making in early-stage PCa.

TABLE. Treatment decision-making themes - early-stage prostate cancer

Prostatectomy = "getting the whole thing out", observable, usually considered best option for cure.

Radiation = "killing the prostate cancer", reasonable, but cancer could remain hidden inside.

Watchful waiting = "doing nothing", unacceptable.

Poor/vague understanding of treatment side effects is common.

Challenged by uncertainty, some men strive to make quick, firm decisions, downplaying possible treatment side effects.

Emotional reaction at great odds with intellectual understanding of PCa.

Knowledge of others' cancer experiences (not always PCa) strongly influences men's ideas about their own prognoses.

Many men who express preferences to make their own decisions desire a sense of control over their situation; oftentimes, they actually want urologists to make decisions on their behalf.

Men often attribute symptoms of BPH to PCa.

Spouses often take the lead in seeking out information about PCa, sharing it very selectively with their husbands.

Many men are ambivalent about accepting decision-making advice from spouses.

5 YEARS LATER, NO CHANGE IN THE WEATHER: GENDER DIFFERENCES IN PHYSICIAN WORKLIFE AND ORGANIZATIONAL CLIMATE. B.A. Homer-Ibler¹; J.E. McMurray²; M. Mundi²; M. Linzer²; E. Riska³. ¹University of Wisconsin-Madison, Brookfield, WI; ²University of Wisconsin-Madison, Madison, WI; ³University of Helsinki, Helsinki, Finland. (Tracking ID #134674)

BACKGROUND: In 2000, we reported on a national survey showing substantial gender differences in physician worklife (JGIM 2000;15:372-80). 5 years later, we sought to determine whether the steady influx of women into Primary Care fields has changed the climate in any way, such that women physicians' stress or burnout is reduced.

METHODS: The MEMO study is a longitudinal cohort study of primary care physicians, their offices and their patients in 3 different states. New surveys assessed women and men physicians. Data collection tools gathered organizational information about culture. Clinic manager observations of physicians and clinic atmosphere corroborated physician reports in ambulatory practices. To assess gender differences in work life, we analyzed physician survey data on climate, ambulatory practice patterns, and organizational culture. For these surveys, we used modified scales from Kravetski's work on organizational culture, a single item for self-rating burnout, and 4 and 5 item job stress and satisfaction scales ranked from 1=low levels to 5=high levels. For an organizational assessment by clinic managers, we used a new tool developed by MEMO investigators.

RESULTS: We studied 420 general internists and family practice physicians (188 women=44%). These 420 represented 61% of those surveyed, and 84% of our target sample of 500. Clinics in which women physicians work cared for more uninsured or Medicaid/Medical Assistance patients (37%) compared to those where men worked (30%, p=0.04). As in our prior studies, women reported more time pressure in office visits than men with women asking for 30% more than the time allotted for visits; men requested only 19% more than time

allotted. Women also noted significantly more stress (3.45 vs. 3.23; $p < 0.003$) and burnout (2.32 vs 2.07; $p < 0.001$). Work control, measured with 14 different items, was on average significantly less for women (2.35 vs. 2.69; $p < 0.001$). Organizationally, women acknowledged less trust (2.47 vs. 2.69; $p < 0.001$) and less alignment between their values and those of the administration (1.97 vs 2.36; $p < 0.001$). In regression analyses, controlling for age, years in practice, and full time/part time status, the impact of stress for women on outcomes such as organizational climate, pace of the office and burnout was nearly twice that of men. In those same regression analyses, stress was no longer of any statistical significance for men in areas of trust, cohesion, and alignment, whereas for women, stress remained a notable variable.

CONCLUSIONS: Despite widespread awareness of prior similar data, workplace and climate for women physicians continues to be lower compared to that of male colleagues. Lack of alignment with their organizations may explain the lower perceptions of work control for women. The experience of stress is far more significant for women physicians than for men. Women physicians and their patients remain at higher risk for the side effects of stress and poor working climate.

A POPULATION-BASED STUDY OF PHYSICIAN RESPONSE TO POORLY CONTROLLED HYPERTENSION, DYSLIPIDEMIA, AND DIABETES MELLITUS: THE KAISER PERMANENTE 3-D STUDY. N. Rodondi¹; T.Y. Peng²; D.C. Bauer¹; S. Tang³; D. Pettitt⁴; E.A. Kerr⁵; J.V. Selby⁶. ¹University of California, San Francisco, San Francisco, CA; ²Kaiser Permanente of Northern California, Oakland, CA; ³Pfizer Global Outcomes Research, New York, NY; ⁴Pfizer Outcomes Research, New York, NY; ⁵University of Michigan, Ann Arbor, MI; ⁶Kaiser Permanente Division of Research, Oakland, CA. (Tracking ID #132672)

BACKGROUND: Recent reports continue to show relatively poor levels of control for blood pressure (BP) in hypertension, LDL-cholesterol (LDL-C) levels in patients with dyslipidemia, and hemoglobin A1c (HbA1c) in diabetes. However, assessing only the proportions with poor control without evaluating whether physicians responded appropriately to poor control does not necessarily reflect quality of clinical care.

METHODS: We identified 253,238 adult patients from Kaiser Permanente Northern California with diabetes, hypertension or dyslipidemia and poor control. Poor control was defined as HbA1c values $\geq 8.0\%$; BP levels of $> 160/100$ mm Hg ($> 140/90$ in persons with diabetes or prior target organ disease); and LDL-c > 190 mg/dL (or risk-specific definitions if high cardiovascular risk). During 2-year observation, we determined the proportion of these patients who had appropriate physician response, defined as a change in pharmacotherapy or return to control without a change in pharmacotherapy within 6 months. We assessed patient factors associated with appropriate physician action using multivariate logistic regression.

RESULTS: Among poorly controlled patients, 64% experienced a change in therapy for poorly controlled systolic BP, 71% for diastolic BP, 56% for LDL-cholesterol, and 66% for HbA1c. Most frequent changes were increases in number of drug classes (70-84%) and increased dosage (15-40%). About 11% of those with poorly controlled BP, but only 3% of those with elevated LDL-C or HbA1c, returned to control without a change in pharmacotherapy. Co-occurrence of the three conditions, higher baseline values, and target organ damage (except for diabetes) were significant predictors of appropriate physician action in response to poor control. In general, non-white patients (African-American, Asian, Latino) were significantly more likely to have appropriate physician actions for BP control, but significantly less likely than whites to have appropriate actions for HbA1c control. Latino and Asian patients, but not African Americans were more likely than whites to have appropriate responses for LDL-C control.

CONCLUSIONS: Physicians respond more quickly to poor risk factor control in the presence of higher cardiovascular disease risk. Room for improvement remains in rates of appropriate physician action to high risk levels. Evaluating quality by examining appropriateness of clinical actions represents an additional opportunity for improving quality and decreasing cardiovascular risk.

A PREDICTION RULE TO IDENTIFY LOW-RISK PATIENTS WITH PULMONARY EMBOLISM. D. Aujesky¹; D.S. Obrosky¹; R.A. Stone²; T.E. Auble¹; M.J. Fine². ¹University of Pittsburgh, Pittsburgh, PA; ²VA Pittsburgh Healthcare System, Pittsburgh, PA. (Tracking ID #132921)

BACKGROUND: Although outpatient care with low-molecular-weight heparin is effective and safe for many patients with pulmonary embolism (PE), physicians are reluctant to treat such patients in an outpatient setting, in part due to uncertainty in quantifying severity of illness at presentation. Our aim was to develop a clinical prediction rule that accurately identifies low-risk patients with PE as potential candidates for outpatient care.

METHODS: This retrospective cohort study used administrative and clinical data, collected by the Pennsylvania Health Care Cost Containment Council and MediQual Atlas for 15,531 inpatients discharged with an ICD-9-CM diagnosis of PE from 186 Pennsylvania acute care hospitals between January 2000 and November 2002. Baseline data included demographics and over 30 potential clinical predictors of short-term mortality. The primary outcome was 30-day all-cause mortality ascertained from the National Death Index; secondary outcomes were nonfatal cardiogenic shock or cardiorespiratory arrest, and major bleeding during the index hospitalization based on ICD-9-CM codes. We randomly split our study cohort into a 70% derivation ($n=10,354$) and 30% validation ($n=5,177$) sample. We derived our rule using a classification tree model to predict 30-day mortality. We developed models with and without laboratory parameters that identified a low-risk group with a membership of $\geq 20\%$ of the derivation sample and a 30-day mortality of $< 1\%$. We internally validated our

rule by comparing the proportion of low-risk patients in the derivation and validation samples who died or had an adverse outcome.

RESULTS: Our final model consisted of 10 routinely available patient factors: age ≥ 70 years, 5 comorbid conditions (cancer, heart failure, chronic lung and renal disease, cerebrovascular disease), and 4 clinical findings (altered mental status, pulse $> 100/\text{min.}$, systolic blood pressure < 100 mmHg, and oxygen saturation $< 90\%$). Patients with none of these factors were defined as low-risk by our rule. Overall, our rule classified 21.6% of patients as low-risk in both the derivation and validation sample ($P=0.86$). Low-risk patients had a 30-day mortality of 0.6% (95% CI: 0.3-1.0%) in the derivation sample and 1.5% (95% CI: 0.9-2.4%) in the validation sample ($P=0.01$). The rates of nonfatal cardiogenic shock/cardiorespiratory arrest and major bleeding among low-risk patients were $< 1\%$ in the derivation and validation samples.

CONCLUSIONS: This simple prediction rule identified one fifth of patients at low-risk of 30-day death and other adverse outcomes. Independent external validation is important prior to the clinical application of this rule.

A RANDOMIZED CONTROLLED TRIAL OF DIFFERENT METHODS OF ASSESSING STUDY-RELATED SIDE EFFECTS. S. Bent¹; A. Padula¹; A.L. Avins¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #135190)

BACKGROUND: Prior studies have suggested that the method of ascertaining adverse-event information in clinical trials may affect estimates of the frequency of such events, but this assertion has never been tested in a randomized trial. We sought to determine whether three different methods of questioning patients about adverse events in a clinical trial would lead to different estimates of the frequency of adverse events.

METHODS: We randomly assigned 214 men who were undergoing a one-month, single-blind, placebo-run in period during an existing clinical trial to three different methods of adverse event assessment at the end of the run-in period in written statements: A) Open-ended, general question: "Did you have any significant medical problem since the last study visit?", B) Open-ended, specific question: "Since the last study visit, have you limited your usual daily activities for more than one day because of a medical problem?", or C) Checklist of 53 common adverse events, asking patients, "Since the last study visit, have you experienced any of the following?" All study participants were enrolled in the Saw Palmetto Treatment for Enlarged Prostates (STEP) study, an NIH-sponsored randomized placebo-controlled clinical trial evaluating the efficacy of saw palmetto in men over the age of 50 with moderate-to-severe symptoms of benign prostatic hyperplasia. The frequency of adverse events reported between groups was compared using ANOVA.

RESULTS: All 214 patients completed the study, and there were no significant differences between the three groups in baseline characteristics. Patients assigned to the checklist (group C) reported a total of 238 adverse events compared with only 11 in group A and 14 in group B. The percentage of patients reporting any adverse event was also much higher in patients assigned to the checklist (77%) compared to groups A (14%) and B (13%). The ten most commonly reported adverse events are shown in the table (all p -values < 0.005).

CONCLUSIONS: Different methods of adverse event collection lead to large differences in the reported rates of adverse events in clinical trials. These varying rates of reported adverse events may affect decisions about drug approval and utilization, and reduce the validity of comparisons between side-effect profiles of drugs. Efforts should be made to standardize the process of adverse event ascertainment in clinical trials.

Table: Ten Most-Reported Adverse Events (n (%))

Adverse Event	A	B	C
Back pain	0 (0)	5 (7)	13 (18)
URI, Flu-like symptoms	1 (1)	2 (3)	15 (21)
Dry Mouth	0 (0)	0 (0)	13 (18)
Fatigue	0 (0)	0 (0)	12 (16)
Cough	0 (0)	0 (0)	11 (15)
Nocturia	0 (0)	0 (0)	11 (15)
Joint pain/swelling	0 (0)	1 (1)	9 (12)
Impotency	0 (0)	0 (0)	10 (14)
Headache	0 (0)	0 (0)	10 (13)
Heartburn/indigestion	1 (1)	0 (0)	9 (12)

A RANDOMIZED, CONTROLLED TRIAL OF A VISUAL MEDICATION SCHEDULE TO IMPROVE ANTICOAGULATION CONTROL. E.L. Machtinger¹; F. Wang¹; L. Chen¹; M. Rodriguez¹; S. Wu¹; D. Schillinger¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #133186)

BACKGROUND: Miscommunication between clinicians and patients is common and may lead to medication-related errors and poor clinical outcomes. The few studies evaluating the effect of communication tools on medication adherence and treatment outcomes have had mixed results, suggesting that either they are ineffective or that the types of patients who may benefit from them have not been identified.

METHODS: We performed a non-blinded randomized controlled trial among 142 chronic warfarin users cared for in an anticoagulant clinic whose most recent International Normalized Ratio (INR) values were outside the therapeutic range. After each scheduled visit at the clinic over 90 days, intervention patients received an updated visual medication schedule, consisting of digitized images

of the weekly warfarin regimen; control patients received standard treatment. Just prior to patient's baseline visit, we recorded patient and clinician reports of the prescribed warfarin regimen to identify patients as "discordant" vs. "concordant" with the clinician. We examined the effect of visual medication schedules on time to achieving target anticoagulation control, using survival models, and examined whether these effects varied with baseline clinician-patient discordance.

RESULTS: At baseline, warfarin regimen discordance between clinician and patient was common in both intervention and control groups (48 percent vs. 52 percent, $P=.70$). Intervention patients achieved anticoagulation control more rapidly than control patients (median 28 days vs. 42 days, hazard ratio 1.42; 95% CI 0.98, 2.05; $P=.06$). The benefit of the intervention was significant among patients with baseline regimen discordance (28 days vs. 49 days, hazard ratio 1.95; 95% CI 1.09, 3.48; $P=.02$) but not among patients with baseline regimen concordance (28 days vs. 35 days, $P=.70$).

CONCLUSIONS: Among patients whose understanding of their warfarin regimen is discordant with that of their providers, receipt of an updated visual medication schedule at each clinic visit improves subsequent anticoagulation control. These results suggest that providing a visual communication tool to those patients with medication discordance may enhance medication safety and efficacy.

ACCURATELY IDENTIFYING PATIENTS LOYAL TO PROVIDERS IN A PRIMARY CARE PRACTICE NETWORK: IMPLICATIONS FOR QUALITY ASSESSMENT AND IMPROVEMENT EFFORTS. S.J. Atlas¹; T.A. Lasko²; H.C. Chueh¹; Y. Chang¹; R.W. Grant¹; M.J. Barry¹. ¹Massachusetts General Hospital, Boston, MA; ²Massachusetts Institute of Technology, Cambridge, MA. (Tracking ID #134043)

BACKGROUND: Despite considerable resources devoted to health care in the United States, major deficiencies in quality of care persist. Quality assessment (QA) reports to primary care providers (PCPs) about "their" patients have been advocated as a means to improve quality of care. However, there is little evidence that the lists used to measure a PCP's performance accurately reflect the patients actually being managed. Inaccurate and incomplete lists may undermine the perceived validity of the QA process. The goal of this study was to develop and validate a simple statistical model for use in an academic primary care practice network to identify patients loyal to a PCP among all those seen by that provider.

METHODS: Eighteen PCPs, representing 9 practice sites, reviewed electronic clinical, billing and appointment records of all 18,529 outpatients billed over a 3-year period, and stated whether they considered the patient to be "my patient" or not. Using this gold standard, a simple recursive partition logistic regression model was created using development and testing sets to assign patients to specific PCPs using readily available billing data, including variables such as "listed PCP", patient age, time since most recent visit encounter, and in-state residence. The model was specifically designed to have a high level of specificity (and an explicit willingness to sacrifice sensitivity), with a positive predictive value (PPV) goal of 90% for any individual provider. Patients who died or were less than 18 years old were excluded. Patient characteristics and preventive cancer screening test rates were compared for patients characterized as loyal (actual or predicted) to the PCP or not using multiple regression models controlling for clustering among PCPs.

RESULTS: Among 16,435 eligible patients, 11,226 (68.3%) were categorized by the PCP as "my patient" (range 15-93%). Overall, the model correctly identified 82.4% of actual PCP patients (sensitivity) with a specificity of 92.8% and a PPV of 96.1% [range 90-100%]. Development and testing sets provided very similar results. Patient age, gender, ethnicity/race, insurance and marital status, and time since last visit were similar among actual and predicted PCP loyal patients. Breast, cervical and colon cancer screening rates were also similar among actual and predicted PCP loyal patients.

CONCLUSIONS: Many outpatients seen by a PCP are not classified by that PCP as being one of his or her patients, and the percentage varied widely among PCPs in a large primary care practice network. A simple statistical model was developed and accurately identified patients loyal to a PCP. For any given PCP, the model could generate a patient list that was at least 90% accurate - pointing to general agreement among PCPs about what defines patient loyalty. To achieve a highly accurate patient list required falsely labeling a small percentage of patients as not being a PCP's patient. Use of models to predict patient loyalty to a PCP may be useful for quality assessment and improvement efforts.

AN ANALYSIS OF CLAIMS OF MISSED OR DELAYED DIAGNOSES IN THE AMBULATORY SETTING: DIAGNOSES, PROCESS OF CARE BREAKDOWNS, AND CONTRIBUTING FACTORS. A. Kachalia¹; T. Gandhi¹; E.J. Thomas²; C. Yoon³; A. Puopolo¹; T.A. Brennan¹; D. Studdert³. ¹Brigham and Women's Hospital, Boston, MA; ²University of Texas Health Science Center at Houston, Houston, TX; ³Harvard School of Public Health, Boston, MA. (Tracking ID #133447)

BACKGROUND: Missed or delayed diagnoses are an important patient safety concern and have become the leading basis for malpractice claims. However, little is known about their nature and origin, especially in the ambulatory setting. We analyzed claims of a missed or delayed diagnosis in the outpatient setting to determine: (1) the types of injuries occurring; (2) the prevalence and types of underlying errors; and (3) the factors that contributed to these errors.

METHODS: We conducted a retrospective review of 494 closed malpractice claims that contained an allegation of a failure in diagnosis. The claims were randomly sampled from four malpractice insurers operating in different regions of the country and covering physicians practicing in academic and community

settings. Physician reviewers examined both the litigation file and the associated medical record to determine whether a claim contained an identifiable adverse outcome and met the Institute of Medicine's definition of error. If both conditions were met, detailed information on the claim was recorded, including where in the diagnostic process (divided into 14 discrete steps) errors occurred and what factors contributed to the errors. To determine reliability of the reviewers' judgments, 10% of the claims were evaluated by two physicians.

RESULTS: Errors in the ambulatory setting (excluding the emergency department) were identified in 183 of the 494 claims reviewed. Of these, 122 (67%) resulted in serious injury or death. Over half of the errors ($n=105$, 57%) involved delayed diagnosis of cancer. The next most frequent diagnoses were infection ($n=10$, 5%), fracture ($n=8$, 4%), myocardial infarction ($n=7$, 4%), and embolism ($n=7$, 4%). Among the cancer diagnoses, breast cancer (43/105, 41%), colorectal cancer (13/105, 12%), and gynecological cancer (7/105, 7%) were the leading types. Errors tended to occur across multiple phases of care, with a mean of 2.9 process errors per claim. The process errors occurring most frequently were: failure to order an appropriate diagnostic test (55%), failure to create a proper follow-up plan (45%), and incorrect interpretation of a diagnostic test (38%). Tests most frequently failed to be ordered were biopsies (25%), lower endoscopies (11%), and ultrasounds (11%). Test most frequently misinterpreted were mammograms (26%), radiographs (15%), and biopsies (8%). Multiple contributing factors, with a mean of 3.3 per claim, were often present. The leading factors were judgment (79%), vigilance/memory (59%), knowledge (47%), patient-related factors (46%), and hand-offs (19%).

CONCLUSIONS: The range of missed or delayed diagnoses in the ambulatory setting is broad; but cancer, especially breast cancer, is particularly prominent. Breakdowns in key steps of the diagnostic process are often in test ordering and creating follow-up plans, and are frequently caused by cognitive contributing factors. The multiple process errors and contributing factors per claim reflect the complexity of the ambulatory diagnostic process and suggest many approaches for intervention. Systems providing support, ideally real-time, to enhance the accuracy and quality of diagnostic decision-making by providers may reduce instances of severe patient injury.

ANTIBIOTIC PRESCRIBING PATTERNS FOR ACUTE RESPIRATORY TRACT INFECTIONS IN ACUTE CARE SETTINGS. R. Gonzales¹; C.A. Camargo, Jr.²; T.D. Mackenzie³; A.S. Kersey⁴; J. Maselli¹; S.K. Levin¹; C.E. Mcculloch¹; J.P. Metlay⁵. ¹University of California, San Francisco, San Francisco, CA; ²Massachusetts General Hospital, Boston, MA; ³Denver Health and Hospital Authority, Denver, CO; ⁴Department of Veterans Affairs, Philadelphia, PA; ⁵University of Pennsylvania, Philadelphia, PA. (Tracking ID #135188)

BACKGROUND: Acute respiratory tract infections (ARIs) account for the majority of antibiotics prescribed each year by US physicians. Although considerable research has increased our understanding of antibiotic prescribing decisions in primary care settings, much less is known about antibiotic prescribing for ARIs in acute care settings.

METHODS: Chart reviews were performed on a random sample of adult ARI visits to 7 academic-affiliated emergency departments (EDs) and 7 Veterans Administration (VA) urgent care centers located in all 4 US Census regions for the period of Nov 2003 - Feb 2004. Visits were limited to those discharged to home, and those with ICD-9 billing codes for antibiotic-responsive (pneumonia, acute exacerbation of chronic bronchitis, pharyngitis, sinusitis, otitis media) and antibiotic-nonresponsive conditions (acute bronchitis, nonspecific upper respiratory tract infection (URI)). Generalized Estimating Equations were used to evaluate independent predictors of antibiotic prescribing while controlling for clustering of antibiotic prescribing patterns by practice site. Results are expressed as adjusted odds ratios (OR) with 95% confidence intervals.

RESULTS: Of 1987 ARI visits, 65% were for antibiotic nonresponsive diagnoses. 75% of acute bronchitis and 39% of URI visits were treated with antibiotics; compared with 83% of remaining ARIs ($P<0.001$). Besides diagnosis, factors associated with antibiotic prescribing included provider type (OR=0.49 [0.37-0.65]) for housestaff compared with attendings, duration of illness (OR=1.68 [1.20-2.37]) for >2 days compared with 1-2 days, and history of asthma (OR=1.72 [1.27-2.34]). Patient age, gender, race, smoking status, recent ARI (within 6 weeks), comorbidities, visit duration and clinical setting (ED vs. VA) were not independently associated with antibiotic prescribing. Stratified analyses show that attendings and housestaff had similar antibiotic prescribing rates for antibiotic-responsive diagnoses, but housestaff prescribed fewer antibiotics for acute bronchitis (60% vs 73%) and URIs (14% vs. 47%) ($p=0.05$ for interaction). Adjusted antibiotic prescription rates for ARIs varied between sites, ranging from 49% to 92%.

CONCLUSIONS: Acute care settings are important targets for reducing inappropriate antibiotic prescribing for adults with ARIs. Housestaff adhere more closely to national recommendations for antibiotic treatment of ARIs compared with attending physicians. Whether housestaff become less judicious prescribers of antibiotics over time, or will retain these judicious prescribing patterns (as a generational effect) merits further study.

ARE DIFFERENCES IN QUALITY BETWEEN MEDICAL GROUPS AND IPA'S EXPLAINED BY ORGANIZATIONAL CHARACTERISTICS OR USE OF QUALITY IMPROVEMENT STRATEGIES? A. Mehrotra¹; M. Rosenthal²; A.M. Epstein². ¹Harvard University, Cambridge, MA; ²Harvard University, Boston, MA. (Tracking ID #135756)

BACKGROUND: Despite continuing evidence demonstrating large variations in health care quality, little is known about what explains these variations. In this study we looked at variations in the quality of primary care across physician groups. We examined the association of organizational characteristics (e.g., size,

use of an electronic medical record (EMR) and the use of quality improvement strategies (e.g., disease registries, reminder systems) with the quality of primary care delivered to beneficiaries of PacifiCare, a large California health maintenance organization.

METHODS: We obtained information on organizational characteristics and quality improvement strategies through structured interviews with PacifiCare-contracted physician groups. Quality data were obtained from PacifiCare's audited quality performance reports (Quality Index and Provider Profile), which are based on administrative data. Quality was measured by determining what percentage of eligible patients received the following: mammography, Pap smear, Chlamydia screening, childhood immunizations, diabetic eye exam, inhaled steroid for asthmatics, beta-blocker after myocardial infarction, and ACE inhibitor for CHF. An overall quality score for each physician group was created by adding standardized individual scores. We tested whether 16 organizational characteristics and 12 quality improvement strategies were associated with quality scores. Those significant in univariate analyses were included in a multivariate linear model. A model was created for each of the 8 individual quality measures as well as the overall quality score.

RESULTS: 119 physician groups who care for over 1.5 million PacifiCare enrollees were used in the final analysis. In univariate analyses, medical groups had statistically significant ($p < 0.05$) higher quality scores on 4 of the 8 individual quality measures as well as in overall quality as compared with IPAs. Medical groups were more likely to be older (36.8 yrs vs. 9.9 yrs), use an EMR (36.8% vs. 1.6%), and use quality improvement programs (e.g., contact patients to remind them of mammograms, 73.7% vs. 27.8%). In multivariate regression, medical groups had significantly higher scores in overall quality and in 3 of the 8 individual quality measures as compared with IPAs (19.5% higher mammogram rate, 20.4% higher Pap smear rate, and 11.4% higher diabetic eye screening rate). The other covariates in the models were not consistently associated with quality scores. Increasing fraction of physician salary based on productivity was predictive of higher overall quality. Higher volume of patients was predictive of higher mammogram and childhood immunization rates and older age of group was predictive of higher Chlamydia screening rates. The use of an EMR, quality bonuses for individual doctors, reminder systems for patients, or feedback of quality performance to individual physicians were not statistically significant predictors of quality in the multivariate model. Overall less than 25% of the quality differences between medical groups and IPAs were explained by organizational characteristics and use of quality improvement strategies.

CONCLUSIONS: As compared to IPAs, medical groups have significantly higher scores on many quality measures. Although medical groups are more likely to be older, use electronic medical records, and use quality improvement strategies, these characteristics and strategies explain only a fraction of the differences in quality.

ASPIRIN USE FOR THE PRIMARY PREVENTION OF CORONARY HEART DISEASE IN A BIRACIAL COHORT OF OLDER ADULTS. N. Rodondi¹; E. Vittinghoff²; J. Cornuz²; J. Butler³; J. Ding⁴; S. Satterfield⁵; A. Newman⁶; T.B. Harris⁷; S.B. Hulley¹; D.C. Bauer¹. ¹University of California, San Francisco, San Francisco, CA; ²University Hospital of Lausanne, Lausanne, ; ³Vanderbilt University, Nashville, TN; ⁴Wake Forest University School of Medicine, Winston-Salem, NC; ⁵University of Tennessee, Memphis, TN; ⁶University of Pittsburgh, Pittsburgh, PA; ⁷National Institutes of Health, Chevy Chase, MD. (Tracking ID #131785)

BACKGROUND: Aspirin for the primary prevention of coronary heart disease (CHD) has a more favorable risk/benefit profile among adults with high CHD risk than among low-risk adults. As there is little information on the current patterns of aspirin use for primary prevention, we determined the prevalence of aspirin use for primary prevention in relation to 10-year CHD risk and changes over time in a population-based cohort of older adults.

METHODS: We studied 2163 community-dwelling older black and white men and women without baseline cardiovascular disease in the Health, Aging and Body Composition Study (Health ABC). Our design was a prospective cohort study from baseline in 1997-98 to 2002-03 to assess changes over time. During this period, three major trials, two meta-analyses and three national guidelines regarding the use of aspirin for primary prevention were published. We calculated 10-year CHD risk using Framingham risk score in 1997-98 to 2002-03. Logistic regression models were used to assess the unadjusted associations between aspirin use and categories of CHD risk, and multivariate models to assess other factors independently associated with aspirin use.

RESULTS: At entry into Health ABC, 42% were black, 55% were women, and 17% reported aspirin use. Aspirin use increased with CHD risk from 13% in persons with 10-year risk <6% (low risk) to 23% in those with 10-year risk >20% (highest risk) (p for trend <0.001). Blacks were less likely to use aspirin (13%) than Whites (20%). In multivariate analysis, Black race was associated with lower aspirin use (OR 0.66, 95% CI 0.49-0.89). Between 1997-98 and 2002-03, aspirin use increased from 17% to 32% among those still free of CHD ($p < 0.001$), and the association with CHD risk continued (p for trend <0.001). Blacks were still less likely to use aspirin (26%) than Whites (36%). Despite their high CHD risk, diabetics were not more likely to use aspirin than non-diabetics, even in 2002-03 (OR 0.89, 95% CI 0.56-1.40). 24% of adults with 10-year CHD risk <6% used aspirin, despite recommendations to target primary prevention with aspirin on subjects with 10-year CHD risk above 6-10%.

CONCLUSIONS: Regular use of aspirin by older adults with no history of CHD has increased in recent years. Individuals at higher CHD risk are more likely to take aspirin, but there is room for considerable improvement in targeting those at high risk, particularly diabetics and Blacks.

ASSOCIATION OF PHYSICIAN CULTURAL COMPETENCY TRAINING AND PATIENT FUNCTIONAL STATUS, BLOOD PRESSURE, AND GLYCEMIC CONTROL. M.A. Shaw¹; E. Liles²; Y.M. Diaz³; J. Bussey-Jones⁴. ¹University of Louisville, Louisville, KY; ²University of North Carolina at Chapel Hill, Chapel Hill, NC; ³University of Miami, Miami, FL; ⁴Emory University, Atlanta, GA. (Tracking ID #134990)

BACKGROUND: In an ongoing effort focused on improving the nation's quality of care, and reducing and eliminating health disparities, attention has been focused on the cultural competency of health care providers and the role of training in cross-cultural issues. However, little data exists on the impact of cultural competency training on clinical outcomes. The purpose of this study is to examine the association between physician cultural competency training and patient health status and clinical outcomes.

METHODS: The CCTOP study (Cultural Competency Training and Outcomes in Patients) is an eleven-center, cross-sectional study of patients presenting for care at geographically diverse outpatient centers and the physicians who care for them. Patients were verbally administered a 76-item survey regarding their health and demographics. Physicians completed surveys regarding their cultural competency training, including format, duration, and timing of training. We examined the association of cultural competency training during medical school, residency, and post residency with the SF-36 Health Survey, blood pressure in hypertensive patients, and diabetic patients' self-reported glycemic control. Multivariate regression analyses were performed to control for age, gender, race, education, and income.

RESULTS: We present preliminary results from 250 matched patient-physician surveys from 6 of the 11 centers. 291 patients responded (70% female) with a mean age of 56 years. 44% were African American, 29% White, 24% Hispanic, and 3% other. 175 physicians (118 residents and 57 attendings) participated in the study. Multivariate analyses revealed only one association between format of training and clinical outcomes. Workshop/small group training was found to be associated with the Role-Physical subscale of the SF-36 Health Survey ($p = 0.038$). No associations were found with lecture or cultural immersion experience formats of training, duration of training (number of hours), or timing of training (year training received in medical school versus residency, or post residency) and the other 7 subscales of the SF-36 Health Survey, systolic blood pressure, or glycemic control ($p > 0.05$).

CONCLUSIONS: Physician cultural competency training, as it currently exists, appears to have minimal association with the patient outcomes that we measured. Further research is needed in the area of cultural competency training to ensure development of a curricula/framework that will work to educate and train health professionals to meet the health care needs of an increasingly diverse population.

CLINICIAN-REPORTED BARRIERS FOR ADMISSION OF LOW RISK AND DISCHARGE OF HIGHER RISK PATIENTS WITH PNEUMONIA. J.B. McCausland¹; T.E. Auble¹; J. Whittle²; D.S. Obrosky¹; D.M. Yealy¹; M.J. Fine¹. ¹University of Pittsburgh, Pittsburgh, PA; ²Kansas University Medical Center, Kansas City, KS. (Tracking ID #134355)

BACKGROUND: Community-acquired pneumonia (CAP) practice guidelines recommend outpatient treatment of low-risk and admission of higher-risk patients based upon the pneumonia severity index (PSI), a previously validated evidence-based prognostic model. We present provider-perceived barriers for discordant admission decisions from the Emergency Department (ED).

METHODS: We examined data from 1317 pneumonia patients enrolled from 12 EDs in the high-intensity arm of a 32-site randomized guideline implementation trial in two states. Emergency providers (EPs) used the PSI and level of oxygenation to identify patients at low risk of 30-day mortality (LR: PSI class I-III without oxygen desaturation) and higher risk patients (HR: PSI class IV-V and I-III, with oxygen desaturation). Clinicians underwent multi-faceted implementation techniques including educational efforts, reminders, audit and feedback and continuous quality improvement to increase the proportion of low-risk patients treated as outpatients. Practitioners making admission decisions in discordance with guideline recommendations were surveyed regarding each decision and reasons for guideline non-adherence were documented. Barriers were grouped into categories; data were summarized as proportions.

RESULTS: Clinicians ($N = 154$) were primarily male (80.5%), physicians (95.5%) with a mean age of 42 ± 6 years. Clinicians included emergency physicians (81.2%), other specialties (e.g. internal medicine, family practice; 14.5%) and nurse practitioners or physician assistants (4.5%). Of the 700 LR patients, 267 (38.1%) were admitted while 20 of the 617 HR patients (3.2%) were discharged. Surveys were completed by clinicians for 260/267 (97.3%) of the LR admitted patients and 15/20 (75%) of the higher-risk discharged patients. In most cases (171; 75%), EPs cited severity of illness (either the pneumonia itself or other comorbid conditions) as a reason for admission. Examples of other barrier categories include abnormal laboratory values, vital signs or symptoms (76; 29.2%), primary care physician request (49; 18.8%), failure of outpatient management (28; 10.8%) and social issues (18; 6.9%). Among the patients judged too ill for outpatient management, EPs most commonly cited pulmonary (37; 14%) and cardiac (36; 14%) comorbidities; abnormal signs and symptoms included wheezing (25; 14.6%), tachycardia (6; 3.5%) and hypotension (4; 2.3%). One category of barriers was present in at least half of the patients (49.6%); two categories (29.6%); three categories (13.5%) and four or more categories (6.5%). Higher-risk patients were discharged due to primary care physician request ($n = 5$), discharge to a nursing home ($n = 3$) and one patient refused admission due to lack of insurance.

CONCLUSIONS: Despite intense implementation techniques, low-risk CAP patients are admitted to the hospital when the clinician judges them to have a serious concomitant illness that they believe warrants hospitalization. Although clinician-perceived barriers and actual barriers may differ, these results can

guide future PSI-based efforts to safely increase the proportion of low-risk pneumonia patients managed as outpatients.

COMMUNICATION FAILURES DURING PATIENT SIGN-OUT: A CRITICAL INCIDENT ANALYSIS. V. Arora¹; J.J. Mohr¹; D. Lovinger¹; H. Humphrey¹; D. Meltzer¹. ¹University of Chicago, Chicago, IL. (Tracking ID #132906)

BACKGROUND: Despite concerns that the handoff or sign-out of inpatient care between physicians is error-prone, few studies have examined failures in sign-out. This study aims to describe failures in communication in patient sign-out and suggest areas for improvement.

METHODS: To solicit communication failures in sign-out that affect patient care, we used critical incident technique (CIT). CIT, first used in aviation, is used to investigate poorly documented or understood events. In CIT-based interviews, internal medicine interns at an academic medical center were asked to describe any near misses or adverse events that occurred due to a poor sign-out from the preceding shift. They also were asked to report the worst event that they were involved in that year due to poor sign-out. Lastly, interns evaluated the written and verbal sign-out received from peers and suggested areas of improvement. To minimize effects on future behavior, interviews were performed on the last weekday of the intern's general medicine inpatient rotation in March and June 2004. Critical incident analysis was performed by constant comparative method, an inductive approach with no a priori hypotheses. All textual data was reviewed, and select incidents compared. Through an iterative approach, categories emerged. The rest of the data was then categorized, allowing for new categories or grouping of like categories. Reliability of categorization was tested through triangulation by three independent reviewers (VA, JJM, DL) with all disagreements resolved by consensus.

RESULTS: 26/30 interns, caring for 82 general medicine inpatients collectively, reported 25 discrete incidents (near misses or adverse events) resulting from failed communication during written or verbal sign-out from the preceding shift. Inter-rater agreement for the categorization of these critical incidents was high (kappa statistic for each category ranged from 0.73 to 1.0, $p < 0.001$). A major category of communication failure was omitted content (e.g. medications, active or anticipated problems, pending tests or consults, reasoning of the primary team). In nearly all incidents, content was either omitted from written sign-out sheets, during verbal communication at the time of sign-out, or both. Factors that impede the communication process (e.g. use of an intervening night-float, illegible notes on sign-out sheets, lack of face to face communication) constituted another major category. In all cases, these failures led to uncertainty during patient care decisions. This uncertainty often resulted in inefficient intern work, such as soliciting information from other sources (e.g. chart, other resident, patient) and in some cases, performing unnecessary tests for patients. In almost all 24 worst events described, an omission resulted in uncertainty in a patient care decision, leading to patient harm. Interns preferred thorough, but relevant, face-to-face verbal sign-outs which review anticipated issues. They favored written sign-out sheets that are legible, relevant, accurate, up-to-date and include the following fields: code status, baseline mental status, pending tests or consults, and active or anticipated medical problems.

CONCLUSIONS: Communication failures during sign-out are characterized by omissions in content or factors that impede the communication process. These failures often result in uncertainty in patient care decisions, which can lead to inefficient intern work, unnecessary tests or patient harm.

CONTRASTING THE PROGRAM CHARACTERISTICS AT FEDERALLY SUPPORTED COMPREHENSIVE WOMEN'S HEALTH CENTERS. B. Bean-Mayberry¹; E. Yano²; N. Judith¹; N. Bayliss³; C. Weisman⁴; S. Scholle⁵. ¹University of Pittsburgh, Pittsburgh, PA; ²VA Greater Los Angeles HSR&D Center of Excellence, Sepulveda, CA; ³VA Pittsburgh Healthcare System, Pittsburgh, PA; ⁴Pennsylvania State University, Hershey, PA; ⁵NCQA & University of Pittsburgh, Washington, DC. (Tracking ID #135824)

BACKGROUND: Fragmentation of women's health care has stemmed from the separation of reproductive care needs from general medical care. To promote the delivery of comprehensive, integrated clinical care for women, the Department of Health and Human Services (DHHS) Office of Women's Health launched the national Centers of Excellence (CoE), while the VA initiated the Comprehensive Women's Health Centers (CWHC). The purpose of this study was to compare the organization, staffing, practice setting, and service availability at specialized women's health centers.

METHODS: 1) Data on the VA women's health programs were obtained from the VHA Survey of Women Veterans Health Programs and Practices: Senior Clinician Questionnaire. VA hospitals selected had either a CWHC (n=8) or a VA-designated Clinical Center of Excellence in Women's Health (CCoE, n=4). 2) The VA tool was adapted for use with clinical directors at DHHS CoEs in operation in June 2003 (n=13). 3) Geographic data and academic affiliation were abstracted from the 2001 American Hospital Association hospital survey. Data were reviewed and entered into a database to make descriptive comparisons.

RESULTS: All VA and DHHS women's health programs were located in urban areas and nearly all had academic partnerships, but DHHS sites had mean caseloads that were triple VA caseloads. Women's health fellowships were common, and all offered educational training to providers in women's health topics. Preventive screening and general reproductive services (e.g., STD treatment, menopause management) were on site at nearly all facilities; however, the DHHS sites offered more extensive reproductive services (e.g., on site obstetrical and infertility services). The VA sites were unique in mental health care with all offering sexual trauma screening, trauma and rape crisis counseling, and most with on-site mental health providers. Staff variations followed these service dif-

ferences with DHHS sites having more gynecologists while VA's had more psychiatrists.

CONCLUSIONS: Despite the separate history and development, the VA and DHHS specialized women's health programs share similar organization, education, and clinical services. The wide range of clinical services available demonstrates the commitment to multidisciplinary care. The presence of these federally recognized centers with common structural components presents an opportunity to create VA and non-VA partnerships to evaluate quality of care for women nationally and benchmark the findings in VA and community standards.

CULTURAL CONSENSUS ANALYSIS SUCCESSFULLY IDENTIFIES SPECIFIC PROBLEMS FOR REMEDIATION AT FIVE VA TEACHING CLINICS. C.S. Smith¹; W. Hill¹; M. Morris¹; C. Francovich¹; J. McMullin¹. ¹VA Medical Center, Boise, ID. (Tracking ID #131665)

BACKGROUND: The chaos confronting patients and physicians has many causes. We believe one significant cause is unrecognized differences in values that lead to conflict. Cultural Consensus Analysis (CCA) is a method used by anthropologists to determine groups with shared values. In an earlier study, we developed a CCA tool from in-depth ethnographic observations, interviews, and focus groups. The current study examined the ability of that CCA to identify value differences that correlate with major problems in five VA clinics.

METHODS: CCA was performed at five VA teaching clinics by asking residents, faculty, patients and administrators to rank order 16 statements about things that might happen during a clinic visit by order of importance. A matrix of matches between subjects was factor analyzed, and "correct" answers for each group were calculated a posteriori using Bayes' theorem. A workgroup was also systematically created at each site. Each workgroup independently identified and prioritized a list of clinic problems using brainstorm and Delphi. The relationship between the CCA results and problems was examined in a three-step process using graphical analysis, factor analysis, and then correspondence analysis.

RESULTS: Some CCA statements had significant between-group differences across all sites. For instance, "Have the same doctor for more than one year" was much more important to patients than to residents, faculty, and administrators. CCA statements corresponding to a site's major problem (for instance, "Use the computer" at a site where a major problem involved the computerized medical record) showed greater differences at these sites. Correspondence analysis revealed five models of the clinic visit that compete for resources and prominence: technical care (doctor is experienced and familiar with the patient's medical history, clinic communicates with patient); education-based care (faculty teach and supervise the development of residents); humanistic care (doctor is concerned about patient as a person and understands their medical choices); guideline-based care (rules based on scientific data that guide the doctor in best medical choices); and efficient care (clinic visit is timely and productive). Patients, residents, faculty and administrators had very different preferences for these models.

CONCLUSIONS: This study had three important findings: 1) Some problems, such as continuity in teaching clinic, are system-wide; 2) CCA is a sensitive indicator of site-specific problems, and 3) competing models of the clinic visit explain many tensions that correlate with recurring problems.

DISCLOSING MEDICAL ERRORS TO PATIENTS: IT'S NOT WHAT YOU SAY, IT'S WHAT THEY HEAR. A.W. Wu¹; I.C. Huang¹; S.L. Stokes¹; P.J. Pronovost¹. ¹Johns Hopkins University, Baltimore, MD. (Tracking ID #134301)

BACKGROUND: There is consensus that physicians are obligated to disclose harmful medical errors to patients and/or their families. Although anecdotal evidence and a few studies suggest that patients want an apology and someone to take responsibility, there are no data on the impact of differing methods of disclosure. In this study, we showed volunteers videotaped vignettes that depicted physicians disclosing medical errors to patients, and surveyed viewers' reactions. We hypothesized that volunteers who viewed a full apology and acceptance of responsibility would rate the physician more favorably and would be less likely to sue.

METHODS: We created video vignettes of disclosure of three adverse events (overlooked mammogram, chemotherapy overdose, and neglectful pediatric surgeon) using actors to portray physicians and patients. Vignettes varied with regard to apology (full, insincere, none), and accepting responsibility (yes, no). Volunteers from Baltimore City were randomized to view vignettes including different combinations of apology and responsibility. They then completed a survey eliciting their emotional response to the video, ratings of the physician's handling of the incident, perception of the physician, trust in the physician, desire to have the physician as their own doctor, inclination to refer, and likelihood they would sue, using a 5-point Likert-type response scale. Analyses pooled responses across vignettes. We dichotomized responses and used a Chi-square test for bivariate analyses and logistic regression for multivariable analyses adjusting for viewer characteristics.

RESULTS: Of 200 volunteers, 50% were <40 years old, 25% were female, 80% were African American, and 50% had completed high school. All demographic variables were balanced across variations of the vignettes ($p > 0.1$). Cronbach's alpha was > 0.70 for multi-item scales. For the designed variations, non-significant trends suggested higher scores for full apology/responsibility, and lower scores for no apology/no responsibility. On the other hand, viewer perceptions that there had been an apology or that the physician had accepted responsibility was significantly related to higher ratings of the physician (e.g., 81% vs. 38% trusted; 56% vs. 27% would refer, $p < 0.05$), though not to decreased inclination to sue (43% vs. 47% would sue). In multivariable analyses adjusting for vignette,

age, gender, race and education, perceived apology and perceived responsibility were independently related to higher ratings for all dimensions. However, proclivity to sue was non-significantly reduced (OR 0.92, 95% CI 0.67-1.25 for apology; 0.87, 0.67-1.13 for responsibility).

CONCLUSIONS: Patients will probably respond more favorably to physicians who apologize and accept responsibility for medical errors. Physicians should be aware that patient perceptions of what is said may be more important than what is actually said. The desire of patients to sue may not be affected despite a disclosure that includes a full apology and accepts responsibility.

DISCONTINUITY OF CHRONIC MEDICATIONS IN PATIENTS DISCHARGED FROM THE INTENSIVE CARE UNIT. P. Rahimi-Darabad¹; A. Omer¹; C.M. Bell¹. ¹University of Toronto, Toronto, Ontario. (Tracking ID #135368)

BACKGROUND: Lifelong medication adherence is essential for continued risk reduction from important adverse patient outcomes in chronic diseases. If chronic medications are unintentionally discontinued, patients may be placed at risk for potentially avoidable adverse events. Transition from community outpatient to hospitalized inpatient and back to the community may be a factor affecting the unintentional discontinuation of medications. Intensive Care Unit (ICU) admission may connote an elevated risk of adverse events because of its focus on acute illnesses and the multiple care transitions. The objective of this study was to identify the proportion and determinants of unintentional discontinuation of selected chronic medications in ICU patients.

METHODS: Hospital charts of consecutive patients discharged from the ICUs of one teaching and two community hospitals from January 1, 2002 to December 31, 2002 were reviewed. Patients transferred from other medical facilities were excluded from the study. Eligible patients were those taking at least one of six selected chronic medications prior to hospital admission (statins, antiplatelets or anticoagulants, bronchodilators or steroid inhalers, acid-suppressing drugs, L-thyroxine, allopurinol). The medications on discharge from ICU as well as on hospital discharge were examined for the presence of the prespecified medications. These charts were thoroughly reviewed with explicit criteria to ensure that medications were not intentionally discontinued. The proportions of patients for whom their selected medications were not prescribed were calculated.

RESULTS: A total of 1,532 charts were reviewed and 1,353 met the criteria. In all, 807 patients were prescribed at least one of the six prespecified chronic medications. There were 297 (37%) patients who had one or more of their chronic medications omitted on discharge from ICU and 269 (33%) patients were discharged from hospital without one or more of their chronic medications. Patients discharged from teaching and community hospitals had similar proportions of medication discontinuation upon discharge from ICU (37% vs 38%) and hospital (32% vs 35%). About one-fifth of all patients on important medications such as L-thyroxine, antiplatelets/anticoagulants, and statins had their medications unintentionally discontinued upon hospital discharge.

CONCLUSIONS: Patients discharged from the ICU often leave the hospital without note of their previously prescribed chronic medications. Careful review of every patient's medications upon ICU discharge could avoid potential adverse patient outcomes related to unintentional discontinuation of important chronic medications on hospital discharge.

Proportion of Discontinued Medications upon Discharge from ICU and Hospital

Medication	ICU Discharge	Hospital Discharge
Acid suppressing drugs	25% (77/303)	25% (77/303)
Allopurinol	56% (29/52)	54% (28/52)
Antiplatelets/Anticoagulants	26% (110/430)	22% (95/430)
L-thyroxine	23% (26/115)	21% (24/115)
Non-PRN bronchodilators/steroid inhalers	19% (46/239)	21% (50/239)
Statins	29% (82/278)	23% (65/278)

EFFECT OF BARCODE TECHNOLOGY ON THE INCIDENCE OF MEDICATION DISPENSING ERRORS AND POTENTIAL ADVERSE DRUG EVENTS IN A HOSPITAL PHARMACY. E.G. Poon¹; J. Cina¹; W. Churchill¹; P. Mitton¹; M. McCrear¹; E. Featherstone¹; J. Rothschild¹; C.A. Keohane¹; D.W. Bates¹; T. Gandhi¹. ¹Brigham and Women's Hospital, Boston, MA. (Tracking ID #133124)

BACKGROUND: Many medication dispensing errors occur in hospital pharmacies and a considerable number of these errors have the potential to harm patients by causing adverse drug events (ADEs). To address this patient safety concern, the Food and Drug Administration has mandated bar-coding of all medications used in hospitals by 2007, and some hospital pharmacies are beginning to deploy bar-code technology. However, empiric data supporting the efficacy of this technology to reduce dispensing errors remain limited.

METHODS: We conducted a prospective study in the hospital pharmacy at a 720-bed tertiary academic medical center. Using direct observation techniques, we measured the incidence of medication dispensing errors and potential ADEs (defined as dispensing errors with the potential to cause patient harm) before and after the deployment of bar-code technology in the dispensing process. Before deployment ('pre' period), the selection of medication and verification of accuracy were performed manually by pharmacy staff. After deployment ('post' period), each medication dose was bar-coded and electronically scanned to verify its accuracy. During both observation periods, a trained observer visually

inspected medications after completion of the dispensing process to identify dispensing errors. Each dispensing error was further reviewed by a 2-physician panel to determine whether it represented a potential ADE. In a sub-analysis, we excluded wrong-quantity dispensing errors since these errors were unlikely to be prevented by bar-code scanning.

RESULTS: In the 5-month 'pre' and 4-month 'post' periods, we observed 140,755 and 253,984 medication doses respectively. In the 'pre' period, 0.75% (1059) of the doses dispensed from the pharmacy had errors, while in the 'post' period, 0.66% (1681) had errors. This represented a 12% reduction in the dispensing error rate (chi-squared=10.8, p=0.001). When wrong-quantity dispensing errors were excluded from the analysis, the dispensing error rate fell by 42% ('pre': 0.40%, 'post': 0.23%; chi-squared=86.9, p<0.001). As for the rate of the potential ADEs, 0.18% (249) of medications dispensed in the 'pre' periods were potential ADEs, compared to 0.10% (265) in the 'post' period. This represented a 41% reduction in the rate of potential ADEs (chi-squared=36.7, p<0.001). Of the potential ADEs in the 'post' period, the two most common medication types were electrolytes (44%) and antibiotics (24%); 77% of these potential ADEs were associated with pre-mixed intravenous medications.

CONCLUSIONS: Bar-code technology significantly reduced the rates of dispensing errors and potential ADEs due to dispensing errors. However, not all potentially harmful errors were eliminated, especially those associated with pre-mixed intravenous medications. Refinements in the scanning software and workflow processes that focus on these remaining errors will likely lead to greater improvements. Our results suggest that bar-code technology can have a substantial impact on serious errors and deserves strong consideration as a tool to improve patient safety.

EFFECTIVENESS AND SAFETY OF USING THE PNEUMONIA SEVERITY INDEX TO TRIAGE PATIENTS WITH PNEUMONIA IN THE EMERGENCY DEPARTMENT: A MULTICENTER EUROPEAN STUDY. B. Renaud¹; E. Coma²; J. Labarere³; J. Hayon⁴; C. Longo⁵; M. Blancher²; M.J. Fine¹; E. Roupie⁶. ¹University of Pittsburgh, Pittsburgh, PA; ²Hospital de San Jaume y Santa Magdalena, Mataro; ³Centre Hospitalier Universitaire de Grenoble, Grenoble; ⁴Hopital de Saint-Germain-en-Laye, Saint-Germain-en-Laye; ⁵Centre Hospitalier Regional Hotel-Dieu, Nantes; ⁶Centre Hospitalier Universitaire Henri Mondor (AP-HP), Creteil. (Tracking ID #135744)

BACKGROUND: Although prior North American trials have demonstrated the effectiveness and safety of using the Pneumonia Severity Index (PSI) to guide the initial site of treatment of patients with community-acquired pneumonia (CAP), similar data do not exist in Europe. The aim of this study was to compare the admission decision and medical outcomes for patients with CAP managed in emergency departments (ED) that routinely used or did not use the PSI to guide the initial site of treatment.

METHODS: Among the 16 participating French EDs, 8 routinely used the PSI to guide the initial site of treatment (PSI-user) and 8 did not. (PSI non-user) Patients were included if they had a clinical diagnosis of CAP and an acute radiographic infiltrate. We excluded patients with immunosuppression, HIV related illness, and prior tracheotomy. We collected baseline demographics, clinical, laboratory, and radiographic data at presentation. Severity of illness was quantified based on PSI risk class; patients in risk classes I-III were defined as low-risk (LR) and those in classes IV or V as high-risk (HR). In agreement with their standard practice, ED physicians at PSI-user hospitals were encouraged to treat LR patients in the outpatient setting at the time of study implementation. The primary outcome was the proportion of LR patients treated as outpatients. Secondary safety outcomes included all cause 28-day mortality, length of stay for inpatients discharged alive, admission to an ICU for inpatients, and subsequent hospital admission for outpatients. Multilevel logistic regression was used to analyze the binary primary outcome at the ED and patient levels, adjusting for variables significantly associated with site of treatment in univariate analyses.

RESULTS: Of the 925 enrolled patients, 472 (51.0%) were treated at PSI-user and 453 (49.0%) at PSI non-user EDs. Overall, 449 (48.5%) patients were LR (47.9% in PSI-user and 52.1% in PSI non-user EDs, p=0.06). The proportions of LR patients treated as outpatients were 42.8% in PSI user and 23.9% in PSI non-user EDs (crude odds ratio (OR)=3.3; 95% CI 1.3-8.1). Adjusted ORs for outpatient treatment for patients in risk classes I and II treated at PSI-user EDs were 7.0 (95% CI 2.0-25.0) and 4.6 (95% CI=1.3-16.2), respectively, compared to those treated at PSI non-user EDs. For patients in risk classes III, IV and V, the adjusted ORs did not differ by PSI-user or non-user ED status. Overall, mortality was 10.6% for all patients and 1.6% for LR patients, with no difference across PSI-user and non-user EDs. No deaths occurred in LR outpatients and no differences in the secondary outcomes were observed across patients treated at PSI-user and non-user EDs.

CONCLUSIONS: This multicenter European study demonstrates that routine use of the PSI in the ED was associated with a higher proportion of LR patients with CAP being treated in the outpatient setting without compromising safety outcomes. These findings expand the generalizability of the PSI as a clinical guide for the outpatient treatment of LR patients with CAP.

EMERGENCY DEPARTMENT STAFFING AND ITS EFFECT ON PATIENT CARE. N.A. Bickell¹; R.M. Anderson¹; C. Barsky¹; M. Rojas¹. ¹Mount Sinai School of Medicine, New York, NY. (Tracking ID #133006)

BACKGROUND: Emergency Departments (EDs) provide a safety net for seriously ill individuals. Overcrowding and understaffing of EDs may adversely affect users' health. Little is known about the relationship of physician, nurse and ED support personnel staffing on time to treatment or health outcomes of pa-

tients with time-sensitive conditions. We studied the relationship between ED staffing and time to care among appendicitis patients.

METHODS: Detailed clinical and time data were abstracted from charts of 242 appendicitis patients presenting to an urban tertiary hospital ED between 6/01 and 11/02. ED census, patient triage and acuity were obtained from ED tracking software. We obtained hourly staffing levels from physician schedules, nursing and clerical payrolls and averaged staffing levels for each study patient's ED stay. ED information was obtained for 237 (98%) appendicitis patients. Bi- & multivariate relationships between ED staffing, patient characteristics with time and perforation were examined. In one model, we assessed factors affecting time between ED arrival and first examination by a physician. We then examined the relationship of staffing with ED length of stay (LOS) and finally, with perforation. **RESULTS:** 9% of appendicitis patients arrived by ambulance; 21% experienced perforation. Patients ≤ 18 yrs old (87% vs 60%, $p < .01$) and those brought by ambulance (85% vs 55%, $p < .01$) were more likely to be seen within an hour of ED arrival compared to their counterparts. Staffing levels were not significantly related to time to 1st examination. Bivariate analyses showed shorter ED LOS with greater numbers of physicians ($r = .26$, $p < .01$), nurses ($r = .19$, $p < .01$) and clerks ($r = .21$, $p < .01$) working in the ED, as well as among patients with rebound and those for whom the first diagnostic impression was appendicitis. Multivariate analysis showed that shorter lengths of stay are associated with greater physician staffing (RR=0.96; 95%CI: 0.93-0.99), a diagnostic 1st impression of appendicitis (RR=0.87; 95%CI: 0.76-1.00) and the presence of rebound (RR=0.84; 95%CI: 0.72-0.98). Patients undergoing CT in the ED had markedly longer LOS (RR=2.2; 95%CI: 1.92-2.53) (model $r^2 = .48$; $p < .01$). ED staffing was not associated with rates of perforation.

CONCLUSIONS: The swiftness of initial ED exams is most affected by transportation mode of arrival even after adjusting for illness severity at initial presentation. Physician staffing levels have a modest effect on time to treatment; the length of time that appendicitis patients spend in the ED is most affected by their illness severity, physicians' diagnostic first impression of a time-sensitive condition and diagnostic testing. Interestingly, neither ED census, nurse nor support personnel staffing was related to ED length of stay.

EMR REGISTRY USE IMPROVES DIABETES CARE IN A GROUP PRACTICE. D. Smith¹, A. Greenglass¹, V. Chang¹, R. Durvasula¹. Yale University, New Haven, CT. (Tracking ID #133674)

BACKGROUND: Translating evidence on the effectiveness of risk factor modification in diabetes into clinical practice is challenging. Traditional paper-based medical records and encounter-based care often fail to make relevant information accessible for diabetes management at each visit, and population-level data is difficult to obtain to permit proactive interventions. Tracking risks over time and weighing complex numerical data are information-intensive endeavors which are especially amenable to an electronic registry approach.

METHODS: Adult diabetics in a group practice were identified through encounter forms and pharmacy records, and a structured chart review established a baseline of "usual care". Processes of care and measures of metabolic control (vs. ADA goals) were tracked using an electronic database. Organized and updated data on individual diabetic patients were made available to clinicians at each clinical encounter, and used to proactively reach out to patients with care reminders. Additional practice interventions included APRN case management of patients identified by the registry as having poor control, and quarterly feedback to clinicians about the status of their diabetes panel. Comparisons were made between outcomes of "usual care" prior to the registry and those of "information-enhanced care", using a χ^2 test.

RESULTS: After 45 months of use, diabetic case identification has increased by 30% to 632 patients (4% prevalence). Processes of care have significantly improved (Table 1), such as an increase in pneumococcal immunization rate from 23 to 88%, yearly dilated retinal exam from 45 to 72%, yearly neurovascular foot exam from 14 to 66%, and use of urine microalbumin from 18 to 66%. Key measures of metabolic control were also significantly improved (Table 1), such as glycosylated hemoglobin (HA1C) $< 7.0\%$ from 26 to 41%, LDL cholesterol < 100 mg/dL from 42 to 60% and blood pressure $< 130/80$ mmHg from 59 to 88%. In addition, fewer patients evidence poor control, with HA1C $> 9.0\%$ decreased from 25.7 to 8.4%, LDL-C > 160 mg/dL decreased from 9.1 to 4.4%, and systolic BP > 170 mmHg decreased from 6.3 to 1.7%. The average weight of patients in the population (201.5 lbs) did not change however.

Table 1. Diabetes Care Processes and Measures of Control

Parameter	Usual Care	Information-Enhanced	χ^2
Pneumovax	23%	88%	$P < 0.01$
Dilated retinal exam	45%	72%	$P < 0.01$
Neurovascular foot exam	14%	66%	$P < 0.01$
Urine microalbumin	18%	66%	$P < 0.01$
HA1C $< 7.0\%$	26%	41%	$P < 0.05$
LDL-C < 100 mg/dL	42%	60%	$P < 0.02$
BP $< 130/80$ mmHg	59%	88%	$P < 0.01$

CONCLUSIONS: By using a dynamic electronic registry to structure information, ADA goals for best practices, and applying a proactive, population-based approach to care, a group practice was able to continuously improve and sustain the quality of care delivered to patients with diabetes. Use of an electronic

registry heightens attention to diabetes care at every clinical encounter regardless of whether diabetes was the primary reason for the visit. The registry facilitates iterative cycles of testing and intensification of treatment, resulting in improved metabolic control of diabetes.

EVIDENCE-DRIVEN POST-MI CARE AND BARRIERS PERCEIVED BY PRIMARY CARE PHYSICIANS. E.F. Tipton¹, M. Pisu², M. Abdolrasulnia¹, L.L. Casebeer¹, T. Houston¹, C.I. Kiefe¹. ¹University of Alabama at Birmingham, Birmingham, AL; ²Center for Outcomes and Effectiveness Research and Education, Birmingham, AL. (Tracking ID #134494)

BACKGROUND: Post-MI patients often present with multiple complex comorbidities. Despite guideline proliferation, diffusion of guidelines into practice is slow in overcoming barriers to optimal care. In preparation for a statewide randomized trial to improve post-MI care via guidelines, we conducted a survey of primary care physicians (PCPs) in adjacent states to assess 1) physician agreement with evidence-based guidelines and 2) the frequency of reported barriers to caring for post-MI patients.

METHODS: The sample included PCPs (family practice, general practice, and internal medicine) who practiced in Georgia, Florida, Mississippi, or Tennessee. A clinical vignette survey instrument was developed to assess prescribing practices and knowledge of guidelines related to post-MI patients with comorbidities. For example, a clinical scenario involving a post-MI with hypertension and diabetes is presented to the physician who is then asked to select the appropriate medications from a list and to specify the goal for blood pressure control. Each physician answered five questions involving three cases. In addition, physicians reported their perceptions of the relative importance of ten recognized patient and practice related barriers to optimal care of the post-MI patient and supplied demographic information about their practice. In October, 2003, the survey was sent by fax to the identified PCPs in the four states. Two follow-up surveys were sent to non-respondents. Physicians answering all five questions correctly were considered concordant with the evidence. Logistic regression was used to characterize responders who disagreed with the guidelines. We also report a frequency analysis of ten patient and practice related barriers.

RESULTS: Of the 1283 survey respondents, 79% were male, 40% were in solo practice, and the sample split evenly between internists and family practitioners. The average number of years since medical school was 20. Fifty-five percent of respondents answered one or more questions in a non-concordant manner. Family practice physicians were more likely to be non-concordant than internists (Odds ratio 1.3; 95%CI 1.04, 1.66) and PCPs in solo practice were more likely to be non-concordant than PCPs in group practice (Odds ratio 1.3; 95%CI 1.00, 1.71). The most frequently cited barrier to optimal care was the cost of medications (88%). In addition, poor patient compliance with advice for lifestyle modification (84%) and poor compliance with prescribed medications (75%) were commonly noted as barriers. Generally recognized barriers less frequently reported by PCPs, included lack of physician counseling skills concerning lifestyle modification (21%), lack of physician time for counseling (39%) and lack of educational materials (39%). 19% of respondents saw no barriers to optimal care.

CONCLUSIONS: Most PCPs recognized significant patient and practice related barriers to optimal care of the post-MI patient. Physicians were more likely to cite patient related barriers than barriers related to their practice situation. Over half of these PCPs indicated disagreement with some aspect of evidence-based guidelines, establishing a need for efforts to improve care that address the physicians' knowledge base as well as behaviors.

FACTORS ASSOCIATED WITH STEP-DOWNS IN ANALGESIC POTENCY AT HOSPITAL DISCHARGE. C. Moore¹, A.L. Siu², C.L. Maroney¹, D. Fischberg¹, A. Litke³, S. Silberzweig¹, R.S. Morrison¹. ¹Mount Sinai School of Medicine, New York, NY; ²Mount Sinai School of Medicine, Albany, NY; ³Mount Sinai Hospital, New York, NY. (Tracking ID #135908)

BACKGROUND: The inadequate treatment of pain may be especially high during the transition of care that occurs when patients with painful medical conditions are discharged from the hospital with prescriptions for pain medication that may be insufficient to provide adequate analgesia during the immediate post-discharge period.

METHODS: We selected a random sample of adult patients admitted to the medical and surgical services of a large urban academic medical center between April 1, 2001 and February 14, 2003 who experienced one or more episodes of moderate or severe pain within 3 days of discharge and whose last administered pain medication prior to discharge was an opioid. We abstracted the discharge pain medications documented in each patient's discharge summary and determined the last inpatient analgesic administered prior to discharge. We then categorized the potency of the hospital and discharge pain medication based on whether or not it was an opioid and, if so, whether or not it was schedule-2 as follows: 0=no analgesic medication, 1=non-opioid (e.g., acetaminophen), 2=non-schedule-2 opioid (e.g., acetaminophen with codeine), and 3=schedule-2 opioid (e.g., morphine). The outcome measured was a "step-down" in the potency of patients' analgesic medication at the time of discharge. A "step-down" was defined as the outpatient pain medication documented in the discharge summary being less potent than the last pain medication administered in the hospital prior to discharge. For example, if a patient was last administered morphine sulfate prior to discharge and was subsequently discharged on codeine with acetaminophen, there is a "step-down" in analgesic potency from 3 to 2. Multivariate logistic regression was used to examine the association between patient and hospital factors, and "step-downs".

RESULTS: 245 patients were included in the study (mean age: 47 years, 44% male, 58% White, 22% Black, and 16% Hispanic). Overall, 60% of patients had a

"step-down", usually from a schedule-2 opioid to a non-schedule-2 opioid (see Table). In multivariate analyses, Hispanic ethnicity (OR: 3.2, CI: 1.2-8.5), undergoing a surgical procedure during the hospitalization (OR: 2.4, CI: 1.2-4.7), and having a schedule-2 opioid as the last administered pain medication prior to discharge (OR: 6.7, CI: 1.02-43.3) were predictors of a "step-down". Conversely, receiving pain medication within 24-hours of discharge (OR: 0.3, CI: 0.10-0.97) was a predictor for not having a "step-down". The last recorded pain score prior to discharge, the number of chronic painful medical conditions, age, gender, insurance, and length of stay were not statistically significant predictors of a "step-down".

CONCLUSIONS: Hospitalized patients frequently have "step-downs" in their pain medications at discharge and Hispanics may be at increased risk of this occurring. Further research is needed to determine if "step-downs" are associated with poor pain outcomes in the immediate post-hospital period.

	Last Inpatient Pain Medication Taken Before Discharge, %(n)	Discharge Pain Medication, %(n)
None	0% (0)	0.4% (1)
Non-Opioid	0% (0)	4.1% (10)
Non-Schedule-2 Opioid	3.3% (8)	57.6% (141)
Schedule-2 Opioid	96.7% (237)	37.9% (93)

FOLLOW-UP OF OUTPATIENT TEST RESULTS: A SURVEY OF HOUSESTAFF PRACTICES AND PERCEPTIONS. J.J. Lin¹; C. Moore¹; A. Dunn¹. ¹Mount Sinai School of Medicine, New York, NY. (Tracking ID #134276)

BACKGROUND: Failure to follow up outpatient test results is a potential patient safety concern; however, data about how housestaff follow up on laboratory and diagnostic tests is sparse. Housestaff are generally only at their outpatient continuity practice site once a week and are therefore in the unique situation of having discontinuity imposed upon their patient care. We sought to assess housestaff practices and perceptions regarding the comprehensiveness and timeliness of following up outpatient tests and identified barriers to timely, consistent follow-up.

METHODS: Anonymous written self-administered survey of internal medicine housestaff at a large urban teaching hospital. The survey asked about housestaff practices for following up lab and radiologic test results, worries about inadequate follow-up, barriers to timely follow-up, and perception of clinical consequences of delayed or inadequate follow-up. Data were analyzed using univariate and multivariate analyses.

RESULTS: Seventy-five of the 111 eligible house staff (68%) responded. There was an equal distribution of respondents per post-graduate year of training (PGY) and respondents did not differ from nonrespondents with respect to gender or PGY level. 74% of respondents reported they were sometimes or frequently unable to follow up their patients' test results. Residents (PGY-2 and PGY-3) were more likely to respond that they were frequently unable to follow-up on patients' test results (23% vs 8%, $p=.033$) when compared with interns (PGY-1). Additionally, 43% were moderately or very worried that test results ordered on their patients will not be followed up and 65% stated that they have seen a patient's medical condition worsen due to a delay in test result follow up. Barriers to timely follow-up included lack of a reminder system (40%), difficulty accessing test results (24%), too many competing demands on time (27%), and uncertainty about who should follow up on results (16%).

CONCLUSIONS: Inadequate follow-up of outpatient test results was perceived to be a common problem with an adverse impact on patient safety. Whether these concerns reflect suboptimal care in real-world practice remains to be confirmed. Future interventions should focus on developing system-based reminder systems to ensure appropriate follow-up and/or ambulatory patient safety curricula for housestaff.

GLOBAL RATINGS OF HEALTHCARE BY PATIENTS ARE NOT ASSOCIATED WITH TECHNICAL QUALITY OF CARE. J.T. Chang¹; R.D. Hays¹; P.G. Shekelle²; C. Maclean¹; D.H. Solomon²; D.B. Reuben¹; C.P. Roth²; C.J. Kamber³; R.T. Young¹; N.S. Wenger¹. ¹University of California, Los Angeles, Los Angeles, CA; ²RAND, Santa Monica, CA; ³RAND, Arlington, VA. (Tracking ID #134497)

BACKGROUND: Patient global ratings of care are commonly used to assess healthcare. However, the extent to which these assessments correlate with technical quality of care is not well understood. Using a set of process of care quality assessment indicators developed by the Assessing Care of Vulnerable Elders (ACOVE) project, we examined the relationship between patient-reported global ratings of health care and technical quality of care.

METHODS: ACOVE developed a set of quality indicators (QIs) for vulnerable community-dwelling older adults at increased risk for functional decline and death. Patients' global ratings of healthcare received over the prior 12 months and ratings of provider communication were measured by interview using CAHPS® 2.0 survey items. Technical quality of care scores were calculated as the percentage of care processes received for all QIs for which each patient was eligible. Technical quality of care data for a 13-month period derived from medical

records and interviews, the global healthcare rating, and communication scores were available for 236 patients of the random sample of 420 vulnerable older adults selected from two senior managed care plans. We performed bivariate and multivariate regression analyses to identify significant correlates of the global healthcare rating.

RESULTS: Patients' mean global rating of healthcare was 8.9 on a 0-10 scale and mean rating of communication was 0.85 on a 0-1 scale. Patients received 55% of recommended care processes contained in the quality indicators. Global ratings of health care were highly correlated with reports about provider communication ($r=0.56$, $p<0.001$). In a multivariate regression model including patient and clinical factors, reports about communication, physical health status, and female gender were positively associated with global ratings of healthcare. Technical quality of care, age, vulnerability, and number of comorbidities were not significantly related to global ratings of care.

CONCLUSIONS: Vulnerable elders' global ratings of care should not be used as a marker of technical quality of care. Assessments of care should include both patient ratings and independent assessments of technical quality.

HOSPITAL FACTORS PREDICTING QUALITY OF CARE: HOW STRONG IS OUR SAFETY-NET? L.E. Goldman¹; R.A. Dudley¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #135501)

BACKGROUND: Recent initiatives to report publicly hospital quality of care have improved the available information about hospital performance. With these data, we can assess quality of care at safety net hospitals to determine if quality differences help explain observed health care disparities.

METHODS: We conducted a cross-sectional analysis of the National Voluntary Hospital Reporting Initiative (NVHRI) data, in which hospitals publicly report on evidence-based process measures for myocardial infarction (MI), congestive heart failure (CHF) and community acquired pneumonia (CAP). MI measures include aspirin (ASA) and beta blocker use on admission and at discharge, and ACE inhibitor use. The CHF indicators are assessment of cardiac function and ACE inhibitor use. The CAP measures are antibiotic timing, oxygenation assessment, and pneumovax administration. We linked the California Office of Statewide Health Planning and Development financial database to the NVHRI to obtain hospital information including teaching status, nursing staff ratios, emergency room visits, nursing hours, number of discharges by payer, and ownership type. For each quality indicator, we included all California acute care hospitals that reported at least 25 observations for the particular indicator on the NVHRI website. As 'safety-net hospitals' have been defined differently in the literature, we defined safety-net hospitals as public hospitals (county, city, district, or state owned) and subsequently as hospitals having 50% or more discharges with Medicaid or no insurance.

RESULTS: 306 hospitals participated in NVHRI, of which 53 were public hospitals. In the univariate analysis, public hospitals performed significantly ($p<0.05$) worse than private investor or non-profit hospitals in: prescribing ASA on arrival (93% versus 96%) and discharge (79% versus 95%), administering beta-blockers on arrival (84% versus 90%) and discharge (83% versus 91%). There were no significant differences between public hospitals and others in the use of ACE inhibitor for MI, ACE inhibitor or cardiac function assessment for CHF, or the three CAP measures. Compared to other institutions, hospitals with >50% of discharges with Medicaid or no insurance performed worse on the CAP measures antibiotic timing and oxygenation (57% versus 70% and 95% versus 99%, respectively). In multivariate analyses controlling for hospital ownership, teaching status, nursing staff ratios, emergency room visits, staffed beds, and nursing hours, the significant relationships between safety-net hospitals and quality of care persisted.

CONCLUSIONS: In California acute care hospitals publicly reporting greater than 25 cases per outcome in NVHRI, public hospitals appear have a lower performance on most MI quality process measures. Having >50% Medicaid or uninsured patients appears to be associated with lower CAP performance. Thus, by both definitions of safety-net, hospitals serving vulnerable patients report below average performance. Further studies are needed to understand whether reporting biases may influence these findings.

HOSPITAL INCIDENT REPORTS PROVIDE LIMITED INSIGHT INTO PATIENT SAFETY EVENTS AND THEIR CAUSES. T.K. Nuckols¹; D.S. Bell¹; H. Liu¹; L.H. Hilborne¹. ¹University of California, Los Angeles, Los Angeles, CA. (Tracking ID #135317)

BACKGROUND: Increasing hospital incident reporting is a priority nationwide. However, even large numbers of reports may be insufficient if they provide little insight into adverse events and their causes. This study attempted to evaluate the insights hospital incident reports provide as stand-alone documents, and to examine report characteristics associated with greater insight.

METHODS: This retrospective cohort study evaluated incident reports for 18,958 randomly selected patients discharged from two hospitals in 2001, over-sampling those with multiple discharges. From structured answers and free text, physician reviewers judged event type; preventability; whether reports addressed adverse outcomes or aberrant care processes; harm; professions of reporters and responsible providers; how explicitly reports identified responsible providers; and contributing system and human factors. Statistical tests involved chi-square, weighted for sampling.

RESULTS: We identified 2,292 reports. Nurses filed 87.9%, physicians 1.9%, other providers 8.9%, and unknown providers 1.3%. The four most common types were medication events (28.7%), operative events (14.7%), falls (13.5%), and miscellaneous events (15.8%); 59.1% were preventable, 32.2% of uncertain preventability, and 8.7% not preventable. Reports described only aberrant proc-

esses in 50.0%, only adverse outcomes in 35.3%, both in 10.3%, and neither in 4.4%. Reports indicated no medical harm occurred in 16.5% and omitted whether it occurred in 35.5%. When medical harm occurred, reporters indicated it was mild to moderate in 9.5%, and severe or fatal in 4.7%; harm was unknown or unspecified in the rest. For reports indicating possible preventability, nurses appeared partly responsible in 43.3%, physicians in 15.7%, other providers in 18.8%, and unidentifiable providers in 29.8%. These reports implicated system factors in 34.9% and human factors in 50.1% but only 26.3% provided enough detail to classify human factors. Report characteristics associated with a greater likelihood ($p < 0.0001$) of depicting preventability, system factors, and classifiable human factors included: explicitly addressing aberrant processes rather than only adverse outcomes; being filed by providers other than nurses; implicating providers of a different profession than the reporter; and naming the providers responsible.

CONCLUSIONS: Hospital incident reports appear to provide somewhat limited insight into patient safety events and their causes. Nurses file over 85% of them, and about 40% of reports address nursing events. In contrast, prior studies found nurses responsible for 1.7% of adverse events identifiable via chart review. Consequently, falls and miscellaneous events are common incidents despite being rare adverse events in previous studies. About a third of incident reports omit information on preventability or severity and only a minority address contributing system or specific human factors. Reports contain greater insight into preventability and contributing factors when reporters describe aberrant care processes. Reporters appear more likely to divulge such information when they are not nurses, are of a different profession than the providers responsible, or are willing to identify the responsible providers by name. Future research should examine strategies for increasing the insights provided by hospital incident reports.

HOSPITAL OVERCROWDING IS ASSOCIATED WITH INCREASED SEVEN DAY EMERGENCY ADMISSION MORTALITY: A NEW IMPERATIVE FOR PATIENT SAFETY. P. Sprivilis¹. ¹Brigham and Women's Hospital, Boston, MA. (Tracking ID #136031)

BACKGROUND: High hospital occupancy (90% or greater) causes emergency department dysfunction/overcrowding due to impaired access to inpatient beds for patients requiring emergency hospital admission. This study examines the relationship between hospital occupancy and the short term mortality of emergency hospital admissions.

METHODS: Analysis was performed to link health data and death records for 68437 emergency hospital admissions at three tertiary metropolitan hospitals in Perth, Western Australia between July-2000 and June-2003. Two, seven and 30 day mortality were evaluated after adjustment for age, diagnosis, and day of attendance. Additional variables assessed for inclusion in the model: gender, referral source, mode of arrival to ED, urgency (Australian triage scale), diagnosis (ICD-10), disposition, month, ED and hospital occupancy rates, waiting time, total ED time and hospital bed turnover.

RESULTS: High hospital occupancy on admission day was associated with a 32% (95%CI 10-60) relative increase in 7 day mortality from 1.5% (95%CI 1.3-1.7) for occupancy below 90% to 2.0% (95%CI 1.9-2.1%). After adjusting for potential confounders, the 7 day hazard ratio was 1.31 (95%CI 1.14-1.50). The 30 day cumulative hazard attributable to high occupancy was estimated to be 6

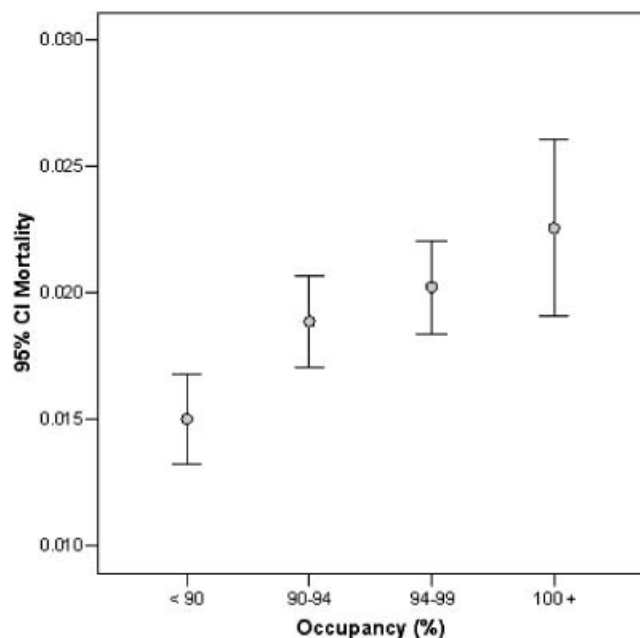


Figure 1 Association of 7 day mortality of emergency admissions with hospital occupancy

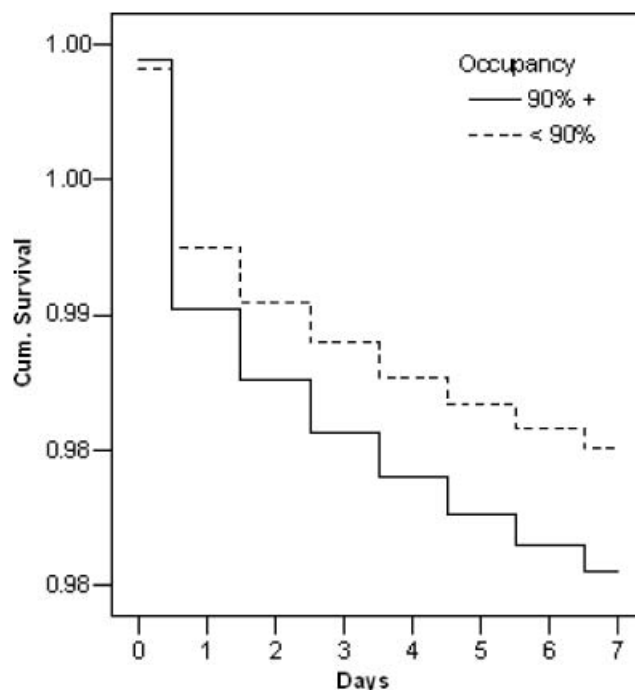


Figure 2 Cox regression 7 day survival function for emergency tertiary hospital admissions for day of admission hospital occupancies above and below 90%

deaths/1000 emergency admissions (95%CI 4-8). Patients who died following admission during periods of high occupancy had longer emergency department waiting times, longer emergency department lengths of stay and were more likely to occur when the emergency department had greater than 15% of cubicles occupied by other patients awaiting admission to an inpatient bed.

CONCLUSIONS: High hospital occupancy is associated with an increased mortality rate and may represent an immediately identifiable and remediable cause of fatal iatrogenic injury. Overnight emergency department inpatient boards are an indicator of high hospital occupancy. Reducing high hospital occupancy may improve outcomes for patients requiring hospital admission.

HOW SHOULD WE MONITOR THE QUALITY OF CARE?: A COMPARISON OF THREE METHODS TO ASSESS QUALITY DIFFERENCES ACROSS SITES OF CARE. E.A. Kerr¹; T.P. Hofer¹; R.A. Hayward¹; M.M. Hogan²; J.L. Adams³; S. Asch⁴. ¹University of Michigan and VA Ann Arbor Healthcare System, Ann Arbor, MI; ²VA Ann Arbor Healthcare System, Ann Arbor, MI; ³RAND, Santa Monica, CA; ⁴VA Greater Los Angeles Healthcare System, UCLA and RAND, Los Angeles, CA. (Tracking ID #134420)

BACKGROUND: Many performance measurement (PM) systems, including those endorsed by VA and the NCQA, attempt to distinguish quality provided by health plans or facilities. Despite their expense, we know little about how different methods of PM vary in their ability to discriminate among facilities. We therefore examined the quality of care in 26 VA facilities using three different standard PM approaches to determine how closely the three agreed on quality and how many patient charts must be reviewed to produce reliable facility-level quality profiles.

METHODS: We examined processes of care for 621 patients in 2 VISNs using three different PM tools: a focused explicit tool (41 measures for 6 conditions/prevention), a global explicit tool (376 measures for 26 conditions/prevention), and a structured implicit review instrument (physician-rated care). Trained nurse abstractors and physicians reviewed all medical records. Correlations between facility-level scores from the three PM tools were adjusted for the patient-level reliability of each tool using multilevel regression models.

RESULTS: Intercorrelations of facility-level scores were generally high across all three tools. For example, overall quality was correlated at 0.5 between the implicit and global tools, 0.34 between implicit and focused, and 0.72 between global and focused. Correlations were even higher for conditions with large evidence base (e.g., diabetes, preventive care). However, only 26 records per facility would be needed to produce an overall quality estimate with 0.8 reliability using the implicit tool, while 70 records would be needed using the focused tool and 290 using the global tool.

CONCLUSIONS: To our knowledge, this is the largest study that has compared three different methods of quality assessment for patients with chronic conditions. We found high agreement in facility-level scores across the three PM tools for most clinical areas, indicating that all three tools were tapping into a similar quality domain. However, the implicit tool reliably discriminated quality among facilities with many fewer records than the two explicit tools. Although often

criticized for its low reliability in distinguishing between individual cases, structured implicit review may be an excellent tool for detecting quality differences across facilities. While explicit review tools have other advantages (e.g., more actionable results), the number, type and weighting of measures within the tools may influence their ability to reliably and efficiently classify facilities on quality.

IMPLEMENTING A VOLUNTARY PRESCRIBING ERROR REPORTING SYSTEM IN OUTPATIENT PRIMARY CARE PRACTICES. A.G. Kennedy¹; B. Littenberg¹. ¹University of Vermont, Burlington, VT. (Tracking ID #133437)

BACKGROUND: Approximately 2% of outpatient prescriptions contain prescribing problems. If undetected by community pharmacists, prescribing errors may cause patient harm. At present, detected errors are expensive to correct, usually involving pharmacist, prescriber, and office staff time. Interventions to reduce outpatient prescribing error need to be developed and tested. However, there first needs to be a system to collect errors, enabling a better understanding of the scope, type, and causes of outpatient prescribing errors. Our objectives were to design a voluntary outpatient prescribing error reporting system for primary care practices, implement the system in a sample of practices, analyze the reports, and provide feedback to prescribers about reported errors.

METHODS: We asked triage nurses and office staff to submit copies of all phone or fax communications with community pharmacists about prescribing errors for inclusion in our database. Reports could be telephoned, mailed, or collected weekly by a courier. We did not specify a standard reporting form because we wished to minimize office intrusion and encourage participation. Although the reporting system was designed to be a job function of the triage nurses and office staff, prescribers were encouraged to report their own errors.

RESULTS: Seven outpatient primary care practices agreed to participate since enrollment began in June 2004. All 7 practices have contributed reports and the majority of practices have submitted reports via courier. To date, 143 prescription error reports have been submitted to the database. Only 6 reports were contributed directly by prescribers. Preliminary analysis of the reports reveals 43 omission errors, 52 commission errors, 17 clarifications due to illegibility, 10 pharmacy errors, and 14 pharmacist clarifications that were not errors. 7 reports could not be categorized due to missing data. Reports most frequently concerned antidepressants (33/143) and antihypertensives (22/143). 43 reports involved problems with strength, including 11 prescriptions written for strengths not commercially available. At least 10 reports described errors with the potential to harm patients if not detected or corrected by pharmacists prior to dispensing.

CONCLUSIONS: Implementing a voluntary outpatient prescribing error reporting system using triage nurses and office staff is feasible. Alternative methods for submitting reports need to be explored, as courier services are expensive. Striking a balance between data collection and office burden requires continued assessment. Many of the errors described in the reports can be prevented using inexpensive, available, and accessible tools. Deeper analysis of specific reports and the development of appropriate mechanisms for feedback to prescribers are in progress.

INCREASING MEDICATION LIST ACCURACY IN THE ELECTRONIC HEALTH RECORD. R. Tsuniko¹; L. Volk¹; L. Pizziferri¹; M. Lippincott¹; A. Kittler¹; T. Wang¹; J. Wald¹; D.W. Bates¹. ¹Partners HealthCare System, Inc., Wellesley, MA. (Tracking ID #136170)

BACKGROUND: Patients often take different medications from those that their physicians believe they are taking. Many recent studies using techniques such as home visits and pill counts demonstrate that patients take different drugs, different doses and regimens than those recorded by the physician. Patients often take additional medications or stop medications without physicians' approval. For this and other reasons, medication lists in electronic health records (EHRs) are often inaccurate, limiting the usefulness of the information, and making it harder to provide appropriate decision support. Secure web-based patient portals may provide an opportunity to improve communication between patients and providers about medications. In this study, we assessed the current state of medication lists in an EHR and the impact of a secure, web-based patient portal (Patient Gateway) by comparing medication lists to patient reported information between Patient Gateway (PG) users and non-users.

METHODS: A paper survey was sent to a sample of PG users and non-users within one practice. The survey was pre-populated from patients' EHRs with the following information on each medication: name, regimen, frequency, and dose. Participants were asked to review that information and indicate whether they took each medication the way it was listed or, whether they had stopped taking this medication. If they were taking a different regimen, patients were asked to specify the new dose, number of times per day, and how they were taking it. When a patient reported any changes, researchers entered this information into the EHR in note form and e-mailed a clinical message to the primary care physician to notify them.

RESULTS: PG users and non-users had an average of 4.7 and 3 prescription medications identified as active in their EHR, respectively. PG users reported 53% of those medications to be accurate, while non-PG users reported 63% as accurate (Table 1). A greater percentage of PG users (31%) reported they had stopped taking a medication listed compared to non-users (22%). PG users and non-users were more similar in the percent of medications for which they were taking a different regimen (14% for users, 12% for non-users). Comparable percentages of PG and non-PG users reported taking additions prescription and non-prescription medications.

CONCLUSIONS: Medications listed in participants' EHRs were consistent with patient report only a little over half the time. Being enrolled in a secure web-

Value	PG Enrollees (n=108)	Non-PG Enrollees (n=55)	Total (n=163)
LMR Medications			
Average number of LMR	4.7	3	4.1
% medications - Yes, accurate. Taking with same regimen	53% (avg 2.5 meds)	63% (avg 2.0 meds)	56% (avg 2.3 meds)
% medications - No, not accurate. Stopped taking them	31% (avg 1.4 meds)	22% (avg 0.7 meds)	28% (avg 1.2 meds)
% medications - No, not accurate. Taking with different regimen	14% (avg 0.7 meds)	12% (avg 0.4 meds)	14% (avg 0.6 meds)
Additional Medications			
% patients with additional prescriptions	62%	60%	61%
Average number of additional prescriptions per patient	0.47	0.53	0.49
% patients with additional OTC / herbs	49%	45%	48%
Average number of additional OTC / herbs per patient	1.3	0.96	1.19

Table 1. Medication List Accuracy by Patient Gateway Enrollment Status

based patient portal did not appear to improve the medication list accuracy even though patients were able to review their EHR medication list. Even though medication information from patients was provided electronically to physicians, it may not have been reviewed or incorporated into the medication list by physicians outside of visits.

INTERNAL MEDICINE RESIDENTS' PERSPECTIVES ON CONTRIBUTORS TO PATIENT CARE ERRORS. K.E. Fletcher¹; V. Parekh²; L. Halasyamani³; S.R. Kaufman⁴; S.K. Saint⁴. ¹Milwaukee VAMC, Milwaukee, WI; ²University of Michigan, Ann Arbor, MI; ³St. Joseph Mercy Medical Center, Ann Arbor, MI; ⁴Ann Arbor VAMC, Ann Arbor, MI. (Tracking ID #132810)

BACKGROUND: The "swiss cheese model" of systems accidents has been commonly applied to patient safety, implying that many "holes" must align before an adverse event occurs. The ACGME work hour rules served to plug one of these holes by reducing fatigue in residents caring for patients. We conducted focus groups with internal medicine residents in order to determine the impact of the ACGME rules on patient safety from the residents' perspectives.

METHODS: We conducted three focus groups with internal medicine residents: two at a large university-based program that included both a university and a VA hospital and one at a community-based program. The focus groups were audiotaped and transcribed verbatim. Using grounded theory to analyze the transcripts, we worked collaboratively to design a coding system based on the themes identified in the transcripts. Two of us independently applied the coding scheme to all three transcripts. We also reviewed 10% of each transcript together to establish a rough estimate of agreement. Axial coding was accomplished through discussions of the interrelationships of the codes.

RESULTS: A total of 22 residents participated (interns through third year residents), 14 women and 8 men. Fourteen were from the university and 8 were from the community program. Agreement between the coders was 49%, but all differences were resolved through discussion with no changes to the coding scheme. A model of contributors to patient care errors emerged that included the following as direct contributors: fatigue, entropy (or the overall chaos in the system), workload, unfamiliarity with patients, inexperience, sign-outs and having the work hour rules become the goal. Many of these facets coalesced to cause errors, especially during the midnight to 6AM time period. Attending supervision was rarely mentioned, even though resident inexperience was often discussed as a contributor. Residents described the tension between their own fatigue and continuity of patient care. Residents were also overwhelmingly happy with the positive effect that new work hour rules had on their personal sense of well-being, but were uncertain whether the rules would actually improve patient safety. **CONCLUSIONS:** Internal medicine residents identified several existing holes that contribute to patient care errors. The sense of well-being felt by the residents as a consequence of the ACGME work hour rules mitigated the effects of fatigue as a contributor to patient care errors. However, some strategies used to comply with these rules may increase problems in other areas by decreasing the number of supervising residents available and by increasing the workload on the residents who are in the hospital, especially at night.

MEASURES OF DRUG PRESCRIBING QUALITY FREQUENTLY DISAGREE. M. Steinman¹; P. Kaboli²; C. Landefeld¹; G.E. Rosenthal². ¹San Francisco VA Medical Center, San Francisco, CA; ²Iowa City VA Medical Center, Iowa City, IA. (Tracking ID #133731)

BACKGROUND: Efforts to assess the quality of care for elders depend on measures of drug prescribing quality. Several measures of prescribing are commonly used, but are often applied independently and have not been compared to one another. In this study, we determined the concordance between three common-

ly-used measures of prescribing quality by applying and comparing them in the same group of elderly patients.

METHODS: Using baseline data from the VA Enhanced Pharmacy Outpatient Clinic Study, we applied measures of prescribing quality to 700 drugs taken by 86 patients age 65 and older. These measures included: (1) the criteria of Beers et al, (2) polypharmacy (use of ≥ 9 medications), and (3) the Medication Appropriateness Index (MAI), which assesses 10 attributes of prescribing in a structured implicit review. For the MAI, we used a score of ≥ 3 to indicate an inappropriate prescription. The Beers and MAI criteria were first applied at the level of the drug and then at the level of the patient, such that any patient taking one or more problem drugs was classified as having inappropriate prescribing. Polypharmacy was applied only at the level of the patient. Vitamins and minerals, herbal drugs, and topical agents were excluded from all analyses.

RESULTS: Mean age was 74.6 years (± 5.1), $\geq 98\%$ were white and male, and a median of 8 drugs were used by each patient (IQR range 6,9). The proportion of drugs judged inappropriate was 7% by Beers criteria and 20% by MAI. Beers criteria and MAI agreed for 80% of drugs (inappropriate in 3.3% and appropriate in 76.3%), but this agreement was little more than expected by chance (κ , 0.16). When quality measures were applied at the level of the patient, the proportion of subjects with one or more inappropriate drugs was 37% by Beers criteria and 76% by MAI; 40% had polypharmacy of ≥ 9 drugs. Prescribing was classified as inappropriate by all three metrics in 17% of patients and as appropriate by all three metrics in 15%; the remaining 68% of patients received discordant quality scores. Thus, in only 32% of patients did all three metrics agree that drug prescribing was appropriate or inappropriate, a level of agreement slightly better than chance (3-way κ , .10). When the three patient-level measures were compared in pairwise fashion, 40–50% of patients received discordant results in each possible comparison, an agreement that remained little better than chance (κ for each pairwise combination, .12 to .18).

CONCLUSIONS: Commonly-used measures of drug prescribing quality yield widely discordant results and correlate poorly with one another. Researchers and the public need be aware of the limitations of these metrics when studying prescribing quality, and better measures are needed to bridge the differences between these metrics.

NATIONAL PATTERNS OF COX-2 INHIBITOR USE. M. Steinman¹; K. Kovinsky¹. ¹San Francisco VA Medical Center, San Francisco, CA. (Tracking ID #133296)

BACKGROUND: Until the spate of recent safety concerns, COX-2 inhibitors were widely touted for their lower risk of gastrointestinal complications. However, rates of NSAID-associated bleeding are low in younger adults, while the elders who may benefit most from lower bleeding risk are at highest risk of potential cardiovascular toxicity. We investigated national trends in COX-2 inhibitor use, with a particular emphasis on rates of use in different age groups.

METHODS: We used the National Ambulatory Medical Care Survey (NAMCS), a nationally-representative survey of outpatient visits in community-based settings. Medications, diagnoses, and other patient data were recorded by the treating clinician or office staff on a standard form at or near the time of the visit. Using data for adults age 18 and older from 1998 (baseline year) through 2002, we calculated national estimates of COX-2 inhibitor use using sampling weights to adjust for the probability of sampling and design weights to adjust for the complex survey design.

RESULTS: Between 1998 and 2002, the prevalence of recorded NSAID use ranged from 6.7% to 8.7% of adult visits ($P=.04$). Over this same period, COX-2 inhibitor use rose from 0% of recorded NSAID uses in 1998 to 52% of recorded uses in 2002 ($P<.001$). Use of proton pump inhibitors or misoprostol remained relatively low throughout, accompanying 6.6% of recorded NSAID uses in 1998 and 3.5% in 2002 ($P=.01$). COX-2 inhibitor use rose sharply in all age groups, by 2002 accounting for 33% of recorded NSAID uses in 18–44 year olds, 54% in 45–64 year olds, and 66% in elders age 65 and older ($P<.001$). On multivariable analysis of NSAID-users in 2001–02 that adjusted for patient and provider characteristics, patient age remained the strongest predictor of COX-2 inhibitor use (adjusted OR 2.1 for age 45–64, 95% CI 1.6–2.8; OR 3.3 for age 65 and over, 95% CI 2.3–4.8, compared with age 18–44). During this same period, COX-2 inhibitors accounted for 86% of recorded NSAID uses in elders with documented ischemic heart or cerebrovascular disease, compared with 65% in elders without documented vascular disease ($P<.001$).

CONCLUSIONS: COX-2 inhibitor use has risen rapidly in all age groups, particularly elders age 65 and older, and appears to have added to rather than substituted for older gastroprotective measures such as proton pump inhibitors and misoprostol. While recent safety concerns are likely to impact current prescribing habits, our data highlight the need for more stringent drug utilization evaluation, both in younger patients for whom the risk of complications from NSAID use are low, and for older ones who may experience both the highest benefit and risk from COX-2 inhibitors.

PATIENT INVOLVEMENT IMPROVES BLOOD PRESSURE CONTROL IN VETERANS: RESULTS FROM A RANDOMIZED CONTROLLED TRIAL. C.L. Rounie¹; T.A. Elasy¹; R. Greevy²; M.R. Griffin¹; V. Alvarez³; R.S. Dittus¹; W.J. Stone¹; K.A. Wallston²; T. Speroff¹. ¹Veterans Affairs Tennessee Valley Healthcare System, Vanderbilt University, Nashville, TN; ²Vanderbilt University, Nashville, TN; ³Veterans Affairs- VISN9, Nashville, TN. (Tracking ID #135593)

BACKGROUND: Hypertension is one of the most common reasons for patients to visit their primary care physician. Despite guidelines for care of hypertension only 34% of the population has a blood pressure (BP) at goal. The purpose of this

quality improvement project was to compare the relative contribution of provider, system, and patient level interventions in the treatment of patients with uncontrolled hypertension.

METHODS: We identified 4667 hypertensive patients (ICD-9 401.1 or 401.9) who were prescribed one antihypertensive medication, between January and June 2004 in the TN Valley VA-VISN database. 1847 (39.6%) of these patients had 2 or more BPs in that 6 month period that were $> 140/90$. The final cohort consisted of 1351 patients, cared for by 182 providers who consented to have their charts reviewed. Primary care providers were randomized to one of 3 groups: 1) Provider education- 54 providers received an e-mail with a web-based link to the JNC 7 guidelines; 2) Provider education and System alerts- 62 providers received #1 and a one time patient-specific chart alert with a summary of JNC 7 recommendations, the patient's last 3 recorded BP readings, and a suggestion to re-evaluate the antihypertensive regimen 3) Provider education, System Alerts and Patient education- 66 providers received #2 and their eligible patients were sent a letter advocating compliance with antihypertensive medications, behavioral strategies to improve BP control, and a recommendation to discuss medication changes with their provider. Changes to the regimen of antihypertensive medications and improvement in blood pressure (SBP < 140 or DBP < 90) were primary outcomes. The data were analyzed using Stata 7.0 (College Station, Texas).

RESULTS: The mean BP pre-intervention of the 1351 patients was 157/83 with no differences across the three groups ($p=0.6$). At 6 months post intervention, 983 patients (73%) had at least 1 recorded BP measurement and there was no difference in the frequency of BP measurement between the 3 groups. The providers randomized to group 3, who received system alerts, provider education and patient education, had better post intervention BP control (137/75), compared to either group 2 - system alerts and provider education (143/76) or group 1 - provider education alone (143/78) ($p<0.05$). More patients in group 3, reached the systolic goals of < 140 , compared to patients in either group 1 or 2 (53% vs 36% vs 45%) respectively ($p<0.001$). There was, however, no difference in antihypertensive regimen intensification between the three groups. 404 patients (30%) had intensification of their antihypertensive regimen: providers intensified medications for 107 (32%) patients in the group 1, 159 patients (29%) in group 2, and 138 (29%) patients in group 3 ($p=0.9$).

CONCLUSIONS: These data suggest that involvement of the patient as key stakeholders in the management of hypertension leads to improved BP outcomes compared to provider directed interventions alone. Patient directed hypertension management resulted in improved BP control without a differential change in medication regimen compared to other experimental conditions. The mechanisms by which the patient education group realized improved BP control (increased adherence vs other behavioral changes) are the subject of ongoing investigation.

PATIENT LOYALTY WITHIN A PRIMARY CARE PRACTICE NETWORK: A POPULATION-BASED APPROACH TO QUALITY ASSESSMENT AND IMPROVEMENT. S.J. Atlas¹; R.W. Grant²; Y. Chang¹; T.G. Ferris¹; M.J. Barry¹. ¹Massachusetts General Hospital, Boston, MA; ²Massachusetts General Hospital, Boston, MA. (Tracking ID #134013)

BACKGROUND: In the United States, major deficiencies in quality of care persist despite considerable resources devoted to health care. Quality improvement (QI) strategies in primary care practice networks (PCPNs) must address multiple barriers that arise at the clinical systems, individual provider, and patient levels. To be successful, such strategies require the ability to correctly identify all eligible patients cared for in a given PCPN. Because primary care providers (PCPs) are the key clinical decision makers in a PCPN, we sought to examine the potential impact of categorizing patients seen based on loyalty to a specific PCP. For patients without a strong linkage to any PCP but receiving care in a specific practice, we also examined the value of defining patient loyalty to that practice.

METHODS: A recursive partitioning algorithm with logistic regression models, previously developed and validated using PCP classification of his/her patients as the gold standard, was used to identify PCP loyal patients in a hospital-affiliated PCPN. For patients without a strong linkage to any PCP, a local consensus approach was used to identify patients loyal to a practice based upon visit frequency and patient age. The result was all patients with at least 1 outpatient visit in the PCPN over a 3-year period being categorized as loyal to a specific provider, a specific practice or neither. A total of 161,031 patients were identified. Patients were excluded if they had a listed PCP who was not in one of the 13 PCPN practices ($n=7377$). Patient characteristics and preventive cancer screening test rates were compared for patients classified as PCP loyal, practice loyal, or loyal to neither using multiple regression models controlling for clustering among PCPs and practices.

RESULTS: Among 153,697 eligible patients, 97,266 (63%) were classified as PCP loyal, 41,006 (27%) as practice loyal, and 15,425 (10%) as loyal to neither. The proportion of patient loyal to a specific provider, the practice or neither varied significantly among practices (for example, the proportion loyal to a specific provider varied among practices from 52% to 74.5%, $p<=.001$). Patient characteristics varied significantly according to provider and practice loyal category. Cancer screening test rates were significantly higher for patients identified as provider loyal compared to patients identified as practice loyal or loyal to neither. For example, mammography screening rates were 83%, 64.4% and 26.3% for eligible women 52–69 years old who were PCP loyal, practice loyal or loyal to neither, respectively ($p<=.001$).

CONCLUSIONS: Many patients cared for in a large PCPN could not be classified as being loyal to a specific PCP. Patients without close linkage to a PCP were less likely to be white, more likely to have Medicaid, free care or self pay insurance, and were less likely to have undergone recommended cancer screening tests. Use of patient loyalty classification in a PCPN when designing QI interventions may help identify those for whom the PCP may be the most effective catalyst for

change, and those requiring alternative means—such as the use of centralized practice managers.

PHYSICIANS' REASONS FOR UNDERUSE OF EFFICACIOUS CANCER TREATMENT. N.A. Bickell¹; F. Lepar¹; S. Cuellar¹; H. Leventhal². ¹Mount Sinai School of Medicine, New York, NY; ²Rutgers, The State University of New Jersey, New Brunswick, NJ. (Tracking ID #133457)

BACKGROUND: There is both variability and racial disparity in underuse of efficacious adjuvant breast cancer treatments. Previously, we identified 680 women with stage 1 or 2 breast cancer of whom 151 experienced underuse; either they did not receive radiotherapy following lumpectomy, tamoxifen for estrogen receptor positive or chemotherapy for estrogen receptor negative tumors. Surgeons usually provide care and referrals for adjuvant therapy to women with breast cancer. We undertook this study to understand why the 151 patients who could have benefited from adjuvant therapy did not receive appropriate treatment by surveying their surgeons.

METHODS: 37 surgeons operated on the 680 women at 6 NYC hospitals in 1999-2000. We asked surgeons about the care they provided and each patient's specific circumstance of underuse. Two surgeons were unreachable leaving a total of 35 surgeons and 146 patients (97%) for evaluation. We conducted interviews assessing the reason for underuse while each surgeon reviewed their patient's chart.

RESULTS: Our survey revealed 20 women had been treated resulting in a final patient population of 126 women experiencing 159 episodes of underuse: 62 with no radiotherapy, 68 with no hormonal therapy and 29 with no chemotherapy. The reasons for underuse were: system failure - episodes for which the physician thought the patient should be treated and did not know she wasn't (36%); patient refusal (19%); lack of patient follow-up (16%); advanced patient age or frailty (12%); presence of comorbid or concomitant condition precluding adjuvant therapy (10%); small tumor size (4%); or lack of physician knowledge of appropriate care (3%). System failures were more common among minority than white patients (42% vs 24%; $p < .05$) as was patient refusal of adjuvant therapies (26% vs 12%; $p < .05$). Nonminority women were more likely not to follow-up (26% vs 6%; $p < .01$).

CONCLUSIONS: System failures are a common cause of underuse of adjuvant therapy for early-stage breast cancer. Among breast cancer patients, minority women may be more likely to experience underuse caused by system failures, a potentially remediable systems-level problem.

PREDICTORS OF TIMELY FOLLOW-UP OF SEVERELY ELEVATED BLOOD POTASSIUMS IN AN AMBULATORY SETTING. C.R. Moore¹; J.J. Lin¹. ¹Mount Sinai School of Medicine, New York, NY. (Tracking ID #135876)

BACKGROUND: Ensuring that outpatients receive timely follow-up for abnormal test results is a growing patient safety concern. Because many medications prescribed in the ambulatory setting can potentially cause severe hyperkalemia which can lead to fatal arrhythmias, we sought to investigate the patient and system factors that predict timely follow up of severely elevated blood potassium (K+) levels detected in an ambulatory setting.

METHODS: We reviewed the medical records of all patients seen at the adult primary care practice of a large urban teaching hospital who had at least one blood K+ result of > 5.8 mEq/L between September 1, 2003 and August 31, 2004. We used multivariable logistic regression to examine the relationship between patient and system factors (including patient age, gender, ethnicity, insurance, blood K+ > 6.0 , specimen hemolysis, day of week results are first posted on the computerized lab system, and provider type) and timely follow-up of severely elevated K+ levels (defined as a repeat K+ within 1 week).

RESULTS: There were 168 episodes of severely elevated potassium levels in 139 patients. The mean patient age was 61 years. 60% of patients were female, 44% Hispanic, 40% Black, 6% White, and 10% of other race. Hypertension (80%), diabetes (55%), renal insufficiency (37%), and congestive heart failure (16%) were the most common medical diagnoses. In 73% of episodes, patients were taking medications that can cause hyperkalemia (ACE inhibitor: 60%, NSAID or Cox-2: 15%, A2RB: 10%). The median K+ was 6.1 mEq/L (range: 5.8-24.8), 63% of K+s were > 6.0 , and 40% of specimens were hemolyzed. The median time that a repeat K+ was performed was 6 days (range: 0-445) and in 41% of episodes, patients did not have repeat K+ s within a week. Multivariate analyses showed that if the K+ was > 6.0 , the likelihood that a repeat blood test was performed within a week was increased (OR=3.45, $P=.001$). Conversely, a K+ result posted on the computerized lab system on Friday decreased the likelihood that a repeat blood test was performed within a week (OR=0.25, $P=.005$). Specimen hemolysis, as well as age, gender, ethnicity, insurance, and provider type were not associated with the likelihood of having a repeat blood test within a week.

CONCLUSIONS: Timely follow-up of severely elevated K+ results in the ambulatory setting may be problematic, particularly when results return just before the weekend. Given the potentially fatal consequences of severe hyperkalemia, it is crucial that we develop better systems to assure timely follow-up of abnormal test results in the outpatient setting. Further research is needed to determine if the results of this study are generalizable to other institutions and to other commonly ordered outpatient tests.

QUALITY OF ANTICOAGULATION MANAGEMENT IN A LARGE PRIMARY CARE PRACTICE: PHYSICIAN ADHERENCE TO GUIDELINES. D.L. Schwab¹; S. Crocetti²; M.J. McGuire¹. ¹Johns Hopkins University, Baltimore, MD; ²Johns Hopkins Community Physicians, Baltimore, MD. (Tracking ID #135743)

BACKGROUND: Treatment with vitamin K antagonists presents a challenge to clinical practice due to complications when adherence to a goal INR is not achieved. Various paradigms for monitoring patients exist including anticoagulation management services (AMS) and usual care with physicians adjusting doses (UC). Each of these may employ point-of-care (POC) testing or central lab (CL) testing. Assessing the quality of anticoagulation management is necessary to assure goals are met. Time in therapeutic range (TTR) is often used as a quality measure although many system issues may cause suboptimal TTRs. It is generally agreed that use of guidelines, such as those developed by the American College of Chest Physicians (ACCP), improves outcomes.

METHODS: We performed a quality review of anticoagulation management in a 15-office medical group with a population of > 950 anticoagulation patients. We evaluated dose adjustment and follow-up based on adherence to a practice guideline. This tool has been in place in these practices since 1997, is updated annually to reflect ACCP guidelines, and provides forms and a dosing algorithm. It is available to all practitioners through our intranet, and is used by our practice's AMS. We sampled 55 patients and 844 INRs; representing 4-5 patients from each practice, and compared practices using nurse-run AMS vs. usual care with POC monitoring (UC-POC) vs. usual care with CL testing (UC-CL).

RESULTS: Despite availability of an evidence-based guideline, significant variability existed in management of anticoagulation. Of 12 practices evaluated, 3 used AMS, 2 used UC-POC, and 7 used UC-CL. Results for studied indicators for AMS vs. UC-POC vs. UC-CL were: % INRs in goal/TTR (60%, 53%, 46%), % INR within 10% goal (77%, 73%, 66%), % patients informed of recommended dosing within 2 days of INR (100%, 100%, 74%), indication documented (96% in all groups), goal INR documented (100%, 86%, 85%), goal consistent with guidelines (94%, 57%, 81%). In UC practices, 9 of 19 physicians had quality issues identified; none of our AMS practices had quality issues. The physician quality issues included "medical inertia" (repeated failure to modify doses for persistently non-therapeutic INRs), failure to repeat INRs in a timely fashion after a dose change, and dose changes not consistent with algorithm. AMS adhered to the guidelines for more than 90% of the INRs evaluated and documented reasons for deviation. Physicians often did not adhere to guidelines and often did not provide documentation to explain non-adherence.

CONCLUSIONS: In a large group practice which adopted an anticoagulation management guideline, we found significant variability in implementation of the guideline and different styles of anticoagulation management. Nurse-managed AMS obtained better TTR than physician managed anticoagulation (UC). We noted a significant amount of clinical inertia in UC-CL practices. While the small sample size did not allow for statistical analysis, this study provides insight into coagulation management and quality issues in primary care practices. This study confirms previous observations that AMS clinics provide better TTR than UC and highlights the need for (1) exploring physician barriers to guideline implementation, (2) understanding reasons for clinical inertia among physicians.

QUALITY OF IN-HOSPITAL CARE AND EVOLUTION OF N-TERMINAL PRO-BNP LEVELS AMONG PATIENTS HOSPITALIZED WITH HEART FAILURE. M.P. Kossovsky¹; M. Louis-Simonet¹; P. Sigaud¹; P.F. Keller²; F. Mach²; T.V. Perneger³; J.M. Gaspoz¹. ¹Department of Internal Medicine and Groupe de Recherche et d'Analyse en Systèmes et Soins Hospitaliers (GRASSH), Geneva University Hospitals, Geneva; ²Geneva University Hospitals, Geneva; ³Quality of Care Unit and Groupe de Recherche et d'Analyses en Systèmes et Soins Hospitaliers (GRASSH), Geneva. (Tracking ID #132487)

BACKGROUND: Elevated serum levels of the N-terminal precursor of Brain Natriuretic Peptide (pro-BNP) have been recognized as predictors of poor outcomes in patients with congestive heart failure (CHF): CHF patients discharged with an elevated level of serum pro-BNP are more likely to be readmitted or to die within 1 year. We evaluated the relationship between compliance with explicit criteria of quality of hospital care for CHF and the evolution of serum pro-BNP during the hospital stay.

METHODS: A set of explicit quality of care criteria for CHF was adapted from Ashton C. and al. (Ashton et al., Med. Care 1994;32:755-70) and updated according to the most recent guidelines published by the American Heart Association and the European Society of Cardiology. The level of adherence to these criteria was evaluated by retrospective chart review after hospital discharge. Three scores, assessing admission work-up (admission score), evaluation and treatment during the stay (treatment score) and readiness for discharge (discharge score), were created, measuring the proportion of fulfilled criteria. Quality of care was deemed acceptable if at least 75% of the criteria were fulfilled. Serum level of pro-BNP was measured within 48 hours of admission and at discharge.

RESULTS: Between November 2001 and November 2002, 322 patients were admitted to the general internal medicine wards with symptoms and signs of CHF. A diagnosis of CHF was not confirmed in 36 patients, medical charts could not be retrieved in 10 patients, and 9 patients died during hospital stay. Thus, the relationship between quality of hospital care criteria and pro-BNP evolution could be analyzed in 268 patients. Acceptable level of quality of care was attained in 32 patients (12%) for admission, 201 (75%) for evaluation and treatment and 240 (90%) for discharge. There was a significantly larger decrease in serum pro-BNP levels in patients for which an acceptable level of quality was attained when compared with patients with a lower level for admission score core (-692 vs. -250 pmol/ml, $p=0.04$) and for discharge score (-360 vs. +155 pmol/ml; $p=0.03$) but not for evaluation and treatment score (-341 vs. -189 pmol/ml; $p=0.30$).

CONCLUSIONS: Compliance with explicit quality of care criteria upon hospital admission and at discharge in patients with CHF was associated with a favorable decline during hospitalization of a biological marker known to predict poor outcomes.

SURVEILLANCE TESTING AFTER EARLY-STAGE BREAST CANCER. N.L. Keating¹; M.B. Landrum¹; E. Guadagnoli¹; E.P. Winer²; J.Z. Ayanian¹. ¹Harvard Medical School, Boston, MA; ²Dana Farber Cancer Institute, Boston, MA. (Tracking ID #135165)

BACKGROUND: National guidelines in the U.S. recommend that women with a history of breast cancer undergo annual mammography but advise against routine bone scans, chest x-rays, tumor markers, and computed tomography (CT) or ultrasonography (US). We compared use of these tests over a 3-year period in a matched cohort of early-stage breast cancer survivors and women who never had breast cancer.

METHODS: We used SEER-Medicare data from 11 U.S. regions to identify a population-based cohort of 37,452 women aged >65 who were diagnosed with stage I/II breast cancer in 1992-1999 and underwent primary surgical therapy who were followed through 2001 or until 6 months before death or recurrence. We matched these women to 37,452 women in the 5% Medicare file with no history of breast cancer by age, geographic region, and comorbidity. We assessed the effect of a breast cancer diagnosis on the relative rate (RR) of mammography, bone scans, tumor markers, chest x-rays, and abdominal and chest CT/US in the 3 years after cancer diagnosis (or the same date for non-cancer patients). Among breast cancer patients, we also explored the association between visits with cancer specialists and receipt of these tests using logistic regression with GEE for repeated measures.

RESULTS: The mean age of the cohort was 76 years. Breast cancer survivors were more likely than women without breast cancer to undergo annual mammography (45% vs. 7%, RR 6.6) but were also more likely to receive annual chest x-rays (54% vs. 27%, RR 3.2) and tumor markers (15% vs. 0.3%; RR 46.6) (all $P < .001$). Over a 3-year period, breast cancer survivors were also more likely than women without cancer to receive at least 1 bone scan (24% vs. 4%; RR 6.1) or CT/US (23% vs. 12%; RR 4.1) (both $P < .001$). In multivariable models among the cancer survivors, women who continue to visit a medical oncologist remain substantially more likely than other women to undergo follow-up chest x-rays (OR 1.4, 95% CI 1.4 to 1.5), tumor markers (OR 4.0, 95% CI 3.9 to 4.2), bone scans (OR 2.0, 95% CI 1.9 to 2.1), and CT/US (OR 1.6, 95% CI 1.5 to 1.7).

CONCLUSIONS: These findings demonstrate a high rate of testing associated with the diagnosis of early-stage breast cancer diagnosis, suggesting these tests may be overused. Whether the high rates of testing among women seeing medical oncologists are due to more symptoms or excessive surveillance is an important question for further study.

SURVEY OF INCOMING INTERNS' RESPONSES AND ATTITUDES TOWARDS MEDICAL ERRORS AND DISCLOSURE. N. Varjavand¹; D.H. Novack¹; D.G. Cohen¹; E. Gracely¹. ¹Drexel University College of Medicine, Philadelphia, PA. (Tracking ID #132467)

BACKGROUND: A variety of ethical and professional guidelines assert that physicians have a responsibility to disclose errors to patients and to other professionals. We wanted to explore residents' attitudes, disclosure practices, and barriers to disclose in order to inform future educational efforts.

METHODS: We surveyed 304 incoming interns (47% internal medicine, 14% emergency medicine, 11% surgery, and 28% other) about their attitudes towards medical errors. We asked them to respond to two scenarios involving an error: one, a "near miss" scenario, and the other with a more serious (adverse) outcome.

RESULTS: Most (87%, $n=263$) indicated they had no prior training about medical mistakes. In both scenarios, less than 40% fully admitted their fault (38%, $n=115$ in the near miss scenario; 28%, $n=85$ in the adverse outcome scenario.) Interns cited a variety of reasons for choosing different degrees of disclosure. In both scenarios, at best, interns prioritized informing their attending 61% of the time. When asked about attitudes, in general, half agreed with these three statements: "mistakes are preventable if physicians know enough," "to be viewed as competent, physicians must keep their uncertainties/emotions to themselves," and "I would leave medicine if I caused serious harm or death to a patient." Reflecting on barriers to error disclosure to colleagues, interns' responses reflected personal embarrassment (40%, $n=122$), while disclosure to patients was limited most frequently by worry about malpractice (26%, $n=79$). When asked how they would feel if they made a mistake, interns said they would feel angry at themselves (27%, $n=82$), or sad (18%, $n=56$).

CONCLUSIONS: At a time when there is a national focus on honest and full disclosure of medical errors, a majority of interns surveyed would not disclose their mistake and they recognize certain barriers to complete disclosure. Educators teaching about medical errors should focus not only on addressing communication behaviors, but also on attitudes and beliefs that could affect house staffs' responses to errors.

TAXONOMY OF AMBULATORY MEDICATION ERRORS. L.B. Meade¹; M.W. Heelon²; M.B. Rothberg²; E. Benjamin²; G. Kerr²; P.S. Pekow³; D.N. Rose². ¹Tufts University, Springfield, MA; ²Baystate Medical Center, Springfield, MA; ³University of Massachusetts, Amherst, MA. (Tracking ID #134756)

BACKGROUND: In 1995, United States Pharmacopeia established a standardized taxonomy for medication errors in the hospital setting. While this taxonomy serves as the basis for clinical and research reporting on hospital medication errors, no similar standardized taxonomy exists in the ambulatory setting. Ambulatory medication errors (AMEs) lead to adverse drug events (ADEs) at a rate of 5% per year and a third of those ADEs are considered serious, life-threatening or fatal. Given the known mortality and morbidity of AMEs, a panel of physicians and clinical pharmacists characterized type and severity of AMEs using six years of clinical experience from a pharmacotherapy clinic.

METHODS: From October 1997 to October 2003, a clinical pharmacist identified AMEs while providing a consultation service to 124 patients at a pharmacotherapy clinic in a busy urban academic health center. During one-on-one visits, the pharmacist identified problems related to medications, educated patients about their medication, supported adherence, and identified insurance/cost barriers. Through case presentations from the pharmacotherapy clinic a panel of four physicians and two clinical pharmacists developed an extensive taxonomy of AMEs.

RESULTS: The panel identified a total of 28 types of ambulatory medication errors in four categories: prescribing, dispensing, adherence, and monitoring. Subcategories of errors include: (1) Prescribing: incorrect dose, frequency, or duration; drug interaction; treatment duplication; contraindication; precaution; allergy to medication; and illegible prescription. (2) Dispensing: incorrect medication, dose, or frequency; drug interaction not identified; and pharmacy medication profile not correlating with the health center medication profile. (3) Adherence: self administering incorrect medication or correct medication inappropriately; self discontinuation of medication; unable to identify medication; and home medications not correlating with health center medication profile. (4) Monitoring: insufficient lab ordering or follow up; failure to identify medication side effect; failure to act on patient reported side effect; and inadequate self monitoring. Severity measures were categorized into five groups: no potential for adverse outcome; potential for adverse outcome (near miss); predictable ADEs; unpredictable ADEs; and adverse chronic disease outcome. An example of this last group would be a patient taking metoprolol once a day and having poorly controlled hypertension. Application of the scheme for 100 patients in the pharmacotherapy clinic identified a mean (SD) of 3.1 (0.2) AMEs per patient; 73% were errors of adherence, 13% monitoring, 10% prescribing, and 3% dispensing. Thirty percent of the AMEs were associated with either ADEs or adverse chronic disease outcome.

CONCLUSIONS: A panel of physicians and pharmacists developed an extensive taxonomy of AMEs by type and severity. This taxonomy not only provides a practical framework for categorizing AMEs but can also serve as the basis for evaluating strategies to lessen the frequency and related morbidity and mortality of AMEs.

TEST RESULTS THAT RETURN AFTER HOSPITAL DISCHARGE: A PATIENT SAFETY CONCERN DURING TRANSITION OF CARE. C.L. Roy¹; E.G. Poon¹; A.S. Karson²; Z. Ladak-Merchant¹; S. Maviglia¹; T.K. Gandhi¹. ¹Brighton and Women's Hospital, Boston, MA; ²Massachusetts General Hospital, Boston, MA. (Tracking ID #134005)

BACKGROUND: Failure to relay information about tests that are pending at hospital discharge and to act upon test results that return after discharge may represent a significant patient safety problem. Few data are available regarding the prevalence, characteristics, and actionability of test results that return after discharge (post-discharge results), or physician awareness of these results.

METHODS: We prospectively reviewed the medical records of patients being discharged from the hospitalist services at two major tertiary care centers in Boston, and collected data on pending laboratory and radiology tests. As abnormal results returned after discharge, we surveyed inpatient physicians and primary care physicians about those that were potentially clinically actionable. We also surveyed inpatient physicians regarding their satisfaction with their current result follow-up systems. The main outcomes of interest were the prevalence and characteristics of actionable post-discharge results, physician awareness of these results, and inpatient physician satisfaction with their current result follow-up systems.

RESULTS: Of 2644 patients discharged, 1095 (41%) had pending tests at discharge. Of these, 191 (9%) had potentially actionable results. Physicians were unaware of 62% of potentially actionable results; 37% of these results changed the diagnostic or therapeutic plan, and 12% of these results were urgent. Among actionable results of which physicians were unaware, microbiology results were most common, followed by radiology results. Physicians were frequently unaware that tests that returned after discharge had been ordered. Inpatient physicians were generally dissatisfied with their current systems for following up on post-discharge results.

CONCLUSIONS: Patients are frequently discharged with test results that are still pending and physicians are often unaware of post-discharge results that are potentially actionable. Further work needs to be done to design better follow-up systems for post-discharge results.

THE CHAOTIC OFFICE ENVIRONMENT: ROLE OF PATIENT ETHNICITY AND IMPACT ON PHYSICIAN STRESS AND BURNOUT. M.D. Schwartz¹; B. Man²; L. Manwell³; M. Mundt³; A.B. Varkey²; E. Williams⁴; M. Linzer³. ¹NYU School of Medicine, New York, NY; ²Rush University Medical Center, Chicago, IL; ³University of Wisconsin, Madison, WI; ⁴University of Alabama, Tuscaloosa, AL. (Tracking ID #135594)

BACKGROUND: A chaotic office environment may diminish physician well being and lead to a lower quality of care. Practices where minority patients seek care may be more chaotic. In the MEMO Study (Minimizing Errors, Maximizing Outcomes) we sought to investigate these questions through a model relating working conditions to quality of care, mediated by physicians.

METHODS: MEMO is a study of 420 primary care physicians, their office working conditions, and their patients in NYC, Chicago, Milwaukee, Madison and rural Wisconsin. Physicians and practice managers assessed their office environment by survey. We asked subjects to describe the overall atmosphere in their office on a single 5-point question with 1=calm, 3=busy, but reasonable, and 5=hectic/chaotic. Chaotic practices were defined as those with mean

scores of at least 4. Physician stress was measured by a 4-item scale ($\alpha=0.84$), and single items assessed physician burnout and intention to leave the practice within the next two years. We compared characteristics of practices described as chaotic with those described as only calm or busy. We determined the association of a chaotic office environment with physician stress, burnout, likelihood of leaving the practice, physician estimate of past medical errors and likelihood of future error (the OSPRE: Occupational Stress and PReventable Error measure).

RESULTS: Of 420 general internists and family physicians (61% of those surveyed, 84% of the target sample of 500), 48% described their office environment as chaotic. Ratings of the office environment was similar for physicians and practice managers (mean=3.4 for both). Compared with practices rated as merely busy or calm, practices rated as chaotic had more minority patients (48% vs. 25%, $p<0.001$), minority physicians (21% vs. 12%, $p=0.02$), minority staff (34% vs. 19%, $p=0.004$), and fewer exam rooms per physician (2.3 vs. 2.6, $p<0.001$). Patients at chaotic practices were more likely to have Medicaid (18% vs. 12%, $p=0.001$) and less likely to have commercial insurance (24% vs. 37%, $p<0.0001$). Sixty-one percent of physicians said their work was stressful; 27% experienced burnout; and 31% were likely to leave their jobs within two years. Physicians at chaotic practices experienced even greater stress and burnout than those at less hectic practices ($p<0.0001$). However, the likelihood of leaving the practice and physicians' estimates of past and future medical errors were similar.

CONCLUSIONS: Many physicians experienced their practice environment as hectic or chaotic. Offices with larger percentages of minority patients, physicians, or staff were viewed as more chaotic still. The increased stress and burnout among physicians at these practices could contribute to health care disparities experienced by minority patients. If chaos is indeed linked to patient outcomes, we will face an urgent need to develop and evaluate strategies to reduce and cope with the current frenzied pace of medical practice.

THE EFFECT OF EVIDENCE-BASED CARDIAC MEDICATION USE ON HOSPITAL READMISSION FOR POST ACUTE CORONARY SYNDROME PATIENTS. A.B. Olomu¹; Z. Yang²; W.D. Corser¹; D. Rovner¹; M.M. Holmes - Rovner¹. ¹Michigan State University, East Lansing, MI; ²University of Florida, Gainesville, FL. (Tracking ID #134992)

BACKGROUND: The importance of outpatient cardiac medication use for patients has been widely recognized, and is a primary focus of quality improvement. While previous research has suggested that hospital discharge is a critical time to promote the initial and continued use of cardiac medications for Acute Coronary Syndrome (ACS) patients, little research has investigated the effects of ongoing outpatient drug use patterns after hospital discharge. Our objective was to analyze patterns of outpatient cardiac medication use after initial hospitalization for ACS, and investigate the impact on the probability of subsequent hospital readmissions.

METHODS: We enrolled and reviewed the charts of 720 patients prospectively identified with ACS in five mid-Michigan community hospitals during the period January 2002 to April 2003. 433 patients that completed follow up survey at two-weeks, 3 months and 8 months were included in this analyses. Study measures were derived from the chart abstract and phone survey including patients' demographic characteristics, medication use, hospital readmission, emergency department (ED) visits, enrollment in rehabilitation services, and health related behaviors. We conducted a descriptive analysis to examine the dynamic features of post-acute ACS patient care, particularly changing patterns of outpatient prescription drug use, post-discharge hospital readmissions, and ED visits. Multivariable logistic regression models were used to estimate the association between post ACS medication use and hospital readmission and ED visits by the 3 and 8 months telephone surveys.

RESULTS: Most changes to medication regimens occurred within 3 months after discharge, with fewer changes in the subsequent 5 months. Taking Beta-Blockers (odds ratio, 0.46 [95% CI, 0.21 to 0.99]) or Angiotensin-Converting Enzyme Inhibitors (ACEI) (odds ratio, 0.31 [95% CI, 0.13 to 0.77]) significantly reduced the probability of hospital readmission within 3 months after discharge ($P<0.05$). Higher adherence to ACEI after discharge contributed to even lower hospital readmission rate (odds ratio, 0.55 [95% CI, 0.34-0.89], $p<0.1$). Re-hospitalized within 3 months after discharge was a strong predictor of later hospital readmission up to 8 months after discharge (odds ratio 0.74 [95% CI 0.73 to 1.74] $p<0.05$).

CONCLUSIONS: In addition to the prescription of effective discharge medications, timely and appropriate medication adjustment in outpatient settings appears to improve health outcomes in ACS patients.

THE QUALITY OF PHARMACOLOGIC CARE FOR ADULTS IN THE UNITED STATES. W.H. Shrank¹; S.M. Asch²; J. Adams³; C. Setodji⁴; E.A. Kerr⁴; J. Keeseey⁵; S. Malik⁶; E. Mcglynn². ¹UCLA/Greater Los Angeles VA Healthcare System, Los Angeles, CA; ²RAND, Los Angeles, CA; ³RAND, Santa Monica, CA; ⁴University of Michigan, Ann Arbor, MI; ⁵University of California, Irvine, Irvine, CA. (Tracking ID #135741)

BACKGROUND: Annual spending for prescription drugs in the United States topped \$168 billion dollars in 2004, and recent Medicare legislation will expand the federal government's role in providing prescription drug access to seniors. Despite this investment in prescription drugs, little systematic information is available concerning the quality of pharmacologic care for adults in the United States. We evaluated how frequently appropriate pharmacologic care is ordered in a national sample.

METHODS: We interviewed a random sample of adults living in 12 metropolitan areas in the United States by telephone, and received consent to evaluate their medical records for the most recent two-year period. We abstracted medical records and supplemented this with information from telephone surveys to evaluate performance on 133 pharmacologic quality of care indicators. We evaluated four domains of the prescribing process that encompassed the entire pharmacologic care experience: appropriate medication prescribing (underuse), avoidance of inappropriate medications (overuse), medication monitoring, and education, documentation and continuity. We constructed aggregate scores and studied whether patient, insurance and community factors impact quality.

RESULTS: In total, 3466 patients (52 percent of our sample) were eligible for at least one pharmacologic care quality indicator. Participants received 61.6 percent of recommended pharmacologic care overall (95 percent CI 60.0-63.2 percent). Performance was lowest in education, documentation and continuity (46.7 percent); medication monitoring (55.1 percent) and underuse of appropriate medications (62.6 percent) was somewhat better. Performance was best for avoiding inappropriate medications (83.5 percent). Patient race, health status, education and geography were associated with modest quality differences, while insurance status was not.

CONCLUSIONS: Significant deficits in the quality of pharmacologic care were seen for adults in the United States, with larger deficits in the underuse of appropriate medications than overuse of inappropriate medications. While efforts to minimize medication errors are important, interventions to address underuse of appropriate medications may go farther to improve the quality of prescribing. In addition, the provision of prescription drug insurance coverage is not sufficient to ensure high quality pharmacologic care.

Strategies to measure and improve pharmacologic care quality ought to be considered, especially as we initiate a federally funded prescription drug benefit for seniors.

UNDERSTANDING WHY PHYSICIANS DO NOT PROVIDE RECOMMENDED CARE. T. Higashi¹; J.L. Adams²; C. Kamberg³; P.G. Shekelle²; D.B. Reuben¹; C. Maclean²; R.T. Young¹; J.T. Chang¹; K. Ransohoff⁴; M. Hoffing⁵; N.S. Wenger¹. ¹University of California, Los Angeles, Los Angeles, CA; ²RAND Corporation, Santa Monica, CA; ³RAND, Arlington, VA; ⁴Sansum-Santa Barbara Medical Foundation Clinic, Santa Barbara, CA; ⁵Desert Medical Group, Palm Springs, CA. (Tracking ID #132765)

BACKGROUND: Studies show that the quality of medical care for older persons needs improvement. Although many factors have been proposed to explain why physicians may not be providing quality care, few studies have assessed whether these factors are related to actual provision of care.

METHODS: We surveyed physicians in two large medical groups to ask about the priority of and obstacles to providing 67 care processes judged by experts to be recommended care for older patients covering 12 conditions. Priority for each care process was rated on a 1-7 scale (1=lowest priority, 7=highest priority) and physicians indicated the obstacles in their practice that impeded care. We then linked physicians' reports of priority and obstacles to their actual provision of care over a 13-month period prior to the survey as assessed by medical record abstraction and patient interview. Physicians were blinded to their performance. **RESULTS:** Thirty-four primary care physicians, who cared for 548 patients, returned a survey (response rate, 84%). Physicians' mean priority rating was 5.9 (standard deviation 1.3), ranging from 3.9 for "Screening for Pain at least every 2 years" to 6.9 for "Measure Hb1Ac annually for diabetic patients." On average, physicians noted obstacles to care as follows: 21% lack of time; 7% lack of resources; 7% patient preference; 6% difficulty remembering periodic care; and 3% lack of expertise. Care processes most commonly cited by providers related to each obstacle were gait evaluation for fallers (74% cited lack of time), asking about falls annually (38% cited difficulty remembering), educating caregivers for dementia patients (32% cited lacking resources), quit smoking advice and colorectal cancer screening (32% cited patient preference) and prescribing an exercise program/assistive device for patients with imbalance (21% cited lack of expertise). Physicians provided 66% of recommended care. Multivariate analysis adjusting for physician, patient and quality indicator characteristics and clustering revealed that low physician ratings (5 or less) of care priority were associated with providing 7% less of the recommended care ($P<0.01$). However, among potential obstacles, only lack of expertise was significantly related to actual care provision (16% reduction in performance, $P=0.02$).

CONCLUSIONS: Physicians' prioritization of care process had a small, but significant, role in whether patients received recommended care. Lack of expertise was not common, but had a substantial effect on quality. Although physicians generally rated recommended care as a high priority, they often did not provide this care, suggesting the need for systems to facilitate provision of recommended care.

UTILIZING PAIR-WISE COMPARISONS TO DETERMINE RELATIVE IMPORTANCE OF DIABETES GUIDELINES: A COMPARISON OF PATIENT AND PHYSICIAN PERSPECTIVES. M. Long¹; R.M. Centor¹. ¹University of Alabama at Birmingham, Birmingham, AL. (Tracking ID #134963)

BACKGROUND: Experts and physicians continue to debate quality of care assessment. All parties would like clear accepted guidelines for quality assessment. Most current "report cards" take several measurable quality indicators and average physician performance to provide a single overall score. We asked 3 physician groups and 1 patient group to use the pair-wise comparison technique from the analytic hierarchy process (AHP) to evaluate the relative importance of four commonly used diabetes quality indicators. We postulated that both physicians and patients would value indicators differently. This technique

allows us to develop a weighted scoring system of overall quality. We also postulated that patients would weight quality indicators differently than physicians.

METHODS: We developed an AHP model to compare four indicators of Type II diabetes management: measuring HgBA1c, measuring LDL, performance of dilated eye exam and performance of foot exam. Four groups completed pair-wise comparisons of these guidelines: academic physicians, community physicians, endocrinologists and Type II diabetic patients. Each subject performed the pair-wise comparisons on a web-based model with either oral or written instructions from a moderator.

RESULTS: The table gives the relative weighting for each group. Each row adds to 1.00.

CONCLUSIONS: All three physician groups had similar rankings of the 4 indicators. Consistently, physicians ranked HgBA1c as most important, followed by LDL, eye exam and foot exam. The patients in this study weighted eye exams as the second most important indicator, delegating LDL to 3rd. We are considering several possible explanations for these results. First, patients value eye sight as a most important component of quality of life. Second, physicians caring for these patients may not emphasize the importance of lipid control. When evaluating overall quality of care which physicians provide, weighting may alter results. Physicians should receive a detailed report card which includes both physician weighting and patient weighting.

Comparison of Relative Weights

Group	HgbA1c	LDL	Eyes	Feet
Academic GIM	.472	.263	.142	.123
Endocrine	.452	.198	.175	.173
Community	.437	.294	.139	.131
Patients	.515	.124	.261	.100

WHAT IS THIS PILL? RESULTS FROM THE COOK COUNTY-HARVARD SOLID ORAL DOSAGE IDENTIFICATION (SODID) STUDY. G. Schiff¹; S. Kim²; A.C. Seger³; J. Bult¹; D.W. Bates⁴. ¹Cook County Hospital, Chicago, IL; ²University of Illinois at Chicago, Oak Park, IL; ³Brigham and Women's Hospital, Boston, MA; ⁴Brigham and Women's Hospital, Watertown, MA. (Tracking ID #135922)

BACKGROUND: To safely use medications, health professionals and patients need easy-to-use and reliable methods to identify medications. However we currently lack a standardized imprint identification system—a status quo defended by the pharmaceutical industry claiming that there is no firm evidence that this a problem. We sought to test the ability of primary care physicians and pharmacists to identify “unknown” medications in their actual practice setting.

METHODS: We randomly selected 50 general internists and 50 pharmacists at 2 urban academic medical centers, and presented 3 “unknown” tablets (a brand name drug-ZocorTM 80 mg, a generic-lorazepam 2 mg, an OTC-naproxen sodium 220 mg) to them in their offices or pharmacies, and asked to use all available resources to identify the tablets. Trained observers recorded how long each identification took and recorded all resources used, and determined whether the pill was correctly identified. Participants were also surveyed on their day-to-day experiences in tablet identification.

RESULTS: A total of 300 observations were recorded in the tablet identification exercise. In 190 (63.3%), the tablet was correctly identified. A total of 18 (36%) physicians and 24 (48%) pharmacists correctly identified all 3 tablets. Ten (20%) physicians and 5 (10%) pharmacists were unable to correctly identify a single one of 3 tablets. Zocor was correctly identified in 77%, lorazepam 63%, and naproxen 51% of the observations. The average time subjects took to identify each tablet was 3.65 minutes, (2.57 minutes for Zocor, 4.25 minutes for lorazepam, and 4.15 minutes for naproxen). Mostly commonly used resources for physicians were PDR book (24 physicians, 48%), calling pharmacist or poison control centers (9 physicians, 18%), and online resources (6 physicians, 6%). Pharmacists at Harvard relied heavily on Indentix (16 pharmacists, 32%), whereas Cook County pharmacists, who lack this online reference, most often consult other pharmacists at institution (4 pharmacists, 8%), local drug information or poison control center (4 pharmacists, 8%). 2/3 of participants stated that they were called upon to identify tablets at least once monthly, but overall 77 participants expressed dissatisfaction with the current system and 91 participants favored a more standardized system of pill imprint identification.

CONCLUSIONS: Randomly selected physicians and pharmacists failed in their attempts to correctly identify three commonly prescribed tablets more than 1/3 of the time. The brand name product was significantly more likely to be correctly identified than a generic or OTC product. Problems included limited availability of easy-to-use resources, lack of skills in utilizing resources that were on hand, and inefficient and unproductive searching strategies. Although participants stated that the need to identify pills arose frequently in their practice of medicine and pharmacy, the current system is insufficient to meet their needs, and a more standardized imprint coding system is greatly preferred.

WHO ISN'T PLAYING? HOSPITAL FACTORS ASSOCIATED WITH NON-PARTICIPATION IN PUBLIC REPORTING OF HOSPITAL QUALITY. L.E. Goldman¹; R.A. Dudley¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #135176)

BACKGROUND: Policies requiring or offering incentives to report publicly on hospital quality of care measures are intended to improve hospital performance. If safety-net hospitals choose not to participate, this might worsen disparities in quality of care.

METHODS: We conducted a cross-sectional study using the National Voluntary Hospital Reporting Initiative (NVHRI) database in which hospitals voluntarily and publicly report on ten evidence-based process measures. We linked the NVHRI to the California Office of Statewide Health Planning and Development (OSHPD) 2003 financial database to obtain hospital characteristics. We included all California acute care hospitals that reported to OSHPD in 2003. Hospitals were considered to have participated in NVHRI if they submitted data for at least one patient. We analyzed whether participation in public reporting was associated with three characteristics of safety-net hospitals: 1. public hospital ownership (all state, district, county, and city-owned hospitals), 2. rural location (defined as having a population density of 250 persons or less), or 3. having greater than 50% Medicaid plus uninsured patients. We then created a different multivariate logistic regression model for each of the three mentioned surrogates for safety-net status, controlling for teaching status, bed size, and % Medicare patients.

RESULTS: Of 384 Californian acute care hospitals including 68 rural, 27 teaching, and 69 public hospitals, 293 participated in NVHRI. In the univariate analysis, teaching hospitals (93% versus 75%, $p=0.039$) and larger hospitals (mean bed size 199 versus 133, $p=0.0002$) participated more frequently. Rural hospitals participated less frequently (68% versus 78%, $p=0.06$). Neither hospital ownership (private investor 77%, private non-profit 76%, and public 75%, $p=0.97$) nor greater than 50% Medicaid plus uninsured discharges (75% versus 76%, $p=0.74$) affected participation rate. In the three multivariate analyses, only bed size consistently predicted participation whether safety-net hospital was defined as rural ($p=0.01$), greater than 50% Medicaid plus uninsured discharges ($p=0.002$), or public ownership ($p=0.003$). Percent Medicare reimbursement predicted participation in the model assessing rural hospitals, ($p=0.05$) and approached significance in the other two models ($p=0.06$ and $p=0.06$).

CONCLUSIONS: Teaching hospitals volunteer more frequently and rural hospitals participate less frequently in public reporting programs. Hospital size explains both findings. Using three different definitions of safety-net hospitals—public ownership, rural location, and high percent Medicaid plus uninsured reimbursement—we did not detect a difference in safety-net hospital participation.

A PRACTICE-CENTERED INTERVENTION TO INCREASE SCREENING FOR DOMESTIC VIOLENCE. D. Bonds¹; S.D. Ellis¹; E. Weeks¹; S.L. Palla¹; P. Lichstein¹. ¹Wake Forest University, Winston-Salem, NC. (Tracking ID #134045)

BACKGROUND: Interventions to change practice patterns and improve quality have had mixed success. Often the education modes or tools that constitute interventions are too rigid to integrate with practice needs. A practice-centered intervention that is sensitive to the particular needs of the practice while still remaining true to the underlying principles of quality may be more successful in implementing change. We tested this approach in an intervention designed to increase screening for domestic violence in primary care practices.

METHODS: Primary care practices (PCP: defined as internal medicine, IM; family medicine, FP; or OB-Gyn) with at least two providers were invited to join. Both academic (A) and community-based (C) practices were eligible. Each practice sent two members (one provider and one staff member) from their clinic to attend an all day training session on domestic violence, who in turn conducted a training session at their own clinic with the help of study staff. Practices selected from a menu of validated screening questionnaires, screening implementation methods (written or verbal), patient materials, and screener assignment (nurse or provider) that best suited their practice patterns. Follow-up support was provided for 18 months in the form of academic detailing visits, newsletters, patient education material, and phone support. Screening rates were determined by a telephone survey of randomly selected female patients who were asked if they had been screened for safety or violence in the home in the last year.

RESULTS: Sixteen primary care practices were recruited (4 IM, 10 FP, 2 GYN; 5A, 11C). 1,643 women were surveyed at baseline and 1,534 at follow-up. There were no significant differences in mean age, race, work status, children in the home, medical insurance or number of visits (see Table). More women were married in the follow-up sample; income was slightly higher; fewer had a female PCP; and more saw FM or IM providers. At follow-up 26% of women reported having been screened for violence in the home compared to 16% at baseline. Using GEE modeling to account for clustering by clinics, the odds ratio of being screened after the intervention was 1.8 (95% CI 1.5-2.2). This remained stable after adjustment for patient (age, race, marital status, work status, children in home, income, insurance status, number of visits) and practice (gender and type of PCP) characteristics (OR 1.75, 95% CI 1.4-2.2).

CONCLUSIONS: Although the study was limited by the lack of a comparison group, it appears that using a practice-centered intervention can improve screening rates for domestic violence. Further testing of this concept with other outcomes is needed.

Demographics and screening rate at baseline and follow-up

	Initial (N=1,643)	Follow-up (N=1,534)	P value
Age (mean)	48 (SD=17.37)	48 (SD=16.93)	NS
Race			NS
NHW	64% (1032)	66% (987)	
AA	34% (545)	32% (471)	
Other	2% (26)	2% (33)	
Married	48% (791)	54% (833)	0.003
Not working outside home	52% (842)	50% (773)	NS
Children <18 in home	40% (659)	41% (626)	NS
Annual Income			0.04
< \$25,000	47% (607)	50% (661)	
\$25,000-\$50,000	28% (362)	24% (361)	
\$50,001-\$75,000	14% (186)	14% (188)	
> \$75,000	10% (129)	12% (158)	
Medical Insurance, Yes	91% (1489)	89% (1363)	NS
Visits # last year (mean)	7.5 (SD 9.3)	8.0 (SD 11.2)	NS
PCP Type			<0.0001
Family Medicine	60% (925)	64% (911)	
Internal Medicine	17% (268)	18% (261)	
OB/Gyn	9% (131)	8% (111)	
Other	14% (214)	9% (131)	
Female PCP	53% (844)	48% (716)	0.05
Screened for domestic violence in last 12 months	16% (251)	26% (398)	<0.0001

A SIMPLE TEST TO HELP PRIMARY CARE CLINICIANS APPROPRIATELY MANAGE URINARY INCONTINENCE IN WOMEN. D.G. Grady¹; C. Bradley²; F. Lin¹; L.L. Subak³; H. Richter⁴; S. Kraus⁵; L. Brubaker²; J.S. Brown¹. ¹University of California, San Francisco, San Francisco, CA; ²University of Iowa, Iowa City, IA; ³University of Alabama at Birmingham, Birmingham, AL; ⁴University of Texas Health Science Center at San Antonio, San Antonio, TX; ⁵Loyola University of Chicago, Maywood, IL. (Tracking ID #133199)

BACKGROUND: Urinary incontinence (UI) is common in women and should be diagnosed and treated by primary care clinicians. Classifying UI type as urge or stress is required to choose proper drug therapy. Currently, classification of type of UI generally requires an extended evaluation (ExE) that is expensive, invasive and restricted to specialists. A simple test that accurately classifies type of UI would allow busy primary care clinicians to appropriately manage women with UI. We conducted a cross-sectional study among 301 women with UI at 5 US sites to determine the accuracy of a brief questionnaire and a 3-day diary.

METHODS: We enrolled community-dwelling, ambulatory women aged 40 or older with at least 3 episodes of UI per week who sought treatment. Women who should be referred to a specialist were excluded (prior incontinence surgery, known neurologic diseases, current UTI or 4 or more UTIs in the prior year). The incontinence questionnaire (3 IQ) included 3 questions and the diary required women to record each UI episode by type over 3 days. All participants underwent an ExE (incontinence, medical, reproductive, and surgical histories, a 3-day diary, pelvic and neurologic examinations, stress cough test, and postvoid residual volume) performed by a urologist or urogynecologist blinded to the 3 IQ responses, but not diary, and who classified each woman as having urge, stress or other type of UI. Women were classified as having urge or stress UI on the 3 IQ if they reported that this type occurred most often and were classified as having urge or stress UI on the diary if they recorded at least 75% of this type of episode. We calculated the accuracy of the two tests using the diagnosis from the ExE as the gold standard.

RESULTS: The mean age of participants was 56 ± 12 years, they were racially diverse (69% White; 13% Black; 12% Latina; and 2% Asian) and most reported good or excellent health (83%). The mean duration of incontinence was 7 ± 7 years and the range of UI severity was broad (mild 5%, moderate 59%, severe 36%). Accuracy of the 3 IQ and 3-day diary for the diagnosis of urge or stress incontinence is presented in the table.

CONCLUSIONS: The 3 IQ is a simple, inexpensive, non-invasive and reasonably accurate test that can be used in primary care practice to determine if a woman has predominantly urge or stress UI. A 3-day diary, which is more cumbersome to use in practice, is more sensitive but much less specific than the questionnaire for the diagnosis of urge UI and somewhat more specific than the simple questionnaire for diagnosis of stress UI. We conclude that a simple questionnaire is a good test for type of UI, especially since the risk associated with missed diagnosis and treatment is low.

Accuracy of 3IQ and Diary

	Sensitivity	Specificity	PPV	LR+
Urge 3IQ	0.75 (0.68-0.81)	0.77 (0.69-0.84)	0.79 (0.72-0.85)	3.26 (2.38-4.48)
Diary	0.98 (0.93-0.99)	0.50 (0.42-0.59)		

1 (continued)

	Sensitivity	Specificity	PPV	LR+
			0.69 (0.63-0.75)	1.96 (1.66-2.33)
Stress 3IQ	0.86 (0.79-0.90)	0.60 (0.51-0.68)	0.74 (0.68-0.80)	2.13 (1.71-2.66)
Diary	0.81 (0.74-0.86)	0.80 (0.72-0.87)	0.85 (0.78-0.90)	4.11 (2.87-5.88)
PPV = positive predictive value;	LR+ = likelihood ratio for a positive test; 3 IQ = 3 Incontinence Questions			

AN EXPLORATION OF FEAR AS RELATED TO MAMMOGRAPHY AMONG LOW-INCOME AFRICAN-AMERICAN WOMEN. M.E. Peek¹. ¹Rush University Medical Center, Chicago, IL. (Tracking ID #136243)

BACKGROUND: Low-income African-American women have the highest rates of breast cancer mortality of any social group in the United States. Disparities in breast cancer screening have been cited as one reason for the poorer breast cancer outcomes among this population. Fear and fatalism within the African-American community has been reported as a prominent cultural theme that may act as a barrier to breast cancer screening, but the concept of fear has not been fully explored. This study sought to further elucidate the nature and reasons for fear as related to breast cancer screening within low-income African-American women.

METHODS: Four focus groups were conducted among 29 low-income African-American recruited from Chicago. Discussions were guided by semi-structured interviews developed from a review of the literature on breast cancer screening in African-Americans and components of the Health Belief Model. Discussions were moderated by experienced African-American health care professionals. All sessions were audiotaped and transcribed verbatim. Transcripts were reviewed through an iterative process and coded using Atlas.ti 4.2 software. Thematic analysis was then performed.

RESULTS: Several themes emerged within the general category of fear. Some women described a fear of cancer and subsequent, imminent death (fatalism) as a primary reason for being afraid of breast cancer screening. However, many women reported a fear of the mammography process itself including concern about pain, radiation and the anxiety-invoking processes of waiting for the films to be processed and for the mailed final mammogram report. Stories of bad experiences with mammography were reportedly commonplace within the community and one woman described getting physically stuck in a mammogram machine. Body image and intimate partner relationships were a prominent theme in these focus groups. Concerns about potential mastectomy—generally assumed to be standard treatment—were primarily discussed within the context of interpersonal relationships. For example, one woman said “so if I had to have [a breast] removed and my husband was still living with me, and he wasn’t educated and he didn’t understand the side effects of this stuff on the mental and emotional level it would cause me to have concern. The first thing to come to mind is ‘Oh he’s going to get another woman now, I’m no longer attractive.’” Many women described a fear arising from a lack of information about what kind of treatment, comprehensive care and support systems existed for them if diagnosed with breast cancer. Spirituality, social support systems and the presence of “positive stories” within the community were reported factors that mitigate the fear associated with breast cancer screening.

CONCLUSIONS: Much of the fear associated with breast cancer screening among low-income African-American women can be addressed by culturally-competent physicians as part of the shared decision making process in the clinical encounter. Additionally, peer educators are an important strategy to address some of the cultural issues specific to this population.

CERVICAL CANCER SCREENING AMONG WOMEN WITH HYSTERECTOMIES BY PROVIDER TYPE. B.G. Lewis¹; E.A. Halm¹; A.D. Federman¹. ¹Mount Sinai School of Medicine, New York, NY. (Tracking ID #134166)

BACKGROUND: Since 1996, the US Preventive Task Force has recommended against cervical cancer screening in women without a cervix. Yet, as recently as 2002, 69% of such women reported getting a pap smear within the previous 3 years and the factors influencing this pattern of care remain vague. Because both gynecologists and generalists perform pap smears, we examined the association between provider type and pap smear rates among post-hysterectomy women.

METHODS: We analyzed data from the 2000 National Health Interview Survey (NHIS), a cross-sectional, nationally representative annual survey of self-reported health and health care utilization among adults (n=32,374). We included post-hysterectomy women ages 18-65. We excluded those with a history of abnormal pap smears, breast, cervical, endometrial or ovarian cancer, current pregnancy and recipients of Medicare. We determined pap smear rates in the past 12 months, stratified by provider type: generalist (gen); gynecologist (gyn) or both. We used multiple logistic regression to assess the relationship between

pap smears and provider type, adjusting for age, race, education, income, insurance status, census region, smoking and health status. SUDAAN statistical software was used to account for the NHIS sampling design.

RESULTS: The analysis included 1,221 women, representing 7.3 million women nationwide. 53% received care from a gyn, 26% from gen, and 20% from both. The mean ages of the women were 49 years in gyn, 53 in gen, and 51 in both. Women in the 3 provider groups differed significantly by black race (41%, 33%, 25%; $p=0.001$), married (77%, 67%, 72%; $p=0.039$), uninsured (57%, 32%, 11%; $p=0.042$), urban residence (46%, 29%, 24%; $p<0.001$), and having ≥ 1 chronic illness (54%, 26%, 19%; $p=.003$). They did not differ significantly by income, immigration status, Census region, or smoking status. Overall, 55% had a pap smear in the previous 12 months. 85% of women in the gyn group underwent Pap smears; 44% in gen, and 73% in both gen/gyn ($p<.001$). In multivariate analysis, women who visited gyn were more likely than those cared for by gen alone to have had a pap smear (Adjusted OR 4.44, 95% CI 2.21 to 8.91). The effect was less pronounced when women visited both (AOR vs. gen: 1.47, 95% CI 0.62 to 3.50).

CONCLUSIONS: Approximately 4 million pap smears are performed each year on women who have had hysterectomies though services are only left in $<5\%$ of cases. Gynecologists perform the majority of these pap smears; however, a large number of such women receiving care from generalists continue to undergo this service. Additional research is needed to identify the physician and patient factors that influence the use of pap smears in post-hysterectomy patients and to prevent unnecessary care.

CHANGES IN HEALTH STATUS EXPERIENCED BY WOMEN WITH GESTATIONAL DIABETES AND PREGNANCY-INDUCED HYPERTENSION. C. Kim¹; P. Brawarsky²; R. Jackson³; E. Fuentes-Afflick³; J.S. Haas². ¹University of Michigan, Ann Arbor, MI; ²Brigham and Women's Hospital, Boston, MA; ³University of California, San Francisco, San Francisco, CA. (Tracking ID #130891)

BACKGROUND: Gestational diabetes (GDM), or glucose intolerance first diagnosed during pregnancy, and pregnancy-induced hypertension (PIH) each affect approximately 5-10% of pregnancies in the U.S and are risk factors for maternal diabetes and hypertension after delivery. The effects of these diseases on maternal health status after delivery are not well described. Therefore, we examined maternal health status from pre-pregnancy to after delivery in women with and without GDM and PIH.

METHODS: We used data from a prospective cohort of pregnant women, Women and Infants Starting Healthy (WISH), which collected patient survey and medical record review data. We examined health status changes from pre-pregnancy to postpartum (12 weeks after delivery) in women with GDM ($n=64$), women diagnosed with PIH (gestational hypertension, pre-eclampsia, eclampsia) ($n=148$), and gravidas without either condition ($n=1,233$). Health status measures included the Short Form-36 scales for physical function, vitality, and self-rated health, and the Center for Epidemiologic Studies-Depression Scale. A significant decline in health status meant that women had physical function scores >85 prior to pregnancy but ≤ 85 in the postpartum; vitality scores >45 to ≤ 45 ; self-rated health that was good, very good, or excellent prior to fair or poor; and depression scores <10 to ≥ 10 . Multivariate logistic regression models controlled for age, parity, race, education, pre-pregnancy body mass index and exercise levels, food insufficiency, and GDM or PIH during a prior pregnancy.

RESULTS: All women reported declines in health status during pregnancy. After adjustment for pre-pregnancy covariates, women with PIH reported greater declines in vitality (OR 1.51, 95% CI 1.02-2.23), self-rated health (OR 1.84, 95% CI 1.11-3.05), and a greater increase in depressive symptoms from pre-pregnancy to postpartum (OR 2.12, 95% CI 1.19-3.77) compared to women with pregnancies unaffected by PIH or GDM. Decline in self-rated health was partially mediated by Cesarean section and pre-term delivery. Women with GDM reported a significant decline in self-rated health from pre-pregnancy to 3rd trimester compared to unaffected women, but these changes did not persist postpartum. **CONCLUSIONS:** PIH, but not GDM, is associated with declines in vitality, self-rated health, and increase in depressive symptoms that persist after delivery compared to unaffected pregnancies. Future studies should explore how long these declines persist, mediators of decline, and whether declines are associated with future maternal diabetes and hypertension.

COMPREHENSIVENESS OF PREVENTATIVE CARE AMONG WOMEN SEEN BY GENERALISTS, GYNECOLOGISTS OR BOTH. B.G. Lewis¹; E.A. Halm¹; A.D. Federman¹. ¹Mount Sinai School of Medicine, New York, NY. (Tracking ID #133546)

BACKGROUND: Both gynecologists and generalists provide preventative care to women. However, the effect of provider type on the delivery of preventive care remains unclear. We examined use of preventive services as recommended by the US Preventative Task Force, among women seen by generalists only (gen), gynecologists only (gyn) or both.

METHODS: Using cross-sectional data from the 2000 National Health Interview Survey (NHIS), we studied self-reported receipt of preventative services for women ages 18-65 ($N=10,347$). We excluded women with a history of breast, colorectal, cervical, endometrial or ovarian cancer, current pregnancy and recipients of Medicare. For women ages 18-65, we analyzed the association of provider type with rates of pap smears (excluding those with a history of hysterectomy), tobacco use screening and exercise/diet counseling; for women 50-65, biannual clinical breast exams (CBE), mammograms, and colon cancer screening were also analyzed. We modeled each outcome using multivariate logistic regression, and total number of services using Poisson regression, and controlled for age,

education, income, race, insurance type, comorbidities. The analysis was weighted to account for the multilevel sampling design of the NHIS.

RESULTS: Women seen by gen (15%), gyn (62%) and both (23%) differed by age (43yrs, 37, 39, respectively, $p<.01$), black race (18%, 48%, 33%, $p<.01$), college-education (12%, 61%, 26%, $p<.01$), and having ≥ 1 chronic disease (16%, 60%, 22%, $p<.01$). The mean number of services by provider type were, for women <50 , 1.9 (gyn), 1.5 (gen) and 1.9 (both) ($p<.01$), and for women 50-65, 4.3 for gyn, 4.0 for gen, 4.2 for both ($p<.01$). These findings remained significant in adjusted Poisson regressions for women <50 (β : gyn, 0.22, $p<.01$; both, 0.24, $p<.01$) and 50-65 (β : gyn, 0.09, $p<.01$; both, 0.11, $p<.01$). Women 18-65 seen by gyn or both were more likely to have pap smears (gyn vs. gen, adjusted OR 11.06, 95% CI 6.87-17.80; both vs. gen: 11.47, 6.25-21.07) and tobacco use screening (gyn vs. gen: 1.15, 0.93-1.42; both vs. gen: 1.28, 1.00-1.64). Gyn patients tended to receive diet/exercise counseling less than gen (diet: 0.88, 0.69-1.12; exercise: 0.91, 0.73-1.14). Among women 50-65, gyn and both gyn/gen patients were more likely than gen patients to have mammograms (gyn vs. gen: 2.95, 1.48-5.87; both vs. gen: 3.59, 1.48-8.70) and CBE (gyn vs. gen: 3.31, 1.81-6.04; both vs. gen: 4.38, 2.13-9.03). There was no significant difference in colon cancer screening. Among women 18-65 with no chronic health problems ($N=1528$), receipt of preventive care did not differ significantly except gyn and both continued to perform significantly more mammograms (AOR, 95% CI: gyn: 3.59, 1.52-8.48; both 5.06, 1.72-14.89) and pap smears (gyn: 10.57, 6.00-18.62; both: 10.77, 5.46-21.24) than gen.

CONCLUSIONS: Women seen by gyn alone or both gyn and generalists receive more preventative services compared to those seen by generalists alone. Collaborative gyn/generalist care may not be necessary to ensure comprehensive preventive care for many women. Moreover, these findings further validate the role of gynecologists as primary care providers for many women.

CONTRACEPTIVE USE IN DIABETIC AND OBESE WOMEN. C.H. Chuang¹; G.A. Chase¹; D.M. Bensyl²; C.S. Weisman¹. ¹Pennsylvania State University College of Medicine, Hershey, PA; ²Centers for Disease Control and Prevention (CDC), Atlanta, GA. (Tracking ID #133826)

BACKGROUND: Women with chronic medical conditions are at increased risk for adverse pregnancy outcomes, yet contraceptive use by these women has not been well described. The purpose of this study was to describe contraceptive use by diabetic and overweight/obese women compared with women without these conditions.

METHODS: Using cross-sectional data from the 11 states participating in the optional Family Planning Module of the Behavioral Risk Factor Surveillance System in 2000, we analyzed contraceptive use among 7,943 sexually active women of reproductive age (18-44) who did not want to get pregnant. Using logistic regression techniques, we modeled the effect of diabetes and overweight/obesity on contraceptive non-use, controlling for age, race/ethnicity, marital status, education, income, and health insurance coverage.

RESULTS: Contraceptive non-use was reported by 1500 (19%) women in the total sample, 31 (26%) diabetic women, 371 (20%) overweight women, and 385 (23%) obese women. Compared to normal weight women, obesity was significantly associated with contraceptive non-use (adjusted OR 1.34, 95% CI 1.16-1.55). Odds of contraceptive non-use was not increased for diabetic women compared to non-diabetic women (adjusted OR 1.23, 95% CI 0.80-1.87) or for overweight women compared to normal weight women (adjusted OR 1.14, 95% CI 0.99-1.31). Older, Black, Hispanic, married, less educated, and women without health insurance were more likely to report contraceptive non-use.

CONCLUSIONS: Among women with need for contraception, obese women were more likely to report contraceptive non-use than normal weight women. Since women with chronic conditions like obesity are at higher risk of pregnancy-related complications and adverse pregnancy outcomes, proper contraceptive use and unintended pregnancy avoidance is a priority.

DETERMINING PREVALENCE OF INTIMATE PARTNER VIOLENCE AMONG WOMEN IN AN URBAN INTERNAL MEDICINE PRACTICE USING THE VIOLENCE EXPOSURE QUESTIONNAIRE - A PILOT STUDY. A.H. Alvanzo¹; D. Svikis¹; W.R. Smith¹. ¹Virginia Commonwealth University, Richmond, VA. (Tracking ID #134271)

BACKGROUND: Intimate partner violence (IPV) is a significant public health problem affecting approximately 1.5 million women annually. Approximately 30-45% of women in primary care practices admit to experiencing IPV within their lifetime, and approximately 5-15% admit to experiencing IPV within the 12 months prior to presentation. However, few of these studies were conducted in settings serving predominantly older minority patients. The purpose of this study was to determine the lifetime and recent (within previous 12 months) prevalence of IPV among women attending an urban primary care clinic serving an older, predominantly Black population.

METHODS: 120 women in an urban, hospital-based internal medicine clinic were screened for IPV using the Violence Exposure Questionnaire (VEQ), a 15-item screening tool that assesses 7 domains: physical abuse, sexual abuse, emotional abuse, perceived safety of patient, presence of weapons in the home, community violence exposure (self and children), and perceived need for help for issues related to intimate partner and/or family violence. Intimate partner violence was defined as a positive response to any of the following questions: "Have you ever been hit, slapped, kicked, or physically hurt by someone?", "Has anyone ever forced you to have sex?", or "Has anyone ever emotionally abused you (e.g. insulting, ridiculing, ignoring, and/or criticizing)?" with the perpetrator being identified as a current or ex-partner. Both lifetime and recent IPV were determined. Respondents were older (mean age 57 years), predominantly Black

(66%), and predominantly of lower socioeconomic status (43% reported an annual household income <\$10,000).

RESULTS: Of the 120 women surveyed, 62 (51.7%) reported a lifetime prevalence of intimate partner violence. Seventeen (14.2%) reported recent IPV. When emotional abuse was excluded from the analysis, 54 (45%) women reported lifetime prevalence of physical and/or sexual IPV. Six (5%) women reported physical and/or sexual IPV within the previous 12 months. Recently abused women were more likely to be married and living with their partner. Women who had a partner who used substances (alcohol and/or street drugs) were also more likely to report recent IPV, but the difference was not statistically significant. In contrast to several earlier studies, women in our sample who had less education and lower household income were less likely to report recent IPV.

CONCLUSIONS: In an urban primary care clinic serving primarily an older, low income, predominantly Black population, 1/2 of female patients sampled had experienced physical, sexual, or emotional intimate partner violence in their lifetime and 1 of every 7 had experienced IPV within the previous 12 months. The results of this pilot study suggest that a substantial portion of female patients in urban primary care clinics have experienced IPV, an epidemic which has been associated with increased rates of depression, anxiety, somatic symptoms, chronic pain and substance abuse. Further results suggest that while the prevalence of recent IPV may decrease as women age, IPV continues to be a pervasive problem for older women.

DISPARITIES IN FOLLOW-UP AFTER AN ABNORMAL MAMMOGRAM. R. Press¹; O. Carrasquillo¹; E.V. Giardina¹. ¹Columbia University, New York, NY. (Tracking ID #134092)

BACKGROUND: Although non-Hispanic white (NHW) women have an increased risk of developing breast cancer, the disease specific survival is lower for African American and Hispanic women compared to NHWs. Moreover, African American and Hispanic women are more likely to experience delays in diagnosis and treatment. Little is known about disparities in follow-up after an initial abnormal mammogram. The goal of this study is to investigate whether disparities in timely follow-up exist after an abnormal mammogram. We hypothesized that African American and Hispanic women have longer times to follow up after an initial abnormal mammogram compared to NHWs.

METHODS: We performed a retrospective cohort analysis using the clinical information system (CIS) at Columbia University Medical Center. Subjects include 6,722 women with an abnormal mammogram based on the American College of Radiology Breast Imaging Reporting and Data System (BIRADS) reported during the period 2000 to 2002. An abnormal mammogram was defined as either BIRAD Category 0: Indeterminate (n=5,394), BIRAD Category 4: Suspicious (n=1,116), or BIRAD Category 5: Highly Suspicious (n=212). The outcome of interest was the number of days between an initial abnormal mammogram and additional imaging or biopsy. Bivariate analyses based on race/ethnicity and zip code of residence were performed. The zip code analysis was performed by dividing the zip codes into 2 groups, either within the hospital service area or outside the hospital service area. The population within the hospital service area is 50% Hispanic and 30% African American; the majority are uninsured or insured by Medicaid and/or Medicare. The median number of days was compared using the Kruskal-Wallis test and linear regression was used for multivariate analyses.

RESULTS: There are 2,143 (32%) NHW women, 3,291 (49%) Hispanic, 915 (14%) African American and 373 (6%) women from other racial/ethnic groups included. The median number of days from initial abnormal mammogram to subsequent test was 13 days for NHW women, 18 days for Hispanic and 16 days for African American women ($p < 0.001$). There was also a significant difference in number of days to follow-up between women living within the hospital service area and those outside the hospital service area (18 vs. 14 days, $p < 0.001$). After controlling for age, BIRAD and zip code of residence, time to follow-up remained longer for both Hispanic and African American women compared NHW women ($p < 0.001$ for Hispanic and African-American). Living within the hospital service area also remained an independent predictor of having a longer time to follow-up ($p < 0.001$).

CONCLUSIONS: After an abnormal mammogram, African American and Hispanic women had longer times to follow-up compared to NHW women, as did women from within the hospital service area compared to those living elsewhere. These disparities in follow-up may contribute to the existing racial/ethnic differences in breast cancer mortality. The etiology of the disparities is likely multifactorial, including healthcare access, as well as provider-level and patient-level barriers. It is essential that these barriers be overcome in order to eliminate disparities in follow-up after abnormal mammograms and, consequently, in breast cancer mortality.

FACTORS RELATED TO UNDERUSE OF SURVEILLANCE MAMMOGRAPHY AFTER BREAST CANCER DIAGNOSIS. N.L. Keating¹; M.B. Landrum²; E. Guadagnoli³; E.P. Winer³; J.Z. Ayanian¹. ¹Brigham and Women's Hospital, Harvard Medical School, Boston, MA; ²Harvard Medical School, Boston, MA; ³Dana Farber Cancer Center, Boston, MA. (Tracking ID #134955)

BACKGROUND: National guidelines strongly recommend that all breast cancer survivors undergo annual mammography, but many elderly breast cancer survivors do not have regular mammograms. We identified sociodemographic and clinical factors associated with underuse of surveillance mammography and examined whether differences in follow-up care explained variation in mammography rates.

METHODS: We used SEER-Medicare data to identify a population-based sample of 51,408 women aged >65 who were diagnosed with stage 1 or 2 breast

cancer in 1992-1999 and underwent primary surgical therapy. We assessed demographic and clinical factors associated with surveillance mammography using repeated measures logistic regression. We also examined whether follow-up care with providers of various specialties explained variation in mammography use.

RESULTS: Women in the cohort had a mean age of 76 years, 6% were black, 4% were Hispanic, and 56% were unmarried. Only three-quarters of women (77%) underwent mammography during months 7-18 after diagnosis, and only 56% of patients had a mammogram in each of the 3 surveillance years. In multivariable analyses, although surveillance mammography rates increased for patients diagnosed more recently ($P < 0.001$), the likelihood of mammography decreased in surveillance years 2 and 3 (vs. year 1; $P < .001$). Older patients, black patients, Hispanic patients, unmarried patients, those living in poor areas, stage 2 (vs. stage 1) patients, and patients in the highest comorbidity quartile were less likely than other patients to undergo surveillance mammography (all $P < .005$). Patients with more visits and those who continued to follow with a medical oncologist, radiation oncologist, or surgeon were most likely to have surveillance mammography (all $P < .001$); however, adjusting for visits with providers did not explain the lower mammography use among older women, black and Hispanic women, unmarried women, and women living in poor areas.

CONCLUSIONS: Many elderly breast cancer survivors do not undergo annual surveillance mammography, particularly women who are older, black, Hispanic, unmarried, and living in poor areas, and this underuse was not explained by access to follow-up care. New strategies are needed to increase use of screening mammography and decrease variations based on factors unrelated to appropriateness of medical care.

FOLIC ACID FOR THE PREVENTION OF NEURAL TUBE DEFECTS: ARE PHYSICIANS FAILING THEIR PATIENTS? N. Phifer¹; P.H. Smith²; D. Duffy². ¹Moses H. Cone Memorial Hospital, Greensboro, NC; ²University of North Carolina at Greensboro, Greensboro, NC. (Tracking ID #136060)

BACKGROUND: Studies show that periconceptual intake of 0.4 mg of folic acid daily significantly reduces a woman's risk of having an infant with a neural tube defect (NTD). Therefore, the USPSTF recommends that all women planning pregnancy take a daily multivitamin with 0.4-0.8 mg folic acid and that all women capable of pregnancy take 0.4 mg folic acid daily ("A" and "B" recommendations, respectively). However, a recent report suggests that only 40% of women in the US age 18-45 follow these recommendations. It is not known if internists routinely discuss this with their patients who are capable of pregnancy. The purpose of this study is to assess compliance with these recommendations in an internal medicine teaching clinic.

METHODS: A one-year retrospective chart review was performed. A sample of 132 women aged 18-45 was randomly selected. For inclusion, patients were required to have a uterus and to have come for 2 visits with their continuity care physician during the one-year study period.

RESULTS: The women enrolled had a mean age of 35 years. Sixty five percent were African American, 26% were HIV positive and 42% were uninsured. Discussions between physician and patient concerning folic acid supplementation were not documented in any chart nor was folic acid noted on any medication list. Contraception counseling was only documented in 16% of charts.

CONCLUSIONS: Physicians in a single internal medicine clinic are not discussing the use of folic acid with their female patients who are capable of pregnancy nor are they discussing contraception methods with them. If this is true in other internal medicine teaching clinics and settings, this may explain, in part, women's lack of participation in this effective yet inexpensive preventive measure. We will expand our study to other sites though we anticipate the results will be similar to the index site. An intervention will be developed to improve physician participation in this practice.

GENDER DIFFERENCES IN HEALTHCARE UTILIZATION FOR CHRONIC PAIN. S. Kaur¹; K.M. Stechuchak²; C.J. Coffman²; L.A. Bastian³. ¹Duke University/Durham VA Medical Center, Durham, NC; ²Durham Veterans Affairs Medical Center, Durham, NC; ³Duke University/Durham VA Medical Center/Duke University/Durham VA Medical Center, Durham, NC. (Tracking ID #135740)

BACKGROUND: A recent study on chronic pain by Kerns et al. reported 48% of veterans at their facility experienced pain regularly and expressed concern over their pain. Use of outpatient medical services in these veterans was significantly higher than those that did not report chronic pain. Women veterans especially are at risk for chronic pain syndromes due to high rates of depression and history of trauma. In this study, we examined gender differences in chronic pain syndromes in the VA and measured VA healthcare utilization related to chronic pain syndromes.

METHODS: Our first step was to identify all women veterans at the Durham VAMC in fiscal year (FY) 2002 between the ages of 21 and 60 that had two visits for the same pain location at least six weeks apart as determined by ICD-9 coding. We then identified a random sample of men with similar criteria for pain visits. Women and men that died in 2002 or 2003 were excluded. Men were then matched in age to the women for a 2:1 ratio. Pain location and number of pain sites for FY 2002 and number of clinic visits related to chronic pain for FY 2003 were determined for this sample through the national outpatient VA database. We compared number of pain related clinic visits between men and women using a two sample t-test. All analyses were conducted using SAS.

RESULTS: We identified 406 women veterans and 812 male veterans for this study. Mean age of both groups was 42.6 years (standard deviation or SD 8.5). The most common chronic pain conditions were as follows: Joint pain (51% men

and 45% women); Back pain (41% men and 23% women); Headaches (8% men and 22% women); Limb pain (5% men and 10% women); and Abdominal pain (3% men and 11% women). On average the women met sample criteria for a chronic pain condition in 1.2 areas (SD 0.5), whereas the men met sample criteria for 1.1 areas (SD 0.4). In comparing the number of pain related clinic visits in FY 2003 between women and men in our sample, women veterans averaged 4.8 clinic visits (SD 6.2) and men averaged 3.8 visits (SD 4.3) ($p < 0.004$).

CONCLUSIONS: To our knowledge, this is the first study to examine gender differences in chronic pain and healthcare utilization in the veteran population. As expected, there appear to be differences in pain locations between male and female veterans. Interestingly, healthcare utilization relating to women veterans with chronic pain is higher than for men. Future studies should explore the types of healthcare visits and the use of therapeutic options for these patients with chronic pain.

GENDER DISPARITIES IN CHOLESTEROL TESTING OF U.S. VETERANS WITH CARDIOVASCULAR DISEASE. R.A. Kaplowitz¹; E.V. Lawler²; D.R. Gagnon²; J.M. Gaziano¹.

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BACKGROUND: Veterans Affairs (VA) clinical practice guidelines recommend at least biannual evaluation of low-density lipoprotein cholesterol (LDL-C) in all U.S. veterans with known cardiovascular disease (CVD). We explored whether female veterans with CVD receive testing for LDL-C at rates equivalent to male veterans.

METHODS: We conducted a cross-sectional analysis of the administrative data of 16,315 male and female U.S. veterans with an ICD-9 diagnosis code for CVD who used VA New England Healthcare System outpatient services at least once yearly between January 1, 1998 and June 30, 2001. We examined the unadjusted association between gender and receipt of at least one LDL-C test during the 42-month study period using the Chi-square test. Logistic regression models determined the odds of cholesterol testing associated with female gender adjusted for outpatient service utilization and age as well as the presence of ICD-9 diagnosis codes for hypercholesterolemia, diabetes mellitus, hypertension, tobacco use, alcohol abuse, substance abuse, major affective disorder, thought disorder, anxiety disorder, and post-traumatic stress disorder (PTSD).

RESULTS: CVD was documented in 403 female veterans (2.5%) during the study period. 74.0% of women received at least one LDL-C test compared to 79.5% of men (unadjusted odds ratio (OR) 0.73, 95% confidence interval (CI) 0.58-0.92). The association of female gender with the likelihood of receipt of an LDL-C test did not substantially attenuate after adjustment for age and outpatient service utilization (OR 0.74, 95% CI 0.58-0.93) or with the addition of medical and psychiatric comorbidities into the logistic regression model (OR 0.75, 95% CI 0.58, 0.97).

CONCLUSIONS: Among U.S. veterans with CVD, women appear to receive LDL-C testing at lower rates than men. Adjustment for age, outpatient service utilization, mental illness, and medical comorbidities did not attenuate the observed disparity. Future studies should examine the patient, provider, and system factors that may explain gender differences in the medical management of CVD.

HEALTH PERCEPTIONS AND BEHAVIORS OF BREAST CANCER SURVIVORS. C.M. Duffy¹; B.E. Bond²; N.F. Col¹. ¹Brown University, Providence, RI; ²Tufts University/New England Medical Center, Boston, MA. (Tracking ID #134107)

BACKGROUND: Breast cancer is increasingly a curable disease, making attention to other chronic health issues important. Breast cancer (BC) survivors are at increased risk of osteoporosis and cardiovascular disease. However, little is known about survivors' risk perceptions for these chronic diseases, determinants of these perceptions, and the relationship between risk perception and health behaviors.

METHODS: Long-term BC survivors (≥ 5 years from diagnosis) were recruited from a breast health clinic in Massachusetts to participate in a survey study exploring attitudes toward menopause. Written questionnaires were given to BC survivors between April of 1999 and April of 2000. Descriptive and bivariate analyses were performed using SPSS.

RESULTS: Seventy-three women responded to the questionnaire. Median age at diagnosis was 49 years. Most women (61%) had early stage disease (DCIS, stage I and II), and most were Caucasian (83%), employed (88%), and married (59%). Over half (66%) had at least some college and 82% were privately insured. Most women reported very good or excellent health (59%). A minority of women reported their doctor discussed their chances of developing heart disease and/or hip fracture (36% and 43%, respectively). Very few women reported having heard of alendronate and raloxifene (8% and 2%, respectively). Well over half of women thought their chances were 1% or less of getting heart disease (55%) or dying from heart disease (61%), and most were either not at all or only slightly worried about getting (59%) heart disease. Women grossly underestimated their chances of breaking a hip: 70% believed their chances were 1% or less, and 83% were not at all or only slightly worried about breaking a hip. Respondents believed their chances of dying from a broken hip were minuscule (82% reported 0.1% or less). Most women (82%) were more concerned about heart disease than hip fracture. We performed bivariate analysis to determine factors associated with women's risk perceptions of heart disease and osteoporosis. Women with a family member affected with heart disease or osteoporosis (versus those without) reported higher perceived risk of these diseases. Having a physician discuss heart disease risk, or survivors' practice of cardioprotective measures (exercise and aspirin) was not associated with perceived risk of heart disease. Similarly, phy-

sician discussion of osteoporosis risk or survivors' practice of preventive measures (exercise and calcium) was not associated with perceived osteoporosis risk. **CONCLUSIONS:** Although more BC survivors will die from cardiovascular disease than breast cancer, BC survivors greatly underestimate their risk of heart disease. Despite BC survivors' elevated risk for osteoporosis, their perceived osteoporosis risk is also grossly underestimated. Family history of these diseases appears to have influenced perceived risk, while physician communication has not. Higher perceived risk was unexpectedly not associated with lifestyle and behavior changes to reduce risk. Attention to educating patients and physicians about screening and risk factor modification in this population is warranted, as are methods to enhance physician communication.

IMPACT ON UTILIZATION AND COST OF AN INTERVENTION TO DECREASE ANXIETY AFTER FALSE-POSITIVE MAMMOGRAMS. K.A. Stewart¹; P.N. Neumann²; S.W. Fletcher¹; M.B. Barton¹. ¹Harvard University, Boston, MA; ²Harvard School of Public Health, Boston, MA. (Tracking ID #135548)

BACKGROUND: Women with false-positive mammograms are at risk for heightened anxiety. We have previously reported results of a trial of a radiology intervention (immediate reading of screening mammograms) meant to decrease anxiety among women with false-positive mammograms, in which the intervention was associated with lower levels of anxiety. Because previous evidence suggested that false-positive mammograms are associated with increased use of medical care in the year following the false-positive mammogram, we investigated outpatient utilization and costs in the year following the false-positive mammogram among participants in this trial. We also evaluated the costs attributable to the study intervention.

METHODS: We identified study subjects from a trial involving 2,390 women aged 39 and older undergoing screening mammography between February 1999 and January 2001 in a large group practice in Massachusetts. Women with false-positive mammograms and continuous enrollment in the year following screening were eligible for the current study cohort. Medical care utilization was obtained from computerized medical record databases. Breast-related utilization was identified from billing and diagnosis codes. Costs included direct medical costs, patient time and travel, and additional time costs associated with the radiology intervention. Medical care costs were based on national averages from Medicare's 2000 Direct Practice Expense File and Resource Based Relative Value Scale for physician and laboratory services. We used non-parametric methods to compare utilization between groups, including the Wilcoxon 2-sided Z-approximation for crude comparisons and negative binomial models of visit counts for multivariable analyses. Cost data were log-transformed due to their highly skewed distribution. Crude and multivariable cost comparisons were based on log-linear regressions.

RESULTS: The radiology intervention was associated with a 15% increase in personnel time compared to usual care. The additional time cost assigned to the intervention was \$3.69 per screening mammogram. The utilization analyses followed 1142 women. Women with immediate reading (the intervention group) had more follow-up mammograms (783 vs. 750, $p = 0.02$) and fewer diagnostic ultrasounds (176 vs. 209, $p = 0.01$) than women in the usual care group. 50% of women in the intervention group had 1 or more return visits for additional imaging compared to 62% of other women ($p < 0.01$). Total breast-related visits and costs in the following year were similar between groups. Non-breast related visits were approximately 14% lower among the intervention group in the multivariable analysis, although no significant differences in total non-breast costs were found between groups. Sensitivity analyses of cost assumptions (e.g. doubling the cost of the radiology intervention, increasing women's time and transportation costs) did not materially affect the multivariable cost estimates.

CONCLUSIONS: Immediate reading of mammograms was associated with different patterns of follow-up testing for women with abnormal mammograms, but overall was cost-neutral with regard to mammography and ensuing breast-related follow-up. At the same time, the intervention, already demonstrated to be associated with lower anxiety levels, may have affected women's health-seeking behavior after the false-positive mammogram as demonstrated by the 14% decrease in non-breast related visits among women in the intervention group.

IMPLEMENTATION OF LIQUID-BASED PAPANICOLOU SMEARS WITH REFLEX TESTING FOR HUMAN PAPILLOMA VIRUS: REDUCING NEED FOR FOLLOW-UP VISITS AMONG HIGH RISK WOMEN. E. Kida¹; J.H. Arntsen¹; H.V. Kunins¹. ¹Albert Einstein College of Medicine, Bronx, NY. (Tracking ID #134369)

BACKGROUND: Cervical cancer screening using liquid-based Papanicolaou (Pap) smears combined with reflex Human Papilloma Virus (HPV) testing has a higher specificity for detecting abnormalities than conventional methods, potentially reducing the need for follow-up Paps or colposcopy. However, among HIV-infected or other women at high risk for HPV, liquid-based Pap smears may only minimally reduce the need for follow-up while still incurring additional cost. We sought to determine the impact of liquid-based Pap smears on need for follow-up Paps or colposcopy among substance-abusing women.

METHODS: Using electronic laboratory data from a nine-clinic substance abuse treatment program, we examined Pap smear results from January, 2002 through October, 2004. Screening for cervical cancer is offered to all female patients, and conventional Pap smear technology was replaced with liquid-based Pap smears with reflex HPV testing for Atypical Squamous Cells of Unknown Significance (ASCUS) in September 2003. Prior to this change, abnormal Pap smears necessitated either repeat Paps within 6 months, or referral to an off-site gynecology clinic for colposcopy. After this change, HIV-uninfected women were referred for colposcopy only if HPV testing was positive for a high-risk

HPV serotype, but only routine Pap smear screening was recommended if HPV testing was negative. Both before and after the change, HIV-infected women with ASCUS were referred for colposcopy. We used chi-square or Fisher exact tests to examine the change in proportion of needed follow-up Paps or colposcopy before and after implementation of liquid-based Pap screening. Because of the high prevalence of HIV among substance-abusing women, we adjusted the rate of needed follow-up colposcopy based on HIV-serostatus, and performed a sensitivity analysis for needed follow-up assuming different HIV prevalences.

RESULTS: Of 1541 Pap smears performed during this period, 986 used conventional technology and 555 used liquid-based methods. 179 of 986 (18.2%) conventional Paps were abnormal, compared to 107 of 555 liquid-based Paps (19.3%, $p=0.63$). The proportion of ASCUS results increased from 9.6% (95/986) with conventional technology to 12.6% (70/555) with liquid-based methods, and the proportion of Pap smears with other abnormal results decreased from 8.5% (84/986) to 6.7% (37/555) ($p=0.10$). Of the 70 ASCUS results detected via liquid-based cytology, 42 (60%) had negative HPV tests and did not require additional follow-up. The need for follow-up Pap or colposcopy was therefore reduced after liquid-based testing was implemented to 11.7% of Paps (65/555), compared to 18.2% (179/986) for conventional technology ($p=.001$). However, because HIV-infected women with ASCUS need colposcopy regardless of HPV status and the prevalence of HIV is 20% in our system, we assumed that 20% (8) of the 42 HPV-negative ASCUS results still needed follow-up colposcopy, changing the needed follow-up rate from 11.7% to 13.2% (v. 18.2% for conventional screening, $p=.01$). Varying HIV prevalence in a sensitivity analysis yielded a statistically lower needed follow-up rate for liquid-based versus conventional Paps, as long as HIV prevalence is less than 33%.

CONCLUSIONS: The increased specificity of liquid-based Pap smears with reflex HPV testing decreased the need for follow-up among women at high risk for HPV.

IMPROVING OSTEOPOROSIS SCREENING THROUGH IMPLEMENTATION OF A STANDARDIZED VISIT TEMPLATE. K.P. Palonen¹, C.I. Kiefe¹. ¹University of Alabama at Birmingham & Birmingham VA National Quality Scholars Program, Birmingham, AL. (Tracking ID #134902)

BACKGROUND: Osteoporosis is the leading cause of fractures among postmenopausal women. Screening women ≥ 65 years is recommended by the U.S. Preventive Services Task Force, and the Veterans Health Administration has added osteoporosis screening as a new performance indicator. We sought to evaluate the impact of a standardized template for a routine visit on osteoporosis screening and on counseling for adequate calcium intake.

METHODS: In a pre-post intervention design with comparison group, we studied women ≥ 65 years old seen at a local VA Medical Center (VAMC) women's clinic and gynecology clinic. Pre-intervention patients were seen Aug 2002–July 2003 and identified retrospectively using the Computerized Patient Record System (CPRS). The intervention consisted of (a) providing literature to the providers, once, on current guidelines for osteoporosis screening and adequate calcium intake (1200 mg/day) and (b) inclusion of a standardized template for routine women's clinic visit in the CPRS. Post-intervention data were collected over a six-month period following the intervention. Data abstracted from charts included past medical history and medications, evidence of osteoporosis screening with dual-energy x-ray absorptiometry (DEXA) within a two-year period and counseling for adequate calcium intake. For comparison, we also obtained baseline and follow-up rates of DEXA screening for all women ≥ 65 years seen at the VAMC, but not in our intervention clinics, in the same time period.

RESULTS: For the intervention clinics, baseline ($N=25$) and follow-up ($N=18$) patients did not differ significantly in age (74.0, 73.3 years, $p=0.71$), weight (167.9, 162.3 lbs, $p=0.66$), or ethnicity (89%, 92% white, $p=0.73$). DEXA screening increased markedly in the intervention ($p=0.003$) but not the comparison group (Table); calcium counseling increased from 36% to 83% ($p=0.002$) in the intervention group. Intervention providers used the standardized template in 89% of post-intervention visits, all of which also showed up-to-date osteoporosis screening.

CONCLUSIONS: Implementation of a standardized template for routine women's clinic visit appears to have markedly improved osteoporosis screening. As the proportion of women among veterans increases and this population ages, implementing a routine visit template may be of use in improving care for this group.

	Intervention		Comparison	
	N	% DEXA	N	% DEXA
Baseline	25	44	114	4.2
Follow-up	18	89	287	3.7

PATTERN OF OSTEOPOROSIS PREVENTION THERAPY AND EXPENDITURE IN MEDICAID BENEFICIARIES. J.A. Udell¹, M.A. Fischer², D.H. Solomon², N.K. Choudhry². ¹Harvard School of Public Health, Boston, MA; ²Brigham and Women's Hospital, Boston, MA. (Tracking ID #135177)

BACKGROUND: Hormone replacement therapy (HRT) has been widely prescribed among postmenopausal women, both for presumed cardiovascular benefit and for osteoporosis prevention, but its use has been decreasing since

adverse cardiac outcomes were found in the Women's Health Initiative (WHI). Little is known about the impact this trial has had on other osteoporosis prevention therapy and costs. Therefore, we evaluated osteoporosis prescribing trends and drug costs in Medicaid.

METHODS: Using quarterly data for all filled prescriptions in 50 state Medicaid programs from 1996 through mid-2004, time-series analysis was performed to measure changes in prescription patterns for osteoporosis prevention after the publication of the WHI. Five medication categories were defined: hormone replacement therapy, bisphosphonates, calcium, calcitonin, and selective estrogen receptor modulators (SERMs), such as raloxifene. Hormone replacement therapy includes both estrogen-only and estrogen-progestosterone combination therapies.

RESULTS: During the study period, osteoporosis prevention therapy increased from approximately 3.5 million prescriptions per year to over 11.1 million per year. Prior to publication of the WHI, HRT use was increasing, reaching a peak of 1.28 million quarterly prescriptions in mid-2002. After dissemination of the WHI results, there was a rapid 39% decline in HRT use to below 800,000 quarterly prescriptions in mid-2004 (Figure 1). Quarterly prescribing for bisphosphonates rose in parallel to HRT between 1996 and mid-2002, with steeper increases from 2000 onward. After the publication of the WHI, bisphosphonate use has continued to rise to over 1 million quarterly prescriptions. Slower rates of growth were observed for calcium supplementation, SERMs and calcitonins. Medicaid spending on these agents also changed dramatically over time (Figure 2). Prior to the publication of the WHI, drug expenditure for each class of medication increased. HRT quarterly expenditure grew from approximately \$12 million to \$37.3 million, while bisphosphonate reimbursements increased from \$2 million to \$44 million. The publication of the WHI was associated with a rapid decline in HRT quarterly expenditure but has had no effect on other treatments as they continue to rise. In particular, bisphosphonate expenditures have continued to rise, reaching over \$81 million per quarter by mid-2004.

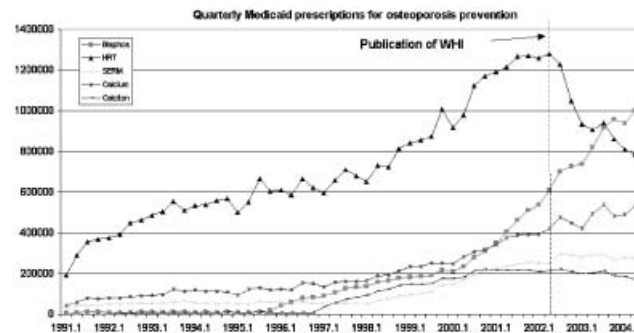


Figure 1



Figure 2

CONCLUSIONS: A substantial transformation in osteoporosis prevention therapy away from HRT towards newer alternatives, including bisphosphonates, has occurred since the publication of the WHI. This trend has had a significant impact on medication spending by state Medicaid programs. Determining whether these shifts are clinically appropriate and cost-effective for osteoporosis prevention will have important implications for the development of future drug-reimbursement programs, especially for elderly patients.

PREDICTORS OF SEEKING TREATMENT FOR URINARY INCONTINENCE AMONG ETHNICALLY-DIVERSE OLDER WOMEN. A.J. Huang¹, A. Kanaya¹, J.S. Brown¹, C. Wassel Fyr², S.K. Van Den Eeden³, A. Ragins³, L. Subak¹, E. Vittinghoff¹, D.H. Thom¹. ¹University of California, San Francisco, San Francisco, CA; ²University of California, San Francisco, SF, CA; ³Kaiser Permanente Division of Research, Oakland, CA. (Tracking ID #133386)

BACKGROUND: Urinary incontinence is a common and burdensome problem among older women, but the majority of women with incontinence do not dis-

cuss their symptoms with a physician. While studies suggest that women with more frequent or severe incontinence are more likely to seek treatment, other determinants of treatment-seeking are not well understood. We sought to identify characteristics associated with treatment-seeking in an ethnically-diverse population of middle-aged and older women with at least weekly urinary incontinence.

METHODS: The Reproductive Risks for Incontinence Study at Kaiser (RRISK) is a population-based cohort of 2109 women aged 40 to 69 years randomly selected from age and race strata from the Northern California Kaiser Permanente Care Program. Demographic and medical characteristics, as well as incontinence symptoms and patterns of treatment, were assessed by self-report questionnaires and in-person interviews for women with weekly incontinence. Severity of incontinence was assessed using the validated Sandvik Severity Score by multiplying the reported frequency and amount of leakage. Multivariable logistic regression analysis was used to identify characteristics independently associated with treatment-seeking.

RESULTS: Over a quarter of participants (28%, n=601) reported weekly incontinence, including 95 Black, 123 Latina, 65 Asian, and 308 White women. Only 41% (n=248) of women with weekly incontinence reported seeking treatment for their symptoms from a physician. Of these women, 42% (n=104) used pelvic floor muscle exercises, 19% (n=47) practiced timed urination, 11% (n=27) took medications, and 11% (n=27) underwent surgery for incontinence. Women were more likely to seek treatment if they had daily rather than weekly leakage, if they had severe rather than moderate incontinence by Sandvik Score (OR=1.9), if they had nighttime in addition to daytime leakage (OR=1.9), if their annual income was over \$100,000 (OR=3.1), if they were employed outside of the home (OR=1.8), or if they took more than 4 medications (OR=1.9) ($p < 0.05$ for each association). Age, race (Black, Latina, or Asian, versus White), clinical type of incontinence (stress versus urge), education, history of diabetes, and overall health status were not significantly associated with seeking treatment.

CONCLUSIONS: Clinicians should question women about symptoms of incontinence, since fewer than half of women who have at least weekly urinary incontinence seek treatment for this problem. Particular attention should be paid to women with lower incomes and those who do not work outside of the home, since they are at significantly higher risk of not seeking treatment, independent of incontinence severity and other demographic characteristics.

PRIMARY CARE PROGRAM DIRECTORS' PERCEPTIONS OF WOMEN'S HEALTH EDUCATION. A.L. Spencer¹; L.M. Kern¹. ¹Cornell University Medical College, New York, NY. (Tracking ID #135296)

BACKGROUND: Every year, women make over 125 million visits to primary care physicians. Previous research from the early 1990's showed that primary care program directors viewed women's health as important but felt that their residents were not adequately mastering the relevant skills. National guidelines have since been published emphasizing the importance of women's health in primary care medical education. The impact of these guidelines is unknown.

METHODS: We conducted a cross-sectional study of all primary care internal medicine residency program directors in the United States, as listed in the 2003 Society of General Internal Medicine Directory. We distributed surveys by e-mail. Using Likert scales, respondents rated how strongly they agreed that all primary care residents should have the knowledge and skills needed to care for each of 16 common clinical conditions (13 of which specifically reflect national guidelines for women's health competencies). They also estimated the proportion of their residents who actually master each of these topics by the end of residency. The topics were incontinence, evaluation of wet mount slides, pelvic examination, domestic violence, pre-conception counseling, amenorrhea, polycystic ovarian syndrome, hormonal and non-hormonal birth control, hot flashes, depression, osteoporosis, cancer screening, hypertension, diabetes, and cholesterol. Using McNemar's test for correlated proportions, we compared for each topic the proportion of program directors who somewhat or strongly agreed that residents should master a topic with the proportion who estimated that the large majority (75%) of residents actually master that topic. We also collected data on program demographics and characteristics of their women's health curriculum.

RESULTS: Of 82 program directors, 42 (51%) responded. Ninety-five percent of respondents reported that they were familiar with the national guidelines for women's health education. While 95% of programs have an ambulatory gynecology rotation available, only 67% of program directors reported that the large majority (> 75%) of their residents experience this rotation during residency. For each of the 16 topics, more than 75% of program directors agreed that it was important for residents to master. The program directors estimated that the proportion of residents who actually mastered these topics was much lower; only 3 of 13 women's health topics (6 of 16 topics overall) were mastered by the large majority (75%) of residents by the end of residency. Thus, there were significant discrepancies between what program directors thought residents should master and what they estimated residents actually master for 10 of 13 women's health topics ($p < 0.001$).

CONCLUSIONS: Despite the publication of national guidelines defining recommended clinical competencies in women's health, a large discrepancy persists between what program directors believe their residents should master and what they believe their residents actually master. These findings underscore the need for systematic improvement in women's health medical education in order to ensure the delivery of comprehensive health care to women by primary care residents.

QUALITY OF LIFE IN LATINA CERVICAL CANCER SURVIVORS. I. De Alba¹; A. Dogan-Ates¹; F.A. Hubbell¹; L.B. Wenzel¹. ¹University of California, Irvine, Irvine, CA. (Tracking ID #136174)

BACKGROUND: Cervical cancer disproportionately affects Latinas. Due to the increase in disease free intervals and survival of cervical cancer patients in the past decades, the number of Latina survivors is expected to increase. However, very little is known about the quality of life (QoL) and survivorship challenges, like coping efforts and sexual dysfunction, in Latina cervical cancer survivors.

METHODS: We used validated quality of life instruments to describe impact of cervical cancer in a group of 30 Latinas diagnosed 1-5 years earlier identified through the local cancer registry. General QoL was assessed with the Functional Assessment of Cancer Therapy (FACT-G) questionnaire; total and subscale scores in our group of Latinas were compared to standardized scores. Sexual functioning was assessed with the Sexual Activity Questionnaire (SAQ) and coping efforts with the Coping Orientations to Problems Experienced (COPE) scale. As an additional marker of coping, we also inquired about current desire for counseling. In addition, we assessed spiritual wellbeing with the FACT spirituality subscale. We compared the SAQ, COPE and FACT spirituality scores in our group to previously published mean scores among non-Hispanic cancer patients.

RESULTS: The median age in our group was 52 years, 73% were Mexican, 83% spoke only or mostly Spanish, 56% were married or living with a partner and 80% had less than high school education. The mean age at diagnosis was 47 years. All participants had a hysterectomy as part of the treatment, 60% also received radiation therapy and 33% received chemotherapy. The total scale score (81.9, SD 16.4) and the physical (22.0, SD 6.7), social (20.1, SD 4.7), emotional (18.2, SD 4.4) and functional (21.6, SD 4.9) subscale scores in our sample were similar to mean published FACT-G standardized scores (82.0, 20.5, 21.9, 14.8 and 18.0 respectively). The SAQ pleasure and discomfort subscale scores in our sample (12.2 and 4.4) were also similar to mean scores in non-Hispanic populations (13.6 and 4.1, respectively). Women in our sample had much higher COPE adaptive scores (42.2 vs. 26.4) as compared to previously published scores among non-Hispanic women; the COPE maladaptive scores (14.1 vs. 12.7) were similar in both groups. However, Latinas in our study expressed greater desire for counseling at the time of diagnosis (93% vs. 69%) and at the present time (83% vs. 59%) as compared to non-Hispanic survivors in recently published data. The FACT spiritual well-being scores in our group was equivalent to those published in previous validation studies (39.3 vs. 38.5).

CONCLUSIONS: Despite low income and low acculturation, Latina cervical cancer survivors in our sample had general quality of life, sexual functioning and spiritual well-being comparable to that of non-Hispanic cervical cancer survivors. Moreover, Latinas had considerably higher coping scores. To our knowledge, this is the first study that assesses quality of life in Latina cervical cancer survivors; further research is needed to confirm these findings and to assess the potential impact of coping strategies on survivorship experience among Latinas with cervical cancer.

RAPE AND ASSAULT OF WOMEN IN THE U.S. MILITARY—PREVALENCE, TREATMENT, AND HEALTH CONSEQUENCES. D. Washington¹; B. Simon²; S. Sun¹; E. Yano³.

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BACKGROUND: Fifteen percent of U.S. military troops currently deployed in Iraq and Afghanistan are female. Periodic media reports and research in VA patient populations consistently identify military sexual assault and trauma (MST) as a risk to which these women are exposed. However, the prevalence, treatment and health consequences of MST among veteran women in the general population are unknown.

METHODS: We conducted a telephone survey of a population-based stratified random sample of 2,174 women veterans residing in the southern California and southern Nevada areas, to determine MST prevalence, associated co-morbidity, and subsequent treatment. Stratification was by VA use (VA user versus nonuser) and age group (younger versus older women), to enroll sufficient numbers of smaller groups to increase precision of estimates. Bivariate and multivariate analyses adjusting for socio-demographic and health-related characteristics were conducted to examine MST prevalence, physical and mental health, and treatment. All analyses applied weights to account for disproportional allocation of the population by strata, so that resulting estimates are representative of the general women veteran population.

RESULTS: 572 women (32.2% of VA users, 15.5% of nonusers, 16.9% overall) reported physical assault that was not a result of combat, or being raped during military service. MST prevalence by war era was 6.5% for pre-Vietnam, 20.4% for Vietnam, and 20.0% post-Vietnam. Of women reporting MST, 15.9% currently use VA healthcare, however, 16.8% avoid or delayed using the VA because of their MST experience. 84.1% of women reporting MST only use services outside the VA. Compared to those without MST, women who experienced MST had lower SF-12 physical component scores (6.3 point mean difference) and mental component scores (5.4 point mean difference), and a greater likelihood of screening positive for depression (OR=2.8), anxiety disorder (OR=3.0), post-traumatic stress disorder (OR=18.1), and hazardous alcohol use (OR=1.3) (all $p < .05$). 6.4% of women received MST counseling or treatment through the VA; 21.7% received care in non-VA settings; 71.9% received no care for these incidents. Receipt of MST care was more likely among women who screened positive for a mental disorder, have a VA service-connected disability, and fair or poor health.

CONCLUSIONS: MST is highly prevalent, associated with significant co-morbidity, but is often untreated. While MST prevention efforts should obviously be at the forefront, the degree to which MST remains under-reported and untreated

demands that greater attention be paid to improved detection in the military to facilitate access to needed care. After military service, the majority of women veterans see primary care providers who practice outside of the VA and may be unaware of their veteran status and MST risk. Given the high prevalence of MST and its health consequences, interventions to help providers outside the VA improve detection and early treatment are also warranted. While the VA mandates MST screening and counseling, these services can only reach the tip of this proverbial iceberg.

SOCIAL SUPPORT AND PHYSICAL HEALTH: GENDER AND AGE EFFECTS. S.M. Frayne¹; V.A. Parker²; C.L. Christiansen²; S. Loveland²; M.R. Seaver³; L. Kazis². ¹VA Palo Alto Health Care System, Menlo Park, CA; ²CHQOER, VA Hospital, Bedford, MA; ³VA Healthcare System, Boston, MA. (Tracking ID #136260)

BACKGROUND: Social support is an important determinant of health outcomes. Since social roles vary by gender and age, we examined the effect of social support upon physical health status as a function of patient gender and age.

METHODS: Data came from the 1999 Large Health Survey of Veteran Enrollees, mailed in 1999 to a random sample of 1.5 million veteran Veterans Health Administration (VA) enrollees nationally (response rate 63%) and designed to characterize their health status. The 28,048 women and 651,811 men who reported that they had used VA care in the prior three years constituted our analytic cohort. The dependent variable was Physical Component Summary (PCS) score, a well-validated measure of physical health status from the Veterans Short Form-36 instrument. Measures of social support were originally developed for the Veterans Health Study and included whether the patient was married and whether he/she had someone who could take him/her to the doctor if needed.

RESULTS: Women tended to be younger than men; 26.4% versus 54.6%, respectively, had age ≥ 65 years. Across every age stratum, women were less likely than men to have social support. This effect was largest in older patients; for example, among patients 65-75 years old, 70.8% of women versus 32.2% of men were not married, and 18.5% versus 10.6% had no one who could take them to a doctor. As the Table shows, in most subgroups, being married was associated with worse PCS score; however, having someone to take the patient to the doctor was associated with better PCS scores, especially in women.

CONCLUSIONS: The consistently lower levels of social support seen across the age spectrum in women compared to men in VA is cause for concern, especially since better social support translates into better physical health status in women (except for being married in the youngest women). The paradoxical adverse effect of marriage upon health in men and in the youngest women deserves more inquiry. Generalists should be at the forefront of efforts to integrate care using a biopsychosocial model, taking the distinct needs of women and men in different age groups into account.

TABLE: Difference in mean PCS in patients with vs. without social support; a positive difference means more social support is associated with better physical health status.

	Married		Someone Could Take to Doctor	
	Female	Male	Female	Male
Age 18 - 44	-1.9*	-2.9*	1.8*	1.0*
Age 45 - 64	0.1	-1.2*	2.0*	0.1
Age 65 - 98	2.0*	-0.8*	0.9	0.0

*P < .01

THE EFFECTIVENESS OF A PRIMARY CARE EXERCISE INTERVENTION FOR RURAL WOMEN. B.J. Sherman¹; G. Gilliland²; J. Speckman²; K.M. Freund¹. ¹Boston University, Boston, MA; ²Missouri Community Center of Excellence in Women's Health, Kirksville, MO; ³Boston Medical Center, Boston, MA. (Tracking ID #133698)

BACKGROUND: Physical inactivity and poor diet together are the second leading cause of death in the US. Rural women experience greater obstacles to engaging in healthy behaviors, due to higher rates of poverty, chronic disease and illness, lower levels of education and traveling further distances for health care. Lack of formal exercise programs, less walking for transportation, and little opportunity to incorporate physical exercise into daily activity has resulted in epidemic obesity levels among rural women. The purpose of this study is to assess the feasibility of a primary care-based walking intervention for rural women.

METHODS: A pre-post exercise intervention was conducted with women recruited through the Northeast Missouri Health Council Primary Care Clinic, or local advertisements. At intake, demographics, physical activity level, vital signs and blood chemistries were collected and women were given an exercise videotape, pedometer and exercise pamphlets. Participants kept daily step count logs and were called at set intervals for their step counts from the preceding 3 days. Average 3-day pedometer readings were calculated assessing activity level between time intervals. Women were categorized based upon calculated body mass index (BMI) as normal weight, overweight or obese and dichotomized by self-reported baseline physical activity. We compared pre and post intervention pedometer readings. Intervention success was assessed by two tailed paired t-test of pre and post pedometer scores. Individual paired t-tests were conducted for each BMI groups and by baseline physical activity group.

RESULTS: Of the 75 women enrolled, 61 (80%) completed the study protocol. One participant was removed as her data qualified as an outlier. Of the 60

women, the average age was 42.5 years. All were Caucasian. Average length of time in the study was 146.1 days. Mean BMI was 30.6 at intake, with 37 women engaging in regular physical activity at baseline. The mean pedometer reading at intake was 6337 steps. Significant difference was found in starting step counts between the BMI groups. Overall end step count was 8910. The overall difference between pre and post counts was significant for each individual ($p < .001$). Significant improvements in step counts were found in each BMI group. The mean pedometer reading at baseline was statistically different for women reporting physical activity (mean 6972) and those who did not (mean 5314; $p < .0005$). An increase in both groups was seen after the intervention (9899 and 7318 respectively; $p < .0005$).

CONCLUSIONS: A brief intervention using educational materials and pedometers, organized through a community health center can achieve short-term increases in physical activity in rural women. All groups of participants benefited from the intervention, regardless of BMI or self-reported exercise status at intake. Longer term studies are needed to determine if long-term benefit can be derived from instituting such an intervention in the primary care setting, with intermittent reinforcement during other primary care visits. These initial findings are promising that primary care providers can begin to address obesity through specific intervention targeted at modifying physical activity.

THE ROLE OF NUMERACY ON BREAST AND COLORECTAL CANCER SCREENING AMONG URBAN WOMEN. A. Aggarwal¹; M.K. Passche-Orlow¹; K.S. Roloff¹; J. Speckman²; T. Battaglia¹. ¹Boston University, Boston, MA; ²Boston Medical Center, Boston, MA. (Tracking ID #134109)

BACKGROUND: Though many clinicians use quantitative information in discussing cancer risks and screening benefits with patients, the association between patient numeracy (facility with numerical concepts) and cancer screening practices is unknown. We hypothesized that innumerate patients would have lower awareness of breast cancer and colorectal cancer screening guidelines and lower rates of being up-to-date with screening than patients with basic numeracy skills.

METHODS: A self-administered survey was given to consecutive female patients > 40 years presenting for primary care at several sites affiliated with an urban academic medical center. Subjects were excluded if they did not speak English, had a personal history of cancer, or cognitive impairment. Numeracy was measured using a test question adapted from a validated instrument: 'Imagine that you flip a coin 1000 times. How many times will the coin come up heads?' Subjects were categorized as innumerate if they failed to answer the question or did so incorrectly, as unaware of breast cancer and colorectal cancer screening guidelines if they answered questions incorrectly on the age to start screening, and up-to-date if they had a mammogram within two years or (for those age > 50) had fecal occult blood test in past year OR ever had routine lower endoscopy. Chi-square tests of independence were used to compare awareness and being up-to-date with numeracy status.

RESULTS: At interim 150 women completed the survey: mean age 55 (40-82) years; 45% white, 45% Black, and 11% others; while 21% had not graduated high school, 24% completed > 3 years of college; 51% had private insurance and 86% had a regular primary care physician; 25% of those who answered the question had income < \$20,000 per annum. Fully, 56% of women were innumerate. While 67% were unaware of breast cancer screening guidelines, 78% were up-to-date with mammography; 65% were unaware of colorectal cancer screening guidelines and 51% were up-to-date on colorectal cancer screening. Being innumerate was associated with: Older age (100% > 70 vs. 52% < 70 years, $p < 0.01$); minority race (64% Black vs. 42% White, 81% Other $p = 0.03$); lack of education (87% not graduated high school, 32% > 3 years college, $p < 0.01$); lack of insurance (44% private vs. 67% no private, $p < 0.01$); lower income (68% < \$20,000 vs. 40% > \$20,000, $p < 0.01$); and being unaware of breast cancer screening guidelines (63% unaware vs. 42% aware, $p = 0.02$). Numeracy was not found to be associated with being up-to-date with breast or colorectal cancer screening.

CONCLUSIONS: In an urban group of women with diverse educational attainment, more than half of the subjects could not correctly answer a simple question relating to risk. Being innumerate was associated with age, ethnicity, education, insurance, income and poor knowledge of breast cancer screening guidelines but was not associated with differences in screening practices for breast and colorectal cancer. Our data suggest numeracy and knowledge of cancer screening guidelines are not key determinants of cancer screening among urban women in primary care settings.

USE OF COMPLEMENTARY AND ALTERNATIVE MEDICAL THERAPIES IN BREAST CANCER SURVIVORS IN THE NURSES HEALTH STUDY. C. Buettner¹; C.H. Kroenke¹; R.S. Phillips¹; D.M. Eisenberg¹; M.D. Holmes¹. ¹Harvard Medical School, Boston, MA. (Tracking ID #133538)

BACKGROUND: Complementary and alternative medicine (CAM) is used commonly in the general population and even more frequently by breast cancer survivors, but little is known about reasons for use and perceptions of efficacy. Use of CAM among nurses who are breast cancer survivors has not been examined.

METHODS: We surveyed breast cancer survivors about their use of 25 CAM therapies over the previous 2 years as part of the Nurses' Health Study and Nurses' Health Study II. We asked about reasons for using CAM (to treat cancer or its symptoms, to treat other illnesses, or for general wellness). If the reason was to treat breast cancer or its symptoms, we asked if it was helpful (very, somewhat, not very, not at all). To investigate factors associated with CAM use,

we used [chi]2 and t-tests to analyze data on age, menopausal status, religious affiliation, breast cancer stage 0-IV (confirmation of stage available for 69% of respondents), cancer treatments, smoking status, self-perceived health, and bodily pain. Variables with significance levels of $p < 0.15$ on univariate tests were included in a multivariate logistic regression model.

RESULTS: Of 8,168 breast cancer survivors (mean age 70, 91% white), 6,637 (81%) returned questionnaires. Other than self-prayer (used by 75%) and diet therapies (used by 50%), 59% of respondents reported CAM use over the past 2 years. The most frequently reported CAM therapies were relaxation therapies (28.8%), massage (19.7%), high-dose vitamins (18.2%), herbs (16.3%) and imagery (12.7%). Of those who used CAM, 20.2% used it to treat cancer or its symptoms, 35.1% used it to treat illnesses other than cancer, and 73.4% used it for general wellness. Among women using it for cancer or its symptoms, the most common therapies used were self-help group (28.7%), spiritual healing by others (26.0%), imagery (25.4%), relaxation (24.5%) and herbs (21.5%). Therapies rated "very helpful" for treating cancer or its symptoms were massage (64.2%), spiritual healing by others (64.1%), self-help group (60.4%) and relaxation (52.1%). Factors independently associated with a higher likelihood of CAM use included having had reconstructive surgery (OR=1.8, 95% CI: 1.6, 2.1), chemotherapy (OR=1.6, 95% CI: 1.4, 1.8), lumpectomy (OR=1.2, 95% CI: 1.1, 1.4) and pain (OR=1.2, 1.1, 1.2).

CONCLUSIONS: This study confirms earlier studies suggesting that use of CAM is very common among breast cancer survivors. Treatment factors and the experience of pain were associated with CAM use. The majority of users of massage, spiritual healing by others, self-help groups, and relaxation reported finding these therapies very helpful for treating cancer or its symptoms.

UTILIZATION OF VETERAN'S HEALTH ADMINISTRATION SERVICES AMONG WOMEN VETERANS WITH DIABETES. K.P. Palonen¹; S.G. Kertesz²; M.M. Safford²; S.A. Rose²; K. Kirk²; J.H. Halanych²; C.I. Kiefe¹; U. Alabama Birmingham & Birmingham VA National Quality Scholars, Birmingham, AL; ²Birmingham VA Medical Ctr. & U. Alabama Birmingham, Birmingham, AL. (Tracking ID #135679)

BACKGROUND: Women represent an increasing proportion of patients served by the Veterans Health Administration (VHA); hence delivering high quality of care for women with chronic diseases, such as diabetes, is an important priority. We assessed gender differences in utilization of diabetes-care related services.

METHODS: We identified patients with diabetes (≥ 2 visits associated with 250.xx diagnostic codes) from a 5% sample of VHA National Patient Care Database for FY2003. We evaluated gender differences in the proportions of patients with diabetes seen in Primary Care clinic (≥ 4 visits per year associated with diabetes care), Diabetes/Endocrine clinic (≥ 1 per year), Optometry/Ophthalmology clinic (≥ 1), and Podiatry/foot-care clinic (≥ 1). In order to evaluate diabetes care specifically, only visits that had an attached ICD-9 diabetes code were included. Logistic regressions adjusted for age and race.

RESULTS: Women were 2.4% ($n=870$) of the 35,721 persons with diabetes. Compared to men, more women utilized Primary Care (25.2% of women, 20.1% of men, $p<0.01$) and Diabetes/Endocrinology clinic services (25.2%, 16.5%, $p<0.01$). Women and men used Optometry/Ophthalmology (15.3%, 14.4%, $p=0.5$) and Podiatry/Foot-care services (19.8%, 19.8%, $p=1.0$) similarly. In multivariable analysis, women were more likely than men to utilize primary care clinic services only in two age groups: 45-54 years (Odds ratio (OR) 1.7, 95%CI 1.1-2.8) and ≤ 65 years (OR 4.0, 95%CI 1.9-11.4). In all age groups, women were more likely to utilize Diabetes/Endocrinology services (OR 1.4, 95%CI 1.2-1.6). In addition, a sex-by-race interaction was significant for Podiatry/Foot-care ($p=0.01$): white women were more likely to utilize this service than white men (OR 1.4, 95%CI 1.2-1.8). No statistically significant differences were seen between the sexes in non-whites.

CONCLUSIONS: Overall, utilization of potentially relevant services was low for this representative sample of patients with diabetes who used the VHA. After adjusting for age and race, men and women were equally likely to utilize Eye and Foot care, with women more likely to utilize the Primary Care and Diabetes/Endocrine clinic services. Understanding the factors that lead to higher utilization among women may lead to improved access for all.

CLINICAL VIGNETTES

ACQUIRED HEMOPHILIA. R. Sanghani¹; R. Dhokar¹; W. Hadid¹; J. Cabello¹; S.B. Glick¹. ¹John Stroger Hospital of Cook County, Chicago, IL. (Tracking ID #133303)

LEARNING OBJECTIVES: 1. Recognize acquired hemophilia as an important cause of spontaneous hemorrhage, especially in otherwise healthy elderly adults; 2. Recognize the clinical presentation of acquired hemophilia; 3. Diagnose coagulation factor inhibitors.

CASE: A 77 year-old man presented with spontaneous swelling and pain of the forearms bilaterally. There was no physical abuse or other trauma. The patient had prolonged bleeding following dental extraction one month prior. Before that, he'd never had easy bruising or bleeding, even following appendectomy in his youth. The family history was remarkable for a daughter with factor XI deficiency and inhibitors to factors VIII and IX. On physical examination, there was swelling, ecchymosis and erythema of the forearms bilaterally consistent with bleeding into the soft tissues. Range of motion was restricted at the elbows. The radial pulses were normal. Laboratory studies revealed hemoglobin=8.5, hematocrit=25.5 and platelet count=320. The PT was normal; PTT was 69.2 (24.6-36.2). The PTT improved immediately with 50:50 mix; after incubation at 37 degrees the PTT was again elevated. Factor VIII level was 2 (50-149). Factors IX

and XI levels were normal. Acquired hemophilia due to factor VIII inhibitor was diagnosed. The patient was transfused fresh frozen plasma and packed red blood cells. Treatment with cyclophosphamide and prednisone was begun. At the time of discharge, the forearm swelling had improved; the patient had no other bleeding.

DISCUSSION: Though hemophilia is most often congenital, it may also be acquired. Patients with acquired hemophilia typically present with spontaneous bleeding into the soft tissues or muscles. Severe or life-threatening bleeding is seen in 80% of patients. Hemarthroses are rarely seen. Acquired hemophilia is caused by antibodies (inhibitors) that alter clotting factors and impair their function. Most often, these are alloantibodies in patients with congenital hemophilia exposed to infused clotting factor. Autoantibodies to clotting factors may also occur, usually in otherwise healthy older adults. Factor VIII inhibitor is the most common of these. Laboratory studies reveal prolongation of the PTT; the PT and bleeding time are normal. Mixing study with 50% patient plasma and 50% normal plasma may correct the PTT initially suggestive of factor deficiency; however, because factor VIII antibodies are time and temperature-dependent, incubation of the mixed sample at 37 degrees results in prolongation of the PTT indicative of inhibitor. Intrinsic pathway factor levels are measured to show the inhibitor is specific. Factor VIII antibodies are usually idiopathic, though in 40-50% of patients they are related to the post-partum period or associated with another disease, including malignancy, autoimmune disease, or drug exposure. The goals of treatment are to abort the bleeding episode and eradicate the antibody. Acute bleeding may be treated with factor VIII (human or porcine) or desmopressin. Patients with high inhibitor titers may require agents that bypass factor VIII in the coagulation cascade, such as activated prothrombin complex concentrate or recombinant factor VIIa. Immunotherapy is used to eliminate the antibody. Prednisone and cyclophosphamide result in complete remission in 60-100% of patients.

AN UNUSUAL SUSPECT. K.H. Trivedi¹; S. Dea¹; M. Rotblatt¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #134560)

LEARNING OBJECTIVES: 1) Recognize that HIV/AIDS is not uncommon in the elderly population 2) Assess the causes of chronic diarrhea in an elderly and immunocompromised patient.

CASE: The patient is an 82 year-old Hispanic man brought in by his daughter for 2 months of persistent watery diarrhea, nausea, vomiting, anorexia and a 20-pound weight loss. He also complained of increasing generalized weakness over the past 3 weeks, during which time he had fallen on multiple occasions. He denied hematemesis, melena or hematochezia. The patient has a past medical history of hypertension and aortic valve insufficiency. He denied IVDA, occasionally drank alcohol and had a remote history of tobacco use. The patient's last travel was to El Salvador in 2003. On admission, the patient was afebrile and normotensive without tachycardia. On physical exam, he was a thin, elderly man with dry mucous membranes and a flat JVP. He had a 2/6 diastolic murmur heard at the left sternal border. His abdominal exam was unremarkable, and a bedside hemocult was negative. He had an unsteady gait on neurological exam but was not orthostatic. On laboratory evaluation, the patient had a slight normocytic anemia, but his serum chemistries, liver tests and LDH were normal. A chest x-ray revealed cardiomegaly and a tortuous aorta. His head CT demonstrated generalized atrophy with calcifications but no acute bleed. The patient was admitted for syncope, dehydration and a work-up of chronic diarrhea. Stool occult blood was positive but studies for fecal fat, stool WBCs and C. difficile toxin were negative. The patient's serum iron saturation was also low. When a fasting trial did not resolve the patient's diarrhea, our GI consultants performed a colonoscopy, revealing only internal hemorrhoids. Although the colonic mucosa appeared normal, staining of random mucosal biopsies identified CMV in the lamina propria. An upper endoscopy demonstrated candida esophagitis, and a subsequent HIV screening test was positive. The patient started treatment with fluconazole and gancyclovir and began HAART therapy soon thereafter.

DISCUSSION: Nearly 900 individuals over the age of 65 were diagnosed with AIDS in 2003, most contracting the virus through transfusions. Clinicians may not suspect HIV in their older patients, even when they present with an AIDS-defining illness. HIV-related infections secondary to pathogens such as MAC, CMV, Cryptosporidium and Isospora frequently involve the GI tract and can lead to chronic diarrhea. In the geriatric population, diarrhea can be a serious problem as the physiologic changes secondary to aging may lead to severe dehydration, renal failure and cardiovascular collapse. A careful, systematic work-up can help avoid these sequelae. Although we did not initially consider HIV in this patient, a thorough evaluation of his chronic diarrhea led to the correct diagnosis.

THE ROLE OF THE GENERALIST IN DIAGNOSING AND PREVENTING IRREVERSIBLE BLINDNESS IN GIANT CELL ARTERITIS. D. Mukherjee¹; M.H. Pillinger¹. ¹New York University, New York, NY. (Tracking ID #135800)

LEARNING OBJECTIVES: 1. Recognize that in giant cell arteritis (GCA), non-specific systemic symptoms can precede vision loss. 2. Recognize the importance of a detailed review of systems and a high index of suspicion in the elderly. 3. Identify steroid-induced myopathy as an adverse effect of high dose steroids used to treat GCA.

CASE: An 82 year-old male presented C/O sudden partial loss of vision in his left eye, without pain, photophobia or discharge. In the 2 weeks prior to presentation he reported 5 lb weight loss, anorexia, fatigue, chills, and his jaw "getting tired" after chewing. Attributing his symptoms to the flu, as well as ill-fitting dentures, he did not seek medical attention until the vision loss. PT denied scalp

or temporal pain, headaches, limb-girdle pain or stiffness. Physical exam revealed low-grade fever and temporal field vision loss of the L eye. The L optic disc was swollen, pale and elevated with blurry margins. He had no scalp or temporal tenderness, carotid or temporal artery bruit, or painful or decreased range of motion in the shoulders/hips. Both ESR (104 mm/hr) and CRP (2.25) were elevated. He had normocytic anemia Hb/Hct (12/36), MCV (83). He refused temporal artery biopsy, and was treated for presumptive GCA with high dose methyl-prednisolone, 250mg IV Q6H for 3 days, followed by prednisone 60mg PO QD. His jaw pain and systemic symptoms resolved, with normalization of ESR (10 mm/hr) and CRP (<0.10) in 2 weeks. Loss of vision persisted, without progression. At one month follow-up, the patient was cushingoid, and complained of generalized weakness, especially of the lower extremities, starting 2 weeks after initiating therapy. He had difficulty rising from a chair or climbing stairs. He denied pain or stiffness. Physical exam showed mild muscle atrophy, 4/5 symmetric muscle strength without tenderness, no limb girdle stiffness, normal sensation and deep tendon reflexes. He had normal TSH (1), ruling out hypothyroidism, and very low CPK (23), ruling out myositis. He was diagnosed with steroid myopathy. A steroid taper was planned, with close monitoring of his symptoms and ESR.

DISCUSSION: GCA affects 0.2% of people over age 50. Symptom onset is usually gradual, and may be non-specific, or flu like as in this case. Patients typically experience headaches, though one-third present without headache. Pain may, but does not always, affect the temples, jaw and tongue. Maintaining a high index of suspicion in patients without visual symptoms, may permit early diagnosis and treatment to prevent irreversible blindness. Temporal artery biopsy before initiation of treatment is advisable. Patients usually are treated with prednisone (1 mg/kg/day, PO) for 1 month prior to taper. In the setting of blindness, higher IV doses are initiated. Steroid myopathy is a potential complication that is increased in the elderly, and with doses exceeding 40mg/day for >1 month, as used for GCA. Polymyalgia rheumatica frequently accompanies GCA, causing limb-girdle stiffness and pain instead of weakness that is predominant in steroid myopathy. TSH and CPK should be checked to rule out hypothyroidism and inflammatory myositis. Onset of weakness after steroid initiation, cushingoid features, normal or low muscle enzymes, normal TSH, and absence of pain or stiffness favor steroid myopathy. Diagnosis is confirmed by return of muscle strength 3 to 4 weeks after marked dose reduction (typically ≤ 10 mg/day), or discontinuation of steroids.

THROMBOSED SACCULAR OUTPOUCHING OF THE AORTIC ROOT SIMULATING AN INTRACARDIAC MASS. C.C. Onunkwo¹; J. Shah¹; A. Wickramasinghe¹; A. Ameen¹; M. Reisner¹. ¹Jersey City Medical Center, Jersey City, NJ. (Tracking ID #135964)

LEARNING OBJECTIVES: 1. Recognize that an unruptured aneurysm of the sinus of Valsalva may present as an intracardiac mass. 2. Delineating the nature of a cardiovascular mass may necessitate multimodality imaging studies. 3. Review diagnosis and management of unruptured aneurysm of the sinus of Valsalva.

CASE: We report the case of a 77 year old woman admitted to our facility with worsening exertional dyspnea. Trans-thoracic echocardiography revealed a large echo dense shadow in the right atrium suggestive of an atrial myxoma versus an atrial thrombus. Dipyridamole stress test was consistent with inferior left ventricular ischemia. A transesophageal echocardiogram showed a sessile mass of mixed tissue reflectance with a predominant cystic component, attached to the interatrial septal annulus of the tricuspid valve extending from the superior vena cava-right atrial junction and inferiorly up to the coronary sinus os. Color flow Doppler did not show any communication between the mass and adjacent structures. The differential diagnosis was expanded to include, an atrial myxoma versus a necrotic tumor versus a saccular aneurysm versus an echinococcal cyst. Cardiac catheterization showed 90 percent stenosis of the midsegment of the right coronary artery, aortic root aneurysm and an ill defined posterior para-aortic density. CT Scan and MRA of the chest revealed a 7.7 cm aneurysm of the ascending aorta and a saccular outpouching from the aortic root in the region of the non-coronary sinus impacting on the right atrium. The aneurysm was almost completely thrombosed. The patient underwent successful surgical repair of the aneurysm and the outpouching. The Right coronary was bypassed with a Saphenous venous graft. The postoperative course remained uneventful.

DISCUSSION: Aneurysms of the sinus of Valsalva rarely present with symptoms unless rupture occur. They often remain undiagnosed. There have been a few case studies reported of unruptured aneurysm of the sinus of Valsalva, which have presented with conduction disturbances, coronary ischemia from the compression of the coronary arteries, hemodynamic compromise and arrhythmia. The 3 sinuses of Valsalva are located in the most proximal portion of the aorta, just above the cusps of the aortic valve. They are named, the right sinus of Valsalva, the left sinus of Valsalva and the Non-coronary sinus. The sinuses correspond to the individual cusps of the aortic valve. Aneurysm of the sinus of Valsalva is a rare congenital cardiac defect but may also be acquired. This case underscores, the diligent multimodality imaging approach to diagnose cardiovascular masses, and is unique due to the rarity of unruptured sinus of Valsalva in the Geriatric age group.

AN UNCOMMON RASH WITH A COMMON CAUSE: A CASE OF CHRONIC ERYTHEMA MULTIFORME. C. Finney¹; A. Cooperman¹; L. Powell¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #133366)

LEARNING OBJECTIVES: 1) Identify the clinical presentation of erythema multiforme (EM). 2) Recognize EM as a presentation of Herpes Simplex Virus (HSV) infection. 3) Discuss treatment of EM minor.

CASE: A 48 year old woman presented to clinic with longstanding, extensive bullous skin lesions. The patient reported painful, pruritic blisters involving her extremities, torso, and oral mucosa for six years. New lesions erupted on a near daily basis, evolving from papules to blisters and target lesions, eventually healing without scarring. The patient was previously seen at a dermatology office where a skin biopsy revealed EM. She received a trial of hydrocortisone, clotrimazole and tacrolimus creams, none of which improved the lesions. She was diagnosed with depression and anxiety, which she attributed to this extensive, unsightly and painful rash that persisted despite abovementioned treatment. Physical examination was only significant for skin lesions at various stages of development. Dusky red papules, vesicles, bullae and target lesions were present on her extremities and torso. Healing lesions left areas of hyperpigmented macules. Oral exam revealed minimal blistering of her buccal mucosa. Acyclovir was initiated empirically for chronic HSV suppression. On follow up the patient happily reported that she had not developed any new lesions.

DISCUSSION: EM minor is a self-limited, often recurring hypersensitivity reaction characterized by symmetric, polymorphous cutaneous and sometimes ocular and mucous membrane eruptions. Cutaneous target-like lesions are pathognomonic for the disease. In EM minor, eruptions are confined to cutaneous lesions with involvement of no more than one mucosal surface. In severe forms of EM, termed EM major or Stevens-Johnson syndrome, two or more mucous membrane surfaces, eyes, and internal organs are involved. Many cases of EM are now believed to be the result of a cell-mediated immune reaction to HSV antigen. HSV DNA has been identified within affected keratinocytes by PCR amplification. What was described as idiopathic EM in the past may be associated with subclinical HSV infection. The remaining cases are caused by multiple other infectious agents, drugs, irritant contacts, or are associated with systemic illnesses. EM may account for up to 1% of dermatology referrals. Internists need to be familiar with initial approaches to managing this condition. While EM minor is usually self-limited, chronic or recurrent EM minor should elicit a search for provoking factors. Associated infections should be ruled out or treated. Offending drugs must be discontinued. When no obvious inciting factor can be elicited, an empiric course of acyclovir should be instituted to suppress subclinical HSV infection. Acyclovir prophylaxis can then be continued if the patient responds to this treatment or if EM recurs more than five times per year.

AN UNUSUAL CAUSE OF PURPLE TOES. A.K. Gupta¹; M.A. Mcneil¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #132603)

LEARNING OBJECTIVES: 1. To recognize the differential diagnosis of cardiac valvular masses. 2. To review the clinical manifestations and natural history of a mobile intracardiac calcific mass. 3. To understand the best course of therapy for such masses generated by systemic calcification.

CASE: A 54 year old male with IDDM and ESRD presented with the sudden onset of gangrenous toes in his right foot along with intermittent fevers. Physical examination revealed normal cardiac auscultation, feeble peripheral pulses and dry gangrene of 4 toes of the right foot. Initial work up revealed leukocytosis with a left shift. Serial blood cultures remained negative. A transesophageal echocardiogram (TEE) revealed a freely mobile, pedunculated, globular mass, 1.4 cm by .4 cm in size attached to the inferior surface of the mitral valve leaflet. An echocardiogram done 1 month ago had revealed a normal mitral valve. A clinical diagnosis of infective endocarditis was made, and the patient was treated with antibiotics for 4 wks along with mitral valve debridement surgery. The mass was composed of mitral valve tissue, calcific deposits, inflammatory cells and fibrin exudates on histopathology exam. Gram stain, special stains and cultures of the mass were negative. 2 months later he again developed gangrene of the big toe of the left foot, along with intermittent fevers, joint pains and constitutional symptoms. Work up revealed leukocytosis with a left shift. Blood cultures were again negative, and a TEE revealed the recurrence of a similar pedunculated, freely mobile mass at the mitral valve leaflet. Since primary cardiac tumors rarely recur after surgery and cultures from blood and the debrided valve were negative, it was inferred that the mass represented a recurrent systemic calcification secondary to ESRD. The average calcium phosphate product (CPP) for 2 months prior to the first occurrence was 76, and at the time of recurrence it was 70. The patient was aggressively treated with phosphate lowering agents. A repeat TEE is pending at this time.

DISCUSSION: The differential diagnosis for a mass attached to a mitral valve consists of infective endocarditis, papillary fibroelastoma or a calcification. Systemic embolic events are commonly associated with mitral annular calcifications (MAC)* due to thromboembolic or calcific emboli. In a few of these cases, a mobile component associated with MAC has been described*. These patients can be with erroneously treated for infective endocarditis. An isolated calcific mobile mass attached to the mitral valve, without concomitant annular calcification has not been described to the best of our knowledge. With poor control of the CPP, these masses can develop and progress rapidly*. Rapid recurrence of such a mass after surgery is another feature hitherto unreported in literature. The best form of therapy of these patients is aggressive use of phosphate binding agents.* Howard et al, Echocardiography. Vol 20:4; 363-367.

CMV COLITIS IN AIDS PATIENTS. R. Bouajram¹. ¹UCLA-VA Greater Los Angeles Internal Medicine Training Program, Los Angeles, CA. (Tracking ID #135515)

LEARNING OBJECTIVES: 1. Review the clinical presentation and diagnosis of CMV colitis. 2. Recognize the usefulness of CT scan and endoscopy. 3. Explore indications for referring patient for further testing including sigmoidoscopy, colonoscopy, and EGD.

CASE: 55 yo male with AIDS (CD4=5) and a hx of multiple opportunistic infections admitted in 9/2004 complaining of 1 wk of abdominal pain that is

sharp, not associated with meals, watery diarrhea 12-13x/day, abdominal distension, and spiking fevers. The patient has an 8 month hx of recurrent diarrhea requiring multiple hospitalizations with negative stool studies. Pt was placed at that time on anti-diarrhea medications with some improvement in symptoms. CT scan revealed ascites and a diffusely thickened colon. He received a course of flagyl for C. diff pancolitis (stool toxin positive) but diarrhea did not completely resolve despite subsequent negative stool C. Diff toxin tests. Pt was discharged but returned 1 week later with recurrence of same symptoms. The patient was seen by GI and colonoscopy with EGD was done on 10/22/04. Appearance was consistent with pseudomembranous colitis and biopsy results were positive for CMV. PO Vancomycin for presumed C.diff and Valgancyclovir for CMV colitis were started. The patient's WBC count trended downward. Pt's diarrhea showed marked improvement and patient was discharged.

DISCUSSION: Chronic Diarrhea is commonly a manifestation of AIDS and a leading cause of morbidity and mortality. For those with AIDS and CD4 <50 cells/mm3, diarrhea developed in 49% of patients within 1 year and 96% by the end of 3 years of follow-up. Microsporidia, CMV, cryptosporidia, and MAC are the most common opportunistic pathogens seen in patients with chronic diarrhea and CD4 <200 cells/mm3. CMV colitis occurs with CD4 <50 cells/mm3. Patients present with watery diarrhea (80%), abdominal pain (50%), and fever (39%-80%). Diagnosis is made by colonic biopsy with intranuclear CMV inclusions. Although more invasive than flexible sigmoidoscopy, colonoscopy is able to detect a considerable number of patients with CMV localized only in the proximal colon. In one study, colonoscopy was more cost-effective than flexible sigmoidoscopy in the workup of HIV-associated diarrhea. It is unclear whether the addition of EGD is cost-effective. In conclusion, stool studies should be the initial test in patients with HIV-associated diarrhea. If these investigations fail to identify a causative pathogen, colonoscopy with biopsy of the left colon, right colon, and terminal ileum is a reasonable next step, especially in those individuals with a CD4 lymphocyte count below 100 cells/mm3. The clinical profile of HIV-associated diarrhea is changing rapidly in the era of highly active antiretroviral therapy and with the introduction of protease inhibitors. The incidence of noninfectious causes of AIDS-associated diarrhea has increased while opportunistic infectious causes have decreased. Given the clinical importance of HIV-related chronic diarrhea, further assessment on the best and most cost-effective approach is indicated for this challenging clinical problem.

THE DILEMMA OF ANTICOAGULATION IN END-OF-LIFE CARE. L. Carlstrom¹; J. Hauser¹. ¹Northwestern University, Chicago, IL. (Tracking ID #136184)

LEARNING OBJECTIVES: Manage venous thromboemboli in hospice patients with malignancy.

CASE: A 64 yo woman with metastatic endometrial cancer, a new lower extremity deep vein thrombus (DVT) and chronic renal insufficiency (CRI) who recently enrolled in a home hospice program was faced with the decision of whether to continue her anticoagulation. Prior to the decision to enter hospice, the patient's DVT was being treated with warfarin and her INR was increasingly difficult to control, requiring monitoring three times per week. Her functional status had deteriorated to where she could no longer travel to clinic and the home hospice program was only able to provide weekly INR monitoring. The DVT was still causing symptoms of edema and discomfort even with supportive treatments of compression stockings, warm compresses, and analgesics. The patient was concerned about the possibilities of worsening DVT symptoms and bleeding complications of warfarin treatment. The family expressed their desire to limit symptoms but also concern over financial difficulties providing for the additional INR monitoring. Treatment with low molecular weight heparin (LMWH) was excluded option because of her CRI. Given the perceived high risk of creating supratherapeutic INR levels and subsequent risk of bleeding, the decision was made to discontinue warfarin. The patient did not have any further laboratory tests, bleeding episodes, worsening DVT symptoms, or symptomatic pulmonary embolus before she died 4 weeks later.

DISCUSSION: The correct balance of providing comfort care and preventing further complications when treating venous thromboemboli (VTE) in the hospice cancer patient is difficult to achieve. VTE can be asymptomatic, cause lower extremity symptoms, or cause shortness of breath and chest pain. On the other hand, the correct degree of anti-coagulation can be difficult to achieve, labor intensive, cause bleeding complications, impose dietary limitations, and be costly. Although there are few data in hospice cancer patients with VTE, non-hospice cancer patients treated with long-term warfarin have an increased risk of both major hemorrhage and recurrent VTE compared to those without malignancy (Gitter et al. Mayo Clin Proc 1995; Hutten, et al, J Clin Oncol 2000; Prandoni, et al, Blood 2002). The majority of cancer patients cannot be maintained at therapeutic INR goals, even when managed by anti-coagulation clinics (Bona, et al, Thromb Haemost 1995). Alternatives, such as anticoagulating patients with VTE at lower INR goals or using aspirin have not been shown to provide adequate protection against recurrence (Krauth, et al, Cancer 1987; Clagett, et al, Chest 1992). Inferior vena cava filter placement may address short-term prevention of pulmonary embolus with avoidance of bleeding complications but it cannot affect DVT recurrence (Hyers, et al. Chest 2001). LMWH may be more effective than warfarin in preventing recurrent VTE without increasing the risk of bleeding in patients with cancer, however one must take into account its renal clearance, associated heparin induced thrombocytopenia, subcutaneous delivery and cost (Lee, et al, NEJM 2003). Since there is no strong evidence-based literature regarding VTE treatment in hospice patients, this presentation will review the above literature in cancer patients and consider the decision-making process of how to treat VTE in the hospice patient with malignant disease.

TWO ARE BETTER THAN ONE: DISCREPANCIES IN HEMOGLOBIN A1C AND GLYCEMIC CONTROL. D. Wobbleton¹; R.M. Malone¹; J.B. Buse¹; B. Bryant¹; M. Pignone¹. ¹University of North Carolina at Chapel Hill, Chapel Hill, NC. (Tracking ID #135896)

LEARNING OBJECTIVES: 1) Understand the generation, metabolism and measurement of hemoglobin A1c (A1C), a commonly used laboratory test for evaluating glycemic control in the management of patients with diabetes mellitus; 2) Recognize the implications of erythrocyte life span on A1C values; 3) Recognize the potential for ribavirin (RBV) to decrease erythrocyte survival. **CASE:** A 46 year old female with a past medical history significant for type 2 diabetes, chronic hepatitis C virus (HCV) since 1993, bipolar disorder, depression, sarcoidosis and arthritis presented to clinic August 2004. At that time, she was monitoring her blood glucose (BG) approximately four times a day and reported a range of 250-350 mg/dl. The patient denied any symptoms suggestive of hypoglycemia. Pertinent medications included: peginterferon alfa-2a 180 mcg once weekly (started 03/04), RBV 600mg daily (started 03/04), epoetin alfa 20,000 units every other week (started 07/04), insulin glargine 50 units at bedtime and insulin lispro 40 units with meals. Labs included: A1C 6.0%, random glucose 215 mg/dl, hemoglobin 11.7 g/dL, hematocrit 35.5%. Prior A1C values included 8.4% in March 2004, without documented self-monitored BG values, and 6.7% in May 2004, with reported BG values in the range of 100 mg/dl to the low 200's.

DISCUSSION: Hemoglobin is a protein that transports oxygen from the lungs to peripheral tissue and is the main component of erythrocytes. Glucose diffuses freely through the cell membrane of erythrocytes and subsequently is non-enzymatically, covalently bound to hemoglobin in proportion to the average glucose concentration and the life-span of the erythrocyte. The A1C molecule is formed as a result of the glycosylation of hemoglobin A and is used as an indicator of average BG for the past 90-120 days. Guidelines recommend targeting A1C as a technique of assuring adequate glycemic control in patients with diabetes. In theory, any process that shortens erythrocyte survival should falsely lower A1C results. Unfortunately, clinical evaluation of the erythrocyte life span is difficult and is predominantly assessed by examining the reticulocyte percentage. A recent study evaluating erythrocyte survival in patients treated with RBV demonstrated a decrease by as much as 60% despite a decrease in mean hemoglobin concentrations by only 20%. Furthermore, erythrocyte survival, evaluated by the reticulocyte count overestimated that detected by a carbon monoxide based method. Providers should be cautious of this potential interaction and other causes of falsely low A1C levels. Patients should be managed based on both documented self-monitored BG values and A1C versus A1C measurements alone.

A 40-YEAR-OLD MAN PRESENTING WITH GENERALIZED LESIONS PROGRESSING FROM PAPULES TO NECROTIC ULCERS. T. Tehrani¹; S.T. Erickson¹; M.D. Wang¹. ¹University of California, Irvine, Orange, CA. (Tracking ID #133355)

LEARNING OBJECTIVES: -Distinguish pyoderma gangrenosum from cutaneous lymphomas, such as Peripheral T-cell Lymphoma (PTCL), which may present with pyoderma-gangrenosum-like (PG) lesions. -Emphasize the role of the generalist in recognizing cutaneous abnormalities requiring biopsy and expediting appropriate diagnosis.

CASE: A 40-year-old otherwise healthy Caucasian male presented with a 3 month history of a sacral ulcer that began as a non-tender, non-pruritic erythematous papule, progressing to a large necrotic ulcer with near bone extension. Subsequent well-circumscribed lesions followed involving the face, scalp, neck, trunk, back, extremities including soles, and oral cavity. The patient was initially seen in a community clinic and prescribed a course of oral antibiotics and topical antifungal ointment for presumed infection without improvement. He denied any recent illness, HIV risk factors, travel, infectious contacts, environmental exposures, family history of cancer or autoimmune disorders. He denied fevers, chills, anorexia, weight loss, night sweats or other constitutional symptoms. He had no known allergies. Physical exam revealed a 15x12 cm purulent, malodorous, deep sacral ulcer invading to muscle, with necrotic debris at its base and surrounding violaceous induration. Skin examination revealed many lesions in various stages of progression from pinpoint erythematous papules to eroded eschars ranging from 1-10 cm in diameter, each with circumferential erythema. No lymphadenopathy or hepatosplenomegaly was appreciated. Cardiac examination revealed no murmur. Routine screening laboratory tests were remarkable only for mild leukocytosis and thrombocytosis. Bacterial, fungal, and viral studies, including HIV, peripheral blood cultures, and autoimmune markers were all negative. Initial skin biopsies demonstrated non-specific inflammation. CT scans of the chest and abdomen revealed bilateral, ill-defined pulmonary nodules with atypical lymphocytes on biopsy. Repeat skin biopsies revealed atypical lymphocytes consistent with stage IVa PTCL. Chemotherapy was initiated with CHOP (cytoxan, doxorubicin, vincristine, prednisone) with clinical improvement of the patient's skin lesions after completion of two cycles.

DISCUSSION: PTCL refers to any T-cell lymphoma with post-thymic differentiation. It accounts for 10-12% of all lymphoid neoplasms and is predominantly found in males, median age 61, with 66% of patients presenting at Stage IV. Long-term prognosis is poor, with frequent relapse and superinfection. PTCL must be distinguished from benign dermatoses and other cutaneous neoplasms by biopsy and immunophenotyping. As this case demonstrates, PTCL can mimic PG, a significant cause of skin ulcerations, which classically presents as rapidly progressive, painful necrotic ulcers with irregular undermined borders. Major criteria for PG require exclusion of other possible diagnoses, which is performed through biopsy. It is important to note that cutaneous neoplasms may require more than one biopsy for correct diagnosis. Initial specimens may be non-representative, inadequate, or malignant cells may be obscured by a reactive inflammatory infiltrate from superimposed infection. Differentiating PG from

PTCL is critical, because therapy for one would prove ineffective or even deleterious for the other. The generalist has the first opportunity to distinguish benign dermatoses from initial presentations of malignant processes, including PTCL.

A 76-YEAR-OLD WOMAN WITH AN ATYPICAL COLITIS. M.M. Yeboah¹; S.L. Kraus¹.
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LEARNING OBJECTIVES: 1. Include atypical colitis in the differential diagnosis in elderly patients with chronic, unexplained diarrhea. 2. Know the diagnostic workup of the microscopic colitides. 3. Know the current treatment options for collagenous colitis.

CASE: A 76-year-old woman was seen at the emergency room following a fall at home. She was admitted on account of significant dehydration and electrolyte derangements. On direct questioning, she reported several weeks of persistent and worsening, nonbloody, watery diarrhea. Her medical history included hypertension, anemia, gastroesophageal reflux disease, alcohol dependence, lung cancer (status post lobectomy), kidney cancer (status post right nephrectomy). She had no history of celiac sprue nor inflammatory bowel disease. Her medications were hydrochlorothiazide and topol XL. Physical and neurologic examinations were significant for lethargy and dehydration. Abnormal laboratory tests included hemoglobin 6.8, hematocrit 21, sodium 121, potassium 2.9, chloride 75, blood urea nitrogen 21. Stool studies, including fecal leukocytes, clostridium difficile toxin, ova, parasites, bacteria cultures, fecal fat were all negative. TSH, iron studies, folate, vitamin B12 levels were normal. Following initial stabilization, including fluid and electrolyte replacement and blood transfusion, colonoscopy was performed. This showed chronic appearing colitis with ecchymosis and aphthous ulceration, edema, loss of haustral markings, some narrowing of the lumen and multiple inflammatory polyps. The histologic features were consistent with collagenous colitis with no evidence of chronic inflammatory bowel disease. She was treated with loperamide and within a few days, stool frequency and volume improved markedly. Two months post-hospitalization, her diarrhea has resolved and she declines to follow up in the clinic.

DISCUSSION: Collagenous colitis (CC), together with lymphocytic colitis is classified under the microscopic colitides. The relationship between the two however, remains unclear. CC is characterized by chronic nonbloody watery diarrhea. It usually occurs in middle-aged patients but can affect children. There is female predominance with the peak incidence in women 70 to 79. The pathogenesis of CC is unknown but the association with autoimmune diseases (e.g., gluten-sensitive enteropathy, thyroiditis, rheumatoid arthritis, or Sjogren's syndrome) suggests that CC may be the result of an immune-mediated phenomenon. CC may also coexist with other gastrointestinal diseases including Crohn's disease, ulcerative colitis and celiac sprue. Stool frequency varies from 4 to 9 stools per day but can exceed 20 stools per day and significantly affect the patient's quality of life. The diagnosis of CC requires colonoscopy with biopsies. Normal colonic mucosa is the most common finding at endoscopy, but non-specific abnormalities such as edema, erythema or an abnormal vascular pattern can be seen in approximately 30% of cases. There is no specific treatment for this entity, although sulfasalazine, mesalamine, and local and systemic steroids have been reported to provide relief. High-dose bismuth subsalicylate has been shown to be useful in some studies. As in our patient, symptomatic therapy with anti-diarrheal agents may be the treatment of choice.

A CASE OF A BAD TO THE BONE COUGH. C.E. Landaverde¹; S. Lundberg¹.
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LEARNING OBJECTIVES: 1) Recognize that primary lung adenocarcinoma can occur in a non-smoking, young woman and should be considered in the right clinical setting. 2) Recognize when to expedite the work-up for a chronic cough and related symptoms not responding to conventional therapy.

CASE: A 22 year-old woman with no significant PMH presented with four months of cough, progressively worsening shortness of breath, and hemoptysis for two weeks. Two months prior to admission, she had been seen at a community clinic and given azithromycin for a presumed community acquired pneumonia; in addition, a PPD was placed and read as negative. Her symptoms persisted and a CT of the chest done at the outside clinic revealed a left lower lobe consolidation with mediastinal lymphadenopathy. The patient was prescribed levofloxacin and referred to our Chest Clinic. In the interim, she developed hemoptysis, severe back pain, and increasing SOB with worsening of her cough. In addition, the patient reported subjective fevers, chills, night sweats, and a 20-pound unintentional weight loss over four months. She denied TB exposures, recent travel, or sick contacts. She had no history of oral ulcers, hematuria, or epistaxis. She denied tobacco, alcohol or intravenous drug use and had no HIV risk factors. Significant vital signs included T 38.1, HR112, the rest of her VS were within normal limits including oxygen saturation of 97% on RA. Significant physical exam findings included decreased breath sounds and dullness to percussion on the left lower lung field, tachycardia, distant heart sounds and bony tenderness over lumbar spine. Laboratory examination revealed a normal WBC, platelet count, and creatinine, but a decreased hemoglobin of 11.8. A repeat CT of the chest revealed similar findings to the outpatient CT but also a new moderate pericardial effusion. A transthoracic echocardiogram demonstrated a moderate to large pericardial effusion with diastolic right atrial collapse. Subsequent pericardial fluid analysis revealed malignant cells and all cultures were negative. The pulmonary team performed a bronchoscopy with biopsy, which revealed mucin positive, poorly differentiated, primary lung adenocarcinoma. The spine MRI revealed metastatic disease at L3-L5. The patient is undergoing combination chemotherapy, but her prognosis remains poor this being less than 2% five-year survival.

DISCUSSION: Adenocarcinoma is the most common type of lung cancer and it is the most frequent histologic type in women and in non-smokers of either gender. Most of these tumors are peripheral in origin, arising from alveolar surface epithelium or bronchial mucosal glands, they can also arise from areas of previous infections, which are called scar tumors. Histologically, these tumors form glands and produce mucin, intracellularly. The incidence of lung cancer in young women is 3.4/100,000. A detailed history and physical examination are the most important steps in evaluating a patient with lung cancer with a focus on risk factors such as smoking history, environmental exposures and family history. Patients with multiple symptoms such as cough, dyspnea, decreased appetite, weight loss and/or who develop new symptoms such as hemoptysis or have history of recurrent respiratory infections should have a prompt evaluation for lung cancer. This is important even in younger patients as it is illustrated in this case.

A CASE OF BEZARRE BELLY-ACHE. K. Trinh¹; M. Rotblatt¹; W. Soma¹; S. Dea¹; M. Mattar¹.
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LEARNING OBJECTIVES: 1. Recognize the diagnosis and management of bezoars. 2. Recognize that ingestion of unripe persimmons is a common cause of phytobezoars.

CASE: A 51 y.o. woman presented with complaints of nausea and vomiting for 7 weeks, and one episode of coffee ground emesis that prompted her visit to the E.D. She had no previous medical history. She complained of nausea and vomiting with ingestion of solids more than liquids, and noted emesis of food she had ingested several days prior. She also complained of feeling like there was a "ball" in her stomach. Just before the symptoms began 7 weeks previously, she had consumed 4-5 hotdogs, 13-16 raw peeled persimmons, and 5-6 cans of soda in one day. She reported feeling upset by family news, and the emotional stress prompted her to food binge. ROS was significant for constipation, with small pencil-like stools. Her physical exam was significant for sinus tachycardia at 114 and a large firm palpable epigastric mass. Labs were notable for a potassium of 2.7. CXR, flat/upright KUB and barium enema were normal. However, EGD revealed a massive fibrous mass located in the fundus of the stomach. During endoscopy, the mass was partitioned and removed. The largest piece measured 10.2 x 6.5 x 1.8 cm and weighed 62 grams. More of the mass was removed over 2 subsequent EGD's, requiring 2-4 hours for each procedure. Pathology noted necrotic portions of partially digested food material consistent with a bezoar. The patient was discharged from the hospital after 4 days, when most of the bezoar had been removed and when she was tolerating po intake. She was scheduled for outpatient EGD for removal of the remainder of the bezoar but did not follow up. She contacted us later and reported that she could no longer afford the medical bills and planned to return to China to have the remainder of her bezoar removed.

DISCUSSION: This case demonstrates the unusual presentation of a bezoar by persimmon ingestion. Bezoars can be categorized as phytobezoars (vegetable matter, with persimmons being one of the most common causes), trichobezoars (hair), and pharmacobezoars (drugs). Other reported causes include milk curd, tissue paper, styrofoam cups, cement and vinyl gloves. "Disopyrobezoars," or bezoars caused by ingesting unripe persimmons, are formed as the high concentration of tannin in the unripe fruit polymerizes in the stomach forming a sticky precipitate. Diagnosis is largely dependent on thorough history taking and recognition of bezoar forming materials. Treatment requires removal of the bezoar. Successful pharmacologic methods to help dissolve or expel the mass have included cellulose, papain, acetylcysteine, and metoclopramide. However, for large bezoars as in our case, EGD is required for manual removal. Rarely, patients fail these treatments and may require surgery.

A CASE OF ENDOPHTHALMITIS FROM MRSA SEPTICEMIA. THE CLUE IS IN THE EYE OF THE BEHOLDER. J.L. Hadam¹; D. Mcadams¹.
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LEARNING OBJECTIVES: 1. To recognize that endophthalmitis is an ophthalmologic emergency. 2. To recognize that finding a hypopyon or endophthalmitis should prompt a systemic infectious evaluation.

CASE: A 62 year-old male with diabetes, renal disease on hemodialysis (HD), and recently treated line-related bacteremia presented with severe bilateral shoulder pain. He was diagnosed with bursitis, given a steroid injection, and discharged. One week later he developed lethargy and hypotension during HD and was admitted. Early in his hospitalization he noticed a sudden onset of bilateral eye pain, redness, and blurred vision. Physical exam revealed bilateral warm and tender shoulders with decreased ROM secondary to pain. Ocular exam was significant for right eye blindness and left eye visual perception of light only, as well as bilateral reactive pupils, and bilateral hypopyons. The rest of his exam was unremarkable. Labs revealed a WBC=24 with an 8% bandemia. An urgent ophthalmology consult confirmed the diagnosis of endophthalmitis and initial treatment included vitreous sampling and intraocular injection of vancomycin, gentamicin, and amikacin. The patient was also empirically started on systemic antibiotics. Investigation revealed 4/4 blood cultures positive for methicillin-resistant Staph aureus (MRSA), and acromioclavicular joint aspiration was also positive for MRSA. Endocarditis work-up with echocardiography was negative. The previously infected line was determined to be the source of septicemia. Despite clearance of blood and joint cultures after treatment, the patient's vision minimally improved and he developed additional complications of retinal detachment.

DISCUSSION: A hypopyon is a collection of pus in the anterior chamber of the eye. The triad of red eyes, ocular pain and hypopyon should prompt immediate consideration of endophthalmitis, an infection of the vitreous. Endophthalmitis

is an ophthalmologic emergency as the vitreous is in direct contact with the retina and infection can lead to permanent vision lost if not recognized and treated immediately. Despite the advent of intravitreal antibiotics about 10% will still have visual acuity of 5/200 or worse after treatment. In one retrospective study, only 40% of individuals preserved vision of counting fingers or better. Sources can be defined as exogenous (trauma and cataract surgery) or endogenous (endocarditis, line-infection, and the like). Although endogenous bacterial endophthalmitis is a rare form, accounting for less than 2-8% of all cases, the association with bacteremia and sepsis has been well documented. Work-up should therefore focus on causes of sustained bacteremia, such as endocarditis, UTI, abscesses, and line and catheter-related infections. Gram-positive organisms are the most common pathogen, with *Staph aureus* being the most common bacterial cause. Gram stain is only positive in 45% of cases and negative cultures should not exclude the diagnosis. Treatment includes systemic antibiotics to treat the underlying source and either vitreous aspiration or vitrectomy along with intravitreal antibiotics.

A CASE OF FATIGUE IN A YOUNG WOMAN. I. Guessous¹; J. Cornuz¹. ¹University Hospital of Lausanne, Lausanne. (Tracking ID #133642)

LEARNING OBJECTIVES: 1. Recognize fatigue as a clinical presentation of celiac disease. 2. Manage fatigue symptom with an evidence based approach. **CASE:** A 36-year old woman presented to her primary care physician's office with a complaint of fatigue. She reported a progressive onset of unusual fatigue over 8 months, exacerbated by physical and mental activities and not relieved by rest. She had a history of severe anemia during pregnancy without a specific clinical diagnosis. She slept well, an average seven hours per night and took no medications. She had neither weight loss nor diarrhea nor steatorrhea. The patient reported no polyuria-polydipsia and no cold intolerance. Cardiovascular risk factors were absent. On examination, she did not show diminished level of alertness or poor grooming, she was well colored, anicteric, apyretic, and had a normal heart rate. Lymphadenopathy and hepatosplenomegaly were absent. Bowel sounds were normal as was abdominal palpation. Initial laboratory results were normal except for elevated transaminasemia (ALT 83 U/L; AST 51 U/L). Serologies for HIV, Hepatitis C and B were negative and the depression screening instrument was not suggestive of a depressive disorder. Because of the elevated transaminasemia, she had an abdominal ultrasound that revealed a normal liver appearance and no biliary tract dilatation. Usual tests for elevated liver enzymes were all in the normal range (antinuclear antibody, smooth muscle antibody, serum iron and total iron-binding capacity, alpha-1antitrypsin and ceruloplasmin). Because of the lack of any explanation for persistent fatigue and elevated transaminases, a test for IgA antiendomysial antibody was requested and the result came back positive. A small-bowel biopsy revealed lesion of celiac sprue confined to the proximal small intestine.

DISCUSSION: Although a typical presentation for celiac disease in childhood is steatorrhea, weight loss, and failure to thrive, only a third of adults with celiac disease report such symptoms. In fact, a frequent presenting symptoms of adult celiac disease is fatigue. Fatigue is a common complaint in primary care settings with a prevalence between 11 and 33 percent in general practice patient populations. In the case of our patient with fatigue and elevated transaminasemia without other features of celiac sprue, the clinical index of suspicion (pretest probability) of celiac sprue was estimated by the physician in charge of the patient to be between 20 to 30 percent. Given the fact that the test for IgA antiendomysial antibody has a high sensitivity (around 90%) and specificity (around 95%), the positive likelihood ratio (sensitivity/1-specificity) has been calculated about 30. By using the Pagan nomogram we found that the positive predictive value (posttest probability) for this woman was around 90%. This case illustrates the fact that in a patient with fatigue as the unique complaint and a positive screening procedure for hepatic disorders, a test for antiendomysial antibodies should be obtained to determine the presence or absence of celiac sprue.

A CASE OF GROUP A STREPTOCOCCAL FASCIITIS LEADING TO RHABDOMYOLYSIS AND ACUTE RENAL FAILURE. J. Miller¹; S. Spencer¹. ¹Temple University, Philadelphia, PA. (Tracking ID #135421)

LEARNING OBJECTIVES: 1) Recognize Group A Streptococcus (GAS) as an important pathogen in soft tissue infections. 2) Understand the importance of differentiating cellulitis from fasciitis because of the need for early debridement in addition to antibiotics in the treatment of fasciitis to prevent progression and potentially fatal outcome. 3) Recognize the advantages of clindamycin over penicillin in treating severe GAS infections.

CASE: A 47 yo man with HIV (CD4 530), HTN, and Type 2 DM presented with swelling and pain of his left arm. He reported cutting his left thumb while changing his car battery 3 days ago. Since then, he noted swelling, pain and erythema of his thumb which extended to his entire hand and forearm over the last day. On exam: T 102.2°; BP 146/82; HR 111. He was nontoxic appearing. He had diffuse swelling and erythema of left hand extending above the wrist. There was a small laceration on his thumb with serosanguinous drainage. Labs were notable for a WBC 22.5 (89% segs, 1% bands), BUN 42, Cr 2.6, CK 14,490. Blood cultures were obtained and he was started on piperacillin-tazobactam. Within 6 hours of presentation he underwent fasciotomies and debridement of the dorsum of his hand. Over the next 48 hours, he developed skin necrosis on his thumb which progressed to the palm. He remained febrile but hemodynamically stable. Surgical cultures revealed GAS and his antibiotics were changed to clindamycin. The region of necrosis demarcated and the surrounding tissue appeared healthy. During this time his creatinine rose precipitously to 6.5 at 48

hours to 10.6 at 72 hours with a bland urine sediment. He was felt to have acute renal failure due to rhabdomyolysis superimposed on chronic renal insufficiency. He was started on hemodialysis with no improvement in his renal function during the hospitalization. He defervesced on clindamycin and was discharged home to complete a 3 week course of oral clindamycin. He was to follow up with Plastic Surgery for likely amputation of his thumb.

DISCUSSION: Necrotizing fasciitis is a deep infection of subcutaneous tissues classified as Type 1 or 2. Type 1 is a polymicrobial infection while type 2 is caused by GAS. Early differentiation of necrotizing fasciitis from cellulitis is important because of the potential for rapid evolution to gangrene, myonecrosis and extension along fascial planes. While antibiotics are sufficient treatment for cellulitis, necrotizing fasciitis requires early surgical debridement. Type 2 necrotizing fasciitis can be associated with shock and organ failure, a condition termed the toxic shock syndrome which carries a 30-60% mortality. Although GAS is exquisitely sensitive to beta-lactam antibiotics, severe GAS infections are associated with high morbidity and mortality despite penicillin therapy. Beta lactams are most effective against rapidly growing bacteria and exert their effect through their interaction with penicillin-binding proteins (PBP). Severe deep-seated infections have large inoculums of organisms with consequent slowed bacterial growth and decreased production of PBPs, leading to decreased effectiveness of the beta-lactams. Clindamycin has several advantages over penicillins: its efficacy is not affected by inoculum size, it suppresses synthesis of bacterial toxins, facilitates phagocytosis of GAS and suppresses protein synthesis. Clindamycin use has been associated with better outcomes in severe GAS infections as compared to penicillins and is therefore the treatment of choice.

A CASE OF INTRADURAL EXTRAMEDULLARY TUBERCULOMA OF THE SPINAL CORD. L. Luo¹; J. Pino². ¹Coast AHEC (New Hanover Reginal Medical Center), Wilmington, NC; ²New Hanover Regional Medical Center, Wilmington, NC. (Tracking ID #136159)

LEARNING OBJECTIVES: Recognize the neurological complications of Mycobacterium Tuberculosis. Realize the potential causes for late complications in the therapy for TB.

CASE: A 51-year-old African-American male presented weight loss, low grade fever, and severe back pain over five-month duration. He denied a history of tobacco or illicit drug use. A PPD placed upon admission was positive and chest radiograph was negative. A CT of the chest revealed a profusion of tiny nodules consistent with military TB. An HIV ELISA was non-reactive. Because of the severe back pain, an MRI of the spine was performed that revealed destructive bony changes at T12-L1 with cord compromise. A CT guided biopsy was performed that demonstrated fragments of bone with necrotizing granulomatous inflammation. AFB stain was positive for mycobacterium and both DNA probe and culture confirmed *M. tuberculosis*. The patient was started on four drug therapy with ethambutol, rifampin, isoniazid, and streptomycin. He tolerated the therapy well, defervescing by the fourth day of treatment. He was discharged home to complete 9-12 months of directly observed therapy. Three months later, the patient presented to our clinic with complaints of severe back pain associated with lower extremity weakness and numbness for 1 month. He denied bowel or bladder incontinence. On exam, he was afebrile. His lungs were clear. The neurological exam was significant for 2/5 strength in the lower extremities bilaterally, with 0/2 hyporeflexia noted of deep tendon reflexes of bilateral lower extremities. Sensation to light touch was diminished. Cerebellar function was intact. An MRI of the thoracic, lumbar and sacral spine was performed revealing a 1.8 cm intradural extramedullary enhancing oval mass at T3 level, with associated compression of the spinal cord. The patient underwent a dorsal thoracic laminectomy at T3 region, with removal of a 4 cm tan-colored mass. Pathology revealed foci of ill defined granulomatous inflammation, suspicious for a tuberculoma. The AFB culture was eventually negative. Following the surgical procedure, four-drug therapy for tuberculosis was resumed. The patient had an uneventful postoperative course. After the completion of the therapy, the patient experienced a near full recovery of function.

DISCUSSION: Mycobacterium Tuberculosis remains a major cause of morbidity and mortality across the world. Tuberculous osteomyelitis (Pott's Disease) is an uncommon complication of this infection, especially in the U.S. Spinal tuberculomas are even more uncommon as causes of spinal compression. There are reports of paradoxical expansion of tuberculomas during antituberculous treatment described in the literature. In military tuberculosis, there is a state of overall immune suppression. Once antituberculous treatment is initiated, there is a heightened inflammatory response, presumably from release of intracellular antigenic substances. Lesions that had been silent can become symptomatic. Our case is unusual, not only in terms of the rarity of the extramedullary tuberculoma itself, but also in regards to it developing in a patient being treated with adequate antituberculous drugs for military TB and Pott's Disease. This progression of disease may have been caused by some of the factors mentioned above, most notably the suggestion that tuberculous lesions may expand after the initiation of treatment, as the result of inflammatory properties of the tuberculin and other breakdown products of the bacilli.

A CASE OF MULTIPLE ENDOCRINE NEOPLASIA TYPE 2A, IN A 33 YEAR OLD MAN. M. El Fatmi¹; J. Taw¹. ¹Englewood Hospital and Medical Center, Englewood, NJ. (Tracking ID #135174)

LEARNING OBJECTIVES: 1- Recognize atypical presentation of pheochromocytoma 2- Link pheochromocytoma tumors to the associated syndrome.

CASE: A 33 year old Caucasian man with a history of sociopathic behavior, frequent fits of rage, multiple incarcerations for homicide attempts and psychiatric

follow up since age 18 when he was separated from his family, presented with a knee trauma complicated by a septic bursitis. His blood pressure on admission was 140/86. He underwent surgery to drain a pre-patellar bursa pus collection; his blood pressure was noted to rise up to 200/110 under general anesthesia. Further workup revealed a plasma metanephrine level of 1796 pgc/ml (normal range 12-81 pgc/ml), the plasma normetanephrine was 2456 pgc/ml (18-112 pgc/ml). An abdominal CT scan revealed bilateral adrenal masses with central necrosis, measuring 5 × 5 cm and 2.4 × 4 cm respectively; MIBG scan showed large uptake in both adrenals. The diagnosis of pheochromocytoma was confirmed and the patient was started on phenoxybenzamine then later switched to atenolol. After reunification with his family members, further inquiry revealed an extensive family history of multiple endocrine neoplasia type (MEN) 2A with an autosomal dominant transmission pattern. Affected family members were: his mother, two maternal aunts, two sisters and 3 cousins, all treated for pheochromocytomas, and/or medullary thyroid carcinomas ± parathyroid hyperplasia. Imaging of the thyroid and parathyroid was negative. Serum calcitonin, and parathyroid hormone were within normal limits. The patient underwent laparoscopic right adrenalectomy and left dissection of pheochromocytoma with normal portion of adrenal left intact. He had persistently low cortisol post-operatively and was started on replacement hydrocortisone. He recovered and his fits of rage disappeared. He is asymptomatic and metastasis free one year after treatment.

DISCUSSION: Pheochromocytoma, a tumor arising from chromaffin cells mostly of adrenal origin, is classically diagnosed in patients with symptoms consisting of hypertensive crises, accompanied by headaches, palpitations and profuse sweating. Less common presentations include psychiatric manifestations such as anxiety, tremors or psychosis, which all have been reported in the literature. Classically 10 % of the cases are part of a syndrome such as MEN 2, Von Hippel Lindau's disease or Von Recklinghausen's neurofibromatosis. 10% are malignant; 10 % are not associated with hypertension; 10 % are extra adrenal; 10% are bilateral; and 10% occur in children. Treatment of pheochromocytoma is essential to avoid premature mortality from cardiovascular complications, and diagnosis of associated conditions such as MEN 2 is equally important, because of the risk of developing medullary thyroid carcinoma, and for genetic counseling purposes.

A CASE OF POLYGLANDULAR AUTOIMMUNE SYNDROME TYPE II. A.L. Misakian¹, J.B. Tyrrell¹, A. Schlegel¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #133019)

LEARNING OBJECTIVES: Review the clinical presentation, diagnostic work-up, and treatment of adrenal insufficiency. Recognize when to screen for co-existing endocrine conditions and in particular, the polyglandular autoimmune syndromes (PAS).

CASE: A 57 year-old white woman with Graves' disease presented to dermatology clinic for evaluation of dark skin. She was found to have symptomatic hypotension so she was transferred to the medical urgent care clinic. Further questioning revealed that she had nausea, weight loss, fatigue, and salt craving for two years. These findings were previously attributed to menopause, over-replacement of thyroid hormone, and occult malignancy. Physical exam revealed a thin, chronically ill-appearing woman with alopecia and marked hyperpigmentation of the skin, particularly of the back, elbows, eyelids and areolae. Laboratory evaluation was notable for an undetectable serum cortisol, markedly elevated adrenocorticotrophic hormone, normal prolactin, suppressed thyroid stimulating hormone, and mildly elevated free thyroxine. Her symptoms resolved after starting glucocorticoid and mineralocorticoid replacement.

DISCUSSION: Addison's disease (primary adrenal insufficiency) is an uncommon disorder, occurring in less than 0.1% of the population. With the improved control of tuberculosis, autoimmune adrenalitis has emerged as the leading cause of primary adrenal insufficiency in adults. Fatigue, weight loss, and hyperpigmentation are among the most common symptoms, occurring in greater than 90% of patients.

Approximately half of the patients with autoimmune adrenal insufficiency have one or more additional autoimmune diseases including type 1 diabetes mellitus, thyroiditis, celiac disease, and pernicious anemia. Specific clinical syndromes include PAS type I (rare; characterized by chronic candidiasis, hypoparathyroidism, and Addison's disease) and PAS type II (characterized by Addison's disease, type I diabetes mellitus, and thyroid dysfunction). The etiology of these syndromes is thought to be genetic; it is therefore appropriate to screen first degree relatives if indicated by history or physical exam. While the role of autoantibodies in screening is not established, patients with one autoimmune endocrine disorder who have antibodies to an unaffected gland are at greater risk for developing a second endocrinopathy.

The patient described here has PAS type II. The onset of PAS type II can be insidious; in fact, there are case reports describing diagnostic delay until the onset of symptomatic hypotension in patients with mild symptoms. While we found evidence for thyroid and adrenal dysfunction in this patient, further testing for associated conditions (diabetes, celiac disease, and pernicious anemia) was unremarkable. We were unable to evaluate her family members; however, by her report, there is strong family history of thyroid disease.

This case highlights the fact that adrenal insufficiency may have a subtle and insidious clinical presentation and that it may coexist with other endocrine conditions.

A CLINICAL HISTORY THAT IS HARD TO SWALLOW. J. Kenny¹, J. Wiese¹. ¹Tulane University, New Orleans, LA. (Tracking ID #135789)

LEARNING OBJECTIVES: 1. Understand the diagnostic evaluation of dysphagia. 2. Recognize the clinical presentation of amyotrophic lateral sclerosis.

CASE: A 31 year-old woman presented with four months of difficulty swallowing. She described difficulty swallowing liquids and solids, and noted associated coughing and choking. She also described excessive drooling, weakness and a thirty-pound weight loss. One month earlier she had a non-diagnostic upper endoscopy. She was extremely cachectic with normal vital signs. There was 2/5 strength in all extremities with significant muscle wasting. Pooling of saliva in the pharynx was noted. She had a prominent thyroid, but no lymphadenopathy. She demonstrated a weak cough and a depressed gag reflex. Reflexes were normal. Fasciculations were present in all extremities as well as the tongue. The presence of both upper and lower motor neuron deficits prompted an EMG that revealed motor axon loss in three limbs; a deltoid muscle biopsy was diagnostic for Amyotrophic Lateral Sclerosis (ALS).

DISCUSSION: Dysphagia is due to either difficulty transferring food from the oropharynx to the upper esophagus (oropharyngeal dysphagia) or difficulty transporting the bolus through the esophagus (esophageal dysphagia). Our patient's history of drooling and coughing following eating suggested an oropharyngeal dysphagia. Neurologic abnormalities such as stroke, amyotrophic lateral sclerosis, or peripheral nerve disease are the most common causes of oropharyngeal dysphagia. Our patient had several unique clues from her history and physical examination that led to the correct diagnosis. Fasciculations and lower motor neuron signs (weakness) should direct the clinician to focus the diagnostic evaluation on a neurological workup rather than gastrointestinal. The combination of upper and lower motor neuron signs is characteristic of amyotrophic lateral sclerosis. In our patient, this prompted a muscle biopsy and EMG as the initial diagnostic tests.

A DIAGNOSIS OF POLYCYSTIC KIDNEY DISEASE IN A YOUNG MAN PRESENTING WITH FLANK PAIN AND HEMATURIA. J. Baez-Escudero¹, H. Shishodia¹, B. Taqui¹. ¹Temple University, Philadelphia, PA. (Tracking ID #132954)

LEARNING OBJECTIVES: 1. Review epidemiology and clinical manifestations of autosomal dominant polycystic kidney disease (ADPKD) 2. Review screening recommendations for ADPKD 3. Review screening recommendations for cerebral aneurysms in patients with ADPKD.

CASE: 32 year old African American male with no prior medical history presented with 2 days of intermittent left flank pain and hematuria. He denied a history of nephrolithiasis, but reported family history of end stage renal disease in his father. His exam was remarkable for BP 179/89 and an S4 on cardiac exam. Labs revealed Na 141, K 4.5, Cl 106, bicarbonate 22, BUN 67, creatinine 8.2, Ca 9.3, phosphorous 6.5. His urinalysis was packed with RBCs (no casts), and showed protein >300 mg/dL. A retroperitoneal ultrasound showed multiple cortical and medullary cysts in bilaterally enlarged kidneys. Abdominal CT showed polycystic kidney disease with some hemorrhagic cysts and diverticulosis. The patient was educated about his disease and referred to a nephrologist for hemodialysis and evaluation for kidney transplant. He was also specifically told to have his family members screened for the disease.

DISCUSSION: Autosomal dominant polycystic kidney disease (ADPKD) affects 0.5 million persons in the United States and accounts for end-stage renal disease (ESRD) in about 10% of the chronic dialysis population. It can be familial (90%) or due to spontaneous mutations (10%). The kidneys are enlarged, with multiple, spherical evenly distributed cysts of varying sizes. The remaining parenchyma may reveal varying degrees of tubular atrophy, interstitial fibrosis, and nephrosclerosis. Patients usually present in the third or fourth decade with flank pain due to infection, urinary tract obstruction due to stones/clot, or sudden hemorrhage into a cyst. Nephrolithiasis occurs in 15-20% and hypertension in 75%. Progressive decline in renal function is common, with approximately 50% of patients developing ESRD by age 60. Hypertension, recurrent infections, male sex and early age of diagnosis are associated with early onset renal failure. At least three to five cysts in each kidney are the standard diagnostic criteria for ADPKD. Ultrasound or CT scan imaging is necessary to screen asymptomatic family members. Genetic testing is available but is reserved for cases where radiographic imaging is negative and the need for definitive diagnosis is critical, such as screening family members for potential kidney donation. Cysts can also occur in the liver, spleen, pancreas, and ovaries. Colonic diverticular disease is the most common extrarenal abnormality. Patients can also present with mitral valve prolapse (25%) and cerebral aneurysm (10%). It has been estimated that 65-75% of patients with cerebral aneurysms and ADPKD will suffer an intracranial bleed due to rupture. Nevertheless, the role for radiologic screening to look for aneurysms in patients with ADPKD is controversial. The decision is based upon weighing the risks of aneurysm surgery versus the benefits of possibly preventing aneurysm rupture. At present, radiologic screening is recommended only for high risk patients, such as those with a previous rupture, a positive family history of an intracranial bleed, a high risk occupation, and prior to surgery that is likely to be associated with hemodynamic instability with hypertension.

A GRAVE SITUATION. J. Baker¹, R. Granieri¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133046)

LEARNING OBJECTIVES: 1. Recognize Myasthenia Gravis (MG) as an etiology of acute respiratory failure 2. Recognize typical presentations of bulbar Myasthenia Gravis 3. State current treatment options for Myasthenia Gravis.

CASE: HF was a 71 year old man who presented with a 2-3 week history of general weakness, worsening dysarthria and dysphagia. His past medical history was significant for emphysema, diabetes mellitus, and tongue cancer status post anterior glossectomy. Initial examination revealed no new anatomic abnormalities. The patient was then sent for a CT scan. While in the CT scanner he

had a cardiopulmonary arrest. He was resuscitated, intubated without difficulty and after a few days met extubation requirements. However, several minutes after extubation he developed marked stridor, respiratory distress and required reintubation. An extended work-up revealed equivocal electromyography results, no acute oropharyngeal or CNS changes or thymic abnormalities, and elevated anti-acetylcholine receptor antibodies. The diagnosis of MG was made. Treatment was begun with mestinon, prednisone, azathioprine, and plasmapheresis. A second extubation attempt was complicated by a central line sepsis. He failed to improve and was given a tracheostomy and IV immunoglobulin therapy. The patient stabilized and he was sent to an outside hospital for continued treatment and rehabilitation.

DISCUSSION: Myasthenia gravis is a relatively common autoimmune disorder of the neuromuscular junction, affecting 1 in 7500 people. Clinical symptoms result from auto-antibodies directed against the post-synaptic acetylcholine receptor. It is often associated with thymus abnormalities. A small subset of patients is anti-Ach R antibody negative. There is a bi-modal age distribution, affecting mostly females in the 20–30 year age group and with a male predominance in patients over 60 years old. Commonly affected muscle groups include bulbar and ocular muscles. Patients can also present in myasthenic crisis with respiratory distress; this manifestation is secondary to respiratory and/or pharyngeal muscle fatigue. Diagnosis of myasthenia gravis is based on testing with edrophonium, determining the presence of Ach-receptor antibodies, and electromyography. Mainstays of treatment include anticholinesterase therapy, thymectomy, and immune suppression (corticosteroids, azathioprine, plasmapheresis, and intravenous immune-globulin.) Our case illustrates a potentially life-threatening presentation of this common disorder, and the diagnostic and therapeutic approaches used in patients with this disease.

A HORMONAL HEARTACHE. J.C. Lin¹; K. Yu¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #136216)

LEARNING OBJECTIVES: 1) Recognize the cardiac manifestations of pheochromocytoma. 2) Manage cardiovascular instability from pheochromocytoma. 3) Identify pheochromocytoma as an etiology of hypertensive crisis, paroxysmal HTN, and HTN that responds poorly to conventional treatment.

CASE: An 80-year-old Hispanic man with history of HTN, type 2 diabetes, BPH presented with one day of confusion, fevers, shaking chills, and incontinence. He was well until two days prior when he complained of intermittent chest pain. Vital signs were significant for T 40.3 °C, BP 187/64, HR 119, RR 26, and O2 sat of 98% on 6L NC. Physical exam revealed respiratory distress, diaphoresis, orientation to name only, normal JVD, tachycardic but regular rhythm, and decreased breath sounds bilaterally. WBC was 14,000 with 92% neutrophils, lactate level was 8.4, and TSH was normal. CSF evaluation and urinalysis showed no evidence of infection. CXR revealed only atherosclerotic changes of the aorta. Brain CT showed microvascular ischemic changes and cerebral atrophy, but no evidence of bleed, infarct, or mass. Troponin was 0.4 on admission and peaked at 78 on day 2. EKG showed a dynamic 2 mm ST-segment elevation anteriorly. We initiated therapy for an acute MI and started empiric antibiotic therapy. The patient was subsequently intubated. He developed alternating hypertensive and hypotensive crises with BP cycling between 240/85 and 32/22, lasting less than 24 hours. An echocardiogram revealed small chamber sizes and decreased EF of 25% with global hypokinesis. 24-hour urine metanephrine, normetanephrine, and VMA were all significantly elevated at 4023mcg, 1554 mcg, and 15.1 mg, respectively. Abdominal CT revealed a 3x3 cm mass on the right adrenal gland. Blood cultures returned positive for *E. coli*. The patient was controlled on medical management and is currently undergoing further evaluation for his adrenal mass and coronary artery disease.

DISCUSSION: Pheochromocytoma is a rare neuroendocrine tumor that presents as sustained HTN or hypertensive paroxysms and occurs in 0.1% of the hypertensive population. Awareness of its presentations is critical, because it is a curable disease with severe consequences if untreated. Classic features include sweating, palpitations, flushing, and chest pain. EKG show nonspecific ST-T wave changes, prominent U wave, bundle branch blocks, and left ventricular strain patterns. Acute MI can occur in absence of CAD due to increased α -adrenergic stimulation, causing coronary vasospasm. Catecholamine toxicity causes irreversible damage to the intima and vasa vasorum, leading to cellular necrosis and cardiomyopathy. Pheochromocytoma and atherosclerotic CAD can both be present during an acute MI, as in our patient, so interventions for both must be addressed, although sepsis precluded further evaluation of the coronary arteries in our patient. Pheochromocytoma is treated with phenoxybenzamine to induce α -receptor blockade by gradual titration. β -adrenergic receptor blockers should be given only after β -blockade has been induced, as they can cause a paradoxical increase in BP. Adrenalectomy is the definitive treatment, with a >95% 5-year survival rate and resolution of HTN in 75% of the patients. Unresectable or malignant tumors require long-term therapy with adrenergic blockers and metyrosine to decrease catecholamine production. Early suspicion and appropriate therapy are the key in preventing devastating consequences of pheochromocytoma.

A HUMBLING CASE OF PROGRESSIVE PULMONARY INFILTRATES. P. Jain¹; N. Aggarwal¹; R.M. Schapira¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #134000)

LEARNING OBJECTIVES: 1) To recognize the clinical and radiographic features of bronchoalveolar carcinoma 2) To review differential diagnosis of hemoptysis. **CASE:** A 66 year old male presented with cough which had worsened over a week. The cough was productive of copious sputum streaked with blood and was

accompanied by dyspnea on exertion. The patient had an insignificant smoking history and was treated with amiodarone for arrhythmia related to cardiomyopathy. He was initially evaluated for hemoptysis and productive cough more than a year prior to admission. This evaluation included a chest CT scan and bronchoscopy, both of which were normal. On admission, the patient was afebrile and vital signs were normal. His resting respiratory rate was 16/minute and room air SpO2 was 95%. Auscultation revealed faint bibasilar crackles. Chest CT scan revealed multiple bilateral randomly distributed nodules (all <10 mm), interlobular septal thickening and an area of focal consolidation in the right lower lobe. These non-specific findings suggested a broad range of diagnoses, including those related to infectious, neoplastic and inflammatory processes. Bronchoscopy showed no endobronchial abnormalities. Analysis of bronchoalveolar lavage fluid demonstrated no malignant cells or infectious organisms. The nodular infiltrates progressed from those seen on admission and respiratory failure developed. A second bronchoscopy or an open lung biopsy were considered but deemed to be too high risk. It was felt that patient may have an interstitial lung disease related to amiodarone toxicity and he was treated with prednisone. Infectious or neoplastic causes were thought to be less likely given the non-diagnostic results from the bronchoscopy. Despite the difficult clinical course, the patient accepted the uncertainty of diagnosis, declined mechanical ventilation and expired few weeks after admission. He requested an autopsy to solve the puzzle that took his breath away, which revealed diffuse multilobar bronchoalveolar carcinoma.

DISCUSSION: Non-massive hemoptysis has a broad differential diagnosis including infectious, neoplastic, structural, autoimmune, cardiovascular and iatrogenic etiologies. In the US, most cases of hemoptysis are caused by bronchitis and bronchogenic cancer while internationally, bronchiectasis and tuberculosis are more common. Bronchoalveolar carcinoma (BAC) is a subtype of adenocarcinoma, accounting for 2-6% of primary lung cancers. BAC arises from type II pneumocytes and grows along alveolar septa without disrupting the parenchymal structure. The most common presenting feature of BAC is a productive cough in up to 70% of cases. Hemoptysis is present in 16%. BAC may present as a localized disease (solitary peripheral nodule) or a diffuse disease. Diffuse BAC is characterized by the presence of nodules or consolidation within a lobe or multiple lobes. The nodules may be well circumscribed similar to pulmonary metastases. Less commonly, BAC undergoes lymphangitic spread and mimics an interstitial lung disease. The admission chest CT scan demonstrated all three patterns (interstitial and nodular densities and focal consolidation) which along with the non-diagnostic bronchoscopic findings, contributed to the elusiveness of the diagnosis.

A MIXED-UP PRESENTATION OF ATRIAL MYXOMA. M. Kuo¹; H. Guu¹. ¹St. Mary Medical Center, Long Beach, CA. (Tracking ID #135511)

LEARNING OBJECTIVES: 1. Recognize different clinical presentations of atrial myxomas. 2. Review treatment of atrial myxomas.

CASE: 48yo male with no significant PMH and on no medications presented to the ED after a witnessed new-onset tonic-clonic seizure. After recovering from a post-ictal state, patient reported a recent history of intermittent exertional left substernal chest pain with radiation to the left arm and spontaneous resolution with rest. He also described associated shortness of breath. The patient denied recent viral illness, headaches, head trauma, fevers, palpitations, nausea/vomiting, or changes in vision or smell prior to the seizure. He reported no family history of cardiac disease. On examination, the patient was alert and oriented. Cardiovascular exam revealed regular rate and rhythm without murmurs, gallops, rubs or extra heart sounds. Lung and abdominal exams were unremarkable. Skin was intact without any lesions. Neurological exam was nonfocal. Workup for seizure, including electrolytes, urine toxicology, alcohol level, head CT and MRI were negative. EKG and cardiac markers were unremarkable. Treadmill stress test was negative. Echocardiogram revealed no focal hypokinesis and a 4 x 4 cm mobile, flopping echodensity occupying the left atrium with intermittent obstruction of the mitral valve consistent with atrial myxoma. The atrial myxoma was surgically removed. The patient recovered well without any complications and was discharged from the hospital in stable condition.

DISCUSSION: Myxomas are the most common primary heart tumors. The majority are benign and sporadic with 75-85% occurring in the left atrium. They can be found in patients of all ages, especially between the third and sixth decades of life. Approximately 10% are familial with autosomal dominant inheritance, which tend to be diagnosed at younger ages. They can also be termed syndrome myxoma, as in the NAME syndrome (nevi, atrial myxoma, myxoid neurofibroma, and ephelides) or the LAMB syndrome (lentiginos, atrial myxoma, and blue nevi). Myxomas can be polypoid, villous or papillary. Clinical presentation depends on the consistency, extent of attachment and length of the stalk of the myxoma, and position of the patient. They can present with fever, malaise, weight loss, symptoms of systemic embolization, intracardiac or valvular obstruction, or without symptoms at all. Our patient presented with new-onset seizure, a likely result of systemic embolization. His chest pain and dyspnea are due to mitral valve obstruction as evident by echocardiogram. Physical examination can reveal systolic or diastolic murmurs and tumor plops. Electrocardiograms and chest xrays can be helpful in revealing hemodynamic cardiac changes or left atrial enlargement. 2-D echocardiogram is the imaging modality of choice. Other noninvasive modalities include cardiac CT and MRI. The mainstay of treatment is surgery. Due to the risk of systemic embolism or sudden death, myxomas should be removed immediately. Patients, in general, have favorable long-term prognoses after surgery. Periodic follow-up echocardiograms are indicated for possible recurrence.

A MYSTERIOUS CASE OF SHRINKING TESTICLES: CRANIOPHARYNGIOMA IN A YOUNG ADULT. S. Kang¹; J. Blank¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #134086)

LEARNING OBJECTIVES: 1) Recognize the clinical presentation of hypopituitarism. 2) Diagnose and treat hypopituitarism secondary to craniopharyngioma. **CASE:** A previously healthy 19 year old man presented with complaints of inability to achieve an erection, shrinking of his penis and testicles, decreased libido, and loss of axillary and pubic hair. The patient also reported fluctuations in body weight, increased thirst and urination, lightheadedness, and fatigue. He denied headache or visual changes. Vital signs included T 36.0, HR 60, and BP 86/44. Notable physical exam findings included a lack of facial, axillary, and pubic hair; and non-palpable testes. Laboratory tests were significant for a serum Na 163, total testosterone <20 (300-950), LH <1.0 (1.0-8.0), TSH 8.64, free thyroxine 0.6, prolactin 75.6 (0.0-20.0), and cortisol <1.0. We initiated hormone replacement therapies, and corrected dehydration and electrolyte abnormalities with IV fluids. MRI of the brain revealed a large suprasellar craniopharyngioma compressing the optic chiasm. The patient was transferred to neurosurgery for sight-preserving resection.

DISCUSSION: Craniopharyngiomas are slow growing, benign tumors that arise in the pituitary stalk. The most common presenting symptoms are due to endocrine dysfunction from hypopituitarism (66-90%). Patients may also complain of headache (55-86%) and visual disturbances (37-68%). This patient presented with symptoms of hypogonadism, antidiuretic hormone (ADH) deficiency, adrenocorticotropic hormone (ACTH) deficiency, and hypothyroidism. Gonadotropin deficiency leads to complaints of sexual dysfunction in men and amenorrhea in women. ADH deficiency manifests as central diabetes insipidus, with polyuria and polydipsia. This leads to hypernatremia when oral fluid intake falls behind fluid losses. ACTH deficiency causes secondary adrenal insufficiency, which can be differentiated from primary adrenal insufficiency by the lack of hyperpigmentation. Secondary hypothyroidism from thyrotropin deficiency produces symptoms that are usually less severe than those seen in primary hypothyroidism. Hypothalamic-pituitary stalk tumors cause hyperprolactinemia, rather than prolactin deficiency, due to disruption of the normal inhibitory effect of hypothalamic dopamine on pituitary prolactin secretion. Treatment of hypopituitarism involves hormone replacement therapy with testosterone or estrogen-progestin, a synthetic analogue of ADH, prednisone, and levothyroxine. In children, low growth hormone (GH) is replaced to prevent short stature. However, in adults GH deficiency presents with non-specific symptoms such as weakness or poor wound healing and is typically not replaced. Definitive treatment of craniopharyngioma involves surgical resection, followed by radiation therapy for residual tumor. Five year survival rates range from 99% in children to only 38% in the elderly.

A NONCASEATING PAIN IN THE SHOULDER. T. Wassenaar¹; J.M. Sosman². ¹University of Wisconsin Medical School, Madison, WI; ²University of Wisconsin-Madison, Madison, WI. (Tracking ID #134068)

LEARNING OBJECTIVES: 1). To identify the epidemiology of sarcoidosis. 2). To recognize the common and uncommon extrapulmonary presentation of sarcoidosis. 3). To review the the diagnostic, prognostic and therapeutic quandaries of this disease.

CASE: A 41 y/o caucasian man with a history significant for poorly controlled hypertension and bilateral sensorineural hearing loss presents for his annual physical examination complaining of a four week history of right shoulder pain. He describes a sharp pain without radiation that began while golfing and is worse with movement. He denies any traumatic events. Plain films of the left shoulder were performed and were suspicious for a rotator cuff injury so a MRI of the left shoulder was performed. The MRI revealed suspicious lytic lesions in the proximal left humerus as well as the left scapula. CT scans of the chest, abdomen and pelvis were performed and identified bilateral apical pulmonary nodules and adenopathy in the mesentery and retroperitoneal regions. No hepatosplenomegaly was noted. The patient denied any constitutional symptoms such as fatigue, fever, night sweats or weight loss. A CBC was performed and showed mild leucopenia (3.7K/ul) with a normal differential. An ESR was 18 mm/hr, CRP 1.0mg/dl, calcium 9.8mg/dl, total bilirubin 0.7 mg/dl, alk phos 108U/l, AST 38U/l, ALT 55U/l and LDH 172U/l. Further workup included an SPEP with a total protein of 8.3gm/dl and a pattern suggesting a polyclonal gammopathy. Additional workup for lymphadenopathy and bone lesions included a bone marrow biopsy, which identified noncaseating granulomas with negative AFB and fungal stains. Sputum AFB times three, blastomycosis antibody, coccidiomycosis antibody, urinary histoplasmosis antigen, and bone marrow cultures were all negative. Despite a normal ACE level at 28IU/l, a presumptive diagnosis of sarcoidosis was made.

DISCUSSION: Sarcoidosis is a multisystem disorder of unknown etiology that affects individuals worldwide and is characterized by the presence of noncaseating granulomas. It has a preference for adults under the age of 40, and is more prevalent and severe in African Americans and those of African descent. The disease may occur in genetically susceptible individuals that are exposed to specific environmental agents. Pulmonary sarcoidosis is the most common manifestation, and the most common presenting symptoms are cough, dyspnea and chest pain. Skin and ocular lesions are also common. Extrapulmonary manifestations can affect any organ system including: erythema nodosum, keratoconjunctivitis, uveitis, hepatosplenomegaly, lymphadenopathy, CNS involvement including cranial nerve palsies, arrhythmias, and polyarthritides. Bone involvement which is less common, results from periosteal bone resorption that appear as cystic and occasionally lytic lesions. A variety of laboratory abnormalities can be seen in patients with sarcoidosis including pancytopenia, eosinophilia, hypergammaglobulinemia, a positive rheumatoid factor, hypercalcemia, and an elevated ACE level. Tissue biopsy revealing noncaseating

granulomas remains the gold standard for diagnosis. An ACE level is elevated in less than 75% of cases. Corticosteroids have traditionally been the treatment of choice, although the optimal dose and duration are unknown. Studies with cytotoxic and anti-malarial agents have shown conflicting results. Our patient was treated with a NSAID with good symptom control. He is being monitored to assess disease progression.

A PAUCI-ETIOLOGY OF ANEMIA AND RENAL FAILURE IN THE ELDERLY: A CASE OF MICROSCOPIC POLYANGIITIS. K.C. Harned¹. ¹University of Kentucky, Lexington, KY. (Tracking ID #134495)

LEARNING OBJECTIVES: 1) Recognize the value of urinalysis in acute renal failure (ARF) and the potential subtle presentation of vasculitis.

CASE: A 74 year-old man presented for transfusion and evaluation of anemia. He reported two-months of fatigue, dyspnea and night sweats. He denied blood loss. He was previously diagnosed with seronegative polyarthritides. Vital signs were normal except BP 175/94. Pale conjunctiva, suprapubic tenderness and a right prostatic nodule were noted. Skin was normal. Laboratory data revealed normocytic, normochromic anemia, reticulocyte count 0.3%; all other anemia labs were normal. Creatinine 3.5 mg/dL, UA with 2+protein; >50 rbc's/HPF; scant fine-granular and hyaline casts. Renal biopsy demonstrated pauci-immune crescentic glomerulonephritis. p-ANCA was positive (1:160) and anti-myeloperoxidase was 4x upper normal limits. He was diagnosed with microscopic polyangiitis.

DISCUSSION: Microscopic polyangiitis is classified on a continuum including Wegner's Granulomatosis (WG), Idiopathic Rapidly Progressive GN (RPGN), and Polyarteritis Nodosa. It primarily involves arterioles, capillaries and venules WITHOUT immune deposits (hence, "pauci-immune" descriptor). Glomerulonephritis is virtually universal, and pulmonary capillaritis is quite common (>50%). Onset is rapid to insidious (years of constitutional symptoms). Treatment is similar to WG. Efficacy of treatment is inversely proportional to degree of disease and recurrence is common (20%). A nephritic UA pointed toward the renal parenchyma as the etiology. The absence of dysmorphic RBC's and RBC casts does not diminish the post-test probability due to cast degradation unless a fresh sample is processed quickly. There has been little improvement of his GFR or anemia despite cyclophosphamide, steroids, erythropoietin and iron.

A RARE BUT PRECARIOUS COMPLICATION OF CORONARY ARTERY STENTING. I.G. Zarraga¹; L. Lu¹. ¹Baylor College of Medicine, Houston, TX. (Tracking ID #133874)

LEARNING OBJECTIVES: (1) To recognize situations where coronary artery stent thrombosis should be highly suspected, and (2) To learn how stent thrombosis differs from in-stent restenosis in terms of pathophysiology, prevention, management, and prognosis.

CASE: A 52-year-old man with hypertension, diabetes, hyperlipidemia, and a 10-pack-year history of cigarette smoking presented to the emergency room with chest pain. His ECG showed ST-segment elevations in the inferior leads and his cardiac enzymes were elevated. The coronary arteriogram and echocardiogram suggested that the posterolateral branch of the left circumflex artery was the infarct-related vessel, but it was too small for mechanical intervention. On the same arteriogram, a significantly stenotic segment was found in the mid-left anterior descending artery and was stented. The patient's angina eventually resolved with medical treatment, and he was discharged on aspirin and clopidogrel, among other cardiac medications. One week later, he returned to the emergency room complaining of severe chest pain again. His repeat ECG showed ST-segment elevations in leads V2 to V6. Stent thrombosis was suspected, and he was emergently taken to the cardiac catheterization lab where the diagnosis was confirmed. He underwent balloon angioplasty and placement of two new stents. The rise in cardiac enzymes beyond the levels during his previous hospitalization and the new antero-septal wall akinesis seen on a repeat echocardiogram indicated that he had sustained an antero-septal wall infarction. Nonetheless, with mechanical and pharmacologic support, he survived this precarious event.

DISCUSSION: Eighteen years after their first use in clinical practice, coronary artery stents are now utilized in more than 85% of percutaneous coronary interventions. Intracoronary stent restenosis and thrombosis are two complicating events that lead to in-stent lumen narrowing, but are very different phenomena. In-stent restenosis results from neointimal hyperplasia after mechanical injury to the artery during angioplasty and stenting, leading to a reduction in luminal diameter in the first six months post-procedure. The use of drug-eluting stents has significantly reduced the incidence of this complication. In contrast, stent thrombosis is a much more dangerous complication that can occur with both bare-metal and drug-eluting stents. It can occur within the first 24 hours, or more often, subacutely within the first 2 weeks after stent placement. Risk factors involved in its occurrence include stent thrombogenicity, procedure-related factors, and subtherapeutic anticoagulation and antiplatelet therapy. With the advances in stent designs, deployment techniques, and antithrombotic therapy, stent thrombosis has become rare, with a reported incidence of 0.5 to 2.5% at 1 month post-procedure and <0.1% once stent endothelialization has occurred. However, it remains an extremely morbid event because stents are generally placed in proximal segments of large coronary arteries. It results in short-term mortality in up to 20 to 25% and transmural myocardial infarction in 60 to 70% of cases. From the standpoint of a non-interventional cardiologist, it is critical to educate patients about the need to adhere to the recommended antiplatelet regimen after stent placement. Similarly, it is important to recognize the possibility of stent thrombosis when one witnesses angina and ECG features of myocardial injury corresponding to a recently stented coronary artery.

A RARE CASE OF NEUROLOGICAL DETERIORATION- WHIPPLE'S DISEASE. B. Kuriya¹; S. Shadowitz². ¹Sunnybrook & Women's College Health Sciences Centre, University of Toronto, Toronto, Ontario; ²Sunnybrook and Women's Health Sciences Centre, University of Toronto, Toronto, Ontario. (Tracking ID #133009)

LEARNING OBJECTIVES: 1) Review clinical manifestations of Whipple's disease 2) Recognize pathognomonic features of CNS involvement 3) Describe the evaluation and treatment of Whipple's disease.

CASE: A 45-year-old man of Italian descent presented to hospital with a 9-month history of progressive insomnia, personality change, gait disturbance and diplopia. Arthralgia of the hands and 30-pound weight loss accompanied the neurological symptoms. Physical examination revealed convergence nystagmus of both eyes synchronous with a chin tremor and myoclonus of the left hand, limited vertical gaze, mild weakness of the left upper extremity and impaired tandem gait. Cognitive function was preserved. The remainder of the physical examination was benign. Subsequent investigations revealed normal biochemistry as well as negative blood cultures and HIV serology. Cerebrospinal fluid (CSF) contained 15 WBC/mm³ (73% lymphocytes) with low glucose (41 mg/dl) and normal protein (33 mg/dl). No bacterial, mycobacterial or fungal elements were isolated from the CSF. Magnetic resonance imaging of the brain and electroencephalogram were non-contributory. The constellation of features, namely the oculo-motor findings, raised the suspicion of Whipple's disease (WD) prompting a diagnostic test to be performed. Duodenal biopsy revealed clusters of foamy histiocytes in the lamina propria and intracellular periodic acid-Schiff (PAS) positive granules, compatible with the histopathology of WD. Electron microscopy later identified Tropheryma whipplei confirming the diagnosis. The patient was started on a 14 day course of streptomycin and ceftriaxone followed by trimethoprim-sulfamethoxazole for one year.

DISCUSSION: Whipple's disease, first described in 1907 by GH Whipple, is a rare infectious disease caused by the gram-positive bacillus, Tropheryma whipplei. It greatly mimics other disorders owing to its widespread organ involvement. The most commonly affected systems are the musculoskeletal and intestinal tract. Cardinal symptoms include arthralgias, weight loss, diarrhea and abdominal pain. Rarely, the heart, lung and skin are involved. Infection of the central nervous system (CNS) is observed in 20-40% of cases. Cognitive dysfunction is the most common abnormality but two findings are considered pathognomonic for WD: oculomasticatory myorhythmia and oculo-facial-skeletal myorhythmia (continuous rhythmic movements of eye convergence with concurrent contractions of masticatory muscles or other body parts). These findings are usually accompanied by supranuclear vertical gaze palsy but dementia, myoclonus, hemiparesis, peripheral neuropathy, seizures and upper motor neuron disorders have also been described. Diagnostic evaluation is rendered difficult by the inability to culture the organism from affected tissues. Diagnosis is based on the demonstration of PAS positive particles in duodenal biopsies and/or the presence of bacteria by electron microscopy. More recently, PCR amplification of the 16S rRNA gene of Tropheryma whipplei has allowed confirmation of the diagnosis with greater sensitivity and specificity. While untreated disease may be fatal, treatment is often able to eradicate the organism with resolution of symptoms. At present, therapy involves parenteral antibiotic for the first two weeks (to allow penetration of the blood brain barrier in CNS disease) followed by one year maintenance therapy with oral trimethoprim-sulfamethoxazole. Prognosis is favorable with extended antibiotic therapy.

A RARE CAUSE OF UPPER GASTRO-INTESTINAL BLEEDING. S.C. Reddymasu¹; S. Konduru¹; M. Singh¹; A. Sheth¹; B. Upton¹; R.M. Anees¹. ¹Louisiana State University Medical Center at Shreveport, Shreveport, LA. (Tracking ID #135399)

LEARNING OBJECTIVES: 1. Splenic vein thrombosis (SVT) is a rare cause of upper gastro-intestinal bleed. 2. Management of SVT.

CASE: We present a 70 year old white male with diabetes, hypertension, gastric ulcer who presented to the ER with hematemesis, melena and abdominal pain. Physical exam was otherwise benign. CBC, Liver function tests and coagulation studies were normal. Rectal guaiac test was positive. A computerized tomography (CT) scan of the abdomen revealed splenic vein thrombosis and gastric varices. No evidence of intra abdominal malignancies, pancreatic cancer or cirrhosis of the liver was found. Esophagogastroduodenoscopy revealed isolated gastric varices and a healed duodenal ulcer. Workup for hypercoagulable states and Poxysmal nocturnal hemoglobinuria were negative. A diagnosis of portal hypertension secondary to idiopathic SVT and upper GI bleed from gastric varices was made. Propanolol and anticoagulation with coumadin was started. Patient was referred for splenectomy by splenic artery embolization.

DISCUSSION: Splenic vein thrombosis (SVT) is a rare cause of upper GI bleeding characterized by a triad of isolated gastric varices, splenomegaly, and normal hepatic function. This condition was first described in 1922. Most commonly SVT is associated with repeated attacks of pancreatitis or to a mass effect from adenocarcinoma of the pancreas. Other etiologies include retroperitoneal fibrosis, non pancreatic intra-abdominal neoplasms and peptic ulcer disease. Rare causes are hypercoagulable states such as polycythemia vera or myelofibrosis, Protein C and S deficiency, antiphospholipid antibody syndrome and paroxysmal nocturnal hemoglobinuria. The gold standard of diagnosis continues to be mesenteric angiography. Other imaging modalities such as ultrasound and CT may be used and are diagnostic if they reveal occlusion of the splenic vein. Splenectomy either surgically or by splenic artery embolization remains the treatment of choice for hemorrhage secondary to SVT. Anticoagulation is another treatment option if splenectomy is not feasible. Streptokinase is used in acute events and warfarin can be used in recent onset and chronic cases. Risk of variceal bleeding is not increased with anticoagulation especially with small varices. Nonselective B-blockers should also be used in all patients unless contraindicated to reduce the risk of bleeding. We conclude that SVT with secondary sinistral portal hypertension is a relatively infrequent cause of upper

gastrointestinal hemorrhage. It is one cause, however, that can be treated with uniform success if the diagnosis is made correctly. A high index of suspicion should exist while diagnosing this condition when patients present with either active or occult upper gastrointestinal bleeding.

A RARE DRUG REACTION MASQUERADING AS CHOLEDOCHOLITHIASIS. J.R. Dwaiby¹; R. Pinto-Powell². ¹Dartmouth Medical School, Hanover, NH; ²Dartmouth-Hitchcock Medical Center, Lebanon, NH. (Tracking ID #135873)

LEARNING OBJECTIVES: 1) Recognize cholestasis as a rare side effect of trimethoprim-sulfamethoxazole 2) Recognize this diagnosis is difficult to make in the presence of gallstones 3) Review the differential diagnosis of jaundice.

CASE: 78 year-old male presented with painless jaundice after taking trimethoprim-sulfamethoxazole (TMP-SMX) for 2 days for a UTI. He complained of pruritis, dark urine and pale stools. He denied abdominal pain, nausea, vomiting and fever. He was admitted for ERCP with extraction of a common bile duct (CBD) stone and sphincterotomy. Labs included: WBC 5.3, total bilirubin 10 mg/dl, direct bilirubin 6.7 mg/dl, alkaline phosphatase 212 IU/L, GGT 179, AST 30, ALT 50, amylase 28, lipase 36. Two days later, a repeat ERCP was done for persistent jaundice and a CBD stent was placed for duct narrowing. On discharge, the patient developed generalized weakness and melena. He was readmitted to the hospital with a Hgb of 9.6. Upper GI endoscopy did not identify a source of bleeding but showed a missing stent. Repeat ERCP showed bleeding from the CBD ampulla. The stent had migrated into the common hepatic duct and was removed. Bile duct cytology was negative for malignancy. The patient was admitted one month later for persistent jaundice. Labs included: WBC 6.3, total bilirubin 26.2 mg/dl, direct bilirubin 16.7 mg/dl, alkaline phosphatase 320 IU/L, AST 48, ALT 32, CA19 normal, negative hepatitis serologies. EUS showed probable cholelithiasis but no pancreatic masses. Repeat ERCP showed multiple gallstones without biliary obstruction. CT showed dilatation of the CBD and cholelithiasis. Liver biopsy showed centrilobular intracanalicular cholestasis with an acute inflammatory infiltrate compatible with drug-induced cholestasis. Hepatic bile ducts were present. There were no signs of fibrosis or extrahepatic biliary obstruction. Six weeks after the onset of jaundice, the patient was diagnosed with TMP-SMX-induced cholestasis. 10 weeks after stopping TMP-SMX, his jaundice improved.

DISCUSSION: Hepatotoxicity and cholestasis are rare side effects of TMP-SMX. Reactions include reversible cholestasis, permanent bile duct loss, hepatocellular necrosis and fulminant hepatic failure. Trimethoprim and sulfamethoxazole have both been implicated in reversible cholestasis without ductopenia. Symptoms include painless jaundice, pruritis, fever, rash and eosinophilia. Onset of symptoms can occur from days to months after beginning therapy and may last weeks to years. The prognosis of TMP-SMX-induced cholestatic jaundice is good and death has only been documented in patients with hepatocellular toxicity. Treatment depends on the severity of disease, but the usual course is watchful waiting. TMP-SMX-induced cholestasis is a difficult diagnosis to make because it is rare. A drug-induced reaction was the likely etiology here given the temporal relationship of TMP-SMX exposure to the onset of jaundice, the histology and exclusion of other diagnoses. Confusion with choledocholithiasis made the diagnosis particularly difficult. This lead to a considerable time, cost and procedure burden (four ERCPs). Other causes of jaundice that were excluded were: ductal cancer (negative cytology), pancreatic cancer (no masses on imaging and a low CA19), hepatitis (negative serologies), and cirrhosis, auto-immune hepatitis and hemochromatosis (ruled out with histology). Primary sclerosing cholangitis was unlikely due to the acute nature of the illness.

A RECURRENT PAIN IN THE NECK. S. Kahlon¹; M.D. Landry¹. ¹Tulane University Health Sciences Center, New Orleans, LA. (Tracking ID #132563)

LEARNING OBJECTIVES: 1. Establish differential diagnoses for recurrent meningitis 2. Identify diagnostic studies of Mollaret's meningitis 3. Recognize possible treatments for Mollaret's meningitis

CASE: A 42 year-old female presented with a frontal headache, nausea, vomiting, photophobia, and diplopia. The headache was gradual in onset; she denied fever or chills. She took over-the-counter medications without relief. She reported a history of aseptic meningitis 3 years prior. Physical examination revealed normal vital signs and photophobia. Her neurological exam revealed positive Kernig's and Brudzinski's signs. Her physical exam was otherwise unremarkable. Diagnostic studies revealed a normal serum white blood cell count with neutrophil predominance and normal serum glucose. Lumbar puncture revealed an elevated white blood cell count with lymphocyte predominance, an elevated protein and low glucose. Gram stain and CSF cultures were negative. The CSF herpes simplex virus PCR was positive while serum HSV PCR was negative. The patient was admitted and treated symptomatically.

DISCUSSION: Differential diagnoses of recurrent meningitis include infectious, non-infectious and idiopathic causes. Mollaret's syndrome is recurrent aseptic meningitis with associated symptoms often in young adults. The etiology is thought to be latent HSV-2. Patients may have extended symptom-free intervals between episodes. CSF studies show low to normal glucose, elevated protein, and lymphocytic predominance of white blood cells. The classic finding in the CSF is Mollaret's "endothelial" cells. Conflicting data exists regarding treatment with anti-viral agents. Acyclovir prophylaxis can be used to combat recurrence; however, no clear indications for anti-viral agents exist uniformly since the virus is often self-limiting.

A SERIES OF UNFORTUNATE EVENTS: A CASE OF DRUG-INDUCED ACUTE INTERSTITIAL NEPHRITIS. M. Lim¹; K. Parana². ¹University of Connecticut, Farmington, CT; ²University of Connecticut, Hartford, CT. (Tracking ID #135336)

LEARNING OBJECTIVES: Diagnose and manage acute interstitial nephritis as a complication of antibiotic use.

CASE: A 46 year old male with no known history valvular disease and intravenous drug use was empirically treated with 4 weeks each of ceftriaxone and vancomycin and 2 weeks of gentamycin for culture negative infectious endocarditis. He had no known drug allergies and tolerated the antibiotics well until his 27th infusion of vancomycin when he reported a pruritic and erythematous macular rash that progressed cephalocaudally with associated anasarca. He complained of a subjective fever but denied shortness of breath. Vancomycin was discontinued. He was started on oral diphenhydramine and topical steroid cream. Ten days (D10) after its first appearance, the rash had improved with residual erythema involving the face and trunk and some weeping blisters on both lower extremities. He continued to have good urine output and remained afebrile. However, his serum creatine increased to 1.7 mg/dl (baseline of 0.7 mg/dl) and WBC was noted to be 19.8 with 14% eosinophils. Urine microscopy showed no casts, 5–10 WBCs and rare eosinophils. Ultrasound showed normal sized kidneys with diffusely increased renal echogenicity. Fractional excretion of sodium was 0.04%. After adequate hydration, his creatinine continued to increase. Despite initiation of prednisone (1 mg/kg/day) on D12, his renal function continued to decline, eventually requiring dialysis on D17 (BUN of 109 and Scr of 6.7). At this point, biopsy results confirmed the clinical suspicion of allergic interstitial nephritis. He eventually regained kidney function and prednisone was tapered off over two months.

DISCUSSION: Acute interstitial nephritis (AIN) is a common renal syndrome most often associated with drug therapy (accounting for up to 70% of all reported cases of AIN). Although numerous medications have been implicated in AIN, only a few have been reported with frequency. These include methicillin and other b-lactam antibiotics, NSAIDs, rifampin, sulfonamides, cimetidine, all-opurinol, proton-pump inhibitors and quinolones. Our patient received three different antibiotics, and although ceftriaxone (cephalosporins being one of the most common causes of AIN) would be a likely culprit, vancomycin has been reported in rare instances to cause fatal AIN and cannot be easily discounted. The temporal relationship of drug exposure to clinical presentation varies from three to five days with a second exposure and to as long as several weeks with a first exposure. The classic triad of rash, fever and eosinophilia is now seen in only 10% of patients presenting with AIN. More often patients will present with only one of these: fever (27%), eosinophilia (23%), or rash (15%). Interestingly, four of the five cases of vancomycin-induced AIN reported in literature presented with rash, eosinophilia and renal dysfunction. Laboratory findings include an acute rise in serum creatinine, eosinophilia or eosinophiluria and active urine sediment with white cells, red cells, or red cell casts. Biopsy confirms the diagnosis. Once suspected, withdrawal of the offending drug is prudent and has been shown in most cases to be effective in reversing renal dysfunction. Occasionally however, as seen in this case, symptoms may persist despite removal of the immune stimulus. In such cases, oral steroids has been shown to be helpful.

A SEVERE COMPLICATION OF UPPER LIP CELLULITIS IN A WELL CONTROLLED DIABETIC. W.A. Golgert¹; L. Moraski¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #133198)

LEARNING OBJECTIVES: 1. Recognize that methicillin resistant *Staphylococcus aureus* is an emerging pathogen in communities around the world. 2. Identify diabetes as a condition which infers relative immunosuppression even when well controlled. 3. Consider cellulitis of the nasolabial folds as a possible cause of bacteremia.

CASE: A 68-year old Caucasian female with well controlled type 2 diabetes (A1C5.9) and 3 weeks of rhinorrhea presents with malaise. Four days prior, she developed cellulitis near the philtrum of the left upper lip. Within 24 hours, the patient began to experience chest congestion and fatigue associated with dull right-sided back pain. She had no recent hospitalizations or sick contacts. On examination, the patient was noted to have a temperature of 101.5 F. An 6 mm erythematous, warm lesion on the upper lip with a small, yellow crusted center was biopsied and cultured. Multiple dental repairs were noted on oral exam; all without signs of infection or abscess. Lab studies were remarkable for an elevated WBC of 14.4 with 17 bands and 80% neutrophils. A right hilar prominence and small right pleural effusion was seen on chest x-ray. On helical CT scan multiple pulmonary emboli were noted. Her blood cultures grew methicillin resistant *Staphylococcus aureus* (MRSA), as did biopsy/culture specimen. A trans-esophageal echocardiogram showed no evidence of endocarditis. Dental x-rays showed no indication of occult caries/abscesses. A magnetic resonance venogram did not show any emboli within the veins of the head or neck. The cellulitis was thus presumed to be the source of the septic emboli.

DISCUSSION: While septic embolization caused by a neck infection involving the jugular vein has been well described, this case did not involve a major vein. In review of the literature, there are 3 other reports of septic emboli caused by upper lip cellulitis without involvement of adjacent major vessels. In each case, *Staphylococcus aureus* was the causative organism. The incidence of community acquired MRSA infections has risen dramatically since the early 1990's. In 2001 Approximately 30% of MRSA infections requiring hospitalization were community acquired (Bukharie et al., 2001). Diabetics are known to have increased susceptibility to pyogenic organisms due to micro vascular disease, diabetic neuropathy, and immune dysfunction. While good control of blood sugars and meticulous skin care greatly reduces the risk of infections, well controlled diabetics are still at increased risk vs. non-diabetics. This case illustrates the potential for life threatening consequences as antibiotic resistance increases and

the prevalence of diabetes reaches epidemic proportions in the US. Patient and provider awareness of this problem needs to be on par with that of the cardiac, ophthalmologic, renal, and vascular consequences of diabetes.

A SOB STORY. J. Reeder¹; M.A. Mcneil¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #135470)

LEARNING OBJECTIVES: 1) To recognize the clinical presentation and pathophysiology of nephrotic syndrome 2) To identify shortness of breath as a presenting symptom of nephrotic syndrome 3) To recognize the importance of the temporal relationship between the onset of diabetes and its clinical complications

CASE: DW is a 54 year-old woman with a history of diabetes mellitus diagnosed 2 years prior to presentation, hypertension, stroke, and hypercholesterolemia who presented with a one-month history of shortness of breath (SOB). She denied chest pain, cough, fever or chills. Her physical exam was remarkable for diabetic retinopathy, decreased breath sounds and dullness to percussion at the lung bases bilaterally, a normal cardiac exam, bilateral pitting edema of her lower extremities, and a normal sensory exam. Laboratory studies were significant for a creatinine of 1.4, a BNP of 259, a total cholesterol of 257, and an albumin of 1.9. An electrocardiogram showed a normal sinus rhythm with evidence of left ventricular hypertrophy and a chest radiograph demonstrated bilateral pleural effusions. A chest CT confirmed the pleural effusions and failed to show any evidence of malignancy or primary lung disease. The patient underwent a thoracentesis which revealed an unremarkable transudative effusion. An echocardiogram revealed an ejection fraction of 40%. A presumptive diagnosis of congestive heart failure was made. At a follow-up appointment, the patient was found to have a 24-hour urine protein of 7173 mg. This degree of proteinuria along with her hypoalbuminemia, edema, and hyperlipidemia confirmed the diagnosis of nephrotic syndrome. Because the etiology of the patient's nephrotic syndrome remained in doubt, a renal biopsy was performed. The pathology showed nodular and diffuse diabetic glomerulosclerosis.

DISCUSSION: Nephrotic syndrome is a clinical syndrome characterized by abnormal permeability of the glomerular membrane leading to massive proteinuria. The loss of protein in the urine leads to the typical components of the syndrome which include hypoalbuminemia, edema, hyperlipidemia, and hypercoagulability. Diabetic nephropathy is known to be one of the most common causes of nephrotic syndrome in adults and is typically believed to occur approximately 10–15 years after the onset of diabetes. In this case, the diagnosis of diabetic nephropathy was initially missed because the patient had only a two-year history of diabetes. Once the biopsy showed the presence of diabetic glomerulosclerosis, it became clear that the patient's diabetes had been undiagnosed for many years. Another important clue that may have led to the diagnosis was the patient's retinopathy. Retinopathy is another known complication of diabetes believed to occur about 10–15 years after the initial onset of the disease. The patient had significant retinopathy on exam which indicated that she had already begun to develop the microvascular complications of diabetes. The treatment of nephrotic syndrome is generally aimed at controlling the underlying etiology. Tight glycemic control and blood pressure control with angiotensin-converting enzyme inhibitors have been shown to be important in the prevention of the progression of diabetic nephropathy.

A SORE THROAT GONE AWRY. Y. Ng¹; H. Tawbi¹; R. Granieri¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #132589)

LEARNING OBJECTIVES: 1) To recognize the clinical presentation of Lemierre's syndrome 2) To state the investigations needed to establish the diagnosis 3) To state the recommended antibiotic regimen and the role of anticoagulation

CASE: T.R. is a 17 year-old, previously healthy white male who presented to his local hospital with 3 days of fever, sore throat, nausea, vomiting and diarrhea. He was a daily marijuana user but denied intravenous drug use, promiscuous sexual activity, recent travel or antibiotic use. Patient was febrile at 39.5 °C, tachycardic and hypotensive. Laboratory findings were significant for creatinine of 2, platelet count of 106,000, frank hematuria and hemocult positive stool. While his creatinine improved with hydration, his platelet count dropped precipitously to 53,000. Patient was transferred to our institution for further management. On transfer, he had mild anemia and stable platelet count at around 40,000. Peripheral smear was negative for schistocytes. Patient continued to be febrile, hypoxicemic and tachycardic after adequate hydration. A spiral chest CT scan was done. It revealed multiple peripheral wedge-shaped infiltrates in the lungs suggestive of septic embolization and splenic infarcts. 2 days after admission, two anaerobic blood cultures grew *Fusobacterium* species. The finding of septic embolization coupled with the growth of *Fusobacterium* prompted a Doppler ultrasound of the neck that was positive for thrombosis of his left internal jugular vein confirming the diagnosis of Lemierre's syndrome. Antibiotic management was focused on adequate anaerobic coverage, initially with ampicillin/sulbactam later changed to metronidazole for 6 weeks.

DISCUSSION: Lemierre's Syndrome is characterized by thrombophlebitis of the internal jugular vein due to direct extension from adjacent acute pharyngitis or tonsillitis. If left untreated, it may progress to systemic septic embolization with the most commonly affected organ being the lungs. Renal failure, hepatic abscess, osteomyelitis and septic arthritis have all been reported. The most common organism involved is gram-negative obligate anaerobe, *Fusobacterium necrophorum*, although others have also been implicated. High clinical suspicion is key. Ancillary tests that help support the diagnosis include 1) blood culture positive for anaerobes especially *Fusobacterium*, 2) ultrasound showing thrombosis of the internal jugular vein, 3) CT scan showing septic emboli in di-

stal organs. Treatment includes antibiotics with good anaerobic coverage, adjusted to microbe sensitivity, for 4–6 weeks and supportive treatment for associated organ damage. Anticoagulation is controversial and is reserved for patients with extensive thrombosis. Mortality was high in the pre-antibiotic era. However, with the widespread use of antibiotics, the incidence and mortality of this disease had dropped dramatically. Recent retrospective studies suggest resurgence due to trends to limit antibiotic use in the treatment of upper respiratory tract infections. As a result, recognition of this syndrome and early intervention is important to lower mortality.

A THORACIC CAUSE OF THE ACUTE ABDOMEN. J.A. Reider¹; J. Sethi¹. ¹University of Pittsburgh Medical Center, Pittsburgh, PA. (Tracking ID #133365)

LEARNING OBJECTIVES: 1. Recognize the clinical manifestations of acute aortic dissection and its complications. 2. Recognize the importance of a high level of clinical suspicion for aortic dissection in patients with Turner's syndrome. 3. Recognize mesenteric ischemia as a rare and potentially fatal complication of aortic dissection.

CASE: A 30 year old woman with Turner's syndrome, hypertension and a bicuspid aortic valve presented to an outside institution with crushing chest pain. EKGs, cardiac enzymes, and transthoracic echocardiogram were reportedly normal, and she was sent home with a diagnosis of musculoskeletal pain. Four days later she developed progressively severe abdominal pain, nausea, vomiting and diarrhea, for which she was admitted. She had a leukocytosis with 50% bands and an ileus of the entire small and large bowel. Deterioration of her condition prompted transfer to our hospital the next day for further evaluation. She arrived mildly tachycardic and hypertensive with an acute abdomen. Following immediate surgical consultation, exploratory laparotomy only hours later revealed ischemic bowel of the distal ileum and right colon. A right hemicolectomy and partial small bowel resection were performed. An emergent aortogram then showed a Type A aortic dissection extending from the aortic valve to the abdominal aorta, with marked aortic regurgitation. The left subclavian and celiac artery notably originated from the false lumen. Within 48 hours she developed multi-system organ failure requiring respiratory and vasopressor support and was not felt to be an operative candidate. She developed ventricular tachycardia and, following failed resuscitative measures, expired. The autopsy confirmed aortic dissection extending from the aortic root to six centimeters proximal to the aortic bifurcation, underlying coarctation of the aorta, and necrosis of remaining large and small bowel. Review of hospitalization records indicated that the aortic dissection was missed on the initial echocardiogram.

DISCUSSION: Aortic dissection is increasingly recognized as a cause of death in patients with Turner's syndrome. Hypertension and cardiovascular malformations, such as a bicuspid aortic valve or aortic coarctation, predispose patients to dissection. Common presenting features include chest pain, aortic regurgitation, a pulse or blood pressure deficit, and a widened mediastinum on chest radiograph. More ominously, though rarely, patients may present primarily as a consequence of the propagating dissection leading to occlusion of involved arteries and subsequent ischemia of the brain (hemiplegia), heart (myocardial infarction), spinal cord (paraplegia), extremities (limb ischemia), kidneys (renal failure), and bowel (mesenteric ischemia). Thus, aortic dissection with secondary occlusion of a visceral aortic branch leading to mesenteric ischemia must be included in the differential diagnosis of the acute abdomen. In our case, the diagnosis of aortic dissection was missed, as it was assumed that the cause of the acute abdomen lay within the peritoneum. We also assumed that diffuse ileus was a result of acute peritonitis, when it actually suggested the cause (i.e. celiac arterial occlusion). This case illustrates the importance of having a high level of clinical suspicion for aortic dissection in patients with Turner's syndrome. Clinicians should also be aware of the complications of aortic dissection, which may assist in earlier diagnosis and prevention of catastrophic consequences.

A TRAVELER'S DILEMMA. A.T. Lee¹; A. Desai¹; P. Basaviah¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #136284)

LEARNING OBJECTIVES: Recognize the importance of hepatitis A vaccination and the risk of potentially serious complications from hepatitis A infection.

CASE: A 27-year-old woman with no significant past medical history presented 2 weeks after returning from a trip to India with a 4-day history of fevers, nausea, non-bloody emesis, and severe right quadrant abdominal pain. She also had complaints of fatigue, malaise, anorexia, and myalgias. She reported taking 1500mg of acetaminophen for fevers 1 day prior to presentation, but did not take any medications or herbal supplements on a regular basis. She denied IV drug use or the ingestion of new foods. She did not receive any vaccines before her trip. Born in India, she had relocated to the United States at the age of 16 and had traveled frequently to India with no complications. On physical examination, the patient was afebrile and had mildly jaundiced skin, right quadrant tenderness, and hepatosplenomegaly. She was alert and oriented and did not demonstrate asterixis. Laboratory analysis revealed an elevated AST of 5995 U/L, ALT of 5217 U/L, total bilirubin of 4.1 mg/dL, and INR of 1.9. She was evaluated by the liver transplant team given the severity of her presentation. Upon admission, frequent mental status checks were performed that revealed clear sensorium and lack of asterixis. On hospital day 2 the test for Hepatitis A IgM antibodies was positive and she was diagnosed with severe Acute Hepatitis A as all other work-up was negative. The patient's AST and ALT slowly declined while her INR reached a peak of 2.1 and her T. Bili continued to rise. Her symptoms gradually improved and she was discharged on day 5 when her INR level returned within normal range and she was tolerating some solid foods. Her jaundice and bilirubin level continued to increase as expected after discharge while

her symptoms were resolving. Her husband received postexposure prophylaxis treatment with immune globulin and also hepatitis A vaccine. Two months later the patient still reported mild intermittent right quadrant abdominal pain. Her transaminases remained around 100 U/L.

DISCUSSION: Hepatitis A still remains one of the most frequently reported vaccine-preventable diseases in the country. In the United States, cyclic communitywide outbreaks of hepatitis A have occurred every 5 to 10 years, with the last nationwide increase in 1995. In 1999, the CDC approximated that each year in the United States an estimated 100 persons die as a result of acute liver failure due to hepatitis A. Rates of hepatitis A infection have gradually decreased over the past several decades, most likely reflecting improvements in hygiene and the use of the vaccine. However, more than 7,500 cases were reported in this country in 2003 with the actual number of new infections, including asymptomatic cases, estimated to be 8 times that reported. Groups at increased risk for hepatitis A include travelers to highly endemic areas, men who have sex with men, illegal-drug users, persons who have occupational risk for infection, and persons who have clotting-factor disorders. This vignette demonstrates the importance of vaccinating high risk groups such as travelers to highly endemic areas, regardless of immigration status. This precaution will help avert preventable morbidities, mortalities, and financial costs.

A TWIST ON TORSADE. P.A. Lamont¹; S. Hunt¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #132582)

LEARNING OBJECTIVES: 1. Recognize high dose methadone is associated with a prolonged QTc interval and progression to torsade de pointes (tdp). 2. Appreciate the cause of tdp is often multifactorial. 3. Identify risk factors for the development of tdp in patients on methadone therapy.

CASE: A 50 year old woman with a history of intravenous drug use and HIV infection presented to the emergency department with complaint of several recent episodes of sudden shaking and sweating followed by lapses in consciousness. She reported that she had recently been using heroin and cocaine. Her medications included methadone 145mg daily for heroin addiction, abacavir/lamivudine/zidovudine, tenofovir, alprazolam, and trazodone. Physical exam was remarkable for normal mental status, blood pressure of 142/72 mmHg, pulse of 86 beats/min and regular, absence of cardiac murmurs, and no signs of heart failure. Laboratory studies included potassium of 3.2 mmol/L and magnesium of 1.3 mmol/L. During evaluation, the patient developed sudden loss of consciousness and her cardiac rhythm showed a polymorphic tachycardia identified as torsade de pointes (tdp). She was treated with intravenous lidocaine and magnesium with subsequent conversion to sinus rhythm. On admission, her EKG demonstrated normal sinus rhythm with bigeminy, rate 60 beats/min, and QTc at 560ms. Trazodone and alprazolam were discontinued, and methadone was tapered by 10% daily. Despite temporary pacing, electrolyte correction, and treatment with labetalol, the patient developed additional episodes of tdp, one of which required cardioversion. During her hospitalization, she required additional intravenous infusions of lidocaine and labetalol, and ultimately the placement of a St. Jude implantable cardioverter-defibrillator.

DISCUSSION: This case demonstrates torsade de pointes or "twisting of the points," which is a polymorphic ventricular tachycardia, in the setting of delayed repolarization and a prolonged QTc interval. Methadone is independently associated with a prolonged QTc interval and progression to tdp. However, as was likely in this case, most often the etiology of tdp is multifactorial. Any drug that delays cardiac repolarization, or inhibits the metabolism of another drug that delays cardiac repolarization, increases the risk for tdp. Methadone is metabolized primarily by the cytochrome P450 isoform 2B6, and less so by 2C19 and 3A4. In this case, the patient was taking alprazolam, also metabolized by the P450 isoform 3A4, and the drug-drug interaction between alprazolam and methadone was likely a contributing factor in the development of tdp. In addition, trazodone and cocaine can both independently prolong the QTc interval. Additional risk factors for the development of tdp include female gender, congenital long QT syndrome, HIV infection, ischemic heart disease, hypokalemia, hypomagnesemia, and a wide variety of medications. This case demonstrates the need to obtain a careful medication history, monitor potassium and magnesium levels, and follow EKGs when treating patients with methadone for heroin addiction or for chronic pain.

A TYPICAL PRESENTATION IN AN ATYPICAL PATIENT: A CASE OF MYELOMA BACK PAIN IN A 34 YEAR-OLD WOMAN. D.J. Kim¹; O. Melamed¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #134144)

LEARNING OBJECTIVES: Recognize a typical presentation of multiple myeloma (MM) in an atypical age group.

CASE: A 34 year-old Hispanic female housekeeper with a two-year history of chronic low back pain presented to the emergency department with worsening back pain especially with activity for one month. She denied trauma and any other associated symptoms. Physical examination was significant for skin pallor and an exquisite tenderness over the lumbar spine; her neurological examination was normal. An L-spine x-ray revealed diffuse osteopenia and an L2 compression fracture. Due to her excessive use of over-the-counter ibuprofen, a chemistry panel was ordered and was significant for renal failure (Cr of 3.1) and hypercalcemia. Other significant laboratory findings included normocytic anemia (Hgb of 9.3) with a peripheral smear negative for plasmacytosis; ESR of 90; a 24-hour urine collection showing 4.8 grams of protein; normal SPEP; abnormal UPEP with gamma globulin protein band. Bone marrow biopsy revealed sheets of atypical plasma cells comprising greater than 80% of nucleated marrow elements and positive for kappa light chains and CD138. Skeletal survey

revealed multiple lytic lesions on her skull film. The patient was diagnosed with multiple myeloma (MM) and was started on dexamethasone and thalidomide. **DISCUSSION:** The median age of MM at presentation is 65 years; less than 2% of patients diagnosed with MM are younger than 40 years of age. The patient in this case presented with common features associated with multiple myeloma—bone pain, typically present in 58% of patients at diagnosis; renal failure, manifested as an elevated serum creatinine in nearly 50% of patients at diagnosis; and skin pallor, which is the most frequent physical finding and is secondary to anemia. However, the atypical feature of this case was the unusually young age of the patient. The hallmark of MM is the demonstration of paraprotein on SPEP. Approximately 15% of MM patients have no detectable serum paraprotein as was the case in this patient. The initial differential diagnosis of her back pain prior to full laboratory and radiographic examinations included a wide range of potential etiologies, including musculoskeletal pain, osteomyelitis, osteoporosis, osteomalacia, steroid use, Vitamin D deficiency, hyperthyroidism, hyperparathyroidism, and malignancies such as metastatic cancer. The suspicious findings leading to the diagnosis of MM were the diffuse osteopenia on the initial L-spine film, hypercalcemia in the presence of renal failure, and significant proteinuria. This case demonstrates the importance of approaching common clinical problems with more caution and wider differential diagnosis when certain unexpected results are observed.

A YOUNG WOMAN WITH SEVERE WEAKNESS AND DRY EYES. A.M. Mallouk¹; N. Abdo¹; S.H. Hashmi¹; P. Pham¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #133568)

LEARNING OBJECTIVES: 1. Review the differential diagnosis of non-anion gap metabolic acidosis (MA) 2. Review the types of renal tubular acidosis (RTA) 3. Discuss the management of RTA

CASE: A previously healthy 35-year-old Hispanic woman presented with a 6-month history of progressive generalized weakness to the point where she couldn't get out of bed. She denied diarrhea or use of medications. At presentation, her vital signs were: temperature 36.4 °C, blood pressure 90/50 mmHg, pulse 90, respiratory rate 14. Her exam revealed severe generalized weakness, inability to speak, 0/5 muscle strengths, and absent reflexes. Laboratory findings were: Na+ 144 meq/L, K+ 1.9 meq/L, Cl- 121 meq/L, CO2 8 meq/L, BUN 24 mg/dl, Cr 1.0 mg/dl, glucose 157 mg/dl, arterial blood gas: pH 7.16/pCO2 22 mmHg/pO2 369 mmHg/O2 saturation 100%, urine pH 7.5 otherwise negative, urine anion gap (UAG) 31, and EKG: 1° heart block with nonspecific ST-T changes. The patient was admitted for severe hypokalemia and a predominantly non-anion gap MA. She was treated with intravenous potassium followed by the addition of sodium bicarbonate when her K+ was >3.0 meq/L. Following all electrolyte corrections, she completely recovered. A diagnosis of distal RTA was entertained. A urine pCO2 determined while the patient was receiving intravenous bicarbonate at a serum CO2 of 24 meq/L was 45 mmHg

DISCUSSION: Non-anion gap MA can result from gastrointestinal bicarbonate loss or RTA. A UAG >0 in the presence of MA suggests RTA, while a UAG <0 suggests a gastrointestinal cause. In general, RTA results from either defective renal H+ secretion (distal-dRTA) or bicarbonate reabsorption (proximal-pRTA). In pRTA, patients present with mild-moderate MA and variable urine pH (UpH) depending on the patients' acid-base status (i.e. during severe MA, UpH is <5.5, but may be higher during a bicarbonate load). In dRTA patients typically present with marked MA and, in most cases, UpH >5.5. In both types of RTA defects in various electrolyte transporters have been described. While pRTA typically results in hypokalemia, dRTA may result in different potassium levels depending on the etiology. In dRTA with defective H+-ATPase, hypokalemia may occur. In autoimmune disorders such as Sjogren's, dRTA has been linked to defective H+-ATPase, presumably due to autoantibodies directed against this pump. A diagnosis of dRTA can be suggested by a failure to increase the urine pCO2 with bicarbonate loading. In patients with normal H+ secretion, the excess filtered bicarbonate combines with H+ to form H2CO3 and subsequent pCO2. In dRTA patients, H+ secretion failure precludes an increase in urinary pCO2 production. Our patient met all criteria for dRTA due to defective H+-ATPase. A search for the etiology of the underlying defect revealed significantly elevated ANA, anti-SSA, and anti-SSB titers and a strongly positive Schirmer's test, all suggestive of Sjogren's syndrome. Our patient's treatment consisted of K+ and bicarbonate replacement. In managing severe hypokalemia, bicarbonate and glucose should not be given until the K+ level is at least 2.5 meq/L. Bicarbonate and glucose infusions may induce intracellular K+ uptake hence further reducing the already dangerously low serum K+ level.

ACQUIRED IMMUNE DEFICIENCY SYNDROME (AIDS) WITH AN UNDETECTABLE VIRAL LOAD. D. Mukherjee¹; M.J. Maslow¹; M.D. Schwartz¹. ¹New York University, New York, NY. (Tracking ID #135894)

LEARNING OBJECTIVES: 1. Diagnose CNS lesions in HIV infected patients presenting with alteration in mental status. 2. Identify common CNS lesions that do not produce mass effect or contrast enhancement on MRI scans. 3. Recognize discordance between immunologic and virologic responses to therapy for HIV infections and that patients may be immunosuppressed even with an undetectable viral load.

CASE: A 52 y/o African American male with AIDS (CD4 count of 88) presented to the outpatient clinic with the CC: "I have been forgetting things". He c/o progressive short-term memory loss with no change in long-term memory. He denied fevers, chills, new rash or skin lesions, headaches, visual changes, seizures, falls, dizziness, loss of consciousness, neck stiffness or focal weakness. He was compliant with his highly active antiretroviral therapy (HAART)

and prophylactic Bactrim for a low CD4 count. He had an undetectable viral load for the last two years, denied use of drugs or alcohol, was on methadone maintenance and denied use of new medications. On exam he looked cachectic, was afebrile, without skin rash or lesions and his exam was essentially unchanged from baseline except for his inability to recall the month and the day. His neck was supple with symmetric muscle strength. His admission EKG, CXR, serum glucose, electrolytes and urinalysis were normal. His HIV viral load was undetectable and CD4 count was 88. MRI revealed abnormal periventricular hyperintensity, without contrast enhancement or mass effect, suggestive of progressive multifocal leukoencephalopathy (PML). The patient's CSF was negative for JC viral DNA by PCR. While hospitalized he had global worsening of mental status as expected with PML, and died from complications of hepatic encephalopathy (underlying active hepatitis B) and lactic acidosis from antiretrovirals. An autopsy was not performed.

DISCUSSION: HIV infected patients presenting with mental status changes often have CNS lesions that may be life-threatening. These patients are often in the late stages of the disease and severely immunocompromised even with undetectable viral loads as in our patient. The degree of immunosuppression in the host dictates the differential diagnoses of CNS lesions that are most common in the severely immunosuppressed with CD4 counts <200. Mass lesions usually enhance with contrast. Toxoplasma encephalitis (TE) is the most common cerebral mass lesion in patients with AIDS with ring enhancement and mass effect. Bactrim prophylaxis for PCP pneumonia can help prevent TE. PML is a common CNS lesion without mass effect with MRI showing multiple periventricular hyperintense signals without contrast enhancement or mass effect. In one study, among patients with focal brain lesions without mass effect, the probability of PML was 81% increasing to 99% if JCV DNA was detected in the CSF. With specificity close to 100%, PCR detection of JC viral DNA in CSF is now an established way to ascertain the diagnosis of PML. Variable PCR sensitivity (74-93%) and JC Viral load below the level of detection of the PCR assay may result in a false negative CSF study, and is a possibility in our patient. HAART can result in suppression of HIV viral load without immune restoration and the mechanism of discordant immunologic and virologic responses is unclear though research is ongoing.

ACUTE CHEST PAIN AND ST-SEGMENT ELEVATION: WHEN TO SAY NO (NITRIC OXIDE) IN THE CARDIAC CATHETERIZATION LAB. M. Kochar¹; J.E. Lee¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133357)

LEARNING OBJECTIVES: 1. Identify the clinical characteristics that help to differentiate between angina due to vasospasm versus epicardial coronary artery disease. 2. Recognize how the diagnosis of coronary vasospasm is made and the treatment options employed.

CASE: DK is a 43 year old man with a history of tobacco use and no other cardiac risk factors who presented to the ER with a 2-year history of worsening, non-exertional chest pain. He had been treated with beta-blockade as an outpatient and experienced no improvement in symptoms. In the ER, his cardiovascular exam and 12-lead ECG were normal. Cardiac biomarkers were negative. While in the ER, the patient developed recurrent chest pain. A 12-lead ECG revealed 7 mm ST-segment elevation in the anterior leads. Treatment with IV nitroglycerin (NTG) and heparin was initiated, with prompt resolution of symptoms and ECG changes. Emergent cardiac catheterization revealed a diffuse 99% stenosis of the distal left anterior descending coronary artery. The remaining vessels were normal. Left ventriculography revealed akinesis of the mid and distal anterior wall and apex. Due to the discrepancy between the distribution of atherosclerosis and extent of regional wall motion abnormalities, coronary vasospasm was suspected. Intracoronary nitroglycerin was administered and resulted in reduction of the stenosis to 30-40%. Beta-blockade was discontinued. Diltiazem and nitrates were initiated. The patient was discharged without further event.

DISCUSSION: Prinzmetal's or variant angina, is caused by a sudden increase in vascular tone which can occur in normal coronary arteries or, more commonly, in arteries with underlying atherosclerosis. While vasospasm may be diffuse, it is typically localized within one centimeter of an atherosclerotic plaque. As its name implies, variant angina is not necessarily precipitated by situations where myocardial oxygen demand exceeds coronary blood flow. The exact pathogenesis remains unknown. However, contributing factors include endothelial dysfunction with abnormal synthesis of nitric oxide (NO) and endothelin, as well as impaired response to parasympathetic tone. Clinical factors which appear to be associated with coronary vasospasm include cocaine and tobacco use, hormonal fluctuations during the menstrual cycle, and a history of vasospastic conditions such as migraine headaches and Raynaud's phenomenon. Vasospasm should be suspected in individuals with non-exertional chest pain and concomitant dynamic ST changes, with no hemodynamically significant stenoses on coronary angiography. Provocative methods utilizing hyperventilation and intracoronary acetylcholine are often necessary to confirm the diagnosis. If induced during cardiac catheterization, vasospasm is readily relieved with the administration of intracoronary NTG. Exercise stress testing is of little use, inducing ST depression or elevation in only fifty percent of patients with coronary vasospasm. Coronary vasospasm is best managed with nitrates and calcium channel blockers for their vasodilatory effects. Beta-blockade is not recommended, as it leaves the vasoconstrictive effects of the alpha-adrenergic system unopposed, which may serve to exacerbate symptoms. Inciting factors such as cocaine and tobacco use should be discontinued. Finally, traditional cardiac risk factor modification should be emphasized, as this may improve intrinsic endothelial function of the coronary vasculature.

ACUTE FEMORAL NEUROPATHY FROM ILIOPSOAS HEMATOMA: A COMPLICATION OF ENOXAPARIN. N. Lee¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #136074)

LEARNING OBJECTIVES: 1. Describe the clinical presentation and causes of acute femoral neuropathy. 2. Recognize iliopectus hematoma as a potential complication of anticoagulation therapy. 3. Discuss treatment options for iliopectus hematoma.

CASE: A 72 year old man with history of lymphoma, small cell lung cancer and lung carcinoid presented with shortness of breath. Five days previously he had collapsed due to severe dyspnea. A non-contrast CT at another hospital showed small to moderate bilateral pleural effusions and right perihilar consolidation. These findings were similar to those of a CT scan performed two weeks before. An ultrasound found no evidence of deep venous thrombus; echocardiogram showed an ejection fraction of 45%. The patient was treated for pneumonia with levofloxacin. Dyspnea persisted, and he was admitted to the medicine service. Because he had an atrophic kidney and chronic renal insufficiency (baseline creatinine 1.7), he was admitted for IV hydration and N-acetylcysteine prior to contrast chest CT. Clinical suspicion for pulmonary embolism (PE) was high due to his history of multiple malignancies and persistent dyspnea, so he was treated empirically with enoxaparin. Creatinine remained elevated after hydration and CT was delayed. A VQ scan showed mismatch with low to intermediate probability for PE. On the third hospital day, the patient complained of hip pain and leg weakness. There was no history of trauma or line placement in this area. On exam he had decreased strength of hip flexion and loss of quadriceps reflex. A contrast CT revealed no evidence of PE, but a large right iliopectus hematoma was seen. Enoxaparin was stopped and a percutaneous drain was placed in the iliopectus hematoma, with rapid resolution of the pain and weakness. Thoracentesis resulted in some improvement in the patient's symptom of dyspnea, which gradually improved over the hospitalization.

DISCUSSION: The differential diagnosis for dyspnea in this case was broad, including pleural effusion, pneumonia, congestive heart failure, and pulmonary embolism. Empiric anticoagulation was begun given high clinical suspicion for PE. This was complicated by acute femoral neuropathy from an iliopectus hematoma. The femoral nerve arises from the L2-L4 levels and descends through the psoas muscle, passing between and supplying the psoas and iliacus. Lesions in the iliopectus such as hematoma, abscess, or tumor can cause femoral compression neuropathy. Iliopectus hematoma has been reported in a small percentage of patients on anticoagulation therapy. Patients present with pain in the hip, groin, or thigh, and may have difficulty walking. On exam they have weakness in hip flexion, loss of quadriceps reflex, and sensory loss over the anteromedial thigh. A psoas sign may be present, with pain upon hip extension, and the thigh may be held in flexion. Patients on anticoagulation therapy should be monitored for symptoms and signs of femoral neuropathy due to iliopectus hematoma. Treatment for iliopectus hematoma includes cessation of anticoagulation, bedrest, analgesia, and surgical decompression or percutaneous drainage. Pain is usually relieved quickly after decompression. Nerve recovery can occur even in patients with complete paralysis, but may take up to a year.

ACUTE FULMINANT HEPATITIS DUE TO ISONIAZID. M. Lau¹; L. Lu¹; T.T. Tran¹. ¹Baylor College of Medicine, Houston, TX. (Tracking ID #133563)

LEARNING OBJECTIVES: 1) Recognize the importance of early diagnosis of hepatitis in patients on tuberculosis medications. 2) Review the clinical presentation, pathogenesis, treatment options, and prognosis of isoniazid-induced hepatitis.

CASE: A 40 year-old homeless man was admitted for 6 months of fever, weight loss, and productive cough. Chest x-rays showed bilateral infiltrates consistent with miliary tuberculosis. Acid fast bacilli smear and polymerase chain reaction of the sputum were indicative of an active tuberculosis infection. He was started on isoniazid, pyrazinamide, ethambutol, and rifampin. On treatment day 8, the patient complained of nausea, malaise, fatigue, and diffuse weakness. His liver function tests, which were normal on admission, increased to 753 U/L (AST) and 372 U/L (ALT). All of his anti-tuberculosis medications were discontinued. The next day, the patient developed shortness of breath, right upper quadrant abdominal pain, and liquid tarry stools. His blood pressure was 75/45 with heart rate of 130. His liver transaminases increased to 1379 U/L (AST) and 723 U/L (ALT), and his hemoglobin fell from 9.8 on admission to 4.8 gm/dL. His coagulation panel showed PT 32.7, PTT 81.1, and INR 6.1. The patient was immediately transferred to the MICU, intubated, and aggressively resuscitated. Upper endoscopy revealed a large duodenal ulcer that was treated with epinephrine injection followed by embolization. He required 38 units of red blood cells, 15 units of fresh frozen plasma, 10 units of cryoprecipitate, and 4 units of platelets. His hemoglobin was later stabilized. He was started on N-acetylcysteine, and his liver functions improved and normalized. Ethambutol, amikacin, rifampin, and pyrazinamide were re-started with no development of hepatic dysfunctions.

DISCUSSION: Isoniazid (INH) is a synthetic antibiotic which inhibits the synthesis of mycolic acid, a component of the mycobacterial cell wall. INH is metabolized in the liver to form several hepatic intermediates, and these intermediates bind covalently to cell macromolecules leading to hepatocyte necrosis. There are two reported syndromes of INH-induced hepatotoxicity. The first form is mild INH hepatotoxicity, which occurs in 10-20% of patients. It is characterized by subclinical serum aminotransferase elevations. With frequent monitoring, INH can be continued. The second form is INH hepatitis, which has an incidence of 0.6% and a mortality rate of 10%. Typical symptoms include malaise, nausea, anorexia, and fatigue. The risk of liver injury is increased with age, female gender, HIV, chronic viral hepatitis, alcohol consumption and concurrent use of other potentially hepatotoxic medications. The treatment is largely supportive with discontinuation of INH and other potential hepatotoxins.

Liver failure has been successfully treated with liver transplantation. N-acetylcysteine has been shown to protect rats against isoniazid-induced hepatic injury. Current recommendation when starting INH includes baseline and serial measurements of blood counts, liver and renal functions. Although the time frame for the serial measurements are not specified, patients should be seen at least once a month to check for potential adverse reactions. In conclusion, starting isoniazid requires careful monitoring due to its potential hepatotoxic effects.

ACUTE INTERSTITIAL NEPHRITIS - A TALE OF ANTIBIOTICS. K. Muniyappa¹; H. Friedman¹. ¹St. Francis Hospital, Evanston, IL. (Tracking ID #133942)

LEARNING OBJECTIVES: 1. Recognize acute interstitial nephritis as an important cause of acute renal failure. 2. Recognize acute interstitial nephritis as one of the complications of antibiotic therapy.

CASE: A 68 year-old white man was admitted to the hospital because of acute renal failure. He had undergone a coronary artery bypass graft six weeks before admission to the hospital. Two weeks after the procedure he developed a seroma which was drained and he was prescribed cephalosporin, levofloxacin, rifampin, and doxycycline over the course of four weeks. Seven days after the last antibiotic, he noticed a rash, which started on his thigh initially and spread to involve the whole body in three days, when he was admitted to the hospital. The temperature was 100.1 F, the pulse was 77, and the respirations were 18. The blood pressure was 137/81. On physical examination, the patient was noted to have a diffuse maculopapular rash, and an otherwise benign exam. His renal panel showed a BUN of 118 and Cr of 8, which were 26 and 1.0 respectively six weeks prior. His hemoglobin was 11, which were his baseline, and no eosinophils were noted on the differential. The anticytoplasmic antibody, cryoglobulins, hepatitis panel and the antinuclear antibody titer were negative. The complement levels were normal. The urine was trace positive for protein, and the sediment showed 87 white cells, 200 red cells, and hyaline casts. There were no bacteria and no eosinophils in the urine. A renal ultrasonographic examination showed no hydronephrosis. The patient became progressively uremic and was started on hemodialysis and steroids. He was discharged after three hemodialysis sessions and outpatient hemodialysis was arranged.

DISCUSSION: The initial approach to acute renal failure is to determine whether the cause is prerenal, intrinsic, or post renal. Normal vital signs and no signs of dehydration excluded pre-renal causes in our patient. Renal ultrasound showed no evidence of hydronephrosis, which essentially rules out post renal cause. The cause of the renal failure must therefore be intrinsic, which can be divided into diseases of the tubules, glomeruli, vessels and the interstitium. It would be unusual not to have granular casts in established case of acute tubular necrosis. The absence of dysmorphic red cells, red cell casts, mild proteinuria, and negative ANCA rules out glomerulonephritis as the cause of renal failure. Acute Interstitial Nephritis (AIN): The symptoms of AIN are non-specific and the classic triad of rash, fever and eosinophilia are seen only in 10%. An acute rise in plasma creatinine is temporally related to the start of the offending medication. Urine sediment usually reveals white cells, red cells and white cell casts, but bland urine sediment will not rule out a diagnosis of AIN. A diagnosis can be suspected by history and lab findings alone and a trial of glucocorticoids can be started after discontinuing the offending medication. Although there are no randomized control data, glucocorticoids are recommended. Drug induced AIN has a very good prognosis and 50% of the patients recover normal renal function in one month and 90% of them in three months. We therefore conclude that our patient had AIN secondary to one of the antibiotics that was administered although rifampin is the most likely culprit. Our case also shows us that antibiotics can have deleterious effects and stringent criteria should be used when prescribing them.

ACUTE MYOCARDIAL INFARCTION IN A PATIENT RECEIVING RITUXIMAB. M. Velagapalli¹; A. Kalyanasundaram¹. ¹Geisinger Medical Center, Danville, PA. (Tracking ID #135338)

LEARNING OBJECTIVES: 1) Hypotension is a potential adverse effect of rituximab therapy, and can occur in up to 10% of patients during infusion. 2) Pretreatment with diphenhydramine and acetaminophen might reduce but will not prevent adverse events during rituximab infusion. 3) Myocardial infarction can occur as a rare event during infusion of rituximab, even in patients with no history of coronary artery disease.

CASE: 74-year-old female patient with relapsed non-Hodgkin's lymphoma was admitted to the hospital following an episode of severe hypotension and shortness of breath during infusion of rituximab. She was originally diagnosed with marginal zone non-Hodgkin's lymphoma 5 years earlier, treated with CHOP regimen leading to clinical remission. Recently, she had a recurrent node in the left axilla, diagnosed as relapse of lymphoma. Treatment with rituximab was initiated. She had received three doses after pre-treatment with diphenhydramine and acetaminophen without incident. During her 4th dose of rituximab she became short of breath, fatigued and hypotensive with systolic pressure in the 70s. She was vigorously volume resuscitated and given intravenous steroids, diphenhydramine and epinephrine. Cardiac work up revealed ST-segment elevation in antero-lateral leads. Aspirin, beta-blocker, clopidogrel, heparin and statin were initiated. 2-D echo showed a LV ejection fraction of 40% with antero-lateral wall akinesis at the apical and mid levels. Cardiac catheterization revealed 40% mid RCA disease, luminal irregularities in the LAD and circumflex arteries, moderate LV dysfunction with an ejection fraction of 30%. Troponin-T peaked at 0.73. Peak CK was 128. ACE inhibitor was initiated given her low ejection frac-

tion. She had no further episodes of chest pain during the hospital course and was discharged in a stable condition.

DISCUSSION: Rituximab is a monoclonal antibody directed against the CD20 antigen on B-lymphocytes. It binds to the antigen on the cell surface, activating complement-dependent cytotoxicity and to human Fc receptors, mediating cell killing through an antibody-dependent cellular toxicity. Infusion related reactions have been reported in patients, mostly following the first dose of rituximab including hypotension, bronchospasm, and angioedema. Fluids, antihistamines, and bronchodilators constitute the treatment of these reactions. Myocardial infarction has been reported in one study patient with known coronary disease, 5 days post-infusion of rituximab. The mechanism of myocardial infarction is unclear. In our patient, hypotension and the infarct were most likely secondary to rituximab infusion and not due to underlying coronary artery disease.

ACUTE PANCREATITIS IN A 19-YEAR OLD MALE-BEYOND ALCOHOL AND GALL-STONES. V. Guiral¹; M. Rahim¹; N. Chadda¹; G. Papachristou¹; S. Cooper¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #132572)

LEARNING OBJECTIVES: 1. Recognize when to test for genetic risk factors in a patient with pancreatitis. 2. Recognize genetic variations associated with pancreatitis.

CASE: A 19-year-old male with past medical history of ulcerative colitis with chronic hematochezia presented with abdominal pain of one day duration. The pain was constant, sharp on movement, 6 out of 10 in severity, and relieved by hydrocodone. He drank one to two beers a week and had six drinks four days prior to admission. Family history was significant for an older sister with idiopathic recurrent pancreatitis. Physical examination revealed abdominal tenderness in mid epigastric region. WBC was 14.7, amylase 980 and lipase 9814. Calcium, lipids and toxicology screen were unremarkable. Ultrasound revealed an enlarged pancreas without any hepatic ductal dilatation or gall stones. CT of the abdomen was consistent with pancreatitis. Oral intake was stopped and patient was treated with intravenous fluids and hydromorphone. Molecular studies revealed SPINK1 (Serine protease inhibitor, Kazal type 1) variant. No mutations were identified in the PRSS1 (cationic trypsinogen) or CFTR (cystic fibrosis transmembrane conductance regulator) gene. Patient was discharged to home on day four after marked improvement in pain.

DISCUSSION: More than 80% of cases of pancreatitis are due to alcohol or gallstones. Other causes include hypertriglyceridemia, hypercalcemia, trauma, infections and medications. Molecular studies to identify genetic variations in SPINK1, PRSS1 and CFTR are warranted in patients with family history of pancreatitis or onset of symptoms at a young age (<35 years) without an obvious etiology. SPINK1 is a protease inhibitor that is thought to be an inactivation factor of 20% intrapancreatic trypsin activity. A SPINK1 mutation, as seen in our patient, fails to prevent premature trypsin activation leading to autodigestion and inflammation of the pancreas. Though Teich N et al believe that the mutation is not associated with cancer, Masamune et al have reported pancreatic cancer in a patient with SPINK1 mutation. A PRSS1 mutation disrupts self-protective mechanism due to excess trypsin build up in the pancreas leading to hereditary pancreatitis characterized by more serious complications like diabetes mellitus and pancreatic cancer. Prophylactic inhibition of pancreatic trypsin appears to be a rational strategy to prevent pancreatitis in these patients, but no trypsin inhibitors are available to reliably interrupt the pathogenic process. Lastly, pancreatitis may be the earliest manifestation of cystic fibrosis and therefore, CFTR testing should be performed in these patients. Patients should be strongly advised against ethanol and tobacco use as smoking and alcohol consumption of 80 ml/day may increase the risk of pancreatitis by 20- to 80-fold in patients with above mutations. Witt H. et al reported that mutations may be inherited in a dominant, recessive or multigenetic manner. Therefore, genetic counseling should also be provided if molecular studies reveal any mutations.

ACUTE RENAL FAILURE DUE TO BILATERAL URETERIC CALCULI- AN UNUSUAL PRESENTATION OF CROHN'S DISEASE. I. Mohammed¹. ¹Mercy Hospital of Pittsburgh, Pittsburgh, PA. (Tracking ID #134409)

LEARNING OBJECTIVES: 1. To recognize acute renal failure caused by bilateral ureteric calculi as an atypical presentation of Crohn's disease. 2. To recognize the importance of prevention of recurrence of calcium oxalate calculi.

CASE: A 28-year-old woman presented with acute onset of abdominal pain. She also had few episodes diarrhea and mild crampy abdominal pain in last three months. She was afebrile, respiratory rate 18-breaths/min, heart rate 126-beats/min and blood pressure was 134/72 mm Hg. Physical examination was unremarkable except for diffuse abdominal tenderness. She had normal bowel sounds with no guarding or rigidity. Laboratory tests showed white cell count 14.4, blood urea nitrogen 116, creatinine 7.4 and -HCG was negative. She was anuric for the last 4-6 hours even after aggressive intravenous hydration. Non-contrast CT scan of abdomen showed bilateral ureteric calculi with hydronephrosis. She underwent an emergent transurethral ureteroscopic lithotomy, and both calculi were successfully removed. The urine oxalate concentration was elevated. The stones were mainly composed of oxalate calcium monohydrate. Colonoscopy and colonic biopsy were characteristic of Crohn's disease. Treatment was aggressive hydration with subsequent resolution of acute renal failure, preventive management for recurrence of calcium oxalate calculi and medical management of Crohn's disease. At a nine-month follow-up, she did not have recurrence of ureteric calculi.

DISCUSSION: Acute renal failure caused by bilateral ureteric calculi as a first manifestation of Crohn's disease has never been reported in the literature. Al-

though patients with Crohn's disease are at high risk of developing calcium oxalate stones, it is not a common manifestation at the time of presentation of the disease. Malabsorption associated with ileal disease causes increased oxalate absorption by increasing oxalate solubility in the intestinal lumen and permeability of the colonic mucosa, and a reduced citrate excretion is associated in relation to mild acidosis due to the loss of bicarbonate in the liquid stool. Lithogenic risk factors include decreased urinary volume, decreased urinary excretion of stone inhibitors (citrate, magnesium and sulfates) and increased urinary excretion of stone promoters (phosphate, oxalate, uric acid and calcium). The formation of calcium oxalate calculi depends on the state of urinary supersaturation with respect to calcium and oxalate, and the action of urinary inhibitors of crystal nucleation, aggregation and growth. Other causes of calcium oxalate calculi are: intestinal bypass operation or ostomy surgery, hyperparathyroidism and other hypercalcemic states, type I and II hyperoxaluria, sarcoidosis and tumors of kidney. Ureteric calculi should be considered as a differential diagnosis of abdominal pain in patients with Crohn's disease. Diet therapy (low oxalate, low salt, low fat and adequate hydration) and administration of stone inhibitors (citrate, magnesium and sodium bicarbonate) may prevent the recurrence of calcium oxalate stones. This case serves to illustrate how Crohn's disease can present in an unusual manner and it emphasizes the importance of prevention of recurrence of calcium oxalate calculi in high-risk diseases like Crohn's disease.

ADRENOCORTICAL CARCINOMA IN A HYPERTENSIVE PREGNANT PATIENT. S.L. Kraus¹; S. Kidambi¹; S. Davids¹; D. Maas¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #135396)

LEARNING OBJECTIVES: 1. Diagnose endocrine causes of hypertension (HTN) in pregnancy. 2. Evaluate an incidentally identified adrenal mass. 3. Review the diagnosis and treatment of adrenocortical carcinoma (ACC).

CASE: A 21 year-old G1P0 woman presented at 19-2/7 weeks gestation with complaint of facial and bilateral ankle swelling and was found to have a blood pressure of 170/100. The pregnancy had been uncomplicated with regular prenatal care. She was admitted with possible early pre-eclampsia. Ultrasonography performed to assess the fetus revealed an eight cm adrenal mass that was later confirmed by MRI. In retrospect, the patient stated that she had noticed unintentional weight gain (32 kg), striae, excessive hair growth on the face and arms, acne and depression for at least one year prior to pregnancy. Physical exam showed facial plethora, acne on the face and chest, lanugo hair on the face, dorsocervical and supraclavicular fat pads, and wide purplish striae on the abdomen, thighs and breasts. Her abdomen was more distended than the gestational age. A 24-hour urinary free cortisol level was markedly elevated at 2949 mcg. ACTH was suppressed at <5 pg/mL. Laboratory evaluation for pheochromocytoma, hyperaldosteronism, and androgen secreting tumor were negative. A diagnosis of Cushing's syndrome was made, and the patient underwent uncomplicated open adrenalectomy with removal of a 13 cm, 370 g mass. Surgical pathology was highly suggestive of ACC. Abdominal exploration, MRI of the abdomen and CT scan of the chest were negative for metastases. Postoperatively glucocorticoids were administered to prevent Addisonian crisis. After discussion with the patient and other consultants, the pregnancy was continued.

DISCUSSION: Signs of pregnancy confound the presentation of Cushing's syndrome in a pregnant woman. Hypercortisolism, up to three times the normal, is physiologic in the 2nd & 3rd trimesters of pregnancy due to increases in corticotropin-releasing hormone of placental origin and cortisol binding globulin. Undiagnosed excess pathologic secretion of cortisol may result in significant morbidity for both mother and fetus. ACC is an exceedingly rare cause of Cushing's syndrome with an overall incidence of 1.7 per million persons. ACC may be secretory or non-secretory; the most common hormone secreted is cortisol. Regardless, surgical excision is recommended for any solid adrenal mass greater than 5 cm due to an increased incidence of ACC. ACC usually responds poorly to conventional chemotherapy. Mitotane, an adrenolytic drug, may decrease steroid production but does not improve survival. Both conventional chemotherapy and mitotane are contraindicated in pregnancy. Despite current treatment modalities, the overall 5-year survival for ACC is 10-25%. Conclusion: HTN diagnosed before 20 weeks of gestation is usually chronic and not pregnancy-induced. This unusual case represents the importance of considering secondary causes of HTN in pregnant women, especially when they present before 20 weeks of gestation.

ADULT STILL'S DISEASE. S. Bertisch¹. ¹Beth Israel Deaconess Medical Center, Boston, MA. (Tracking ID #135479)

LEARNING OBJECTIVES: 1. To review the differential diagnosis of polyarthralgia and monoarthritis 2. To review diagnostic criteria for Adult Still's disease

CASE: A 28-year old sexually active white female with no previous medical history was admitted for fever and monoarthritis. Three weeks prior to admission, she experienced low-grade fevers, headache, sore throat, and lower extremity myalgias. In outpatient clinic she was diagnosed with a viral syndrome and was treated with NSAIDs (Non-steroidal anti-inflammatories) without improvement. She developed new migratory joint aches, predominantly involving her wrists and knees, with subsequent right wrist swelling, associated with daily spiking fevers. Her physical exam was notable for temperature of 100.1, HR of 106 and right wrist swelling, warmth, and pain, though with full range of motion. Her skin, pelvic and lymph node examination were unremarkable. Laboratory evaluation revealed a leukocytosis of 24 K (85% poly) and an ESR of 108. She was

admitted to the hospital. Further work-up revealed an ANA 1:320 with a speckled pattern, and negative rheumatoid factor, ds-DNA and anti-Smith antibodies. Serum chemistry, renal, liver function tests, urinalysis, bacterial and cervical cultures were negative. A chest radiograph was normal. Tests for parvovirus and HIV were negative. While hospitalized, she continued to spike daily temperatures to 101°F, with mild improvement of her joint symptoms on celecoxib. Based on her clinical history and diagnostic workup, she was diagnosed with Adult Still's disease. Three weeks after discharge, her fevers were well controlled on NSAIDs, but she developed migratory arthritis despite various therapeutic regimens with prednisone, plaquenil and methotrexate. One year after discharge, she was started on infliximab with significant improvement of her chronic arthritis.

DISCUSSION: Adult Still's disease is an inflammatory disorder characterized by high fever and polyarthritis. It may appear in young adults or in older adults either as a new illness or a recrudescence of a dormant childhood disease. The diagnosis of Still's disease is based on clinical criteria alone, with later confirmation from the evolution of chronic polyarthritis. The criteria proposed by Yamaguchi, et al have the greatest sensitivity with fever, arthralgia, typical rash, and leukocytosis as major criteria, and sore throat, lymphadenopathy and/or splenomegaly, liver dysfunction, and the absence of rheumatoid factor and antinuclear antibody as minor criteria. Infectious, oncologic, and other rheumatologic diseases must be excluded. Fever in excess of 104°F is present in half of cases and usually has a quotidian or at times double quotidian pattern, often accompanied by shaking chills, peaking in the afternoon. The classic rash of Still's disease is an evanescent, salmon-colored, maculopapular rash seen usually on the extremities, and exhibits dermatographism and Koebner phenomenon. The course of Still's disease can be divided into three major patterns: self-limited; polycyclic; and a chronic course characterized by persistently active disease. Treatment of Still's disease is based on disease severity and organ involvement. NSAIDs and aspirin are the first line; glucocorticoids and methotrexate are second line. No controlled trials evaluate the efficacy of immune modulation drugs.

ALL THAT EMBOLIZES TO THE CORONARIES IS NOT PLAQUE. P.A. Lamont¹; P. Bulova¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #132585)

LEARNING OBJECTIVES: 1. Recognize coronary artery embolization as a potential complication of infective endocarditis. 2. Identify risk factors which may increase the likelihood of embolization.

CASE: A 50 year old man with a history of bovine aortic valve replacement and end stage renal disease on hemodialysis, presented to the emergency department with complaint of malaise, fatigue and anorexia. Physical examination was remarkable for tachycardia with a systolic murmur best appreciated at the right sternal border and second left intercostal space, decreased breath sounds at lung bases bilaterally, and trace edema. Admission laboratory values included a WBC count of 12,000 with 88% neutrophils and blood cultures which grew coagulase negative staphylococcus resistant to oxacillin. A transesophageal echo revealed a mobile vegetation on the bioprosthetic valve with abnormal thickening around the aortic root consistent with inflammation. The patient was started on vancomycin and gentamicin for treatment of bacterial endocarditis. On hospital day #7 the patient developed chest pain and an electrocardiogram revealed acute ST segment elevation consistent with anterior wall injury. He was taken emergently for cardiac catheterization where angiography revealed no evidence of atherosclerotic disease, but total occlusion at the ostium of the LAD. He underwent successful percutaneous transluminal angioplasty of the LAD with extraction of large amounts of organized material composed of gram positive cocci, fibrin, and platelets. His troponin peaked at 1983 mg/dL. The myocardial infarction was complicated by cardiogenic shock from which the patient died several days later.

DISCUSSION: This case demonstrates acute myocardial infarction resulting from coronary embolization as a complication of bacterial endocarditis. The prevalence of embolic events associated with infective endocarditis is estimated to be between 30-40%, and commonly involves the central nervous system and systemic sites such as the spleen, liver, and kidneys. The coronary arteries are rare, but potentially devastating sites of embolization. Anticoagulation, including the use of aspirin or fibrinolytic therapy, has not been shown to be of benefit in reducing the incidence of or morbidity from thromboembolism. Since there are no prophylactic measures to prevent cardiac embolic events during treatment for endocarditis, it is imperative to act with extreme vigilance in the setting whereby a patient develops hypotension or chest pain, as these may serve as signs of acute infarction which is unlikely to respond to medical therapy. There is no clear model to predict which patients have a higher risk for embolization. Recent research, however, has indicated several factors that may predict a higher likelihood for embolization, including vegetation size > 10 mm, C-Reactive Protein levels greater than 30 mg/L, patient age under 40, and the organism involved, specifically with fungal and Staphylococcus aureus vegetations exhibiting a high frequency of embolization. At present, the only effective treatment to prevent embolic complications is early surgical valve replacement.

ALLOPURINOL HYPERSENSITIVITY SYNDROME IN RENAL INSUFFICIENCY. J.M. Schwab¹; M.C. Gilligan¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #133275)

LEARNING OBJECTIVES: To recognize that allopurinol hypersensitivity, a potentially life-threatening condition, can occur in renal insufficiency even when an appropriately renal-adjusted dose of allopurinol is used.

CASE: A 63-year-old African-American female with a history of hypertension, diabetes mellitus, and gout presented with a 2-day history of low grade fevers

and acute mental status changes in the setting of baseline mild cognitive decline. Temperature on admission was 100.8°F with no localizing symptoms other than a mild cough. Serum creatinine was 2.1 mg/dL (baseline 1.8 mg/dL) and WBC count was 12,000/CMM. The patient was treated with IV hydration and azithromycin. On hospital day 3 the patient developed a maculopapular rash on her upper chest consistent with a viral exanthem. By hospital day 5 the rash had progressed to include her torso and extremities. Azithromycin was discontinued and all bedding was changed to non-bleach. Folstein Mini-Mental Status Exam (MMSE) score on this day was 8/30. The rash persisted and on hospital day 7 laboratory studies revealed AST 161U/L (normal 10-45), ALT 289 (normal 4-35), alkaline phosphatase 539U/L (normal 35-125), total bilirubin 6.2 mg/dL (normal 0.1-1.0), and WBC count 27,000/CMM with 18% eosinophils. On further investigation of her medication list we learned she had been started on allopurinol six weeks prior to admission. Review of the literature on drug-induced rashes suggested a diagnosis of Allopurinol Hypersensitivity Syndrome (AHS). Allopurinol was discontinued and within 24-48 hours the rash had essentially resolved and laboratory values started to normalize. On day 11, MMSE score was 24/30 which is her baseline. The patient's chart was marked ALLERGIC TO ALLOPURINOL.

DISCUSSION: While most patients tolerate allopurinol well, a rare complication, with less than 300 cases in the literature, is Allopurinol Hypersensitivity Syndrome (AHS). AHS is characterized by allopurinol exposure with a subset of the following symptoms: rash (toxic epidermal necrolysis, exfoliative dermatitis, erythema multiforme, or a diffuse maculopapular rash), worsening renal function, acute hepatocellular injury, marked eosinophilia, leukocytosis, and fever. Onset of symptoms usually occurs 4-6 weeks following first exposure. AHS is fatal in 25% of cases, and treatment consists of discontinuing the drug followed by supportive therapy. Allopurinol should be dosed according to creatinine clearance, but cases of AHS have been reported with doses as low as 100 mg/day. In this vignette, we believe that the patient's baseline mental status may have resulted in intermittent dosing which may have predisposed a hypersensitivity reaction. This case reminds clinicians that even appropriate renal dosing of drugs such as allopurinol does not completely protect against adverse events and reinforces the need to aggressively pursue drug reactions in the clinical picture of rash.

AN ABDOMINAL PAIN ZEBRA: PORPHYRIA. J. Clemmer¹. ¹Hospital of the University of Pennsylvania, Philadelphia, PA. (Tracking ID #133119)

LEARNING OBJECTIVES: 1. Recognize the clinical features of acute intermittent porphyria. 2. Select appropriate laboratory tests for diagnosis of porphyria. 3. Identify treatment and prevention strategies for porphyria attacks.

CASE: A 27-year-old female with a history of intermittent, often premenstrual abdominal pain presented with severe abdominal pain, nausea, and vomiting for two days. In the setting of job stress, she had similar more severe than usual symptoms plus constipation one month earlier which resolved spontaneously. She denied dietary changes or use of medications, tobacco, alcohol, or illicit drugs. Physical exam revealed hypertension and diffuse abdominal tenderness. The sodium was 133. The white blood cell count, abdominal CT scan, and urine culture were normal. Despite analgesics and anti-emetics, the patient's symptoms worsened. The sodium decreased to 117 although she was adequately hydrated with normal saline. On the third hospital day the patient's father revealed a family history of acute intermittent porphyria. Dextrose containing intravenous fluids were instituted. The patient's symptoms improved, and the sodium normalized over the next 48 hours. Urine porphobilinogen and aminolevulinic acid levels were measured and were markedly elevated.

DISCUSSION: The porphyrias are a heterogeneous group of seven genetic disorders caused by partial enzymatic deficiencies in the heme synthesis pathway. Acute intermittent porphyria is the most common and most severe of these disorders, though only 10-15% of gene carriers develop symptoms. Abdominal pain is the most common symptom with vomiting, constipation, hypertension, and tachycardia also occurring frequently. Neurologic symptoms such as muscle weakness and sensory neuropathy occur in up to two thirds of patients, but delirium, areflexia, seizures, cranial nerve palsies, depression, psychosis, respiratory paralysis, postural hypotension, and coma are less common. Hyponatremia occurs because of dehydration or the syndrome of inappropriate antidiuretic hormone secretion. Attacks can be triggered by medications, smoking, alcohol, illicit drugs, infection, stress, menstruation, fasting, and pregnancy. Initial diagnostic studies should be measurement of 24-hour urinary excretion of porphobilinogen and aminolevulinic acid, both of which are high during attacks. Stool and erythrocyte porphyrins can be measured to confirm the diagnosis and distinguish among the types of porphyria. Treatment is with oral or intravenous carbohydrates. Hemin can be used if there is no response to carbohydrates or in severe attacks. Porphyrigenic drugs and triggers must be avoided. Patients should be counseled to avoid triggers, and family members should be screened for the gene defect.

AN ENDOCRINE DEMENTIA. M.A. Waxman¹; N. Mikhail². ¹Olive View - UCLA Medical Center, Los Angeles, CA; ²Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #132731)

LEARNING OBJECTIVES: 1. To recognize the clinical features and treatment of myxedema coma. 2. Recognize severe hypothyroidism as a cause of weakness and confusion in the elderly patient.

CASE: An 83 year old female was brought to the emergency department by her family with complaints of lethargy, cough and severe constipation. The patient had not seen a physician in over 40 years. Her sons had been taking her to a homeopathic clinic for high colonic because of constipation for the past six

months. The patient was able to provide some history, but was profoundly weak. She was living with her two unmarried sons, who stated that one year ago she was running a business, six months ago able to complete her activities of daily living, and for the past month had become bed bound and lethargic. There was no significant past medical history and she had never taken prescription medications. Examination revealed a pale white female, with dry skin and bilateral coarse crackles. Her face, hands and feet were puffy. She was normotensive, with a pulse of 70, and a rectal temperature of 35.5. Chest radiography showed a left upper lobe infiltrate and bilateral pleural effusions. Laboratory values were significant for sodium of 122, and a normocytic anemia (Hgb 9.3). Initially, the hyponatremia was thought to be secondary to SIADH. On hospital day 2, the TSH was 46.7, and free T4 undetectable. The patient was transferred to the intensive care unit and 50 mcg intravenous Levothyroxine and hydrocortisone were initiated. After two doses of intravenous Levothyroxine, and passive re-warming the patients mental status had not improved. Levothyroxine was increased, and intravenous T3 was begun whereupon the patient expired after a bradycardic arrest.

DISCUSSION: This elderly female presented with classic features of myxedema coma in the setting of pneumonia. Myxedema coma is defined as altered mental status, hypothermia, and a precipitating illness associated with low serum levels of thyroid hormones. This extreme presentation of hypothyroidism is associated with a mortality rate of greater than 75% in patients older than 80. Myxedema coma is most common in the elderly, and can be precipitated by a wide variety of illnesses including pneumonia, GI bleeding, and congestive heart failure. The hyponatremia, hypothermia, constipation, skin findings, and bilateral pleural effusions in this case have all been associated with profound hypothyroidism. Hyponatremia associated with myxedema coma is dilutional and responds to thyroid replacement and water restriction. The initial treatment of myxedema coma is IV thyroxine, 50 mcg to 500 mcg initially, and then 50 mcg-100 mcg daily thereafter. Intravenous T3 has been suggested because the peripheral conversion of T4 to T3 is impaired in myxedema coma. The intravenous thyroid replacements are associated with life threatening arrhythmias and should be given under with caution, especially in the elderly, under the supervision of an endocrinologist. This case demonstrates that chronic complaints such as constipation and weakness, and a seemingly typical presentation of a confused, hyponatremic elderly female with pneumonia may be a classic presentation of an endocrine emergency.

AN INFECTION THAT CAN BURN: A CASE OF STREPTOCOCCAL TOXIC SHOCK SYNDROME. E. Chang¹; A. Blatt¹; S. Chance¹; A. Cutney¹; D. Cywinski¹; E. Caiola¹. ¹University of Rochester, Rochester, NY. (Tracking ID #135625)

LEARNING OBJECTIVES: 1. Recognize the various presentations of streptococcal toxic shock syndrome 2. Review treatment options including antibiotics and intravenous immunoglobulins 3. Recognize potential sequelae of the syndrome **CASE:** A 45-year-old man presented to his physician's office one month after treatment of a right hand cellulitis with fever, rigors, diffuse myalgias and dizziness with standing. The patient's past medical history was significant for severe eczema, asthma and history of alcohol abuse. On examination, he was febrile to 39 °C and had a BP of 80/50. He had extreme tenderness of his left buttock without erythema or fluctuance. He was hospitalized for further evaluation and stabilization. He was transferred to the ICU due to profound hypotension despite vigorous fluid resuscitation and broad-spectrum antibiotics. Blood cultures grew Group A Streptococcus within 12 hours of admission. The patient required ventilatory and vasopressor support and was given intravenous immunoglobulin for adjuvant treatment of his streptococcal sepsis. CT scan was positive for myonecrosis of the left biceps femoris. The patient went on to develop multi-system organ failure with ARDS and generalized desquamation of his skin. He was transferred to the burn unit for wound care and remained hospitalized for 5 months. He was discharged home with continuing physical and occupational therapy care needs.

DISCUSSION: Streptococcal toxic shock syndrome is an uncommon but rapidly progressive and often fatal invasive infection. Necrotizing fasciitis and spontaneous gangrenous myositis can be early manifestations of infection. Infection may start from seemingly minor trauma to complications from surgical procedures. Symptoms include abrupt onset of fever, pain, and tenderness and may lack obvious physical findings. Patients with diabetes and alcohol abuse are at highest risk. Multi-organ system failure and ARDS are common sequelae. Early treatment with clindamycin and intravenous immunoglobulins has been shown to increase survival. Both agents block pathogenic streptococcal pyrogenic exotoxins. Aggressive surgical debridement is often concurrently required. Even with optimal care mortality is 30-60%.

AN OUNCE OF PREVENTION WORTH A POUND OF CURE: A CASE OF DOXORUBICIN CARDIOTOXICITY. A. Koli¹; M. Seidman¹; B. Taqui¹; H. Shishodia¹. ¹Temple University, Philadelphia, PA. (Tracking ID #132951)

LEARNING OBJECTIVES: 1. Recognize cardiotoxicity as a potentially irreversible, dose dependent side effect of doxorubicin therapy 2. Review steps in making diagnosis of doxorubicin cardiotoxicity 3. Recognize importance of strategies to prevent doxorubicin cardiotoxicity

CASE: 64 year old white female with hypertension and breast cancer presented with 2 days of dyspnea, persistent chest pain, orthopnea and leg swelling. Five months prior, patient had been treated for recurrent breast cancer with mastectomy, radiation therapy and 5 cycles of doxorubicin/docetaxal (total dose 300 mg/m²). Echocardiogram and MUGA obtained prior to treatment showed normal ejection fraction. At current presentation, exam revealed JVD with hepatjugular reflex, bibasilar crackles, S3, 2+ bilateral lower extremity edema. Labs revealed brain natriuretic peptide 1857. She had a normal EKG. CXR

showed pulmonary vascular congestion. Patient was diagnosed with new onset congestive heart failure and started on diuretics and ace inhibitor. Echocardiogram revealed EF 10-20%, akinesis of septal, periaipical and midinferior wall and anteroseptal wall thrombus. Cardiac catheterization showed minor luminal irregularities and pressures consistent with volume overload. Patient improved with medical management, was discharged with a diagnosis of doxorubicin induced cardiomyopathy and instructed to follow up with oncology and cardiology. **DISCUSSION:** Anthracyclines, especially doxorubicin, are among the most effective and widely used chemotherapeutic agents. They are also the most likely cause of cardiovascular complications in cancer patients. With the advent of colony stimulating factors, myelosuppression is no longer a dose limiting factor and cardiotoxicity is becoming more prevalent. The incidence of congestive heart failure has been reported as 4% for cumulative doses of 500-550 mg/m² and as high as 18% for doses greater than 550. In addition to cumulative dose, risk factors for cardiotoxicity include age greater than 70, combination therapy, mediastinal radiotherapy, previous cardiac disease, hypertension, liver disease and hyperthermia. Complications can occur acutely (during administration), early (days to months) and later (years to decades). Oxidative stress has been implicated as a cause of myocyte damage, but the exact mechanism remains unclear. EKG lacks sensitivity and specificity. Echocardiogram and radionuclide imaging may show focal or diffuse wall motion abnormalities. Reliable diagnosis can be made when the fall in ejection fraction is temporally linked to anthracycline exposure. Endomyocardial biopsy demonstrates loss of myofibrils, distension of sarcoplasmic reticulum, and vacuolization of the cytoplasm, but invasiveness, cost, sampling bias, and interpreter expertise limit its use. Since cardiotoxicity is irreversible and potentially fatal, management goals focus on prevention. Strategies using doxorubicin analogues, modifications in dose/delivery methods, cardioprotective agents have had limited success. Thus, the new focus has been on noninvasive monitoring to detect early (and maybe even late) toxicity. Our patient received doses less than 500 mg/m², but had underlying hypertension and received concomitant radiotherapy which may have put her at higher risk.

AN UNUSUAL CASE OF REVERSIBLE SYNCOPE: ATRIAL STANDSTILL DUE TO HYPERKALEMIA. V. Subbiah¹; S. Subbiah¹; R. Raina¹; T. Mohyuddin¹; D.C. Kaelber¹; D. Mansour¹. ¹Case Western Reserve University, Cleveland, OH. (Tracking ID #136103)

LEARNING OBJECTIVES: 1. To review an atypical cause of reversible syncope 2. To discuss atypical presentation of hyperkalemia

CASE: We report a case of a 53 year old type 1 diabetic female who presented to the ED following an episode of syncope. She was at the hospital pharmacy when she reported sudden onset of dizziness, dyspnea with diaphoresis followed by loss of consciousness lasting less than 2 mins. She coded and had a pulse of 26/min and systolic blood pressure was 60 mmHg and a diastolic blood pressure was not palpable. She was transported to the ED and EKG showed junctional escape rhythm (narrow complex rhythm with a HR of 30 and no atrial activity). She did not respond to atropine. She subsequently responded to a combination of IV fluids and dopamine. BMP revealed hyperkalemia (6.7 mmol/L). She was treated with calcium gluconate, insulin, and dextrose and was admitted to the medical intensive care unit. She was treated with Kayexalate. A repeat EKG after a couple of hours showed normal sinus rhythm and her baseline LVH strain pattern. Repeat potassium was 4.3. She was not on any ACE inhibitor or other drugs or diet rich in potassium that would contribute to her hyperkalemia. Her transtubular potassium gradient was 4.6 which, in face of hyperkalemia, was thought to indicate hypoaldosteronism. Since the patient is a known diabetic for 20 yrs and also has renal insufficiency it was concluded that she had Type 4 RTA. During her hospital stay she remained stable with her baseline EKG. She was started on fludrocortisone for her hyporeninemic hypoaldosteronism and was discharged home with outpatient cardiology follow up. She was readmitted to another hospital with similar presentation attributable to her non-compliance. The patient did not have any further episodes after she started taking fludrocortisone regularly.

DISCUSSION: Atrial standstill is a rare arrhythmogenic condition characterized by the absence of electrical and mechanical activity in the atria. Atrial standstill can be transient or persistent, and complete or partial. The diagnosis relies on the EKG demonstration of bradycardia, absence of P waves, and junctional narrow complex escape rhythm. Hyperkalemia is a common problem associated with acid-base and electrolyte disturbances. The most common clinical manifestations are EKG abnormalities and neuromuscular symptoms, such as weakness. The latter are usually overshadowed by the more urgent cardiac instability. Atrial standstill due to hyperkalemia has been reported rarely. Hyperkalemia depresses cardiac contractility, automaticity, conductivity and excitability, leading to EKG changes which must be recognized. Cardiac dysrhythmias due to hyperkalemia, and the resulting EKG changes, arise due to partial depolarization of resting membrane potentials due to a higher extra cellular-to-intracellular potassium gradient. Our patient, presented with bradycardia, junctional in origin that would have rapidly progressed to asystole if treatment was delayed. Her EKG showed a narrow QRS complex, without T-wave peaking. The potassium level does not always correlate with the EKG, as life-threatening hyperkalemia can exist without usual EKG findings. Hyperkalemia must be recognized before clinical cardiovascular deterioration progresses to asystole, if successful treatment is to be given to reverse this potentially fatal condition.

AN UPPER GI BLEED 30 YEARS IN THE MAKING. J.S. Go¹; J. Derby-Porter¹; J. O'Brien¹. ¹Creighton University, Omaha, NE. (Tracking ID #134121)

LEARNING OBJECTIVES: 1. Recognize jejuno-ileal bypass (JIB) as a risk factor for the development of liver cirrhosis; 2. Recognize and appropriately monitor

patients at risk for this complication; 3. Manage patients with liver cirrhosis caused by JIB.

CASE: A 59 y/o previously healthy Caucasian female is transferred from a nearby community hospital following massive hematemesis. The patient denied abdominal pain, nausea, or previous episodes of vomiting. Initial evaluation with an upper GI endoscopy revealed the presence of bleeding esophageal varices. Her physical exam did not demonstrate any stigmata of chronic liver disease. She was non-alcoholic and was not on any medications. She did not have any history of viral hepatitis, autoimmune disease, or any hereditary condition that predisposed to liver cirrhosis. Labs were negative for viral hepatitis, autoimmune markers, or enzyme deficiencies. CT Scan of the abdomen and ultrasound revealed a liver with irregular contour, peri-hepatic and peri-splenic ascites, splenomegaly, and an enlarged gallbladder. No masses were noted and there was no sign of Budd-Chiari. Further review of her history was remarkable for a jejuno-ileal bypass done in 1975. She subsequently underwent a cholecystectomy and a liver biopsy. Biopsies revealed micronodular cirrhosis. She was subsequently discharged and is being carefully followed. A plan for liver transplant along with reversal of her JIB is being considered.

DISCUSSION: This case illustrates that a patient with a jejuno-ileal bypass (JIB) is at risk for the development of liver cirrhosis and its complications. These procedures were common in the 1960's-1970's for weight reduction but were largely abandoned in the early 80's following recognition that it predisposed to a long-term risk of liver failure or cirrhosis. Literature reports the risk for liver disease following JIB to be anywhere from 6 to as much as >30% following a 15-year period. The pathogenesis of liver disease following JIB is complicated and not fully understood. The proposed mechanisms include malabsorption leading to NASH, production of reactive oxygen species, and bacterial overgrowth with production of endogenous hepatotoxins. To date a number of articles have been published showing evidence that bacterial overgrowth in the bypassed loop cause the formation of endogenous hepatotoxins and reactive oxygen species which may cause inflammation and liver cell injury. This has led to some treatment strategies that target bacterial overgrowth with metronidazole which may help prevent development of liver cirrhosis. Other strategies involve reversal of the JIB with or without liver transplantation. More evidence is needed to determine the efficacy of these therapies. Regardless, all patients with a history of a JIB need to be followed closely for signs of liver disease, and when possible, a reversal of the JIB may be necessary to prevent progressive liver injury. Given the fact that most of these procedures were done 20-30 years ago, a review of this case is timely.

APPROACH TO HYPOGLYCEMIA: A LESSON FROM A PATIENT WITH INSULINOMA. M. Laguna¹; P. Knudson¹; K. Pfeifer¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #134574)

LEARNING OBJECTIVES: Recognize the Whipple's triad to assess the diagnostic approach and treatment of suspected insulinomas.

CASE: We present the case of an 82 year-old retired female nurse who 10 years prior to the current admission developed infrequent and short episodes of lightheadedness, dizziness, tremors, sweats, generalized weakness and perioral paresthesias relieved by food intake. Within that period she was hospitalized twice with no clear diagnosis. On presentation to our institution the patient had a documented blood sugar of 26 mg/dL associated with her symptoms and both were relieved by glucose intake. A 72-hour fasting test showed a glucose level of 42 mg/dL, insulin of 32 microU/mL and C-peptide of 1300 pmol/L with neuroglycopenic symptoms (lightheadedness and disorientation.) Urinary sulfonylureas were negative; cortisol was 16.8 mg/dL (normal), TSH 1.85 U/mL (normal) and HbA1c 5.1% (normal). Abdominal CT, non-stimulated angiogram, EGD, ERCP, MRI and octreoscan were unable to localize the abnormal source of insulin. An intraoperative ultrasound demonstrated a 1.5 cm mass in the tail of the pancreas that was excised. The pathology report was consistent with insulinoma.

DISCUSSION: Hypoglycemia can present with neuroglycopenic symptoms or symptoms secondary to sympathetic discharge. These symptoms not only are typical for the presentation of insulinoma, but also for many other causes of hypoglycemia. It can take years before the correct diagnosis of insulinoma is made unless a systematic approach is followed. The Whipple's triad consists of neuroglycopenic/vasomotor symptoms, documented low blood sugars (<45 mg/dL), and symptom relief with glucose administration. If present, a 72-hour fasting test is indicated to document hyperinsulinemia as the cause of hypoglycemia. If positive, one must choose an imaging study to localize the abnormal source of insulin. It is prudent to begin with a less invasive and more sensitive study, such as CT, MRI or endoscopic ultrasound (sensitivity range of 82-92%). If negative, an intraoperative pancreatic ultrasound (sensitivity 83-100%) should be done followed by excision of the mass. If intraoperative ultrasound is negative, termination of the surgery and later performance of a post-stimulated calcium angiogram should be done, followed by surgical resection of the zone of the pancreas with the highest levels of insulin.

ARE FAD DIETS FOR EVERYONE? SYMPTOMATIC HYPONATREMIA WITH A LOW-CARBOHYDRATE DIET. E.N. Johnson¹; A. Truesdell¹; T. Villines¹. ¹Walter Reed Army Medical Center, Washington, DC. (Tracking ID #135561)

LEARNING OBJECTIVES: Employ an algorithm to diagnose the cause(s) of hyponatremia. Recognize potential risks associated with popular diets and weight loss plans.

CASE: A 77-year-old woman with a history of hypertension treated with hydrochlorothiazide presented to the emergency room following progressive fatigue

and nausea culminating in a loss of postural tone. She recently initiated a high-protein, low-carbohydrate diet, losing 3.6 kg of body weight over 2 weeks. On presentation she was mildly orthostatic; physical exam was otherwise normal. Labs revealed a serum sodium of 119 mmol/L and a serum osmolality of 245 mmol/L. Urine osmolality was 212 mmol/kg, urine sodium was 39 mEq/L, and urinalysis was positive for ketones (50 mg/dL). On admission, the patient's thiazide diuretic was discontinued and she was resuscitated with normal saline. Over 24 hours her serum sodium improved to 126 mmol/L. At one-week follow-up her serum sodium was 135 mmol/L—and remained stable during subsequent follow-up, even after reinitiation of a thiazide diuretic for hypertension. The patient did not restart her ketogenic diet.

DISCUSSION: High-protein, low-carbohydrate diets have gained popularity as a fast and effective means of weight loss. Until recently there was limited randomized data demonstrating the efficacy of such diets compared to traditional low-fat, low-calorie diets. More importantly, comprehensive safety data for ketogenic diets is lacking—particularly in vulnerable populations. This case report of symptomatic hyponatremia in an elderly woman on a high-protein, low-carbohydrate diet highlights a potential risk of the ketone-induced diuresis associated with these diets. High-protein, low-carbohydrate diets cause glycogen catabolism and free water loss. Ketogenesis also precipitates urinary excretion of cations along with ketones. We propose that hyponatremia in our patient resulted from the combined effects of a thiazide diuretic and diet-induced ketosis—with ensuing loss of sodium in excess of free water. While high-protein, low-carbohydrate diets remain an attractive weight loss option for many patients, unanswered questions remain. We recommend that practitioners caution patients—particularly elderly patients on multiple medications—about the potential risks before initiating a ketogenic diet.

ASCITES IN AN HIV MAN. M. Lau¹; L. Lu¹; H. Majid¹. ¹Baylor College of Medicine, Houston, TX. (Tracking ID #133569)

LEARNING OBJECTIVES: 1. Recognize the clinical presentation and symptoms of tuberculous ascites. 2. Review the diagnosis of tuberculosis peritonitis. 3. Learn the appropriate treatment of tuberculosis peritonitis.

CASE: A 67 year-old African-American gentleman with HIV diagnosed in 1999 (CD4 of 190) on anti-viral therapy was admitted with a two month history of 20 lbs weight loss with concomitant increase in abdominal girth 2 weeks prior to admission. Patient denied fever, chills, night sweats, shortness of breath, orthopnea and paroxysmal nocturnal dyspnea. Three years ago, the patient had a positive purified protein derivative (PPD) test (14 mm) and was treated with prophylactic isoniazid. He denied a history of alcohol intake and contracted HIV from blood transfusion. Physical exam was significant only for massive ascites and mild diffuse abdominal tenderness. Laboratory results revealed hemoglobin 9.6 with MCV 107, platelets 78,000, WBC 4,000/liter; total protein 3.8, albumin 1.8, AST 11, ALT 23; and PT/PTT 15/58 with INR 1.2. Hepatitis serologies were negative. Ascitic fluid had a cell count of 300 WBC with 83% macrophages and 17% lymphocytes, no segmented cells, 900 RBCs, and a low serum-ascites albumin gradient (SAAG) of 0.5 g/dl and a total protein >4 mg/dl suggestive of tuberculosis peritonitis, peritoneal carcinomatosis, serositis, etc. Ascitic fluid gram-stain, cultures, and cytology were negative on initial paracentesis. Repeat PPD was anergic, and chest x-ray was unremarkable. Computerized tomography (CT) of thorax/abdomen showed minimal bibasilar fibrosis of lungs and a normal size liver with ascites. On repeat large volume ascitic fluid analysis, acid-fast bacilli (AFB) were identified and adenosine deaminase level was elevated at 36 U/L (0-7.6 U/L). Mycobacterium cultures for ascitic fluid have been negative to date (4 weeks). With the positive AFB and elevated adenosine deaminase, patient was diagnosed with TB peritonitis and initiated his anti-tuberculosis treatment.

DISCUSSION: TB peritonitis is a rare cause for ascites (2% of ascites cases in USA). Common symptoms include abdominal pain, fever, and weight loss. Risk factors for the disease include cirrhosis, HIV, diabetes mellitus, malignancy, and peritoneal dialysis. Only 20-30% of chest x-rays show evidence of old TB and 30% have a negative PPD. The diagnosis of TB peritonitis often requires a peritoneal biopsy as yield of ascitic fluid culture is very low (<20% sensitivity). Analysis of large volume ascitic fluid enhances the yield (as high as 80% with >1 L fluid) as shown in this patient. Ascitic fluid typically reveals a SAAG <1.1 g/dL and protein >3.0 mg/dL which was present in this patient. One available test is adenosine deaminase (ADA), an enzyme that is necessary for the maturation of lymphoid cells and is high in TB ascites. It has a high sensitivity and specificity (100% and 97% respectively if >33U/L) in non-cirrhotic patients. In cirrhotic patients, the sensitivity is lower (<30%) due to poor immune response. Treatment includes a multi-drug regimen of isoniazid, rifampin, pyrazinamide, and ethambutol for 9-12 months; steroids are also recommended by some authors for 2-3 months to prevent formation of adhesions in the peritoneal cavity. With the growing prevalence of tuberculosis, one should suspect TB peritonitis when the preliminary work-up of new-onset ascites reveals a low SAAG, especially in a high-risk setting.

AUTOIMMUNE HEMOLYTIC ANEMIA AND IMMUNE THROMBOCYTOPENIC PURPURA: A CASE OF EVAN'S SYNDROME. R.H. Orakzai¹; S.H. Orakzai¹; M.T. Rahim¹; S. Kurup¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133224)

LEARNING OBJECTIVES: 1. Recognize the presentation of Evan's syndrome. 2. Recognize the management of Evan's syndrome. 3. Appreciate the association of Evan's syndrome with other medical conditions.

CASE: A 77 year old female with a history of Coomb's positive auto-immune hemolytic anemia (treated with steroids and splenectomy), Hodgkin's disease

status post chemotherapy, hypertension, coronary artery disease and deep venous thrombosis presented with a 2 day history of epistaxis and oral ulcers. Patient received an influenza vaccine 3 weeks prior after which she developed diffuse "body aches" and sinus congestion. On exam, she was normotensive and afebrile. She had several ulcers on her buccal mucosa and petechiae on her legs. Initial workup revealed a hemoglobin of 12.9mg/dl, white blood cell count of 10,900 and platelet count of 6,000. Peripheral smear showed decreased platelets, nucleated red blood cells and no schistocytes. Patient was transfused 6 packs of platelets and started on prednisone 2mg/kg/day. Her platelet count 1 hour post transfusion was 77,000. Her platelet count the next day was 66,000. Prednisone was decreased to 40mg per day. She remained stable during her hospital stay and her petechiae decreased. Lupus anticoagulant profile was negative. At the time of discharge patient's platelet count was 80,000.

DISCUSSION: The combination of direct Coomb's positive hemolytic anemia and immune thrombocytopenic purpura in the absence of underlying etiology defines Evan's syndrome. The anemia and thrombocytopenia in Evan's syndrome are variable with respect to time of onset, course and duration. Thrombocytopenia and anemia are the presenting features in 76% and 67% patients respectively. In addition neutropenia has been noted in 14%. Evan's syndrome is seen in adults and children. Adult cases are associated with systemic lupus erythematosus, aplastic anemia and multisystem disease manifestations but most pediatric cases are idiopathic. Etiology remains speculative at this time. Non cross reacting autoantibodies are directed against antigens specific to red blood cells and platelets. Most patients with Evan's syndrome also have decreased levels of IgG, IgM and IgA along with decreased T helper and increased T suppressor lymphocyte populations supporting the concept of aberrant immunoregulation. Management of patients with Evan's syndrome is a challenge. The clinical course is characterized by periods of remission and exacerbation with variable and often disappointing responses to therapy. Most patients are treated with corticosteroids, intravenous immunoglobulin (IVIG), splenectomy or immunosuppressive agents. Corticosteroids are commonly used as initial therapy. Splenectomy is utilized in combination with corticosteroid therapy. The role of splenectomy in patients with Evan's syndrome has never been clearly established; remissions last only 1-2 months without corticosteroid support. Anecdotal reports suggest that high dose IVIG may be useful in management of patients who have not responded to corticosteroids or splenectomy. Immunosuppressive therapy with agents such as cyclosporine, cyclophosphamide, vincristine and danazol has also been utilized in the treatment of refractory cases of Evan's syndrome. Given the rarity of this syndrome, it would be of value to collect data prospectively through a central registry which would allow better understanding of the pathophysiology of the disease and assessment of the efficacy of different therapeutic approaches.

BODY BUILDING TO THE POINT OF NAUSEA. J. Schieffelin¹; J. Wiese¹. ¹Tulane University, New Orleans, LA. (Tracking ID #136026)

LEARNING OBJECTIVES: 1. Review the differential diagnosis of hyperkalemia 2. Review prevalence of diuretic abuse

CASE: A 37 year-old female body-builder presented with three days of vomiting, diarrhea and weakness. Minutes before she was scheduled to compete at a local competition, she collapsed but did not lose consciousness. She reported that her vomitus was non-bloody, non-bilious and there was no blood or mucous in the stool. She denied any allergies, medication or past medical history. Her physical exam was normal with the exception of signs of dehydration. A CBC was normal. Her chemistry was significant for a sodium of 143 mEq/l, potassium: 7.1 mEq/l, chloride: 105 mEq/l, total CO₂: 24, urea 30 mEq/l and a creatinine of 1.4. Her creatinine kinase was normal. An EKG showed peaked T waves. Upon further questioning the patient admitted that she used aldactone in the days prior to her competitions to lose "water weight."

DISCUSSION: Hyperkalemia can be caused by decreased excretion or increased intake of potassium. Decreased excretion can be due to drugs, renal failure and hypoadosteronism. Increased production can be caused by a trauma or tumor lysis. Spironolactone is commonly abused in competitive body-builders in an effort to lose body fluid to augment the appearance of muscle tone. By antagonizing aldosterone at the distal convoluted tubule, it inhibits the absorption of sodium and the excretion of potassium. When used alone and in high doses, as in our patient, it can cause acute hyperkalemia. Physicians should be aware of the serendipitous drug use in competitive athletes in order to accurately diagnosis the cause of electrolyte abnormalities and to preclude invasive testing.

BRONCHORRHEA IN A PATIENT WITH NONRESOLVING PNEUMONIA: CLUE TO THE DIAGNOSIS OF BRONCHOALVEOLAR CARCINOMA. J. Diamond¹; J. Clemmer¹. ¹Hospital of the University of Pennsylvania, Philadelphia, PA. (Tracking ID #135748)

LEARNING OBJECTIVES: 1. Recognize bronchorrhea as an unusual symptom of bronchoalveolar carcinoma. 2. Identify the indications for further evaluation of nonresolving pneumonia.

CASE: A 53-year-old previously healthy male stone mason presented to his primary care physician with two weeks of cough and progressive shortness of breath. He was prescribed levofloxacin for presumed pneumonia. Despite therapy, his symptoms worsened over the next three weeks to include dyspnea while climbing stairs and marked bronchorrhea of about two cups per day. Therefore, outpatient bronchoscopy was performed which revealed negative stains or cultures for bacteria, fungi, and acid-fast bacilli and normal cytology. He was admitted to the hospital for continued worsening symptoms. Physical exam showed a well-nourished man utilizing accessory muscles of respiration and

requiring high-flow oxygen to maintain oxygen saturations greater than 88%. Lung exam revealed bibasilar rales. Admission chest x-ray was consistent with multilobar pneumonia, and antibiotics were changed to cefepime and metronidazole. Tests for human immunodeficiency virus, legionella, respiratory viral pathogens, erythrocyte sedimentation rate, IgE, antinuclear antibodies, and rheumatoid factor were negative or normal. Serial chest computed tomography scans showed progression of multifocal infiltrates over one week. The patient's respiratory status continued to deteriorate in spite of antibiotics, so he underwent video assisted thoracic surgery with lung biopsy. Pathology revealed mucinous subtype bronchoalveolar carcinoma. He was offered treatment with chemotherapy, but elected homeopathic eastern medicine and hospice care instead.

DISCUSSION: Bronchoalveolar carcinoma (BAC), a form of lung adenocarcinoma, has three subtypes: mucinous, nonmucinous, and sclerotic. Bronchorrhea occurs in 6% of patients with the mucinous type. Hypoxemia can be profound because the alveoli are filled with mucus and malignant cells, resulting in shunt physiology. Compared to other lung cancers, patients with BAC are more likely to be young, female, or non-smokers. Infections or occupational exposures resulting in scar are hypothesized to effect epithelial proliferation, promoting tumorigenesis. A single peripheral mass is the most common finding in BAC overall, but diffuse infiltrates often occur in the mucinous subtype. Attempts at resection are possible for localized BAC, but in most cases the cancer is diffuse. Chemo and radiotherapeutic efficacy in such cases is poor. The differential diagnosis of nonresolving pneumonia includes resistant bacterial or nonbacterial infections, inflammatory diseases, drug-induced lung toxicity, pneumoconioses, and neoplasms. Further evaluation needed after four to eight weeks includes imaging with computed tomography and microbiologic and cytologic sampling via bronchoscopy. If inconclusive, thorascopic or open lung biopsy is required, especially for worsening symptoms or radiographic findings. As illustrated by this case, the diagnosis of BAC should be entertained in patients with pulmonary infiltrates who do not respond to antibiotic therapy and have associated symptoms such as bronchorrhea and refractory hypoxia.

CAN'T WORM OUT OF THIS ONE. R.M. Treger¹; R.A. Cader². ¹Olive View - UCLA Medical Center, Sylmar, CA; ²Sepulveda VA Ambulatory Care Center, North Hills, CA. (Tracking ID #135913)

LEARNING OBJECTIVES: 1. Recognize the varied clinical presentations of Strongyloides infection. 2. Recognize bacterial complications associated with Strongyloides infection.

CASE: A 77 year-old Hispanic male veteran with prednisone-dependent COPD and allergic bronchopulmonary aspergillosis presented with a one day history of headache, fever/chills, nausea, and lethargy. The patient denied any history of ill contacts or recent travel. Of note, he was hospitalized twice in the past six months for apparent bacterial meningitis after presenting in a similar manner. Exam revealed an afebrile, lethargic male, with evidence of meningismus but a nonfocal neurologic exam. Labs were significant for a WBC count of 26 (1.8% eosinophils) and CSF with 13,000 WBC (95% neutrophils), glucose 48 mg/dl and protein 308 mg/dl. Blood cultures grew Streptococcus Bovis. Given the recurrent meningitis and transient eosinophilia, the work-up was expanded. Transesophageal echocardiogram demonstrated no evidence of valvular vegetations. Stool O&P showed Strongyloides stercoralis with a positive serum IgG antibody. Strongyloides hyperinfection syndrome was diagnosed and ivermectin followed by thiabendazole was given with clinical improvement.

DISCUSSION: Strongyloides stercoralis is an intestinal nematode endemic to tropical and subtropical regions. In the US, highest infection rates are found in Appalachia and the southeastern states, in immigrants from endemic areas and in those with occupational exposures, such as military veterans. Strongyloides is unique in that it can complete its life cycle within the human host. Although this process of autoinfection is kept in check by an intact immune system, it still occurs at a low rate, permitting the organism to persist for decades after the initial infection. Our patient's exposure probably occurred in Guam during World War II and manifested later due to steroid use. Common clinical manifestations include pulmonary, gastrointestinal, or cutaneous symptoms that are typically chronic and may linger for years, although eosinophilia may be the only clue in asymptomatic patients. A hyperinfection syndrome consisting of massive dissemination of larvae to multiple organs may result in secondary Gram-negative bacteremia and meningitis due to carriage of bacterial pathogens from the colon by circulating larvae. It is generally associated with conditions that cause depressed immunity, such as HIV or corticosteroids. In our case, the S. bovis isolated from blood was likely the causative organism of the meningitis. S. bovis meningitis complicating Strongyloidiasis has been reported in the literature. Strongyloides is treated with ivermectin or thiabendazole. Immunocompromised patients with disseminated disease may require prolonged or repeat therapy.

CARDIAC AMYLOIDOSIS. P. Garg¹; A. Nautiyal¹; M. Disalle¹; D. Hsi¹. ¹Unity Health System, Rochester, NY. (Tracking ID #135405)

LEARNING OBJECTIVES: 1)Recognize Cardiac Amyloidosis and its clinical presentation. 2)Suspect cardiac amyloidosis in unexplained diastolic heart failure.

CASE: A 70-year-old white female presented with new syncope and symptoms of biventricular congestive heart failure for the past three months. She had a prior history of hypertension and transient ischemic attacks. Medications included clopidogrel and furosemide with a notable penicillin allergy. She was married and had a 40-pack year smoking history. She denied alcohol consumption. On examination her vital signs were stable, without orthostasis. Pertinent

clinical findings were increased jugular venous distension, bibasilar lung crackles, nontender hepatomegaly with 20 cm liver span and bilateral pitting pedal edema. Laboratory data revealed normal hematocrit, electrolytes and renal functions. Globulins were increased and the A/G ratio was reversed at 0.6. Though urine dipstick was negative for protein she had nephritic range proteinuria of 2gm/24 hrs. Urine electrophoresis and immunofixation were positive for IgG Kappa light chains. Troponins were mildly increased. Electrocardiogram showed sinus rhythm and low voltage complexes. Echocardiogram revealed increased ventricular septal thickness, granular "sparkling" of the myocardium with a preserved left ventricular systolic function and ejection fraction of 55-60%. Cardiac catheterization and coronary angiography revealed a normal coronary anatomy. An endomyocardial biopsy was positive for cardiac amyloidosis. She developed complete heart block requiring a permanent pacemaker and died 2 weeks later due to end-stage congestive heart failure.

DISCUSSION: Our patient had primary amyloidosis with hepatic, renal and cardiac involvement. The symptoms of amyloidosis depend upon the precursor protein, tissue distribution and amount of amyloid deposited. Primary amyloidosis is a type of plasma cell dyscrasia associated with deposition of amyloid (AL) light chains. Diagnosis is made by demonstration of amyloid protein on Congo red staining where it shows green birefringence under a polarizing microscope. Tissue biopsy can be obtained from dysfunctional organs such as kidney, nerve or clinically uninvolved organs such as subcutaneous fat, minor salivary glands or rectal mucosa. Cardiac involvement occurs in up to 50% of the patients with Primary Amyloidosis and is frequently fatal. Amyloid can infiltrate the ventricular myocardium, conducting system, intramyocardial or epicardial coronary arteries, heart valves and pericardium. The mechanism of syncope is often multifactorial, a combination of tachy-brady arrhythmia with associated postural hypotension. Clinical clues to the presence of amyloid heart disease are heart failure with normal left ventricular systolic function and ECG findings of low voltage complexes, AV block, atrial fibrillation or a pseudoinfarct pattern. Diastolic dysfunction is the earliest and commonest abnormality. Endomyocardial biopsy is indicated in patients with suspected cardiac amyloidosis and unexplained diastolic heart failure and clinical manifestation of restrictive cardiomyopathy, especially when extra cardiac tissue is negative for amyloid. Mean survival with cardiac involvement is 4-6 months.

CARDIAC TAMPONADE SECONDARY TO METHICILLIN RESISTANT STAPHYLOCOCCUS AUREUS AND MYCOBACTERIUM AVIUM COMPLEX POSITIVE PERICARDIAL EFFUSION. L. Samal¹; L. Martin¹; V. Bead¹; R. Boonyasai¹. ¹Johns Hopkins Bayview Medical Center, Baltimore, MD. (Tracking ID #133163)

LEARNING OBJECTIVES: 1. Assess pericardial effusions in HIV-positive patients. 2. Recognize the prognostic significance of pericardial effusion in HIV/AIDS patients.

CASE: A 45 year old man with advanced HIV infection (CD4 =3) presented with severe chest pain and low-grade fever for three days. There was no pericardial rub, but history and electrocardiogram (ECG) were suggestive of pericarditis. A transthoracic echocardiogram showed a small to moderate pericardial effusion. He was presumed to have a viral pericarditis associated with HIV and was treated with Indomethacin. Three weeks later, he presented with a one-week history of progressive dyspnea, leg edema, and chest pain. On exam, he was in moderate respiratory distress. BP 96/70 mmHg, HR 122/min, RR 26/min, T 98.2oF. He had a pulsus paradoxus of 24 mm Hg. The JVP was distended to 11 cm above the sternal angle, heart sounds were distant, lung fields were clear to auscultation, and there was 2+ bilateral leg edema. An echocardiogram showed a large pericardial effusion, with diastolic impingement of right ventricular and right atrial filling. A pericardiocentesis was performed with removal of 1300 cc of purulent fluid. Due to persistent drainage, a procedure was performed to create a pericardial window into the pleural space. A blood culture from the previous admission became positive for Mycobacterium Avium Intracellulare (MAI). Cultures of the pericardial fluid revealed MRSA and MAI. He was treated with Vancomycin, Ethambutol, Rifampin and Azithromycin and slowly improved.

DISCUSSION: Pericardial effusions occur in about 20% of patients with HIV/AIDS. Most effusions are small and asymptomatic, and often resolve spontaneously. No further diagnostic testing is indicated in these cases, especially given that pericardiocentesis has a higher complication rate and lower diagnostic accuracy for small effusions. Thus, further evaluation and management is limited to symptomatic effusions, or to larger effusions in patients with signs of systemic illness. Reported case series suggest that culture and microscopic examination of the pericardial fluid and tissue diagnose infection or neoplasm as the cause of two-thirds of symptomatic effusions. Treatment can then be directed at the underlying etiology, most commonly tuberculosis and other mycobacteria, Staphylococcus aureus, Streptococcus Pneumoniae, lymphoma and Kaposi's sarcoma. Cardiac tamponade is the most serious complication of pericardial effusion, and clinicians must suspect this complication even with smaller effusions, as "low-pressure" tamponade may occur in HIV/AIDS patients with severe dehydration and wasting. History, physical examination, and ECG are insensitive to diagnose tamponade; echocardiography is required. Findings such as right atrial and right ventricular diastolic collapse and respiratory variation of mitral inflow and aortic outflow are characteristic. Pericardial effusions, even if asymptomatic, portend a poor prognosis in patients with HIV/AIDS. Although rarely a direct cause of mortality, as cardiac tamponade and constrictive pericarditis account for only a small percentage of deaths in this population, pericardial effusions are considered markers of advanced HIV infection.

CARNITINE DEFICIENCY PRESENTING AS RECURRENT HYPOGLYCEMIA IN A NON DIABETIC HEMODIALYSIS PATIENT. F.A. Noor¹; M. Nashat¹; M. Altaf¹. ¹Mercy Catholic Medical Center, Aldan, PA. (Tracking ID #135192)

LEARNING OBJECTIVES: 1. Recognize carnitine deficiency as a cause of recurrent hypoglycemia in hemodialysis patients. 2. Learn the symptoms and treatment of carnitine deficiency.

CASE: 66yr-old Male was brought by Paramedics to E.R for an episode of acute confusion. There was no h/o head trauma, fever, vomiting or drug abuse. He had a history of HTN and ESRD on hemodialysis for 7 years. Of note he had no h/o diabetes. He was taking Cardizem, Lisinopril and Renagel. He denied alcohol or IV drug abuse. Vitals were normal except for an Accu-check of 45 mg/dl. Patient was oriented to name alone, rest of the physical exam was unremarkable. His mentation improved with an amp. of D-50. Serum alcohol and UDS was negative. During his hospital stay, he continued to have early morning hypoglycemia. Serum TSH 1.44 mIU/ml, A.M Cortisol level 29 µg/dl (8.7-22), LFTs and coagulation profile were normal. His serum Insulin 2 µU/ml (<17 µU/ml), Prolinsulin 3.7 pmol/L (<19 pmol/L), C- Peptide 3.0 ng/ml (0.8-3.2 ng/ml), insulin growth factor II 396 ng/ml (28-444 ng/ml), serum sulfonurea screen all drawn during hypoglycemic episode were normal except for low serum keto acids. Substrate deficiency was suspected and was later found to have low serum total carnitine 14 µmol/l (42-81 µmol/l) and free carnitine of 12 µmol/l (35-67 µmol/l). He received intravenous carnitine at dialysis, after which he had no hypoglycemic episodes and had improvement in his over all well being.

DISCUSSION: Carnitine is an important intermediary in fat metabolism, required to shuttle long-chain fatty acids such as acylcarnitine into mitochondria for β-oxidation and is crucial for energy production in tissues dependent upon fatty acid oxidation such as cardiac and skeletal muscle. Though primary carnitine deficiency manifests in teen age years and is caused by functional anomalies in a specific organic cation/carnitine transporter, secondary deficiency is usually seen in patients with liver cirrhosis, renal tubular disorders which cause increased excretion of carnitine and in hemodialysis patients where carnitine is lost through dialysis membranes leading to carnitine depletion and is associated with clinical problems such as skeletal myopathies, intradialytic symptoms, reduced cardiac function, hypoglycemia and erythropoietin resistant anemia. Treatment is intravenous administration of high doses of carnitine at the end of each hemodialysis session to ensure adequate replacement of depleted muscle stores. **Conclusion:** This case emphasizes the fact that carnitine deficiency though uncommon in healthy adults, should be considered as a cause of recurrent hypoglycemia in dialysis patients.

CASE OF ULCERATIVE COLITIS PRESENTING WITH THROMBOSIS. S.R. Ganesh¹; D. McAdams². ¹University of Pittsburgh Medical Center, Pittsburgh, PA; ²University of Pittsburgh, Pittsburgh, PA. (Tracking ID #134515)

LEARNING OBJECTIVES: 1) To recognize the presentation of Ulcerative Colitis (UC), including extra-intestinal manifestations. 2) To recognize causes of venous thrombosis (VT). 3) To recognize VT as an initial presentation in UC.

CASE: An 18-year-old Caucasian male was admitted with a three-day history of bloody diarrhea, fever, and fatigue. He had 10-12 bloody bowel movements daily, with associated lower abdominal cramps and tenesmus. He denied any sick contacts, recent travel, hematuria, sinusitis, arthralgias, urethritis, or rash. On physical examination he was febrile to 38.6 C, had pale conjunctiva, LLQ tenderness, and frank blood on DRE. Labs included WBC =24.6, Hgb =7.4, ALT =1030, AST =746, Alkaline phosphatase =218, GGTP =1910, and ESR =47. CT of the abdomen and pelvis revealed multiple, non-occlusive thromboses of the superior mesenteric vein, inferior mesenteric vein, portal vein, and splenic vein, as well as a diffusely edematous small bowel, and colonic wall thickening. At this point, he underwent a hypercoagulable work up which included Factor V Leiden (FVL) mutation, Protein C, Protein S, Antithrombin (AT) III, methylenetetrahydrofolate reductase (MTHFR) gene mutation, and lupus anticoagulant (LAC) all of which were normal. The possibility of underlying Ulcerative Colitis (UC) as a cause of venous thrombosis (VT) was considered. Colonoscopy revealed an erythematous, granular, friable colonic mucosa, and biopsies were consistent with UC. The diagnosis of UC presenting with venous thrombosis was made, and he was started on anticoagulation and steroids. His signs and symptoms resolved over 4-5 days.

DISCUSSION: UC is characterized by uniform, continuous, mucosal inflammation with shallow ulcers extending proximally from the rectum. A bimodal age distribution (15-40 and 50-60) is present. Clinical course and degree of involvement are widely variable and range from mild colitis, with minimal symptoms, to severe colitis with abdominal pain, BRBPR, and extra-intestinal manifestations (EIM). EIM etiology is not entirely understood, but felt to be an autoimmune process. EIM of UC can include musculoskeletal findings such as ankylosing spondylitis, sacroiliitis, and polyarthritis; ocular findings such as iritis, episcleritis, and uveitis; skin lesions including erythema nodosum and pyoderma gangrenosum; primary sclerosing cholangitis; and VT. VT pathophysiology is complex and its differential diagnosis is broad. The various causes include hypercoagulable states such as FVL, protein C or S deficiencies, AT III deficiency, prothrombin variant, LAC, and MTHFR gene mutation; major acquired states such as pregnancy, smoking, surgery, prolonged immobilization, and IV catheters; and other disease states such as malignancies, myeloproliferative disorders, paroxysmal nocturnal hemoglobinuria, retroperitoneal fibrosis, vasculitides, and UC. Patients with UC have a three-fold increased risk of developing VT, but risk is decreased when disease is controlled. VT in UC is usually associated with younger patients, active disease and flares and steroid use, and can occur in any venous system. Pathophysiology of VT in UC are abnormal platelet aggregation, abnormal fibrinolysis, increased cytokines, thrombocytosis, circulating immune complexes, increased factors V and VIII, increased

plasminogen activator inhibitor, and increased activated protein C. The treatment for VT in UC is anticoagulation for at least 6 months.

CAVITARY PULMONARY SARCOIDOSIS. R. Abrams¹; S. Bhawan¹. ¹Rush University Medical Center, Chicago, IL. (Tracking ID #136099)

LEARNING OBJECTIVES: 1. Recognize and distinguish the unusual feature of cavitary lesions in sarcoidosis. 2. Manage cavitary pulmonary sarcoidosis.

CASE: A 33-year-old African American male patient presented with complaints of progressive dyspnea on exertion, a non-productive cough, night sweats and a 30 lb weight loss over the preceding three months. His past medical history was significant for morbid obesity, HTN, smoking, right-sided CHF, and obstructive sleep apnea. Ten months earlier a diagnosis of pulmonary embolus was made, but no confirmatory test could be done because of his weight. He was treated with Coumadin. Six months later, after successful planned weight loss, a CT scan of his chest showed significant hilar and mediastinal adenopathy with multiple lung nodules including at least one that was cavitating. An autoimmune panel, ACE level and pulmonary function tests were normal. He did not return for four months, when he presented to the Pulmonary Clinic with the above complaints. On physical examination, he had a low grade fever, RR 22, BP 110/80. Oxygen saturation on room air was 97%. Lung exam revealed bibasilar crackles. The remainder of his physical exam was unremarkable. A repeat CT scan showed that the dominant cavitary lesion had increased in size, and two new cavitary lesions had appeared. There was continued lymphadenopathy, and numerous reticulonodular densities. The ACE level was now elevated. PPD was negative, and sputum smears were negative for AFB. Bronchoalveolar lavage smears and culture were negative for AFB and fungi. Transbronchial biopsy revealed non-caseating granulomas. Prednisone treatment was initiated. A repeat chest CT scan four months later showed disappearance of all cavitary lesions, and a decrease in size of the mediastinal adenopathy and nodular lesions. The patient noted resolution of all his symptoms.

DISCUSSION: Sarcoidosis is a relatively common disease which usually has an insidious onset, a benign course and a diagnosis that is made by exclusion. Symptomatic patients may complain of cough, dyspnea and chest pain. CXR typically shows bilateral hilar lymphadenopathy and/or pulmonary interstitial infiltrates. Cavitary lesions are extremely rare and are thought to occur in only 1-3% of all cases. Therefore, sarcoidosis patients with cavitary lesions at presentation may be easily misdiagnosed. While the most common differential diagnosis for cavitary pulmonary lesions is neoplastic disease, an infectious process, autoimmune disease, and pulmonary infarction secondary to septic or pulmonary emboli, sarcoidosis should be considered. In order to diagnose cavitary sarcoidosis, cultures for AFB and fungi should be negative (in addition to ruling out other potential causes) and the hallmark finding of non-caseating granulomas should be demonstrated on biopsy. Cavitary lesions may develop any time during the course of the disease. As seen with our patient, cavitary sarcoidosis may respond to steroid therapy, both clinically and radiologically. Keeping the diagnosis in mind can avoid a delay in administering the appropriate treatment.

CENTRAL NERVOUS SYSTEM LYMPHOMA: A RARE CAUSE OF HYPERVENTILATION. T.K. Khoo¹; T. Tan¹. ¹Mayo Clinic, Rochester, MN. (Tracking ID #130958)

LEARNING OBJECTIVES: 1) Learn that central neurogenic hyperventilation due to an intracranial tumour is rare. However, in the absence of a central nervous system (CNS) injury, lymphoma becomes the most likely cause of the hyperventilation. 2) Know that treatment of hyperventilation may include respiratory paralysis and early chemotherapy

CASE: A 29 year old male presented with episodic dyspnea of one month's duration, associated with nausea as well as a mild occipital headache. This was occurring several times a day lasting 15 minutes with spontaneous resolution. Several episodes appeared to be triggered by emotional stress. Physical exam revealed a fairly anxious gentleman with good vital signs but a tachypnea of 24 breaths per minute. Oxygen saturation remained normal. Otherwise he had no abnormal findings. His laboratory studies were remarkable only for a respiratory alkalosis. This was initially suspicious for a panic disorder. However, his symptoms progressed to include diplopia, sleep tachypnea and confusion. A head MRI was done, showing widespread meningeal enhancement and increased T2 signal at the mid and hindbrain. A spinal tap was performed and this showed an atypical lymphocytosis. Immunohistochemical studies provided the diagnosis of malignant large cell type B-cell lymphoma. Treatment with methotrexate and leucovorin was initiated. His confusion and hyperventilation improved and he was subsequently dismissed after a stay of 2 weeks.

DISCUSSION: Central neurogenic hyperventilation (CNH) is a rare syndrome consisting of decreased PaCO₂, elevated PaO₂ and respiratory alkalosis in the absence of pulmonary congestion, sepsis or use of a respiratory stimulant. Usual etiologies of CNH include CNS injuries from trauma, cerebrovascular accidents and infections. CNS tumors may also cause this syndrome. While primary CNS lymphoma is not uncommon, contributing to 5-7% of intracranial neoplasms, most patients with this present with headaches, visual, motor or sensory changes. Some patients may even present with a cognitive or personality change. Leptomeningeal involvement may produce a meningitis-like picture. However, as seen in our patient, they occasionally may present with a centrally-mediated respiratory disorder. To date, only 18 cases of tumor-induced CNH have been reported in the literature. Of these, 9 were cases of primary CNS lymphoma with the pons being the most common site of involvement. The exact mechanism of hyperventilation is unclear, but it is thought that increased levels of hydrogen ions in the cerebrospinal fluid bathing the chemosensitive areas of

the brainstem may play a role. Treatment consists of respiratory paralysis and mechanical ventilation until a more specific therapy can take effect. The options include corticosteroids, whole-brain irradiation and chemotherapy. Because the standard chemotherapeutic agents used for systemic lymphoma appear to be less efficacious in CNS lymphoma, other agents such as high-dose methotrexate and leucovorin are frequently used. Prognosis is poor, with a mean survival of 1.5 months without, and 44 months with treatment.

CEREBROVASCULAR THROMBOSIS IN A 33 YEAR OLD FEMALE WITH SYSTEMIC LUPUS ERYTHEMATOSIS: A CASE OF SNEDDON'S SYNDROME. S.H. Orakzai¹; R.H. Orakzai¹; M.T. Rahim¹; S. Kurup¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133221)

LEARNING OBJECTIVES: 1. Recognize the presentation of antiphospholipid syndrome and its complications. 2. Develop an understanding of the management of antiphospholipid syndrome and its complications. 3. Appreciate the association of antiphospholipid syndrome with other medical conditions.

CASE: A 33 year old female with systemic lupus erythematosus (SLE), hypertension and antiphospholipid syndrome (APS) presented to the ED with sudden onset of nausea, vomiting and headache associated with dizziness, vertigo and sensation of falling towards the left side. Patient was prescribed enoxaparin sodium for multiple thrombotic events in the past but discontinued this medication due to vaginal bleeding ten months ago. On exam she was afebrile with pulse of 90 and BP 159/86. She was somnolent but arousable. She had a lacy vasculitic rash on her upper extremities indicating livedo reticularis. Neurological examination revealed slight dysmetria of left upper extremity on finger to nose test and dysidiadokinesis. Head CT scan showed no evidence of intracerebral hemorrhage. MRI showed a stroke in the distribution of left superior cerebellar artery. Angiogram documented occlusion of distal left superior cerebellar artery but no signs of vasculitis. Transesophageal echocardiogram and Holter monitor were negative. Patient was started on heparin and coumadin maintaining PTT between 50-70. After initial treatment patient was transferred to rehabilitation center.

DISCUSSION: APS is characterized by arterial and/or venous thrombosis with possible pregnancy complications in association with antiphospholipid antibodies (aPL). Most commonly detected aPL are lupus anticoagulant (LA), anticardiolipin antibody (aCL) and anti B2 glycoprotein 1 antibody. LA is more specific for APS while aCL is more sensitive. Despite its name, LA is associated with thromboembolic events rather than bleeding. Primary APS occurs in patients without evidence of other autoimmune diseases while secondary APS occurs in association with other autoimmune diseases specially SLE. APS can cause both arterial and venous thrombosis. Most common site of deep venous thrombosis is in the calf but renal, hepatic, axillary, subclavian and retinal veins or vena cava may be involved. Most common site of arterial thrombosis is in the cerebral vessels causing transient ischemic attack or stroke but coronary, renal, mesenteric arteries and arterial bypass graft occlusions have also been noticed. Venous thrombosis and pulmonary embolism are more common in patients with LA while coronary, cerebrovascular and peripheral arterial events are more common in patients with aCL. The combination of APS, livedo reticularis and ischemic cerebrovascular disease is known as Sneddon's syndrome. Treatment falls into four main areas: prophylaxis, treatment of acute thrombosis, prevention of thrombosis of large vessels and management of pregnancy in association with APS. Patients with APL but no history of thrombosis or fetal death should receive prophylaxis with aspirin. Short term therapy with unfractionated or low molecular weight heparin in combination with warfarin as long term therapy have been successful in treating and preventing further vaso/veno occlusive complications. In pregnancy, intravenous immunoglobulin may be utilized if a patient is refractory with heparin and/or aspirin therapy. Duration of therapy is individually based and recurrence may occur if there is cessation of standard anticoagulation.

CHRONIC URTICARIA/THE 1-YEAR ITCH. D.L. Howell¹; J.V. Blair-Elortegui¹. ¹University of South Alabama, Mobile, AL. (Tracking ID #135252)

LEARNING OBJECTIVES: To recognize acute versus chronic urticaria and to differentiate urticaria from vasculitides To recognize the underlying causes of urticaria To become familiar with the treatment of chronic urticaria

CASE: The patient is a 14 year old white man who presented to clinic with wheals covering approximately 80% total body surface area with extreme pruritus and one episode of angioedema of lips without breathing difficulty. Lesions lasted less than 24 hours and resolved without residual skin discoloration. Patient was unable to relate any triggers to the onset of urticaria including foods, contact, heat or cold, pressure or medications. He stated he had no recent infections or illness. Diphenhydramine provided minimal relief. Follow-up examinations revealed recurrent urticaria and a reported history of worsening lip swelling and airway compromise requiring multiple visits to local emergency departments without necessitating hospitalization. Past medical history includes depression treated with escitalopram. Patient's mother and maternal aunt have thyroid disease and there is no history of familial urticaria. Physical exam was unremarkable except for multiple wheals of varying sizes (1 x 1 cm to 5 x 5 cm) with excoriations and dermatographism. Patient had no angioedema or stridor during any office visits. Labs: Erythrocyte sedimentation rate 1, WBC 12.2 with normal differential and no eosinophilia; TSH 1.84; Free T4 7.5, ANA negative, Rheumatoid factor negative; C1 esterase inhibitor 16 mg/dL (range 10-25), C4 Complement 22 mg/dL (range 90-207) Allergen skin prick negative to all indoor and common food antigens

DISCUSSION: Urticaria presents the general internist and the patient with a diagnostic and treatment challenge. Chronic urticaria is defined as wheals that have recurred at least twice during a minimum 6 week period. Individual urticarial lesions lasting greater than 24 hours are suspicious for vasculitis and should be investigated as such with punch biopsy, immunofluorescence and histopathology. Urticarial vasculitis may often appear as wheals that are palpable and purpuric in nature. These lesions are typically more prominent on the lower extremities and more painful than true chronic urticaria. Physical factors, medications, infectious etiologies, foods, hormonal effects, autoimmune diseases, psychological, or systemic causes should be evaluated. Initial laboratory testing after complete history and physical in patients with suspected chronic urticaria should include CBC with differential, ESR, urinalysis, thyroid studies, and liver function tests. Special attention should be paid to thyroid disease as 28% have antithyroid antibodies and 19% have abnormal thyroid function. However, a cause for chronic urticaria is identifiable in only 5-20% of patients. Treatment consists of eliminating known triggers, such as nonspecific agents which are well established triggers such as NSAIDs, alcohol, opiates, and aspirin. A patient's history should be investigated for physical stimuli such as cold, heat, and deep pressure which are common triggers and can be partially eliminated. Finally the treatment of reversible diseases known to cause urticaria such as infections and thyroid disease should be attempted. Symptomatic treatment consists of first and second generation antihistamines, H2 anti-receptor antihistamines, Leukotriene inhibitors, tricyclic antidepressants (doxepin), glucocorticosteroids, cyclosporine, and hydroxychloroquine.

CLINICALLY DIAGNOSED NEUROBORELLIOSIS IN A YOUNG WOMAN. M. Gabrilovich¹; B. Keefer¹; K. Pfeifer¹; J. Bellizzi¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #132167)

LEARNING OBJECTIVES: 1. Describe limitations of available immunological tests for Lyme disease. 2. Recognize importance of early treatment of neuroborreliosis

CASE: A 24 y.o. female presented with 8-week history of weakness, fatigue and paresthesias. She took a biking trip in northern Wisconsin in July 2004, following which she noticed a target-shaped lesion on her leg, which was warm to touch, nontender and lasted for a week. In September she developed a constant tingling sensation in her hands, feet and lips followed by progressively worsening episodes of disorientation, confusion and extremity weakness. The patient was seen by her internist and a neurologist. A brain MRI was performed, which revealed three small areas of low intensity enhancing seen in both hemispheres, interpreted as nonspecific. Lyme ELISA and other blood tests were normal. She works as a pharmaceutical representative and attends graduate school. On admission she was afebrile and fully oriented. However, she was slow to respond to questions and had photophobia. Her strength was equally decreased to 4/5 in her extremities and 3/5 with feet dorsiflexion; deep tendon reflexes were brisk but symmetric. She was unable to walk without support or squat due to weakness. Her neurological exam was otherwise unremarkable. Differential diagnoses considered included vasculitis, multiple sclerosis, aseptic meningitis, autoimmune disease, and conversion disorder. TSH, Lyme ELISA, HIV, RPR, ANA, C-spine MRI and EKG were within normal limits, ESR 36. LP revealed an opening pressure of 21 cm and normal CSF glucose, protein, cell count, negative bacterial and viral cultures, cryptococcal antigen, VRDL, HSV PCR and multiple sclerosis panel. A presumed diagnosis of neuroborreliosis (NB) was made and empiric therapy started. Her symptoms quickly improved and she was discharged home with 1 month of IV antibiotics. After 1 month the only residual was transient tingling in the finger tips and low energy level. She completed the curriculum for master's degree. CSF Lyme WB was positive for two bands (IGG 41 and 18), which did not meet CDC criteria of 5 positive bands.

DISCUSSION: NB remains a diagnostic challenge because the symptoms may not be supported by objective findings, yet early treatment can lead to resolution of the deficits. As many as one third of patients remain seronegative after a year of infection, even though erythema migrans is highly specific. Detection of antibodies is time dependent because IgG production in the CNS increases over months and there is gradual evolution from a humoral to cell mediated immune response to the infection. Our patient's treatment was based on sudden onset of symptoms months after a good description of erythema migrans and abnormal inflammatory markers pending CSF test results. Treatment was continued because of response and suboptimal sensitivity of the available tests. Treatment for NB has sometimes been based on a presumptive diagnosis that is not supported by serologic testing, according to CDC guidelines. We are looking forward for the results of current research to identify the nature of the immune response to infection with this pathogen.

CLOSE COUSINS: TB AND HISTOPLASMOSIS. C. Donald¹; J. Wiese¹. ¹Tulane University, New Orleans, LA. (Tracking ID #135647)

LEARNING OBJECTIVES: 1. Recognize the clinical presentation of histoplasmosis 2. Understand that histoplasmosis, like tuberculosis, can reactivate

CASE: A 49 year-old man with HIV (CD4: 58) presented with four weeks of painful swallowing, and difficulty swallowing liquids and solids. The pain had progressively increased despite an attempt at using analgesics. He also noted a non-productive cough for one week. There was no hemoptysis, fevers, or night sweats. He was a non-smoker, and did not consume alcohol or illicit drugs. His blood pressure was 113/69 mmHg; heart rate 115 beat/min; temperature 38 °C. He had white plaques in the oropharynx. The CBC and electrolytes were normal. There were bilateral air-space opacities in the upper lobes, but no interval

change from five years earlier. Blood and sputum cultures were obtained and he was started on fluconazole for the thrush. A bronchoscopy revealed an edematous epiglottis with thrush. Marked tissue swelling of arytenoids, normal esophagus, and duodenal ulcerated lesions were noted on endoscopy. A laryngoscopy revealed soft tissue swelling of the arytenoids with a friable material in the anterior esophagus. Sputum cultures and broncho-alveolar lavage yielded Histoplasmosis. Laryngeal, duodenal biopsy had granulomatous inflammation consistent with Histoplasmosis. He was started on a seven-day course of amphotericin B followed by an indefinite course of Itraconazole.

DISCUSSION: Histoplasmosis has a worldwide distribution. Although it is more prevalent in patients exposed to soil, most cases have an unknown source of exposure. Spores are inhaled causing localized or patchy pneumonitis. Like tuberculosis, cellular immunity controls infection; reactivation occurs during immunosuppression. Disseminated disease occurs in immunodeficient patients; severity varies with degree of immune deficiency. Severe manifestations occur if treatment is delayed or if corticosteroids are administered for a mistaken diagnosis of sarcoidosis or vasculitis. Symptoms include fever, night sweats, weight loss, and respiratory symptoms. Oropharyngeal lesions with GI involvement occur as painful ulcerations. 85% of disseminated or chronic histoplasmosis will have positive cultures. Cultures are not typically positive in subacute, localized or pulmonary histoplasmosis. Other means of diagnosis are antigen detection, fungal stains and serologic tests. Treatment for severe or disseminated disease includes amphotericin B followed by itraconazole for life.

COCAINE INHALATION INDUCED PNEUMOMEDIASTINUM. M. Nashat¹; T. Ali²; F. Noor²; S. Dabral². ¹Mercy Catholic Medical Center, Aldan, PA; ²Mercy Catholic Medical Center, Darby, PA. (Tracking ID #134646)

LEARNING OBJECTIVES: 1. Recognize the dangers of cocaine inhalation 2. Learn the clinical features and management of spontaneous Pneumomediastinum

CASE: 23 yr. old Afro-American male presented to the ER with sudden onset of sharp substernal pleuritic chest pain which started four hours after snorting cocaine and radiated to his left jaw. He denied fever, shortness of breath, cough or hemoptysis. He had no significant past or family history. He smoked cigarettes 1pk/day for past four years and admitted snorting cocaine few hours before presentation. Physical exam revealed BP 150/95, Temp. 98.6F, heart rate 95/min, pulse ox of 99% on room air. He had no JVD but palpable subcutaneous air in the neck. Cardiac exam revealed normal heart sounds and presence of crunching sound in systole at left sternal border. Lungs were clear with equal and bilateral breath sounds. Rest of the physical exam was unremarkable. Initial labs showed normal EKG with no ST/T changes. His CBC, chemistry panel and cardiac enzymes were all normal. UDS was positive for cocaine. Chest x-ray revealed air in the mediastinum. Subsequent chest CT confirmed pneumomediastinum. Retrospectively obtained history revealed the patient's habit of valsalva while snorting cocaine. 2D echo showed normal systolic function and no evidence of pneumopericardium. Barium swallow ruled out esophageal tear. The patient's symptoms improved with supplemental oxygen and he was discharged after two days with counseling to avoid his risky behavior. F/u chest x-ray two weeks later revealed resolution of pneumomediastinum.

DISCUSSION: Spontaneous pneumomediastinum is an uncommon disorder representing between 1/12,000 to 1/30,000 hospital admissions. The diagnosis generally implies the absence of other causes of air in the mediastinum, such as those caused by gas-producing organisms, rupture of oropharynx or esophagus, post-op ventilation and blunt trauma to the chest. Recently, pneumomediastinum after cocaine inhalation has been frequently reported. Cocaine inhalation with valsalva maneuver causes an increase in intra-alveolar pressure with alveolar over distention resulting in rupture of alveolar walls, allowing air to travel through the pulmonary interstitium along perivascular sheaths to the lung hilum and mediastinum. The most frequent symptoms are retrosternal pain, dyspnea, dysphagia and neck pain. 73% have detectable subcutaneous emphysema and 50% have Hamman's sign; an unusual systolic crunch heard over the cardiac apex and the left sternal border and is usually transient. A parallel line of lucency along the cardiac border is seen on radiographs, representing air between the heart and the mediastinal pleura. Treatment of spontaneous pneumomediastinum includes bed rest and observation if the patient is stable. Breathing high concentrations of oxygen may allow the air in the mediastinum to be absorbed more quickly. Surgical management is necessary only in severe cases of spontaneous pneumomediastinum with hemodynamic compromise or in patients with esophageal involvement, tracheal rupture or a concomitant significant pneumothorax. Thus, pneumomediastinum, though uncommon should be included in the differential diagnoses of chest pain in young people whose daily activities include unusual frequency of valsalva maneuvers or increased intrathoracic pressure.

COMMUNITY ACQUIRED METHICILLIN-RESISTANT STAPHYLOCOCCUS AUREUS PYOMYOSITIS. J. Miller¹; S. Khan¹. ¹Temple University, Philadelphia, PA. (Tracking ID #135710)

LEARNING OBJECTIVES: 1. To recognize the clinical features of pyomyositis. 2. To recognize the increasing incidence of tropical pyomyositis in temperate climates. 3. To gain awareness of the rising incidence of community acquired methicillin-resistant Staphylococcus Aureus (MRSA) infections.

CASE: A 32 yo woman with chronic HCV and remote intravenous drug use presented with 4 days of left thigh pain without preceding history of trauma. Over the last day she had also noted redness and swelling of her anterior thigh and subjective fevers. On exam, she was afebrile and nontoxic appearing. She had

erythema and tenderness of her left medial thigh without fluctuance. She was discharged home on TMP-SMX for presumed cellulitis. Over the next day, the pain intensified and the area of erythema expanded prompting her to return to the hospital. On exam T 102.5, HR 121, BP 97/60. There was swelling of the left thigh extending from the inguinal ligament to the knee with warmth, induration and exquisite tenderness. Labs revealed WBC 25.5 (64% segs, 24% bands). MRI revealed a 15 × 15 cm multiloculated abscess in the adductor muscles of the left thigh. She underwent serial debridements over a 5 day period. Cultures from the wound grew MRSA and pathology revealed extensive muscle inflammation and necrosis. She received 3 weeks of intravenous Vancomycin and was discharged home.

DISCUSSION: Pyomyositis is a suppurative infection of the muscle with one or multiple abscesses. Originally described in the tropics, it has been increasingly recognized in temperate climates throughout the world. The most common organism cultured is *Staph aureus*, accounting for 75-90% of cases. Although MRSA is a known major cause of hospital-acquired infection worldwide, there has been an increased incidence of MRSA infections in individuals with no apparent risk factors for hospital acquisition. Community-acquired strains are distinguished from nosocomial strains by several features: 1) They are non-multidrug resistant 2) methicillin resistance is conferred by the *mecA* gene and 3) they carry the gene for Pantone-Valentine leucocidin which is a virulence factor associated with more severe infections. Organisms are thought to reach muscle tissue during transient bacteremia, but blood cultures are sterile in the majority of cases. The pathogenesis of abscess formation remains unclear. Predisposing factors are IVDA, HIV infection, and other immunocompromised states. In healthy subjects, other risk factors include muscle trauma, nutritional deficiencies and viral or parasitic infection. Pyomyositis typically affects large muscles of the thigh, calf and gluteal region. There are 3 clinical stages: 1) Invasive (diffuse muscle infection) characterized by localized pain, edema and low-grade fevers; 2) Suppurative (abscess formation) characterized by high spiking fevers and more systemic symptoms; and 3) Sepsis with disseminated infection and multi-organ dysfunction. Most patients present at stage 2. Definitive diagnosis is made by aspiration of pus from the abscess or muscle biopsy with culture. MRI and CT are the best imaging techniques with MRI having the advantage of increased ability to distinguish abscess from muscle edema, leading to earlier diagnosis. Stage 1 can be treated by antibiotics alone but stages 2 and 3 require complete drainage of infected muscle with antibiotic therapy. Despite advances in diagnosis and treatment, pyomyositis still carries a mortality of 0.5-2.0%.

CONSIDER THIS: A CASE OF GROUP B STREPTOCOCCUS PYOMYOSITIS. J. Miller¹; R. Khadilkar¹. ¹Temple University, Philadelphia, PA. (Tracking ID #135735)

LEARNING OBJECTIVES: 1. Review the clinical presentation of pyomyositis. 2. Review the medical and surgical management of pyomyositis.

CASE: A 72 yo woman with Type 2 DM and HTN presented with left shoulder pain. She was discharged with ibuprofen for symptomatic relief of presumed osteoarthritis. 3 days later she returned with worsening pain, nausea and vomiting. She denied fever but noted malaise and poor PO intake and had stopped taking her insulin for the last 3 days. On exam T 99.0, HR 111, BP 133/68. She was lethargic. Her left shoulder was unremarkable. Labs revealed Na 130, K 5.4, HCO3 10, anion gap 32, glucose 572; WBC 23.9 (89% segs, 1% band). UA revealed pyuria and bacteriuria and she was empirically started on Ciprofloxacin. She was admitted to the ICU for management of DKA. Urine culture grew Group B *Streptococcus* and antibiotics were changed to Clindamycin. After 48 hours, she was more alert and complained of severe shoulder pain. Exam now revealed diffuse swelling and erythema of her shoulder extending into her upper arm with and exquisite tenderness. MRI showed multiple multiloculated rim-enhancing fluid collections in the rotator cuff, deltoid, biceps, and triceps consistent with pyomyositis. Fluid aspirated from one of the abscesses grew Group B *Streptococcus* on culture. While initially contemplated, surgical drainage was not required and she improved on antibiotics.

DISCUSSION: Pyomyositis is a deep-seated infection of muscle, first described by Scriba in 1885. Most early case reports are from tropical areas; the first case of non-tropical pyomyositis in the United States was described by Levin in 1971 and cases of pyomyositis in temperate climates are now common throughout the world. Predisposing factors include HIV, DM, malignancy, rheumatologic conditions, cirrhosis, renal disease, and injection drug use. Pyomyositis is most commonly caused by *Staphylococcus aureus* but other infectious agents are cultured with increasing frequency including gram negative organisms, atypical mycobacteria and fungus. Although invasion of muscle by bacteria is thought to occur when transient bacteremia seeds a site of prior muscle damage, only 25-50% of patients with pyomyositis have a history of trauma. The presentation of pyomyositis is divided into three stages. Stage 1 is characterized by muscle pain, fever, and swelling; the affected muscle has a "woody," indurated texture. Ninety percent of patients present with Stage 2 disease, which is characterized by fever, edema, and intense pain, as well as leukocytosis. Stage 3 is characterized by bacteremia, systemic toxicity and muscle fluctuance. MRI and CT scan may demonstrate abscess formation, although the former is more sensitive. The definitive diagnosis is made by aspiration of surgical drainage of the abscess. Treatment varies by stage: Stage 1 disease is initially treated with antibiotics having activity against *S. aureus* and therapy is then tailored based on culture results. In immunocompromised patients, antibiotics such as gentamicin or clindamycin should be used for empiric gram-negative coverage until culture results are available. Patients with Stage 2 or 3 disease usually require surgical drainage of the infected muscle. In conclusion, pyomyositis, while uncommon, should be included in the differential diagnosis of muscle pain, fever, and swelling, as the disease is potentially fatal but curable if diagnosed early.

CORTICOSTEROID INDUCED BRONCHOSPASM. S.C. Reddymasu¹; A. Sequeira¹; S. Konduru¹; R. Gonzales¹; S. Thomas¹; R. Jackson¹. ¹Louisiana State University Medical Center at Shreveport, Shreveport, LA. (Tracking ID #135359)

LEARNING OBJECTIVES: 1. Recognize drug reaction to methyl prednisolone sodium succinate (Solumedrol). 2. Suspect corticosteroid induced bronchospasm (CIB) in patients who do not improve or decompensate on steroid therapy. 3. Cautious use of Solumedrol in the presence of nasal polyps and aspirin allergy.

CASE: A 64 year old African American male with severe persistent asthma/COPD, nasal polyps, aspirin induced worsening of dyspnea presented to the Emergency Department with a 2-week history of worsening dyspnea, wheezing and productive cough. He was diagnosed with COPD exacerbation and admitted to the hospital. After admission he was given nebulized albuterol, ipratropium and Solumedrol 125 mg I.V. Q6 hours. Five minutes after I.V. Solumedrol was begun patient reportedly developed diaphoresis, worsening dyspnea, wheezing and tachycardia. Stopping Solumedrol infusion terminated the event and symptoms resolved. Examination during the event revealed heart rate of 140, blood pressure of 100/50 mm Hg, and diffuse loud polyphonic wheezing throughout a prolonged expiratory phase. After recovering from this event, the patient was put on oral prednisone with no adverse reactions and discharged home shortly after. Review of labs during the event showed a WBC count of 6,800 with 14% eosinophils, which was new, compared to previous labs. Chest radiograph revealed marked hyperinflation without any infiltrates. CT scan of the nasal sinuses showed polyposis and pan-sinusitis. A diagnosis of bronchospasm induced by intravenous Solumedrol was made and the patient was referred for an intradermal skin test with Solumedrol.

DISCUSSION: Review of literature showed that with systemic administration, the prevalence of anaphylactic reactions attributable to corticosteroids is approximately 0.3%. CIB is more common with Solumedrol and hydrocortisone sodium succinate. Most authors believe that bronchospasm and anaphylactic reactions are due to an IgE-mediated mechanism against the succinate component of the drug and not to the steroid component as similar reactions are not seen with prednisone or free methyl prednisolone or methyl prednisolone acetate. This reaction occurs within five minutes of starting the drug and is more common in patients with aspirin allergy and nasal polyps. The reaction is often dose-dependent and some experts believe that boluses of drug can provoke this reaction as opposed to slow infusion. Intradermal skin test and demonstration of specific IgE antibodies against the succinate component of the drug are accepted modalities for diagnosing this condition. CIB should also be in the differential when any asthmatic fails to improve or decompensates on corticosteroid therapy. Whenever corticosteroids are required for such patients, alternate options like oral prednisone should be considered. It is also prudent to avoid boluses of systemic steroids, especially in patients with nasal polyps and aspirin allergy.

CYANOSIS FOLLOWING TRANSESOPHAGEAL ECHOCARDIOGRAM. A. Chang¹; L. Lu¹. ¹Baylor College of Medicine, Houston, TX. (Tracking ID #132627)

LEARNING OBJECTIVES: 1. Review clinical presentation and symptoms of methemoglobinemia (MHb). 2. Recognize precipitating factors of methemoglobinemia 3. Learn the treatment options for methemoglobinemia

CASE: A 45 year-old man with a past medical history of hypertension and Hepatitis C was admitted for treatment of multiple abscesses on his arms, legs, and abdomen. The cultures from these abscesses grew methicillin resistant *Staphylococcus aureus*. Patient subsequently underwent a 2-D echocardiogram revealing a mobile structure on his aortic valve suggestive of vegetations, and a transesophageal echocardiogram (TEE) was performed the following morning. About 30 minutes after the procedure, he developed acute respiratory distress and became tachypneic, cyanotic, and agitated. Vitals signs were BP 144/85, HR 105, Temp 98.8, RR 24 and O2 saturation of 83% on a 100% non-rebreather mask with clear lungs on physical examination. ABG done on 100% oxygen at that time showed pH 7.45, pCO2 27, pO2 223, HCO3- 18.8, SO2 100%. The TEE did not reveal vegetation. Within the next 30 minutes, patient's cyanosis resolved and he returned to baseline. A repeat ABG on room air later revealed pH 7.46, pCO2 29.9, pO2 86.7. Given the discrepancy between the initial pulse oximetry reading and the ABG, a methemoglobin level was drawn and revealed a level of 40.5% (normal <1%). Patient was diagnosed to have methemoglobinemia caused by the benzocaine spray used to anesthetize his upper airway for the TEE. He recovered uneventfully without any intervention.

DISCUSSION: Methemoglobinemia can be caused by exposure to many chemicals and drugs and is occasionally severe or life-threatening. This results from the oxidation of the iron moiety of hemoglobin which is changed from the normal ferrous state to the ferric state. This state is incapable of binding oxygen while at the same time the release of oxygen is impaired, and the oxyhemoglobin-dissociation curve is shifted to the left. Normally, methemoglobin is about 1% of the total hemoglobin, and cyanosis begins to be clinically evident when the level reaches about 10%. Drugs that have been shown to cause methemoglobinemia include local anesthetics (benzocaine, lidocaine), amyl nitrite, chloroquine, dapsone, nitroglycerine, nitroprusside, and sulfonamides. Several case reports of benzocaine-induced methemoglobinemia have been documented in the medical literature; however, the incidence is difficult to estimate. One academic institution reported an estimated incidence of 0.115% (95% CI 0.037 to 0.269). Between November 1997 through March 2002, out of 198 cases of benzocaine adverse events reported by the US Food and Drug Administration, 132 cases (66.7%) involved definite or probable MHb. Of the 69 cases that reported a dose, 53.6% indicated that a single spray was used. Treatment involves methylene blue IV 1-2 mg/kg for severe cases or signs and symptoms of hypoxia over 5 minutes. This acts as a cofactor that increases the rate of NADPH-dependent

methemoglobin reduction. Exchange transfusion or hyperbaric oxygen chambers have also been used for refractory or life threatening cases. Since Internists are responsible for many patients that have procedures such as TEE and endoscopies performed that may involve benzocaine spray, physicians must be able to recognize the clinical presentation and symptoms of methemoglobinemia and understand how to manage the adverse reactions.

CYTOCHROME P450 INDUCTION - DRUGS ARE NOT THE ONLY CULPRIT. D. Sharma¹; K. Pfeifer¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #132823)

LEARNING OBJECTIVES: 1. Identify uncommon causes of cytochrome p450 enzyme system-mediated medication problems. 2. Recognize potential causes for subtherapeutic drug levels.

CASE: A 29 year-old Caucasian male with a history of cerebral palsy and seizure disorder was admitted to the medical ICU with community-acquired pneumonia. Shortly after admission, the patient was intubated and mechanical ventilation initiated for respiratory failure. Among other medications, the patient was on sodium valproate 750 mg PO BID. The valproate levels were confirmed to be therapeutic (90 mg/L) at the time of admission, and the patient was continued on his home dose via nasogastric tube. The patient remained on mechanical ventilation and sedation for 6 days. Following extubation he developed seizures, and valproate levels were found to be subtherapeutic (6 mg/L). The patient's active medications were screened by the ICU team and three different pharmacists, and no potential drug interactions or administration errors/incompatibility were identified. The antibiotic was changed from eropenem to ampicillin/sulbactam, and the patient was given sodium valproate 1 g IV daily in addition to increasing his oral dose to 750 mg PO QID. In addition, he was also given a daily bolus of sodium valproate 1 g IV. Following 5 days of this treatment the valproate level rose to 26 mg/L, and IV therapy was discontinued while the higher oral dose was continued. After 10 days of increased dosing, the patient's valproate level was 90 mg/L, so IV therapy was discontinued and a gradual tapering back to his previous oral dose of 750 mg PO BID was begun. At the time of discharge, his valproate level was stable and therapeutic on his home regimen, and he was free of seizures and respiratory complaints.

DISCUSSION: This case walks us through the approach to a patient with subtherapeutic medication levels. The important causes to consider are: non-compliance; enzyme induction and other drug interactions; non-compatibility with coadministered medications; effects of gastric pH change (due to antacid use); defective delivery method (jejunal tube for medication which requires gastric pH for absorption); non-compatibility with delivery material; and excessive excretion. In our case, we were not able to find any of the above mentioned etiologies of subtherapeutic medication levels. Acute illness was also considered as a possible cause for alteration of medication metabolism in this patient, but in general, suppression of cytochrome p450 and resultant supertherapeutic drug levels is more common in this setting. Reviewing the literature, we found animal studies indicating that transient high oxygen levels, which are possible with mechanical ventilation, can induce the cytochrome p450 enzyme system. Thus, we concluded that the patient's subtherapeutic valproate levels were most likely the result of hyperoxemia from his mechanical ventilation and suggest that this case provides evidence of the validity of these animal models.

"DANCING EYES AND FEET SYNDROME": A PARANEOPLASTIC CASE OF OPSOCLONUS-MYOCLONUS-SYNDROME IN PATIENT WITH BREAST CANCER. J. Baez-Escudero¹; J. Glass²; B. Taqui¹; H. Shishodia¹. ¹Temple University, Philadelphia, PA; ²Fox Chase Cancer Center, Philadelphia, PA. (Tracking ID #132971)

LEARNING OBJECTIVES: 1. Recognize opsoclonus-myoelonus syndrome (OMS) as a rare neurological paraneoplastic syndrome associated with breast cancer 2. Recognize the role of anti-Ri antibody in the pathophysiology and diagnosis of OMS 3. Review management of OMS

CASE: 53 year old female with recently diagnosed breast cancer metastatic to lungs, liver and bone, presented with 4 weeks of worsening blurred vision, limb jerks, ataxia, and memory loss. On examination she was highly irritable, had chaotic eye and pupil movements, severe frequent myoclonic jerks of her extremities, and generalized tremor. CBC, electrolytes, LFT, TSH, cultures/serological tests for infections were all normal. CT scan of the chest/abdomen/pelvis and EEG were normal. Brain MRI was negative for metastases or meningeal carcinomatosis. Lumbar puncture was negative for oligoclonal bands and did not show evidence of infection and malignancy. CSF (and blood) was sent for tumor markers including anti-Ri, anti-Yo, and anti-Hu antibodies. CSF anti-Ri level, specific for opsoclonus-myoelonus syndrome (OMS), was found to be markedly elevated. Patient developed rhabdomyolysis, diffuse paralysis and required mechanical ventilation. She did not respond to high dose dexamethasone and intravenous immunoglobulin. Plasmapheresis was planned but the patient acquired a severe nosocomial infection and expired. On autopsy, microscopic examination of the cerebellum revealed atrophy of the granular layer and marked loss of Purkinje cells, and a loss of neurons in the cerebellar nuclei. Neurons of the nucleus raphe interpositus, responsible for opsoclonus generation under pathological conditions, were intact.

DISCUSSION: Opsoclonus-myoelonus syndrome (OMS) or "dancing eyes-dancing feet" syndrome is a rare, autoimmune neurological disorder. It is characterized by progressive opsoclonus, myoclonus, cerebellar dysfunction and severe hypotonia. It can be idiopathic but has been described as a paraneoplastic syndrome in adults with breast, ovarian, uterine, and lung cancers as well as children with CNS malignancies. It may also be a manifestation of brainstem stroke or infection. The precise immunological mechanism is not well understood. However, specific autoantibodies, such as anti-Ri, may be detected in the serum

and CSF of up to 40% of affected patients. The anti-Ri antibody is thought to cross-react with two antigens, Nova-1 and Nova-2 that are widely expressed within the CNS. Treatment with intravenous immunoglobulin and high-dose steroids has limited benefit in the majority of cases. Removal of autoantibodies through plasmapheresis has been reported to improve symptoms in a few patients. Adrenocorticotrophic hormone, azathioprine and rituximab have been used with mixed success in children. Our case emphasizes the importance of including paraneoplastic syndrome as a potential etiology for neuromuscular symptoms in patients with cancer.

DIABETIC MUSCLE INFARCT IN A NEW DIABETIC. Y. Lou¹; S. Wali². ¹Olive View - UCLA Medical Center, Sylmar, CA; ²University of California, Los Angeles, Sylmar, CA. (Tracking ID #136111)

LEARNING OBJECTIVES: 1. Recognize the symptoms of diabetic muscle infarct even in a new diabetic. 2. Diagnose diabetic muscle infarction by magnetic resonance imaging. 3. Manage diabetic muscle infarct with supportive care and strict glucose control.

CASE: A 36 year old man with no past medical history presented to the Olive View Medical Center ER with bilateral, atraumatic inner thigh pain for one week. He had difficulty ambulating due to the pain. During this time, he also noted polydipsia, polyuria, and polyphagia. On admission his temperature was 36.6. Physical exam of his lower extremities revealed no open wounds, swelling, or erythema. He had exquisite bilateral inner thigh tenderness. Laboratory values showed white blood cell count 8700 cells/dL, blood sugar 739 mg/dL, anion gap 18, serum ketones positive, and hemoglobin A1C 15.1%. Ultrasound revealed a 5.9 x 2.9 x 3.5 centimeter hypoechoic mass in the right proximal thigh adjacent to the groin without evidence of aneurysm or cystic components. Plain films were normal. MRI showed increased signal intensity of right adductor longus and left adductor magnus suggesting edema consistent with a myocytic process. Blind FNA showed normal muscle. Based on these findings, we diagnosed the patient with new diabetes and diabetic muscle infarction. We treated him with supportive care and analgesia. His symptoms resolved and he was discharged home.

DISCUSSION: Diabetic muscle infarction (DMI) is a rare complication of long-standing diabetes. Patients present with acute muscular pain not attributable to trauma or infection. The average duration from onset of diabetes to the first episode of DMI is 17 years. In our review of the literature, we found only one reported case of DMI in a new diabetic. Typically, patients who develop DMI also have other diabetic vasculopathies such as retinopathy, nephropathy or neuropathy. Therefore, the pathogenesis is thought to be related to vascular disease. MRI is the most sensitive and specific diagnostic tool. In fact, clinical presentation and MR findings make muscle biopsy unnecessary. However, when it is done pathology reveals pale, non-hemorrhagic muscle in various stages of muscle infarction. Treatment includes supportive care with bed rest and analgesics. DMI is self-limited and resolves over weeks to months. Long term prognosis is usually poor considering its association with other diabetic complications. There is a growing body of literature concerning diabetic muscle infarction; however, it remains a seldom recognized entity in clinical practice. This case reminds us that DMI is an important etiology to consider in the differential diagnosis of any diabetic, even a new diabetic, who presents with unusual muscular pain. This will not only aid in more rapid diagnosis and prevention of invasive tests, but may also prove to have significant prognostic implications.

DIABETIC MUSCLE INFARCTION: AN UNCOMMON FINDING IN AN EXTREMELY COMMON DISEASE. C.M. Giesler¹; V.T. Martin¹. ¹University of Cincinnati, Cincinnati, OH. (Tracking ID #135362)

LEARNING OBJECTIVES: 1) Recognize the clinical features of diabetic muscle infarction, 2) Distinguish the presentation of diabetic muscle infarction from common causes of single muscle pain, 3) Expand the differential diagnosis of extremity pain in diabetic patients to include diabetic muscle infarction.

CASE: A 42 year-old African American woman with poorly-controlled, non-insulin dependent diabetes mellitus, diabetic neuropathy, and hypertension presented with a 5 day history of severe left leg pain. She stated that the pain was localized in her calf, was constant, and was worse with movement. Pt denied any trauma to the area or any constitutional symptoms. Lower extremity Doppler studies were negative for DVT. CBC, renal panel, CPK and blood cultures were all normal. Her ESR was elevated to 110. An MRI of the calf showed a 2x3.6x5 cm fluid collection in the lateral head of the gastrocnemius and increased muscle edema surrounding the area. A CT-guided biopsy of the fluid collection did not yield any organisms or evidence of infection, but a few PMNs were noted. The diagnosis of diabetic muscle infarction was made after exclusion of DVT, infection, and tumor. The patient was initially treated supportively with rest and analgesics followed by physical therapy and long term strict glycemic control. Her symptoms resolved gradually over the following weeks.

DISCUSSION: Diabetic muscle infarction is a rare complication of diabetes mellitus found most often in poorly-controlled diabetics or those with end-organ damage. Patients present with acute onset, severe muscle pain most commonly on the thigh or calf that is worse with movement. Several diagnoses must be excluded before the diagnosis of diabetic muscle infarction is made. The differential includes deep venous thrombosis, soft tissue sarcoma or lymphoma, cellulitis, pyomyositis, osteomyelitis and necrotizing fasciitis. With the extensive differential diagnosis and relative rarity of this disease, diabetic muscle infarction is a difficult diagnosis to make and one that is not always considered. It remains a diagnosis of exclusion and should be considered in the differential of all

diabetic patients with severe pain in the calf or thigh and a sterile fluid collection on radiological imaging.

DIAGNOSING PULMONARY EMBOLISM (PE): LOOKING BEYOND THE SPIRAL CTA AND VENTILATION-PERFUSION SCANNING, TO INCLUDE EXERCISE STRESS ECHOCARDIOGRAPHY. V.S. Ramanath¹; W.E. Katz². ¹University of Pittsburgh, Pittsburgh, PA; ²University of Pittsburgh Medical Center, Pittsburgh, PA. (Tracking ID #132547)

LEARNING OBJECTIVES: 1) To recognize that acute pulmonary hypertension and right heart dilatation or dysfunction on stress echocardiogram may signal the diagnosis of PE. 2) To recognize the resting echocardiographic findings suggestive of PE, including intraluminal thrombi in the pulmonary artery, RV dilatation and severe pulmonary hypertension.

CASE: A 32 y/o female presented with a 2-month history of dyspnea on exertion that had worsened a few days before presentation. The patient denied chest pain. Her family history was notable for her mother dying at the age of 56 and her father having hypercholesterolemia. Her physical examination was normal. The EKG showed normal sinus rhythm with T wave inversions in leads III, and V1 to V6. The CXR was normal. Given the clinical history and EKG, she was referred for an exercise treadmill echocardiogram to evaluate for cardiac ischemia. The baseline echocardiogram (TTE) showed a small left ventricle with normal wall function. The exercise test was terminated after 4 minutes due to severe dyspnea. The posttreadmill echocardiogram revealed normal left ventricle (LV) segmental function, a dilated hypokinetic right ventricle (RV), and marked ventricular septal shift. Pulmonary artery systolic pressure was 58 mmHg. It was suspected that these findings were secondary to pulmonary embolism (PE). A chest CTA subsequently showed a large thrombus in the left interlobar pulmonary artery with extension into the left main pulmonary artery. All laboratory tests for hypercoagulability were negative.

DISCUSSION: In addition to chest CTA, ventilation-perfusion scintigraphy, and pulmonary angiography, EKG and TTE can also provide valuable information toward diagnosis. T-wave negativity in leads V1 through V4 indicating RV strain is part of the criteria used to diagnose pulmonary embolism. TTE findings secondary to pulmonary embolism are a result of the development of acute pulmonary hypertension. As pulmonary vascular resistance increases, RV wall tension increases, resulting in RV dilatation and dysfunction. Consequently, there is a shift of the ventricular septum into the LV. However, due to its low sensitivity, TTE is not recommended as a primary means for diagnosing PE. In addition, exercise stress echocardiogram is not routinely performed or recommended on patients with suggested or known PE, as it can be potentially detrimental to the patient. However, in patients presenting with dyspnea on exertion and with findings of acute right heart dilatation and dysfunction and significant pulmonary hypertension on exercise stress echocardiogram, pulmonary embolism should be considered as a possible etiology.

DISPOSING OF OCKHAM'S RAZOR IN PATIENTS WITH HIV: A CASE OF ABDOMINAL ACTINOMYCESES AND TUBERCULOSIS. M.J. Richman¹; A. Jeng². ¹Olive View - UCLA Medical Center, Los Angeles, CA; ²Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #132086)

LEARNING OBJECTIVES: 1) Recognize that in patients with HIV, multiple organisms may simultaneously infect one organ system. Clinicians must vigilantly search for secondary and tertiary diagnoses. 2) Diagnose and treat Actinomycosis.

CASE: A 24-year-old previously healthy man presented with 1 month of abdominal pain, scleral icterus, and a sensation of a mass in his abdomen. 2 weeks prior to admission (PTA), he had 3 days of brown, watery diarrhea. 3 days PTA, he had fever and chills. He had decreased appetite and energy, but no nausea, consumption of unprepared food, or sick contacts. He had recently come from Mexico. He denied high-risk sexual behavior, IV drug use, or blood transfusions. Initial vital signs: T 39.8, BP 95/58, P 123, RR 22, Room air SaO2 97%. He was cachectic, jaundiced, and dehydrated, with cervical lymphadenopathy (LAN) and a soft, tender abdomen with a 2 x 2 cm mass to the right of the umbilicus. Initial labs: WBC 10, HCT 32 (MCV 85), Plts 102, Na 128, K 3.1, BUN 3, AST 314, ALT 127, Alk Phos 285, D Bili 3.8, INR 1.6, LDH 726. Hepatitis B serology, blood/stool cultures, CXR, and 3 sputum AFB were (-). HIV test was (+) with CD4=97. Abdomen/pelvis CT showed a 6 x 8 x 14 cm intraperitoneal, necrotic mesenteric, nodal mass suspicious for lymphoma. CT-guided biopsy revealed not lymphoma, but sulfur granules staining Gram (+) for branching rods. A modified acid-fast stain was (-) for Nocardia, but revealed 3 (+) M. tuberculosis. The team initiated ampicillin for Actinomycetes, and rifampin, INH, pyrazinamide, and ethambutol for TB.

DISCUSSION: In patients with HIV, the principle of "Ockham's razor" does not apply. Multiple organisms may simultaneously infect one organ system. The clinician must vigilantly pursue multiple diagnoses until they have been reasonably excluded. In this case, M. TB acted as a "companion" organism, facilitating intra-abdominal invasion by Actinomycetes. Actinomycetes is a natural colonizer of the oropharynx and GI tract. Invasive disease infects the oral-cervico-facial area (55%), thorax (15%), abdomen (20%), and pelvis. Often part of a polymicrobial infection, Actinomycetes acts as a co-infecting agent, most frequently in odontofacial disease processes, although it can cause disease as the sole pathogen. Intra-abdominal infection occurs after the organism seeds the abdomen through a breach in the GI mucosa either mechanically (fish bone, trauma/surgery, IUD, perforated viscus) or, uncommonly, facilitated by a "companion" organism. The infection progresses indolently over months to years as a growing mass, and is commonly mistaken for malignancy. Diagnosis is best made through CT- or US-guided biopsy or through FNA, avoiding unnecessary surgery. The finding of sulfur granules containing Gram (+) branching rods ("clubs" of Actinomycetes) in non-mucosal sites, with a (-) modified acid-fast

stain for Nocardia, confirms the diagnosis. Treatment is with pen/amp/amoxicillin, erythromycin, or clindamycin. Rare case reports have described Actinomycetes causing intra-thoracic disease in pre-existing TB cavities, but never in association with intestinal TB. This is the first reported case of a patient with abdominal co-infection by Actinomycetes and M. TB.

DRUG INDUCED MYXEDEMA MADNESS. C.S. Almeida¹; A.P. Burger¹. ¹Yale University, Waterbury, CT. (Tracking ID #136013)

LEARNING OBJECTIVES: 1) To recognize "Myxedema Madness" as a presentation of hypothyroidism. 2) To identify medications especially amiodarone as a cause of hypothyroidism.

CASE: A 92yo female with a history of myocardial infarction, coronary-bypass surgery, atrial fibrillation and hyperthyroidism presented with complaints of hand tremor, visual hallucinations, memory loss, confusion, delusions, and sleepiness which began five months prior to presentation. The patient described her hallucinations as frequent visions of a person dressed as a "bunny" being in the room. She also had vivid dreams that would continue after awakening. During the interview the patient saw the "bunny" in the room. Paranoid feelings of harm from her family were strong. She was afebrile with stable vital signs. Medications included amiodarone started seven months prior, methimazole which she had been on for many years, furosemide, aspirin, potassium, captopril, metoprolol, metoclopramide, and multivitamin. On physical exam she was a thin appearing woman in no significant distress. She was alert and oriented to self only. Her voice was harsh. Occasionally the patient would fall asleep during the interview. There was no peripheral edema. Her hair and brow line were thin. The rest of the exam was benign. A CT-scan of the head showed chronic atrophy and white matter changes. She had a TSH of 52.8, FT4 of <0.4, and T3 of 39. A toxic metabolic work-up was negative. Her amiodarone and methimazole were held. She was given L-thyroxine 25 mcg daily with gradual improvement interrupted by intermittent episodes of confusion and lethargy. She was later discharged to a skilled nursing facility with follow-up.

DISCUSSION: Psychosis or mental status changes are an unusual initial manifestation of hypothyroidism (5%), commonly referred to as myxedema madness or psychosis. Hypothyroidism is a common malady and it is therefore important to recognize or suspect this entity as it is a rapidly reversible cause of mental illness. The exact cause of this disease has not yet been well established (although there are several postulated mechanisms). Amiodarone, a commonly used drug in arrhythmias, has been implicated in inducing thyroid dysfunction, both hyper and hypo. Our patient was not only on amiodarone but also methimazole, a direct suppressant of thyroid gland function. Either drug could cause this condition on its own. In combination we saw a rapid onset of psychosis and a profoundly elevated TSH. Treatment of myxedema madness involves thyroid replacement and discontinuation of causative agents allowing restoration of a euthyroid state. This reverses the psychosis in a dose dependent fashion. Our patient also demonstrated the paradoxical nature of this disease, in that there is a waxing and waning of mental status during the reversal stage of treatment. It is also important to remember that in a patient with underlying coronary disease replacement therapy should be done with caution and started at a low dose to avoid precipitating further cardiac events. This case demonstrates the need to keep this unusual clinical manifestation in mind when evaluating older patients and to remain cognizant of medication complications as a cause of altered mental status.

DRY AS A BONE, RED AS A BEET, MAD AS A HATTER: A CASE OF ANTICHOLINERGIC TOXICITY DUE TO JIMSONWEED. S.R. Herrie¹; R. Granien¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133042)

LEARNING OBJECTIVES: 1. Recognize the signs and symptoms of anticholinergic poisoning. 2. State the appropriate treatment for patients presenting with an anticholinergic toxidrome. 3. Recognize the Jimsonweed plant and understand the pharmacology of the plant.

CASE: A seventeen year old male was brought to the Emergency Department after being found by his parents to be agitated and confused. Earlier in the evening, the patient had been in a wooded area in western Pennsylvania with friends who reported that he had consumed a handful of seeds. Upon presentation, the patient was noted to be extremely agitated with garbled speech. Examination revealed that he was afebrile with a heart rate of 134 and regular and a blood pressure of 134/56. He was in no respiratory distress. His pupils were bilaterally dilated to 8 mm and were minimally reactive. Mucous membranes were dry. Skin was diffusely erythematous and warm to the touch but without any diaphoresis. Bowel sounds were decreased. A Foley catheter was placed and drained 450 cc of clear yellow urine. The patient was given both activated charcoal and physostigmine 2 mg intravenously with temporary improvement in his mental status. A urine toxicology screen was negative. The patient was then transferred to a monitored bed. He was treated only with intravenous lorazepam on an as needed basis for agitation. Within 48 hours, the patient had completely recovered and was discharged from the hospital. During his hospitalization, the patient's friends brought in a sample of the seeds that he had ingested. These seeds were identified as Datura Stramonium.

DISCUSSION: Datura Stramonium, commonly known as the Jimsonweed plant, is a potent source of the alkaloids atropine and scopolamine. The fully mature plant can grow to 5 feet in height with capsules each containing approximately one hundred seeds. Each seed, measuring less than 5 mm in diameter, possesses the equivalent of nearly 6 mg of atropine. Datura Stramonium is indigenous to the continental United States. The seeds, leaves, and flower nectar can be either smoked or eaten. Both atropine and scopolamine function

by binding to and blocking both central and peripheral muscarinic cholinergic receptors. Peripheral receptors are responsible for carrying out the effects of the parasympathetic nervous system. Blockage results in midriasis, decreased lacrimation, decreased bowel motility, inhibition of bladder contraction, and tachycardia. Blockage of the central receptors leads to impairment of short-term memory and hallucinations. Together, these signs constitute the anticholinergic toxidrome. Treatment consists of gastric lavage with activated charcoal if consumption occurs within one to two hours of ingestion. Physostigmine is useful for confirming the diagnosis and is indicated for symptomatic supraventricular tachycardias, severe agitation, coma, and intractable seizures. Its use does carry the risk of seizures. Agitation should primarily be treated with intravenous lorazepam on an as needed basis. Hyperthermia can be treated with cooling blankets and with antipyretics such as acetaminophen.

DYNAMIC MITRAL REGURGITATION: AN ATYPICAL PRESENTATION OF RECURRENT ACUTE DYSPNEA. J.P. Lynch¹; J.J. Teuteberg¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133373)

LEARNING OBJECTIVES: 1. Recognize the clinical manifestations of dynamic mitral regurgitation. 2. Describe the pathophysiology of dynamic mitral regurgitation.

CASE: A 59 year old man with a non-ischemic cardiomyopathy and an ejection fraction of 20% presented in transfer to the coronary care unit from an outside hospital in cardiogenic shock. During the prior six months, he had five episodes of acute dyspnea with mild exertion requiring intubation. Angiography performed during one such decompensation demonstrated no significant coronary stenoses, but did reveal moderate to severe mitral regurgitation (MR) with moderate pulmonary hypertension (PHTN). Follow-up echocardiography demonstrated only mild to moderate MR and no PHTN. Two days prior to transfer, the patient presented with another episode of acute dyspnea requiring intubation, dopamine and dobutamine, and eventually an intra-aortic balloon pump to stabilize his hemodynamics. Over the course of the next several days his inotropes were weaned, the balloon pump was discontinued and he was extubated. Echocardiography demonstrated normal mitral valve morphology with only mild to moderate MR. However, upon ambulation the patient had recurrence of his acute dyspnea nearly requiring intubation. His symptoms were transient and not associated with hypertension or arrhythmias. However, echocardiography demonstrated that his mild to moderate MR became severe during semi-supine bicycle stress and coincided with an increase in his pulmonary arterial systolic pressure from 45 mmHg to 75 mmHg.

DISCUSSION: Recurrent acute dyspnea in patients with cardiomyopathy is frequently due to acute ischemia, an arrhythmia, or a hypertensive crisis. Progression of chronic valvular disease can also cause acute dyspnea; however, recurrence of exercise induced dyspnea should prompt consideration of dynamic MR. Dynamic MR is a transient, exercise-induced increase in the severity of MR with a resultant increase in pulmonary arterial pressure. To coapt properly, the mitral valve leaflets form a complex three-dimensional structure that depends not only on normal leaflet morphology, but on a precise relationship of the annulus, tendinous cords, papillary muscles, and left ventricular walls. Mitral regurgitation in patients with cardiomyopathy is not usually due to leaflet pathology, but rather ventricular dilation and the resultant dilation of the mitral annulus and apical tethering of the leaflets. During exercise there is an increase in venous return and a concomitant elevation in afterload. These changes exacerbate the already abnormal ventricular geometry and increase the severity of MR, abruptly elevating the left atrial pressure and resulting in acute dyspnea. The transient nature of dynamic MR prevents the development of compensatory mechanisms, as seen in chronic valvular disease, and accounts for the acuity of the symptoms. The degree of dynamic MR is independent of the degree of MR at rest, and dynamic MR strongly correlates with a poor prognosis. Therefore patients with cardiomyopathy and exercise symptoms out of proportion to their underlying valvular disease warrant evaluation for dynamic MR. Evaluation consists of an assessment of the valve with exercise, either non-invasively with echocardiography or invasively with a Swan-Ganz catheter.

DYSFIBRINOGENEMIA: A COMMON CAUSE OF BLEEDING IN PATIENTS WITH LIVER DISEASE. R.F. Charles¹; S. Cheng¹; S.D. Sisson². ¹Johns Hopkins Hospital, Baltimore, MD; ²Johns Hopkins University, Baltimore, MD. (Tracking ID #134236)

LEARNING OBJECTIVES: 1) To become familiar with bleeding disorders in patients with liver disease. 2) To learn the differential diagnosis of an increased activated partial thromboplastin time (aPTT) 3) To understand the role of mixing studies in the evaluation of coagulopathies

CASE: A 47-year-old man with hepatitis C and end-stage liver disease (ESLD) was admitted with encephalopathy and acute renal failure. Dialysis catheter placement was complicated by persistent oozing and a drop in hematocrit. Laboratory values included: platelet count 63,000, prothrombin time (PT) 18.7 seconds (INR 1.9), activated thromboplastin time (aPTT) >200 seconds (ratio >5.9), normal fibrinogen and haptoglobin levels, d-dimer 10.01 mg/L (upper normal limit 2.24 mg/L), and lactate dehydrogenase 3131 IU/L (upper normal limit 273 IU/L). Thrombin time was prolonged at 50.3 seconds (upper normal limit 20.7 seconds). Despite treatment with fresh frozen plasma (FFP) and vitamin K, the PT and aPTT did not correct. Mixing studies were performed, and clotting times did not correct with addition of clotting factors. No factor inhibitors or lupus anticoagulant were detected. Dysfibrinogenemia was diagnosed, and treatment with cryoprecipitate and FFP corrected the coagulopathy.

DISCUSSION: Hemostasis requires vascular integrity, normal coagulation factors, platelets, and proper fibrinolysis. Dysregulation of any component can re-

sult in bleeding. Initial evaluation of abnormal hemostasis includes the platelet count, PT, and aPTT. In this case, the most significant abnormality was a prolonged aPTT. The differential for a prolonged aPTT includes coagulation factor deficiency, presence of a factor inhibitor, lupus anticoagulant, heparin administration, fibrinogen deficiency/dysfunction, or disseminated intravascular coagulation. Mixing studies differentiate between a factor deficiency and a clotting inhibitor. In mixing studies, the patient's blood is mixed in equal ratios with normal plasma, replacing deficient clotting factors. Correction of the aPTT with mixing suggests clotting factor deficiency, and individual factor assays are used to identify the deficient factor. Failure of the aPTT to correct with mixing suggests presence of an inhibitor, such as heparin contamination, a factor inhibitor, or lupus anticoagulant. In this patient, the aPTT did not correct although no inhibitors, heparin, or lupus anticoagulant were detected. Other causes of abnormal hemostasis include disseminated intravascular coagulation (DIC) or abnormalities with fibrinogen. In these instances, both the aPTT and PT are prolonged. However, review of the peripheral blood smear and normal haptoglobin levels excluded the diagnosis of DIC. In this patient, in addition to a non-correcting mixing study, the other abnormality was a prolonged thrombin time (with normal fibrinogen levels), leading to the diagnosis of dysfibrinogenemia. Dysfibrinogenemia, characterized by abnormal polymerization of fibrin monomers, leads to prolonged aPTT, PT, and thrombin time despite normal fibrinogen levels. Dysfibrinogenemia is the most common clotting abnormality in liver disease, occurring in 60-70%, although it is easily overlooked due to its often mild severity and the presence of other bleeding disorders common in ESLD. In addition to thrombocytopenia, vitamin K deficiency, fibrinolysis, and clotting factor deficiency, dysfibrinogenemia should be considered in patients with cirrhosis presenting with a coagulopathy.

ELEVATED FERRITIN IN FEVER OF UNKNOWN ORIGIN: AN OFTEN SUBTLE DIAGNOSIS (AOSD)? M. Velagapalli¹; A. Kalyanasundaram¹. ¹Geisinger Medical Center, Danville, PA. (Tracking ID #135322)

LEARNING OBJECTIVES: 1) Serum ferritin values above 3000 ng/mL, in the right clinical scenario should prompt the physician to strongly consider AOSD, particularly with no evidence of a bacterial or viral infection.

CASE: 56-year-old Filipino lady presented with a six-week history of scratchy sore throat, submandibular adenopathy and intermittent high-spiking fevers that coincided with a transient rash. The rash was light red, slightly raised, and distributed throughout her body. She had systemic symptoms including weight loss, malaise and generalized musculoskeletal pain along with swelling and pain of wrists and knees. She was admitted due to the debilitating nature of her illness. On exam, she was ill appearing with tender wrists, bilateral knee effusions and some erythematous areas on chest and back. No periorbital edema, heliotrope or periungual changes were seen. Extensive lab work revealed a hemoglobin of 9.6, AST of 303, ALT of 250, lactate dehydrogenase of 941, an elevated white blood cell count of 32,000 with 97% granulocytes and platelets of 725. Multiple blood cultures and viral cultures for Epstein-Barr, CMV, and hepatitis were negative. Her transthoracic echocardiogram and CT scan of the chest, abdomen and pelvis were unremarkable. Rheumatologic investigation revealed an elevated erythrocyte sedimentation rate of 116 but negative ANA, RF and ANCA studies. Ferritin was elevated at 18,000 and a repeat value was over 25000. Association of marked leukocytosis, rash, fever, arthritis and abnormal LFTs coupled with markedly elevated ferritin level in the absence of an alternate diagnosis satisfied the criteria for diagnosis of Adult Onset Still's Disease. Treatment was initiated with steroids with satisfactory response.

DISCUSSION: We have presented a patient with Adult Onset Still's disease (AOSD) manifesting as fever of unknown origin with a remarkably high ferritin level. In general, fever of unknown origin (FUO) is most likely due to one of three general categories of illness - infections, malignancies and collagen-vascular diseases (eg, vasculitis, rheumatoid arthritis). AOSD is an inflammatory disorder characterized by spiking high fevers, arthritis, and an evanescent rash. In a study of 130 Turkish adults with FUO, AOSD was the second most identifiable cause of fever after occult infection. Other diseases such as infections, neoplasia or immune diseases can mimic AOSD. Diagnosis of AOSD should be made only after the other more common diagnoses are eliminated after extensive investigation, as was the case with our patient. It is to be noted that hyperferritinemia is usually not observed in this range with other rheumatic diseases. Serum ferritin correlates with disease activity, and therefore suggested as a serologic marker to monitor treatment response.

ELEVATED TROPONINS AND THE CHURG-STRAUSS SYNDROME. J. Zaky¹; R. Yu²; C. Caraang¹; A. El-Bialy¹. ¹Olive View - UCLA Medical Center, Sylmar, CA; ²UCLA School of Medicine, Los Angeles, CA. (Tracking ID #132648)

LEARNING OBJECTIVES: 1) Recognize an atypical presentation of Churg-Strauss syndrome (CSS). 2) Treat CSS early to prevent irreversible organ damage.

CASE: A 65-year-old woman presented to our ED several times over a period of six months with complaints of chest pain. She had a history of asthma, and her only cardiac risk factor was age. On each presentation, the patient had elevated troponins, while the EKG was unremarkable. Coronary angiography with ventriculography revealed normal coronaries, and normal left ventricular function with a LVEF of 65%. On her fourth admission, she again presented with multi-organ abnormalities including asthma and vasculitis. Troponins were elevated again, up to 5.4 g/L, and she was noted to have a significant eosinophilia (31.7%). Bilateral infiltrates were seen on CXR, and bronchial lavage was suggestive of eosinophilic pneumonitis. Histopathology of tissues obtained via

transbronchial biopsy discovered intra-alveolar and interstitial eosinophils. An EMG study for evaluation of leg pain revealed a mononeuritis complex and was later confirmed with sural nerve biopsy. These findings led to the diagnosis of CSS and with corticosteroid therapy, both the eosinophilia and troponins immediately normalized and the patient's presenting symptoms resolved.

DISCUSSION: A history of asthma, eosinophilia (> 10% on differential), mononeuropathy, biopsy findings, and pulmonary infiltrates are diagnostic for CSS with 99.7% specificity according to the criteria set forth by the American College of Rheumatology. Although not typically recognized as causing cardiac pathology, eosinophilia from CSS can lead to multi-organ damage, which may include the heart. Cardiac involvement is an unusual but major cause of morbidity and mortality and presents as a variety of conditions; most commonly, myocarditis. Eosinophil-mediated heart damage evolves through three stages that include an acute necrotic stage, an intermediate phase with thrombus formation along damaged endocardium, and a fibrotic stage. Fibrosis can lead to a restrictive cardiomyopathy or entrapment of the chordae tendineae, which may result in mitral and tricuspid incompetence. In a review of 50 cases of CSS, damage to the heart is responsible for 48% of all deaths. The mainstay of treatment in CSS focuses on the use of corticosteroids, and in severe cases cyclophosphamide and high-dose IVIG. Doses of corticosteroids usually range from 0.5 to 1.5 mg/kg given over six to 12 weeks, but may be continued as necessary until the active disease is resolved. The higher doses of corticosteroids are usually necessary for patients with cardiac, renal, or neurological symptoms. In the more severe cases not responsive to corticosteroids alone, addition of cyclophosphamide or azathioprine may aid in control. High dose IVIG has been proposed for cases of CSS in patients who do not respond to conventional treatment. Poorer prognosis is associated with cardiac, CNS, and gastrointestinal involvement. Treatment efficacy can be monitored using serum eosinophil levels and cardiac markers, such as troponins. For cases with coronary artery or diffuse myocardial involvement, cardiac catheterization and serial echocardiography may be useful. Internists should consider CSS in the differential diagnosis of myocarditis or elevated troponin levels in the setting of eosinophilia, as early detection and treatment may be critical in decreasing morbidity and mortality.

ENTERIC FEVER IN TRAVELERS. P. Warrier¹; A. D'Aria¹; R.W. Lyons². ¹University of Connecticut, Farmington, CT; ²University of Connecticut, Hartford, CT. (Tracking ID #135398)

LEARNING OBJECTIVES: Recognize enteric fever as a cause of febrile illness in travellers to endemic regions Recognize the importance of blood cultures in diagnosis Recognize that immunizations against typhoid fever do not protect against *Salmonella* paratyphi

CASE: An Indian couple, presented to the Emergency Room 5 days after returning from travel to Hyderabad, India. The husband and wife were both healthy 31 year olds with no significant past medical history. They both complained of high grade fevers and chills for 3 days and nausea. They were unable to tolerate oral intake. The wife had a dry cough and the husband had had 10 episodes of emesis. On admission they were febrile, and other vital signs were normal. Of note, there was no relative bradycardia. Physical exam was unremarkable and revealed no rose spots. A working diagnosis of Typhoid fever and Malaria were considered. Peripheral smears for Malaria were negative. CBC showed no leukocytosis. Blood cultures revealed *Salmonella* paratyphi in both patients and ciprofloxacin was started. Stool cultures were negative. Our patients did not suffer from any complications of the infection.

DISCUSSION: Enteric fever, also called typhoid fever or paratyphoid fever, is a systemic febrile illness that is most commonly caused by *Salmonella* typhi; less frequently by *S. paratyphi* A, B and C. Typhoid fever, a common illness in developing countries, is seen only sporadically in developed countries, usually in travelers returning from endemic regions especially the Indian Subcontinent. A high index of suspicion is required as patients usually have nonspecific illness. High fevers, headache, poorly localized abdominal discomfort and a dry cough are common presenting features. In 2002, there were 321 cases of typhoid fever in the USA, and of these 6 were in Connecticut (CDC data). Classically three stages are described in typhoid fever. Stage 1 consists of stepwise rising fevers. Stage 2 consists of abdominal pain and the blanching 2-4 mm erythematous maculopapular lesions called rose spots seen in 5-30% of cases. Stage 3 includes hepatosplenomegaly and complications like intestinal perforation. Other common manifestations include a dry cough, arthralgias and myalgias. Paratyphoid fever is usually milder, with less complications. Diagnosis is made by isolating the organism in blood, stool, urine or duodenal contents. Stool cultures are positive only in 30-40% of patients, whereas blood cultures are positive in 40-80% of patients. However, if these remain negative, bone marrow cultures should be obtained which are highly sensitive and positive about 80-95% of the time, remaining so even after antibiotic therapy is begun. Older agents for treatment were chloramphenicol, ampicillin and trimethoprim-sulfamethoxazole, however, due to drug resistant strains, fluoroquinolones are preferred. Recently resistance to fluoroquinolones has also been reported. Immunizations available for travelers include Ty21a, an oral live attenuated vaccine and the Vi vaccine, a capsular polysaccharide vaccine for parenteral use. Interestingly neither vaccine protects against *Salmonella* paratyphi.

EPSTEIN-BARR VIRUS-ASSOCIATED HEMOPHAGOCYTIC LYMPHOHISTIOCYTOSIS IN A 21-YEAR-OLD MALE. A. Wadhwa¹; S.D. Zucker¹; L. Coberly¹. ¹University of Cincinnati, Cincinnati, OH. (Tracking ID #135567)

LEARNING OBJECTIVES: 1) Recognize that hemophagocytic lymphohistiocytosis (HLH) is a syndrome characterized by fever, hepatosplenomegaly, hypertriglyceridemia, and cytopenia. 2) Recognize that proliferation of Epstein-Barr virus (EBV) infected immune cells is prominent in the pathogenesis of EBV-

associated HLH. 3) Recognize that effective treatment of EBV-associated HLH requires chemotherapy and immunotherapy, and successfully treated patients can have good outcomes and can remain disease free.

CASE: A 21-year-old college student with Crohn's disease presented with fever, abdominal pain and jaundice. A presumptive diagnosis of EBV mononucleosis was made based on a positive heterophile antibody test. Physical exam showed scleral icterus, erythematous pharynx without exudate, cervical lymphadenopathy, icteric skin, and no hepatosplenomegaly. Labs revealed Hb 10.9 g/dL, WBC 2,800/ μ L, platelets 47,000/ μ L, INR 1.3 U, AST 245 U/L, ALT 227 U/L, total bilirubin 6.2 mg/dL, direct bilirubin 5.1 mg/dL, fibrinogen 36 mg/dL, ferritin 11,112 ng/mL, and triglycerides 483 mg/dL. Viral hepatitis A, B and C serologies were negative. EBV IgM was positive. Serum ceruloplasmin was normal. Serum ANA, ASMA, and ANCA were negative. An abdominal ultrasound was normal. During the hospital course, he developed pancytopenia. A bone marrow biopsy was consistent with hemophagocytosis, and liver biopsy was consistent with EBV-hepatitis. Based on these findings, the diagnosis of EBV-associated HLH was established. Treatment with the HLH-94 protocol (IV etoposide and dexamethasone) was initiated. Ganciclovir, initiated to target EBV replication, was discontinued in favor of foscarnet due to possible myelosuppression. On this regimen, the patient's liver enzymes and blood counts improved and he recovered.

DISCUSSION: HLH occurs in both children and adults; an annual incidence of 1.2 cases per million children has been reported. It is believed to result from the unregulated proliferation and activity of benign macrophages, natural killer cells, and type I T-cells. It is postulated that EBV-infected cells initiate HLH through uncontrolled production of cytokines, including interferon-gamma and TNF-alpha. Phagocytosis of erythrocytes, platelets, and their precursors leads to fever, splenomegaly, jaundice, and hemophagocytosis in bone marrow, liver, and lymph nodes. Laboratory findings include hypofibrinogenemia, anemia, thrombocytopenia, elevated transaminases, hyperbilirubinemia, hypertriglyceridemia, hyponatremia, elevated ferritin, and elevated LDH. HLH is categorized into genetic and acquired forms, but most cases are thought to be initiated by acute infections (often EBV). EBV-associated HLH is almost universally fatal if untreated. Initiation of etoposide within 4 weeks of diagnosis has been shown to substantially improve 2.5 year survival (85% vs 10%). In the present case, histopathologic evidence and multidisciplinary management facilitated a propitious outcome for this patient.

ERYTHEMA MULTIFORME ASSOCIATED WITH CRANIAL IRRADIATION AND PHENYTOIN. S.K. Thambidorai¹; R. Gudavalli¹; B. Sankarandian¹; T. Slattery¹; J. Derby-Porter¹. ¹Creighton University, Omaha, NE. (Tracking ID #135904)

LEARNING OBJECTIVES: 1. To distinguish the spectrum of disorders included under Erythema Multiforme (EM), and to recognize the association of EM with cranial irradiation and phenytoin.

CASE: A 55 year old male with malignant melanoma received cranial irradiation and phenytoin following seizures secondary to brain metastasis. Patient developed an erythematous macular eruption on the scalp which was felt to represent radiation dermatitis. 4 weeks after starting phenytoin, the skin lesions progressed into a generalized maculopapular eruption involving the trunk, upper and lower extremities including palms and soles. Associated mucosal involvement with oral blistering and desquamation was noted. Marked interface dermatitis, vacuolization of the basal cell layer and rare necrotic keratinocytes on histologic specimen was consistent with the clinical impression of EM. Discontinuation of phenytoin and administration of systemic steroids led to resolution of the skin lesions.

DISCUSSION: Erythema Multiforme is a spectrum of disorders ranging from a self limited cutaneous eruption (EM minor) to a progressive mucocutaneous process (EM major). EM major has been further categorized into 5 subtypes: 1. Bullous EM; 2. Stevens - Johnson syndrome (SJS); 3. overlap Stevens-Johnson syndrome toxic epidermal necrolysis (SJS-TEN); 4. TEN with spots; and 5. TEN without spots. Several medications and infectious agents have been implicated as causative agents in these disorders. Phenytoin in conjunction with whole brain irradiation is commonly used in intracranial malignancies complicated by seizure activity. There have been case reports of patients who received cranial irradiation and phenytoin therapy, developing a scalp eruption which subsequently extended to involve mucocutaneous sites and diagnosed as EM major. Discontinuation of phenytoin, with administration of systemic steroids led to the resolution of the mucocutaneous lesions. This case illustrates the need to promptly evaluate any skin lesions including those that develop at the radiation site in patients on phenytoin therapy with a high index of suspicion.

EVOLVING CRANIAL NERVE DEFICITS: DIFFERENTIAL DIAGNOSIS IN A PATIENT WITH CRYPTOCOCCAL MENINGITIS. N. Maruthur¹; D. Cosgrove¹; S.D. Sisson². ¹Johns Hopkins Hospital, Baltimore, MD; ²Johns Hopkins University, Baltimore, MD. (Tracking ID #135573)

LEARNING OBJECTIVES: 1. To learn the differential diagnosis and evaluation of a patient presenting with cranial nerve deficits. 2. To recognize cryptococcal meningitis as a cause of cranial nerve deficits.

CASE: A 48-year-old male with a history of AIDS (CD4 = 6) presented with complaints of headache, lightheadedness, and visual changes. On physical examination, he was afebrile, and maximum blood pressure was 233/108; neurologic examination and CT scan of the brain were normal. Blood pressure was treated with IV nicardipine, oral nifedipine, and spironolactone, and the patient's blood pressure normalized over five hours. Headache, lightheadedness, and visual changes resolved. Four hours after admission, the patient complained of blind-

ness and hearing loss. On physical examination, pupils were fixed and dilated and then became asymmetric over the next several hours. Blood pressure was normal, and the remainder of the neurologic exam was nonfocal. Repeat imaging of the brain was negative. CSF obtained from a lumbar puncture was positive for cryptococcal antigen. Treatment with amphotericin and 5-flucytosine was implemented, and a lumbar drain was placed for CSF removal. Cranial nerve abnormalities resolved.

DISCUSSION: The differential diagnosis of causes for new cranial nerve deficits is broad. Metabolic, infectious, non-infectious and vascular etiologies should be considered. Metabolic encephalopathies that can cause cranial nerve deficits include Wernicke's encephalopathy (i.e. thiamine deficiency resulting in ophthalmoplegia, bilateral cranial nerve VI palsies, pupillary abnormalities, and horizontal nystagmus), hyponatremia and hypernatremia (due to swelling or shrinkage of brain tissue), profound hypophosphatemia (due to muscle weakness), hepatic encephalopathy (due to CNS swelling), and hypoglycemia. Medication-related metabolic causes of cranial nerve deficits include cyclosporine-induced encephalopathy, anticholinergic intoxication, and opioid ingestion. Infectious causes include bacterial meningitis, cryptococcal meningitis, tuberculous meningitis, Lyme disease and syphilis. Non-infectious etiologies include carcinomatous meningitis, neurosarcoidosis, pseudotumor cerebri, brain trauma, CNS tumors, and diabetic neuropathy (usually involving cranial nerves III, IV, VI or VII). Vascular causes include hypertensive encephalopathy, stroke, intracranial hemorrhage, and vasculitis. The diagnostic evaluation includes a complete history, neurologic examination, assessment of electrolytes, glucose, and liver function, neuroimaging, and a lumbar puncture. The patient presented developed cranial nerve deficits due to cryptococcal meningitis and responded to antifungal treatment. Cryptococcal meningitis, which rarely occurs in HIV-infected patients whose CD4 count is greater than 100, causes cranial nerve deficits in 20% of cases, even in the absence of classic signs of meningitis such as fever and headache. The course of cryptococcal meningitis is often insidious, with waxing and waning symptoms and signs. Cryptococcal meningitis should be considered in any patient with HIV and a low CD4 count who presents with cranial nerve deficits.

FENCING ZEBRAS: A TRAGEDY OF THE COMMON. G. Wishik¹; G. Stallings¹; J. Wiese¹. ¹Tulane University, New Orleans, LA. (Tracking ID #136087)

LEARNING OBJECTIVES: 1. Recognize the clinical presentation of thrombocytopenia. 2. Recognize that severe iron deficiency can cause thrombocytopenia. **CASE:** A 48-year-old woman presented with two weeks of a lower extremity rash and mouth bleeding. She noted no preceding viral illness and no gastrointestinal bleeding. She had hemorrhagic bullae on the oral mucosa and lower extremity petechia. The spleen was not palpable. Her hemoglobin was 5 g/dl; the platelet count was 4,000; and the WBC was normal. Her stool was guaiac-positive. The Coombs test was negative, the serum LDH and bilirubin were normal. There were no schistocytes on peripheral smear; there were no spherocytes. Her ferritin was 2.3. Her bone marrow had normal erythrocytic and granulocytic lines, but rare megakaryocytes. Prednisone was started for presumed ITP. A total of twenty-four units of platelets were administered. Each time, the platelet count would rise and sustain for two days, and then decline. Intravenous iron supplementation was initiated on day four of hospitalization. After the seventh day, the platelet count rose to a normal range. Coagulation studies were normal throughout hospital stay, as was her creatinine.

DISCUSSION: Thrombocytopenia commonly presents with mucosal bleeding and petechiae in dependent regions. Diseases that accelerate the consumption or destruction of platelets are the most common etiologies of thrombocytopenia: splenomegaly, DIC, TTP and ITP. Our patient had no evidence of splenomegaly; she had no fever, schistocytes, or renal or neurologic dysfunction to suggest TTP. The normal PT/PTT excluded DIC. ITP seemed likely but did not explain the anemia. The absence of hemolysis excluded Evan's syndrome, a rare variant of ITP, in which red cells are also destroyed by IgG antibodies. A bone marrow biopsy is critical in distinguishing ITP from primary marrow failure. The finding of amegakaryocytosis in this case was consistent with pure platelet aplasia, or platelet aplasia secondary to severe iron deficiency. Although thrombocytopenia due to iron deficiency is rare, iron deficiency is not. The severity of iron deficiency (ferritin of 2.3) suggested that her iron stores had declined below the threshold needed for normal platelet maturation.

FEVER AND ABDOMINAL PAIN IN A NEUTROPENIC PATIENT. T.L. Krisko¹; L. Lu¹; M.M. Abuzahra¹. ¹Baylor College of Medicine, Houston, TX. (Tracking ID #135596)

LEARNING OBJECTIVES: 1. Distinguish among the various causes of fever and abdominal pain in neutropenic patients. 2. Recognize the association of typhlitis with neutropenia. 3. Properly treat and manage typhlitis.

CASE: A 58-year old gentleman with a history of metastatic lung adenocarcinoma presented with complaints of fever and abdominal pain one week after his first cycle of chemotherapy. The pain was cramping, localized to the suprapubic region without radiation, and worsened with defecation. On the day prior to admission, he developed a fever of 101.2 °C. In the ED, the patient had a large bloody bowel movement and was found to have an absolute neutrophil count (ANC) of 770. His vital signs revealed temperature of 101.6 °C, blood pressure of 100/80, heart rate of 100, and RR of 18. Pertinent physical findings included a soft, tender suprapubic region without rebound tenderness or guarding. Rectal exam was deferred due to low ANC. His absolute neutrophil count (ANC) of 770 dropped to 380 on the following day. Computerized tomography of the abdomen revealed "marked bowel-wall edema" of the cecum extending up into the ascending colon at the hepatic flexure. Based on his neutropenia and the radiographic findings of cecal enterocolitis, the diagnosis of typhlitis was made. The

patient was treated with intravenous cefepime and metronidazole and put on bowel rest. His ANC then began to recover, and the patient was advanced to a liquid diet on hospital-day five.

DISCUSSION: Typhlitis (from the Greek Word "typhlon," or cecum) is a necrotizing enterocolitis of the cecum occurring mainly in immunosuppressed patients with neutropenia. It is a relatively rare GI disease in which normal and pathogenic organisms penetrate the lining of the bowel wall, leading to inflammation, pain, and eventually necrosis. Typhlitis occurs with the confluence of three factors which include a damaged gut mucosal barrier from cytotoxic drugs, an altered GI flora, and profound neutropenia. Thus, it is most frequently observed in the immunocompromised patient, most notably in cancer patients 10-14 days post chemotherapy. Rates of prevalence and incidence are poorly studied, though examination of pediatric leukemia cases demonstrates evidence of typhlitis in up to 46% at autopsy. Patients often complain of abdominal pain, fever, and possibly nausea, vomiting, and watery or bloody diarrhea. The differential diagnosis of such a presentation should include appendicitis, appendiceal abscess, pseudomembranous enterocolitis, diverticulitis, and ischemic colitis. Diagnosis is often aided by abdominal imaging via ultrasound or computerized tomography, which demonstrates mural thickening and edema of the cecum that may extend into the ascending colon and terminal ileum. CT is the preferred test since it has a lower false-negative rate of diagnosis (15%) as compared to ultrasound (23%). If the patient is unstable with peritonitis or shock, management should be immediate surgery to resect the affected portion. Otherwise, conservative medical therapy is recommended with bowel rest, broad spectrum antibiotic coverage, and intravenous fluids. Mortality can reach 40% in patients with typhlitis, with the majority of deaths occurring due to bowel wall perforation and subsequent sepsis. If perforation can be prevented, the prognosis is much improved. In conclusion, typhlitis must be in the differential diagnosis in a neutropenic patient presenting with fever and abdominal pain.

FLAPPING TOILET SEATS AND FLAPPING HANDS: AN INTERESTING CAUSE OF MENTAL STATUS CHANGES. N. Milojkovic¹; E. Anish¹; D. Zalsenski¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #134316)

LEARNING OBJECTIVES: 1.To recognize hyperammonemic encephalopathy as a complication of valproate therapy. 2.To diagnose valproate-related hyperammonemic encephalopathy 3.To treat valproate-related hyperammonemic encephalopathy and understand the rationale behind the use of oral carnitine therapy

CASE: A 68-year-old male with a history of cluster headaches, iron deficiency anemia, B12 deficiency, tremors, dyslipidemia, and hypertension was admitted with progressive confusion for a month. His wife found the patient in the bathroom extremely confused, after she heard him repeatedly banging the toilet seat up and down. The patient demonstrated little insight into the change in mental status reported by his wife. Additionally, he denied any fever, chills, headache, cardiopulmonary symptoms, urinary symptoms, and no weakness or paresthesias in his extremities. His medications included: valproate, topiramate, loperamide, gemfibrozil, vitamin B12, iron and vitamin B6. Physical exam revealed a lethargic male with normal vital signs. His speech was slow, but not dysarthric. Mini-mental status exam score was 17/30. His cranial nerves were intact. He had no focal weakness or sensory deficits. A resting tremor was present in both upper extremities and bilateral asterixis was present. Diagnostic studies including: CBC, serum chemistries, RPR, B12, TSH, urinalysis, toxicology screen, CXR, brain CT and brain MRI were all unremarkable. His valproate level was therapeutic at 64.8 (50-100). An EEG demonstrated generalized slowing. The only abnormality found was a serum ammonia level of 127 umol/L. Having excluded other causes of hyperammonemia, this abnormality was attributed to his valproic acid use. The patient was diagnosed with valproate-related hyperammonemic encephalopathy (VHE). His valproate was stopped and he was started on oral L-carnitine supplementation. Over the next several days, his confusion cleared and his level of alertness improved dramatically.

DISCUSSION: The differential diagnosis of mental status changes in a patient can be quite broad. In this case, the presence of asterixis led to the measurement of a serum ammonia level. Additional testing helped to exclude more common causes of hyperammonemia, such as liver disease. Symptomatic hyperammonemia with valproate therapy is termed VHE. Patients often display confusion, lethargy, vomiting, and increased seizure frequency. Progression to stupor, coma, and death may occur. The severity of encephalopathy does not correlate with valproate levels, which may be in the normal range. The elevation in serum ammonia with valproate use has been attributed to a metabolite of the drug - propionic acid, which inhibits an enzyme essential for the elimination of ammonia through the urea cycle. Valproate may also interact with carnitine, which is critical for mitochondrial long-chain fatty acid metabolism. The actual mechanism of hyperammonemia-induced neurotoxicity has not been clearly elucidated. Theories implicating alterations in GABA and glutamine transmission in the CNS have been proposed. If VHE is suspected the valproate therapy should be stopped immediately. Additionally, oral carnitine is recommended given the concern that carnitine deficiency may play a role in the development of valproate-induced hyperammonemia.

FOCAL SEGMENTAL GLOMERULOSCLEROSIS WITH ANASARCA IN MORBID OBESITY. M. Singh¹; S. Reddymasu¹; J. Huang¹; J. Blondin¹. ¹Louisiana State University Medical Center at Shreveport, Shreveport, LA. (Tracking ID #136187)

LEARNING OBJECTIVES: 1) To recognise obesity as the cause of focal segmental glomerulosclerosis in primary care practice. 2) To recognise obesity related

focal segmental glomerulosclerosis (O-FSGS) as the cause of hypoalbuminemia and anasarca in African American.

CASE: A 35 yo African-american female with no significant past medical history presented with generalized anasarca and nocturia for 5 months. Her outpatient medications were Lasix and KCl. Physical exam revealed normotensive, morbidly obese patient with BMI of 60 and generalized anasarca. The cardiopulmonary exam was unremarkable. Pertinent lab data: BUN 11 mg/dl, Cr 1.1 mg/dl, glucose 90 mg/dl, HbA1c 5.7%, total protein 4.4 mg/dl, albumin 1.4 mg/dl, total cholesterol 248 mg/dl, triglycerides 160 mg/dl, HDL 70 mg/dl, LDL 138 mg/dl and proteinuria of 27.13 grams/24 hrs. CXR and EKG were normal. Hepatitis panel and HIV tests were negative. Renal biopsy demonstrated focal segmental glomerulosclerosis with no immunohistochemical evidence of immune complex deposits.

DISCUSSION: FSGS in obesity is a consequence of glomerular hyperfiltration and glomerular hypertension. The contributing factors include excess tissue turnover, functional leptin deficiency, reduced NaCl delivery to macula densa, lipotoxicity and insulin resistance with hyperinulinemia. This entity is more common in middle aged caucasians. O-FSGS can cause nephrotic range proteinuria but rarely causes hypoalbuminemia and edema. Early diagnosis of this entity for early interventions is important because 50% of these cases progress to renal failure. The key to management is weight reduction. ACE inhibitors and statins. Overt evidence of metabolic syndrome cannot be substantiated, though suspected, in this case because of the common occurrence of hyperlipidemia in the Nephrotic syndrome. This case was unique in that FSGS occurred in morbidly obese African-american with massive proteinuria, hypoalbuminemia and edema.

FROM MRSA TO VISA TO VRSA - THE AGE OF VANCOMYCIN RESISTANCE. B. Tellwala¹; H. Shishodia¹; B. Taqui¹. Temple University, Philadelphia, PA. (Tracking ID #135073)

LEARNING OBJECTIVES: 1. To review the definition of vancomycin-intermediate Staphylococcus aureus (VISA) and vancomycin-resistant Staphylococcus aureus (VRSA). 2. To discuss the risk factors associated with the emergence of methicillin resistant S. aureus (MRSA) infections resistant to vancomycin. 3. To review the current guidelines for prevention and treatment for VISA and VRSA. **CASE:** A 77-year-old African American male was admitted with generalized weakness, fever and body aches. The patient complained of neck and shoulder pain. On exam T 98.3°F, P 105 bpm, BP 136/67, RR 18/min. The patient's physical exam was notable only for tenderness of his right shoulder. His WBC count was 17.5. Overnight the patient spiked a fever 102.4°F. Blood cultures drawn were positive for MRSA. He was started on vancomycin but he continued to spike fevers with persistent MRSA bacteremia. Gentamicin was added for 7 days with no response. An echocardiogram found no evidence for endocarditis. An arthrocentesis of his right shoulder demonstrated turbid fluid, RBCs 372,500, WBCs 62,500 with 94% segs, cultures were positive for MRSA. Despite arthroscopic debridement and wash out of his shoulder, he continued to have fevers and MRSA bacteremia. Repeat antibiotic sensitivity demonstrated MRSA with vancomycin MIC of <0.5 µg/ml but he continued to be bacteremic. Rifampin was added to his antibiotic regimen and vancomycin was discontinued. The patient improved and was discharged with a 4-week regimen of daptomycin and rifampin.

DISCUSSION: Vancomycin-intermediate Staphylococcus aureus (VISA) is defined by having a MIC of 8-16 µg/ml, and vancomycin-resistant Staphylococcus aureus (VRSA) has a MIC of >32 µg/ml. There are also strains of MRSA that are referred to as "heteroresistant." These strains are susceptible to vancomycin with a MIC <4 µg/ml, but they contain subpopulations of organisms for which the MIC of vancomycin is in the intermediate range. VISA and VRSA infections are rare. Risk factors associated with isolation of VISA and VRSA are prolonged vancomycin use, hemodialysis dependence and indwelling catheters. Patients who are infected with or colonized with MRSA and who receive vancomycin frequently are at highest risk. Detection of VISA and VRSA isolates can be difficult. VISA are slow growing and may not appear as the primary strains on culture until >2 days of incubation. In addition, laboratories currently may fail to reliably recognize VISA strains. There is no standardized therapy for VISA and VRSA infections. Removal of the infected indwelling hardware and debridement of infected site is indicated. Infections have been treated with fluoroquinolones, trimethoprim-sulfamethoxazole, clindamycin and minocycline. Rifampin can be added to therapy if the isolates are susceptible. In addition to these older drugs, new drugs with antistaphylococcal activity such as quinupristin-dalfopristin, linezolid, and daptomycin can be used. Given the history of rapid spread of MRSA and VRE in hospitals, prevention plays a crucial role in management. Recommendations include contact precautions, rapid isolation of infected patients, and the education of health care workers, the patient and the patient's family. Our patient did not have VISA but may have had heteroresistant strains of MRSA. Our case provides an example of the clinical dilemmas of caring for patients who do not respond to vancomycin therapy at a time when the emergence of VISA and VRSA as pathogens have the potential to become more prevalent.

GALACTORRHEA: DOUBLE TROUBLE. A. Himme¹; C. Lai¹. University of California, San Francisco, San Francisco, CA. (Tracking ID #135493)

LEARNING OBJECTIVES: 1) Recognize that hyperprolactinemia is the most common cause of galactorrhea. 2) Review the diagnosis and management of antipsychotic-induced hyperprolactinemia. 3) Recognize that the cause of hyperprolactinemia may be multifactorial.

CASE: A 31 year-old woman with schizophrenia presented to her PCP complaining of milky white discharge from both breasts for 1 month. Her menstrual

cycles were regular, and she denied change in vision or mood. She had been on a stable regimen of daily risperidone 5mg, diphenhydramine 50mg, and citalopram 20mg for at least one year. Physical examination was normal except for expressible bilateral milky white breast discharge. A prolactin (PL) level was 197 µg/L (normal range 1.4-24.2 µg/L). TSH, creatinine, LH, and FSH levels were normal. Urine pregnancy test was negative. A brain MRI revealed a 4 mm lesion in the pituitary gland. Her psychiatrist was consulted to assess the possibility of discontinuing risperidone; however, given the difficulty controlling her psychosis in the past with failure to respond to several other antipsychotics, it was strongly advised that she continue on the current regimen. Her medications were not adjusted, and 3 months later she reported some spontaneous improvement but not full resolution of the galactorrhea. A repeat PL level at that time was 98 µg/L.

DISCUSSION: The differential diagnosis for hyperprolactinemia, which is the most common cause of galactorrhea, includes causes that are physiologic (stress, nipple stimulation, pregnancy) and pathologic (pituitary adenoma, medications, chronic renal failure, hypothyroidism). Conventional antipsychotic medications and some atypical antipsychotics such as risperidone and amisulpride elevate PL levels in up to half of patients. In contrast, clozapine, olanzapine, quetiapine, aripiprazole, and ziprasidone are considered PL-sparing agents. Features that suggest medication-induced hyperprolactinemia include use of a commonly implicated antipsychotic, onset of symptoms shortly after starting the drug or increasing the dose (median of 3 weeks), a PL level less than 70 µg/L, and lack of sellar space-occupying signs or symptoms. Definitive proof of causality occurs when the PL level normalizes after discontinuation of the offending medication. The decision to treat antipsychotic-induced hyperprolactinemia is based on the risks and benefits to the individual patient. Treatment options include reducing the dose of the drug, switching to a PL-sparing agent, adding bromocriptine or cabergoline (which has not caused worsening psychosis in case reports and a pilot study), or using oral contraceptive pills if amenorrhea is the primary symptom. If discontinuation of the medication is not possible and the PL level is markedly elevated, additional causes of hyperprolactinemia should be considered and a brain MRI is warranted to rule out a prolactinoma. Several case reports have suggested a possible relationship between the use of antipsychotics and growth of prolactinomas. Treatment indications for a prolactinoma include infertility, hypogonadism, and size >1 cm (macroadenoma). Careful follow-up is an appropriate strategy for our patient, who has two reasons for her hyperprolactinemia: risperidone and a prolactinoma.

GROUP B STREPTOCOCCUS: FROM THE HEART TO THE BRAIN. A. Kalyanasundaram¹; M. Velagapalli¹. Geisinger Medical Center, Danville, PA. (Tracking ID #135265)

LEARNING OBJECTIVES: 1) To present an unusual manifestation of a rare clinical entity - Group B bacterial endocarditis manifesting as meningitis. 2) TEE might be indicated in patients with meningitis with rare isolates.

CASE: 55-year-old female presented at an outside hospital with headache and new-onset right-sided weakness. She was awake and in no acute distress. She was febrile at 39 degrees C, had nuchal rigidity and had right hemiparesis on exam. Her initial CT scan was negative. Lumbar puncture showed gram-positive cocci in chains, protein - 129 mg/dL, glucose - 60 mg/dL, white cells - 1,890/cu mm. She was given ceftriaxone and ampicillin and transferred to our hospital where she received ceftriaxone, vancomycin, and steroids. MRI showed multiple embolic infarcts in the brain, especially in the left frontal region. Transesophageal echocardiogram (TEE) showed a 10 mm vegetation on the anterior mitral valve leaflet along with 2+ mitral regurgitation. CSF grew Group B Streptococcus and patient was felt to have group B Streptococcal mitral valve endocarditis with multiple CNS emboli causing meningitis. Antibiotics were switched to penicillin G and gentamicin. Multiple blood cultures remained negative. Her condition gradually improved and eventually, she was discharged to a nursing home in stable condition.

DISCUSSION: Group B Streptococcus endocarditis is a rare clinical entity, generally characterized by an acute onset, large vegetations and a high incidence of complications, including embolization. Meningitis due to Group B streptococcus (GBS) is extremely uncommon in adults. However, an increase in the incidence of Group B streptococcal meningitis in South East Asia has been recently reported. Bacterial meningitis as a presenting manifestation of infective endocarditis is very unusual but potentially a major complication. Although evidence of a stroke mandated a TEE in our patient, a literature review suggested that association of endocarditis and meningitis is not infrequent. One study quoted that up to 7% of patients with infective endocarditis developed meningitis as a complication. Group B streptococcal endocarditis leading to an embolic stroke with concomitant meningitis has not been reported to the best of our knowledge. We suggest that a TEE might be indicated whenever bacterial meningitis is caused by an uncommon organism. This hypothesis needs further validation by larger studies.

HOW GOOD IS COLONOSCOPY? N. Alim¹; L. Lu¹. Baylor College of Medicine, Houston, TX. (Tracking ID #133033)

LEARNING OBJECTIVES: Learning Objectives: 1. Review the clinical presentation of colon cancer. 2. Recognize the methods of diagnosis of colon cancer and their limitation. 3. Discuss the use of virtual colonoscopy as a modality of colon cancer screening.

CASE: A 52 year old Hispanic male with history of GERD and benign polyps on a colonoscopy done in the previous year presented with an eight month history of cramping and burning epigastric and lower abdominal pain and a 35 lb weight

loss. Patient was initially evaluated in clinic, where EGD showed chronic gastritis with *Helicobacter Pylori* infection. Despite H. Pylori antibiotic therapy, his symptoms persisted. He then sought medical care in Mexico where a repeat colonoscopy was unremarkable; however, a computerized tomography of the abdomen revealed multiple hepatic lesions consistent with metastases and a 60 mm mass adjacent to the splenic flexure. CEA level was elevated at 35.3 (normal 0-3ng/mL) and AFP was normal at 1.19 (normal 0-9). He was advised to undergo exploratory laparotomy, but his family refused and brought him back to the U.S. for further evaluation. On admission to our facility, his physical examination was significant for a distended and diffusely tender abdomen. Rectal exam was guaiac positive. Lab results revealed hemoglobin of 9.6 with MCV 75. Patient underwent a CT-guided liver biopsy that showed metastatic adenocarcinoma consistent with colonic primary. Patient was offered surgery but given the overall poor prognosis of his metastatic cancer, patient and his family opted for home hospice.

DISCUSSION: Colonoscopy is the current gold standard for screening of colorectal cancer with sensitivity and specificity both reported as 99-100%. Along with polypectomy, it has been shown to reduce the incidence and mortality of colorectal cancer. As demonstrated by our patient, there have been reported cases of colonic cancers within 1-4 yrs after a "normal" colonoscopy. The miss rate of colonoscopy varies from 24% up to 40% depending on the operator and the nature or size of the lesions. The risk of missing a lesion is inversely related to size of the lesions, especially ones <1 cm. The prevalence of cancer in adenomas <1 cm is reported to be minimal although some recent studies show the prevalence of neoplasm in such diminutive polyps to vary from 49.2 to 60.4%. In addition, flat lesions can easily be missed and require scopes with high magnification and resolution. Based on the National Polyp Study (New England Journal of Medicine, 1993), a single colonoscopy performed in patients with previous colonoscopic polypectomy missed approximately 24% of patients bearing recurrent adenomas. CT colonography, also known as virtual colonoscopy, is being investigated for colon cancer screening. This method uses spiral or helical CT or MRI to image an air-filled distended colon. Currently, there is no definite data regarding the sensitivity and specificity of this technique, but some studies suggest sensitivity ranging from 34-94% and specificity 95-96% for adenomatous polyps >1 cm. It does appear to be relatively successful at detecting polyps 6 mm or larger. However, virtual colonoscopy is still in its investigational state. Even though colonoscopy remains the gold standard for screening of colorectal cancer, other imaging modalities must be utilized concurrently when the clinical suspicion is high for colorectal cancer despite a negative colonoscopy.

HYPERCALCEMIA AND PENILE CANCER. J.N. Wainaina¹; T. Thenappan¹; N. Awadallah²; H. Friedman³. ¹St. Francis Hospital, Evanston, IL; ²Saint Francis Hospital Resurrection Health Care, Northbrook, IL; ³St. Francis Hospital of Evanston, Evanston, IL. (Tracking ID #133360)

LEARNING OBJECTIVES: 1. Recognize that hypercalcemia can occur in association with any squamous cell carcinoma 2. Identify humoral hypercalcemia of malignancy as one of several mechanisms of hypercalcemia.

CASE: A 75 year old Caucasian man was admitted directly from his primary physician's office with week's history of lethargy and increased drowsiness as reported by his wife and daughter. This was associated with constipation, mood changes and unsteadiness of gait. His past medical history was significant for metastatic squamous cell carcinoma of the penis. Physical examination revealed a well-built elderly gentleman who was alert and in no apparent distress at the time. He was afebrile, mildly tachypneic with a tachycardia and elevated blood pressure. Neurologically he was oriented to place, person and time but had short term memory deficits. There were no focal neurological deficits. His entire right lower extremity was swollen and he had a large erythematous, firm and indurated groin mass that was moderately tender. The rest of his examination was normal. Laboratory studies revealed an elevated WBC count of 17.7 with a left shift and a hypercalcemia of 13.7. Other electrolytes, BUN/creatinine, liver function tests, alkaline phosphatase isoenzymes, leucocyte alkaline phosphatase, PTT and PT/INR were all within normal range. Subsequent investigations revealed a normal PTH and elevated PTHrP of 11.9 (normal is <1.3). The patient responded well to intravenous fluids, pamidronate and furosemide with gradual lowering of the calcium levels to normal. He also received antibiotics for cellulitis of the right thigh. FNA and later biopsy of the inguinal lymph nodes confirmed metastatic squamous cell carcinoma of the penis.

DISCUSSION: Humoral hypercalcemia of malignancy can occur with virtually any squamous cell cancer. It has been reported most commonly with squamous cell carcinomas of the lung as well as the liver, esophagus and renal pelvis. The predominant cause is PTH-related peptide. It resembles PTH in terms of genetic sequence, structure and action. Unlike PTH, which is purely a circulating hormone, PTHrP is a local messenger within tissues. It functions as a polypeptide that gives rise to several biologically active peptides each of which may have its own receptor. It has intracrine, juxtacrine, paracrine and possibly endocrine effects. As a result, while primary hyperparathyroidism is a relatively mild disease, hypercalcemia in patients with cancer often occurs abruptly, is severe and portends a poor prognosis. Definitive management requires treatment of the causative malignancy.

HYPERPARATHYROIDISM PRESENTING AS QUADRICEPS TENDON RUPTURE. A. Chekuri¹; S.B. Glick¹. ¹John Stroger Hospital of Cook County, Chicago, IL. (Tracking ID #134810)

LEARNING OBJECTIVES: 1. Recognize the clinical presentation of quadriceps tendon rupture; 2. Recognize quadriceps tear as a possible initial presentation

of hyperparathyroidism; 3. Identify diseases that may cause quadriceps tendon rupture.

CASE: A 25 year-old man presented with a three month history of bilateral knee pain and swelling following a fall onto the left knee while playing soccer. He had no past medical history. On examination, the knees were swollen; the patient was unable to perform straight leg raise bilaterally. There was no palpable suprapatellar defect. Left knee x-ray revealed subperiosteal bone resorption, but no fracture. Right knee x-ray was unremarkable. Quadriceps tendon rupture was suspected based on the inability to actively extend the knees against gravity. Left knee MRI revealed complete quadriceps tendon tear. Right knee MRI revealed a high grade partial tear of the distal quadriceps tendon. Bloodwork showed creatinine 2.8 mg/dL, calcium 12.8 mg/dL (8.5-10.5), and alkaline phosphatase 2593 units/L (50-120). Neck ultrasound revealed a large parathyroid adenoma. The patient underwent parathyroidectomy. Surgical repair of the quadriceps tendons is planned.

DISCUSSION: Rupture of the quadriceps tendon most often occurs when the quadriceps contracts forcefully while the knee is flexed with the foot fixed in place. Patients present with a painful, swollen knee; they are unable to bear weight on the affected limb. The inability to actively and completely extend the knee against gravity with contraction of the quadriceps muscle (extensor lag) suggests the diagnosis. Passive range of motion of the knee is normal. A soft tissue defect may be palpable superior to the patella (suprapatellar gap), though knee swelling and effusion may prevent its palpation. The patient is unable to stand. Knee x-ray may reveal patella baja, a low riding patella. When the diagnosis is uncertain, MRI can confirm it. Rupture of the quadriceps tendon is usually seen in patients over the age of 40. Two-thirds of patients with unilateral tendon rupture have no underlying etiology; one-third have an underlying medical disease such as hyperparathyroidism, chronic renal failure, systemic lupus erythematosus, rheumatoid arthritis, diabetes or gout. Quadriceps tendon rupture may be the initial manifestation of this disease. Virtually all patients with non-traumatic bilateral rupture of the quadriceps tendons have an underlying medical disease. Chronic renal failure is seen in 36% of these patients; hyperparathyroidism is seen in 33%. Hyperparathyroidism causes dystrophic calcifications and subperiosteal bone resorption that weaken the junction between tendon and bone. Quadriceps tendon tears are frequently misdiagnosed due to inability to palpate the suprapatellar defect due to knee swelling and lack of knowledge of the presentation of tendon rupture. An accurate initial diagnosis is important, because timely repair of the tendon improves surgical outcome. Treatment of quadriceps tendon rupture includes immobilization of the knee in extension to prevent knee instability, ice, compression, analgesia, and prompt referral to an orthopedic surgeon. Evaluation for underlying medical disease is particularly important in patients with bilateral, non-traumatic rupture of the quadriceps tendons.

HYPOTHYROIDISM PRESENTING AS HOFFMANN'S SYNDROME. L. Subramanyam¹; M. Eapen¹; H. Friedman²; B. Singh³. ¹St. Francis Hospital, Evanston, IL; ²Skokie, IL; ³St. Francis Hospital, Evanston, IL. (Tracking ID #136189)

LEARNING OBJECTIVES: This case report emphasizes the importance of recognizing muscle cramps and myalgias as one of the presenting manifestations of profound hypothyroidism. Diagnosis can be made clinically without any EMG studies or muscle biopsy. Prompt treatment with thyroid supplements leads to complete resolution of symptoms.

CASE: Our patient is a 48-year-old man who presented with one-month history of profound myalgias, weakness of lower extremities and limitation in his daily activities. He denied other symptoms like depression, constipation or cold intolerance. Physical examination showed significant hypertrophy of the calf muscles bilaterally and delayed relaxation of the deep tendon reflexes. Cardiac, lung, abdomen and thyroid exam was essentially benign. Initial laboratory tests showed a mildly elevated serum creatinine of 1.5 mg/dL. Serum Creatinine Phosphokinase (CPK) enzyme level was 7147 IU/L. He was treated with intravenous hydration for the rhabdomyolysis. A serum Thyroid Stimulating Hormone (TSH) level obtained showed a value of 199.04 mIU/ml. A repeat test confirmed the TSH level. His serum free Thyroxine (T4) was <0.4 ng/dL. He was started on levothyroxine 100 mcg once a day. The CPK level and the serum creatinine decreased with hydration and he was discharged home. At follow-up 6 weeks later, his symptoms were barely noticeable, his serum TSH was 98 mIU/ml and serum free T4 was 0.76 ng/dL. The patient was advised to continue the levothyroxine and to have regular follow-up.

DISCUSSION: Hoffmann's syndrome is a constellation of clinical symptoms and signs, which include muscle hypertrophy, stiffness, weakness and associated muscle cramps. There are case reports, though rare, of hypothyroidism presenting as Hoffmann's syndrome. Thyroid hormone influences cellular energy production. Lack of this hormone causes impaired glycogenolysis leading to post exertional rhabdomyolysis. The symptoms and signs of Hoffmann's syndrome respond promptly to thyroid replacement. This case stresses the importance of considering hypothyroidism as one of the causes of muscle weakness and myalgias, as the diagnosis is straightforward and the treatment is very simple.

I JUST CAN'T STAND ANYMORE. E.L. Garcia¹; H. Whelan¹. ¹Oliver View - UCLA Medical Center, Sylmar, CA. (Tracking ID #134566)

LEARNING OBJECTIVES: 1) Recognize diabetic peripheral neuropathy as a cause of profound, mixed sensory and motor neuropathy. 2) Recognize fecal incontinence as a possible complication of diabetic neuropathy.

CASE: A 62 year old man with a past medical history of diabetes mellitus, presented with the gradual inability to ambulate or stand due to subjective, bilat-

eral progressive proximal leg weakness as well as a gradual loss of reliable bowel control for the preceding few months. He specifically described difficulty knowing when he needed to have a bowel movement, and found that he was often taken quite by surprise after soiling himself. His past medical history was significant for a relatively new diagnosis of diabetes mellitus 11 months earlier, when he required amputation of a gangrenous toe. HbA1C was 11.7 at initial diagnosis, however, all subsequent measurements revealed a HbA1C less than 6. Physical exam was significant for notable wasting of his proximal leg muscles, with a modest, but concomitant reduction in strength. Strength was intact in his distal lower and upper extremities. He had reduced sensation to pinprick and light touch in bilateral lower extremities. He had reduced patellar reflexes, and absent ankle reflexes. His labs were significant for anemia, with hemoglobin 8.8 and hematocrit 25.8, correction of which did not result in improvement of his symptoms. Chemistry was unremarkable, both creatine kinase and aldolase were not elevated. MRI of his spine did not reveal any cord compressing lesion. EMG and nerve conduction studies revealed widespread chronic denervation in all muscle groups, more pronounced in distal muscle groups, with evidence of sensory and motor mixed axonal neuropathy, all characteristics of severe diabetic neuropathy. There was no evidence of myopathy.

DISCUSSION: Diabetic neuropathy is a well known, and well described complication of diabetes that encompasses both peripheral sensory and motor, as well as autonomic neuropathy. The natural progression of peripheral sensory neuropathy is typically gradual and insidious, developing in a "stocking glove" type distribution. Motor neuropathy can manifest as diabetic amyotrophy, a proximal neuropathy causing pain, weakness and muscle wasting. Autonomic neuropathy can manifest as tachycardia, gastroparesis, and orthostatic hypotension. Less often described is fecal incontinence, which is estimated to occur in approximately 20% of diabetic patients. The etiology of fecal incontinence in diabetes is complex and poorly understood, likely involving both autonomic and peripheral neuropathy. This type of incontinence may not necessarily be limited to profoundly neuropathic patients, so may be overlooked by the physician. It is likely that this condition is underreported by patients, particularly in its early stages.

I LOST 185 POUNDS BUT MY JOINTS STARTED TO HURT: INFLAMMATORY POLY-ARTHRITIS AFTER GASTRIC BYPASS. T.J. Hamilton¹; J. Hefner¹; J. Potter². ¹Beth Israel Deaconess Medical Center, Boston, MA; ²Harvard University, Boston, MA. (Tracking ID #132579)

LEARNING OBJECTIVES: 1) Review the differential diagnosis and evaluation of a patient with asymmetric polyarthropathy; 2) Describe the association of gastric bypass with arthritis.

CASE: PP is a 38-year-old Puerto Rican man who presented with pain in his neck, shoulders, low back, hips, knees, and ankles. He had low-grade fevers, morning stiffness lasting two hours, but no rash. He developed knee and ankle effusions. PMH is notable for Roux-en-Y gastric bypass for morbid obesity in 1995, with weight reduction from 350 to 165 lbs. From 1995 to 2002, he underwent multiple abdominal surgeries but at no time did pathologic specimens reveal evidence of inflammatory bowel disease. CBC, chemistries, liver and thyroid function, vitamin D and parathyroid hormone levels were unremarkable. Hepatitis and Lyme serologies, SPEP, RF, ANA, ANCA, HIV antibody, and tests for gonorrhea and chlamydia were negative. Complement levels were high normal, ESR 118, and C-reactive protein 24. MRI of the cervical and lumbar spine and pelvis revealed a small right hip effusion, with edema in the sacroiliac joints, sacrum and pubic symphysis. Analysis of the right knee effusion revealed 30,000 white blood cells with 82% neutrophils, no crystals, and gram stain and culture were negative. A TTE revealed no evidence of endocarditis. Blood cultures were repeatedly negative. Upper GI series revealed post-gastric bypass anatomy, with no filling of the excluded gastric remnant. He did not improve with high dose prednisone and was subsequently treated with doxycycline. Colchicine has thus far been effective in controlling his pain.

DISCUSSION: Diseases of the gastrointestinal tract are often linked to arthritis syndromes: ulcerative colitis and Crohn's disease are well-known examples. Less often appreciated is the fact that a "bypass-arthritis syndrome" is a potential complication of jejunoileal bypass surgery, performed frequently in the 1960s and 1970s as a treatment for morbid obesity. The responsible pathophysiologic mechanism is felt to be bacterial overgrowth and mucosal changes in the blind loop of small bowel. This leads to increased absorption of bacterial antigens. Immune complexes form in the circulation, and have been found in the synovial fluid and synovial membranes of affected joints. Surgical excision of the blind loop, or reversal of the bypass has been reported to lead to complete cure of arthritis. Antibiotics to treat the bacterial overgrowth have also reduced joint symptoms in some cases. PP has a gastric and duodenal blind loop, which may contain flora which differs from an intestinal blind loop. This may explain why he did not respond to doxycycline. Inflammatory arthritis is a well-documented complication of jejunoileal bypass, mediated by immune complex deposition from bacterial overgrowth in the blind loop. A similar entity may also develop after gastric bypass surgery; this possibility should be entertained after other causes of inflammatory arthritis are excluded.

IATROGENIC HYPERMAGNESEMIA CAUSING PARALYSIS AND RESPIRATORY FAILURE IN A HOSPITALIZED ELDERLY WOMAN. C.R. Ghaly¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #133376)

LEARNING OBJECTIVES: To recognize serious clinical manifestations of hypermagnesemia. To learn therapeutic options available for treating hypermagnesemia.

CASE: An 85 year-old woman was hospitalized in the intensive care unit for symptomatic bradycardia after an accidental overdose of diltiazem, metoprolol, and amiodarone. After failing medical therapy, she was taken to the cardiac catheterization laboratory for placement of a temporary transvenous pacing wire. On return to the ICU, nursing staff found her to be lethargic and attributed this to the 2 mg of ativan and 100 mcg of fentanyl she had received during the procedure. Over the next two hours, the patient was increasingly somnolent. Physical exam revealed an obtunded elderly woman, unresponsive to basic commands. She was afebrile, blood pressure 105/65, pulse 70 and paced, respiratory rate 6, and oxygen saturation 100% on 4 liters oxygen. Pupils were 4 mm and reactive; her gag reflex was weak. She did not respond to painful stimuli and was hyporeflexic. Heart and lung examination was unremarkable. Abdomen had hypoactive bowel sounds. Review of the patient's morning labs were notable for a normal CBC and electrolytes except for a magnesium (Mg) of 1.3 mg/dL, which was repleted with 3g Mg sulfate, and a BUN and creatinine of 43 mg/dL and 1.8 mg/dL respectively. Stat arterial blood gas was pH 7.14, PCO2 74, PO2 110. Her electrocardiograph, with the pacing wire temporarily disabled, revealed sinus bradycardia at a rate of 26, PR interval of 0.25s, QRS of 0.12s, and QTc of 0.458s. Chest radiograph was clear. The patient was intubated for hypercapnia and airway protection. Labs sent before intubation were unchanged from the morning except for a Mg of 3.9 mg/dL. The patient was treated with IV calcium gluconate and aggressive intravenous fluids. After 24 hours, her Mg level had fallen to 2.8 mg/dL and her paralysis, lethargy, and hypoventilation had resolved sufficiently to be extubated. She was discharged home in good condition three days later, after resolution of her other medical issues.

DISCUSSION: Hypermagnesemia is commonly found among routinely screened hospitalized patients. As Mg is eliminated through the urine, hypermagnesemia frequently occurs in patients with renal insufficiency after large doses of oral (e.g. laxatives) or intravenous Mg-containing medications. Mg acts as a calcium channel blocker and causes neuromuscular blockade with effects on smooth, cardiac, and skeletal muscle as well as on peripheral and central nerves. Correlation between serum Mg levels and clinical symptoms is extremely variable and this patient demonstrated severe symptoms at Mg concentrations lower than typically described. The first manifestations of hypermagnesemia, beginning normally around 5 mg/dL, include nausea, lethargy, hyporeflexia, and flushing. Obtundation, areflexia, and cardiovascular instability including hypotension and bradycardia occur classically at levels from 7-12 mg/dL. Finally, paralysis (including the diaphragm) and complete heart block are typically seen at concentrations above 15 mg/dL. Cardiac arrest generally occurs above 20 mg/dL though patients have been reported to survive with higher values. The mainstay of therapy is supportive care including intubation for hypoventilation and airway protection as well as intravenous fluids to enhance elimination, calcium to counteract Mg's action as a calcium channel blocker, and dialysis in extreme cases for direct removal of Mg from the blood.

ICANCER. G.R. Berger¹; J. Perras¹. ¹Dartmouth-Hitchcock Medical Center, Lebanon, NH. (Tracking ID #135728)

LEARNING OBJECTIVES: 1) Recognize an uncommon presentation of metastatic lung cancer and 2) Recognize the role and limitations of percutaneous CT-guided biopsy in diagnosing lung cancer

CASE: A 77 year old Caucasian female with a history of diabetes, congestive heart failure, atrial fibrillation, and a recent diagnosis of giant-cell arteritis was admitted to the hospital with dyspnea and confusion. She had been seen recently by her primary care physician with right eye pain and was started on prednisone for presumed giant-cell arteritis. Before her scheduled temporal artery biopsy she was admitted to the hospital with dyspnea due to heart failure. She complained of increasing lower extremity edema, orthopnea, lethargy, anorexia, weight loss, night sweats, and right-sided headaches. She denied cough, hemoptysis, chest pain, palpitations, visual changes, or bone pain. Her mother had gastric cancer and her father died of pneumonia. She had a 15 pack year smoking history with her last cigarette being over 30 years ago. She had not traveled outside the area recently and had never been exposed to tuberculosis. On admission she was afebrile and hemodynamically stable. There was a discrete subcentimeter white lesion on the right sclera with significant conjunctival injection. Breath sounds were decreased and there was egophony over the right upper lobe and crackles at both bases. She had dependent edema. Admission labs were significant for a white blood cell count of 16,000, a hemoglobin of 11.5, and a sodium of 130. In addition to mild pulmonary edema, a chest x-ray showed a right upper lobe mass. Chest CT further defined the mass as 12 cm, pleural-based, and extending to the right hilum. A CT-guided biopsy showed necrotic lung tissue and rare fungal elements, notable for the absence of malignant cells. A CT Scan of the head was normal and a bone scan was negative for lesions suspicious for metastases. A detailed ophthalmologic evaluation of the ocular lesion was most consistent with metastatic cancer with resultant iritis. A repeat CT-guided biopsy confirmed squamous cell carcinoma of the lung, and an MRI of the orbits and brain revealed a lone ring-enhancing lesion of the right ciliary body extending to the sclera and lacrimal gland. Her ocular symptoms were treated with steroid eye drops and improved significantly. She was discharged to a skilled nursing facility closer to home with plans for oncologic care at a local hospital.

DISCUSSION: The question of disseminated fungal infection was raised during this patient's hospital course based on her prednisone use, ocular lesion, fungus recovered from her initial biopsy, absence of malignant cells, and negative bone scan. The correct diagnosis was made after repeating the biopsy based on suspicion generated from an ophthalmologic exam. The sensitivity of percutaneous CT-guided biopsy in diagnosing lung cancers has been reported as 74%. Ocular

metastases are rare and most commonly seen in malignant melanoma, central nervous system lymphoma, and choroidal tumors in the pediatric patient population. A Medline search did not reveal any prior case reports of lone ocular metastasis with primary squamous cell cancer of the lung.

IDIOPATHIC MYELOFIBROSIS CAUSING INTRACRANIAL TUMOR BY EXTRAMEDULLARY HEMATOPOIISIS. L.C. Murillo¹; A. Mirza¹. ¹Geisinger Medical Center, Danville, PA. (Tracking ID #135153)

LEARNING OBJECTIVES: 1. Recognize that extramedullary hematopoiesis (EMH) is a characteristic feature of idiopathic myelofibrosis. 2. Identify that in extreme form EMH can cause thoracic, abdominal and intracranial masses. 3. Include EMH in differential diagnosis of intracranial masses in patients with known conditions that predispose them to EMH

CASE: A 65-year-old female presented to emergency department with sudden onset of headache, nausea, vomiting and mental status change. She had past history of Idiopathic Myelofibrosis diagnosed 5 years prior to presentation, which had been treated with Hydroxyurea. On presentation patient was comatose with Glasgow Coma Scale of 8/15, and she had a dilated right pupil that was nonreactive to light. Patient had prominent hepatosplenomegaly and rest of the physical examination was unremarkable. Initial laboratory results revealed White Cell Count of 18,400 cells/mm3 (normal range 4-10.8) and Platelet Count of 601,000 platelets/mm3 (normal value 150,000-300,000). An urgent Magnetic Resonance Imaging scan of brain was obtained which showed multiple left frontal, right frontoparietal and left temporal hyper-dense lesions with gadolinium enhancement. There was right-sided deviation of the frontal horn of the left lateral ventricle. Patient underwent urgent decompressive surgery of brain. A reddish mass measuring 6 × 5 cm with a shiny congested surface was discovered under the Dura mater. The mass was soft and no necrotic or pustular material was identified. Histopathology evidenced pleomorphic clumps of erythroid and myeloid precursors together with megakaryocytes in the leptomeninges with cerebral parenchyma infiltration through the Virchow-Robin spaces. These findings were consistent with extramedullary leptomeningeal hematopoiesis. Decompressive surgery resulted in significant improvement of patient's symptoms. **DISCUSSION:** Idiopathic Myelofibrosis is uncommon disease but it commonly causes extramedullary hematopoiesis of spleen, liver and lymph nodes. This occurs to compensate for the fibrous tissue replacement of bone marrow. In rare instances EMH can result in formation of tumor like soft tissue masses in abdominal, thoracic or intracranial cavities or viscera. Extramedullary hematopoiesis of meninges is extremely rare and may trick unwary clinicians into mistaking it for a primary or secondary brain malignancy. It is essential to make this differentiation since management of these conditions differs significantly. Thus physicians should consider extramedullary hematopoiesis in differential diagnosis of intracranial masses in patient with known condition that may predispose them to extramedullary hematopoiesis.

IMMUNE THROMBOCYTOPENIC PURPURA FOLLOWING MMR VACCINATION. C. Rajpal¹; A. Nautiyal¹; T. Woodlock¹; M. Disalle¹. ¹Unity Health System, Rochester, NY. (Tracking ID #135551)

LEARNING OBJECTIVES: To understand Immune Thrombocytopenic purpura, a rare complication of MMR vaccination.

CASE: A 34-year-old healthy white man developed sudden onset of extensive gum bleeding and fine rash on his arms and legs. He received measles mumps and rubella (MMR) combined vaccine 2 months prior. This was his first MMR vaccination, a requirement for college admission. He had no constitutional symptoms to suggest a viral infection and no history of familial hemorrhagic disease. He had petechiae present in a gravitationally dependent fashion with an otherwise unremarkable exam. His platelet count was strikingly low at 1 X10³/mm with a hemoglobin 15.7 gm/dl, hematocrit 45.6%, WBC 7.8 × 10³/mm3 with a normal differential count. Peripheral smear revealed a complete absence of platelets with no schistocytes or RBC fragments. Additional laboratory data included normal metabolic, thyroid, and hepatitis panels. Also notable were a negative Epstein Barr antigen, HIV 1 and negative antiplatelet IgG antibody but a positive CMV IgG antibody. Chest radiograph and CT scan of the abdomen and pelvis were normal. MRI of the left humerus was consistent with an old benign osteochondroma. In view of active bleeding, platelet transfusion was attempted with no improvement in platelet count. He was treated with intravenous methylprednisolone and oral danazol and showed some improvement with a pre-discharge platelet count of 42 × 103/cu mm. Regular outpatient follow up has confirmed complete recovery with normal counts and resolution of thrombocytopenia in 2 months, consistent with the acute form of immune thrombocytopenia.

DISCUSSION: The reported occurrence of acute ITP following MMR vaccine appears rare; using MEDLINE we found a total of 47 cases of acute ITP following MMR vaccination in the literature.(1) The vaccine safety committee of the Institute of Medicine (IOM) concluded in 1994 that a causal relationship does exist between live virus MMR vaccination and thrombocytopenia.(2) There is a temporal association between acute thrombocytopenic purpura and MMR vaccine.(3) Two mechanisms postulated for this effect are bone marrow suppression (4) and increased platelet destruction (4). This vaccine-associated case further adds fuel to the biologic plausibility of a relationship. This case highlights a serious complication related to a beneficial vaccination. Fortunately, all prior cases of acute ITP following MMR vaccination have had a benign course with full recovery. Because ITP is believed to be an immunologic disorder, it seems logical that an attenuated live virus vaccine would have the antigenic capacity to initiate ITP in a susceptible patient.

IT IS NOT JUST A WHITE WOMAN DISEASE! A. Kolpakchi¹; L. Lu¹. ¹Baylor College of Medicine, Houston, TX. (Tracking ID #135562)

LEARNING OBJECTIVES: 1. Review the diagnostic criteria for autoimmune hepatitis (AIH). 2. Recognize the occurrence of autoimmune hepatitis in young African American males. 3. Learn the treatment of autoimmune hepatitis.

CASE: A 36 year-old African American man presented to the outpatient clinic for evaluation of abnormal liver function tests. Patient was in good health with no specific complaints, but on review of systems, he reported easy bruisability and intermittent gum bleeding. He denied history of illegal drug use, heavy alcohol drinking, and multiple sexual partners. Physical examination was significant only for chest wall telangiectasias, bilateral gynecomastia, and splenomegaly. Pertinent laboratory studies revealed AST 640 IU/L (10-42), ALT 868 IU/L (10-63), albumin 3.1 g/dl, PT 19, INR 1.62, and platelets 87K/cmm. Further tests showed negative hepatitis serologies; negative anti-mitochondrial antibodies (AMA); normal ceruloplasmin, alpha1-antitrypsin, and iron studies; positive ANA with 1:320 nucleolar pattern (<1:40); and positive anti-smooth muscle antibodies (SMA) with 1:320 (1<20). Computed tomography of the abdomen showed cirrhotic morphology of the liver, splenomegaly, and evidence of portal hypertension. Patient was diagnosed with autoimmune hepatitis and started on prednisone with improvement of his liver function tests. **DISCUSSION:** Autoimmune hepatitis (AIH) is a chronic inflammation of the liver of unknown etiology. It affects women (female to male ratio of 8:1) between third and fifth decade of life. It is most common in those geographic regions where the HLAB8 and DR3 phenotypes are prevalent and the carriage frequency of chronic viral hepatitis is low. HLAB8 and DR3 phenotypes are most common in Northern Europe. Among Caucasian Northern Europeans, the mean annual incidence of AIH is 1.9 per 100,000. Data on autoimmune hepatitis in African Americans is limited; from the literature search, there was only one retrospective series of 27 patients. It is rare in African Americans males with male to female ratio of 2:25. African Americans tend to present at earlier age with more advanced disease, usually with liver cirrhosis. There are no conventional immunologic markers pathognomonic for this disease. Codified by international panel, diagnostic criteria for definite AIH include normal alpha 1- antitrypsin phenotype; normal ceruloplasmin level; normal serum iron and ferritin levels; no markers of active infection with hepatitis A, B, and C viruses; daily alcohol <25 g/day and no recent use of hepatotoxic drugs; predominant serum aminotransferase abnormality; ANA, SMA or anti-LKM1 > 1:80; no AMA; and no biliary lesions, granulomas, or prominent changes suggestive of other diseases. Treatment of AIH is with immunosuppression with corticosteroid. If corticosteroid does not produce a remission, azathioprine may be added. Cyclosporine has been used in patients who fail corticosteroid therapy. The course of AIH is fluctuant, marked by episodes of deterioration. The ultimate effect is inevitably cirrhosis with rare exception. The mean survival is 12.2 years. Mortality is greatest during the first two years when the disease is most active. After an initial remission following 2 years of corticosteroid therapy, 1/3 achieves a 5-year remission, but 2/3 relapses and has to be retreated. Despite the late presentation and more advanced AIH, African Americans respond well to higher dose of immunosuppressive therapy. The prognosis in African American is unknown.

IT'S JUST PHARYNGITIS, OR IS IT? E. Yoon¹; J. Morial¹. ¹Tulane University, New Orleans, LA. (Tracking ID #136104)

LEARNING OBJECTIVES: 1. Recognize clinical presentation and pathogenesis of Lemierre's syndrome. 2. Understand the treatment and the role of anticoagulation in Lemierre's syndrome.

CASE: A 21 year-old man presented with nine days of fever, chills and sore throat. He reported three days of shortness of breath and a tender right-neck mass. He had a history of pharyngitis two months earlier that had resolved without treatment. His temperature was 103.5 F; blood pressure 128/66 mmHg; heart rate 120; pulse oximetry 100% on room air. He had bilateral tonsillar enlargement with bilateral white exudates. There was a one cm immobile mass at the right cervical area. The lungs examination was normal; there were no heart murmurs or hepatosplenomegaly was appreciated. His WBC was 12.3 with 20 percent bands. HIV and monospot tests were negative. The CXR revealed multiple pulmonary nodules; a subsequent chest CT showed cavitation of the nodules. A CT of the neck revealed a thrombosis of the right external jugular vein. His blood cultures were positive for *Fusobacterium necrophorum*. He was treated with clindamycin and imipenem for three weeks, and underwent thoracotomy with drainage of the empyema and pulmonary abscesses.

DISCUSSION: Lemierre's syndrome is characterized by septic thrombophlebitis of the internal jugular vein(IJV) and metastatic infection caused by an oropharyngeal infection. Invasion of the carotid sheath can result in thrombophlebitis of the IJV, sepsis and distant metastatic infections. The causative organism is *Fusobacterium necrophorum* or *Fusobacterium nucleatum*. The lungs are the most common metastatic target, presenting as pulmonary infiltrates, pleural effusion, abscess, and empyema. Involvements of joints, bones, liver, spleen, skin kidneys and meninges have been reported. The diagnosis of Lemierre's syndrome should be considered when a previously healthy individual presents with symptoms of oropharyngeal infection, fever, and signs of metastatic infection. The diagnosis is confirmed by blood cultures, and radiographic evidence of internal jugular vein thrombophlebitis. Treatment includes three to six weeks of intravenous beta-lactamase resistant penicillins. Close observation for signs of continued sepsis and metastatic infection is essential. The role of anticoagulation is controversial, and there are no controlled studies to assess its value. However, the outcome of most patients is good without it. Surgical ligation or excision of the IJV should be considered in refractory cases, although it is rarely required.

JUST LISTEN TO ME. J. Schieffelin¹; J. Wiese¹. ¹Tulane University, New Orleans, LA. (Tracking ID #136059)

LEARNING OBJECTIVES: 1. Recognize avascular necrosis as a complication of sickle cell trait. 2. Understand that sickle cell trait may cause similar symptoms as sickle cell SS disease.

CASE: A 49 year-old African-American man presented to the clinic following an admission for cellulitis and a right forearm abscess two weeks earlier. On that admission, he had presented with left knee pain and was diagnosed with bursitis. When he presented in the clinic, his arm had improved, but his knee pain was still severe. He had a past medical history of sickle cell trait and rheumatoid arthritis for which he had been taking prednisone 60 mg a day for several years. His knee was edematous but there was no erythema. An arthrocentesis revealed a small amount of sero-sanguinous fluid; the knee was injected with steroids. Two months later, he reappeared in clinic. He had missed his last appointment because he had been hospitalized for severe abdominal pain. An exploratory laparotomy had been performed, but was benign. His knee pain had persisted, and knee radiographs showed severe degenerative joint disease. His hemoglobin was 10 g/dl. An MRI revealed extensive bone infarction of the distal femur, proximal tibia and patella. His electrophoresis revealed Hemoglobin SC disease. A steroid taper and folic acid were administered. He was referred to orthopedics for total knee arthroplasty.

DISCUSSION: The patient's chief complaint must be adequately addressed even when more severe conditions arise. Our patient's chief complaint on initial presentation, and on follow-up examination, was ignored in light of his forearm abscess, thereby delaying treatment that might have obviated his knee replacement. Similarly, past medical history must be evaluated in the context of the chief complaint. Sickle cell trait does not cause avascular bone necrosis with the same frequency as sickle cell SS disease, but it may still cause this complication, illustrating the tautology: "where sickle cell goes, so goes trait." Steroids can exacerbate these complications especially if used for long periods of time.

LEAD AVR-THE NOT SO IGNORED LEAD ANYMORE! V. Gujral¹; A. Lopez-Candales¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #132578)

LEARNING OBJECTIVES: 1. Recognize the importance of Lead aVR in clinical cardiology. 2. Distinguish between Left Main versus Left Anterior Descending versus Right Coronary Artery occlusion using Lead aVR.

CASE: A 69-year-old white male with hypertension was referred for a stress test after complaining of exertional dyspnea while golfing and climbing stairs. He denied associated chest pain, diaphoresis, palpitations, nausea or vomiting. During the test, the patient demonstrated 3mm ST segment depressions in leads II, III, aVF, V5 and V6. In addition, 2mm ST segment elevations in lead aVR and V1 were noted. Cardiac catheterization revealed a 70% left main (LM) stenosis, a 70% mid left anterior descending (LAD) stenosis, a 50% proximal first obtuse marginal stenosis, and a 50% proximal and a 70% distal right coronary artery (RCA) stenosis. Left ventricular function was preserved. Patient underwent a four vessel coronary artery bypass graft. Post-operative course was unremarkable and patient was discharged to home on day seven.

DISCUSSION: Lead aVR reflects information about the right ventricular outflow tract and the basal part of the septum. In practice, however, most electrocardiographers consider lead aVR as giving reciprocal information from the left lateral wall, being already covered by leads aVL, V5 and V6. Therefore, lead aVR has largely been ignored. However, lead aVR can be very helpful in diagnosing a number of different clinical disease entities like acute pulmonary embolism, arrhythmias and acute coronary syndromes. ST-segment elevation in lead aVR can be used to distinguish between LM, LAD and RCA occlusion. YU Fu et al showed that aVR ST elevation predicts LM occlusion, as opposed to LAD/RCA occlusion, with 80.2%/93.7% sensitivity, 87.4%/89.5% specificity, 89.0%/89.0% positive predictive value and 76.0%/94.0% negative predictive value respectively. Yamaji et al distinguished LM group from LAD group looking at aVR ST elevation greater than or equal to that in lead V1. Based on these studies, the ECG changes observed in our patient (2mm aVR ST elevation and aVR ST elevation equal to that in V1) were indicative of LM occlusion that lead to immediate intervention. Although the patient had marked ST depressions, Reddy et al have reported that ST depressions in inferior and lateral leads have about 50% predictive ability for significant coronary artery disease and do not provide any more discriminative ability than pretest probability. It is well known that LM occlusion is one of the most severe lesions associated with coronary artery disease. Furthermore, severe LM disease is nearly always associated with disease of two or three remaining major coronary arteries. Therefore, noninvasive identification of significant LM stenosis using lead aVR is undoubtedly crucial with regards to patient management, prognosis and survival.

LEVOFLOXACIN CAUSING ACHILLES TENDON RUPTURE. S. Kumar¹; M. Rehmetullah¹; G. Gleeson¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #132568)

LEARNING OBJECTIVES: To recognize an uncommon complication from a common medication: achilles tendon rupture as a complication of fluoroquinolone therapy.

CASE: A 61 year old male with past medical history of hypertension and chronic obstructive lung disease presented to the emergency department with right calf pain, swelling and difficulty in walking which had progressively worsened over the last week. He was recently treated with levofloxacin 500 mg orally twice a day for 7 days for acute bronchitis. The symptoms started one day after stopping the antibiotic, when he noticed a snapping sound while getting out of bed. There was

no history of trauma. Physical examination was remarkable for swelling of the right calf and a palpable gap with tenderness in the achilles tendon area. The patient was unable to stand up on toes. Squeezing the calf of right leg while in prone position did not flex the foot. The left leg examination was noticeable for mild tenderness in the achilles tendon area. He had good peripheral pulses and neurological examination was normal. The differential diagnosis included an achilles tendonitis or rupture and deep venous thrombosis. Laboratory studies revealed electrolytes and complete blood count within normal limits. A lower extremity doppler was negative for deep venous thrombosis. MRI of the right leg revealed achilles tendon rupture. He was treated symptomatically with analgesics for pain. The right leg was immobilized and an urgent repair was performed with subsequent physical rehabilitation and recovery.

DISCUSSION: Achilles tendon rupture is very common after trauma while spontaneous achilles tendon rupture usually occurs in elderly with underlying systemic disease or in patients on chronic steroid therapy. Other predisposing factors include aging, rheumatoid arthritis, systemic lupus erythematosus, gout, hyperparathyroidism, chronic renal failure and fluoroquinolone use. The presenting complaint range from insidious onset of pain in lower leg, gait instability to sudden onset of excruciating pain and complete inability to walk. Fluoroquinolones have been shown to cause cartilage damage in immature laboratory animals but the pathophysiology of fluoroquinolone-induced tendinitis and rupture is poorly understood. A systemic reaction to fluoroquinolone is suggested, as involvement of other tendons such as long head of biceps, extensor pollicis longus and supraspinatus has been described. The achilles tendon seems to be more vulnerable due to its weight bearing role and being relatively avascular. The management relies on keeping the casual association of fluoroquinolones and achilles tendinitis or rupture in mind. If symptoms develop fluoroquinolones should be stopped. After splinting, orthopedic consultation is needed for reconstruction options. Physical therapy plays a major role in subsequent rehabilitation efforts. Prognosis is generally good if recognized early and appropriate therapy though results vary in elderly.

LIFE THREATENING HYPOGLYCEMIA INDUCED BY TRIMETHOPRIM/SULFAMETHOXAZOLE. L.C. Murillo¹; A. Mirza¹. ¹Geisinger Medical Center, Danville, PA. (Tracking ID #135183)

LEARNING OBJECTIVES: 1. Recognize that Trimethoprim/Sulfamethoxazole (TMP/SMX) a commonly used antibiotic can cause life-threatening hypoglycemia. 2. Understand the mechanism of TMP/SMX induced hypoglycemia. 3. Identify predisposing conditions for drug-induced hypoglycemia (DIH).

CASE: 86yrs-old female with prior history of congestive heart failure, atria fibrillation and chronic lymphocytic lymphoma was transferred from nursing home (NH) to emergency department (ED) at our tertiary care medical center. She was found by NH staff to have altered mental status and finger stick blood sugar test revealed hypoglycemia with blood glucose level of 40mg/dl (normal range 80-120) prompting her transfer to ED. Patient was being treated with TMP/SMX for urinary tract infection. Physical examination in ED revealed pulse 82/minute, blood pressure 131/82 mmHg, and arterial oxygen saturation 95% on room air. Neurological examination showed patient to be very lethargic and unresponsive to verbal commands, however no focal neurological deficit could be demonstrated. Rest of physical examination was unrevealing. Laboratory tests revealed hemoglobin of 9.7grams/dl (12-15 mg/dl), serum creatinine level 1.0 mg/dl (0.5-0.7 mg/dl) and blood sugar 39 mg/dl. Other tests including serum electrolytes, liver function test, thyroid function test and serum cortisol level were within normal range. Urine test did not detect any sulfonyleureas or their byproducts. Patient had very high serum insulin level 48 uU/ml (3-17) and pro-insulin level 223 Pmol/L (<8.8) despite hypoglycemia suggesting pancreatic islet cells stimulation. TMP/SMX was held and patient was treated with intravenous high concentration dextrose with dramatic improvement in mental status. For first 12 hours she required several bolus doses of 50% dextrose in addition to 10% intravenous infusion to maintain blood sugar in the normal range.

DISCUSSION: Polypharmacy is one of the commonest causes of delirium in elderly patients. DIH is rare in general population but not so uncommon in elderly population. Antidiabetic drugs are the commonest cause however other drugs that can decrease serum blood sugar level are thiazide diuretics, ACE inhibitors, beta-blockers, and some lipid lowering agents. Sulfomethoxazole element of TMP/SMX combination has structural similarity to sulfonyleureas thus TMP/SMX can stimulate insulin release from pancreas with resultant hypoglycemia. In most cases patients have associated precipitating factor like altered renal function, malnutrition or conditions causing high metabolic state like infection. Our case demonstrates that TMP/SMX a commonly used antibiotic can cause life-threatening hypoglycemia regardless of patient's predisposition for hypoglycemia.

LONGUS COLLI TENDINITIS: A REAL PAIN IN THE NECK. A.M. Sanders¹; H. Day¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #132246)

LEARNING OBJECTIVES: 1) Generate a differential diagnosis of neck pain; 2) Recognize the presentation of this unusual pathology as a cause of common neck pain; 3) Distinguish this cause of neck pain from others via its diagnostic radiographic features

CASE: 42 y/o female with treated hypothyroidism presented to the ED complaining of sore throat, progressive neck pain, stiffness and swelling over the preceeding 3 days. Her symptoms quickly progressed to odynophagia and difficulty breathing described as feeling "cut off" from air. She denied headache, fever, chills, or injury. Her symptoms did not improve with intermittent NSAIDs,

antihistamines or narcotics. On exam, oximetry was 100% on room air, and she maintained an impressively fixed, slightly flexed neck position. Her anterior neck was symmetrically enlarged. She was unable to fully open her mouth, but did not have an apparent retropharyngeal bulge. She was diffusely tender along her paraspinal and upper trapezius muscles, but no spasm was noted. She had severely limited range of motion of her neck in all directions due to pain. Laboratory data was unremarkable including a WBC of 9.5 with 80% neutrophils. Initial interpretation of neck CT was that of a retropharyngeal fluid collection. She was started on antibiotics, but had no symptom relief. After her CT was re-evaluated, the diagnosis of longus colli tendinitis was made. She was given a short course of steroids and high-dose NSAIDs with marked symptom improvement.

DISCUSSION: The differential diagnosis of neck pain with stiffness includes cervical strain, cervical spondylosis, degenerative arthritis, meningitis, retropharyngeal abscess, mastoiditis, infectious spondylitis, c-spine injury and post-traumatic muscle spasm. Although unusual, longus colli tendinitis (a.k.a. acute calcific retropharyngeal tendinitis) is a cause of neck pain of which the internist should be aware. The longus colli muscle is a paired neck flexor anterior to the vertebral column from C1-T3. Inflammation from hydroxyapatite deposits in the muscle tendon results in the above typical presentation with progressive neck pain severely limiting range of motion in the absence of causative etiologies. The diagnosis is made radiographically with lateral c-spine x-ray and/or neck CT revealing pathognomonic calcifications along the longus colli tendon commonly associated with soft tissue swelling or a fluid collection in the retropharyngeal space anterior to the calcifications. This finding can be easily confused with a retropharyngeal abscess and lead to unnecessary treatments. Treatment is conservative management with high-dose NSAIDs. Symptoms typically resolve over 1-2 weeks. Self-limited symptoms along with relatively conservative management may make longus colli tendinitis an underrecognized diagnosis. This patient was discharged home after a 3-day hospitalization and had marked improvement of her symptoms within 7 weeks on naproxen and complete resolution of radiographic findings within 2 months.

LOW VOLTAGE REQUIRES A HIGH INDEX OF SUSPICION. S. Makani¹; A. Khan¹; M.P. Hudson². ¹Henry Ford Hospital Detroit, Detroit, MI; ²Henry Ford Heart & Vascular Institute, Detroit, MI. (Tracking ID #135640)

LEARNING OBJECTIVES: 1. Formulate a differential diagnosis of left ventricular hypertrophy. 2. Recognize clinical and electrocardiogram (ECG) characteristics associated with infiltrative cardiomyopathy/cardiac amyloidosis.

CASE: A 42 year old female with no significant past medical history was diagnosed with hypertrophic obstructive cardiomyopathy by an outside physician based on echocardiographic findings. She was started on treatment with torsemide and verapamil. Two months after initial diagnosis, she presented with decompensated heart failure leading to admission in the cardiac intensive care unit for refractory hypotension, pulmonary edema, and hypoxia. Physical exam showed severe hypotension, echymosis, diffuse pulmonary crackles, II/VI mitral regurgitation murmur, and lower extremity edema. ECG showed atrial fibrillation, possible anterior infarct, and low QRS voltage. These ECG findings prompted consideration of a possible infiltrative cardiomyopathy in this young patient with rapidly worsening heart failure. Transthoracic echocardiogram showed an LV ejection fraction of 30%, severe concentric ventricular hypertrophy, and global hypokinesia. Right and left heart catheterization was performed and demonstrated cardiogenic shock, severely elevated right-sided heart pressures consistent with advanced restrictive cardiomyopathy, and no coronary artery disease. A myocardial biopsy was performed which was consistent with cardiac AL amyloidosis. A bone marrow biopsy was also performed. The patient was started on oral melphalan and prednisone. Following three days of chemotherapy, she suffered a bradycardic arrest and expired.

DISCUSSION: Primary AL amyloidosis demonstrates clinical evidence of cardiac involvement in up to 50% of patients. Cardiac involvement in any type of amyloidosis carries a poor prognosis with a median survival of approximately eight months compared to forty months without cardiac involvement. Patients present with signs and symptoms of advanced congestive heart failure with either systolic dysfunction or restrictive cardiomyopathy hemodynamics. ECG features of low QRS voltage and "pseudo-infarction" are important diagnostic features which may distinguish cardiac amyloidosis and is seen in greater than 50% and 83% of cases, respectively. A low voltage electrocardiogram along with concentric hypertrophy should prompt a clinician to consider myocardial biopsy and/or blood and urine protein studies. Treatment with oral melphalan and prednisone has favorable response rates after one year of therapy. Caution should be used with digoxin and calcium channel blockers, due to case reports of clinical decline after administration of such medications. Heart transplantation and allogeneic bone marrow transplantation can be offered only to selected patients. With response to treatment occurring after one year and a median survival of eight months, early definitive diagnosis by biopsy and subsequent directed therapy is essential.

LYMPHEDEMA TARDA. S. Fordan¹; L. Lu¹. ¹Baylor College of Medicine, Houston, TX. (Tracking ID #135163)

LEARNING OBJECTIVES: 1). Review the causes of bilateral leg edema. 2). Recognize the diagnosis and prognosis of primary lymphedema. 3). Learn the treatment options available for primary lymphedema.

CASE: A 51 year old African-American male, with a history of hypertension controlled on hydrochlorothiazide, presented with a four month history of progressive bilateral lower extremity swelling. He reported that his leg swelling had initially improved with tight stockings, but then continued to worsen. He denied

any systemic symptoms including shortness of breath, orthopnea, paroxysmal nocturnal dyspnea, weight gain, or temperature instability. Physical examination was significant only for ++ non-pitting edema bilaterally below his knees, left greater than right. Outpatient work up revealed a normal venous Doppler exam of both lower extremities and normal laboratory studies including electrolytes, urinalysis, CBC, thyroid and liver function tests. Due to his progressive edema, he was admitted for further work-up. An echocardiogram and abdominal and pelvic computerized tomography were unremarkable. Lymphoscintigraphy was then performed, which revealed lymphosclerosis in both legs, left greater than right, consistent with primary lymphedema. The patient was treated with compression stockings, a pneumatic compressive device, and physical therapy.

DISCUSSION: The differential diagnosis of lower extremity edema includes deep vein thrombosis, chronic venous insufficiency, lipidemia, nephrotic syndrome, congestive heart failure, protein-losing enteropathies, myxedema, hepatic dysfunction, secondary lymphedema (from infections, malignant obstruction, lymph node dissection, and radiation), and primary lymphedema. Primary lymphedema classically presents with unilateral non-pitting edema. Bilateral extremity presentation is rare. Primary lymphedema is categorized by the age of onset. Lymphedema tarda develops in adulthood. While its etiology is unknown, some patients are thought to have presentations of late onset hereditary lymphedema, while others are believed to have an autoimmune destruction of lymphatic channels. Although lymphedema has a wide prognostic spectrum, it is usually a chronic disease that will worsen over time. Lymphoscintigraphy, a non-invasive nucleotide test with 73%-100% sensitivity and 100% specificity, has become the gold standard for diagnosis and should be performed in any patient with unexplained edema, as demonstrated in our patient. Treatment modalities are focused on controlling swelling and usually consist of a combination of non-fatiguing exercise/massage (to improve lymphatic drainage), compressive garments, and intermittent pumps. There are no definitive studies suggesting any pharmacological benefit, including diuretics or coumarin. Surgical removal of the lymphedematous tissue and subcutaneous fat or the creation of a lymphatic anastomosis bypassing the stenosis is reserved for extreme cases.

MALIGNANCY MASQUERADING AS A RHEUMATOLOGICAL DISEASE. N. Aggarwal¹; P. Jain¹; R. Schapira¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #134006)

LEARNING OBJECTIVES: 1. To recognize the rheumatologic manifestations of malignancy 2. To consider malignancy in geriatric patients presenting with musculoskeletal symptoms.

CASE: An 83 year-old man presented with low back pain, lower extremity weakness, generalized aches with morning stiffness and right temporal headache for 6 weeks. He had several outpatient visits and was treated with analgesics. The ESR during one of these visits was noted to be > 100 mm/hr and a rheumatological disorder was a primary diagnostic consideration. Past medical history was remarkable for hypertension, diabetes mellitus, coronary artery disease and significant tobacco use. The patient was admitted for further evaluation of a disorder such as polymyalgia rheumatica (PMR) or temporal arteritis. Exam showed good pulses in temporal artery with no tenderness to palpation. There was focal point tenderness in the cervical and lumbar vertebrae. Lower extremity strength was slightly diminished (4-/5) L>R. Admission laboratory results were significant for normocytic anemia, WBC of 6200/μl, alkaline phosphatase of 209 U/L and calcium of 11.3 mg/dl. The ESR remained > 100 mm/hr and C-reactive protein (CRP) was 11.4 mg/dl. ANA, rheumatoid factor and CPK were normal. On admission, rhonchi were noted in the right lower and middle lung fields and a chest radiograph showed a right hilar prominence with a faint interstitial infiltrate in the right base. Chest CT-scan revealed a 7 cm mass in the right lower lobe and lytic lesions in the lumbar spine. A bronchoscopic biopsy was positive for non-small cell carcinoma. A bone scan showed multiple metastatic lesions in the thoracic and lumbar vertebrae, femur and ribs.

DISCUSSION: Musculoskeletal symptoms are common in the elderly and many specific disorders need to be considered. PMR, mostly seen adults over 50 years of age, is a clinical syndrome with subacute or chronic onset of symmetric aching discomfort and stiffness of neck, shoulders and hips, low grade fever, weight loss and lethargy. Muscle strength is usually normal. Lab studies usually show an ESR of > 40 mm/h, elevated CRP and normocytic anemia. Dramatic response to a single dose of steroids helps in diagnosis. The diagnosis of PMR requires the exclusion of other specific diseases like rheumatoid arthritis, polymyositis, hypothyroidism, chronic infection and malignancy. Malignancy in the elderly can present with various rheumatological symptoms which may be the sole presenting symptoms. A malignancy-associated illness mimicking PMR causes asymmetric involvement at sites typical of PMR, has characteristic abnormalities on radioisotope scanning and shows no significant response to steroids. Other rheumatologic manifestations of malignancy include dermatomyositis, polymyositis, hypertrophic osteoarthropathy with digital clubbing and polyarthralgia, peripheral arthritis, proximal muscle weakness due to Lambert-Eaton syndrome and back or bone pain from skeletal metastases. The combination of weakness and elevated ESR and CRP focused the diagnostic efforts on a primary rheumatological disorder, delaying the diagnosis of lung cancer. Malignancy can masquerade as a rheumatological disorder.

MEDICAL AND ETHICAL ISSUES IN CONSIDERING VALVE REPLACEMENT FOR AN IV DRUG USING PATIENT WITH INFECTIOUS ENDOCARDITIS. S. Fernando¹; S.S. Vasudevan¹; G.H. Tabas¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #132588)

LEARNING OBJECTIVES: 1. Evaluate ethical implications of non-surgical management of infectious endocarditis (IE). 2. Examine optimal treatment strategies

in a patient with advanced endocarditis with multiple organ system involvement.

CASE: A 38 year old woman with a 23-year history of intravenous (IV) drug use presented two days after she had received two months of antibiotics for IE at another hospital. She complained of pleuritic chest pain and had 2/6 systolic and 2/6 diastolic murmurs but no signs of heart failure. We began vancomycin and gentamycin when gram-positive cocci grew from the blood. An echocardiogram revealed 1 cm and 0.4 cm vegetations on the tricuspid and aortic valves respectively, severe tricuspid regurgitation, an ejection fraction of 40% and no evidence of para-aortic abscess. CT revealed multiple septic pulmonary emboli and a splenic infarction. MRI of the brain showed an early abscess in the right frontal lobe. Due to her failure to respond to antibiotics and a large mobile vegetation with persistent septic embolization, she met the usual criteria for surgical intervention. The ethical dilemma was whether to continue medical management or proceed with surgical valve replacement in this patient with active in-hospital substance abuse and high risk of re-infecting a new prosthetic valve. We obtained an ethics consult to help us examine the implications of not operating on this patient. Due to her high-predicted surgical mortality and risk of re-infecting the prosthetic valve, she was managed both medically and psychosocially for the remainder of the hospitalization.

DISCUSSION: Our vignette demonstrates that in patients with endocarditis, ethical considerations including patient lifestyle choices, can significantly impact the management. Usual indications for valve surgery in IE include valvular dysfunction with CHF, aortic abscess, fungal IE and failure to respond to antibiotics. Although IV drug use is not an absolute contraindication for valve replacement/repair, mortality is directly related to patients' continued use. In such patients the mortality is as high as 100% for tricuspid valve replacement, and 30% for tricuspid valvulotomy. In our patient, mortality for valve surgery was estimated at 5-6%, but her risk of re-infection was estimated at >50%. Because of our patient's requirement for simultaneous aortic valve replacement, her expected mortality would increase to >10-15% and her risk of re-infection would be much greater than 50%. After considering our patient's active IV drug use and the associated risk of re-infection, her age and her opposition to prolonged hospitalization, the ethics committee agreed with medical treatment for our patient.

MENINGOCOCCEMIA WITHOUT MENINGITIS IN A 20-YEAR-OLD PREGNANT WOMAN. W. Gellad¹; G. Ruiz¹; D. Li¹. ¹Brigham and Women's Hospital, Boston, MA. (Tracking ID #132507)

LEARNING OBJECTIVES: Recognize the clinical features of meningococcal disease in the absence of meningitis.

CASE: A 20-year-old woman who was 16 weeks pregnant presented with 1 day of fevers and headache. She felt well until the evening prior to admission, when she developed sore throat and headache. Overnight, she further developed fever to 102°F, nausea, and diffuse arthralgias involving her right thumb, right knee, and left ankle. She presented to her ob/gyn and was noticed to have a petechial rash, so she was sent to the emergency department for evaluation. She has a son who recently had a fever and rash and was diagnosed with a viral illness. In the ED she had a temperature of 99.7°F, blood pressure 103/64, pulse 115. She appeared tired but non-toxic. Her neck was supple and her fundoscopic and neurological exams were normal. Her joint exam was notable for a swollen right thumb, and a tender, swollen right knee without clear effusion. Her skin exam revealed a diffuse pink blanching maculopapular rash on her trunk and extremities, with scattered petechiae involving her palms and soles. Laboratory data were notable for a WBC count of 18,000 with 84% polys and 11% bands. Blood cultures were drawn and she received antibiotics. Her lumbar puncture had 1 WBC in both tubes with no RBCs, and gram stain and culture were negative. Considering her clinical appearance, the negative LP, and her sick contact, she was admitted to the floor with the presumed diagnosis of a viral exanthema. She remained well on the floor. The next morning, blood cultures returned with gram negative diplococci, at which point she was placed on droplet precautions and started on ceftriaxone. The cultures eventually returned with *Neisseria meningitidis*, of intermediate susceptibility to penicillin. The patient continued to do well with negative follow up cultures and was sent home after two days with a 14-day course of ceftriaxone. Her close contacts, including her son, were prophylaxed. No hospital employees were prophylaxed. Her baby suffered no ill effects.

DISCUSSION: The clinical manifestations of meningococcal disease can be varied, ranging from occult bacteremia to fulminant disease. Between 30-40% of patients with meningococcal disease have positive blood cultures without evidence of meningitis. This patient had the classic findings of meningococemia with fever, nausea, arthralgia, and the characteristic rash. However, in the absence of objective findings of meningitis, the rash and clinical symptoms could be consistent with other entities, including viral exanthema and Rocky Mountain Spotted Fever. In cases such as these, with the patient's son recently being diagnosed with a 'viral illness' and the patient's relatively benign appearance, a high degree of clinical suspicion for meningococemia needs to be maintained. Patients with meningococemia may progress to purpura fulminans, which can be associated with the rapid onset of hypotension and multiorgan failure. In this particular patient and her son it might be interesting to investigate for complement deficiency, which plays a role in increased susceptibility to meningococcal infection and is often associated with milder forms of the disease.

MENTAL STATUS CHANGES & DKA IN A T4 PARAPLEGIC WHO DIDN'T RESPOND TO INITIAL THERAPIES: A CASE OF NON-CONVULSIVE STATUS EPILEPTICUS. C.L. Skagen¹; A.I. Wilk¹. ¹University of Wisconsin-Madison, Madison, WI. (Tracking ID #136078)

LEARNING OBJECTIVES: 1. Recognize non-convulsive status epilepticus (NCSE) in patients with mental status changes. 2. Recognize prognosis of patients with NCSE.

CASE: A 45 y.o. man with T4 paraplegia, type 2 diabetes, narcotic-dependent low back pain and recent suicidality presented after being found somnolent and lying in bed in his own feces. He received naran en route and became more alert. On exam, he was lethargic but he opened his eyes to his name and occasionally answered yes-or-no questions. He was afebrile and tachycardic with spastic paralysis in the lower extremities. He exhibited subtle facial grimaces and repetitively raised his arms to his head. No tongue trauma was noted. Laboratory studies revealed diabetic ketoacidosis, hypernatremia, prerenal azotemia, and UTI. CXR, toxicology screen and non-contrast CT scan of the head were unremarkable. Despite 12 hours of IV hydration, insulin and antibiotic therapy, his depressed mental status persisted. He was inconsistently oriented to person and place, and answered only simple questions appropriately. An EEG demonstrated diffuse, episodic epileptiform discharges that coincided with his waxing and waning mental status and confirmed non-convulsive status epilepticus (NCSE). IV lorazepam and fosphenytoin led to resolution of mental status changes and patient's status by EEG.

DISCUSSION: NCSE is defined by multiple or continuous seizures by EEG with nonconvulsive clinical symptoms, most typically cognitive or behavioral changes and minimal facial or eye myoclonus. As in our patient, NCSE may present with abnormalities as subtle as aphasia or confusion and include metabolic derangements such as elevated BUN/creatinine ratio, leukocytosis, and hyperglycemia. A prospective study addressing clinical features of NCSE reports that ocular movement abnormalities (nystagmus, eye deviation) have a specificity of 86% and sensitivity of 50%. When these abnormalities are combined with remote risk factors for seizures (stroke, tumor, previous neurosurgery, dementia, meningitis), the sensitivity improves to 100% but the specificity declines to 55%. Although the mortality associated with NCSE is much debated, a retrospective study of 100 patients with NCSE reported a mortality rate of 27% in those with an acute medical illness, compared with 3% in patients with a history of epilepsy and no concomitant illness. This may be attributed to the high mortality rate of the medical illness itself in combination with the respiratory complications of depressed mental status. Additionally, patients with severe mental status impairment had a 39% mortality rate versus 7% for those with mild impairment. In conclusion, it is imperative that individuals with mental status changes who do not respond to initial therapies be evaluated for status epilepticus. Treatment should be tailored to the urgency and severity of the patient's medical illness.

METHICILLIN-RESISTANT STAPHYLOCOCCUS AUREUS INFECTION IN A MEDICAL RESIDENT. J. Miller¹; M. Costello¹. ¹Temple University, Philadelphia, PA. (Tracking ID #135698)

LEARNING OBJECTIVES: 1. Recognize the risk of colonization of healthcare workers caring for methicillin-resistant *Staphylococcus Aureus* (MRSA) infected patients. 2. Understand the treatment necessary for eradication of MRSA carriage in a colonized individual.

CASE: Patient is a 26 yo white man who was previously healthy with no chronic medical problems. The patient works as a medical resident in an urban university hospital and had recently cared for two patients with an invasive MRSA soft tissue infections. He presented to his primary doctor complaining of a painful erythematous region on his anterior left thigh, painful swelling in his perirectal region and subjective fevers. The lesions had developed over approximately ten days with the area on his thigh beginning with what looked like a small pimple. On exam he was afebrile and nontoxic appearing. He had an 7x7 cm area of erythema and warmth on his left thigh, and an area of tenderness and fluctuance in the perirectal region. He was diagnosed with cellulitis and a pilonidal cyst and sent him home with TMP-SMX. Later that night he presented to a local hospital after the abscess in his perirectal region ruptured with bloody purulent drainage. On exam he had multiple kissing lesions - areas of erythema approximately a half centimeter in diameter running down the medial aspects of his legs corresponding to areas of contact. It was also noted the lesion on his left thigh had developed central fluctuance. He underwent incision and drainage of the abscess on his left thigh as well as on what remained on the abscess in the perirectal region. Cultures from the wounds grew MRSA with an antibiotic resistance pattern identical to the organisms isolated from his patients. He was admitted to the university hospital where he worked and was treated with intravenous Vancomycin for 4 days. Multiple blood cultures were collected which were negative. Patient was discharged home and completed a course of Rifampin and TMP-SMX. Patient was prescribed Mupirocin ointment to be applied to the nares twice daily for 5 days and he was instructed to shower with Hibiclenz soap for 5 days. He was also advised to clean all linens and clothes upon arrival home. **DISCUSSION:** MRSA is thought to spread within hospitals by direct contact between patients and hospital care providers. One study demonstrated a 17% transmission rate between infected patient and healthcare worker gloves. A statistic which is made more pertinent given that glove use compliance is only 75% among healthcare workers as a whole, being lowest in doctors with a compliance rate of 27%. Currently many hospitals, including the one in which the patient in this case worked, do not implement contact precautions for patients with known MRSA infection. The current recommendation for eradication of MRSA in healthcare workers is mupirocin nasal ointment and antiseptic soap use for the whole body for five days. Research has been done as to the treatment necessary for eradication of MRSA in healthcare workers colonized with MRSA. It has demonstrated that eradication with topical antibiotics/antiseptics such as Mupirocin was successful in most cases. However there is evidence that instructing colonized workers to clean and disinfect their households increases the success rate for eradication.

MICHAEL'S THYROID FACTOR, MORE THAN JUST A STORM: A CASE AND REVIEW OF THE LITERATURE. L. Perez¹; L.A. Corsino¹; K. Grullon¹; E. Montesino¹; A. Diez¹; W. Wiense¹. ¹Wayne State University, Detroit, MI. (Tracking ID #131684)

LEARNING OBJECTIVES: To recognize the development of Multiple Organ System Failure as a complication of thyroid crisis. To enhance awareness of potentially deleterious effects of over the counter herbal supplements.

CASE: A 46-year-old female office worker presented with dyspnea and diarrhea for a week. Over the preceding eight months, she had intermittently noticed palpitations, hand tremors, heat intolerance, orthopnea, gradual protrusion of her eyes and involuntary weight loss of more than 20 pounds. She had been using an herbal supplement called "Michael's Thyroid Factor", which contains 300% of the recommended daily value of iodine. Examination revealed a diaphoretic woman with temperature of 100.2°F and regular pulse of 150 beats/min. Exophthalmos, without lid lag or stare and a non-enlarged, non-tender thyroid gland without bruits were noted. There was pretibial pitting edema in the lower extremities bilaterally. Her neurological examination demonstrated bilateral hand resting tremor and normal deep tendon reflexes. EKG showed sinus tachycardia. Her TSH level was <0.002uIU/mL and total T3 508ng/dL. Basic metabolic and hematological profiles were normal. Standard treatment for hyperthyroidism with antithyroid drugs and propranolol was initiated. Five hours after being admitted, she became agitated, developed marked dyspnea and hypoxemia accompanied by hypotension that required vasopressors and intravenous fluids. She was transferred to the MICU and mechanical ventilation was started. She developed lactic acidosis with an arterial pH level of 7.152, PCO2 22.2torr, bicarbonate 7.7 meq/L and lactic acid 6.4 mmol/L. Despite continuing antithyroid treatment and adrenergic agents to maintain hemodynamic status, her course was complicated by multiple organ system failure with striking elevation in liver enzymes, persistent lactic acidosis, and azotemic anuria. She developed disseminated intravascular coagulation with aPTT 63.2secs, PT 47.4secs, platelets 20K/mm3, D-dimer 11.19 mg/L and fibrinogen 60mg/dL. A 2-D echocardiogram revealed global hypokinesis and EF of 10-15%. Aggressive treatment of her condition included multiple FFP transfusions and glucocorticosteroids. Her condition degenerated to anoxic encephalopathy. She was transferred to palliative care and subsequently died.

DISCUSSION: Thyroid storm is an uncommon complication of hyperthyroidism characterized by cardiovascular, thermoregulatory and central nervous system dysfunction and high mortality typically incited by an underlying illness, surgery or infection. We present a unique case in which accelerated hyperthyroidism lead to multiple organ system failure. This patient had been consuming iodine-containing tablets and was hyperthyroid for eight months before presentation. We believe an overwhelming release of thyroid hormone during hospitalization failed to respond to standard management and the consequential elevation in basal metabolic rate created widespread derangement of organ function. Thyroid storm-associated multiorgan system failure has been reported in only two cases. Furthermore, physicians need to recognize that over-the-counter supplements carry an inherent risk of precipitating medical conditions.

MILK, DOES A BODY GOOD? - A CASE OF HYPERCALCEMIA DUE TO THE MILK-ALKALI SYNDROME. M. Nystrom¹; P.P. Balingit². ¹University of California, Los Angeles, Los Angeles, CA; ²Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #134666)

LEARNING OBJECTIVES: 1) Recognize the milk-alkali syndrome as an etiology of hypercalcemia. 2) Discuss treatment of the milk-alkali syndrome.

CASE: A 45 year old Caucasian woman with past history of peptic ulcer disease and partial gastrectomy presented to the emergency room with 3 weeks of progressively worsening burning bilateral upper quadrant pain. The pain was worsened by eating, and the patient had taken over-the-counter antacids and milk for relief. During the two weeks prior to admission, the patient took at least 12 tablets of calcium carbonate and drank approximately 1/2 gallons of milk daily. The patient reported having multiple episodes of nausea, vomiting, and melanic stools in the week prior to admission. She denied bone pain, edema, altered mental status, constipation, steatorrhea, fatigue, depression, or confusion. Physical exam was significant for mild bilateral upper quadrant tenderness upon palpation. Initial laboratory data included calcium 15.3, phosphate 4.7, HCO3 30, BUN 61, and Cr 2.4. Chest x-ray, renal ultrasound, CBC, and PTH level were all normal. EKG showed a normal QT interval. The patient was admitted to the hospital, and famotidine, furosemide, and intravenous saline infusion were ordered. All antacids and milk were discontinued. Over the next 24 hours, the patient's abdominal pain subsided and calcium level decreased to 10.7. The hypercalcemia was attributed to the milk-alkali syndrome.

DISCUSSION: The milk-alkali syndrome refers to hypercalcemia and alkalosis stemming from the ingestion of large amounts of calcium and absorbable alkali. It was first recognized in the 1920s as a potential side effect of the Sippy regimen for peptic ulcer disease, which involved taking large amounts of milk and bicarbonate. However, with the advent of nonabsorbable alkali and histamine-2 blockers for treatment of peptic ulcer disease, milk-alkali syndrome became a much less common cause of hypercalcemia. However, since the early 1990s there has been a resurgence of the milk-alkali syndrome coincident with the increased popularity of calcium carbonate for dyspepsia and calcium supplementation. In a 1993 study of patients hospitalized for emergent hypercalcemia, milk-alkali syndrome was the third most common cause (12%), behind hyperparathyroidism and solid malignancy in frequency. Women are more commonly affected than men, and post-menopausal women are at higher risk. Treatment of hypercalcemia due to the milk-alkali syndrome includes cessation of exogenous calcium supplementation. Additionally, stimulation of calciuresis may be achieved with administration of intravenous saline infusion and loop diuretics. In evaluating a patient presenting with hypercalcemia, obtaining a

careful medication and dietary history may lead the physician to entertain this diagnosis.

MISTAKING APPLES FOR ORANGES: A CASE OF PATENT FORAMEN OVALE AND CRYPTOGENIC STROKE. S. Kang¹; J. Blank¹; A. Cooperman¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #134053)

LEARNING OBJECTIVES: 1) Recognize the differential diagnosis of stroke in a younger patient. 2) Diagnose and treat patent foramen ovale (PFO) associated with cryptogenic stroke.

CASE: A previously healthy 39 year old woman presented to the emergency department after an episode of sudden onset right arm numbness and weakness. This event lasted several minutes two days prior to admission. The patient subsequently developed several hours of difficulty with speech and naming objects. She referred to family members by the wrong name and called an apple an "orange" and a corn-on-the-cob a "potato". On review of systems, she reported three days of frontal headache and one episode of nausea and vomiting without hematemesis. She took no medications and denied tobacco, alcohol, or illicit drug use. Family history was significant for a brother who had died of a brain tumor three weeks prior to her symptoms. Vital signs were normal. She was alert and fully oriented. Heart exam was normal and a neurological exam did not reveal any focal abnormalities. Laboratories were unremarkable except for a microcytic anemia with low iron indices. Cerebrospinal fluid analysis was normal. CT of the brain showed nonspecific calcifications in the right parietal lobe. MRI of the brain revealed a left fronto-parietal infarct consistent with a left middle cerebral artery stroke. MR angiogram and carotid dopplers were unremarkable. TEE revealed a PFO. Lower extremity dopplers were negative for deep venous thrombosis (DVT) and a hypercoagulability work-up revealed no abnormalities. The patient was diagnosed with a cryptogenic stroke due to a paradoxical embolus associated with her PFO. We started anticoagulation with warfarin.

DISCUSSION: Cryptogenic strokes with no identified cardioembolic or large vessel source and in a distribution that is not consistent with small vessel disease, are more common in patients who are younger than 55 years of age. Strokes in younger patients can also be caused by intracerebral or subarachnoid hemorrhage, vascular malformations, tumors, illicit drug use, venous sinus thrombosis, fibromuscular dysplasia and vasculitis. The most common cause of cryptogenic stroke is a paradoxical embolus associated with a PFO. PFO has a high prevalence of 25% in the general population but is usually asymptomatic. However, PFOs may function as valve-like structures allowing a right to left shunting when right atrial pressure increases. Paradoxical emboli from pelvic or lower extremity veins can traverse the PFO and cause infarction in the cerebral arterial circulation. Guidelines for the treatment of patients with cryptogenic stroke associated with PFO call for anticoagulation to prevent further strokes. Aspirin may be used as first line therapy. For prolonged neurologic deficits or when a DVT is found, warfarin should be used as initial therapy. For recurrent cryptogenic stroke, percutaneous or surgical closure should be strongly considered.

MOLD INDUCED HYPERSENSITIVITY PNEUMONITIS CAUSING RESPIRATORY FAILURE. S.S.Salib¹; A. Mirza¹. ¹Geisinger Medical Center, Danville, PA. (Tracking ID #135940)

LEARNING OBJECTIVES: 1. Recognize hypersensitivity pneumonitis (HP) as a cause of respiratory failure. 2. Understand the importance of history taking in exploring the cause of shortness of breath. 3. Recognize the importance of eliminating the offending agent in treatment of HP.

CASE: 62-years old male with history of coronary artery disease and 45 years of tobacco abuse presented to emergency department of our tertiary care hospital with shortness of breath of 3-months duration that had been worsening over 4 days prior to hospital visit. He also noticed worsening of chronic smoker's cough and new symptoms of fatigue, loss of appetite, fevers, night sweats and weight loss. Patient had taken several courses of antibiotics without much help. Social history revealed that he spends most of the time in the basement of his house where he had a wood shop. Further questioning revealed that patient noticed black mold on the wallboard in his basement. On physical examination patient was hypoxic on room air (80% oxygen saturation) and chest examination revealed dry coarse crackles bilaterally. Laboratory investigations showed erythrocyte sedimentation rate of 105 mm/hr (0-15), normal complete blood count, renal and liver functions tests. Chest X-ray showed increased interstitial markings bilaterally and high resolution CT scan of the chest showed diffuse interstitial lung disease suggestive of fibrosis. These findings were consistent with the diagnosis of acute interstitial pneumonitis. Based on these findings and the patient's history diagnosis of HP was made. Patient was treated with supplemental oxygen and systemic corticosteroids with remarkable improvement within 48 hours of treatment. Subsequently Stachybotrys species were cultured on fungal culture of filter of humidifier at patient's home basement. Patient was discharged home on supplemental oxygen that he used only with exertion and prednisone was tapered to be stopped in 3 months. Offending Mold was removed by changing humidifier and replacing the wallboard where mold was growing.

DISCUSSION: HP is a complex syndrome of varying severity and clinical presentation. Many patients have sub-clinical disease and escape detection or are misdiagnosed as suffering from viral illness or asthma. It represents an immunologic reaction to a usually inhaled, offending agent. Over 300 etiologies of HP have been identified depending on the exposure setting including those affecting farmers, vegetable and dairy cattle workers, bird and poultry handlers and as in our patient ventilation related contamination. Symptoms of overt HP include fever, cough and chest tightness and later dyspnea and fatigue as pulmonary fibrosis develops. Laboratory investigations are usually unrevealing except

erythrocyte sedimentation rate which is often elevated. Identification of the serum fungal antibody is not helpful as their elevation indicates exposure but not necessarily HP. Diagnosis is established by history of exposure supplemented by findings of ground glass appearance of lungs on CT scan of chest and hypoxemia on arterial blood gases analysis in some instances. Steroids and removal of offending agent often result in dramatic improvement of symptoms.

MONONUCLEOSIS A RED HERRING AND UNLIKELY HARBINGER. W. Pease¹; K. Karamchandani¹. ¹University of Illinois at Chicago, Chicago, Peoria, IL. (Tracking ID #135618)

LEARNING OBJECTIVES: 1. Recognize the presentation of postanginal sepsis coexisting with a viral illness. 2. Management of a parapneumonic effusion. 3. Treatment of a septic jugular vein thrombus.

CASE: A 17-year-old white female presented to her primary care physician with sore throat, fever, chills, and malaise. Throat culture was negative and heterophile antibody testing was positive. Prednisone was initiated for severe symptoms of mononucleosis. She initially improved but then seven days later developed pleuritic right sided chest pain with a dry cough and shortness of breath. Chest x-ray showed a right lower lobe infiltrate and the patient was admitted for intravenous antibiotics. Laboratory data revealed mildly elevated white blood cell count with ninety four percent neutrophils. Peripheral smear showed toxic granulation and Dohle bodies. Hyponatremia, hyperglycemia, and a mild elevation in total bilirubin were present as well. Treatment was initiated with intravenous ceftriaxone and oral azithromycin. Despite antibiotic treatment the patient became more dyspneic, continued to spike fevers, and repeat chest x-rays showed an advancing right pleural effusion. At that point a Pulmonology consult was sought. Physical exam revealed a temperature of 104.1, a blood pressure 110/60, a pulse of 137, and a respiratory rate of 20. There was mild tonsillar hyperemia, a palpable cord in the left lateral neck, decreased breath sounds and dullness to percussion in the lower right lung field. The patient also had a diffuse maculopapular rash on the anterior thorax. Blood cultures drawn on admission had returned positive for *Fusobacterium necrophorum*. The diagnosis of Lemierre's Syndrome was made. A chest tube was placed to evacuate the parapneumonic effusion. A CT of the neck revealed a 1.8-cm left paratonsillar abscess and a 7-cm thrombus in the left external jugular vein. The abscess was drained. Antibiotics were changed to ampicillin with sulbactam. The patient was started on heparin and bridged to oral treatments with coumadin and amoxicillin with clavulanic acid as her clinical status improved.

DISCUSSION: The clinical deterioration of the patient as her pleural effusion grew highlights the expression that the sun should never set on an untapped parapneumonic effusion. Use of anticoagulation is generally not recommended in Lemierre's. Our patient was started on anticoagulants for precautionary purposes. Shortly after discharge from the hospital she developed alopecia from coumadin and it was discontinued. Follow up CT scan showed complete resolution of the thrombus. While still rare the incidence of Lemierre's Syndrome, first described by Andre' Lemierre in 1936 has seen resurgence at least in the literature. There are several speculations regarding the etiology of this resurgence including the hesitance to treat non-Streptococcal sore throats with antibiotics, increasing antibiotic resistance, and better isolation techniques for anaerobic bacteria. Whatever the cause it is important for primary care physicians to be aware of this potentially life threatening cause of a culture negative sore throat. This particular case highlights the potential for this syndrome to hide itself like a time bomb in the shroud of mononucleosis.

MORE THAN A RASH: ADULT STILL'S DISEASE OR PARVOVIRUS B-19 INFECTION? L. Acharya¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #133188)

LEARNING OBJECTIVES: 1. Recognize the clinical manifestations and diagnostic criteria of Adult Still's disease. 2. Differentiate the clinical manifestations of Adult Still's disease and Parvovirus B-19 infection.

CASE: Mrs. P is a 40 y/o woman presenting with three weeks of fevers that peaked in the late afternoon, arthralgias of the feet and hands and anemia. She transferred to us from an outside hospital where she had undergone a workup including: negative blood and throat cultures, rheumatoid factor, ANA, abdominal CT, HIV antibody test and PCR for Parvovirus B19. She developed an erythematous, confluent, macular rash on her lower extremities which was mildly puritic, and a reticulated rash on the upper extremities. Initial laboratory tests were significant for a white blood cell count of 18,000, hemoglobin of 28, AST of 120, ALT of 100 and an LDH of 1466. She underwent a bone marrow biopsy which showed severe iron deficiency anemia. She was treated empirically for Still's disease with prednisone which improved her arthralgias moderately. Subsequently, her hemoglobin decreased to a nadir of 5, making Parvovirus infection a more likely diagnosis. She is a Jehovah's Witness and refused blood transfusion, however, with the permission of the Jehovah's counsel she was able to receive IVIG. PCR for Parvovirus was sent upon transfer and subsequently came back positive. She was given the diagnosis of Parvovirus B-19 infection and her hemoglobin eventually rose and stabilized at 9.

DISCUSSION: While Adult Still's Disease and Parvovirus B-19 infection have common clinical features there are distinct characteristics of the fever, rash and arthralgias that can help clinicians differentiate between these two entities. The rash associated with Still's disease is non-puritic and evanescent whereas the rash of Parvovirus infection is puritic and alters with changes in temperature and emotional stress. The knees and wrists are affected in Still's disease while Parvovirus affects the wrist and hands more frequently. The diagnosis of Still's disease is a clinical one, assessed by major and minor criteria with a sensitivity

of 93% based on the Yamaguchi criteria. Parvovirus infection can be diagnosed with PCR, IgM and IgG serologies. Differentiating between these clinical entities is important for the initiation for appropriate therapy. Parvovirus infection is usually a self limited infection, however in chronically infected patients or patients with aplastic crisis, intravenous immunoglobulin may be helpful. Patients with Still's disease may require immunosuppressant agents in addition to systemic steroids and non-steroidal anti-inflammatory agents. Approximately one third of Still's disease patients will have a devastating chronic course while Parvovirus infection can be dangerous during pregnancy.

MORE THAN SKIN DEEP: A CASE OF DERMATOMYOSITIS AND BREAST CANCER. F.H. Lin¹; J. Frank¹. ¹University of Pittsburgh Medical Center, Pittsburgh, PA. (Tracking ID #132484)

LEARNING OBJECTIVES: 1) Recognize the association between dermatomyositis and malignancy. 2) Conduct appropriate screening for malignancy in a patient diagnosed with dermatomyositis.

CASE: A 62 year old woman presented to the hospital with bilateral thigh hematomas while taking warfarin for a deep vein thrombosis. Four months ago, she had developed progressive proximal muscle weakness and a rash over her face, arms, legs, and in a shawl-distribution over her shoulders. A rheumatologist diagnosed her with dermatomyositis and treated her with prednisone. Because she failed to improve, a workup for an occult malignancy was started. A CT scan was negative for malignancy, but a mammogram revealed a lump in the right breast. The lump was resected and pathology showed mucinous adenocarcinoma. A bone scan was normal. At our hospital, an EMG confirmed the diagnosis of dermatomyositis. However, the patient did not respond to IVIG and pulse steroid treatment. After stabilization of her thigh hematomas, the patient underwent right segmental mastectomy with clear margins and sentinel node biopsy. The sentinel node was negative for metastatic carcinoma.

DISCUSSION: This case is an example of the association between dermatomyositis and malignancy. Since 1916, when the first case of dermatomyositis and coexisting gastric carcinoma was published, many case reports have suggested an association between dermatomyositis or polymyositis and underlying cancer. In 1992, a population-based Swedish study found that the incidence of cancer was 15% among 392 patients with dermatomyositis, with a relative risk of 3.4 for women and 2.4 for men, respectively. Additional studies have also supported the link between dermatomyositis and occult malignancy. This was particularly notable in patients that were resistant to standard steroid therapy. The diagnosis of dermatomyositis should prompt age-appropriate screening for malignancy. In the absence of clinical signs and symptoms that indicate a diagnosis, all patients with newly diagnosed dermatomyositis should have a complete physical examination, including rectal, breast, and pelvic examinations. Basic laboratory tests should be obtained, including CBC, urinalysis, LFTs, ESR, chest radiograph, and fecal occult blood. In women, mammography, pelvic ultrasonography, and CA-125 levels are appropriate. Patients with a personal or family history of malignancy should undergo more extensive testing. When initial screening does not reveal a malignancy, there is disagreement regarding how long to continue annual screening. A study from the Danish Cancer Registry showed an increased incidence of malignancies in the first two years after dermatomyositis was diagnosed. One reasonable approach would be to screen annually for the first 2-5 years.

MPO-ANCA-ASSOCIATED VASCULITIS FOLLOWING PROPYLTHIOURACIL THERAPY. E. Chuong¹; R. Castle¹; N. Mikhail¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #133059)

LEARNING OBJECTIVES: 1. Recognize drug-induced small-vessel vasculitis. 2. Identify MPO-ANCA-associated vasculitis as rare adverse reaction to PTU therapy.

CASE: A 58 year-old man without any significant past medical history presented in atrial flutter (a-flutter) with a rapid ventricular rate (RVR) of 150 bpm. Pharmacological agents controlled his tachycardia, and warfarin was started for anticoagulation. He was diagnosed with underlying hyperthyroidism, and he was started on propylthiouracil (PTU) therapy. Seven days later, he experienced worsening tachycardia and hemoptysis. His physical exam was significant for a BP of 92/52 mmHg, tachycardia at 117 bpm, tachypnea at 22 bpm, O2 saturation at 99%, an irregularly irregular rhythm, and bibasilar crackles. EKG re-confirmed a flutter, and CT scanning demonstrated new infiltrative changes in multiple lung lobes. Labs revealed an interval increase in creatinine from 1.5 to 2.3 mg/dL, and urinalysis detected 91 RBCs. Studies for ANA, rheumatoid factor, c-ANCA, infections and malignancy were negative. Studies for myeloperoxidase antineutrophil cytoplasmic antibodies (MPO-ANCA) were positive, and a renal biopsy demonstrated acute crescentic glomerulonephritis (GN) with 50% active involvement. PTU and anticoagulation were discontinued. His tachycardia was pharmacologically controlled, and methimazole was substituted for PTU. The patient received steroids and cyclophosphamide for his acute crescentic GN, and his condition improved.

DISCUSSION: MPO-ANCA, also known by its immunofluorescent staining pattern in neutrophils as perinuclear ANCA (p-ANCA), is an important diagnostic marker for systemic vasculitic disorders such as microscopic polyangiitis, polyarteritis nodosa, Churg-Strauss syndrome, Wegener granulomatosis, and necrotizing and crescentic GN. On rare occasions, MPO-ANCA is associated with small vessel vasculitis reported in patients treated with PTU and other drugs. Some investigators suggest that PTU or its metabolites cause vascular damage by cross-reacting with a neutrophilic cytoplasmic antigen or directly altering myeloperoxidase structure or function, but the exact role of MPO-ANCA in

vasculitis is unknown. The clinical presentation of MPO-ANCA-associated vasculitis following PTU therapy may occur days to years after the initiation of PTU and includes non-specific findings such as fevers, arthralgias, hematuria, proteinuria, alveolar hemorrhage, pericarditis, mucosal ulcers, and skin lesions. Crescentic GN and, less often, mesangial proliferating GN and acute interstitial nephritis are diagnosed by renal biopsy. Treatment involves withdrawing the causative drug; some patients may require steroids and cyclophosphamide. Methimazole can be cautiously substituted for PTU as it is less strongly linked to MPO-ANCA-associated vasculitis, or patients may prefer surgical or radioactive correction of their thyrotoxicosis.

MUCOCELES WITH INTRACRANIAL AND INTRAORBITAL EXTENSION: A REPORT OF TWO CASES. G.G. Pendell¹; J.C. Guarderas¹. ¹Mayo Clinic, Jacksonville, FL. (Tracking ID #135281)

LEARNING OBJECTIVES: 1. Recognize clinical signs and symptoms that may suggest the presence of invasive mucocoeles. 2. Diagnose patients with invasive mucocoeles and make appropriate referrals in a timely fashion prior to the development of life-threatening complications.

CASE: Case 1: A 59 year-old male with a history of allergic rhinitis and nasal polyposis presented with worsening symptoms of continuous clear nasal discharge, bifrontal headache, and hyposmia. For several years his symptoms were seasonal and well controlled with steroids and oral antihistamines. Over the past two years his symptoms had become more perennial and he also noticed more frequent episodes of purulent nasal discharge. Physical examination was significant for marked nasal polyps and a clear nasal discharge. His right eye was mildly proptotic and displaced inferiorly, though the patient reported no difficulties with his vision. CT imaging showed bilateral frontal mucocoeles extending intracranially through the frontal sinus wall compressing the frontal lobe. There were also mucocoeles extending into the right orbit displacing the globe inferiorly. The patient underwent surgical intervention successfully without any complications. Case 2: A 62 year-old male with a history of asthma, allergic rhinitis, nasal polyposis, and recurrent ear infections presented with symptoms of purulent nasal discharge, hyposmia, periorbital swelling, frequent eye tearing, and a pressure sensation of the right eye. Physical examination revealed marked nasal polyps with a clear nasal discharge, bilateral periorbital swelling, and mild conjunctival injection. His right eye was mildly proptotic and inferiorly displaced, though the patient reported no difficulties with his vision. CT imaging showed bilateral frontal mucocoeles extending intracranially through the frontal sinus wall compressing the frontal lobe. The mucocoeles also extended into both orbits causing inferior displacement of the right globe. The patient was referred to surgery which he elected to have performed in his home country.

DISCUSSION: Physical ailments involving the eyes, ears, nose, and throat are extremely common and make up a significant number of cases encountered by the generalist—especially in the ambulatory setting. Mucocoeles, in particular, are benign, expansile, locally destructive lesions that occur secondary to the obstruction of sinus ostia - which may occur with nasal polyps, chronic infection, or even trauma. Their course is progressive and unless found and treated may result in complications such as visual deficits, facial deformity, abscess formation, seizures, and meningitis. Key signs and symptoms, which may be very subtle, include proptosis, eye pain, vertical diplopia, headache, and increased tearing. These two cases bring attention to the critical role of the generalist in recognizing subtle clinical signs and symptoms and making appropriate specialist referrals in a timely fashion prior to the development of life-threatening complications. While the diagnosis can usually be suggested by plain films, CT and MR imaging can provide more information necessary for diagnosing and guiding therapy. Referrals to allergy, ophthalmology, otorhinolaryngology, and/or neurosurgery may be appropriate. For the destructive lesions found in these two cases, surgical intervention was critical to prevent the development any life-threatening complications.

MULTIPLE MYELOMA IN A 30-YEAR-OLD MAN PRESENTING WITH GRADUALLY WORSENING FATIGUE. N.K. Goraya¹; E. Ortiz¹. ¹Washington DC VA Medical Center, Washington, DC. (Tracking ID #131679)

LEARNING OBJECTIVES: 1) Recognize multiple myeloma as a possible cause of anemia in an otherwise healthy young patient 2) Review the diagnosis and treatment of multiple myeloma.

CASE: This 30-year-old African American male with no significant past medical or family history presented with complaints of gradually worsening fatigue over the previous three years. He had been a competitive athlete in college but could no longer run 1/4 of a mile without becoming short of breath. He denied chest pain, wheezing, night sweats, fevers or weight loss. Physical exam was unrevealing. A normocytic anemia with a hematocrit of 30% appeared to explain his symptoms, but the etiology was unclear. Iron studies were normal, HIV test was negative, and creatinine was 0.9mg/dL. The only other abnormality was his urinalysis, which showed heavy albuminuria. A follow-up urine protein electrophoresis showed 2.0g/24hr of albumin excretion and a monoclonal protein of 1.7g/24hr, which immunofixation confirmed to be a Bence-Jones protein. Serum electrophoresis and calcium were normal, and a skeletal x-ray series was negative for lytic lesions. The patient underwent a bone marrow biopsy, which confirmed light chain multiple myeloma. Treatment with dexamethasone and thalidomide resulted in improvement in the hemoglobin and clearance of the Bence Jones proteins and albuminuria. He was referred for autologous bone marrow transplantation.

DISCUSSION: Multiple myeloma, a malignancy of plasma cells, has an annual incidence of 0.004%, accounts for 1% of all malignant diseases in the United

States, and occurs twice as frequently in blacks as whites. The median age at diagnosis is 66, with a peak incidence occurring in the 7th decade of life. In patients younger than 30, it is exceedingly rare. In a Mayo Clinic series, only 10 cases were documented over a 36-year period. It commonly presents with bone pain from osteolytic lesions, fatigue, weakness, hypercalcemia, anemia, elevated total protein, renal insufficiency, weight loss, and recurrent infections. This patient's young age and the relative absence of classic findings made it easy to miss the diagnosis, reinforcing the need for clinicians to follow-up unexplained abnormalities, even in apparently healthy young patients. The cause of multiple myeloma is unknown, although exposure to radiation and other chemicals may play a role. Treatment includes standard chemotherapy with prednisone and melphalan. Most recently, thalidomide in combination with dexamethasone has shown equivalent efficacy with possibly less risk of myelodysplasia. Autologous stem cell transplantation is the standard of care in patients younger than 65 years with good renal function. Supportive therapy includes bisphosphonates and erythropoietin.

MYXOMA PRESENTING AS FLASH PULMONARY EDEMA. M. Rehmetullah¹; S. Kumar¹; M. Elnicki¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #132482)

LEARNING OBJECTIVES: 1. To recognize atrial myxoma as a cause of flash pulmonary edema. 2. To review the presentation, diagnosis and management of atrial myxoma.

CASE: A 55 year-old male, who underwent an uneventful radical prostatectomy, developed sudden onset of shortness of breath on the second post-operative day. His past medical history was significant for prostatic carcinoma, but no history of hypertension, coronary artery disease, stroke or diabetes mellitus. Physical exam revealed a caucasian male in mild distress with dyspnea. He was afebrile with a blood pressure - 126/78 mmHg, heart rate - 88 beats per minute and oxygen saturation at 97% on 2 liters. He had crackles at both lung bases and his cardiac examination was remarkable for a prominent diastolic rumble with an opening snap. No S3 or S4 was appreciated. The remainder of his exam was unremarkable except for jugular venous distension of 8 cms. The differential diagnosis included pulmonary embolus, myocardial infarction, mitral valve disease and volume overload. Initial diagnostic tests were significant for pulmonary edema on chest x-ray and normal electrocardiogram. Troponin-I was normal at <0.08. The complete blood count and serum chemistries were within normal limits. Brain natriuretic peptide was elevated at 560. Furosemide 40 mg intravenously was given to which he responded with diuresis and resolution of his symptoms. An echocardiogram revealed a large, pedunculated, mobile mass attached to atrial septum, freely moving in the mitral valve outflow and good left ventricular function. He underwent surgery with removal of the mass, which proved to be an atrial myxoma.

DISCUSSION: Atrial myxoma is the most common primary benign cardiac tumor. Of them, 75% arise in left atrium in proximity to fossa ovalis while the right atrium and left ventricle are the origin of remainder. It presents as a triad including- mitral valve obstruction, embolization and constitutional symptoms like fever, chills and weight loss. They should be kept in the differential diagnosis while evaluating fever of unknown origin. Rapidly evolving heart failure is a common presentation, particularly in patients without previous evidence of heart disease. Diagnosis is usually by a transthoracic echocardiogram, but rarely a transesophageal echocardiogram is required to delineate the boundaries clearly. Myxomas have a propensity for embolization. If no contraindications exist, they should be excised surgically. The results of surgery are excellent and only the familial variety requires annual transthoracic echocardiogram for five to six years to look for recurrence.

NAUSEA, VOMITING AND CONSTIPATION: MUNDANE OR ARCANE. N.S. Murali¹; G. Lumer²; M. Liebow¹; A. Ghosh¹. ¹Mayo Clinic, Rochester, MN; ²US Securities and Exchange Commission, Washington, DC. (Tracking ID #136286)

LEARNING OBJECTIVES: 1. Recognize clinical presentation and EKG changes in hypercalcemic medical emergencies 2. Discriminate benign from malignant parathyroid disease and the latter from Humoral Hypercalcemia of Malignancy 3. Recognize the most cost effective approach to assessment and management of hypercalcemia

CASE: A previously healthy 29-year-old Caucasian male was admitted for persistent nausea and vomiting of 2 weeks duration. His symptoms had progressively worsened despite around the clock regimen of anti-emetics and proton pump inhibitors prescribed by his primary physician. On specific inquiry he admitted to increasing constipation, fatigue and polyuria but denied any abdominal pain, melena or hematochezia. Except for a history of 30-lb weight loss rest of the review of systems, past medical, social and family history and clinical exam was non-contributory. Admission labs revealed a hemoglobin 10.2g/dL (12-15.5g/dL) with a normal cell count and differential. The chemistry revealed potassium of 3.0 (3.6-4.8mEq/L), a BUN of 18 (6-21 mg/dL) and a creatinine 2.2mg/dL (0.7-1.2mg/dL). Ionized calcium was 9.6 (4.65 to 5.3mg/dL), serum calcium was 18mg/dL and phosphorous was 3.8mg/dL (2.5-4.5mg/dL). Alkaline phosphatase was 986U/L (114-312U/L) with the bone fraction contributing 818U/L (24-146U/L). An electrocardiogram revealed a normal sinus rhythm with first degree heart block and ST elevation in V2 and V3 leads. Cardiac biomarkers were normal. An esophagogastroduodenoscopy done elsewhere was normal. A diagnostic blood test was performed-intact PTH of 206 (1-52pmol/L). He was temporized with aggressive saline diuresis, intravenous bisphosphonate administration and recovered completely after a surgical removal of the parathyroid carcinoma.

DISCUSSION: Primary hyperparathyroidism and malignancy account for at least 90% of cases with hypercalcemia. Parathyroid carcinomas are very rare and account for less than 1% of all cases of hyperparathyroidism. Determining the etiology of hypercalcemia is very difficult in the absence of histopathology. However, the intact PTH assay is an extremely valuable test to rule in primary parathyroid disorders and rule out non parathyroid disorders like Humoral Hypercalcemia of Malignancy. Furthermore, it is very unusual to identify an adenoma or hyperplasia of the parathyroid when calcium levels are greater than 13 mg/dL and even more so, if levels of intact PTH are 30 pmol/L or higher. Our patient had an iPTH of 206 pmol/L in the setting of hypercalcemia—virtually diagnostic of parathyroid carcinoma even in the absence of histology. The constellation of symptoms and signs attributed to hypercalcemia are broad. Barring history or presence of urolithiasis in primary hyperparathyroidism, all other signs and symptoms so attributed are neither specific nor sensitive in discriminating the major etiologies. EKG changes can be fairly dramatic and can mimic myocardial injury. Serum calcium greater than 13.5 mg/dL in a symptomatic patient is usually a medical emergency. The most effective approach is to assay intact PTH followed by other tests. Intravenous bisphosphonates are the first line treatment that effectively lower serum calcium in three to five days. Oral bisphosphonate are not effective in an urgent situation.

NICK OF TIME: CASE OF MARFAN'S SYNDROME. J.E. Cho¹; A.M. Ichiuji²; D. An³. ¹Olive View - UCLA Medical Center, Sylmar, CA; ²Olive View/UCLA-SFVP, Los Angeles, CA; ³University of California, Los Angeles, Los Angeles, CA. (Tracking ID #133354)

LEARNING OBJECTIVES: 1) Recognize Marfan's syndrome (MFS) as a risk factor for aortic dissection 2) Describe diagnostic criteria for MFS 3) Recognize treatment options for MFS

CASE: A previously healthy 32 year old Hispanic male presented to emergency room with acute onset of 10/10, sub-sternal chest pain without radiation to the back, arm, and neck. Sharp, epigastric abdominal pain was associated with the chest pain. He denied diaphoresis, shortness of breath, or palpitations. Review of systems was significant for nausea and vomiting. On examination, he was afebrile, blood pressure 135/64, pulse 69, respiratory rate 16, and oxygen saturation 100% on room air. General appearance revealed a tall and thin gentleman in mild distress. Patient was noted to have high-arched palate, striae atrophicans over his sides, pectus excavatum, and kyphosis. Equal bilateral upper extremity blood pressures and pulses were present. There was no jugular venous distension, cardiac murmurs, pulmonary wheezes, or crackles. Abdominal examination was unremarkable. Chest radiograph did not show infiltrates or widened mediastinum. Chest CT angiogram revealed an aortic dissection below the left subclavian artery extending to left renal artery and celiac axis. He was admitted to the coronary care unit for medical management of a distal aortic dissection. His blood pressure and chest pain were controlled with intravenous esmolol and nitroprusside. The following day, transthoracic echocardiogram showed a normal ejection fraction and a significant aortic root dilation of 4.7 cm. Transesophageal echocardiogram revealed a possible ascending aortic dissection with a flap near the right coronary ostium. Subsequently, patient was emergently taken to surgery for mediastinal exploration with opening of the pericardium. Ascending aortic dissection was not found on surgical examination. This was confirmed with epiaortic ultrasound and later, MRI of the thorax. Aortic root repair was deferred for six months given high risk of morbidity and mortality.

DISCUSSION: Marfan syndrome (MFS) is an autosomal dominant inherited condition that involves a mutation of the fibrillin-1 (FBN1) gene. Common skeletal phenotypic expressions of MFS include reduced upper to lower body segment ratio, arm span exceeding height, arachnodactyly of fingers and toes with positive thumb and wrist signs, ectopia lentis, dural ectasia, and scoliosis > 20 degrees or kyphosis. Common cardiac abnormalities include aortic root dilation involving the sinuses of Valsalva and aortic dissection. Criteria for establishing the diagnosis of MFS must involve two organ systems including one major criteria. Major criteria includes four of eight skeletal abnormalities, ectopia lentis detected on slit lamp, aortic root dilatation or aortic dissection, and lumbosacral dural ectasia by CT or MRI. The prognosis has significantly improved with the use of beta-blockers, restriction of vigorous physical exercise, monitoring of aortic root size, and elective surgical repair of aortic root. Total aortic root replacement with a composite valve graft should be considered with aortic root dilatation greater than 50 mm or with a rapid increase in root diameter. Primary care physician should also be aware of other complications including dislocated lens, scoliosis, and arthropathy due to joint laxity.

NOT JUST ANOTHER HEADACHE. J. Bunploog¹; K.J. Pfeifer¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #132924)

LEARNING OBJECTIVES: 1. Distinguish malignant hypertension from thrombotic thrombocytopenic purpura. 2. Recognize atypical symptomatic and hematologic presentations of malignant hypertension

CASE: JC is a 43 year-old woman with a history of hypertension, noncompliance with medications and prior hospitalizations for hypertensive urgencies. She had not taken any medications for three weeks when she presented with a three-day history of headaches, blurred vision, nausea, vomiting and abdominal pain. She denied recent use of illicit drugs or over-the-counter medications. Her physical examination was remarkable for a blood pressure of 230/120 mmHg and for diffuse abdominal tenderness. Of note, she was afebrile, alert and completely oriented. She did not have papilledema, nuchal rigidity, skin lesions or any neurological deficits. Laboratory results were significant for blood urea nitrogen of 120 mg/dL, creatinine 9.5 mg/dL, hemoglobin 9 g/dL, hematocrit

26%, mean corpuscular volume 107 fl and platelet count 47,000/uL. Schistocytes were noted on a peripheral smear. Haptoglobin was less than 6 mg/dL, and lactate dehydrogenase was elevated at 795 U/L. Her hepatic profile, creatinine kinase, and DIC panel were within normal limits. The patient's urinalysis was positive for blood and for significant proteinuria. Fenoldopam IV infusion followed by labetalol were initiated for blood pressure control. This improved her symptoms. Concerned that the patient may have thrombotic thrombocytopenic purpura (TTP), she was also started on plasmapheresis. She proceeded to have plasmapheresis four more times until the lab value for ADAMTS 13 finally returned as negative. She was able to maintain an adequate blood pressure after changing to oral medications. However, her renal function never improved and was severe enough that she had to be placed on long-term hemodialysis.

DISCUSSION: Because of the similar presentations of malignant hypertension and TTP, the final diagnosis of malignant hypertension was difficult to make. Malignant hypertension is more common in noncompliant patients and has been shown to cause a hemolytic anemia, thrombocytopenia, and renal failure besides other end-organ damages. Patients with TTP typically present with the pentad of thrombocytopenia with platelet counts less than 15,000/uL, hemolytic anemia, renal insufficiency, fever and neurological complaints such as headaches or mental status changes. However, TTP is often diagnosed with the classic triad composed of the first three symptoms. Although ADAMTS 13 is not a diagnostic test, it can further support the clinical diagnosis. In this case, the significantly elevated blood pressure, the severe renal failure, and the lack of a more profound thrombocytopenia favored malignant hypertension. The diagnosis of malignant hypertension was further supported over TTP by the improvement in her symptoms with better blood pressure control even before starting plasmapheresis.

NOT JUST YOUR AVERAGE CASE OF ABDOMINAL PAIN. S.J. Herring¹; P.M. Cocks¹; V.N. Hayashi¹. ¹New York University, New York, NY. (Tracking ID #133389)

LEARNING OBJECTIVES: 1) Recognize the clinical features of mesenteric venous thrombosis 2) Identify the renal and extrarenal manifestations of nephrotic syndrome 3) Manage symptomatic thromboembolic complications of nephrotic syndrome with heparin and/or thrombolytic therapy

CASE: We describe a rare case of nephrotic syndrome (NS) complicated by portal, splenic and mesenteric vein thromboses. A 57 year-old man sought medical attention in December 2004 due to the acute onset of mid-abdominal pain and diarrhea. Approximately three months prior to the onset of symptoms, he presented to urgent care clinic for medication refills related to his known thyroid disease, at which time laboratory tests revealed normal serum protein levels, a normal free thyroxine level, and a normal urine analysis. He was found to have only mild hypercholesterolemia and was advised to follow-up with his primary care physician concerning treatment. He felt well until two days prior to presentation in December, when he developed intermittent mid-abdominal pain and bloating. His symptoms worsened over the next day and were coupled with two episodes of bloody diarrhea, causing him to come to the emergency room for evaluation and treatment. On presentation, he was found to have a diffusely tender abdomen with laboratory studies now revealing profound hypoalbuminemia, hypoalbuminemia and proteinuria. CT scan of the abdomen showed extensive portal and mesenteric venous thromboses along with colonic thickening from the cecum to the splenic flexure secondary to bowel wall edema. He was admitted to the medical service, immediately started on unfractionated heparin due to his large clot burden, and treated with pulse steroids for NS by day 12 of his inpatient course. After three weeks of anticoagulation and steroid therapy, the patient's abdominal pain and proteinuria slowly improved. Plans were for discharge on warfarin and prednisone along with follow-up in nephrology clinic for renal biopsy and possible treatment with cytotoxic medications.

DISCUSSION: While the most common clinical features of NS are proteinuria, hypoalbuminemia, hyperlipidemia and generalized edema, patients are at risk of developing other problems, such as bacterial infections, electrolyte abnormalities, and venous thromboses. Although thromboembolic disease is a known complication of NS, only five cases of portal vein thrombosis have been reported in patients with this disease. Our patient is particularly unique in that his initial presentation of NS occurred in the context of acute abdominal pain related to his mesenteric venous thrombosis and associated bowel ischemic congestion. The etiology of hypercoagulability in NS has not been fully elucidated, but data suggest that multiple factors may be involved, including increased urinary loss of antithrombin III and enhanced platelet aggregability. In patients with symptomatic thromboembolic complications, heparin should be given, although its effect is attenuated because antithrombin III levels are decreased. Local infusion of intravenous thrombolytics can also provide benefit, but given the high risk of bleeding complications, they are typically reserved for patients with persistent symptoms despite heparin therapy. This case highlights not only the importance of performing a complete evaluation of patients with abdominal pain, but the necessity of recognizing and appropriately managing patients with acute nephrotic syndrome and its complications of thromboembolic disease.

NOT YOUR TYPICAL LIVER CANCER. M. Said¹; N. Gupta¹; D. Corry¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #136262)

LEARNING OBJECTIVES: 1) Exploring the presentation of a rare variant of hepatocellular carcinoma in a young female patient. 2) Discussion of fibrolamellar hepatocellular carcinoma.

CASE: A 19 y.o. woman presented with right upper quadrant abdominal pain and inability to tolerate oral food (solids and liquids) of one day in duration. She

also had one and a half years of progressive nausea, bilious vomiting, fatigue, and severe subjective weight loss. The patient denied any fevers, chills, hematemesis, diarrhea, constipation, melena, hematochezia, urinary, and vaginal symptoms. She was sexually monogamous with one partner for 5 years, and denied any IVDU, ETOH use, past transfusions, tattoos, or history of hepatitis. She had no family history of any cancers, and denied use of medicines, herbs, or supplements. Upon admission, the patient was noted to have elevated liver transaminases (AST: 97; ALT: 58; alkaline phosphatase: 170; GGT: 230; total bilirubin: 0.4), elevated coagulation factors (PT: 15.7; PTT: 46.3; INR: 1.47). Other significant labs: WBC: 9.2; Hgb: 7.1; Hct: 23.4; platelets: 624, normal automated differential; chem 7 normal; albumin: 2.3; calcium: 7.6; lipase: 22; urinalysis normal; urine beta HCG negative; peripheral blood smear normal; reticulocyte count: 1.4; ferritin: 200; serum iron: 86; TIBC: 209; and Iron/TIBC of 41% suggesting anemia of chronic disease. Mixing studies illustrated an inherent deficiency of coagulation factors. Hep A antibody/Hep Bs Ag/Hep Bs Ab/Hep Be Ab/Hep C Ab negative. CEA control. <0.5 (normal control. <0.5); AFP: 13 (normal: 0-20); hemolysis labs negative; PPD negative. The patient initially received an abdominal ultrasound which showed splenomegaly/hepatomegaly and multiple liver masses in the right lobe of the liver, largest: 9 × 9 cm; no ascites; no biliary distention; no gallstones. Later, an MRI of the abdomen/pelvis revealed multiple solid heterogeneously enhancing liver masses involving both lobes of the liver, with the largest, 9 × 9 cm. There was no indication of nodal involvement/cirrhosis/portal hypertension. The patient underwent IR guided liver core biopsy which confirmed the diagnosis of fibrolamellar hepatocellular carcinoma. She was subsequently referred for liver transplant evaluation and sent home with anti-emetics and pain control.

DISCUSSION: Fibrolamellar hepatocellular carcinoma (FHCC) is a malignant neoplasm of the liver which tends to occur in young adults (mean age of diagnosis: 23, range: 5-69), with no racial or gender predominance. HCC accounts for 2-3% of all cancers diagnosed in the USA. Of these, FHCC accounts for fewer than 10% of all cases in adults > 50 years of age but can comprise approximately 35% of HCC's in people < 50 years of age without cirrhosis. It has features distinct from hepatocellular carcinoma (HCC) with regards to the demographics, the constitution of the affected liver, tumor markers, and prognosis. The typical risk factors for HCC such as viral hepatitis, alcohol abuse, and metabolic disease, are not present in FHCC. In addition, cirrhosis and elevated tumor markers (AFP/CEA) levels are not seen in FHCC. Unlike HCC, cirrhosis, vascular invasion, or multifocal involvement is uncommon in FHCC. Surgical resection has produced five-year survival rates of greater than 50%. In the event that the lesion is nonresectable, liver transplantation, and/or chemotherapy, is a consideration. The outcome for liver transplantation for FHCC is better than that observed in HCC.

NOT YOUR USUAL PAIN IN THE BACK. Y. Ng¹; H. Jasti¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133147)

LEARNING OBJECTIVES: 1) Identify the clinical presentation of Guillain-Barre Syndrome (GBS) 2) Recognize the indications for Intensive Care Unit (ICU) admission in GBS 3) Review the treatment options for GBS

CASE: Ms. D is an 80 year old, Caucasian female, who presented to the ER with a one-day history of worsening back pain and bilateral lower extremity weakness. She endorsed some sensory loss on her lower extremities but no urinary or stool incontinence. She denied any trauma or falls. Physical exam was significant for an elderly lady in extreme distress. Neurological exam revealed 3/5 muscle strength in her distal upper and lower extremities with absent reflexes. Sensation was intact to light touch although significant hyperalgesia was noted. A lumbar X-ray showed a stable compression fracture at the L4 level with no new fractures or dislocation. A diagnosis of possible cord compression was entertained and a dose of steroids was administered. An MRI was planned to confirm the diagnosis. However, four hours later, her weakness had rapidly progressed to the point where she was unable to move all four extremities or lift her head off the bed. GBS was suspected and she was transferred to the ICU and intubated for respiratory failure. A diagnostic lumbar puncture was normal and electromyographic studies revealed evidence of demyelination. Based on the clinical picture, the patient was given intravenous immunoglobulin (IVIg). Three days later, she was extubated and her muscle strength improved dramatically. She was subsequently transferred to a nursing facility for further rehabilitation.

DISCUSSION: Guillain-Barre syndrome is currently the most common cause of acute flaccid paralysis, occurring at an annual incidence of 1-2 cases per 100,000 population. Patients usually report a gradual onset of muscle weakness, progressing from distal to proximal over 2-4 weeks, occasionally accompanied by sensory loss. Most patients have a benign course with gradual recovery of muscle strength. However, it is vital to recognize the 30% of patients who require intensive care during the course of their illness. Indications for ICU admission include: 1) Significant reduction in vital capacity (<20 mL/Kg); 2) Rapid progression (<7 days) resulting in inability to raise the head against gravity and bulbar or autonomic dysfunction. Supportive care is usually sufficient for most patients. Indications for more aggressive treatment include: 1) Inability to walk unassisted; 2) Patients requiring ventilatory support. Both IVIg and plasmapheresis have been shown to be equally effective in shortening the duration of mechanical ventilation and expediting clinical recovery. Although most recover completely, there is still a 5% mortality rate with another 15% of patients who require assistance with walking one year later. Therefore, it is vital to recognize the early signs and symptoms of GBS and initiate treatment to prevent permanent damage.

NUMEROUS COMPLICATIONS OF MRSA ENDOCARDITIS IN A SINGLE PATIENT. N. Firooz¹; S. Vazirani¹. ¹Greater Los Angeles VA Healthcare System, Los Angeles, CA. (Tracking ID #133233)

LEARNING OBJECTIVES: 1. Review the clinical presentation and complications of infective endocarditis. 2. Recognize prevalence of multiple complications in one patient.

CASE: A 48-year-old African American male with known intravenous drug use was admitted with polyarthralgia, fever, and pleuritic chest pain. Pertinent physical findings included temperature of 101°F, II/VI systolic murmur, spinal tenderness, and edema of bilateral knees and left wrist. Significant laboratory data included leukocytosis, elevated ESR and rheumatoid factor, and negative ANA. An EKG revealed global ST segment elevation and PR depression consistent with pericarditis. Blood cultures grew methicillin-resistant Staph aureus (MRSA), and treatment with intravenous Vancomycin, in conjunction with rifampin and gentamicin, was initiated. As fever persisted, first a trans-thoracic echocardiogram, and subsequently a trans-esophageal echocardiogram, were performed to rule out endocarditis. These studies revealed a large, circumferential pericardial effusion with right atrial diastolic collapse consistent with cardiac tamponade. No vegetations were noted on either study, but diagnosis of infective endocarditis (IE) was made based on Duke's Criteria. Arthrocentesis of left knee and left wrist were performed; analysis of the synovial fluid showed mild leukocytosis and positively birefringent rhomboidal crystals but no organisms. Pseudogout was treated with colchicine. MRI of lumbar spine demonstrated extensive discitis and osteomyelitis at L4-5 and a left psoas abscess. The abscess was drained via CT-guided needle aspiration, the culture also growing MRSA. The patient developed acute renal failure, which resolved after adjustment of medications and resolution of infection. Antibiotic treatment was continued for 6 weeks, with resolution of the patient's symptoms.

DISCUSSION: The clinical syndrome of IE is highly variable, with an array of possible cardiac and extra-cardiac manifestations. Some complications such as congestive heart failure, paravalvular abscesses, neurologic complications, embolization, and renal failure are relatively common; while conditions such as pericarditis, tamponade, pseudogout, metastatic abscesses, osteomyelitis, and septic arthritis are infrequently reported. Presenting with renal failure, arthritis, pericarditis, tamponade, pseudogout, osteomyelitis, and psoas abscess, our patient was unique in developing a number of uncommon as well as common complications of IE. Involvement of the pericardium in patients with subacute IE is more often due to deposition of immune complexes than due to infection. Large volume pericardial effusion leading to tamponade and collapse of the right atrium developed in our patient within one week of hospitalization. Metastatic abscesses in IE occur as a sequela of septic embolization, affecting organs like kidneys, spleen, brain, or soft tissue. Nevertheless, reports of metastatic abscesses of the psoas muscles as seen in this patient are extremely rare. The suspected cause of acute renal failure in our patient was tubulointerstitial nephritis due to nephrotoxic drugs, versus immune complex deposition glomerulonephritis. Infection was thought to be the cause of pseudogout. In summary, our patient developed multiple and rare complications of IE. We may see patients present with numerous and less common complications, as patients with IE may live longer due to advances in diagnosis and treatment.

ON BEING AN "AIRHEAD": PNEUMOCEPHALUS AND MENINGITIS AS COMPLICATIONS OF OTITIS MEDIA. S. Ketha¹; S. Gupta¹; H. Friedman¹. ¹St. Francis Hospital, Evanston, IL. (Tracking ID #135046)

LEARNING OBJECTIVES: 1) Recognize pneumocephalus as a very rare but important complication of meningitis. 2) Recognize meningitis as an important complication of acute otitis media. 3) Recognize the absence of classic findings of meningeal irritation in the elderly.

CASE: A 69-year-old man presented with right earache for two days and headache for one day. The patient was treated with amoxicillin for two days as an outpatient for acute otitis media. However, the patient returned to the ED after two days with worsening headache. Findings on physical examination were normal except for a temperature of 101°F and tenderness over the right mastoid process. His neck was supple and he had no focal neurological deficits. The patient was admitted to the hospital with a presumptive diagnosis of meningitis. Computed tomography (CT) showed opacification of the right mastoid air cells and fluid in the right middle ear. Diffuse intracranial air was seen bilaterally. Lumbar puncture revealed purulent cerebrospinal fluid with 12000 WBCs of which 66% were neutrophils. These findings were suggestive of a diagnosis of meningitis and pneumocephalus secondary to otogenic causes. Culture of the cerebrospinal fluid grew streptococcus, viridans group. Antibiotic therapy was started with vancomycin and ceftriaxone. The patient's fever subsided, and his earache and headache resolved.

DISCUSSION: Intracranial manifestations of otitis media are limited, the most common being meningitis. Pneumocephalus, characterized by the presence of air in the cranial cavity occurs due to a breach in the integrity of the cranium or the duramater. It is frequently caused by trauma and surgery. It can also occur after diagnostic procedures like a lumbar puncture. Meningitis is a very rare cause of pneumocephalus. We report a patient who had meningitis as a cause for his pneumocephalus. The meningitis itself was a result of acute otitis media. This patient was diagnosed to have pneumocephalus on CT scan study and was treated with antibiotics to which he responded well. To the best of our knowledge, this is the second reported case of pneumocephalus and meningitis as a complication of otitis media. In summary, it is important to be aware of pneumocephalus as a rare but important complication of meningitis. This case also illustrates the absence of classic findings of meningeal irritation in the elderly.

OOPS, I'M PARALYZED AGAIN. E. Chuong¹; M. Nguyen¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #133061)

LEARNING OBJECTIVES: 1. Recognize the clinical presentation of hypokalemic periodic paralysis 2. Identify hyperthyroidism as a secondary cause of periodic paralysis.

CASE: A 50 year-old man presented after an episode of profound weakness upon waking in the morning. He was unable to move his arms and legs for several hours, but he denied altered mental status, loss of consciousness, convulsions, fevers, chills, headaches, dizziness, visual changes, dysarthria, tongue biting, neck stiffness, dyspnea, palpitations, incontinence, numbness, or pain. He had worked the previous day on a physically demanding construction project, and his last dinner included tortillas. He denied taking medications, dietary supplements, licorice, ethanol, or controlled substances, and he had no significant family history. He had experienced two or three similar episodes of profound weakness beginning at 12 years of age. All previous episodes occurred upon waking in the morning, but all resolved within 6 hours and were never medically evaluated. His physical exam, including a comprehensive neurological assessment, was unremarkable. Significant lab abnormalities included potassium 2.7 mmol/L, creatinine kinase 1730 units/L, TSH <0.01 units/L, and Free T4 2.2 ng/dL. His EKG was without hypokalemic changes. The patient's symptoms completely resolved with a single dose of KCl 40 meq, and his potassium levels have since been normal.

DISCUSSION: Periodic paralysis (PP) are a heterogeneous group of muscle disorders characterized by episodes of flaccid muscle weakness at irregular intervals. Common clinical features include inexcitability of affected muscle membranes, eyelid myotonia, fixed proximal weakness, and diminished stretch reflexes. Sensation, cranial nerves, and respiratory muscles are usually spared, and muscle strength may be normal between attacks. Primary hypokalemic PP has a prevalence of 0.001%, and onset is usually prior to 16 years of age. The principal defect may involve a mutation in a calcium channel gene. Weakness often presents in the morning following a day of stress, strenuous exercise, or high carbohydrate meals, and generally lasts less than 72 hours. Alterations in serum potassium levels are a consequence of this disorder and are normal between attacks. Attacks are treated with potassium supplements and averted with carbonic anhydrase inhibitors or potassium-sparing diuretics. Muscle weakness and wasting may become permanent in the course of severe cases. Thyrotoxic PP is a more common secondary variant of hypokalemic PP with a prevalence as high as 13-14%. Onset is usually between 20-50 years of age, but the clinical presentation and sequelae are similar to that of primary hypokalemic PP. The principal defect seems to be increased Na-K-AT-Pase activity, and treatment involves controlling the underlying thyrotoxicosis. Nerve conduction studies, EMGs, exercise testing, and exogenous glucose or epinephrine administration may substantiate the diagnosis of hypokalemic PP.

OPPORTUNITIES FOR EARLY DIAGNOSIS OF MULTIPLE MYELOMA. H.D. Venters¹; E.H. Green¹. ¹Montefiore Medical Center, Bronx, NY. (Tracking ID #133258)

LEARNING OBJECTIVES: (1) Recognize bone pain and anemia as early signs of multiple myeloma (2) Highlight the importance of pursuing pending test results for inpatients after their hospital discharge.

CASE: A 53 year old woman with a past medical history of hypertension presented with 2 days of 10/10 mid back pain. She described the back pain as sharp and non-radiating, and described at least one month of more mild pain in the same area. The pain was not associated with lower extremity weakness or neurologic symptoms. She denied any recent trauma, fevers, or history of cancer. The patient had recently been told she was "anemic" by her primary physician. She had been started on iron supplements but stopped secondary to gastrointestinal side effects. She was a non-smoker and had no known history of osteoporosis or bone diseases. On her initial exam she exhibited mid-spinal tenderness at T-10 but had no other bone tenderness or other abnormalities. X-rays revealed a compression fracture of the T-10 vertebral body. The patient's initial laboratory studies included a hematocrit of 26 and an erythrocyte sedimentation rate (ESR) of 145. Three months earlier, during a previous admission, she had a hematocrit of 27 and an ESR of 140. When we reviewed her inpatient chart from that admission we found that she had almost the identical complaint of back pain. At that time she had been treated with ibuprofen and discharged after two days. A serum protein electrophoresis (SPEP) was sent at that time as part of an anemia work-up. This test was completed after her discharge and showed a monoclonal spike in the IgA region consistent with multiple myeloma. It was unclear if this result had ever been reviewed by either her primary care physician or any of the physicians who treated her during her hospitalization.

DISCUSSION: Patients with multiple myeloma frequently present with bone pain, anemia, renal failure or hypercalcemia. Although the classic presentation of multiple myeloma involves an elderly patient with vertebral compression fracture, the percentage of multiple myeloma patients presenting in this manner has fallen from almost 70% in the 1960's to less than 40% today. In contrast, over 90% of multiple myeloma patients have bone pain (without fracture) as part of their primary presentation. In addition, anemia is present in roughly 90% of patients at the time of multiple myeloma diagnosis. Early diagnosis is important to minimize pathological fractures as well as preventing renal failure. In addition, advances in multiple myeloma treatment will likely increase the importance of early diagnosis. The presentation of this patient several months before her first compression fracture created an opportunity to begin therapy, and her initial team correctly included multiple myeloma in their differential diagnosis. However, multiple obstacles in our medical system prevented her primary care physician from learning of key test results. Prompt recognition of the abnormal SPEP might have delayed or prevented her pathologic fracture.

PAINFUL EAR: CASE OF RELAPSING POLYCHONDritis. J.E. Cho¹; J. Blank¹; J. An²; M.A. Dulay¹. ¹Olive View - UCLA Medical Center, Sylmar, CA; ²University of California, Los Angeles, Los Angeles, CA. (Tracking ID #133353)

LEARNING OBJECTIVES: 1) Recognize relapsing polychondritis (RPC) as an etiology of painful ear. 2) Describe the typical presentation of RPC. 3) Recognize diagnostic criteria and treatment options of RPC.

CASE: A previously healthy 65 year old woman with a history of hypertension presented to urgent care with one week history of left ear pain and swelling. She denied hearing loss, dizziness, vertigo, tinnitus, head trauma, voice changes, ear fullness and discharge. Pain was relieved with motrin and cold packs over the affected ear. The patient noted a history of bilateral ear pain and edema two years ago. At that time, she was diagnosed with otitis externa and treated with antibiotics without symptomatic improvement. On presentation, she was afebrile, blood pressure 130/80, pulse 80, and respiratory rate of 12. Physical exam revealed diffusely edematous left ear with mild erythema and tenderness to palpation with sparing of the ear lobe. Ear canal examination showed edema, intact tympanic membrane and light reflex without discharge. Hearing was intact to finger rub and no cervical or axillary lymphadenopathy appreciated. Biopsy of the auricle revealed perichondrial lymphatic infiltration. She was diagnosed with relapsing polychondritis and treated with a short course of oral prednisone.

DISCUSSION: Relapsing polychondritis (RPC) is a systemic inflammatory disease process that compromises the structural integrity of the cartilage. Patients typically present with unilateral or bilateral external ear inflammation, pain, tenderness, and patchy erythema sparing the ear lobes. In severe cases, the external auditory meatus may remain edematous for weeks with hearing compromise. Recurrent episodes of auricular RPC cause permanent structural changes resulting in a floppy ear or cauliflower appearance. RPC may affect other cartilage based structures including eyes, nose, airways, heart, vascular system, skin, joints, kidney, and nervous system resulting in scleritis, episcleritis, conjunctivitis, and saddle nose deformity. Diagnosis may be made using McAdam's criteria with at least three of the following: bilateral auricular chondritis, non-erosive seronegative polyarthritis, nasal chondritis, ocular inflammation, respiratory tract chondritis, cochlear or vestibular dysfunction. Other diagnostic criteria include chondritis of two or more separate anatomic locations responsive to steroid treatment. Tissue biopsy can be obtained for histologic confirmation of RPC, and to rule out other causes of inflammation including tuberculosis, fungal disease, syphilis, or leprosy. Laboratory tests are non-specific with elevated erythrocyte sedimentation rate, eosinophilia (10%) and anti-type II collagen antibodies (>50%). In cases without visceral involvement, NSAIDs can be prescribed for 7 to 10 days, followed by dapsone or prednisone if an adequate response is not achieved. However, prednisone 30-60 mg daily is commonly regarded as first-line therapy, with subsequent tapering as disease activity remits. Severe, life-threatening cases require high dose steroids in addition to methotrexate, cyclosporine, azathioprine, cyclophosphamide, or anti-tumor necrosis factor antibody (infliximab).

PARADOXICAL RESPONSE SYNDROME: ENIGMATIC PRESENTATION OF AN OLD FOE. A. Sequeira¹; J. Nijjar¹; D. Sula¹; S.C. Reddymasu¹; N. Koshy¹; C. Mitchell¹. ¹Louisiana State University Medical Center at Shreveport, Shreveport, LA. (Tracking ID #135340)

LEARNING OBJECTIVES: Recognize paradoxical growth of intracranial tuberculomas during initiation of antituberculous therapy.

CASE: A 23-year old Indian male was admitted for evaluation of a frontal headache and intractable vomiting present for two weeks. Contrast tomography (CT) of the brain demonstrated a left frontal, rim enhancing lesion with surrounding edema suggestive of either malignancy or abscess. Chest radiograph was normal. Antimicrobial therapy for a pyogenic brain abscess was initiated, and the mass was urgently resected with no remaining residual tissue. Histopathology revealed a necrotic granuloma. Skin testing for tuberculosis was positive; hence, antituberculous therapy (ATT) and steroids were initiated. He subsequently improved and was discharged on only therapy for tuberculosis and a steroid taper (unbeknownst to the infectious diseases service). Two days later, he returned with fever, headache and vomiting. Pertinent readmission lab data: Hb 10 g/dL, WBC 8.9 k/cmm, ESR 52 mm/hr with normal liver function tests. Lumbar puncture (LP) revealed a normal opening pressure with 174 cells/mm³ (lymphocyte predominance), protein 50 mg/dL and glucose 57 mg/dL. Treatment for potential nosocomial central nervous system infection was initiated in addition to ATT. High-dose steroids were restarted but again weaned by the primary team after, over the ensuing days, he became drowsy. Blood and CSF cultures were sterile. Repeat LP after continued decompensation was unchanged. Brain magnetic resonance imaging (MRI) demonstrated the presence of a new left frontal, rim enhancing mass which was subsequently resected to reveal necrotic tissue with gliosis. Special stains for acid fast bacilli (AFB) were negative. By this time, the initial resected tissue was growing an acid-fast bacillus in rapid culture media. Nucleic acid probe for Mycobacterium tuberculosis complex RNA was positive. Based on this and the present features, paradoxical growth of an intracranial tuberculoma was entertained as the etiology of the decompensation. He was restarted on high doses of intravenous (IV) steroids along with ATT, and the antibiotics were discontinued. His symptoms improved, and he was discharged without sequelae.

DISCUSSION: Intracranial tuberculomas occur as a result of haematogenous dissemination from a primary site. The "paradoxical response" is believed to be an immunological phenomenon characterized by the enlargement of preexisting or the appearance of new tuberculomas during therapy, which usually occur when steroids are being tapered. Symptoms may be present. CT appearances include solid enhancing, ring enhancing or mixed forms. The former may develop surrounding edema or ring enhancement. Treatment includes ATT for at

least a year (or even two) with high dose adjunctive steroids that are tapered slowly over months. This case emphasizes that in patients suspected of having CNS tuberculosis, the growth of tuberculomas during initiation of therapy does not indicate a treatment failure. All patients require high dose steroids for an extended duration to ameliorate symptoms and improve outcomes.

PERFORATED TYMPANIC MEMBRANE AND EXPRESSIVE APHASIA. K.A. Steiling¹; A.H. Jackson¹. ¹Boston Medical Center, Boston, MA. (Tracking ID #132670)

LEARNING OBJECTIVES: 1) Recognize life threatening complications of middle ear disease in adults. 2) Discuss the imaging modality of choice for detecting intracranial complications of middle ear disease. 3) Discuss morbidity arising from this condition.

CASE: A 51 year old woman in otherwise good health was treated with oral and otic antibiotics after rupturing her left tympanic membrane (TM) with a cotton swab and subsequently experiencing left ear drainage. She presented three days later with headache, lethargy, and fevers. On exam, she was ill appearing and febrile. Left TM was ruptured, and there was a white exudate in the canal. The patient was arousable to voice. Speech was fluent. The remainder of her neurologic exam was nonfocal. A contrast enhanced CT of the head showed multifocal collections of gas along the inner table of the left side of the skull, fluid within the left middle ear cavity, and near complete opacification of the left mastoid air cells. There was no intracranial fluid collection or mass effect, and osseous structures were intact. Lumbar puncture was consistent with bacterial meningitis, and culture revealed streptococcus pneumoniae. A bedside myringotomy was performed. Gram stain of the aural exudate revealed gram positive diplococci. One week later, the patient represented with intermittent right arm numbness, expressive aphasia and fevers. Repeat CT of the head showed interval reabsorption of the left pneumocephaly and no defined fluid collection. Opening pressure during lumbar puncture was 27 mm of water. CSF was clear and colorless, and cell count was consistent with a resolving bacterial meningitis. MRI of the head revealed a left parietal subdural empyema (SDE) which resolved with an extended course of antibiotics. She suffers from persistent left conductive hearing loss.

DISCUSSION: Traumatic perforation of the TM is a problem commonly encountered in the primary care setting that can often be managed expectantly. However, it can be complicated by middle ear infection. SDE is a rare but potentially lethal complication of middle ear disease. In the largest case series of supratentorial SDEs, 74% of patients initially presented with meningismus and 40% had no focal findings on exam (Nathoo, 1999). A high index of suspicion is critical in making a timely diagnosis. Although no studies directly compare the sensitivity of MRI to that of contrast enhanced CT for the detection of SDEs, several case series suggest that findings on CT are subtle and that initial CT may even appear normal. As this case demonstrates, MRI may be superior to contrast enhanced CT in detecting intracranial suppuration. This diagnostic study should be considered in a patient presenting with meningismus, seizures or focal neurologic findings. While the incidence of intracranial complications from middle ear disease has markedly decreased following the widespread use of antibiotics, the prompt diagnosis and treatment of these complications remain paramount in minimizing long term sequelae. Patients presenting with supratentorial SDEs appear to have a favorable neurologic outcome. In one series, two-fifths of patients experienced a full neurologic recovery (Nathoo, 1999). Although over half of patients with SDEs present with hemiparesis, all but 13% regain function. Speech deficits, present in 20% of patients initially, resolve in 75% of survivors. (Cowie, 1983) Long term sequelae can include seizures, hearing loss, and persistent focal deficits.

PINK BLOOD CAUSES HEADACHE. M. Rehmetullah¹; S. Kumar¹; G. Gleeson¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #132571)

LEARNING OBJECTIVES: 1. To recognize acute presentation of carbon monoxide poisoning 2. Review the clinical manifestations, diagnosis and management of carbon monoxide poisoning.

CASE: A 23 year-old male, presented to the emergency department with four-hour history of headache, nausea, vomiting and lethargy. He was working in the basement of an old house which had a kerosene heating system and poor ventilation. He c/o headache 4 hours after the start of his work, which he described as throbbing and diffuse. It steadily progressed and was associated with nausea, vomiting and lethargy. He did not have any major past medical history. Physical exam revealed a young male in acute distress with severe headache. He was afebrile with a BP-146/84 mmHg and HR-78 beats/minute. The remainder of his exam was unremarkable with no focal neurological deficits. The differential diagnosis included subarachnoid hemorrhage and carbon monoxide(CO) poisoning. On drawing blood for lab work it appeared unusually "pink". His CBC and serum chemistries were within normal limits. A 12 lead electrocardiogram and CT scan of brain were normal. Arterial blood gas revealed carboxyhemoglobin(CO-Hgb) level of 22. He was placed on 100% oxygen via a high flow face mask and was treated symptomatically for headache & nausea. His symptoms resolved in the first few hours of initiating oxygen therapy and, he was admitted for observation. A repeat CO-Hgb level was less than 5. He was discharged the next day in stable condition

DISCUSSION: Smoke inhalation is the most common cause of accidental CO poisoning, working in poorly ventilated areas is also increasingly common. CO poisoning causes a multitude of effects due to inhibition of cellular oxidation, resulting in tissue hypoxia and cellular poisoning. The clinical presentation of CO poisoning is highly variable. Symptoms of mild poisoning may mimic those of a nonspecific viral illness with headache being most common, others include

vomiting, malaise and weakness. Severe CO toxicity can produce seizures, syncope or coma. Neuropsychiatric effects can appear several days after exposure. The diagnosis of CO poisoning is based upon a compatible history & physical exam in conjunction with an elevated CO-Hgb level measured by arterial blood gas. Pulse oximetry is not a reliable estimate of oxyhemoglobin saturation. CO-Hgb levels above 25 percent (0.25) are usually considered toxic, although CO-Hgb levels do not correlate well with clinical severity. Management is by administering 100% oxygen by tight-fitting face mask to reduce the biological half-life of CO. Hyperbaric oxygen therapy should be considered for severely poisoned patients presenting with neurologic abnormalities, myocardial ischemia and in pregnant females. Prognosis is generally excellent with early recognition and oxygen therapy.

PNEUMOCEPHALUS OR AIR HEAD : "PNEUMA"-"KEPHALE". S. Kumar¹; M. Rehmetullah¹; M. Elnicki¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #132483)

LEARNING OBJECTIVES: 1) To recognize an uncommon complication of a common procedure: pneumocephalus as a complication of epidural analgesia. 2) To review the clinical presentation of pneumocephalus.

CASE: A 79 year old female had a significant past medical history of hypertension, chronic obstructive lung disease, hypo-thyroidism and chronic back pain due to spinal stenosis. She presented to the emergency department immediately after undergoing an epidural steroid injection for chronic back pain. Her presenting complaints were nausea, vomiting, dizziness and vertigo. Her physical examination, including neurological examination was normal. The differential diagnosis included cerebrospinal fluid leakage and pneumocephalus. A computed tomography (CT) scan was done which revealed multiple areas of intracranial gas adjacent to the clivus and sella turcica, extending to the frontal convexities and suggestive of pneumocephalus. She was admitted and treated symptomatically with prochlorperazine for nausea and meclizine for vertigo. Neurologic exam was performed regularly and her symptoms resolved with symptomatic therapy. The patient gradually improved and was discharged the next day without any complications. She has been regularly observed as an outpatient and is doing well. A repeat CT scan done 6 months later showed resolution of the air.

DISCUSSION: Pneumocephalus is air within the intracranial cavity. It comes from the Greek words "pneuma" meaning "air" and "kephale" meaning "head." The air or gas within the intracranial cavity (e.g., epidural space, subdural space, intracerebral etc.) may result from traumatic events, fistulous tract formation, erosions of the skull from neoplasms or infection, neurosurgical procedures and iatrogenically. The latter is usually from inadvertent intrathecal or subdural injection of the air used to localize the epidural space and rarely after a lumbar puncture. It may present simply as headache with neck, shoulder, and back discomfort or it may present as confusion, deteriorating mental status, unconsciousness, and seizure. A brow-up lateral x-ray film of the skull or CT scan of head confirms the diagnosis of pneumocephalus. CT scan of head is preferred as it can help to differentiate between headache due to pneumocephalus and that by stretching of the meninges due to cerebrospinal fluid leakage. If the clinical features are deteriorating, a tension pneumocephalus should be suspected and immediate neurosurgical consultation is needed along with decompression. There are no specific guidelines regarding outpatient vs. inpatient management of neurologically stable patients, and clinical judgment remains vital. Most cases of iatrogenically produced pneumocephalus resolve spontaneously and no further intervention is required. It has been suggested that using saline instead of air to find the epidural space and a smaller caliber needle might be associated with fewer incidence of pneumocephalus.

PNEUMOCOCCAL PERICARDITIS AND THE VAGARIES OF VACCINATION. J.A. Hardman¹; C.K. Bates¹. ¹Beth Israel Deaconess Medical Center, Boston, MA. (Tracking ID #133995)

LEARNING OBJECTIVES: 1. Review the presentation and management of pneumococcal pericarditis. 2. Review operating characteristics of available pneumococcal vaccinations and their potential limitations

CASE: A healthy 64 year-old woman presented with fever to 102.9, myalgias and pleuritic chest pain. Her past medical history was notable for a single episode of presumed bacterial pneumonia three years earlier. Exam on admission revealed a woman in distress with pulse of 115, pulsus paradox of 15 mmHg, and JVD. EKG showed diffuse ST-elevations and electrical alternans. An echocardiogram revealed a large echodense pericardial effusion consistent with blood or inflammation. Cardiac catheterization showed equalization of pressures. Emergent pericardiocentesis yielded 680cc of thick fibrinous fluid. Cultures of blood and pericardial fluid grew pan-sensitive pneumococcus. Pleural effusions were present on admission and enlarged over several days with CT evidence of loculation. Video-assisted thorascopic surgery with bilateral decortication was done. She was evaluated for immunodeficiency after recuperation from the acute illness. Analyses of IgG levels and mononuclear cell type were normal. The patient received a dose of Pneumovax, the 23-valent polysaccharide vaccine, on the day of discharge from the hospital. Assays for pneumococcal specific antibodies three months later demonstrated a robust response to a single pneumococcal serotype. A second vaccination with Pneumovax did not result in protective antibody for any additional serotypes. Tetanus antibody level was found to be protective. Three doses of Prevnar, the heptavalent conjugated vaccine, were administered and repeat serologies demonstrated a significant immunologic response to five additional serotypes.

DISCUSSION: The prevalence of purulent pericarditis has declined significantly in the antibiotic era. Given its high mortality rate, it remains an important clin-

ical diagnosis to identify. This patient exhibited many of the commonly seen clinical manifestations of purulent pericarditis, including the acute onset of fever and pleuritic chest pain, with tachycardia and pulsus paradoxus on physical exam. Useful diagnostic studies included EKG, echocardiogram and ultimately cardiac catheterization. With aggressive intervention of pericardiocentesis and pleural evacuation, she had a complete recovery. She has had no evidence of constrictive pericardial physiology in long-term followup. Given the severity of her illness and absence of an identifiable conventional immunodeficiency, specific serotyping was done to understand the degree to which vaccination would protect the patient from repeat infection. Assays of pneumococcal specific antibodies after vaccination with Pneumovax revealed protective levels for a single pneumococcus serotype that we believe to have been the serotype of her infective strain. Absence of other serotype protection suggested an inadequate response to polysaccharide antigens. Her response to tetanus vaccine suggests that her response to protein antigens is robust. Heptavalent conjugated vaccine was thus administered and resulted in protective antibody to five additional serotypes. Our patient has remained well, but we presume remains at risk for life-threatening pneumococcal disease. We suggest that other patients with serious pneumococcal infections might be tested for pneumococcal antibodies following vaccination and that conjugated vaccine be used if the conventional polysaccharide vaccine is ineffective.

POLYMYOSITIS COMPLICATED BY CONGESTIVE HEART FAILURE. N. Saxena¹; F. Crock¹; R. Granieri¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133570)

LEARNING OBJECTIVES: 1. Identify the manifestations of polymyositis and potential consequences of its treatment. 2. Recognize the relationship between polymyositis and clinically apparent cardiomyopathy.

CASE: A 48 year old female with a history of Anti-Jo positive polymyositis and obesity presented with one week of increased shortness of breath, dyspnea on exertion, and dry cough. She denied fevers, chest pain, sputum production, and hemoptysis. Home medications included prednisone, folic acid, seroquel and ambien. She occasionally drank alcohol, but never smoked or used illicit drugs. On physical exam, the patient was obese and ill-appearing with JVP 10 cm above the right atrium, expiratory wheezes, tachycardia with distant heart sounds, and bilateral pitting edema. EKG showed atrial flutter with 2:1 AV conduction, right bundle branch block, and left anterior fascicular block. Laboratory studies were significant for negative cardiac enzymes and a BNP of 2130. CXR showed cardiomegaly and interstitial lung disease. Transthoracic echo showed biatrial enlargement and decreased biventricular function worse on the right. Coronary catheterization revealed normal coronary arteries, pulmonary hypertension, and biventricular dysfunction. Simultaneous recordings of the right and left ventricular pressures were equal suggesting restrictive physiology. The patient was diagnosed with cardiomyopathy secondary to polymyositis and was started on treatment with intravenous diuretics. She improved, her shortness of breath soon resolved, and she was discharged on steroids and oral diuretics. One month later, she remained stable and began evaluation for heart and lung transplantation.

DISCUSSION: Polymyositis is a systemic autoimmune-mediated inflammatory myopathy which may present with a wide variety of manifestations. The diagnosis is said to be certain when all four of the following are present: symmetric proximal muscle weakness which evolves over weeks to months, elevation of muscle enzymes, myopathic changes on EMG, and biopsy consistent with myositis. Anti-Jo antibodies are present in 5-30 percent of cases. Cardiopulmonary involvement with polymyositis is common, occurring in up to 70% of patients, and is one of the most significant causes of death in patients with the disease. Cardiac abnormalities include atrioventricular conduction disturbances, arrhythmias, pericarditis, and myocardial involvement manifesting as congestive heart failure. Pulmonary complications include weakness of the muscles involved in respiration, and interstitial lung disease with pulmonary hypertension either as a result of pneumonia, medication or autoimmune destruction from the disease process itself. The cause of polymyositis is unknown. Treatment is aimed at controlling the underlying disease process including immunosuppression with corticosteroids and methotrexate. There have been case reports describing successful transplantation in selected patients with polymyositis and cardiomyopathy, although the data is limited. Physicians should suspect polymyositis in patients with proximal muscle weakness which evolves over weeks to months and should understand that it can have a broad range of manifestations, not only limited to skeletal muscle.

POORLY CONTROLLED HYPERTENSION: WHEN MEDICAL MANAGEMENT JUST ISN'T ENOUGH. S.R. Greenhill¹; M.F. Muldoon¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133338)

LEARNING OBJECTIVES: Recognize when and how to screen for primary hyperaldosteronism

CASE: A 57 year-old white man was seen in clinic for routine follow-up for hypertension. His blood pressure (BP) had ranged from 140-165/90-100 mmHg, despite use of several anti-hypertensive medications. Noncompliance was suspected but unconfirmed. His family history was positive for hypertension but negative for coronary artery disease and endocrinopathies. He is a lifetime non-smoker but drinks 3-4 beers per day and more on the weekends. His medications included atenolol 50 mg qd, lisinopril 40 mg qd, HCTZ/triamterene 1 tablet qd, hydralazine 50 mg tid and potassium chloride 40 meq qd. On exam, his HR was 71, BP 142/83, and BMI 29.3. He had no murmurs, extra heart tones, or displaced PMI on cardiac exam, no pedal edema, and no carotid, abdominal or femoral bruits. The patient's potassium was low at 3.0 mmol/L, sodium

140 mmol/L, chloride 106 mmol/L, bicarbonate 22 mmol/L, BUN 15 mg/dL, creatinine 1.0 mg/dL. His potassium had been 3.3 mmol/L 6 months previously at which time oral supplementation was begun. An echocardiogram in 2003 did not find evidence of cardiac enlargement or dysfunction, or significant valvular abnormalities. Due to inadequate BP control and hypokalemia, the patient underwent screening for primary hyperaldosteronism (PA). The patient's erect, plasma renin activity (PRA) was <0.15 ng/mL/hr (normal 1.3-3.95) and plasma aldosterone concentration (PAC) was 40.7 ng/dL (erect, unrestricted dietary sodium normal range 4-31). His PAC/PRA ratio was calculated at 271 ng/dL (normal <30). A 24-hour urine collection for aldosterone contained 2.2 g creatinine and 46 µg of aldosterone (normal 6-25). A contrasted CT scan of the abdomen tailored for adrenal pathology revealed a 1.4 cm nodule in the posterior limb of the right adrenal gland compatible with an adenoma. The adrenal glands were otherwise unremarkable. Surgery was consulted for possible right adrenalectomy.

DISCUSSION: According to some experts, PA is now considered to be the leading cause of secondary hypertension, affecting 5-13% of all hypertensive patients. Indications for screening include: high blood pressure with hypokalemia, high blood pressure with adrenal incidentaloma, treatment-resistant hypertension, or whenever considering a workup for secondary hypertension. Frank hypokalemia need not be present. Screening involves measurement of PAC and PRA and is positive if the PAC is > 15 ng/dL and the PAC/PRA ratio > 30 ng/dL. This measurement can be affected by certain medications including beta-blockers which may decrease PRA (increasing the aldosterone-renin ratio) and ACE-inhibitors which may increase PRA (decreasing the aldosterone-renin ratio). If screening is positive, confirmatory tests include a 24-hour urine aldosterone level > 30 µg/24hrs. Finally, CT scanning can differentiate between the two most common types of PA, aldosterone-producing adenoma (APA) and idiopathic hyperaldosteronism (IHA) with bilateral adrenal hyperplasia. Generally, unilateral adrenalectomy is recommended for APA while medical management with receptor antagonists (spironolactone or eplerenone) is effective for IHA.

POST-POLIO SYNDROME PRESENTING AS RIGHT HEART FAILURE. A. Leadford¹; P. Radhakrishnan². ¹University of Arizona, Phoenix, AZ; ²St. Joseph's Hospital, Phoenix, AZ. (Tracking ID #134569)

LEARNING OBJECTIVES: 1) Recognize the late effects of polio. 2) Review the manifestations of post-polio syndrome.

CASE: An 84 year old Hispanic male presented with a two month history of shortness of breath, poor sleep, fatigue and lower leg edema. Patient described episodes of wheezing and trouble speaking during the day, as well as associated orthopnea. PMHx includes polio, which mainly affected the left side of his body. He is currently wheelchair bound because of a fall four years ago. The patient denies tobacco use, history of asthma, recent fevers, chills or cough. Physical exam revealed a centrally obese gentleman with stable vital signs. Lung exam revealed scattered wheezes and crackles. CV exam was significant for elevated JVP with hepato-jugular distention, along with 3+ pitting edema in both legs, extending up to the knees. Neuro exam revealed intact cranial nerves, intact sensation bilaterally, but diminished strength in the left lower extremity. Fasciculations were noted in both upper extremities as well as contractures in both hands. EKG revealed sinus arrhythmia. ECHO showed EF=70%, with elevated pulmonary artery pressures and dilated atrial chambers. PEFR=200's. Patient was started on diuretic and ACE-inhibitor and referred for evaluation for CPAP. **DISCUSSION:** Post-polio syndrome is a neurologic and musculoskeletal disorder, typically manifesting 10-40 years after the initial disease. It is estimated that over a million polio survivors are currently living in the U.S. and that approximately 50% could be affected by post-polio symptoms. Post-polio typically manifests with new progressive weakness, fatigue, joint pain, and muscle atrophy. Other symptoms include breathing and swallowing difficulty, sleep apnea and cold intolerance. The etiology of post-polio syndrome is unclear, but theories include an autoimmune reaction and the persistence of the polio virus. Severity of symptoms typically correlates with severity of the original disease. Diagnosis is made on clinical suspicion. Routine blood work is usually normal. EMG studies and muscle biopsy results are inconclusive and nonspecific and therefore unreliable. Management of post-polio is symptomatic. No treatment is available to reverse the underlying disorder. Our patient presented with symptoms of right-heart failure, with no evidence of left-heart failure or valvular disorder and no history of lung problems before this episode. Presumably his heart failure is caused by pulmonary hypertension caused by generalized hypoventilation due to weakening respiratory muscles and sleep apnea from post-polio syndrome. It is important for internists to be aware of this condition because although polio was officially eradicated in the U.S. in 1979, there are potentially hundreds of thousands of people who will be affected by post-polio syndrome in this country alone and millions of people worldwide.

POST TRANSFUSION PURPURA: A RARE TYPE OF TRANSFUSION REACTION. B. Telivala¹; B. Taqui¹; H. Shishodia¹. ¹Temple University, Philadelphia, PA. (Tracking ID #132948)

LEARNING OBJECTIVES: 1. Recognize post transfusion purpura (PTP) as a rare, but life threatening transfusion reaction 2. Review pathophysiology and clinical manifestations of PTP 3. Review diagnostic testing and management of PTP.

CASE: 66 year old African American female presented with intractable nausea and vomiting. She underwent exploratory laparotomy and was diagnosed with cholangiocarcinoma which was surgically resected. She developed postoperative hypotension, sepsis and thrombocytopenia of 42,000 (from 256,000). She was

transfused one pool of single donor platelets and developed a febrile non hemolytic transfusion reaction. Her exam was remarkable for diffuse purpura on the trunk and all extremities. She had multiple subsequent reactions to platelets and developed persistent profound thrombocytopenia (2,000). DIC panel and HIT Elisa test were negative, as were HIV and hepatitis panel. Anti-platelet antibody screen came back positive for anti-PIA1, anti-HLA class 1, anti-glycoprotein 1A2A. Patient was started on high dose steroids and intravenous immunoglobulin. Her platelet count remained low and plasmapheresis was started. Her symptoms and platelet count (260,000) improved dramatically. The patient relapsed after intravenous immunoglobulin, plasmapheresis and steroids were stopped. She responded to reinitiation of therapy, but expired soon afterwards due to unknown reasons.

DISCUSSION: Transfusion reactions may be classified as immune mediated and non-immune mediated. Immune mediated reactions include acute/delayed hemolytic, nonhemolytic/allergic and delayed transfusion related thrombocytopenia (post transfusion purpura). Post Transfusion Purpura (PTP) is a rare, but life threatening complication which results in a sudden dramatic thrombocytopenia developing 5 to 10 days after transfusion of whole blood, RBC, FFP or platelets. Patients have developed an anti-platelet antibody (usually against PIA1 antigen). Approximately 98% of white females and their spouses are PIA1 antigen positive. The remaining 2% develop antibody to this "foreign antigen" during pregnancy or prior transfusions. Subsequent transfusion triggers immune mediated destruction of the patient's own platelets resulting in a profound thrombocytopenia (<10,000). PTP manifests in adults, primarily women, with mucousal, gastrointestinal and genitourinary bleeding. Nine percent of patients develop intracranial hemorrhage. Bone marrow aspirates reveal abundant megakaryocytes and platelets coated with large amounts of IgG. Definitive diagnosis involves the demonstration of both circulating platelet specific alloantibody and the absence of appropriate platelet antigen following recovery. Several serological tests, such as immunoblotting, antigen capture ELISA and the monoclonal antibody immobilization of platelet antigen (MAIPA) are available. Spontaneous platelet recovery occurs within two weeks. Glucocorticoids reduce mucosal bleeding, but do not immediately improve platelet count. Intravenous immunoglobulin and plasmapheresis shorten the duration of thrombocytopenia (4-5 days). Infusion of PIA1 negative platelets is not effective because even antigen negative platelets are destroyed. Patients with diagnosed PTP should receive only frozen/thawed/washed or filtered blood products.

PRIMARY PULMONARY COCCIDIOIDOMYCOSIS CAUSING ACUTE RESPIRATORY DISTRESS SYNDROME (ARDS): AN UNCOMMON PRESENTATION OF COMMUNITY-ACQUIRED PNEUMONIA (CAP). S.Y. Chien¹; G.E. Mathisen¹. ¹University of California, Los Angeles, Sylmar, CA. (Tracking ID #136012)

LEARNING OBJECTIVES: 1. Increase awareness of *Coccidioides immitis* as a cause of severe respiratory failure in the community setting.

CASE: A 59 year-old Caucasian female was in excellent health until 2 weeks prior to admission when she developed upper respiratory tract symptoms. Five days later, she complained of pleuritic chest pain. An outpatient chest X-ray then demonstrated a left lower lobe infiltrate and she was given oral levofloxacin. Despite treatment, her condition did not improve and she was admitted to the hospital because of profound hypoxemia. Her lung examination showed egophony and crackles. Laboratory tests were significant for marked leukocytosis with left shift, mild eosinophilia, and thrombocytosis. Repeat chest X-ray showed a new infiltrate in the right upper lobe. A chest CT scan revealed diffuse interstitial, ground-glass opacification. Clinical impression at this time was CAP caused by resistant *Streptococcus pneumoniae* and possibly atypical organisms. She was started on broad-spectrum antibiotics including linezolid, azithromycin, cefotaxime, and aztreonam. Despite this therapy, her respiratory status deteriorated and she developed ARDS requiring mechanical intubation. At this time, ELISA IgM/IgG antibodies and sputum culture for *Coccidioides immitis* became positive and the patient was started on liposomal amphotericin B and fluconazole. Upon further questioning, her husband stated that the patient went hiking and horseback riding in San Luis Obispo and returned to Los Angeles via the San Joaquin Valley 3 weeks prior to admission. Unfortunately, the patient continued to be febrile and died one week later of progressive respiratory failure.

DISCUSSION: Although ARDS is rarely seen in primary pulmonary *Coccidioides immitis* infection, severe respiratory distress may be seen in patients with military or disseminated coccidioidomycosis. These patients usually present with several weeks of fever and cough prior to the onset of ARDS. Laboratory tests often show leukocytosis with an associated eosinophilia. Although our patient was an immunocompetent Caucasian female, studies suggest that severe coccidioidomycosis cases, especially disseminated disease, are more common in dark-skinned individuals. In a patient with CAP, diagnosis of coccidioidomycosis requires a high index of suspicion. A travel history to endemic areas—especially when patients participate in "high-risk" outdoor activities—along with the presence of peripheral eosinophilia are clues to the diagnosis. Unfortunately, coccidioidomycosis-related ARDS has a high mortality rate despite appropriate antifungal therapy. In a patient with CAP, aggressive efforts to identify and treat coccidioidomycosis may prevent the development of more serious complications such as the ARDS seen in this case.

PRIMARY SPLENIC HAIRY CELL LEUKEMIA - AN UNUSUAL CASE. T. Thenappan¹; S. Parikh¹; A. Radhakrishnan¹; H. Friedman¹; U. Muthyala¹. ¹St. Francis Hospital, Evanston, IL. (Tracking ID #134533)

LEARNING OBJECTIVES: 1. Recognize isolated splenomegaly without bone marrow involvement as an initial manifestation of hairy cell leukemia (primary

splenic hairy cell leukemia). 2. Differentiate primary splenic hairy cell leukemia from splenic lymphoma with villous lymphocytes.

CASE: A 51-year-old male presented for an evaluation of fatigue and generalized weakness of three weeks duration. He did not have hematemesis, nausea, vomiting, fever, night sweats, or bone pain. The patient had a history of well-controlled hypertension and diabetes. He denied smoking and drinking alcohol. The physical examination was consistent with splenomegaly; the spleen tip was palpable 3 cm below the left costal margin. There was no lymphadenopathy and hepatomegaly. The rest of the examination was unremarkable. The hemoglobin was 11.8 gm%, the white-cell count was 1460/mm³ with 20% segmented neutrophils, 67% mature lymphocytes, 10% monocytes and 1% eosinophils and the platelet count was 51,000/mm³. The complete metabolic profile and coagulation studies were normal. The peripheral-blood smear revealed normocytic and normochromic erythrocytes, marked neutropenia, and a decreased platelet count. No "hairy" or atypical lymphocytes were present. An abdominal CT scan showed an enlarged spleen, mild hepatomegaly and no lymphadenopathy. A bone marrow examination revealed hypercellular erythroid, myeloid and megakaryocytic series with adequate maturation. The flow cytometry of the bone marrow showed no evidence of clonal lymphoproliferative disorder. A diagnosis of hypersplenism was made based upon the splenomegaly, peripheral pancytopenia and hypercellular bone marrow. An explorative laprotomy with splenectomy and liver biopsy were performed. Microscopic examination of the spleen revealed a diffuse infiltration of atypical lymphocytes in the red pulp and marked atrophy of the white pulp. Lymphocytic infiltration was noted in the liver biopsy also. Flow cytometry performed on the splenic and hepatic tissues was consistent with hairy cell leukemia (CD 19, 20, 22, 11c, 25, 103, FMC7 and monoclonal kappa surface light chain). The patient's cell counts improved remarkably after splenectomy and he was subsequently started on cladribine.

DISCUSSION: Hairy cell leukemia (HCL) is a chronic lymphoproliferative disorder of B-cell origin that manifests primarily in the blood, bone marrow and spleen. Characteristic bone marrow infiltration by hairy cells is a consistent feature in all patients with this disorder. Primary splenic HCL without bone marrow involvement, like our case, is rare; there being only isolated case reports. It has been suggested that the spleen is the initial site of leukemic transformation in HCL with subsequent invasion of the bone marrow by the leukemic process. The onset of this invasion may vary from patient to patient. Hence patients can present with isolated splenomegaly without marrow involvement. Splenectomy is both diagnostic and therapeutic. Primary splenic HCL closely resembles splenic lymphoma with villous lymphocytes (SLVL). Flow cytometry and involvement of red pulp of the spleen by HCL aids in the differentiation. In conclusion, internists should be aware that bone marrow may not necessarily be involved in HCL and primary splenic HCL should therefore be considered as a diagnostic possibility in patients with unexplained splenomegaly, with or without pancytopenia.

PROFOUND ALCOHOLIC STEATOFIBROSIS VEILED AS METASTATIC CANCER. G.R. Berger¹; K.A. Kieffer¹. ¹Dartmouth-Hitchcock Medical Center, Lebanon, NH. (Tracking ID #135645)

LEARNING OBJECTIVES: 1) Generate a differential diagnosis of steatosis and 2) Recognize the pathology of alcoholic liver disease and the importance of alcohol abstinence in disease regression

CASE: A 51 year old Caucasian female presented with one month of progressive dyspnea and lower extremity edema. She denied fevers, hemoptysis, weight loss, abdominal pain, jaundice, or a change in bowel habits. Past medical history was significant for exposure to viral hepatitis, a remote history of injection drug use, and ongoing alcohol abuse. She was a retired emergency room nurse. Members of her paternal grandparents' family had been diagnosed with primary cancers of the colon, esophagus, stomach, lung, breast, and kidney. Exam was notable for tachycardia, hypoxia, scleral icterus, decreased breath sounds half way up the right chest, hepatomegaly, and edema. The patient was admitted to the hospital. Laboratory work up included: hemoglobin 12.5, MCV 108, total bilirubin 6.8, direct bilirubin 3.6, alkaline phosphatase 345, AST 185, and ALT 44. Chest x-ray revealed a moderate right pleural effusion. CT scan of the chest, abdomen, and pelvis showed multiple liver lesions suspicious for metastases and focal cecal wall thickening. Subsequent testing included positive hepatitis B surface and core antibodies and hepatitis C antibody, with absent hepatitis B surface antigen and undetectable hepatitis C virus by PCR. ANA, anti-smooth muscle antibody titer, and HIV were negative. CA-125 was markedly elevated at 409. Pleural fluid was transudative and negative for malignancy. Transvaginal ultrasound was normal. Colonoscopy was normal, though limited due to tortuosity; a complete barium enema was normal. CT-guided biopsy of the liver revealed severely active steatohepatitis with steatofibrosis, extensive Mallory's hyaline, neutrophils, and ballooning degeneration. Due to ongoing concern for possible malignancy, she underwent laparoscopic exploration. Abdominal and pelvic organs were grossly normal, other than an enlarged, nodular liver. Four biopsies of representative nodules were taken, all showed focal steatofibrosis, consistent with the first biopsy but far less severe. Taken together, her presentation of dyspnea secondary to hepatic hydrothorax in the setting of steatofibrosis, which improved dramatically after a period of alcohol abstinence, supports the diagnosis of alcoholic liver disease. The patient was subsequently discharged home, has remained abstinent, and has continued to improve clinically.

DISCUSSION: The differential diagnosis for fatty liver includes alcoholic steatosis, non-alcoholic fatty liver disease (NAFLD), effects from drugs, HIV infection, malnutrition, and lipid storage diseases. In this patient, the differential can be narrowed to alcohol or NAFLD, the two being indistinguishable on biopsy. A marked improvement in the degree of disease between biopsies and while abstaining from alcohol is a strong argument against NAFLD and in support of

alcohol as the culprit. A spectrum of severity can be seen in alcoholic liver disease ranging from steatosis to hepatitis to cirrhosis, with alcohol abstinence remaining the cornerstone of therapy and prevention of disease progression.

PROSTHETIC VALVE ENDOCARDITIS: AN IV ORIGIN IN GI DISGUISE. B.M. Kaplan¹; E. Hoffman¹. ¹University of Pittsburgh Medical Center, Pittsburgh, PA. (Tracking ID #134099)

LEARNING OBJECTIVES: To diagnose prosthetic valve endocarditis (PVE); To state the medical treatment of PVE; To state when surgical intervention is indicated.

CASE: 55 yo man with a history of IV heroin abuse, Hepatitis C, and a porcine aortic valve replacement secondary to endocarditis presents with 3 weeks of sharp, intermittent left upper quadrant and epigastric pain radiating straight to the back. Pain is worse when supine and not associated with eating. He reports recent post-prandial nausea and emesis, early satiety, lower extremity myalgias, fevers, chills, night sweats, productive cough with clear sputum, and 30 pound weight loss over the past year. He last used intravenous heroin 11 days prior to admission. Medications include aspirin, simvastatin and lisinopril. His last incarceration was 2 years ago. Physical exam: Sitting BP 82/55 Pulse 79; Standing BP 85/64, Pulse 92; Temp. 97.3; RR 18; O2 Sat 100% RA; Cachectic male; no scleral icterus, normal fundus, dry mucous membranes; clear lungs; II/VI systolic murmur at RUSB radiating to carotids; remainder of exam was normal. EKG normal. Hemocult negative stool. Laboratory values significant for WBC 18.9 (88 PMNs, 3 Bands), Hg 11, Plt 286; Na 127, K 4.2, Cl 91.8, HCO3 24.5, BUN 41, Cr 1.9, Glucose 339, Ca 8.5; liver enzymes, coagulation studies, amylase and lipase were normal. CT abdomen revealed multiple peripheral hypodense splenic lesions and small, wedge-shaped, bilateral lower pole kidney defects, compatible with infarcts. On second hospital day, patient developed fever, hypotension and IV/VI RUSB murmur radiating to the carotids. EKG showed first degree AV block. 4/4 blood cultures positive for MSSA. TEE revealed globular vegetation obstructing the AV bioprosthetic valve, severe AI with an abscess outside the valve ring, and possible fistula in the RV inflow area. Pt began a course of nafcillin and gentamycin and underwent a prosthetic valve replacement.

DISCUSSION: Staph. aureus, the most common cause of PVE, accounts for almost half of all cases. Recognizing MSSA PVE is difficult as the classic peripheral findings - Janeway lesions, splinter hemorrhages and Osler's nodes - are often absent on presentation. Additionally, bacteremia may be intermittent and may not cause fevers. The lesson from this case is two-fold: PVE can present as a painful abdominal syndrome due to splenic and renal infarcts caused by emboli showered from left sided valve vegetations; a new murmur, resulting from perivalvular infection and valve dehiscence, usually becomes evident late in the disease process. Early recognition of endocarditis in patients with atypical manifestations is therefore important. Diagnosis is made with appropriately obtained blood cultures and echocardiographic confirmation in conjunction with the Duke's criteria. Medical treatment should begin immediately with triple antibiotic therapy - nafcillin/oxacillin or vancomycin if concern for methicillin resistance exists, gentamicin and rifampin - and continue for at least 6 weeks. Indications for surgery include staphylococcal PVE unresponsive to antibiotic therapy, heart failure with prosthetic valve dysfunction, annular/aortic abscess, fistula formation, new-onset conduction abnormalities, and any vegetation on or near the prosthesis. When treated with antibiotics alone, MSSA PVE is associated with a 70% mortality rate; with prompt surgical intervention in combination with antibiotic therapy, a 20% reduction in this mortality rate can be achieved.

PSEUDOTUMOR CEREBRI IN A 24 YEAR OLD WOMAN PRESENTING TO AMBULATORY MEDICINE CLINIC. J. Pirkle¹; W. Moran¹; R. Velez¹. ¹Wake Forest University, Winston-Salem, NC. (Tracking ID #133156)

LEARNING OBJECTIVES: 1. When evaluating a patient with headaches, a careful history and physical examination are essential. 2. An adequate fundoscopic exam is critical in making the correct diagnosis. 3. Pseudotumor cerebri should be in the differential diagnosis of new headache in young obese females.

CASE: A 24 year-old woman presented to our outpatient clinic complaining of intermittent headaches for several months. The patient described the headaches as diffuse and throbbing with occasional blurriness of vision. She denied any nausea, vomiting, fever, chills, or neck stiffness. She reported gaining weight over the past year. On physical examination, her temperature was 97.2° Fahrenheit, weight 217 pounds, and height 65 inches. Blood pressure was 163/83 mmHg with a pulse of 88. She was an obese female in no distress. Neurological exam revealed normal visual acuity with no visual field defects. Cranial nerve exam, motor, and sensory exams were normal. Fundoscopic exam was attempted with a handheld ophthalmoscope, but the optic discs were difficult to visualize. The pupils were then dilated with tropicamide ophthalmic 0.5% solution, allowing clear visualization of the optic discs. There was blurring of the disc margin with engorged retinal veins. The remainder of the physical examination was unremarkable. Our chief concern was for elevated intracranial hypertension causing this patient's headaches. Contrast MRI with MR venography showed no evidence of venous thrombosis, hydrocephalus, mass, or other cause for elevated intracranial pressure. LP revealed an opening pressure of greater than 55 cm H2O. A large amount of cerebrospinal fluid was removed to achieve a closing pressure of 20 cm H2O. CSF analysis was normal. Her headache subsequently resolved. The patient was started on acetazolamide 250mg twice a day for pseudotumor cerebri and hydrochlorothiazide 25mg once a day for hypertension. She was discharged with a normal neurological exam and headache free.

DISCUSSION: Headache is one of the most common complaints presenting to practitioners in the primary care setting. The key to diagnosis of headache is a detailed, systematic history. Physical examination rarely reveals new information if the history is adequate, but measurement of blood pressure, neurological examination, and fundoscopic exam are key elements which can elicit "important headaches" that can be associated with severe morbidity. Pseudotumor cerebri is a rare cause of headache that should not be missed and can result in visual loss if not treated. A successful fundoscopic examination was the key to diagnosis in our patient, who may have gone undiagnosed if we had not dilated the pupils. Internist should consider pharmacologic dilation when evaluating headache in appropriate patients.

PULMONARY EDEMA, A PRESENTING MANIFESTATION OF PULMONARY EMBOLISM. R. Sher Ali¹; S. Parikh¹; P. Amleshi¹; K. Gujral¹; H. Friedman¹. ¹St. Francis Hospital, Evanston, IL. (Tracking ID #134359)

LEARNING OBJECTIVES: 1. Recognize pulmonary edema as the presenting manifestation in patients with massive pulmonary embolism. 2. Consider the diagnosis of pulmonary embolism in patients who have a high likelihood of having this condition.

CASE: A 69-year-old woman presented with signs and symptoms of diabetic ketoacidosis. She was appropriately managed in the intensive care unit. On the fourth day of admission, she complained of shortness of breath and was noted to be hypotensive, tachycardic and tachypneic. She was saturating 85% on room air. Her chest examination revealed bilateral crackles. The chest x-ray showed bilateral interstitial infiltrates consistent with pulmonary edema. A 12-lead EKG showed new T wave inversions in leads V2, V3 and V4. Cardiac markers and B-type natriuretic peptide levels were elevated and a diagnosis of myocardial ischemia was entertained. An echocardiogram was performed on an emergent basis; it showed normal wall motion, severely dilated right ventricle and bulging of the interventricular septum into the left ventricle. The pulmonary artery pressure was estimated to be 45 mmHg. An urgent infused CT scan of the chest revealed a saddle embolus in the main pulmonary artery. The patient was heparinized and 100mg of alteplase was administered intravenously. A subsequent echocardiogram demonstrated complete resolution of the right ventricular dilatation and the patient was discharged home in stable condition on warfarin therapy.

DISCUSSION: The most common cause of pulmonary edema resulting from massive pulmonary embolism is left ventricular dysfunction. Increased pulmonary vascular resistance causes right ventricular dilatation. The interventricular septum therefore bulges into and compresses an intrinsically normal left ventricle leading to diastolic dysfunction. In other cases, there is extensive precapillary obstruction, which leads to overperfusion of patent microvasculature, increase their permeability and lead to subsequent pulmonary edema. Our patient's increased cardiac markers and EKG changes suggestive of myocardial ischemia could adequately explain her symptoms. However, the patient's hospital course, relative dehydrated state due to diabetic ketoacidosis and the results of the echocardiogram led us to consider an alternate diagnosis of pulmonary embolism. Our case emphasizes the importance of considering the diagnosis of pulmonary embolism in patients who have a high likelihood of having this condition. It also underscores the importance of diagnosing massive pulmonary embolism in patients presenting with pulmonary edema.

RAPIDLY PROGRESSIVE GLOMERULONEPHRITIS WITH UNUSUAL CLINICAL PRESENTATION. F.M. Harji¹. ¹University of New Mexico, Albuquerque, NM. (Tracking ID #135585)

LEARNING OBJECTIVES: 1. To recognize that urinalysis is the most important noninvasive test in the diagnostic evaluation of renal failure-Burton Rose 2. The importance of accurate and timely diagnosis in the setting of rapidly progressing renal failure.

CASE: Early diagnosis and timely treatment of rapidly progressive glomerulonephritis (RPGN) are critical for the preservation of renal function. We present a case of RPGN, in which the diagnosis was delayed because of an unusual clinical presentation. A previously healthy 59-year old man was seen in the Emergency Department (ED) of one hospital with a two week history of anorexia, weakness, low grade fevers, and night sweats followed by one week of severe bilateral flank pain and red urine. The episode was diagnosed as a urinary tract infection (UTI) without a urine culture, and he was placed on a 10-day course of levofloxacin. The pain worsened the first four days after the ED visit but improved with ibuprofen. Anorexia and fevers persisted, and he was seen in the ED of a second hospital, where laboratory tests revealed a serum creatinine of 3.6 mg/dL, WBC 17.3 k/mm3 with a left shift, hematocrit 41.7%, and urinalysis (UA) with 2+ protein, 32 RBCs, 7-10 WBCs, and a few bacteria (urine cultures turned out negative). He was admitted with the diagnosis of pyelonephritis and renal failure. Repeat UA revealed RBC casts and dysmorphic RBCs, characteristic of glomerular injury. Serologic work-up revealed normal serum complement levels, ASO, ANA, RPR, hepatitis B and C serologies, antiphospholipid antibody, anti-glomerular basement membrane antibody, and cryoglobulins. C-ANCA (anti-proteinase-3) was highly positive (6.44 ISR, normal range 0-0.9 ISR), rheumatoid factor titer was 146 IU/mL (normal 0-20 IU/mL), C-reactive protein was 12.7 mg/dL (normal < 1 mg/dL), and erythrocyte sedimentation rate was 89 mm/hr. A percutaneous renal biopsy revealed pauci-immune crescentic glomerulonephritis, consistent with the positive C-ANCA serology. He was treated with intravenous (IV) hydrocortisone followed by oral prednisone and IV cyclophosphamide. His symptoms rapidly improved, and his serum creatinine, which rose to 5.6 mg/dL, decreased to a value of 1.3 mg/dL six months later.

DISCUSSION: RPGN may present clinically imitating a UTI, and careful urinalysis is invaluable for its diagnosis. Rapidly progressing glomerulonephritis is characterized by extensive crescent formation - degree of which determines severity of the disease and more importantly, rapid progression to end-stage renal disease in most untreated patients. Accurate and timely diagnosis aid in initiating therapy early before the disease progresses to an irreversible stage. It is important to recognize differences in the types of rapidly progressing glomerulonephritis as therapy may vary in some cases.

RECURRENT STROKES IN A YOUNG MALE: MOYAMOYA DISEASE. R. Satpathy¹; C. Satpathy²; S. Pasupaleti¹; R.P. Warrior³. ¹Creighton University Medical Center, Omaha, NE, Omaha, NE; ²SCB Medical College, Cuttack, Orissa; ³Creighton University, Omaha, NE. (Tracking ID #133857)

LEARNING OBJECTIVES: Learn the differential diagnoses and diagnostic work-up of recurrent strokes or TIAs in a young male. Recognize moyo moyo disease, a narrowing of the basal intracranial vessels around the circle of Willis accompanied by the development of a cloud of collateral 'moyamoya' vasculature causing cerebral ischemia and stroke.

CASE: A 28-year-old white male without significant history of hypertension, diabetes, smoking or atherosclerotic heart disease presented to the ER with tingling, weakness and numbness of his right arm. This was associated with slurring of speech and difficulty in swallowing. The episode lasted for about one to two hours. Patient denied any chest pain, shortness of breath, palpitation, diplopia, headache, seizure or loss of consciousness. However, patient admitted to having these similar events for last six months. The frequency had increased lately. On admission, his BP was high. A detailed neurological examination did not reveal any focal neurological deficit. The CT scan of head, EEG, carotid Doppler, EKG and echocardiogram were normal. However, the MRI and MRA showed focal acute ischemic infarct in the left lentiform nucleus with considerable irregularity, narrowing and beaded appearance of bilateral middle and anterior cerebral arteries and proximal stenoses more on the left than the right. Vasculitis work-up was normal. Eventually the cerebral angiogram confirmed the above findings and showed evidence of formation of collaterals, but no aneurysms. Distal branches and the major vessels in the neck were normal bilaterally.

DISCUSSION: "Moyamoya" is a Japanese word meaning puffy, obscure, or hazy like a puff of smoke in the air. This term describes the smoky angiographic appearance of the vascular network of small arteries, which is the hallmark of Moyamoya disease. The common clinical presentations include ischemic events, hemorrhagic stroke and epilepsy. Ischemic events (TIA and stroke) and epilepsy are more common in the children, where as adult Moyamoya disease presents with intracerebral hemorrhage due to rupture of fragile abnormal vessels. The reason for our patient presenting with ischemic events, not hemorrhages could probably be explained by the early collaterals, which are not yet very fragile. Since the cause of Moyamoya disease is not known, treatment is mainly symptomatic. The natural history of this disease tends to be progressive, although this may be more benign in United States compared with Asian population.

RECURRENT LEG PAIN: A CASE OF KAPOSI'S SARCOMA MIMICKING CELLULITIS. Z. Szep¹; B. Taqui¹; H. Shishodia¹. ¹Temple University, Philadelphia, PA. (Tracking ID #132958)

LEARNING OBJECTIVES: 1. Recognize Kaposi's sarcoma (KS) as the most common AIDS associated cancer in the United States 2. Review clinical presentation of KS, which may mimic cellulitis 3. Review management of KS

CASE: 39 year old African American male with HIV and past history of IVDA presented with third episode of right lower leg redness, pain and swelling over the past 4 months. Patient had presented with similar complaints two prior times and had been diagnosed and treated for cellulitis. Both times, his symptoms improved but did not completely resolve. Exam revealed T 100.6 F, diffusely erythematous and tender right lower extremity, edematous right foot with sausage toes, onychomycosis, skin breakdown and crusting between toes. Track marks were present on both lower extremities. Labs revealed CD4 204. MRI of the right lower extremity showed circumferential soft tissue edema of the ankle and foot without any evidence of osteomyelitis. Patient was started on vancomycin for presumed, recurrent cellulitis. Dermatology was consulted for a skin biopsy which showed a vascular tumor consistent with Kaposi's sarcoma. Vancomycin was stopped and patient was discharged with outpatient follow-up at HIV clinic. His symptoms markedly improved with initiation of highly active anti-retroviral therapy (HAART).

DISCUSSION: Kaposi's sarcoma (KS) is a vascular tumor associated with human herpesvirus 8. There are four clinical variants of KS: classic (affecting elderly men of Mediterranean/Eastern European descent), African endemic, transplant/immunosuppression associated and epidemic/AIDS associated. Although the incidence of KS in HIV population has decreased markedly, it remains the most common AIDS associated cancer in the United States. It can affect people at all stages of HIV infection and is unlikely to be serious (with internal organ involvement) as long as the CD4 cell count is above 250. Skin lesions are most characteristic of KS but extracutaneous sites include lungs, gastrointestinal tract, oral cavity. Skin lesions usually occur on lower extremities, face, genitalia or oral mucosa and may be accompanied by lymphedema. They are multi-colored and usually papular, though KS lesions on the feet and thigh may appear plaque like. Extensive plaque formation and edema in the lower extremities may take on the appearance of cellulitis, and in dark-skinned persons, the lesions of the neoplasm may not be noticeable. Biopsy confirms the diagnosis. Patients are treated with HAART, with or without treatment directed

against the sarcoma. Other modalities include irradiation, intralesional or systemic chemotherapy or cryotherapy. There is no known cure for KS. The CD4 count and viral load are important prognostic factors in KS. Our case highlights the importance of searching for alternative diagnoses in patients with persistent/recurrent/refractory symptoms. We also learned to pursue a diagnosis of KS in dark skinned AIDS patients who present with recurrent cellulitis.

REEXPANSION PULMONARY EDEMA: TALE OF TWO LUNGS. A. Sequeira¹; S.C. Reddymasu¹; R. Gonzales¹; I. Merlos¹; D. Barfield²; J. Huang¹. ¹Louisiana State University Medical Center at Shreveport, Shreveport, LA; ²Overton Brooks Medical center, Shreveport, LA. (Tracking ID #132606)

LEARNING OBJECTIVES: Recognize risk factors and utilize preventive strategies in the management of reexpansion pulmonary edema (RPE).

CASE: A 46-year old male with a history of cirrhosis, ascites and chronic right sided pleural effusion was admitted for worsening dyspnea, orthopnea and ascites. Examination was remarkable for an afebrile tachypneic patient using accessory muscles of respiration. Chest exam revealed stony dullness of the entire right hemi thorax with decreased breath sounds. His pulse oximetry on 1 liter of oxygen was 93%. Lab data: Hb 12.8 g/dL, WBC 10.7k/cmm, PLT 124k/cmm, PT 16s, ammonia 27 umol/L, BUN 47 mg/dL, creatinine 1.8mg/dL, Na 124 mmol/L, albumin 2.2 mg/dL, AST 89 U/L and ALT 43 U/L. CXR showed complete whitening of the right lung field. Two liters of transudative fluid was removed by thoracentesis with mild improvement in symptoms. The next day after an additional 2.5 liters of pleural fluid was removed, the patient started coughing, became hypoxemic, tachycardic and hypotensive. The procedure was immediately terminated. A repeat CXR showed bilateral pulmonary edema. He was then transferred to the MICU where he received albumin and furosemide in addition to 50% oxygen via venturi mask. His oxygenation status improved gradually over the next few days. He however, continued to require oxygen to maintain a saturation of 90% and was therefore discharged on home oxygen.

DISCUSSION: RPE is a rare condition that is known to occur after release of a pneumothorax, large volume thoracentesis and release of atelectasis from airway obstruction. Majority of cases are unilateral with only 6.7% being bilateral. The clinical features of RPE vary from asymptomatic radiographic findings to severe cardiopulmonary decompensation. Symptoms usually appear within the first hour after reexpansion. The etiology is believed to be a combination of increased micro vascular permeability and increased hydrostatic pressure from vascular flooding of the reexpanded lung. Risk factors include: chronicity of effusion (greater than 3 days), volume of fluid present (greater than 1500cc) and underlying cardio-pulmonary dysfunction which hinders compensation. Treatment includes use of diuretics and cardio-respiratory support with positive end expiratory pressure (PEEP) and inotropes. Despite these measures, RPE carries a mortality rate of 20%. Therefore, preventive strategies are of clinical importance. These include assessing patients for the above mentioned risk factors, using continuous oximetry and limiting thoracentesis to less than a liter with slow drainage.

REFEEDING SYNDROME. S.C. Reddymasu¹; S. Konduru¹; M. Singh¹; S. Ankur¹; N. Koshy¹; J. Blondin¹. ¹Louisiana State University Medical Center at Shreveport, Shreveport, LA. (Tracking ID #135382)

LEARNING OBJECTIVES: 1. Understanding Refeeding syndrome. 2. Importance of measuring phosphate levels periodically during refeeding after prolonged starvation.

CASE: A 50-year-old African-American man presented to the Emergency department with complaints of decreased oral intake and weight loss for the past 2 years. Patient reported he did not "feel like eating". On exam we found a poorly groomed, dehydrated, malnourished man with markedly decreased muscle mass and BMI around 16. Otherwise his exam was within normal limits. Psychiatric diagnosis was major depressive disorder with psychosis. Lab work showed hemoglobin of 11.2g/dl. Electrolytes, BUN/Creatinine, phosphate and magnesium were normal. Serum albumin was 1.8g/dl. Thyroid function tests were within normal limits. Serology's for Hepatitis A, B, C and HIV were negative. Active infection was not documented on admission. Work up for malignancy including Gastrointestinal endoscopy was negative. Patient was aggressively refeed with nasogastric tube feeds because oral intake was very poor. Parenteral thiamine was also administered. On day 5 there was a rapid decline in the patients' status. He developed pulmonary edema and urosepsis. Phosphate level at that time was 1.3mg/dl (Normal 2.3-4.7). He had a weight gain of 5kg since admit. Chest radiograph was read as pulmonary edema. EKG revealed no acute changes. Cardiac enzymes and echocardiogram were within normal limits. Patient did not respond to diuresis and antibiotics and clinical improvement correlated with replenishment of phosphorus levels. The diagnosis of refeeding syndrome was obvious.

DISCUSSION: Refeeding syndrome was first described in Far East prisoners of war after the Second World War, when starting to eat again after a period of prolonged starvation seemed to precipitate cardiac failure. The pathophysiology of refeeding syndrome is that in starvation the secretion of insulin is decreased in response to a reduced intake of carbohydrate. Catabolism of fat and protein produce energy in this state. This results in intracellular loss of phosphate. Phosphate is necessary for the generation of adenosine triphosphate and other crucial phosphorylation reactions. In Malnourished patients, intracellular phosphate stores are depleted despite normal serum phosphate concentrations. Aggressive feeding after a period of prolonged starvation causes a sudden shift from fat to carbohydrate metabolism with resultant increase in insulin secretion and increased cellular uptake of phosphate leading to profound hypo-

phosphataemia. This phenomenon usually occurs within four days of starting to feed again. Serum phosphate concentrations of less than 1.5 mg/dl (normal range 2.5–4.5 mg/dl) can produce the clinical features of refeeding syndrome which is essentially secondary to cellular dysfunction including rhabdomyolysis, leukocyte dysfunction, respiratory failure, cardiac failure, hypotension, arrhythmias, seizures, coma, and sudden death. Patients at risk for developing refeeding syndrome should be treated with awareness that phosphate levels will very likely decline. Periodic measurement of phosphate and treating abnormal values should be prioritized. Intravenous phosphate is often necessary because oral supplementation may be inadequate. In efforts to establish feeding in prolonged starvation states, a weight increase of not more than 1 kg per week should be attempted. Increase by more than 1 kg suggests fluid retention and could be a pointer for refeeding syndrome.

REFEEDING SYNDROME: CLINICAL VIGNETTE AND KEYS TO RECOGNITION AND MANAGEMENT. A.M. Shah¹; A.A. Donato¹. ¹Reading Hospital and Medical Center, West Reading, PA. (Tracking ID #135274)

LEARNING OBJECTIVES: 1. Recognize patients at risk for refeeding syndrome and screen appropriately. 2. Treat vitamin and electrolyte deficiencies while assessing for cardiac decompensation in the refeeding syndrome patient.

CASE: A 34-year old male prisoner with paranoid schizophrenia was admitted with history of avoiding food for 56 days with 9-kg (16%) weight loss and intention to begin feeding again. Symptoms at presentation included nausea and vomiting for 2 days and symptoms of orthostasis. The patient's physical examination revealed a malnourished, otherwise healthy oriented male, with weight of 49 kg (BMI 16.7). Vital signs showed hypotension in the supine position with tachycardia, but was otherwise within normal limits. Initial laboratory analysis revealed serum potassium 3.4 meq/l, phosphorus 4.3 meq/l, magnesium 2.6 meq/l, and albumin of 4.2 g/dl. Due to concerns related to refeeding syndrome, thiamine was given intravenously, and oral repletion of phosphorus and potassium was instituted. Dextrose-containing fluids and a liberalized diet was given. The patient's serum phosphorus level dropped to 0.5 meq/l at hospital day 2 without symptoms, requiring 50 millimoles of intravenous phosphorus over 2 days to replete his phosphorus to normal. His potassium and magnesium levels also dropped during refeeding, and were appropriately supplemented. Diarrhea developed during oral feeding, lasting 4 days but resolved by time of discharge. No volume overload was detected, and the patient's tachycardia resolved by day 6.

DISCUSSION: Since its first description following refeeding starving prisoners in World War II, the refeeding syndrome has been reported in patients with alcoholism, anorexia, short bowel syndrome receiving total parenteral nutrition and in prisoners undergoing prolonged voluntary fasts. Thiamine deficiency must be addressed before feeding is begun. Total body stores of phosphorus, potassium and magnesium are significantly depleted, however serum levels are initially normal to slightly low due to renal adjustments in excretion. Rapid, high-carbohydrate feeding, as seen in this case, causes intracellular shifts of glucose, phosphorus and potassium as a massive insulin secretion is stimulated. Severe hypophosphatemia can quickly develop and remain asymptomatic until causing acute myocardial dysfunction. This along with the antidiuretic effect of insulin can precipitate acute heart failure and death. Very close monitoring of electrolytes with aggressive phosphorus repletion, and slow initiation of caloric intake can avert this iatrogenic hazard. Clinicians that care for patients that may have had prolonged caloric interruption should be aware how to recognize this syndrome and navigate its pitfalls.

RETRIEVAL OF 25 CM LONG INTRACARDIAC FOREIGN BODY. S.G. Khurshid¹; U. Ahmed¹; O. Marzouki²; M. Akhtari²; H. Friedman²; S. Dadkhah². ¹Saint Francis Hospital at Evanston, Evanston, IL; ²St. Francis Hospital, Evanston, IL. (Tracking ID #134971)

LEARNING OBJECTIVES: 1) To recognize the presence of retained catheter fragments in the cardiac chambers as a complication of intravenous catheterization. 2) Foreign bodies in the heart can lead to serious complications but may also remain asymptomatic until found incidentally. 3) Percutaneous retrieval of intracardiac foreign bodies is the standard method of treatment.

CASE: An 80-year-old Spanish-speaking male presented to the ED complaining of shortness of breath. CT scan of chest was done to rule out pulmonary embolism given the history of esophageal cancer. It revealed bilateral infiltrates and right side pleural effusion consistent with pneumonia and no pulmonary embolism. An incidental finding was also noted on the CT scan. A curvilinear foreign body extending from the right atrium through the right ventricle all the way into the left pulmonary artery was found. This foreign body was thought to be a retained catheter and a transesophageal echocardiogram confirmed its presence. Patient underwent per cutaneous trans-femoral retrieval of the foreign body. It was found to be a 25 cm long intertwined catheter with an adherent thrombus. On further probing it was found that the patient had history of multiple hospitalizations in past few years including at least one admission to the intensive care unit when central intravenous access was inserted.

DISCUSSION: An intracardiac foreign body is an iatrogenic complication that occurs during arterial and venous catheterization or interventional procedure. Although foreign bodies have been described as being in the heart more than 20 years without complications, many serious complications have also been described. Fischer and Ferreiro found 71% incidence of serious complications caused by catheter fragments. These include: septicemia, myocardial perforation and tamponade, bacterial endocarditis, thrombus formation, dysrhythmias and sudden death. Therefore, it is generally recommended that catheter fragments should be removed as soon as possible. Percutaneous retrieval of intra-

cardiac catheter is the standard method of treatment that can obviate the need for major surgical procedures. Open thoracotomy is indicated if the migrated foreign body is adherent to the myocardium or not retrievable by percutaneous approach. We report a case of a broken central venous catheter that dislodged and migrated to the right heart and pulmonary artery with unknown duration. The patient remained asymptomatic and the finding on CAT scan was incidental. TEE revealed that the catheter was not embedded in the myocardium. Therefore decision was made to remove the catheter percutaneously. Several techniques have been used successfully to retrieve foreign body from heart and great vessels. The most important instruments are Loop snares, grasping forceps, basket retrieval and Koch catheter. Our patient underwent percutaneous trans femoral vein approach using loop snare and hockey stick guider's catheter. Femoral venous cutdown had to be performed to remove the catheter from the access site given the size of the catheter. To our knowledge only three cases of asymptomatic intracardiac foreign body have been reported in the literature. All of them underwent percutaneous transvenous retrieval of the foreign body. Therefore surgery should only be contemplated when percutaneous techniques have failed. Asymptomatic foreign bodies in the heart that are treated conservatively require close follow up, anticoagulation and antibiotic prophylaxis.

RHABDOMYOLYSIS ASSOCIATED WITH LEVOFLOXACIN: A YOUNG MAN WITH HEMATURIA AND BODY ACHES. S.R. Pulagam¹; B. Khanna¹; R.D. Crook¹; L. Kirchner¹. ¹Canton Medical Education Foundation, Canton, OH. (Tracking ID #134536)

LEARNING OBJECTIVES: 1. To recognize quinolones as a potential cause of rhabdomyolysis. 2. To discuss differential causes of rhabdomyolysis.

CASE: A 22-year-old black male was admitted to the hospital with a complicated urinary tract infection. He was treated with Levofloxacin IV (intravenous) and subsequently discharged home on oral Levofloxacin (7 days total treatment). After the third day of treatment, he returned with proximal thigh tenderness and pain, followed by a change in urine color to reddish-brown. His urinalysis showed a pH of seven, protein 500 mg/dL, blood 250 ery/μL, and 2–5 RBC/HPF, but no casts. Leukocyte esterase and nitrite were negative. His aspartate aminotransferase was 1,979 IU/L, alanine aminotransferase level 388 IU/L, alkaline phosphatase 63 IU/L, total bilirubin 0.8 mg/dL, and prothrombin time 10.3 s. His hemoglobin was 16.6 mg/dl and platelets 209,000/mL. Creatine kinase (CK) was 106,000 IU/L, serum myoglobin 6,792 IU/L, and lactate dehydrogenase 11,590 IU/L. CK levels peaked the following day at 363,000 IU/L. Sedimentation rate was 29. He was aggressively hydrated with alkalized fluids and his creatinine remained stable at 0.9 mg/dL. A urine drug screen was negative for cannabinoids, cocaine, and acetaminophen. The only medication taken was Levofloxacin. He denied taking over-the-counter or herbal medications or dietary supplements. Anti-nuclear antigen, anti-liver kidney microsome antibody, hepatitis panel, anti-smooth muscle antibody, alpha-1 antitrypsin, ceruloplasmin were negative. Antibody titers for HIV, EBV, and CMV all were negative.

DISCUSSION: Rhabdomyolysis is an uncommon problem, but can be fatal. Rhabdomyolysis is a syndrome characterized by skeletal muscle necrosis and elevated creatinine kinase levels, sometimes greater than 100,000 IU/L. The most common precipitating factor is crush injury to muscle. Some common causes include trauma, genetic myopathies, seizures, drugs, alcohol and toxins, extreme exertion, endocrine disorders, and metabolic abnormalities. Other causes include cocaine intoxication, amphetamines, alcohol, phenothiazines, and statin drugs. This is a unique case with marked elevations in CK, which appears related to Levofloxacin use. To our knowledge, this is the first case of rhabdomyolysis associated with Levofloxacin to be reported.

RING-ENHANCING LESIONS. J. Schieffelin¹. ¹Tulane University, New Orleans, LA. (Tracking ID #136034)

LEARNING OBJECTIVES: 1. Review the evaluation of neurosensory changes 2. Review the differential diagnosis of ring-enhancing lesions.

CASE: A 50 year-old man with a history of right and left-sided strokes presented with one day of right arm numbness. The numbness extended from his fingertips to his shoulder in a glove-like pattern. He denied decreased strength or coordination. He denied any difficulty walking, dysarthria, dizziness or incontinence. His wife reported that he had two days of slurred speech. He lived on a farm with cows and pigs, but denied any recent travel, fishing or hunting. He was afebrile and normotensive. He had decreased right peripheral vision, a mild right facial droop, and decreased sensation in the right fifth cranial nerve distribution. He had decreased sensation in the right arm. His strength was normal in all extremities, and his gait was normal. A head CT with contrast demonstrated ring-enhancing lesions adjacent to the left lateral ventricle and extending into the corpus callosum; a second lesion was in the left temporal lobe with edema. An HIV test was negative. Cryptococcal antigen, blood cultures, and toxoplasma titers were all negative. A subsequent brain biopsy of one of the lesions was consistent with hemangioblastoma.

DISCUSSION: The differential diagnosis of neurosensory abnormalities includes electrolyte disturbances, infection, malignancy, transient ischemic attacks and strokes. After a thorough exam is performed, a head CT should be obtained to exclude an acute bleed, ischemia or mass. Ring-enhancing brain lesions are rare, but are more common in immunocompromised patients. The differential diagnosis for these lesions includes toxoplasmosis, cysticercosis, tuberculosis, lymphoma, metastatic cancer and glioblastoma multiforme. Cysticercosis is common in endemic areas of Asia and Latin America among pig farmers, but not in non-endemic areas. Metastatic cancer is more likely when

there are multiple lesions as well as an identified primary tumor. Lymphoma and glioblastoma must be evaluated by biopsy when other causes have been excluded. Hemangioblastoma is a rare brain tumor that is sometimes associated with von Hippel-Lindau disease, an autosomal dominant inherited disorder.

RULING IN A DIAGNOSIS OF EXCLUSION: IDENTIFYING ADULT STILL'S DISEASE. C.G. Reed¹; Y. Min¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #131989)

LEARNING OBJECTIVES: 1. Recognize Adult Still's disease (ASD) as an etiology of fever of unknown origin 2. Distinguish ASD from other causes of persistent fever, rash, and polyarthritis

CASE: A 26 year old previously healthy Filipina woman complained of ten days of fevers as high as 40.3° C, myalgias, and a rash on her thighs. She endorsed sore throat and arthralgias but denied cough, dysuria, flank pain, or diarrhea. On exam she was uncomfortable, diaphoretic, and tachycardic to 116. The oropharynx was clear and there was no lymphadenopathy. She had severe diffuse muscle tenderness without true weakness. There was a salmon-colored maculopapular rash on her bilateral anterior thighs and chest. She had persistent daily fever spikes as high as 39.4° C throughout her 11 days in the hospital. The rash was persistent, but became more prominent during fever spikes. She developed a migratory polyarthritis involving the ankles, wrists, and MCPs. LP and CXR were unremarkable, and blood and urine cultures remained negative. Laboratory exams were significant for a WBC of 13,000 with 87% neutrophils; mildly elevated transaminases with normal bilirubin and alkaline phosphatase; and ESR 89. CT abdomen/pelvis revealed only splenomegaly, without occult abscess or lymphadenopathy. Infectious workup included negative serologies for HIV, CMV, EBV, hepatitis B/C, West Nile virus, coccidioides, and parvovirus B19. A PPD was negative. Rheumatologic evaluation included negative antibodies for ANA, SCL-70, RNP, centromere, lupus anticoagulant, cardiolipin, SSA/SSB, anti-streptolysin O, and rheumatoid factor. Complement levels were normal. A ferritin level was very high at 5,675 µg/l. By using established criteria, a diagnosis of Adult Still's disease was made. Prednisone was started at 40 mg/day, and her symptoms completely resolved.

DISCUSSION: The differential diagnosis of fever of unknown origin includes infection, malignancy, and autoimmune disease. ASD is a systemic inflammatory disorder of unknown etiology. Patients typically go through extensive work-up to exclude occult infection or malignancy. Major criteria developed by Yamaguchi for ASD include fever > 39° C for 1 week or longer, arthralgias or arthritis for 2 weeks or longer, a characteristic maculopapular nonpruritic rash over the trunk and extremities during febrile episodes, and leukocytosis with >80% neutrophils. Minor criteria include sore throat, lymphadenopathy, hepatosplenomegaly, abnormal LFTs, and negative tests for ANA and rheumatoid factor. Five or more criteria, including at least two of the major criteria, must be fulfilled for the diagnosis of ASD. The case patient displayed all four of the major criteria and four of the five minor criteria. Rash in this patient was somewhat atypical in that it was persistent, although it did worsen with fever spikes. Although very high ferritin level (>3,000 µg/l) is not one of the criteria, it is found in as many 70% of ASD patients and can be helpful in making the diagnosis. Once the diagnosis is made, treatment can be initiated. Milder cases respond to high dose NSAIDs, whereas more severe cases require corticosteroid therapy (0.5-1 mg/kg/day of prednisone equivalent) with gradual taper. Chronic or recurrent symptoms may require immunomodulating therapy such as methotrexate or azathioprine.

SALICYLATE TOXICITY AND CENTRAL HYPOTHYROIDISM? THE DRAMATIC EFFECT OF SALSALATE ON THYROID FUNCTION TESTS. S.R. Adams¹; H. Chen²; M. Rotblatt³. ¹UCLA Medical Center Internal Medicine Residency, Los Angeles, CA; ²David Geffen UCLA School of Medicine, Los Angeles, CA; ³Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #132754)

LEARNING OBJECTIVES: 1. Diagnose salicylate toxicity based on the classic 'toxidrome' of symptoms, exam, and laboratory findings. 2. Recognize that multiple medications, including salicylates, can alter thyroid function tests (TFTs). 3. Recognize the specific effect that salicylate-containing medications, especially salsalate, have on TFTs.

CASE: A 22-year-old man presented to the ED with 3 weeks of fatigue, itching, and stomach pain. During the past 4 days, he developed "ringing" in his ears, blurry vision, N/V, sweats, and fevers. He injured his right knee in an accident 8 months prior, and was being treated at a rehab clinic. He had taken various NSAIDs for knee pain, but due to "stomach ache", was recently switched to salsalate 500 mg BID. He denied over-using salsalate, taking other medications, supplements or illicit drugs. VS were T 36.1, BP 119/47, P 94, RR 24, and O2 Sat 98% on RA. Exam revealed an obese, diaphoretic male with tachypnea and mild abdominal tenderness. Labs: WBC 13.1, Na 149, K 3.7, Cl 111, HCO3 19, Cr 1.0 (anion gap 19); ABG: pH 7.44, pCO2 30, pO2 101, HCO3 20; salicylate level 67 (therapeutic:<20 mmol/L). His TFTs were all low: TSH 0.36 (0.4-4.76 µIU/ml), free T4 (FT4) 0.44 (0.80-1.45 ng/dl), and total T3 (TT3) 0.47 (0.96-2.08 ng/dl). Our primary diagnosis was salicylate toxicity and the patient was treated with IV NaHCO3. Tests of pituitary function and an MRI of the brain were ordered for further evaluation. By hospital day 4, the salicylate level was 9 and his symptoms had improved. Repeat TFTs showed a normal TSH (1.68), but the FT4 (<0.4), and TT3 (0.68) were still low.

DISCUSSION: The presentation and labs (respiratory alkalosis and gap metabolic acidosis) are classic for salicylate toxicity. However, the patient also had low TFTs suggesting central hypothyroidism. Do we need further diagnostic evaluation? The answer is no. Many medications, including salicylates, can result in misleading TFTs in euthyroid patients. Thyroid physiology offers many

targets for drug interactions. After secretion, >99% of thyroid hormone is tightly bound to transport proteins. The small amount of unbound hormone (<0.1%) determines the action in tissues. Various medications, including salicylates, displace thyroid hormones from transport proteins. Salsalate, a salicylic acid dimer, appears to be the most potent salicylate that can interfere with protein-binding. The initial effect is a transient increase in the serum FT4 that suppresses TSH. This leads to decreased T4 and T3 values, although the patient remains euthyroid for unclear reasons. The TSH normalizes rapidly, while the T4 and T3 may remain low, as was the case with our patient. The literature suggests that salsalate often significantly alters TFTs even in therapeutic doses, and that TSH is the most useful thyroid test in patients receiving salicylates. This case demonstrates the importance of considering salicylate (especially salsalate) ingestion when evaluating patients with abnormal TFTs in order to avoid unnecessary diagnostic and therapeutic interventions.

SELECTIVE SEROTONIN REUPTAKE INHIBITORS AND ABNORMAL BLEEDING: IS THERE AN ASSOCIATION? A. Chang¹; G.M. Harper¹; K.E. Covinsky¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #134332)

LEARNING OBJECTIVES: 1. Recognize reported associations between selective serotonin reuptake inhibitors (SSRIs) and abnormal bleeding 2. Apply criteria for linking a treatment to an adverse event

CASE: Mr. A was an 89 year-old man with advanced Alzheimer's dementia receiving palliative care at a nursing home. His past history included depression in remission, osteoarthritis, and remote total hip arthroplasty. His medications included acetaminophen, calcium, docusate, multivitamin, senna, and bisacodyl. After a period of time, he had a recurrence of depressive symptoms and paroxetine 10 mg/day was initiated. Four weeks later his right thigh was edematous and tender. Ultrasound confirmed a 7 × 4 × 3 cm fluid collection consistent with a hematoma. He was treated conservatively resulting in resolution of pain. Eight weeks later his hematoma recurred. Laboratory studies indicated a normal platelet count and recently normal prothrombin time but a hematocrit of 18, a decrease from his baseline of 25. Paroxetine was discontinued. He experienced no further bleeding until his death from an unrelated cause 3 months later.

DISCUSSION: Depression affects 10-14 million people and is increasingly treated by primary care providers. Many consider SSRIs first-line treatment for depression due to their safety profile. Its widespread use implies that even rare adverse events are relevant. The proposed association between SSRIs and abnormal bleeding has been well documented in the psychiatric literature, but only a few case reports exist in the medical literature. Forty-four case reports have linked SSRI therapy to abnormal bleeding events such as bruising, epistaxis, hematoma, gastrointestinal bleed, petechiae, excessive menstrual flow, bloody sputum, gum bleed, and subhyaloid bleed threatening vision. Until 1998, the WHO International Drug Surveillance Program included 3512 cases of bleeding reported in association with SSRIs. Two population-based studies showed a 3-fold increased risk of upper gastrointestinal bleeding with SSRI use alone, and up to a 15-fold increased risk when SSRIs are used with aspirin or other NSAIDs. One retrospective cohort study showed the greatest increase in upper GI bleeding in patients over age 80 and those with a previous history of upper GI bleeding. The only randomized study is in rats and showed that exposure to fluoxetine in utero increases frequency of skin hematomas in offspring at birth from 1.8% to 29.3%. Plausible biologic mechanisms to explain this association have been described. When a vascular tear is detected, platelets release serotonin which markedly potentiates platelet aggregation. By the same mechanism seen in neurons, SSRIs inhibit the reuptake of serotonin into platelets and decreases plasma levels of serotonin. In assessing the probability that the hematoma in our patient was due to paroxetine therapy, we applied criteria proposed by Sackett and Levine for deciding whether a treatment has caused an adverse effect in a particular patient. Factors favoring a link include the timing of the therapy and event, the lack of event recurrence off therapy, the lack of a likely alternative cause, and previously reported cases of similar events. These factors, along with the biologic mechanism and prior epidemiologic evidence, lead us to conclude that this patient's hematoma was probably caused by his paroxetine use.

SEVERE TRIMETHOPRIM/SULFAMETHOXAZOLE INDUCED THROMBOCYTOPENIA IN A HEALTHY ADULT. L.F. Zegada¹; S. Koya¹; W. Many¹. ¹University of Alabama at Birmingham, Montgomery, AL. (Tracking ID #132868)

LEARNING OBJECTIVES: * Recognize the rare and serious side effects of the frequently used medication: trimethoprim/sulfamethoxazole. * Diagnose severe trimethoprim/sulfamethoxazole induced thrombocytopenia in a timely fashion to avoid fatal complications. * Treat severe trimethoprim/sulfamethoxazole induced thrombocytopenia effectively.

CASE: A 53-year-old Hispanic man, without known past medical history, was seen at a local prison infirmary because of multiple bruises in different body areas. The patient had been well and healthy previously except for the presence of testicular pain of one-week duration which grew in intensity and severity slowly. He was diagnosed with epididymitis and started on trimethoprim/sulfamethoxazole double strength twice a day and acetaminophen PRN. The next day the patient noted the sudden onset of multiple bruises in his shoulders, arms, abdomen, thighs and legs. On further questioning, the patient denied any history of trauma, drug allergies or constitutional symptoms. Physical examination revealed a well-developed man in no acute distress. He was afebrile, with stable vital signs. His mouth showed numerous petechiae in the hard palate. Lungs were clear to auscultation and his heart had regular beats with no mur-

murs, gallops or rubs. The abdomen was negative to exam. The skin findings were positive for purpuric patches in shoulders and arms, multiple ecchymotic areas were seen in the abdomen and extremities. Laboratory data: WBC 12, hemoglobin 15, hematocrit 44, platelets 6, normal differential. BMP and liver enzymes were all within normal limits. PT 12, PTT 26, INR 1.04. A CXR was negative for infiltrates.

DISCUSSION: Trimethoprim/sulfamethoxazole (TMP/SMX) is a widely used medication initially approved for medical use in 1973. TMP/SMX is usually well tolerated by the immunocompetent patient in whom reactions occur in approximately 6 percent of cases. The most feared adverse reactions include agranulocytosis, hemolytic anemia (especially in those with G6PD deficiency), toxic epidermal necrolysis, erythema multiforme, anaphylaxis and thrombocytopenia that has led sometimes to intracranial bleeding and other severe hemorrhages. Cases of severe TMP/SMX induced thrombocytopenia are scarce with only a few cases reported since 1975 up to date. The pathogenesis underlying this disorder is thought to be an immune mediated process, where platelets are damaged by the formation of drug-antibody complexes. The treatment is based on discontinuing the offending agent which results in normalization of platelet counts within a few days, transfusion of pheresed platelets and supportive measures. Corticosteroids, even though do not hasten or alter recovery, are used by many practitioners.

IMPLICATIONS/CONCLUSION: The clinician should always be aware of all the serious side effects of the most frequently used medications. Elucidating the cause of thrombocytopenia in this case was of utmost importance to prevent serious side effects and guide proper treatment and future prevention of fatalities.

SLAM-DUNK CHRONIC LYMPHOCYTIC LEUKEMIA – LOOK AGAIN! S. Parikh¹; A. Serrano¹; O. Ajayi¹; T. Thenappan¹; H. Friedman¹. ¹St. Francis Hospital, Evanston, IL. (Tracking ID #134523)

LEARNING OBJECTIVES: 1. Recognize the importance of flow cytometry in making an accurate diagnosis of different types of leukemias.

CASE: A 45-year-old male with no significant past medical history presented with complaints of fever, chills and generalized weakness for five days. He also reported a weight loss of 20 pounds over the past six months. The patient was a non-smoker and denied intravenous drug abuse. He worked in a construction company since his emigration from Iraq 12 years ago. On physical examination, his temperature was 101°F. The cardiac examination revealed a systolic ejection murmur and rest of physical examination was unremarkable. The white cell count was 234,000/cmm (97% lymphocytes and 3% blasts), Hb was 7.7 gm/dl (MCV 91 fl) and the platelet count was 75,000/cmm. The peripheral smear showed smudge cells and numerous abnormal lymphocytes. The basic metabolic profile, liver function tests and coagulation parameters were within normal limits. The platelet associated antibodies and Direct Coomb's test were negative. His chest X-ray was normal and the transthoracic echocardiogram did not show any vegetations. The bone marrow biopsy revealed a monotonous population of immature cells. A preliminary diagnosis of Rai's Stage IV Chronic Lymphocytic Leukemia (CLL) was made. However, immunophenotypic analysis showed that most of these cells strongly expressed CD-13 and CD-34, findings compatible with Acute Myelogenous Leukemia (AML, FAB M1). A repeat evaluation of his peripheral smear and bone marrow aspirate disclosed that these mature-looking lymphocytes were actually micromyeloblasts. Induction therapy with cytarabine and idarubicin was started immediately, following which the patient was referred for bone marrow transplantation.

DISCUSSION: Flow cytometry is a method in which measurements are performed on cells in liquid suspension, which flow one at a time through a focused laser beam at rates up to several thousand particles per second. Light scattered and fluorescence emitted by the cells is collected, filtered, digitized and sent to a computer for analysis. Our patient's symptoms, WBC count with differential, presence of atypical lymphocytes and smudge cells on the peripheral smear collectively pointed towards a diagnosis of CLL. However, his young age, absence of hepatosplenomegaly and lymphadenopathy on physical examination and the presence of blast cells on the peripheral smear suggested an alternate diagnosis. Flow cytometry confirmed our suspicion and revealed the myeloid origin of these cells. Our case emphasizes the pitfalls in relying solely on the morphologic characteristics of cells in the peripheral smear to characterize different types of leukemia. Flow cytometry can be an extremely useful adjunct in making an accurate diagnosis, which has therapeutic implications.

STEAMED SHRIMP AS A RED HERRING: A CASE OF AUTOIMMUNE HEPATITIS. M. Bohning¹; B. Taqui¹; H. Shishodia¹. ¹Temple University, Philadelphia, PA. (Tracking ID #134821)

LEARNING OBJECTIVES: 1. Recognize epidemiology and clinical manifestations of autoimmune hepatitis (AH) 2. Recognize diagnostic criteria for AH 3. Review management of AH

CASE: 25 year old female presented with 2 weeks of progressive nausea, vomiting, jaundice, dark urine and general malaise. She denied history of travel, sick contacts, alcohol, intravenous drug or medication use. She had two blood transfusions during vaginal deliveries in 1996 and 1998. She also reported eating steamed shrimp from a Chinese food restaurant 2 weeks prior to symptom onset. Her exam was only remarkable for scleral icterus and mild right upper quadrant tenderness. Labs revealed AST 980, ALT 530, total bilirubin 8.3, direct bilirubin 6.9, alkaline phosphatase 154. She improved with supportive treatment (hydration/antimetotics) for presumed hepatitis A. However, hepatitis panel (A, B, C, E) was negative. ANA, AMA, ceruloplasmin, ferritin, RPR, HIV were

negative. CMV, toxoplasmosis, varicella titers were negative. SPEP revealed mild hypergammaglobulinemia. ASMA (1:40) and anti-LKM (2:2) were mildly abnormal and deemed inconclusive. Right upper quadrant ultrasound with dopplers was normal. Liver biopsy revealed marked lobular disarray, with peri-portal and zone 3 dropout, mixed lobular/portal inflammatory cell infiltrate with some plasma cells, and suggestion of bridging necrosis. Over the course of the next three weeks her symptoms recurred, labs worsened (transaminases > 1000, bilirubin > 22), and she developed coagulopathy (INR 3.8). She was evaluated for and received emergent liver transplant for probable autoimmune hepatitis. **DISCUSSION:** Autoimmune hepatitis (AH) predominantly occurs in women who are 15-40 years old. Exposure of genetically predisposed individuals to an environmental agent triggers autoimmune process directed at liver antigens, causing progressive necroinflammatory process leading to cirrhosis. The clinical spectrum includes asymptomatic with normal labs, abnormal labs found incidentally, mildly symptomatic and rarely, acute severe disease. Labs reveal hyperglobulinemia and circulating autoantibodies. There are 3 subtypes of AH, differentiated mainly by their associated lab abnormalities (ANA and ASMA for 1, anti-LKM for 2 and SLA and LP for 3). Pathology reveals a portal mononuclear-cell infiltrate that invades the sharply demarcated hepatocyte boundary surrounding portal triad and permeates the surrounding lobule. Occasionally, there is also a dense plasma cell infiltrate and in advanced disease bridging necrosis. Characteristic serologic and histologic findings may not be present in all cases. Because of this and the heterogeneous presentation of AH, definitive diagnosis requires exclusion of other etiologies, positive titers of characteristic antibodies and classic interface hepatitis on biopsy. If there is a high clinical suspicion for AH but some of the diagnostic criteria are absent, a scoring system has been proposed by the American Association of Study of Liver Diseases. The system incorporates many environmental and laboratory factors to stratify patients into definite and probable diagnosis. Our patient met the criteria for probable AH based on this system. Diagnosis is essential because treatment with immunosuppressants is very effective. However, AH with acute liver failure is rare and medical treatment at this stage has not been extensively studied. Thus, patients may require a liver transplant. Our case emphasizes the importance of early diagnosis and treatment of AH.

STOOL TO THE HEART: AND WHO'S TO BLAME? C. Sapp¹; C. Feddock¹. ¹University of Kentucky, Lexington, KY. (Tracking ID #135861)

LEARNING OBJECTIVES: (1) To recognize that the presentation of myocarditis can be similar to that of an Acute Coronary Syndrome (2) To emphasize that myocarditis should be included in the differential diagnosis of acute chest pain, particularly in young patients (3) To recognize that myocarditis can be associated with both bacterial and viral infections and can often be differentiated on clinical grounds

CASE: An 18-year-old white male with no significant past medical history presented to the emergency room after a 30 minute episode of nonradiating, sharp, substernal chest pain. The pain was associated with diaphoresis and nausea. There was no associated dyspnea. The chest pain had completely resolved spontaneously at the time of arrival to the hospital. On review of systems the patient complained of a 3 day history of 4-5 loose bloody bowel movements. On physical exam the patient was chest pain free and was otherwise noncontributory. EKG revealed 2 mm ST elevation in the inferior leads. Initial labs revealed CK of 722, CK-MB of 82, and Troponin I of 10.36. Initial echocardiogram was normal. Cardiac catheterization demonstrated normal coronaries and mild global hypokinesia. An Ergonovine provocation test was negative for coronary artery spasm. A right ventricular biopsy revealed rare necrotic myofiber and a single focus of lymphocytic inflammation. After 2 days stool cultures grew *Campylobacter jejuni*. The diagnosis of *Campylobacter jejuni* enteritis and related myocarditis was made based on clinical and laboratory results. The patient was discharged home in stable condition to complete a course of Ciprofloxacin. Follow up echocardiogram was made for 6 weeks.

DISCUSSION: Myocarditis as a complication of infectious disease should be considered when a patient presents with unexplained chest pain and elevated cardiac enzymes in the absence of coronary disease. Myocarditis is most commonly linked to enterovirus infections but has also been associated with bacterial infections. Salmonellosis and shigellosis are the most frequently reported bacterial causes; however there are general case reports in the literature describing myocarditis associated with *C. jejuni*. In our patient and in other bacterial cases previously described, the cardiac symptoms occurred 2 to 3 days after the first gastrointestinal complaints. It has been suggested that the temporal relationship between bacterial infections and myocarditis is secondary to the direct effect of infection by the bacteria itself or is perhaps toxin related. This presentation is different from enterovirus infections when myocarditis may occur several weeks after the infection suggesting a post-viral immune-mediated mechanism.

SUCCESSFUL TREATMENT OF VIBRIO VULNIFICUS SEPTICEMIA. H. Khalid¹; I. Vassileva¹. ¹New Hanover Regional Medical Center, Wilmington, NC. (Tracking ID #135355)

LEARNING OBJECTIVES: 1. Recognize *Vibrio Vulnificus* as a cause of cellulitis. 2. Acknowledge the risk factors for *Vibrio* infection. 3 Implement quick and aggressive pathogen-specific treatment.

CASE: Patient is 42 year old male with past medical history significant for diabetes and hypertension, not on medications currently who presented with one day history of rapidly progressive left lower leg pain, swelling and redness. He experienced shaking chills, fevers, nausea and vomiting. Patient's social history is significant for extensive alcohol abuse and smoking. He is a commercial fish-

erman on the Atlantic coast. Physical exam on admission revealed very tender, edematous and erythematous left lower extremity with multiple skin blisters. Vital signs were stable, only low grade fever was detected. Labs showed leukocytes of 18,300 with 30% Bands. Serum creatinine was 1.9 and blood glucose 227. Doppler ultrasound of the leg was negative for deep venous thrombosis. Due to a high index of suspicion for Vibrio infection, patient was started on broad spectrum antibiotics plus Doxycycline. The next day patient became hypotensive, transferred to the intensive care unit, managed as a septic shock. Blood cultures grew Vibrio Vulnificus. Surgery was consulted to assess the need for debridement of the bullae or if there is a necrotizing fasciitis. The patient recovered very well, was discharged home in a few days on oral antibiotics but required full-thickness skin graft later on as an outpatient.

DISCUSSION: Vibrio vulnificus is a Gram negative rod that can cause infections and septicemia usually in patients with underlying illnesses such as chronic liver disease, diabetes mellitus, alcoholism and other immunocompromised states. This pathogen should be considered in the differential diagnosis of wound/skin infections and septicemia in the appropriate group of patients with history of contact with sea water. Prompt initiation of treatment with tetracyclines is the mainstay of therapy. It has been shown to decrease mortality particularly if the therapy is initiated before the onset of septicemia. Surgical debridement and fasciotomy are often necessary to control infection.

SUPERIOR MESENTERIC VENOUS THROMBOSIS: AN UNCOMMON CAUSE OF INTESTINAL VASCULOPATHY. P.S. Abadeer¹; L. Cation¹. ¹University of Illinois at Peoria, Peoria, IL. (Tracking ID #136196)

LEARNING OBJECTIVES: Recognize the presentation, various etiologies, proven methods of diagnosis, and established treatment of superior mesenteric venous thrombosis (SMVT).

CASE: A 54 year old man with a past history of hypertension and hypercholesterolemia presented with constant, sharp, diffuse, lower abdominal pain that started 5 days prior while shopping. He had occasional nausea but no emesis. He was eating less but eating didn't worsen the pain. He denied fever, chills, constipation, diarrhea, or hematochezia. On physical exam, he was afebrile, abdomen was soft with mild tenderness in bilateral lower quadrants. No discoloration, distension, guarding, or rebound. Labs were unremarkable. Abdominal film showed a normal gas pattern and no free air. Computed tomography (CT) of the abdomen showed a thrombus in the ileocolic branch of the superior mesenteric vein with phlebitis and thickening of a segment of the terminal ileum. Surgery and gastroenterology were consulted. Lactate remained normal and the patient remained stable. A heparin drip was started. Colonoscopy showed ileitis. Biopsies returned negative. Further workup revealed no evidence of hypercoagulable state, infectious bowel etiology, malignancy, or myeloproliferative disorder. Symptoms resolved and the patient was discharged home on coumadin.

DISCUSSION: Mesenteric venous thrombosis is an uncommon cause (5%) of intestinal vasculopathy. Mean age is 47-60 years old, and younger than mesenteric artery thrombosis. Causes are identified in most cases and include: inherited and acquired hypercoagulable states, abdominal infections, malignancy, previous surgery or trauma, portal hypertension, pancreatitis, and hyperviscosity syndromes. Clinical findings are variable at presentation and less dramatic than mesenteric artery thrombosis. They include: abdominal tenderness, distension, hypoperistalsis, anorexia, diarrhea, nausea, upper and lower gastrointestinal bleeding, and pyrexia. Pain often is out of proportion to exam and diffuse. Labs frequently are unremarkable. On plain films, ileus is an early finding. Thumbprinting, and intramural and gas are late. CT scan is diagnostic in 70-90% of cases. Angiography is less sensitive for venopathy than for arteriopathy, particularly in distal venous thromboses. Therapy is surgical if there are peritoneal signs or other evidence of intestinal infarction. Medical management involves intravenous fluid, intravenous heparin initially if no gross bleeding, broad spectrum antibiotics often, and warfarin chronically, depending on the etiology. This case demonstrated that in patients with SMVT, history, physical exam, routine lab testing, and radiographic studies often don't suggest a diagnosis. Instead, diagnosis relies on advanced imaging techniques, in our case CT, and a high index of suspicion. Furthermore, a desirable outcome is obtained with early diagnosis and prompt anticoagulation.

SYNCOPE IN A YOUNG WOMAN. D. Takahashi¹; K.J. Pfeifer¹; J.F. Kleczka¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #135400)

LEARNING OBJECTIVES: 1. Describe common mechanisms of ventricular tachycardia. 2. Identify ventricular tachycardia as an important etiology of syncope. 3. Describe treatments available for ventricular tachycardia.

CASE: A 21 year-old healthy Caucasian woman presented to the emergency department complaining of palpitations, chest discomfort, and near syncope. The patient has been experiencing these symptoms intermittently for nearly five years but never sought medical attention. Her most recent episode began the day prior to admission with chest discomfort and intermittent palpitations and then progressed to include lightheadedness. An electrocardiogram taken immediately on arrival at the emergency department revealed a wide-complex tachycardia with left bundle branch block morphology and an extreme inferior axis. These electrocardiographic findings are most consistent with right ventricular outflow tract ventricular tachycardia (RVOT VT). Echocardiogram and exercise myocardial perfusion scanning ruled out structural heart disease and myocardial ischemia as potential causes of the patient's ventricular tachycardia. After medical stabilization with metoprolol, the patient underwent an electrophysiology study where she was found to have an aberrant conduction pathway in the anteroseptal right ventricle outflow tract immediately adjacent to the anterior

septal cusp of the pulmonic valve. After ablation, ventricular tachycardia (VT) was no longer inducible. One month later, the patient was doing well without symptoms.

DISCUSSION: Cardiac arrhythmia remains one of the most serious etiologies of syncope. Most general internists know VT as a serious, potentially life-threatening arrhythmia but the mechanisms of VT and various treatments available are often less understood. VT is divided into monomorphic and polymorphic varieties and can occur in the presence or absence of structural heart disease. Most often, monomorphic VT is due to healed myocardial infarction or idiopathic VT syndrome. Idiopathic VT syndrome is classified into repetitive monomorphic VT (RMVT), paroxysmal sustained VT, and idiopathic left VT. RMVT is further classified into RVOT and left ventricular outflow tract VT. Most forms of idiopathic ventricular tachycardia syndrome are amenable to cardiac ablation therapy. Polymorphic VT is more serious and can occur as a result of myocardial infarction or a prolonged QT interval. Rarely, polymorphic VT is seen in a familial disease called catecholaminergic polymorphic VT. RVOT VT, such as in the case described above, is the most common form of idiopathic VT and accounts for 70% of all idiopathic VTs. It is characterized by left bundle branch block morphology and inferior axis on electrocardiogram and is most often treated by ablation with success rates from 80 to 100 percent. Various anti-arrhythmic medications can also be used for emergent stabilization and when ablation is contraindicated or declined. With greater knowledge of VT, general internists can more efficiently recognize signs and symptoms of ventricular arrhythmia and determine appropriate interventions based on its probable cause.

TAKOTSUBO CARDIOMYOPATHY: A HEART SYNDROME MIMICKING ACUTE MYOCARDIAL INFARCTION. C. Kasapis¹; A. Kapetanopoulos². ¹University of Connecticut, Farmington, CT; ²Hartford Hospital, Connecticut, Hartford, CT. (Tracking ID #133963)

LEARNING OBJECTIVES: (1) Recognize Takotsubo cardiomyopathy as a transient disorder often similar to that of an acute myocardial infarction (MI), with normal epicardial coronary arteries. (2) Recognize the clinical presentation and the characteristic echocardiographic and/or ventriculographic features of this syndrome. (3) Recognize the potential complications and the natural history of this syndrome.

CASE: A 72-year-old Indian female presented with a severe episode of substernal chest pressure, shortness of breath and diaphoresis after an emotional argument with her family at home. Her past medical history was significant for hypertension and asthma. Her initial electrocardiogram (ECG) was consistent with anteroseptal ST-elevation MI. She also developed transient left-bundle-branch block on repeat ECGs. Cardiac enzymes were elevated, indicating an acute MI. The patient was treated for ST-elevation MI with aspirin, heparin, beta-blockers and referred for urgent cardiac catheterization. Coronary angiography revealed minimal non-obstructive coronary artery disease. Left ventriculography and two-dimensional echocardiography demonstrated severe LV systolic dysfunction with dyskinesis of the apex (apical ballooning) and hyperdynamic function at the base. These findings were consistent with Takotsubo cardiomyopathy and the patient improved clinically with symptomatic treatment. A repeat echocardiogram, 10 days after the acute episode, revealed remarkable improvement of the left ventricular systolic function with an estimated EF of 55-60% and mild apical hypokinesis. An exercise stress test with technetium-99m Sestamibi SPECT imaging, three months later, showed normal ventricular perfusion and function with an estimated EF of 65-70%.

DISCUSSION: Takotsubo cardiomyopathy or transient left ventricular apical ballooning is a disorder characterized by transient dysfunction of the apical portion of the left ventricle, with compensatory hyperkinesis of the basal wall, producing a ballooning of the apex in systole. This condition was first described in Japan and is much more common in women. The onset is typically associated with acute medical illness or with physical or emotional stress. The most common presenting symptom is chest pain while other patients may present with dyspnea, shock or electrocardiographic abnormalities alone. ECG changes, more frequently ST elevation or depression, T wave inversion and abnormal Q waves as well as elevations of cardiac enzymes, especially troponin T, may be demonstrated. Coronary angiography reveals normal epicardial coronary arteries. Left ventriculography or echocardiography can be used to demonstrate the characteristic apical ballooning with a reduction in the global ejection fraction. Complications include tachyarrhythmias or bradyarrhythmias, pulmonary edema, and cardiogenic shock as well as apical thrombus formation and stroke. However, Takotsubo cardiomyopathy is a transient disorder managed with supportive therapy and patients who survive the acute episode recover normal left ventricular function in three to four weeks. The pathogenesis of this syndrome is unclear but it may be related to catecholamine-induced microvascular spasm and possible discrepancy of sympathetic innervation between the apical and basal region, which causes the characteristic apical ballooning.

TB MENINGITIS WORSENS WITH TREATMENT? N. Gupta¹; D. Corry¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #135184)

LEARNING OBJECTIVES: 1) Exploring the clinical presentation of immune reconstitution inflammatory syndrome (IRIS) with TB meningitis. 2) Discussion of IRIS.

CASE: 26 y/o woman presented with 4 days of febrile headache & altered mental status. Initial vital signs were notable for a temperature of 39.0°. She appeared to be in moderate distress, confused, and disoriented to person, place, & time, but had an otherwise normal exam. CBC with diff., Chem. 10, TSH, UA, urine cultures, blood cultures x 2, CXR, B12, folate, RPR, ABG, PPD, HIV, CT

head (w/o contrast), & drug screens were all normal. LP was performed with results as follows: glucose < 32; protein: 247; WBC: 126; RBC: 110; normal diff. India Ink, gram stain/bacterial/fungal cultures, AFB smear, HSV PCR, and serum/CSF cocci titers were all normal. MTB DNA PCR+, and AFB cultures 3+ for TB. Antibiotics were initiated & her sensorium cleared 3 days later, with persistently decreasing symptoms. Antibiotics were continued with a steroid taper in-house for 9 days and she was eventually sent home with TB meningitis treatment. One week later, she returned to the hospital for recurrent worsening headache with subjective fevers. She had stable vital signs. The team re-admitted the patient for meningitis treatment. All serum tests and imaging were normal. LP results: glucose < 32; protein: 225; WBC: 537; RBC: 4; with a lymphocyte predominance on the diff.; MTB DNA PCR+, and AFB cultures 1+ TB. Infectious disease was consulted and the patient's symptoms were attributed to immune reconstitution inflammatory syndrome. She was kept on her initial TB regimen with IV steroids/prednisone added for a total of a 4 week outpatient taper. Her symptoms decreased drastically. Upon follow up in clinic, repeat LP showed: glucose: 32; protein: 268; WBC: 150; RBC: 117; PMN predominant diff., with negative AFB Smear/MTB DNA PCR. She had a repeat positive 1+ AFB culture. Clinically, pt had improving residual minimal intermittent headaches, and was alert and oriented at baseline levels.

DISCUSSION: IRIS, or immune reconstitution inflammatory syndrome, describes a collection of inflammatory disorders associated with paradoxical worsening of infectious processes and symptoms following the initiation of treatment. It is a diagnosis of exclusion, and one must rule out subclinical opportunistic infections, drug reactions, patient noncompliance, persistently active infection and/or drug resistance prior to diagnosing a patient with IRIS. The pathophysiology is not well understood, but with regards to TB infections, appears to be immune mediated, with some changes in the ability to respond to tuberculin proteins after starting treatment. The time interval from initiation of TB treatment to the onset of IRIS, tends to occur within the first few weeks after starting treatment. IRIS is considered to be a self-limited process, although several controlled studies that adjunctive corticosteroids, given for a total of 4-8 weeks after starting TB treatment, results in faster short term symptomatic improvement. The incidence of IRIS in TB patients is unknown. Generally, treatment should be continued in patients with IRIS, and steroids can be added to decrease inflammation. Interruption of TB treatment should only be considered if patients have life threatening IRIS, permanent neurologic or other sequelae. In the absence of well-performed clinical trials, recommendations for managing IRIS are based on clinical experience, case reports, case series, and expert opinion.

TETANUS PRESENTING WITH RIGHT UPPER EXTREMITY WEAKNESS. A. Mitchell¹; E. Warm¹. ¹University of Cincinnati, Cincinnati, OH. (Tracking ID #135324)

LEARNING OBJECTIVES: 1) Identify the common presentations for tetanus infections, 2) Demonstrate common strategies for treating and preventing tetanus infections.

CASE: Tetanus is a neuromuscular disease caused by an exotoxin from *Clostridium tetani*. Although its incidence in the United States has declined because of widespread vaccination, with less than fifty cases annually, tetanus remains epidemic in developing countries. We report a case of a patient with tetanus who initially presented with right arm weakness. An 88 year-old female with a history of diet-controlled diabetes presented to the hospital with acute onset of right arm weakness and paresthesias. She also complained of mild dysphagia and jaw pain with mastication. Her symptoms were unremitting since onset that morning. She fell while gardening approximately one week prior to admission, and had a minor abrasion on her right forearm. Neurological exam was significant for decreased gag reflex and mildly decreased strength on her right upper extremity. CT Scan of her head revealed old lacunar infarcts in the basal ganglia and an age-indeterminate left pontine infarct, though MRI and MRA showed no evidence of acute cerebrovascular disease. On hospital day number three, the patient developed trismus and hypertonicity of her right arm. Despite tetanus immune globulin and metronidazole, the patient's condition continued to decline. She became hypotensive requiring vasopressors, and had an episode of asystole. Per discussion with her family, care was withdrawn and the patient died on hospital day six.

DISCUSSION: Tetanus toxin binds to neurons in the brain and spinal cord, resulting in painful muscle spasm and autonomic instability. Classically, patients present with trismus, occurring in over half of all cases. Patients with generalized tetanus may also present with labile hypertension, diaphoresis, and cardiac arrhythmias. In our case, the patient presented with localized tetanus involving her right arm, with signs and symptoms similar to a stroke. Often, localized tetanus will progress into generalized tetanus. Tetanus is a clinical diagnosis, requiring a careful history and high degree of suspicion. Goals for therapy include eradication of unbound toxin with wound debridement, immune globulin, and antibiotics, as well as control of symptoms with sedatives and neuromuscular blocking agents. As the toxin binds irreversibly, care is supportive and may last several weeks. Three-fourths of all fatalities occur in individuals over age sixty, because of decreased levels of protective antibodies. Nearly all cases can be prevented with booster vaccination every ten years.

THE BRAIN CHALLENGING OB CONSULT. P. Frey¹; R. Wilson¹; S.D. Sisson². ¹Johns Hopkins Hospital, Baltimore, MD; ²Johns Hopkins University, Baltimore, MD. (Tracking ID #134384)

LEARNING OBJECTIVES: 1) To recognize a preventable and reversible etiology of mental status change 2) Understand the clinical sequelae of Wernicke's encephalopathy and the distinction from Korsakoff's syndrome

CASE: A 17 year-old female, G1P0, pregnant at 19 weeks presented to the hospital with persistent nausea and vomiting. In the emergency department, an abdominal ultrasound diagnosed intrauterine fetal demise. The patient was admitted to the OB/GYN service for induction of labor, as well as management of dehydration and starvation ketosis secondary to nearly two months of hyperemesis gravidarum. On hospital day 2, following delivery of the fetus and volume resuscitation with D5-1/2 NS, the patient developed mental status changes. On physical exam she was oriented to person, place and time but had a markedly blunted affect, with decreased spontaneous speech. Extraocular exam was significant for bilateral lateral gaze palsies and horizontal nystagmus. Gait was stumbling and ataxic. A T2 weighted brain MRI showed bilateral hyperintensities of the medial thalami, and a diagnosis of Wernicke's encephalopathy was made. After five days of parenteral thiamine supplementation, the patient's neuropsychiatric symptoms and lateral gaze deficit had improved; however, ataxia remained.

DISCUSSION: Thiamine (vitamin B1) is a cofactor necessary for the proper activity of several enzymes, including transketolase, pyruvate dehydrogenase, and alpha-ketoglutarate dehydrogenase. While the exact pathophysiologic mechanism of Wernicke's encephalopathy is unclear, abnormal carbohydrate metabolism from thiamine deficiency results in neuronal damage producing the classic clinical triad of encephalopathy, oculomotor dysfunction and ataxia. In the hospital setting, Wernicke's can be precipitated by the administration of glucose in a thiamine deficient patient. In the case presented above, administration of dextrose-containing intravenous fluids to the patient malnourished due to hyperemesis gravidarum resulted in acute thiamine deficiency and the onset of Wernicke's encephalopathy. While thiamine levels and enzymatic activity can be measured, the diagnosis remains clinical. MRI findings, if present, characteristically show increased signal intensities on T2 images around the 3rd ventricle, the dorsomedial thalami, and the periaqueductal regions of the mid-brain. Commonly associated with alcoholism, the diagnosis should be considered for any patient presenting with a global confusional state in which nutritional compromise is suspected (e.g. hyperemesis gravidarum, gastric bypass, anorexia nervosa, malabsorptive disorders, HIV, or chronic dialysis). Korsakoff's syndrome, often confused with Wernicke's encephalopathy, is characterized by a more long-term condition of anterograde and retrograde amnesia, apathy, confabulation, and intact long-term memory. Risk factors for Korsakoff's syndrome are similar to those of Wernicke's

THE CONDUCTOR WHO COULDN'T CONDUCT. J. Rubright¹; J. Ramsey¹; S. Kahlon¹; M.D. Landry¹. ¹Tulane University Health Sciences Center, New Orleans, LA. (Tracking ID #132564)

LEARNING OBJECTIVES: 1. Establish differential diagnoses for ataxia. 2. Recognize rapid progression of syphilis in HIV. 3. Evaluate and treat neurosyphilis in HIV co-infection

CASE: A 48 year-old bisexual choir conductor presented with progressive ataxia and left arm weakness. Three weeks prior, a rash developed on his trunk that progressed to his palms and soles. Physical examination revealed a maculopapular rash on his trunk and bilateral palms and soles. Mucocutaneous lesions were noted on his tongue. Neurologic testing revealed ataxic gait, positive Romberg sign, decreased strength and coordination in the left upper extremity. Laboratory testing including toxicology was normal. Cultures were negative; lumbar puncture results were normal including CSF VDRL. CT and MRI of the head were normal. Serum RPR and HIV Western Blot later returned positive. Penicillin G was begun intravenously. Within twenty-four hours, his ataxia and left arm weakness improved; his rash decreased. On day three, his symptoms had nearly resolved. A fourteen-day course of IV penicillin G was completed.

DISCUSSION: Differential diagnoses for ataxia in HIV include pre-syncope, vertigo, common and opportunistic CNS infections, HIV associated etiologies and drug side effects. With a concomitant maculopapular rash involving the trunk, hands, and feet, treponemal co-infection with rapid progression to secondary and tertiary syphilis must be considered. Diagnosing neurosyphilis in HIV is complicated because ataxia, CSF pleocytosis and CSF protein elevation may be seen in both diseases. VDRL-CSF testing lacks sensitivity and does not exclude neurosyphilis if negative. Negative serology in the context of clinical neurosyphilis warrants treatment with penicillin G.

THE FILIPINO MAN FROM SAUDI ARABIA. C. Sather¹; A.M. Mallouk¹; E. Wasson¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #134018)

LEARNING OBJECTIVES: 1) Recognize the most common causes of marked peripheral eosinophilia. 2) Recognize the manifestations of hyper eosinophilic syndrome (HES). 3) Manage HES and its complications.

CASE: A 60 year old Filipino man with no known PMH presented with two months of nonproductive cough and accelerating dyspnea on exertion. He also reported rhinorrhea, sinus pressure, anorexia, and easy bruising. He denied fever, chest pain, and hemoptysis. He had a forty pack-year smoking history and had worked in a Saudi Arabian cement manufacturing facility. Physical exam was significant for heart rate of 115 bpm, O2 sat 92% room air, mild dyspnea, and multiple ecchymoses on the extremities. Lab work revealed a WBC count of 27,900 of which 66% were eosinophils, BUN 38 and creatinine 8.8. UAC showed eosinophiluria. CXR was unremarkable. Renal biopsy revealed acute interstitial nephritis with marked eosinophilic infiltration, and bone marrow biopsy showed marked eosinophilic hyperplasia without blasts. Additional imaging showed opacification of the sinuses, a few small right lung opacities, and mild bilateral hydronephrosis. Echocardiogram showed a hyperdynamic left ventricle with EF 70%. ANA, ANCA, stool O&P, HIV, cocci serologies, and malarial smears were

negative. AFB from sputum, urine, and bone marrow were negative. Parasitic serologies were negative with the exception of a borderline positive paragonamiasis titer which was treated with praziquantel. After ruling out infectious etiologies, methylprednisolone treatment was initiated for idiopathic hypereosinophilic syndrome. After one day of treatment, the peripheral eosinophilia was suppressed to near-normal levels. Throughout the patient's stay his creatinine remained elevated and he was started on hemodialysis before discharge. Two weeks later his creatinine had decreased to 1.6, and WBC had decreased to 13,000 with 0% eosinophils.

DISCUSSION: Peripheral eosinophilia may be caused by parasitic infections (the most common cause worldwide), allergic disorders (the most common cause in industrialized countries), drug reactions, hematologic or neoplastic disorders, and connective tissue diseases. Hypereosinophilic syndrome (HES) is a clinical diagnosis with three features: 1) blood eosinophilia $> 1500/\mu\text{L}$ for six months; 2) absence of other apparent etiologies of eosinophilia; 3) signs and symptoms of end-organ dysfunction. Clinical and biological features are heterogeneous, as eosinophilic infiltration and mediator release may variably affect organ systems. The most commonly affected systems are cardiac (myocarditis and mural thrombi), neurologic (cerebral thromboemboli, encephalopathy, peripheral neuropathy), pulmonary (non-productive cough, fibrosis, emboli), and dermatologic. Renal failure from eosinophilic infiltration, as in the above case, is rare. Treatment of HES includes supportive care for end-organ complications and corticosteroids to suppress eosinophilia. Biologic response modifiers such as interferon alpha, chemotherapeutic agents, and tyrosine kinase inhibitors have also been used. Reports of prognosis have been mixed, however, in a recent series of forty patients with HES, survival was 80 percent at five years and 42 percent at 15 years.

THE NON-SURGICAL ACUTE ABDOMEN – EPIPLOIC APPENDAGITIS. G.H. Tabas¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID # 132729)

LEARNING OBJECTIVES: 1. Recognize an important CT diagnosis in patients with acute abdominal pain. 2. Manage epiploic appendagitis.

CASE: A 35-year-old previously healthy woman presented to the office with left lower abdominal pain accompanied by nausea without vomiting. Pain was worse while lying on her back or rolling over in bed, better in the sitting position and occasionally radiated to the left anterior thigh. Bowel movements were unchanged and she denied dysuria, frequency, or history of kidney stones. She was on day three of menstrual bleeding. She appeared in moderate pain, BP was 164/68, pulse was 64 and weight was 145 lbs. Movement onto the exam table was slow because of pain. The left lower quadrant of the abdomen was tender without rebound tenderness but palpation caused the patient to wince with pain. No masses were palpable. Pelvic exam revealed tenderness in the region of the left ovary without palpable mass. Urinalysis revealed 16-30 RBC's and 3-5 WBC's. Quantitative HCG was $<2.0\text{ mIU/mL}$. Same day outpatient abdominal and pelvic CT revealed epiploic appendagitis in the sigmoid colon. She was sent home with ibuprofen and her pain resolved over the next few days.

DISCUSSION: Epiploic appendagitis is an inflammatory condition involving the fatty tissue protruding from the surface of the colon (epiploic appendix) due to torsion or thrombosis of the vessels within the fat. It occurs equally in men and women between the ages of 20 and 50 (mean age 38). It is found during laparotomy in 7% of patients suspected of having diverticulitis. Symptoms mimic other causes of acute abdomen and include abdominal pain without nausea or vomiting. The differential diagnosis includes diverticulitis, ruptured or hemorrhagic ovarian cyst, ovarian torsion, ectopic pregnancy, left-sided appendicitis and Crohn's disease. Patients usually have mild fever and leukocytosis. The CT scan is distinctive and shows a fat density mass adjacent to the colon surrounded by a higher density ring representing the edematous serosa covering the fatty mass. CT may also show a small high density area in the center representing the thrombosed vein. The most common location is the sigmoid colon. The disease is benign and self-limited and pain can be treated with NSAIDs or narcotic medication. Resolution typically occurs in 3 to 14 days. Before CT scanning, the diagnosis was made at the time of laparotomy when other conditions were suspected. It is important for physicians to be aware of epiploic appendagitis because with timely CT scanning and diagnosis of this condition, needless exploratory laparotomy and even hospitalization can be avoided.

THE UNEXPECTED THYROTOXIC PERIODIC PARALYSIS. R. Shah¹; L. Cation². ¹University of Illinois at Peoria, East Brunswick, NJ; ²University of Illinois at Peoria, Peoria, IL. (Tracking ID # 131919)

LEARNING OBJECTIVES: Recognize thyrotoxic periodic paralysis, understand the pathophysiology of the disease, know the proper treatment of the disease which will prevent the life-threatening consequences of this syndrome.

CASE: A 32y/o Salvadorian male presented with sudden onset of profound lower extremity muscle weakness. The night prior to his admission, the patient experienced a brief episode of dizziness and some mild lower back pain. He awoke the next day and fell as he attempted to rise from bed. Furthermore, the patient was unable to proceed up his staircase. After three additional episodes of falling at work, he was brought to a local emergency room where the admitting diagnosis was transverse myelitis. Following transfer to our hospital, he was diagnosed with Guillen-Barre Syndrome. Examination revealed bilateral exophthalmos, lid lag upon downward gaze, moderate thyromegaly, muscle tenderness in both thighs, and diminished muscle strength in deltoids and proximal lower extremities bilaterally. The patient immediately received two grams of IV methylprednisolone followed by plasmapheresis. At this point, laboratory findings of interest were: TSH $< .01\text{ IU/mL}$, free T4 = 2.6 ng/dL , T3 = 5.14 ng/dL , K = 1.3 mEq/L , Mg = 1.3 mg/dL , PO4 = 1.3 mg/dL ,

CPK = 1,183. An endocrinology consultant established the diagnosis of thyrotoxicosis periodic paralysis and treatment was immediately initiated with propylthiouracil 150mg t.i.d. and potassium replacement, which resulted in rebound hyperkalemia of 7.7 mEq/L . The patient's hypophosphatemia and mild hypomagnesemia were managed appropriately, and he was placed on a low carbohydrate diet. Symptom resolution was achieved the next day. Subsequently, the patient was scheduled for an outpatient thyroid scan. Upon stabilization of thyroid function, the patient was to receive radioactive iodine therapy to render him hypothyroid.

DISCUSSION: Thyrotoxicosis Periodic Paralysis (TPP) is a syndrome characterized by intermittent episodes of muscle weakness often accompanied by thyrotoxicosis and hypokalemia. Despite the increasing immigration of Asian and Latin American males, who are the primary victims of TPP, unfamiliarity with this disorder in the United States frequently results in an initial misdiagnosis. Despite the classic clinical and laboratory findings of TPP, the above case demonstrates the difficulty in recognizing this clinical entity. Our patient had typical thyrotoxic findings, but a possible confounding factor in diagnosing TPP can be the presentation of periodic paralysis prior to the classic signs of hyperthyroidism. Therefore, it is important to consider TPP as a differential in lower extremity weakness even in the absence of overt hyperthyroidism. Proper identification and management of this curable disorder can prevent the potentially life-threatening rebound hyperkalemia.

THERE'S A FUNGUS AMONG EOS: COCCIDIOIDOMYCOSIS AND MARKED EOSINOPHILIA. M.J. Richman¹; A. Jeng². ¹Olive View - UCLA Medical Center, Los Angeles, CA; ²Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID # 132590)

LEARNING OBJECTIVES: 1) Review the differential diagnosis for pulmonary-eosinophilia syndromes. 2) Recognize coccidioidomycosis as a common cause of eosinophilia, albeit an uncommon cause of marked eosinophilia.

CASE: A 38 year-old previously healthy Korean man presented with 4 weeks of cough (dry, then clear-yellow sputum), SOB, fever to 103, chills, malaise, anorexia, night sweats, and weight loss. He did not improve with a 5-day outpatient course of a fluoroquinolone. He denied chest pain, hemoptysis, GI symptoms, sick contacts, or recent travel. He was born in South Korea, lived in South Dakota for 3 years, Missouri for 7 years, and, most recently, Los Angeles for 1 year. He had no occupational exposures. Initial vital signs: T 36.9, BP 100/55, P 94, RR 28, and SaO₂ on 2L NC 92%. His exam was otherwise normal. Initial lab studies showed a serum WBC 21.5 (peaking at 27 his 2nd day of admission), with 33% eosinophils (peaking at 46% his 4th day of admission). ESR was 46. ABG and chemistry panel were normal. Blood cultures were (-). CXR showed bilateral patchy infiltrates. Chest CT revealed multiple areas of ground-glass opacity and nodules (the largest, in the left apex, was 3 cm and irregular), mediastinal lymphadenopathy, and a small left pleural effusion. Thoracentesis revealed 890 WBC and 850 RBC. The team initiated Ceftriaxone and Azithromycin, and admitted the patient to an isolation bed until 3 AFB smears were (-). His PPD and HIV tests were (-). The patient's condition did not change during his first 5 days of hospitalization, and the diagnosis was in question. Prior to bronchoscopy, the patient coughed up a sputum sample which revealed coccidioides spherules on H & E stain. This was confirmed by DNA probe, though sputum fungal culture was negative. His serum cocci complement fixation antibody was positive at 1:4. To investigate for disseminated cocci, the team performed a lumbar puncture, which was (-). A bone scan showed abnormal activity in the right superolateral orbital wall, right maxillary sinus, and mid-posterior right 11th rib, suggesting dissemination. The team discharged the patient on Fluconazole.

DISCUSSION: Eosinophilia occurs in several diseases with pulmonary infiltrates, posing a dilemma for clinicians. These include: parasitic pneumonia (Tropical Filarial Pulmonary Eosinophilia, and Loeffler's syndrome from Strongyloides, Ascariasis, Paragonimiasis, and Toxocariasis), Coccidioides immitis, Acute and Chronic Eosinophilic Pneumonia, ABPA, connective tissue diseases (Sarcoidosis, Churg-Strauss), Eosinophilic Leukemia/Lymphoma, Hypereosinophilic Syndrome, drug reactions (sulfasalazine), and toxins (contaminated L-tryptophan). Strongyloides and Toxocariasis are the most common infectious etiologies presenting with marked eosinophilia. Cocci is the only fungus associated with eosinophilia. Although uncommon for cocci to present with eosinophils $> 26\%$, the literature reveals rare cases of eosinophil counts reaching as high as 89%. This case illustrates that cocci still must be considered in the differential diagnosis for pulmonary infiltrates with marked eosinophilia.

THIS IS YOUR GUT ON DRUGS. B.D. Shah¹; S. Dea¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID # 136273)

LEARNING OBJECTIVES: 1. Recognize the presentation of ischemic colitis in younger patients. 2. Understand the damage that cocaine can cause to the gastrointestinal tract

CASE: A 35 year old man was brought to the emergency room by police after being subdued by force. He was found breaking glass in a liquor store and the police restrained him using several Taser shots. Patient had no past medical history, and social history was significant for cocaine and methamphetamine use. The patient passed four bloody bowel movements in the emergency room. Physical examination revealed temperature 39.2°C, heart rate 160, blood pressure 120/59, and respiratory rate 33. Pupils were dilated to six millimeters and reactive. Cardiac exam revealed tachycardia with no murmur present. Multiple lacerations and abrasions were noted on his arms. Laboratory data included hemoglobin 14, initially, with repeat level 7. Urine drug screen was positive for cocaine and methamphetamines. CT abdomen did not demonstrate evidence of

acute infectious process. Blood transfusion was ordered for acute gastrointestinal hemorrhage. Colonoscopy was subsequently performed, showing an acute inflammatory colitis limited to the recto-sigmoid area. Although the endoscopic impression was consistent with both inflammatory bowel disease and infectious colitis, the location and focal nature were more typical of ischemic colitis. Biopsies were consistent with ischemia. Hemoglobin level eventually stabilized without further intervention.

DISCUSSION: The clinical manifestations of colonic ischemia vary depending upon the clinical setting and the extent and duration of the ischemia. Patients with acute colonic ischemia usually present with rapid onset of mild abdominal pain and tenderness over the affected bowel, most often involving the left side. Mild to moderate amounts of rectal bleeding or bloody diarrhea usually develops within 24 hours of the onset of abdominal pain. Common etiologies of ischemic colitis in younger patients include hypercoagulable state, sickle cell disease, long-distance running, pancreatitis, vasculitis, and strangulated hernia. There have been 28 reported cases of cocaine-induced ischemic colitis. Symptoms usually occur within 1 hour to 2 days of cocaine use. Acute bleeding episodes are usually self-limited and do not require intervention. Histologic studies in chronic cocaine users demonstrate the presence of colonic lesions suggestive of recurrent ischemic episodes likely secondary to repeated injury from the cocaine. Unfortunately the long-term outcomes of chronic cocaine use is still unknown. Though an uncommon etiology of ischemic colitis, cocaine use must be considered in the differential diagnosis of young patients with abdominal pain and bloody diarrhea.

THYROTOXIC PERIODIC PARALYSIS. W. Hadid¹; R. Dhokarh¹; R. Sanghani¹; J. Cabello¹; S.B. Glick¹. ¹John Stroger Hospital of Cook County, Chicago, IL. (Tracking ID #132531)

LEARNING OBJECTIVES: 1. Recognize periodic paralysis as a possible initial manifestation of thyrotoxicosis, especially in Asian men; 2. Understand the clinical presentation of thyrotoxic periodic paralysis; 3. Understand the treatment of thyrotoxic periodic paralysis.

CASE: A 30 year-old man from Thailand presented to the ER after his third episode of paralysis. The first and second episodes occurred while the patient was walking. His arms and legs became suddenly weak and he fell without loss of consciousness. The third episode occurred when the patient awoke from sleep to find his arms and legs too weak to move. On each occasion, the symptoms resolved spontaneously in 10-15 minutes. In between episodes, the patient experienced occasional proximal myalgias, but the strength was normal. The patient had no sensory, bladder or bowel symptoms. There was no significant past medical or family history. On examination, the vital signs were BP 156/90, HR 135, RR 18, T 98.6. There was no exophthalmos, lid lag, or stare. The thyroid was not tender to palpation; there was no thyromegaly. Cardiac examination was remarkable for tachycardia. On neurological examination, the strength was 5/5 in all extremities. Sensation and reflexes were normal. A fine resting tremor was present. The potassium, calcium and creatinine were normal. EKG revealed sinus tachycardia. TSH was < 0.003 (normal range 0.4-6.0); free T4 was 5 (normal range 0.6-1.7). Thyrotoxic periodic paralysis was diagnosed based on the clinical presentation and thyroid function tests. Treatment with propranolol and methimazole was begun. The patient was discharged home on a low-carbohydrate diet. Two weeks later he is doing well.

DISCUSSION: Thyrotoxic periodic paralysis is seen in up to 13% of Asian men with hyperthyroidism. For 82% of Thai patients with thyrotoxic periodic paralysis, paralysis is the initial manifestation of thyroid disease. Other symptoms and signs of thyrotoxicosis may be subtle or absent entirely. Paralysis affects proximal muscles more often than distal. Attacks frequently occur at night and are precipitated by high-carbohydrate meals and by periods of rest following heavy exercise. Patients may experience proximal muscle myalgias prior to attacks. Attacks are usually self-limited, though occasionally may be accompanied by respiratory muscle paralysis or ventricular fibrillation. Hypokalemia is often seen during attacks due to intracellular shifts in potassium; potassium levels are normal in between attacks. Thyrotoxic periodic paralysis should be suspected in Asian men with onset of paralysis after age 30. The diagnosis is confirmed by thyroid function tests. Treatment of the thyrotoxicosis results in resolution of the periodic paralysis. Paralysis can recur with relapses of hyperthyroidism.

THYROTOXICOSIS AFTER A STANDARD RADIOGRAPHIC STUDY?. I.G. Zarraga¹; L. Lu¹. ¹Baylor College of Medicine, Houston, TX. (Tracking ID #133876)

LEARNING OBJECTIVES: Learning objectives: (1) To recognize that iodine-containing agents used in clinical practice have the potential to disrupt normal thyroid function, and (2) To learn the mechanisms by which such disruption occurs in susceptible individuals.

CASE: A 62-year-old woman with diabetes and hypertension presented to the emergency room with shortness of breath. One month prior to presentation, she received intravenous radiographic contrast for a computed tomography scan of the chest. The scan was unremarkable except for areas of attenuation within the thyroid gland. A thyroid ultrasound confirmed the presence of bilateral thyroid nodules, but thyroid-stimulating hormone (TSH) and free thyroxine levels were both normal. On presentation in the emergency room one month later, she reported worsening exertional dyspnea, palpitations, increased perspiration, and weight loss of 3.5 kg since the radiographic scan. Her blood pressure was 160/84 with heart rate 110 despite being on a beta-blocker for hypertension, and her temperature was 99.6 degrees Fahrenheit. She had no palpable anterior neck mass. Cardiac auscultation revealed a regular rhythm without a murmur or gallop. Breath sounds were decreased in both lung bases. A chest radiograph revealed small pleural effusions bilaterally. An echocardiogram showed a left ventricular ejection fraction of >70% with impaired relaxation. A work-up for infection, including blood cultures, was negative. Her repeat thyroid function

tests showed a suppressed TSH level and elevated free thyroxine and triiodothyronine levels. The clinical scenario was consistent with thyrotoxicosis following exposure to radiographic contrast iodine, otherwise known as the Jod-Basedow phenomenon. She was treated with beta-blockade, an anti-thyroid medication, and a brief period of diuresis, after which her symptoms improved dramatically. A fine-needle aspiration of one of the thyroid nodules demonstrated a colloid adenoma. A few months later, a radioactive iodine uptake study demonstrated persistence of a hyperfunctioning thyroid gland, so she elected to undergo radioactive iodine ablation of the gland.

DISCUSSION: Iodine plays a vital role in the maintenance of normal thyroid function. However, exposure to excessive doses can also disrupt thyroid function. Agents such as radiographic contrast, amiodarone, iodoquinol, and certain expectorants contain amounts of iodine that are large compared to the minimum daily requirement. One effect of excessive doses of iodine is the acute Wolff-Chaikoff effect, which results in blockage of iodide organification and thyroid hormone synthesis. In rare instances, excessive doses of iodine can cause the opposite effect, known as iodine-induced thyrotoxicosis or the Jod-Basedow phenomenon. When it occurs, the thyrotoxicosis begins weeks to months after iodine administration. In the United States, the Jod-Basedow phenomenon is more commonly seen in patients with nodular thyroid disease. It is believed to result from stimulation of thyroid hormone synthesis in areas of the gland that possess autonomous nodular activity. Eliciting a history of iodine ingestion or administration is always relevant in thyrotoxic patients with a known thyroid nodule. Likewise, recognizing the possibility of thyrotoxicosis in patients who had a recent exposure to iodine is important, especially in the geriatric population in which the incidence of thyroid nodules is higher and the manifestations of thyrotoxicosis are more subtle.

TRANSIENT LEFT VENTRICULAR SYSTOLIC DYSFUNCTION FOLLOWING ABRUPT CLONIDINE WITHDRAWAL. M. Desai¹; A. Atieno¹. ¹Johns Hopkins University, Baltimore, MD. (Tracking ID #135524)

LEARNING OBJECTIVES: 1) Recognize abrupt discontinuation of sympatholytic antihypertensives as a possible cause of acute myocardial insult. 2) Recognize the electrocardiographic (ECG) and echocardiographic findings and temporal course suggestive of adrenergic stress-induced cardiomyopathy (CM).

CASE: A 53 year old postmenopausal woman was hospitalized for constipation secondary to adynamic ileus. Her stage II hypertension was being controlled with nifedipine and a transdermal clonidine patch of 0.2 mg/day. During this hospitalization, she accidentally dislodged the clonidine patch one day. A few hours later, she developed 8/10 severity substernal chest pain radiating to the left arm while at rest, along with dyspnea, nausea, diaphoresis, and blood pressure of 160/105. The pain lasted fifteen minutes and was controlled with sublingual nitroglycerin and morphine. Deep T-wave inversions developed on her ECG, in leads V2-V5, and II, III, aVF; along with 1-2 mm ST elevations in V2-V3, and an incomplete left bundle-branch block. Troponin I rose from 1.02 initially to a peak of 2.38, confirming myocardial infarction. Echocardiogram revealed depressed left ventricular systolic function with an ejection fraction (EF) of 25%, and septal and mid-inferior akinesis with basal sparing. Arteriography revealed no epicardial coronary artery disease. Repeat echocardiogram three days later revealed an EF of 45%, with full recovery of systolic function (EF of 55%) two weeks later.

DISCUSSION: Significant transient left ventricular systolic dysfunction has been described in the setting of marked adrenergic stress, including that due to emotional stress, critical illness, alcohol withdrawal, hyperthyroidism, iatrogenic epinephrine overdose, and acute beta-blocker withdrawal. These causes are thought to be mediated by a hyperadrenergic state causing myocardial stunning. This report suggests that acute clonidine withdrawal should be added to the list of causes of adrenergic stress-induced CM. Several series of case reports have now established some characteristic features of adrenergic stress-induced CM, many of which are also exhibited by this patient. They include: postmenopausal females, diffuse deep T-wave inversion on the ECG, absence of coronary disease or vasospasm, wall motion abnormalities and severe left ventricular dysfunction with full recovery over days-to-weeks. While 'rebound hypertension' and acute myocardial infarction are known side effects of abrupt discontinuation of alpha-2 agonist sympatholytics, transiently depressed systolic function and diffuse deep T-wave inversion on the ECG in this setting has not previously been described. Additionally, previous reports of rebound hypertension from abrupt clonidine discontinuation have been for the oral formulation of high dose clonidine. The transdermal patch is thought to allow more sustained release and thus avoid this problem, although this report does not support such an argument. Catecholamines have been implicated in the mechanism of myocardial stunning and transient left ventricular dysfunction in adrenergic stress-induced CM. While a hyperadrenergic state following abrupt clonidine withdrawal has been described, this report indicates that clonidine withdrawal may also produce a transient form of adrenergic stress-induced CM.

UNEXPLAINED WEAKNESS: AN UNFORTUNATE CASE OF AMYTROPHIC LATERAL SCLEROSIS. E. Daveiga¹; B. Taqui¹; H. Shishodia¹. ¹Temple University, Philadelphia, PA. (Tracking ID #132952)

LEARNING OBJECTIVES: 1. Recognize clinical manifestations of amyotrophic lateral sclerosis (ALS) 2. Review diagnostic testing for ALS 3. Recognize importance of multidisciplinary approach to management of ALS

CASE: 68 year old African American male who was being evaluated for 1 year of progressive generalized weakness was admitted for 3 weeks of worsening dyspnea, left arm weakness and 20 pound weight loss. He reported past tobacco/alcohol abuse and denied family history of malignancy. Exam revealed cachexia with bilateral wasting, diffuse muscle atrophy, 3-4/5 strength in

left upper/lower extremities, deep tendon reflexes (1+ bilateral upper extremity and 4+ left patellar). Cranial nerves and sensation were intact. Fasciculations, which had been noted on prior exams, were absent. Labs revealed elevated CPK (1172) and aldolase (13.3). TSH, B12, folate, RPR, HIV, SPEP, ANA, Anti-JO, RF, PSA were normal. Negative inspiratory force was -50 cm (normal -100). MRI/MRA brain was negative. MRI spine showed diffuse radiculopathy. CT thorax/abdomen/pelvis revealed small bowel attenuation. Since he was poor candidate for enteroscopy/colonoscopy, he had a small bowel follow which was normal. Thus, paraneoplastic syndrome was deemed less likely. EMG/NCS showed severe denervation of upper/lower extremities as well as facial nerve denervation. Left deltoid biopsy revealed demyelinating changes consistent with ALS. Riluzole was initiated without significant improvement. After extensive discussions with patient and family, referral was made to an ALS center and then to hospice. The patient expired 1 month later.

DISCUSSION: Amyotrophic lateral sclerosis (ALS) is a rare progressive neurodegenerative disorder involving upper and lower motor neurons that culminates in weakness, disability and death within 3-5 years. Lower motor neuron findings of weakness, atrophy, and fasciculations result from muscle denervation. Upper motor neuron findings of hyperreflexia and spasticity result from degeneration of lateral corticospinal tracts. Bulbar involvement with dysphagia and dysarthria may occur. Risk factors for ALS are age and family history. Proposed mechanisms for both sporadic and familial forms of ALS include superoxide dismutase type 1 mediated toxicity, glutamate mediated excitotoxicity, heavy metal toxicity, cytoskeletal derangements, mitochondrial dysfunction, viral infections and apoptosis. The correct diagnosis is established in 95% by clinical criteria in association with electrodiagnostic examination. Muscle biopsy is not routinely performed but may be important to exclude inclusion body myositis. Riluzole, a glutamate antagonist, has demonstrated slightly prolonged survival, with minor symptomatic improvement. Other agents are under investigation. Treatment is multidisciplinary focusing on physical therapy, respiratory care, nutritional support, psychiatric care and end of life issues. Education and support for patient and family is the primary goal.

UNILATERAL LEG SWELLING: WHERE'S THE CLOT? S. Homsi¹; E. Anish¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #134315)

LEARNING OBJECTIVES: 1-To appreciate the potential causes of unilateral leg swelling in the absence of venous thrombosis. 2-To perform a diagnostic evaluation of a patient with unilateral leg swelling and negative lower extremity venous Dopplers. 3-To recognize the clinical manifestations of an internal iliac artery pseudoaneurysm.

CASE: An 85-year-old male with a PMH significant for PVD, s/p AAA repair with aorto-bi-iliac and R ilio-femoral bypasses was admitted with painless R lower extremity swelling of eight months duration. He denied claudication symptoms. The swelling had increased over the past month. Physical exam revealed RLE + 2 pitting edema from the foot to the hip. His R femoral pulse was decreased compared to the L. Abdominal exam was benign without masses or bruits. Lower extremity venous Dopplers were negative for DVT. An arterial Doppler revealed no evidence of arterial insufficiency. An abdominal/pelvic CT was then performed revealing a 9 cm in diameter, saccular, R internal iliac artery aneurysm with a rim of mural thrombus. The aneurysm was compressing the R iliac vein resulting into extraluminal obstruction. The aneurysm also compressed the R ureter, creating hydronephrosis. An angiogram then demonstrated the aorto-bi-iliac bypass with the R iliac limb terminating in a large aneurysm at its insertion on the R internal iliac artery. The graft from the R limb of the aorto-iliac bypass to the femoral artery was patent. The patient first underwent cystoscopy with stent placement in the R ureter. In the OR, the aneurysm was actually found to be a pseudoaneurysm. Most of the false aneurysm sac was resected, although a portion adherent to the iliac vein could not be entirely removed. Ligation of the R limb of the aorto-iliac graft was also performed.

DISCUSSION: Unilateral LE edema may result from LE DVT, venous insufficiency, lymphedema, a ruptured Baker's cyst, cellulitis, venous thrombosis above the level of the iliac veins (e.g., IVC thrombosis), or extrinsic compression of the venous system. External compression of the iliac veins may be created by a pelvic or abdominal mass, lymphadenopathy, a markedly distended bladder, or as demonstrated in this case, an arterial pseudoaneurysm. If LE thrombosis has been excluded by Doppler ultrasound, or a venogram shows extrinsic compression of the venous system within the pelvis or abdomen, then a CT scan should often be performed. In this case, a CT demonstrated a large, internal iliac artery pseudoaneurysm. A pseudoaneurysm, also termed a false aneurysm, is created when blood leaks from an artery into the surrounding tissue with persistent communication between the originating artery and the terminating blood filled cavity. The leakage may result from the artery being punctured during cardiac catheterization, angiography, or arterial puncture by injection drug users. They may also occur in a surgical graft or at the anastomosis site of an arterial bypass graft. Pseudoaneurysms may also result from arterial wall infection and they are seen more frequently in patients taking anticoagulants or anti-platelet agents. Pseudoaneurysms are often detected only after patients develop symptoms from compression of adjacent structures. Although rare, an internal iliac artery pseudoaneurysm should be considered as a potential cause of unilateral leg edema in a patient without LE DVT, especially if the patient has a history of prior arterial bypass grafting.

UNUSUAL CAUSE OF CHEST PAIN AND DYSPNEA. B. Arya¹; J. Kleczka¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #131912)

LEARNING OBJECTIVES: 1) Consider Takotsubo cardiomyopathy as a differential diagnosis of Unstable coronary syndromes. 2) Recognize the clinical,

echocardiographic features & natural history of this disorder. 3) Highlight the importance of making this diagnosis & the implications on long term management.

CASE: A 58 year-old white female with a history of bipolar disorder was admitted with a 12-hour history of confusion and 3-hour history of dyspnea & chest pain radiating to her back & left arm which started while she was filling out her tax return forms. Clinical exam revealed stable vital signs, a non-focal neurological exam & normal cardiac exam with no murmurs and no signs of failure. ECG showed diffuse T-wave inversion in V2-V6, I, Avl & prolonged QTc (512 msec). Troponin was elevated at 1.0. CXR showed normal heart size & clear lung fields. The patient was admitted with a diagnosis of non-ST elevation myocardial infarction (NSTEMI). Coronary angiography revealed normal epicardial coronary arteries with no obstructive lesions. Left ventriculography showed decreased ejection fraction (EF) of 30-35% & regional wall motion anomalies (WMA) with hyperkinetic basal segments & akinetic mid and apical segments. Brain MRI, EEG, and CT angio were all normal. The patient was managed with ACE-inhibitors & beta blockers. Echocardiography on day 4 documented normal LV function (EF 55-60%) & complete resolution of WMA. At 3-month follow up, the patient is doing well with no signs of failure & resolution of the widespread T wave inversions on the EKG.

DISCUSSION: Takotsubo cardiomyopathy is a condition that mimics acute MI with clinical features of chest pain, ECG changes & elevation of cardiac enzymes. The name is derived from the characteristic shape of the LV on ventriculography which resembles the Japanese octopus pot (Takotsubo) with a narrow neck & a broad base. As in our case, the characteristic feature of the LV on angiography & echocardiography is hyperkinesis of the basal segments & akinesis/dyskinesis of the mid and apical segments often with apical ballooning as well. Unlike acute MI, in this condition, the coronary arteries are normal & the WMA resolve in a short time. The condition has a female predominance & commonly occurs in the 40-70 years age group. Classically these patients have an intense emotional or physical stressor at the onset of the illness. The pathogenesis of this syndrome is unclear. It is thought that the intense sympathetic overdrive that accompanies emotional or physiological stress induces spasm of the coronary microvasculature, leading to the LV dysfunction. Unlike acute coronary syndromes due to atherosclerotic disease, thrombolysis, primary angioplasty or antiplatelet therapy with GIIb/IIIa antagonist are not indicated in the management. The natural history of the disease is one of early resolution of the LV dysfunction with very good short & long term prognosis & a very low incidence of recurrence.

USE OF RADIOCONTRAST MEDIA AND HYPERTHYROIDISM: EFFECTS ON DISEASE PROGRESSION AND TREATMENT. M. Villano¹; E. Warm¹. ¹University of Cincinnati, Cincinnati, OH. (Tracking ID #135083)

LEARNING OBJECTIVES: 1) Discuss a case of contrast media related hyperthyroidism. 2) Review basic aspects of thyroid gland pathophysiology.

CASE: The use of iodinated contrast media in high-risk individuals has been reported to cause or aggravate hyperthyroidism. We report a case of previously undiagnosed hyperthyroidism in a patient whose hospital course and treatment were complicated by the use of iodinated contrast media. A 45-year-old obese African American male with no past medical history presented to the emergency room with a one-month history of palpitations and dyspnea on exertion. Initial evaluation showed atrial fibrillation in rapid ventricular response. A chest x-ray was consistent with mild pulmonary edema and the patient had an elevated D-dimer (720 mcg/ml). Given these findings, a computed tomography pulmonary angiogram (CTPA) was done for possible pulmonary embolism. The study did not reveal an embolism but did show an enlarged thyroid gland not noted on physical exam. TSH level was < 0.02 mIU/L and free T4 was 4.4 mcg/dL. An endocrine consult was called and propylthiouracil and hydrocortisone were ordered. However, the patient soon became hemodynamically unstable, requiring intubation and vasopressor support. After a complicated MICU stay, patient was eventually stabilized and discharged. Two months after hospital discharge the patient received radioactive thyroid ablation.

DISCUSSION: With the millions of computed tomography (CT) procedures done each year, it is important to recognize that radiocontrast dyes have been documented to cause or worsen hyperthyroidism. In normal individuals, such a large iodine load would cause an inhibition of thyroid hormone synthesis (Wolff-Chaikoff effect). However, with defective thyroid autoregulation, increased hormone synthesis may lead to thyrotoxicosis, and in some cases, thyroid storm. Moreover, there is often a delay in definitive treatment with radioactive iodine ablation due to a need for a washout period. Thus, it is important to identify those patients at risk. While some may present with a history of hyperthyroidism, many are undiagnosed. The frequency of previously unidentified hyperthyroidism as a cause of atrial fibrillation is approximately 3%. In this subset of patients (to which our patient belonged), it is important to weigh the risks and benefits of exposure to contrast media as this will likely complicate the treatment of their underlying illness.

WEAKNESS IN A HYPERTHYROID PATIENT. A.F. Domingo¹; K.J. Pfeifer¹. ¹Medical College of Wisconsin, Milwaukee, WI. (Tracking ID #135449)

LEARNING OBJECTIVES: 1. Recognize thyrotoxicosis as a cause of neuromuscular weakness in selected populations. 2. Distinguish hypokalemic periodic paralysis from other causes of hypokalemia in order to avoid overly aggressive potassium supplementation. 3. Manage thyrotoxic periodic paralysis with beta-blockers

CASE: A twenty year old Hispanic male was admitted for a one-day history of sudden-onset generalized weakness. Physical exam revealed symmetric predominantly lower extremity muscular weakness and absent deep tendon reflexes. Laboratory exams revealed severe hypokalemia and an undetectable TSH. Administration of a non-specific beta-blocker resulted in rapid correction of hypokalemia. A euthyroid state was achieved with propylthiouracil, and by the third hospital day the patient had recovered completely.

DISCUSSION: Expanding Asian and Hispanic populations may be the cause of the increasing incidence of thyrotoxic periodic paralysis in the United States. Transtubular potassium gradient and potassium creatinine ratio may be used to differentiate hypokalemic periodic paralysis from other causes of hypokalemia. Early diagnosis is important because the ultimate goal of management is to achieve a euthyroid state. General internists need to be able to recognize hyperkalemic periodic paralysis because they are often the first health care providers seen by these patients. In up to a fourth of patients with hypokalemic periodic paralysis, a hyperthyroid state may be present. Emergent treatment involves prompt administration of high-dose propranolol. A non-selective beta-blocker should be the first-line drug and may obviate the need for potassium supplementation. Reflex administration of potassium may lead to rebound hyperkalemia requiring correction. Higher doses of potassium do not lead to faster recovery.

WEST NILE VIRUS: A DIFFERENTIAL DIAGNOSIS AMONG IMMUNOCOMPROMISED PATIENTS. J. Chuang¹; P. Balingit¹. ¹Olive View - UCLA Medical Center, Sylmar, CA. (Tracking ID #135424)

LEARNING OBJECTIVES: 1) Recognize the population that is more at risk of being affected by the West Nile Virus (WNV). 2) Recognize the signs and symptoms of West Nile viral meningitis, which may be atypical meningitis.

CASE: A 56 year old woman with stage IV colon cancer with metastasis to the liver who was treated with a sigmoid colectomy and one cycle of chemotherapy presented with fevers, chills, nausea and vomiting, which eventually led to collapse and syncope. Her last dose of chemotherapy was given 15 days ago. Initial vital signs were stable, but the patient appeared fatigued and somnolent. She was alert and oriented to person, place, and time and showed no signs of meningismus. Her lungs were clear, heart was regular, and abdomen was soft and non-tender. No lymphadenopathy was noted. However, the patient had non-pruritic rash along her right cheek, neck, upper torso, and bilateral lower extremities from the ankles to the knee. She was not neutropenic on initial presentation with a white count of 8.3 and normal differential. Her urine analysis was negative for infection, and her chest X-ray was normal. Initial blood cultures were drawn with negative smears and cultures. Abdominal CT with contrast revealed no evidence of abscess. The patient continued to be somnolent and was unable to get out of bed without assistance. A lumbar puncture was performed revealing glucose 70, protein 68, and WBC 50 with 4 neutrophils and 95 lymphocytes. India ink was negative as well as Gram stain and cultures. Viral cultures of Herpes Simplex Virus (HSV), Varicella Zoster Virus (VZV), and Epstein-Barr Virus (EBV) were sent as well as serologies for WNV. The CSF was positive for WNV IgM.

DISCUSSION: WNV is a flavivirus that was first discovered in Uganda in 1937 and emerged in North America in 1999. It is a member of the Japanese encephalitis virus family and is maintained in a bird-mosquito-bird cycle where birds are the primary amplifying hosts. Humans are incidental hosts infected through mosquito bites and infected transplanted organs or blood products. Peak incidence tends to be during the end of August and early September. Most people infected with WNV are asymptomatic or present with upper respiratory complaints. However, the elderly and the immunocompromised patients have a more pronounced risk of neurologic infection. Systemic manifestations include fevers, non-pruritic maculopapular rash, polyarthralgias, and anorexia. Nausea, vomiting, and abdominal pain can also occur. Fatigue may persist for weeks, and acute flaccid paralysis may develop. Diagnosis includes testing for IgM antibody to WNV within the serum or CSF. Treatment is supportive care. However, the efficacy of ribavirin, interferon alpha 2b, steroids, antiepileptic medications, intravenous Ig and osmotic agents are currently being studied. Advanced age is the most important predictor of death.

WHAT'S GOOD FOR THE GOOSE IS GOOD FOR THE GANDER. N.A. Abdo¹; A. Gomez². ¹Olive View - UCLA Medical Center, Sylmar, CA; ²David Geffen School of Medicine at UCLA, Sepulveda, CA. (Tracking ID #133947)

LEARNING OBJECTIVES: 1. Recognize the rising clinical significance of breast cancer in males. 2. Recognize the challenges in the treatment of breast cancer in males.

CASE: A 65 year old man with medication-controlled schizophrenia was referred by his psychiatrist for evaluation of a left breast lump. Two months prior, the patient noticed a small non-tender mass behind his left nipple, then measuring "less than one inch". Although the mass doubled in size, it was not until it became tender that he sought medical attention. He reported no associated skin changes, other masses or any other symptoms. On examination he had a left 4-5 cm retro-areolar breast mass that was tender to touch. There was no nipple discharge, skin changes, axillary adenopathy or hepatosplenomegaly. A breast ultrasound and mammogram showed bilateral suspicious masses. A core biopsy confirmed the diagnosis of bilateral invasive ductal carcinoma. After a PET scan, he was staged as T2 N1 Mx, found to be ER/PR+ and Her-2/neu+. He underwent bilateral modified radical mastectomies. His chemotherapy has been delayed due to a cellulitis he developed at surgical site seromas. He has been

started on tamoxifen with anticipation of adjuvant chemotherapy once the infection is resolved.

DISCUSSION: Breast cancer in males (BCM) accounts for 1% of cancer in males and less than 1% of all diagnosed breast cancer. Despite the lower incidence, breast cancer specific death rate is comparably higher in males. Risk factors for BCM include advanced age, alcohol use, obesity, positive family history, Jewish ancestry, black race, history of irradiation, exposure to excessive female hormones and higher socio-economic status. The clinical course may vary from indolent to aggressively metastatic. The tumor is usually sub-areolar and in advanced cases bloody and painful ulcerations may appear. Contrasted with females, BCM has a unimodal age of presentation, later age of onset, advanced stage at diagnosis, higher incidence of nodal metastasis and higher expression of hormonal markers. Overall, BCM carries a worse prognosis. Unlike breast cancer in females, the incidence of BCM in patients under-forty is increasing. Though necessary, a treatment strategy specifically targeting BCM is elusive; to date there have been no randomized prospective treatment trials. As done with our patient, modified radical mastectomy appears to be preferred in case controlled and retrospective studies. There is no conclusive data supporting the use of adjuvant therapy in BCM. Because of the preponderance of evidence from female studies, adjuvant chemotherapy is strongly recommended. Probably related to quicker invasion in less tissue, men fair less well with adjuvant radiation therapy. Though not clear why, men are more likely to receive suboptimal care; it may relate to lack of experience with BCM. Application of breast cancer treatment guidelines, well studied in females, to BCM may prove helpful; so, until further evidence to prove the contrary, what's good for the goose is "hopefully" good for the gander.

WHEN 2 AGONISTS ARE CONTRAINDICATED IN AN ASTHMATIC ATTACK. H.A. Younes¹; J. Sethi². ¹University of Pittsburgh, Pittsburgh, PA; ²University of Pittsburgh Medical Center, Pittsburgh, PA. (Tracking ID #133385)

LEARNING OBJECTIVES: 1) To learn about the different acid base abnormalities in asthmatic patients, 2) To recognize β_2 agonists as a cause of lactic acidosis in patients with asthmatic attacks

CASE: A 34 y.o. man, was admitted to the hospital for an asthmatic exacerbation and was treated over the next two days with steroids, and albuterol and ipratropium nebulizations, with marked improvement. On the third hospital day, his dyspnea and tachypnea worsened despite several treatments with Albuterol and Ipratropium aerosol. Upon admission to the ICU, the patient was in severe respiratory distress, with pronounced Kussmaul's respirations, and tachycardia. Lung exam, however, showed good air entry without wheezing. Arterial blood gas (ABG) on 6L/min oxygen by face mask revealed a PH of 7.40, PCO2 of 23, PO2 of 181, HCO3 of 14, and a lactate level of 10.4. A β_2 agonist induced lactic acidosis was suspected. Albuterol was discontinued, and the patient was treated with Ipratropium inhalations alone. Over the next few hours, the patient's dyspnea and respiratory rate dramatically improved, and his lactic acidosis resolved. He was transferred to the floor the next morning.

DISCUSSION: While respiratory alkalosis is the most common acid base disturbance in patients with asthmatic attacks, lactic acidosis is being increasingly described in the literature. Initially thought to result from the decreased oxygen supply to the overworked respiratory muscles, the cause of lactate overproduction in these patients is apparently multifactorial. One of the largely underdiagnosed causes is the over use of β_2 agonists, commonly used as an essential component in breathing treatment in patients with asthmatic attacks. The mechanism of this β_2 agonist induced lactate acidosis is still undetermined. One hypothesis relates to changes in glycolysis caused by β_2 agonists, with a resultant increase in anaerobic metabolism of the muscular cells. Another potential mechanism depends on the fact that β_2 receptor activation is known to increase the rate of lipolysis. The increased free fatty acids are known to decrease the conversion of pyruvate to acetyl coenzyme A with a consequent increase in lactic acid production. While our patient was in clear respiratory distress, he was not hypoxic and his lung exam showed a good air entry that strongly argued against the obvious initial thought that his asthmatic attack was worsening. The ABG confirmed he was not hypoxic, and so his lactic acidosis was not due to increased work of breathing. The worsening respiratory status in the face of repetitive β_2 agonist aerosol treatments, and the fact that his acute lactic acidosis developed during the resolution of his asthmatic attack raised our suspicion of this unusual diagnosis, with its totally different treatment approach.

WHEN ADDITIONAL HISTORY SOLVES THE MYSTERY: A CASE OF BIRD FANCIER'S LUNG DISEASE. G. Silberman¹; B. Taqui¹; H. Shishodia¹. ¹Temple University, Philadelphia, PA. (Tracking ID #132964)

LEARNING OBJECTIVES: 1. Recognize the importance of occupational/environmental history in patients with respiratory symptoms 2. Review epidemiology and clinical presentation of hypersensitivity pneumonitis (HP) 3. Review management of HP

CASE: 59 year old female presented with 1 week of increasing dyspnea, wheezing, and nonproductive cough. She had similar symptoms 2 years prior and was diagnosed with asthma. She required hospitalization 1 month prior and was treated for asthma exacerbation and atypical pneumonia. She was discharged on steroids, but presented to us with symptom recurrence after taper. At this time, she was asked about occupational/environmental exposures and reported having 30 parakeets in her home for several years. Exam revealed tachypnea, oxygen saturation 88% RA, bilateral wheezing and bibasilar crackles. CBC, electrolytes, ANA, RF, ANCA, ACE level, and HIV were negative. ABG: pH 7.40

pCO₂ 40 pO₂ 77 HCO₃ 24, 95% RA. CXR had mild interstitial edema. Pulmonary function tests showed moderate obstructive and severe restrictive defects, moderate decrease in diffusion capacity. High resolution CT scan of thorax showed bilateral diffuse ground-glass densities. Bronchoscopy with BAL/trans-bronchial biopsy showed thickened alveolar walls with increased collagen deposition, chronic inflammatory changes, fibroblastic proliferation of airspaces. Staining for PCP/CMV, cultures for legionella/fungus/bacteria/AFB were negative. She was diagnosed with bird fancier's lung, treated with steroids and advised to remove the birds from her home. Approximately 8 months after removing the birds, the patient was asymptomatic without any treatment.

DISCUSSION: Bird fancier's lung, a hypersensitivity pneumonitis involving alveolar walls/terminal airways, is caused by an immune response to antigens present in bird droppings/feathers. Hypersensitivity pneumonitis (HP) is associated with a variety of organic dusts including thermophilic actinomycetes, "moldy" hay, silage/grain, and water systems. Its prevalence varies with intensity of exposure, season, geography and host risk factors. Tobacco smoking is associated with a decreased risk of HP. The prevalence of bird fancier's lung in breeders is 5-15%, but the disease can also occur (with uncertain prevalence) in others exposed to parakeets, owls and feather pillows. Patients with HP present with acute, subacute or chronic symptoms of dyspnea, cough, fever, and malaise. PFT may reveal a mixed restrictive and obstructive pattern with decreased diffusion capacity, and exercise induced hypoxemia. High resolution CT may reveal ground-glass opacities, interstitial and reticulonodular pattern. BAL may reveal lymphocytosis with low CD4/CD8 ratio. Pathology may reveal interstitial alveolar inflammatory mononuclear infiltrate, varying degrees of fibrosis, bronchiolitis, noncaseating granulomas. However, these findings are often nonspecific. Thus, history of exposure (and symptom resolution with avoidance) is the crux of making the diagnosis, with inhalation challenge recommended in some cases. The definitive diagnosis is provided by a 2 week trial of avoidance with symptom resolution within 24-48 hours (acute form) or several weeks (subacute, chronic form). Management is exposure avoidance, with steroids reserved for refractory cases. Prognosis varies with age, duration and intensity of exposure. Our case highlights the importance of obtaining an occupational/environmental history in patients presenting with respiratory symptoms.

WHEN PREEXCITATION IN A MAN IS NOT GOOD: A CASE OF WOLFF PARKINSON WHITE WITH A RARE COMPLICATION OF RADIOABLATION. A. Koka¹; S. Toh¹; B. Taqui¹. ¹Temple University, Philadelphia, PA. (Tracking ID #132942)

LEARNING OBJECTIVES: 1. Recognize Wolff Parkinson White (WPW) as a rare, but well established cause of syncope 2. Review clinical and EKG criteria for WPW 3. Review indications and complications of therapy for WPW

CASE: 29 year old male with no past medical history presented with sudden onset of lightheadedness. He denied chest pain, palpitations, dyspnea, diaphoresis. Witnesses noted that the patient did not respond to verbal stimuli for 1-2 minutes. Physical exam was unremarkable. EKG revealed short PR interval and delta waves. Electrophysiology study revealed inducible atrial fibrillation and rapid antegrade conduction down an accessory pathway, which was subsequently ablated using radiofrequency waves. The day after, the patient demonstrated intermittent runs of sinus tachycardia and 2:1 atrioventricular (AV) block. A treadmill stress test performed to assess AV conduction revealed 2:1 AV conduction with sinus rates higher than 80-90 bpm. However, the patient was asymptomatic during the stress test and did not require further intervention. One month later, he continued to deny symptoms but has since failed to follow up.

DISCUSSION: Wolff Parkinson White (WPW) describes a rare EKG pattern with two major features. The first feature is a short PR interval (less than .12 sec) due to rapid atrioventricular conduction (AV) through an accessory pathway which bypasses the AV node and results in preexcitation of the ventricles. The second is a slurred QRS upstroke (delta wave) due to fusion between this early ventricular activation and later activation due to normal AV nodal conduction. The prevalence of WPW is .15-.25%. WPW syndrome, even less common, consists of EKG findings in association with a tachyarrhythmia, palpitations, lightheadedness, syncope or sudden death. There are two types of tachycardias. AV reentrant tachycardias require the accessory pathway for initiation and maintenance. For other tachycardias, including atrial fibrillation, the bypass tract acts as a "bystander" providing alternative route of conduction. Atrial fibrillation occurs in 10-30% of WPW syndrome and is more likely when accessory pathways have short refractory periods and rapid antegrade impulse conduction. These findings are also risk factors for sudden death and require ablation of bypass tract. Other risk factors include multiple accessory pathways and inducible AV reentrant tachycardia and/or atrial fibrillation. Although surgery has almost a 100% cure rate, less invasive radiofrequency ablation with 90% cure rate is preferred. Complications of ablation include bleeding, infection, clots, perforation, arrhythmias, and very rarely conduction abnormalities. Mortality risk is .1-.3%. Medications such as verapamil, digoxin, adenosine, beta blockers and lidocaine may increase ventricular rate and should be avoided. Our case taught us the importance of recognizing WPW on EKG in patients with (near) syncope, avoiding treatment with medications that increase ventricular rate, risk stratifying for sudden death and radiofrequency ablation, and recognizing ablation complications.

WHEN VIOLET EYES ARE NOT DESIRED: AN UNFORTUNATE CASE OF DERMATOMYOSITIS. A. Kalra¹; B. Taqui¹; H. Shishodia¹. ¹Temple University, Philadelphia, PA. (Tracking ID #132945)

LEARNING OBJECTIVES: 1. Recognize dermatomyositis (DM) as a rare, but treatable cause of muscle weakness 2. Recognize that diagnosis of DM is pre-

dominantly clinical, with adjunctive roles for EMG and muscle biopsy 3. Review management of DM

CASE: 35 year old African American female presented with 1 week of bilateral muscle pain and weakness in her thighs, calves, arms and neck. She had difficulty walking and carrying her purse. She also reported 1 year of bilateral eyelid rash. She denied drug, alcohol, medications. Exam revealed violaceous eruption over upper eyelids, bilateral 4/5 proximal muscle weakness. Three weeks later, she presented to rheumatology with inability to do overhead activities, difficulty climbing stairs, and dysphagia. Labs: CK 16,200 AST 500 ALT 342 ESR 34 Aldolase 143. CBC/electrolytes/TSH were normal. Anti-Jo, anti-mitochondrial IgG, RF, anti-SM/RNP were negative. EMG was deferred in favor of MRI and muscle biopsy. MRI of thighs showed increased signal in multiple muscles, but particularly in right rectus femoris making it a good site for biopsy. Pathology revealed myopathic changes, scattered necrotic fibers, regenerating fibers, but no active inflammatory infiltrates. Patient was diagnosed with dermatomyositis and started on prednisone. Malignancy workup including UA, CXR, mammogram, pap smear and pelvic ultrasound was negative. Two weeks later, she was admitted for worsening symptoms, at which point the CK was 21,000. She was treated with pulse steroids, intravenous immunoglobulin and methotrexate. CK levels dropped to 3,000, but there was only minor improvement in symptoms. She was discharged to a rehab facility.

DISCUSSION: Dermatomyositis (DM) is a rare idiopathic inflammatory myopathy predominantly affecting females in the fifth decade of life. It is an immune mediated process triggered by environmental factors in genetically predisposed individuals. Patients have characteristic skin findings including Gottron's sign (slightly elevated papules/plaques on bony prominences), violaceous erythema in shawl and v-neck distribution and heliotrope rash (reddish-violaceous eruption over upper eyelids). Both Gottron's sign and the heliotrope rash are very specific for DM. Patients also have symmetric proximal muscle weakness which may progress to dysphagia and respiratory failure. Elevation of CK, LDH, aldolase, AST and ALT occur in most patients. Nonspecific autoantibodies occur in 80%, but myositis specific autoantibodies (Anti-Jo) occur in only 20-30%. EMG shows evidence of increased membrane irritability. Muscle biopsy is the definitive test, but its indications are controversial. MRI guidance and open needle approach improve diagnostic yield. Histologic features include muscle fiber necrosis, degeneration, regeneration and inflammatory cellular infiltrate (predominantly perifascicular or perivascular, with B lymphocytes and increased CD4+ cells). Diagnosis is predominantly clinical, but also based on EMG and biopsy findings, as well as ruling out other disorders. Corticosteroids and/or immunosuppressive agents are the mainstay of treatment. Other considerations are avoiding sun, exercising, preventing osteoporosis, and treating dysphagia/respiratory illness. DM is associated with malignancy and all patients should be screened based on their specific risk factors. Our patient had severe weakness and dysphagia, which along with malignancy are indicators of poor prognosis.

WHY BOTHER? IT'S JUST A RASH! DERMATOMYOSITIS IN A 56-YEAR-OLD WOMAN. M.T. Rahimi¹; V. Gujral¹; R. Orakzai¹; S.H. Orakzai¹; E. Campbell¹. ¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #132782)

LEARNING OBJECTIVES: 1. Recognize the presentation of dermatomyositis (DM). 2. Diagnose and initiate management of DM early. 3. Appreciate the close association of DM with malignant disease.

CASE: A 56-year-old female with hypertension and hypothyroidism presented with a 3-week history of fever, chills and flushing. She also noticed a flat, erythematous rash initially on the back of her neck, spreading to her chest, dysphagia and difficulty raising both arms. Her examination was remarkable for diffuse erythema over the chest and shoulders with proximal muscle weakness and tenderness. Initial workup was significant for an ESR of 53, ALT of 523, AST of 323, and CPK of 11825. Her immunological workup revealed ANA of 1280 with homogenous pattern, and negative results for RF, SSA, SSB, scl-70, anti-centromere, anti-smooth muscle and anti-LKMB antibodies. Anti-Jo1 and anti-KU antibodies were positive. Her EMG and muscle biopsy was consistent with DM. A diagnosis of DM/systemic sclerosis(SS) overlap syndrome was made and the patient was treated with prednisone and methotrexate(MTX). Evaluation for underlying malignancy included colonoscopy, CT chest, abdomen and pelvis, serum AFP and CA-125, results of which were negative. After 20 days, her rash was gone, muscle strength partially regained and CPK trending down. However her LFTs were high, so MTX was discontinued. Subsequently, she stopped taking prednisone on her own as it was making her feel "goofy". Within several weeks, the rash and myalgias reappeared. The patient was started on mycophenolate.

DISCUSSION: The incidence of DM is estimated at 5.5 cases per million people. The cause is unknown however immunological abnormalities are usually implicated. This case fulfills Bohan and Peter's diagnostic criteria (Bohan et al. NEJM 1975) for DM:systemic proximal muscle weakness, typical rash, elevated CPK, myopathic changes on EMG and muscle biopsy abnormalities. The rash here depicts the classic Shawl sign. Muscle weakness is typically symmetric and proximal. DM may overlap with other CT diseases particularly SS and SLE. Early diagnosis of the disease is important, as response is best achieved with early initiation of therapy. There is correlation between the severity of the weakness and the height of elevation in plasma muscle enzymes. High ANA titers are suggestive and anti-Jo1 antibody is present in 20% of cases. Anti-KU antibodies suggests DM/SS overlap syndrome. EMG may be normal in 11% of cases. Muscle biopsy is the definitive test and establishes the diagnosis. Corticosteroids are the mainstay of management. Azathioprine, MTX and mycophenolate are tried as steroid sparing agents. There are reports of the use of cyclosporine, immunoglobulin and TNF inhibitors in resistant cases. An increased risk of malignant disease has been found with DM including cervical, lung, ovarian, pancreatic,

and gastric carcinoma. Along with age appropriate cancer screening, chest, abdomen, and pelvic CT scans have been recommended for newly diagnosed patients. It is also appropriate to screen women with a serum CA-125 level at least twice yearly, and perform annual pelvic and intravaginal ultrasound for up to five years after the diagnosis of DM.

WHY HAS SHE GONE MAD? T. Wilcox¹. ¹University of Chicago, Chicago, IL. (Tracking ID #132468)

LEARNING OBJECTIVES: 1. Recognize clinical features of Cat Scratch-Disease 2. Recognize B. Henselae infection as a potential cause of fever and behavioral changes

CASE: A 55 year old female with a history of hypertension presented with fever, headache, and left groin pain for 1 week. She developed behavioral changes including wandering, insomnia, and hallucinations. The medical history was negative for psychiatric illness. The patient was febrile (38.6°C) and had a tender 4 cm left inguinal lymph node. Neurologic exam, including orientation, was normal but on psychiatric exam the patient demonstrated easy distractibility, pressured speech, paranoia and hallucinations. Laboratory work up revealed anemia (8.0 g/dl) but a normal WBC (7.5 k/ul). EEG, lumbar puncture, RPR, TSH, PPD, HIV antibody and viral load were negative. MRI of the brain showed nonspecific white matter abnormalities. Upon further questioning, the patient reported being scratched on her left leg by a recently adopted stray kitten. Indirect fluorescent antibodies for Bartonella henselae were then sent and were elevated with an IgG level of 1:1024. The left inguinal node was removed due to increasing size, pain, and suppuration. Pathology revealed necrotizing granulomas as are classically seen in cat-scratch disease. The patient was treated with doxycycline, rifampin and olanzapine. At discharge one month later, she was afebrile and her mental status and behavior had returned to baseline.

DISCUSSION: Bartonella henselae is a fastidious gram-negative bacillus and is the primary etiologic agent of cat-scratch disease (CSD). Transmission is linked to exposure to young cats. Estimates of incidence range from 3.7-9 cases/100,000 persons and 80-90% of infections occur in patients under the age of 21. The disease is typically self-limited, presenting with a cutaneous lesion at the site of inoculation and followed by painful regional lymphadenopathy. There may be associated fever and malaise. Neurologic involvement, ocular disease, and visceral organ involvement are atypical manifestations. Though rare, a wide range of neurologic manifestations has been described. Encephalopathy is the most common and usually presents with headache and delirium, though can be complicated by coma or seizures. EEG is usually abnormal and CSF may show mild mononuclear pleocytosis. Imaging abnormalities are rare but include nonspecific white matter and basal ganglia lesions. In this case the neurologic manifestations of CSD were atypical. The patient had marked behavioral changes but preserved cognition and a normal LP and EEG. Studies have suggested a role for B. henselae infection in the development of psychosis and neurocognitive decline in HIV patients. Psychosis is also seen in patients with bacillary angiomatosis, a unique form of B. henselae infection characterized by proliferation of blood vessels. This is the first described case of psychosis in an immunocompetent patient with CSD.

ZINC DEFICIENCY IN A PATIENT WITH DEPRESSION. R.M. Haddad¹. ¹Mayo Clinic, Rochester, MN. (Tracking ID #135297)

LEARNING OBJECTIVES: 1. Distinguish the characteristic rash of zinc deficiency 2. Recognize the dermatologic manifestations of mineral deficiencies 3. Expand the differential diagnosis of a rash

CASE: A 61 year old woman was referred to our tertiary medical center by her brother because of his concerns that she was severely depressed. She was seen in the emergency department and admitted to the hospital service when found to be anemic, hypokalemic, and hypocalcemic. She had suffered from depression for three years since the death of her husband. At the time of admission she reported decreased appetite for approximately one year with very minimal oral intake in the preceding eight to twelve weeks. Her major complaint was a pruritic, red, scaling generalized rash. She had been treated by her local physician with oral antibiotics and a short course of oral steroids with no resolution or her rash. Additional history was significant for loose stools during the month before admission and that she had been drinking four to five beers a day. Physical examination revealed a cachectic woman weighing 38 kilograms. There was a generalized desquamative rash with erythematous erosions in the flexural areas of the elbows and knees. Laboratory studies revealed a low zinc level, and skin biopsies showed epidermal acanthosis with parakeratosis, hypogranulosis and mixed dermal inflammation consistent with zinc deficiency. Further evaluation revealed that her malabsorption was secondary to chronic pancreatitis.

DISCUSSION: Etiology: In its purest form, zinc deficiency has been described in a genetic disorder, acrodermatitis enteropathica. It is an autosomal recessive inherited trait in which intestinal absorption of zinc is decreased. It is due to mutations in the SLC39A4 gene on chromosome 8q24.3, which codes for a protein thought to be involved in zinc transportation (5). Acquired zinc deficiency is now known to occur in several clinical scenarios, such as during the administration of chronic total parenteral nutrition (6). Other causes of acquired zinc deficiency include malabsorption caused by inflammatory bowel disease, pancreatitis, and intestinal or gastric bypass. If the status of this mineral is not addressed then it can potentially lead to a poor outcome. Manifestations of Zinc Deficiency: The clinical presentations of zinc deficiency are many and varied. In young patients, deficiency can lead to decreased growth retardation and delayed sexual maturation. In older patients other symptoms such as impotence, hypogonadism, oligospermia, dysgeusia, immune system dysfunction, diarrhea, im-

paired wound healing and night blindness are potential consequences of zinc deficiency. Dermatologic Findings: Dermatologic manifestations include patches and plaques of dry, scaly, distinctly marginated eczematous dermatitis which evolves to vesiculobullous, pustular, and crusted lesions. Initially the dermatitis is noticed in the perioral and anogenital area but later the scalp face, hands, feet, acral surfaces of the upper and lower extremities and trunk are affected. When assessing such dermatologic findings the differential includes atopic dermatitis, seborrheic dermatitis, and mucocutaneous candidiasis. The rash can also be confused with rare disorders such as glucagonoma (7). The differentiation between such varied diseases and the zinc deficient state can be made on the clinical context of the presentation as well as a high level of suspicion.

DON'T DRINK THE WATER. M. Glass¹; J. Wiese¹. ¹Tulane University, New Orleans, LA. (Tracking ID #135686)

LEARNING OBJECTIVES: 1. Recognize the clinical presentation of psychogenic polydipsia. 2. Understand the relationship between psychogenic polydipsia and mental illness.

CASE: A 51-year-old woman was brought to the hospital for "talking like a prophet." She had no history of trauma and described no symptoms of infection, including no headache, visual changes or fever. Her medications consisted of carbamazepine and hydrochlorothiazide. Except for a pressured speech pattern, her physical exam and vital signs were normal. Her sodium was 125 mEq/l and her glucose was normal. Urine and serum toxicology screens, a urinalysis, and a CT of the head without contrast were normal. Her serum osmolality was 264, the urine osmolality was 54, and the urine sodium was 9 mEq/l. She denied excessive water consumption, but after being admitted to the psychiatric ward with a diagnosis of mania, she was noted to consume massive quantities of water. Once restricted to less than one liter of free water per day, her laboratory values normalized and her mental status improved.

DISCUSSION: Hyponatremia is frequently encountered in psychiatric patients. The two most common causes of hyponatremia in a euvolemic patient are SIADH and psychogenic water drinking. Internists must be aware of the value of the urine osmolality in distinguishing between the two when performing medical consultations. Since ADH concentrates the urine, a low urine osmolality excludes the diagnosis of SIADH and therefore confirms psychogenic water drinking as the diagnosis. Both disorders are common in psychiatric patients; the juxtaposition of the hypothalamus (rich in dopamine receptors) to the supra-optic nucleus (the source of ADH and the thirst center) makes SIADH and psychogenic water drinking common findings in psychiatric patients, especially those receiving neuroleptics. Establishing the diagnosis is important in the medical consultation, since SIADH may prompt reduction in neuroleptic therapy, while psychogenic water drinking may prompt increasing neuroleptic therapy. The first-line treatment for psychogenic polydipsia is free-water restriction and treatment of the primary psychiatric disease. If psychiatric symptoms persist after slow correction of the hyponatremia, primary anti-psychotic therapy should be instituted.

FEEL MY THIGH, SAVE MY LIFE. J. Towbin¹; J. Wiese¹. ¹Tulane University, New Orleans, LA. (Tracking ID #136150)

LEARNING OBJECTIVES: 1. Recognize the importance of a thorough physical examination in medical consultation. 2. Recognize the clinical presentation of necrotizing fasciitis

CASE: A 55 year-old man was admitted to the neurology service with bilateral lower extremity weakness. On MRI he was found to have had a mass compressing his spinal cord, presumed to be a metastasis from a previously diagnosed laryngeal cancer. He received eight doses of radiation that improved the strength of his lower extremities. On hospital day #16, he again began to lose lower extremity strength. The medicine service was consulted. He had bilateral lower extremity weakness (2/5), and a three cm stage-two sacral decubitus ulcer. The remainder of the exam was normal. The following day his white blood cell count dropped to 1,500/mm³ and his platelets declined to 31,000/mm³. He was started on vancomycin and cefepime. Skeletal survey X-rays were obtained to exclude possible bony metastatic lesions. On review of the films, it was noted that he had subcutaneous air spreading across his pelvis and down his right thigh. On returning to the bedside, we noted subcutaneous crepitance over his right thigh. Surgery was consulted. At the time of their consultation 15 minutes later, crepitance has spread to his mid-thorax. His antibiotics were switched to penicillin G and clindamycin. However, he expired prior to surgical intervention.

DISCUSSION: This case illustrates the importance of prompt recognition of necrotizing fasciitis, and the importance of a thorough physical examination with each patient assessment. Necrotizing fasciitis is often observed in post-surgical patients, but can be seen in any patient with a portal into the skin, including decubitus ulcers. Patients at higher risk include diabetics, intravenous drug abusers and the malnourished. Crepitance of the skin, bullae formation, purpuric discoloration, and anesthesia result as the infection tracks along fascial planes, destroying sub-cutaneous vessels and nerves. Common pathogens include Clostridium, Strep and Staph. Parenteral antibiotics should target polymicrobial aerobic and anaerobic organisms, although definitive therapy requires surgical debridement. Bed-bound patients frequently receive limited physical examinations, especially on non-medical services where patient care is usually focused on one predominate problem. Our patient illustrates that a thorough physical examination may be one of the most important contributions a medical consultant can offer. Immediate recognition of the necrotizing fasciitis by physical examination, instead of discovering it serendipitously by X-ray, may have offered our patient a chance at survival.

MICROCOCOCCUS ENDOCARDITIS IN A PATIENT WITH A PACEMAKER. E.T. Salvana¹, D. Wagner². ¹Medical College of Wisconsin, Milwaukee, WI; ²Department of Veterans Affairs, Milwaukee, WI. (Tracking ID #135299)

LEARNING OBJECTIVES: 1. Recognize that patients with pacemakers are at higher risk for endocarditis 2. Diagnose endocarditis in a patient with a pacemaker 3. Manage a patient with a pacemaker with vegetations from Micrococcus endocarditis

CASE: An eighty year old Caucasian male with a history of pacemaker implantation was admitted from an outside hospital with fever, chills, nausea, and vomiting and multiple blood cultures positive for Micrococcus. The patient had been admitted to the same hospital a month prior with the same symptoms. He had grown out Micrococcus in blood cultures which cleared with administration of IV levofloxacin and clindamycin. Transthoracic echocardiogram at that time was negative for any vegetations. Transesophageal echocardiogram on the day of admission revealed a large soft tissue mass consistent with a large vegetation on his pacemaker atrial lead occasionally passing through the tricuspid valve. Infectious disease, cardiology and cardiothoracic surgery services were consulted. Pacemaker explantation was offered but was declined initially. The patient was treated with intravenous vancomycin, and oral linezolid and rifampin for several weeks with clearing of blood cultures and disappearance of the mass on transesophageal echocardiography. There was however note of persistent swelling in the dentopectoral groove on the echocardiogram. The pacemaker and generator were subsequently explanted and Micrococcus was recovered from the pacemaker lead, generator and pacemaker pocket. The patient was then restarted on intravenous vancomycin and discharged on linezolid with no further complications.

DISCUSSION: The increasing prevalence of pacemaker recipients makes the recognition of endocarditis by the general internist in this subgroup of patients increasingly relevant. While the management of endocarditis is generally straightforward, the treatment of such in pacemaker recipients is more difficult and requires coordination of subspecialty care. This is true particularly since the causative organisms may be different. Our case involves Micrococcus which is a part of the normal skin flora but has been implicated in only seventeen cases of bacterial endocarditis. Three of these cases involved prosthetic valves. This is the only case associated with a pacer wire. Despite a long course of intravenous antibiotic therapy, Micrococcus was still recovered from the explanted device. This illustrates the need to involve subspecialty services early since cure may only be achieved with prolonged antibiotic treatment and possible device explantation.

THE METABOLIC IMPACT OF SECOND GENERATION ANTIPSYCHOTICS. A.W. Dollahide¹, J.M. Meyer¹. ¹University of California, San Diego, San Diego, CA. (Tracking ID #133207)

LEARNING OBJECTIVES: 1) Recognize the unique metabolic impact of second-generation antipsychotic medications. 2) Diagnose and manage the metabolic syndrome, seen commonly in patients with schizophrenia.

CASE: A 46 year old African-American male was diagnosed with late-onset chronic paranoid schizophrenia. He was started on a second-generation antipsychotic (SGA) risperidone, which was effective in controlling his target symptoms of auditory hallucinations and paranoia. Average one year weight gain on risperidone is 5.5 lbs. Pre-antipsychotic baseline: weight 145 lbs, BMI 21, BP 130/80 mm Hg, cholesterol 194 mg/dL, triglycerides 103 mg/dL, HDL 79 mg/dL, LDL 94 mg/dL, glucose 98 mg/dL. During the ensuing 36 months of treatment, the patient gained 137 lbs resulting in a peak weight of 283 lbs (BMI 42), nearly doubling his baseline body weight. He became hypertensive (BP 160/96 mm Hg), with a relative dyslipidemia (cholesterol 260 mg/dL, triglycerides 109 mg/dL, HDL 42 mg/dL, LDL 196) and hyperglycemia (glucose 162 mg/dL, HgA1c 8.10%). The patient was counseled extensively about diet, exercise and weight control during this treatment. His antipsychotic prescription was changed from risperidone to more weight neutral atypical antipsychotics: ziprasidone, then aripiprazole. Despite these interventions, he continued to meet criteria for the metabolic syndrome.

DISCUSSION: The **metabolic syndrome** is a constellation of metabolic abnormalities that confers dramatically increased cardiovascular risk. In 2002, the third report of the National Cholesterol Education Program (ATP III) identified clinical criteria including **abdominal obesity, elevated triglycerides, low HDL cholesterol, hypertension and impaired fasting blood glucose**. Three of five criteria must be met to diagnose this syndrome. Although debate continues on the individual syndrome criteria, consensus has emerged that this cluster of metabolic abnormalities is an important predictor of cardiovascular disease. The widespread use of SGAs over the last decade has led to concern regarding their metabolic adverse effects. There are significant differences in the liability for weight gain, dyslipidemia and hyperglycemia among the SGAs. Clozapine and olanzapine have greatest risk, while aripiprazole and ziprasidone are relatively neutral, and quetiapine and risperidone have modest weight gain risk with discrepant data regarding effects on lipids and glucose. Both obesity and the metabolic syndrome are overrepresented among patients with schizophrenia, with the prevalence of the metabolic syndrome in one published study of 37%, more than double the age-matched prevalence for the surrounding geographic area. Psychiatrists are increasingly aware of the metabolic consequences of antipsychotics, and the FDA has recently mandated that all patients treated with an SGA receive glucose monitoring. In February 2004, **Diabetes Care** published multi-disciplinary consensus guidelines addressing monitoring and treatment for patients developing obesity and metabolic syndrome on SGAs. Treatment recommendations include considering switch of SGA if a patient gains > 5% of his or her initial weight during treatment. For patients developing worsening glycemia or dyslipidemia while on therapy, a switch in SGA should be considered.

The cooperative efforts of psychiatrists and general internists will be critical to effectively monitor and manage cardiovascular risk in this unique patient population.

THROMBOEMBOLISM IN BRAIN CANCER PATIENTS UNDERGOING NEUROSURGERY. A. Swift¹, B. Taqui¹, H. Shishodia¹. ¹Temple University, Philadelphia, PA. (Tracking ID #132961)

LEARNING OBJECTIVES: 1. Recognize high incidence of thromboembolism in patients undergoing neurosurgery for glioblastoma multiforme 2. Recognize the need for thromboembolism prophylaxis in this population 3. Review treatment options for this population

CASE: 65 year old male was admitted for surgical evaluation of left temporal mesial tumor, which was subsequently diagnosed as glioblastoma multiforme. On post operative day 5 after craniotomy, medicine team was consulted to evaluate patient for chest pain. On exam, he had P 110, BP 104/60, RR 20-24 and oxygen saturation 78% room air. He also had elevated JVP and decreased breath sounds at bases. ABG: pH 7.46 pCO2 37 paO2 74 HCO3 25.8 on 95% NRB. EKG revealed new findings of sinus tachycardia 103, right bundle branch block, right ventricular strain pattern. Bedside echocardiogram showed new findings of right ventricular and atrial dilatation. Presumptive diagnosis of pulmonary embolism was made. Patient had only been receiving subcutaneous heparin for prophylaxis. Neurosurgery attending was reluctant to allow initiation of anticoagulation treatment and patient was sent for urgent inferior vena cava filter placement. The patient continued to deteriorate, requiring mechanical ventilation and expired on post operative day 16 due to multi organ failure.

DISCUSSION: Deep vein thrombosis (DVT) occurs in 24-43% of patients undergoing elective neurosurgery for central nervous system tumors. The incidence of pulmonary embolism (PE) in this population is 1.5 - 5%, with an associated mortality of 9-50%. In addition, cerebral neoplasia has the second highest incidence of thromboembolism of all malignancies. DVT in neurosurgical patients are more likely to be clinically silent, with only 10-17% producing symptoms. Risk factors for thromboembolism in this population include leg paresis, age, and location/histology of tumor (more likely with suprasellar/supratentorial lesions and meningiomas). However, in one study glioblastoma multiforme was associated with a fivefold risk of DVT. The pathophysiology of malignancy associated thromboembolism is poorly understood and remains an active area of research. Prophylaxis is essential in patients undergoing brain tumor surgery. Historically, clinicians have favored compression devices (SCD) due to concern of bleeding with anticoagulation. However, most studies of low molecular weight heparins (LMWH) have demonstrated reduced risk of thromboembolism (compared to unfractionated heparin) without significant bleeding. LMWH in conjunction with SCD has been shown to be particularly effective. Inferior vena cava (IVC) filters were traditionally the mainstay of therapy for DVT and/or PE in this population. However, they have been associated with high complication rates. In addition, several studies have shown that anticoagulation in patients with CNS malignancy is associated with a fairly low risk of bleeding. Therefore, in patients without absolute contraindications (previous intracranial hemorrhage, profound thrombocytopenia), treatment with LMWH is recommended by many experts. In spite of this, the best strategy remains controversial. The reluctance to treat with anticoagulation was evidenced in one study by a low rate of LMWH usage and high rate of IVC filter insertion. Our case highlights the importance of educating physicians about using LMWH in prophylaxis and treatment of thromboembolic complications in neurosurgical patients with glioblastoma multiforme.

WHEN A GOOD THING GOES BAD. B. Delgado¹, J. Wiese¹. ¹Tulane University, New Orleans, LA. (Tracking ID #135676)

LEARNING OBJECTIVES: 1. Recognize suppurative thrombophlebitis as an iatrogenic complication of vascular catheterization and an etiology of sepsis. 2. Develop an empiric antibiotic approach to the septic patient.

CASE: A 57 year-old woman with diabetes was admitted for elective abdominal surgery for gastroparesis and chronic vomiting. A gastrojejunostomy was performed without complications. On post-operative day four, a right internal jugular catheter was placed for vascular access. By the ninth day of hospitalization, she was ambulating and tolerating a clear liquid diet. The following day, however, she began to develop shortness of breath, tachycardia, and fever. A spiral CT was negative for pulmonary embolism, but it did reveal a vascular occlusion in the right internal jugular vein where the central line had been placed. The right internal jugular vein central line was removed, and a left internal jugular central line was placed. Blood cultures were drawn, and empiric antibiotics were started. Immediately following removal of the central line, she became hypotensive and her dyspnea worsened. A CT scan revealed multiple sub-pleural lung nodules with central cavitation consistent with septic emboli. A transesophageal echocardiogram was negative for vegetations, though blood cultures were positive for methicillin-resistant Staphylococcus aureus. The following day the right neck became suppurative, requiring drainage of an abscess. A presumptive diagnosis of septic thrombophlebitis of the right internal jugular vein was made. She completed six weeks of antibiotics with good recovery.

DISCUSSION: Thrombosis of a central catheter is a common finding in patients with catheter-related sepsis. A biofilm layer consisting of fibrin and microbial glycoproteins is universally present on the surface of indwelling vascular catheters, commonly in association with asymptomatic microbial colonization. This fibrin deposition can extend to the vascular wall forming a mural venous clot. In the presence of venous thrombosis, there is a high association of suppurative thrombophlebitis (Lemierre's Syndrome). Suppurative thrombophlebitis is sug-

gested by persistently positive cultures despite the removal of the catheter. The treatment of choice is surgical excision of the culprit vein with long-term antibiotic therapy, since thrombolysis increases the risk for septic embolization. Internists consulting on patients with this presentation should be aware of the up-front risk of embolization immediately following removal of the catheter. Patients should be closely monitored in the twenty-four hours following the catheter removal. The best approach is prevention, with removal of all central catheters as soon as possible.

INTERNATIONAL TRAVEL AS LIFE CLOSURE TASK AND GOAL OF CARE IN PALLIATIVE MEDICINE. C. Deamant¹; K. Paruchuru¹. ¹Cook County Hospital, Chicago, IL. (Tracking ID #135406)

LEARNING OBJECTIVES: 1. Identify international travel to country of origin as a life closure task and goal of care. 2. Identify factors influencing the decision to return to country of origin. 3. Assess and manage symptoms and feasibility to travel.

CASE: Thirty-two year old Hispanic man recently diagnosed with metastatic gastric cancer was admitted for intermittent bowel obstruction, with uncontrolled nausea, vomiting and pain. Palliative care was consulted to evaluate for home hospice care. Eight months prior to consultation, the patient came to the U.S. to work and provide money to care for his family in Ecuador. Diagnosed with the cancer shortly after arriving, he underwent surgical resection and chemotherapy. In reviewing life closure tasks, the patient identified seeing his two children, 4 & 8 years old, and his parents prior to his death. His wife had died from hemorrhage during the birth of his youngest child. He identified lack of health care as a reason for her death and feared he would suffer if he returned. His only support in the U.S. was his sister, who was unable to provide 24-hour care for him at home. His care would be provided at the public hospital's long term care facility. His sister did not have the funds to return his body to Ecuador if he died in the U.S. He chose to return to Ecuador to spend his remaining weeks of life with family. He was able to travel safely once his symptoms of bowel obstruction were controlled with antiemetics, anticholinergic agents and opioids. A supply of medications for symptom control was provided to him for use in Ecuador.

DISCUSSION: Palliation and hospice care are integral aspects of a dying patient's needs. Patients often wish to die at home and it is important to recognize that home may be across international borders. Our palliative care program is often consulted to establish goals of care and evaluate for supportive services. As one of the largest public hospital systems in the U.S., a significant percentage of the patients seeking care at our hospital include immigrants, both documented and undocumented. Patients who lack insurance are often diagnosed with diseases already in the advanced stages. In reviewing goals of care, life closure tasks, and prognosis, many of them express a desire to return to their country of origin to be reunited with family. In addition, dying in the U.S. and then transferring the body can be prohibitively expensive. Burial in one's own country is often a high priority. Cremation may not be culturally acceptable as a less expensive alternative for returning a body's remains. It is also important to assess the adequacy of usual medical care in the country of origin. For some patients, remaining in the U.S. to ensure their death is comfortable is more important than returning. For those who choose to return, it is important to provide an adequate supply of medications to control symptoms and ensure comfort. Determining the medical clearance requirements for the designated airline and assessing stability of condition for travel is essential. Early in the care of patients with life limiting illness, especially if treatment is palliative, it is critical to identify if return to country of origin prior to death is an eventual goal.

RETURN OF THE VIVAX. S. Oh¹; G. Orshansky¹. ¹UCLA-WLA VA Internal Medicine Residency Program, Los Angeles, CA. (Tracking ID #134521)

LEARNING OBJECTIVES: 1. Review the diagnosis and treatment of malaria. 2. Recognize malaria as a frequent infection in travelers, including military personnel, traveling to endemic areas. 3. Recognize etiologies of treatment failure. **CASE:** Mr. C is a 21 year old Caucasian Army Ranger who presented with shortness of breath and was found to have a pneumothorax. A chest tube was placed and the patient was admitted for further management. He recently returned from a military tour in Afghanistan and Iraq where he was hospitalized for malarial sepsis with acute respiratory distress syndrome complicated by nosocomial pneumonia and adrenal insufficiency. He had been discharged about one month earlier and was undergoing cardiopulmonary rehabilitation until this admission. During his first night in the hospital the patient developed high fevers to 103°F with marked rigors, nausea and non-bloody vomitus. He appeared to be in moderate distress with vital signs significant for fever, heart rate of 133, blood pressure of 97/52 and an oxygen saturation of 95% breathing ambient air. Chest examination revealed decreased sounds on the left with the chest tube site having no evidence of infection. Cardiac examination was significant for a regular tachycardia without significant murmurs or jugular venous distension. Abdomen was diffusely tender but without peritoneal signs and bowel sounds were present. Portable chest radiograph revealed no significant changes with a moderate left pneumothorax without infiltrates. Electrocardiogram revealed sinus tachycardia. Laboratory values were significant for hemoglobin of 10.3 and total bilirubin of 2.3 of which 0.3 was direct. Coombs test was negative. Thick smear obtained during this febrile episode revealed a relapse of plasmodium vivax despite prior treatment.

DISCUSSION: Malaria continues to be a major cause of disease in military personnel and other travelers visiting endemic areas. 205 cases of malaria were reported in 2003 by the armed forces RMES (Reportable Medical Events System)

and in 2002 the NMSS (National Malaria Surveillance System) reported 910 cases in US travelers. Of concern is the rising incidence of malaria attributed to parasite resistance to antimalarials as well as resistance of the anopholes mosquito to insecticides combined with increased global travel. Diagnosis requires a high clinical suspicion to obtain thick and thin smears to identify and quantify parasitemia even with appropriate prophylaxis or prior treatment in travelers to endemic areas presenting with fever. Evidence of hemolysis is often reflected in more routine labs as was the case with this patient. Treatment typically entails oral chloroquine or intravenous quinidine for severe disease with an additional 2 weeks of primaquine to eradicate hepatic hypnozoites with plasmodium vivax or ovale infections. G6PD deficiency should be evaluated prior to therapy when possible to avoid drug precipitated hemolysis. The most common errors in management are due to a delay in diagnosis, not treating for hepatic hypnozoites and prophylaxis failure. Some consider infection with plasmodium falciparum an indication for inpatient management given the potential for decline from symptoms to death within 72 hours. In summary a high clinical suspicion, identification of species and evaluation for resistance are essential for the proper management of malaria.

CULTURE IS A USEFUL CONCEPTUAL TOOL IN CLINICAL ETHICS CONSULTATIONS. H.S. Perkins¹; H.P. Hazuda¹. ¹University of Texas Health Science Center at San Antonio, San Antonio, TX. (Tracking ID #132970)

LEARNING OBJECTIVES: Attendees will learn to a. Define culture and b. Contrast the cultures of physician and patient.

CASE: An elderly African-American woman with emphysema and lung cancer was hospitalized for new dyspnea and confusion. Convinced she was dying, her Euroamerican physicians recommended comfort care only. The family demanded full life support. Extensive discussions failed to resolve the impasse. As the patient slid toward respiratory failure, the physicians requested an ethics consultation from me. I suspected that unidentified value conflicts blocked effective communication and decision-making. A likely source for such conflicts was culture—the shared values, beliefs, and behaviors a group uses to interpret life events. Culture surely frames patients' understanding of illness. Because much literature suggests that African Americans distrust medical care by Euroamericans, I initially attributed the family's demands to that distrust. But interviews failed to support this hypothesis. The family expressed only great confidence in the physicians. Abandoning that hypothesis, I tried another based on professional, not ethnic, culture. Each profession, including medicine, has a culture different from its clients' culture. For example, the physicians here saw science alone as undergirding medical care, focused on diagnosing and treating the patient's disease, made decisions based on "facts," and felt that physicians controlled the outcome. In contrast, the family saw religious faith as also undergirding medical care, focused on improving the patient's well-being, made decisions based on the patient's strong will to live, and felt that God alone controlled the outcome. This new hypothesis guided my consultative approach. Defining the value conflicts enabled me to discuss the case with each side on their terms and to explain the other side's view. Defining the value conflicts also enabled me to avoid two management extremes—all comfort care or all curative care. Survival data from many similar patients convinced me that cardiopulmonary resuscitation (CPR) would not save this patient in a cardiac arrest. So I recommended the physicians withhold CPR. I suggested that, when communicating this decision, the physicians explain to the family that God alone would have to determine the outcome from a cardiac arrest. Further, because the patient had always wanted to live and never refused treatments, I recommended that the physicians allow the family great discretion to request aggressive treatments for her. But I also urged the physicians to attend to comfort measures as possible. The physicians eventually intubated and ventilated the patient, diagnosed and treated her pneumonia, and discharged her home at baseline function. She lived 18 more months.

DISCUSSION: We all belong to multiple cultures. Thus, resolving some clinical ethics issues hinges on identifying the key cultures in conflict. Those cultures in this case were not Euroamerican and African-American, but physician and patient. Disagreements with patients or families should prompt physicians to explore cultural differences based on distinctions such as profession as well as ethnic group.

HONORING PATIENT AUTONOMY: THE REFUSAL OF CARE ON RELIGIOUS GROUNDS. D.M. Swiderski¹. ¹Montefiore Medical Center, Yonkers, NY. (Tracking ID #135708)

LEARNING OBJECTIVES: 1. Recognize the difficulties in practicing appropriate medical care when patient values conflict with accepted medical practice. 2. Assess the limits of patient autonomy.

CASE: Ms. M is a 54 year old woman who presents with a left breast mass. She states that she first noticed it four months ago and she believes it is getting larger. On examination she has a hard, 1 centimeter nodule in the left breast. Urgent mammography and fine needle aspiration are arranged and the nodule proves to be malignant. The patient is referred to Surgery. She does not return to the internist for a visit scheduled post-operatively. One month later the surgeon notifies the internist that the final pathology report on the patient's lumpectomy specimen showed tumor cells in the margin. The patient was scheduled for definitive surgery but did not appear on the day of the procedure and could not be reached. The internist reaches the patient who states that she understands her condition but believes that she does not need surgery. She states that she is a devout Christian and she is certain that God will cure her. After several telephone calls she agrees to come to the office to discuss her situation in person. She is well groomed, fully oriented, and has understanding and insight about

the choice she is making. She declines the doctor's request to contact members of her family or her church. After this visit the patient stops returning the doctor's calls. Putting herself in the place of a possible family member, the doctor worries that perhaps the patient's judgment was inappropriately affected by her religious preoccupations and the stress of a cancer diagnosis. She remains concerned that she should have notified family members or friends over the patient's objections to give them the opportunity to be involved in the patient's decision.

DISCUSSION: Patient autonomy is generally honored in capacitated patients even when it conflicts with other ethical standards such as beneficence. However, when a patient is refusing potentially curative care for a condition that may be life threatening, a full appreciation of the patient in her life context is imperative. Family structure and dynamics undoubtedly play a role in this patient's decision, as does her involvement with her religious community. The needs and values of family members should be understood by the physician if at all possible, as well as an understanding of how those close to the patient might affect her decision. This level of information would allow the doctor, and patient, to have a greater comfort level with a difficult decision. This case also highlights the potential difficulties for practitioners in honoring patients' religious beliefs in such a clinical setting.

TEACHING PROFESSIONALISM IN AN UNPROFESSIONAL ENVIRONMENT. R.C. Jones¹; N.A. Desbiens²; H. Gadadhar³. ¹University of Tennessee College of Medicine, Chattanooga Unit, Chattanooga, TN; ²University of Tennessee, Chattanooga, TN; ³University of Tennessee College of Medicine, Chattanooga, TN. (Tracking ID #136116)

LEARNING OBJECTIVES: 1. Recognize unprofessional behavior in academic medical centers 2. Understand the need to discuss unprofessional behavior with trainees

CASE: A 21 year old non-English speaking, indigent, Mexican illegal alien presented to a rural emergency room with a two hour history of palpitations and chest pressure. He gave a history of 10 to 15 similar episodes since 1998, at least one of which was associated with syncope. His EKG was read as showing atrial fibrillation with aberrancy at a ventricular rate of at least 200 beats per minute. He was treated with Cardizem, Digoxin, Amiodarone, and Adenosine. He was then defibrillated with 100 joules but his rhythm deteriorated to ventricular fibrillation. He was then successfully cardioverted with 360 joules, transferred to our hospital, and admitted to an internal medicine teaching service. Cardiology consultant diagnosed Wolf-Parkinson-White Syndrome and referred him to an electrophysiologist who declined to schedule an EP study stating that the hospital had a policy prohibiting the scheduling of invasive cardiac studies for patients lacking social security numbers. The attending contacted the hospital's chief medical officer who indicated that there was no such policy. Seven days after admission the patient underwent unsuccessful radiofrequency ablation of an accessory pathway. After the procedure, the electrophysiologist did not return to see the patient in the hospital. The attending reconsulted the initial cardiologist who stated that the patient did not need another procedure but could be safely discharged on antiarrhythmic therapy. The attending consulted an electrophysiologist from another group who agreed that a repeated attempt to ablate the accessory pathway was indicated. Eight days later, a left ventricular accessory pathway was successfully ablated utilizing a transseptal approach. The patient was discharged the next day on 325 mg of aspirin daily.

DISCUSSION: When a presumed lack of payment appears to affect professional judgment, teaching physicians must be extra vigilant to ensure that appropriate patient care occurs. In addition, teaching physicians have an added obligation to identify professional breaches and to discuss them with trainees so that unprofessional behavior is not incorporated into trainees' practice patterns.

WHEN DOCTORS CAN'T AGREE - DID WE DO THE RIGHT THING? L. Selassie¹; D. Shroff¹; E. Ortiz². ¹Washington DC Veterans Affairs Medical Center, Washington, DC. (Tracking ID #135043)

LEARNING OBJECTIVES: 1) Understand how to assess and deal with conflicts arising from a patient's right to self-determination and the provision of medically indicated care vs. physician autonomy and the allocation of costly health care resources.

CASE: 33 y.o. female with h/o IVDA and PTSD presented to the VA hospital on multiple occasions with fevers, heart murmur and bacteremia. She was initially admitted and treated for infective endocarditis in July 2004 and later readmitted with similar symptoms in August & November, which required treatment with 6 weeks of IV antibiotics for multiple bacterial and fungal organisms. Despite therapy, an echocardiogram revealed new mild to moderate tricuspid regurgitation and an enlarging valve vegetation. She developed osteomyelitis, septic pulmonary emboli, and pulmonary artery thrombosis from embolic complications. Throughout her stays, she was verbally abusive and uncooperative, and we suspected she was still using injection drugs based on multiple extended disappearances from the medical ward. During her previous two admissions, she left the hospital before completing treatment, and she admitted to using her central line for injecting drugs while she was out of the hospital. Despite surgical indications for valve replacement, surgery refused to perform the procedure, citing her noncompliance with treatment, her hospital disappearances and continued IVDA, and the likelihood of a poor surgical outcome. The medicine service continued antibiotic treatment for 6 weeks, while attempting to enroll her in a drug rehab program.

DISCUSSION: This case illustrates the problems that arise when a patient's non-compliant, self-destructive behavior leads providers to believe that a medically indicated treatment should be withheld. A moral dilemma thus ensues,

where patient self-determination and societal beliefs that Americans are entitled to appropriate medical care clashes with physician autonomy and the belief that costly resources should be used judiciously. Managing such patients can be clinically difficult and ethically challenging, as internal conflicts arise between our attempts to provide high-quality, compassionate care and our frustration with patients who repeatedly sabotage their own care. Many of these patients also have substance abuse or mental health problems, making it difficult to blame them entirely for their behavior. Disagreements also arose between the surgical and medical services, with surgery refusing to do the procedure, while some (but not all) medicine physicians believing that surgery should be performed. We attempted to set limits on this patient's privileges by establishing a verbal contract, which prohibited her from leaving the hospital if not accompanied by staff and subjected her to random drug testing. Other mechanisms for potential resolution include involvement of a multidisciplinary ethics committee, evaluation by surgeons from an outside facility, and agreeing that if the valve was replaced and subsequently infected secondary to IVDA, she would not be eligible through the VA system for further valve replacement.

A DEATH SENTENCE: THE ULTIMATE MEDICAL ERROR. A.L. Misakian¹; A.D. Auerbach¹. ¹University of California, San Francisco, San Francisco, CA. (Tracking ID #133017)

LEARNING OBJECTIVES: (1) Raise awareness of the types and consequences of communication breakdowns between providers in different healthcare settings. (2) Review the existing literature on ways to improve interprovider communication and reduce adverse events.

CASE: A 77 year-old male with a history of localized prostate cancer was admitted to our medicine service for severe fatigue and dark urine. After an extensive work-up (including CT scan of the chest/abdomen/pelvis and bone marrow biopsy with flow cytometry), he was thought to have an idiopathic autoimmune hemolytic anemia. His anemia was responsive to steroids and he was discharged with instructions to complete a steroid taper. One week later, the patient's sputum cultures grew a mycobacterium species, whereupon an attempt was made to contact the patient. We learned that the patient had been admitted to another hospital with a similar presentation. At the outside hospital, the patient was diagnosed with metastatic prostate cancer and was transitioned to comfort/hospice care. No attempt was made to contact our facility. It is unclear whether the treating physician was aware of the patient's recent hospitalization; the patient was a poor historian, was altered at the time of admission, and likely could not provide details about his medical history. After contacting the treating physician, steroid therapy was initiated and the patient was transferred back to our medical center. Following multiple transfusions and steroid therapy, the patient recovered and no longer required hospice services. His prostate specific antigen level was undetectable.

DISCUSSION: The transition of care between providers (e.g. during hospital discharge) is a time when patients are particularly vulnerable to medical error and adverse events. This issue becomes more salient for complex patients such as the one whose case we present here. At the core of this problem are gaps in the transfer of information between providers. There is a growing literature that suggests that improving communication may favorably affect outcomes such as re-hospitalization rates, adverse drug events, patient satisfaction, and possibly cost. While many mechanisms have been proposed to address communication breakdowns between providers (e.g. better education of patients and their families, timely discharge summaries, comprehensive discharge planning, case managers, involvement of the primary care physician via the "continuity of care" visit, and electronic medical records), how to implement these strategies and which are more effective in which setting remain a subject of debate. This case illustrates a potentially fatal communication gap at a critical junction of our patient's care. Adequate communication between providers is a complex issue that requires heightened awareness and further study.

A SWAN SONG FOR THE PULMONARY ARTERY CATHETER. D.S. Kazi¹; L. Lu¹. ¹Baylor College of Medicine, Houston, TX. (Tracking ID #134125)

LEARNING OBJECTIVES: 1. To recognize Swan-Ganz catheter-induced pulmonary artery rupture as a rare, potentially fatal, yet treatable cause of massive hemoptysis in the critically ill patient. 2. To recognize risk-factors associated with increased risk of iatrogenic pulmonary artery rupture. 3. To learn various treatment modalities available for the management of pulmonary artery pseudoaneurysms.

CASE: A 60 year-old African American female with a history of severe dilated cardiomyopathy, AICD placement, paroxysmal atrial fibrillation and prosthetic mitral valve replacement on anticoagulation with warfarin underwent an elective right-heart catheterization to determine eligibility for participation in a novel research protocol. However, her cardiac hemodynamic parameters did not qualify for the clinical trial, and a pulmonary artery (PA) catheter was placed to evaluate inotropic response to intravenous milrinone. The following morning, the patient reported a small amount of hemoptysis and increasing shortness of breath. The hemoptysis initially ceased spontaneously, but returned within a couple of hours, at which time copious hemoptysis was noted and the patient's clinical status deteriorated rapidly. A significant drop in her oxygen saturation and somnolence aroused serious concern about her ability to protect her airway, and an endotracheal tube was placed. The PA catheter was promptly removed. A repeat chest xray was unchanged from the one obtained at admission. Bronchoscopy through the endotracheal tube demonstrated a significant amount of blood lining the tracheobronchial tree but no obvious site of ongoing bleeding. After receiving three units of packed red cells, her condition stabilized clinically over the next twenty-four hours and she was extubated uneventfully.

She underwent pulmonary angiography to determine the source of the massive hemoptysis, which revealed a pseudoaneurysm of the right pulmonary artery at a location that corresponded to the position of the tip of the PA catheter that had been removed the previous day. The pseudoaneurysm was successfully treated with transcatheter coil-embolization with the assistance of digital subtraction angiography.

DISCUSSION: The pulmonary artery catheter frequently provides invaluable hemodynamic information in the critically ill patient. However, it can cause a number of complications and there is no clinical evidence suggesting improved clinical outcomes with its use. Pulmonary artery rupture occurs in 0.03% of all catheter placements, usually resulting in massive hemoptysis. Anticoagulation, age greater than 60 years, catheter manipulation and female gender are associated with a significantly increased risk. Diagnostic modalities include contrast-enhanced CT scanning and pulmonary angiography. Of note, an imaging study must be performed if the diagnosis is suspected even if the bleeding has stopped, as asymptomatic pseudoaneurysms have a 30-40% risk of delayed bleeding and an attendant 40-70% mortality if untreated. With the advent of percutaneous transcatheter embolization of the pseudoaneurysm or its feeding vessel, surgical options such as lobectomy, blood vessel ligation and repair are infrequently used. Despite these technological advances, pulmonary artery rupture is associated with considerable mortality due to aspiration and asphyxiation from the intrapulmonary hemorrhage, thus mandating early recognition and a prompt response to this rare but potentially lethal complication.

CHOOSE THE RIGHT STATIN! M. Eapen¹; P. Matmani¹; A. Rajamanickam¹; H. Friedman¹. ¹St. Francis Hospital, Evanston, IL. (Tracking ID #134563)

LEARNING OBJECTIVES: 1. Recognize rhabdomyolysis as an adverse reaction to concurrent use of simvastatin and cyclosporine 2. Learn that pravastatin is the statin of choice for concurrent use with cyclosporine 3. Emphasize the importance of a thorough review of the medication list

CASE: A 44 year old African American female was admitted to the hospital with vomiting, fatigue and weakness of her legs. She had no history of fever, trauma or prolonged exertion. She had a history of focal segmental glomerulosclerosis and Evans Syndrome and was taking prednisone, prnivil, furosemide, valsartan and cyclosporine 100 mg orally once daily. Her physical examination was significant for bilateral lower extremity weakness. On admission, laboratory data revealed BUN 12 mg/dL, Creatinine 2.5 mg/dL (five months ago, 1.1 mg/dL), AST 221 U/L, ALT 207 U/L and Creatinine Phosphokinase (CPK) of 21,476 IU/L. Urinalysis revealed positive urine myoglobin. Ultrasound of the kidneys revealed no obstruction. Her TSH was normal. The lisinopril, furosemide, valsartan and cyclosporine were held and the patient received intravenous fluids. She had an electromyogram which was suggestive of a metabolic myopathy. Four days later, the patient on repeated review of her medication list, said that she had been seeing multiple physicians and she was started on 40 mg of simvastatin by one of them. Over the course of a week, her symptoms resolved and her CPK, transaminases and creatinine normalized. She was discharged home in stable condition.

DISCUSSION: Myopathic syndromes associated with statin therapy, range from myalgias to overt rhabdomyolysis. The increase in susceptibility to myopathy is substantially greater in patients receiving concurrent therapy with a number of drugs, particularly those that inhibit CYP3A4. These include cyclosporine, gemfibrozil, macrolide antibiotics, itraconazole and HIV protease inhibitors. Inhibition of the CYP3A4 pathway leads to increased levels of statin metabolites, which increases the risk of rhabdomyolysis. Pravastatin is the only statin approved by the Food and Drug Administration for patients on cyclosporine. The risk of rhabdomyolysis is dose dependent. This case report also highlights the importance of a thorough review of the medication list so as to avoid expensive tests to evaluate the etiology of rhabdomyolysis. As this clinical vignette suggests, patients who change physicians frequently may be at a much higher risk for adverse drug interactions.

DRUG ELUTING CORONARY STENT ALLERGY: A NEW AND UNIQUE PATIENT SAFETY ISSUE. E. Martorell¹; J. Guarderas¹; J. Trejo¹; K. Calamia¹. ¹Mayo Clinic, Jacksonville, FL. (Tracking ID #132539)

LEARNING OBJECTIVES: 1. Recognize sirolimus eluting coronary stent allergy as a new and unique patient safety issue. 2. Learn the pharmacokinetics of the sirolimus eluting coronary stent 3. Manage the symptoms of patients with possible drug eluting stent allergy

CASE: A 56-year-old woman underwent placement of a sirolimus-eluting Cypher stent for treatment of unstable angina. The next day, she developed urticaria and pruritus around the right groin. She was started on a histamine antagonist and discharged. The urticaria and pruritus progressed and her symptoms persisted despite treatment with histamine antagonists and the discontinuation of all newly prescribed medications. Her physical examination only revealed diffuse urticarial rash and a left wrist synovitis. Laboratory results indicated an increased C-reactive protein concentration of 46 mg/L (0.2 - 8 mg/L) and an increased eosinophil count of 880 cells/mL (0.05 - 0.50 cells/mL). Complements C3 and C4, creatinine, and hemoglobin levels, leukocytes, platelets, and sedimentation rate were within normal limits. The patient was started on prednisone 20 mg twice daily with a rapid taper down to 5 mg daily and continued until symptoms and eosinophilia resolved about 3 1/2 months later.

DISCUSSION: Sirolimus-eluting Cypher stents were approved in April 2003 for the prevention of coronary artery restenosis. Hypersensitivity reactions have been associated with these stents; about 70 cases had been reported to the US Food and Drug Administration (FDA) as of November 2003. In most cases the

reactions were minor, but some rare severe reactions occurred, including deaths from anaphylaxis. Pharmacokinetic studies show that approximately 50% of the total drug (sirolimus) is eliminated from the stent within the first 10 days of implantation. About 90% of sirolimus is eliminated from the stent after 60 days, and the drug is completely dissipated by about 90 days after implantation. In the case of this patient, urticarial symptoms appeared shortly after stent placement and persisted for approximately 3 1/2 months despite treatment and the discontinuation of medications, which suggests the possibility of an allergic reaction to the drug eluted from the stent. This finding contrasts with the FDA statement that most of the reported reactions to sirolimus-eluting stents were related to periprocedural medications. The increasing frequency of both placement of drug-eluting coronary stents and the allergic reactions associated with them, including anaphylaxis, creates a new and unique patient-safety issue because removing the stent is not feasible. In the absence of anaphylaxis, patients with a suspected drug-eluting stent allergic reaction may be discharged safely with medications for symptomatic relief and close outpatient observation.

FATAL COLCHICINE OVERDOSE. F. Quratul Ain¹; A. Radhakrishnan¹; S. Parikh¹; H. Friedman¹. ¹Saint Francis Hospital, Evanston, IL. (Tracking ID #134633)

LEARNING OBJECTIVES: 1.To recognise the potential dangers of colchicine overdose. 2.To emphasize the need for patient education regarding side effects and maximum dose of any drug being newly prescribed.

CASE: A 30-year-old male presented to his primary care physician with the chief complaint of a painful swollen red right big toe. He was diagnosed with gout, and was given a prescription for sixty tablets of colchicine 0.6 mg. He was instructed to take 1-2 tabs every hour till the pain subsided or he developed diarrhea. The patient started to take 2 tabs every hour for the next 16 hours after which he developed abdominal discomfort and diarrhoea. He then continued to take 2 tabs every 2 hours for the next 14 hours until he developed fever, worsening of abdominal pain and diarrhea. The patient was brought to the emergency department for evaluation. His temperature was 102.5 F, pulse was 129/min and blood pressure was 105/84 mmHg. He was in severe distress due to abdominal pain. Physical exam was essentially unremarkable except for mild abdominal distention. His total WBC count was 22,000/cmm with 13% bands, urinalysis was normal, total protein was 8.7 gm/dL, total bilirubin was 0.7 mg/dL, alkaline phosphatase was 225 IU/dL, AST was 147 IU/L, ALT was 82 IU/L, PT was 16.6 seconds and PTT was 36.9 seconds. EKG showed sinus tachycardia. KUB revealed a few dilated loops of small bowel. Our patient was admitted to the ICU, was rehydrated with intravenous fluids and supportive measures were instituted. During the next six hours, his condition rapidly deteriorated and he went into asystolic cardiac arrest. Despite aggressive resuscitative measures, the patient died approximately 42 hours after initiation of colchicine therapy.

DISCUSSION: Colchicine overdose is potentially life threatening if ingested in doses that exceed the recommendations. Colchicine is almost certain to cause death if the amount ingested is more than 0.8 mg/kg body weight. Our patient ingested 27 mg (0.5 mg/kg body wt) of colchicine over a period of 30 hours. The clinical manifestations include nausea, vomiting, diarrhea, abdominal pain, anorexia, bone marrow hypoplasia, cardiac arrhythmias, cardiovascular collapse, respiratory distress, oliguric renal failure, electrolyte abnormalities, seizures and ascending paralysis. Management includes gastric lavage, supportive care and cardiac monitoring. The large volume of distribution of colchicine and the fact that 50% of its plasma concentration is linked to proteins make hemodialysis ineffective in clearing the drug. This case highlights the importance of patient education and proper physician patient communication as to the possible side effects of any drug which is newly being prescribed and to advise patients on the maximum possible amount that they may consume over a period of time. One important way to limit the number of pills that patients may consume is to prescribe only the maximum number of pills that the patient can safely take. This will also keep a check on the possibility of accidental overdose especially for drugs with such narrow therapeutic index like colchicine.

GATIFLOXACIN INDUCED HYPOGLYCEMIA - CHARACTERIZING THE AT-RISK PATIENT. P.W. Helgeson¹. ¹Stanford University, Palo Alto, CA. (Tracking ID #135489)

LEARNING OBJECTIVES: 1) Recognize gatifloxacin as a potential cause of hypoglycemia. 2) Identify risk factors for this potentially severe drug reaction.

CASE: An 85 year-old man was evaluated for a two day history of dysuria and hematuria without fever or flank pain. Past medical history was significant for longstanding type 2 diabetes mellitus and chronic renal insufficiency with a baseline creatinine of 4.0. Medications included furosemide, clonidine, simvastatin, amlodipine, and erythropoietin. Glipizide 2.5 mg daily had been discontinued two days prior to presentation due to concerns over poor renal clearance. Vital signs were normal. Urinalysis revealed full fields of red and white blood cells. The patient was diagnosed with a UTI and prescribed a seven day course of gatifloxacin 400 mg daily. Two days later, the patient contacted his primary care physician noting episodes of hypoglycemia to approximately 50 mg/dL each evening after starting gatifloxacin. These were symptomatic with diaphoresis and mild confusion. The episodes resolved with self administration of orange juice. Blood sugars had ranged from 100-140 prior to starting the medication, and the patient had not had symptomatic hypoglycemia in years. Gatifloxacin was discontinued, and cephalexin 500 mg BID prescribed as urine culture had subsequently revealed a pansensitive *Proteus* species. The patient experienced no further hypoglycemic episodes. The Naranjo probability scale indicates a "possible" drug-related event.

DISCUSSION: To date, at least nine previous cases of hypoglycemia thought related to gatifloxacin have been described in the literature. As in this case,

gatifloxacin cannot be proven to be the cause of the hypoglycemia; each event rates "possible" to "probable" on the Naranjo scale. Causality is difficult to establish when the adverse event, hypoglycemia, can also be a consequence of the infection that gatifloxacin is used to treat. Worsening renal failure or poor PO intake during acute illness may also act as confounders. This problem will be overcome only with analysis of larger data sets generated by more widespread reporting of such events. In all cases examined, the patient has been elderly (age range 71-94) and had type 2 diabetes mellitus. Further, each has been on a sulfonylurea up to 1-2 days prior to the administration of the gatifloxacin. Five of the seven patients for whom renal function was reported had significant renal insufficiency. In most, the dose of gatifloxacin was not reduced. The mechanism through which hypoglycemia is induced remains incompletely understood. Increased insulin secretion by inhibition of pancreatic beta-cell K (ATP) channels by gatifloxacin has been shown in one study. This may explain the synergy with a sulfonylurea in bringing about clinically significant hypoglycemia. Several clear lessons emerge: clinicians must exercise extreme caution in prescribing gatifloxacin to patients who are elderly, diabetic, and on sulfonylureas. Additionally, as in all cases of prescribing to the geriatric patient, renal function must be carefully assessed and doses adjusted accordingly.

LEAPING WITHOUT LOOKING. D.S. Gloss¹; J. Willis¹; J. Wiese¹. ¹Tulane University, New Orleans, LA. (Tracking ID #135775)

LEARNING OBJECTIVES: 1. Understand the importance of critically evaluating clinical trials before changing clinical practice. 2. Recognize the potential consequences of consulting services instituting therapy without involvement of the primary team.

CASE: A 76 year-old man with severe aortic stenosis presented with one week of dyspnea. The dyspnea was exacerbated with exertion and associated with orthopnea and a dull, interscapular pain. He had no fever or cough, and he was compliant with his medications and dietary advice. His JVP was 9 cm and he had bibasilar crackles. There was no S3 or S4; the PMI was laterally displaced and he had a late peaking systolic murmur at the base. His troponin was 1.4 and creatinine of 2.3. He had LVH on EKG, but no acute ST/T wave changes. Operating on the results of a recent New England Journal of Medicine article, the cardiology consultant instituted a nitroprusside drip. He immediately became dyspneic and diaphoretic. The nitroprusside was stopped and his symptoms abated. Five hours later, he became unresponsive, slipping into PEA. An emergent 2-D echo revealed a pericardial effusion. He was resuscitated by pericardiocentesis. Twenty-five minutes later, he went back into PEA and was not successfully resuscitated.

DISCUSSION: Nitroprusside has been shown to improve symptoms in patients with aortic stenosis. (NEJM 348; 18, 1756-1763). However, the study that changed management in this case was comprised of only twenty-five patients, commanding only a level-four degree of confidence (analogous to a case series). Further, the patients in this trial did not approximate our patient's clinical presentation, particularly with respect to his inter-scapular pain. Our patient was presenting with an acute aortic dissection, and while this was being considered by the primary team, it was not considered by the cardiology consultative service. The PEA induced by pericardial tamponade suggests a type A retrograde dissection into the pericardial sack. The administration of nitroprusside lowered system vascular resistance and lowered preload to the left ventricle. With the aortic stenosis limiting cardiac output, the heart rate suddenly increased, increasing the shear force on the aortic dissection, thereby extending the dissection. This case emphasizes the importance of critically evaluating clinical trials before implementing them into practice. Further, the patient's clinical presentation must always be compared to that of those in the trial to insure that the study results are generalizable to a particular patient. This case also illustrates the importance of communication between the primary and consultative teams.

NOT JUST A DRUG RASH: ALLOPURINOL HYPERSENSITIVITY SYNDROME. K.J. Kim¹. ¹UCSF Division of General Internal Medicine, San Francisco, CA. (Tracking ID #135533)

LEARNING OBJECTIVES: 1) Recognize a potentially fatal reaction to allopurinol. 2) Describe the main components of the allopurinol hypersensitivity syndrome. 3) Recognize the importance of allopurinol dose adjustment in patients with renal insufficiency.

CASE: An 85 year-old Cambodian male with a history of congestive heart failure, chronic renal insufficiency, and COPD, presented to his community hospital with fever, and diffusely erythematous and flaking skin. The patient reported pruritus, sore throat, and dysphagia. The daughter noted that the patient recently took some allopurinol, a medication he does not take on a regular basis. The patient had a history of two similar episodes of erythroderma in response to amoxicillin and shellfish for which he was hospitalized. Previous skin biopsies showed interface dermatitis consistent with generalized fixed drug reaction. Physical exam was notable for mild tongue swelling, wheezing without stridor, and erythema of his entire skin surface with small areas of desquamation on the extremities. Laboratory studies showed acute renal failure (serum creatinine 3.8 mg/dL, baseline 2.5 mg/dL) and new transaminitis (AST 484 U/L, ALT 551 U/L). WBC count and differential were normal, and blood cultures and hepatitis serologies were negative. The dermatology consult service felt the patient's fever, rash, renal failure and hepatitis were consistent with the allopurinol hypersensitivity syndrome. The patient was started on oral prednisone and topical triamcinolone to which he had a good clinical response. Over the next week the patient remained afebrile, erythroderma improved, creatinine dropped to 2.3 mg/dL, and transaminases steadily decreased (AST 41 U/L, ALT 182 U/L).

DISCUSSION: Allopurinol, a xanthine oxidase inhibitor, is commonly used for the treatment of gout. Approximately 1 in 260 patients may develop a severe or fatal adverse reaction to allopurinol, called the allopurinol hypersensitivity syndrome (AHS). The diagnostic criteria for AHS are: 1) exposure to allopurinol; 2) lack of exposure to another drug which may cause a similar clinical picture; 3) (a) at least two of the following major criteria: worsening renal function, acute hepatocellular injury, rash (ie. diffuse maculopapular or exfoliative dermatitis, erythema multiforme, or toxic epidermal necrolysis (TEN), or (b) one major and at least one of the following minor criteria: fever, eosinophilia, leukocytosis. AHS can develop within days to weeks of exposure to allopurinol, occurring more frequently in patients with pre-existing renal insufficiency and those on thiazide diuretics. Mortality has been reported to be as high as 25%, often related to TEN or sepsis. AHS is related to an excess of the active allopurinol metabolite, oxypurinol, which is renally excreted. The development of AHS is likely autoimmune, but the exact mechanism of hypersensitivity is unknown. In case reports, prolonged courses of steroids have resulted in significant clinical improvement, however, the role of steroids in treatment is not proven. Studies have found that many patients who developed AHS were prescribed allopurinol for asymptomatic hyperuricemia. Given the potential lethality of AHS, it is important to avoid allopurinol in patients without symptomatic gout, and to adjust dosing in patients with renal insufficiency.

WHOSE LINE IS IT ANYWAY? H. Dehghani¹; J. Aliota¹; J. Wiese¹. ¹Tulane University, New Orleans, LA. (Tracking ID #135656)

LEARNING OBJECTIVES: 1. Recognize the importance of prompt removal of indwelling catheters and hand-washing precautions in preventing nosocomial infections. 2. Understand the importance of being aware of a patient's roommate's medical condition when considering occult infections.

CASE: A 77 year-old man with steroid-dependent obstructive lung disease was transferred out of the ICU with a resolving community-acquired pneumonia. He was transferred to the floor and improved for the next six days. He was ready for discharge when his platelet count dropped to 90,000. The following day his mental status deteriorated and his platelets fell to 50,000. He remained afebrile and had no change in his physical exam. His low-molecular weight heparin was stopped due to a concern for heparin-induced thrombocytopenia; all medications were discontinued for fear of drug-induced thrombocytopenia. Blood cultures were positive with gram-positive cocci, that were initially thought to be a contaminate. This prompted a more thorough examination for possible portals of infection. A central venous catheter was found covered in the folds of his groin; it was pulled and cultured. The following day, his hospital roommate also experienced a decline in his platelet count and altered mental status. His cultures were positive for vancomycin-resistant enterococcus. Subsequent cultures from our 77 year-old patient's catheter tip and blood revealed the same organism. D-dimer and fibrinogen were elevated in both patients, suggesting a shared diagnosis of DIC due to vancomycin-resistant enterococcus. Both patients were immediately placed in contact isolation; appropriate antibiotics were initiated. Both patients improved, with normalization of their platelet counts.

DISCUSSION: Hospital-acquired infections are the most common complications affecting hospitalized patients, with catheter-related infections being the most frequent source. In one study, sixty percent of physicians were unaware that their patient has an indwelling catheter. The twenty-percent compliance of physicians with hand-washing guidelines may explain the high rate of cross-infection of patients. Our patients illustrate the importance of expeditious removal of central catheters, and the importance of hand washing in caring for patients who share a hospital room. Vancomycin-resistant enterococcus is a growing concern among hospitalized patients and is a highly transmissible pathogen. Where occult infections are suspected, physicians should be aware of the medical condition of other patients that occupy the same hospital room.

"DOC, I HAVE PAIN ALL OVER!" A CASE OF VITAMIN D DEFICIENCY. M. Kai¹; C. Risinger¹. ¹Providence Portland Medical Center, Portland, OR. (Tracking ID #133525)

LEARNING OBJECTIVES: 1. Consider vitamin D deficiency in any patient with persistent, nonspecific musculoskeletal pain. 2. Recognize the clinical manifestations of the osteomalacia syndrome. 3. Learn to manage the medical complications of bariatric surgery including vitamin D deficiency.

CASE: A 57 year old African-American woman with morbid obesity, prior gastric bypass, diabetes mellitus, hyperlipidemia, poliomyelitis, and chronic non-malignant pain. The main complaint at her monthly visits was chronic "all over body" pain requiring hydrocodone/acetaminophen, carisoprodol, diazepam and eventually methadone with minimal relief. The patient was never able to localize her pain until she developed rib pain. Plain radiographs showed multiple fractures in the left 5th, 6th and 7th and right 2nd and 3rd ribs. Labs were notable for an elevated alkaline phosphatase and a low serum calcium. A limited differential included Paget's disease, osteomalacia, and malignancy. A 1, 25 dihydroxy vitamin D level was drawn, which came back low at 6 pg/mL (15-75). She was started on vitamin D supplementation. Her calcium, vitamin D level, and alkaline phosphatase all normalized and no further fractures have been reported.

DISCUSSION: Vitamin D deficiency leads to osteomalacia, with its characteristic clinical features of diffuse bone pain and muscle weakness. Other manifestations include low or normal serum calcium, low serum phosphate, an elevated alkaline phosphatase, radiologic pseudofractures, and real fractures. Gastrointestinal disease is the leading cause of vitamin D deficiency in the United States with gastrectomy and celiac sprue being the two leading causes in this group. African Americans are even more prone as evidenced by the Third Na-

tional Health and Nutrition Examination Survey reporting that 42% of US African American women aged 15-49 years are vitamin D deficient. In addition, Rucker et al reported that as many as 34% of healthy people can develop vitamin D deficiency during fall and winter. Patients with osteomalacia often experience a dull unrelenting aching sensation in their bones. These symptoms are often dismissed or mistakenly diagnosed as fibromyalgia by many physicians. However, a physical examination that includes pain with minimal pressure with the thumb or forefinger on the sternum, anterior tibia, radius or ulna, should raise suspicion for the osteomalacia syndrome. Other symptoms include muscle weakness and hyperesthesia involving the extremities. Plotnikoff reported a prevalence of vitamin D deficiency of 93% in patients with persistent nonspecific musculoskeletal pain presenting to a community health center in Minneapolis. It was once believed that only housebound, elderly patients were at risk for hypovitaminosis and osteomalacia. However, as this case describes, patients who are status post bariatric surgery, dark skinned, and live in northern latitudes are also at high risk. Even without any risk factors, it is highly prevalent in the group of patients with chronic non-specific pain. Furthermore, as the prevalence of both morbid obesity and bariatric surgeries increase in the United States, primary care physicians will need to become more aware of its medical consequences. Vitamin B12 is the most commonly recognized deficiency, but other nutrients are also affected including iron, calcium, folate, and the lipid soluble vitamins A, D, and K.

FROM IUD TO DKA: ACRONYMS AND ACTINOMYCES. S. Tsai¹; I. Chibisov¹; E. Anish¹.
¹University of Pittsburgh, Pittsburgh, PA. (Tracking ID #133598)

LEARNING OBJECTIVES: 1) To recognize the importance of a thorough gynecological history and examination in a patient with diabetic ketoacidosis and pelvic symptoms 2) To appreciate the relationship between the presence of an IUD and the development of pelvic actinomycosis 3) To recognize the clinical manifestations and treatment of pelvic actinomycosis

CASE: A 54-year-old woman with a history of type 2 diabetes mellitus and obesity was admitted with a several day history of abdominal pain, nausea, and vomiting. The patient presented with similar symptoms when she was hospitalized 2 weeks earlier and found to be in mild DKA. Her symptoms resolved with fluids and insulin, and she was discharged. During that admission, she complained of irritation in the genital region and was prescribed topical clotrimazole for a presumed candidal infection. On this admission, the genital symptoms persisted, and she also reported a several week history of drainage from the pelvic area. Physical exam showed mild lower abdominal tenderness but no masses, organomegaly, or peritoneal signs. The genital and groin regions were erythematous with fluctuant ulcerations and sinus tracts draining copious, brown, foul discharge. Gynecological exam revealed no ulcers in the vagina, and the cervix appeared unremarkable. Tests for GC and Chlamydia via PCR were negative. Due to a penicillin allergy, empiric treatment was begun with levofloxacin, metronidazole, doxycycline, and fluconazole. Admission labs were notable for an anion-gap acidosis with serum and urine ketones, and the patient's GI symptoms resolved with IV fluids and insulin that corrected her metabolic derangements. A CT scan to evaluate for deeper abscess formation demonstrated only soft tissue swelling in the left groin and, surprisingly, an intrauterine device in the uterus. The patient had actually forgotten about the IUD placement over 10 years earlier. She subsequently underwent drainage of the abscesses, excision of the sinus tracts, and removal of the IUD. An endometrial curettage was performed, and pathology revealed chronic endometritis with actinomyces colonies. The patient's antibiotics were narrowed to doxycycline, and she completed a 4 week course of antibiotics.

DISCUSSION: Actinomyces are anaerobic or microaerophilic bacteria that colonize the mouth, colon, and vagina. Disruption of normal mucosal barriers may lead to actinomycosis, an indolent, slowly progressive infection. Starting with local inflammation, infection extends through sinus tracts and abscesses into adjacent soft tissues. Actinomycosis has been documented in all areas of the body, including the oropharynx, thorax, abdomen, bone, and brain. Pelvic actinomycosis is most commonly associated with IUD use. The risk of actinomycosis increases starting one year after IUD insertion and continues until months after removal. Fever, abdominal pain, and vaginal discharge may indicate endometrial infection. However, abdominal and perineal wall sinus tracts or non-healing fistulae indicate local extension of actinomyces. Unfortunately, a diagnosis is often delayed and, in fact, actinomycosis has been called "the most misdiagnosed disease." Either microscopic identification, immunofluorescence testing, or culture can confirm actinomycosis. Treatment of pelvic actinomycosis requires high dose and prolonged antibiotic therapy; penicillin is the first-line choice. Surgical intervention for abscesses and sinus tracts and removal of the IUD, if present, are also essential.

HARMFUL CRAVINGS OF PREGNANCY: A CASE OF PICA INDUCED SODIUM BICARBONATE TOXICITY. S. Domskey¹; B. Taqui¹; H. Shishodia¹.¹Temple University, Philadelphia, PA. (Tracking ID #132959)

LEARNING OBJECTIVES: 1. Recognize pica as an underdiagnosed illness of pregnancy 2. Review proposed etiologies and complications of pica 3. Recognize importance of asking pregnant women about pica, and doing so in nonjudgmental manner

CASE: 21 year old Latina female at 31 weeks gestation presented with 2 weeks of progressive fatigue, myalgias, numbness and muscle cramps/weakness. She reported two prior uncomplicated vaginal deliveries. Exam was unremarkable except for 4/5 motor strength in all extremities. Fundal height was 31 cm, with a normal bedside ultrasound of the fetus. Labs revealed K 1.7, MG 1.4, bicarbo-

nate 35, Hgb 9.4 (MCV 75), CPK 32750, ALT 415, AST 762, ABG: pH 7.53 pCO₂ 40 pO₂ 92 HCO₃ 32, 96% RA. Upon further questioning, she admitted to baking soda ingestion (1 box/day) during prior and current pregnancies. She was admitted for aggressive electrolyte repletion and cardiac monitoring. She was found to have severe iron deficiency and was started on iron supplementation. Her symptoms markedly improved and she was told to avoid further ingestions. She was discharged, had normal labs on follow up and uncomplicated vaginal delivery at 38 weeks.

DISCUSSION: Pica is the craving for and ingestion of nonnutritive substances. It has been practiced for centuries worldwide and may involve an interplay of biochemical, hematological, psychological, and cultural factors. Commonly ingested substances include soil, clay, starch, soap, ashes, chalk, paint, matches, toilet paper, pebbles, dust, tires, ice and baking soda. The prevalence of pica during pregnancy has ranged from 14-73% in different populations. It is often under-reported due to patient embarrassment and/or lack of inquiry by health-care providers. In the United States, pica during pregnancy has been linked with prior/childhood history of pica, low socioeconomic status, African American heritage, and family/cultural views that link pica to fertility and femininity. Conflicting evidence exists about effect of pica on mother and fetus. There have been reports of iron and calcium deficiencies, colon abnormalities, helminthic infections, lead toxicity, preclampsia, low birth weights, fetal death. Association between pica and anemia is well established but poorly understood. There have been several case reports regarding pica and pregnancy including one about baking soda ingestion leading to maternal hypertension, hypokalemia, and elevated liver tests. These symptoms resolved when ingestion was stopped. Sodium bicarbonate ingestion is particularly concerning because it can cause a profound metabolic alkalosis leading to hypokalemia and hypomagnesemia. Hypokalemia may lead to rhabdomyolysis. Hypomagnesemia may lead to pre-term labor. Our case emphasizes the importance of asking pregnant women about pica, doing so in a nonjudgmental manner and exploring further research opportunities in this area.

MEDIASTINAL NEOPLASM PRESENTING DURING PREGNANCY. B. Alexander¹; T.E. Vettese².¹St. Joseph Mercy Hospital, Ann Arbor, MI; ²St. Joseph Mercy Hospital, Ypsilanti, MI. (Tracking ID #133696)

LEARNING OBJECTIVES: 1) To recognize the unique challenges in diagnosis, staging and treatment in the pregnant patient with cancer

CASE: A previously healthy 34 year-old G3 P2 presented at 22 weeks gestation with progressive chest pain, cough and shortness of breath. Her history was significant for outpatient treatment with antibiotics for presumed community acquired pneumonia the week prior to admission and a 10 pack-year cigarette smoking history, including continuing to smoke throughout the pregnancy. At admission, her vital signs were normal, including an oxygen saturation of 98% on room air. Physical exam was unremarkable except for a gravid uterus. Fetal heart tones were normal. EKG was normal. CXR revealed an anterior mediastinal mass with displacement of the trachea and left mainstem bronchus. Chest computerized tomography (CT) confirmed a 7 cm x 5.6 cm anterior mediastinal mass compressing the main and left pulmonary arteries. CT guided biopsy was performed. Pathological and immunohistochemical evaluation revealed a poorly differentiated adenocarcinoma from an unknown primary site. MRI of the abdomen and pelvis revealed two lesions in the liver suspicious for metastases. After appropriate counseling, the patient decided to proceed with treatment with paclitaxel and carboplatin prior to delivery and planned surgical resection of the mass post-delivery by Caesarian section at 32 weeks. Unfortunately, at 32 weeks the patient developed worsening compressive symptoms despite chemotherapy. A follow-up CXR revealed an interval increase in size of the mass. A healthy baby boy was delivered by Caesarian section and the patient was scheduled for radiotherapy.

DISCUSSION: Cancer complicating pregnancy occurs in approximately 1/1000 live births. The majority of these cases are made up of cancers of the cervix, breast and thyroid. With advancing maternal age, previously rare cancers presenting during pregnancy, including lung cancer, are being seen with greater frequency. Patients presenting during pregnancy with cancer present unique challenges to the clinician in terms of diagnosis, staging and treatment. Whereas in all patients, the risk of diagnostic and treatment modalities must be balanced with potential therapeutic gain, in pregnant patients risk to the fetus must be also considered. In regard to diagnostic testing in these patients, ionizing radiation (CXR, CT) at doses less than 5 rads during pregnancy is considered to be safe to the fetus. MRI poses no known risk to the fetus. In regard to treatment modalities, chemotherapy should be avoided during the first trimester when possible, given the increased risk for fetal malformation and spontaneous abortion. Chemotherapy given in the second and third trimesters is not associated with a significant fetal risk, but long-term effects to prenatal exposure are unknown. Radiation therapy (with abdominal shielding) during pregnancy in women with Hodgkin's disease has showed good neonatal outcomes. Given the emotional and ethical issues involved when treating pregnant patients with cancer, all treatment plans should be individualized.

METASTATIC BREAST CANCER IN A PATIENT WITH A HISTORY OF DUCTAL CARCINOMA IN SITU AFTER PROPHYLACTIC BILATERAL MASTECTOMY AND OOPHRECTOMY. D.A. Bradley¹; L. Coberly¹; E. Lower¹.¹University of Cincinnati, Cincinnati, OH. (Tracking ID #135119)

LEARNING OBJECTIVES: 1) Recognize the importance of a BRCA1 mutation in a woman diagnosed with breast cancer, 2) Recognize the significance of ER/PR receptor status in breast cancer.

CASE: A 35 year old white female with a strong family history for breast cancer (including a sister diagnosed at age 32) developed a suspicious lesion on screening mammogram. She underwent needle guided excisional biopsy which revealed ductal carcinoma in situ (DCIS). Although not standard at the time, the patient requested ER/PR staining of the DCIS specimen and sentinel node biopsy. Staining was ER -/PR+. Sentinel node biopsy was negative. Given the patient's family history and her recent diagnosis of DCIS, the patient underwent genetic testing which confirmed the patient and her sister to be positive for a BRCA1 mutation. The patient therefore elected prophylactic, bilateral mastectomy and oophorectomy. Pathology revealed residual DCIS in her right breast but was otherwise negative for malignancy. Given the ER- status of her DCIS lesion, the patient started estrogen replacement for post-operative hot flashes. Two years later, the patient presented to her doctor with complaints of back pain. X-ray of her spine was suspicious for a lytic lesion. Bone scan showed focal areas of increased activity in her thoracic spine, several ribs, sternum, skull, sacrum, and femoral neck. Open iliac crest biopsy revealed ER/PR positive, malignant cells consistent with metastatic breast cancer. Complete staging also revealed multifocal brain lesions consistent with metastatic disease. The patient was taken off estrogen therapy and started on Arimidex and Zometa for treatment.

DISCUSSION: DCIS is a preinvasive breast tumor known to have an excellent prognosis. Mastectomy affords a 98% cure rate. Recurrent disease is typically local. This is the only case in the literature where DCIS recurred with extensive metastatic disease. Adding complexity to the case, the patient carries a BRCA1 mutation. However, even with this mutation, presentation with extensive metastatic after bilateral mastectomy is unreported. We suspect that this patient likely had an occult microinvasive tumor. The PR positive characteristic of the original DCIS specimen was indicative of an intact ER pathway. It is possible that estrogen use in this patient fed the occult tumor so that metastatic disease was able to progress despite prophylactic measures. This case illustrates the need for a high index of suspicion for metastatic disease in any patient with a history of breast cancer, even those thought to be cured. This case also calls into question the wisdom of prescribing HRT in patients with a history of breast cancer regardless of receptor status.

PERSISTENT POST-OPERATIVE BREAST ERYTHEMA AND SWELLING- IS IT LYMPHEDEMA? K. Ghosh¹; A.C. Degnim¹; K.R. Brandt¹; G.L. Gamble¹. ¹Mayo Clinic, Rochester, MN. (Tracking ID #134827)

LEARNING OBJECTIVES: 1) To recognize that breast lymphedema is a cause for a red, swollen breast following breast or axillary surgery. 2) To manage breast lymphedema with complex decongestive therapy and recognize that patient education and institution of appropriate therapy improves overall outcome.

CASE: A 65 year-old lady with history of left breast cancer presented with persistent redness and swelling of the left breast for over a year, with onset of symptoms since the first week after surgery. The patient was initially noted to have a mammographic abnormality in the left upper outer breast on a screening evaluation. A stereotactic biopsy confirmed invasive ductal carcinoma and she had "partial mastectomy" and sentinel lymph node biopsy (SLNB) revealing a 1.7 cm tumor, ER positive, and lymph nodes were negative for metastases. A week later, she noted redness, warmth and mild tenderness of the central and inferior left breast that persistent unchanged despite multiple courses of antibiotics. In view of persistent breast discomfort and concern about these changes, the patient contemplated mastectomy and visited the Breast Clinic for advice regarding management. Clinical examination revealed a healthy appearing, afebrile lady with erythema and swelling involving the central and inferior left breast and mild peau d'orange changes. Palpation revealed mild induration and skin thickening but no dominant mass; no lymphadenopathy was noted. Mammography suggested an increase in breast density involving the left breast. Ultrasound evaluation showed thickened skin and edema corresponding with the area of erythema, shadowing at the lumpectomy site and no fluid collection. MRI of the breast was negative for focal lesions and mild diffuse enhancement of the skin was suggestive of lymphedema. The patient was reassured of the absence of breast cancer recurrence and referred to the Physical Medicine and Rehabilitation- Lymphedema Clinic. She was evaluated by the specialist and received complex decongestive therapy including manual lymphatic drainage, compression bandaging, fitting of breast compression garment, and skin care education.

DISCUSSION: Lymphedema following breast-related surgery most commonly involves the ipsilateral arm, but breast and chest wall lymphedema can occur after axillary dissection or radiation therapy to the axilla. With increasing adoption of SLNB for axillary evaluation of women with invasive breast cancer, the incidence of arm lymphedema has decreased but the incidence of breast lymphedema is not well known. Breast lymphedema presents as cutaneous erythema, induration and fullness, usually occurring up to a year after a surgical procedure involving the breast/axilla, or radiation therapy to the axilla. The potential risks of lymphedema include infection, functional impairment, pain, and poor cosmetic outcome that may result in significant psychological distress. Breast lymphedema can present a clinical conundrum as it mimics mastitis, dermatitis, or inflammatory breast cancer, resulting in a variety of interventions such as multiple long-term use of antibiotics and even breast biopsy. This clinical entity must be included in the differential diagnosis of post-operative erythema/swelling of the breast. Breast lymphedema is treated by complex decongestive therapy with dramatic reduction of symptoms. Accurate diagnosis of breast lymphedema can enable institution of appropriate therapy with a satisfactory outcome for the patient.

THE PRE-PHARMACY STUDENT WITH MORE THAN A RASH. D.L. Howell¹; J.V. Blair-Elortegui¹. ¹University of South Alabama, Mobile, AL. (Tracking ID #135223)

LEARNING OBJECTIVES: Objectives: • Recognize early manifestations of HIV infection in the outpatient setting • Recognize diffuse, severe new onset seborrhea as a marker of HIV infection • Discuss the need for prevention counseling and screening in young heterosexual females that are at otherwise low risk for HIV infection.

CASE: A previously healthy twenty-four year old black woman presented to clinic with two-weeks of pruritic rash on her posterior neck. Partial relief was achieved using topical triamcinolone. The patient denied exposures for contact dermatitis. This patient's social history revealed pre-pharmacy college enrollment, work in health care without occupational HIV exposures, a current monogamous relationship with one prior sexual partner, intermittent condom use, no known HIV exposure, and no intravenous drug use. Examination revealed severe scalp seborrhea and two 4 x 6 centimeter erythematous lesions with peripheral scaling on the posterior neck. The patient was prescribed ketoconazole shampoo and continuation of triamcinolone. Two-weeks later resolution of neck lesions and scalp improvement were noted. However, the patient had now developed numerous 0.2-3.0 centimeter erythematous, slightly raised lesions with fine scales over trunk and extremities. Diagnostic studies revealed superficial perivascular dermatitis on punch biopsy, a negative ANA, and positive HIV serology. The patient was referred to HIV clinic for further treatment. The patient made a return visit to Internal Medicine clinic approximately 4 months after beginning treatment. At that time she was completely free of symptoms and doing well.

DISCUSSION: HIV is frequently diagnosed when a high-risk patient requests or is recommended testing or when the patient presents with AIDS defining or serious illness. Young heterosexual females with otherwise low risk continue to be a growing population at risk for HIV infection. Recognizing subtle presentations of HIV and considering HIV in low risk patients can lead to early diagnosis and treatment. This can dramatically improve the disease course and prevent unknowing transmission. Early findings may include thrush, flu-like symptoms, refractory vaginal candidiasis, seborrhea and other skin findings. Additionally, improved efforts at HIV prevention counseling in this population can help reverse this growing trend.

ERRATA

Due to a technical problem, the following abstracts were omitted from the 2004 abstracts supplement

PATIENT-PROVIDER COMMUNICATION TOWARD COLORECTAL CANCER SCREENING: A REVIEW OF AUDIOTAPED CLINIC VISITS. B. Ling¹; J. Trauth¹; J. Weissfeld¹; J. Jernigan¹; J. Whittle¹; R. Schoen¹; D. Arndt¹; M. Mor¹; M.J. Fine¹. ¹University of Pittsburgh, Pittsburgh, PA; ²Kansas University, Kansas City, KS. (Tracking ID #117032)

BACKGROUND: Although recent guidelines call for shared decision making (SDM) in colorectal cancer (CRC) screening which requires detailed patient-provider discussion, the current state of such communications is not known. The goal of this study is to describe existing patient-provider communications toward CRC screening.

METHODS: At the VA Pittsburgh's primary care clinic, visits from 212 enrolled patients (males aged 50-74) were audiotaped. In a preliminary review of 27 patients, a discussion on CRC screening was found in audiotapes from 21 patients (10 providers). From these 21 audiotapes with CRC screening discussions, we determined the presence or absence of the fundamental elements of SDM including: (1) invitation for patient participation in decision making, (2) discussion of options, (3) discussion of pros-cons of the options, and (4) elicitation of patient preference. In addition, the audiotapes were coded for discussions addressing domains of the Health Belief Model (HBM) which included: (1) susceptibility to CRC, (2) severity of CRC, (3) benefits of screening, (4) effectiveness of early treatment, (5) cues to action for screening, and (6) self-efficacy to complete screening. The HBM has been associated with CRC screening adherence. Lastly, we also examined the audiotapes for the types of tests discussed, descriptions of tests, recommendations for these tests, and addressing barriers for completing tests.

RESULTS: With regard to the fundamental elements of SDM, there was only one instance (4.8%) of an invitation for patient participation in decision making, 13 instances (61.9%) where more than one test option was discussed, no instances of a discussion on the pros-cons of these various options, and one instance (4.8%) of an elicitation of patient preference. From the HBM, benefits of CRC screening (19%) and cues to action (66.7%) were the only domains addressed. There were no instances of communication addressing the other HBM domains. Fecal occult blood testing was most often discussed (90.5%) with mention of colonoscopy (57.1%) and flexible sigmoidoscopy (33.3%) occurring much less often. Description of tests occurred in five discussions (23.8%), recommendations for CRC screening in 18 instances (85.7%), and barriers to completing testing addressed in two instances (9.5%).

CONCLUSION: Communications incorporating SDM for CRC screening are lacking. Furthermore, discussion addressing pertinent issues associated with adherence, such as the HBM domains and barriers to competing tests, are generally uncommon. This preliminary report suggests that interventions are needed to improve patient-provider communication toward CRC screening.

“DO WE REALLY NEED ALL THAT EQUIPMENT?” FACTORS INFLUENCING COLORECTAL CANCER SCREENING DECISIONS. S.B. Wackerbarth¹; J.C. Peters¹; S.A. Haist¹. ¹University of Kentucky, Lexington, KY. (Tracking ID #T17252)

BACKGROUND: The lifetime risk of developing colorectal cancer is 2.5 to 5% in the general population. Colorectal screening can prevent mortality and there are a number of screening methods, ranging from non-invasive to highly invasive. Although colorectal screening recommendations have been widely disseminated in the media and scientific journals, screening utilization is extremely poor. This project examines the factors that influence patients' decisions regarding colorectal screening.

METHODS: We conducted semi-structured interviews with thirty people (13 men and 17 women) selected to provide a maximum variation sample.

RESULTS: Factors that emerged were categorized into the following themes: concern for one's personal well-being (“it's worth it to know you're healthy”), competing demands (“this is too much right now”), preparing for the procedure (“I read the prep and was not going to do it”), the actual screening procedure (“Do

really need all that equipment?”), gender concerns (“It's hard for macho guys”), fear of having cancer (“Cause you don't want to hear the news”), feeling healthy (“I feel healthy, and I don't feel like I have to worry”), cost (“I did not have it because my insurance didn't cover it”), the experiences of others (“My friends didn't have good luck with it, so I didn't want to go through it”), and turning 50 years old (“I needed to act my age and have it scheduled”).

CONCLUSION: While a majority of participants suggested that numerous factors influenced their colorectal screening decision, others indicated that their decision was based on a single factor. In addition, participants that identified numerous factors often suggested that several conditions had to be met to successfully push them to be screened whereas it only took a single concern to pull them away from being screened. In addition, a factor that pushes some people toward screening will pull others from screening. It is likely that effective educational campaigns will need to be highly tailored to address the specific concerns of each decision maker.